WEBSITE BOOK EDITION

Pixels → Perception

From Pixels to Understanding

Foundations • Vision • Language • Learning • Theory

Volume 3 of 4

Machine Learning and Deep Learning for Vision

English Edition • Volume 3

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https://blog.shyandram.dev/

About this edition

This is volume 3 of 4 of the Pixels → Perception course, *Machine Learning and Deep Learning for Vision*. It contains 41 published tutorials and 0 blog posts, in the order of the website's learning tracks. This is a typeset edition of the website, not a separately peer-reviewed publication.

The prose, code, equations, and references retain the website's content. Exercise hints are expanded for print. Static figures remain in the book, image comparisons show their input and output images, and interactive demonstrations link to the website. Reference numbers are local to each tutorial.

The website is still under review. AI tools assisted its production, and errors or omissions may remain. Read it alongside the original sources and consult the website for subsequent corrections.

English and Traditional Chinese, and each volume, are separate PDFs. Links to tutorials in this volume navigate within the PDF. Links to tutorials in other volumes, to specific web sections, interactive demonstrations, notebooks, and external sources remain online. Content snapshot: 2026-11-01

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Part I

Machine Learning

ML 01 • Introduction to Machine Learning

What it means for a computer to learn from data: the learning paradigms, linear and logistic regression, cost functions, gradient descent, overfitting and evaluation, built from scratch in NumPy.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-01-introduction/

What you'll learn

- What "learning from data" means, and how it differs from writing rules by hand, using small images of handwritten digits as the running example.
- The main learning paradigms: supervised (regression and classification), unsupervised (clustering, dimensionality reduction, anomaly detection), self-supervised, and reinforcement learning, including how reinforcement learning from human feedback (RLHF) tunes chat models.
- The vocabulary of a learning problem: model $f_{w,b}(x)$, parameters, features, targets, predictions, training/validation/test sets and cost functions.
- Linear regression with the mean squared error cost, and gradient descent to minimize it: the update rule, the learning rate, convergence, and batch versus mini-batch versus stochastic updates.
- Why feature scaling matters, how logistic regression turns a line into a classifier with the sigmoid and cross-entropy loss, and how to detect and fix overfitting with validation data and L2 regularization.
- How to report results honestly with accuracy, precision, recall and MSE.

The big picture

Most programs are recipes: a person thinks hard about a problem and writes down every step. Machine learning flips this. We write a *flexible* program with many adjustable numbers, show it many examples of the job done correctly, and let an algorithm adjust the numbers until the program does the job well. The examples replace the recipe.

> **Explain like I'm five**
>
> Think about how you learned to tell cats from dogs. Nobody gave you a rulebook ("pointy ears plus whiskers minus barking equals cat"). Grown-ups pointed and said "cat", "dog", "cat" many times, and you made mistakes and got corrected until you just *knew*. Machine learning is the same: we show the computer lots of examples with the right answer, it guesses, we tell it how wrong it was, and it nudges its guesses a tiny bit each time until it is rarely wrong.

Why this matters: almost every modern computer-vision system, from phone face unlock to medical image analysis, is a learned model, not a hand-written program. The ideas in this tutorial (a model with parameters, a cost that measures error, gradient descent that lowers the cost, and separate data to check the result) are the same ideas that train today's largest neural networks. Only the size of the model changes. Once these few ideas are clear, the rest of the series (ML 02 and the deep-learning tutorials) is mostly about *which* model and *which* cost.

About this tutorial

This tutorial is an expanded, formal version of the earlier notes Introduction to Machine Learning [1], which followed Andrew Ng's Machine Learning Specialization [4] and the *Deep Learning* book [3].

What is machine learning?

Plain version. A machine-learning system improves at a task by looking at data, instead of having every decision written by a programmer.

Learning instead of programming

The idea is old. In 1959, Arthur Samuel described a program that learned to play checkers better than he did, by playing many games and adjusting how it scored board positions [2]. His point was that a computer which learns from experience removes much of the detailed, hand-written programming the task would otherwise need. The famous one-line definition, machine learning as the "field of study that gives computers the ability to learn without being explicitly programmed", is a later paraphrase of this idea, usually attributed to Samuel, rather than a sentence from his paper.

A more operational definition, common in textbooks, has three parts: a **task** (for example, recognizing digits), a **performance measure** (the fraction recognized correctly), and **experience** (a collection of labeled digit images). A program *learns* if its performance on the task improves with experience [3], [7].

An image example: rules versus learning

Our running example is the scikit-learn digits dataset [5], [6]: 1,797 grayscale images of handwritten digits, each only 8×8 pixels with values from 0 to 16. As in Images as Arrays, each image is just a small array of numbers, and flattening it gives a vector of 64 numbers.

Suppose we only need to tell a **0** from a **1**. A programmer might write a rule: "a 1 is thin, a 0 is wide, so count how many columns contain ink; five columns or fewer means 1." This rule is easy to read, and it is right for most upright 1s. But people write 1s with slants, serifs and bases, and

a slanted 1 covers as many columns as a 0 (Figure 1). The rule gets about 79% of the cases right. Fixing it means more rules ("unless it slants..."), and each new rule creates new exceptions.

The machine-learning approach writes *no* digit-specific rule. It picks a flexible family of functions (here, logistic regression, which we build later), and lets an algorithm choose the 65 numbers inside it from examples. On held-out images, it separates 0s from 1s perfectly, and exactly the same code handles all ten digits with no new rules:

```
import numpy as np
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

digits = load_digits()                        # 1,797 images, 8x8 pixels, values 0..16
X, y = digits.data, digits.target             # X: (1797, 64) flattened pixels; y: 0..9

# Hand-written rule for "0 versus 1": a 1 is thin, a 0 is wide.
keep = (y == 0) | (y == 1)
imgs, labels = digits.images[keep], y[keep]
width = (imgs.max(axis=1) > 4).sum(axis=1)   # how many columns contain ink
rule = np.where(width <= 5, 1, 0)
print("hand rule, 0 vs 1:", round((rule == labels).mean(), 3))           # 0.789

# Learned rule for the same task, judged on images it never saw.
X01 = imgs.reshape(len(imgs), -1)
Xa, Xb, ya, yb = train_test_split(X01, labels, test_size=0.3, random_state=0)
print("learned, 0 vs 1:", LogisticRegression(max_iter=1000).fit(Xa, ya).score(Xb, yb))   #
 ↪  1.0

# The same recipe scales to all ten digits with no new rules.
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.25, random_state=0)
print("learned, 10 digits:",
      round(LogisticRegression(max_iter=5000).fit(X_tr, y_tr).score(X_te, y_te), 3))  # 0.953
```

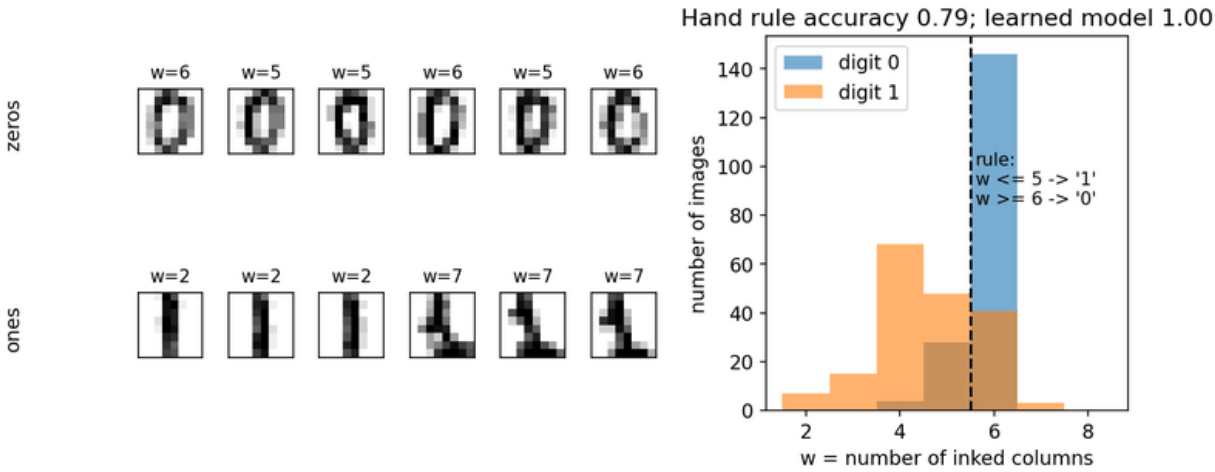


Figure 1 — Why hand-written rules are brittle. Left: zeros and ones from the digits dataset; slanted ones are as wide as zeros. Right: the 'count inked columns' feature overlaps heavily between the two classes, so any threshold on it makes many mistakes. A learned model uses all 64 pixels at once.

This small experiment contains the whole argument for machine learning. When the rule is hard to put into words but examples are cheap, learn the rule from examples. It also shows the price: the learned model is a list of 65 numbers that is much harder to read than "count the columns."

The learning paradigms

Plain version. Learning problems differ mainly in *what feedback the learner gets*: the right answer for every example, no answers at all, answers it can manufacture from the data itself, or only an occasional score for how well it did.

Paradigm	What the data looks like	What is learned	Typical example
Supervised	pairs (x, y): input and the "right answer"	a function from x to y	digit image → digit label
Unsupervised	inputs x only	structure: groups, directions, outliers	grouping similar digits
Self-supervised	inputs x only, but part of x is hidden and used as the target	a general-purpose representation	predicting a masked word or image patch
Reinforcement	an agent's actions and the rewards they earn	a policy: which action to take in each situation	game playing, tuning chat models

My original notes listed the first, second and fourth; self-supervised learning has since become the main way large models are pre-trained, so it gets its own row [9], [10].

Supervised learning

In supervised learning, the learner is given the right answers. The data is a set of examples $\{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$, where $x^{(i)}$ is an input, $y^{(i)}$ is the desired output (the **label** or **target**), and m is the number of examples. The goal is a function f such that $f(x)$ is close to y, *also for new inputs not in the data*.

That last part, called **generalization**, is the real goal; memorizing the examples is easy.

Unsupervised learning

In unsupervised learning there are no answers, only inputs $\{x^{(i)}\}$. The goal is to find something interesting in the data: groups of similar examples, a compact description, or examples that do not fit. There is no single "correct" output, which makes these methods useful for exploration and harder to evaluate.

Self-supervised learning

Self-supervised learning uses unlabeled data but turns it into a supervised problem by hiding part of each example and asking the model to predict the hidden part. In text, hide a word and predict it from the rest of the sentence. In images, hide most of the patches and reconstruct them: masked autoencoders hide 75% of an image's patches and train a network to fill them in [12]. LeCun and Misra describe self-supervised learning as a way for models to acquire broad background knowledge from raw observation, because the labels come for free [9].

The point is not the fill-in-the-blank task itself; it is that a model which can fill in blanks well must have learned a lot about the structure of the data. The learned internal **representation** can then be adapted to many other tasks with few labels. This is the recipe behind **foundation models**: models trained on broad data at scale (usually self-supervised) and then adapted to a wide range of downstream tasks [10]. GPT-3, trained simply to predict the next word on a large text corpus, was able to perform many language tasks from a few examples given in the prompt, without further training [11].

Reinforcement learning

In reinforcement learning (RL), an **agent** interacts with an **environment**. At each step it observes a **state**, chooses an **action**, and receives a **reward** (a number) and a new state. Nobody tells it the right action; it must discover which actions lead to high total reward over time, often when the reward arrives long after the action that earned it. What is learned is a **policy**, a rule for choosing actions in each state [8]. Samuel's checkers player is an early example of this setting [2].

Today, RL is best known for tuning large language models. In **reinforcement learning from human feedback** (RLHF), as described by Ouyang et al. [13], a pre-trained language model is first fine-tuned on demonstrations written by people (supervised learning). People then rank several model answers to the same prompt, and a separate **reward model** is trained to predict those rankings. Finally, the language model is optimized with RL to produce answers that the reward model scores highly. The authors report that labelers preferred the outputs of their 1.3-billion-parameter tuned model over those of the 175-billion-parameter GPT-3 [13]. More recently, DeepSeek-R1 used large-scale RL with automatically checkable rewards (such as whether a math answer is correct) to improve step-by-step reasoning [14].

Notice that RLHF chains three paradigms: self-supervised pre-training, supervised fine-tuning, and reinforcement learning. Real systems mix paradigms freely; the table above is a map, not a set of walls.

Supervised learning: regression and classification

Plain version. Supervised learning comes in two main kinds. **Regression** predicts a number; **classification** predicts a category.

Regression

In regression the target y is a real number, so there are infinitely many possible outputs. Examples: predicting a stock price, a student's final grade from earlier test scores, or a person's age from a face image. The last one shows that the input can be an image while the output is still a single number.

Classification

In classification the target is one of a small, fixed set of categories (classes). Examples: spam versus not spam (two classes, **binary classification**), the shape category of a face, or a tumor type such as benign, malignant type 1 or malignant type 2 (**multi-class classification**). Classes are names, not quantities: "class 2" is not twice "class 1". A classifier usually outputs a probability for each class and then picks the most likely one.

```
from sklearn.datasets import make_regression, make_classification
from sklearn.linear_model import LinearRegression, LogisticRegression

# Regression: the target is a real number.
X_r, y_r = make_regression(n_samples=100, n_features=1, noise=10.0, random_state=0)
reg = LinearRegression().fit(X_r, y_r)
print(reg.predict(X_r[:3]).round(1), y_r[:3].round(1))   # numbers vs numbers

# Classification: the target is one of a few labels.
X_c, y_c = make_classification(n_samples=100, n_features=2, n_redundant=0,
                               n_clusters_per_class=1, random_state=0)
clf = LogisticRegression().fit(X_c, y_c)
print(clf.predict(X_c[:5]), y_c[:5])                     # labels vs labels
print(clf.predict_proba(X_c[:2]).round(3))               # class probabilities
```

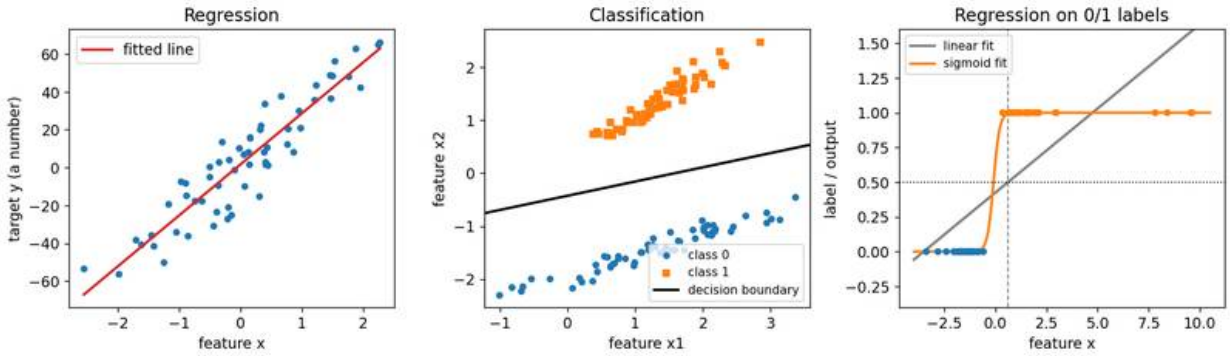


Figure 2 — Left: regression fits a curve through numeric targets. Middle: classification separates labeled points with a decision boundary. Right: fitting a straight regression line to 0/1 labels (gray) gives outputs outside $[0, 1]$, and far-away points drag its 0.5 crossing; a sigmoid (logistic regression, orange) is the right tool.

Why not use regression for classification?

It is tempting to code the classes as 0 and 1, fit a regression line, and call anything above 0.5 a 1. This can work on easy data, but it usually performs badly, for two reasons visible in Figure 2 (right). First, the line outputs values like -0.4 or 1.7, which are not probabilities. Second, the squared error punishes a point that is *correctly* classified but far from the boundary, so a few extreme but easy examples pull the line and move its 0.5 crossing to the wrong place. Logistic regression (later in this tutorial) fixes both problems.

Unsupervised learning

Plain version. Without any answers, we can still ask: which examples belong together, how can each example be described with fewer numbers, and which examples are strange?

Clustering

Clustering groups similar examples. The classic algorithm is **k-means** [15]. Choose the number of groups K. Start with K center points, then repeat two steps until nothing changes: assign each example to its nearest center, and move each center to the mean of the examples assigned to it. Each step can only lower the total squared distance from examples to their centers,

$$\sum_{i=1}^{m} \|x^{(i)} - \mu_{c^{(i)}}\|^2,$$

where μ_k is the center of cluster k and $c^{(i)}$ is the cluster that example i is assigned to. So the algorithm always stops, but possibly at a poor local solution, which is why it is run from several

random starts. Jain's review of fifty years of clustering research [15] stresses that there is no best clustering algorithm in general: the "right" groups depend on what similarity means for the data.

A common confusion: **k-nearest neighbors** (KNN) is *not* a clustering method. KNN classifies a new example by a vote among the k most similar *labeled* training examples, so it is a supervised method [5]. The "k" in k-means counts clusters; the "k" in KNN counts neighbors.

Dimensionality reduction

Dimensionality reduction describes each example with fewer numbers while keeping as much useful information as possible. **Principal component analysis** (PCA) finds the directions along which the data varies most and projects the data onto the top few of them; the new coordinates are uncorrelated and successively capture as much variance as possible [16]. Two PCA coordinates can turn 64-pixel digit images into points on a plane that we can look at (Figure 3, middle).

Anomaly detection

Anomaly detection finds examples that do not fit the pattern of the rest: a fraudulent transaction, a defective part, a corrupted image. Chandola, Banerjee and Kumar's survey [17] organizes the many techniques by the assumption each one makes about what "normal" looks like. A simple version uses PCA: fit a few components on the data, rebuild each example from them, and flag those with a large reconstruction error, because the components only describe what is common.

```
import numpy as np
from sklearn.datasets import load_digits
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA

X, y = load_digits(return_X_y=True)             # y is used only to check, never to fit

# Clustering: ten groups, found without looking at y.
km = KMeans(n_clusters=10, n_init=10, random_state=0).fit(X)
purity = sum(np.bincount(y[km.labels_ == k]).max() for k in range(10)) / len(y)
print("cluster purity:", round(purity, 3))      # 0.792: most clusters are one digit

# Dimensionality reduction: 64 numbers per image -> 2 numbers.
Z = PCA(n_components=2).fit_transform(X)        # shape (1797, 2)

# Anomaly detection: images that a 20-component PCA rebuilds badly are unusual.
pca20 = PCA(n_components=20).fit(X)
err = ((X - pca20.inverse_transform(pca20.transform(X))) ** 2).sum(axis=1)
print("most unusual images:", np.argsort(err)[-5:])
```

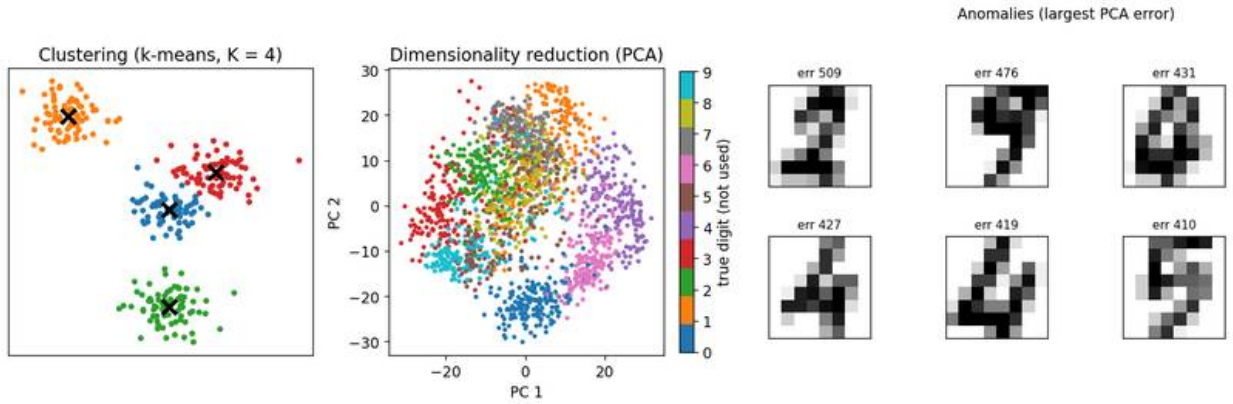


Figure 3 — Unsupervised learning. Left: k-means groups points around K centers. Middle: PCA compresses each 64-pixel digit to two numbers (colors show the true digits, which PCA never saw). Right: the digits with the largest PCA reconstruction error are the oddest-looking ones.

The purity check in the snippet uses the labels only *after* clustering, to see whether the discovered groups match the digits. About 79% of the images fall in a cluster dominated by their own digit, even though k-means never saw a label.

The vocabulary of a learning problem

Plain version. Before any math, we need names for the parts: what goes in, what comes out, what the right answer is, what the model is, and how we measure its mistakes.

Inputs, outputs and targets

- **Features** (inputs), x: the numbers that describe one example. For a digit image, the 64 pixel values; for a house, its size and number of rooms. With n features we write $x = (x_1, \dots, x_n)$, and x_j is the j-th feature.
- **Target**, y: the correct output, also called the label or **ground truth**. Only supervised learning has targets.
- **Prediction**, $\hat{y}$ ("y-hat"): what the model outputs for an input. The hat means "estimated".
- **Training example** i: the pair $(x^{(i)}, y^{(i)})$. The superscript in parentheses is an index, not a power. $x_j^{(i)}$ is feature j of example i.

The model and its parameters

The **model** is the function that maps features to a prediction. We write it $f_{w,b}$ to show that it depends on adjustable numbers called **parameters**, here w (the **weights**) and b (the **bias**):

$$\hat{y} = f_{w,b}(x).$$

Choosing the *form* of f (a line, a curve, a neural network) is a design decision. Choosing the *values* of w and b is what learning does. Numbers we set by hand, such as the learning rate or polynomial degree, are called **hyperparameters** to distinguish them from the learned parameters.

Cost function

A **loss** $L(\hat{y}, y)$ measures how wrong one prediction is. The **cost function** $J(w, b)$ averages the loss over the training set, so it is a single number that says how wrong the whole model is for a given choice of parameters. The model is f; the cost is J. Keeping these two apart avoids a lot of confusion: $f_{w,b}(x)$ takes an input and returns a prediction, while $J(w, b)$ takes parameters and returns an error.

The **objective** of training is then precise: find the parameters that make J as small as possible, so that $\hat{y}^{(i)}$ is close to $y^{(i)}$ for all training examples.

Training, validation and test sets

The data is split into three parts with different jobs:

- the **training set** is used to fit the parameters;
- the **validation set** is used during development to compare models and choose hyperparameters;
- the **test set** is used once, at the end, to estimate how the final model will do on new data.

The test set must not influence any decision, or the estimate becomes optimistic. We come back to *why* this split matters in the section on overfitting.

Linear regression

Plain version. Fit a straight line through the data. The line has two knobs: its slope and its height. Learning means turning the knobs until the line passes as close as possible to the points.

The model

With one feature, the linear regression model is

$$f_{w,b}(x) = wx + b,$$

where x is the feature, w is the slope (how much the prediction changes when x grows by 1) and b is the intercept (the prediction when $x = 0$). Changing w tilts the line; changing b moves it up and down (Figure 4, left).

With n features, the same model is a weighted sum,

$$f_{\mathbf{w},b}(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b = \sum_{j=1}^{n} w_j x_j + b,$$

where $\mathbf{w} = (w_1, \dots, w_n)$ holds one weight per feature. A common trick appends a constant feature $x_0 = 1$, so the bias becomes just another weight w_0 and the model shortens to $\mathbf{w} \cdot \mathbf{x}$. Either way, "linear" means linear in the *parameters*; we will later feed it squared or cubed features and it remains a linear regression.

The cost function: mean squared error

For each example, the **error** (residual) is $f_{w,b}(x^{(i)}) - y^{(i)}$. Squaring makes errors positive and punishes big ones more than small ones. Averaging over the m training examples gives the **mean squared error** (MSE) cost:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)} \right)^2,$$

where m is the number of training examples and $(x^{(i)}, y^{(i)})$ is the i-th example. The extra factor $\frac{1}{2}$ is a convenience: it cancels the 2 that appears when we differentiate the square, and it does not change where the minimum is.

Picture the cost as a landscape over the parameters. If we fix b and vary only w, J is a parabola: lines that are too flat or too steep both have large error, and one slope in between is best (Figure 4, middle). With both w and b, J is a bowl, which we draw as a contour map of ellipses (Figure 4, right). The best line sits at the bottom of the bowl.

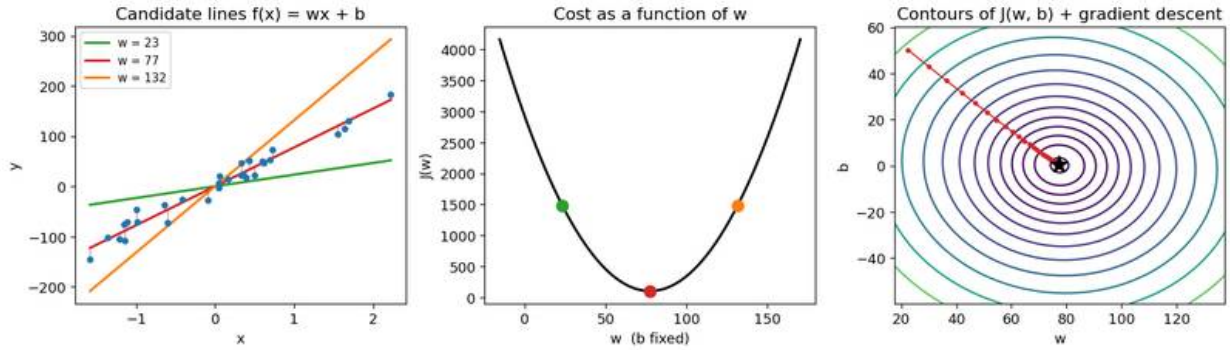


Figure 4 — From lines to costs. Left: three candidate lines; vertical segments are the residuals of the best one. Middle: with b fixed, $J(w)$ is a parabola, and each line is one point on it. Right: $J(w, b)$ is a bowl (contours), and gradient descent walks downhill to its bottom.

Why this bowl has one bottom

For linear regression, the MSE cost is a **convex** function of (w, b): any straight segment between two points on its surface lies on or above the surface. A convex function has no separate local minima; every minimum is the global one. This is a special, friendly case. For many other models, including the neural networks of ML 02, the cost landscape has many valleys, and an optimizer can stop in one that is not the lowest. The original notes make exactly this point: in general, the best we can hope for is a good local minimum, close to the global one.

For linear regression we could even solve for the bottom directly with a formula (the normal equations, which `np.polyfit` uses). We use gradient descent anyway, because it is the method that still works when no formula exists.

Gradient descent

Plain version. Stand somewhere on the cost landscape in fog. Feel which way the ground slopes, take a small step downhill, and repeat. When the ground is flat, you are at the bottom.

The update rule

The **derivative** $\partial J/\partial w$ tells us how J changes when w increases a little: positive means "uphill to the right", negative means "uphill to the left". Its sign gives the direction, and its size says how steep the slope is. Gradient descent moves each parameter a small step *against* its derivative:

$$w \leftarrow w - \alpha \frac{\partial J(w,b)}{\partial w},$$
$$b \leftarrow b - \alpha \frac{\partial J(w,b)}{\partial b},$$

where $\alpha > 0$ is the **learning rate** (step size) and $\leftarrow$ means "replace with". Both derivatives are computed at the *current* (w, b) before either is changed; this is called a **simultaneous update**. The vector of all partial derivatives is the **gradient** ∇J, which points in the direction of steepest increase, so the update moves in the direction of steepest decrease [3].

For the MSE cost of linear regression, the chain rule gives

$$\frac{\partial J}{\partial w} = \frac{1}{m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)x^{(i)}, \qquad \frac{\partial J}{\partial b} = \frac{1}{m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right).$$

Read them in words: the gradient for w is the average error times the feature; the gradient for b is the average error. If predictions are too high on average, b goes down. If predictions are too high mainly where x is large, w goes down. With n features, $\partial J/\partial w_j$ uses $x_j^{(i)}$ in place of $x^{(i)}$.

Implementing it from scratch

```
import numpy as np
from sklearn.datasets import make_regression

X, y = make_regression(n_samples=100, n_features=1, noise=10.0, random_state=0)
x = X[:, 0]                                  # one feature, shape (m,)
m = len(x)

def cost(w, b):
    return ((w * x + b - y) ** 2).sum() / (2 * m)

w, b, alpha = 0.0, 0.0, 0.1
for step in range(200):
    err = w * x + b - y                      # f_wb(x_i) - y_i for every i
    dj_dw = (err * x).sum() / m
    dj_db = err.sum() / m
    w, b = w - alpha * dj_dw, b - alpha * dj_db   # simultaneous update
    if step % 50 == 0:
        print(f"step {step:3d}  J = {cost(w, b):8.2f}")

print("gradient descent:", round(w, 3), round(b, 3))   # 42.619 -0.814
print("closed form     :", np.polyfit(x, y, 1).round(3))  # [42.619 -0.814]
```

The cost drops from about 800 to 57 within 50 steps and then stops changing. Gradient descent and the exact formula agree to three decimals. The remaining cost is not zero, because the data contains noise that no straight line can explain.

Convergence: where the derivative is zero

"Repeat until convergence" needs a definition. At a minimum, the landscape is flat, so the **derivative is zero** there, and the update $w - \alpha \cdot 0$ leaves w unchanged. Note that it is the *derivative*, not the cost, that becomes zero: the cost at the minimum is just the smallest value available, 57 in the example above. In practice we stop when the cost decreases by less than a small tolerance between iterations, or when the gradient's size falls below a threshold, or after a fixed budget of steps.

A useful consequence: as we approach a minimum, the derivative shrinks, so even with a *fixed* α the steps $\alpha\, \partial J/\partial w$ get smaller automatically. Gradient descent naturally slows down near the bottom.

Choosing the learning rate

The learning rate is the most important hyperparameter of gradient descent (Figure 5):

- **Too small:** every step is tiny. The algorithm does go downhill, but it may need a very large number of steps.
- **About right:** the cost falls quickly and settles.
- **Too large:** a step jumps past the bottom to the other side of the valley, possibly higher than where it started. The parameters bounce back and forth, may never settle (**overshoot**), and can fly off to infinity (**divergence**).

The standard diagnostic is a **learning curve**: plot J against the iteration number. With a good α, J decreases at every step. If J ever goes *up*, α is too large (or there is a bug in the gradient). A practical recipe is to try values spaced by about a factor of 3, such as 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, and pick the largest one whose curve decreases smoothly.

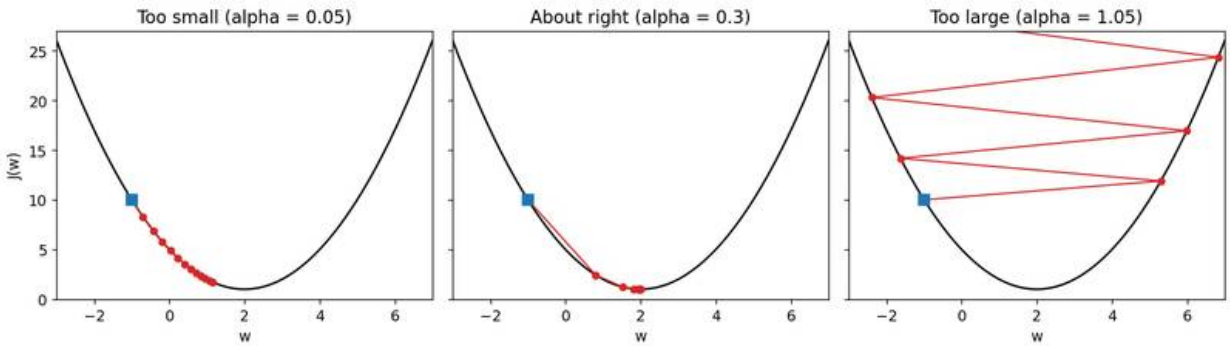


Figure 5 — The same cost $J(w)$, starting from the same point, with three learning rates. Too small: slow progress. About right: fast convergence. Too large: each step overshoots further, and the iterates diverge.

Many optimizers adapt the step size during training, either by a **schedule** that lowers α over time or by scaling each parameter's step using running statistics of its past gradients, as Adam does [19]. These make "the steps become smaller near the minimum" an explicit design choice rather than just a side effect.

Batch, mini-batch and stochastic gradient descent

The gradient formula sums over all m examples. That is fine for 100 examples and a problem for 100 million. The key insight, explained clearly in [3] and analyzed in depth by Bottou, Curtis and Nocedal [18], is that the gradient is an *average*, that is, an expectation over the data, and an average can be estimated from a sample.

- **Batch gradient descent** uses the *whole* training set for every update. Each step is exact and expensive.
- **Mini-batch gradient descent** shuffles the data and, for each update, uses a small random subset (a **mini-batch**) of B examples, typically 32 to a few hundred. Each step is a noisy but unbiased estimate of the full gradient, and costs B/m as much.
- **Stochastic gradient descent** (SGD), in the strict sense, uses one example per update ($B = 1$). In practice, "SGD" often refers to the mini-batch version too.

One full pass through the training data is an **epoch**. Beware of naming in code: the `batch_size` argument in deep-learning libraries means the *mini-batch* size, not the batch-gradient-descent "whole set".

The noise in mini-batch gradients is not only tolerable but often helpful: in the same number of epochs, mini-batch descent makes many more updates than batch descent. Bottou et al. argue that this is why stochastic methods dominate large-scale learning: they reach a useful accuracy with far less computation than exact methods [18].

```
import numpy as np
from sklearn.datasets import make_regression

X, y = make_regression(n_samples=10_000, n_features=5, noise=10.0, random_state=0)
m, n = X.shape
rng = np.random.default_rng(0)

def minibatch_gd(batch_size, alpha=0.05, epochs=5):
    w, b = np.zeros(n), 0.0
    for epoch in range(epochs):
        order = rng.permutation(m)                    # shuffle once per epoch
        for start in range(0, m, batch_size):
            idx = order[start:start + batch_size]
            err = X[idx] @ w + b - y[idx]
```

```
            w -= alpha * X[idx].T @ err / len(idx)    # gradient estimated on the batch
            b -= alpha * err.mean()
    return ((X @ w + b - y) ** 2).mean() / 2

for bs in [m, 64, 1]:                                 # batch, mini-batch, stochastic
    print(f"batch size {bs:5d}: J = {minibatch_gd(bs, alpha=0.05 if bs > 1 else 0.005):.2f}")
# batch size 10000: J = 6127.00   (only 5 updates in 5 epochs)
# batch size    64: J = 49.93     (785 updates)
# batch size     1: J = 50.64     (50,000 noisy updates)
```

With the same five passes over the data, batch descent has made only five steps and is still far from the bottom, while mini-batch descent has essentially converged (the noise level of this data puts the best achievable J near 50). Single-example SGD gets there too, but needs a smaller learning rate because its gradients are noisier, and it cannot use vectorized arithmetic efficiently.

Feature scaling

Plain version. If one feature is measured in thousands and another in fractions, the cost landscape becomes a long, narrow valley, and gradient descent zigzags. Putting all features on similar scales turns the valley into a round bowl that is easy to descend.

Why scale matters

Suppose feature x_1 (say, house area in square feet) ranges over thousands and feature x_2 (number of bedrooms) ranges from 1 to 5. A small change in w_1 changes the predictions a lot; a similar change in w_2 barely matters. So J is extremely steep along w_1 and very flat along w_2: the contours are thin ellipses. A learning rate small enough not to diverge along the steep direction makes almost no progress along the flat one (Figure 6, left).

Standardization and min–max scaling

The most common fix is **standardization** (z-score normalization):

$$x'_j = \frac{x_j - \mu_j}{\sigma_j},$$

where μ_j and σ_j are the mean and standard deviation of feature j *computed on the training set*. Each scaled feature then has mean 0 and standard deviation 1. An alternative is **min–max scaling**, $x'_j = (x_j - \min_j)/(\max_j - \min_j)$, which maps the training range to $[0, 1]$ and is the usual choice for pixel intensities (divide by 255 for 8-bit images). Both are available in scikit-learn as `StandardScaler` and `MinMaxScaler` [5].

Two rules prevent subtle bugs. Compute μ_j, σ_j on the training set only, then apply the *same* numbers to validation, test and future data. And remember that scaling changes the meaning of the learned weights: a weight now says how much the prediction changes per standard deviation of the feature.

```
import numpy as np
from sklearn.datasets import make_regression

X, y = make_regression(n_samples=200, n_features=2, noise=5.0, random_state=1)
X = X * np.array([1000.0, 1.0])               # feature 1 now has a ~1000x wider range

def gd(X, y, alpha, steps=500):
    m, n = X.shape
    w, b = np.zeros(n), 0.0
    for _ in range(steps):
        err = X @ w + b - y
        w -= alpha * X.T @ err / m
        b -= alpha * err.mean()
    return ((X @ w + b - y) ** 2).mean() / 2

print("raw,    alpha=1e-7:", round(gd(X, y, 1e-7), 2))    # 158.01; alpha=3e-6 diverges
mu, sigma = X.mean(axis=0), X.std(axis=0)
Xs = (X - mu) / sigma                          # z-score standardization
print("scaled, alpha=0.1 :", round(gd(Xs, y, 0.1), 2))    # 13.47
```

On the raw features, any learning rate above about 2×10^{-6} diverges, and with a safe one the cost is still stuck near 158 after 500 steps (it is still near 158 after 5,000), because the second weight barely moves. After standardization, $\alpha = 0.1$ reaches the minimum (about 13, the noise level) easily.

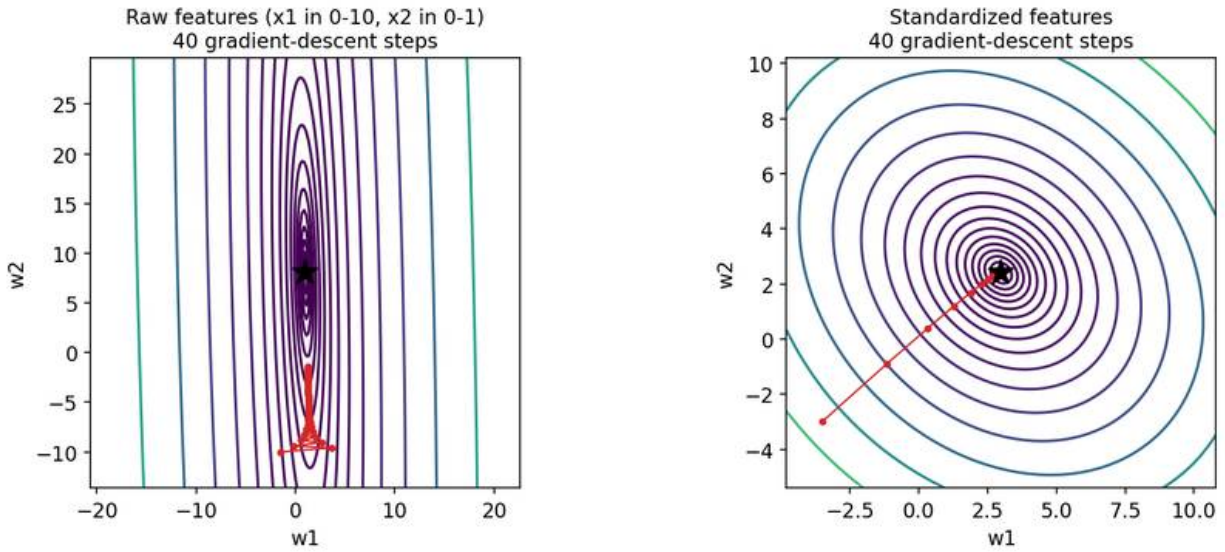


Figure 6 — Feature scaling reshapes the cost landscape. Left: features on very different scales give a narrow

valley, and gradient descent zigzags. Right: after standardization, the contours are nearly circular, and the same number of steps reaches the minimum.

Logistic regression for classification

Plain version. Keep the weighted sum from linear regression, but squash its output into the range 0 to 1 so it can be read as a probability. Then choose the weights so that the probability is high for the true class.

The sigmoid function

The **sigmoid** (logistic) function is

$$g(z) = \frac{1}{1 + e^{-z}},$$

where z is any real number. It maps $-\infty$ to 0, $+\infty$ to 1 and 0 to exactly 0.5, with an S-shaped curve in between (Figure 7, left). It is also symmetric, $g(-z) = 1 - g(z)$, and has the convenient derivative $g'(z) = g(z)\,(1 - g(z))$.

The model and the decision boundary

Logistic regression feeds the linear model into the sigmoid:

$$f_{\mathbf{w},b}(\mathbf{x}) = g(\mathbf{w} \cdot \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}\cdot\mathbf{x}+b)}},$$

and interprets the output as the estimated probability that the label is 1, $f_{\mathbf{w},b}(\mathbf{x}) = P(y = 1 \mid \mathbf{x})$. Despite the name, it is a classification method. To make a decision, predict 1 when this probability is at least 0.5. Since $g(z) \geq 0.5$ exactly when $z \geq 0$, the rule is

$$\hat{y} = 1 \quad \text{if} \quad \mathbf{w} \cdot \mathbf{x} + b \geq 0, \qquad \hat{y} = 0 \text{ otherwise.}$$

The set where $\mathbf{w} \cdot \mathbf{x} + b = 0$ is the **decision boundary**. With two features it is a straight line; in general it is a flat hyperplane, with $\mathbf{w}$ perpendicular to it. Points far from the boundary get probabilities near 0 or 1; points near it get probabilities near 0.5. To obtain a curved boundary, feed the model nonlinear features such as x_1^2 or $x_1 x_2$; the model stays linear in its parameters.

The cost: cross-entropy (log loss)

Why not reuse squared error? Combined with the sigmoid, the squared-error cost is *not* convex in $(\mathbf{w}, b)$: it develops flat plateaus and multiple valleys. Instead, logistic regression uses the **log**

loss, also called **binary cross-entropy**:

$$L(f, y) = -y \log f \; - \; (1 - y) \log(1 - f),$$

where $f = f_{\mathbf{w},b}(\mathbf{x})$ is the predicted probability and $y \in \{0, 1\}$ is the true label. Only one of the two terms is active for each example. If $y = 1$, the loss is $-\log f$: zero when $f = 1$, and growing without bound as $f \to 0$. If $y = 0$, the loss is $-\log(1 - f)$, the mirror image (Figure 7, middle). Confident wrong answers are punished very heavily; confident right answers cost almost nothing.

The cost is the average over the training set:

$$J(\mathbf{w}, b) = \frac{1}{m} \sum_{i=1}^{m} \Big[- \, y^{(i)} \log f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - (1 - y^{(i)}) \log \left(1 - f_{\mathbf{w},b}(\mathbf{x}^{(i)})\right) \Big].$$

This choice is not arbitrary. If we treat each label as a coin flip with probability $f_{\mathbf{w},b}(\mathbf{x}^{(i)})$ of coming up 1, then J is exactly the negative log-likelihood of the training labels, divided by m. Minimizing it is **maximum likelihood estimation** [3]. And with the sigmoid, this cost is convex, so gradient descent again finds the single global minimum.

Gradient descent, again

Using $g' = g(1 - g)$, the derivative of the log loss simplifies remarkably:

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^{m} \left(f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)} \right) x_j^{(i)}, \qquad \frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^{m} \left(f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)} \right).$$

This has *exactly the same form* as the linear-regression gradient: average error times feature. The only difference is inside f. This is not a coincidence; it holds for a whole family of models that pair an output function with its matching likelihood, and it is one reason gradient descent code is so reusable.

```
import numpy as np
from sklearn.datasets import make_classification

X, y = make_classification(n_samples=200, n_features=2, n_redundant=0,
                           n_clusters_per_class=1, class_sep=1.5, random_state=3)

def sigmoid(z):
    return 1.0 / (1.0 + np.exp(-z))

def log_loss(p, y, eps=1e-12):
    p = np.clip(p, eps, 1 - eps)              # avoid log(0)
    return -(y * np.log(p) + (1 - y) * np.log(1 - p)).mean()

w, b, alpha = np.zeros(2), 0.0, 0.5
for step in range(1000):
    p = sigmoid(X @ w + b)                    # predicted P(y = 1 | x)
    w -= alpha * X.T @ (p - y) / len(y)       # same form as linear regression
```

```
    b -= alpha * (p - y).mean()

p = sigmoid(X @ w + b)
print("log loss:", round(log_loss(p, y), 3))  # 0.06
print("accuracy:", ((p >= 0.5) == y).mean())  # 0.99
print("boundary: %.2f*x1 + %.2f*x2 + %.2f = 0" % (w[0], w[1], b))
```

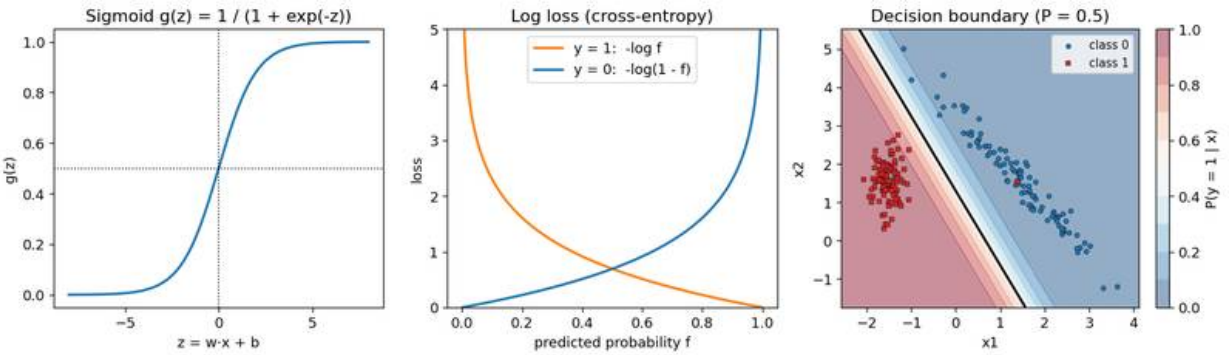


Figure 7 — Logistic regression. Left: the sigmoid squashes any score into $(0, 1)$. Middle: the log loss for $y = 1$ and $y = 0$ punishes confident mistakes heavily. Right: a model trained from scratch; shading is the predicted probability, and the black line is the decision boundary $w \cdot x + b = 0$.

More than two classes

For $K > 2$ classes, the standard generalization gives each class its own weight vector and score $z_k = \mathbf{w}_k \cdot \mathbf{x} + b_k$, and turns the scores into probabilities that sum to 1 with the **softmax** function, $P(y = k \mid \mathbf{x}) = e^{z_k} / \sum_{l=1}^{K} e^{z_l}$. The cost is again the average negative log probability of the true class (categorical cross-entropy) [3]. This is what scikit-learn's `LogisticRegression` did for the ten digits at the start of this tutorial, and it is exactly the output layer of most classification networks in ML 02.

Overfitting, underfitting and regularization

Plain version. A model that is too simple misses the pattern (**underfitting**). A model that is too flexible memorizes the training examples, noise included, and fails on new ones (**overfitting**). We detect the difference with data the model did not train on, and we tame an over-flexible model by penalizing large weights.

Seeing it happen

Take 25 noisy samples of a smooth curve and fit polynomials of different degrees, which is linear regression on the features $x, x^2, \ldots, x^d$ (Figure 8):

- **Degree 1 (underfitting, high bias).** A straight line cannot follow the curve. The error is large on the training set *and* on new data.
- **Degree 3 (a good fit).** The curve follows the trend and ignores the noise. Errors are small on both.
- **Degree 15 (overfitting, high variance).** The curve passes almost exactly through every training point, so the training error is nearly zero. Between and beyond the points it swings wildly, and the error on new data explodes.

The training error alone cannot tell the good model from the overfitted one; it even prefers the overfitted one. Only data that the model did not see can reveal the difference. This is why we keep a validation set.

Using the validation and test sets

The workflow is:

1. Fit each candidate model (each degree, each regularization strength) on the **training set**.
2. Compare the candidates on the **validation set** and keep the best one.
3. Report the chosen model's error on the **test set**, once.

Why three sets and not two? Because picking the best of many candidates on the validation set is itself a form of fitting: the winner is partly the one that got lucky on those particular examples.

The untouched test set gives an unbiased final estimate. When data is scarce, **cross-validation** replaces the single validation set: split the training data into k folds, train k times, each time validating on a different fold, and average.

The pattern to look for: if training and validation errors are both high, the model underfits (use a more flexible model or better features). If training error is low but validation error is much higher, the model overfits (get more data, simplify the model, or regularize).

L2 regularization

Regularization keeps a flexible model but discourages extreme parameter values. The wild curve of the degree-15 fit needs huge, finely balanced weights. **L2 regularization** adds a penalty on the

squared size of the weights to the cost:

$$J_{\text{reg}}(\mathbf{w}, b) = J(\mathbf{w}, b) + \frac{\lambda}{2m} \sum_{j=1}^{n} w_j^2,$$

where J is the original cost (MSE or log loss), $\lambda \geq 0$ is the **regularization strength** (a hyperparameter), and the sum runs over the weights but conventionally not the bias b. With $\lambda = 0$ we recover the original model; as λ grows, the weights are pulled toward zero and the fitted function becomes smoother; too large a λ underfits. For linear regression, this is **ridge regression**, introduced by Hoerl and Kennard to stabilize least-squares estimates when features are strongly correlated [20].

The gradient-descent update shows what the penalty does:

$$w_j \leftarrow w_j \left(1 - \alpha \frac{\lambda}{m}\right) - \alpha \frac{\partial J}{\partial w_j}.$$

Before each ordinary step, every weight is multiplied by a number slightly less than 1. This is why L2 regularization is also called **weight decay** in neural-network training [3].

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.metrics import mean_squared_error

rng = np.random.default_rng(0)
x = rng.uniform(0, 1, 25)
y = np.sin(2 * np.pi * x) + rng.normal(0, 0.25, 25)      # true curve + noise
X = x[:, None]

# 60% train, 20% validation, 20% test
X_tmp, X_te, y_tmp, y_te = train_test_split(X, y, test_size=0.2, random_state=0)
X_tr, X_va, y_tr, y_va = train_test_split(X_tmp, y_tmp, test_size=0.25, random_state=0)

def fit(degree, lam=0.0):
    reg = Ridge(alpha=lam) if lam > 0 else LinearRegression()   # sklearn calls lambda
    ↪  "alpha"
    model = make_pipeline(PolynomialFeatures(degree), StandardScaler(), reg)
    return model.fit(X_tr, y_tr)

for degree, lam in [(1, 0), (3, 0), (15, 0), (15, 1e-2)]:
    model = fit(degree, lam)
    tr = mean_squared_error(y_tr, model.predict(X_tr))
    va = mean_squared_error(y_va, model.predict(X_va))
    print(f"degree {degree:2d}, lambda {lam:g}: train MSE {tr:.3f}, val MSE {va:.3g}")
```

```
best = fit(15, 1e-2)                                    # chosen on validation only
print("test MSE:", round(mean_squared_error(y_te, best.predict(X_te)), 3))
```

The output tells the whole story: degree 1 has train/validation MSE of 0.144/0.292 (underfit), degree 3 has 0.031/0.052 (good), and unregularized degree 15 has a training MSE of 0.003 but a validation MSE in the millions, because some validation points fall in wide gaps between training points (for example between $x \approx 0.27$ and $x \approx 0.61$), where the polynomial swings wildly. Adding a mild L2 penalty ($\lambda = 0.01$) to the same degree-15 model gives 0.041/0.035, and its test MSE is 0.075.

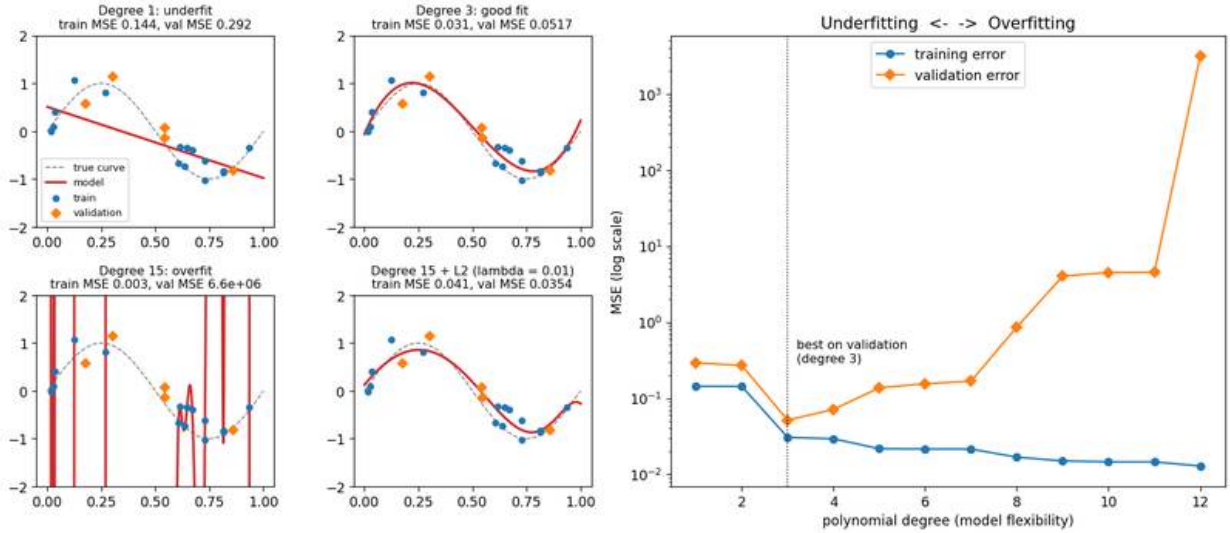


Figure 8 — Underfitting, a good fit, overfitting, and overfitting tamed by L2 regularization (training points in blue, validation points in orange). Right: as the degree grows, training error keeps falling while validation error falls and then rises; the bottom of the validation curve is the model to choose.

Evaluating a model

Plain version. A single number never tells the whole story. Choose a metric that matches what a mistake costs, and always compare against a trivial baseline.

Regression metrics

For regression, the most common metric is the MSE, $\frac{1}{m}\sum_i(\hat{y}^{(i)} - y^{(i)})^2$, here without the $\frac{1}{2}$ that was only a training convenience. Its square root, the **RMSE**, is in the same units as y and is easier to interpret ("off by about 3 points on average"). The **mean absolute error**, $\frac{1}{m}\sum_i |\hat{y}^{(i)} - y^{(i)}|$, is less sensitive to a few large errors [5].

Classification metrics

For classification, **accuracy** is the fraction of correct predictions. It is easy to understand and easy to be fooled by. In the digits data, about 10% of the images are 9s. A "model" that always answers "not a 9" is 90% accurate and completely useless.

The **confusion matrix** counts the four outcomes for a binary problem, with "positive" meaning the class we care about:

	predicted positive	predicted negative
actually positive	true positive (TP)	false negative (FN)
actually negative	false positive (FP)	true negative (TN)

From it,

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \qquad \text{precision} = \frac{TP}{TP + FP}, \qquad \text{recall} = \frac{TP}{TP + FN}.$$

Precision answers "when the model says positive, how often is it right?" **Recall** answers "of all the real positives, how many did the model find?" They trade off: lowering the 0.5 decision threshold of logistic regression finds more positives (higher recall) but raises more false alarms (lower precision). Which matters more depends on the cost of each mistake: a cancer screen should not miss tumors (high recall), while a spam filter should not hide real mail (high precision). The **F1 score**, the harmonic mean of precision and recall, combines both into one number [5].

```
import numpy as np
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score

X, y = load_digits(return_X_y=True)
y = (y == 9).astype(int)                  # "is it a 9?": only about 10% positives
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.3, stratify=y, random_state=0)

lazy = np.zeros_like(y_te)                # always answers "not a 9"
model = LogisticRegression(max_iter=5000).fit(X_tr, y_tr)
pred = model.predict(X_te)

print("lazy accuracy :", round(accuracy_score(y_te, lazy), 3), " recall:",
 ↪  recall_score(y_te, lazy))
print("model accuracy:", round(accuracy_score(y_te, pred), 3),
      " precision:", round(precision_score(y_te, pred), 3),
      " recall:", round(recall_score(y_te, pred), 3))
print(confusion_matrix(y_te, pred))       # rows: true 0/1, columns: predicted 0/1
# lazy accuracy : 0.9   recall: 0.0
# model accuracy: 0.974  precision: 0.87  recall: 0.87
```

The lazy baseline reaches 90% accuracy with zero recall. The real model's 97.4% accuracy sounds only a little better, but its recall of 0.87 shows that it actually finds most of the 9s. Always report the baseline, and always look past accuracy when classes are imbalanced. The same metrics reappear, with more care about thresholds and multiple classes, in the DIP chapter on pattern classification.

Modern view

The mechanics in this tutorial (a parameterized model, a cost, gradient-based optimization, and held-out evaluation) have not changed in decades, and every system discussed below still uses them. What has changed is *where the features come from*, *where the labels come from*, and *how large the models are*. A handful of reviews map these shifts well.

Machine learning as a field. Jordan and Mitchell's review in *Science* [7] frames machine learning around a single question, how to build computers that improve automatically through experience, and places it at the intersection of computer science and statistics, at the core of artificial intelligence and data science. They attribute its rapid spread across science, medicine, manufacturing, education and finance to three things arriving together: new learning algorithms and theory, an explosion in the amount of available data, and cheap computation. That framing is a useful antidote to hype: the method is only as good as the data and the experience it is given.

From feature engineering to representation learning. For a long time, the success of a classical model depended mostly on hand-designed features, the "count the inked columns" step of Figure 1 done by experts: edge histograms, texture descriptors, keypoint descriptors (see IP 12). Bengio, Courville and Vincent [21] argue that this dependence is a weakness, because the right representation is what makes a task easy or hard, and propose learning the representation itself from data. Their review covers the tools for this (probabilistic models, autoencoders, manifold learning and deep networks) and the general priors that make a representation good, such as disentangling the separate explanatory factors behind the data. In the language of this tutorial, a deep network is still "features followed by logistic regression", except that the features are produced by earlier layers whose weights are learned by the same gradient descent. That is the subject of ML 02.

Self-supervised learning. Learning representations needs data, and labels are the expensive part. Jing and Tian's survey of self-supervised visual feature learning [22] sorts the label-free "pretext tasks" into four families: generation-based (for example, reconstructing or colorizing images), context-based (exploiting spatial or temporal structure, such as the arrangement of image patches), free-semantic-label-based (using labels that come automatically, for example from rendering engines or classical methods), and cross-modal (checking whether two signals, such as video and audio, belong together). It also sets out the evaluation protocol that is still standard: judge a representation by how well it transfers to downstream tasks such as classification, detection and segmentation. The later survey by Gui et al. [23] reorganizes the field around context-based methods, contrastive learning (with negative examples, self-distillation, or feature decorrelation), generative methods dominated by masked image modeling, and hybrids of contrastive and generative objectives, reflecting the shift toward masked prediction after masked autoen-

coders [12]. For practitioners, Balestriero et al.'s *Cookbook of Self-Supervised Learning* [24] groups modern methods into a deep metric learning family, a self-distillation family, a canonical correlation analysis family and masked image modeling, and is candid about how much the outcome depends on details such as data augmentation and the small "projector" network used during training.

Foundation models. The report by Bommasani et al. [10] named the resulting paradigm: models trained on broad data at scale, usually self-supervised, then adapted to many downstream tasks. It highlights two consequences. **Emergence**: capabilities such as following instructions from a few examples in the prompt [11] appear with scale rather than being designed in. **Homogenization**: many applications are now adapted from the same few base models, which gives great leverage but means that a defect in the base model is inherited by everything built on it. The report also stresses how limited our understanding is of how these models work and when they fail.

Reinforcement learning from human feedback. RL has moved from games to the final stage of training language models [13], [14]. Casper et al. [25] survey what can go wrong. They follow the three steps of RLHF (collecting human feedback, fitting a reward model, and optimizing the policy) and catalog problems at each: feedback from people is noisy, inconsistent and can be manipulated; a reward model is only an imperfect proxy for what people want; and a policy optimized hard against that proxy can exploit its errors. They separate problems that better methods within RLHF could solve from fundamental limitations that would require a different approach, and they argue for transparency and for combining RLHF with other safety measures rather than relying on it alone.

Optimization at scale. Bottou, Curtis and Nocedal [18] explain why the humble stochastic gradient method, and not more sophisticated optimizers from classical numerical analysis, became the workhorse of machine learning: when data is plentiful, a cheap noisy step beats an expensive exact one. Adaptive methods such as Adam [19] add per-parameter step sizes on top. Every model mentioned in this section, however large, is trained with a descendant of the mini-batch loop in this tutorial.

Open problems. Despite the scale, several questions from this beginner's tutorial remain the hard ones. *Generalization*: very large models fit their training data almost perfectly yet still generalize, which does not match the simple overfitting picture of Figure 8, and why this happens is not fully understood. *Evaluation*: when a model is trained on a large fraction of the public internet, it is hard to guarantee that the test set was not in the training data, so a clean train/test separation, the foundation of honest evaluation, is difficult to verify. *Objectives*: the cost function is the only thing the model optimizes, and RLHF shows how hard it is to write a cost that captures what people actually want. And *data*: what a model learns is bounded by the data and labels it sees, with all their gaps and biases.

For computer vision specifically, the classical route (hand-made features plus a simple classifier) is developed in the DIP chapter on pattern classification, and the learned route continues in ML 02 and Deep Learning for Image Processing.

Key takeaways

- Machine learning replaces hand-written rules with a flexible model whose parameters are fitted to examples; it shines when the rule is hard to state but examples are easy to get.
- The paradigms differ in feedback: supervised (right answers; regression for numbers, classification for categories), unsupervised (no answers; clustering, dimensionality reduction, anomaly detection), self-supervised (answers made from the data itself), and reinforcement learning (rewards for actions). KNN is supervised; k-means is clustering.
- Keep the model $f_{w,b}(x)$ and the cost $J(w,b)$ apart: the model maps inputs to predictions; the cost maps parameters to an error. Training means minimizing J.
- Gradient descent repeats $w \leftarrow w - \alpha\, \partial J/\partial w$. At a minimum the *derivative* is zero. The learning rate must be small enough to avoid overshooting and large enough to make progress; mini-batches make each step cheap.
- Scale features before gradient descent, using statistics from the training set only.
- Logistic regression applies a sigmoid to a linear score and minimizes cross-entropy; its decision boundary is $\mathbf{w} \cdot \mathbf{x} + b = 0$, and its gradient has the same form as linear regression's.
- Judge models on data they did not train on: choose with the validation set, report once on the test set, regularize (L2) when the model overfits, and look beyond accuracy when classes are imbalanced.

Exercises

1. **A derivative by hand.** For the one-feature MSE cost $J(w,b) = \frac{1}{2m}\sum_i (wx^{(i)} + b - y^{(i)})^2$, derive $\partial J/\partial w$ and $\partial J/\partial b$. Then, for the three points $(1,2)$, $(2,4)$, $(3,6)$ starting at $w = 0, b = 0$, compute one gradient-descent step with $\alpha = 0.1$.

   Hint

   The chain rule on each square gives $2(wx^{(i)}+b-y^{(i)})\cdot x^{(i)}$, and the 2 cancels the $\frac{1}{2}$, giving the formulas in the gradient-descent section. At $w = b = 0$ the errors are $-2, -4, -6$. So $\partial J/\partial w = (-2\cdot 1 - 4\cdot 2 - 6\cdot 3)/3 = -28/3$ and $\partial J/\partial b = -12/3 = -4$. After one step, $w = 0.1 \cdot 28/3 \approx 0.933$ and $b = 0.4$.

2. **Find the divergence threshold.** For $J(w) = \frac{1}{2m}\sum_i (wx^{(i)} - y^{(i)})^2$ (no bias), show that one gradient step multiplies the distance to the optimum w^* by $1 - \alpha s$, where $s = \frac{1}{m}\sum_i (x^{(i)})^2$. For which α does gradient descent converge? Check your answer with the `gd` function from the feature-scaling section.

   Hint

   $\partial J/\partial w = s\,(w - w^*)$ because J is a parabola with curvature s and minimum at w^*. So $w_{\text{new}} - w^* = (1-\alpha s)(w - w^*)$. It converges when $|1-\alpha s| < 1$, that is $0 < \alpha < 2/s$; at $\alpha = 1/s$ it lands on the minimum in one step. With several features, s is replaced by the largest eigenvalue of $\frac{1}{m}X^\top X$, which is why one huge-scale feature forces a tiny α.

3. **Regression for classification.** Generate 1-D data with `make_classification(n_features=1, n_informative=1, n_redundant=0, n_clusters_per_class=1)`, then add five extra examples of class 1 with very large x. Fit both `LinearRegression` (thresholding at 0.5) and `LogisticRegression`. Compare their accuracy before and after adding the extra points, and explain the difference.

   Hint

   The extra points are easy to classify, but the squared error still wants the line to pass near them. The line's slope drops, its 0.5 crossing moves toward class 1, and some ordinary class-1 points near the boundary become misclassified. The log loss of a correctly classified, far-away point is already close to 0, so logistic regression barely moves.

4. **Scaling the digits.** Train the from-scratch logistic regression of this tutorial on "9 versus not 9" with the digits data, once on raw pixels (0 to 16) and once on pixels divided by 16. Use the same number of steps and find the largest learning rate that works for each. Does scaling change the final accuracy, the speed, or both?

   Hint

   All pixels share one unit here, so the effect is milder than in Figure 6, but the raw inputs are 16 times larger, which multiplies the gradient's size and the landscape's curvature. The safe learning rate on raw pixels is roughly $16^2 = 256$ times smaller. With properly tuned learning rates, both reach similar accuracy; the scaled version is simply easier to tune.

5. **Choosing lambda.** Using the polynomial example, fix degree 15 and try $\lambda \in \{10^{-6}, 10^{-4}, 10^{-2}, 1, 100\}$. Plot training and validation MSE against λ on a log axis. Which end underfits, which overfits, and which λ would you pick? Why must you not use the test set to make this choice?

   Hint

   Tiny λ behaves like no regularization (overfitting: low training error, high validation error). Huge λ flattens the curve toward a constant (underfitting: both errors high). Pick the λ with the lowest validation error. If the test set guided the choice, it would no longer be an unbiased estimate of performance on new data.

6. **Which paradigm?** Classify each problem as supervised (regression or classification), unsupervised, self-supervised or reinforcement learning, and name the target or reward: (a) estimating the remaining battery life of a phone from usage logs with recorded outcomes; (b) grouping a million unlabeled product photos for browsing; (c) pre-training an image model by predicting the missing quarter of each photo; (d) teaching a robot arm to stack blocks with a score given only when the stack stands; (e) flagging unusual readings in a factory sensor stream with no labeled failures.

Hint

(a) Supervised regression; target: remaining hours. (b) Unsupervised clustering. (c) Self-supervised; the target is the hidden pixels, made from the data itself. (d) Reinforcement learning; reward: whether the stack stands, delayed until the end. (e) Unsupervised anomaly detection.

Continue through ML/DL

If the mathematics is new, take linear algebra, probability and information theory, and numerical computation before neural networks.

The rest of the track connects image filtering and CNNs with recurrent/recursive networks, transformers and representation learning. It then builds from factor models and autoencoders to VAEs, GANs, normalizing flows, diffusion/flow matching, autoregressive models and reusable foundation models. Finish with evaluation and deployment. For the statistical decision and inference viewpoint, follow Pattern Recognition; for task-specific systems, follow Deep Learning for Computer Vision.

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ML 01A • Linear Algebra for Learning

Vectors, matrices and tensors, norms, eigendecomposition, the SVD, the pseudo-inverse, trace and determinant, and a step-by-step derivation of PCA, with tested NumPy code.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-01a-linear-algebra/

What you'll learn

- How scalars, vectors, matrices and tensors describe images, batches and learned transformations, and how to keep their shapes straight.
- Four ways to read a matrix product, and what identity, inverse, span, rank and null space mean for a system of equations.
- How norms measure size, how special matrices (diagonal, symmetric, orthogonal) simplify computation, and how dot products define similarity.
- What eigendecomposition and the singular value decomposition (SVD) reveal about a matrix, and why the SVD gives the best low-rank approximation.
- How the Moore–Penrose pseudo-inverse solves least-squares and under-determined problems, and why conditioning and ridge regularization matter.
- How the trace and determinant work, and how to derive principal component analysis (PCA) from scratch using only these tools.

The big picture

Almost every operation inside a neural network is a linear map followed by a simple nonlinear function. A fully connected layer multiplies by a weight matrix. A convolution is a structured matrix product. Attention computes dot products between vectors and then mixes vectors with a matrix of weights. Linear algebra is the language for all of these. It tells us what a map keeps, what it throws away, what it stretches, and how to compute it without losing accuracy.

> **Explain like I'm five**
>
> A vector is a list of measurements, like the height, weight and age of one person. A matrix is a machine that takes one list and mixes it into a new list. Some machines just turn the list around (rotation), some make some parts bigger and others smaller (stretching), and some squash parts flat so the information is gone forever. Linear algebra is the instruction manual for these machines: it tells you which way each machine stretches, by how much, and whether you can undo it.

This chapter follows the "Applied Math" part of Goodfellow, Bengio and Courville's *Deep Learning* [1] and the linear algebra chapters of *Mathematics for Machine Learning* [2]. Strang's textbook [3] is the gentlest full course if you want more practice, and Golub and Van Loan [4] and Trefethen and

Bau [5] are the standard references for how these computations are done reliably on a computer. Read this chapter before ML 02 if matrix notation is unfamiliar. The companion chapters are ML 01B ·Probability and Information Theory and ML 01C ·Numerical Computation.

The chapter has a single thread. We start with objects and shapes, move to products and solving equations, then measure size with norms. Eigendecomposition and the SVD then explain the "personality" of a matrix. The pseudo-inverse, trace and determinant are tools built on top of them. Finally, we use every one of these tools to derive PCA, the classic method for compressing data, exactly as it is derived in [1].

Scalars, vectors, matrices and tensors

Plain version. Every number in machine learning lives in an array with named axes. Name every axis before you multiply anything.

A **scalar** is a single number, written in italics: $s \in \mathbb{R}$. A **vector** is an ordered list of n numbers, written in bold lowercase or plain lowercase in this chapter: $x \in \mathbb{R}^n$, with entries $x_1, \dots, x_n$. By convention a vector is a column. A **matrix** is a 2-D grid of numbers: $A \in \mathbb{R}^{m \times n}$ has m rows and n columns, and A_{ij} is the entry in row i and column j. A **tensor** is an array with any number of axes.

These objects map directly onto data:

- A grayscale image of height H and width W is a matrix in $\mathbb{R}^{H \times W}$. Flattened, it becomes a vector $x \in \mathbb{R}^{HW}$. (See Images as Arrays.)
- A dataset of n examples with d features each, stored one example per row, is a **design matrix** $X \in \mathbb{R}^{n \times d}$.
- A fully connected layer with weights $W \in \mathbb{R}^{d \times m}$ and bias $b \in \mathbb{R}^m$ maps the whole batch at once: $Y = XW + \mathbf{1}b^\top \in \mathbb{R}^{n \times m}$, where $\mathbf{1} \in \mathbb{R}^n$ is a vector of ones.
- An image batch in PyTorch is usually a 4-D tensor of shape $B \times C \times H \times W$ (batch, channels, height, width). A batch of token embeddings in a Transformer is usually $B \times N \times D$ (batch, sequence length, embedding width).

The **transpose** $A^\top$ swaps rows and columns: $(A^\top)_{ij} = A_{ji}$. It satisfies $(A^\top)^\top = A$ and $(AB)^\top = B^\top A^\top$. Note the reversed order.

Broadcasting

Writing $\mathbf{1}b^\top$ is mathematically honest but tedious, so libraries let you add a vector to every row of a matrix directly. This is called **broadcasting**: when two arrays have different shapes, axes of length 1 (or missing leading axes) are stretched to match. Broadcasting is convenient and dangerous,

because a wrong shape often still runs. The most common bug is subtracting a column vector of shape $(n, 1)$ from a flat vector of shape $(n,)$, which silently produces an $n \times n$ matrix instead of n residuals.

```
import numpy as np

rng = np.random.default_rng(0)
X = rng.normal(size=(32, 64))      # batch of 32 examples, 64 features
W = rng.normal(size=(64, 10))      # linear map 64 -> 10
b = rng.normal(size=10)            # one bias per output
Y = X @ W + b                      # b broadcasts over the 32 rows
assert Y.shape == (32, 10)

# A classic broadcasting bug: (32,) - (32, 1) silently becomes (32, 32)
y_true = rng.normal(size=32)
y_pred = rng.normal(size=(32, 1))
print("buggy residual shape:", (y_true - y_pred).shape)
print("fixed residual shape:", (y_true - y_pred[:, 0]).shape)

# Four equivalent views of the same matrix product
A, B = rng.normal(size=(3, 4)), rng.normal(size=(4, 2))
entry = np.array([[A[i] @ B[:, j] for j in range(2)] for i in range(3)])
cols = np.stack([A @ B[:, j] for j in range(2)], axis=1)
outer = sum(np.outer(A[:, k], B[k]) for k in range(4))
assert np.allclose(A @ B, entry) and np.allclose(A @ B, cols) and np.allclose(A @ B, outer)
print("all views agree")
```

The output shows `(32, 32)` for the buggy line and `(32,)` for the fixed one. A mean over a 32×32 array still returns a single number, so a loss computed this way trains without errors while optimizing the wrong thing. Put `assert` statements on shapes in any code you care about.

Multiplying matrices and vectors

Plain version. A matrix product chains two linear machines into one. Each output number is a dot product between a row of the first matrix and a column of the second.

For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, the **matrix product** $C = AB \in \mathbb{R}^{m \times p}$ is

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}.$$

Here i indexes rows of A (and of C), j indexes columns of B (and of C), and k runs over the shared inner dimension n. The inner dimensions must match; that is the first thing to check when a shape error appears.

The same product can be read in four ways, and each one is useful:

1. **Entry view.** C_{ij} is the dot product of row i of A with column j of B. This is how attention scores $QK^\top$ are read: every query is compared with every key.
2. **Column view.** Column j of C is A times column j of B. Each column of C is a combination of the columns of A.
3. **Row view.** Row i of C is row i of A times B. In $Y = XW$, each output example is its input row pushed through the same weights.
4. **Outer-product view.** $C = \sum_k a_{:k} b_{k:}$, a sum of n rank-one matrices, each formed from column k of A and row k of B. This view explains low-rank approximation and LoRA later in the chapter.

The code above checks that all four views give the same answer.

Matrix multiplication is **associative**, $(AB)C = A(BC)$, and **distributive**, $A(B + C) = AB + AC$, but **not commutative**: $AB \neq BA$ in general, and often one of the two products is not even defined. The **elementwise** (Hadamard) product $A \odot B$ multiplies matching entries instead and needs equal shapes. In NumPy and PyTorch, `A @ B` is the matrix product and `A * B` is elementwise. The **dot product** of two vectors $x, y \in \mathbb{R}^n$ is the special case $x^\top y = \sum_i x_i y_i$, and it is symmetric: $x^\top y = y^\top x$.

Identity, inverse and systems of equations

A system of m linear equations in n unknowns can be written compactly as

$$Ax = b, \qquad A \in \mathbb{R}^{m \times n},\ x \in \mathbb{R}^n,\ b \in \mathbb{R}^m.$$

Each row of A holds the coefficients of one equation, x holds the unknowns, and b holds the right-hand sides.

The **identity matrix** I_n has ones on the diagonal and zeros elsewhere; it leaves every vector unchanged, $I_n x = x$. A square matrix A is **invertible** if there is a matrix A^{-1} with $A^{-1}A = AA^{-1} = I$. Then the system has exactly one solution, $x = A^{-1}b$.

That formula is excellent for derivations and poor for computation. Explicitly forming A^{-1} costs more and loses more accuracy than solving the system directly with an LU, Cholesky or QR factorization [4], [5]. In NumPy, write `np.linalg.solve(A, b)`, not `np.linalg.inv(A) @ b`. In this chapter, A^{-1} is a thinking tool.

Linear dependence, span and rank

Plain version. Rank counts how many genuinely different directions a matrix can produce. Repeated or redundant columns add nothing new.

A **linear combination** of vectors $v^{(1)}, \dots, v^{(k)}$ is $\sum_i c_i v^{(i)}$ for scalars c_i. The **span** of a set of vectors is the set of all their linear combinations. Because $Ax = \sum_j x_j a_{:j}$, the vector Ax is a linear combination of the columns $a_{:j}$ of A. So $Ax = b$ has a solution exactly when b lies in the span of the columns, called the **column space** (or range) of A.

A set of vectors is **linearly independent** if no vector in it is a linear combination of the others; equivalently, $\sum_i c_i v^{(i)} = 0$ only when every $c_i = 0$. The **rank** of A is the number of linearly independent columns, which always equals the number of linearly independent rows. For $A \in \mathbb{R}^{m \times n}$, $\operatorname{rank}(A) \le \min(m, n)$; if equality holds, A has **full rank**.

The **null space** of A is the set of vectors z with $Az = 0$. Those are the input directions that A destroys. The **rank–nullity theorem** says

$$\operatorname{rank}(A) + \dim \operatorname{null}(A) = n,$$

where n is the number of columns and dim is the number of independent directions in the null space. This one equation explains the three regimes of $Ax = b$:

Shape and rank	What happens	Typical ML case
square, full rank	exactly one solution	a well-posed change of coordinates
tall ($m > n$), full column rank	usually no exact solution; solve in the least-squares sense	regression with more examples than features
wide ($m < n$) or rank-deficient	infinitely many solutions (if any)	more parameters than constraints, as in deep networks

For a square matrix, "invertible", "full rank", "columns linearly independent" and "null space contains only zero" are all the same statement. A square matrix that fails them is called **singular**.

Norms: measuring size

Plain version. A norm turns a vector into one non-negative number that says how big it is. Different norms disagree about what "big" means.

A **norm** is any function f with $f(x) = 0 \Rightarrow x = 0$, $f(x + y) \le f(x) + f(y)$ (the triangle inequality), and $f(\alpha x) = |\alpha| f(x)$ for every scalar α. The most common family is the L^p norm,

$$\|x\|_p = \Big(\sum_i |x_i|^p \Big)^{1/p}, \qquad p \ge 1.$$

- $\|x\|_2$, the **Euclidean** norm, is ordinary length; we often write $\|x\|$. Its square $\|x\|_2^2 = x^\top x$ is easier to differentiate and appears in mean-squared-error losses and weight decay.

- $\|x\|_1 = \sum_i |x_i|$ grows at the same rate everywhere, even near zero. That makes it useful when the difference between exactly zero and slightly nonzero matters, which is why L^1 penalties encourage sparse solutions.
- $\|x\|_\infty = \max_i |x_i|$, the **max norm**, is the size of the largest entry. Adversarial-robustness work often bounds perturbations in this norm.
- The count of nonzero entries is sometimes called the "L^0 norm", but it is not a norm: scaling x by 2 does not change it.

The dot product connects to the Euclidean norm through angles: $x^\top y = \|x\|_2 \|y\|_2 \cos\theta$, where θ is the angle between x and y. The **cosine similarity**

$$\cos(x, y) = \frac{x^\top y}{\|x\|_2 \|y\|_2}$$

is defined for nonzero vectors and ignores scale. That is useful for some embeddings (contrastive learning normalizes features for this reason) and wrong when magnitude carries meaning, such as image brightness.

Matrix norms

For matrices we use two norms constantly. The **Frobenius norm** treats the matrix as one long vector,

$$\|A\|_F = \sqrt{\sum_{i,j} A_{ij}^2}.$$

The **spectral norm** (operator 2-norm) measures the largest factor by which A can stretch a unit vector,

$$\|A\|_2 = \max_{\|x\|_2 = 1} \|Ax\|_2 = \sigma_{\max}(A),$$

where $\sigma_{\max}(A)$ is the largest singular value, defined in the SVD section below. In terms of all singular values $\sigma_1 \geq \sigma_2 \geq \ldots$, we have $\|A\|_F^2 = \sum_i \sigma_i^2$ while $\|A\|_2 = \sigma_1$. So many small singular values can make the Frobenius norm large without changing the largest amplification. The spectral norm bounds how much a layer can magnify a perturbation of its input, which is why it appears in robustness and GAN-stability work [17].

```
import numpy as np

x = np.array([3.0, -4.0, 0.0, 1.0])
for p in [1, 2, np.inf]:
    print(f"L{p} norm:", np.linalg.norm(x, p))
print("L0 'norm' (count of nonzeros):", np.count_nonzero(x))

A = np.array([[3.0, 0.0], [4.0, 5.0]])
```

```
s = np.linalg.svd(A, compute_uv=False)
print("singular values:", s.round(4))
print("Frobenius:", np.linalg.norm(A, "fro"), "=", np.sqrt((s**2).sum()))
print("spectral :", np.linalg.norm(A, 2), "=", s[0])
assert np.isclose(np.linalg.norm(A, "fro")**2, np.trace(A.T @ A))
```

For $x = (3, -4, 0, 1)$ this prints $L^1 = 8$, $L^2 = \sqrt{26} \approx 5.10$, $L^\infty = 4$ and three nonzeros. For the 2×2 matrix the singular values are $3\sqrt{5} \approx 6.71$ and $\sqrt{5} \approx 2.24$, giving $\|A\|_F = \sqrt{50} \approx 7.07$ and $\|A\|_2 \approx 6.71$.

Special kinds of matrices and vectors

Plain version. Some matrices are much easier to work with than others. Learning to spot them saves computation and explains many design choices.

- A **diagonal** matrix $D = \operatorname{diag}(v)$ has nonzero entries only on its main diagonal. Multiplying by it scales each coordinate: $\operatorname{diag}(v)x = v \odot x$. Its inverse, when every $v_i \neq 0$, is $\operatorname{diag}(1/v_1, \dots, 1/v_n)$. Per-channel scaling in normalization layers is a diagonal matrix in disguise. Rectangular diagonal matrices also exist; they appear in the SVD.
- A **symmetric** matrix satisfies $A = A^\top$. Covariance matrices, Gram matrices $X^\top X$, Hessians of smooth functions and distance matrices are symmetric.
- A **unit vector** has $\|x\|_2 = 1$. Two vectors are **orthogonal** if $x^\top y = 0$; if they are also unit vectors, they are **orthonormal**.
- An **orthogonal matrix** Q is square with orthonormal rows and columns, so $Q^\top Q = QQ^\top = I$ and $Q^{-1} = Q^\top$. Its inverse is free to compute. It preserves lengths and angles, $\|Qx\|_2 = \|x\|_2$, so geometrically it is a rotation, a reflection, or both.

If $Q \in \mathbb{R}^{n \times k}$ has $k < n$ orthonormal columns, then $Q^\top Q = I_k$ but $QQ^\top \neq I_n$. The matrix $P = QQ^\top$ is the **orthogonal projection** onto the span of the columns of Q. The projection Px is the closest point to x in that subspace, and the residual $x - Px$ is perpendicular to every column of Q. Projecting twice changes nothing: $P^2 = P$. PCA, below, is exactly the search for the best such Q.

Eigendecomposition

Plain version. Most vectors change direction when a matrix acts on them. Eigenvectors are the special directions that only get stretched or shrunk, and the eigenvalue says by how much.

An **eigenvector** of a square matrix $A \in \mathbb{R}^{n\times n}$ is a nonzero vector v with

$$Av = \lambda v,$$

where the scalar λ is the corresponding **eigenvalue**. If v is an eigenvector, so is cv for any $c \neq 0$, so we usually normalize eigenvectors to unit length.

Suppose A has n linearly independent eigenvectors $v^{(1)}, \dots, v^{(n)}$ with eigenvalues $\lambda_1, \dots, \lambda_n$. Stack them as columns of V and put the eigenvalues in $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Then $AV = V\Lambda$, so

$$A = V\Lambda V^{-1}.$$

This is the **eigendecomposition**. It says: change to eigenvector coordinates (V^{-1}), scale each coordinate independently (Λ), and change back (V). Not every matrix has one. The shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has only one independent eigenvector. A rotation by 90° has eigenvalues $\pm i$, which are complex, because no real direction is left unturned.

The symmetric case

Symmetric matrices are the friendly case, and they are the ones ML meets most. The **spectral theorem** says that every real symmetric matrix can be written as

$$A = Q\Lambda Q^{\top},$$

where Q is orthogonal (its columns are orthonormal eigenvectors) and Λ is real and diagonal [3], [4]. Because $Q^{-1} = Q^{\top}$, no inversion is needed. Geometrically, A stretches space by λ_i along the direction $q^{(i)}$, the i-th column of Q. The decomposition is unique up to the order of the eigenvalues and the choice of basis within any repeated eigenvalue; by convention we sort eigenvalues from largest to smallest.

For a symmetric A, the **quadratic form** $f(x) = x^{\top}Ax$ restricted to unit vectors is bounded by the extreme eigenvalues:

$$\lambda_{\min} \leq x^{\top}Ax \leq \lambda_{\max} \qquad \text{for } \|x\|_2 = 1,$$

with the maximum reached at the top eigenvector and the minimum at the bottom one. The ratio $x^{\top}Ax/x^{\top}x$ is called the **Rayleigh quotient**. This fact is the engine of the PCA derivation later.

Positive definite matrices

The signs of the eigenvalues classify symmetric matrices:

Name	Eigenvalues	Equivalent condition
positive definite (PD)	all $\lambda_i > 0$	$x^\top A x > 0$ for every $x \neq 0$
positive semidefinite (PSD)	all $\lambda_i \geq 0$	$x^\top A x \geq 0$ for every x
negative (semi)definite	all $\lambda_i < 0$ (or ≤ 0)	the reverse inequalities
indefinite	both signs present	$x^\top A x$ takes both signs

Every covariance matrix and every Gram matrix $X^\top X$ is PSD, because $v^\top X^\top X v = \|Xv\|_2^2 \geq 0$. A Gaussian density needs a PD covariance so that it can be inverted. The Hessian of a loss is PD at a strict local minimum and indefinite at a saddle point; ML 01C uses this to explain why optimization can be slow. A PD matrix has a **Cholesky factorization** $A = LL^\top$ with L lower triangular, which is the fastest stable way to solve systems with it and to sample from a Gaussian [4].

```
import numpy as np

rng = np.random.default_rng(1)
M = rng.normal(size=(4, 4))
S = M @ M.T + 0.1 * np.eye(4)        # symmetric positive definite
lam, Q = np.linalg.eigh(S)           # ascending eigenvalues, orthonormal Q
assert np.allclose(Q @ np.diag(lam) @ Q.T, S)
assert np.allclose(Q.T @ Q, np.eye(4))
print("eigenvalues:", lam.round(3), "all positive:", bool((lam > 0).all()))

# The quadratic form x^T S x is bounded by the extreme eigenvalues
x = rng.normal(size=4); x /= np.linalg.norm(x)
print("Rayleigh quotient in range:", lam[0] <= x @ S @ x <= lam[-1])

# A non-symmetric matrix can have complex eigenvalues
R = np.array([[0.0, -1.0], [1.0, 0.0]])  # 90-degree rotation
print("rotation eigenvalues:", np.linalg.eigvals(R))
```

Use `np.linalg.eigh` for symmetric matrices: it is faster, returns real eigenvalues in ascending order and orthonormal eigenvectors. `np.linalg.eig` handles the general case and may return complex numbers, as the rotation shows.

Singular value decomposition

Plain version. Every matrix, of any shape, does the same three things in order: it rotates, it stretches along perpendicular axes, and it rotates again. The SVD tells you those rotations and stretch factors.

Eigendecomposition needs a square matrix and may not exist. The **singular value decomposition** always exists, for every real matrix $A \in \mathbb{R}^{m \times n}$:

$$A = U \Sigma V^\top.$$

- $U \in \mathbb{R}^{m \times m}$ is orthogonal. Its columns are the **left singular vectors**, directions in the output space.
- $\Sigma \in \mathbb{R}^{m \times n}$ is rectangular diagonal. Its diagonal entries $\sigma_1 \geq \sigma_2 \geq \ldots \geq 0$ are the **singular values**.
- $V \in \mathbb{R}^{n \times n}$ is orthogonal. Its columns are the **right singular vectors**, directions in the input space.

Read from right to left, $Ax = U\Sigma V^\top x$ first expresses x in the V coordinates (a rotation or reflection), then scales coordinate i by σ_i, then places the result along the U directions. Figure 1 shows this for a 2×2 matrix: the unit circle becomes an ellipse whose semi-axes have lengths σ_1 and σ_2.

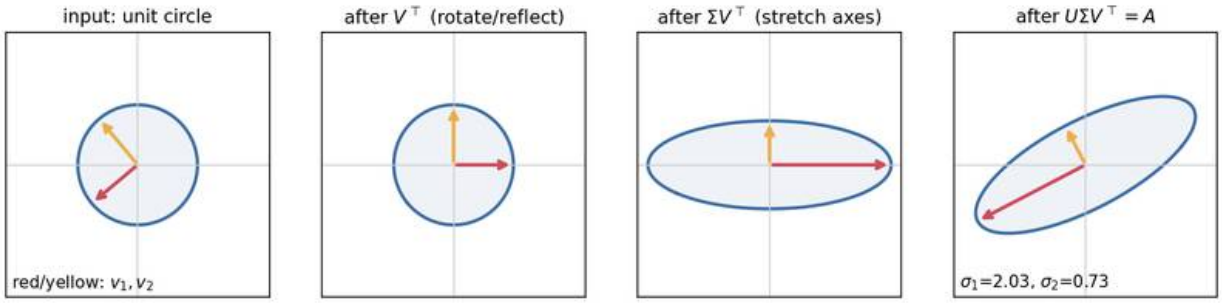


Figure 1 — The SVD as rotate, stretch, rotate. The right singular vectors v1 and v2 (red, yellow) are first rotated onto the axes by V-transpose, stretched by the singular values, then rotated by U. The final ellipse is the image of the unit circle under A.

The SVD and eigendecomposition are closely related. Since $A^\top A = V\Sigma^\top \Sigma V^\top$ and $AA^\top = U\Sigma\Sigma^\top U^\top$, the right singular vectors are eigenvectors of $A^\top A$, the left singular vectors are eigenvectors of $AA^\top$, and $\sigma_i = \sqrt{\lambda_i(A^\top A)}$. For a symmetric PSD matrix, the SVD and the eigendecomposition coincide.

The number of nonzero singular values equals the rank r. Keeping only those gives the **compact** (thin) SVD $A = U_r \Sigma_r V_r^\top = \sum_{i=1}^{r} \sigma_i u^{(i)} v^{(i)\top}$, a sum of r rank-one matrices. That is the outer-product view of a matrix product again.

The best low-rank approximation

Truncating that sum after k terms gives $A_k = \sum_{i=1}^{k} \sigma_i u^{(i)} v^{(i)\top}$. The **Eckart–Young theorem** [6] says that A_k is the best rank-k approximation of A in the Frobenius norm:

$$\min_{\operatorname{rank}(B) \leq k} \|A - B\|_F^2 = \|A - A_k\|_F^2 = \sum_{i>k} \sigma_i^2.$$

The same A_k is also optimal in the spectral norm, with error $\|A - A_k\|_2 = \sigma_{k+1}$ [4]. In words:

throw away the smallest singular values, and the energy you lose is exactly what you threw away. Storing A_k needs $k(m + n + 1)$ numbers instead of mn.

Figure 2 applies this to a 256 × 256 grayscale photograph. Natural images have rapidly decaying singular values, so a rank-50 approximation already keeps the main structure. The streaky artifacts at low rank come from the fact that each rank-one term is an outer product of a column pattern and a row pattern.

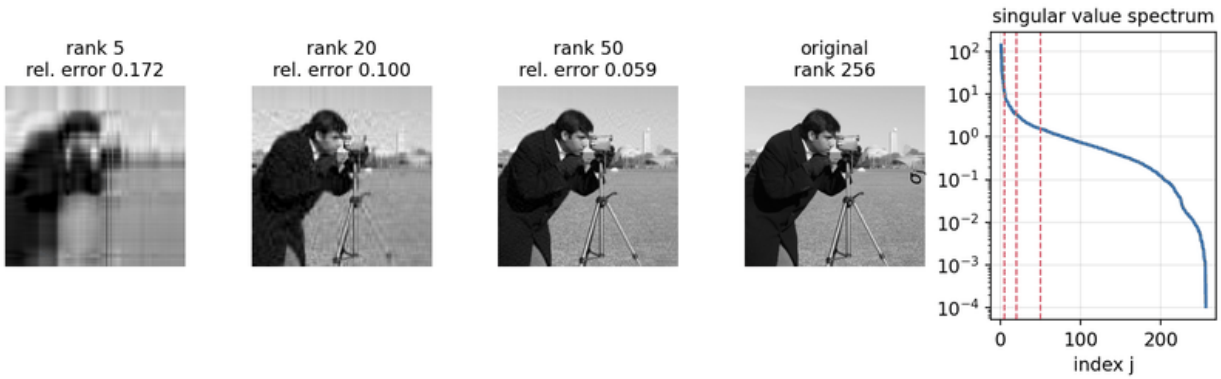


Figure 2 — Rank-k approximations of the 256×256 'camera' image from scikit-image, with the relative Frobenius error in each title. Right: singular values on a log scale; dashed lines mark $k = 5, 20, 50$. The error of each approximation is set by the singular values to the right of its line.

```
import numpy as np

rng = np.random.default_rng(0)
A = rng.normal(size=(12, 8))
U, s, Vt = np.linalg.svd(A, full_matrices=False)
assert np.allclose(U @ np.diag(s) @ Vt, A)
k = 3
Ak = (U[:, :k] * s[:k]) @ Vt[:k]
err_F = np.linalg.norm(A - Ak, "fro")**2
err_2 = np.linalg.norm(A - Ak, 2)
assert np.isclose(err_F, (s[k:]**2).sum()) and np.isclose(err_2, s[k])
print("rank-k squared Frobenius error:", round(err_F, 3))
print("rank-k spectral error = sigma_{k+1}:", round(err_2, 3), round(s[k], 3))
# Eigenvalues of A^T A are squared singular values of A
assert np.allclose(np.sort(np.linalg.eigvalsh(A.T @ A))[::-1], s**2)
print("storage: full", A.size, "numbers; rank-3 factors", k * (12 + 8 + 1))
```

`full_matrices=False` returns the thin SVD, which is what you almost always want. `(U[:, :k] * s[:k])` scales the columns of U by broadcasting instead of building a diagonal matrix. Both Eckart–Young identities hold to machine precision.

One caveat: the directions with the largest singular values carry the most energy, not necessarily the most useful information. For classification, a small-variance direction can be the one that separates the classes. PR 05 contrasts PCA with discriminant methods for exactly this reason.

The Moore–Penrose pseudo-inverse

Plain version. When a matrix has no inverse, the pseudo-inverse gives the next best thing: the answer that fits the data as closely as possible, and among equally good answers, the smallest one.

Only square, full-rank matrices have inverses. For any $A \in \mathbb{R}^{m\times n}$, the **Moore–Penrose pseudo-inverse** $A^+ \in \mathbb{R}^{n\times m}$ is the unique matrix satisfying the four Penrose conditions [7]:

$$AA^+A = A, \quad A^+AA^+ = A^+, \quad (AA^+)^\top = AA^+, \quad (A^+A)^\top = A^+A.$$

It is easiest to compute from the SVD. If $A = U\Sigma V^\top$, then

$$A^+ = V\Sigma^+U^\top,$$

where $\Sigma^+ \in \mathbb{R}^{n\times m}$ is formed by transposing Σ and replacing each nonzero singular value σ_i by $1/\sigma_i$ (zeros stay zero). Goodfellow et al. give the equivalent limit $A^+ = \lim_{\alpha\searrow 0}(A^\top A + \alpha I)^{-1}A^\top$ [1], which already hints at ridge regression.

The pseudo-inverse gives the right answer in every regime of the rank table:

- **Tall, full column rank** $(m > n)$. $Ax = b$ usually has no exact solution. Then $x = A^+b = (A^\top A)^{-1}A^\top b$ is the **least-squares** solution: it minimizes $\|Ax - b\|_2^2$.
- **Wide, full row rank** $(m < n)$. There are infinitely many solutions. Then $x = A^+b = A^\top(AA^\top)^{-1}b$ is the solution with the smallest $\|x\|_2$, the **minimum-norm** solution. Any other solution adds a null-space vector, which can only increase the norm.
- **Rank-deficient.** A^+b is both: among all least-squares minimizers, the one with the smallest norm.

The minimum-norm property is a small window into a modern question. Gradient descent on an under-determined least-squares problem, started from zero, converges to the minimum-norm solution, because every gradient $A^\top(Ax - b)$ lies in the row space of A. Over-parameterized networks are not linear, but this simple case is the usual starting point for thinking about which of many perfect fits a training algorithm picks.

Least squares, the normal equations and conditioning

For a regression design matrix $X \in \mathbb{R}^{n\times d}$ and targets $y \in \mathbb{R}^n$, least squares minimizes $\|Xw - y\|_2^2$.

Setting the gradient to zero gives the **normal equations**

$$X^\top Xw = X^\top y.$$

These are correct but numerically risky. The sensitivity of a linear problem is measured by the **condition number** $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$. Roughly, relative errors in the data can be amplified by up to a factor κ in the solution [5]. Forming $X^\top X$ squares the condition number, $\kappa(X^\top X) = \kappa(X)^2$, so a mildly ill-conditioned problem becomes a severely ill-conditioned one. Library least-squares solvers (`np.linalg.lstsq`, `torch.linalg.lstsq`) work with QR or SVD factorizations of X directly and avoid that squaring [4].

When two features are almost identical, $\sigma_{\min}$ is tiny, and the individual coefficients become extremely sensitive even when predictions are fine. **Ridge regression** [8] adds a penalty $\lambda\|w\|_2^2$ with $\lambda > 0$:

$$w_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y = \sum_i \frac{\sigma_i}{\sigma_i^2 + \lambda} \, (u^{(i)\top} y) \, v^{(i)}.$$

The SVD form on the right shows what ridge does. Plain least squares uses the factor $1/\sigma_i$, which explodes for small σ_i. Ridge uses $\sigma_i/(\sigma_i^2+\lambda)$, which stays bounded by $1/(2\sqrt{\lambda})$. Large-σ directions are nearly untouched; small-σ directions are suppressed. The price is bias. In neural networks the same penalty is called **weight decay**.

```
import numpy as np

rng = np.random.default_rng(0)
# Over-determined: more equations than unknowns -> least squares
x = np.linspace(-1, 1, 40)
X = np.c_[np.ones_like(x), x]
y = 2 + 3*x + rng.normal(0, .05, len(x))
w, residuals, rank, sv = np.linalg.lstsq(X, y, rcond=None)
print("intercept and slope:", w.round(3), "rank:", rank)
assert np.allclose(w, np.linalg.pinv(X) @ y)

# Under-determined: fewer equations than unknowns -> minimum-norm solution
A = rng.normal(size=(2, 5)); b = rng.normal(size=2)
w_min = np.linalg.pinv(A) @ b
null = np.linalg.svd(A)[2][-1]             # a direction with A @ null = 0
w_other = w_min + 0.7 * null
print("both solve Aw=b:", np.allclose(A @ w_min, b), np.allclose(A @ w_other, b))
print("norms:", np.linalg.norm(w_min).round(3), "<", np.linalg.norm(w_other).round(3))

# Near-collinear features: condition number and ridge
z = rng.normal(size=200)
F = np.c_[z, z + 1e-4 * rng.normal(size=200)]
t = z + 0.01 * rng.normal(size=200)
print("cond(F) = %.1e, cond(F^T F) = %.1e" % (np.linalg.cond(F), np.linalg.cond(F.T @ F)))
w_ls = np.linalg.lstsq(F, t, rcond=None)[0]
lam = 1e-2
w_ridge = np.linalg.solve(F.T @ F + lam * np.eye(2), F.T @ t)
print("least squares:", w_ls.round(2), " ridge:", w_ridge.round(3))
```

The line fit recovers intercept ≈ 2.00 and slope ≈ 3.01, and `pinv` agrees with `lstsq`. The underdetermined system has two exact solutions; the pseudo-inverse one has the smaller norm (0.885 versus 1.128). In the collinear case, $\kappa(F) \approx 2 \times 10^4$ while $\kappa(F^\top F) \approx 4 \times 10^8$, its square. Least squares returns wild coefficients $(-2.63, 3.63)$ whose sum is still about 1, so its predictions are fine. Ridge returns the stable split $(0.5, 0.5)$. The intercept column in the first example lets the line have a nonzero offset. In a real pipeline, fit any centering and scaling on the training set only, then apply it unchanged to validation and test data.

The trace operator

Plain version. The trace adds up the diagonal. It looks too simple to matter, but it lets you rewrite awkward sums as matrix expressions and move matrices around inside them.

The **trace** of a square matrix is

$$\operatorname{Tr}(A) = \sum_i A_{ii}.$$

Its useful properties are:

- **Frobenius norm.** $\|A\|_F = \sqrt{\operatorname{Tr}(AA^\top)} = \sqrt{\operatorname{Tr}(A^\top A)}$.
- **Transpose.** $\operatorname{Tr}(A) = \operatorname{Tr}(A^\top)$.
- **Cyclic property.** $\operatorname{Tr}(ABC) = \operatorname{Tr}(CAB) = \operatorname{Tr}(BCA)$ whenever the products are defined, even when the matrices are not square and the intermediate products have different shapes. In particular $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$ for $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times m}$. You may cycle, but not arbitrarily reorder: $\operatorname{Tr}(ABC) \neq \operatorname{Tr}(ACB)$ in general.
- **Scalars.** A scalar is its own trace, $a = \operatorname{Tr}(a)$. This turns a quadratic form into a trace: $x^\top Ax = \operatorname{Tr}(x^\top Ax) = \operatorname{Tr}(Axx^\top)$.
- **Eigenvalues.** $\operatorname{Tr}(A) = \sum_i \lambda_i$. For a covariance matrix, that is the total variance, the sum of the variances along every principal direction.

The scalar trick is how many expectations are evaluated. For a random vector x with mean zero and covariance Σ, $\mathbb{E}[x^\top Ax] = \mathbb{E}[\operatorname{Tr}(Axx^\top)] = \operatorname{Tr}(A\Sigma)$, because trace and expectation are both linear.

The determinant

Plain version. The determinant of a square matrix says how much it scales area (in 2-D) or volume (in higher dimensions). If it is zero, the matrix flattens space and cannot be undone.

The **determinant** $\det(A)$ of a square matrix is a single number with these properties:

- $\det(A) = \prod_i \lambda_i$, the product of the eigenvalues.
- $|\det(A)|$ is the factor by which A scales volume. The unit cube maps to a parallelepiped of volume $|\det(A)|$.
- The sign records orientation. A negative determinant means A includes a reflection.
- $\det(A) = 0$ exactly when A is singular: some direction is collapsed to zero, so information is lost.
- $\det(AB) = \det(A)\det(B)$, $\det(A^\top) = \det(A)$, and $\det(A^{-1}) = 1/\det(A)$.

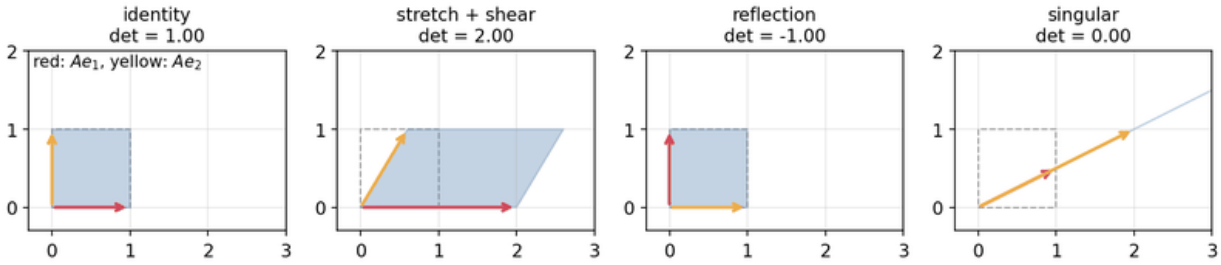


Figure 3 — The determinant as signed area. Dashed: the unit square. Shaded: its image, spanned by Ae1 (red) and Ae2 (yellow). A shear does not change area; a reflection flips the sign; a singular matrix collapses the square onto a line (det = 0).

> **Interactive demonstration (online)**: open the online edition of this tutorial to use this demonstration.Open demonstration

Determinants appear in machine learning in two main places. The multivariate Gaussian density contains $\det(\Sigma)^{-1/2}$, the volume normalization of the covariance; ML 01B uses it. Normalizing flows (ML 06) need $\log|\det J|$ of a Jacobian matrix J to account for how an invertible network stretches probability mass. In both cases you need the logarithm, never the raw value: the determinant of a 500×500 diagonal matrix with entries 10^{-3} is 10^{-1500}, which underflows to zero in float64.

```
import numpy as np

rng = np.random.default_rng(2)
A, B, C = rng.normal(size=(3, 4)), rng.normal(size=(4, 5)), rng.normal(size=(5, 3))
print("cyclic trace:", np.isclose(np.trace(A @ B @ C), np.trace(C @ A @ B)),
      np.isclose(np.trace(A @ B @ C), np.trace(B @ C @ A)))

S = rng.normal(size=(4, 4))
lam = np.linalg.eigvals(S)
print("trace = sum of eigenvalues:", np.isclose(np.trace(S), lam.sum().real))
print("det = product of eigenvalues:", np.isclose(np.linalg.det(S), np.prod(lam).real))

# log-determinant of a covariance without overflow/underflow
Sigma = np.diag(np.full(500, 1e-3))
print("det underflows:", np.linalg.det(Sigma))
sign, logdet = np.linalg.slogdet(Sigma)
print("slogdet:", sign, round(logdet, 2))
```

```
L = np.linalg.cholesky(Sigma)
print("via Cholesky:", round(2 * np.log(np.diag(L)).sum(), 2))
```

The direct determinant prints `0.0`, while `slogdet` and the Cholesky route both give $\log\det = 500\log 10^{-3} \approx -3453.88$. For a PD matrix, $\det(A) = \prod_i L_{ii}^2$ when $A = LL^\top$, so $\log\det A = 2\sum_i \log L_{ii}$.

Worked example: deriving principal component analysis

Plain version. PCA looks for the flat subspace that the data cloud lies closest to. Projecting onto it compresses each point to a few numbers while losing as little as possible.

PCA goes back to Pearson's "lines and planes of closest fit" [9] and Hotelling's analysis of "principal components" [10]; Jolliffe and Cadima give a modern review [11]. We derive it in the style of [1], using only the tools above.

Setup. We have n points $x^{(1)}, \dots, x^{(n)} \in \mathbb{R}^d$ and want to store each one with only $l < d$ numbers. First subtract the mean $\mu = \frac{1}{n}\sum_i x^{(i)}$ from every point, so we can assume the data are centered. Stack the centered points as rows of $X \in \mathbb{R}^{n\times d}$.

Step 1: choose the form of the code. We use a linear **decoder** $g(c) = Dc$ with $D \in \mathbb{R}^{d\times l}$. To make the problem well-posed we require the columns of D to be orthonormal, $D^\top D = I_l$. (Without this, we could shrink c and enlarge D forever.)

Step 2: find the best code for a fixed decoder. For a given x, choose c to minimize the reconstruction error:

$$c^* = \arg\min_c \|x - Dc\|_2^2 = \arg\min_c \left(-2x^\top Dc + c^\top D^\top Dc\right) = \arg\min_c \left(-2x^\top Dc + c^\top c\right).$$

We expanded the square, dropped $x^\top x$ (it does not depend on c) and used $D^\top D = I$. The gradient with respect to c is $-2D^\top x + 2c$. Setting it to zero gives the **encoder**

$$f(x) = D^\top x.$$

So encoding is a matrix–vector product, and the reconstruction is $r(x) = g(f(x)) = DD^\top x$, the orthogonal projection of x onto the span of D.

Step 3: find the best decoder. Now choose D to minimize the total reconstruction error over all points:

$$D^* = \arg\min_D \|X - XDD^\top\|_F^2 \quad \text{subject to } D^\top D = I_l.$$

Start with $l = 1$, so D is a single unit vector d. Write the Frobenius norm as a trace and expand:

$$\begin{aligned}\|X - Xdd^\top\|_F^2 &= \operatorname{Tr}\left((X - Xdd^\top)^\top (X - Xdd^\top)\right) \\ &= \operatorname{Tr}(X^\top X) - 2\operatorname{Tr}(X^\top Xdd^\top) + \operatorname{Tr}(dd^\top X^\top Xdd^\top) \\ &= \operatorname{Tr}(X^\top X) - 2\,d^\top X^\top Xd + d^\top X^\top Xd\,(d^\top d) \\ &= \operatorname{Tr}(X^\top X) - d^\top X^\top Xd.\end{aligned}$$

The cyclic property moved $d^\top$ to the front in the second and third terms, and $d^\top d = 1$ simplified the last line. The first term does not depend on d. So minimizing the error is the same as

$$d^* = \arg\max_d \; d^\top X^\top Xd \quad \text{subject to } d^\top d = 1.$$

That is a Rayleigh quotient of the symmetric PSD matrix $X^\top X$. By the eigendecomposition section, it is maximized by the **eigenvector of $X^\top X$ with the largest eigenvalue**.

Step 4: more than one component. Repeating the argument for general l (for example, by induction, requiring each new direction to be orthogonal to the previous ones) gives the answer: the columns of D are the eigenvectors of $X^\top X$ for the l largest eigenvalues [1]. The minimum error equals the sum of the discarded eigenvalues.

Step 5: the variance view and the SVD link. Dividing by n turns $X^\top X$ into the sample covariance $S = \frac{1}{n} X^\top X$ (some texts use $n - 1$). The quantity $d^\top Sd$ is the variance of the projected data Xd. So "minimize reconstruction error" and "maximize retained variance" select the same directions: Pearson's and Hotelling's views agree. Finally, if the centered data matrix has thin SVD $X = U\Sigma V^\top$, then $X^\top X = V\Sigma^2 V^\top$. The principal directions are the right singular vectors V, the eigenvalues are σ_i^2, and the codes are $XV_l = U_l\Sigma_l$. In practice PCA is computed with the SVD of X rather than by forming the covariance, for the conditioning reason we met in least squares.

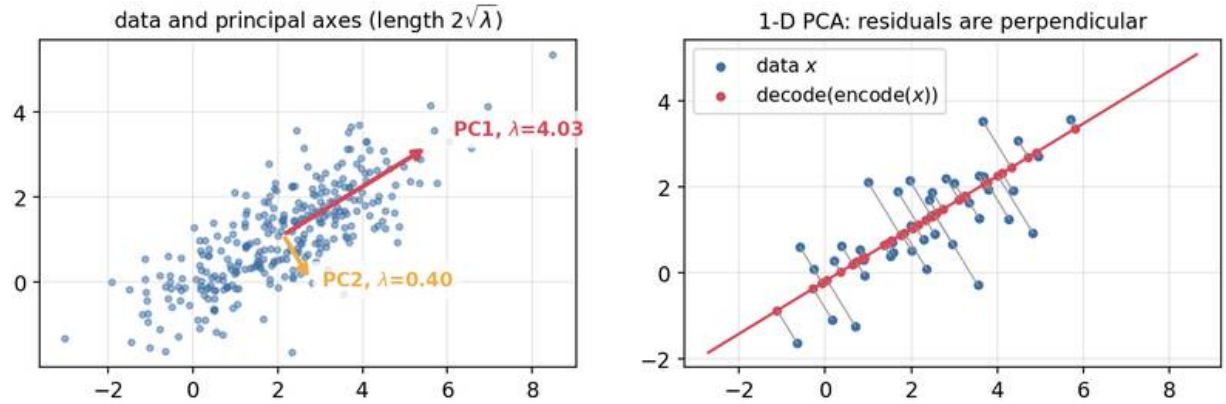


Figure 4 — PCA on 300 correlated 2-D points. Left: principal axes from the eigendecomposition of the sample covariance, drawn with length $2\sqrt{\lambda}$. Right: a 1-D PCA reconstruction $\text{decode}(\text{encode}(x))$; each gray residual meets the red line at a right angle, because the reconstruction is an orthogonal projection.

```
import numpy as np
```

```
rng = np.random.default_rng(0)
n, d, l = 500, 5, 2
Z = rng.normal(size=(n, l))
X = Z @ rng.normal(size=(l, d)) + 0.1 * rng.normal(size=(n, d)) + 3.0

mu = X.mean(axis=0)
Xc = X - mu                                  # centre first
S = Xc.T @ Xc / n                            # sample covariance (d x d)
lam, Q = np.linalg.eigh(S)
lam, Q = lam[::-1], Q[:, ::-1]               # descending
D = Q[:, :l]                                 # decoder matrix, orthonormal columns

codes = Xc @ D                               # encode: f(x) = D^T (x - mu)
X_hat = codes @ D.T + mu                     # decode: g(c) = D c + mu
mse = np.mean(np.sum((X - X_hat)**2, axis=1))
print("explained variance ratio:", (lam[:l].sum() / lam.sum()).round(4))
print("reconstruction MSE:", mse.round(5), " sum of discarded eigenvalues:",
 ↪  lam[l:].sum().round(5))
assert np.isclose(mse, lam[l:].sum())

# Same subspace from the SVD of the centred data matrix
U, s, Vt = np.linalg.svd(Xc, full_matrices=False)
assert np.allclose(s[:l]**2 / n, lam[:l])
assert np.allclose(np.abs(Vt[:l] @ D), np.eye(l), atol=1e-8)
print("SVD and eigendecomposition agree")
```

The synthetic data are 5-D points that really live near a 2-D plane, plus small noise. Two components explain 99.36% of the variance, and the average squared reconstruction error equals the sum of the three discarded eigenvalues, exactly as Step 4 predicts. The SVD route finds the same directions (up to sign, which is arbitrary for eigenvectors).

Three practical notes. Always center the data, or the first "component" just points at the mean. Standardize features if they have incomparable units, because PCA is not scale-invariant. And fit the mean and directions on training data only. PCA is the linear ancestor of autoencoders; ML 05A develops the probabilistic version and its nonlinear successors.

Matrix calculus you will meet

Plain version. Training means following gradients, and most gradients in ML are a handful of matrix formulas used over and over.

For a scalar function $f(x)$ of a vector $x \in \mathbb{R}^n$, the **gradient** $\nabla_x f \in \mathbb{R}^n$ collects the partial derivatives $\partial f / \partial x_i$. For a vector function $f : \mathbb{R}^n \to \mathbb{R}^m$, the **Jacobian** $J \in \mathbb{R}^{m \times n}$ has entries $J_{ij} = \partial f_i / \partial x_j$. The rules you will use most are:

Function	Gradient	Note
$a^\top x$	a	linear function
$x^\top A x$	$(A + A^\top)x$	becomes $2Ax$ when A is symmetric
$\frac{1}{2}\lVert Xw - y\rVert_2^2$	$X^\top(Xw - y)$	least-squares loss
$\lVert x\rVert_2^2$	$2x$	weight decay
$\operatorname{Tr}(AB)$ w.r.t. A	$B^\top$	used with the trace tricks above

Each can be verified by writing the expression as a sum over indices and differentiating term by term [2]. The **chain rule** for vector functions multiplies Jacobians: if $y = g(x)$ and $L = f(y)$ is a scalar, then $\nabla_x L = J_g^\top \nabla_y L$. Backpropagation applies this rule layer by layer, multiplying transposed local Jacobians by the incoming gradient without ever building the full Jacobians, which would be enormous. ML 02 builds this up in detail, and ML 01C adds the Hessian, the matrix of second derivatives.

Setting the least-squares gradient to zero recovers the normal equations from the previous section. The Hessian of that loss is $X^\top X$, which is PSD, so the loss is convex and every stationary point is a global minimum. A quick finite-difference check confirms the formula:

```
import numpy as np

rng = np.random.default_rng(3)
X, y = rng.normal(size=(30, 4)), rng.normal(size=30)
w = rng.normal(size=4)
L = lambda w: 0.5 * np.sum((X @ w - y)**2)
grad = X.T @ (X @ w - y)                    # analytic gradient
eps = 1e-6
num = np.array([(L(w + eps*e) - L(w - eps*e)) / (2*eps) for e in np.eye(4)])
print("max gradient mismatch:", np.abs(grad - num).max())
# Setting the gradient to zero gives the normal equations
w_star = np.linalg.lstsq(X, y, rcond=None)[0]
print("gradient at solution ~ 0:", np.abs(X.T @ (X @ w_star - y)).max() < 1e-10)
```

The analytic and numerical gradients agree to about 10^{-8}, the size expected from finite-difference rounding.

Modern view

The definitions in this chapter are a century old, but the way they are used in machine learning keeps changing. Four lines of work are worth knowing about.

Randomized numerical linear algebra. Exact SVDs of large matrices are expensive, roughly $O(mn \min(m, n))$ operations. Halko, Martinsson and Tropp's SIAM Review survey [12] made a simple idea standard: multiply A by a thin random matrix Ω to "sketch" its range, orthonormalize the result to get a basis Q, and compute an exact SVD of the small matrix $Q^\top A$. With a

little oversampling and a few power iterations, the result is accurate for matrices whose spectrum decays. Martinsson and Tropp's *Acta Numerica* survey [13] covers the theory and a wide range of algorithms: norm estimation, random embeddings, regression, low-rank approximation, Nyström approximation of PSD matrices, streaming algorithms and kernel matrices from machine learning. Murray et al. [14] (LAPACK Working Note 299, 2023) take the software view. They note that RandNLA methods have entered MATLAB, the NAG Library, NVIDIA's cuSOLVER and scikit-learn, and they argue for standard "RandBLAS" and "RandLAPACK" libraries. scikit-learn's randomized PCA solver is a direct descendant. The core algorithm fits in a few lines:

```
import numpy as np

def randomized_svd(A, k, oversample=10, power_iters=2, seed=0):
    rng = np.random.default_rng(seed)
    Omega = rng.normal(size=(A.shape[1], k + oversample))  # random test matrix
    Y = A @ Omega                                          # sample the range of A
    for _ in range(power_iters):                           # sharpen a slow spectrum
        Y = A @ (A.T @ Y)
    Q, _ = np.linalg.qr(Y)                                 # orthonormal basis
    B = Q.T @ A                                            # small (k+p) x n matrix
    Ub, s, Vt = np.linalg.svd(B, full_matrices=False)
    return (Q @ Ub)[:, :k], s[:k], Vt[:k]

rng = np.random.default_rng(1)
m, n, r = 2000, 500, 20
A = rng.normal(size=(m, r)) @ rng.normal(size=(r, n)) + 0.01 * rng.normal(size=(m, n))
U, s, Vt = randomized_svd(A, k=r)
s_exact = np.linalg.svd(A, compute_uv=False)[:r]
print("max relative error in top singular values:", np.max(np.abs(s - s_exact) / s_exact))
rel = np.linalg.norm(A - (U * s) @ Vt) / np.linalg.norm(A)
print("relative reconstruction error:", round(rel, 5))
```

On this nearly rank-20 matrix the top 20 singular values match the exact ones to about 10^{-15} relative error, and the rank-20 reconstruction error (about 0.2%) is set by the added noise. The power iterations are what make it work when the spectrum decays slowly. For a matrix with a flat spectrum, no rank-k method can do well, randomized or not.

Low-rank structure in large models. The outer-product view of matrix products has become a practical tool for adapting large networks. LoRA [15] freezes a pretrained weight $W_0 \in \mathbb{R}^{d\times k}$ and learns an update $\Delta W = BA$ with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$ and $r \ll \min(d,k)$. Only $r(d+k)$ numbers are trained instead of dk; for $d = k = 4096$ and $r = 8$, that is 65,536 instead of about 16.8 million per matrix. B starts at zero, so training begins exactly at the pretrained model. GaLore [16] (ICML 2024) keeps all weights trainable but projects the *gradients* onto a low-rank subspace, which shrinks the optimizer's memory. Its authors argue that restricting the weight update itself to a low-rank subspace, as LoRA does, can underperform full-rank training. Both rely on the Eckart–Young intuition that much of a matrix's useful content lives in a few singular directions. Whether that holds for a given task is an empirical question, not a theorem.

Controlling the spectral norm. A network's sensitivity to input perturbations is bounded by the product of its layers' spectral norms (for 1-Lipschitz activations such as ReLU). Spectral normalization [17] divides each weight matrix by an estimate of its largest singular value, obtained cheaply with one step of power iteration per training step. It was introduced to stabilize the discriminator in GANs (ML 06), and the same idea appears wherever a Lipschitz bound is wanted.

Beyond matrices. Convolution kernels and video features are naturally tensors with three or more axes. Kolda and Bader's SIAM Review survey [18] is the standard introduction to tensor decompositions. It covers the CP decomposition (a sum of rank-one tensors, generalizing the outer-product view) and the Tucker decomposition (a small core tensor multiplied by a matrix along each mode, generalizing the SVD). Unlike matrices, computing the rank of a tensor is hard in general, and best low-rank approximations need not exist. The clean Eckart–Young picture does not carry over. These decompositions are used to compress layers and to analyze multi-way data.

Open questions. Why do trained weight matrices often have rapidly decaying spectra, and how does that relate to generalization? How low can the rank of an adaptation be before quality drops, and does that depend on the task or the model? How should randomized algorithms be combined with low-precision hardware, where rounding errors are much larger than in float64? None of these have settled answers. The tools in this chapter are what you need to read the papers that study them. For broader textbook treatments that connect linear algebra to deep learning, see [1], [2] and [19].

Key takeaways

- Name every axis. Most silent bugs are shape or broadcasting errors, and `assert` statements catch them.
- A matrix product has four readings (entries, columns, rows, sums of outer products); choose the one that makes the problem clear.
- Rank and null space decide whether $Ax = b$ has zero, one or infinitely many solutions. Never form A^{-1} in code; use a solver.
- Symmetric matrices have real eigenvalues and orthonormal eigenvectors; the signs of the eigenvalues say whether a quadratic form is a bowl, a saddle or a dome.
- The SVD exists for every matrix: rotate, stretch, rotate. Truncating it gives the best low-rank approximation, with error equal to the discarded singular values.
- The pseudo-inverse gives least-squares and minimum-norm solutions. Small singular values amplify noise; the condition number measures how much, and ridge regularization tames it.
- Trace and determinant summarize eigenvalues (sum and product). Use the cyclic trace property in derivations and log-determinants in code.
- PCA is a projection onto the top eigenvectors of the covariance, equivalently the top right singular vectors of the centered data. Minimizing reconstruction error and maximizing variance give the same answer.

Exercises

1. **Shapes and bias.** A batch has shape 32×64 and a weight matrix has shape 64×10. What shape should the bias have, and what goes wrong if you use a bias of shape $(32, 1)$ instead?

   Hint

   A bias of shape $(10,)$ broadcasts over the 32 rows: one offset per output unit, shared by every example. A $(32, 1)$ bias broadcasts across the 10 columns, giving one offset per *example*. The code runs, but the "bias" now depends on the position of an example in the batch, which is meaningless and changes with batch size.

2. **Projection.** Let $Q \in \mathbb{R}^{n \times k}$ have orthonormal columns and $P = QQ^\top$. Show that $P^2 = P$, $P^\top = P$, and that $x - Px$ is orthogonal to every column of Q. What are the eigenvalues of P?

   Hint

   $P^2 = Q(Q^\top Q)Q^\top = QIQ^\top = P$. For orthogonality compute $Q^\top(x - Px) = Q^\top x - Q^\top QQ^\top x = 0$. From $P^2 = P$, any eigenvalue satisfies $\lambda^2 = \lambda$, so $\lambda \in \{0, 1\}$: eigenvalue 1 with multiplicity k (the subspace) and 0 with multiplicity $n - k$ (its orthogonal complement). Hence $\operatorname{Tr}(P) = k$.

3. **SVD by hand.** Find the singular values of $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$ by computing the eigenvalues of $A^\top A$. Check that $\|A\|_F^2 = \sigma_1^2 + \sigma_2^2$ and $|\det A| = \sigma_1 \sigma_2$.

   Hint

   $A^\top A = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$ has eigenvalues 45 and 5, so $\sigma_1 = 3\sqrt{5}$, $\sigma_2 = \sqrt{5}$. Then $\sigma_1^2 + \sigma_2^2 = 50 = 9 + 16 + 25$, and $\sigma_1 \sigma_2 = 15 = |\det A| = |3 \cdot 5 - 0 \cdot 4|$. In general $|\det A| = \prod_i \sigma_i$ for a square matrix, because $|\det U| = |\det V| = 1$.

4. **Conditioning and ridge.** Starting from the least-squares code, replace the second feature by an exact copy of the first. What does `np.linalg.lstsq` return, and why is it the minimum-norm solution? Then derive the SVD form of the ridge solution and show that each factor $\sigma/(\sigma^2 + \lambda)$ is at most $1/(2\sqrt{\lambda})$.

   Hint

   With identical columns, the rank is 1 and the solution set is a line $w_1 + w_2 = c$. `lstsq` uses the SVD and returns the pseudo-inverse solution, which splits the weight equally, $w_1 = w_2 = c/2$, because that point on the line is closest to the origin. For the bound, maximize $g(\sigma) = \sigma/(\sigma^2 + \lambda)$: $g'(\sigma) = 0$ at $\sigma = \sqrt{\lambda}$, giving $g = 1/(2\sqrt{\lambda})$.

5. **PCA error.** Using the derivation in this chapter, show that the total reconstruction error of l-component PCA equals $\sum_{i>l} \lambda_i(X^\top X)$, and that the fraction of variance retained is $\sum_{i\le l} \lambda_i / \sum_i \lambda_i$. Then run PCA on `sklearn.datasets.load_digits()` and find the smallest l that keeps 90% of the variance.

   Hint

   Write $\|X - XDD^\top\|_F^2 = \mathrm{Tr}(X^\top X) - \mathrm{Tr}(D^\top X^\top X D)$ (use $D^\top D = I$ and the cyclic property). With D the top-l eigenvectors, the second trace is $\sum_{i\le l} \lambda_i$, and $\mathrm{Tr}(X^\top X) = \sum_i \lambda_i$. For the digits, center the 1797×64 matrix, compute its singular values, and take the cumulative sum of σ_i^2. Report your own number rather than trusting a quoted one.

6. **Determinants and volume.** A 2×2 matrix maps the unit square to a parallelogram with vertices 0, $(2,1)$, $(1,3)$ and $(3,4)$. Write down a matrix that does this, compute its determinant, and explain why a normalizing flow must track $\log|\det J|$ for every layer.

   Hint

   Put the edge vectors in the columns: $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, with $\det A = 5$, so areas grow by a factor of 5. If a map stretches volume by 5, probability density must shrink by the same factor to keep total probability equal to 1. Stacking layers multiplies these factors, so their logarithms add, which is why flows sum log-determinants layer by layer.

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ML 01B • Probability and Information Theory

Random variables, common distributions, Bayes' rule, expectation and covariance, likelihood, entropy, cross-entropy, KL divergence and structured models, connected to how neural networks are trained and evaluated.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-01b-probability-information/

What you'll learn

- Why machine learning needs probability, and the two common ways of reading a probability (frequency and degree of belief).
- How random variables, probability mass functions and densities work, and how marginal, conditional and joint probabilities relate through the sum rule, the product rule and the chain rule.
- How expectation, variance, covariance and correlation summarize a distribution, and why zero covariance does not mean independence.
- The distributions you will meet constantly in deep learning — Bernoulli, categorical, Gaussian, exponential, Laplace, Dirac, empirical and mixtures — plus the sigmoid and softplus functions that parameterize them.
- How Bayes' rule, the change-of-variables formula, maximum likelihood and MAP estimation work, with tested NumPy code.
- What entropy, cross-entropy, KL divergence and mutual information measure, why the classification loss is a cross-entropy, and how structured (graphical) models factor a large joint distribution.

The big picture

A neural network never sees the whole world. It sees a finite training set, measured by imperfect sensors, labelled by busy people, and then it is asked about images it has never seen. Every step of that story involves **uncertainty**. Probability theory is the mathematics of uncertainty: it tells us how to describe what we do not know, how to combine evidence, and how to make predictions that admit doubt. Information theory, its close relative, measures *how much* uncertainty there is and how costly it is to use the wrong distribution. Together they explain almost every loss function in this site's ML and DL tracks.

> **Explain like I'm five**
>
> Imagine a bag of marbles you cannot see into. **Probability** is your best guess about what you will pull out next: "probably red, maybe blue, almost never green." Every time you pull a marble, you learn a little more about the bag and update your guess — that updating is **Bayes' rule**. **Entropy** is how surprised you expect to be: a bag of all-red marbles never surprises you; a bag with ten colours in

equal amounts surprises you a lot. A machine learning model is a very fancy guesser, and training means making its guesses less surprised by the real marbles.

This chapter follows the "Applied Math" part of *Deep Learning* by Goodfellow, Bengio and Courville [1], with additional material from Bishop and Bishop [2], Murphy [3] and *Mathematics for Machine Learning* [4]. It sits between ML 01A ·Linear Algebra and ML 01C ·Numerical Computation. If you are heading to the Pattern Recognition track, PR 01 ·Bayes Decisions builds directly on the Bayes-rule section here.

Why probability?

Plain version. Computer science usually deals with certain facts: a bit is 0 or 1, a program either halts or does not. Machine learning cannot work that way because the data, the world and the model are all imperfect.

Goodfellow et al. [1] list three sources of uncertainty:

1. **Inherent stochasticity.** Some systems are genuinely random — photon arrival at a camera sensor (shot noise), the roll of a die, or the exact moment a pedestrian steps off a curb.
2. **Incomplete observability.** Even a deterministic system looks random if we cannot see all of its variables. A single photo of a car does not show whether its brake lights are about to turn on.
3. **Incomplete modelling.** A model must discard information to be useful. A classifier that sees 32×32 thumbnails cannot recover details that were lost when the image was downsampled, so its predictions about fine-grained classes must remain uncertain.

It is often more practical to use a simple rule with uncertainty ("most birds fly") than a complex rule that is always true ("birds fly, except penguins, ostriches, injured birds, ...").

Two readings of a probability

A probability can be read in two ways, and both obey the same mathematical rules.

- **Frequentist probability** describes long-run frequencies of repeatable events: "if we toss this coin many times, about 70% of tosses land heads".
- **Bayesian probability** describes a *degree of belief* about something that might not be repeatable: "given this X-ray, I am 40% confident the patient has the disease".

If degrees of belief are required to satisfy a few common-sense consistency properties, they turn out to obey exactly the same rules as frequencies, which is why one calculus serves both readings [1], [3]. In practice, deep learning mixes the two: the data are treated as samples from a distribution (frequentist), while the model's output is often read as a belief about the label (Bayesian).

Random variables and probability distributions

Plain version. A random variable is a quantity whose value is not known in advance. A probability distribution says how likely each value is.

We write random variables in roman letters (for example x in [1]); in this tutorial we use X for the variable and x for one of its values. A random variable can be **discrete** (a finite or countable set of values, such as a class label in $\{1, \dots, K\}$) or **continuous** (real values, such as a pixel intensity treated as a real number).

Probability mass functions

A discrete variable has a **probability mass function** (PMF) $P(X = x)$, often shortened to $P(x)$. It must satisfy

$$0 \le P(x) \le 1 \text{ for every } x, \qquad \sum_{x} P(x) = 1.$$

Here the sum runs over every possible value x of X. The first condition says no event is more than certain; the second says that *something* always happens. A joint PMF $P(X = x, Y = y)$ gives the probability that both events happen together.

Probability density functions

A continuous variable has a **probability density function** (PDF) $p(x)$, which must satisfy

$$p(x) \ge 0, \qquad \int p(x)\, dx = 1.$$

A density is *not* a probability. The probability that X falls in an interval $[a, b]$ is the area under the density:

$$P(a \le X \le b) = \int_a^b p(x)\, dx.$$

The probability of any single exact value is zero. Because only areas must be at most one, a density can exceed one: a narrow Gaussian with standard deviation 0.25 peaks at about 1.6 (Figure 1, top right). A common mistake is to report "probability 1.6" for a density value.

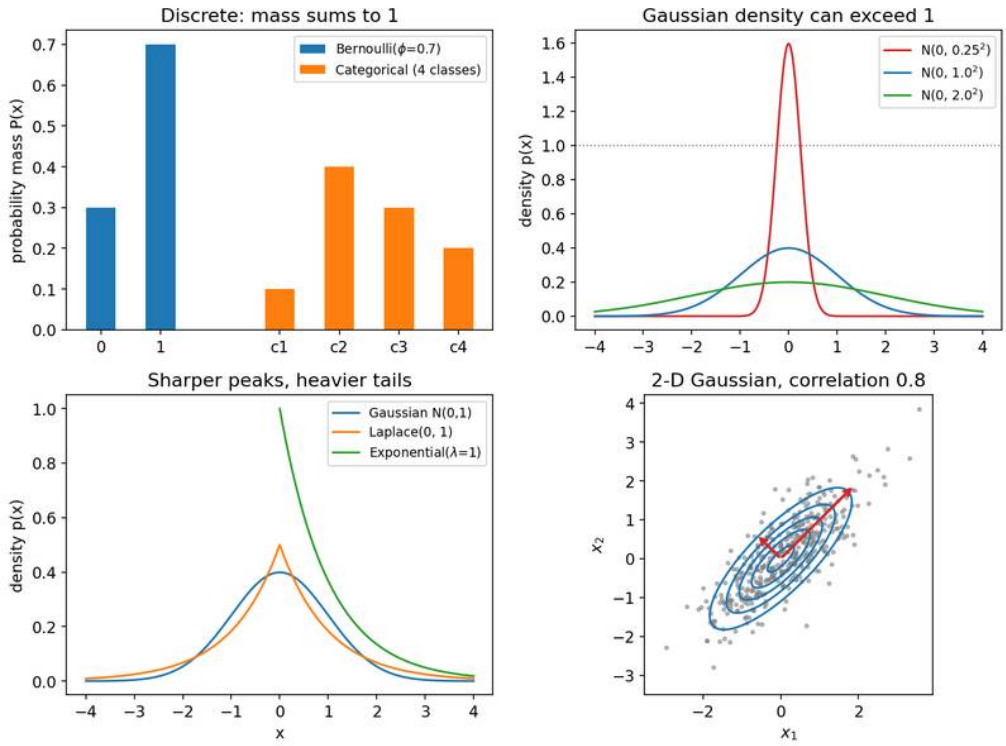


Figure 1 — Common distributions. Top left: discrete masses sum to one. Top right: a density can exceed one because only its area must equal one. Bottom left: the Laplace density has a sharp peak and heavier tails than the Gaussian; the exponential lives on $x \geq 0$. Bottom right: a 2-D Gaussian with correlation 0.8; red arrows show the eigenvectors of its covariance, scaled by two standard deviations.

Marginal probability

Plain version. If you know how two things vary together, you can recover how one of them varies on its own by "summing out" the other.

Given a joint distribution over X and Y, the **marginal** distribution of X is

$$P(x) = \sum_{y} P(x, y) \quad \text{(discrete)}, \qquad p(x) = \int p(x, y)\, dy \quad \text{(continuous)}.$$

This is the **sum rule**. If the joint is written as a table with rows for x and columns for y, the marginal of x is the row sums — historically written in the margin of the page, hence the name.

Conditional probability

Plain version. A conditional probability answers "how likely is this, *given that* I already know that?"

$$P(y \mid x) = \frac{P(x, y)}{P(x)}, \qquad P(x) > 0.$$

$P(y \mid x)$ is the probability of $Y = y$ after we have observed $X = x$. It is only defined when the condition can actually happen. Rearranging gives the **product rule** $P(x, y) = P(y \mid x)P(x)$.

Do not confuse conditioning with intervention. "The probability that the street is wet given that people carry umbrellas" is high, but *forcing* people to carry umbrellas will not wet the street. Questions about interventions belong to causal inference, which needs more than a joint distribution.

The chain rule of probability

Applying the product rule repeatedly factorizes any joint distribution over n variables:

$$P(x^{(1)}, \dots, x^{(n)}) = P(x^{(1)}) \prod_{i=2}^{n} P(x^{(i)} \mid x^{(1)}, \dots, x^{(i-1)}).$$

Here $x^{(i)}$ is the value of the i-th variable. This identity is *exact* — no assumption is involved. It is the backbone of autoregressive models: a language model or an autoregressive image model predicts one token or pixel at a time given all previous ones (ML 08, NLP 01). The modelling choice lies in *how* each conditional is parameterized, not in the factorization itself.

Independence and conditional independence

Two variables are **independent**, written $X \perp Y$, if

$$p(x, y) = p(x)\, p(y) \quad \text{for all } x, y.$$

They are **conditionally independent** given Z, written $X \perp Y \mid Z$, if

$$p(x, y \mid z) = p(x \mid z)\, p(y \mid z) \quad \text{for all } x, y, z.$$

Ice-cream sales and sunburn are dependent, but roughly independent once you condition on the weather. Conditional independence is the main tool for making high-dimensional models tractable — naive Bayes assumes the features are independent given the class; a hidden Markov model assumes the future is independent of the past given the present state. These are *assumptions* [3]: they should be justified by the problem, not adopted only because they make the algebra easy.

Code: a joint table, marginals and conditionals

The following example builds a small joint table for weather and umbrella use, computes marginals and conditionals, and tests independence. It also shows a case where two variables are clearly dependent yet have (almost) zero covariance.

```
import numpy as np

# Joint table P(weather, umbrella): rows = sunny/rainy, columns = no/yes
P = np.array([[0.56, 0.14],
              [0.06, 0.24]])
assert np.isclose(P.sum(), 1)

p_weather = P.sum(axis=1)            # marginal over umbrella
p_umbrella = P.sum(axis=0)           # marginal over weather
p_umb_given_w = P / p_weather[:, None]   # each row sums to 1
print("P(weather):", p_weather)
print("P(umbrella):", p_umbrella)
print("P(umbrella | weather):\n", p_umb_given_w)
print("independent?", np.allclose(P, np.outer(p_weather, p_umbrella)))

# Zero covariance does not imply independence
rng = np.random.default_rng(0)
x = rng.uniform(-1, 1, 200_000)
y = x**2                              # y is a deterministic function of x
print("cov(x, y) ~", np.cov(x, y)[0, 1].round(4))
```

Output: the marginals are $[0.7, 0.3]$ and $[0.62, 0.38]$; the conditional table is $[[0.8, 0.2], [0.2, 0.8]]$; the variables are not independent; and the sample covariance of x and x^2 is about -0.0007, essentially zero, even though y is completely determined by x.

Expectation, variance and covariance

Plain version. The expectation is the long-run average; the variance is how far values typically spread from that average; the covariance says whether two quantities tend to go up and down together.

The **expectation** of a function f under a distribution is

$$\mathbb{E}_{x\sim P}[f(x)] = \sum_{x} P(x) f(x), \qquad \mathbb{E}_{x\sim p}[f(x)] = \int p(x) f(x)\, dx.$$

The subscript $x \sim P$ names the distribution we average over; we drop it when it is obvious. Expectation is **linear**: $\mathbb{E}[\alpha f(x) + \beta g(x)] = \alpha \mathbb{E}[f(x)] + \beta \mathbb{E}[g(x)]$ for constants α, β. Almost every training loss is an expectation over the data distribution, estimated by an average over a mini-batch.

The **variance** measures spread around the mean $\mu = \mathbb{E}[f(x)]$:

$$\mathrm{Var}(f(x)) = \mathbb{E}[(f(x) - \mu)^2] = \mathbb{E}[f(x)^2] - \mu^2.$$

Its square root is the **standard deviation**, which has the same units as $f(x)$.

The **covariance** of two quantities measures how they vary together:

$$\mathrm{Cov}(f(x), g(y)) = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])\,(g(y) - \mathbb{E}[g(y)])].$$

A positive covariance means both tend to be above (or below) their means at the same time; a negative covariance means one tends to be high when the other is low. Its magnitude depends on the units, so we often normalize it into the **correlation**

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} \in [-1, 1].$$

Independence implies zero covariance, but **zero covariance does not imply independence**. Covariance measures only *linear* dependence. In the code above, $y = x^2$ with x uniform on $[-1, 1]$: knowing x tells you y exactly, but the covariance is zero because the relationship is symmetric and nonlinear. Mutual information, introduced later, captures every kind of dependence.

The covariance matrix

For a random vector $\mathbf{x} \in \mathbb{R}^n$ with mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}]$, the **covariance matrix** is

$$\Sigma = \mathrm{Cov}(\mathbf{x}) = \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top], \qquad \Sigma_{ij} = \mathrm{Cov}(x_i, x_j).$$

Its diagonal holds the variances $\Sigma_{ii} = \mathrm{Var}(x_i)$. Σ is symmetric and positive semidefinite, so (by ML 01A) it has real nonnegative eigenvalues and orthogonal eigenvectors. The eigenvectors are the principal axes of the data cloud and the eigenvalues are the variances along them — the red arrows in Figure 1 — which is exactly what PCA computes. For an affine map $\mathbf{y} = A\mathbf{x} + \mathbf{b}$, $\mathbb{E}[\mathbf{y}] = A\boldsymbol{\mu} + \mathbf{b}$ and $\mathrm{Cov}(\mathbf{y}) = A\Sigma A^\top$.

Common probability distributions

Plain version. A handful of distributions appear over and over. Choosing one is choosing an assumption about your data or your target.

Bernoulli

A single binary variable $X \in \{0, 1\}$ with parameter $\phi \in [0, 1]$:

$$P(X=1)=\phi, \quad P(X=0)=1-\phi, \quad P(X=x)=\phi^{x}(1-\phi)^{1-x}.$$

Its mean is ϕ and its variance $\phi(1-\phi)$. A binary classifier with a sigmoid output predicts the parameter ϕ of a Bernoulli distribution over the label.

Categorical (multinoulli)

A single discrete variable with K states, parameterized by a probability vector $\mathbf{p} \in [0,1]^K$ with $\sum_k p_k = 1$:

$$P(X=k)=p_k.$$

If the label is encoded one-hot as $\mathbf{y} \in \{0,1\}^K$, then $P(\mathbf{y}) = \prod_k p_k^{y_k}$. A softmax classifier outputs exactly such a vector. (The *multinomial* distribution counts outcomes over several draws; for one draw it reduces to the categorical, which Goodfellow et al. call "multinoulli" [1].)

Gaussian (normal)

The most widely used continuous distribution:

$$\mathcal{N}(x;\mu,\sigma^2)=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big).$$

μ is the mean (and the peak), σ^2 the variance. Sometimes we use the **precision** $\beta = 1/\sigma^2$ instead, which is cheaper to evaluate when the variance changes often.

Two reasons make the Gaussian the default. The **central limit theorem** says that the sum of many independent variables with finite variance tends towards a Gaussian, so measurement noise built from many small effects is often near-Gaussian. And among all distributions with a given mean and variance on the real line, the Gaussian has the **maximum entropy**: it builds in the least extra structure.

In n dimensions,

$$\mathcal{N}(\mathbf{x};\boldsymbol{\mu},\Sigma)=\frac{1}{\sqrt{(2\pi)^n \det\Sigma}}\exp\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\top}\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big),$$

where $\boldsymbol{\mu} \in \mathbb{R}^n$ is the mean vector and $\Sigma \in \mathbb{R}^{n\times n}$ is a symmetric positive-definite covariance matrix. The **precision matrix** $\Lambda = \Sigma^{-1}$ has a neat meaning: $\Lambda_{ij} = 0$ exactly when x_i and x_j are conditionally independent given all other coordinates. An **isotropic** Gaussian has $\Sigma = \sigma^2 I$ — the noise distribution used by diffusion models (ML 07).

A fundamental link to the losses you will use: if a regression network predicts the mean of a Gaussian with fixed variance, maximizing the likelihood is the same as minimizing the **mean squared error**.

Exponential and Laplace

Sometimes we want a sharp point at a specific location.

$$p(x;\lambda) = \lambda\,\mathbf{1}_{x\geq 0}\,e^{-\lambda x}, \qquad \text{Laplace}(x;\mu,\gamma) = \frac{1}{2\gamma}\exp\Big(-\frac{|x-\mu|}{\gamma}\Big).$$

$\lambda > 0$ is the rate, $\mathbf{1}_{x\geq 0}$ is the indicator that is 1 when $x \geq 0$ and 0 otherwise, μ is the location and $\gamma > 0$ the scale. Compared with the Gaussian, the Laplace density has a sharper peak and heavier tails (Figure 1, bottom left). Predicting the location of a Laplace distribution by maximum likelihood is equivalent to minimizing the **absolute error** (L_1 loss), which is why L_1 is more tolerant of outliers than L_2; a Laplace *prior* on weights gives L_1 regularization (sparsity).

Dirac and empirical distributions

The **Dirac delta** $\delta(x-\mu)$ puts all of its mass at the point μ: it is zero everywhere else yet integrates to one. It is a generalized function, defined by what it does inside integrals: $\int \delta(x-\mu)f(x)\,dx = f(\mu)$.

A training set $\{\mathbf{x}^{(1)},\dots,\mathbf{x}^{(m)}\}$ defines the **empirical distribution**

$$\hat{p}(\mathbf{x}) = \frac{1}{m}\sum_{i=1}^{m}\delta(\mathbf{x}-\mathbf{x}^{(i)}),$$

which puts mass $1/m$ on each of the m examples. Training a model by averaging a loss over the dataset is the same as taking an expectation under $\hat{p}$. It is also the distribution that maximizes the likelihood of the training data — which is precisely why minimizing training loss alone overfits.

Mixtures and latent variables

Plain version. A mixture first picks a "type" at random, then draws a sample from the distribution belonging to that type.

$$P(\mathbf{x}) = \sum_{k=1}^{K} P(z=k)\,P(\mathbf{x}\mid z=k).$$

z is a categorical **latent variable** — a variable we never observe directly — chosen with probabilities $\pi_k = P(z=k)$, and $P(\mathbf{x}\mid z=k)$ is the k-th component. A **Gaussian mixture model** (GMM) uses Gaussian components $\mathcal{N}(\mathbf{x};\boldsymbol{\mu}_k,\Sigma_k)$; with enough components it can approximate any smooth density to any desired nonzero error [1]. Given an observation $\mathbf{x}$, Bayes' rule gives the **posterior responsibility** $P(z=k\mid\mathbf{x})$: how likely it is that component k generated that point. Fitting a GMM with expectation–maximization is the subject of PR 07.

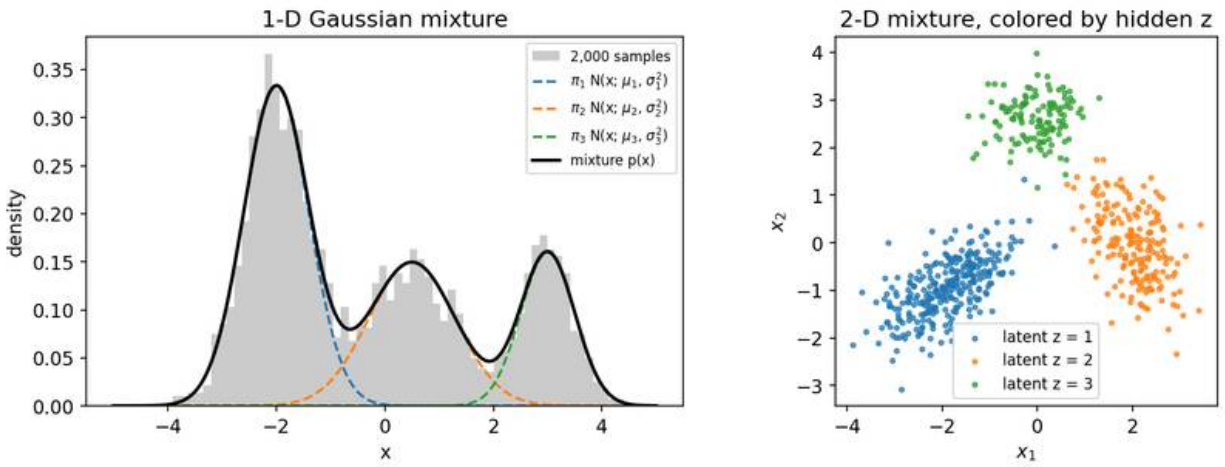


Figure 2 — A Gaussian mixture. Left: the black density is the weighted sum of three components; the grey histogram is 2,000 samples drawn by first picking a component and then sampling from it. Right: in two dimensions, colour shows the hidden component z, which a real dataset would not reveal.

Latent variables are central to modern generative models. A VAE (ML 06) is, informally, a mixture with a *continuous* latent $\mathbf{z}$ and a neural network for $p(\mathbf{x} \mid \mathbf{z})$ [8]; the marginal $p(\mathbf{x}) = \int p(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z})\, d\mathbf{z}$ is then intractable and must be approximated.

```
import numpy as np

rng = np.random.default_rng(0)
pi = np.array([0.5, 0.3, 0.2])      # mixing weights P(z = k)
mu = np.array([-2.0, 0.5, 3.0])     # component means
sd = np.array([0.6, 0.8, 0.5])      # component standard deviations

# Ancestral sampling: first the hidden component z, then x given z
z = rng.choice(3, size=100_000, p=pi)
x = rng.normal(mu[z], sd[z])

def log_normal(x, m, s):
    return -0.5*((x - m)/s)**2 - np.log(s) - 0.5*np.log(2*np.pi)

def mixture_logpdf(x):
    # log sum_k pi_k N(x; mu_k, sd_k^2), computed with log-sum-exp
    a = np.log(pi) + log_normal(x[:, None], mu, sd)
    m = a.max(axis=1, keepdims=True)
    return (m + np.log(np.exp(a - m).sum(axis=1, keepdims=True))).ravel()

grid = np.linspace(-6, 6, 4001)
print("density integrates to",
 ↪  (np.exp(mixture_logpdf(grid)).sum()*(grid[1]-grid[0])).round(4))
print("sample mean", x.mean().round(3), "vs exact", (pi*mu).sum().round(3))
# Posterior responsibility P(z | x) for one observation
a = np.log(pi) + log_normal(np.array([0.0])[:, None], mu, sd)
resp = np.exp(a - a.max()); resp /= resp.sum()
```

```
print("P(z | x=0):", resp.round(3))
```

The density integrates to 1.0, the sample mean (-0.255) matches the exact mean $\sum_k \pi_k \mu_k = -0.25$, and the point $x = 0$ is assigned to the middle component with probability 0.99. The log-sum-exp trick used here is explained in ML 01C.

Useful functions: sigmoid and softplus

Plain version. The sigmoid squashes any real number into a probability between 0 and 1; the softplus is a smooth version of $\max(0, x)$ that is always positive.

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \qquad \zeta(x) = \log\left(1 + e^{x}\right).$$

The **logistic sigmoid** σ produces the parameter ϕ of a Bernoulli distribution; it **saturates** (becomes flat) for large $|x|$, which makes gradients small there. The **softplus** ζ produces a positive number, such as a standard deviation σ or a precision β; its name comes from being a softened "positive part" $x^+ = \max(0, x)$.

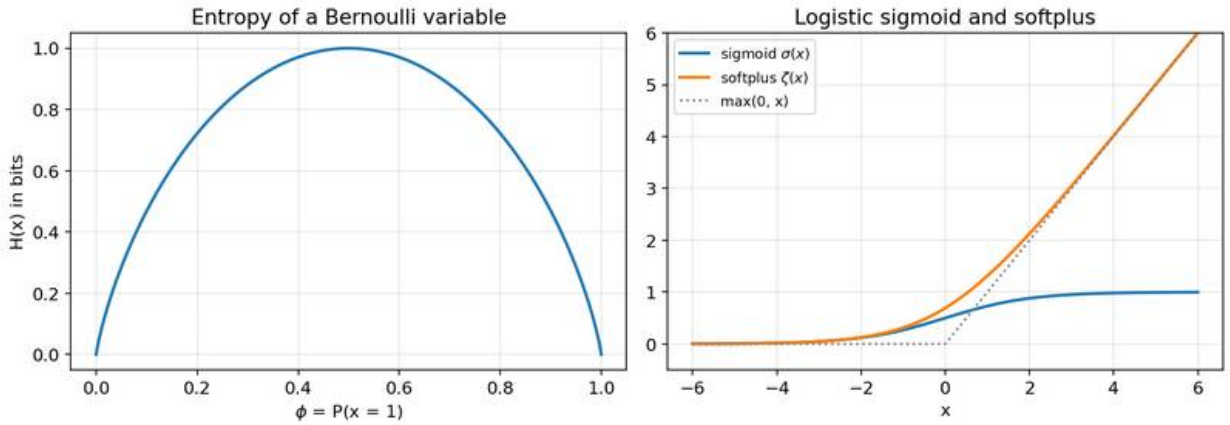


Figure 3 — Left: entropy of a Bernoulli variable, maximal (1 bit) at $\varphi = 0.5$ and zero when the outcome is certain. Right: the logistic sigmoid $\sigma(x)$ and the softplus $\zeta(x)$, a smooth version of $\max(0, x)$.

These identities are worth memorizing, because they appear every time a loss is simplified [1]:

$$1 - \sigma(x) = \sigma(-x), \qquad \frac{d}{dx}\sigma(x) = \sigma(x)(1 - \sigma(x)),$$

$$\log \sigma(x) = -\zeta(-x), \qquad \frac{d}{dx}\zeta(x) = \sigma(x),$$

$$\sigma^{-1}(p) = \log \frac{p}{1-p}, \qquad \zeta(x) - \zeta(-x) = x.$$

σ^{-1} is the **logit** (log-odds) function. The identity $\log \sigma(x) = -\zeta(-x)$ shows why binary cross-entropy should be computed from logits: $-\log \sigma(x) = \zeta(-x)$, and softplus can be evaluated without overflow.

```
import numpy as np

def sigmoid(x):
    return 1/(1 + np.exp(-x))

def softplus(x):
    return np.logaddexp(0, x)          # log(1 + exp(x)) without overflow

x = np.linspace(-30, 30, 13)
assert np.allclose(1 - sigmoid(x), sigmoid(-x))
assert np.allclose(np.log(sigmoid(x)), -softplus(-x))
assert np.allclose(softplus(x) - softplus(-x), x)
h = 1e-6
assert np.allclose((softplus(x+h) - softplus(x-h))/(2*h), sigmoid(x), atol=1e-6)
p = sigmoid(x[4:9])
assert np.allclose(np.log(p/(1-p)), x[4:9])   # logit inverts sigmoid
print("naive log(1+exp(1000)):", np.log1p(np.exp(np.float64(1000))))
print("stable softplus(1000):", softplus(1000.0))
```

All identities pass. The naive expression returns `inf` (with an overflow warning) while `np.logaddexp` returns 1000.0.

Bayes' rule

Plain version. Bayes' rule turns "how likely is the evidence if the hypothesis is true?" into "how likely is the hypothesis now that I have seen the evidence?" — and it insists that you account for how common the hypothesis was in the first place.

From the product rule written both ways, $P(a, b) = P(a \mid b)P(b) = P(b \mid a)P(a)$, so

$$P(a \mid b) = \frac{P(b \mid a)\, P(a)}{P(b)}, \qquad P(b) = \sum_{a'} P(b \mid a')\, P(a').$$

- $P(a)$ is the **prior**: how plausible a is before seeing the evidence.
- $P(b \mid a)$ is the **likelihood** of the evidence b under hypothesis a.
- $P(a \mid b)$ is the **posterior**: the updated plausibility.
- $P(b)$ is the **evidence** (or marginal likelihood), the normalizer obtained by summing over all hypotheses a'.

A worked example: the base-rate effect

Consider a hypothetical defect detector with **sensitivity** $P(+ \mid \text{defect}) = 0.9$, **false-positive rate** $P(+ \mid \text{no defect}) = 0.05$, and a **prevalence** (base rate) $P(\text{defect}) = 0.01$. The probability that an item flagged as positive really is defective is

$$P(\text{defect} \mid +) = \frac{0.9 \cdot 0.01}{0.9 \cdot 0.01 + 0.05 \cdot 0.99} \approx 0.154.$$

Only about 15% of alarms are real, even though the detector catches 90% of defects. Counting makes this obvious: among 10,000 items, 100 are defective and 90 of them raise an alarm, but 9,900 are fine and 495 of them also raise an alarm (Figure 4, right). The same detector at 10% prevalence gives $0.09/(0.09 + 0.045) = 2/3$. This is a mathematical illustration, not a performance claim about any real system. Its lesson carries over to every deployed classifier: **sensitivity and false-positive rate alone do not determine how trustworthy a positive prediction is; the deployment prevalence matters.** ML 10 returns to this when discussing precision under distribution shift.

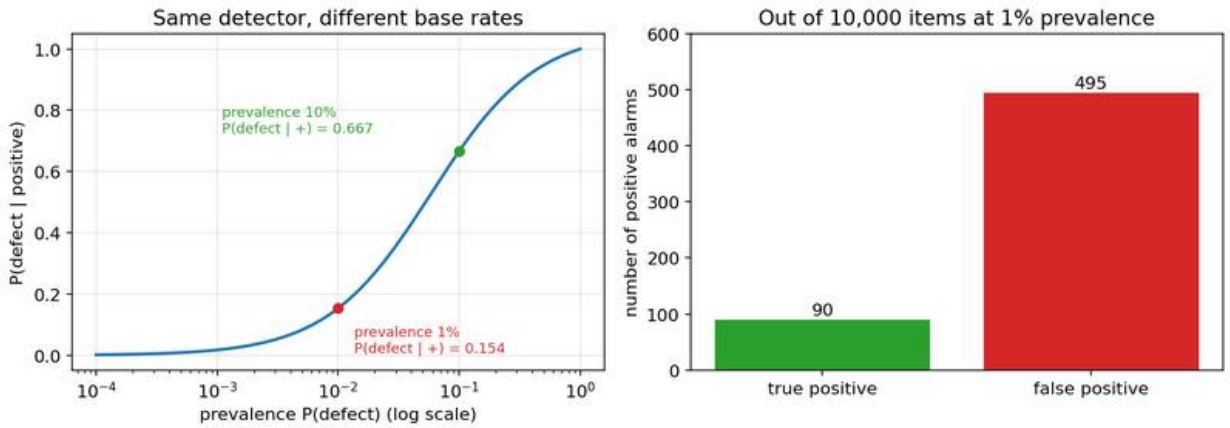


Figure 4 — The base-rate effect for a hypothetical detector with sensitivity 0.9 and false-positive rate 0.05. Left: the posterior $P(\text{defect} \mid +)$ depends strongly on prevalence. Right: at 1% prevalence, false alarms outnumber true alarms more than five to one.

```
import numpy as np

def posterior_positive(prevalence, sensitivity=0.9, false_pos=0.05):
    evidence = sensitivity*prevalence + false_pos*(1 - prevalence)   # P(+)
    return sensitivity*prevalence / evidence                          # P(defect | +)

for prev in [0.001, 0.01, 0.1, 0.5]:
    print(f"prevalence {prev:>5}: P(defect | +) = {posterior_positive(prev):.3f}")

# Simulation check with 1,000,000 synthetic items
```

```
rng = np.random.default_rng(0)
defect = rng.random(1_000_000) < 0.01
alarm = np.where(defect, rng.random(defect.size) < 0.9, rng.random(defect.size) < 0.05)
print("simulated:", defect[alarm].mean().round(3))
```

The formula gives 0.018, 0.154, 0.667 and 0.947 for the four prevalences, and a one-million-item simulation reproduces 0.154.

Bayes' rule also underlies the **Bayes classifier**: predicting the class k with the largest posterior $P(y = k \mid \mathbf{x}) \propto p(\mathbf{x} \mid y = k)P(y = k)$ minimizes the expected error rate. PR 01 develops this decision theory in depth.

Change of variables for densities

Plain version. If you stretch a rubber sheet, the ink on it gets thinner. When a random variable is transformed, its density must be rescaled by how much the transformation stretches space.

A tempting mistake: if $y = g(x)$ and x has density p_x, then $p_y(y) = p_x(g^{-1}(y))$. This is wrong, because it does not integrate to one. The correct rule for an invertible, differentiable g in one dimension is

$$p_y(y) = p_x(g^{-1}(y)) \left| \frac{d\, g^{-1}(y)}{dy} \right|.$$

The probability in a small interval must be preserved: $|p_y(y)\,dy| = |p_x(x)\,dx|$. In n dimensions, with $\mathbf{y} = g(\mathbf{x})$,

$$p_x(\mathbf{x}) = p_y(g(\mathbf{x})) \left| \det \frac{\partial g(\mathbf{x})}{\partial \mathbf{x}} \right|,$$

where $\partial g / \partial \mathbf{x}$ is the $n \times n$ **Jacobian matrix** of g (entry (i, j) is $\partial y_i / \partial x_j$) and $|\det(\cdot)|$ is the absolute value of its determinant — the factor by which g scales a small volume (see ML 01A for determinants). This formula is the entire foundation of **normalizing flows**, which stack invertible networks with cheap Jacobian determinants to obtain an exact likelihood [9].

```
import numpy as np

rng = np.random.default_rng(0)
x = rng.normal(0, 1, 500_000)        # x ~ N(0, 1)
y = np.exp(x)                        # y = g(x), invertible on (0, inf)

def p_x(x):
    return np.exp(-0.5*x**2)/np.sqrt(2*np.pi)

def p_y(y):
```

```
    # p_y(y) = p_x(g^{-1}(y)) * |d g^{-1}(y) / dy| = p_x(log y) / y
    return p_x(np.log(y))/y

counts, edges = np.histogram(y, bins=np.linspace(0.05, 4, 80), density=False)
centers = 0.5*(edges[:-1] + edges[1:])
empirical = counts/(y.size*np.diff(edges))
print("max |histogram - formula|:", np.abs(empirical - p_y(centers)).max().round(3))
print("wrong formula (no Jacobian) error:", np.abs(empirical -
 ↪  p_x(np.log(centers))).max().round(3))
```

With the Jacobian factor $1/y$ the histogram matches the formula to within 0.011 (sampling noise); without it the error is 0.46.

Likelihood, maximum likelihood and MAP

Plain version. Hold the observed data fixed and ask which parameter values make that data most plausible. If you also have prior knowledge about the parameters, weigh that in too.

Suppose a model $p_{\boldsymbol{\theta}}(\mathbf{x})$ has parameters $\boldsymbol{\theta}$, and we observe a dataset $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}\}$ of independent and identically distributed (i.i.d.) examples. The **likelihood** is

$$L(\boldsymbol{\theta}) = p_{\boldsymbol{\theta}}(\mathcal{D}) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}).$$

As a function of $\boldsymbol{\theta}$, this is *not* a probability distribution over parameters: it need not integrate to one over $\boldsymbol{\theta}$. The **maximum likelihood estimate** (MLE) is

$$\boldsymbol{\theta}_{\mathrm{ML}} = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{m} \log p_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) = \arg\min_{\boldsymbol{\theta}} \; \mathbb{E}_{\mathbf{x}\sim\hat{p}}[\,-\log p_{\boldsymbol{\theta}}(\mathbf{x})\,].$$

We take logarithms because a product of many numbers below one underflows, while a sum of logs is numerically safe and has the same maximizer. The right-hand form says MLE **minimizes the average negative log-likelihood** (NLL) under the empirical distribution $\hat{p}$. For a classifier $p_{\boldsymbol{\theta}}(y \mid \mathbf{x})$ the same idea gives the conditional NLL $-\log p_{\boldsymbol{\theta}}(y \mid \mathbf{x})$, which is the cross-entropy loss we meet below.

Two classic results: for a Bernoulli coin with k heads in n flips, $\phi_{\mathrm{ML}} = k/n$; for a Gaussian, μ_{ML} is the sample mean and $\sigma^2_{\mathrm{ML}} = \frac{1}{m}\sum_i (x^{(i)} - \mu_{\mathrm{ML}})^2$ — a slightly *biased* estimate (dividing by $m-1$ removes the bias).

MAP estimation

Bayesian inference treats $\boldsymbol{\theta}$ itself as a random variable with a **prior** $p(\boldsymbol{\theta})$ and computes the **posterior**

$$p(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(\mathcal{D})} \;\propto\; p(\mathcal{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}).$$

The full posterior is often intractable, so a common shortcut is its mode, the **maximum a posteriori** (MAP) estimate:

$$\boldsymbol{\theta}_{\mathrm{MAP}} = \arg\max_{\boldsymbol{\theta}} \left[\log p(\mathcal{D} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta}) \right].$$

With a Gaussian prior $p(\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \frac{1}{\lambda} I)$, the term $-\log p(\boldsymbol{\theta})$ equals $\frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2$ plus a constant: **weight decay is MAP with a Gaussian prior**. A Laplace prior gives L_1 regularization. This links probabilistic assumptions to the regularizers in ML 02. But a MAP point estimate is still a single parameter vector; it does not tell you how uncertain the parameters are. Approximating the whole posterior is the topic of PR 08 and of variational inference [10].

```
import numpy as np

rng = np.random.default_rng(0)
flips = rng.random(10) < 0.7          # 10 coin flips, true P(heads) = 0.7
k, n = flips.sum(), flips.size

phi = np.linspace(1e-3, 1 - 1e-3, 999)
log_lik = k*np.log(phi) + (n - k)*np.log(1 - phi)
a, b = 5.0, 5.0                        # Beta(5, 5) prior, centered on 0.5
log_prior = (a - 1)*np.log(phi) + (b - 1)*np.log(1 - phi)

print(f"{k} heads in {n} flips")
print("MLE  grid:", phi[log_lik.argmax()].round(3), " closed form:", k/n)
print("MAP  grid:", phi[(log_lik + log_prior).argmax()].round(3),
      " closed form:", round((k + a - 1)/(n + a + b - 2), 3))

# Gaussian MLE: sample mean and the 1/m (biased) variance
x = rng.normal(2.0, 3.0, 50)
print("Gaussian MLE mean, var:", x.mean().round(3), x.var().round(3),
      " unbiased var:", x.var(ddof=1).round(3))
```

With this seed, 6 heads appear in 10 flips. The MLE is 0.6; the Beta$(5, 5)$ prior pulls the MAP estimate towards 0.5, giving 0.556, matching the closed form $(k + a - 1)/(n + a + b - 2)$. With more data the likelihood dominates and the two estimates converge. The Gaussian MLE variance (7.71) is smaller than the unbiased estimate (7.87), as expected.

Information theory

Plain version. Information theory measures surprise. A rare event carries a lot of information when it happens; a certain event carries none. Averaging surprise gives entropy, and the extra surprise from using the wrong distribution gives KL divergence.

Claude Shannon founded the field to study how many bits are needed to send messages over a noisy channel [5]; the textbook treatment is Cover and Thomas [7]. For machine learning, the useful idea is to quantify how different two distributions are.

Self-information and entropy

The **self-information** of an outcome x is

$$I(x) = -\log P(x).$$

With the natural logarithm the unit is the **nat**; with $\log_2$ it is the **bit** (one nat $= 1/\ln 2 \approx 1.443$ bits). Certain events ($P = 1$) carry zero information; unlikely events carry a lot; and information from independent events adds, because logs turn products into sums.

The **Shannon entropy** is the expected self-information:

$$H(P) = \mathbb{E}_{x\sim P}[I(x)] = -\sum_x P(x)\log P(x).$$

We use the convention $0\log 0 = 0$, justified by the limit $p\log p \to 0$ as $p \to 0$. Entropy is the minimum average number of bits (or nats) needed to encode outcomes drawn from P. It is zero for a deterministic outcome and maximal, $\log K$, for a uniform distribution over K outcomes. For a Bernoulli variable with parameter ϕ, $H = -\phi\log\phi - (1-\phi)\log(1-\phi)$, which peaks at one bit when $\phi = 0.5$ (Figure 3, left).

For continuous variables, $h(p) = -\int p(x)\log p(x)\,dx$ is the **differential entropy**. Treat it carefully: it can be negative (a narrow uniform distribution on $[0, \frac{1}{2}]$ has $h = -\log 2$) and it changes when you rescale the coordinates. It is not a limiting version of the discrete count of bits.

KL divergence

The **Kullback–Leibler divergence** [6] measures how different a distribution Q is from a reference P:

$$D_{\mathrm{KL}}(P\,\|\,Q) = \mathbb{E}_{x\sim P}\Big[\log\frac{P(x)}{Q(x)}\Big] = \sum_x P(x)\log\frac{P(x)}{Q(x)}.$$

In coding terms, it is the *extra* number of nats needed to encode samples from P when the code was designed for Q. Its key properties:

- **Nonnegative**: $D_{\mathrm{KL}}(P\|Q) \geq 0$, with equality if and only if $P = Q$ (for discrete variables; "almost everywhere" for continuous ones). This follows from Jensen's inequality, because $-\log$ is convex.
- **Infinite under unsupported mass**: if $Q(x) = 0$ for some x with $P(x) > 0$, the divergence is infinite.
- **Asymmetric**: in general $D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$, and it does not satisfy the triangle inequality. KL is therefore *not* a distance metric, even though it is often described as "the distance between distributions".

Forward versus reverse KL

The asymmetry has visible consequences when the approximating family is too simple. Figure 5 fits a single Gaussian q to a target p with two well-separated modes.

- Minimizing **forward KL** $D_{\mathrm{KL}}(p\|q)$ heavily penalizes q for being small where p has mass. The best Gaussian therefore **covers both modes**, placing its mean between them ($\mu = 0$, $\sigma \approx 2.62$) and putting a lot of mass in the empty valley. For Gaussian q, the forward-KL optimum simply matches the mean and variance of p.
- Minimizing **reverse KL** $D_{\mathrm{KL}}(q\|p)$ heavily penalizes q for being large where p is small. The best Gaussian therefore **locks onto one mode** ($\mu \approx 2.49$, $\sigma \approx 0.81$) and ignores the other.

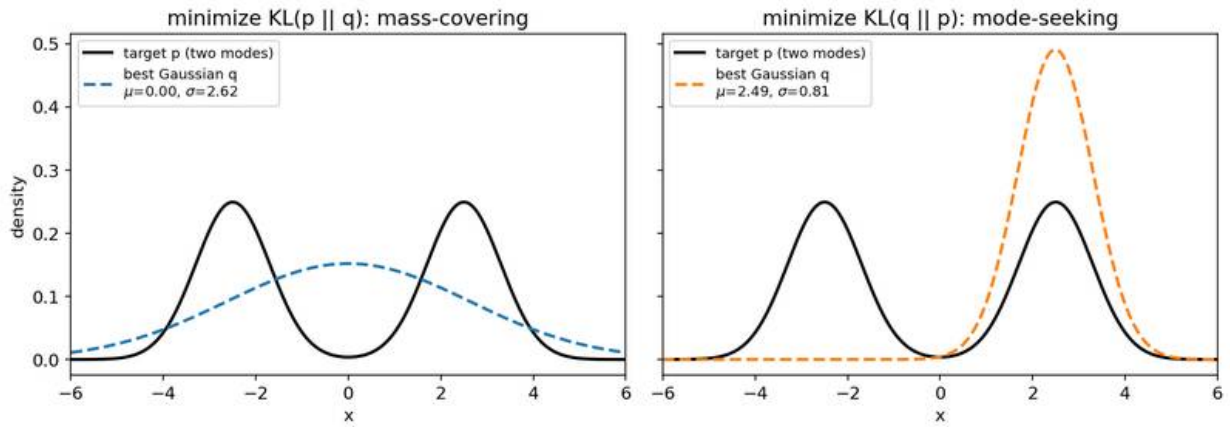


Figure 5 — KL is asymmetric. Fitting one Gaussian q to a two-mode target p: forward $KL(p\|q)$ gives a mass-covering fit centred between the modes; reverse $KL(q\|p)$ gives a mode-seeking fit on one peak (the other peak gives an equally good mirror solution).

Which direction you minimize is a design choice. **Maximum likelihood minimizes forward KL** from the data distribution to the model, so likelihood-trained models (autoregressive models, flows, VAEs) tend to spread probability over all the data, sometimes producing blurry samples.

Variational inference usually minimizes reverse KL from the approximation to the true posterior [10], and so tends to underestimate posterior uncertainty.

Cross-entropy

The **cross-entropy** of Q relative to P is

$$H(P, Q) = -\mathbb{E}_{x\sim P}[\log Q(x)] = H(P) + D_{\mathrm{KL}}(P \,\|\, Q).$$

Because $H(P)$ does not depend on Q, minimizing cross-entropy with respect to Q is the same as minimizing forward KL. In classification, P is the empirical label distribution — a one-hot vector for each example — and Q is the model's softmax output $\hat{\mathbf{y}}$. For one example with true class y,

$$H(\mathbf{e}_y, \hat{\mathbf{y}}) = -\sum_k [\mathbf{e}_y]_k \log \hat{y}_k = -\log \hat{y}_y,$$

where $\mathbf{e}_y$ is the one-hot vector for class y. Averaged over a dataset, this is exactly the conditional negative log-likelihood. So "cross-entropy loss", "log loss" and "negative log-likelihood of a categorical model" are three names for the same objective. Any loss consisting of a negative log-likelihood is a cross-entropy between the empirical distribution and the model — which is why the mean squared error is also "the cross-entropy" for a Gaussian model [1].

Mutual information

Mutual information measures how much knowing one variable reduces uncertainty about another:

$$I(X; Y) = D_{\mathrm{KL}}(p(x, y) \,\|\, p(x)p(y)) = H(X) - H(X \mid Y) = H(X) + H(Y) - H(X, Y).$$

Here $H(X \mid Y) = -\mathbb{E}_{x,y}[\log p(x \mid y)]$ is the conditional entropy and $H(X, Y)$ the joint entropy. $I(X; Y) \geq 0$, with equality if and only if X and Y are independent — unlike covariance, it detects any form of dependence. It is symmetric in X and Y.

Code: information quantities

```
import numpy as np

p = np.array([.7, .2, .1])
q = np.array([.5, .3, .2])
entropy = -(p*np.log(p)).sum()
cross_entropy = -(p*np.log(q)).sum()
```

```
kl = (p*np.log(p/q)).sum()
reverse_kl = (q*np.log(q/p)).sum()
assert np.allclose(cross_entropy - entropy, kl) and kl >= 0
print("H(p), H(p,q), KL(p||q), KL(q||p) in nats:",
      np.round([entropy, cross_entropy, kl, reverse_kl], 4))
print("H(p) in bits:", (entropy/np.log(2)).round(4))

# Mutual information from a joint table (same weather/umbrella table)
P = np.array([[0.56, 0.14],
              [0.06, 0.24]])
px, py = P.sum(1), P.sum(0)
mi = (P*np.log(P/np.outer(px, py))).sum()
H = lambda t: -(t[t > 0]*np.log(t[t > 0])).sum()
assert np.isclose(mi, H(px) + H(py) - H(P.ravel()))
print("I(weather; umbrella) =", mi.round(4), "nats")

# Cross-entropy loss on one-hot labels equals mean -log q(y)
probs = np.array([[0.7, 0.2, 0.1], [0.1, 0.8, 0.1], [0.3, 0.3, 0.4]])
labels = np.array([0, 1, 2])
print("mean cross-entropy:", (-np.log(probs[np.arange(3), labels])).mean().round(4))
```

Output: $H(p) = 0.8018$, $H(p,q) = 0.8869$, $D_{\mathrm{KL}}(p\|q) = 0.0851$ and $D_{\mathrm{KL}}(q\|p) = 0.0920$ nats — the two directions differ. $H(p)$ is 1.1568 bits. The weather–umbrella mutual information is 0.1637 nats, and both formulas for it agree. The mean cross-entropy of the three predictions is 0.4987 nats. In real training code, compute the cross-entropy from logits with a fused function rather than from probabilities (see ML 01C).

Structured probabilistic models

Plain version. A joint distribution over many variables is far too big to write down as a table. Structured models draw a graph saying which variables directly influence which, and then the joint splits into small pieces.

A joint table over 100 binary variables would need $2^{100} - 1$ numbers. **Structured probabilistic models**, also called **graphical models**, exploit conditional independence to factorize the joint into small factors [1], [3]. There are two main kinds.

Directed models (Bayesian networks) use a directed acyclic graph $\mathcal{G}$. Each variable depends only on its parents $Pa_{\mathcal{G}}(x_i)$:

$$p(\mathbf{x}) = \prod_i p(x_i \mid Pa_{\mathcal{G}}(x_i)).$$

For example, with $a \to b \to c$ and $a \to c$ we have $p(a,b,c) = p(a)\,p(b \mid a)\,p(c \mid a,b)$; deleting the edge $a \to c$ asserts $c \perp a \mid b$ and shortens the last factor to $p(c \mid b)$. Directed graphs are natural for

generative stories ("first choose the class, then the pixels"), and a mixture model is the simplest example: $z \to \mathbf{x}$.

Undirected models (Markov random fields) use an undirected graph. Each **clique** $C^{(i)}$ (a set of mutually connected nodes) has a nonnegative **factor** $\phi^{(i)}$, and

$$p(\mathbf{x}) = \frac{1}{Z} \prod_i \phi^{(i)}(C^{(i)}), \qquad Z = \sum_{\mathbf{x}} \prod_i \phi^{(i)}(C^{(i)}).$$

The **partition function** Z normalizes the product; computing it is often the hard part. Undirected models suit symmetric interactions, like neighbouring pixels in an image prior or a conditional random field for segmentation.

A graph is a *description* of a distribution, not a different kind of distribution: any distribution can be written either way, although one may be much more compact. Deep learning often uses these ideas implicitly — an autoregressive model is a fully connected directed model, and a VAE is a two-node directed model $\mathbf{z} \to \mathbf{x}$ whose conditional is a neural network [8].

Modern view

The probability and information theory in this chapter is classical. What has changed in the deep-learning era is *how* it is used and what goes wrong in practice. This section surveys the review literature and recent directions; it summarizes their framing rather than reproducing benchmark numbers.

Calibration: are predicted probabilities honest?

A classifier is **calibrated** if, among all inputs for which it predicts confidence 0.8, about 80% are correct. Guo et al. [11] (ICML 2017) found that modern deep networks are often **overconfident**, and identified depth, width, weight decay and batch normalization as factors that affect calibration. Their main recommendation, **temperature scaling** — dividing all logits by a single scalar T fitted on a validation set — is a one-parameter variant of Platt scaling and remains a strong baseline because it changes confidence without changing the predicted class. Wang's survey [12] organizes calibration methods into four families: post-hoc calibration, regularization during training, uncertainty estimation, and compositions of these; its revised version also covers calibration of large language models. The probabilistic lesson: a softmax output is a model of $P(y \mid \mathbf{x})$, and like any model it can be wrong in its uncertainty, not only in its argmax.

Uncertainty quantification

Two major reviews map the field of uncertainty in deep learning. Abdar et al. [13] (*Information Fusion*, 2021) review Bayesian approximations, ensembles and their applications across many domains. Gawlikowski et al. [14] (*Artificial Intelligence Review*, 2023) distinguish **reducible model (epistemic) uncertainty** from **irreducible data (aleatoric) uncertainty** — roughly the third and first sources of uncertainty from the start of this chapter — and compare deterministic single networks, Bayesian methods, ensembles and test-time augmentation, together with calibration measures.

Two primary studies are especially influential. **Deep ensembles** [15] train several networks from different random initializations and average their predictive distributions; this simple, parallel method was competitive with approximate Bayesian neural networks and expressed higher uncertainty on out-of-distribution inputs. Ovadia et al. [16] (NeurIPS 2019) benchmarked uncertainty methods under **dataset shift** and found that post-hoc calibration on in-distribution validation data did not carry over well, while methods that marginalize over models (such as ensembles) held up better.

A complementary, assumption-light approach is **conformal prediction**. Angelopoulos and Bates [17] give an accessible introduction: using a held-out calibration set, conformal methods turn any pre-trained model's scores into prediction *sets* that contain the true label with a user-chosen probability, assuming the calibration and test data are exchangeable. It does not require the model's probabilities to be calibrated, which makes it attractive for vision and language systems that are used as black boxes.

Mutual information in representation learning

Mutual information became a popular objective for self-supervised learning (ML 05). Contrastive Predictive Coding [18] introduced the **InfoNCE** loss, which classifies a positive pair among negatives and can be shown to maximize a lower bound on mutual information. Poole et al. [19] (ICML 2019) unified many variational MI bounds and showed that they degrade when the true MI is large, trading high bias against high variance. Tschannen et al. [20] (ICLR 2020) argued that the success of these methods "cannot be attributed to the properties of MI alone": the encoder architecture and the parameterization of the estimator matter at least as much. The modern consensus is to treat InfoNCE-style objectives as well-behaved contrastive losses with an information-theoretic motivation, not as precise MI estimators.

Divergences and latent variables in generative models

Generative modelling is applied probability. VAEs [8] maximize an evidence lower bound that combines a reconstruction likelihood with a KL term; Blei et al. [10] review variational inference,

which approximates intractable posteriors by minimizing (reverse) KL within a tractable family. Normalizing flows [9] are built entirely on the change-of-variables formula. Diffusion and flow-matching models (ML 07) combine Gaussian noise, conditional expectations and likelihood bounds. The choice of divergence (forward KL, reverse KL, or something else) shapes whether a model covers all the data or concentrates on its modes — the same picture as Figure 5, at scale.

Open problems

- **Calibration under shift.** Calibration that holds on a validation set can fail on new cameras, hospitals or prompts [16]; detecting when this happens is an open question.
- **Separating uncertainty types.** Decomposing predictive uncertainty into epistemic and aleatoric parts is conceptually clear but hard to do reliably for large networks [14].
- **Estimating information in high dimensions.** Variational MI bounds have fundamental bias–variance limits [19], so information-theoretic claims about deep representations need care.
- **Uncertainty for foundation models.** Applying calibration and conformal methods to large vision–language and language models, whose outputs are open-ended, is an active research area [12], [17].

Key takeaways

- Probability describes uncertainty from randomness, partial observation and simplified models; frequentist and Bayesian readings share the same rules.
- A PMF assigns probabilities; a PDF assigns densities, which can exceed one. Probabilities of continuous events are areas.
- The sum rule (marginalization), product rule (conditioning) and chain rule are exact; independence and conditional independence are modelling assumptions.
- Covariance measures linear co-variation only — zero covariance does not imply independence; mutual information detects any dependence.
- Choosing a distribution is choosing a loss: Bernoulli/categorical give cross-entropy, Gaussian gives squared error, Laplace gives absolute error; Gaussian and Laplace priors give L_2 and L_1 regularization.
- Bayes' rule combines evidence with base rates; the same detector can be trustworthy at one prevalence and misleading at another.
- Transforming a random variable rescales its density by the absolute Jacobian determinant.
- Cross-entropy equals entropy plus KL; minimizing it is maximum likelihood. KL is asymmetric: forward KL covers modes, reverse KL seeks one.

Exercises

1. **Units and certainty.** (a) Convert a loss of 1 nat into bits. (b) Compute the entropy, in bits, of a fair coin, a certain outcome, and a fair 8-sided die.

   Hint

   (a) Divide by $\ln 2$: about 1.443 bits. (b) 1 bit, 0 bits and $\log_2 8 = 3$ bits. A uniform distribution over K outcomes has entropy $\log K$, the maximum for K outcomes.

2. **Prevalence shift.** Using the detector in the Bayes section, find the prevalence at which a positive result is exactly a coin flip, $P(\text{defect} \mid +) = 0.5$.

   Hint

   Set $0.9\pi = 0.05(1 - \pi)$, where π is the prevalence. Then $\pi = 0.05/0.95 \approx 0.053$, about 5.3%. Below that, most alarms are false.

3. **Dependent but uncorrelated.** Let X take values $-1, 0, 1$ with equal probability and $Y = X^2$. Show that $\mathrm{Cov}(X, Y) = 0$ and compute $I(X; Y)$ in bits.

   Hint

   $\mathbb{E}[X] = 0$ and $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0$, so the covariance is zero. Because Y is a function of X, $H(Y \mid X) = 0$ and $I(X; Y) = H(Y)$. $Y = 1$ with probability $2/3$ and $Y = 0$ with probability $1/3$, so $I \approx 0.918$ bits.

4. **Gaussian MLE equals squared error.** Show that maximizing $\prod_i \mathcal{N}(y^{(i)}; f_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}), \sigma^2)$ over $\boldsymbol{\theta}$ (with σ fixed) is equivalent to minimizing the mean squared error. What changes if σ is also predicted by the network?

   Hint

   The negative log-likelihood is $\sum_i \frac{(y^{(i)} - f_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}))^2}{2\sigma^2} + m \log \sigma + \text{const}$. With σ fixed only the squared error depends on $\boldsymbol{\theta}$. If the network also predicts $\sigma(\mathbf{x})$ (for example through a softplus), each example's error is weighted by $1/\sigma^2$ and a $\log \sigma$ penalty prevents the variance from growing without bound — a heteroscedastic regression model.

5. **Change of variables.** If $X \sim \mathcal{U}(0, 1)$ (uniform) and $Y = -\frac{1}{\lambda} \log X$ with $\lambda > 0$, what is the density of Y? Check your answer by sampling in NumPy.

   Hint

   $g^{-1}(y) = e^{-\lambda y}$ and $|dg^{-1}/dy| = \lambda e^{-\lambda y}$, while $p_X = 1$ on $(0, 1)$. So $p_Y(y) = \lambda e^{-\lambda y}$ for $y \geq 0$: the exponential distribution. This is the inverse-transform sampling trick.

6. **KL direction.** For $p = (0.5, 0.5, 0)$ and $q = (0.4, 0.4, 0.2)$, compute $D_{\mathrm{KL}}(p\|q)$ and $D_{\mathrm{KL}}(q\|p)$. Which one is infinite, and what does that tell you about training a model by maximum likelihood?

Hint

$D_{\mathrm{KL}}(p\|q) = \log(0.5/0.4) \approx 0.223$ nats (the third term is $0 \log 0 = 0$). $D_{\mathrm{KL}}(q\|p) = \infty$ because q puts mass where p has none. Maximum likelihood minimizes $D_{\mathrm{KL}}(p_{\mathrm{data}}\|q_{\mathrm{model}})$, which becomes infinite if the model assigns zero probability to any observed example — so likelihood-trained models must spread some probability everywhere the data might be.

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ML 01C • Numerical Computation and Gradient Checks

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What you'll learn

- How computers store real numbers, what machine epsilon means, and why float16, bfloat16, float32 and float64 behave differently.
- Why exact formulas overflow, underflow or lose digits, and how to compute softmax, log-sum-exp and logistic losses stably.
- What a condition number measures, and how ill-conditioning differs from an unstable algorithm.
- How gradients, Jacobians and Hessians describe a loss surface, why gradient descent zigzags, and what Newton and quasi-Newton methods change.
- How to handle constraints with projections, the generalized Lagrangian and the KKT conditions.
- How to check derivatives, debug non-finite values and use mixed precision safely.

The big picture

Machine learning is written in the language of real numbers, but computers do not have real numbers. They have a finite set of floating-point numbers, and every arithmetic operation rounds its result to one of them. Most of the time this rounding is invisible. Sometimes it is catastrophic: an exponential overflows to infinity, a tiny probability becomes exactly zero and its logarithm becomes minus infinity, or two nearly equal numbers are subtracted and almost all their significant digits disappear.

The second half of the story is optimization. Training a model means minimizing a function of millions or billions of parameters, and we can only do that numerically, by taking many small steps guided by derivatives. How fast and how reliably those steps make progress depends on the shape of the loss surface — its slopes and its curvatures — and on how we take the steps. This chapter follows the "Numerical Computation" chapter of Goodfellow, Bengio and Courville [1], adds the floating-point details that modern low-precision training relies on, and connects each idea to small NumPy experiments.

Explain like I'm five

> Imagine a calculator that can show only seven digits. If you ask it for one million plus 0.1, it shows one million: the 0.1 fell off the screen. If you then subtract one million, you get zero instead of 0.1. Good numerical code arranges its steps so the important digits never fall off the screen. Optimization is a different game: you are blindfolded on a hilly field and want to reach the lowest point. You can feel the slope under your feet (the gradient) and, with more effort, how the ground curves (the Hessian). Numerical computation is about walking down safely without your calculator lying to you.

Read ML 01B first if softmax and cross-entropy are new, and keep ML 01A nearby for eigenvalues and the singular value decomposition. ML 02 then uses everything here to train neural networks.

Floating-point numbers

Plain version. A floating-point number stores a sign, a few significant digits and a power of two. The *relative* spacing between neighbouring numbers is roughly constant, so a big number has a big absolute rounding error and a small number has a small one.

How a float is stored

The IEEE 754 standard [3] defines binary floating-point formats used by essentially all CPUs and GPUs. A normal (non-special) number is stored as

$$x = (-1)^s \times \left(1 + \sum_{i=1}^{t} b_i\, 2^{-i}\right) \times 2^{e},$$

where $s \in \{0,1\}$ is the sign bit, $b_1, \dots, b_t \in \{0,1\}$ are the t stored fraction bits of the significand (the leading 1 is implicit), and e is an integer exponent restricted to a fixed range. Two numbers matter most:

- **Machine epsilon** $\varepsilon_{\text{mach}} = 2^{-t}$ is the gap between 1 and the next larger representable number.
- **Unit roundoff** $u = \varepsilon_{\text{mach}}/2$ bounds the relative error of rounding to nearest: for any real x in the normal range, the stored value is $\mathrm{fl}(x) = x(1+\delta)$ with $|\delta| \le u$.

The standard also requires that each basic operation $(+, -, \times, \div, \sqrt{\cdot})$ is computed as if exactly and then rounded once: $\mathrm{fl}(a \circ b) = (a \circ b)(1+\delta)$, $|\delta| \le u$. This one rule is the foundation of rounding-error analysis [2], [4].

Below the smallest normal number the format uses **subnormal** numbers, which give up leading precision to fill the gap down to zero gradually. Special values complete the picture: $\pm\infty$ for results that are too large, and **NaN** ("not a number") for invalid operations such as $0/0$, $\infty - \infty$ or $\infty \times 0$. Once a NaN appears, almost every operation that touches it returns NaN, which is why a single bad value can poison a whole training run.

The four formats you will meet

The table below was computed with `numpy.finfo` (and `torch.finfo` for bfloat16, which NumPy lacks). "Bits" lists sign/exponent/fraction.

Format	Bits	$\varepsilon_{\text{mach}}$	Largest finite	Smallest normal	Smallest subnormal
float16 (IEEE half)	1/5/10	9.77×10^{-4}	65504	6.10×10^{-5}	5.96×10^{-8}
bfloat16	1/8/7	7.81×10^{-3}	3.39×10^{38}	1.18×10^{-38}	9.18×10^{-41}
float32 (single)	1/8/23	1.19×10^{-7}	3.40×10^{38}	1.18×10^{-38}	1.40×10^{-45}
float64 (double)	1/11/52	2.22×10^{-16}	1.80×10^{308}	2.23×10^{-308}	4.94×10^{-324}

Read the table in two directions. The **exponent bits** set the *range*: float16 overflows above 65504,

while bfloat16 keeps float32's 8 exponent bits and therefore its range. The **fraction bits** set the *precision*: float32 carries about 7 significant decimal digits, float64 about 16, float16 about 3, and bfloat16 only about 2. That trade-off — bfloat16's wide range with coarse precision versus float16's narrow range with finer precision — is the reason the two half-precision formats need different training recipes (see the mixed-precision section below and [5], [12]).

```
import numpy as np

for dtype in (np.float16, np.float32, np.float64):
    info = np.finfo(dtype)
    print(f"{info.dtype}: eps={float(info.eps):.3e}  max={float(info.max):.3e}  "
          f"tiny={float(info.tiny):.3e}  subnormal={float(info.smallest_subnormal):.3e}")

# Rounding: 1 + eps/2 rounds back to 1 in float32
e = np.finfo(np.float32).eps
print(np.float32(1) + np.float32(e/2) == np.float32(1))   # True
print(np.float32(1) + np.float32(e) == np.float32(1))     # False

# Addition is not associative
a, b, c = np.float32(1e8), np.float32(-1e8), np.float32(1.0)
print((a + b) + c, a + (b + c))                           # 1.0 0.0
```

The last line is worth pausing on. In exact arithmetic both orders give 1. In float32, $-10^8 + 1$ rounds back to -10^8 because the spacing between float32 numbers near 10^8 is 8, so the 1 is lost. Consequences in practice: summing the same numbers in a different order — for example on a different number of GPUs, or with a different reduction kernel — can change the result in the last digits. This is a major source of run-to-run nondeterminism, and it is not a bug in your code [13].

Overflow, underflow and cancellation

Plain version. Overflow means a result is too large to store; underflow means it is too small and becomes zero (or loses precision as a subnormal). Cancellation means subtracting nearly equal numbers, which keeps the rounding error but throws away the true signal.

Overflow and underflow

The exponential is the usual culprit in machine learning. Because $\exp(z)$ exceeds the largest float32 once $z > \ln(3.40 \times 10^{38}) \approx 88.7$, a logit of 100 overflows in float32 (and a logit of about 11.1 already overflows in float16, since $\ln 65504 \approx 11.09$). In the other direction, $\exp(-110) \approx 1.7 \times 10^{-48}$ is below the smallest float32 subnormal and becomes exactly 0. Neither event raises an error by default; they produce `inf` and `0`, and the next operation turns them into `nan` or $-\infty$.

```
import numpy as np

with np.errstate(over="ignore", under="ignore", invalid="ignore"):
    print(np.exp(np.float32(100.0)))        # inf: overflow
    print(np.exp(np.float32(-110.0)))       # 0.0: underflow
    big = np.exp(np.float32(100.0))
    print(big / big)                        # nan: inf / inf

# Catastrophic cancellation and its cure
x = 1e-10
print(np.log(1 + x), np.log1p(x))           # roundoff vs accurate
print(np.exp(x) - 1, np.expm1(x))

# Variance: the one-pass formula cancels badly when the mean is large
rng = np.random.default_rng(0)
data = (1e4 + rng.normal(0, 1, 100_000)).astype(np.float32)
one_pass = np.mean(data**2) - np.mean(data)**2
two_pass = np.mean((data - data.mean())**2)
print("one-pass:", one_pass, " two-pass:", two_pass)
```

On this machine the one-pass variance comes out as -8.0 — a negative variance — while the two-pass formula gives the correct value of about 1.0. The data have mean 10^4 and variance 1, so $\mathbb{E}[X^2] \approx 10^8$ and $(\mathbb{E}X)^2 \approx 10^8$; float32 can only resolve differences of about 8 at that magnitude, so the true difference of 1 is pure noise.

Why subtraction is the dangerous operation

Suppose $\hat{a} = a(1 + \delta_a)$ and $\hat{b} = b(1 + \delta_b)$ are stored with relative errors $|\delta_a|, |\delta_b| \le u$. Even if the subtraction itself is exact,

$$\frac{|(\hat{a} - \hat{b}) - (a - b)|}{|a - b|} = \frac{|a\delta_a - b\delta_b|}{|a - b|} \le u \, \frac{|a| + |b|}{|a - b|}.$$

Here a, b are the true values and u is the unit roundoff. The factor $(|a| + |b|)/|a - b|$ is huge when $a \approx b$: the subtraction does not create error, it *reveals* the error already present in the inputs by removing the digits they shared. The cures are algebraic: rewrite the formula so the subtraction never happens. Examples you will use:

- `log1p(x)` computes $\log(1 + x)$ and `expm1(x)` computes $e^x - 1$ accurately for small x, where the naive forms lose almost all digits (see the output above).
- Variance: subtract the mean first (two-pass), or use Welford's streaming update.
- Softmax and log-sum-exp: subtract the maximum first, as the next section shows.
- Distances: $\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2x^\top y$ is fast but cancels badly when $x \approx y$; clamp it at zero or compute $x - y$ directly when accuracy matters.

Stable softmax and log-sum-exp

Plain version. Subtract a shared constant before exponentiating, and compute logarithms of probabilities directly instead of first creating tiny probabilities and then taking their logarithm.

The softmax turns a vector of logits $z \in \mathbb{R}^K$ into probabilities:

$$\operatorname{softmax}(z)_i = \frac{\exp(z_i)}{\sum_{j=1}^{K} \exp(z_j)}, \qquad i = 1, \ldots, K.$$

It is invariant to adding the same constant c to every logit, because the factor $\exp(c)$ cancels between numerator and denominator. Choose $c = -m$ with $m = \max_i z_i$:

$$p_i = \frac{\exp(z_i - m)}{\sum_j \exp(z_j - m)}.$$

Now every exponent is ≤ 0, so nothing overflows, and the largest term is exactly $\exp(0) = 1$, so the denominator is at least 1 and cannot underflow to zero. The same trick stabilizes the **log-sum-exp** function,

$$\operatorname{LSE}(z) = \log \sum_j \exp(z_j) = m + \log \sum_j \exp(z_j - m), \qquad \log p_i = z_i - \operatorname{LSE}(z).$$

LSE is a smooth version of the maximum: $\max_i z_i \le \operatorname{LSE}(z) \le \max_i z_i + \log K$. Its gradient is exactly the softmax, $\partial \operatorname{LSE} / \partial z_i = p_i$.

For a target class y, the cross-entropy loss is $-\log p_y = \operatorname{LSE}(z) - z_y$, and its gradient with respect to the logits is the famously simple $p - e_y$, where e_y is the one-hot vector of the target. Computing `softmax` first and then `log` can still fail: if p_y underflows to 0, its logarithm is $-\infty$ even though $\operatorname{LSE}(z) - z_y$ is a perfectly ordinary finite number. This is why frameworks provide fused `log_softmax` and `cross_entropy` functions that take logits.

Code: logits that overflow naive exponentiation

This standalone NumPy example checks stable probabilities, log probabilities and invariance to shifting logits.

```
import numpy as np

def logsumexp(z):
    m = np.max(z)
    return m + np.log(np.exp(z-m).sum())
def logsoftmax(z):
    return z-logsumexp(z)

z = np.array([1000., 1001., 999.], dtype=np.float64)
logp = logsoftmax(z)
p = np.exp(logp)
assert np.allclose(p.sum(), 1)
assert np.allclose(logsoftmax(z+12345), logp)
print("probabilities:", p)
print("cross-entropy for class 1:", -logp[1])
with np.errstate(over="ignore", invalid="ignore"):
    naive = np.exp(z) / np.exp(z).sum()
print("naive softmax:", naive)
```

The stable version prints probabilities $[0.245, 0.665, 0.090]$ and a cross-entropy of 0.408 nats for class 1; the naive version prints `[nan nan nan]`, because $\exp(1000)$ overflows even in float64 (whose limit is $\exp(709.78)$).

Stable logistic (binary cross-entropy) loss

Binary classification has the same issue in a different form. With logit z, label $y \in \{0, 1\}$ and sigmoid $\sigma(z) = 1/(1 + e^{-z})$, the loss $-y \log \sigma(z) - (1 - y) \log(1 - \sigma(z))$ simplifies algebraically to

$$\ell(z, y) = \operatorname{softplus}(z) - yz, \qquad \operatorname{softplus}(z) = \log(1 + e^{z}) = \max(z, 0) + \log\bigl(1 + e^{-|z|}\bigr).$$

The right-hand form never exponentiates a positive number, and `log1p` keeps the small correction accurate. Prefer `binary_cross_entropy_with_logits` to a hand-built sigmoid followed by a logarithm.

```
import numpy as np

def bce_naive(z, y):
    s = 1 / (1 + np.exp(-z))
    return -(y*np.log(s) + (1-y)*np.log(1-s))

def bce_stable(z, y):
    # softplus(z) - y*z, with softplus(z) = max(z,0) + log1p(exp(-|z|))
    return np.maximum(z, 0) + np.log1p(np.exp(-np.abs(z))) - y*z
```

```
z = np.array([-800., -30., 0., 30., 800.])
y = np.array([1., 1., 0., 0., 0.])
with np.errstate(over="ignore", divide="ignore", invalid="ignore"):
    print("naive :", bce_naive(z, y))
print("stable:", bce_stable(z, y))
```

The naive version returns `inf` for the two confident mistakes ($z = \pm 800$) and 30.001 instead of 30.000 at $z = 30$, because $1 - \sigma(30)$ is computed by cancellation. The stable version returns 800, 30, $\log 2$, 30 and 800 — the correct values.

Poor conditioning

Plain version. Some problems are inherently sensitive: a tiny change in the input produces a large change in the answer, no matter how carefully you compute. The **condition number** measures that sensitivity.

Condition number of a linear system

Consider solving $Ax = b$ for a square invertible matrix A. If the data change to $b + \delta b$, the solution changes to $x + \delta x$ with

$$\frac{\|\delta x\|}{\|x\|} \le \kappa(A)\,\frac{\|\delta b\|}{\|b\|}, \qquad \kappa(A) = \|A\|\,\|A^{-1}\| = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} \ \text{(2-norm)}.$$

Here $\|\cdot\|$ is the Euclidean vector norm and its induced matrix norm, and $\sigma_{\max}, \sigma_{\min}$ are the largest and smallest singular values of A. For a symmetric positive-definite matrix, singular values equal eigenvalues and $\kappa = \lambda_{\max}/\lambda_{\min}$. Goodfellow et al. state the same idea for the function $f(x) = A^{-1}x$ with an eigendecomposition: its condition number is $\max_{i,j} |\lambda_i/\lambda_j|$ [1]. A practical rule of thumb: with $\kappa \approx 10^k$, expect to lose about k decimal digits of accuracy relative to the precision of your inputs [4].

```
import numpy as np

A = np.array([[1.0, 1.0],
              [1.0, 1.0001]])
print("condition number:", np.linalg.cond(A))

b = np.array([2.0, 2.0001])
x = np.linalg.solve(A, b)
db = np.array([0.0, 0.0001])                # tiny change in the data
x2 = np.linalg.solve(A, b + db)
rel_in = np.linalg.norm(db) / np.linalg.norm(b)
rel_out = np.linalg.norm(x2 - x) / np.linalg.norm(x)
```

```
print("x:", x, " x after perturbation:", x2)
print(f"relative change in b: {rel_in:.1e}, in x: {rel_out:.1e}, "
      f"amplification: {rel_out/rel_in:.0f}")
```

The two columns of A are almost identical, $\kappa(A) \approx 4 \times 10^4$, and a relative change of 3.5×10^{-5} in b moves the solution from $(1, 1)$ to $(0, 2)$ — a 100% change, an amplification of about 28,000, within the bound κ. Nothing was computed wrongly. The *problem* is ill-conditioned.

Ill-conditioned problem versus unstable algorithm

Keep two ideas apart:

- An **ill-conditioned problem** amplifies small changes in its input. No algorithm can fix that; you need better data, regularization (which changes the problem) or a reformulation.
- An **unstable algorithm** adds avoidable error to a problem that may be perfectly well conditioned. The naive softmax and the one-pass variance are unstable algorithms; changing the algorithm fixes them.

Least squares shows both. If features are nearly collinear, the coefficients are ill-conditioned (the problem). Solving the normal equations $X^\top X w = X^\top y$ makes things worse than necessary because $\kappa(X^\top X) = \kappa(X)^2$ (the algorithm); a QR- or SVD-based solver works with X directly. Ridge regression, which adds $\lambda \|w\|^2$, changes the problem to $(X^\top X + \lambda I)w = X^\top y$, whose condition number $(\sigma_{\max}^2 + \lambda)/(\sigma_{\min}^2 + \lambda)$ is smaller — a theme ML 01A develops further.

Gradient-based optimization

Plain version. To minimize a function, look at its slope and step downhill. The gradient points uphill most steeply, so step in the opposite direction, with a step size small enough that the slope you measured is still roughly valid.

Derivatives, gradients and directional derivatives

We minimize an **objective** (also called cost, loss or error) function $f : \mathbb{R}^n \to \mathbb{R}$. For one variable, the derivative $f'(x)$ gives the first-order approximation $f(x+\epsilon) \approx f(x) + \epsilon f'(x)$, so moving a small step against the sign of $f'(x)$ decreases f. For many variables, the **gradient** $\nabla_x f(x) \in \mathbb{R}^n$ collects the partial derivatives $\partial f / \partial x_i$. The **directional derivative** in a unit direction u is the slope along u:

$$\frac{\partial}{\partial \alpha} f(x + \alpha u)\Big|_{\alpha=0} = u^\top \nabla_x f(x) = \|\nabla_x f(x)\|_2 \cos\theta,$$

where θ is the angle between u and the gradient. It is most negative when $\theta = \pi$, that is, when u points along $-\nabla_x f(x)$. This is the **method of steepest descent**, or **gradient descent**:

$$x \leftarrow x - \eta \nabla_x f(x),$$

with **learning rate** (step size) $\eta > 0$. You can pick η as a small constant, try several values and keep the one with the lowest objective (a **line search**), or follow a schedule. Gradient descent converges when every element of the gradient is (close to) zero.

Critical points: minima, maxima and saddles

Points where $\nabla_x f(x) = 0$ are **critical** or **stationary points**. A **local minimum** is lower than all neighbouring points, a **local maximum** is higher, and a **saddle point** is neither: it is a minimum along some directions and a maximum along others. A **global minimum** has the lowest value anywhere; a function can have several global minima and many local minima that are not global. For deep networks we rarely find a true global minimum, and we do not need to. We settle for parameters with a low enough loss that generalize well. In high dimensions, saddle points rather than poor local minima are the more common critical points of many random-like functions, and they can slow gradient-based optimizers because the gradient becomes small near them [8]. Figure 1 shows the three kinds of critical point.

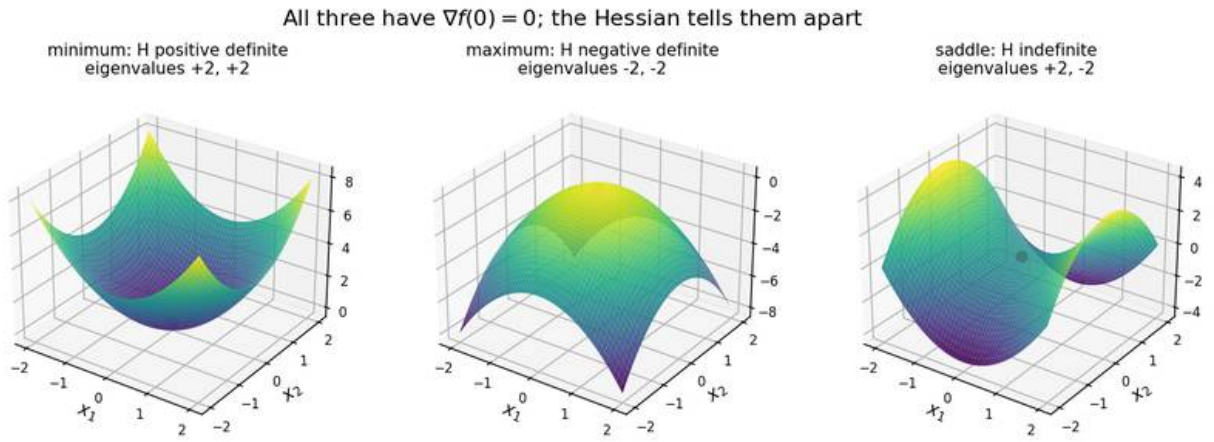


Figure 1 — Three functions with zero gradient at the origin. The Hessian's eigenvalues distinguish them: all positive (minimum), all negative (maximum), mixed signs (saddle).

Jacobian and Hessian matrices

Plain version. The Jacobian collects all first derivatives of a vector-valued function. The Hessian collects all second derivatives of a scalar function: it tells you how the slope itself changes, which is curvature.

Definitions

For a function $f : \mathbb{R}^m \to \mathbb{R}^n$, the **Jacobian** $J \in \mathbb{R}^{n \times m}$ has entries

$$J_{ij} = \frac{\partial f_i(x)}{\partial x_j}, \qquad i = 1, \dots, n,\ j = 1, \dots, m.$$

Row i is the gradient of output f_i. Backpropagation is repeated multiplication by Jacobians; reverse-mode automatic differentiation computes vector–Jacobian products $v^\top J$ without ever forming J, which is why a gradient costs only a small constant multiple of one function evaluation [11].

For a scalar function $f : \mathbb{R}^n \to \mathbb{R}$, the **Hessian** $H(f)(x) \in \mathbb{R}^{n \times n}$ is the Jacobian of the gradient:

$$H(f)(x)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \, \partial x_j}.$$

Wherever the second partial derivatives are continuous, the order of differentiation does not matter, so $H_{ij} = H_{ji}$: the Hessian is symmetric. A real symmetric matrix has real eigenvalues $\lambda_1, \dots, \lambda_n$ and an orthonormal basis of eigenvectors $v_1, \dots, v_n$. The **second derivative in a unit direction** d is $d^\top H d$. If d is an eigenvector v_i, it equals λ_i; for other directions it is a weighted average $\sum_i \lambda_i (d^\top v_i)^2$ of the eigenvalues, with weights between 0 and 1 that sum to 1. So $\lambda_{\max}$ is the largest curvature in any direction and $\lambda_{\min}$ the smallest.

What a gradient step really does: second-order Taylor expansion

Write $g = \nabla_x f(x^{(0)})$ and $H = H(f)(x^{(0)})$. The second-order Taylor approximation around $x^{(0)}$ is

$$f(x) \approx f(x^{(0)}) + (x - x^{(0)})^\top g + \tfrac{1}{2}(x - x^{(0)})^\top H (x - x^{(0)}).$$

Substituting a gradient step $x = x^{(0)} - \eta g$ gives

$$f(x^{(0)} - \eta g) \approx f(x^{(0)}) - \eta\, g^\top g + \tfrac{1}{2}\eta^2\, g^\top H g.$$

The three terms are the current value, the improvement predicted by the slope, and a correction for curvature. If $g^\top H g \le 0$, the approximation keeps decreasing as η grows (until the approximation itself stops being valid). If $g^\top H g > 0$, the curvature term eventually wins, and the approximation is minimized at

$$\eta^* = \frac{g^\top g}{g^\top H g}.$$

In the worst case, g is aligned with the eigenvector of $\lambda_{\max}$ and $\eta^* = 1/\lambda_{\max}$. Large curvature forces small steps.

The second derivative test

At a critical point ($g = 0$) the Taylor expansion reduces to the quadratic term, so the Hessian decides what kind of point it is:

- H **positive definite** (all $\lambda_i > 0$): local minimum.
- H **negative definite** (all $\lambda_i < 0$): local maximum.
- H has at least one positive and at least one negative eigenvalue: **saddle point**.
- Otherwise (some $\lambda_i = 0$ and the rest of one sign) the test is inconclusive.

In one dimension a saddle cannot exist; in n dimensions it needs only one eigenvalue of the "wrong" sign, which is why saddles become so common as n grows [8].

Condition number of the Hessian: why gradient descent zigzags

On the quadratic $f(w) = \frac{1}{2} w^\top H w$ with H symmetric positive definite, gradient descent is linear: $w \leftarrow (I - \eta H) w$. Expanding w in the eigenvectors of H, the component along v_i is multiplied by $(1 - \eta\lambda_i)$ at every step. Therefore:

- The iteration is **stable** only if $|1 - \eta\lambda_i| < 1$ for every i, that is, $0 < \eta < 2/\lambda_{\max}$.
- The **slowest** direction is the flat one, contracting by $|1 - \eta\lambda_{\min}|$ per step.
- The best fixed step, $\eta = 2/(\lambda_{\max} + \lambda_{\min})$, contracts the error by $(\kappa - 1)/(\kappa + 1)$ per step, with $\kappa = \lambda_{\max}/\lambda_{\min}$ the condition number of the Hessian.

So a large condition number means one direction wants tiny steps (or it diverges) while another direction needs many steps to make progress. Gradient descent wastes its time bouncing across the steep walls of a narrow valley — the zigzag in Figure 2 — while creeping along the valley floor. Badly scaled input features produce exactly this situation, which is one reason standardization and normalization layers speed up training.

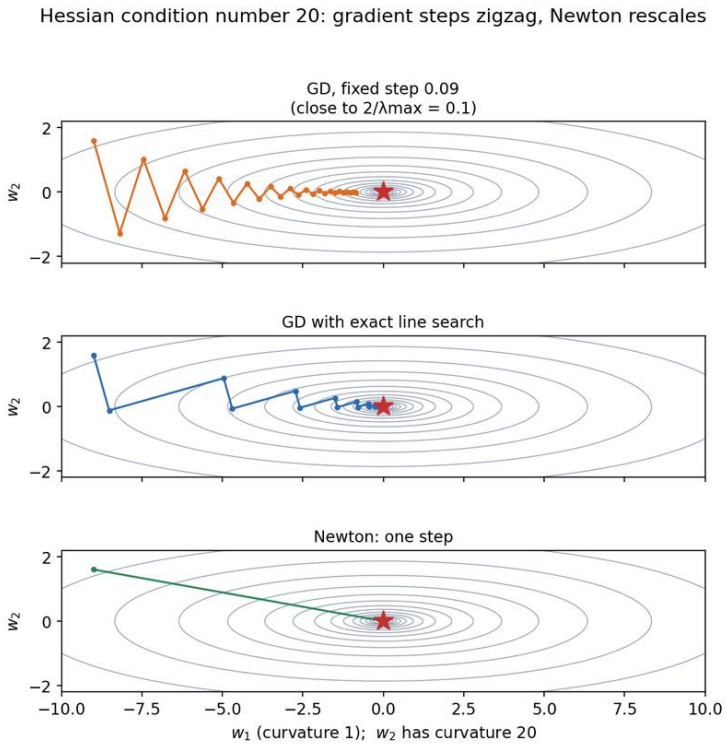


Figure 2 — Minimizing a quadratic with Hessian condition number 20. Gradient descent with a step close to the stability limit oscillates across the steep direction; exact line search makes successive steps orthogonal; Newton's method rescales by the inverse Hessian and reaches the minimum of a quadratic in one step.

Figure 3 shows the stability limit numerically. Below $2/\lambda_{\max}$ the loss decreases; just above it the steep component grows by a factor $|1 - \eta\lambda_{\max}| > 1$ every step and the loss explodes.

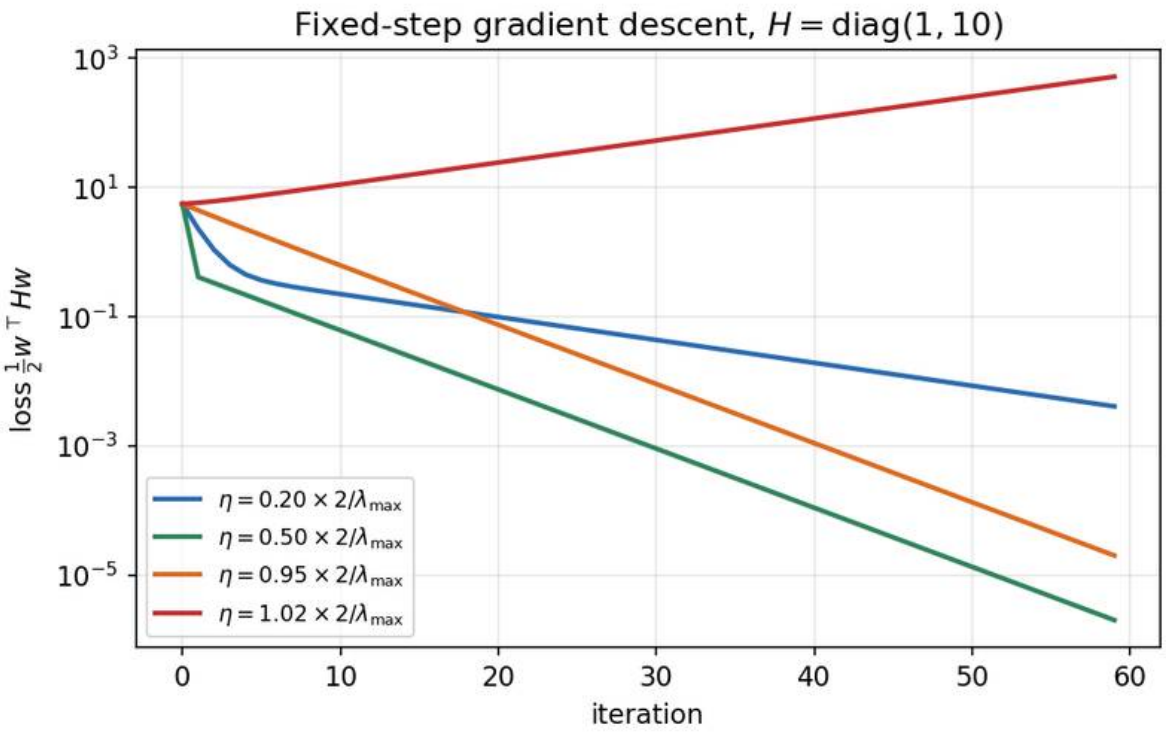


Figure 3 — Fixed-step gradient descent on a quadratic with eigenvalues 1 and 10. Step sizes below 2/λmax converge (fastest near 2/(λmax+λmin)); a step just 2% above the limit diverges.

```
import numpy as np

H = np.diag([1.0, 25.0])                    # Hessian: condition number 25
f = lambda w: 0.5 * w @ H @ w
grad = lambda w: H @ w

def run_gd(eta, steps=50):
    w = np.array([10.0, 1.0])
    for _ in range(steps):
        w = w - eta * grad(w)
    return f(w)

lam_max = np.linalg.eigvalsh(H).max()
for eta in (0.01, 0.07, 0.079, 0.081):       # stability limit 2/25 = 0.08
    print(f"eta={eta:<6} loss after 50 steps: {run_gd(eta):.3e}")

# Exact line search along -g for a quadratic: eps* = g'g / g'Hg
w = np.array([10.0, 1.0])
g = grad(w)
eps_star = (g @ g) / (g @ H @ g)
print("optimal step along the gradient:", round(eps_star, 4))

# One Newton step lands on the minimizer of a quadratic
w_newton = w - np.linalg.solve(H, g)
```

```
print("after one Newton step:", w_newton)
```

The output shows the whole story: $\eta = 0.01$ is stable but slow (loss 18.3 after 50 steps), $\eta = 0.07$ reaches 0.035, $\eta = 0.079$ is barely stable and the steep component still oscillates (loss about 1.0), and $\eta = 0.081$ diverges (loss 148). The optimal step along the first gradient is 0.046, and one Newton step lands exactly on $(0, 0)$.

Second-order methods

Plain version. Instead of following the slope alone, use the curvature as well: divide the step by how sharply the function curves in each direction, so steep directions get small steps and flat directions get large ones.

Newton's method

Setting the gradient of the second-order Taylor approximation to zero gives the **Newton step**

$$x^* = x^{(0)} - H(f)(x^{(0)})^{-1} \nabla_x f(x^{(0)}).$$

On a positive-definite quadratic this jumps to the minimum in one step (Figure 2, bottom). On a general function that is locally close to a positive-definite quadratic, iterating the update converges much faster than gradient descent — quadratically near the solution, under standard assumptions [6]. The cost is the Hessian: storing it takes $O(n^2)$ memory and solving with it $O(n^3)$ time, impossible for a network with 10^8 parameters.

There is also a more fundamental problem for non-convex objectives. Newton's method finds a point where the *gradient* vanishes, and it treats a saddle the same way as a minimum: near a saddle, the Newton step points *toward* it. In a landscape full of saddles, plain Newton is attracted to exactly the points we want to escape. Dauphin et al. proposed a "saddle-free Newton" method that rescales by the absolute values of the Hessian eigenvalues to avoid this [8]. Gradient descent, by contrast, is not attracted to saddles in the same way, although it can move slowly near them.

Quasi-Newton methods and L-BFGS

Quasi-Newton methods build an approximation $B_k \approx H$ (or of its inverse) from gradients alone. After a step $s_k = x_{k+1} - x_k$ with gradient change $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, they require the **secant condition** $B_{k+1} s_k = y_k$, which says the model's curvature along the step matches the observed change in slope. BFGS is the most widely used update of this kind; **L-BFGS** (limited-memory BFGS) stores only the last m pairs (s_k, y_k) and applies the inverse-Hessian approximation implicitly in $O(mn)$ time [6], [10]. L-BFGS is excellent for smooth, deterministic, medium-sized

problems (logistic regression, style transfer on one image, fitting a few thousand parameters); its curvature estimates suffer when gradients are noisy mini-batch estimates.

Hessian-vector products

Many curvature-aware methods never need H itself, only products Hv. These can be computed exactly at roughly the cost of a couple of gradient evaluations, because $Hv = \nabla_x(\nabla_x f(x)^\top v)$ is the gradient of a scalar; Pearlmutter showed how to do this with automatic differentiation [9]. A finite-difference version, $Hv \approx [\nabla f(x + rv) - \nabla f(x - rv)]/(2r)$, is a useful check. Power iteration with Hessian-vector products estimates $\lambda_{\max}$, which is how researchers track the sharpness of the loss during training.

```
import numpy as np

rng = np.random.default_rng(0)
X = rng.normal(size=(50, 4))
y = (rng.random(50) < 0.5).astype(float)

def grad(w):                                    # logistic regression, mean loss
    p = 1 / (1 + np.exp(-(X @ w)))
    return X.T @ (p - y) / len(y)

def hessian(w):
    p = 1 / (1 + np.exp(-(X @ w)))
    return (X.T * (p * (1 - p))) @ X / len(y)

w = rng.normal(size=4)
v = rng.normal(size=4)
r = 1e-5
hvp_fd = (grad(w + r*v) - grad(w - r*v)) / (2*r)   # never forms H
print("Hv (finite diff):", hvp_fd.round(6))
print("Hv (explicit)   :", (hessian(w) @ v).round(6))
print("eigenvalues of H:", np.linalg.eigvalsh(hessian(w)).round(4))
```

Both products agree to six decimals, and all Hessian eigenvalues are positive: logistic regression is convex, so its Hessian is positive semidefinite everywhere.

Lipschitz continuity and convexity

Deep-learning objectives come with few guarantees, but two properties explain much of the theory. A function is **Lipschitz continuous** with constant L if $|f(x) - f(y)| \le L\|x - y\|_2$ for all x, y: its rate of change is bounded. When the *gradient* is L-Lipschitz (the function is "L-smooth"), the curvature is bounded by L, and a gradient step with $\eta \le 1/L$ is guaranteed not to increase the function:

$$f(x - \eta\nabla f(x)) \le f(x) - \tfrac{\eta}{2}\|\nabla f(x)\|_2^2 \qquad (0 < \eta \le 1/L).$$

This is the general form of the "$\eta < 2/\lambda_{\max}$" rule above, and it is the starting point of most convergence proofs [7], [14].

A function is **convex** if its Hessian is positive semidefinite everywhere (for twice-differentiable functions). Convex functions have no saddles and no spurious local minima: every local minimum is global. Least squares, logistic regression and support vector machines are convex; deep networks are not. Convex optimization is the source of many tools, such as duality and the KKT conditions below, that deep learning borrows [7].

Constrained optimization

Plain version. Sometimes the answer must lie in an allowed region — weights with bounded norm, probabilities that sum to one, a physical quantity that must be positive. Either step and then project back into the region, or add a penalty for each constraint whose strength is chosen automatically.

Projected gradient descent

We want to minimize $f(x)$ over a feasible set $\mathbb{S} \subseteq \mathbb{R}^n$. When projecting onto $\mathbb{S}$ is easy, take a normal gradient step and then return to the nearest feasible point:

$$x \leftarrow \text{Proj}_{\mathbb{S}}\left(x - \eta \nabla f(x)\right), \qquad \text{Proj}_{\mathbb{S}}(v) = \arg\min_{x \in \mathbb{S}} \|x - v\|_2.$$

For the Euclidean ball $\{x : \|x\|_2 \leq r\}$ the projection is $v \cdot \min(1, r/\|v\|_2)$: rescale if too long. Max-norm constraints on weights, clipping parameters into a box, and projecting onto the probability simplex all follow this pattern. Another option is to reparameterize so the constraint holds automatically, for example $x = r\,(\cos\theta, \sin\theta)$ for a vector on a circle, or a softmax to produce probabilities.

The generalized Lagrangian

A more general approach describes the feasible set with m equality constraints $g^{(i)}(x) = 0$ and k inequality constraints $h^{(j)}(x) \leq 0$:

$$\mathbb{S} = \{x \mid g^{(i)}(x) = 0 \ \forall i,\ h^{(j)}(x) \leq 0 \ \forall j\}.$$

Introduce one **KKT multiplier** for each constraint, λ_i for equalities and α_j for inequalities, and define the **generalized Lagrangian**

$$L(x, \lambda, \alpha) = f(x) + \sum_{i=1}^{m} \lambda_i g^{(i)}(x) + \sum_{j=1}^{k} \alpha_j h^{(j)}(x).$$

The constrained problem is equivalent to the unconstrained min–max problem

$$\min_{x} \max_{\lambda} \max_{\alpha,\, \alpha \geq 0} L(x, \lambda, \alpha).$$

Why does this work? If x violates a constraint, the inner maximization can drive L to $+\infty$ by choosing a large multiplier, so infeasible points are never chosen. If x is feasible, the best the inner maximization can do is set the terms to zero (equalities are already 0; for $h^{(j)}(x) \leq 0$ the best non-negative α_j makes $\alpha_j h^{(j)}(x) = 0$), so the value is exactly $f(x)$ [1].

The KKT conditions

At a solution x^* (under mild regularity conditions called constraint qualifications), there exist multipliers satisfying the **Karush–Kuhn–Tucker (KKT) conditions** [1], [7]:

1. **Stationarity:** $\nabla_x L(x^*, \lambda^*, \alpha^*) = 0$.
2. **Primal feasibility:** all constraints hold at x^*.
3. **Dual feasibility:** $\alpha^* \geq 0$.
4. **Complementary slackness:** $\alpha_j^* \, h^{(j)}(x^*) = 0$ for every j.

Complementary slackness says each inequality is either **active** ($h^{(j)}(x^*) = 0$: the solution sits on the boundary and the constraint pushes back with $\alpha_j \geq 0$) or **inactive** ($h^{(j)}(x^*) < 0$: the constraint does not matter locally and $\alpha_j = 0$). Geometrically, stationarity says that at an active constraint the downhill direction $-\nabla f$ points straight out of the feasible set, parallel to ∇h; any feasible move would go uphill (Figure 4). For convex problems the KKT conditions are also sufficient for optimality [7].

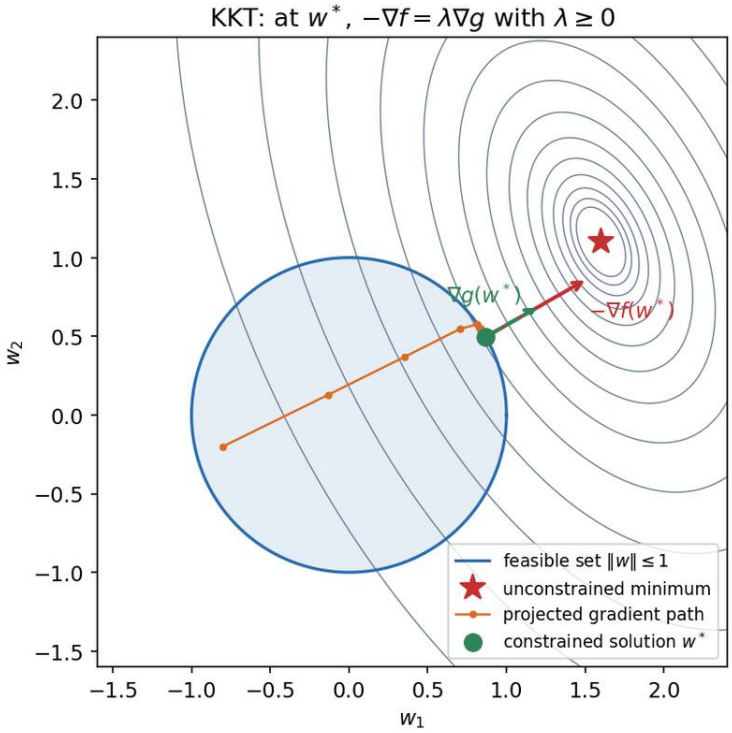


Figure 4 — Minimizing a quadratic over the unit disk. The unconstrained minimum (star) is infeasible. Projected gradient descent stops on the boundary at w, where $-\nabla f$ is parallel to the outward normal ∇g of the constraint: the KKT stationarity condition with a positive multiplier.*

Worked example: least squares with a norm constraint

Following the example in [1], minimize the squared error $f(w) = \|Aw - b\|_2^2$, with $A \in \mathbb{R}^{N \times n}$ and $b \in \mathbb{R}^N$, subject to $h(w) = w^\top w - r^2 \leq 0$. The Lagrangian is

$$L(w, \alpha) = \|Aw - b\|_2^2 + \alpha\,(w^\top w - r^2).$$

Stationarity gives $2A^\top(Aw - b) + 2\alpha w = 0$, so

$$w(\alpha) = (A^\top A + \alpha I)^{-1} A^\top b.$$

This is ridge regression with penalty α. If the ordinary least-squares solution $w(0)$ already satisfies $\|w\| \leq r$, the constraint is inactive, $\alpha^* = 0$, and we are done. Otherwise the constraint is active, and we increase α until $\|w(\alpha)\| = r$; because $\|w(\alpha)\|$ decreases monotonically in α, a bisection finds it. This is also how the inner maximization over α behaves: the multiplier grows while the constraint is violated and stops when it is just satisfied. The code solves the problem both ways and checks all four KKT conditions.

```
import numpy as np
```

```
# minimize ||Aw - b||^2  subject to  ||w||^2 <= r^2
rng = np.random.default_rng(1)
A = rng.normal(size=(30, 5))
b = rng.normal(size=30)
r = 0.2

def w_of(lam):                                  # stationarity of the Lagrangian
    return np.linalg.solve(A.T @ A + lam * np.eye(5), A.T @ b)

w_ls = w_of(0.0)
print("unconstrained norm:", np.linalg.norm(w_ls).round(3))

lo, hi = 0.0, 1e6                               # find lambda with ||w|| = r
for _ in range(200):
    mid = 0.5 * (lo + hi)
    if np.linalg.norm(w_of(mid)) > r:
        lo = mid
    else:
        hi = mid
lam = hi
w = w_of(lam)
print("lambda:", round(lam, 3), " ||w||:", np.linalg.norm(w).round(6))

# KKT checks
stationarity = 2 * A.T @ (A @ w - b) + 2 * lam * w
print("stationarity residual:", np.abs(stationarity).max())
print("primal feasible:", w @ w <= r**2 + 1e-9, " dual feasible:", lam >= 0)
print("complementary slackness:", lam * (w @ w - r**2))

# Projected gradient descent reaches the same point
L = 2 * np.linalg.eigvalsh(A.T @ A).max()     # Lipschitz constant of the gradient
v = np.zeros(5)
for _ in range(5000):
    v = v - (1 / L) * 2 * A.T @ (A @ v - b)
    n = np.linalg.norm(v)
    if n > r:
        v = v * (r / n)                         # projection onto the ball
print("projected GD matches KKT solution:", np.allclose(v, w, atol=1e-6))
```

The unconstrained solution has norm 0.219, larger than $r = 0.2$, so the constraint is active. Bisection finds $\alpha^* \approx 1.81$, the stationarity residual is about 10^{-15}, complementary slackness holds, and projected gradient descent with step $1/L$ converges to the same vector. Note how L, the Lipschitz constant of the gradient, sets a safe step size, exactly as in the previous section.

Checking derivatives

Plain version. Check that the gradient your code computes matches how the loss actually changes when you nudge the parameters.

A central finite difference estimates a partial derivative,

$$\frac{\partial f}{\partial x_j} \approx \frac{f(x + \epsilon e_j) - f(x - \epsilon e_j)}{2\epsilon},$$

where e_j is the j-th standard basis vector and $\epsilon > 0$ a small step. Its **truncation error** (from ignoring higher-order Taylor terms) is $O(\epsilon^2)$, better than the $O(\epsilon)$ of the forward difference $[f(x + \epsilon e_j) - f(x)]/\epsilon$. But the numerator subtracts two nearly equal numbers, so its **roundoff error** grows like $u\,|f|/\epsilon$. The total error is smallest at an intermediate step — roughly $\epsilon \approx u^{1/3}$ for central differences and $u^{1/2}$ for forward differences, when f and its derivatives have unit scale. Figure 5 shows the resulting V-shaped curves, and why float32 gradient checks are much less precise than float64 ones.

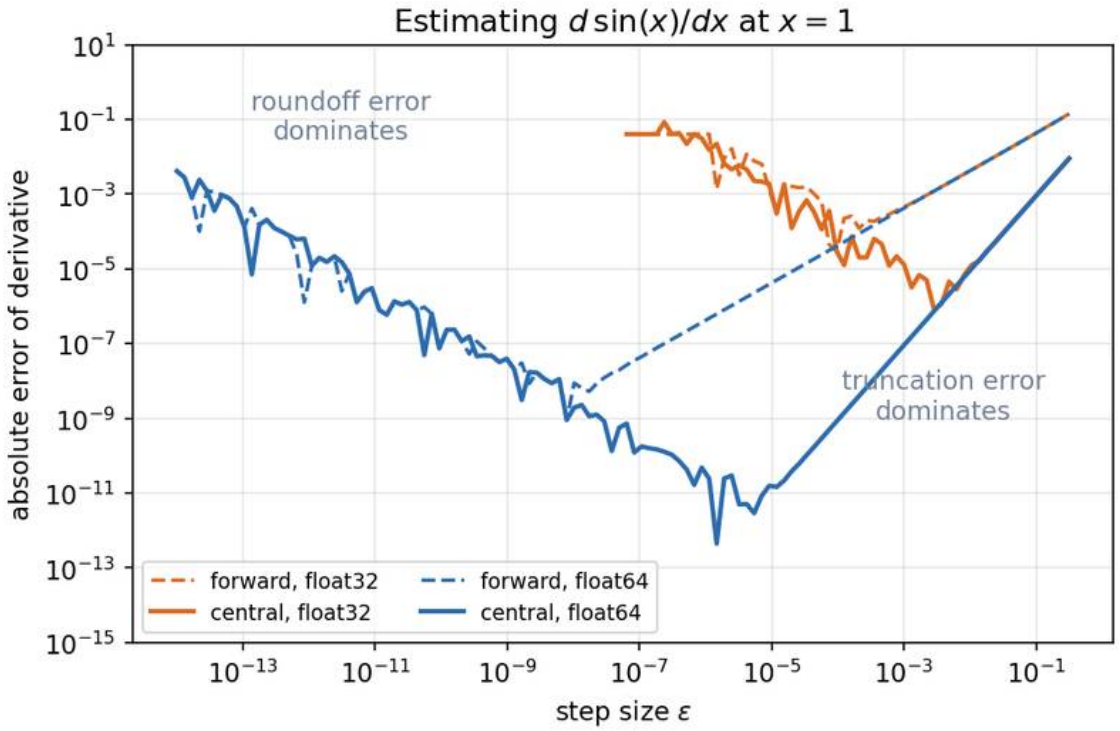


Figure 5 — Error of finite-difference derivatives of $\sin(x)$ at x=1. Right of each minimum, truncation error dominates (slope 1 for forward, 2 for central differences); left of it, roundoff dominates. Central differences bottom out around 1e−6 in float32 but around 1e−11 or lower in float64.

Practical rules: use float64 for checks; try several step sizes; compare with a relative error such as $|a - n|/\max(|a|, |n|, \text{tiny})$; and test at smooth points. ReLU exactly at zero is not differentiable, so there is no single derivative to match there. For many parameters, a **directional check** is cheaper: choose a random unit vector v and compare $\nabla f(w)^\top v$ with a finite difference along v — one pair of function evaluations instead of $2n$.

```
import numpy as np
```

```
rng = np.random.default_rng(0)
X = rng.normal(size=(20, 5))
y = rng.normal(size=20)
w = rng.normal(size=5)
v = rng.normal(size=5)
v /= np.linalg.norm(v)
def loss(w):
    return .5*np.mean((X@w-y)**2)
grad = X.T@(X@w-y)/len(y)
eps = 1e-5
numeric = (loss(w+eps*v)-loss(w-eps*v))/(2*eps)
analytic = grad@v
assert np.allclose(numeric, analytic, rtol=1e-6, atol=1e-8)
print("directional derivative:", numeric, analytic)
```

The two numbers agree to about nine significant digits. Remember what a passing check means. Automatic differentiation computes derivatives of the *recorded program*, not of the program you intended [11]. A detached tensor, a wrong reduction, accidental broadcasting or target leakage produces a perfectly differentiable but incorrect objective, and a gradient check will happily confirm its gradient. Check the loss values themselves on cases where you know the answer.

Mixed precision

Plain version. Doing most arithmetic in 16-bit numbers halves memory traffic and uses fast hardware units, but 16 bits are not enough for everything. Keep a few sensitive quantities in 32 bits.

Mixed-precision training, as introduced by Micikevicius et al. [12], combines three ideas:

1. Keep a **float32 master copy of the weights**. An update smaller than about 2^{-11} times the weight it is added to rounds away entirely in float16 (the same effect as $1 + \varepsilon_{\text{mach}}/2 = 1$ above), and many updates are that small.
2. Use **loss scaling** for float16: multiply the loss by a factor S before backpropagation so small gradients do not underflow below float16's smallest subnormal (6×10^{-8}), then divide the gradients by S before the update. Dynamic loss scaling reduces S when an overflow (inf or NaN) appears and skips that step.
3. **Accumulate** dot products and reductions in float32 even when inputs are float16.

bfloat16 has the same 8-bit exponent as float32, so it rarely needs loss scaling; Kalamkar et al. found bfloat16 training reached float32 results across a range of models without changing hyperparameters [5]. The price is precision: with $\varepsilon_{\text{mach}} \approx 0.0078$, bfloat16 cannot represent $1 + 0.003$ distinctly from 1, so long sums and weight updates still need float32 accumulation.

Loss scaling protects small float16 gradients; it cannot repair an invalid objective or an overflow elsewhere in the network (an attention logit or a normalization statistic, for example). Changing

precision can also change optimization trajectories and outputs, especially for reductions and long training runs. Validate on the target hardware; a smaller dtype does not guarantee that every operation becomes faster [13].

A debugging order for non-finite values

When a loss becomes `nan` or `inf`, resist the urge to clip everything. Find the *first* operation that produced a non-finite value, then ask which of this chapter's failure modes it is. A useful order:

1. **Inputs:** check shapes, ranges, dtypes and missing values. A single NaN pixel or an un-normalized 0–255 image changes everything downstream.
2. **Forward pass:** check that activations and the loss are finite. Look for `exp`, `log`, division, `sqrt` and normalization layers with near-zero variance.
3. **Loss:** use logits-based losses (`cross_entropy`, `binary_cross_entropy_with_logits`) rather than probabilities followed by `log`.
4. **Gradients:** run a gradient check on a tiny model; inspect per-layer gradient norms.
5. **Updates:** compare the update size with the weight size; a learning rate above the curvature limit $2/\lambda_{\max}$ causes oscillation and divergence even with perfect arithmetic.
6. **Precision:** reproduce in float32 (or float64) before adding mixed precision or compilation.

Overfit a tiny batch first to establish that the pipeline can learn. Gradient-norm clipping bounds the gradient passed to the optimizer; it does not directly bound an adaptive optimizer's update, and it does not make NaN gradients meaningful. ML 02 develops optimizers and normalization in more detail.

Modern view

The core ideas of this chapter are decades old, but they are at the centre of current research because models are larger, training runs are longer, and hardware keeps pushing precision downward.

Optimization theory for large-scale learning. Bottou, Curtis and Nocedal's SIAM Review survey [14] is the standard map of the field. It explains why stochastic gradient methods dominate large-scale machine learning (cheap noisy steps beat expensive exact ones when data are redundant), analyses their convergence under smoothness assumptions like the Lipschitz condition above, and organizes improvements into two families: *noise reduction* (larger or growing batches, variance reduction, iterate averaging) and *second-order* methods (Hessian-free Newton, Gauss–Newton, quasi-Newton, natural gradient and diagonal scaling). Nocedal and Wright [6] and Boyd and Vandenberghe [7] remain the reference textbooks for the deterministic and convex theory, and Baydin et al. [11] survey the automatic differentiation that makes gradients cheap.

What the loss surface actually looks like. Dauphin et al. [8] argued that saddle points, not poor local minima, are the main obstacle in high-dimensional non-convex optimization. Later, Cohen et al. [15] observed that full-batch gradient descent on neural networks typically operates at the "edge of stability": the largest Hessian eigenvalue rises until it hovers just above $2/\eta$ — precisely the classical stability limit derived above — while the loss keeps decreasing over long time scales, though not monotonically. The classical quadratic analysis predicts divergence there, so explaining this regime remains an open theoretical problem.

Curvature-aware optimizers. Full Newton is impossible at scale, but structured approximations are practical. K-FAC [16] approximates the Fisher information matrix (a positive-semidefinite curvature matrix closely related to the Gauss–Newton approximation of the Hessian) with Kronecker products of small per-layer matrices, so inverting it is cheap. Shampoo [17] keeps one preconditioning matrix per tensor dimension. More recently, Muon applies an approximate orthogonalization to each weight matrix's update; Liu et al. [18] report that, with suitable weight decay and per-parameter update scaling, it trains large language models with roughly twice the computational efficiency of their adaptive-optimizer baseline in their experiments. These methods all revisit the question this chapter asks with the Hessian: how should a step be rescaled direction by direction?

Lower precision. The format story continues from float16 with loss scaling [12] and bfloat16 [5] to 8-bit floating point. Micikevicius et al. [19] proposed two FP8 encodings, E4M3 (more precision) and E5M2 (more range), and showed training results comparable to 16-bit baselines across image and language models. The range–precision trade-off in the table above is exactly what these designs balance, and the IEEE 754 rules [3] are what they relax. Per-tensor or per-block scaling factors play the role that loss scaling played for float16.

Stable softmax at scale. The max-subtraction trick has a streaming version. Milakov and Gimelshein [20] showed that a softmax normalizer can be computed in one pass by keeping a running maximum and rescaling the running sum whenever the maximum changes. FlashAttention [21] builds on this idea: it computes exact attention block by block in fast on-chip memory, never materializing the full attention matrix, and FlashAttention-3 [22] extends it to FP8 with block quantization and "incoherent processing" to reduce quantization error. The core of the online trick fits in a few lines:

```
import numpy as np

def online_logsumexp(chunks):
    m, s = -np.inf, 0.0                 # running max and rescaled running sum
    for c in chunks:
        m_new = max(m, c.max())
        s = s * np.exp(m - m_new) + np.exp(c - m_new).sum()
        m = m_new
    return m + np.log(s)

rng = np.random.default_rng(0)
z = rng.normal(0, 50, size=10_000)
```

```
chunks = np.array_split(z, 37)          # seen one block at a time
m = z.max()
reference = m + np.log(np.exp(z - m).sum())
print(online_logsumexp(chunks), reference)
assert np.isclose(online_logsumexp(chunks), reference)
```

Both values print as 174.0918752579 ..., identical to the last digit shown, although the online version never sees all logits at once. ML 04 covers attention itself, and ML 10 covers quantization for deployment.

Open problems. Why gradient methods succeed at the edge of stability; how to make low-precision (8-bit and below) training as robust as float32 without hand-tuned scaling; how to make curvature-aware optimizers cheap and robust enough to replace first-order defaults; and how to obtain reproducible results when parallel reductions change summation order.

Key takeaways

- Floating-point numbers have a fixed relative precision ($\varepsilon_{\text{mach}}$) and a fixed range; exponent bits set range, fraction bits set precision.
- Overflow, underflow and cancellation are predictable. Fix them algebraically: subtract the max before `exp`, use `log1p`/`expm1`, use logits-based losses, subtract means before squaring.
- Conditioning is a property of the problem; stability is a property of the algorithm. A condition number of 10^k costs about k digits.
- The Hessian's eigenvalues give curvature. Gradient descent is stable only for $\eta < 2/\lambda_{\max}$ and slows down by the condition number $\lambda_{\max}/\lambda_{\min}$.
- Newton's method rescales by curvature and converges fast near minima, but it is expensive and attracted to saddles; quasi-Newton methods and Hessian-vector products are the practical compromises.
- Constraints can be handled by projection or by the generalized Lagrangian; the KKT conditions characterize the solution.
- Gradient checks need float64, sensible step sizes and smooth test points, and they only test the program you wrote.
- Mixed precision needs float32 master weights and accumulation, and loss scaling for float16; validate rather than assume identical behaviour.

Exercises

1. **Softmax shift.** Prove that softmax is invariant under $z_i \mapsto z_i + c$, and show that with $c = -\max_i z_i$ the denominator lies in $[1, K]$.

Hint

The shared factor $\exp(c)$ cancels between numerator and denominator. After the shift, every term $\exp(z_j - m)$ is in $(0, 1]$ and the largest equals 1, so the sum of K terms is between 1 and K. It can therefore neither overflow nor underflow to 0.

2. **Finite-difference sweep.** For $f(x) = e^x$ at $x = 0$, compute forward and central differences for $\epsilon = 10^{-1}, \ldots, 10^{-12}$ in float32 and float64. At which ϵ is each error smallest? Compare with $u^{1/2}$ and $u^{1/3}$.

Hint

For float64, $u \approx 1.1 \times 10^{-16}$, so $u^{1/2} \approx 10^{-8}$ (forward) and $u^{1/3} \approx 5 \times 10^{-6}$ (central). For float32, $u \approx 6 \times 10^{-8}$, giving about 2×10^{-4} and 4×10^{-3}. The observed minima should be within an order of magnitude of these, and the error at the minimum is roughly $u^{1/2}$ and $u^{2/3}$ respectively.

3. **Stability limit.** For $f(w) = \frac{1}{2} w^\top H w$ with $H = \mathrm{diag}(2, 50)$, find the largest stable fixed learning rate and the step size that minimizes the worst-case contraction factor. How many iterations does that best step need to reduce the error by a factor of 10^6?

Hint

Stability requires $\eta < 2/50 = 0.04$. The best step is $2/(\lambda_{\max} + \lambda_{\min}) = 2/52 \approx 0.0385$, with contraction $(\kappa - 1)/(\kappa + 1) = 24/26 \approx 0.923$. Solve $0.923^k \leq 10^{-6}$: $k \geq \ln 10^6 / (-\ln 0.923) \approx 173$ iterations. A Newton step would need one.

4. **Classify critical points.** Find and classify the critical points of $f(x, y) = x^3 - 3x + y^2$.

Hint

$\nabla f = (3x^2 - 3,\ 2y) = 0$ gives $(1, 0)$ and $(-1, 0)$. The Hessian is $\mathrm{diag}(6x,\ 2)$: at $(1, 0)$ it is $\mathrm{diag}(6, 2)$, positive definite, a local minimum; at $(-1, 0)$ it is $\mathrm{diag}(-6, 2)$, indefinite, a saddle. Neither is a global minimum because $f \to -\infty$ as $x \to -\infty$. Starting Newton's method near $(-1, 0)$ converges to the saddle.

5. **KKT on the simplex.** Minimize $f(x) = \frac{1}{2}\|x - c\|_2^2$ subject to $\sum_i x_i = 1$ and $x_i \geq 0$, for $c = (0.8,\ 0.6, -0.4)$. Write the Lagrangian and the KKT conditions, and find the solution.

Hint

$L = \frac{1}{2}\|x - c\|^2 + \lambda(\sum_i x_i - 1) - \sum_i \alpha_i x_i$ with $\alpha_i \geq 0$. Stationarity gives $x_i = c_i - \lambda + \alpha_i$; complementary slackness gives $x_i = \max(c_i - \lambda, 0)$. Guess that only the first two are positive: $(0.8 - \lambda) + (0.6 - \lambda) = 1$, so $\lambda = 0.2$ and $x = (0.6, 0.4, 0)$. Check: $c_3 - \lambda = -0.6 < 0$, so $x_3 = 0$ with $\alpha_3 = 0.6 \geq 0$. This is the Euclidean projection onto the probability simplex.

6. **A correct gradient of the wrong loss.** Change a mean reduction into a sum in the gradient-check example. Does the check still pass? Is autograd wrong? What else changes in training?

Hint

If you change both the loss and the analytic gradient consistently, the check passes: the derivative is correct for the new objective. Nothing is wrong with autograd. But the gradient is now N times larger, so the largest stable learning rate shrinks by N, and in a multi-term loss the balance between terms changes. A gradient check cannot tell you that you chose the wrong objective.

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ML 02 • Neural Networks from Perceptron to Deep Networks

From a single perceptron to deep networks: activations, normalization, output layers and losses, backpropagation, optimizers, and the habits that make a network train and generalize.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-02-neural-networks/

What you'll learn

- What a perceptron is, how its learning rule works, and exactly how it differs from logistic regression.
- How stacking layers gives a multilayer perceptron (MLP), what the universal approximation theorem does and does not promise, and what "deep" really means.
- The activation functions in use today (sigmoid, tanh, softmax, ReLU, Leaky ReLU, GELU, SiLU/Swish, SwiGLU) and the normalization layers (BatchNorm, LayerNorm, RMSNorm), with their formulas and derivatives.
- How to pair the output layer with the right loss, and how backpropagation computes every gradient, derived by hand for a two-layer network and checked numerically in NumPy.
- How to train in practice: initialization, SGD, momentum, Adam and AdamW, learning-rate schedules, batch size, gradient clipping, and how to read a loss curve.
- How to diagnose underfitting and overfitting, the regularization tools that fix overfitting, and how residual connections lead to the Transformer.

The big picture

A neural network is a function with many adjustable knobs (its **parameters**). It takes an input, such as the pixels of an image or two coordinates of a point, and produces an output, such as a class label. Training means turning the knobs, a tiny amount at a time, so the outputs on known examples get closer to the right answers. Everything in this tutorial is about one of three questions: what shape the function has (the architecture), how to measure "closer" (the loss), and how to turn the knobs well (the optimization).

> **Explain like I'm five**
>
> Think of a team of people passing notes down a line. The first person looks at a picture and writes a few simple observations ("there is a curve here", "this part is bright"). The next person reads those notes and writes slightly smarter ones ("this looks like an eye"). The last person says "cat" or "dog". When the answer is wrong, a teacher walks back up the line telling each person how much their note

was to blame, and each one changes their habits a little. That walk backward is **backpropagation**, and doing it many thousands of times is **training**.

Why it matters: almost every modern system in vision, speech and language is built from the pieces in this tutorial. Convolutional networks, Transformers and diffusion models all use the same ingredients: linear layers, nonlinear activations, normalization, residual connections, a loss, backpropagation and an adaptive optimizer. Once these are clear, the deep-learning track is about arranging them for images.

About this tutorial

This tutorial is expanded from the author's earlier study notes on machine learning and neural networks [1], which are written in Traditional Chinese: 機器學習及類神經網路筆記. It assumes the ideas of ML 01: data, features, a loss, and gradient descent.

The perceptron

Plain version. A perceptron is the simplest artificial neuron. It multiplies each input by a weight, adds them up with a bias, and answers "yes" if the total is positive and "no" otherwise. It learns by nudging its weights only when it gets an example wrong.

The model

Rosenblatt introduced the perceptron in 1958 as a model of how a brain might store and recognize patterns [4]. For an input vector $\mathbf{x} \in \mathbb{R}^d$ it computes

$$z = \mathbf{w}^\top \mathbf{x} + b, \qquad \hat{y} = \mathbf{1}[z > 0],$$

where $\mathbf{w} \in \mathbb{R}^d$ is the weight vector, $b \in \mathbb{R}$ is the bias, z is the **pre-activation** (a weighted sum), $\mathbf{1}[\cdot]$ is the indicator that equals 1 when its condition is true and 0 otherwise, and $\hat{y} \in \{0, 1\}$ is the predicted class. The indicator is the **step function**. Geometrically, $\mathbf{w}^\top \mathbf{x} + b = 0$ is a hyperplane (a line in 2-D), and the perceptron labels points by which side of it they fall on.

The perceptron learning rule

Go through the training examples $(\mathbf{x}_i, y_i)$ one at a time. For each one, compute $\hat{y}_i$ and update

$$\mathbf{w} \leftarrow \mathbf{w} + \eta\,(y_i - \hat{y}_i)\,\mathbf{x}_i, \qquad b \leftarrow b + \eta\,(y_i - \hat{y}_i),$$

where $\eta > 0$ is the **learning rate** (step size) and $y_i \in \{0, 1\}$ is the true label. If the prediction is right, $y_i - \hat{y}_i = 0$ and nothing changes. If the perceptron said 0 but the answer was 1, the weights

move *toward* $\mathbf{x}_i$, which raises z for this input next time; the opposite mistake moves them away.

When the two classes can be separated by a hyperplane (the data are **linearly separable**), this rule stops making mistakes after a finite number of updates. When they cannot, it never settles.

```
import numpy as np

def perceptron_train(X, y, epochs=20, eta=1.0):
    """X: (N, d), y in {0, 1}. Returns w, b."""
    w, b = np.zeros(X.shape[1]), 0.0
    for _ in range(epochs):
        mistakes = 0
        for x_i, y_i in zip(X, y):
            y_hat = float(w @ x_i + b > 0)       # step function
            if y_hat != y_i:                     # update only on a mistake
                w += eta * (y_i - y_hat) * x_i
                b += eta * (y_i - y_hat)
                mistakes += 1
        if mistakes == 0:
            break
    return w, b

# AND is linearly separable; XOR is not
X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]], dtype=float)
for name, y in [("AND", [0, 0, 0, 1]), ("XOR", [0, 1, 1, 0])]:
    w, b = perceptron_train(X, np.array(y, float))
    print(name, (X @ w + b > 0).astype(int))
# AND [0 0 0 1]
# XOR [1 1 0 0]   <- wrong: no single line can do it
```

Perceptron versus logistic regression

The two models are easy to confuse because they share the same first step, $z = \mathbf{w}^\top \mathbf{x} + b$. They are different methods:

	Perceptron	Logistic regression
Output function	step: $\hat{y} = \mathbf{1}[z > 0]$	sigmoid: $\hat{p} = \sigma(z) = \dfrac{1}{1+e^{-z}}$
Output meaning	a hard 0/1 decision	a probability $P(y = 1 \mid \mathbf{x})$
Training	perceptron rule, updates only on mistakes	minimize cross-entropy (maximum likelihood) with gradient descent
Non-separable data	never converges	converges to a well-defined best fit

The gradient of the logistic-regression loss for one example is $(\hat{p}_i - y_i)\,\mathbf{x}_i$, so its gradient-descent step is $\mathbf{w} \leftarrow \mathbf{w} + \eta\,(y_i - \hat{p}_i)\,\mathbf{x}_i$. It looks almost identical to the perceptron rule, but $\hat{p}_i$ is a probability between 0 and 1, so *every* example contributes, and confidently correct ones contribute very little. That smoothness is what makes logistic regression trainable by gradient descent and what lets it

be stacked into larger differentiable networks. The step function, in contrast, has derivative zero almost everywhere, so gradients cannot flow through it.

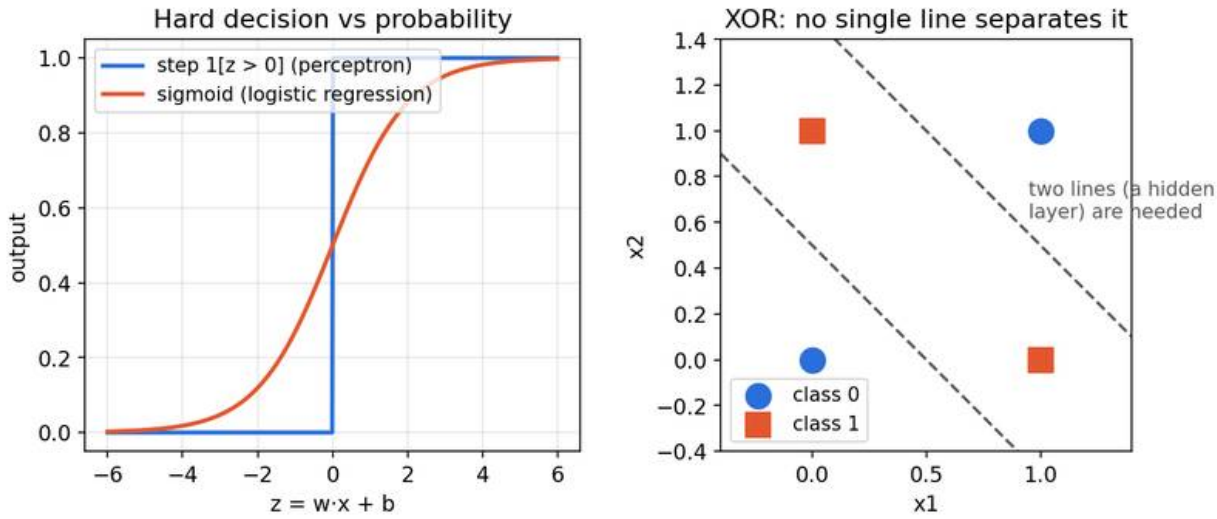


Figure 1 — Left: the perceptron's step output versus logistic regression's sigmoid probability. Right: XOR, the classic problem that no single linear boundary can solve; two boundaries (a hidden layer) are needed.

The multilayer perceptron (MLP)

Plain version. One neuron draws one straight line. Put many neurons side by side to get many lines, then feed their outputs into another layer that combines them. The combination can bend, so the network can carve out curved regions.

Stacking layers

An MLP with one hidden layer computes

$$\mathbf{h} = f(W_1\mathbf{x} + \mathbf{b}_1),$$
$$\hat{\mathbf{y}} = g(W_2\mathbf{h} + \mathbf{b}_2),$$

where $\mathbf{x} \in \mathbb{R}^d$ is the input, $W_1 \in \mathbb{R}^{m\times d}$ and $\mathbf{b}_1 \in \mathbb{R}^m$ are the first layer's weights and biases, m is the **width** (number of hidden units), f is a nonlinear **activation function** applied element-wise, $\mathbf{h} \in \mathbb{R}^m$ is the hidden representation, $W_2 \in \mathbb{R}^{K\times m}$ and $\mathbf{b}_2 \in \mathbb{R}^K$ map it to K outputs, and g is the output function chosen for the task (see below). Each hidden unit is one "logistic-regression-like" neuron; the name multilayer *perceptron* is historical, since modern MLPs use smooth activations, not the step function.

The nonlinearity f is essential. Without it,

$W_2(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 = (W_2W_1)\mathbf{x} + (W_2\mathbf{b}_1 + \mathbf{b}_2)$ is again a single linear map: any number of linear layers collapses to one, and the network can still only draw one hyperplane. For XOR, two hidden units can each draw one of the dashed lines in Figure 1, and the output unit then answers "between the two lines".

Universal approximation

Cybenko proved in 1989 that a network with **one** hidden layer of sigmoidal units can approximate any continuous function on a bounded box (the unit hypercube) to any desired accuracy, provided the hidden layer is wide enough [5]. Later work extended the result to other activations, including ReLU. The theorem is reassuring, but read it carefully:

- It is about **existence**. It says suitable weights exist; it says nothing about whether gradient descent will find them.
- It says nothing about **how wide** "wide enough" is. For some functions the required width grows extremely fast with the input dimension.
- It says nothing about **generalization**: fitting the training points well does not mean predicting new points well.

So the theorem explains why MLPs are flexible enough in principle. The rest of this tutorial is about the practical questions it leaves open.

Deep neural networks: depth and width

Plain version. "Deep" just means many layers stacked in sequence. There is no official number of layers at which a network becomes deep.

A **deep neural network** (DNN) is an MLP, or any other layered network, with several hidden layers: $\mathbf{h}_\ell = f(W_\ell\mathbf{h}_{\ell-1} + \mathbf{b}_\ell)$ for $\ell = 1, \dots, L$, with $\mathbf{h}_0 = \mathbf{x}$. There is no agreed threshold for "deep". A common informal usage calls any network with more than one hidden layer deep, while other texts count total layers or hidden layers differently, so "three layers" means different things in different books. What matters is the idea, not the count: a deep network builds its answer as a **composition** of many simple steps.

The two ways to add capacity are **width** (more units per layer) and **depth** (more layers). They are not equivalent. A wider layer adds more pieces that are combined once; a deeper network re-uses the features of earlier layers to build more abstract ones. The review by LeCun, Bengio and Hinton [2] explains the success of deep learning through exactly this point: each layer turns the representation from the previous layer into a slightly more abstract one, and composing many

such layers lets the network learn very intricate functions from raw data, with the features learned rather than hand-designed.

Historically, the 1990s were dominated by **shallow** methods. Support vector machines, a kernel method rather than a wide neural network, were the main tool, and deep networks were hard to train. From the mid-2000s onward, more data, GPUs and better training techniques made depth practical, and deep networks began to win benchmark after benchmark [2]. Depth alone is not the whole story, though: simply stacking more plain layers eventually makes training *worse*, a problem that residual connections solved (see the last main section).

Figure 2 shows width and depth at work on the two-moons dataset from scikit-learn. With two or eight hidden units in one layer, the boundary is made of only a few straight pieces and cannot follow the moons. Wider or deeper networks bend the boundary to fit; the widest and deepest one even wraps around single noisy points, a first hint of overfitting.

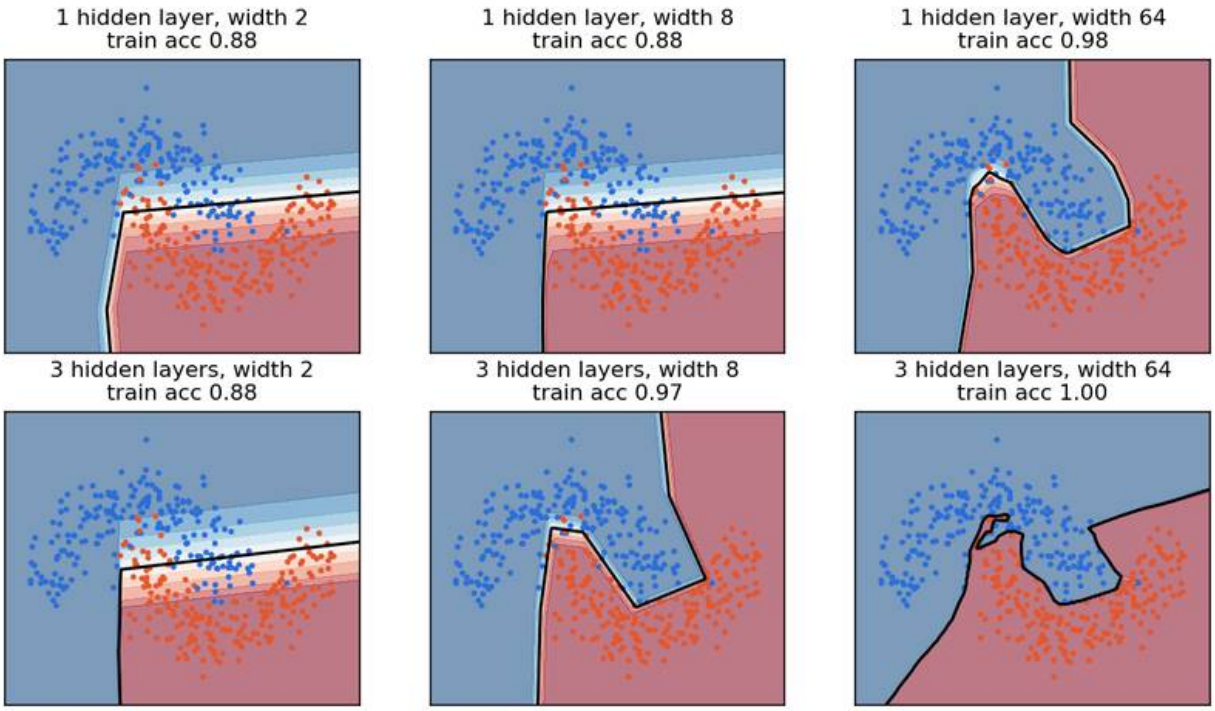


Figure 2 — Decision boundaries of ReLU MLPs on make_moons (400 points, noise 0.2) as width (columns) and depth (rows) change, each trained with Adam for 1,500 full-batch steps. The black curve is the 0.5 probability contour.

Activation functions

Plain version. The activation function is the "bend" applied after each weighted sum. Without it, the whole network would be a straight-line model. Different bends trade off smoothness, speed

and how well gradients pass through many layers.

Figure 3 plots the common choices and their derivatives. The derivative matters as much as the function, because backpropagation multiplies derivatives layer after layer (see the training section).

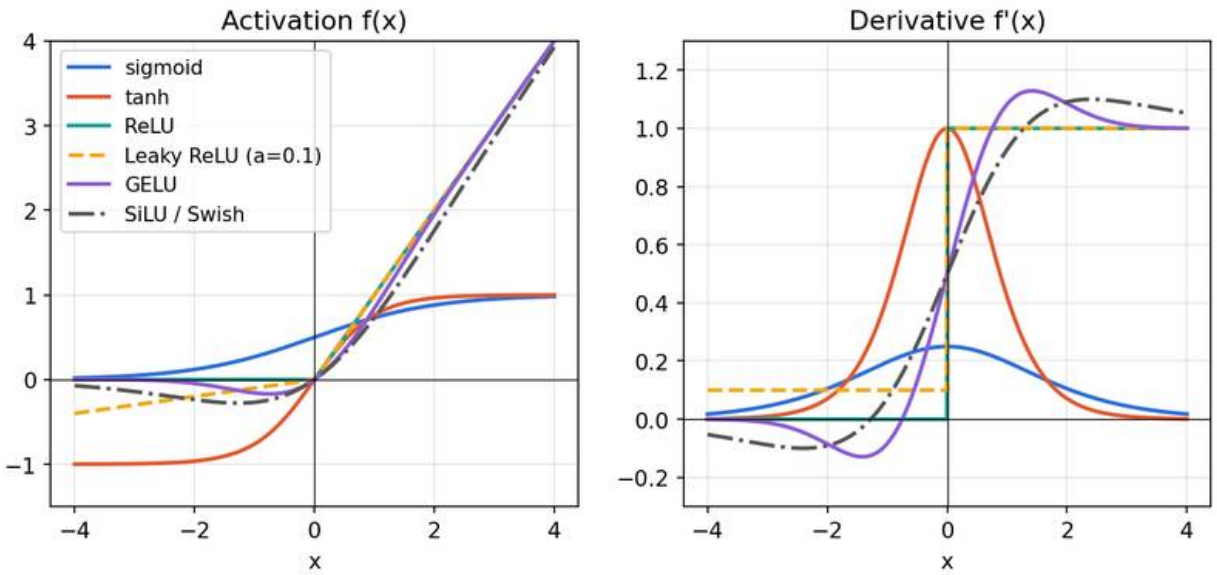


Figure 3 — Common activation functions (left) and their derivatives (right). Note how small the sigmoid's derivative is everywhere (at most 0.25) and how flat both sigmoid and tanh become for large inputs.

Sigmoid and tanh

$$\sigma(x) = \frac{1}{1+e^{-x}}, \quad \sigma'(x) = \sigma(x)(1-\sigma(x)); \qquad \tanh(x) = 2\sigma(2x) - 1, \quad \tanh'(x) = 1 - \tanh^2(x).$$

The sigmoid maps any real number to $(0, 1)$; tanh maps it to $(-1, 1)$ and is centered at zero. Both **saturate**: for large $|x|$ the curve is flat and the derivative is nearly zero, so almost no gradient passes through. The sigmoid's derivative is at most $1/4$ (at $x = 0$). These two facts made deep sigmoid networks hard to train, which Glorot and Bengio analyzed in detail [17]. Today sigmoid is used mainly at the **output** of binary classifiers and inside gates (as in SwiGLU below), and tanh in some recurrent networks.

Softmax

Softmax turns a vector of K real scores $\mathbf{z}$ (called **logits**) into a probability distribution:

$$p_k = \mathrm{softmax}(\mathbf{z})_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, \qquad \frac{\partial p_k}{\partial z_j} = p_k(\delta_{kj} - p_j),$$

where δ_{kj} is 1 if $k = j$ and 0 otherwise. Each p_k is positive and they sum to 1. Unlike the others in this section, softmax acts on a **whole vector**, not element by element, so it is used almost only in the output layer of multi-class classifiers (and inside attention), not as a hidden activation. In code, subtract $\max_j z_j$ before exponentiating; it does not change the result and avoids overflow.

ReLU and Leaky ReLU

$$\mathrm{ReLU}(x) = \max(0, x), \quad \mathrm{ReLU}'(x) = \mathbf{1}[x > 0]; \qquad \mathrm{LeakyReLU}_a(x) = \max(ax, x), \quad 0 < a < 1.$$

The rectified linear unit passes positive inputs unchanged and zeroes the rest. Its derivative is exactly 1 for all positive inputs, so gradients do not shrink as they pass through active units, and it is very cheap to compute. ReLU was one of the changes that made deep supervised networks trainable without unsupervised pre-training [2]. Its weakness is the **dying ReLU**: a unit whose input is negative for every example has zero gradient and stops learning. Leaky ReLU keeps a small slope a (for example 0.01) on the negative side so some gradient always flows. The activation survey by Dubey, Singh and Chaudhuri [7] groups these and many later variants into a ReLU-based family.

GELU and SiLU/Swish

These are smooth versions of ReLU that let small negative values through:

$$\mathrm{GELU}(x) = x\,\Phi(x), \qquad \mathrm{GELU}'(x) = \Phi(x) + x\,\varphi(x),$$

where Φ is the standard normal cumulative distribution function and φ its density [8]. Instead of gating by sign as ReLU does, GELU weights the input by how large it is: $\Phi(x)$ is the probability that a standard normal variable is below x. A common fast approximation is $0.5x(1 + \tanh[\sqrt{2/\pi}\,(x + 0.044715x^3)])$ [8]. GELU is the default activation in many Transformer models.

$$\mathrm{Swish}_\beta(x) = x\,\sigma(\beta x), \qquad \mathrm{SiLU}(x) = x\,\sigma(x), \qquad \mathrm{SiLU}'(x) = \sigma(x)(1 + x(1 - \sigma(x))).$$

The sigmoid-weighted linear unit (SiLU) was proposed by Elfwing, Uchibe and Doya for reinforcement learning [9]. Ramachandran, Zoph and Le found the same shape with a learnable or fixed β by automated search and called it Swish [10]; with $\beta = 1$, Swish is SiLU. GELU and SiLU look very similar (Figure 3). Both are non-monotonic: they dip slightly below zero for moderate negative inputs.

SwiGLU: a gated feed-forward block

A **gated linear unit** (GLU) multiplies two linear projections element-wise, one of them passed through a nonlinearity that acts like a gate. Shazeer tested several variants in the Transformer's feed-forward block [11]. The SwiGLU version, written in the paper's row-vector notation and without biases, is

$$\mathrm{FFN}_{\mathrm{SwiGLU}}(\mathbf{x}) = \left(\mathrm{Swish}_1(\mathbf{x}W) \odot \mathbf{x}V\right) W_2,$$

where $\mathbf{x}$ is a row vector of size d_{model}, W and V are $d_{\text{model}} \times d_{\text{ff}}$ matrices, W_2 is $d_{\text{ff}} \times d_{\text{model}}$, and $\odot$ is element-wise multiplication. In Shazeer's experiments the GLU variants, SwiGLU and GEGLU among them, gave better results than the ReLU and GELU versions of the block. Because SwiGLU has three weight matrices instead of two, d_{ff} is reduced by a factor of $2/3$ to keep the parameter count equal [11]. LLaMA, for example, replaced ReLU with SwiGLU using a hidden size of $\frac{2}{3} \cdot 4d$ [12], and SwiGLU has become a common default in large language models.

Activation	Range	Smooth	Typical use today
sigmoid	$(0, 1)$	yes	binary / multi-label output, gates
tanh	$(-1, 1)$	yes	some recurrent networks
softmax	probability vector	yes	multi-class output, attention weights
ReLU / Leaky ReLU	$[0, \infty)$ / $\mathbb{R}$	no (kink at 0)	CNNs, MLPs, the safe default
GELU, SiLU	about $[-0.17, \infty)$, $[-0.28, \infty)$	yes	Transformers, modern CNNs
SwiGLU	$\mathbb{R}$	yes	Transformer feed-forward blocks

Normalization layers

Plain version. As training goes on, the numbers flowing through a network can drift to very large or very small scales, which makes learning unstable. A normalization layer rescales them to a standard size at each step, then lets the network choose the final scale it prefers.

All three common layers share one template. For a set of values $x_1, \ldots, x_n$,

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2, \qquad y_i = \gamma\,\frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta,$$

where μ and σ^2 are the mean and variance, ϵ is a small constant for numerical safety, and γ, β are a learnable scale and shift (so the layer can undo the normalization if that helps). The layers differ in **which values** form the set (Figure 4).

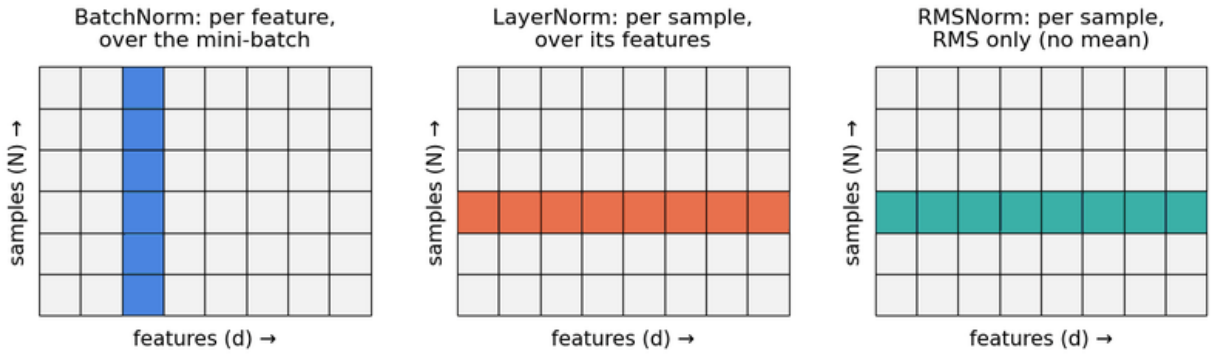


Figure 4 — Which entries share statistics. BatchNorm normalizes each feature over the mini-batch; LayerNorm and RMSNorm normalize each sample over its own features.

- **BatchNorm** (Ioffe and Szegedy [13]) computes μ, σ^2 for each feature (each channel, in a CNN) **across the mini-batch**. It speeds up training and became standard in CNNs. Because the statistics depend on the batch, it behaves differently in training and inference: at inference it uses running averages collected during training. It also degrades when batches are very small, since the batch statistics become noisy.
- **LayerNorm** (Ba, Kiros and Hinton [14]) computes μ, σ^2 **over the features of a single sample**. It does not depend on the batch, behaves identically at training and test time, and works naturally for sequences. It is the standard normalization in Transformers.
- **RMSNorm** (Zhang and Sennrich [15]) drops the mean subtraction and divides by the root mean square, $y_i = \gamma_i \, x_i / \sqrt{\frac{1}{n}\sum_j x_j^2 + \epsilon}$. The authors hypothesize that LayerNorm's re-centering is dispensable and report comparable quality with lower running time [15]. LLaMA uses RMSNorm and applies it to the **input** of each Transformer sub-layer ("pre-normalization") rather than to its output [12].

```
import torch

x = torch.randn(4, 8) * 5 + 3                       # 4 samples, 8 features
ln = torch.nn.LayerNorm(8, elementwise_affine=False)
bn = torch.nn.BatchNorm1d(8, affine=False)
rms = x / torch.sqrt(x.pow(2).mean(dim=1, keepdim=True) + 1e-6)

print(ln(x).mean(dim=1))     # ~0 for every sample (row)
print(bn(x).mean(dim=0))     # ~0 for every feature (column), in train mode
print(rms.pow(2).mean(dim=1))  # ~1 for every sample: unit RMS, mean not removed
```

Huang et al. [16] survey this family and decompose any normalization method into three choices: which values are pooled together (the *normalization area*), what operation is applied (standard-

izing, centering, scaling or whitening), and how the representation is recovered afterward (the learnable affine step). BatchNorm, LayerNorm and RMSNorm differ mainly in the first two.

Designing the output layer and choosing the loss

Plain version. The last layer must speak the language of the answer: any real number for a quantity, a probability for yes/no, a probability distribution for "which of K". Each kind of output has a natural loss, and the pairs below are the ones to use.

Task	Output layer	Output range	Loss
Regression	linear (no activation)	$\mathbb{R}$	mean squared error (MSE, L2) or mean absolute error (MAE, L1)
Binary classification	1 unit + sigmoid	$(0, 1)$	binary cross-entropy
Multi-label (several yes/no)	K units + sigmoid each	$(0, 1)^K$	binary cross-entropy per label
Multi-class (exactly one of K)	K units + softmax	probability vector	cross-entropy

A common mistake is to put a sigmoid on a regression output. The sigmoid's range is the open interval $(0, 1)$, and its job is to output a probability for binary or multi-label classification. A regression head is normally **linear**, so it can produce any real value. A sigmoid on a regression output is reasonable only when the target itself is known to lie between 0 and 1, such as a normalized pixel intensity or a fraction.

The losses, for one example, are

$$\begin{aligned} \text{MSE:} \quad & \ell = \tfrac{1}{2}\|\hat{\mathbf{y}} - \mathbf{y}\|^2, \\ \text{binary cross-entropy:} \quad & \ell = -[\, y \log \hat{p} + (1 - y) \log(1 - \hat{p}) \,], \\ \text{cross-entropy:} \quad & \ell = -\sum_{k=1}^{K} y_k \log p_k = -\log p_c, \end{aligned}$$

where $\mathbf{y}$ is the target, $\hat{p} = \sigma(z)$, $\mathbf{p} = \text{softmax}(\mathbf{z})$, y_k is the one-hot label and c is the correct class. (The factor $\frac{1}{2}$ in MSE is a convenience; PyTorch's `MSELoss` averages without it.) MAE, $\|\hat{\mathbf{y}} - \mathbf{y}\|_1$, is less sensitive to outliers than MSE. Each pair is a maximum-likelihood estimate: MSE corresponds to Gaussian noise on the target, binary cross-entropy to a Bernoulli label, cross-entropy to a categorical label [3].

The pairs are also chosen for a beautiful practical reason. In all three cases the gradient of the loss with respect to the **pre-activation** of the output layer is simply

$$\frac{\partial \ell}{\partial \mathbf{z}} = \hat{\mathbf{y}} - \mathbf{y} \quad \text{(prediction minus target)},$$

with $\hat{\mathbf{y}}$ meaning the linear output, $\sigma(z)$ or $\text{softmax}(\mathbf{z})$ respectively. The saturating sigmoid or softmax derivative cancels out, so a confidently wrong prediction still produces a large gradient. (Exercise 3 asks you to verify this for softmax.) This is why frameworks fuse the last activation into the loss: PyTorch's `nn.CrossEntropyLoss` and `nn.BCEWithLogitsLoss` take **logits**, so the model itself must not end with a softmax or sigmoid during training.

Training I: the forward pass and backpropagation

Plain version. The forward pass computes the prediction and the loss, layer by layer from input to output. Backpropagation then walks back from the loss to the input, using the chain rule to find how much each weight contributed to the error. It reuses intermediate results, so all gradients cost about as much as one extra forward pass.

Backpropagation was popularized for training multi-layer networks by Rumelhart, Hinton and Williams [6]. It is nothing more than the chain rule of calculus, organized so that no work is repeated. Figure 5 shows the computation graph for the network we will differentiate.

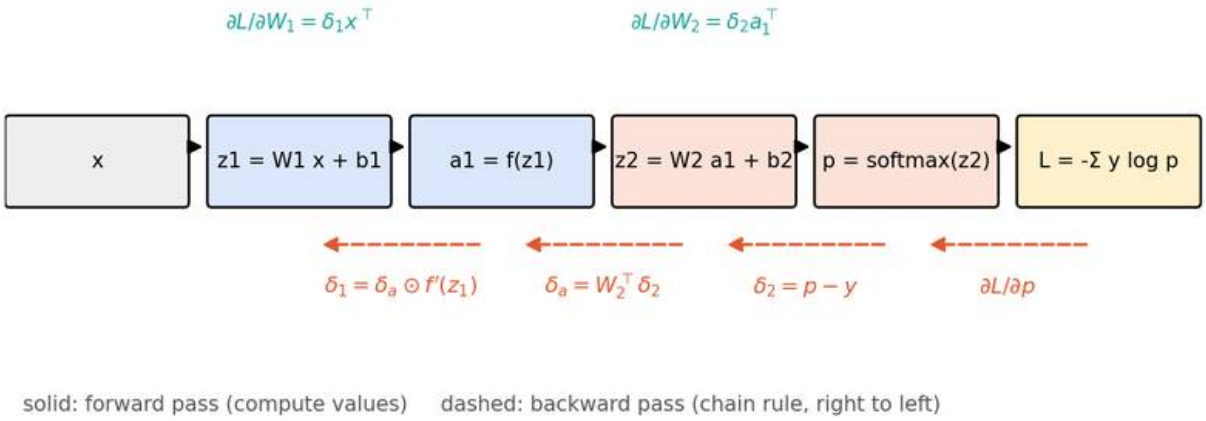


Figure 5 — Computation graph of a two-layer MLP with softmax cross-entropy. Solid arrows: forward pass. Dashed arrows: the backward pass, which carries the error signal δ from the loss back toward the input.

Derivation for a two-layer MLP

Take one example $(\mathbf{x}, \mathbf{y})$ with $\mathbf{x} \in \mathbb{R}^d$ and a one-hot label $\mathbf{y} \in \{0,1\}^K$. The forward pass is

$$\begin{aligned} \mathbf{z}_1 &= W_1\mathbf{x} + \mathbf{b}_1 \in \mathbb{R}^m, \quad \mathbf{a}_1 = f(\mathbf{z}_1) \in \mathbb{R}^m, \\ \mathbf{z}_2 &= W_2\mathbf{a}_1 + \mathbf{b}_2 \in \mathbb{R}^K, \quad \mathbf{p} = \text{softmax}(\mathbf{z}_2), \qquad L = -\textstyle\sum_k y_k \log p_k, \end{aligned}$$

with $W_1 \in \mathbb{R}^{m\times d}$, $W_2 \in \mathbb{R}^{K\times m}$, and f an element-wise activation. Define the **error signal** of a layer as the gradient of the loss with respect to its pre-activation, $\delta_\ell = \partial L/\partial \mathbf{z}_\ell$.

Step 1, output layer. From the previous section, softmax with cross-entropy gives

$$\delta_2 = \frac{\partial L}{\partial \mathbf{z}_2} = \mathbf{p} - \mathbf{y} \in \mathbb{R}^K.$$

Step 2, output weights. Since $z_{2,k} = \sum_j W_{2,kj}\, a_{1,j} + b_{2,k}$, we have $\partial z_{2,k}/\partial W_{2,kj} = a_{1,j}$, so

$$\frac{\partial L}{\partial W_2} = \delta_2\, \mathbf{a}_1^\top \in \mathbb{R}^{K\times m}, \qquad \frac{\partial L}{\partial \mathbf{b}_2} = \delta_2.$$

The gradient of a weight is (error at its output) times (activation at its input). The shapes match W_2, which is a useful check.

Step 3, back through the linear layer. Each hidden activation $a_{1,j}$ affects every output, so its gradient sums over them: $\partial L/\partial a_{1,j} = \sum_k \delta_{2,k} W_{2,kj}$, or in matrix form

$$\frac{\partial L}{\partial \mathbf{a}_1} = W_2^\top \delta_2 \in \mathbb{R}^m.$$

Step 4, back through the activation. Because f acts element by element, its Jacobian is diagonal, and the chain rule becomes an element-wise product:

$$\delta_1 = \frac{\partial L}{\partial \mathbf{z}_1} = (W_2^\top \delta_2) \odot f'(\mathbf{z}_1).$$

Step 5, first-layer weights. Exactly as in Step 2,

$$\frac{\partial L}{\partial W_1} = \delta_1\, \mathbf{x}^\top \in \mathbb{R}^{m\times d}, \qquad \frac{\partial L}{\partial \mathbf{b}_1} = \delta_1.$$

For a deeper network, Steps 3 to 5 simply repeat:
$\delta_\ell = (W_{\ell+1}^\top \delta_{\ell+1}) \odot f'(\mathbf{z}_\ell)$ and $\partial L/\partial W_\ell = \delta_\ell\, \mathbf{a}_{\ell-1}^\top$. For a mini-batch of N examples, the loss is the average, so each gradient is the average of the per-example gradients.

From scratch in NumPy, with a gradient check

The code below stores examples as **rows** (`X` has shape $N\times d$), which is the usual convention in code. Then `X @ W1` with `W1` of shape $d \times m$ plays the role of $W_1\mathbf{x}$ in the math, and every formula above appears transposed: for example `dW2 = A1.T @ dZ2` is $\delta_2\mathbf{a}_1^\top$ stacked over the batch. The **gradient check** compares one analytic derivative with a central finite difference, $(L(w+\epsilon) - L(w-\epsilon))/2\epsilon$. If

they agree to many digits, the backward pass is correct. Always do this when you write gradients by hand.

```
import numpy as np
from sklearn.datasets import make_moons

rng = np.random.default_rng(0)
X, y = make_moons(n_samples=200, noise=0.2, random_state=0)
N, d, h, K = X.shape[0], 2, 16, 2
Y = np.eye(K)[y]                                  # one-hot targets, (N, K)

# He-style initialization (std = sqrt(2 / fan_in))
W1 = rng.normal(0, np.sqrt(2 / d), (d, h)); b1 = np.zeros(h)
W2 = rng.normal(0, np.sqrt(2 / h), (h, K)); b2 = np.zeros(K)

def forward(X, W1, b1, W2, b2):
    Z1 = X @ W1 + b1                              # (N, h)
    A1 = np.maximum(Z1, 0)                        # ReLU
    Z2 = A1 @ W2 + b2                             # (N, K) logits
    Z2 = Z2 - Z2.max(axis=1, keepdims=True)       # numerical stability
    P = np.exp(Z2) / np.exp(Z2).sum(axis=1, keepdims=True)
    loss = -np.mean(np.sum(Y * np.log(P + 1e-12), axis=1))
    return Z1, A1, P, loss

def backward(X, Z1, A1, P, W2):
    dZ2 = (P - Y) / N                             # delta at the output
    dW2 = A1.T @ dZ2;  db2 = dZ2.sum(axis=0)
    dA1 = dZ2 @ W2.T
    dZ1 = dA1 * (Z1 > 0)                          # ReLU'(z) = 1[z > 0]
    dW1 = X.T @ dZ1;   db1 = dZ1.sum(axis=0)
    return dW1, db1, dW2, db2

# Gradient check on one entry of W1 (central difference)
Z1, A1, P, loss = forward(X, W1, b1, W2, b2)
dW1, db1, dW2, db2 = backward(X, Z1, A1, P, W2)
eps = 1e-6; E = np.zeros_like(W1); E[0, 3] = eps
num = (forward(X, W1 + E, b1, W2, b2)[3] - forward(X, W1 - E, b1, W2, b2)[3]) / (2 * eps)
print(f"analytic {dW1[0, 3]:.8f}  numeric {num:.8f}")

# Plain gradient descent
lr = 0.5
for step in range(2000):
    Z1, A1, P, loss = forward(X, W1, b1, W2, b2)
    dW1, db1, dW2, db2 = backward(X, Z1, A1, P, W2)
    W1 -= lr * dW1; b1 -= lr * db1; W2 -= lr * dW2; b2 -= lr * db2
acc = (P.argmax(axis=1) == y).mean()
print(f"final loss {loss:.3f}, train accuracy {acc:.3f}")
# analytic -0.01941581  numeric -0.01941581
# final loss 0.063, train accuracy 0.970
```

In practice you never write the backward pass yourself: frameworks such as PyTorch record the forward computation and run this exact procedure automatically when you call `loss.backward`

`()`. Writing it once by hand is still the best way to understand what that call does, and why it fails when something in the graph has a zero derivative.

> **Interactive demonstration (online)**: open the online edition of this tutorial to use this demonstration.Open demonstration

Training II: making optimization work

Plain version. Knowing the gradient is not enough. You also need a sensible starting point (initialization), a rule for turning gradients into steps (the optimizer), a step size that changes over time (the schedule), a choice of how many examples to look at per step (the batch size), and a guard against occasional huge steps (clipping).

Vanishing and exploding gradients

Unroll the recursion $\delta_\ell = (W_{\ell+1}^\top \delta_{\ell+1}) \odot f'(\mathbf{z}_\ell)$ across L layers and the gradient at the first layer contains a **product** of L weight matrices and L activation derivatives. If the typical factor is below 1, the product shrinks exponentially with depth and the early layers barely learn: **vanishing gradients**. With the sigmoid, every f' is at most 0.25, so ten layers can shrink the signal by a factor of up to $0.25^{10} \approx 10^{-6}$. If the typical factor is above 1, for example because the initial weights are too large, the product grows exponentially, updates become huge, and the loss turns into `inf` or `NaN`: **exploding gradients**.

The remedies, each covered below, attack the product from different sides:

- activations whose derivative is 1 over a wide range (ReLU and its smooth relatives);
- initialization that keeps signal size constant from layer to layer;
- normalization layers that re-standardize activations;
- **residual connections**, which give the gradient a path with factor 1 (the last main section);
- **gradient clipping**, the standard emergency brake for explosions.

Other practices sometimes listed, such as fine-tuning a pre-trained model or using a smaller network, make training easier overall, but they do not directly fix the gradient problem.

Weight initialization

If all weights start equal, every hidden unit computes the same thing and receives the same gradient, and they stay identical forever. So weights start random. The scale matters: we want the

variance of activations (forward) and of gradients (backward) to stay roughly constant from layer to layer. For a linear layer with n_{in} inputs and n_{out} outputs:

- **Xavier (Glorot) initialization** [17] balances the forward and backward conditions for symmetric activations such as tanh: $\operatorname{Var}(W_{ij}) = \dfrac{2}{n_{\text{in}} + n_{\text{out}}}$, for example uniform on $[-\sqrt{6/(n_{\text{in}} + n_{\text{out}})}, \sqrt{6/(n_{\text{in}} + n_{\text{out}})}]$.
- **He (Kaiming) initialization** [18] accounts for ReLU zeroing about half its inputs, which halves the variance at every layer, by doubling the scale: $\operatorname{Var}(W_{ij}) = \dfrac{2}{n_{\text{in}}}$, for example $W_{ij} \sim \mathcal{N}(0, 2/n_{\text{in}})$.

Biases usually start at zero. He et al. showed that this initialization lets very deep ReLU networks be trained from scratch [18]. The NumPy example above uses it.

Optimizers

Every optimizer below is a variant of **stochastic gradient descent** (SGD): estimate the gradient on a small random mini-batch, take a step against it, repeat. Write $\mathbf{g}_t = \nabla_\theta L_{\text{batch}}(\theta_t)$ for the mini-batch gradient at step t, θ for all parameters and η for the learning rate.

SGD. $\theta_{t+1} = \theta_t - \eta\,\mathbf{g}_t$. Simple and memory-free, but it zigzags in narrow valleys and moves slowly along flat directions.

SGD with momentum. Keep a running velocity: $\mathbf{v}_{t+1} = \mu\,\mathbf{v}_t + \mathbf{g}_t$, $\theta_{t+1} = \theta_t - \eta\,\mathbf{v}_{t+1}$, with μ around 0.9. Like a heavy ball rolling downhill, it averages out the zigzag and builds speed along consistent directions.

Adam (Kingma and Ba [19]) keeps two running averages per parameter, of the gradient and of its square:

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\,\mathbf{g}_t, \quad \mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)\,\mathbf{g}_t^2,$$
$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1-\beta_1^t}, \qquad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1-\beta_2^t}, \qquad \theta_t = \theta_{t-1} - \eta\,\frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon},$$

where squares, square roots and division are element-wise, $\beta_1 \approx 0.9$ and $\beta_2 \approx 0.999$ are decay rates, and ϵ is a small constant. The first average is momentum. Dividing by the root of the second gives every parameter its own step size, which is the idea of RMSprop that Adam builds on [19]: parameters with consistently large gradients take smaller steps, rarely updated ones take larger steps. The **bias correction** $1/(1-\beta^t)$ fixes the fact that both averages start at zero and would otherwise be too small in the first steps.

AdamW (Loshchilov and Hutter [20]). Weight decay shrinks weights toward zero at every step.

For plain SGD this is the same as adding an L2 penalty $\frac{\lambda}{2}\|\theta\|^2$ to the loss. For Adam it is not: an L2 term enters $\mathbf{g}_t$ and is then divided by $\sqrt{\hat{\mathbf{v}}_t}$, so weights with large gradients are regularized *less*. AdamW **decouples** the decay from the adaptive step:

$$\theta_t = \theta_{t-1} - \eta \left(\frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} + \lambda\, \theta_{t-1} \right).$$

The authors showed this improves Adam's generalization [20], and AdamW is now the most common optimizer for Transformers and large language models. In PyTorch, `torch.optim.Adam(..., weight_decay=λ)` is the coupled L2 version and `torch.optim.AdamW` is the decoupled one.

Newer optimizers (brief notes). Two recent alternatives are worth knowing by name. **Lion** was found by an automated program search [21]: it keeps only a momentum buffer (so it uses less memory than Adam) and updates every parameter by the **sign** of an interpolated momentum, so all updates have the same magnitude; it needs a smaller learning rate than AdamW. **Muon** [22] applies to the 2-D weight matrices of hidden layers: it takes the momentum update and approximately orthogonalizes it with a few Newton–Schulz iterations, while embeddings, output layers, biases and other scalar or vector parameters are still trained with AdamW. Liu et al. report that, with weight decay and per-parameter update scaling added, Muon reached about twice the compute efficiency of AdamW in their compute-optimal large-language-model training experiments [23]. These are active research topics; AdamW remains the safe default.

Learning-rate schedules

The learning rate is usually the most important hyperparameter. A rate that is fine early in training is often too large later, when the model needs small, careful steps, so the rate is varied over time (Figure 6).

- **Step decay**: multiply the rate by a factor such as 0.1 at fixed epochs. Traditional for CNN training with SGD.
- **Cosine annealing**: $\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})(1 + \cos(\pi t/T))$, a smooth decay from $\eta_{\max}$ to $\eta_{\min}$ over T steps; SGDR [24] combined it with periodic warm restarts.
- **Warmup**: increase the rate linearly from near zero over the first few hundred or thousand steps. Early in training, Adam's second-moment estimates are noisy and the weights are far from any good region, and a full-size step can be destabilizing.

As a concrete recipe from a published large model, LLaMA was trained with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$), weight decay 0.1, gradient clipping at 1.0, 2,000 warmup steps, and a cosine schedule ending at 10% of the peak rate [12].

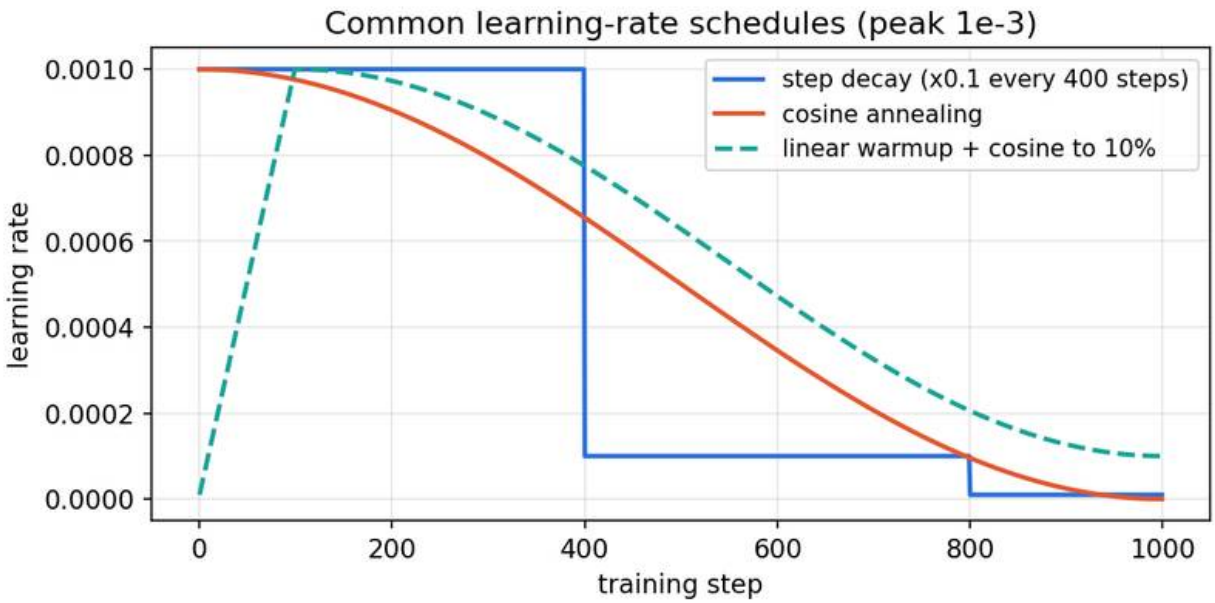


Figure 6 — Three common learning-rate schedules over 1,000 steps with a peak of 1e-3.

Mini-batches, iterations and epochs

A **mini-batch** (often just "batch") is the small random subset of B examples used for one gradient estimate. One **iteration** is one parameter update, using one mini-batch. One **epoch** is one full pass over the training set, which takes $\lceil N/B \rceil$ iterations for N examples. For example, $N = 50{,}000$ and $B = 128$ give 391 iterations per epoch.

The batch size B affects both speed and the result, and bigger is **not** always better:

- **Small batches** give noisy gradient estimates (the variance of the mini-batch mean falls roughly like $1/B$), so the loss curve is jumpy and each epoch takes many steps with poor hardware utilization.
- **Large batches** give stable gradients and run efficiently in parallel on GPUs, but each epoch has fewer updates, and they tend to generalize worse when other settings are kept fixed. Keskar et al. found that large-batch training tends to converge to **sharp minima**, where the loss rises steeply around the solution, while small-batch training, thanks to its gradient noise, tends to find **flat minima**, which generalize better [25].

In practice, choose the largest batch that fits comfortably in memory *and* still trains well, and retune the learning rate when you change B; the two interact strongly.

Gradient clipping

Even in a well-designed network, a rare batch can produce a huge gradient. **Gradient-norm clipping** rescales the whole gradient when its norm exceeds a threshold c:

$$\mathbf{g} \leftarrow \mathbf{g} \cdot \min\left(1, \frac{c}{\|\mathbf{g}\|}\right),$$

where $\|\mathbf{g}\|$ is the norm of all parameter gradients concatenated. It keeps the direction but limits the step size, which is the standard defense against exploding gradients, especially in recurrent networks and Transformers [3]. A threshold of 1.0 is a common choice.

Putting it together in PyTorch

The snippet trains a small MLP on `make_moons` with mini-batches, AdamW, a cosine schedule and gradient clipping. It runs on a CPU in seconds.

```
import torch, torch.nn as nn
from sklearn.datasets import make_moons
from sklearn.model_selection import train_test_split

torch.manual_seed(0)
X, y = make_moons(n_samples=1000, noise=0.25, random_state=0)
Xtr, Xva, ytr, yva = train_test_split(X, y, test_size=0.3, random_state=0)
Xtr, Xva = torch.tensor(Xtr, dtype=torch.float32), torch.tensor(Xva, dtype=torch.float32)
ytr, yva = torch.tensor(ytr), torch.tensor(yva)

model = nn.Sequential(
    nn.Linear(2, 64), nn.ReLU(),
    nn.Linear(64, 64), nn.ReLU(),
    nn.Linear(64, 2),               # logits: no softmax here
)
loss_fn = nn.CrossEntropyLoss()     # applies log-softmax internally
opt = torch.optim.AdamW(model.parameters(), lr=1e-2, weight_decay=1e-2)
sched = torch.optim.lr_scheduler.CosineAnnealingLR(opt, T_max=200)

for epoch in range(200):
    model.train()
    perm = torch.randperm(len(Xtr))
    for i in range(0, len(Xtr), 64):                # mini-batches of 64
        idx = perm[i:i + 64]
        loss = loss_fn(model(Xtr[idx]), ytr[idx])
        opt.zero_grad()
        loss.backward()                              # backpropagation
        torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)
        opt.step()
    sched.step()

model.eval()
```

```
with torch.no_grad():
    acc = (model(Xva).argmax(dim=1) == yva).float().mean().item()
print(f"validation accuracy: {acc:.3f}")   # about 0.94
```

Note `model.train()` and `model.eval()`: they switch layers such as dropout and BatchNorm between their training and inference behavior. Forgetting `eval()` at test time is a classic bug.

Generalization: underfitting, overfitting and regularization

Plain version. The goal is not to do well on the training data; it is to do well on data the model has never seen. A model can fail in two opposite ways: it can be too weak to learn even the training data (underfitting), or it can memorize the training data, noise included, and fail on new data (overfitting).

Capacity and the two failure modes

A model's **capacity** is, informally, how complex a function it can fit; it grows with the number of parameters, depth and width, and training time. As capacity increases, training error keeps falling, while validation error first falls and then rises again once the model starts fitting noise.

The gap between the two is the **generalization gap** [3].

- **Underfitting**: the **training** error itself is high. The model, or the training procedure, cannot capture the pattern.
- **Overfitting**: the training error is low but the validation or test error is much higher.

What to do about underfitting, in order:

1. **Check the code and data first.** A bug, mismatched inputs and labels, wrong preprocessing, or a loss applied to the wrong tensor all look like underfitting.
2. **Increase capacity**: a wider or deeper model, or a better architecture.
3. **Improve optimization**: train longer, tune the learning rate and schedule, switch optimizer, check initialization and normalization.

Making the model **smaller** never fixes underfitting; it lowers capacity further. A smaller model can be a reasonable choice for speed or memory once accuracy is adequate, but that is an efficiency decision, not a cure.

What to do about overfitting: use more or more varied data, add regularization (below), reduce capacity, and stop early. It is also worth checking that the training and validation sets really come

from the same distribution; a mismatch between them looks like overfitting but needs a data fix, not a regularizer.

Reading a loss curve

Plot both the training and the validation loss against iterations or epochs. Four patterns cover most situations:

- **A. Both curves flat from the start.** If the loss is high, the model is not learning at all: suspect a bug, a far too small or large learning rate, or a model/data mismatch. If the loss is already low, the task may be very easy, or each logged point covers so many iterations that the curve's shape is hidden; log more often.
- **B. Training loss keeps falling, validation loss turns upward.** Classic overfitting. Keep the parameters from the point of lowest validation loss (early stopping) and add regularization.
- **C. Both curves fall and level off close together.** Either the model is near its best, which is common when training and validation data are very similar, or both are still high, which is underfitting: train longer or increase capacity.
- **D. Validation loss above training loss, but both still decreasing steadily.** The normal, healthy situation. Keep training and keep the checkpoint with the lowest validation loss.

Regularization tools

Figure 7 shows overfitting and two cures on purpose. A wide MLP (two hidden layers of 256 units) is trained on only 100 noisy two-moons points. Without regularization the training loss falls toward zero while the validation loss reaches its minimum early and then climbs steeply. Weight decay and dropout both slow the climb dramatically.

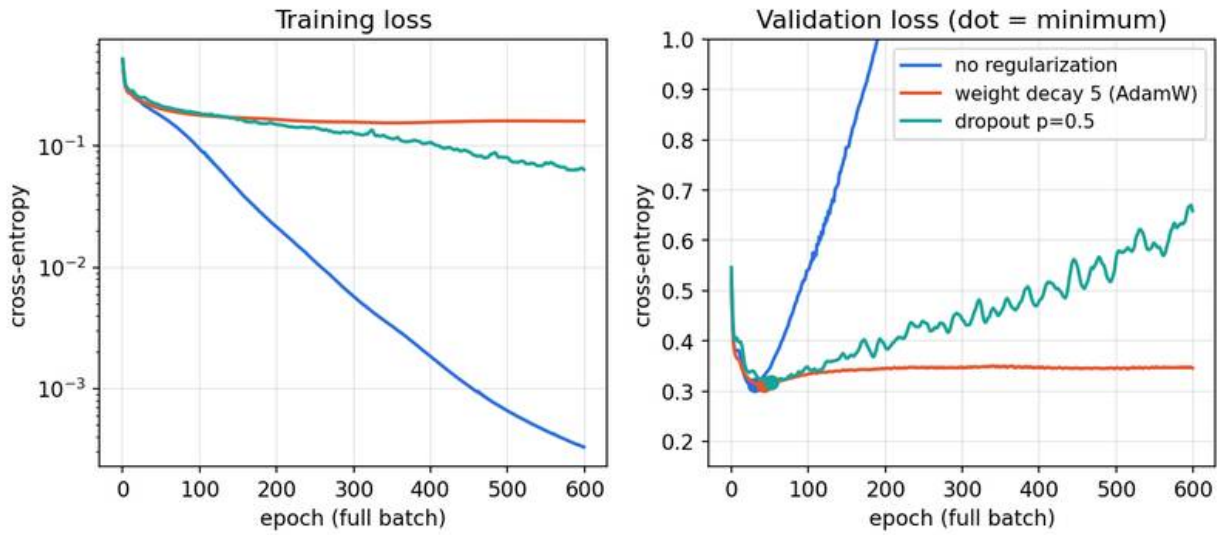


Figure 7 — Overfitting on 100 training points from make_moons (noise 0.35), evaluated on 2,000 validation points. Full-batch AdamW, learning rate 3e-3. Dots mark each run's minimum validation loss, which is where early stopping would keep the model.

Weight decay (L2 regularization). Penalize large weights by adding $\frac{\lambda}{2}\|\theta\|^2$ to the loss, or, with Adam, use decoupled decay (AdamW). Large weights let a network make sharp, wiggly boundaries around individual points; keeping them small favors smoother functions. In Figure 7, decoupled decay with $\lambda = 5$ keeps the validation loss almost flat after its minimum.

Dropout (Srivastava et al. [27]). During training, set each hidden unit to zero with probability p, independently for every example and step, and scale the survivors by $1/(1-p)$ so the expected activation is unchanged ("inverted dropout"). At test time, use all units with no scaling. Each step thus trains a different thinned sub-network; units cannot rely on specific partners and must learn features that are useful on their own, and the full network at test time behaves like an average of many thinned networks [27]. This is why `model.eval()` matters.

Data augmentation. Create new training examples by applying transformations that do not change the label: flips, crops, small rotations, color jitter and noise for images. It is often the single most effective regularizer in vision, because it teaches the invariances directly. It must respect the task: a horizontal flip is fine for cats, not for reading text.

Early stopping. Monitor the validation loss during training and keep the parameters from the epoch where it was lowest; stop when it has not improved for a set number of epochs (the "patience"). It costs nothing, needs no change to the model, and acts as a regularizer by limiting how far the parameters can travel from their initialization [3].

More data and better strategy. The other levers the original notes list are worth repeating: more training data is the most reliable cure for overfitting; a stronger architecture (deeper, or better

structured, so that it converges well) helps when the model underfits; and a better training strategy (optimizer, learning-rate schedule, batch size) helps both.

Residual connections and the road to Transformers

Plain version. If you stack many layers, each must pass the signal on intact as well as improve it, which turns out to be hard. A residual connection gives every block an express lane: the block only has to learn a *correction* that is added to its input.

He et al. observed that adding more plain layers to an already deep network made even the **training** error worse, which is an optimization problem rather than overfitting [28]. Their **residual block** computes

$$\mathbf{y} = \mathbf{x} + F(\mathbf{x}),$$

where $\mathbf{x}$ is the block's input and F is a small stack of layers (in ResNet, convolutions with BatchNorm and ReLU). If the best thing a block can do is nothing, it only has to push F toward zero, which is easy. The backward pass shows why it trains well:

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = I + \frac{\partial F}{\partial \mathbf{x}},$$

where I is the identity matrix. The identity term passes the gradient straight through every block, so the product over many blocks no longer has to shrink to zero. With residual connections, He et al. trained networks with up to 152 layers on ImageNet and won several ILSVRC 2015 tasks [28].

The **Transformer** (Vaswani et al. [29]) combines every idea in this tutorial. Each block has two sub-layers, multi-head self-attention (which uses softmax to mix information between positions) and a position-wise MLP, and each sub-layer is wrapped in a residual connection with LayerNorm. The original paper normalized after the addition; many recent models, LLaMA among them, normalize the sub-layer **input** instead (pre-normalization) and use RMSNorm and SwiGLU [12]:

$$\mathbf{h} = \mathbf{x} + \mathrm{Attention}\,(\,\mathrm{Norm}(\mathbf{x})), \qquad \mathbf{y} = \mathbf{h} + \mathrm{FFN}\,(\,\mathrm{Norm}(\mathbf{h})).$$

Since 2017 this design has spread from machine translation to language modeling, vision, speech and more, and it is the basis of today's large language models. The trained recipe is the one from this tutorial too: AdamW, warmup and cosine decay, gradient clipping, and weight decay [12]. The deep-learning track picks up from here with convolutional networks and Transformers for images, starting with image classification.

Modern view

The basic toolkit above has been stable for several years; research continues on why it works and on better versions of each piece. The following reviews are good guides, each from a different angle.

The big picture: LeCun, Bengio and Hinton [2] (Nature, 2015). This short review is still the clearest statement of what deep learning is: representation learning with many layers, where each layer transforms the previous representation into a more abstract one and the features are learned from data with backpropagation rather than engineered. It walks through supervised learning with SGD, the role of ReLU, convolutional networks for images, recurrent networks for sequences, and distributed representations, and it closes by predicting that unsupervised learning and the combination of representation learning with reasoning would matter more in the future. For a textbook treatment of every topic in this tutorial, including the maximum-likelihood view of loss functions, regularization and optimization, the book by Goodfellow, Bengio and Courville [3] remains the standard reference.

Optimization: Sun [26]. Sun's survey asks "when and why can a neural network be successfully trained?" and organizes the answer in three parts: problems with the gradient (explosion, vanishing) and their fixes through careful initialization and normalization; the practical algorithms (SGD, adaptive methods such as Adam, and distributed training) with their convergence theory; and the global picture of the loss landscape, including bad local minima, mode connectivity, the lottery-ticket hypothesis and the infinite-width limit. It is a good bridge from the practice above to theory.

Normalization: Huang et al. [16] (IEEE TPAMI, 2023). This survey gives a unified view of normalization methods from the perspective of optimization, and a taxonomy that splits any method into its normalization area, normalization operation and representation recovery. It covers the history from BatchNorm on, the analyses of why normalization speeds training and helps generalization, and applications in different domains. Read it to see BatchNorm, LayerNorm, group-wise and weight-based normalization as points in one design space.

Activation functions: Dubey, Singh and Chaudhuri [7] (Neurocomputing, 2022). This survey classifies activation functions into logistic sigmoid/tanh-based, ReLU-based, ELU-based and learning-based families, characterizes them by properties such as output range, monotonicity and smoothness, and benchmarks 18 of them across network types and data. Its practical message is that no single activation is best everywhere; the choice interacts with the architecture and the data.

Where things stand (2023–2026). Several convergences are visible in published model recipes:

- **Architecture.** The pre-normalized Transformer block with residual connections is the dominant architecture for language and increasingly for vision; RMSNorm and SwiGLU, as in

LLaMA [12], are common defaults in large language models.

- **Optimization.** AdamW [20] with linear warmup, a cosine (or similar) decay, and gradient clipping is the standard recipe. Optimizers that look beyond per-coordinate scaling are an active area: Lion [21] uses sign updates and less memory, and Muon [22] orthogonalizes matrix updates, with reported compute-efficiency gains at scale [23]. How robust such gains are across tasks and scales is still being studied, which is why AdamW remains the safe choice.
- **Generalization.** The sharp-versus-flat-minima view [25] is a useful intuition rather than a complete theory, and why heavily over-parameterized networks generalize as well as they do is still an open question; Sun's survey [26] summarizes the main lines of attack.

Open problems. A reliable theory of generalization for deep networks; principled rules for choosing the learning rate, batch size and schedule together, especially when scaling models up; understanding exactly how normalization, residual connections and adaptive optimizers interact; and making training stable at very large scale without the many small tricks that current recipes rely on.

Key takeaways

- A perceptron is a linear score followed by a step function and trained by a mistake-driven rule; logistic regression uses a sigmoid probability and cross-entropy. Similar structure, different methods, and only the smooth one can be stacked and trained by gradients.
- Hidden layers with nonlinear activations are what make networks more than linear models. One wide hidden layer can in principle approximate any continuous function, but that is an existence result; depth makes useful functions easier to represent and learn. There is no fixed layer count that defines "deep".
- ReLU-family activations (ReLU, Leaky ReLU, GELU, SiLU) keep gradients alive; sigmoid and softmax belong mostly at the output; SwiGLU is the modern feed-forward choice in Transformers. BatchNorm normalizes over the batch, LayerNorm and RMSNorm over the features of each sample.
- Match the head to the task: linear plus MSE for regression, sigmoid plus binary cross-entropy for binary or multi-label, softmax plus cross-entropy for multi-class. In each case the output gradient is prediction minus target.
- Backpropagation is the chain rule applied from the loss backward: $\delta_\ell = (W_{\ell+1}^\top \delta_{\ell+1}) \odot f'(\mathbf{z}_\ell)$ and $\partial L/\partial W_\ell = \delta_\ell \mathbf{a}_{\ell-1}^\top$. Check hand-written gradients numerically.
- Practical training: He or Xavier initialization, AdamW with warmup and a decaying schedule, gradient clipping, and a batch size chosen with the learning rate, not "as large as possible".
- Underfitting means high training error and calls for more capacity or better optimization, never a smaller model; overfitting calls for more data, augmentation, weight decay, dropout and early stopping.
- Residual connections make very deep networks trainable, and the Transformer is these ideas combined.

Exercises

1. **Perceptron by hand.** Run the perceptron rule on the AND data with $\eta = 1$, $\mathbf{w} = \mathbf{0}$, $b = 0$, visiting the points in the order $(0,0), (0,1), (1,0), (1,1)$. Write down $\mathbf{w}$ and b after each update until an epoch passes with no mistakes. Then explain in one sentence why the same procedure on XOR never stops.

   Hint

   With the strict test $z > 0$, the first point $(0,0)$ has $z = 0$, so it is predicted 0 and is correct. The first mistake is $(1,1)$, which gives $\mathbf{w} = (1,1)$, $b = 1$. Keep going: each mistake on a 0-labeled point subtracts $\mathbf{x}$ and 1, each mistake on the 1-labeled point adds them. With this order there are ten updates over five epochs, and the sixth epoch is error-free with $\mathbf{w} = (2,1)$, $b = -2$, that is, the line $2x_1 + x_2 = 2$. A different visiting order gives a different, equally valid line. For XOR, no $(\mathbf{w}, b)$ classifies all four points correctly, so some point is always wrong and an update always happens.

2. **How fast do sigmoid gradients vanish?** Show that $\sigma'(x) = \sigma(x)(1 - \sigma(x)) \leq 1/4$. Then consider a chain of 20 sigmoid layers with all weights equal to 1. What is the largest factor by which the gradient can be multiplied on its way back? Repeat the reasoning for ReLU with active units.

   Hint

   $s(1-s)$ with $s \in (0,1)$ is a downward parabola with maximum $1/4$ at $s = 1/2$. Twenty factors of at most $1/4$ give at most $4^{-20} \approx 9 \times 10^{-13}$. For an active ReLU unit, $f' = 1$, so the activations contribute no shrinkage at all; only the weights do, which is why initialization matters.

3. **The prediction-minus-target gradient.** Using $\partial p_k / \partial z_j = p_k(\delta_{kj} - p_j)$, prove that for $L = -\sum_k y_k \log p_k$ with a one-hot $\mathbf{y}$, $\partial L / \partial z_j = p_j - y_j$. Then show the same result for the sigmoid with binary cross-entropy.

   Hint

   $\partial L / \partial z_j = -\sum_k (y_k / p_k)\, p_k(\delta_{kj} - p_j) = -\sum_k y_k \delta_{kj} + p_j \sum_k y_k = -y_j + p_j$, since $\sum_k y_k = 1$. For the sigmoid, $\partial \ell / \partial \hat{p} = -y/\hat{p} + (1-y)/(1-\hat{p})$ and $\partial \hat{p} / \partial z = \hat{p}(1-\hat{p})$; multiply and simplify to $\hat{p} - y$.

4. **Extend the gradient check.** In the NumPy example, check every entry of `W2`, `b1` and `b2` (not just one entry of `W1`) against finite differences, and report the maximum relative error $|a-n| / \max(|a|, |n|, 10^{-12})$. Then deliberately introduce a bug, for example drop the `* (Z1 > 0)` factor, and see how large the error becomes.

   Hint

Loop over `np.ndindex(param.shape)`, perturb one entry by $\pm 10^{-6}$, and recompute the loss. A correct implementation gives relative errors around 10^{-7} or smaller (a few entries next to ReLU kinks can be larger). Removing the ReLU derivative makes errors of order 1 appear in `W1` and `b1`, while `W2` and `b2` stay correct, because the bug is only in the backward path below the hidden layer.

5. **Iterations, epochs and batch size.** A dataset has 60,000 training images. (a) How many iterations make one epoch with $B = 32$, and with $B = 1{,}024$? (b) If you train both for 20 epochs, how many parameter updates does each make? (c) Based on the batch-size discussion, give two reasons the $B = 1{,}024$ run might reach a worse validation accuracy with the same learning rate, and one change that could help.

Hint

(a) $\lceil 60000/32 \rceil = 1875$ and $\lceil 60000/1024 \rceil = 59$. (b) 37,500 versus 1,180 updates. (c) Far fewer updates in the same number of epochs, and less gradient noise, which the sharp-minima findings associate with worse generalization. Tuning the learning rate for the new batch size (typically larger, with warmup) and training for more epochs are the usual adjustments.

6. **Diagnose the curve.** A colleague's network has a training loss that drops quickly at first and then stays flat at a high value; the validation loss tracks it closely. They propose halving the number of hidden units "to make it converge faster". What is happening, what would you check first, and what would you suggest instead?

Hint

Both losses are high and close together: this is underfitting (pattern C with high values), not overfitting. First check for bugs: labels aligned with inputs, the loss applied to logits, data normalized, the learning rate sensible. Then increase capacity or improve optimization (wider or deeper model, a learning-rate schedule, longer training). Shrinking the model lowers capacity and will make underfitting worse.

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ML 03 • Convolution, Image Filtering and CNNs

From blur and Sobel kernels to learned filters: convolution versus correlation, border handling, padding, stride, dilation, pooling, receptive fields, backpropagation through a convolution, and the classic CNN architectures.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-03-convolutional-networks/

Notebook: https://github.com/Shyandram/cv-blog/blob/743a40e/notebooks/image-processing/convolution.ipynb

What you'll learn

- What a kernel is, how it slides over an image, and why the same operation can blur, sharpen or find edges.
- The exact difference between **correlation** and **convolution**, and which one NumPy, SciPy, OpenCV and PyTorch actually compute.
- How to write 2-D filtering from scratch in NumPy, reproduce it in OpenCV, SciPy and PyTorch, and handle image borders deliberately.
- How a convolutional layer generalizes a fixed filter: channels, weight sharing, padding, stride, dilation, 1×1 convolutions and pooling.
- How receptive fields grow with depth, and how gradients flow backward through a convolution.
- What the classic architectures (LeNet, AlexNet, VGG, GoogLeNet, ResNet, U-Net, MobileNet, EfficientNet, ConvNeXt) each contributed, and where CNNs stand in 2026.

The big picture

An image is a grid of numbers (see Images as arrays). Many of the most useful things we can do to it are **local**: whether a pixel sits on an edge, inside a smooth region or in a noisy patch depends mostly on its neighbors. **Convolution** is the operation that formalizes "look at a small neighborhood, combine it with fixed weights, and repeat everywhere." It is the workhorse of classical image processing, and it is also the building block of **convolutional neural networks** (**CNNs**), which learn their weights from data instead of having them designed by hand.

> **Explain like I'm five**
>
> Imagine you have a small cardboard window with nine holes, and behind each hole you write a number. You place the window on a picture, multiply each visible dot of color by the number behind its

hole, add everything up, and write the answer on a new sheet. Then you slide the window one step and do it again, until you have covered the whole picture. If all nine numbers are equal, the new picture is a blurry average. If the numbers on the left are negative and those on the right are positive, you get big answers only where the picture changes from dark to light: an edge finder. A CNN is a machine that slides thousands of such windows and **learns** which numbers to write behind the holes.

This tutorial has two halves. The first half (filtering) is the classical toolbox and needs only NumPy. The second half (CNNs) builds on ML 02, where you met layers, losses and backpropagation. If you want more depth on fixed filters afterward, IP 03 covers spatial filtering and IP 04 explains the frequency-domain view through the convolution theorem. For the deep-learning side, DLCV 02 studies modern classification backbones in detail, ML 03A turns to sequences, and ML 04 introduces attention, the main alternative to convolution. Standard textbook treatments are chapter 9 of Goodfellow et al. [1] and chapter 10 of Prince [10].

Part I • Filtering with a fixed kernel

The sliding weighted sum

Plain version. Every output pixel is a weighted sum of the input pixels around the same position. The small grid of weights is called a **kernel** (also *filter* or *mask*).

Precise version. Let I be a grayscale image, with $I(x, y)$ the intensity at column x and row y. Let K be a kernel of size $(2a + 1) \times (2b + 1)$, indexed from $-a$ to a vertically and $-b$ to b horizontally, so that $K(0, 0)$ is its center. The filtered image G is

$$G(x, y) = \sum_{i=-a}^{a} \sum_{j=-b}^{b} K(i, j)\, I(x + j,\ y + i).$$

Here i indexes kernel rows and j kernel columns, $G(x, y)$ is the output at the same position, and the double sum visits every tap of the kernel once. For a 3×3 kernel, $a = b = 1$ and each output pixel costs nine multiplications and eight additions.

Figure 1 works through one output value by hand. The orange window covers the top-left 3×3 neighborhood of the padded input. Multiply element-wise by the kernel and add: $1 \cdot 1 + 0 \cdot 2 + (-1) \cdot 0 + 2 \cdot 0 + 0 \cdot 1 + (-2) \cdot 3 + 1 \cdot 2 + 0 \cdot 0 + (-1) \cdot 1 = -4$. That number becomes G at the window's center. Sliding the window one step to the right and repeating fills in the next output.

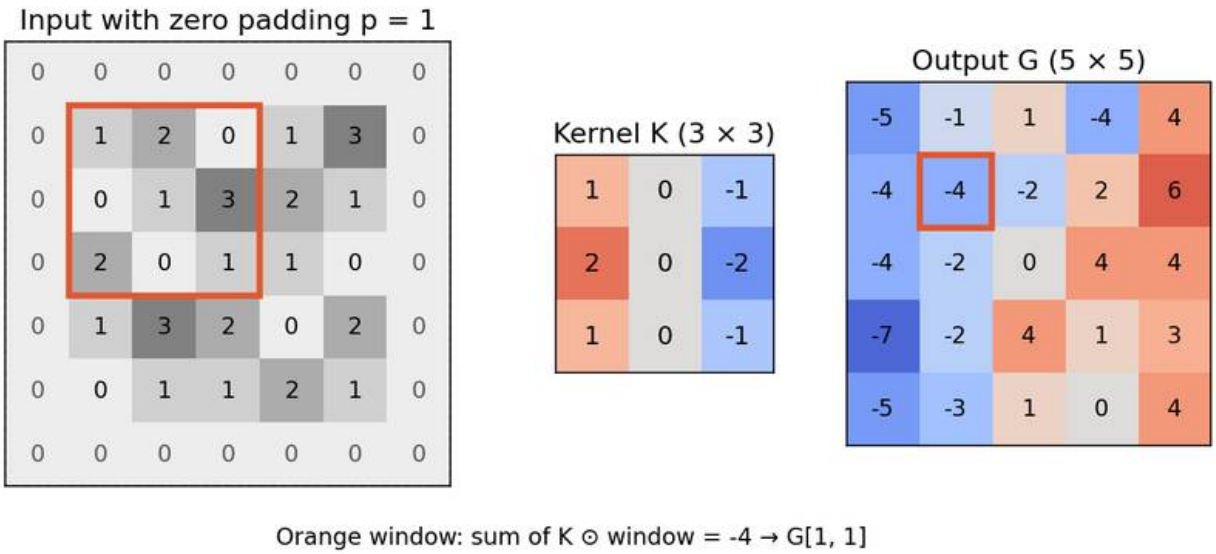


Figure 1 — One step of the sliding weighted sum. The orange 3 × 3 window, multiplied element-wise by K and summed, gives −4 at the matching output position. Zero padding (gray) lets the output keep the input size.

Two observations already matter for everything that follows. First, the same nine weights are used at every position: the operation is **shift-invariant** (also called *space-invariant*). Second, the operation is **linear**: filtering $\alpha I_1 + \beta I_2$ gives $\alpha G_1 + \beta G_2$. Linear shift-invariant (LSI) operations are exactly the ones a convolution can express, which is why convolution is so central in signal processing.

Correlation versus convolution

Plain version. The formula above is technically **cross-correlation**. True **convolution** does the same thing with the kernel rotated by 180 degrees. For symmetric kernels the two agree; for asymmetric ones they differ.

Precise version. Mathematical convolution, written $K * I$, is

$$(K * I)(x, y) = \sum_{i=-a}^{a} \sum_{j=-b}^{b} K(i, j)\, I(x - j,\ y - i) = \sum_{i,j} K(-i, -j)\, I(x + j,\ y + i).$$

The second form shows that convolution equals correlation with the flipped kernel $K(-i, -j)$. A clean way to see the difference is to filter a single bright pixel (an **impulse**). Correlation leaves a copy of the kernel rotated by 180 degrees; convolution leaves an exact copy of the kernel. That copy is the **impulse response**, and it is why signal processing prefers convolution: the output for

any image is the sum of shifted, scaled impulse responses.

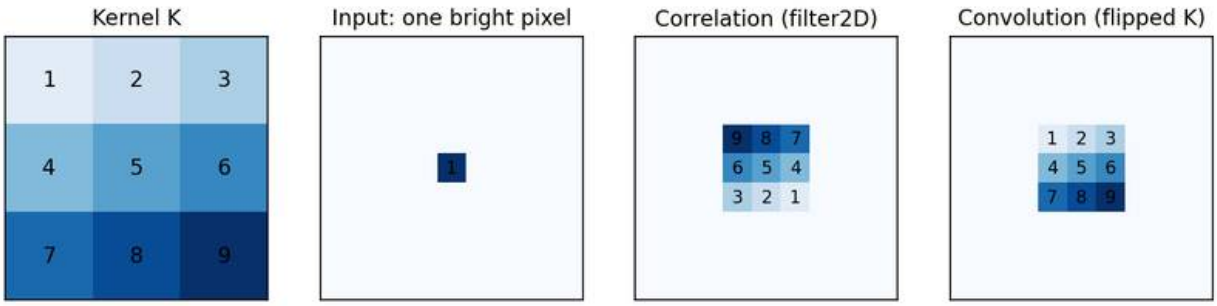


Figure 2 — Filtering an impulse. Correlation (what cv2.filter2D computes) stamps a 180°-rotated kernel; convolution stamps the kernel itself.

Convolution has algebraic properties that correlation lacks. It is **commutative** ($K * I = I * K$) and **associative** ($K_2 * (K_1 * I) = (K_2 * K_1) * I$), so two filters applied in a row equal one bigger filter. Applying a 3×3 box blur twice, for instance, is a single 5×5 kernel. Associativity also explains **separable** kernels: a $k \times k$ Gaussian equals a $k \times 1$ column filter followed by a $1 \times k$ row filter, cutting the cost per pixel from k^2 to $2k$ operations. The **convolution theorem** says convolution in space is multiplication in the Fourier domain; IP 04 develops that view.

Which libraries compute which?

Library call	Computes	Default border
`cv2.filter2D`	correlation [11]	reflect-101 (`BORDER_DEFAULT`)
`scipy.ndimage.correlate` / `convolve`	correlation / convolution	`reflect` (edge sample repeated)
`scipy.signal.convolve2d`	convolution	zero fill
`torch.nn.functional.conv2d`, `nn.Conv2d`	correlation [31]	zero padding

For hand-designed asymmetric kernels (Sobel, emboss, motion blur) the flip changes the sign or direction of the result, so flip deliberately when you need true convolution, for example with `cv2.flip(K, -1)`. For learned kernels the distinction disappears: a network that "should" use the flipped kernel simply learns it. That is why deep-learning frameworks call their correlation "convolution" without harm.

Blur, edges and sharpening

Plain version. Positive weights that sum to one average neighbors and blur. Weights that sum to zero cancel flat regions and keep only changes, so they find edges. Adding a bit of "the image minus its blur" back to the image sharpens it.

A **box blur** gives every neighbor equal weight; the 3×3 version has nine entries of $1/9$. A **Gaussian blur** samples

$$K(i,j) = \frac{1}{Z}\exp\!\left(-\frac{i^2+j^2}{2\sigma^2}\right),$$

where σ (in pixels) sets the blur width and Z normalizes the weights to sum to one. Nearby pixels count more than distant ones, which avoids the blocky artifacts of a box. A practical rule is to use a kernel radius of about 3σ.

The **sum of the weights** tells you what happens in a flat region where every pixel equals c: the output is $c\sum K$. A blur with sum one preserves brightness; a kernel with sum two doubles it (and saturates an 8-bit image); a kernel with sum zero outputs zero.

The **Sobel** kernels estimate horizontal and vertical derivatives while smoothing in the orthogonal direction:

$$S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}, \qquad S_y = S_x^{\top}.$$

S_x responds to intensity changes along x, so it highlights **vertical** boundaries; S_y highlights **horizontal** ones. The outputs G_x, G_y are signed: dark-to-light and light-to-dark edges have opposite signs. The **gradient magnitude** $\sqrt{G_x^2+G_y^2}$ is a nonnegative edge strength, and $\operatorname{atan2}(G_y, G_x)$ is the edge orientation. The factorization shows that Sobel is separable: a $[1,2,1]^{\top}$ smoothing column times a $[-1,0,1]$ central difference.

The **Laplacian** kernel $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ approximates $\partial^2 I/\partial x^2 + \partial^2 I/\partial y^2$; it responds to thin lines and to both sides of an edge. **Unsharp masking** sharpens with

$$I_{\text{sharp}} = I + \lambda\,(I - G_\sigma * I),$$

where $G_\sigma * I$ is a Gaussian-blurred copy, $I - G_\sigma * I$ is the high-frequency detail, and $\lambda > 0$ controls strength. Large λ amplifies noise and creates bright and dark halos along edges.

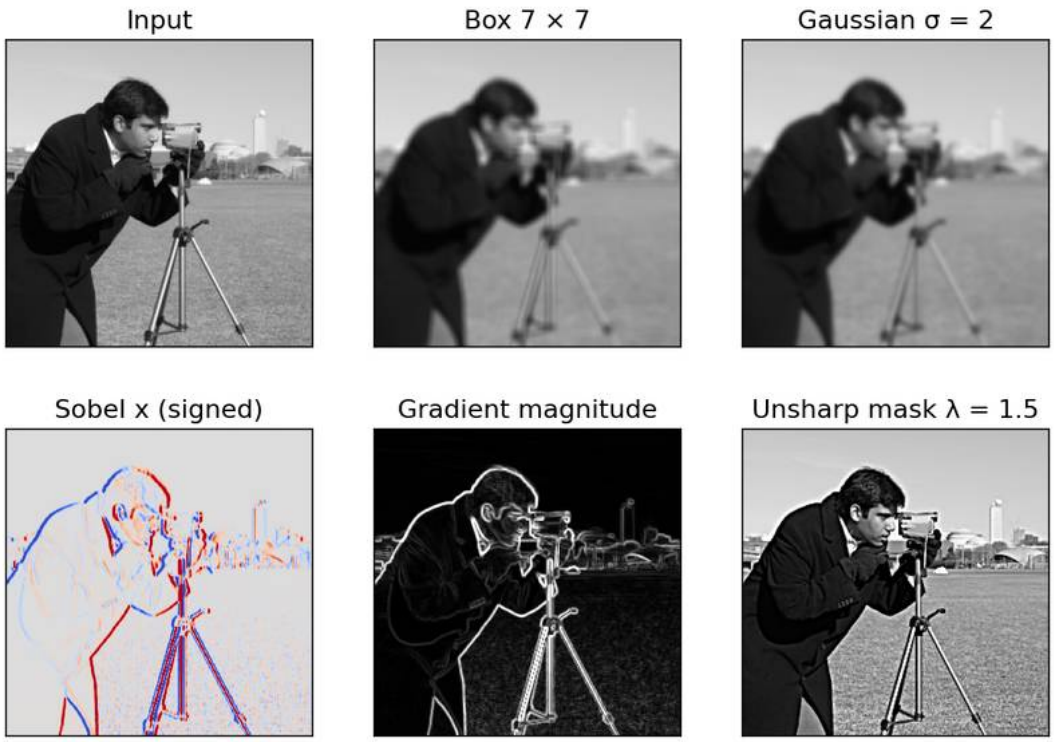


Figure 3 — The same image through six kernels. Box and Gaussian blur suppress fine detail; Sobel x is signed (red = dark-to-light, blue = light-to-dark); the gradient magnitude outlines objects; unsharp masking adds back amplified detail.

Try it: the kernel playground

Edit the kernel cells below and watch the output update. Start from Sobel x, then transpose it to detect horizontal edges instead of vertical ones. Make every weight of the blur preset larger and notice that flat regions brighten, exactly as the sum-of-weights rule predicts. Try the Laplacian and emboss presets and predict their kernel sums before you look.

> **Interactive demonstration (online)**: open the online edition of this tutorial to use this demonstration.Open demonstration

The slider below compares a synthetic test image (a disc, a square, a checkerboard and stripes) with a 5×5 Gaussian blur, $\sigma = 1.4$. The fine stripes and the checkerboard disappear first because a blur is a **low-pass filter**: it removes fast intensity variations while leaving slow ones.

> **Interactive demonstration (online)**: open the online edition of this tutorial to use this demonstration.Open demonstration

Code: from scratch, then OpenCV, SciPy and PyTorch

The loop below is the whole algorithm. Instead of looping over every pixel (slow in Python), it loops over the kernel taps and adds one shifted copy of the padded image per tap, which NumPy vectorizes. The script then checks that four libraries agree once you match the definition (correlation versus convolution) and the border rule. It needs `numpy`, `scipy`, `opencv-python` and `torch`.

```
import numpy as np
import cv2
import torch
import torch.nn.functional as F
from scipy import ndimage, signal

def correlate2d(img, K, mode="edge"):
    """Cross-correlation of a 2-D float image with an odd-sized kernel.
    mode is any np.pad mode: "constant" (zeros), "edge", "reflect", "wrap"."""
    kh, kw = K.shape
    ph, pw = kh // 2, kw // 2
    padded = np.pad(img, ((ph, ph), (pw, pw)), mode=mode)
    H, W = img.shape
    out = np.zeros((H, W), dtype=np.float64)
    for i in range(kh):              # loop over kernel taps, not pixels:
        for j in range(kw):          # each step adds one shifted copy of the image
            out += K[i, j] * padded[i:i + H, j:j + W]
    return out

def convolve2d(img, K, mode="edge"):
    return correlate2d(img, K[::-1, ::-1], mode)     # convolution = correlation with
     ↪  flipped K

rng = np.random.default_rng(0)
img = rng.random((32, 40))
K = rng.standard_normal((3, 5))                      # deliberately asymmetric

ours_corr = correlate2d(img, K, "reflect")
ours_conv = convolve2d(img, K, "reflect")
# OpenCV: filter2D computes correlation; BORDER_REFLECT_101 is np.pad's "reflect"
cv_corr = cv2.filter2D(img, cv2.CV_64F, K, borderType=cv2.BORDER_REFLECT_101)
# SciPy: ndimage.correlate / ndimage.convolve with mode="mirror" (= reflect-101)
sp_corr = ndimage.correlate(img, K, mode="mirror")
sp_conv = ndimage.convolve(img, K, mode="mirror")
# PyTorch: conv2d is correlation; pad first with the same rule
t = torch.tensor(img)[None, None]
t = F.pad(t, (2, 2, 1, 1), mode="reflect")
pt_corr = F.conv2d(t, torch.tensor(K)[None, None])[0, 0].numpy()

for name, a in [("OpenCV", cv_corr), ("SciPy", sp_corr), ("PyTorch", pt_corr)]:
    print(f"{name:8s} correlation matches:", np.allclose(ours_corr, a))
print("SciPy    convolution matches:", np.allclose(ours_conv, sp_conv))
print("conv == corr for this kernel?", np.allclose(ours_conv, ours_corr))
# zero padding: our convolution equals signal.convolve2d with mode="same"
same = signal.convolve2d(img, K, mode="same", boundary="fill")
```

```
print("zero-padded 'same' convolution matches:", np.allclose(convolve2d(img, K, "constant"),
 ↪  same))
```

All checks print `True` except `conv == corr`, which is `False` because the random kernel is asymmetric. Note the naming trap: NumPy's `"reflect"` padding (`dcb|abcd|cba`, edge sample not repeated) is OpenCV's `BORDER_REFLECT_101` and SciPy's `"mirror"`, while SciPy's `"reflect"` repeats the edge sample (`dcba|abcd|dcba`).

The second snippet shows the two bugs that cause most beginner frustration with real images: integer saturation and the kernel-sum rule.

```
import numpy as np
import cv2
from skimage import data

img_u8 = data.camera()                                  # 512 x 512 uint8 test image
img = img_u8.astype(np.float32) / 255.0
sobel_x = np.array([[-1, 0, 1], [-2, 0, 2], [-1, 0, 1]], np.float32)

gx = cv2.filter2D(img, -1, sobel_x)                     # float in, float out: negatives kept
gy = cv2.filter2D(img, -1, sobel_x.T)
mag = np.hypot(gx, gy)
print("float Sobel x range:", gx.min().round(3), gx.max().round(3))

gx_u8 = cv2.filter2D(img_u8, -1, sobel_x)               # uint8 in, uint8 out: saturated
print("uint8 Sobel x range:", gx_u8.min(), gx_u8.max(),
      "| pixels clipped to 0:", f"{(gx_u8 == 0).mean():.0%}")

blur = cv2.GaussianBlur(img, (5, 5), 1.4)               # 5 x 5 Gaussian, sigma 1.4
print("mean brightness kept by blur:", np.isclose(img.mean(), blur.mean(), atol=1e-3))
K2 = 2 * np.ones((3, 3), np.float32) / 9                # weights sum to 2
print("mean after sum-2 kernel:", round(float(cv2.filter2D(img, -1, K2).mean() /
 ↪  img.mean()), 3))
```

In floating point the Sobel response ranges from about −3.4 to 3.3. With `uint8` input and output, every negative response is clipped to zero (more than half the pixels here), so all light-to-dark edges vanish. Always filter in `float32` and convert back only for display.

Border handling

Plain version. Near the edge of the image, part of the kernel hangs outside. You must decide what values to pretend are there.

The common choices are: **zero** (constant) padding, which pretends the world outside is black; **replicate**, which repeats the edge pixel; **reflect**, which mirrors the image at its border; and **wrap**, which treats the image as periodic, as the discrete Fourier transform implicitly does. OpenCV's

default, `BORDER_REFLECT_101`, mirrors without repeating the edge pixel [11]. PyTorch's `nn.Conv2d` uses zero padding unless you set `padding_mode` to `'reflect'`, `'replicate'` or `'circular'` [31].

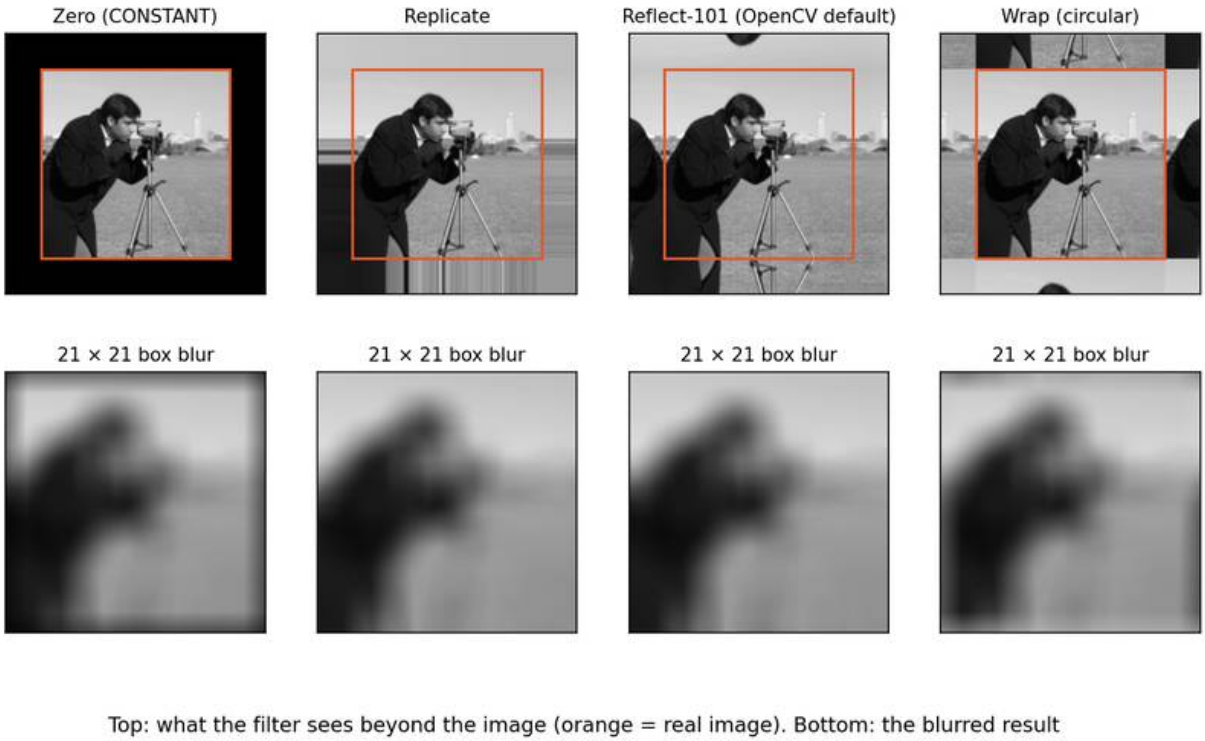


Figure 4 — Four border rules and their effect on a large 21 × 21 box blur. Zero padding darkens every border; wrap blends opposite edges; replicate and reflect keep flat borders flat.

Zero padding darkens the borders of a blurred image because black pixels are averaged in. Wrap mixes the top of the image into the bottom. Replicate and reflect are usually the least visible choices for photographs. With small kernels the difference touches only a thin frame, but in a deep CNN every layer adds its own frame: after many zero-padded layers, a large fraction of the feature map has "seen" the artificial border, which lets networks infer absolute position from padding.

Part II • Convolutional neural networks

From a fixed filter to a learned layer

Plain version. A convolutional layer is a bank of filters whose weights are learned by gradient descent. Each filter looks at all input channels at once and produces one output channel, called a **feature map**.

Precise version. Let the input be $X \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ (channels, height, width) and the weights $K \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k \times k}$ with bias $b \in \mathbb{R}^{C_{\text{out}}}$. A stride-one layer with zero padding p computes

$$Y_{o,u,v} = b_o + \sum_{c=1}^{C_{\text{in}}} \sum_{i=0}^{k-1} \sum_{j=0}^{k-1} K_{o,c,i,j}\, X_{c,\, u+i-p,\, v+j-p}.$$

o indexes output channels (filters), c input channels, (u, v) output positions and (i, j) kernel taps; positions outside the image read the padding value. A nonlinearity such as ReLU is then applied element-wise. An RGB filter has three $k \times k$ slices, one per color channel, and their responses are **summed** into a single output number per position: a filter does not produce three separate images.

Two structural assumptions are built into this formula:

- **Local connectivity**: each output depends only on a $k \times k$ neighborhood, not on the whole image.
- **Weight sharing** (parameter sharing): the same K is used at every position (u, v).

Together they are the CNN's **inductive bias**, an assumption about which functions are likely. Its payoff is efficiency. A layer with $C_{\text{in}} = 3$, $C_{\text{out}} = 32$ and $k = 3$ has $32 \times (3 \cdot 9 + 1) = 896$ parameters, whatever the image size. A fully connected layer from a flattened $224 \times 224 \times 3$ image to just 32 numbers needs $150{,}528 \times 32 \approx 4.8$ million weights, and it would have to learn the same edge detector separately at every position.

Goodfellow et al. [1] summarize the motivation as sparse interactions, parameter sharing and equivariance. Figure 5 shows what a tiny CNN learns in its first layer when trained on the 8×8 digits dataset used in the classifier code later on. The learned 3×3 filters look like oriented derivative and blob detectors: the network rediscovered Sobel-like kernels because they are useful, not because anyone told it to. Zeiler and Fergus [30] showed the same phenomenon at scale by visualizing the filters and feature maps of large ImageNet networks.

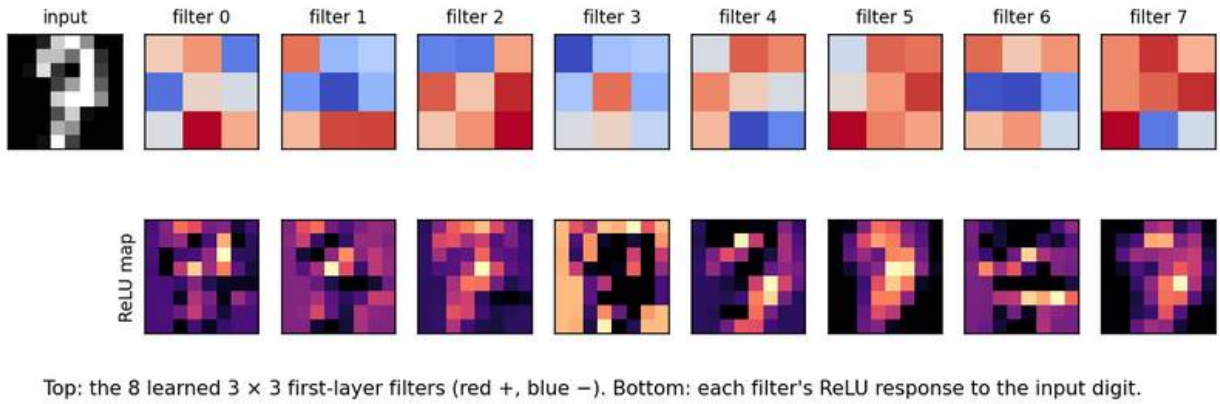


Figure 5 — First-layer filters learned on 8×8 digits (top) and their ReLU responses to one test digit (bottom). Many look like oriented edge or stroke detectors.

Padding, stride and dilation

Plain version. Padding controls what happens at the border and whether the output keeps its size. Stride skips positions to shrink the output. Dilation spreads the kernel taps apart to see farther without adding weights.

For input size n (height or width), kernel size k, padding p on each side, stride s and dilation d, the output size is [12]

$$n_{\text{out}} = \left\lfloor \frac{n + 2p - d(k-1) - 1}{s} \right\rfloor + 1.$$

$d(k-1)+1$ is the **effective kernel size**: dilation d inserts $d-1$ gaps between taps. Three padding conventions have names: **valid** ($p = 0$, output shrinks by $k-1$), **same** ($p = (k-1)/2$ for odd k and $s = 1$, output keeps its size) and **full** ($p = k-1$, every partial overlap counted, output grows by $k-1$). Stride two roughly halves each spatial dimension and is a common alternative to pooling for downsampling.

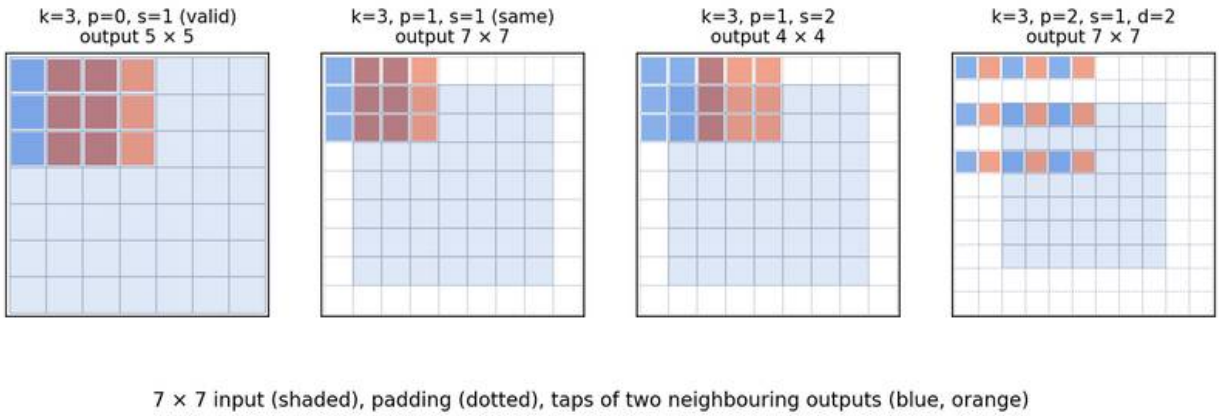


Figure 6 — Taps of two neighboring outputs (blue and orange; overlap appears brown) under four settings. Stride moves the window by s; dilation spaces the taps d apart.

Dilated (atrous) convolutions were popularized for dense prediction by Yu and Koltun [18]: stacking layers with dilations 1, 2, 4, ... grows the field of view exponentially while keeping full resolution. A single dilated layer sees only a sparse lattice of inputs, so dilations should be mixed with care to avoid "gridding" artifacts. A **transposed convolution** runs the forward map of a strided convolution backward to upsample; it reappears in the backpropagation section below. The convolution arithmetic guide by Dumoulin and Visin [12] has diagrams for every combination of these settings.

The snippet checks the formula against PyTorch, counts parameters, shows that a 1×1 convolution is a per-pixel linear layer, and measures translation equivariance:

```
import torch
from torch import nn

def out_size(n, k, p=0, s=1, d=1):
    return (n + 2 * p - d * (k - 1) - 1) // s + 1

x = torch.randn(1, 3, 64, 64)                          # one RGB 64 x 64 image
for k, p, s, d in [(3, 1, 1, 1), (5, 2, 2, 1), (3, 2, 1, 2), (1, 0, 1, 1)]:
    conv = nn.Conv2d(3, 16, k, stride=s, padding=p, dilation=d)
    y = conv(x)
    n_params = sum(q.numel() for q in conv.parameters())
    assert y.shape[-1] == out_size(64, k, p, s, d)
    assert n_params == 16 * (3 * k * k + 1)
    print(f"k={k} p={p} s={s} d={d}: output {tuple(y.shape[1:])}, parameters {n_params}")

# A 1x1 convolution is the same linear layer applied at every pixel
conv1 = nn.Conv2d(16, 4, kernel_size=1)
lin = nn.Linear(16, 4)
lin.weight.data = conv1.weight.data[:, :, 0, 0].clone()
lin.bias.data = conv1.bias.data.clone()
z = torch.randn(2, 16, 8, 8)
same = torch.allclose(conv1(z), lin(z.permute(0, 2, 3, 1)).permute(0, 3, 1, 2), atol=1e-6)
print("1x1 conv == per-pixel Linear:", same)

# Translation equivariance away from borders (stride 1, no padding effects in the centre)
conv = nn.Conv2d(1, 4, 3, padding=1, bias=False)
img = torch.zeros(1, 1, 32, 32); img[..., 10:14, 8:12] = 1
shift = lambda t: torch.roll(t, shifts=(3, 5), dims=(-2, -1))
with torch.no_grad():
    a, b = conv(shift(img)), shift(conv(img))
print("shift then conv == conv then shift:", torch.allclose(a, b, atol=1e-6))
conv2 = nn.Conv2d(1, 4, 3, padding=1, stride=2, bias=False)
with torch.no_grad():
    y0, y1 = conv2(img), conv2(torch.roll(img, 1, dims=-1))   # shift by one pixel
print("stride 2, 1-pixel shift: max output change =", round((y0 - y1).abs().max().item(), 3))
```

It prints output shapes $16 \times 64 \times 64$, $16 \times 32 \times 32$, $16 \times 64 \times 64$ and $16 \times 64 \times 64$ with 448, 1216, 448 and 64 parameters, confirms the 1×1 and equivariance checks, and shows that a one-pixel shift changes a stride-two output by a clearly nonzero amount; the exact value depends on the randomly initialized weights.

Channels and 1×1 convolutions

Plain version. Channels are the "what" dimension of a feature map; height and width are the "where". A 1×1 convolution mixes channels at each pixel without looking at neighbors.

A 1×1 convolution with C_{in} inputs and C_{out} outputs multiplies the channel vector at every pixel by the same $C_{\text{out}} \times C_{\text{in}}$ matrix, as the snippet verified. Lin et al.'s Network in Network [17] used

this idea as a tiny per-pixel MLP and introduced **global average pooling** as a classifier head. GoogLeNet [16] used 1×1 layers to reduce channels before expensive 3×3 and 5×5 branches, and ResNet's bottleneck block [5] wraps a 3×3 layer between a 1×1 reduction and a 1×1 expansion. In every case the motivation is the same: spatial filtering costs k^2 times more than channel mixing, so do the cheap part with 1×1 layers and keep the expensive part narrow.

Pooling and downsampling

Plain version. Pooling summarizes each small block of a feature map by a single number, usually the maximum or the average. It shrinks the map and makes the representation less sensitive to small shifts.

Max pooling with a 2×2 window and stride two keeps the largest value in each block: "was this feature present anywhere here?". **Average pooling** keeps the mean. **Global average pooling** averages each channel over all positions, producing one number per channel regardless of input size, which is why modern classifiers can accept different image sizes. Pooling has no learnable parameters. Its gradient is simple: max pooling routes the whole upstream gradient to the winning input of each window; average pooling spreads it equally over the window.

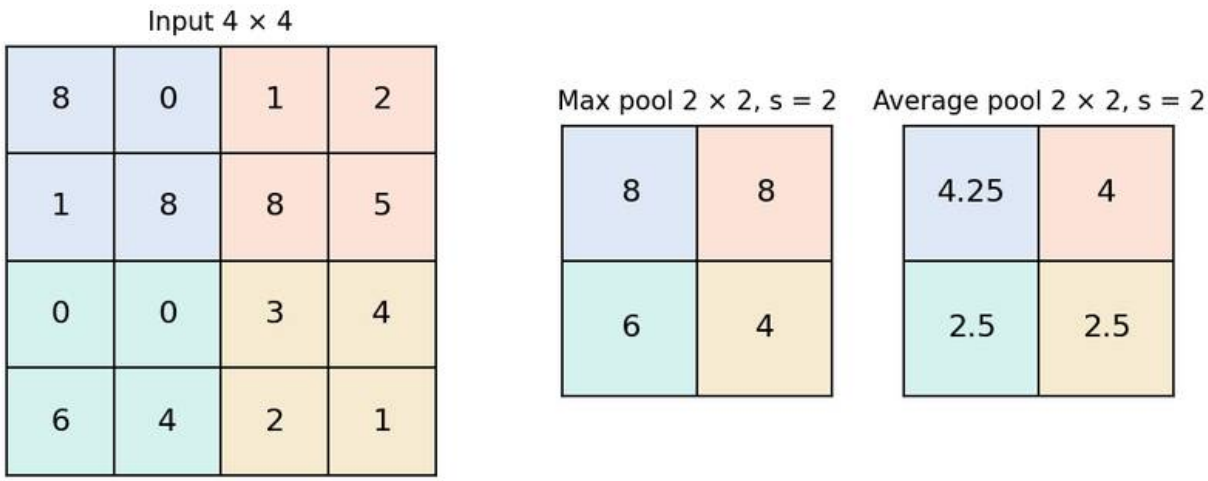


Figure 7 — Max and average pooling with a 2×2 window and stride 2. Each colored block becomes one number.

Downsampling, by pooling or by strided convolution, saves computation and enlarges later receptive fields, but it is also **subsampling** without a proper low-pass filter. High frequencies alias, and the output can change noticeably when the input moves by a single pixel, as the snippet above measured. Zhang [20] showed that inserting a small blur before subsampling ("anti-aliased" or

"blur pooling") makes standard CNNs measurably more shift-invariant: a neat case of a classical signal-processing rule fixing a deep network.

Equivariance is not invariance

Plain version. Shift the input, and a convolutional feature map shifts with it. That is not the same as the final prediction staying unchanged.

Let T_Δ shift an image by Δ pixels. Away from the borders, a stride-one convolution is **translation-equivariant**: $f(T_\Delta X) = T_\Delta f(X)$. A classifier wants **invariance**: $g(T_\Delta X) = g(X)$. CNNs get approximate invariance by stacking equivariant layers and then pooling, especially global average pooling, which discards position. Padding, strides, cropping and the finite image break exact equivariance. Rotation and scale equivariance are **not** built in; data augmentation or specially designed group-equivariant layers must supply them.

The distinction matters in practice. A detector must be equivariant: moving a person should move the predicted box. A classifier should be invariant: moving the person should not change the label. The same backbone serves both, with different heads.

Receptive field

Plain version. Early layers see small pieces of the image; deeper layers combine those pieces and see larger regions.

The **receptive field** of a unit is the set of input pixels that can influence it. Let r_ℓ be its width after layer ℓ, and j_ℓ the **jump**, the distance in input pixels between adjacent units of layer ℓ. Starting from $r_0 = j_0 = 1$, a layer with kernel size k_ℓ, dilation d_ℓ and stride s_ℓ gives

$$r_\ell = r_{\ell-1} + (d_\ell(k_\ell - 1))\, j_{\ell-1}, \qquad j_\ell = s_\ell\, j_{\ell-1}.$$

Three stacked stride-one 3×3 layers see 7×7 pixels, the same area as one 7×7 layer but with $3 \cdot 9C^2 = 27C^2$ instead of $49C^2$ weights and three nonlinearities instead of one; this was the central design argument of VGG [15]. A stride-two layer doubles the jump, so every later layer grows the field twice as fast.

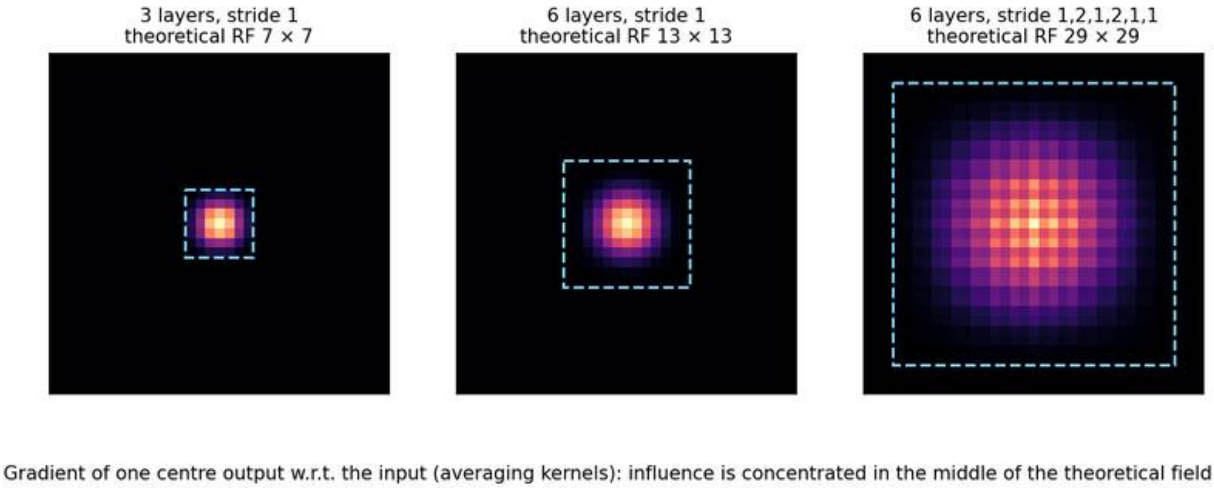


Figure 8 — Theoretical receptive field (dashed) versus actual influence, measured as the input gradient of one central output through stacks of 3 × 3 averaging layers. Influence is Gaussian-like and concentrated in the center; strides add a checkerboard pattern.

The theoretical field says which pixels **could** matter. Luo et al. [3] measured the **effective receptive field** via input gradients, as in Figure 8, and found that it is roughly Gaussian and occupies only a fraction of the theoretical field, growing like the square root of depth for uniform layers. Two practical lessons follow: a network may need more context than its theoretical field suggests, and downsampling, dilation and large kernels are ways to enlarge the effective field without unbounded depth.

> **Interactive demonstration (online)**: open the online edition of this tutorial to use this demonstration.Open demonstration

Backpropagation through a convolution

Plain version. A convolution is linear, so its backward pass is also a convolution: the gradient with respect to the kernel correlates the input with the upstream gradient, and the gradient with respect to the input convolves the upstream gradient with the kernel.

Precise version. Take one channel, no padding, stride one: $Y_{u,v} = \sum_{i,j} K_{i,j} X_{u+i,v+j}$. Let L be the loss and $G_{u,v} = \partial L / \partial Y_{u,v}$ the **upstream gradient** delivered by backpropagation, as in ML 02. Each weight $K_{i,j}$ is used at every output position, so by the chain rule its gradient sums over them:

$$\frac{\partial L}{\partial K_{i,j}} = \sum_{u,v} G_{u,v}\, X_{u+i,v+j}.$$

That is a valid correlation of the input X with G, which is why weight sharing does not complicate training: shared weights simply accumulate gradients from every location. Each input pixel $X_{m,n}$ contributes to every output whose window covers it, at tap $(i, j) = (m - u, n - v)$:

$$\frac{\partial L}{\partial X_{m,n}} = \sum_{i,j} G_{m-i,\,n-j}\, K_{i,j},$$

with G taken as zero outside its range. This is a **full convolution** of G with K, equivalently a correlation of the zero-padded G with the flipped kernel. It is exactly what a **transposed convolution** computes, which is how transposed convolutions got their name. Bias gradients are just $\sum_{u,v} G_{u,v}$ per output channel. With multiple channels, the input gradient also sums over output channels, and the kernel gradient also sums over the batch.

```
import torch
import torch.nn.functional as F

torch.manual_seed(0)
X = torch.randn(1, 1, 6, 7, requires_grad=True)     # input
K = torch.randn(1, 1, 3, 3, requires_grad=True)     # kernel
Y = F.conv2d(X, K)                                   # 'valid' correlation: 4 x 5
G = torch.randn_like(Y)                              # pretend upstream gradient dL/dY
(Y * G).sum().backward()                             # so that dL/dY = G exactly

# 1) dL/dK: correlate the input with the upstream gradient
dK = F.conv2d(X.detach(), G)
# 2) dL/dX: 'full' convolution of G with K = correlation of zero-padded G with flipped K
dX = F.conv2d(F.pad(G, (2, 2, 2, 2)), K.detach().flip(-1, -2))
# 3) the same thing is what a transposed convolution computes
dX_t = F.conv_transpose2d(G, K.detach())

print("dL/dK matches autograd:", torch.allclose(dK, K.grad, atol=1e-5))
print("dL/dX matches autograd:", torch.allclose(dX, X.grad, atol=1e-5))
print("conv_transpose2d gives the same:", torch.allclose(dX_t, X.grad, atol=1e-5))
```

All three lines print `True`. The trick `(Y * G).sum().backward()` makes autograd use a chosen upstream gradient G, a handy way to test any hand-derived backward pass.

Classic architectures in brief

Plain version. Every famous CNN is a stack of convolutions, nonlinearities and downsampling. What changed over the years was depth, how to keep deep stacks trainable, and how to spend computation wisely.

Model (year)	Key idea	What it taught
LeNet-5 (1998) [13]	conv → subsample → conv → subsample → fully connected, trained end to end	Learned convolutional features beat hand-designed ones for digits
AlexNet (2012) [14]	Deeper and wider, ReLU, dropout, data augmentation, GPU training	Large labeled data plus GPUs made CNNs win ImageNet
VGG (2014) [15]	Only 3×3 convolutions, 16–19 layers	Depth with small kernels is simple and effective
GoogLeNet / Inception (2014) [16]	Parallel 1×1, 3×3, 5×5 branches with 1×1 reductions	Multi-scale features at controlled cost
ResNet (2015) [5]	Residual connections $y = x + F(x)$, BatchNorm	Very deep networks (100+ layers) become trainable
U-Net (2015) [4]	Encoder–decoder with skip connections at each resolution	Dense, pixel-aligned outputs from few images
MobileNetV2 (2018) [7]	Depthwise-separable convolutions, inverted residuals, linear bottlenecks	Accurate models for phones
EfficientNet (2019) [8]	Compound scaling of depth, width and resolution	Scale dimensions together, not one at a time
ConvNeXt (2022) [9]	7×7 depthwise kernels, LayerNorm, inverted bottlenecks, transformer-era training	A modernized pure CNN competes with vision transformers

The pattern from LeNet to ResNet is a steady increase in depth. AlexNet [14] showed that a deep CNN trained on ImageNet with GPUs, ReLU and dropout could beat traditional pipelines by a wide margin. VGG [15] argued for uniform stacks of 3×3 layers, as computed in the receptive-field section. GoogLeNet [16] made width cheap with 1×1 reductions. Then depth hit an optimization wall: plain networks with more layers had **higher training error**, which is not overfitting but a failure to optimize. ResNet [5] fixed it with identity shortcuts. DLCV 02 follows this line further, through training recipes, transfer learning and evaluation.

Residual blocks and normalization

Plain version. Instead of asking a block to compute the whole output, ask it to compute a correction to its input.

A residual block computes $y = x + F_\theta(x)$, where F_θ is typically two or three convolutions with normalization and ReLU. Its Jacobian is $\partial y/\partial x = I + \partial F_\theta/\partial x$: even if F_θ contributes little, the identity path carries the signal forward and the gradient backward. If $F_\theta \approx 0$ is the best a block can do, it can learn that easily, so adding blocks no longer hurts optimization. When the shape changes (more channels, stride two), the shortcut becomes a projection $P(x)$, usually a strided 1×1 convolution.

Batch normalization [19] standardizes each channel using the batch mean and variance during training, then applies a learned scale and shift; at evaluation it uses running averages. It makes training faster and less sensitive to initialization, but it misbehaves with very small or correlated

batches, where **GroupNorm** [6] (statistics over groups of channels within each sample) is more reliable. Always call `model.eval()` before validation so BatchNorm uses its running statistics.

```
import torch
from torch import nn

count = lambda m: sum(p.numel() for p in m.parameters())

C = 64
dense = nn.Conv2d(C, C, 3, padding=1, bias=False)
separable = nn.Sequential(
    nn.Conv2d(C, C, 3, padding=1, groups=C, bias=False),   # depthwise: one 3x3 per channel
    nn.Conv2d(C, C, 1, bias=False))                        # pointwise: mix channels
print("dense:", count(dense), "| depthwise+pointwise:", count(separable),
      "| ratio:", round(count(separable) / count(dense), 4), "vs 1/C + 1/9 =",
      round(1 / C + 1 / 9, 4))

class BasicBlock(nn.Module):
    """ResNet-style block: y = relu(x + F(x)), with a 1x1 projection when shapes change."""
    def __init__(self, c_in, c_out, stride=1):
        super().__init__()
        self.f = nn.Sequential(
            nn.Conv2d(c_in, c_out, 3, stride, 1, bias=False), nn.BatchNorm2d(c_out),
            ↪  nn.ReLU(),
            nn.Conv2d(c_out, c_out, 3, 1, 1, bias=False), nn.BatchNorm2d(c_out))
        self.proj = (nn.Identity() if stride == 1 and c_in == c_out else
                     nn.Sequential(nn.Conv2d(c_in, c_out, 1, stride, bias=False),
                                   nn.BatchNorm2d(c_out)))
    def forward(self, x):
        return torch.relu(self.proj(x) + self.f(x))

x = torch.randn(2, 64, 32, 32)
print("same shape :", tuple(BasicBlock(64, 64)(x).shape))
print("downsample :", tuple(BasicBlock(64, 128, stride=2)(x).shape))
```

The output reports 36,864 weights for the dense layer against 4,672 for the separable pair, a ratio of 0.1267 that matches the formula below, and block outputs of shape $64 \times 32 \times 32$ and $128 \times 16 \times 16$.

Efficient CNNs separate spatial filtering from channel mixing

Plain version. Looking around (spatial filtering) and combining features (channel mixing) need not happen in one expensive operation.

A dense $k \times k$ convolution has $k^2 C_{\text{in}} C_{\text{out}}$ weights. A **depthwise** convolution filters each channel separately ($k^2 C_{\text{in}}$ weights, `groups=C` in PyTorch); a following **pointwise** 1×1 convolution mixes channels ($C_{\text{in}} C_{\text{out}}$ weights). With $C_{\text{in}} = C_{\text{out}} = C$,

$$\frac{\text{depthwise} + \text{pointwise}}{\text{dense}} = \frac{k^2C + C^2}{k^2C^2} = \frac{1}{C} + \frac{1}{k^2}.$$

For $k = 3$ and large C this approaches $1/9$.

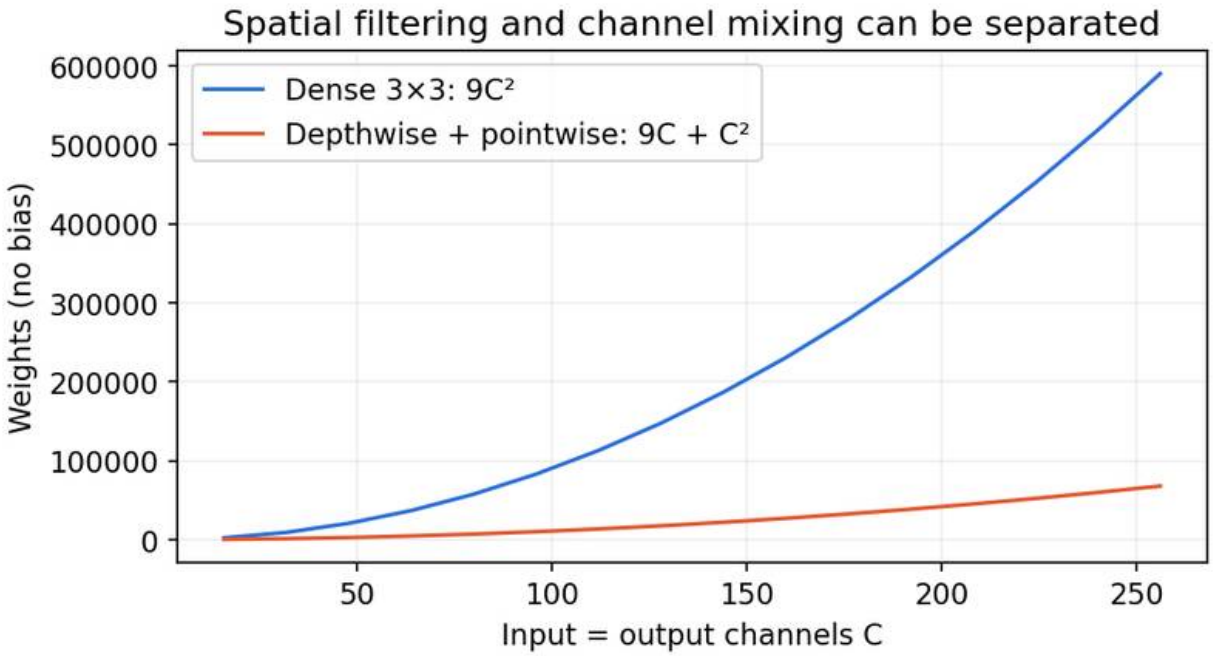


Figure 9 — Analytical weight counts for equal input and output channels. Fewer weights do not automatically mean lower latency: depthwise layers have low arithmetic intensity and may run slowly on some hardware.

MobileNetV2 [7] builds blocks that expand channels with a 1×1 layer, filter with a depthwise 3×3, then project back with a linear (no ReLU) 1×1 bottleneck, with a residual connection between the narrow ends. EfficientNet [8] scales depth, width and input resolution together with a single compound coefficient. ConvNeXt [9] started from a ResNet and applied, one at a time, design choices associated with vision transformers: patchify stem, 7×7 depthwise kernels, inverted bottlenecks, fewer activations, LayerNorm, and longer training with heavy augmentation. The result matched Swin Transformers of similar size, evidence that much of the transformer advantage came from training recipes and macro-design rather than from attention alone.

Code: a complete small-image classifier

This example needs `torch`, `numpy` and `scikit-learn`. It uses the bundled 8×8 digits dataset, so no download or GPU is needed. A stratified split gives separate training, validation and test sets: validation picks the best epoch, and the test set is evaluated once at the end. It is a teaching baseline, not an ImageNet recipe.

```
import copy
import numpy as np
import torch
from torch import nn
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split

torch.manual_seed(0)
torch.set_num_threads(1)
d = load_digits()
X = torch.tensor(d.images[:, None] / 16.0, dtype=torch.float32)
y = torch.tensor(d.target, dtype=torch.long)
train, test = train_test_split(np.arange(len(y)), test_size=.2,
                               stratify=d.target, random_state=0)
train, val = train_test_split(train, test_size=.2,
                              stratify=d.target[train], random_state=1)
model = nn.Sequential(
    nn.Conv2d(1, 16, 3, padding=1), nn.ReLU(),
    nn.Conv2d(16, 32, 3, padding=1), nn.ReLU(),
    nn.MaxPool2d(2),                       # B × 32 × 4 × 4
    nn.Flatten(), nn.Linear(32 * 4 * 4, 10),
)
opt = torch.optim.AdamW(model.parameters(), lr=1e-3, weight_decay=1e-4)
best_loss, best_state = float("inf"), None
for epoch in range(12):
    model.train()
    order = torch.tensor(train)[torch.randperm(len(train))]
    for idx in order.split(64):
        loss = nn.functional.cross_entropy(model(X[idx]), y[idx])
        opt.zero_grad()
        loss.backward()
        opt.step()
    model.eval()
    with torch.no_grad():
        val_loss = nn.functional.cross_entropy(model(X[val]), y[val]).item()
    if val_loss < best_loss:
        best_loss = val_loss
        best_state = copy.deepcopy(model.state_dict())
model.load_state_dict(best_state)
model.eval()
with torch.no_grad():
    accuracy = (model(X[test]).argmax(1) == y[test]).float().mean().item()
print("parameters:", sum(p.numel() for p in model.parameters()))
print("best validation loss:", round(best_loss, 4))
print("held-out accuracy:", round(accuracy, 4))
```

On our CPU run it reports 9,930 parameters and about 96% held-out accuracy (exact numbers vary slightly across PyTorch versions). Cross-entropy receives **logits**, with no softmax in front. The head flattens the $32 \times 4 \times 4$ map rather than global-average-pooling it, because on tiny digits the position of strokes carries information; for variable-size images, adaptive or global pooling

gives a fixed-size head.

Inspect the errors, not just the score. Which digit pairs get confused? Does shifting every test digit by one pixel hurt? Compare against a parameter-matched MLP and a classical SVM, as in IP 13, before crediting every gain to convolution.

Modern view

Survey literature. Several reviews map the CNN landscape, each with a different emphasis. Gu et al. [23] (*Pattern Recognition*, 2018) organize CNN advances by component: convolution variants, pooling, activations, loss functions, regularization and optimization, followed by applications. Khan et al. [22] (*Artificial Intelligence Review*, 2020) propose a taxonomy of architectures by their main innovation: spatial exploitation (kernel sizes), depth, multi-path designs (shortcuts, Inception branches), width, feature-map exploitation (channel attention), and attention-based designs. Li et al. [21] (*IEEE TNNLS*, 2022) cover the history from LeNet onward, compare one-, two- and multi-dimensional convolutions, and survey lightweight networks and applications with rules of thumb for choosing hyperparameters. Alzubaidi et al. [2] (*Journal of Big Data*, 2021) give a broad, application-oriented review of deep learning centered on CNN architectures, with discussion of data scarcity, transfer learning and hardware. These surveys predate ConvNeXt, so read them as maps of the 2012–2020 era.

CNNs after transformers (2022–2026). Vision transformers (ML 04) prompted a re-examination of convolution, and the main finding is that the gap was smaller than early comparisons suggested:

- **Modernized CNNs.** ConvNeXt [9] showed that a carefully modernized ResNet matches Swin. ConvNeXt V2 [24] (CVPR 2023) added masked-autoencoder pretraining with sparse convolutions and a global response normalization layer, bringing self-supervised pretraining (see ML 05) to pure CNNs.
- **Large kernels.** RepLKNet [26] (CVPR 2022) used depthwise kernels as large as 31×31, with re-parameterized small kernels to ease optimization, to enlarge the effective receptive field. UniRepLKNet [27] (CVPR 2024) extended large-kernel design to audio, video, point clouds and time series.
- **Dynamic and deformable convolution.** InternImage [25] (CVPR 2023) scaled deformable convolution, whose sampling locations are predicted from the input, into a large foundation backbone for detection and segmentation.
- **Scale.** Smith et al. [29] (2023) pretrained NFNet CNNs on the large JFT-4B image dataset and found that their performance followed similar compute scaling trends to vision transformers under comparable budgets, arguing that compute and data, more than architecture, dominate at scale.
- **Edge devices.** MobileNetV4 [28] (ECCV 2024) designed blocks and a neural architecture

search for good latency across CPUs, GPUs, DSPs and accelerators, an acknowledgment that parameter counts alone do not predict speed.

Where CNNs fit in 2026. Convolution remains a default for latency-constrained and on-device inference, for dense prediction at high resolution (restoration, segmentation, medical imaging, see DLCV 04), for small datasets where the locality bias helps, and as the stem or early stages of hybrid CNN–transformer models. Large vision foundation models are mostly transformer-based, but they frequently contain convolutions in their patch embeddings, upsampling decoders and diffusion U-Nets.

Open problems. Exact shift equivariance under downsampling (aliasing [20]); building rotation and scale symmetry in without heavy cost; understanding what the effective receptive field really needs to be for a task [3]; and fair comparison across architectures, which requires matching data, augmentation, resolution, schedule and compute before attributing a difference to convolution versus attention.

Key takeaways

- A kernel slides over the image and computes a weighted sum at every position. Libraries usually compute **correlation**; true convolution flips the kernel, which matters only for asymmetric, hand-designed kernels.
- The sum of a kernel's weights sets its response to flat regions: sum 1 blurs while keeping brightness; sum 0 detects changes. Filter in floating point, and choose the border rule on purpose.
- A convolutional layer is a bank of learned filters with **local connectivity** and **weight sharing**. Its parameter count, $C_{\text{out}}(C_{\text{in}}k^2 + 1)$, does not depend on image size.
- Padding, stride and dilation set the output size, $\lfloor (n + 2p - d(k-1) - 1)/s \rfloor + 1$. 1×1 convolutions mix channels; pooling and strides downsample but can alias.
- Convolution is shift-**equivariant**; invariance comes from pooling. Receptive fields grow with depth and stride, but the effective field is much smaller than the theoretical one.
- Backpropagation through a convolution is itself a convolution: kernel gradients correlate input with upstream gradient, and input gradients are a transposed convolution.
- Architecture history is a story of depth (LeNet → VGG → ResNet), efficient channel use (1×1, depthwise-separable) and better training. Modernized CNNs remain competitive with transformers when budgets are matched.

Exercises

1. **Design and verify.** Write a 3×3 kernel that detects only horizontal edges, and another that detects only diagonal edges running from top-left to bottom-right. Verify both in the kernel

playground or with `cv2.filter2D` on a synthetic image containing a horizontal, a vertical and a diagonal line.

Hint

A horizontal edge is a change along y, so use $S_y = S_x^\top$ (rows $-1,-2,-1$ / $0,0,0$ / $1,2,1$). For a diagonal edge, put the derivative across the diagonal, for example $\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \\ -2 & -1 & 0 \end{bmatrix}$. Its weights sum to zero, so flat regions give zero.

2. **Kernel sums.** Why should a blur kernel's weights sum to one? Predict, then check, what happens to a mid-gray `uint8` image filtered with a 3×3 kernel whose weights all equal $2/9$, and what changes if you filter in `float32` instead.

Hint

A flat region of value c maps to $c \sum K$. With sum two, a gray of 128 becomes 256, which saturates at 255 in `uint8`; every brighter-than-mid-gray region turns white. In `float32` the values double without clipping, and you can rescale afterward.

3. **Composition.** Show that applying a 3×3 box blur twice equals one 5×5 kernel, and write that kernel. Verify numerically away from the borders, and explain why the equality can fail at the borders.

Hint

By associativity, the combined kernel is the box convolved with itself: the outer product of $[1, 2, 3, 2, 1]$ with itself, divided by 81. At the border, the first pass already invented padding values, and the second pass pads that result; a single 5×5 pass pads the original image instead, so the two differ in a frame of width two.

4. **Layer arithmetic.** For a 64×64 RGB input and a layer with 16 filters, $k = 5$, $p = 2$, $s = 2$: give the output shape and the number of parameters. Then compute (r, j) after three 3×3 layers with strides 1, 2, 1, and compare with all strides equal to one.

Hint

$\lfloor (64 + 4 - 4 - 1)/2 \rfloor + 1 = 32$, so the output is $16 \times 32 \times 32$ with $16(3 \cdot 25 + 1) = 1216$ parameters. Receptive field: $(3, 1) \to (5, 2) \to (9, 2)$. All strides one gives $(7, 1)$.

5. **Gradient by hand.** For a 1×4 input $X = [x_1, x_2, x_3, x_4]$ and a kernel $K = [k_1, k_2]$ with valid correlation, write Y, then derive $\partial L/\partial K$ and $\partial L/\partial X$ in terms of $G = \partial L/\partial Y$. Check your formulas with the autograd trick from the backpropagation snippet.

Hint

$Y = [k_1x_1 + k_2x_2,\ k_1x_2 + k_2x_3,\ k_1x_3 + k_2x_4]$. Then $\partial L/\partial k_1 = G_1x_1 + G_2x_2 + G_3x_3$ and $\partial L/\partial k_2 = G_1x_2 + G_2x_3 + G_3x_4$ (a correlation of X with G). $\partial L/\partial X = [G_1k_1,\ G_1k_2 + G_2k_1,\ G_2k_2 + G_3k_1,\ G_3k_2]$, the full convolution of G with K.

6. **Ablation on digits.** In the classifier, replace `Flatten` + `Linear(512, 10)` by global average pooling + `Linear(32, 10)`. Compare parameter counts, validation loss and test accuracy, then evaluate both models on test digits shifted by one pixel. Explain the trade-off you observe.

Hint

`nn.AdaptiveAvgPool2d(1)` then `nn.Flatten()`. Pooling removes most head parameters and position information, which can help shift robustness but hurt on 8×8 digits where stroke position matters. Keep everything else fixed, select on validation only, and report both accuracy and parameter count.

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ML 03A • Recurrent and Recursive Networks

Learn from sequences and trees: shared recurrent state, backpropagation through time, vanishing and exploding gradients, LSTM and GRU gates, bidirectional and encoder–decoder models, attention as the bridge to Transformers, recursive networks, and their uses in vision.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-03a-recurrent-recursive-networks/

What you'll learn

- How a recurrent network reuses one set of weights at every time step, so it can read sequences of any length.
- How backpropagation through time (BPTT) trains it, and why gradients over long spans vanish or explode.
- How LSTM and GRU gates build a protected memory path, written out equation by equation and checked against PyTorch.
- How bidirectional, stacked and encoder–decoder (sequence-to-sequence) models are assembled, and how attention removed the encoder bottleneck and led to the Transformer.
- How recursive networks compose representations over trees instead of chains.
- Where recurrence is used in vision (captioning, video, spatiotemporal forecasting) and how Mamba and Mamba-2 use selective state spaces, parallel scans and fixed-size streaming memory.

The big picture

ML 03 used one assumption about images: nearby pixels are related, and the same pattern can appear anywhere, so the same filter is reused across **space**. Sequences invite the same trick across **time**. A video is a sequence of frames, a caption is a sequence of words, a pen stroke is a sequence of positions. A **recurrent neural network** (**RNN**) reads such a sequence one element at a time, carries a summary called the **hidden state** from step to step, and uses the same weights at every step [1]. A **recursive neural network** generalizes the idea from a chain to a tree: it combines parts into wholes by repeatedly applying one shared composition rule.

> **Explain like I'm five**
>
> When you listen to a story, you do not remember every word. You keep a little picture in your head of what has happened so far, and each new sentence updates that picture. A recurrent network does exactly this: it has a small notebook (the hidden state), reads one word or one video frame at a time, and rewrites the notebook after each one using the same rules every time. An LSTM is a notebook with

an eraser and a pen that the network learns to use carefully, so important notes from the beginning are not smudged away by the end. A recursive network is like building with LEGO: snap two small pieces into a bigger piece, then snap that into an even bigger piece, using the same kind of snap each time.

Why study RNNs when Transformers dominate large-scale language and vision? Three reasons. First, every concept in modern sequence modeling — teacher forcing, encoder–decoder, attention, masking, causal generation — was developed on RNNs, and the Transformer is easiest to understand as the answer to RNN limitations. Second, recurrence is efficient for **streaming**: a constant-size state per step suits robots, sensors and on-device video. Third, recurrence is back: modern state-space models such as Mamba are recurrences with clever parameterizations, and ML 04 applies them to images.

A recurrent state

Plain version. At every step, combine the current input with the previous state to make a new state, using the same weights each time. Read outputs off the state whenever you need them.

Precise version. Let the input sequence be $x_1, \dots, x_T$ with $x_t \in \mathbb{R}^d$, and let the hidden state be $h_t \in \mathbb{R}^m$ with a fixed starting state h_0 (usually zeros). The simple (Elman [9]) RNN is

$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b), \qquad \hat{y}_t = g\big(W_y h_t + c\big).$$

$W_x \in \mathbb{R}^{m \times d}$ maps the input into the state space, $W_h \in \mathbb{R}^{m \times m}$ is the **recurrent** weight matrix that carries information from the previous step, b and c are biases, W_y maps the state to outputs, and g is an output function such as softmax. Crucially, none of these parameters depends on t. That is **parameter sharing across time**, the temporal version of a CNN's weight sharing: the model has the same number of parameters whether the sequence has 5 or 5,000 steps, and a pattern learned at step 3 can be recognized at step 300.

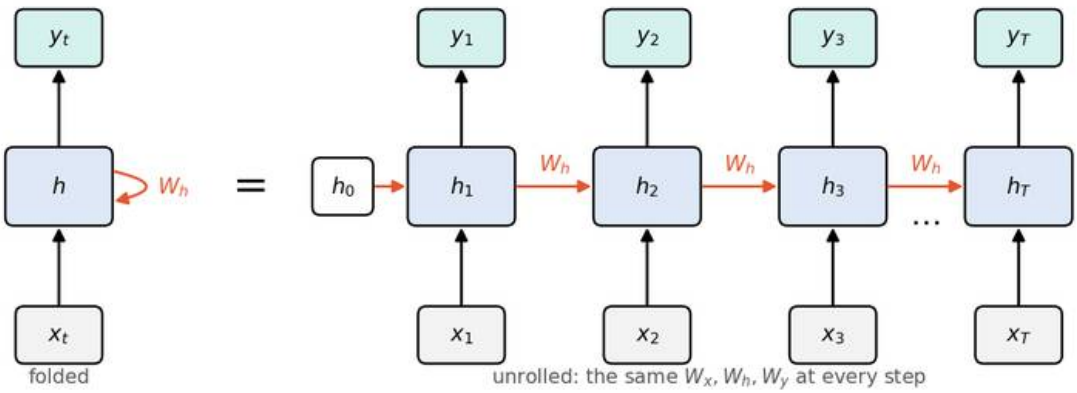


Figure 1 — Folded and unrolled views of an RNN. Unrolling turns the loop into a deep feed-forward network whose layers all share the same weights.

Unrolling (Figure 1) is the key mental move. Writing out the loop for T steps produces an ordinary feed-forward network that is T layers deep, in which every layer has the same weights. Everything you learned about backpropagation in ML 02 applies to this unrolled graph.

The same recurrence supports several input–output patterns:

Pattern	Example	Output read from
Many-to-one	Classify a video clip; sentiment of a sentence	The last state h_T (or a pooled state)
Many-to-many, aligned	Label every video frame; tag every word	Every h_t
One-to-many	Generate a caption from one image vector	A decoder that feeds its own outputs back in
Many-to-many, unaligned	Translate a sentence; describe a video	An **encoder–decoder** (covered below)

An RNN that predicts x_{t+1} from $x_{\le t}$ is an **autoregressive** model, and sampling from it generates new sequences; but an RNN used as an encoder is not automatically a generator. The head and the loss decide the task.

Code: an RNN and its backward pass in NumPy

The snippet below implements the forward pass and backpropagation through time by hand for a loss $L = v^\top h_T$, then checks one gradient entry against a finite difference. Notice the line `dWh += ⋯`: because W_h is used at every step, its gradient is the **sum** of contributions from all steps.

```
import numpy as np

rng = np.random.default_rng(0)
d, m, T = 3, 5, 7                                  # input size, state size, length
Wx = rng.normal(0, 0.5, (m, d))
Wh = rng.normal(0, 0.5, (m, m))
b = np.zeros(m)

def rnn_forward(xs, h0):
    hs = [h0]
    for x in xs:                                    # the SAME Wx, Wh, b at every step
        hs.append(np.tanh(Wx @ x + Wh @ hs[-1] + b))
    return hs

def rnn_backward(xs, hs, dhT):
    """BPTT for the loss L = dhT . h_T; returns dL/dWh and dL/dx_t for every t."""
    dWh, dxs, dh = np.zeros_like(Wh), [None] * len(xs), dhT
    for t in reversed(range(len(xs))):
```

```
        da = dh * (1 - hs[t + 1] ** 2)              # through tanh
        dWh += np.outer(da, hs[t])                  # shared weight: gradients ADD up
        dxs[t] = Wx.T @ da
        dh = Wh.T @ da                              # pass the gradient one step back
    return dWh, dxs

xs = rng.normal(size=(T, d))
hs = rnn_forward(xs, np.zeros(m))
dhT = rng.normal(size=m)
dWh, dxs = rnn_backward(xs, hs, dhT)

# finite-difference check of one entry of dL/dWh
eps, (i, j) = 1e-6, (2, 3)
Wh[i, j] += eps; up = dhT @ rnn_forward(xs, np.zeros(m))[-1]
Wh[i, j] -= 2 * eps; down = dhT @ rnn_forward(xs, np.zeros(m))[-1]
Wh[i, j] += eps
print("BPTT vs finite difference:", np.isclose(dWh[i, j], (up - down) / (2 * eps)))
print("|dL/dx_t| from first to last step:", np.round([np.linalg.norm(g) for g in dxs], 3))
```

It prints `True`, and the input-gradient norms are $[0.009, 0.015, 0.006, 0.029, 0.383, 0.212, 0.433]$: even in a seven-step sequence, the early inputs receive gradients roughly ten to fifty times smaller than the late ones. That is the vanishing-gradient problem in miniature.

Backpropagation through time

Plain version. Unroll the network, run ordinary backpropagation through the unrolled graph, and add up the gradients for each shared weight.

Precise version. Write the pre-activation as $a_t = W_x x_t + W_h h_{t-1} + b$, so $h_t = \tanh(a_t)$. Suppose the loss is a sum over steps, $L = \sum_t L_t(h_t)$. Working backward from $t = T$, the total gradient arriving at h_t is

$$\frac{\partial L}{\partial h_t} = \frac{\partial L_t}{\partial h_t} + \left(\frac{\partial h_{t+1}}{\partial h_t}\right)^{\top} \frac{\partial L}{\partial h_{t+1}}, \qquad \frac{\partial h_{t+1}}{\partial h_t} = \operatorname{diag}(1 - h_{t+1}^2)\, W_h.$$

The first term is the local loss at step t; the second is everything later that h_t influenced, passed back one step. With $\delta_t = (1 - h_t^2) \odot \partial L / \partial h_t$ (the gradient at the pre-activation), the parameter gradients are sums over time:

$$\frac{\partial L}{\partial W_h} = \sum_{t=1}^{T} \delta_t\, h_{t-1}^{\top}, \qquad \frac{\partial L}{\partial W_x} = \sum_{t=1}^{T} \delta_t\, x_t^{\top}, \qquad \frac{\partial L}{\partial b} = \sum_{t=1}^{T} \delta_t.$$

This algorithm, **backpropagation through time** (**BPTT**), was laid out clearly by Werbos [8]. Its cost is linear in T, but so is its memory: every h_t must be stored for the backward pass.

Truncated BPTT

For a very long stream (an hour of video, a continuous sensor log) storing all states is impossible.

Truncated BPTT processes the stream in windows of k steps. It carries the hidden state forward

from one window to the next, so the model still *sees* the whole past, but it **detaches** the state from the graph at each window boundary, so gradients flow back at most k steps. The model can therefore use long-range information only if it learns to preserve it from the gradient signal within windows.

```
import torch
from torch import nn

torch.manual_seed(0)
rnn, head = nn.GRU(1, 16, batch_first=True), nn.Linear(16, 1)
params = list(rnn.parameters()) + list(head.parameters())
opt = torch.optim.Adam(params, lr=1e-2)
t = torch.arange(0, 2000) * 0.05
stream = torch.sin(t) + 0.1 * torch.randn_like(t)   # one long noisy signal
x, y = stream[:-1, None][None], stream[1:, None][None]   # predict the next sample

h, window = None, 50
for start in range(0, x.shape[1] - window, window):
    out, h = rnn(x[:, start:start + window], h)
    loss = nn.functional.mse_loss(head(out), y[:, start:start + window])
    opt.zero_grad()
    loss.backward()                       # gradients flow back at most `window` steps
    nn.utils.clip_grad_norm_(params, 1.0)
    opt.step()
    h = h.detach()                        # keep the state, cut the graph
print("final window MSE:", round(loss.item(), 4), "(noise variance is 0.01)")
```

After a single pass over the stream, the final-window mean squared error is about 0.014, close to the 0.01 floor set by the added noise. Forgetting `h.detach()` makes the second window try to backpropagate into a graph that was already freed, and PyTorch raises an error; resetting `h = None` at each window instead would throw the context away.

Vanishing and exploding gradients

Plain version. The gradient that travels from step t back to step s is multiplied by roughly the same matrix at every step. Repeated multiplication either shrinks it toward zero or blows it up, so plain RNNs struggle to learn dependencies more than a few dozen steps long.

Precise version. Chaining the one-step Jacobians gives

$$\frac{\partial h_t}{\partial h_s} = \prod_{k=s+1}^{t} \frac{\partial h_k}{\partial h_{k-1}} = \prod_{k=s+1}^{t} \operatorname{diag}(1 - h_k^2)\, W_h,$$

an ordered product of $t - s$ matrices (matrices do not commute, so the order matters). Because $|\tanh'| \le 1$,

$$\left\| \frac{\partial h_t}{\partial h_s} \right\| \le \left(\sigma_{\max}(W_h)\right)^{t-s},$$

where $\sigma_{\max}$ is the largest singular value. If $\sigma_{\max}(W_h) < 1$, the gradient from distant steps **vanishes** exponentially. If W_h has large singular values and the units are not saturated, the gradient can **explode**. Bengio, Simard and Frasconi [6] showed in 1994 that this is not a bug of a particular training method: a system that robustly stores information in the presence of noise must be in a regime where gradients of long-range dependencies shrink. Pascanu, Mikolov and Bengio [7] revisited the analysis with dynamical-systems tools and gave a practical remedy for explosions.

The intuition is clearest for a scalar linear RNN, $h_t = \lambda h_{t-1} + x_t$: then $\partial h_t / \partial h_s = \lambda^{t-s}$, which is $0.9^{80} \approx 2 \times 10^{-4}$ for $\lambda = 0.9$ and $1.1^{80} \approx 2 \times 10^{3}$ for $\lambda = 1.1$. Figure 2 shows the same behavior measured in real networks.

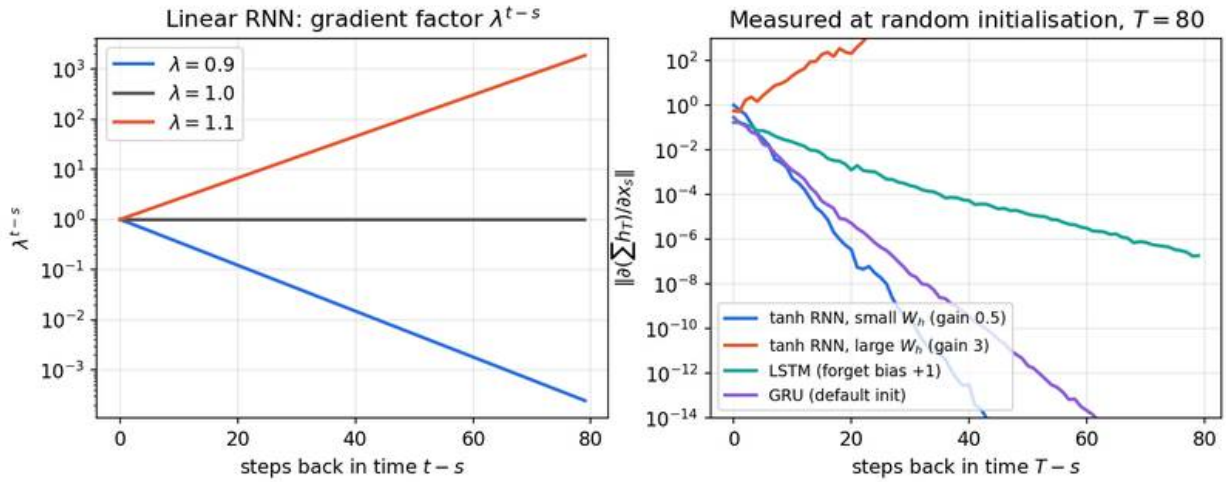


Figure 2 — Left: the scalar factor λ^(t−s). Right: gradient of the final state with respect to each input, measured at random initialization ($T = 80$). The plain tanh RNN vanishes or explodes; the LSTM's additive memory path decays far more slowly.

Remedies. Explosions are handled by **gradient clipping** [7]: if the gradient norm $\|g\|$ exceeds a threshold τ, rescale it to $g \leftarrow \tau\, g / \|g\|$. This keeps the update direction and caps its length. PyTorch's `clip_grad_norm_` does exactly this. Vanishing gradients are harder: they require a different architecture, one whose state update is close to the identity so that gradients can pass through many steps. That architecture is the LSTM.

Long short-term memory (LSTM)

Plain version. Give the network a separate memory lane, the **cell state**, that is updated by addition rather than by repeated matrix multiplication, and let learned **gates** decide what to erase, what to write and what to reveal.

Precise version. With $[x_t, h_{t-1}]$ denoting concatenation and σ the logistic sigmoid, a standard LSTM step is

$$
\begin{aligned}
f_t &= \sigma\big(W_f[x_t, h_{t-1}] + b_f\big) && \text{forget gate} \\
i_t &= \sigma(W_i[x_t, h_{t-1}] + b_i) && \text{input gate} \\
o_t &= \sigma(W_o[x_t, h_{t-1}] + b_o) && \text{output gate} \\
\tilde{c}_t &= \tanh(W_c[x_t, h_{t-1}] + b_c) && \text{candidate memory} \\
c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t && \text{cell update} \\
h_t &= o_t \odot \tanh(c_t) && \text{exposed state}
\end{aligned}
$$

Each gate is a vector of numbers between 0 and 1, and $\odot$ is element-wise multiplication, so each gate acts like a bank of dimmer switches, one per memory unit. The forget gate f_t decides how much of the old memory to keep, the input gate i_t how much of the candidate $\tilde{c}_t$ to write, and the output gate o_t how much of the memory to reveal as h_t.

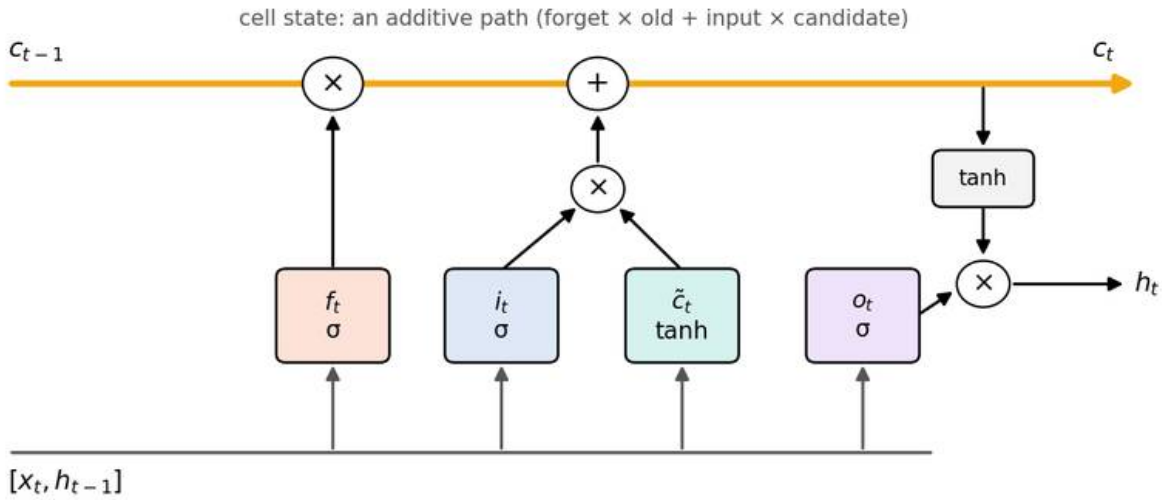


Figure 3 — Inside an LSTM cell. The cell state (gold) changes only by an element-wise scaling and an addition, which gives gradients a protected path through time.

Why it works. Along the cell-state path, $\partial c_t / \partial c_{t-1} = \mathrm{diag}(f_t)$ (ignoring the smaller indirect paths through the gates). There is no recurring weight matrix and no squashing nonlinearity on this path. If the forget gate stays near one, the gradient passes through many steps almost unchanged.

That is why the LSTM curve in Figure 2 decays so slowly. It is also why a common initialization sets the forget-gate bias to a positive value such as +1, so the network starts out remembering. This is not a guarantee of unlimited memory: the forget gate can and does learn to close, and long-range credit assignment still gets harder with distance.

History and attribution. Hochreiter and Schmidhuber introduced the LSTM in 1997 [2] with a constant-error "carousel": the cell state passed on unchanged and there was **no forget gate**. Gers, Schmidhuber and Cummins added the forget gate in 2000 [5] so that the cell could learn to reset its memory, for example between independent segments of a continuous stream. The equations above, with the forget gate, are the version used in practice and in every major framework. Greff et al. [19] compared eight LSTM variants on speech, handwriting and music tasks and found that none significantly beat the standard architecture, while the forget gate and the output activation were its most critical components.

Parameters. Each of the four gate blocks has an $m \times (d + m)$ weight matrix and a bias, so an LSTM layer has about $4m(d + m + 1)$ parameters, four times a simple RNN with the same state size. (PyTorch keeps separate input and recurrent biases, giving $4m(d + m + 2)$.)

Code: an LSTM step from scratch, checked against PyTorch

PyTorch stores the four gate blocks stacked in the order input, forget, cell candidate (g), output. Implementing the step by hand and comparing with `nn.LSTMCell` is the best way to be sure you read the equations correctly:

```
import torch
from torch import nn

torch.manual_seed(0)
d, m = 4, 6
cell = nn.LSTMCell(d, m)                # PyTorch stacks gate weights in order i, f, g, o

def lstm_step(x, h, c):
    z = x @ cell.weight_ih.T + cell.bias_ih + h @ cell.weight_hh.T + cell.bias_hh
    i, f, g, o = z.chunk(4, dim=-1)
    i, f, o = torch.sigmoid(i), torch.sigmoid(f), torch.sigmoid(o)
    g = torch.tanh(g)                   # candidate memory, often written c~_t
    c = f * c + i * g                   # additive cell update
    h = o * torch.tanh(c)
    return h, c

x = torch.randn(2, d)
h0, c0 = torch.randn(2, m), torch.randn(2, m)
h_ref, c_ref = cell(x, (h0, c0))
h_ours, c_ours = lstm_step(x, h0, c0)
print("matches nn.LSTMCell:", torch.allclose(h_ours, h_ref, atol=1e-6),
      torch.allclose(c_ours, c_ref, atol=1e-6))
```

```
print("parameters:", sum(p.numel() for p in cell.parameters()), "= 4 * m * (d + m + 2) =",
      4 * m * (d + m + 2))
```

It prints `True True` and `288 = ⋯ = 288`.

Gated recurrent unit (GRU)

Plain version. A GRU is a slimmer LSTM: it merges the cell and hidden states into one, and uses two gates instead of three.

Precise version. Following Cho et al. [3] and PyTorch's `nn.GRU`,

$$
\begin{aligned}
r_t &= \sigma(W_r x_t + U_r h_{t-1} + b_r) && \text{reset gate} \\
z_t &= \sigma(W_z x_t + U_z h_{t-1} + b_z) && \text{update gate} \\
\tilde{h}_t &= \tanh(W x_t + r_t \odot (U h_{t-1}) + b) && \text{candidate} \\
h_t &= z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t && \text{interpolation}
\end{aligned}
$$

The reset gate r_t decides how much of the old state to use when forming the candidate; the update gate z_t interpolates between keeping the old state ($z_t \approx 1$) and replacing it ($z_t \approx 0$). The interpolation plays the role of the LSTM's additive cell path. Two caveats are worth knowing. Some papers and textbooks swap the roles of z_t and $1 - z_t$; the models are equivalent, but the gate means the opposite. And the original paper applies the reset gate before the recurrent matrix ($U(r_t \odot h_{t-1})$), whereas PyTorch and cuDNN apply it after, as written above. Always check the framework's documentation before reading meaning into a gate.

GRUs have three gate blocks instead of four, so about 25% fewer parameters than an LSTM of the same width. Neither consistently wins across tasks; the GRU is a reasonable default when data or compute is limited.

Code: a recurrent classifier that generalizes to fresh sequences

This complete CPU example learns to classify the sign of a sequence's sum, which requires accumulating information over all twelve steps. A fresh test set checks that the model did not merely memorize its training sequences.

```
import torch
from torch import nn
from torch.nn import functional as F

torch.manual_seed(0)
torch.set_num_threads(1)
rnn = nn.GRU(input_size=1, hidden_size=24, batch_first=True)
head = nn.Linear(24, 2)
```

```
opt = torch.optim.Adam(list(rnn.parameters())+list(head.parameters()), lr=.003)
def batch(n):
    x = torch.randn(n, 12, 1)
    y = (x.sum(dim=(1, 2)) > 0).long()
    return x, y
for step in range(400):
    x, y = batch(64)
    _, h = rnn(x)
    loss = F.cross_entropy(head(h[-1]), y)
    opt.zero_grad(); loss.backward()
    nn.utils.clip_grad_norm_(list(rnn.parameters())+list(head.parameters()), 1.)
    opt.step()
rnn.eval(); head.eval()
with torch.no_grad():
    x, y = batch(512)
    _, h = rnn(x)
    acc = (head(h[-1]).argmax(1) == y).float().mean()
    print("fresh-sequence accuracy:", acc.item())
```

It reaches about 99% accuracy on fresh length-12 sequences. The returned hidden tensor has shape `layers × batch × hidden` even with `batch_first=True`, which affects only the input and output sequence layout. Testing on longer sequences asks a different question, length generalization, and a model can succeed at length 12 and fail at length 100 (Exercise 2).

Deep, bidirectional and streaming RNNs

Stacking. A **deep RNN** feeds the sequence of hidden states of one recurrent layer into another (`num_layers` in PyTorch). Lower layers can capture short-range patterns and higher layers longer ones. Two to four layers are typical; deeper stacks usually need residual connections between layers.

Bidirectional. Schuster and Paliwal [10] proposed running one RNN forward in time and another backward, then concatenating their states, $h_t = [\vec{h}_t; \overleftarrow{h}_t]$. Each position then summarizes both its past and its future. This is ideal for **offline** tasks where the whole input is available (tagging a recorded video, encoding a sentence), but it is **not causal**: it cannot be used for a live prediction that must be made before future frames arrive. Reporting offline bidirectional accuracy for an online application overstates what the deployed system can know.

Variable lengths. Batches contain sequences of different lengths, padded to a common length. Without care, the padding leaks into states and losses. Use per-sequence lengths (`pack_padded_sequence` in PyTorch) or masks so that padded steps neither update the state nor contribute to the loss.

Streaming. For a live stream, keep the state between calls and reset it only between indepen-

dent streams. Accidentally carrying state from one user, document or video into the next leaks information and produces irreproducible results.

Encoder–decoder models and teacher forcing

Plain version. To turn one sequence into another of a different length (a sentence into its translation, a video into a description), use one RNN to read the input into a summary and a second RNN to write the output from that summary, one token at a time.

Precise version. In the **sequence-to-sequence** (seq2seq) model of Sutskever, Vinyals and Le [11] and the closely related encoder–decoder of Cho et al. [3], an **encoder** RNN reads $x_1, \dots, x_T$ and its final state becomes a context vector $c = h_T$. A **decoder** RNN, initialized from c, models

$$p(y_1, \dots, y_U \mid x) = \prod_{u=1}^{U} p(y_u \mid y_{<u}, c), \qquad s_u = \mathrm{RNN}(s_{u-1}, y_{u-1}),$$

where s_u is the decoder state and $p(y_u \mid \cdot)$ is a softmax over the vocabulary. Generation stops when the decoder emits an end-of-sequence token, so input and output lengths are independent.

Teacher forcing. During training, the decoder input at step u is the **ground-truth** previous token y_{u-1}, not the model's own prediction. This makes all steps trainable in parallel against the known target and greatly stabilizes learning. At test time, however, the model must consume its own predictions, which may contain errors it never saw during training. This train–test mismatch is called **exposure bias**. Scheduled sampling [25] gradually replaces ground-truth inputs with model samples during training to reduce it. Decoding usually uses greedy or beam search rather than sampling.

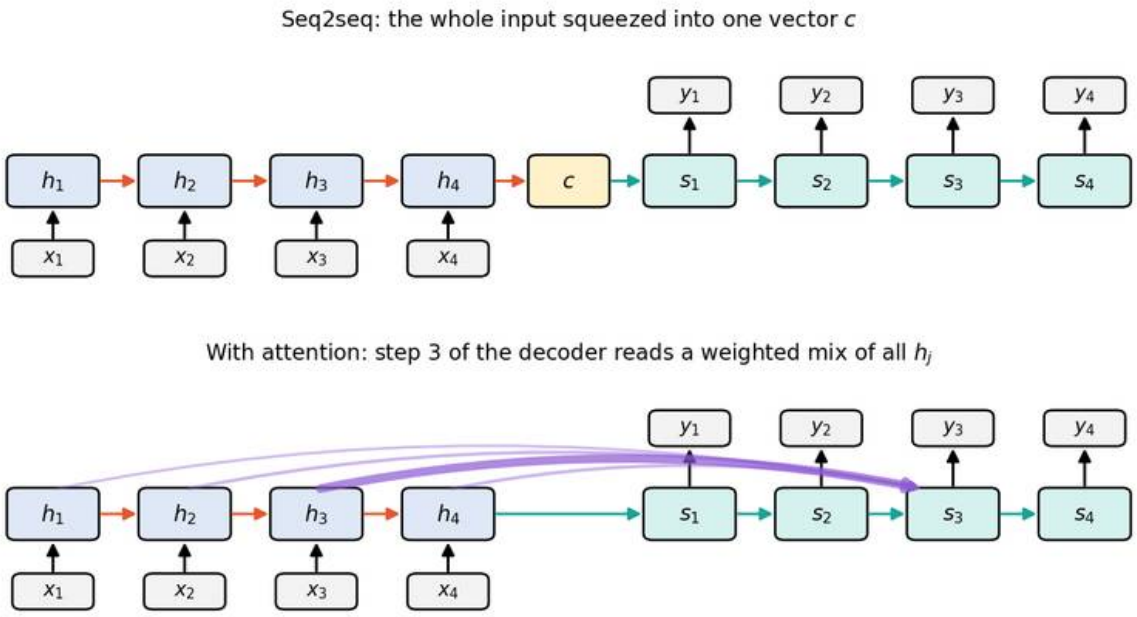


Figure 4 — Top: plain seq2seq squeezes the entire input into one vector c. Bottom: with attention, every decoder step reads a learned, weighted mix of all encoder states.

The weakness is visible in Figure 4 (top): everything the decoder knows about the input must pass through one fixed-size vector c, no matter how long the input. Translation quality of plain seq2seq models dropped on long sentences. Sutskever et al. found that simply **reversing the source sentence** helped, because it shortened the distance between corresponding words, a hint that the bottleneck and long paths were the problem [11].

Attention: the bridge to Transformers

Plain version. Instead of forcing the decoder to rely on one summary vector, let it look back at **all** encoder states at every step and decide which ones are relevant right now.

Precise version. Bahdanau, Cho and Bengio [12] kept every encoder state $h_1, \dots, h_T$ (from a bidirectional encoder) and, at each decoder step u, computed

$$e_{u,j} = v^\top \tanh\big(W s_{u-1} + U h_j\big), \qquad \alpha_{u,j} = \frac{\exp(e_{u,j})}{\sum_{k=1}^{T} \exp(e_{u,k})}, \qquad c_u = \sum_{j=1}^{T} \alpha_{u,j}\, h_j .$$

$e_{u,j}$ is a learned **alignment score** between the decoder's current need (s_{u-1}) and encoder position j; W, U and v are learned parameters. The softmax turns scores into weights $\alpha_{u,j}$ that are positive and sum to one, and the **context vector** c_u is their weighted average of encoder states. The decoder

then uses c_u together with its own state. This form is called **additive attention**. Because c_u changes at every step and has direct access to every input position, the path length between any input and any output becomes one step, regardless of the sequence length.

```
import torch
from torch import nn

class AdditiveAttention(nn.Module):
    """Bahdanau-style scores e_j = v^T tanh(W s + U h_j), then softmax over j."""
    def __init__(self, d_dec, d_enc, d_att):
        super().__init__()
        self.W = nn.Linear(d_dec, d_att, bias=False)
        self.U = nn.Linear(d_enc, d_att, bias=False)
        self.v = nn.Linear(d_att, 1, bias=False)

    def forward(self, s, H, mask=None):
        # s: B x d_dec (decoder state), H: B x T x d_enc (all encoder states)
        e = self.v(torch.tanh(self.W(s)[:, None] + self.U(H))).squeeze(-1)   # B x T
        if mask is not None:                        # True where the input is padding
            e = e.masked_fill(mask, float("-inf"))
        alpha = e.softmax(dim=-1)                   # attention weights, rows sum to 1
        context = (alpha[:, :, None] * H).sum(1)    # B x d_enc weighted average
        return context, alpha

torch.manual_seed(0)
att = AdditiveAttention(d_dec=16, d_enc=32, d_att=8)
H = torch.randn(2, 5, 32)
s = torch.randn(2, 16)
pad = torch.tensor([[False] * 5, [False, False, False, True, True]])   # 2nd sequence has
 ↪  length 3
ctx, alpha = att(s, H, pad)
print("context:", tuple(ctx.shape), "| weights sum to 1:",
      torch.allclose(alpha.sum(-1), torch.ones(2)))
print("weights on padding:", alpha[1, 3:].tolist())
```

It prints a context of shape `(2, 32)`, confirms that weights sum to one, and shows exactly zero weight on padded positions: masking with $-\infty$ before the softmax is the standard way to make attention ignore padding (and, in ML 04, the future).

Attention weights are also interpretable as a soft **alignment**. Figure 5 shows a small GRU encoder–decoder with this attention trained for under a minute on CPU to reverse random digit sequences (about 96% per-token accuracy with teacher forcing on fresh sequences). Without being told, it learned to look at input position $T + 1 - u$ when writing output u: the anti-diagonal.

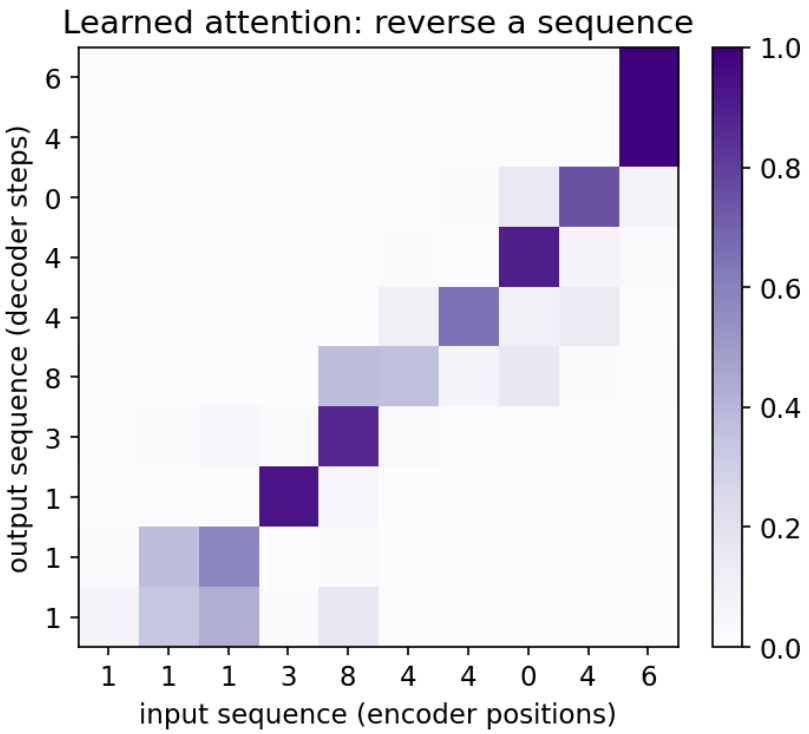


Figure 5 — Attention weights of a toy encoder–decoder trained to reverse a sequence (generated by the figure script). Each output step concentrates on the mirrored input position.

From here to Transformers. Attention fixed the bottleneck, but the encoder and decoder were still recurrent, so computation over a sequence of length T required T sequential steps that cannot be parallelized during training. Vaswani et al. [28] asked what happens if recurrence is removed entirely and **attention is used inside the encoder and decoder as well** (self-attention), with positional encodings to restore order. The answer, the Transformer, trains in parallel across positions and became the dominant architecture for language and, with ViT, for vision. ML 04 picks up the story with scaled dot-product attention, which replaces the additive score with $q^\top k / \sqrt{d_k}$.

Recurrent networks in vision

Image captioning. Show and Tell [13] (Vinyals et al.) is a direct image-to-sequence model: a CNN (ML 03) encodes the image into a vector that initializes an LSTM language model, which then generates the caption word by word, trained with teacher forcing on image–caption pairs. Show, Attend and Tell [14] (Xu et al.) kept the CNN's spatial feature grid instead of a single vector and let the LSTM attend over image regions at each word, producing attention maps that often highlight the object being named. This was one of the first widely seen demonstrations of visual attention.

Video understanding. Long-term Recurrent Convolutional Networks (LRCN) [15] (Donahue et al.) apply a CNN to each frame and an LSTM across frames, trained end to end, for activity recognition, image description and video description. The CNN handles space, the RNN handles time: a common pattern for any per-frame feature sequence, such as pose tracks or per-frame detections in object tracking.

Spatiotemporal forecasting. When the state itself should be an image, fully connected gates waste the spatial structure. The **ConvLSTM** of Shi et al. [16] replaces the matrix products in the LSTM gates with convolutions, $i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + \dots)$, so inputs, states and gates are all 3-D tensors (channels × height × width). It was introduced for precipitation nowcasting from radar maps and is widely used for video prediction and other forecasting on grids.

What changed. For captioning and video classification at scale, Transformers and vision–language models (see ML 09) have largely replaced LSTMs, because they train in parallel and scale better with data. Recurrent and convolutional-recurrent models remain common where frames arrive one at a time and memory must stay constant: online video processing, tracking, robotics and low-power devices.

Recursive networks: composing over trees

Plain version. A recursive network applies one shared combination rule to merge two child representations into a parent, following a tree, until it reaches the root.

Precise version. For a binary tree whose leaves carry vectors (word embeddings, region features), each internal node computes

$$h_{\text{parent}} = \tanh\!\left(W\,[h_{\text{left}}; h_{\text{right}}] + b\right),$$

with the **same** $W \in \mathbb{R}^{D \times 2D}$ and b at every node. The root vector represents the whole structure and can be classified; every node can also have its own label. Training uses **backpropagation through structure**: the same idea as BPTT, but along the tree's edges. A chain of length n is the special case of a completely unbalanced tree; a balanced tree reduces the path from any leaf to the root from $n-1$ to about $\log_2 n$ steps (Figure 6), which shortens the route for gradients.

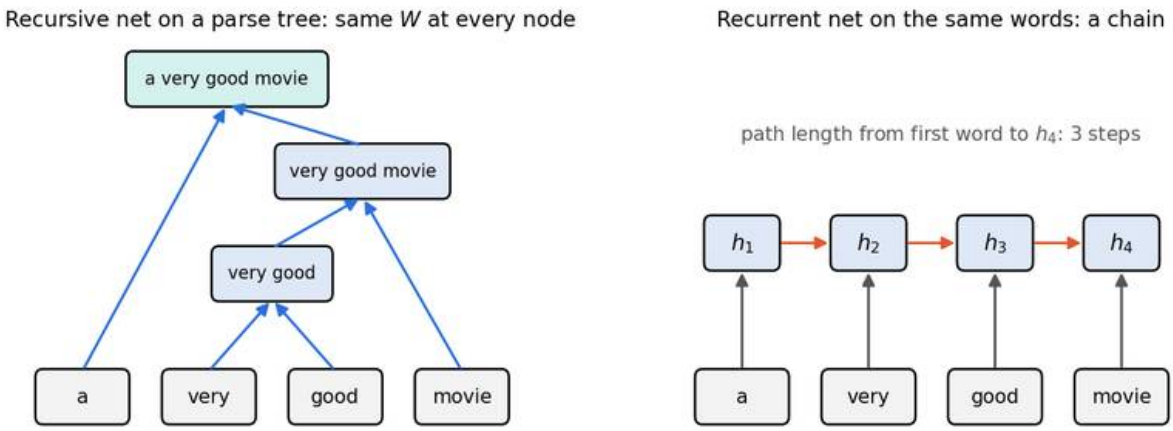


Figure 6 — A recursive network follows a tree (left); a recurrent network follows a chain (right). Both share weights, but the tree mirrors the structure of the input.

Socher et al. [18] used recursive networks in a vision setting first: they segmented an image into superpixels, then learned both to merge adjacent regions into a scene parse tree and to compute region representations, using the same method to parse sentences. Their later Recursive Neural Tensor Network and Stanford Sentiment Treebank [17] showed that composing along parse trees captures phenomena such as negation ("not very good"). Tai, Socher and Manning's **Tree-LSTM** [4] added LSTM-style gates to tree composition, with one forget gate per child, so a parent can selectively keep information from each subtree.

```
import torch
from torch import nn

torch.manual_seed(0)
D = 8
vocab = {"a": 0, "very": 1, "good": 2, "movie": 3}
emb = nn.Embedding(len(vocab), D)
compose = nn.Linear(2 * D, D)            # ONE weight matrix shared by every internal node
classify = nn.Linear(D, 2)               # e.g. negative / positive sentiment

def encode(tree):
    """tree is a word (leaf) or a pair (left, right)."""
    if isinstance(tree, str):
        return emb(torch.tensor(vocab[tree]))
    left, right = tree
    return torch.tanh(compose(torch.cat([encode(left), encode(right)])))

tree = ("a", (("very", "good"), "movie"))          # (a ((very good) movie))
root = encode(tree)
loss = nn.functional.cross_entropy(classify(root)[None], torch.tensor([1]))
loss.backward()                                     # backpropagation through structure
print("root vector:", tuple(root.shape), "| loss:", round(loss.item(), 4))
print("gradient reached the shared W:", compose.weight.grad.abs().sum().item() > 0)
swapped = encode((("movie", ("good", "very")), "a"))
```

```
print("swapping children changes the root:", not torch.allclose(root, swapped))
```

The recursion is ordinary Python, and autograd builds the tree-shaped graph on the fly. Swapping children changes the result because concatenation gives the left and right slots different roles.

The main practical limitation is the tree itself: someone has to provide it (a parser, a segmentation hierarchy), and batching variable trees efficiently is awkward. Learning structure and representation jointly is possible but harder. In current practice, self-attention can learn soft, input-dependent groupings without an explicit tree, which is one reason recursive networks are now uncommon; the idea of hierarchical composition lives on in hierarchical vision backbones and scene graphs.

Mamba: selective state spaces and scan

A useful memory should hold a relevant observation through distractions and overwrite it when the situation changes. Mamba [21] makes the memory update depend on the current input, while keeping that update affine in the previous state. This distinction lets us retain a recurrent interpretation and compute whole sequences with a parallel prefix scan.

From continuous dynamics to selective updates

Start with a small continuous-time memory. A stable transition lets stored information decay; an input projection writes new information; an output projection reads it:

$$\frac{dh(\tau)}{d\tau} = Ah(\tau) + Bu(\tau), \qquad y(\tau) = Ch(\tau).$$

Here τ is continuous time, $u(\tau) \in \mathbb{R}$ is one input channel, $h(\tau) \in \mathbb{R}^N$ is its N-dimensional state, $A \in \mathbb{R}^{N\times N}$ is the transition matrix, $B \in \mathbb{R}^{N\times 1}$ writes the input, $C \in \mathbb{R}^{1\times N}$ reads the state, and $y(\tau) \in \mathbb{R}$ is the output. A full neural block mixes multiple channels with learned projections and includes additional local convolution and gating [21]. This equation isolates its memory mechanism.

Hold the input and its write matrix constant during each interval. Exact zero-order-hold discretization gives

$$\bar{A}_t = \exp(\Delta_t A), \qquad \bar{B}_t = \left(\int_0^{\Delta_t} \exp(sA)\, ds\right) B_t,$$
$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t u_t, \qquad y_t = C_t h_t.$$

Here t indexes discrete tokens, $\Delta_t > 0$ is the interval length, s is an integration variable, u_t is the held input, B_t, C_t are the current write/read matrices, and bars mark discrete coefficients. The integral remains valid when A is singular; an inverse-based formula needs additional assumptions. A diagonal A makes these operations elementwise.

Selectivity means predicting Δ_t, B_t, C_t from the current token. For example, a learned affine projection followed by softplus$(z) = \log(1 + e^z)$ supplies a positive interval. The underlying learned A can remain fixed: $\bar{A}_t$ nevertheless changes through Δ_t. With negative diagonal entries, a larger interval reduces the fraction of old state retained. Selective writes and reads also determine which content enters and leaves memory. The resulting map is nonlinear in its inputs even though each state update is affine. Figure 7 isolates the retention effect with synthetic coefficients; it does not measure a trained model.

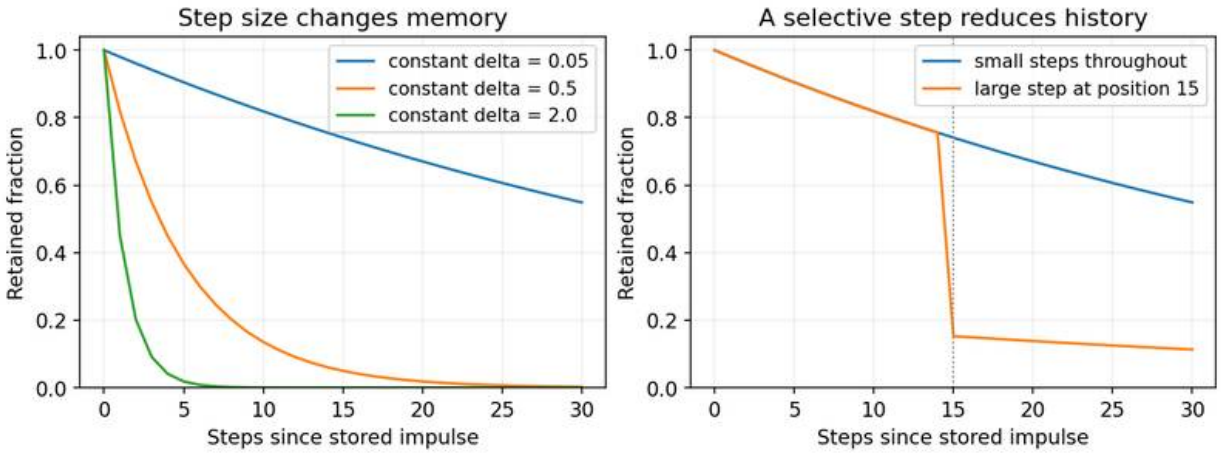


Figure 7 — For a scalar transition $A = -0.4$, retained history is the product of exp$(-0.4\ delta)$. A large selected interval sharply reduces old memory; small intervals preserve it longer.

Why this recurrence can be scanned

Once input-dependent coefficients are computed, composing two affine updates requires only another affine update:

$$(a_2, b_2) \circ (a_1, b_1) = (a_2 a_1,\ a_2 b_1 + b_2).$$

Here pair (a_i, b_i) represents $h \mapsto a_i h + b_i$, $i = 1, 2$ identifies consecutive updates, and $\circ$ means applying the first update and then the second. Composition is associative; for a diagonal transition, products are elementwise. A prefix scan can therefore regroup the operations into a tree while preserving their chronological effect. Ordinary LSTM/GRU gates depend on the previous hidden state, so their coefficients cannot generally be prepared this way before the state is known.

The following standalone CPU example uses exact diagonal discretization and hand-chosen selective coefficients. It checks causality and state continuity across chunks. It is a teaching recurrence, not a pretrained Mamba implementation or a speed benchmark.

```
import numpy as np

rng = np.random.default_rng(44)
u = rng.normal(size=24)
```

```
A = -np.array([0.1, 0.8])  # two stable diagonal entries

def selective_scan(values, state=None):
    h = np.zeros(2) if state is None else state.copy()
    outputs = []
    for x in values:
        delta = np.logaddexp(0.0, 0.5 * x)
        B = np.array([1.0, np.tanh(x)])
        C = np.array([1.0, 0.5 + 0.1 * np.tanh(x)])
        a = np.exp(delta * A)
        b = np.expm1(delta * A) / A * B * x
        h = a * h + b
        outputs.append(C @ h)
    return np.array(outputs), h

y, final_state = selective_scan(u)
changed = u.copy()
changed[12:] += 10.0
changed_y, _ = selective_scan(changed)
assert np.allclose(y[:12], changed_y[:12])
left, boundary = selective_scan(u[:12])
right, resumed_state = selective_scan(u[12:], boundary)
assert np.allclose(y, np.r_[left, right])
assert np.allclose(final_state, resumed_state)
reset_right, _ = selective_scan(u[12:])
assert not np.allclose(right, reset_right)
print('causal prefix and carried-state chunk checks passed')
```

The output confirms both checks. Resetting the state at a batch boundary changes the computation. Conversely, sharing state between independent examples leaks information. A streaming implementation must track which sequence owns each state and reset it at genuine sequence boundaries.

Mamba-2 and the attention connection

Mamba-2 [29] changes both the inner sequence mixer and the surrounding block. Its structured state-space duality uses a scalar-times-identity transition within each head and organizes computation into chunks whose dense matrix multiplications suit accelerator hardware. It also restructures parameter projections and normalization. This is more specific than adding attention to the original Mamba block.

With a zero initial state, the scalar-transition recurrence can be expanded as

$$y_t = \sum_{j=1}^{t} \left[C_t \left(\prod_{k=j+1}^{t} a_k \right) B_j \right] u_j.$$

Here j indexes earlier inputs, k indexes intervening transitions, a_k is the scalar discrete transition multiplying the identity, B_j is the discrete write column, C_t is the read row, and an empty product equals one. The bracketed quantity is a structured causal mixing coefficient. This exposes an

attention-like matrix view without requiring ordinary softmax attention or storing that matrix. Hu et al. [30] (ICML 2026) extend the duality to diagonal transitions and clarify why it does not extend directly to standard softmax attention.

At fixed width and state dimension, recurrent inference carries a fixed-size state, whereas ordinary cached causal attention retains keys and values whose storage grows with context length. Compression saves memory but puts pressure on exact recall of many unrelated details. Linear sequence scaling also does not imply constant training memory: activations, gradients and recomputation policies matter. Evaluate retrieval quality, context length and measured device latency together. The image-specific ordering problem appears next in ML 04.

Modern view

Survey literature. Goodfellow et al.'s chapter 10 [1] remains the clearest unified treatment of recurrent and recursive networks, echo-state ideas, and long-term dependencies. Lipton, Berkowitz and Elkan [20] wrote a critical review of RNNs for sequence learning in 2015: a self-contained, historically ordered account of three decades of work leading to LSTMs and bidirectional RNNs, with conflicting notation reconciled and applications to captioning, translation and handwriting recognition. Greff et al. [19] is less a survey than a large empirical study, but it is the reference for which LSTM components matter. Two recent reviews by Tiezzi et al. reframe recurrence for the Transformer era: an extensive survey [26] places Transformer–RNN hybrids, attention variants with recurrent form, and deep state-space models (a special class of linear RNNs) under the single umbrella of recurrence, and highlights research opportunities in learning online from potentially infinite streams rather than fixed-length sequences; and a shorter perspective in *Nature Machine Intelligence* [27] revisits recurrent processing at the crossroad of Transformers and state-space models.

Recurrence returns (2023–2026). Transformers process all pairs of positions, which costs memory and computation quadratic in length. A family of models revisits recurrence to get linear-time inference with a constant-size state:

- **Linear recurrent units.** Orvieto et al. [24] (ICML 2023) showed that a carefully parameterized **linear** recurrence with diagonal complex-valued state matrices, stable initialization and normalization matches deep state-space models on long-range benchmarks, and that the nonlinearity can live outside the recurrence.
- **Selective state spaces.** Mamba [21] makes the state-space parameters functions of the input, so the model can choose what to store or ignore, as LSTM gates do, while a hardware-aware parallel scan keeps training efficient. Vision variants and the debate about whether they are necessary for images are covered in ML 04.
- **Revived LSTMs and GRUs.** xLSTM [22] (NeurIPS 2024) adds exponential gating and a matrix-valued memory to the LSTM and scales it to language-model sizes. Feng et al. [23] (2024) remove the hidden-state dependence from the gates of LSTMs and GRUs ("minLSTM", "min-

GRU"), which allows a parallel scan during training, and report results competitive with recent sequence models.

The shared lesson is that the **gated, additive state update** invented for the LSTM was right; what limited classical RNNs was mostly sequential training. Making the recurrence linear in the state (so it can be computed with a parallel scan) recovers parallel training while keeping cheap, streaming inference.

Open problems. Length generalization beyond the training horizon; credit assignment over very long streams without full BPTT; how much a fixed-size state can remember compared with attention's growing cache, a real limitation for tasks that need exact recall of distant details; and choosing a scan order for 2-D images and 3-D video, where no single natural sequence order exists.

Key takeaways

- An RNN applies one shared update $h_t = f(x_t, h_{t-1})$ at every step, so it handles any sequence length with a fixed number of parameters. Unrolled, it is a deep network with tied weights.
- BPTT is ordinary backpropagation on the unrolled graph; gradients of shared weights are summed over time. Truncated BPTT carries state forward but limits how far gradients flow back.
- Long-range gradients are products of Jacobians and tend to vanish or explode. Clipping fixes explosions; architectures with near-identity, additive state paths fix vanishing.
- LSTM gates (forget, input, output) and GRU gates (reset, update) control a protected memory path. The forget gate was a 2000 addition to the 1997 LSTM. Check the framework's gate conventions.
- Bidirectional encoders use future context and are not causal; streaming models must manage state carefully.
- Seq2seq maps sequences to sequences with teacher forcing; attention removed its single-vector bottleneck, and self-attention without recurrence became the Transformer.
- Recursive networks compose along trees with one shared rule. In vision, recurrence powered captioning, video models and ConvLSTM forecasting, and gated linear recurrences are back as state-space models.
- Mamba predicts write, read and interval coefficients from inputs while keeping updates affine in state; Mamba-2 uses structured duality for efficient chunk computation. Compression saves cache memory but limits exact recall.

Exercises

1. **Causality.** Your team reports 92% frame-level accuracy for an action-recognition model with a bidirectional LSTM and proposes deploying it for live alerts. What is wrong with this plan, and how would you evaluate the deployable alternative fairly? Run the selective-scan example, then add a backward scan: which causality check should fail?

   Hint

   The backward LSTM at frame t has read frames after t, which a live system has not seen yet. Retrain a unidirectional (causal) model, or allow a fixed look-ahead delay and evaluate with exactly that delay. Report accuracy as a function of latency. A backward scan reads the altered future, so the combined prefix generally changes even when the forward prefix does not.

2. **Length generalization.** Train the GRU sum classifier at length 12, then evaluate on fresh sequences of length 6, 24, 100 and 400 without retraining. Plot accuracy versus length. Then repeat with an LSTM and with training on mixed lengths 6–24.

   Hint

   Generate fresh data per length. The sum's scale grows like $\sqrt{T}$, so the state may saturate on long inputs even if the logic is right. Mixed-length training usually helps more than switching cell type. Success at length 12 says nothing by itself about length 400.

3. **Gradient bound.** For the scalar RNN $h_t = \tanh(wh_{t-1} + x_t)$, show that $|\partial h_t / \partial h_s| \leq |w|^{t-s}$. For what values of w can the gradient explode, and why does saturation of tanh work against explosion?

   Hint

   Each factor is $w(1 - h_k^2)$ with $0 \leq 1 - h_k^2 \leq 1$. Explosion requires $|w| > 1$, but a large $|w|$ also drives h_k toward ± 1, which makes $1 - h_k^2$ small. The gradient explodes only on trajectories that stay in the non-saturated region, which is why explosions are bursty in practice.

4. **LSTM memory.** In the LSTM equations, suppose $f_t = 1$ and $i_t = 0$ for all t from s to t. What is c_t in terms of c_s, and what is $\partial c_t / \partial c_s$ along the cell path? What would happen if the forget gate were fixed at 1 forever, as in the original 1997 design, on a never-ending input stream?

   Hint

   $c_t = c_s$ and $\partial c_t / \partial c_s = I$: perfect memory and perfect gradient flow. With $f_t \equiv 1$ and nonzero writes, the cell state accumulates without bound over an endless stream, which is exactly the problem Gers et al. solved by letting the network learn to forget.

5. **Attention shapes and masks.** In the additive-attention code, a batch has encoder states of shape $8 \times 40 \times 256$ (with some sequences padded) and a decoder state of shape 8×512. Give the shapes of e, α and the context, and explain what would go wrong if padded positions were masked with 0 instead of $-\infty$.

   Hint

   e and α are 8×40; the context is 8×256. A score of 0 is an ordinary score: $e^0 = 1$ would still give padding positive weight, and the context would mix in meaningless padded states.

6. **Recursive versus recurrent.** For a balanced binary tree over $n = 1024$ leaves, how many composition steps lie between a leaf and the root? Compare with a left-to-right RNN over the same 1024 inputs, and relate the answer to the vanishing-gradient bound.

   Hint

   $\log_2 1024 = 10$ steps versus up to 1023 for the chain. The gradient bound decays like $(\text{factor})^{\text{path length}}$, so a shorter path from each input to the output helps optimization. Attention takes this to the extreme: path length one.

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ML 04 • Attention and Vision Transformers

Build self-attention from queries, keys and values; add multiple heads, positional encodings and the Transformer block; turn images into patch tokens with ViT; then study DeiT, Swin, hybrid CNN–Transformer models, efficient attention and state-space alternatives.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-04-vision-transformers/

What you'll learn

- How self-attention computes an input-dependent weighted average, why queries, keys and values play different roles, and why the scores are divided by $\sqrt{d_k}$.
- How multi-head attention, causal masks and cross-attention are built, with code checked against PyTorch.
- Why attention needs positional information, and how sinusoidal, learned, relative and rotary encodings provide it.
- How the Transformer block combines attention, an MLP, LayerNorm and residual connections.
- How the Vision Transformer (ViT) turns an image into patch tokens, why it needs so much data, and how DeiT, Swin and hybrid CNN–Transformer designs respond.
- What makes attention expensive, what FlashAttention, windows and token merging change, and how Mamba and Mamba-2 compare with attention, and how image scan ordering affects global context and directional bias.

The big picture

A CNN (ML 03) moves information through fixed local neighborhoods: a pixel learns about a distant pixel only after many layers. **Attention** lets every location decide, based on its current content, which other locations to read from, in a single step. To recognize a bird, a patch containing its head can directly consult a distant patch containing its wing.

> **Explain like I'm five**
>
> Imagine a classroom where each child holds one piece of a jigsaw puzzle. Each child asks a question about their piece ("I have a blue edge, who has sky?"), and every other child holds up a card describing their own piece. Each child listens most closely to the classmates whose cards best match the question, and updates their idea of what their piece shows. Then everyone does it again with the improved ideas. Attention is the listening rule. A Transformer repeats many rounds of listening, with a little thinking time in between.

The attention idea came from machine translation with recurrent networks: ML 03A shows how it removed the single-vector bottleneck of encoder–decoder RNNs. The Transformer of Vaswani

et al. [1] went further and removed recurrence entirely, building both encoder and decoder from attention and small per-token networks. The Vision Transformer (ViT) of Dosovitskiy et al. [2] applied the Transformer encoder almost unchanged to sequences of image patches. Bishop and Bishop [10] give a modern textbook treatment, and the surveys discussed in the Modern view [3], [11], [12] map the many vision variants. Afterwards, DLCV 02 compares CNN and Transformer backbones for classification in practice, and ML 05 shows how such backbones are pretrained without labels.

Self-attention: queries, keys and values

Plain version. Each token produces three vectors: a **query** ("what am I looking for?"), a **key** ("what do I contain?") and a **value** ("what will I pass on if you listen to me?"). A token's output is the average of all values, weighted by how well its query matches each key.

Precise version. Stack N tokens as the rows of $X \in \mathbb{R}^{N\times D}$, where D is the model width. Three learned matrices project them:

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V,$$

with $W_Q, W_K \in \mathbb{R}^{D\times d_k}$ and $W_V \in \mathbb{R}^{D\times d_v}$. **Scaled dot-product attention** [1] is

$$\text{Attention}(Q,K,V) = \text{softmax}_{\text{row}}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V = AV.$$

$QK^\top$ is an $N\times N$ matrix of **scores**: entry (i,j) is the dot product of query i with key j, a measure of how relevant token j is to token i. The row-wise softmax turns each row into nonnegative weights that sum to one, giving the **attention matrix** A. The output $Y = AV \in \mathbb{R}^{N\times d_v}$ has one row per token: row i is $\sum_j A_{ij}v_j$, a weighted average of value vectors.

Why three different projections? If queries and keys were the same vector, every token would match itself most strongly, and relevance would be forced to be symmetric. Separate W_Q and W_K let "what I seek" differ from "what I offer" (an eye patch can seek a nose without being one), and a separate W_V lets "what I offer" differ from "what I pass on".

Why divide by $\sqrt{d_k}$? If the components of a query and a key are independent with mean zero and variance one, their dot product $q\cdot k = \sum_{l=1}^{d_k} q_l k_l$ has mean zero and variance d_k. Without scaling, larger heads produce larger scores, the softmax saturates (almost all weight on one token), and its gradients become tiny. Dividing by $\sqrt{d_k}$ restores unit variance. It is a scale convention, not a normalization of each query to unit length.

Content, not position, decides who talks to whom. Unlike a convolution, whose weights are

tied to relative positions, attention weights are computed from the data at run time. A related consequence: self-attention is **permutation-equivariant**. Shuffle the input tokens, and the outputs are shuffled in the same way, because nothing in the formula refers to token order. The snippet checks both the shapes and this symmetry.

```
import torch

torch.manual_seed(0)
X = torch.randn(5, 8)                    # N=5, D=8
Wq, Wk, Wv = [torch.randn(8, 4) for _ in range(3)]
def attention(x):
    q, k, v = x @ Wq, x @ Wk, x @ Wv
    a = (q @ k.T / q.shape[-1]**.5).softmax(dim=-1)
    return a @ v, a
y, a = attention(X)
perm = torch.tensor([3, 0, 4, 1, 2])
yp, _ = attention(X[perm])
assert torch.allclose(a.sum(-1), torch.ones(5), atol=1e-6)
assert torch.allclose(yp, y[perm], atol=1e-5)
print("output shape:", tuple(y.shape))
```

This symmetry suits sets, but an image is not a set of patches: a face with its eyes below its mouth is not a face. Positional information, introduced below, breaks the symmetry.

Multi-head attention

Plain version. Run several attention operations in parallel, each with its own smaller projections, so different heads can track different relationships (nearby texture, the same object, symmetric parts). Then concatenate their outputs and mix them.

Precise version. With h heads, head r has its own $W_Q^{(r)}, W_K^{(r)}, W_V^{(r)} \in \mathbb{R}^{D\times d_k}$, usually with $d_k = d_v = D/h$:

$$\text{head}_r = \text{Attention}\big(XW_Q^{(r)}, XW_K^{(r)}, XW_V^{(r)}\big), \qquad \text{MHA}(X) = [\,\text{head}_1, \dots, \text{head}_h\,]\,W_O,$$

where the brackets concatenate the h outputs of width D/h back to width D, and $W_O \in \mathbb{R}^{D\times D}$ mixes them. Because each head has width D/h, multi-head attention costs about the same as one full-width head, and its projections have $4D^2$ weights in total (plus biases). Heads are not guaranteed to learn human-readable roles; some are redundant and can be pruned.

The implementation below splits the projections into heads, computes the scores per head, concatenates and projects, and matches `nn.MultiheadAttention` exactly. It also shows a **causal mask**: setting forbidden scores to $-\infty$ before the softmax gives them exactly zero weight.

```
import torch
from torch import nn

torch.manual_seed(0)
B, N, D, h = 2, 6, 16, 4
mha = nn.MultiheadAttention(D, num_heads=h, batch_first=True, bias=False)
X = torch.randn(B, N, D)

def multi_head(x):
    Wq, Wk, Wv = mha.in_proj_weight.chunk(3)          # each D x D (rows = outputs)
    q, k, v = x @ Wq.T, x @ Wk.T, x @ Wv.T              # B x N x D
    split = lambda t: t.view(B, N, h, D // h).transpose(1, 2)   # B x h x N x d_k
    q, k, v = split(q), split(k), split(v)
    scores = q @ k.transpose(-2, -1) / (D // h) ** 0.5  # B x h x N x N
    A = scores.softmax(dim=-1)
    heads = A @ v                                       # B x h x N x d_k
    concat = heads.transpose(1, 2).reshape(B, N, D)     # put the heads side by side
    return concat @ mha.out_proj.weight.T, A

ours, A = multi_head(X)
ref, _ = mha(X, X, X)
print("matches nn.MultiheadAttention:", torch.allclose(ours, ref, atol=1e-5))
print("attention tensor:", tuple(A.shape), "| parameters:",
      sum(p.numel() for p in mha.parameters()), "= 4 * D^2 =", 4 * D * D)

# Causal mask: query i may only read keys j <= i
causal = torch.triu(torch.ones(N, N, dtype=torch.bool), diagonal=1)   # True = forbidden
out_c, w_c = mha(X, X, X, attn_mask=causal)
print("no weight on the future:", torch.all(w_c[:, causal] == 0).item())
```

It prints `True`, an attention tensor of shape `(2, 4, 6, 6)` (batch, heads, queries, keys), `1024 = 4 * D^2 = 1024`, and `True` for the causal check. In production code, call `torch.nn.functional.scaled_dot_product_attention`, which computes the same thing and dispatches to fused, memory-efficient kernels when the device, data type and shapes allow [7].

Three ways to connect tokens

Plain version. Which tokens a query is allowed to read is part of the model's design, expressed as a mask on the score matrix.

Mechanism	A query reads	Typical use
Bidirectional self-attention	All tokens of the same sequence	ViT and other image encoders
Causal (masked) self-attention	Itself and earlier positions only	Next-token generation, autoregressive image models
Cross-attention	Tokens from a different sequence	A decoder reading an encoder; image tokens reading text

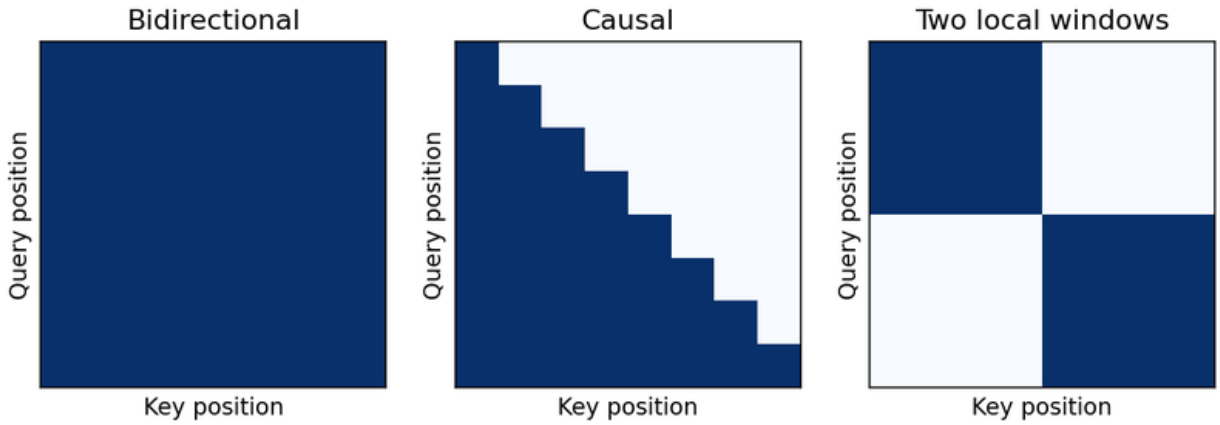


Figure 1 — Attention masks: blue entries are allowed, white entries are forbidden. Causal generation also shifts its inputs by one position so a target cannot simply be copied.

In **cross-attention**, queries come from one sequence X and keys and values from another sequence Z: $Q = XW_Q$, $K = ZW_K$, $V = ZW_V$. The score matrix has shape $N_X \times N_Z$, and the output has one row per query. This is how the original Transformer decoder reads its encoder, how image captioners read image features, and how text-to-image diffusion models read a text prompt (ML 07) without being autoregressive over image patches.

Attention weights are part of a computation, not automatically an explanation. A large weight can multiply an uninformative value, and the residual and MLP paths also shape the output.

Positional information

Plain version. Because attention ignores order, we must tell the model where each token is. The simplest way is to add a position-specific vector to each token before the first layer; more refined ways inject relative position directly into the attention scores.

Sinusoidal encodings. The original Transformer [1] adds a fixed vector $p_{pos} \in \mathbb{R}^D$ to the token at position pos:

$$p_{pos,2i} = \sin\left(\frac{pos}{10000^{2i/D}}\right), \qquad p_{pos,2i+1} = \cos\left(\frac{pos}{10000^{2i/D}}\right), \qquad i = 0, \dots, \frac{D}{2} - 1.$$

Each pair of dimensions $(2i, 2i+1)$ is a point rotating around a circle at angular frequency $\omega_i = 10000^{-2i/D}$, from fast (one radian per position) to very slow. Two properties follow from the angle-addition formulas. First, for any fixed offset k, $p_{pos+k} = p_{pos} R_k$ for a block-diagonal rotation R_k that does not depend on pos; a linear layer can therefore learn to "look k positions ahead". Second, $p_m \cdot p_n = \sum_i \cos(\omega_i (m - n))$ depends only on the offset $m - n$ (Figure 2, middle).

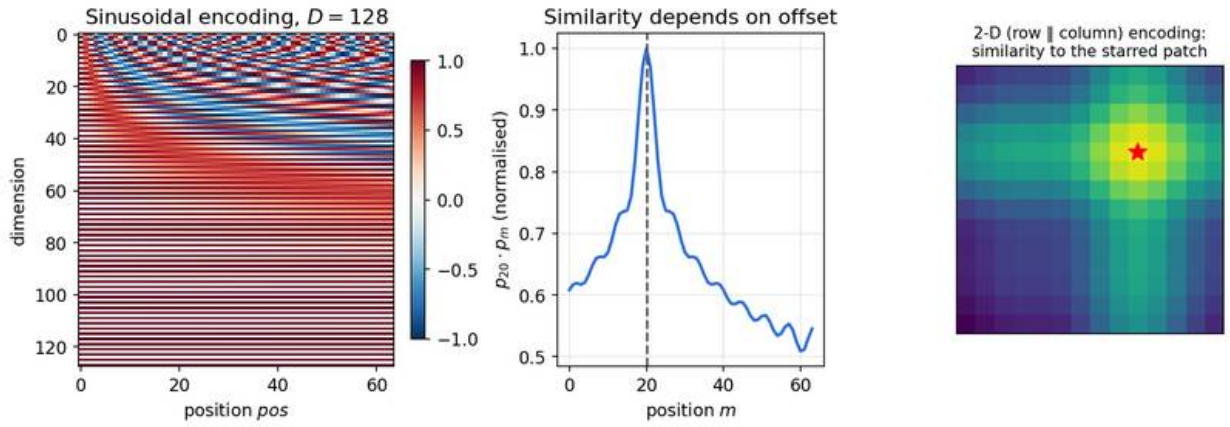


Figure 2 — Sinusoidal positional encodings. Left: each dimension pair oscillates at its own frequency. Middle: similarity between encodings depends on the offset. Right: concatenating row and column encodings gives a 2-D encoding for a 14 × 14 patch grid.

```
import numpy as np

def sinusoidal(n_pos, D):
    pos = np.arange(n_pos)[:, None]
    omega = 1.0 / 10000 ** (np.arange(0, D, 2) / D)    # one frequency per sin/cos pair
    pe = np.zeros((n_pos, D))
    pe[:, 0::2] = np.sin(pos * omega)
    pe[:, 1::2] = np.cos(pos * omega)
    return pe, omega

pe, omega = sinusoidal(100, 16)
k = 7                                                  # a fixed offset
# block-diagonal rotation that maps p_pos to p_{pos+k}, the same for every pos
R = np.zeros((16, 16))
for i, w in enumerate(omega):
    c, s = np.cos(w * k), np.sin(w * k)
    R[2 * i:2 * i + 2, 2 * i:2 * i + 2] = [[c, -s], [s, c]]
print("p[pos+k] == p[pos] @ R for all pos:", np.allclose(pe[:-k] @ R, pe[k:]))
print("dot products depend only on offset:",
      np.isclose(pe[10] @ pe[17], pe[50] @ pe[57]))
```

Both checks print `True`.

Other schemes.

- **Learned absolute embeddings**: a trainable vector per position, as in ViT [2]. ViT's 1-D learned embeddings, indexed by raster order, worked about as well as explicitly 2-D ones; after training, embeddings of patches in the same row or column became similar, so the model learned the grid by itself. At a new resolution, the grid of embeddings is resized by 2-D interpolation before fine-tuning.

- **Relative position representations** (Shaw et al. [17]) add a learned term that depends on the offset $j - i$ to the score between tokens i and j. Swin [5] adds a learned **relative position bias** table to each window's scores.
- **Rotary position embedding (RoPE)** (Su et al. [16]) rotates each query and key by an angle proportional to its position, using the same rotations as the sinusoidal code above. The score $q_m^\top k_n$ then depends on the positions only through $m - n$. RoPE is standard in large language models and increasingly used in vision with separate rotations for the two image axes.
- **No explicit encoding**: convolutions inside the network, as in CvT [19], or zero padding at borders can leak enough position information on their own.

The Transformer block

Plain version. A block alternates two jobs. Attention lets tokens exchange information; a small MLP then processes each token on its own. Residual connections and normalization keep the stack stable.

Precise version. The **pre-norm** block used by ViT and most modern models is

$$U = X + \mathrm{MHA}(\mathrm{LN}(X)), \qquad Y = U + \mathrm{MLP}(\mathrm{LN}(U)), \qquad \mathrm{MLP}(u) = W_2\,\mathrm{GELU}(W_1 u + b_1) + b_2.$$

LN is **layer normalization** [14]: it standardizes each token's D features to zero mean and unit variance, then applies a learned scale and shift; unlike BatchNorm, it does not depend on other samples in the batch. The MLP expands each token from D to typically $4D$ hidden units, applies the GELU nonlinearity [26] ($\mathrm{GELU}(z) = z\,\Phi(z)$, where Φ is the standard normal CDF), and projects back. The residual additions mean each sub-layer only learns a correction, exactly as in ResNet.

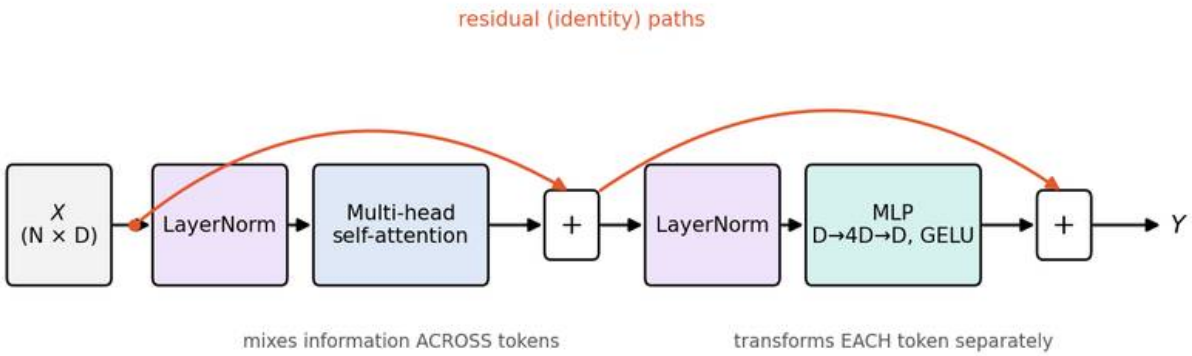


Figure 3 — A pre-norm Transformer block. Attention mixes information across tokens; the MLP transforms each token separately; residual paths (orange) carry the input around each branch.

The original Transformer placed LayerNorm **after** each residual addition (post-norm). Xiong et al. [15] showed that with post-norm the gradients near the output are large at initialization, which is why it needs a learning-rate warm-up, whereas pre-norm keeps gradients well behaved and can train without one. So "a Transformer" is not fully specified until you say where the normalization goes. Counting weights per block: $4D^2$ for attention and $8D^2$ for the MLP, so about $12D^2$, two thirds of it in the MLP.

The Vision Transformer (ViT)

Plain version. Cut the image into square patches, turn each patch into a vector, add position information, and feed the sequence to a standard Transformer encoder. Read the answer from a special extra token.

Patchify. For an image with $H \times W$ pixels and C channels, and patch size P dividing both H and W, there are $N = HW/P^2$ patches. Flatten each into $x_i^{\text{patch}} \in \mathbb{R}^{P^2C}$ and project linearly:

$$z_0 = [\, x_{\text{cls}};\ x_1^{\text{patch}}E;\ \ldots;\ x_N^{\text{patch}}E \,] + E_{\text{pos}},$$

where $E \in \mathbb{R}^{P^2C \times D}$ is the learned **patch embedding**, $x_{\text{cls}} \in \mathbb{R}^D$ is a learned **class token** prepended to the sequence, and $E_{\text{pos}} \in \mathbb{R}^{(N+1)\times D}$ holds learned position embeddings. After L blocks, a LayerNorm and a linear head applied to the class token's output give the logits. Averaging all patch outputs (mean pooling) is an equally common alternative to the class token.

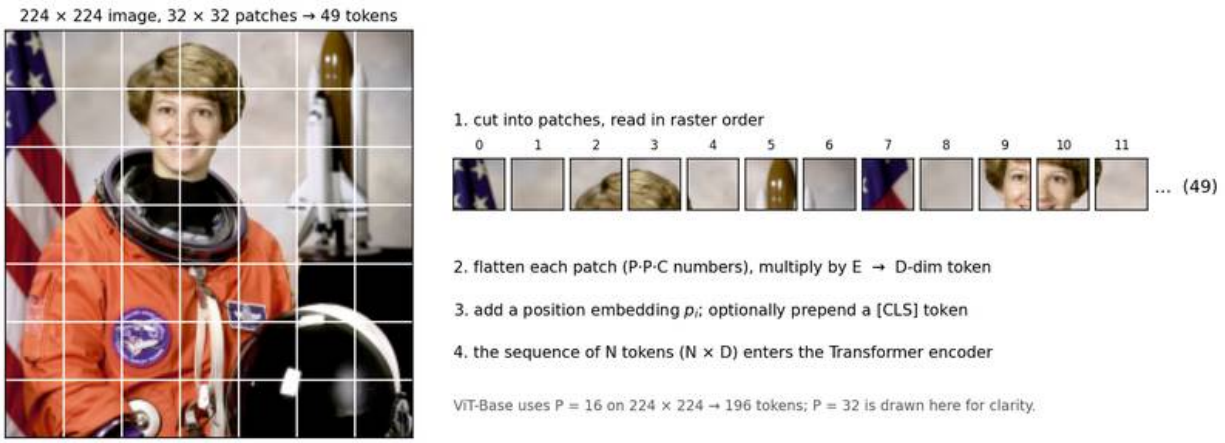


Figure 4 — Patchify. A 32-pixel grid is drawn for readability; ViT-Base uses 16 × 16 patches, which turn a 224 × 224 image into 196 tokens.

A linear map applied to non-overlapping $P \times P$ patches is exactly a convolution with kernel size and stride both equal to P, which is how implementations write it:

```
import torch
from torch import nn

torch.manual_seed(0)
B, C, H, W, P, D = 2, 3, 32, 32, 8, 48
x = torch.randn(B, C, H, W)

# 1) explicit: cut into P x P patches, flatten each, multiply by E
patches = x.unfold(2, P, P).unfold(3, P, P)                 # B, C, H/P, W/P, P, P
patches = patches.permute(0, 2, 3, 1, 4, 5).reshape(B, -1, C * P * P)   # B, N, P*P*C
E = torch.randn(C * P * P, D) * 0.02
tokens_explicit = patches @ E                                # B, N, D

# 2) the same linear map written as a convolution with kernel = stride = P
conv = nn.Conv2d(C, D, kernel_size=P, stride=P, bias=False)
conv.weight.data = E.T.reshape(D, C, P, P)
tokens_conv = conv(x).flatten(2).transpose(1, 2)             # B, N, D

print("tokens:", tuple(tokens_conv.shape), "| identical:",
      torch.allclose(tokens_explicit, tokens_conv, atol=1e-5))
```

It prints 16 tokens of width 48 and `True`.

Sizes. The standard configurations are ViT-Base (12 blocks, $D = 768$, 12 heads, about 86 million parameters), ViT-Large (24 blocks, $D = 1024$, 16 heads) and ViT-Huge (32 blocks, $D = 1280$, 16 heads) [2]. "ViT-B/16" means Base with 16×16 patches. **Patch size is the main compute knob**: halving P quadruples N and multiplies the attention-score count by sixteen. Small patches keep fine detail; large patches are cheap but lose anything finer than the patch, which no later layer can recover.

The data lesson. Dosovitskiy et al. [2] found that ViT trained only on ImageNet-1k (about 1.3 million images) was somewhat worse than comparable ResNets, but when pretrained on ImageNet-21k (14 million) or JFT-300M (300 million) and then fine-tuned, it matched or exceeded them at lower pretraining cost. Their explanation is **inductive bias**: a CNN assumes locality and translation equivariance in every layer, whereas a ViT is local only within the patch embedding and must **learn** spatial relationships from data. Raghu et al. [18] compared the internal representations of ViTs and ResNets and found that ViTs have more uniform representations across layers, that their lower layers attend both locally and globally, and that learning local attention in early layers requires large-scale pretraining.

Code: a tiny ViT trained on digits

This complete example trains a small ViT on the same 8×8 digits as the CNN in ML 03, with 2×2 patches (16 tokens), a class token, learned positions and four pre-norm blocks. It runs on CPU in about half a minute.

```
import copy
import numpy as np
import torch
from torch import nn
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split

torch.manual_seed(0)
torch.set_num_threads(1)

class TinyViT(nn.Module):
    def __init__(self, img=8, patch=2, chans=1, D=64, depth=4, heads=4, classes=10):
        super().__init__()
        N = (img // patch) ** 2
        self.patch = nn.Conv2d(chans, D, kernel_size=patch, stride=patch)   # patchify + E
        self.cls = nn.Parameter(torch.zeros(1, 1, D))                       # [CLS] token
        self.pos = nn.Parameter(torch.randn(1, N + 1, D) * .02)             # learned
         ↪  positions
        layer = nn.TransformerEncoderLayer(D, heads, dim_feedforward=4 * D, dropout=0.1,
                                           activation="gelu", batch_first=True,
                                           norm_first=True)                 # pre-norm block
        self.blocks = nn.TransformerEncoder(layer, depth, enable_nested_tensor=False)
        self.norm, self.head = nn.LayerNorm(D), nn.Linear(D, classes)

    def forward(self, x):
        t = self.patch(x).flatten(2).transpose(1, 2)                         # B, N, D
        t = torch.cat([self.cls.expand(len(t), -1, -1), t], dim=1) + self.pos
        return self.head(self.norm(self.blocks(t))[:, 0])                    # read [CLS]

d = load_digits()
X = torch.tensor(d.images[:, None] / 16.0, dtype=torch.float32)
y = torch.tensor(d.target)
train, test = train_test_split(np.arange(len(y)), test_size=.2, stratify=d.target,
                               random_state=0)
train, val = train_test_split(train, test_size=.2, stratify=d.target[train], random_state=1)
model = TinyViT()
opt = torch.optim.AdamW(model.parameters(), lr=1e-3, weight_decay=0.05)
best = (float("inf"), None)
for epoch in range(30):
    model.train()
    for idx in torch.tensor(train)[torch.randperm(len(train))].split(64):
        loss = nn.functional.cross_entropy(model(X[idx]), y[idx])
        opt.zero_grad(); loss.backward(); opt.step()
    model.eval()
    with torch.no_grad():
        v = nn.functional.cross_entropy(model(X[val]), y[val]).item()
    if v < best[0]:
        best = (v, copy.deepcopy(model.state_dict()))
model.load_state_dict(best[1]); model.eval()
with torch.no_grad():
    acc = (model(X[test]).argmax(1) == y[test]).float().mean().item()
print("parameters:", sum(p.numel() for p in model.parameters()))
```

```
print("best validation loss:", round(best[0], 4), "| held-out accuracy:", round(acc, 4))
```

On our run it uses about 202,000 parameters and reaches about 95% held-out accuracy, while the CNN in ML 03 reached about 96% with fewer than 10,000 parameters and the same split. On a small dataset, the CNN's built-in locality is worth a factor of twenty in parameters. This is the data lesson in miniature, not a verdict on large-scale performance.

Training recipes: DeiT and beyond

Plain version. Much of the early gap between ViTs and CNNs on ImageNet-sized data turned out to be about training, not architecture.

DeiT (Touvron et al. [4]) trained ViTs competitively **on ImageNet-1k alone** using strong data augmentation (RandAugment, Mixup, CutMix, random erasing), heavy regularization (stochastic depth, label smoothing) and long schedules. It also introduced a **distillation token**: an extra learned token, like the class token, whose output is trained to match the predictions of a pre-trained CNN teacher. The CNN's inductive bias is thus transferred through its predictions. DeiT III [27] later showed that a simpler, carefully tuned supervised recipe (a lightweight "3-Augment" scheme, simple cropping and a binary cross-entropy loss) makes plain ViTs stronger still. Steiner et al. [25] ran a large study of data, augmentation and regularization ("AugReg") and found that the right combination of augmentation, regularization and compute can match models trained on roughly ten times more data.

The general lesson, repeated in ML 03's discussion of ConvNeXt: compare architectures only under matched data, augmentation, resolution, schedule and compute.

Swin and hierarchical Transformers

Plain version. Global attention over thousands of tokens is expensive, and plain ViT keeps one coarse resolution throughout, which is awkward for detection and segmentation. Swin restricts attention to local windows, shifts the windows between layers so information still spreads, and merges patches to build a CNN-like pyramid.

Windows. Swin [5] partitions the token grid into non-overlapping windows of $M \times M$ tokens (typically $M = 7$) and computes self-attention inside each window only. For an $h \times w$ token grid with C channels, the paper gives the costs

$$\Omega(\text{global MSA}) = 4hwC^2 + 2(hw)^2C, \qquad \Omega(\text{window MSA}) = 4hwC^2 + 2M^2hwC,$$

where $4hwC^2$ counts the four projections and the second term counts the score and weighting products. Global cost is quadratic in the number of tokens hw; window cost is linear for fixed M.

Shifted windows. With fixed windows, tokens in different windows would never interact. Swin alternates regular windows with windows displaced by $\lfloor M/2 \rfloor$ tokens in each direction. A token near a window boundary at layer ℓ shares a window with tokens from neighboring windows at layer $\ell + 1$, so information crosses boundaries (Figure 5). An efficient implementation cyclically shifts the feature map and uses a mask so that tokens wrapped from opposite edges do not attend to each other.

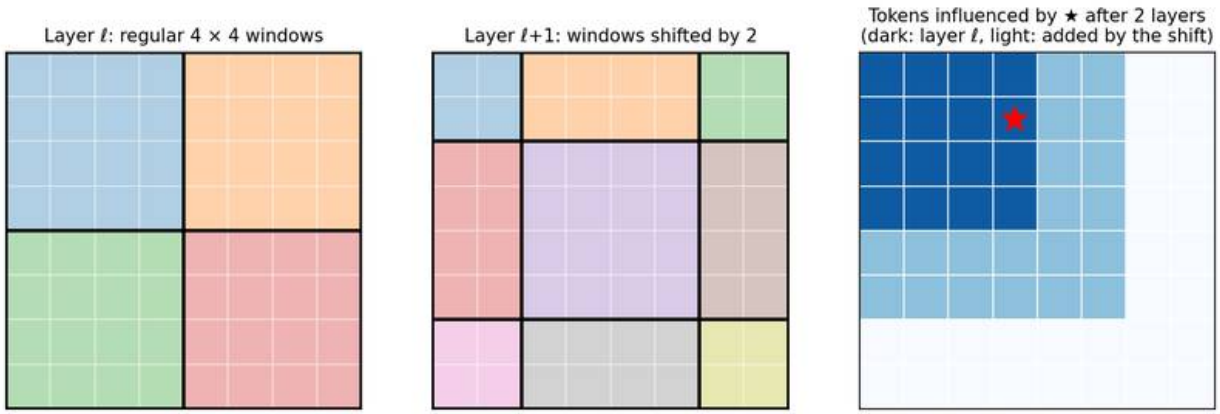


Figure 5 — Swin's regular and shifted windows on an 8 × 8 token grid. After two layers, information from the starred token reaches the dark window (layer ℓ) plus the light band added by the shifted windows (layer ℓ+1).

Hierarchy. Between stages, **patch merging** concatenates each 2 × 2 group of neighboring tokens and projects them linearly, halving the resolution and doubling the channels, as a strided convolution would. The four stages produce feature maps at 1/4, 1/8, 1/16 and 1/32 of the input resolution, the same pyramid a ResNet produces, so Swin can replace a CNN backbone in detection and segmentation frameworks. Each window also adds a learned **relative position bias** to its scores. Swin V2 [32] scaled the design up with residual post-normalization, scaled cosine attention and a continuous, log-spaced position bias that transfers better across window sizes and resolutions.

Hybrid CNN–Transformer models

Plain version. Convolutions are good at local detail with little data; attention is good at long-range interaction. Hybrids use each where it helps.

- **Convolutional stem.** The original ViT paper already tested a "hybrid" in which the patches come from a ResNet feature map instead of raw pixels [2]; it helped at smaller compute budgets.
- **Convolutions inside attention.** CvT [19] uses a convolutional token embedding at each stage

and depthwise-convolutional projections for queries, keys and values, which builds in locality and lets it drop explicit position embeddings.

- **Stacking by stage.** CoAtNet [20] places depthwise-convolution stages early, where resolution is high and local patterns dominate, and relative-attention stages later; it reports strong results across data scales because the convolutional stages supply inductive bias when data is limited.
- **Mobile hybrids.** MobileViT [28] inserts small Transformer blocks into a MobileNet-style network to add global context at mobile cost.
- **The reverse direction.** ConvNeXt (see ML 03) imported Transformer-era design choices into a pure CNN and matched Swin.

Cost and efficient attention

Plain version. Global attention compares every token with every other one, so doubling the image side multiplies the work by sixteen. Efficient methods either make exact attention faster, restrict which tokens interact, or use fewer tokens.

For N tokens of width D, the score matrix costs about N^2D multiply–adds per layer and the projections and MLP about $12ND^2$. A naive implementation stores an $N \times N$ matrix per head. Doubling the image height and width at a fixed patch size multiplies N by four and the pairwise scores by sixteen.

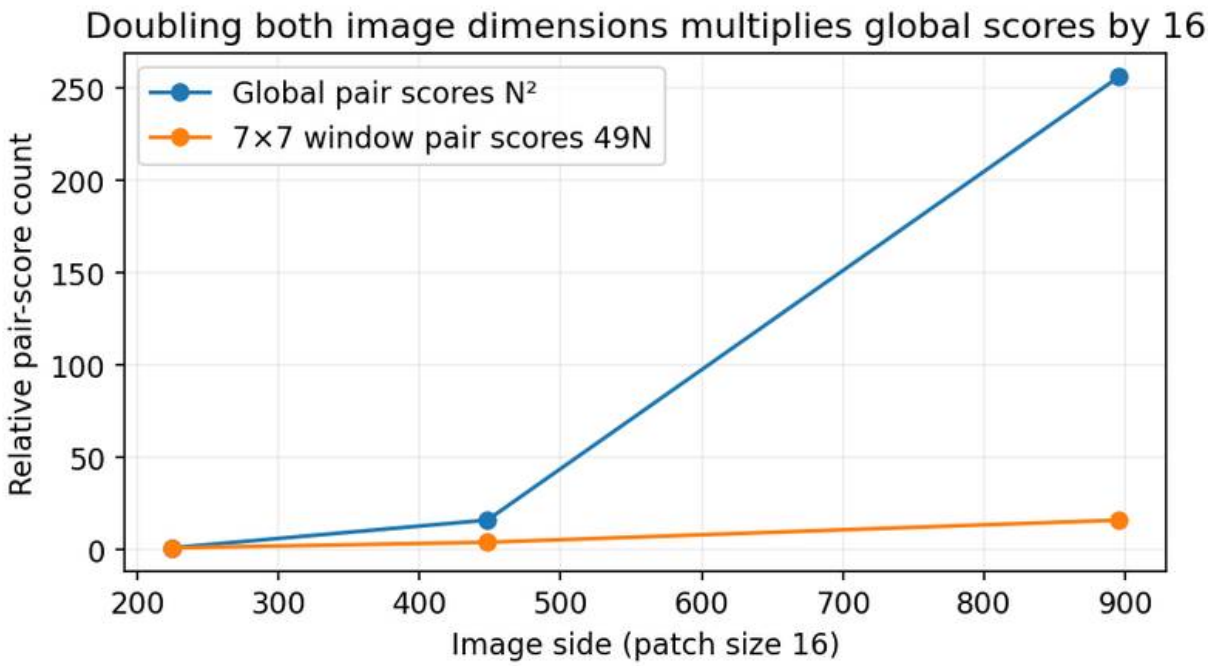


Figure 6 — Pair-score scaling, not a measured runtime. Projections, MLPs, memory traffic and kernel

quality also determine latency.

The main families:

- **Exact but faster.** FlashAttention [6] computes **exact** attention by tiling the computation so that blocks of queries, keys and values stay in fast on-chip memory, and never writes the full $N \times N$ matrix to slow GPU memory; FlashAttention-2 [29] improves the work partitioning further. The arithmetic is still quadratic; the memory is linear and wall-clock time drops substantially. PyTorch's `scaled_dot_product_attention` selects such kernels automatically when it can [7].
- **Restricted connectivity.** Windowed (Swin) or other sparse patterns reduce the number of pairs, at the price of needing several layers for global communication.
- **Fewer tokens.** Token Merging (ToMe) [21] gradually merges similar tokens inside a pretrained ViT using a fast matching step, increasing throughput without retraining. Larger patches, token pruning and pooling stages do the same more bluntly.
- **Smaller models.** Papa et al.'s survey [13] organizes efficient ViTs into compact architectures, pruning, knowledge distillation and quantization, and proposes an "Efficient Error Rate" metric to compare models across parameters, bits, FLOPs and size.

State-space models and hybrids

Plain version. Instead of comparing every pair of tokens, carry a compact state through the sequence, as a recurrent network does (ML 03A). The cost grows linearly with the number of tokens.

A simplified state-space recurrence over a token sequence is

$$h_t = \bar{A}_t\, h_{t-1} + \bar{B}_t\, x_t, \qquad y_t = C_t\, h_t,$$

where h_t is a hidden state of fixed size, and $\bar{A}_t$, $\bar{B}_t$, C_t are (discretized) state-space matrices. In **selective** models such as Mamba, the write/read coefficients and step sizes depend on input x_t; the underlying continuous transition can remain fixed while its discretization varies. Input selection and an affine state update permit a parallel scan [33]. An image has no natural one-dimensional order, so vision variants must choose **scan orders**: Vision Mamba (Vim) [30] scans the patch sequence forward and backward with position embeddings, and VMamba [31] scans along four directions over the 2-D grid. The Visual Mamba survey [8] reviews over 200 papers and identifies scanning as the critical design decision for vision.

A useful counterpoint is MambaOut [9] (CVPR 2025). Its authors argue that Mamba suits tasks with long sequences and causal (autoregressive) structure, and build MambaOut models by removing the state-space component while keeping the gated convolutional blocks. Their models outperform visual Mamba models on ImageNet classification, suggesting that Mamba is unnecessary there, while for detection and segmentation, which involve long token sequences, they find state-space models may still be worth exploring. The broader point is methodological: isolate the mechanism (attention, scan, convolution) from gating, normalization, training recipe and

parameter budget before crediting a result to it.

An image scan is an inductive bias

Mamba [33] supplies an efficient sequence operator; it does not choose an image geometry for us. Vim [30] uses a patch sequence, position embeddings and forward/backward processing. VMamba [31] instead places a four-route selective scan inside a hierarchical visual backbone. Neither label alone specifies receptive-field strength, resolution behavior or runtime. The state dimensions, routes, local operators and downsampling stages all matter.

For a fixed scan order, two adjacent pixels in the image may be many steps apart in the sequence. Conversely, the last pixel of one raster row and the first pixel of the next row become consecutive despite being far apart spatially. A local convolution can supply nearby evidence before the scan carries context farther. Multiple routes offer additional paths, but do not restore every geometric symmetry automatically. Position embeddings can also encode useful location while making transfer to a different grid a separate design problem.

A mathematical way to expose the choice is

$$Y^{(r)} = P_r^{-1} \mathcal{S}_r(P_r X), \qquad Y = \frac{1}{m} \sum_{r=1}^{m} Y^{(r)}.$$

Here X contains flattened image tokens, r indexes one of m routes, P_r permutes tokens into that route, $\mathcal{S}_r$ applies its sequence recurrence, P_r^{-1} restores image coordinates, and Y averages the resulting features. This is a conceptual route-fusion model. A learned vision block can use different projections or fusion rules; averaging is chosen here to keep the experiment interpretable.

CPU experiment: measure directional bias

The next example uses a single scalar state with the same decay on every route. It compares forward/backward raster scans with four raster/column routes, then rotates the input by 90 degrees, processes it and rotates the output back. Everything uses a synthetic 6 × 6 impulse; no weights or images are downloaded. This isolates ordering rather than reproducing a trained Vim or VMamba.

```
import numpy as np

def scan_image(image, four=False):
    ids = np.arange(image.size).reshape(image.shape)
    row, col = ids.ravel(), ids.T.ravel()
    orders = [row, row[::-1]]
    if four:
        orders += [col, col[::-1]]
    outputs = []
    for order in orders:
        h = 0.0
        out = np.zeros(image.size)
```

```
        for i in order:
            h = 0.8 * h + 0.2 * image.ravel()[i]
            out[i] = h
        outputs.append(out.reshape(image.shape))
    return np.mean(outputs, axis=0)

image = np.zeros((6, 6))
image[1, 4] = 1.0
errors = []
for four in [False, True]:
    original = scan_image(image, four=four)
    rotated_back = np.rot90(scan_image(np.rot90(image), four=four), -1)
    error = np.max(np.abs(original - rotated_back))
    errors.append(error)
    print('four routes:', four, 'rotation discrepancy:', round(error, 6))
assert 0.0 < errors[1] < errors[0]
```

The maximum discrepancies are `0.063622` for two routes and `0.026738` for four. More routes reduce directional bias in this example, but the remaining discrepancy is nonzero. Rotating a raster path reverses one traversal axis; that path need not belong to the original four-route collection. Even shared coefficients and averaging therefore do not guarantee rotation equivariance. Learned directional projections, position information and local operations introduce further choices. Test the property you need instead of inferring it from "2-D scan."

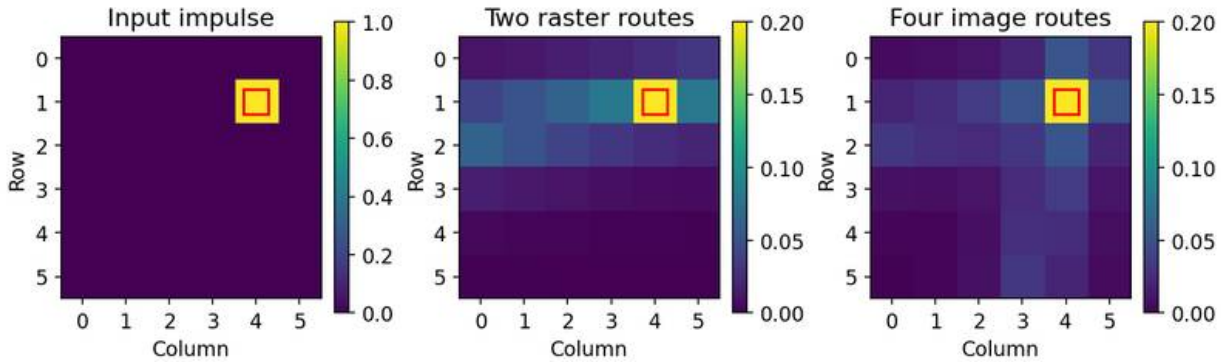


Figure 7 — The red outline marks the input impulse. Two raster routes spread context unevenly; adding column routes changes the influence map. These synthetic responses use the exact recurrence in the example, not learned attention maps or trained VMamba activations.

The recurrence can have a global receptive field because information travels through the entire scan. This says a dependence is possible, not that its influence remains strong at every distance. With scalar decay 0.8, the carried contribution decays by another factor of 0.8 per intervening step. Selective transitions can change this retention, as derived in ML 03A, but training must still learn which information to preserve. Fine boundaries and small objects can be lost through state compression or downsampling even when distant context is available.

What Mamba-2 changes, and how to evaluate it

Mamba-2 [34] reorganizes the selective sequence mixer around structured state-space duality and efficient chunk computation. It changes the transition structure and block parameter projections, rather than making an arbitrary image patch sequence equivalent to a Transformer. The causal structured coefficient matrix has an attention-like interpretation, but standard softmax attention retains different expressive and computational properties. An image application must still specify directions and fusion. The duality therefore answers a computation question; it does not remove the geometry question.

A fair vision comparison should separate three experiments: replace only the mixer inside a shared block; compare complete backbones under a matched training budget; and measure the deployed models on the target hardware. Keep data, augmentation, token resolution and downstream heads controlled. Record state size as well as parameter count, and measure latency and peak memory across image resolutions and batch sizes. A linear token-count term can coexist with expensive projections, route permutations and small-kernel overheads. GPU throughput claims do not follow from the Python loop above.

For dense tasks, inspect errors near boundaries, on small structures and across aspect ratios; for video, distinguish an offline bidirectional encoder from causal live processing. Include rotation/crop sensitivity and longer-grid evaluations rather than only in-distribution classification. MambaOut [9] supplies an important mixer-removal ablation. Its results support a task-dependent comparison, not a theorem that scans are unnecessary for every image problem. Xu et al. [8] organize visual Mamba work by backbones, scan mechanisms and image/video/point-cloud/multimodal applications. Their survey remains a 2024 preprint on the verified record. Its taxonomy is useful for selecting controlled experiments; the number of proposed variants does not establish a universal winner.

Modern view

Survey literature. Khan et al. [3] (*ACM Computing Surveys*, 2022) start from the foundations (self-attention, large-scale pretraining, bidirectional encoding) and then survey Transformers across recognition, generative modeling, multimodal tasks such as visual question answering, video, low-level vision and 3-D analysis, closing with open problems. Han et al. [11] (*IEEE TPAMI*, 2023) organize vision Transformers by task (backbones, high- and mid-level vision such as detection and segmentation, low-level vision, video), and also cover efficient Transformer methods. Liu et al. [12] (*IEEE TNNLS*, 2024) review over a hundred visual Transformers for classification, detection and segmentation on images, point clouds and vision–language data, and suggest three future directions. For efficiency, Papa et al. [13] (*IEEE TPAMI*, 2024) is the most systematic. For state-space vision models, see Xu et al. [8]. Bishop and Bishop [10] provide the textbook derivations.

State of the art, 2023–2026. The plain ViT has become the default **foundation** backbone. ViT-22B [24] (ICML 2023) scaled a ViT to 22 billion parameters, with architectural changes (such

as normalizing queries and keys) needed for stable training at that size. DINOv2 [23] (TMLR 2024) trained ViTs with self-supervision on a large curated image collection and produced general-purpose features that work frozen across classification, segmentation and depth estimation (ML 05). Darcet et al. [22] (ICLR 2024) found that large ViTs repurpose uninformative background patches as internal scratch space, producing high-norm artifact tokens in attention maps, and that adding a few extra **register tokens** removes the artifacts and improves dense predictions. Vision–language foundation models (ML 09) and diffusion Transformers (ML 07) build on the same block. At the same time, hierarchical and hybrid designs remain common for dense, high-resolution tasks, modernized CNNs remain competitive under matched budgets (ML 03), and state-space and linear-attention models continue to be explored for very long token sequences such as video.

Open problems. Quadratic cost at high resolution and long video; the large data and compute appetite of plain ViTs; position encodings that generalize across resolutions and aspect ratios; interpreting attention without over-reading it; and fair, compute-matched comparisons between attention, convolution and state-space mechanisms.

Key takeaways

- Self-attention is a learned, input-dependent weighted average: $\mathrm{softmax}(QK^\top/\sqrt{d_k})V$. The $\sqrt{d_k}$ keeps scores at unit variance so the softmax does not saturate.
- Multi-head attention runs h narrower heads in parallel at about the cost of one; masks (causal, window) and cross-attention control who reads whom.
- Attention is permutation-equivariant, so position must be injected: sinusoidal, learned absolute, relative bias or rotary encodings.
- A pre-norm Transformer block is $X + \mathrm{MHA}(\mathrm{LN}(X))$ then $U + \mathrm{MLP}(\mathrm{LN}(U))$; attention mixes tokens and the MLP transforms each token.
- ViT = patchify (a stride-P convolution) + position embeddings + class token + Transformer encoder. It has weak inductive bias and needs large pretraining or strong recipes (DeiT, AugReg).
- Swin brings locality and hierarchy back with shifted windows and patch merging; hybrids such as CvT, CoAtNet and MobileViT combine convolution and attention.
- FlashAttention makes exact attention faster without changing its quadratic arithmetic; windows, token merging and state-space scans change what is computed. Compare mechanisms only under matched budgets.
- Vision Mamba and VMamba adapt selective state spaces to image sequences. Multiple routes change influence paths; they do not automatically guarantee rotation equivariance or preserve every distant detail.

Exercises

1. **Token budget.** Calculate the number of tokens and attention-score entries per head for a 512×512 image with patches of size 16 and 8. By what factor does the score count grow when the patch size halves?

   Hint

   $N = (512/16)^2 = 1024$ and $(512/8)^2 = 4096$ tokens; $N^2 = 1{,}048{,}576$ and $16{,}777{,}216$ scores. Halving the patch size multiplies N by 4 and the scores by 16.

2. **Scaling check.** Draw random queries and keys with independent standard-normal components for $d_k = 4, 64, 1024$. Measure the variance of $q \cdot k$ with and without division by $\sqrt{d_k}$, and the largest softmax weight over 100 keys in each case. What happens to the softmax without scaling?

   Hint

   The variance of the raw dot product is close to d_k; after scaling it is close to 1. Without scaling, at $d_k = 1024$ the softmax puts nearly all weight on a single key (it becomes almost one-hot), so gradients through the other keys vanish.

3. **Remove position.** In the TinyViT code, set the position embeddings to zero, drop the class token and use mean pooling, then permute the patch tokens before the blocks and compare the logits in evaluation mode. Explain the result, and what it implies for an image classifier without positional information.

   Hint

   The logits agree up to floating-point error, because every block is permutation-equivariant and mean pooling is permutation-invariant. Use dropout off (`eval()`) and permute tokens, not pixels inside a patch. Such a model cannot tell a face from a scrambled face built from the same patches.

4. **Cross-attention shapes.** In a text-to-image model, 256 image tokens of width 320 read 77 text tokens of width 768 with 8 heads of width 40. Give the shapes of W_Q, W_K, W_V, one head's score matrix, and the output.

   Hint

   $W_Q \in \mathbb{R}^{320\times320}$ (8 heads × 40), $W_K, W_V \in \mathbb{R}^{768\times320}$. Each head's scores are 256×77. The output follows the queries: 256×320 after the output projection. Key and value lengths must match each other, not the query length.

5. **Swin arithmetic.** Using the cost formulas, compare global and window attention for a 56×56 token grid with $C = 96$ and $M = 7$. Which term dominates in each case?

Hint

$hw = 3136$. Global: $4 \cdot 3136 \cdot 96^2 \approx 1.16 \times 10^8$ plus $2 \cdot 3136^2 \cdot 96 \approx 1.89 \times 10^9$, so the quadratic term dominates. Window: the same 1.16×10^8 plus $2 \cdot 49 \cdot 3136 \cdot 96 \approx 2.95 \times 10^7$, so the projections dominate. The window version is about 14 times cheaper here.

6. **Fair comparison.** You have 2,000 labeled images and access to pretrained CNN, ViT and visual Mamba backbones. Design an experiment that tells you which backbone to deploy, and list what must be held fixed. Include a mixer-removal ablation and a resolution/rotation test.

Hint

Compare linear probes on frozen features first, then fine-tuning with matched budgets and tuned learning rates per model on a validation split. Record pretraining data, input resolution, parameters, FLOPs and measured latency on the target device. A pretrained ViT against a CNN trained from scratch tests pretraining, not architecture. For the scan backbone, record state size and test whether benefits survive removing the mixer while retaining gating and local convolution. Keep the same preprocessing in the resolution and rotation tests.

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ML 05 • Representation Learning and Self-Supervision

What a representation is, why distributed features transfer, and how SimCLR, MoCo, BYOL, SimSiam, DINO, MAE, JEPA and CLIP learn them without labels, with honest linear-probe and fine-tuning evaluation.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-05-representation-learning/

What you'll learn

- What a **representation** is, why **distributed** representations generalize better than one-hot codes, and which invariances a good representation should (and should not) have.
- How **transfer learning** works: pretrain an encoder once, then reuse it through k-NN, a linear probe or fine-tuning.
- How the main **self-supervised learning** families build a training signal from unlabeled data: contrastive (SimCLR, MoCo, with the InfoNCE loss), teacher–student without negatives (BYOL, SimSiam, DINO), redundancy reduction (Barlow Twins, VICReg), masked modeling (BEiT, MAE) and feature prediction (I-JEPA, V-JEPA 2).
- Why "make two views agree" can **collapse**, and how alignment and uniformity make that precise.
- How **CLIP** uses captions as supervision and enables zero-shot classification.
- How to evaluate a representation honestly, including why fine-tuning can underperform a linear probe and what LP-FT does about it.

The big picture

Suppose you have a million unlabeled images but only 500 labeled examples for the task you care about. Training a large network from scratch on those 500 examples ignores almost all the available data. **Representation learning** splits the job in two. First, learn an encoder $h = f_\theta(x)$ from a broad signal that does not need your labels. Second, train a small task-specific head on top of h using the few labels you have.

> **Explain like I'm five**
>
> Before a child learns the names of every bird, they already notice wings, beaks, colors and how things move. Later, a parent points at three robins and two crows, and that is enough: the child can tell robins from crows from then on. The "noticing" was learned for free, just by looking at the world. A

representation is that reusable way of noticing things, and self-supervised learning is the "just looking" part.

Why it matters: almost every modern vision system starts from a pretrained encoder. Image classifiers, detectors, segmentation models, image–text models and the image encoders inside generative models are built on representations learned this way. The quality of that starting point often matters more than the details of the task head.

This chapter starts with what a representation is and why transfer works, then walks through the self-supervised objectives that dominate practice, and ends with how to measure whether a representation is any good. It assumes you know neural networks (ML 02), CNNs (ML 03) and vision transformers (ML 04). Linear factor models and autoencoders, the classical route to representations, have their own chapter: ML 05A.

What is a representation?

Plain version. A representation is the list of numbers a model uses to describe an input. A good one makes the things you care about easy to read off, and hides the things you do not care about.

Precise version. An encoder is a function $f_\theta : \mathcal{X} \to \mathbb{R}^D$ from inputs x (here, images) to feature vectors $h = f_\theta(x)$, where θ are the learnable parameters and D is the feature dimension. A task head g maps h to a prediction, so the full model is $g \circ f_\theta$. We call h a good representation for a task when a *simple* head, such as a linear classifier, can solve the task from h using *few* labeled examples. Bengio, Courville and Vincent's review [1] is the classic statement of this view; it lists the "priors" that make representations useful, such as smoothness, multiple explanatory factors, a hierarchy of increasingly abstract features, and shared factors across tasks.

Local versus distributed representations

The simplest representation of a category is **one-hot**: one unit per concept, with exactly one unit on. It is easy to read but it cannot generalize. Knowing about "red car" says nothing about "red truck", because they share no active units. A **distributed** representation describes each input with many features that are each shared by many inputs: "red", "blue", "car", "truck". Now "red truck" reuses the "red" unit learned from red cars.

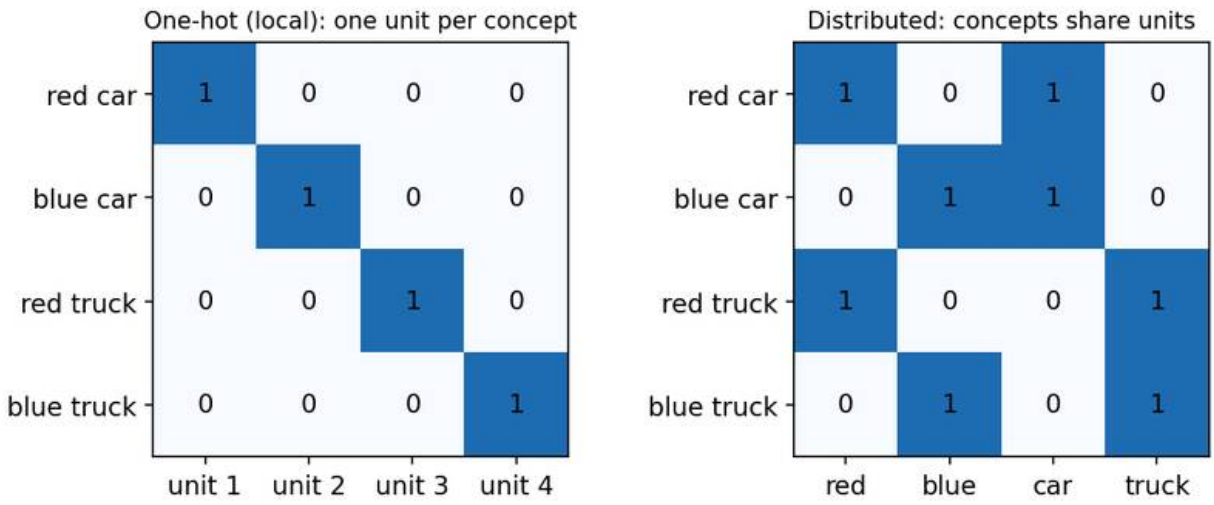


Figure 1 — Local (one-hot) versus distributed codes for four concepts. With distributed codes, a new combination such as a red truck reuses features learned from other examples.

The counting argument is striking. With D binary features, a one-hot code distinguishes D concepts, while a distributed code can in principle distinguish 2^D configurations [1]. Deep networks learn distributed codes automatically: each hidden unit responds to many inputs, and each input activates many units. Most of the practical value of representation learning comes from this sharing.

Invariance and equivariance

A useful representation keeps differences relevant to the task and suppresses differences that distract from it.

- A feature is **invariant** to a transformation T if $f(T(x)) = f(x)$: shifting a cat by two pixels should not change "this is a cat".
- A feature is **equivariant** if it changes in a predictable way, $f(T(x)) = T'(f(x))$: shifting the cat should shift its segmentation mask by the same amount.

"Useful" always depends on the downstream task. Color invariance helps object recognition but hurts fruit-ripeness estimation. Rotation invariance helps for textures but confuses 6 and 9. Dense prediction (segmentation, depth) needs spatially equivariant patch features, not a single invariant vector per image. We will see that self-supervised objectives choose their invariances mostly through the **data augmentations** they use, so this choice is a modeling decision, not a detail.

Learning distances: metric learning

PCA, introduced in ML 01 and developed in ML 05A, keeps the directions of highest variance. But high variance may come from lighting rather than object identity. Supervised learning shapes features around labels; **metric learning** shapes distances around pairs that should match. For an anchor a, a positive p (same identity) and a negative n (different identity), the triplet loss is

$$\mathcal{L}_{\text{triplet}} = \max\left(0,\ \|h_a - h_p\|^2 - \|h_a - h_n\|^2 + m\right),$$

where h_a, h_p, h_n are the three feature vectors and $m > 0$ is a margin. The loss is zero once the positive is closer than the negative by at least m. An easy negative gives zero loss and teaches nothing, so choosing informative ("hard") negatives matters. Contrastive self-supervised learning, below, is metric learning in which the positives are created by augmentation instead of by labels.

Transfer learning: pretrain once, reuse many times

Plain version. Train a network on a big task, keep its feature extractor, and reuse it for a small task. You can freeze the features and only train a new head, or let the whole network keep learning.

Precise version. Pretraining finds encoder parameters $\theta^\star$ on a source dataset or objective. Transfer to a target task with labeled set $\{(x_i, y_i)\}_{i=1}^n$ uses one of three protocols:

1. **k-nearest neighbors (k-NN):** no training. Classify a test feature $f_{\theta^\star}(x)$ by the labels of its k most similar training features.
2. **Linear probe:** keep $\theta^\star$ frozen and fit a linear classifier $\hat{y} = \text{softmax}(Wh + b)$ on $h = f_{\theta^\star}(x)$, where W and b are the only trained parameters.
3. **Fine-tuning:** start from $\theta^\star$ and update the encoder together with the head, usually with a small learning rate.

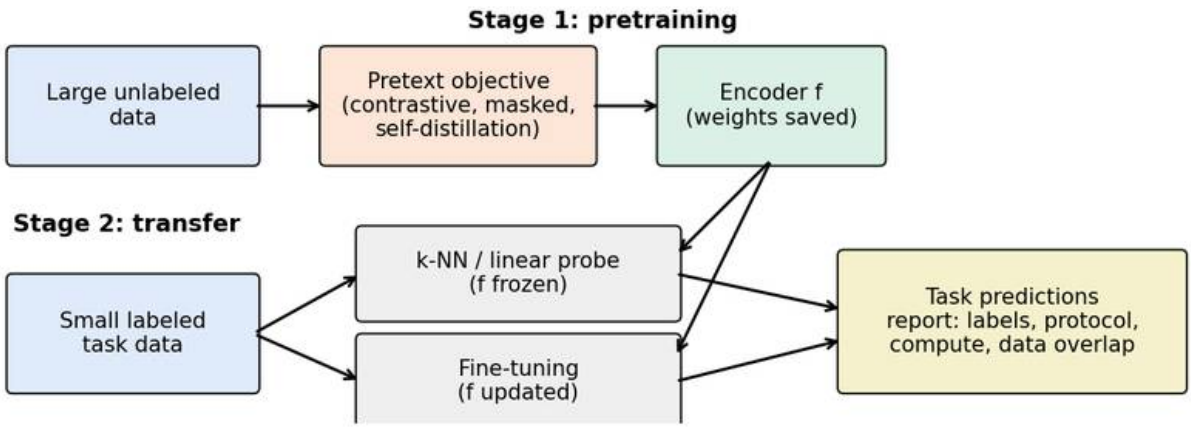


Figure 2 — The two-stage pattern of representation learning. Pretraining is expensive and done once; transfer is cheap and done per task.

Why does transfer work at all? Yosinski et al. [2] measured it layer by layer on ImageNet-trained CNNs. First-layer features (edge and color detectors much like Gabor filters) were general: they transferred well between tasks. Features became progressively more specific to the source task in higher layers, and transferability decreased as the source and target tasks became more different. They also found that initializing from transferred features and fine-tuning improved generalization even on the target task, compared with training from random initialization.

For years, the default pretraining source was supervised ImageNet classification. Labels are expensive, though, and a label like "dog" tells the network nothing about the dog's pose, the grass or the lighting. Self-supervised learning removes the label bottleneck by making the data supervise itself.

Self-supervised learning: let the data make the labels

Plain version. Invent a task whose answer you can compute from the data itself: "are these two crops from the same photo?", "what was under this mask?", "which caption goes with this image?". Solving such a **pretext task** forces the network to learn useful features.

Precise version. A self-supervised objective is a loss $\mathcal{L}(\theta; x)$ computed from unlabeled inputs alone (or from naturally paired data such as image–caption pairs). The surveys by Liu et al. [3] and Gui et al. [4] and the practical "cookbook" by Balestriero et al. [5] organize the field into the families summarized here:

Family	Training signal	Representative methods	Main risk
Contrastive	Pull positive pairs together, push other samples apart	SimCLR, MoCo	Needs many negatives; false negatives
Self-distillation (no negatives)	Student matches a teacher on another view	BYOL, SimSiam, DINO	Collapse without asymmetry or centering
Redundancy reduction	Agreement plus decorrelated, non-constant feature dimensions	Barlow Twins, VICReg	Extra hyperparameters for the regularizers
Masked modeling	Predict hidden pixels or tokens	BEiT, MAE	Features may focus on low-level detail
Feature prediction	Predict features of hidden regions	I-JEPA, V-JEPA 2	Needs anti-collapse design like distillation
Language supervision	Match images with their captions	CLIP, SigLIP	Inherits web-data biases and noise

The families are learning objectives, not architectures: each can train a CNN or a transformer. Modern systems often combine several of them. The sections below take them in turn.

Contrastive learning and the InfoNCE loss

Plain version. Make two augmented versions of the same image agree, and make them disagree with every other image in the batch. It is a multiple-choice question: "out of these N candidates, which one is my partner?"

InfoNCE

Van den Oord, Li and Vinyals introduced the **InfoNCE** loss in Contrastive Predictive Coding (CPC) [6]. Take a query q, one positive key k^+ that "belongs" to it, and $N-1$ negative keys k_1^-, $\dots, k_{N-1}^-$. With a similarity score $s(\cdot,\cdot)$ and a temperature $\tau > 0$,

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp\left(s(q,k^+)/\tau\right)}{\exp\left(s(q,k^+)/\tau\right) + \sum_{j=1}^{N-1} \exp\left(s(q,k_j^-)/\tau\right)}.$$

This is exactly softmax cross-entropy for an N-way classification problem whose correct class is the positive. Two consequences follow.

- **A lower bound on mutual information.** CPC shows that $I(q;k^+) \geq \log N - \mathcal{L}_{\text{InfoNCE}}$, where I is mutual information. Minimizing the loss raises this lower bound, and more negatives (larger N) allow a higher bound. In practice this bound is loose, and the quality of the features depends heavily on the encoder and augmentations, so read it as motivation rather than as the explanation of why the features are good.
- **The temperature controls hardness.** Dividing by a small τ sharpens the softmax, so the loss concentrates on the negatives most similar to the query (the hard negatives).

SimCLR

SimCLR [7] is InfoNCE in its simplest form. Take a batch of B images and produce two augmented views of each (random crop and resize, color distortion, blur), giving $2B$ views. A backbone f maps each view to h, a small MLP **projection head** g maps h to $z = g(h)$, and the similarity is cosine similarity $s(z_i, z_j) = z_i^\top z_j / (\|z_i\|\|z_j\|)$. For view i with partner j,

$$\ell_{i,j} = -\log \frac{\exp(z_i^\top z_j/\tau)}{\sum_{k \neq i} \exp(z_i^\top z_k/\tau)},$$

where the z are ℓ_2-normalized and the sum runs over all $2B-1$ other views. Every other image in the batch is a negative, so no memory bank is needed, but the method benefits from large batches. The total loss averages $\ell_{i,j}$ over both orderings of every positive pair; the paper calls this the NT-Xent loss (normalized temperature-scaled cross-entropy).

Three findings from the SimCLR paper are worth remembering [7]. The *composition* of augmentations matters a lot, with random cropping plus color distortion being especially important. The projection head helps: features *before* the head (h) transfer better than those after it (z), so the head is discarded after pretraining. And contrastive learning benefits from larger batches and longer training than supervised learning.

Code: check the loss and its collapse value

This standalone PyTorch example uses synthetic paired vectors to verify the loss, the self-mask and the gradients. It does not train an encoder.

```
import torch
from torch.nn import functional as F

torch.manual_seed(0)
B, D = 8, 16
base = torch.randn(B, D)
views = torch.cat([base + .1*torch.randn_like(base),
                   base + .1*torch.randn_like(base)]).requires_grad_()
z = F.normalize(views, dim=-1)
logits = z @ z.T / .2
self_mask = torch.eye(2*B, dtype=torch.bool)
logits = logits.masked_fill(self_mask, float("-inf"))
positive = (torch.arange(2*B) + B) % (2*B)
loss = F.cross_entropy(logits, positive)
loss.backward()
assert torch.isfinite(views.grad).all()
print("paired loss:", round(loss.item(), 4))

collapsed = F.normalize(torch.ones(2*B, D), dim=-1)
scores = (collapsed @ collapsed.T / .2).masked_fill(self_mask, float("-inf"))
collapsed_loss = F.cross_entropy(scores, positive)
print("collapsed loss:", round(collapsed_loss.item(), 4))
print("expected:", round(torch.log(torch.tensor(2*B-1.)).item(), 4))
```

It prints a paired loss of 0.1423 and a collapsed loss of 2.7081, equal to $\log 15$. If every representation is identical, the positive gets probability $1/(2B-1)$ and the loss is $\log(2B-1)$, the worst value a uniform guess can get. Contrastive discrimination therefore makes complete collapse a bad solution, although optimization and negative selection can still be difficult.

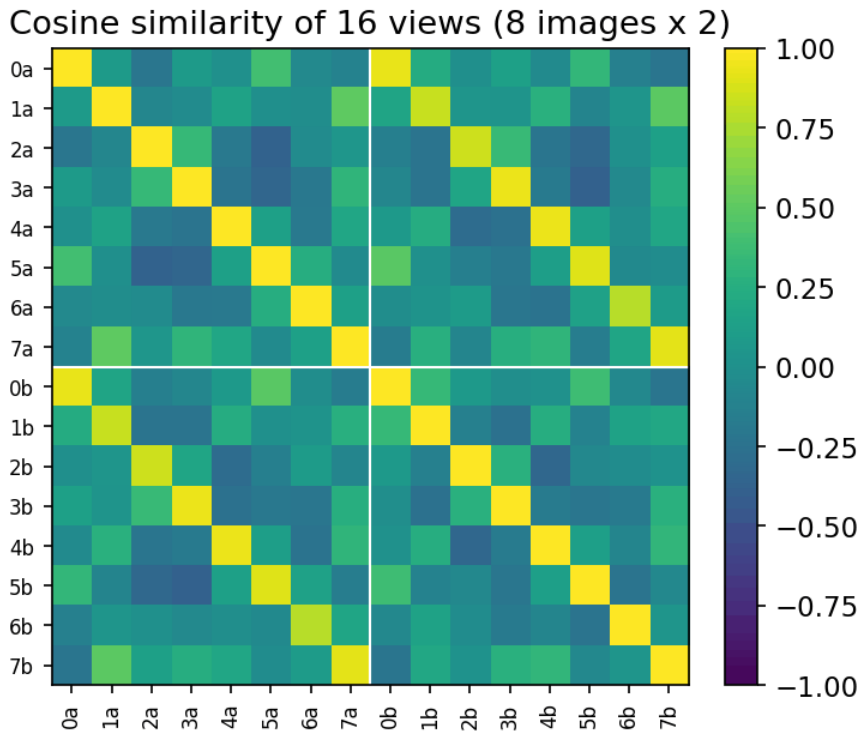


Figure 3 — Cosine similarities between 16 views (8 digit images, two augmentations each) after training the small SimCLR encoder used later in this chapter. The bright off-diagonal stripes are the positive pairs; InfoNCE pushes them up and everything else down.

MoCo: a queue of negatives and a momentum encoder

InfoNCE prefers many negatives, but SimCLR gets them only from the current batch. **Momentum Contrast (MoCo)** [8] decouples the number of negatives from the batch size with two ideas.

1. **A queue.** Keys encoded in recent batches are stored in a first-in, first-out queue of length K and reused as negatives. The query sees K negatives no matter how small the batch is.
2. **A momentum encoder.** Keys in the queue were produced by slightly older networks. If the key encoder changed quickly, old and new keys would be inconsistent. MoCo therefore computes keys with a separate encoder whose parameters θ_k follow the query encoder's parameters θ_q slowly:

$$\theta_k \leftarrow m\,\theta_k + (1-m)\,\theta_q,$$

where $m \in [0,1)$ is the momentum, close to 1 (the paper's default is 0.999). Only θ_q receives gradients. This **exponential moving average (EMA)** teacher idea reappears in BYOL, DINO and

JEPA.

```
import torch
from torch.nn import functional as F

torch.manual_seed(0)
B, D, K = 8, 16, 32          # batch, embedding size, queue length

# --- MoCo-style InfoNCE: one positive key per query, negatives from a queue
q = F.normalize(torch.randn(B, D), dim=1)                 # queries (online encoder)
k = F.normalize(q + 0.1 * torch.randn(B, D), dim=1)       # keys (momentum encoder)
queue = F.normalize(torch.randn(K, D), dim=1)             # keys from past batches
l_pos = (q * k).sum(1, keepdim=True)                      # B x 1
l_neg = q @ queue.T                                       # B x K
logits = torch.cat([l_pos, l_neg], dim=1) / 0.07
loss_moco = F.cross_entropy(logits, torch.zeros(B, dtype=torch.long))
queue = torch.cat([k, queue])[:K]                         # enqueue new, drop oldest
print("MoCo logits:", tuple(logits.shape), "loss:", round(loss_moco.item(), 3))

# momentum (EMA) update of the key encoder's parameters
theta_q, theta_k, m = torch.randn(5), torch.zeros(5), 0.999
theta_k = m * theta_k + (1 - m) * theta_q
print("key params moved by:", round((theta_k.norm() / theta_q.norm()).item(), 4))
```

Each query gets a $1 + K$-way classification problem whose correct answer is always column 0, hence the all-zero targets. The output shows logits of shape `(8, 33)`, and the EMA step moves the key encoder only 0.1% of the way toward the query encoder (from zero, after one step). In a real implementation, q and k come from the two encoders applied to two augmentations of the same images.

Alignment and uniformity

Wang and Isola [9] gave a clean geometric reading of what contrastive losses do on the unit sphere. With normalized features, the loss asymptotically optimizes two properties:

$$\mathcal{L}_{\text{align}} = \mathbb{E}_{(x,x^+)} \|f(x) - f(x^+)\|_2^2, \qquad \mathcal{L}_{\text{unif}} = \log \mathbb{E}_{x,y}\, e^{-t\|f(x)-f(y)\|_2^2},$$

where (x, x^+) is a positive pair, x, y are independent samples, and $t > 0$ (commonly $t = 2$).

Alignment is low when positive pairs map to nearby points. **Uniformity** is low when features spread out over the sphere, so they keep as much information about the input as possible. A collapsed encoder has perfect alignment (0) and the worst uniformity (0); a useful encoder has both small.

```
import torch
from torch.nn import functional as F

def alignment(z1, z2):           # mean squared distance between positive pairs
```

```
    return (z1 - z2).square().sum(1).mean()

def uniformity(z, t=2.0):         # log of mean Gaussian potential over all pairs
    return torch.pdist(z).square().mul(-t).exp().mean().log()

torch.manual_seed(0)
n, D = 512, 16
collapsed = F.normalize(torch.ones(n, D), dim=1)
spread = F.normalize(torch.randn(n, D), dim=1)
noisy_view = F.normalize(spread + 0.1 * torch.randn(n, D), dim=1)
print("collapsed: align", alignment(collapsed, collapsed).item(),
      "uniform", round(uniformity(collapsed).item(), 3))
print("spread:    align", round(alignment(spread, noisy_view).item(), 3),
      "uniform", round(uniformity(spread).item(), 3))
```

The collapsed set scores alignment 0.0 and uniformity 0.0; the spread set scores about 0.138 and −3.51. Figure 4 tracks the same two quantities while training the digit encoder below. At step 0 the random network maps every view to almost the same point, so alignment is near zero only because everything has collapsed. Training first spreads the features out (uniformity drops sharply), then gradually pulls positives back together.

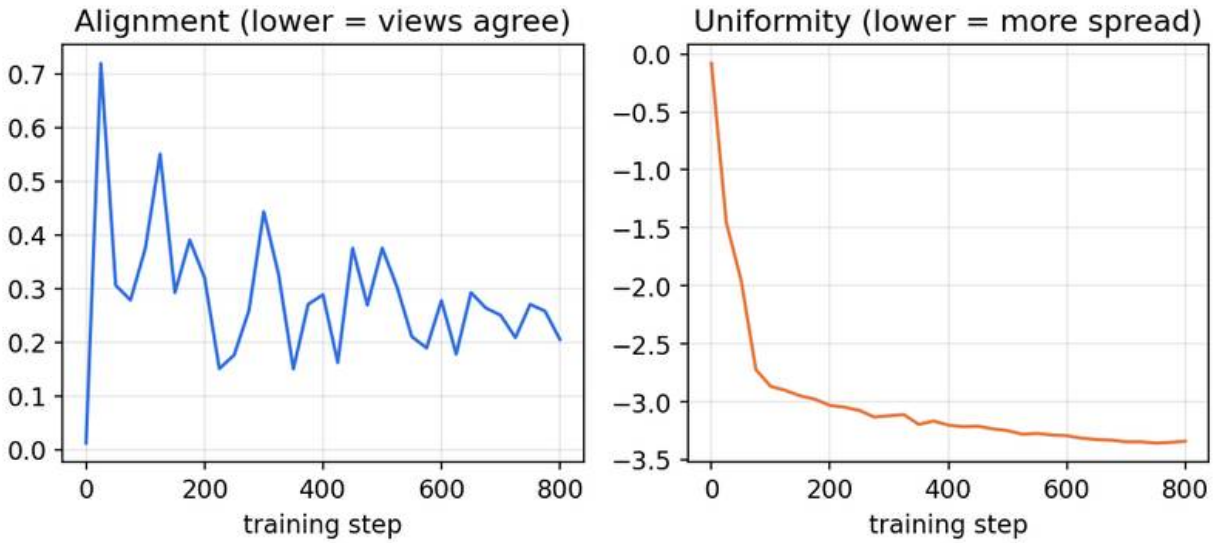


Figure 4 — Alignment and uniformity of the projection features during SimCLR training on 8×8 digits. Low alignment alone is meaningless (a constant encoder has it); the useful regime is low alignment together with low uniformity.

A worked example: SimCLR on 8×8 digits

Plain version. Train a tiny SimCLR encoder on scikit-learn's digits without using any labels, then check whether its features make a nearest-neighbor classifier more robust.

The augmentation here is a random shift of up to one pixel, a random brightness change and pixel noise. We therefore expect the learned features to be roughly invariant to small shifts, something raw pixels are not. The evaluation uses the same labeled training set for every feature type and tests on clean digits and on digits shifted by up to one pixel.

```
import copy
import numpy as np
import torch
from torch import nn
from torch.nn import functional as F
from sklearn.datasets import load_digits
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

torch.manual_seed(0); np.random.seed(0); torch.set_num_threads(1)
d = load_digits()
Xtr, Xte, ytr, yte = train_test_split(d.data / 16., d.target, test_size=.3,
                                      random_state=0, stratify=d.target)
Xtr_t = torch.tensor(Xtr, dtype=torch.float32)

def augment(x):                       # the augmentations define the invariances
    img = F.pad(x.view(-1, 1, 8, 8), (1, 1, 1, 1))
    i, j = np.random.randint(0, 3, size=2)              # shift by up to 1 px
    img = img[:, :, i:i + 8, j:j + 8]
    img = img * (0.7 + 0.6 * torch.rand(len(img), 1, 1, 1))  # brightness
    return (img + 0.1 * torch.randn_like(img)).reshape(len(x), 64)  # noise

backbone = nn.Sequential(nn.Unflatten(1, (1, 8, 8)),
                         nn.Conv2d(1, 32, 3, padding=1), nn.ReLU(),
                         nn.Conv2d(32, 64, 3, padding=1), nn.ReLU(),
                         nn.AdaptiveAvgPool2d(2), nn.Flatten())  # h: 256-d
random_backbone = copy.deepcopy(backbone)                    # baseline
head = nn.Sequential(nn.ReLU(), nn.Linear(256, 64))          # projection g
opt = torch.optim.Adam([*backbone.parameters(), *head.parameters()], lr=1e-3)

def nt_xent(z1, z2, tau=0.2):
    z = F.normalize(torch.cat([z1, z2]), dim=-1)
    n = len(z1)
    logits = (z @ z.T / tau).fill_diagonal_(float("-inf"))
    return F.cross_entropy(logits, (torch.arange(2 * n) + n) % (2 * n))

for step in range(800):                                  # about a minute on CPU
    x = Xtr_t[torch.randint(len(Xtr_t), (256,))]
    loss = nt_xent(head(backbone(augment(x))), head(backbone(augment(x))))
    opt.zero_grad(); loss.backward(); opt.step()

def shift(x, dy, dx):                    # move one 8x8 image, fill with zeros
    img = np.zeros((10, 10)); img[1:9, 1:9] = x.reshape(8, 8)
    return img[1 - dy:9 - dy, 1 - dx:9 - dx].ravel()

moves = np.random.default_rng(0).integers(-1, 2, size=(len(Xte), 2))
Xte_shift = np.stack([shift(x, *m) for x, m in zip(Xte, moves)])
```

```
enc = lambda net: (lambda A: net(torch.tensor(A, dtype=torch.float32)).detach().numpy())
pca = PCA(32, random_state=0).fit(Xtr)
for name, f in [("raw pixels", lambda A: A), ("PCA-32", pca.transform),
                ("random CNN", enc(random_backbone)), ("SimCLR CNN", enc(backbone))]:
    knn = KNeighborsClassifier(5, metric="cosine").fit(f(Xtr), ytr)
    print(f"{name:11s} clean {knn.score(f(Xte), yte):.3f}  "
          f"shifted {knn.score(f(Xte_shift), yte):.3f}")
```

On our machine this prints:

```
raw pixels  clean 0.978  shifted 0.524
PCA-32      clean 0.978  shifted 0.507
random CNN  clean 0.974  shifted 0.524
SimCLR CNN  clean 0.987  shifted 0.919
```

Exact numbers vary slightly across PyTorch versions and hardware. The pattern is the lesson. On clean digits, raw pixels are already excellent: this dataset is easy, and a fancy encoder adds little. On shifted digits, raw pixels, PCA and an untrained CNN of the same architecture all drop to about 50%, while the self-supervised features keep most of their accuracy. The network learned the invariance we asked for through the augmentations, and nothing more: had we tested on rotated digits, which the augmentations never showed, we should not expect the same robustness.

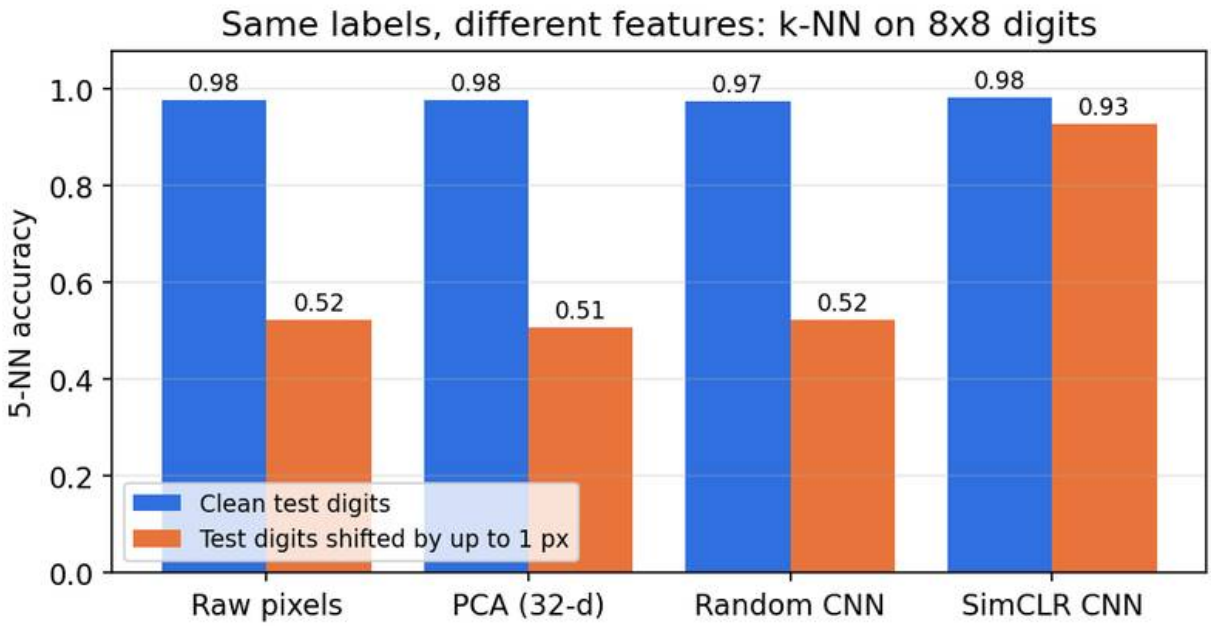


Figure 5 — Same labels, same classifier, different features. The figure script uses a separate random stream, so its numbers differ from the printed output by up to 0.01. The random CNN rules out the explanation that convolution alone produces the robustness.

Without negatives: why matching alone can collapse

Plain version. If the only instruction is "make the two outputs agree", returning the same constant for every image solves it perfectly. Methods without negatives need some other mechanism to stop this.

Minimizing $\|f_\theta(v_1) - f_\theta(v_2)\|^2$ over two views v_1, v_2 has the trivial solution $f_\theta(x) = c$ for a constant c. Agreement alone does not require the features to carry any information. **Complete collapse** maps everything to one point; **dimensional collapse** is subtler: the features vary, but only along a few directions, wasting most of the D dimensions.

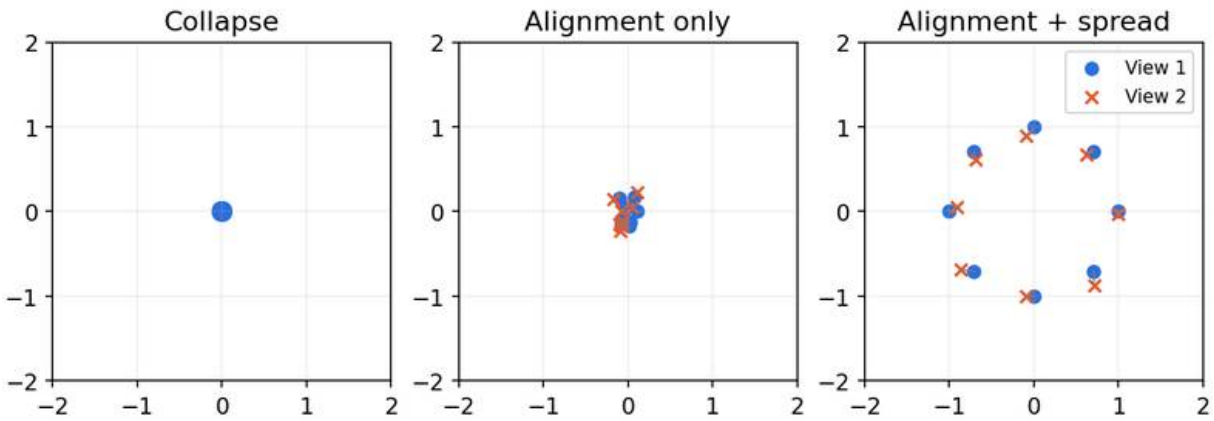


Figure 6 — Schematic of collapse, alignment without spread, and alignment with spread. The points are constructed illustrations, not embeddings learned by a published model.

BYOL: an online network chases a slow target

BYOL [10] uses two networks. The **online** network (encoder, projector and an extra **predictor** q_ϕ) is trained to predict the **target** network's projection of a different view of the same image. The target is not trained by gradients; it is an EMA of the online weights, as in MoCo:

$$\theta_{\text{target}} \leftarrow m\,\theta_{\text{target}} + (1-m)\,\theta_{\text{online}}.$$

The loss is the squared distance between the normalized prediction and the normalized target projection, symmetrized over the two views. The target output is treated as a constant (**stop-gradient**). There are no negatives, yet BYOL does not collapse in practice. The asymmetry (predictor on one side, stop-gradient and EMA on the other) and the full training recipe matter; EMA alone does not prove collapse impossible.

SimSiam: stop-gradient is the key ingredient

Chen and He's SimSiam [11] stripped BYOL down further: one shared encoder for both views, a predictor h on one branch, **no** momentum encoder, **no** negatives and no large batches. With $p_1 = h(f(v_1))$ and $z_2 = f(v_2)$,

$$\mathcal{L} = \tfrac{1}{2}\,\mathcal{D}(p_1, \mathrm{sg}(z_2)) + \tfrac{1}{2}\,\mathcal{D}(p_2, \mathrm{sg}(z_1)), \qquad \mathcal{D}(p, z) = -\frac{p^\top z}{\|p\|\,\|z\|},$$

where sg is stop-gradient and $\mathcal{D}$ is negative cosine similarity. The paper's experiments show that removing the stop-gradient makes the model collapse quickly, while the predictor is also needed. The authors interpret the method as an alternating, expectation-maximization-like optimization. The practical lesson: very small architectural asymmetries decide whether these methods learn or collapse.

DINO: self-distillation with centering and sharpening

DINO [12] turns the agreement problem into classification with soft labels. Teacher and student both output a probability vector over K "prototype" dimensions via a temperature-scaled softmax. The student, seeing a view x', is trained to match the teacher's distribution on another view x:

$$\mathcal{L} = -\sum_{k=1}^{K} P_t(x)_k \,\log P_s(x')_k, \qquad P_t(x) = \mathrm{softmax}\Big(\frac{g_t(x) - c}{\tau_t}\Big),$$

where g_t is the teacher network (an EMA of the student), c is a running mean of teacher outputs (**centering**) and τ_t is a low teacher temperature (**sharpening**). Centering stops one dimension from dominating; sharpening stops the output from becoming uniform. Each alone collapses in a different way; together they balance. DINO also uses **multi-crop**: the teacher sees only large global crops, while the student also sees small local crops, so the student must infer global content from local evidence. With vision transformers, DINO's attention maps were notable for highlighting object boundaries without any segmentation labels.

Barlow Twins and VICReg: regularize the statistics directly

A different route is to forbid collapse explicitly in the loss. **Barlow Twins** [13] computes the $D \times D$ cross-correlation matrix C between the (batch-standardized) embeddings of the two views and pushes it toward the identity:

$$\mathcal{L}_{\mathrm{BT}} = \sum_i (1 - C_{ii})^2 + \lambda \sum_i \sum_{j \neq i} C_{ij}^2,$$

where the first term makes each dimension agree across views and the second, weighted by $\lambda > 0$, decorrelates different dimensions so they do not carry redundant information. **VICReg** [14]

splits the same idea into three named terms: **invariance** (mean squared distance between the two embeddings), **variance** (a hinge loss keeping the standard deviation of every dimension above a threshold, which directly forbids constant outputs) and **covariance** (penalizing off-diagonal covariance entries). "No negatives" does not mean "no anti-collapse design"; these methods simply put the design in the loss instead of in the architecture.

Masked modeling: learn by filling gaps

Plain version. Hide part of the input and predict what is missing. This is how BERT learned language, and it works for images too, once you choose the right amount of hiding.

BEiT [15] brought BERT-style pretraining to vision transformers. An image is split into patches; some are masked, and the network predicts discrete **visual tokens** for the masked positions, where the tokens come from a separately trained image tokenizer.

MAE [16] predicts raw pixels instead. It masks a large fraction of the patches, sends only the visible ones through the encoder, and reconstructs the missing pixels with a lightweight decoder. With the set of masked patch indices $\mathcal{M}$,

$$\mathcal{L}_{\text{mask}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\hat{x}_i - x_i\|^2,$$

where x_i is the true pixel vector of patch i and $\hat{x}_i$ is the prediction. MAE's design choices are deliberate. A high masking ratio (75% by default) makes the task hard enough that the network cannot simply interpolate from neighbors. The encoder only processes visible patches, which cuts pretraining cost. Normalizing each target patch by its own mean and variance improves the features. After pretraining, the decoder is discarded.

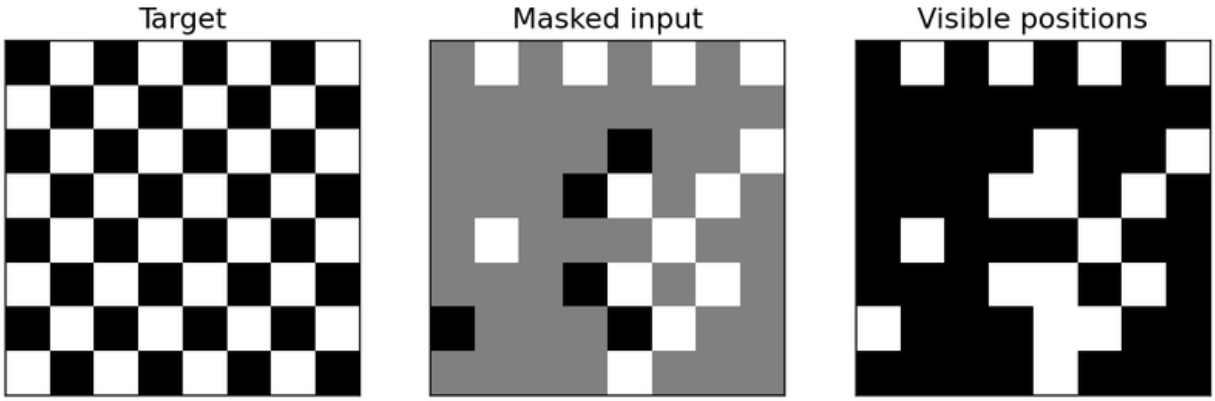


Figure 7 — Masked prediction creates its own targets from the input. The checkerboard illustrates the masking operation, not semantic understanding.

Reconstruction is not the same as generation. A model that predicts masked pixels from visible ones is not trained to sample complete images from scratch. Low reconstruction loss does not guarantee the best semantic features either: background texture can dominate pixel error. Masked-modeling features are typically weaker than contrastive or distillation features under a linear probe but strong after fine-tuning, which is one reason the evaluation protocol must be stated.

Predicting features instead of pixels: JEPA

Plain version. Instead of painting in every uncertain pixel, predict a *description* of the hidden region.

I-JEPA [17] (Image Joint-Embedding Predictive Architecture) encodes a visible **context** block and predicts the encoder features of several **target** blocks from it. The target encoder is an EMA of the context encoder, and a predictor receives the context features plus the positions of the targets. A simplified objective is

$$\mathcal{L}_{\text{pred}} = \left\| g_\phi(f_\theta(x_{\text{context}}), \text{positions}) - \text{sg}\left(f_{\bar{\theta}}(x_{\text{target}})\right) \right\|^2,$$

where f_θ is the context encoder, $f_{\bar{\theta}}$ the EMA target encoder, g_ϕ the predictor and sg stop-gradient. The formula summarizes the idea; the masking strategy (large, semantic-scale target blocks) and the architecture are part of the method, not optional details.

V-JEPA 2 [18] extends feature prediction to video at scale. Its report separates action-free pretraining on video from an action-conditioned model, trained afterward on a small amount of robot data, that is used for planning. Predicting future latent states, recognizing actions and generating photorealistic frames are different capabilities, and a "world model" claim should say which one it makes.

Language as supervision: CLIP

Plain version. The internet is full of images with captions. Train an image encoder and a text encoder so that each image lands next to its own caption. Afterwards you can classify images by writing the class names as sentences.

CLIP [19] trains on image–text pairs collected from the web. In a batch of N pairs, the image encoder gives normalized embeddings $u_1, \dots, u_N$ and the text encoder gives $v_1, \dots, v_N$. The $N \times N$ similarity matrix $S_{ij} = u_i^\top v_j / \tau$, with a learned temperature τ, has the correct pairs on its diagonal. The loss is the average of two InfoNCE losses, one over rows (which caption fits this image?) and one over columns (which image fits this caption?):

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2}\Big[\frac{1}{N}\sum_i -\log \frac{e^{S_{ii}}}{\sum_j e^{S_{ij}}} + \frac{1}{N}\sum_j -\log \frac{e^{S_{jj}}}{\sum_i e^{S_{ij}}}\Big].$$

Zero-shot classification follows directly. Embed one prompt per class, such as "a photo of a dog", and assign an image to the class whose prompt embedding is most similar. No task-specific labels are needed, although prompt wording affects accuracy.

SigLIP [20] replaces the softmax with an independent **sigmoid** loss on every image–text pair: each entry of the matrix becomes a yes/no question, "do these match?". This removes the need to normalize over the whole batch, which simplifies large-scale distributed training.

```
import torch
from torch.nn import functional as F

torch.manual_seed(0)
D = 16

# --- CLIP-style symmetric loss on N matched image-text pairs
N = 6
img = F.normalize(torch.randn(N, D), dim=1)
txt = F.normalize(img + 0.3 * torch.randn(N, D), dim=1)   # caption i matches image i
logit_scale = torch.tensor(1 / 0.07)
sim = logit_scale * img @ txt.T                           # N x N, diagonal = matches
labels = torch.arange(N)
loss_clip = (F.cross_entropy(sim, labels) + F.cross_entropy(sim.T, labels)) / 2
print("CLIP loss:", round(loss_clip.item(), 3))

# zero-shot classification: embed one prompt per class, pick the most similar
class_prompts = txt[:3]                                   # stand-ins for "a photo of a
 ↪  {class}"
pred = (img @ class_prompts.T).argmax(1)
print("zero-shot predictions for images 0-2:", pred[:3].tolist())

# SigLIP-style alternative: an independent yes/no decision for every pair
t, b = 10.0, -10.0                                        # learnable in practice
signs = 2 * torch.eye(N) - 1                              # +1 on matches, -1 elsewhere
loss_sig = -F.logsigmoid(signs * (t * img @ txt.T + b)).sum() / N
print("sigmoid loss:", round(loss_sig.item(), 3))
```

The synthetic captions are noisy copies of the image embeddings, so the CLIP loss is small (0.054) and the zero-shot step recovers the right "classes" `[0, 1, 2]`. In real CLIP the two encoders are a vision transformer or CNN and a text transformer, and the hard part is the data: what the captions describe, and what they leave out, determines what the representation knows. The sigmoid loss value (3.473) is not comparable to the softmax loss; it uses different units and a bias term b.

DINOv2 and DINOv3: features as a product

Plain version. With enough curated data and compute, pretraining can produce an encoder that you freeze and reuse for many different tasks, from classification to depth estimation.

DINOv2 [21] combines image-level self-distillation (as in DINO) with a patch-level masked objective, and pays close attention to **data curation**: building a large, diverse, deduplicated pretraining set from uncurated images by retrieval against curated ones. The result is a family of frozen encoders whose global and patch features work well with simple heads. DINOv3 [22], released in 2025, scales this further and adds **Gram anchoring** to counter the degradation of dense (patch) features during long training. Rather than only matching each feature independently, Gram anchoring constrains the matrix of similarities among patch features toward that of an earlier model. The report also distills a very large teacher into smaller models for deployment.

The practical question is whether frozen global or dense features solve **your** task. A model strong at semantic retrieval may miss a small scratch on a manufactured part. Increasing the input resolution and using patch features can help, but it changes the compute budget and the comparison protocol.

Evaluate transfer honestly

Plain version. A representation has earned its name only when it helps on data and tasks that were not used to build or choose it. How you test it changes the answer.

Protocol	What is trained	What it measures
k-NN	Nothing; similarity to labeled reference features	Local structure of the representation
Linear probe	Linear head; encoder frozen	Information accessible through a simple readout
Fine-tuning	Head plus some or all encoder layers	Adaptability under a stated label and compute budget
Dense transfer	Spatial head on patch maps	Suitability for pixel or region outputs

Linear probing versus fine-tuning

Fine-tuning has more freedom, so it usually wins when the target data is plentiful and similar to the pretraining data. It is not always better. Kumar et al. [23] showed theoretically and empirically that full fine-tuning can **distort** good pretrained features: while a randomly initialized head is still wrong, its large gradients flow into the encoder and change features in directions that suit the training distribution but hurt **out-of-distribution** (OOD) inputs. A linear probe cannot distort the features. Their remedy, **LP-FT**, is two-step: first train the linear head with the encoder frozen, then fine-tune everything starting from that head.

We reproduced the qualitative effect on the digits example. The encoder is the SimCLR CNN from above; the labeled training digits are clean; the test digits are shifted, a mild distribution shift. For each label budget, we average three random draws of the labeled examples.

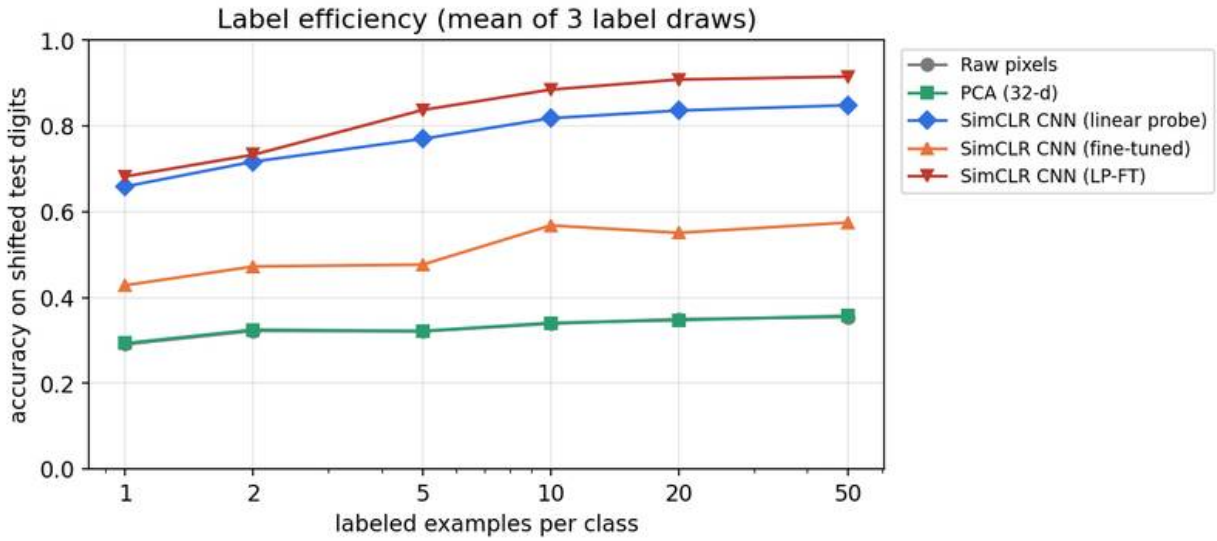


Figure 8 — Label efficiency on shifted test digits. Self-supervised features are far more label-efficient than pixels or PCA. With one fixed recipe, plain fine-tuning from a random head hurt the shifted-test accuracy, while LP-FT improved on the probe. A toy result: with a tuned learning rate or more labels, plain fine-tuning can do better.

Read Figure 8 for its shape, not its exact values. The frozen SSL features with a linear head need only a handful of labels per class to beat raw pixels with 50 labels per class. Plain fine-tuning with a randomly initialized head did worse than the frozen probe here, and LP-FT did better than both, the same ordering Kumar et al. report under distribution shift. Fine-tuning needs its own hyperparameter search, and the head initialization matters.

Rules against leakage

- Normalize features consistently for similarity comparisons (cosine k-NN needs normalized vectors).
- Fit scalers, PCA and classifier hyperparameters on training and validation data only.
- A frozen encoder still has a training history: overlap between pretraining data and evaluation data can inflate results, especially for web-scale models.
- Report the number of labels, backbone, checkpoint, input resolution, augmentations, head, tuning budget and protocol. Choosing the best of many encoders on the test set is test leakage, even if none of them is fine-tuned.

Modern view

Surveys and review papers. Four reviews map the field from complementary angles.

- **Bengio, Courville and Vincent (TPAMI 2013) [1]** frame the whole enterprise. They ask what makes one representation better than another and answer with a list of generic priors: distributed codes, depth and abstraction, disentangling of explanatory factors, manifolds and natural clustering. They review probabilistic models, autoencoders and manifold learning. It predates modern SSL but remains the best statement of *why* we learn representations.
- **Liu et al. (TKDE 2023) [3]** classify self-supervised methods into **generative** (autoencoding and autoregressive reconstruction), **contrastive** (context–instance and instance–instance) and **generative-contrastive** (adversarial) groups across vision, language and graphs, and discuss why contrastive objectives were winning on vision benchmarks at the time.
- **Gui et al. (TPAMI 2024) [4]** survey algorithms more recently, covering contrastive, self-distillation, redundancy-reduction and masked image modeling methods, and discuss applications and open problems such as theory, the choice of pretext task and multimodal SSL.
- **Balestriero et al. (2023) [5]** is a practitioner's guide written by many of the field's authors. It groups methods into a deep-metric-learning family (SimCLR and relatives), a self-distillation family (BYOL, SimSiam, DINO), a canonical-correlation-analysis family (VICReg, Barlow Twins) and masked image modeling, and collects the "tricks" (augmentations, projectors, EMA schedules, batch-size effects) that the papers often bury. Prince's textbook [24] gives a gentle, figure-rich introduction to the same ideas with notebooks.

State of practice (2023–2026). The trends visible in the papers cited above are:

- **Combined objectives.** Leading image encoders mix image-level distillation with patch-level masked prediction (DINOv2 [21], DINOv3 [22]) rather than relying on one pretext task.
- **Data as a first-class design choice.** DINOv2's curation pipeline and the web-scale pair collection behind CLIP [19] and SigLIP [20] show that what goes into pretraining matters as much as the loss.
- **Dense features.** Segmentation, depth and correspondence need good patch features, which motivated objectives like Gram anchoring [22].
- **Prediction in latent space and video.** I-JEPA [17] and V-JEPA 2 [18] predict features instead of pixels, and extend to video and action-conditioned prediction.
- **Language-aligned encoders as the default backbone** for multimodal models (see ML 09).

Open problems. We still lack a predictive theory of which pretext task suits which downstream task; collapse is prevented by recipes more than by guarantees; evaluation depends heavily on protocol and on overlap between pretraining and test data; and web-scale pretraining inherits biases and gaps from its data. Our reading of [3], [4], [5], [21], [22] is that data and target design

deserve as much attention as architecture. That is an interpretation of the literature, not a theorem that one objective wins everywhere.

The next chapters change the question: instead of *describing* an image, how can a model describe and **sample** its whole distribution? See ML 05A for autoencoders and ML 06 for generative models.

Key takeaways

- A representation is useful when a simple head can solve the task from it with few labels. Distributed codes generalize because features are shared across examples.
- The invariances you want depend on the task. In self-supervised learning, the augmentations choose them, as the shifted-digits example showed.
- Contrastive methods (SimCLR, MoCo) are InfoNCE classification: find your positive among negatives. Collapse costs them $\log(2B-1)$.
- Methods without negatives (BYOL, SimSiam, DINO) avoid collapse through asymmetry: predictors, stop-gradient, EMA teachers, centering and sharpening. Barlow Twins and VICReg put the anti-collapse term in the loss.
- Masked modeling (BEiT, MAE) reconstructs hidden content; JEPA predicts hidden *features*. CLIP aligns images with captions and enables zero-shot classification.
- k-NN, linear probes and fine-tuning answer different questions. Fine-tuning can distort features under distribution shift; LP-FT is a simple fix.

Exercises

1. **Temperature and collapse.** In the first code block, change the temperature from 0.2 to 0.05 and to 1.0. Does the collapsed loss change? Does the paired loss?

   Hint

   The collapsed loss stays $\log 15$: all non-self logits are equal, so dividing them by any τ leaves the softmax uniform. The paired loss changes, because temperature rescales the *differences* between scores, and with real differences a smaller τ gives a sharper softmax.

2. **Augmentations are assumptions.** You want features for (a) counting cells in microscopy, (b) recognizing bird species, (c) reading street-sign text. For each, name one standard SimCLR augmentation you would remove and why.

   Hint

   (a) Random cropping can change the number of cells in view, so the "same" label is no longer the same. (b) Strong color jitter can remove plumage color, a key species cue. (c) Horizontal flips turn text into mirror text. Decide what must be preserved before choosing invariances.

3. **MAE compute.** An image has 196 patches and MAE masks 75%. How many patches enter the encoder? By what factor does the number of pairwise attention scores in the encoder shrink?

Hint

49 visible patches enter the encoder. Attention scores scale with the square of the token count: $(49/196)^2 = 1/16$. The decoder still processes all 196 positions, but it is small.

4. **Break the digits experiment.** Modify the SimCLR digits code so that `augment` only adds noise (no shift). Retrain and compare the shifted-test k-NN accuracy with the original. What does this tell you?

Hint

Replace the random `i, j` with `i, j = 1, 1`. On our machine the shifted-test accuracy fell from 0.919 to 0.743, while clean accuracy stayed at 0.981. Without the shift augmentation, nothing in the objective asks for shift invariance; the robustness that remains comes from what a trained convolution-and-pooling encoder learns anyway (the untrained one in the main example had none). The augmentations, not the word "self-supervised", decide most of what is learned.

5. **SimSiam ablation, on paper.** In SimSiam, suppose you remove the stop-gradient so both branches receive gradients. Explain in two sentences why a constant output becomes an easy minimizer, and why the predictor alone is not enough to prevent it.

Hint

With gradients on both sides, the loss $-\cos(p_1, z_2)$ is minimized whenever p_1 and z_2 point the same way, which a constant encoder plus a predictor that outputs that same constant direction achieves for every input. The stop-gradient changes the optimization dynamics so that the target is held fixed during each step; the SimSiam paper reports rapid collapse without it.

6. **Fair probe.** Design an experiment comparing a frozen pretrained encoder with PCA plus logistic regression at 10, 50 and 100 labels per class. List three things you must hold fixed and two things you must report.

Hint

Hold fixed: the labeled subsets (same examples for both), the held-out test split, and the procedure for choosing regularization (on validation data only). Report: the pretrained encoder's training data (and any overlap with your test set) and the number of random label draws with their spread.

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ML 05A • Linear Factor Models and Autoencoders

PCA, probabilistic PCA, factor analysis, ICA, sparse coding, NMF and autoencoders (undercomplete, sparse, denoising, contractive): how compression, noise assumptions and the manifold view lead to VAEs.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-05a-factor-models-autoencoders/

What you'll learn

- How a small number of hidden **factors** can explain many correlated measurements, and why that idea underlies both classical statistics and modern deep generative models.
- How PCA, probabilistic PCA and factor analysis differ, and what each one assumes about noise.
- How independent component analysis (ICA), sparse coding and nonnegative matrix factorization (NMF) pick *different* factors by adding different assumptions.
- How linear and nonlinear autoencoders relate to these models, and how sparse, denoising and contractive penalties change what an autoencoder learns.
- What the **manifold view** says about high-dimensional data, and why it leads naturally from autoencoders to variational autoencoders (VAEs).

The big picture

An 8×8 digit image has 64 pixel values, and a phone photo has millions. Yet the meaningful variation in a collection of such images depends on far fewer underlying quantities: which digit it is, the stroke thickness, the slant, the position, the lighting. A **factor model** tries to explain each observation through a handful of hidden (latent) coordinates. An **autoencoder** is a neural network that learns to squeeze an input into such coordinates and expand it back again.

> **Explain like I'm five**
>
> Imagine describing a friend's face over the phone. You don't read out every freckle; you say "round face, short hair, big smile, light from the left." Those few words are factors: they let the listener rebuild a good picture. A factor model looks for the words that explain the most about many faces at once. An autoencoder is a pair of helpers: one who turns a picture into a few words, and one who turns the words back into a picture. They practise together until the rebuilt pictures look right.

Why spend a whole chapter on this? Because almost every idea in modern representation learning and generative modeling has an ancestor here. Dimensionality reduction for visualization, denoising, compression, anomaly detection, the "tokenizers" that turn images into compact codes

for diffusion models, and the probabilistic machinery of VAEs all start from the same question: *what small set of causes could have produced these observations?* The progression from linear factor models to autoencoders follows Goodfellow, Bengio and Courville [1, ch. 13–14] and Murphy [2]; Bengio, Courville and Vincent's review [3] gives the broader motivation for why learned representations matter.

One template, many models

Every method in this chapter fits the same template. There is an **encoder** that maps an observation $x \in \mathbb{R}^d$ to a code $z \in \mathbb{R}^k$, and a **decoder** that maps a code back to a reconstruction $\hat{x}$. The models differ in three choices:

Choice	Options in this chapter
Shape of encoder/decoder	linear (PCA, factor analysis, ICA, NMF, sparse coding) or nonlinear (deep autoencoders)
Size of the code	**undercomplete** ($k < d$: the bottleneck forces compression) or **overcomplete** ($k \geq d$: something else must stop the model from copying)
Extra assumption or penalty	Gaussian noise (PPCA, FA), independence and non-Gaussianity (ICA), sparsity (sparse coding, sparse autoencoders), nonnegativity (NMF), robustness to corruption (denoising), insensitivity to small input changes (contractive)

Keep this table in mind. When two methods disagree, it is almost always because they made a different choice in one of these rows, not because one of them is "wrong."

PCA: the best linear reconstruction subspace

Plain version. Find a few orthogonal directions such that, if you keep only the coordinates of each example along those directions, you lose as little as possible.

Let $X \in \mathbb{R}^{n \times d}$ hold n centered examples as rows (the column means have been subtracted). PCA chooses a matrix $W \in \mathbb{R}^{d \times k}$ with orthonormal columns that minimizes the squared reconstruction error

$$\min_{W} \ \|X - XWW^{\top}\|_F^2 \qquad \text{subject to} \qquad W^{\top}W = I_k.$$

Here $\|\cdot\|_F$ is the Frobenius norm (the square root of the sum of squared entries), $Z = XW \in \mathbb{R}^{n \times k}$ are the **codes** (also called scores), and $\hat{X} = ZW^{\top}$ is the reconstruction. The encoder is $x \mapsto W^{\top}x$ and the decoder is $z \mapsto Wz$: PCA is already an autoencoder, just a linear one with tied weights.

Where the solution comes from

Two equivalent views lead to the same answer.

Maximum variance. Take a single unit vector w. The variance of the projected data Xw is $w^\top Sw$, where $S = \frac{1}{n-1}X^\top X$ is the sample covariance. Maximize $w^\top Sw$ subject to $w^\top w = 1$ with a Lagrange multiplier λ:

$$\nabla_w(w^\top Sw - \lambda(w^\top w - 1)) = 0 \quad \Longrightarrow \quad Sw = \lambda w.$$

So w must be an eigenvector of S, and the variance it captures is its eigenvalue λ. The best direction is the eigenvector with the largest eigenvalue; the next best direction, constrained to be orthogonal to the first, is the second eigenvector, and so on.

Minimum reconstruction error. Because W has orthonormal columns, the error splits into "total variance minus kept variance." Minimizing the error is therefore the same as maximizing the kept variance, and the residual equals the sum of the discarded eigenvalues (times $n-1$). The Eckart–Young theorem states the same result in matrix language: truncating the singular value decomposition $X = U\Sigma V^\top$ to its k largest singular values gives the best rank-k approximation of X in Frobenius norm. The PCA directions are the first k right singular vectors (columns of V), and the covariance eigenvalues are $\sigma_j^2/(n-1)$.

Code: PCA from the SVD

The snippet fits PCA on the training split of scikit-learn's bundled digits, evaluates reconstruction on held-out images, and checks the result against scikit-learn's own implementation.

```
import numpy as np
from sklearn.datasets import load_digits
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split

X = load_digits().data / 16.
train, test = train_test_split(X, test_size=.2, random_state=0)

mean = train.mean(axis=0)
U, s, Vt = np.linalg.svd(train - mean, full_matrices=False)
var = s**2 / (len(train) - 1)              # eigenvalues of the covariance
ratio = np.cumsum(var) / var.sum()
for k in (2, 10, 20):
    W = Vt[:k].T                            # d x k, orthonormal columns
    rec = (test - mean) @ W @ W.T + mean
    print(f"k={k:2d}  explained={ratio[k-1]:.3f}  "
          f"test MSE={np.mean((test - rec)**2):.4f}")

sk = PCA(n_components=10).fit(train)
print("matches sklearn:", np.allclose(np.abs(sk.components_), np.abs(Vt[:10])))
```

Output:

```
k= 2  explained=0.291  test MSE=0.0546
k=10  explained=0.740  test MSE=0.0202
k=20  explained=0.896  test MSE=0.0084
matches sklearn: True
```

The `np.abs` in the last line is not a hack: each principal direction is only defined up to sign, because w and $-w$ span the same line. Note that the mean is estimated on the training split only and reused for the test split; estimating it on test data would leak information.

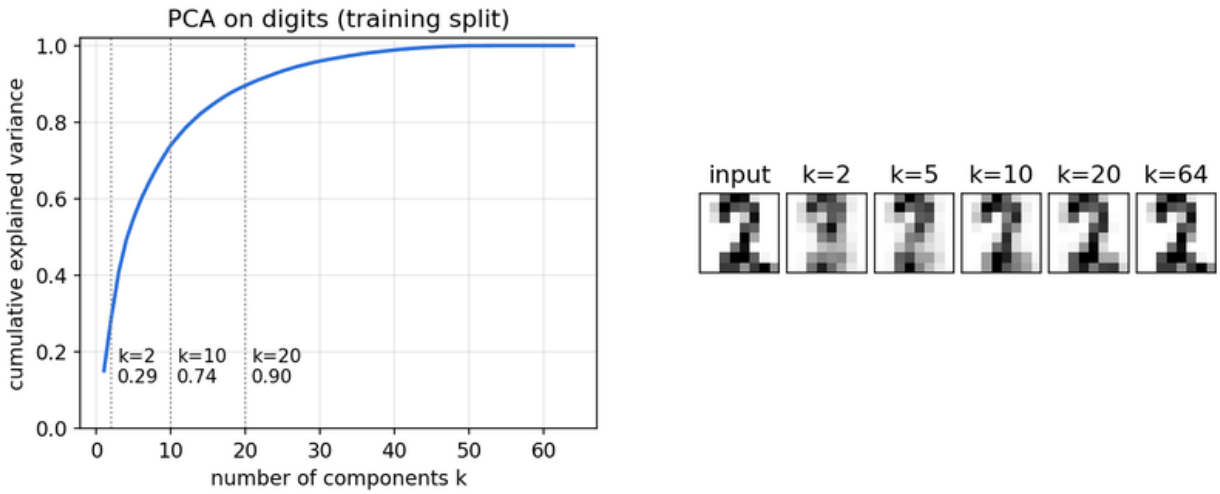


Figure 1 — PCA on the bundled digits. Left: cumulative explained variance (fit on the training split). Right: one held-out digit reconstructed from k components; k = 64 reproduces the input exactly.

> **Interactive demonstration (online)**: open the online edition of this tutorial to use this demonstration.Open demonstration

What PCA does not promise

Three cautions follow directly from the objective.

- **Centering and scaling change the answer.** Without centering, the first "component" mostly points at the mean. Standardizing every feature to unit variance gives each pixel equal weight, so raw-pixel PCA and standardized PCA solve different problems.
- **Variance is not relevance.** The direction of largest variance can be lighting or background rather than the class you care about. PCA is unsupervised; it maximizes retained variance, not class separation (that is the job of methods such as linear discriminant analysis).

- **PCA is linear.** If the data lie on a curved surface, a flat subspace may need many dimensions to describe what is really a low-dimensional shape. We return to this in the manifold section.

Probabilistic PCA and factor analysis

Plain version. Instead of just finding a subspace, write down a story for how the data were generated: draw a few hidden factors, mix them linearly, then add noise. Then fit that story by maximum likelihood.

Both models share the generative equation

$$x = \mu + Wz + \epsilon, \qquad z \sim \mathcal{N}(0, I_k),$$

where $\mu \in \mathbb{R}^d$ is the mean, $W \in \mathbb{R}^{d \times k}$ is the **loading matrix** (how each factor shows up in each measured coordinate), z is the vector of latent factors, and ϵ is noise independent of z. They differ only in the noise.

Probabilistic PCA (PPCA)

In PPCA [4], the noise is isotropic, $\epsilon \sim \mathcal{N}(0, \sigma^2 I_d)$: every coordinate gets the same noise variance. Integrating out z gives a Gaussian marginal

$$x \sim \mathcal{N}(\mu,\, C), \qquad C = WW^\top + \sigma^2 I_d.$$

The covariance has a clear interpretation: a low-rank part $WW^\top$ that the factors explain, plus a constant "floor" σ^2 that they do not. Tipping and Bishop showed that the maximum-likelihood solution is available in closed form [4]:

$$W_{\mathrm{ML}} = U_k(\Lambda_k - \sigma^2 I)^{1/2} R, \qquad \sigma^2_{\mathrm{ML}} = \frac{1}{d-k} \sum_{j=k+1}^{d} \lambda_j,$$

where U_k holds the top k eigenvectors of the sample covariance, Λ_k their eigenvalues $\lambda_1, \dots, \lambda_k$, and R is any $k \times k$ orthogonal matrix. In words: the loadings span exactly the PCA subspace, and the noise variance is the average of the variance PCA throws away. PPCA therefore *is* PCA, plus a likelihood. That likelihood is useful: you can compare models with different k on held-out data, handle missing values with expectation–maximization (EM), and build mixtures of PPCA models.

Given an observation, the posterior over its factors is Gaussian with mean

$$\mathbb{E}[z \mid x] = M^{-1} W^\top (x - \mu), \qquad M = W^\top W + \sigma^2 I_k.$$

This is the probabilistic encoder. As $\sigma^2 \to 0$ it approaches the ordinary PCA projection; for $\sigma^2 > 0$ it shrinks codes toward zero, because part of what we see is attributed to noise.

Factor analysis (FA)

Factor analysis keeps the same structure but allows a diagonal noise covariance, $\epsilon \sim \mathcal{N}(0, \Psi)$ with $\Psi = \text{diag}(\psi_1, \dots, \psi_d)$. Each observed coordinate has its own residual noise level ψ_i, while the latent factors explain the **shared** covariance between coordinates. The marginal is $x \sim \mathcal{N}(\mu, WW^\top + \Psi)$, and the posterior mean of the factors is $\mathbb{E}[z \mid x] = W^\top (WW^\top + \Psi)^{-1}(x - \mu)$.

There is no closed-form solution, so FA is fitted with EM or direct likelihood optimization. The extra flexibility matters when measurements come from sensors with unequal noise, or when some pixels are almost constant (such as the always-blank border pixels of the digits). PCA happily spends components on any high-variance coordinate; FA can instead explain an individually noisy coordinate as "noise," and reserve its factors for correlations.

The rotation problem

Both models share an ambiguity. For any orthogonal R, replacing W by WR and z by $R^\top z$ leaves $WW^\top$, and hence the distribution of x, unchanged, because $R^\top z$ is still standard normal. The *subspace* is identified; the individual axes are not. That is why the factor-analysis literature uses rotations such as varimax to pick more interpretable axes, and why a factor is not automatically a physical cause. To pin down individual factors we must assume something beyond Gaussianity, which is precisely what ICA does.

Independent component analysis (ICA)

Plain version. Several people talk at once in a room, and several microphones record different mixtures of their voices. ICA tries to recover each voice by finding mixtures that are as statistically independent, and as non-Gaussian, as possible.

The noise-free ICA model is

$$x = As, \qquad s = (s_1, \dots, s_k) \text{ with independent components,}$$

where $x \in \mathbb{R}^d$ is the observed mixture, $A \in \mathbb{R}^{d \times k}$ is an unknown **mixing matrix**, and the sources s_j are independent and **non-Gaussian**. The goal is an unmixing matrix $B \approx A^{-1}$ (or a pseudo-inverse) so that $\hat{s} = Bx$ recovers the sources [5].

Why non-Gaussianity is the key

The rotation problem from the previous section explains why ICA needs non-Gaussian sources.

First **whiten** the data: transform x so that its covariance is the identity (PCA followed by rescaling each component to unit variance does this). After whitening, the remaining unknown is just a

rotation. If the sources were Gaussian, every rotation of independent unit Gaussians would again be independent unit Gaussians, so no rotation could be preferred. ICA's identifiability theorem states the converse: if **at most one** source is Gaussian, the sources can be recovered up to three harmless ambiguities [5]:

- **order** (we cannot know which source is "first"),
- **sign**, and
- **scale** (a louder source with a quieter mixing column looks the same).

The central limit theorem gives the intuition for how to find the right rotation: a sum of independent variables tends to be *more* Gaussian than its parts. So the projections of x that are *least* Gaussian are the ones most likely to be single sources rather than mixtures. ICA algorithms therefore maximize a measure of non-Gaussianity. A principled one is **negentropy**,

$$J(y) = H(y_{\text{gauss}}) - H(y) \;\geq 0,$$

where H is differential entropy and y_{gauss} is a Gaussian with the same variance as y. Among all distributions with a given variance, the Gaussian has the largest entropy, so $J(y) = 0$ exactly for Gaussian y. Because entropy is hard to estimate, the **FastICA** algorithm approximates negentropy with contrast functions such as $G(u) = \log\cosh u$, and finds each unmixing direction with a fast fixed-point iteration [5]. Maximizing non-Gaussianity is closely related to minimizing the mutual information between the estimated components and to maximum-likelihood estimation of the ICA model.

Code: unmix three signals

The toy problem uses a sine wave, a square wave and a sawtooth, all non-Gaussian, mixed by a fixed matrix.

```
import numpy as np
from sklearn.decomposition import FastICA, PCA

rng = np.random.default_rng(0)
t = np.linspace(0, 8, 2000)
S = np.c_[np.sin(2*t),                  # smooth sine
          np.sign(np.sin(3*t)),         # square wave
          2*(t % 1) - 1]                # sawtooth
S += 0.05 * rng.normal(size=S.shape)
A = np.array([[1.0, 1.0, 1.0],
              [0.5, 2.0, 1.0],
              [1.5, 1.0, 2.0]])         # unknown mixing matrix
X = S @ A.T                             # three "microphones"

def best_match(est):
    C = np.abs(np.corrcoef(S.T, est.T)[:3, 3:])
```

```
    return np.round(C.max(axis=1), 3)

ica = FastICA(n_components=3, whiten="unit-variance", random_state=0)
print("ICA |corr| with each source:", best_match(ica.fit_transform(X)))
print("PCA |corr| with each source:", best_match(PCA(3).fit_transform(X)))
```

Output:

```
ICA |corr| with each source: [0.997 0.999 0.997]
PCA |corr| with each source: [0.689 0.811 0.735]
```

We compare with absolute correlation and take the best match because ICA returns the sources in arbitrary order, sign and scale. PCA only decorrelates; it finds orthogonal directions of high variance, which here are still mixtures.

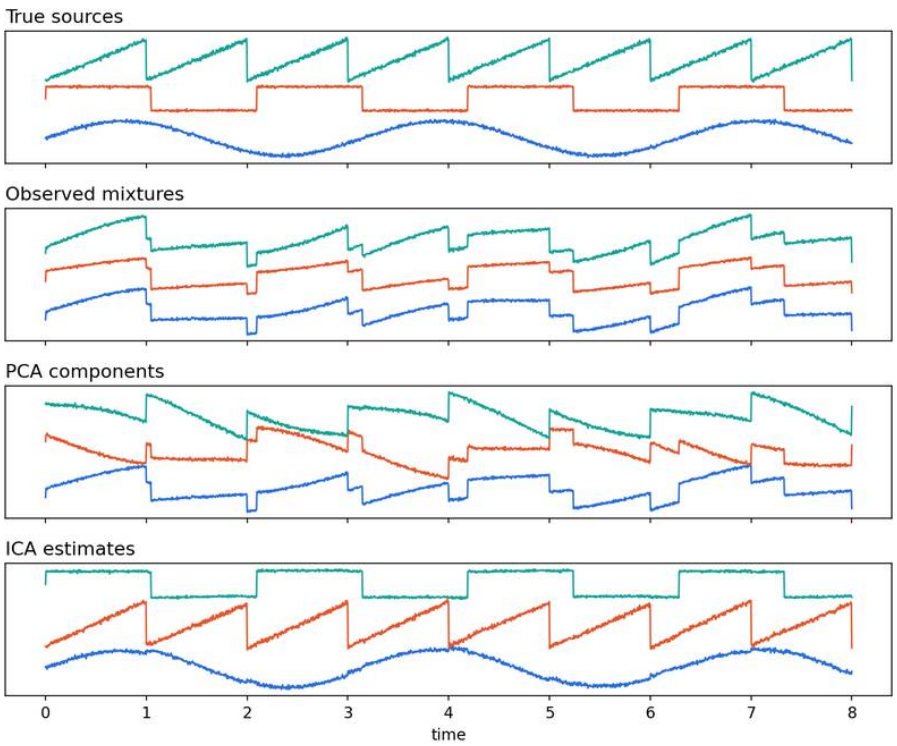


Figure 2 — Blind source separation. PCA components remain mixtures; FastICA recovers the sine, square and sawtooth waves up to order, sign and scale.

ICA is widely used for separating artifacts in EEG and other sensor recordings. Its assumptions also mark its limits: it needs genuinely independent, non-Gaussian sources and a mixing process that is close to linear and instantaneous. On image patches, ICA tends to learn localized, oriented edge-like filters, which brings us to a closely related idea.

Sparse coding

Plain version. Keep a large dictionary of small patterns ("atoms") and describe each input using only a few of them at a time.

Sparse coding learns a dictionary $D = [d_1, \dots, d_m] \in \mathbb{R}^{d \times m}$ and, for each example x_i, a sparse code $\alpha_i \in \mathbb{R}^m$ by solving

$$\min_{D,\,\{\alpha_i\}} \sum_{i=1}^{n} \left(\tfrac{1}{2} \|x_i - D\alpha_i\|_2^2 + \lambda \|\alpha_i\|_1 \right) \qquad \text{subject to} \qquad \|d_j\|_2 \le 1.$$

The first term asks for accurate reconstruction; the ℓ_1 norm $\|\alpha\|_1 = \sum_j |\alpha_j|$ pushes most code entries to exactly zero; $\lambda > 0$ sets the trade-off. The norm constraint on atoms stops the model from cheating by making atoms huge and codes tiny. The dictionary is often **overcomplete** ($m > d$): there are more atoms than pixel dimensions, but only a few are active for any one input.

Olshausen and Field [6] trained this kind of model on patches of natural images and found that the learned atoms were localized, oriented and band-pass, much like the receptive fields of simple cells in the primary visual cortex. The result was influential because it suggested that sparsity alone, without any labels, can explain a known property of biological vision.

Two practical points distinguish sparse coding from the autoencoders later in this chapter:

- **Inference is an optimization.** Given D, finding α for a new input means solving a lasso problem. There is no fixed encoder network; the "encoder" is an algorithm. Training alternates between updating the codes (with D fixed) and updating the dictionary (with codes fixed).
- **Sparsity is a prior.** The ℓ_1 penalty corresponds to a Laplace prior on the codes, so sparse coding can be read as maximum a posteriori inference in a linear factor model with a sparse, non-Gaussian prior, which is the same family that ICA belongs to.

Code: sparsity versus fidelity

```
import numpy as np
from sklearn.datasets import load_digits
from sklearn.decomposition import DictionaryLearning
from sklearn.model_selection import train_test_split

X = load_digits().data / 16.
train, test = train_test_split(X, test_size=.2, random_state=0)
for alpha in (0.1, 1.0):
    dl = DictionaryLearning(n_components=32, alpha=alpha, max_iter=30,
                            transform_algorithm="lasso_lars",
                            transform_alpha=alpha, random_state=0)
    dl.fit(train[:600])
    codes = dl.transform(test)
    rec = codes @ dl.components_
```

```
    active = np.mean(np.abs(codes) > 1e-8, axis=1).mean()
    print(f"alpha={alpha}: active atoms per image={active*32:.1f}/32, "
          f"test MSE={np.mean((test - rec)**2):.4f}")
```

Output (about half a minute on a laptop CPU):

```
alpha=0.1: active atoms per image=17.9/32, test MSE=0.0075
alpha=1.0: active atoms per image=3.0/32, test MSE=0.0401
```

Raising λ (here `alpha`) buys a much sparser description, about three active atoms per image instead of eighteen, at the price of higher reconstruction error. Neither setting is "correct": a compression codec, a denoiser and an interpretability tool would each choose differently. Note that this dictionary does not subtract a mean, so its error is not directly comparable to the PCA numbers above.

Nonnegative matrix factorization (NMF)

Plain version. Describe each example as an *additive* mixture of nonnegative parts: you may add strokes, but you may never subtract one.

For data $X \geq 0$ (pixel intensities, word counts, spectra), NMF finds $Z \geq 0$ and $H \geq 0$ that minimize a reconstruction criterion such as

$$\min_{Z \geq 0,\, H \geq 0} \|X - ZH\|_F^2,$$

where the rows of $H \in \mathbb{R}^{k \times d}$ are nonnegative parts and the rows of $Z \in \mathbb{R}^{n \times k}$ are nonnegative mixing weights. Because nothing can cancel, the parts tend to look like pieces of the object rather than the positive-and-negative "ghost" patterns of PCA. Lee and Seung [7] demonstrated this parts-based behavior on faces and text and popularized simple multiplicative update rules for fitting it.

Two cautions. First, parts-based results are a tendency, not a guarantee: on some datasets the factors are not clean parts. Second, the objective is nonconvex in Z and H jointly, so initialization matters and different runs can give different factors. Scikit-learn's `init="nndsvda"` uses an SVD-based start to make results more stable.

Code: compare factor reconstructions

This CPU example fits each model on training digits only and evaluates reconstruction on held-out images. It compares different model assumptions; it is not a claim that one method wins every task.

```
import numpy as np
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.decomposition import PCA, FactorAnalysis, NMF

X = load_digits().data / 16.
train, test = train_test_split(X, test_size=.2, random_state=0)
models = {
    "PCA": PCA(n_components=12),
    "factor analysis": FactorAnalysis(n_components=12, random_state=0),
    "NMF": NMF(n_components=12, init="nndsvda", max_iter=1000,
               random_state=0),
}
for name, model in models.items():
    model.fit(train)
    z = model.transform(test)
    if name == "factor analysis":
        reconstructed = z @ model.components_ + model.mean_
    else:
        reconstructed = model.inverse_transform(z)
    mse = np.mean((test-reconstructed)**2)
    print(name, "held-out reconstruction MSE:", round(mse, 5))
```

Output:

```
PCA held-out reconstruction MSE: 0.01652
factor analysis held-out reconstruction MSE: 0.02501
NMF held-out reconstruction MSE: 0.02207
```

PCA wins on squared reconstruction error, as it must: among all rank-12 linear reconstructions, the PCA subspace is optimal for this criterion (up to the train/test split). That does not make it the best *model*. The factor-analysis reconstruction above is a posterior-mean reconstruction that deliberately attributes part of each pixel to noise, and FA's real output, a full probability model with per-pixel noise, is not summarized by MSE at all. NMF trades some error for nonnegative, additive parts.

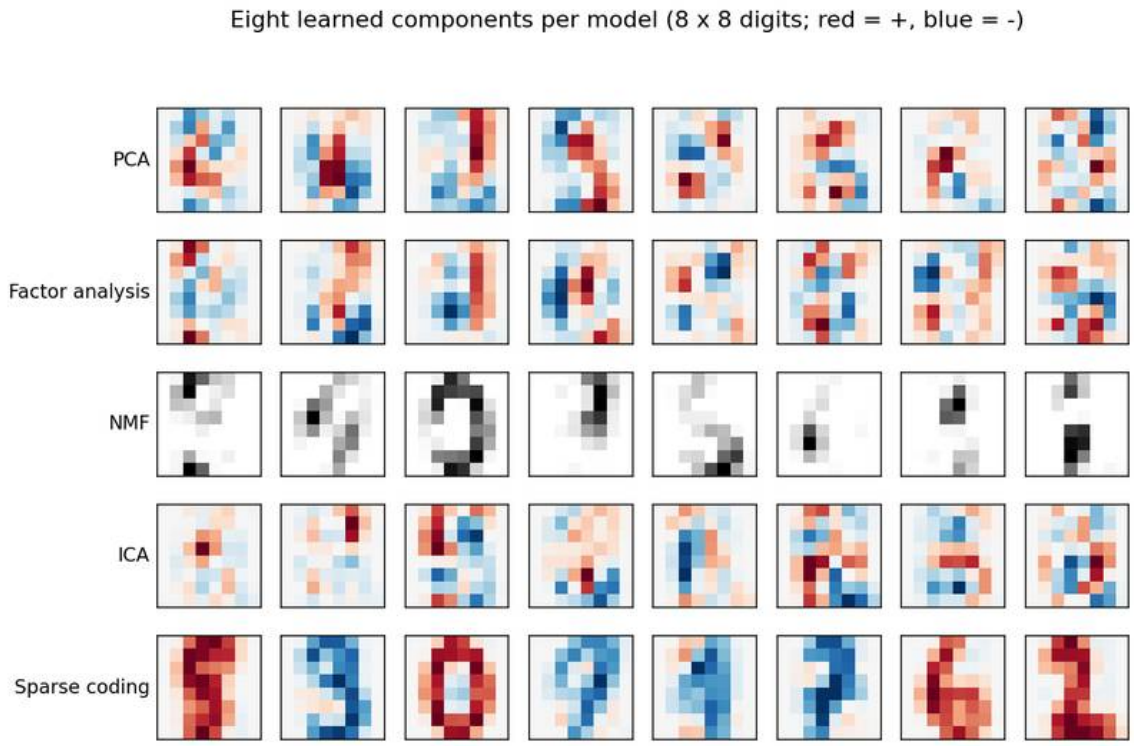


Figure 3 — Eight components from each linear factor model on the same digits. PCA and FA give signed, global patterns; NMF gives nonnegative strokes; ICA gives signed but more localized patterns; sparse-coding atoms resemble prototypes that are combined a few at a time.

Linear autoencoders and PCA

Plain version. If you train a neural network with no nonlinearities to compress and reconstruct its input, it ends up doing PCA, though possibly in rotated or stretched coordinates.

A linear autoencoder with centered input computes

$$z = W_e x, \qquad \hat{x} = W_d z, \qquad \mathcal{L}(W_e, W_d) = \frac{1}{n} \sum_i \|x_i - W_d W_e x_i\|^2,$$

with encoder $W_e \in \mathbb{R}^{k \times d}$ and decoder $W_d \in \mathbb{R}^{d \times k}$, $k < d$. The product $W_d W_e$ has rank at most k, so by Eckart–Young the best possible product is the projection onto the top-k principal subspace. Baldi and Hornik [8] proved more: for this squared-error linear network, every local minimum is a global one, and at the global minimum the network projects onto the principal subspace. The individual weight matrices, however, are not unique. For any invertible $k \times k$ matrix G, replacing W_e by GW_e and W_d by $W_d G^{-1}$ gives the same product, so the code coordinates can be any invertible linear transform of the PCA scores.

Code: a linear autoencoder rediscovers the PCA subspace

```
import numpy as np, torch
from sklearn.datasets import load_digits
from sklearn.decomposition import PCA
from scipy.linalg import subspace_angles

torch.manual_seed(0)
X = load_digits().data / 16.
X = X - X.mean(axis=0)
Xt = torch.tensor(X, dtype=torch.float32)
k = 5
enc = torch.nn.Linear(64, k, bias=False)
dec = torch.nn.Linear(k, 64, bias=False)
opt = torch.optim.Adam([*enc.parameters(), *dec.parameters()], lr=1e-2)
for step in range(3000):
    loss = (dec(enc(Xt)) - Xt).square().mean()
    opt.zero_grad(); loss.backward(); opt.step()

Wd = dec.weight.detach().numpy()            # 64 x k decoder columns
P = PCA(n_components=k).fit(X).components_.T
angles = np.degrees(subspace_angles(Wd, P))
print("largest angle to PCA subspace (deg):", round(angles.max(), 3))
print("decoder columns orthonormal?",
      np.allclose(Wd.T @ Wd, np.eye(k), atol=1e-2))
```

Output:

```
largest angle to PCA subspace (deg): 0.003
decoder columns orthonormal? False
```

The subspace angle (the largest angle between the two k-dimensional subspaces) is essentially zero: the network found the PCA subspace. Yet its decoder columns are not orthonormal, so they are not the principal components themselves. This is the coordinate ambiguity in action, and it is the first hint of a theme that recurs through the chapter: reconstruction alone pins down *what is kept*, not *how it is labeled*.

Nonlinear autoencoders

Plain version. Replace the linear maps with neural networks, so the code can describe curved structure that no flat subspace can capture.

A general autoencoder has an encoder f_ϕ and a decoder g_θ, both neural networks:

$$z = f_\phi(x), \qquad \hat{x} = g_\theta(z), \qquad \min_{\phi,\theta} \frac{1}{n} \sum_i \ell(x_i,\, g_\theta(f_\phi(x_i))),$$

where ℓ is a reconstruction loss, such as squared error for real values or binary cross-entropy for values in $[0, 1]$. The choice of ℓ is itself an assumption: squared error corresponds to a Gaussian likelihood around $\hat{x}$, cross-entropy to a Bernoulli likelihood.

Hinton and Salakhutdinov [9] showed in 2006 that deep autoencoders, initialized with greedy layer-wise pretraining and then fine-tuned end to end, could find low-dimensional codes that reconstructed images much better than PCA with the same number of dimensions. Today, better initializations, normalization and optimizers make the pretraining step unnecessary, but the core lesson stands: nonlinear encoders can compress much more aggressively than linear ones when the data are curved.

Undercomplete is not enough, overcomplete needs a rule

An **undercomplete** autoencoder ($k < d$) is forced to compress. That is useful, but not a full guarantee of good features. A sufficiently powerful encoder and decoder could, in principle, map each training example to its own arbitrary code (say, its index) and memorize the decoding, achieving low training error without learning any reusable structure [1, ch. 14]. An **overcomplete** autoencoder ($k \geq d$) can simply learn the identity function. In both cases, what makes the representation useful is not the bottleneck by itself but the *inductive bias*: limited capacity, the architecture (convolutions, for instance), and explicit penalties. The next section covers the three classic penalties, which together are called **regularized autoencoders**.

Regularized autoencoders: sparse, denoising and contractive

Each penalty answers the question "what should the code ignore?" in a different way.

Sparse autoencoders

Plain version. Allow many code units, but require that only a few of them be active for any given input.

The loss adds a sparsity penalty on the code $h = f_\phi(x)$:

$$\mathcal{L} = \ell(x,\, g_\theta(h)) + \lambda\, \Omega(h), \qquad \Omega(h) = \sum_j |h_j|.$$

Alternatives penalize the deviation of each unit's average activation from a small target rate, or keep only the k largest activations (a "top-k" or k-sparse autoencoder). A sparse autoencoder is the amortized cousin of sparse coding: the encoder network predicts the sparse code in one forward pass, instead of solving a lasso problem for every input. Goodfellow et al. interpret the penalty as an approximate prior over codes rather than an ordinary weight regularizer [1, ch. 14].

As the Modern view explains, sparse autoencoders have recently become a key tool for interpreting the internals of large neural networks.

Denoising autoencoders

Plain version. Show the network a damaged input and ask it to rebuild the clean one. To succeed, it must learn what clean data look like.

A **denoising autoencoder (DAE)** draws a corrupted copy $\tilde{x} \sim C(\tilde{x} \mid x)$ and minimizes

$$\mathcal{L} = \mathbb{E}_x \, \mathbb{E}_{\tilde{x} \sim C(\cdot \mid x)} [\, \ell(x, \, g_\theta(f_\phi(\tilde{x}))) \,],$$

where C is a chosen corruption process: additive Gaussian noise, masking noise (setting a random fraction of inputs to zero), or salt-and-pepper noise [10], [11]. Vincent et al. introduced the idea in 2008 [10] and showed in 2010 that stacking DAE layers gave useful features for classifiers [11]. Because the input differs from the target, copying is no longer a solution, even for an overcomplete code.

The corruption defines what the model learns to be robust to. A DAE trained on Gaussian noise has no reason to fill in missing patches, and one trained with masking noise learns to complete missing parts. Modern masked image modeling, such as the masked autoencoder (MAE) in ML 05, is a direct descendant of masking-noise DAEs.

There is also a deeper geometric reading. For small Gaussian corruption, the vector $g(f(\tilde{x})) - \tilde{x}$ points from a corrupted point back toward regions of high data density; it approximately estimates the **score** $\nabla_x \log p(x)$, the gradient of the log-density [1, ch. 14]. This connection is one of the roots of score-based and diffusion generative models, covered in ML 07.

Code: a denoising autoencoder on digits

This standalone PyTorch example trains a small nonlinear DAE on training digits and evaluates clean reconstruction from noisy held-out input.

```
import torch
from torch import nn
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split

torch.manual_seed(0); torch.set_num_threads(1)
train, test = train_test_split(load_digits().data/16., test_size=.2, random_state=0)
X = torch.tensor(train, dtype=torch.float32)
Xt = torch.tensor(test, dtype=torch.float32)
encoder = nn.Sequential(nn.Linear(64, 32), nn.ReLU(), nn.Linear(32, 12))
decoder = nn.Sequential(nn.Linear(12, 32), nn.ReLU(), nn.Linear(32, 64), nn.Sigmoid())
model = nn.Sequential(encoder, decoder)
opt = torch.optim.Adam(model.parameters(), lr=.003)
for step in range(2000):
```

```
    clean = X[torch.randint(len(X), (64,))]
    noisy = (clean + .2*torch.randn_like(clean)).clamp(0, 1)
    loss = (model(noisy)-clean).square().mean()
    opt.zero_grad(); loss.backward(); opt.step()
model.eval()
with torch.no_grad():
    noisy = (Xt + .2*torch.randn_like(Xt)).clamp(0, 1)
    print("noisy-input MSE:", (noisy-Xt).square().mean().item())
    print("reconstructed MSE:", (model(noisy)-Xt).square().mean().item())
```

Output (about 0.0248 for the noisy input and 0.0197 after the DAE; the last digits depend on the PyTorch build):

```
noisy-input MSE: 0.024781296029686928
reconstructed MSE: 0.01968955807387829
```

The improvement is real but modest, and Figure 4 shows why: with a 12-dimensional code, the DAE removes the speckle but also smooths away some stroke detail. Denoising and faithful compression pull in different directions; a wider code or a convolutional architecture would shift the balance.

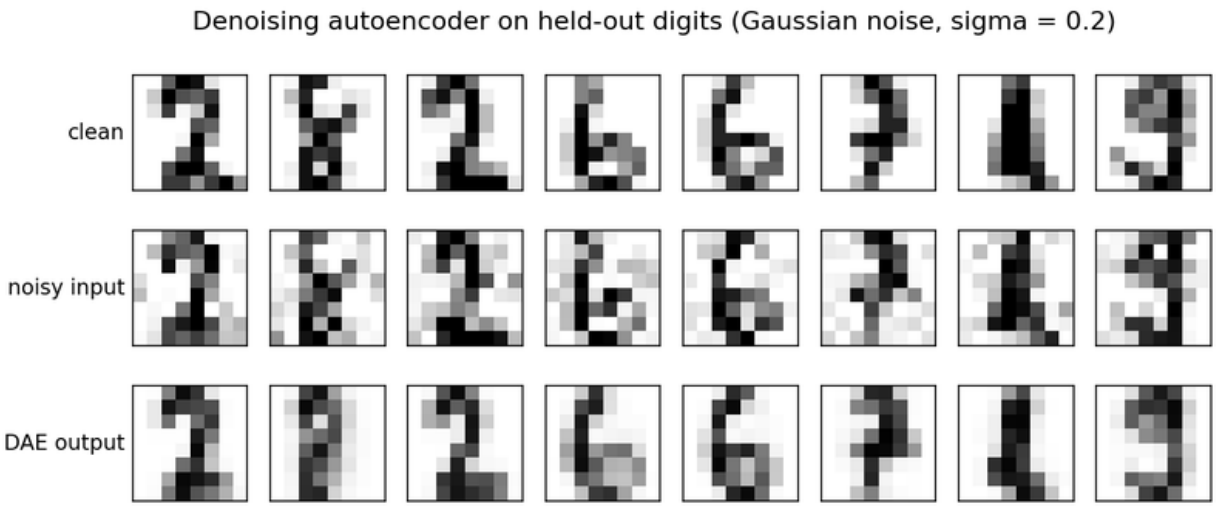


Figure 4 — Denoising autoencoder outputs on held-out digits. The background speckle is removed; some fine stroke detail is lost through the 12-unit bottleneck.

Contractive autoencoders

Plain version. Penalize the encoder whenever a tiny change in the input causes a big change in the code. The code should only react to changes that matter.

Rifai et al. [12] add the squared Frobenius norm of the encoder's Jacobian:

$$\mathcal{L} = \ell(x,\, g_\theta(f_\phi(x))) + \lambda \left\| \frac{\partial f_\phi(x)}{\partial x} \right\|_F^2, \qquad \left\| \frac{\partial f_\phi(x)}{\partial x} \right\|_F^2 = \sum_{j,i} \left(\frac{\partial h_j}{\partial x_i} \right)^2 .$$

Here $h = f_\phi(x)$ and the Jacobian entry $\partial h_j / \partial x_i$ measures how much code unit j moves when input coordinate i moves. For a single sigmoid layer $h = \sigma(Wx + b)$, the entry is $h_j(1 - h_j)W_{ji}$, so the penalty is cheap to compute:

$$\left\| \frac{\partial h}{\partial x} \right\|_F^2 = \sum_j \left(h_j(1 - h_j) \right)^2 \sum_i W_{ji}^2 .$$

The reconstruction term wants the code to change with the input (otherwise it could not reconstruct different inputs), while the penalty wants it to change as little as possible. The compromise is a code that is sensitive only along the few directions in which the training data actually vary. A DAE achieves a related effect implicitly: for small Gaussian noise, denoising training penalizes the sensitivity of the whole reconstruction function [1, ch. 14], whereas the CAE penalizes only the encoder, explicitly.

Code: the contractive penalty in action

```
import torch
from torch import nn
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split

train, test = train_test_split(load_digits().data/16., test_size=.2, random_state=0)
X = torch.tensor(train, dtype=torch.float32)
Xt = torch.tensor(test, dtype=torch.float32)

def fit(lam):
    torch.manual_seed(0)
    W = nn.Linear(64, 16)                    # encoder h = sigmoid(Wx + b)
    dec = nn.Linear(16, 64)
    opt = torch.optim.Adam([*W.parameters(), *dec.parameters()], lr=3e-3)
    for step in range(1500):
        x = X[torch.randint(len(X), (64,))]
        h = torch.sigmoid(W(x))
        rec = (dec(h) - x).square().sum(1).mean()
        # ||J||_F^2 = sum_j (h_j(1-h_j))^2 * ||W_j||^2 for a sigmoid layer
        jac = ((h*(1-h))**2 * W.weight.square().sum(1)).sum(1).mean()
        loss = rec + lam*jac
        opt.zero_grad(); loss.backward(); opt.step()
    with torch.no_grad():
        h = torch.sigmoid(W(Xt))
        mse = (dec(h) - Xt).square().mean().item()
        jac = ((h*(1-h))**2 * W.weight.square().sum(1)).sum(1).mean().item()
    return mse, jac
```

```
for lam in (0.0, 0.5):
    mse, jac = fit(lam)
    print(f"lambda={lam}: test MSE={mse:.4f}, mean ||J||_F^2={jac:.2f}")
```

Output:

```
lambda=0.0: test MSE=0.0141, mean ||J||_F^2=4.79
lambda=0.5: test MSE=0.0194, mean ||J||_F^2=0.70
```

The penalty cuts the encoder's average sensitivity by a factor of almost seven, at the cost of a somewhat higher reconstruction error. Whether that trade is worthwhile depends on the downstream use: a smoother code often helps nearest-neighbor retrieval or classification with few labels, while a codec would prefer the unpenalized model.

The manifold view

Plain version. Real data do not fill their high-dimensional space. They cluster near a thin, curved surface, like a sheet of paper rolled up inside a room. Good representations describe where you are *on the sheet*.

The **manifold hypothesis** says that natural data, such as images, concentrate near a low-dimensional manifold embedded in the high-dimensional input space [3]. A random 64-pixel array almost never looks like a digit; the set of plausible digits is a tiny, structured region.

Locally, around any data point, the manifold looks like a flat plane, the **tangent space**, spanned by the few directions you can move while staying realistic (thicker stroke, slight slant, small shift). Directions orthogonal to the tangent space lead off the manifold into noise.

This view gives a single lens for everything in the chapter:

- **PCA** fits one global flat sheet. It works when the manifold is nearly flat and fails when it is curved.
- **Mixtures of PPCA or FA** fit many local flat sheets.
- **Nonlinear autoencoders** learn a curved chart of the manifold: the encoder gives coordinates on the sheet, and the decoder maps them back into the room.
- **Denoising** teaches the model to move points that fell off the manifold back onto it.
- **Contractive** penalties make the encoder insensitive to directions orthogonal to the manifold, while reconstruction keeps it sensitive along the tangent directions [1, ch. 14], [12].

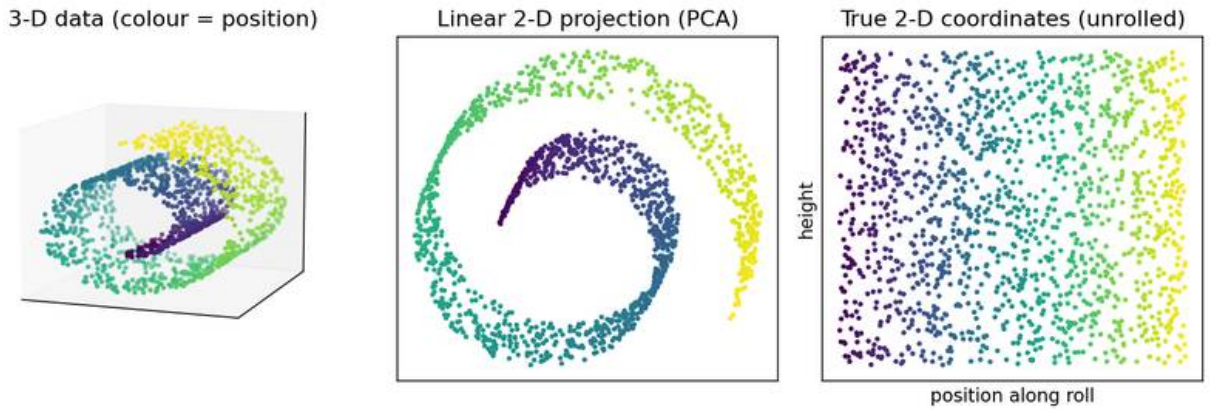


Figure 5 — The Swiss roll is a two-dimensional sheet rolled up in three dimensions. A linear projection (PCA) flattens it into a spiral that does not preserve position along the sheet; the true intrinsic coordinates form a flat rectangle.

Figure 5 makes the limitation of linear methods concrete. The Swiss roll has two intrinsic dimensions, position along the roll and height, but no two-dimensional linear projection can unroll it: the PCA view keeps the spiral, so points far apart along the sheet are not laid out in order. A nonlinear encoder can, in principle, learn the unrolled coordinates. In practice, whether it does depends on capacity, regularization and the amount of data; manifold learning in high dimensions remains hard.

From autoencoders to VAEs

Plain version. An ordinary autoencoder learns to compress and decompress, but it gives no rule for *inventing* new codes. A VAE adds that rule.

Suppose a trained autoencoder maps every training digit to a code in $\mathbb{R}^{12}$. To generate a new digit, we would like to pick a random code and decode it. But which random code? The training codes may occupy a strange, clumpy region, and a random point between them may decode to garbage. Nothing in the reconstruction loss asked the code space to be smooth, filled, or shaped like any known distribution. Excellent reconstruction therefore does not imply good generation.

The **variational autoencoder** [13] fixes this by returning to the probabilistic factor-model story of PPCA and FA, but with a neural decoder:

$$z \sim p(z) = \mathcal{N}(0, I), \qquad x \sim p_\theta(x \mid z),$$

where $p_\theta(x \mid z)$ is a distribution whose parameters are produced by a decoder network. Because the true posterior $p_\theta(z \mid x)$ is now intractable, a second network, the encoder, outputs an approximate posterior $q_\phi(z \mid x)$, typically a diagonal Gaussian. Training maximizes the **evidence lower bound (ELBO)** on $\log p_\theta(x)$:

$$\log p_\theta(x) \;\geq\; \underbrace{\mathbb{E}_{q_\phi(z|x)}[\,\log p_\theta(x \mid z)\,]}_{\text{reconstruction}} \;-\; \underbrace{D_{\mathrm{KL}}(q_\phi(z \mid x)\,\|\,p(z))}_{\text{keep codes near the prior}}.$$

Read the two terms against everything above:

- The **reconstruction** term is exactly an autoencoder loss, now with noise injected into the code. With a Gaussian decoder it becomes squared error, with a Bernoulli decoder binary cross-entropy.
- The **KL** term is a regularizer, like the sparse and contractive penalties, but chosen so that codes from all training examples together fill the prior. That is what makes sampling $z \sim \mathcal{N}(0, I)$ meaningful.
- With a **linear** Gaussian decoder, the model *is* PPCA. Its exact posterior is Gaussian and can be matched by a linear Gaussian encoder, so the bound becomes tight and the optimum coincides with the PPCA maximum-likelihood solution. The VAE is the nonlinear generalization of the factor models we began with.

Kingma and Welling's introduction [14] develops this connection at length. Weighting the KL term by a factor $\beta > 1$ gives the **β-VAE** [15], which was proposed to encourage disentangled factors at the cost of reconstruction quality; note that with $\beta \neq 1$ the objective is no longer the ELBO. The full derivation of the ELBO, the reparameterization trick and failure modes such as posterior collapse are developed in ML 06.

What reconstruction error can and cannot tell you

Before leaving autoencoders, a practical warning. Reconstruction error is often used as an anomaly score: an autoencoder trained on normal data should reconstruct normal inputs well and unusual ones poorly. This can work, but it fails when the autoencoder generalizes too well and reconstructs unusual inputs too, or when anomalies differ only in fine detail that the bottleneck discards anyway. Similarly, a code that reconstructs well may be poor for classification, and a code that classifies well may discard the pixel detail a restoration task needs. Evaluate compression, denoising, downstream usefulness and generation separately.

Modern view

Plain version. The classical models of this chapter did not disappear. They became components: compression stages for generative models, pretraining objectives for vision, and microscopes for looking inside large networks.

Surveys worth reading

- **Bengio, Courville and Vincent (2013)** [3] is the standard review of representation learning. It frames good representations through a list of general priors (smoothness, multiple explanatory factors, a hierarchy of concepts, shared factors across tasks, the manifold hypothesis, sparsity and others), and connects autoencoders, probabilistic models and manifold learning under that lens. It is a good map for *why* the penalties in this chapter exist.
- **Bank, Koenigstein and Giryes (2020)** [16] is a compact survey chapter on autoencoders. It walks through regularized variants (sparse, denoising, contractive), VAEs and disentangled variants, applications (generative modeling, classification, clustering, anomaly detection, recommendation and dimensionality reduction) and hybrids with adversarial training, which makes it a useful companion to this tutorial.
- **Kingma and Welling (2019)** [14] is the authors' own extended introduction to VAEs. It covers the ELBO, the reparameterization trick, richer posteriors and hierarchical models, and places VAEs among other deep latent-variable models.
- **Goodfellow, Bengio and Courville** [1, ch. 13–14] and **Murphy** [2] remain the clearest textbook treatments of linear factor models, the DAE–score connection and the manifold interpretation.

Disentanglement: a sober result

A recurring hope has been that unsupervised factor models would recover *the* true underlying factors, one per code unit ("disentanglement"). Locatello et al. [17] showed theoretically that unsupervised learning of disentangled representations is fundamentally impossible without inductive biases on both models and data, which mirrors the rotation ambiguity of factor analysis at a much larger scale. In a large reproducible study they also found that well-disentangled models could not be identified without supervision, and that more disentanglement did not reliably reduce the sample complexity of downstream tasks. The practical lesson echoes ICA: identifiability always comes from extra assumptions, so state them.

Autoencoders as tokenizers

The biggest modern role of autoencoders is as the first stage of large generative models. **VQ-VAE** [18] replaces the continuous code with a learned codebook of discrete vectors, so images become grids of discrete tokens that an autoregressive model can predict (see ML 08). **Latent diffusion** [19] trains an autoencoder for perceptual compression and then runs the diffusion model in its latent space instead of pixel space, which greatly reduces compute. In both cases, the autoencoder's reconstruction quality sets a ceiling: the generator cannot restore detail the encoder discarded. Designing these tokenizers, balancing compression, reconstruction and how easy the latent space is for the generator to model, is an active research area as of 2026.

Sparse autoencoders for interpretability (2023–2026)

Sparse autoencoders have found a new and very active application: **mechanistic interpretability** of large language and vision models. The motivating observation is that individual neurons are often polysemantic, responding to several unrelated concepts. Cunningham et al. [20] trained overcomplete sparse autoencoders to reconstruct a language model's internal activations, and found sparsely activating features that automated measures rated more interpretable and monosemantic than the neurons or other decompositions. Gao et al. [21] used k-sparse (top-k) autoencoders to control sparsity directly, reported clean scaling laws in autoencoder size and sparsity, proposed new metrics for feature quality, and trained a 16-million-latent autoencoder on GPT-4 activations. Conceptually this is sparse coding (Olshausen and Field's idea) applied to a network's activations instead of image patches.

Open problems

- **Evaluation.** Reconstruction error, sparsity and downstream accuracy disagree, and there is no agreed measure of whether learned features are the "right" ones, either for disentanglement or for interpretability.
- **Identifiability.** When can learned factors be trusted to correspond to real causes? Nonlinear ICA and causal representation learning study which extra assumptions (time structure, auxiliary variables, interventions) make this possible.
- **Tokenizer design.** How much to compress, whether codes should be discrete or continuous, and how to make latents both reconstructive and easy to model remain open engineering and theory questions.

Key takeaways

- All models here share one template: an encoder to a code, a decoder back, and an assumption that decides which code to prefer.
- PCA minimizes linear reconstruction error; its directions are covariance eigenvectors, found with the SVD of centered data.
- PPCA adds isotropic Gaussian noise and recovers the PCA subspace by maximum likelihood; factor analysis allows a separate noise level per coordinate. Both identify the subspace but not the axes.
- ICA resolves the axes by assuming independent, non-Gaussian sources, up to order, sign and scale; sparse coding and NMF pick factors by sparsity and nonnegativity.
- A linear autoencoder recovers the PCA subspace, not unique PCA weights. Nonlinear autoencoders need a bottleneck and/or a penalty (sparse, denoising, contractive) to learn useful structure.

- The manifold view unifies these: good codes move along the data's low-dimensional surface and ignore directions off it.
- Reconstruction does not provide a sampler. The VAE adds a prior and an approximate posterior, and with a linear decoder it reduces to PPCA.

Exercises

1. **Noise assumptions.** You record 20 sensors, five of which are much noisier than the rest, but all respond to the same few underlying causes. Would you start with PPCA or factor analysis? What would PCA tend to do with the noisy sensors?

   Hint

   Factor analysis: its diagonal Ψ can assign large residual variance to the five noisy sensors without spending factors on them. PCA (and PPCA, with a single shared σ^2) may devote components to the high-variance noisy sensors, because high variance is all PCA sees.

2. **Latent ambiguity.** For an invertible $k \times k$ matrix G, show that replacing W_e by GW_e and W_d by $W_d G^{-1}$ leaves a linear autoencoder's reconstruction unchanged. Which matrices G also preserve the PPCA likelihood?

   Hint

   The product is $W_d G^{-1} G W_e = W_d W_e$. For PPCA, the likelihood depends on $WW^\top$; replacing W by WG gives $WGG^\top W^\top$, which is unchanged when $GG^\top = I$, that is, when G is orthogonal.

3. **Why ICA fails on Gaussians.** Generate two independent standard Gaussian sources, mix them with any invertible 2×2 matrix, and run FastICA several times with different random seeds. What do you observe, and why?

   Hint

   After whitening, every rotation of two independent unit Gaussians is again two independent unit Gaussians, so there is no "right" rotation to find. Different seeds give different, equally valid unmixing directions, and the correlation with the true sources varies from run to run.

4. **Sparse codes versus PCA codes.** Using the sparse-coding snippet, compare the number of nonzero code entries per test image with PCA codes of the same dimension (32). Then plot the test MSE of sparse coding as `alpha` varies over 0.05, 0.1, 0.3, 1.0.

   Hint

   PCA codes are dense: essentially all 32 entries are nonzero. As `alpha` grows, the number of active atoms drops and the error rises, tracing a sparsity–fidelity curve. Remember that the dictionary was not given a mean term, so compare curves rather than single numbers with PCA.

5. **Denoising mismatch.** Train the DAE from the tutorial on Gaussian noise, then test it on inputs where a random 3×3 block is set to zero. Then train a second DAE on that masking corruption and test both on both corruptions.

Hint

These are different corruption processes. The Gaussian-trained model has never been asked to fill in missing structure, so it typically leaves holes; the masking-trained model learns completion. Report a 2×2 table (train corruption × test corruption) rather than inferring robustness to one from success on the other.

6. **Contractive penalty by hand.** For a single sigmoid unit $h = \sigma(w^\top x + b)$, derive $\|\partial h / \partial x\|^2$. When is it largest, and what does that say about where a contractive autoencoder allows its code to be sensitive?

Hint

$\partial h / \partial x = h(1-h)w$, so the squared norm is $h^2(1-h)^2\|w\|^2$. It is largest at $h = 1/2$ (the steep middle of the sigmoid) and nearly zero when the unit is saturated. The penalty pushes units to be saturated for most inputs, keeping them in the sensitive regime only where the data require variation.

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ML 06 • Generative Models: Autoencoders, VAEs, GANs and Flows

Learn distributions rather than labels: likelihood-based versus implicit models, the ELBO and reparameterization, adversarial training and Wasserstein GANs, normalizing flows, and how generative models are evaluated.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-06-generative-models/

What you'll learn

- How modeling a distribution $p(x)$ differs from predicting a label, and why maximum likelihood is the same as minimizing a KL divergence.
- The two big families: **likelihood-based** models that write down (or bound) $\log p(x)$, and **implicit** models that only provide a sampler.
- How a VAE turns an intractable integral into a trainable lower bound (the ELBO), and why the reparameterization trick makes it differentiable.
- How GANs learn through a discriminator, what the original objective secretly minimizes, why mode collapse happens, and what Wasserstein GANs change.
- How normalizing flows compute exact likelihoods with the change-of-variables formula, and where energy-based, autoregressive and diffusion models fit.
- How to evaluate generators with likelihood, Inception Score, FID and precision/recall — and what each of these cannot tell you.

The big picture

A classifier asks, "What is in this image?" A **generative model** asks, "What kinds of images could occur?" If it learns a useful distribution, we can draw new samples, fill missing regions, compress data, detect unusual inputs, or condition generation on text, a class label or another image.

> **Explain like I'm five**
>
> A child who recognizes drawings of houses can say "house." A child who understands how houses are drawn can invent a new one with a door, windows and a roof. Generative modeling tries to learn the recipe behind the examples, so it can cook up new ones — and a good cook makes a different, but still tasty, dish every time instead of the same one over and over.

Learning a distribution is harder than learning a label. A label is one number; an image has thousands of pixels whose values depend on one another in complicated ways. Every family

in this chapter is a different answer to the same engineering question: how do you represent a distribution over very high-dimensional data so that you can both **train** it from examples and **sample** from it afterwards?

This chapter uses the probabilistic viewpoint developed in Bishop and Bishop [1] and Prince [2]. The comparative review by Bond-Taylor et al. [3] organizes the same families along the trade-offs we will meet — training speed, sample quality, diversity, and whether a likelihood is available — and the flow review by Papamakarios et al. [4] is particularly helpful for distinguishing a learned distribution from a convenient sampling procedure. Diffusion and autoregressive models get their own chapters (ML 07 and ML 08) because their training and sampling deserve careful treatment; here we place them on the map.

Modeling a distribution

Plain version. Fit a probability model so that the observed examples look likely under it, then use the model to produce new examples that look like they came from the same source.

Maximum likelihood

Suppose the data $x_1, \dots, x_n$ are independent draws from an unknown distribution p_{data}, and p_θ is a model with parameters θ. Maximum likelihood minimizes the average **negative log-likelihood (NLL)**:

$$\mathcal{L}_{\text{NLL}}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} \log p_\theta(x_i).$$

Here n is the number of training examples and $\log p_\theta(x_i)$ is the log-density the model assigns to example i.

Why is this a sensible goal? As n grows, the average approaches an expectation over the data distribution, and

$$\mathbb{E}_{x\sim p_{\text{data}}}[-\log p_\theta(x)] = D_{\text{KL}}(p_{\text{data}} \,\|\, p_\theta) + H(p_{\text{data}}).$$

$D_{\text{KL}}(p\|q) = \mathbb{E}_p[\log p - \log q]$ is the Kullback–Leibler divergence, which is zero only when the two distributions agree, and $H(p_{\text{data}})$ is the entropy of the data, which does not depend on θ. So **maximum likelihood is the same as minimizing the forward KL divergence** from data to model.

The direction of the KL matters. The forward KL heavily punishes a model that assigns near-zero density where real data lives, so likelihood training is "mass-covering": it prefers to spread probability over every mode, even if that means putting some probability between modes. Keep this in mind; we will see that GANs behave differently.

Densities, discrete pixels and bits per dimension

For continuous values, $p_\theta(x)$ is a **density**, not the probability of observing exactly x; a density can exceed 1. Real images, however, store integers such as 0–255. Fitting a continuous density directly to integers is ill-posed: the model can place infinitely tall spikes on the integer grid and drive the likelihood to infinity. The standard fix is **dequantization** — add uniform noise in $[0, 1)$ to each integer value before fitting a continuous density. Theis et al. [5] show that the continuous model's log-likelihood on dequantized data is then a lower bound on the log-likelihood of a corresponding discrete model, which makes the numbers comparable across papers. Alternatively one uses a discrete likelihood directly (a softmax over 256 values, or a Bernoulli for binary pixels, as in our code below).

Image likelihoods are usually reported in **bits per dimension**:

$$\text{bpd} = \frac{\mathcal{L}_{\text{NLL}}}{D \ln 2},$$

where $\mathcal{L}_{\text{NLL}}$ is measured in nats per image and D is the number of pixel values (height × width × channels). Bits per dimension are the average number of bits a lossless compressor built on the model would need per subpixel, which is why likelihood and compression are two sides of the same coin.

Conditional and unconditional generation

A discriminative classifier models $p(y \mid x)$. A conditional generator models $p(x \mid c)$, where c can be text, a class, a low-resolution image or observed pixels. Bayes' rule links related quantities, but a classifier does not by itself provide an image sampler.

Conditioning also explains a familiar failure. Minimizing pixel MSE predicts the **conditional mean** $\mathbb{E}[x \mid c]$. If an ambiguous input has several valid outputs — a gray-scale photo whose shirt could be red or blue — the mean blends them into a brownish average. A generative model instead tries to represent the whole distribution $p(x \mid c)$ and sample one plausible answer at a time.

Two families: likelihood-based and implicit

Every model below needs two abilities: a way to **train** from data and a way to **sample**. They differ in whether they can evaluate $p_\theta(x)$.

- **Likelihood-based (explicit) models** write down a density, or a bound on it, and maximize it. Autoregressive models and normalizing flows give exact likelihoods; VAEs (and diffusion mod-

els, in their variational view) optimize a lower bound; energy-based models define a density only up to an unknown normalizing constant.

- **Implicit models** define a sampler — "push noise through a network" — without any formula for $p_\theta(x)$. GANs are the main example. Because there is no likelihood to maximize, they must be trained by comparing samples with data in some other way.

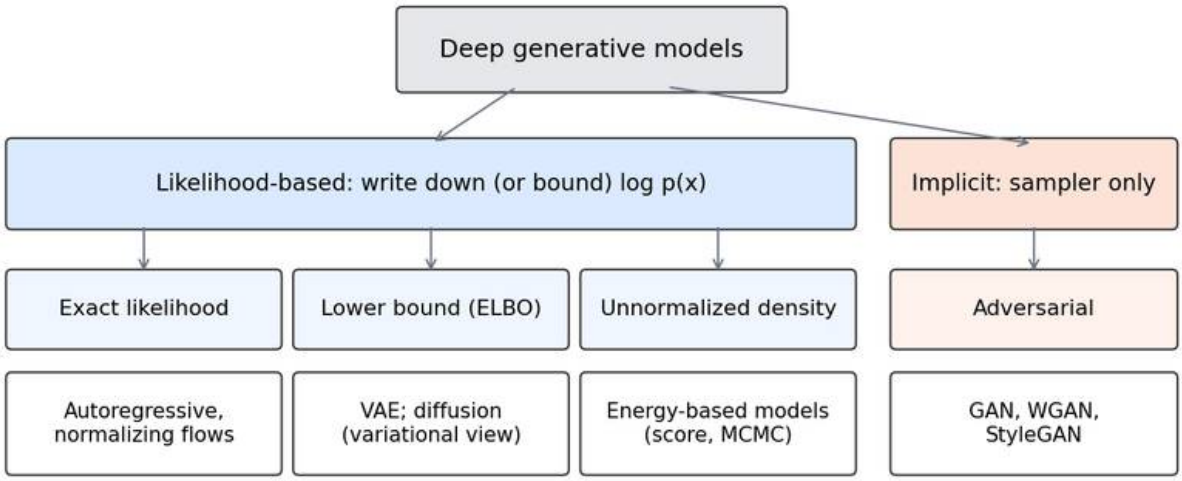


Figure 1 — A teaching taxonomy of generative models, following the explicit/implicit split used in comparative reviews [3]. Real systems often combine several branches.

Ordinary autoencoders: compression without a prior

Plain version. Encode an example into a compact code and decode it back. That is excellent for compression and representation learning, but it does not by itself tell you how to invent new codes.

An encoder $z = f_\phi(x)$ and decoder $\hat{x} = g_\theta(z)$ minimize a reconstruction loss such as $\|x - \hat{x}\|^2$. Autoencoders, their linear relatives (PCA, factor analysis) and their regularized variants are developed in ML 05A. The point for this chapter is narrow: an ordinary autoencoder places no requirement on **where** codes live. The training codes may occupy a thin, oddly shaped region of latent space. If you then sample $z \sim \mathcal{N}(0, I)$ and decode it, you may land in an empty region the decoder never saw, and get garbage. Excellent reconstruction therefore does not guarantee good unconditional generation.

A VAE fixes this by turning the autoencoder into a probability model with a prior over codes.

VAEs: a latent-variable probability model

Plain version. Imagine that each image was made by first picking a few hidden "settings" (pose, thickness, style) at random from a simple distribution, then rendering an image from those settings with a bit of noise. A VAE learns both the renderer (decoder) and a quick guesser (encoder) that reads an image and proposes which settings probably produced it.

The model

Choose a prior $p(z)$ over a latent vector $z \in \mathbb{R}^k$, usually a standard Gaussian $\mathcal{N}(0, I)$, and a decoder $p_\theta(x \mid z)$, a neural network that outputs the parameters of a distribution over images (Bernoulli probabilities for binary pixels, or Gaussian means). The model's density over images is the marginal

$$p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, dz.$$

Sampling is easy: draw $z \sim p(z)$, then draw $x \sim p_\theta(x \mid z)$. Evaluating $p_\theta(x)$ is hard: the integral runs over every possible code, and the decoder is a nonlinear network, so there is no closed form. The true posterior $p_\theta(z \mid x) = p_\theta(x \mid z)p(z)/p_\theta(x)$ is intractable for the same reason.

Deriving the evidence lower bound

Kingma and Welling [6] introduce an **encoder** (or inference network) $q_\phi(z \mid x)$ that approximates the posterior. Here is the derivation step by step. Since $\log p_\theta(x)$ does not depend on z, we can take its expectation under any distribution over z without changing it:

$$\begin{aligned}
\log p_\theta(x) &= \mathbb{E}_{q_\phi(z|x)}\big[\log p_\theta(x)\big] \\
&= \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_\theta(x,z)}{p_\theta(z \mid x)}\right] \\
&= \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_\theta(x,z)}{q_\phi(z \mid x)}\right] + \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{q_\phi(z \mid x)}{p_\theta(z \mid x)}\right] \\
&= \mathrm{ELBO}(x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x)).
\end{aligned}$$

The second line uses $p_\theta(x) = p_\theta(x,z)/p_\theta(z \mid x)$; the third multiplies and divides by $q_\phi(z \mid x)$ inside the log. Because a KL divergence is never negative, the first term is a lower bound on $\log p_\theta(x)$:

the **evidence lower bound (ELBO)**. The **gap** between the bound and the true log-likelihood is exactly the KL divergence between the approximate and true posteriors. A better encoder tightens the bound; a perfect one closes it.

Expanding $p_\theta(x,z) = p_\theta(x \mid z)p(z)$ gives the form used in code:

$$\mathrm{ELBO}(x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)]}_{\text{reconstruction}} - \underbrace{D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z))}_{\text{stay close to the prior}}.$$

The first term rewards codes that let the decoder reproduce x; the second penalizes codes whose distribution wanders away from the prior. The same bound can be reached in one line with Jensen's inequality ($\log \mathbb{E} \geq \mathbb{E} \log$), but the decomposition above is more useful because it names the gap. Training maximizes the average ELBO over the data with respect to both θ and ϕ; equivalently, it minimizes the **negative ELBO**.

For a diagonal Gaussian encoder $q_\phi = \mathcal{N}(\mu, \mathrm{diag}(\sigma^2))$ and a standard Gaussian prior, the KL term has a closed form:

$$D_{\mathrm{KL}}(q_\phi \| p) = \frac{1}{2} \sum_{j=1}^{k} \left(\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1 \right),$$

where μ_j and σ_j^2 are the encoder's mean and variance for latent coordinate j. Each summand is zero exactly when $\mu_j = 0$ and $\sigma_j^2 = 1$.

Reparameterization

The reconstruction term is an expectation over $z \sim q_\phi(z \mid x)$, and we need its gradient with respect to ϕ. Sampling is not differentiable as written: a random draw is a black box. The **reparameterization trick** [6] moves the randomness into a fixed noise variable:

$$\epsilon \sim \mathcal{N}(0, I), \qquad z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon.$$

Here $\odot$ is elementwise multiplication. Now z is a deterministic, differentiable function of ϕ and an independent ϵ, so ordinary backpropagation gives low-variance gradient estimates through μ and σ. In code, the encoder predicts `logvar` $= \log \sigma^2$ and uses `exp(.5 * logvar)` as the standard deviation, which guarantees positivity without constraints.

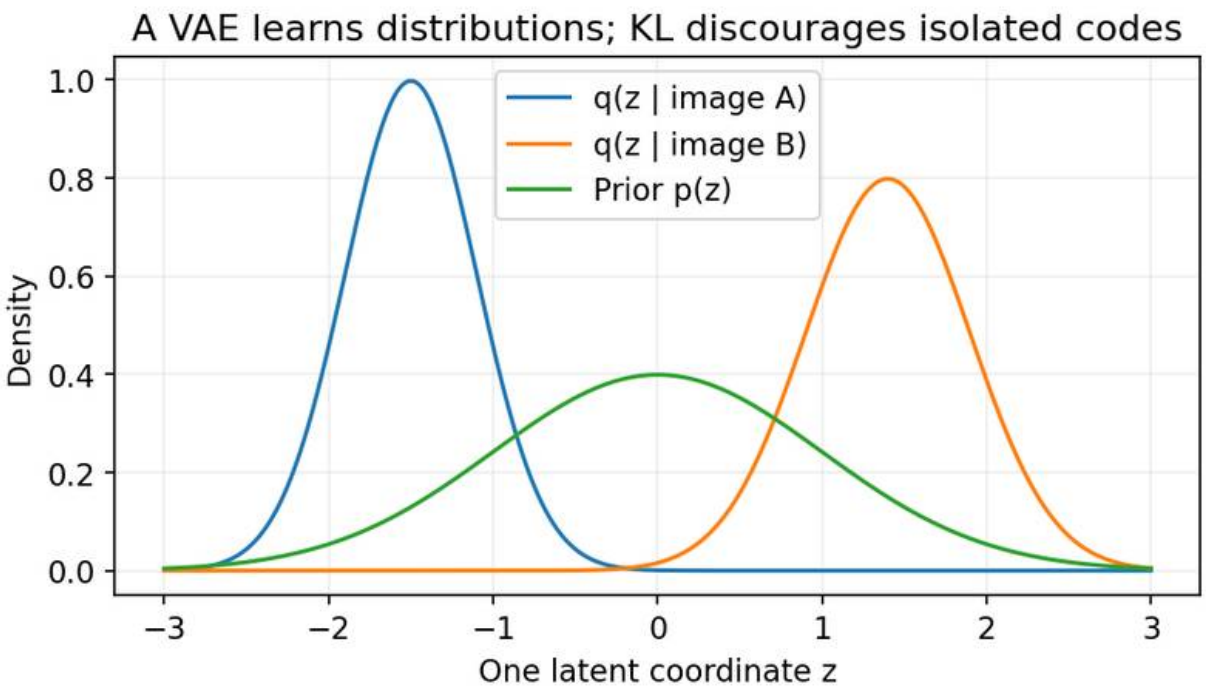


Figure 2 — A VAE represents uncertainty in its code and encourages compatibility with a shared prior. The curves are illustrative.

Code: a small VAE for binary digits

This complete CPU example uses scikit-learn's bundled 8 × 8 digits, binarized so that a Bernoulli decoder is a stated likelihood model. Reconstruction is summed over pixels and KL over latent dimensions, then both are averaged over examples; silently averaging only one term would change their balance.

```
import torch
from torch import nn
from torch.nn import functional as F
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split

torch.manual_seed(0)
torch.set_num_threads(1)
d = load_digits()
train, test = train_test_split(d.data, test_size=.2, random_state=0,
                               stratify=d.target)
X = torch.tensor((train > 8).astype("float32"))
Xt = torch.tensor((test > 8).astype("float32"))
enc = nn.Sequential(nn.Linear(64, 64), nn.ReLU(), nn.Linear(64, 16))
dec = nn.Sequential(nn.Linear(8, 64), nn.ReLU(), nn.Linear(64, 64))
opt = torch.optim.Adam(list(enc.parameters()) + list(dec.parameters()), lr=.002)
```

```
def terms(x):
    mu, logvar = enc(x).chunk(2, dim=-1)
    z = mu + (.5*logvar).exp() * torch.randn_like(mu)
    recon = F.binary_cross_entropy_with_logits(dec(z), x, reduction="none").sum(1)
    kl = .5*(mu.square() + logvar.exp() - logvar - 1).sum(1)
    return recon, kl

for step in range(400):
    x = X[torch.randint(len(X), (64,))]
    recon, kl = terms(x)
    loss = (recon + kl).mean()
    opt.zero_grad()
    loss.backward()
    opt.step()
enc.eval(); dec.eval()
with torch.no_grad():
    recon, kl = terms(Xt)
    samples = dec(torch.randn(4, 8)).sigmoid().reshape(4, 8, 8)
    print("test negative ELBO estimate:", round((recon + kl).mean().item(), 3))
    print("reconstruction / KL:", round(recon.mean().item(), 3),
          round(kl.mean().item(), 3))
    print("sample probability maps:", tuple(samples.shape))
```

On our machine it prints a test negative ELBO of about 20.5 nats per image, split into roughly 17.0 nats of reconstruction and 3.5 nats of KL. Since the bound is a lower bound on log-likelihood, 20.5 nats is an **upper bound** on the test NLL. The output maps are Bernoulli probabilities; to obtain binary samples, draw each pixel with those probabilities. The test estimate is stochastic, so average several latent draws for a more stable number. No test-based checkpoint selection happens here.

Figure 3 shows what a VAE with only a **two-dimensional** latent space learns on the same data (trained longer by the figure script). Even though labels are never used, digits of the same class land near each other, and decoding a regular grid of z values produces smooth transitions between digit shapes. This is the property an ordinary autoencoder lacks: every point near the prior's bulk decodes to something digit-like.

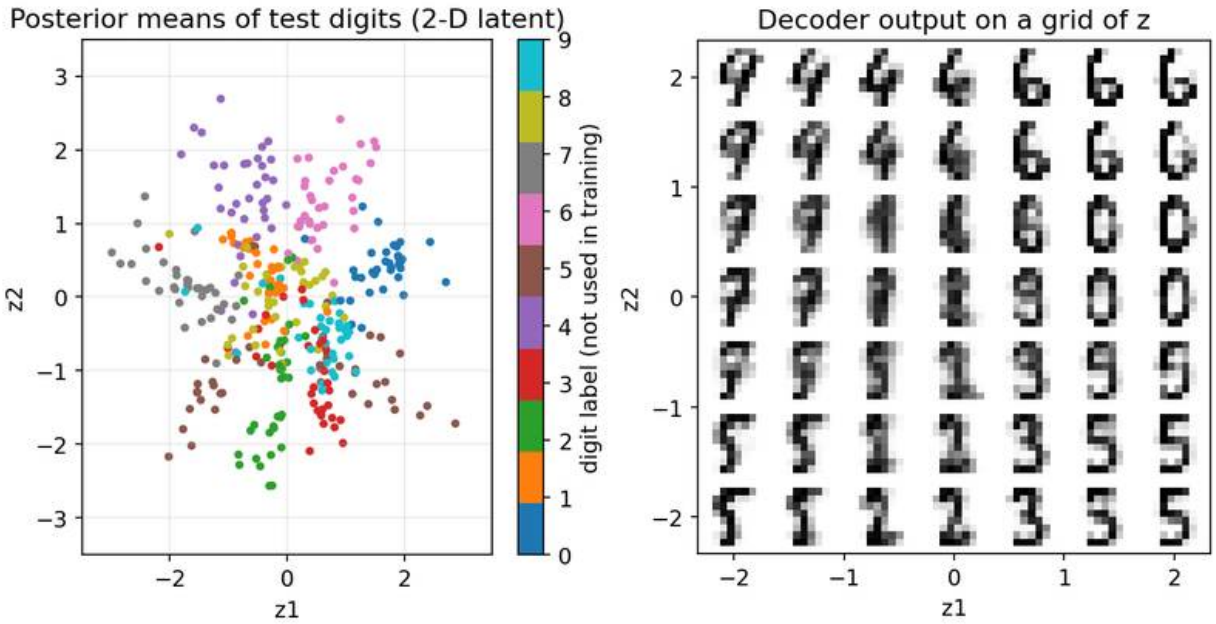


Figure 3 — A VAE with a 2-D latent space on binarized 8 × 8 digits. Left: encoder means for test images (labels used only for coloring). Right: decoder outputs on a grid of latent values.

Posterior collapse, β-VAE and deeper VAEs

If the decoder is very powerful (for example, an autoregressive decoder that can model pixels on its own), the cheapest way to raise the ELBO may be to ignore z entirely: set $q_\phi(z \mid x) = p(z)$ so the KL term is zero, and let the decoder do all the work. This is **posterior collapse**. Bowman et al. [7] observed it when training VAEs on text and proposed **KL annealing**: start with a small weight on the KL term and raise it to 1 during training, so the model learns to use the code before the penalty bites. Monitoring the KL per latent dimension is the simplest diagnostic — dimensions with KL near zero carry no information.

Higgins et al. [8] go the other way: the **β-VAE** multiplies the KL term by $\beta > 1$, pressuring the model toward a more factorized, "disentangled" code at the cost of reconstruction. Be precise about what this means: with $\beta \neq 1$ the objective is no longer the ELBO, so the value is not a bound on $\log p_\theta(x)$, and Kingma and Welling's tutorial review [9] discusses this and many other variants. Finally, single-layer VAEs with factorized Gaussians tend to produce blurry images, partly because the Gaussian decoder averages over uncertainty. **Hierarchical VAEs** stack several layers of latent variables; NVAE [10] showed that a carefully engineered deep hierarchy with stabilizing regularization can produce sharp samples on natural images up to 256 × 256 pixels.

GANs: learning with an adversary

Plain version. Two networks play a game. A **generator** turns random noise into fake examples; a **discriminator** looks at an example and guesses whether it is real or fake. The generator wins by fooling the discriminator, which forces its outputs to look more and more like data.

The minimax game

Let G map noise $z \sim p(z)$ to samples $G(z)$, whose distribution we call p_g. Let $D(x) \in (0,1)$ be the discriminator's estimated probability that x is real. The original GAN objective of Goodfellow et al. [11] is

$$\min_G \max_D \; V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p(z)}[\log(1 - D(G(z)))].$$

The discriminator maximizes V: it is ordinary binary cross-entropy with real data labeled 1 and fakes labeled 0. The generator minimizes V: it wants $D(G(z))$ to be large.

What the game minimizes

For a fixed generator, write V as an integral over x: $V = \int [p_{\text{data}}(x)\log D(x) + p_g(x)\log(1 - D(x))]dx$. For each x separately, the function $a\log D + b\log(1-D)$ with $a, b > 0$ is maximized at $D = a/(a+b)$ (set the derivative $a/D - b/(1-D)$ to zero). So the **optimal discriminator** is

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}.$$

It is the probability that a point came from the data rather than the generator, if both were equally likely a priori. Substituting D^* back in and adding and subtracting $\log 4$ gives

$$V(D^*, G) = -\log 4 + 2\,\text{JSD}(p_{\text{data}} \,\|\, p_g),$$

where $\text{JSD}(p\|q) = \frac{1}{2}D_{\text{KL}}(p\|m) + \frac{1}{2}D_{\text{KL}}(q\|m)$ with $m = \frac{1}{2}(p+q)$ is the **Jensen–Shannon divergence**. So, with a perfect discriminator, the generator minimizes a symmetric divergence between data and model, and the global optimum is $p_g = p_{\text{data}}$, where $D^* = 1/2$ everywhere [11]. Notice that no likelihood was ever computed: the discriminator acts as a learned, adaptive test of "does this look like the data?"

The non-saturating loss and how to implement it

Early in training the generator is poor, $D(G(z)) \approx 0$, and $\log(1 - D(G(z)))$ is flat, so the generator's gradient vanishes. In practice the generator instead minimizes $-\mathbb{E}\log D(G(z))$, the **non-**

saturating loss proposed in the same paper [11], which has strong gradients precisely when the generator is losing. It has the same fixed point but different dynamics.

Updates alternate. When updating D, **detach** generated samples so no gradient reaches G; when updating G, let gradients flow through D into G but step only G's optimizer. Using logits with `binary_cross_entropy_with_logits` is numerically safer than applying a sigmoid and then a log.

Code: a GAN on eight Gaussians

The classic test bed for mode coverage is a ring of eight tight Gaussian clusters in 2-D. It is small enough to train on a CPU in under a minute, and you can **see** whether every mode is covered.

```
import math
import torch
from torch import nn
from torch.nn import functional as F

torch.manual_seed(0)
torch.set_num_threads(1)
angles = torch.arange(8) * (2 * math.pi / 8)
centers = 2.0 * torch.stack([angles.cos(), angles.sin()], dim=1)

def sample_data(n):
    idx = torch.randint(8, (n,))
    return centers[idx] + 0.05 * torch.randn(n, 2)

def mlp(i, o):
    return nn.Sequential(nn.Linear(i, 128), nn.ReLU(),
                         nn.Linear(128, 128), nn.ReLU(), nn.Linear(128, o))

G, D = mlp(8, 2), mlp(2, 1)
opt_g = torch.optim.Adam(G.parameters(), lr=1e-3, betas=(.5, .999))
opt_d = torch.optim.Adam(D.parameters(), lr=1e-3, betas=(.5, .999))
for step in range(3000):
    real = sample_data(256)
    fake = G(torch.randn(256, 8))
    # Discriminator: logits, real -> 1, fake -> 0; detach so G is untouched.
    d_loss = (F.binary_cross_entropy_with_logits(D(real), torch.ones(256, 1))
              + F.binary_cross_entropy_with_logits(D(fake.detach()),
                                                   torch.zeros(256, 1)))
    opt_d.zero_grad(); d_loss.backward(); opt_d.step()
    # Generator: non-saturating loss -log D(G(z)).
    g_loss = F.binary_cross_entropy_with_logits(D(fake), torch.ones(256, 1))
    opt_g.zero_grad(); g_loss.backward(); opt_g.step()

with torch.no_grad():
    x = G(torch.randn(5000, 8))
    dist = torch.cdist(x, centers)
    nearest = dist.min(1)
```

```
    good = nearest.values < 0.15          # within 3 std of some mode
    counts = torch.bincount(nearest.indices[good], minlength=8)
print("fraction near a mode:", round(good.float().mean().item(), 3))
print("modes with >= 1% of samples:", int((counts >= 50).sum()))
print("samples per mode:", counts.tolist())
```

It prints:

```
fraction near a mode: 0.896
modes with >= 1% of samples: 7
samples per mode: [590, 14, 710, 276, 651, 731, 863, 646]
```

About 90% of samples land on a mode, but the mode at 45° receives only 14 of 5,000 samples and another is under-represented. That is **partial mode collapse** caught in the act. Figure 4 shows the generator's journey: an unstructured cloud early on, a ring that covers every mode but also the space between them, and finally sharper clusters with one mode nearly abandoned. Exact numbers depend on the seed and library version; the qualitative picture is typical.

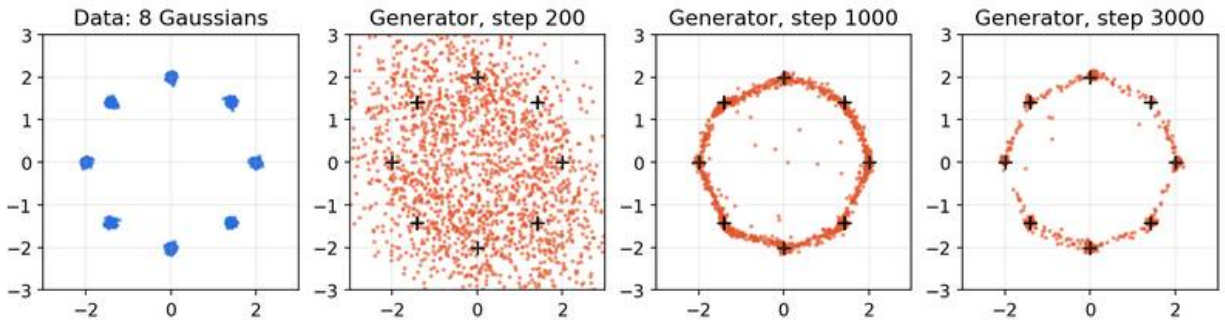


Figure 4 — A small GAN learning a ring of eight Gaussians (crosses mark the true modes). By step 3000 samples are sharp but one mode is nearly dropped: sharpness and coverage are separate achievements.

Mode collapse and why it happens

Mode collapse occurs when the generator produces too few kinds of output. The generator is rewarded for samples the current discriminator accepts, not for diversity. If it finds one region the discriminator likes, it can pile samples there; the discriminator then learns to reject that region, the generator hops to another, and the pair can chase each other around instead of converging.

Unlike maximum likelihood, nothing in the per-sample generator loss directly punishes **missing** a mode. Salimans et al. [12] proposed early remedies such as feature matching and minibatch discrimination, which let the discriminator look at statistics of a whole batch so that a lack of diversity becomes detectable.

A collection of sharp, convincing images can therefore still cover only a small part of the data distribution. Always measure coverage separately from quality — we return to this under evaluation.

Wasserstein GANs

When p_{data} and p_g live on thin, non-overlapping sets (as images do in pixel space), the JSD is stuck at its maximum $\log 2$ and provides no useful gradient about **how far apart** they are. Arjovsky et al. [13] proposed the **Wasserstein-1 (earth mover's) distance** instead: the minimum cost of moving probability mass from one distribution to the other. Its dual form, from the Kantorovich–Rubinstein theorem, is

$$W(p_{\text{data}}, p_g) = \sup_{\|f\|_L \le 1} \mathbb{E}_{x \sim p_{\text{data}}}[f(x)] - \mathbb{E}_{x \sim p_g}[f(x)],$$

where the supremum is over all 1-Lipschitz functions f (functions whose output changes by at most $\|x - x'\|$ when the input changes from x' to x). The network f is now called a **critic**: it outputs an unbounded score, not a probability. The Wasserstein distance shrinks smoothly as the generator's samples move toward the data, even when the supports do not overlap.

The hard part is enforcing the Lipschitz constraint. The original WGAN clips every critic weight to a small interval, which is crude and can make the critic too simple. **WGAN-GP** [14] instead adds a gradient penalty

$$\lambda \, \mathbb{E}_{\hat{x}}\big[(\|\nabla_{\hat{x}} f(\hat{x})\|_2 - 1)^2\big],$$

where $\hat{x}$ is drawn uniformly on straight lines between pairs of real and generated samples and λ is a weight (the paper uses 10). Two other widely used stabilizers act on the discriminator directly: **spectral normalization** [15] divides each weight matrix by its largest singular value, bounding the Lipschitz constant layer by layer, and the **R1 penalty** analyzed by Mescheder et al. [16] penalizes $\frac{\gamma}{2}\|\nabla_x D(x)\|^2$ on real data, which they show gives local convergence where unregularized GAN training can oscillate.

Architectures and where GANs are used now

StyleGAN [17] illustrates how architecture matters as much as the loss: it maps noise to an intermediate latent space and injects that "style" at every resolution, giving controllable, high-quality face synthesis. GANs sample with a single generator pass, which makes them fast at inference.

Adversarial losses also appear as **components** of other systems — for example, inside image tokenizers trained to reconstruct sharp detail. That does not make every model using such a tokenizer a GAN generator; the distribution may be modeled by something else entirely.

Normalizing flows: invertible transformations

Plain version. Start from simple noise, such as a standard Gaussian, and push it through a chain of invertible transformations. Because every step can be undone, you can run any image backwards to the noise that would have produced it — and keep exact track of how each step stretches or squeezes space, which gives an exact likelihood.

Change of variables

Let $z \sim p_0(z)$ be a simple base distribution and $x = f_\theta(z)$ an invertible, differentiable map with the same dimension. Then

$$\log p_\theta(x) = \log p_0(f_\theta^{-1}(x)) - \log \left| \det \frac{\partial f_\theta}{\partial z} \right|_{z=f_\theta^{-1}(x)} .$$

$\partial f_\theta / \partial z$ is the Jacobian matrix of the map, and its determinant measures how much the map scales a small volume around z. Probability mass is conserved, so if a region is stretched to twice its volume, its density must halve. In one dimension, $x = 2z$ doubles every interval and adds $-\log 2$ to the log density. A composition $f = f_K \circ \cdots \circ f_1$ simply adds up the log-determinants of the steps.

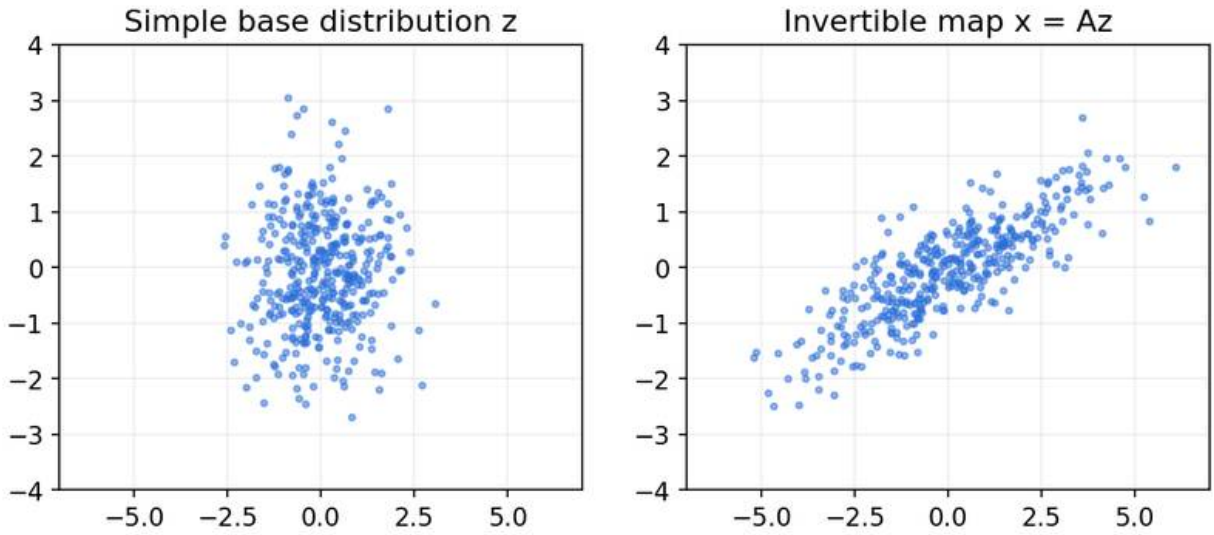


Figure 5 — Change of variables preserves total probability while changing density. This affine map has an analytically tractable determinant.

Coupling layers

The formula is exact, but a general $D \times D$ determinant costs $O(D^3)$ — hopeless for images. Flow layers are designed so that the Jacobian is **triangular**, whose determinant is just the product of the

diagonal. The affine coupling layer of RealNVP [18] splits x into two halves (x_1, x_2) and computes

$$y_1 = x_1, \qquad y_2 = x_2 \odot \exp\big(s(x_1)\big) + t(x_1),$$

where s and t are arbitrary neural networks. The Jacobian is block-triangular with the identity on the first block and $\operatorname{diag}(\exp s(x_1))$ on the second, so $\log|\det J| = \sum_j s_j(x_1)$. The inverse is just as cheap: $x_2 = (y_2 - t(y_1)) \odot \exp(-s(y_1))$. Crucially, s and t never need to be inverted, so they can be as complex as you like. Stacking layers with alternating splits lets every coordinate be transformed. Glow [19] added learned invertible 1×1 convolutions as a generalized channel permutation between coupling layers.

Code: verify a coupling layer's log-determinant

The snippet below builds one affine coupling layer and checks its cheap log-determinant against the full Jacobian computed by autograd.

```
import torch
from torch import nn

torch.manual_seed(0)

class AffineCoupling(nn.Module):
    """RealNVP-style layer: keep x1, scale-and-shift x2 using x1."""
    def __init__(self, dim, hidden=32):
        super().__init__()
        self.d = dim // 2
        self.net = nn.Sequential(nn.Linear(self.d, hidden), nn.Tanh(),
                                 nn.Linear(hidden, 2 * (dim - self.d)))

    def forward(self, x):
        x1, x2 = x[..., :self.d], x[..., self.d:]
        s, t = self.net(x1).chunk(2, dim=-1)
        s = torch.tanh(s)                       # keep scales moderate
        y2 = x2 * s.exp() + t
        return torch.cat([x1, y2], -1), s.sum(-1)   # output, log|det J|

    def inverse(self, y):
        y1, y2 = y[..., :self.d], y[..., self.d:]
        s, t = self.net(y1).chunk(2, dim=-1)
        s = torch.tanh(s)
        return torch.cat([y1, (y2 - t) * (-s).exp()], -1)

layer = AffineCoupling(4)
x = torch.randn(4)
y, logdet = layer(x)
J = torch.autograd.functional.jacobian(lambda v: layer(v)[0], x)
print("log|det J| from layer:   ", round(logdet.item(), 5))
print("log|det J| from autograd:", round(torch.linalg.slogdet(J)[1].item(), 5))
print("upper-right block is zero:", bool(torch.allclose(J[:2, 2:], torch.zeros(2, 2))))
print("inverse error:", (layer.inverse(y) - x).abs().max().item() < 1e-6)
```

```
# Log density of y under a standard-normal base after one layer:
base = torch.distributions.Normal(0., 1.)
z = layer.inverse(y)
log_px = base.log_prob(z).sum() - layer(z)[1]
print("log p(y):", round(log_px.item(), 4))
```

Both log-determinants print 0.14867, the upper-right Jacobian block is exactly zero (the triangular structure), and the inverse recovers the input. The last line is the change-of-variables formula in action. Training a flow simply maximizes this `log p` over data with a stack of such layers.

Strengths, limits and continuous flows

Flows give exact likelihoods, exact inversion (useful for editing and for mapping data to a well-behaved latent space) and one-pass sampling. The price is architectural: every layer must be invertible with a cheap determinant, and the latent must have the **same dimension** as the data, which wastes capacity on representing noise in all pixel directions. Rezende and Mohamed [20] introduced flows as a way to make the approximate posterior of a VAE more flexible, which shows that the same tool serves inference as well as generation.

Continuous normalizing flows replace a discrete stack by an ordinary differential equation $\mathrm{d}x/\mathrm{d}t = v_\theta(x, t)$. Neural ODEs [21] showed that the log-density then evolves through the **divergence** (trace of the Jacobian) of v_θ rather than a determinant, which removes the architectural restriction but makes likelihood evaluation require an ODE solve. **Flow matching** [22], covered in ML 07, trains such a vector field by simple regression instead of maximum likelihood. The word "flow" therefore refers to related transformations, but the training objective need not be likelihood.

Energy-based models

Plain version. Instead of describing how to generate an image, give every possible image a score — an "energy" — where low energy means "looks real." Probability is high wherever energy is low.

An energy-based model (EBM) defines

$$p_\theta(x) = \frac{\exp(-E_\theta(x))}{Z_\theta}, \qquad Z_\theta = \int \exp(-E_\theta(x))\, dx,$$

where E_θ is any network that outputs a scalar and Z_θ is the **partition function** that makes the density integrate to one. EBMs are extremely flexible — any architecture works — but Z_θ is intractable in high dimensions, so neither the likelihood nor exact samples are directly available.

Goodfellow, Bengio and Courville [23] cover the classical tools (the partition-function chapter and approximate inference), and Song and Kingma [24] survey modern training routes: maximum likelihood with MCMC samples (for example, Langevin dynamics), **score matching**, which fits $\nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x)$ and thereby never needs Z_θ, and noise-contrastive methods. Score matching is the bridge to diffusion models.

Autoregressive models, briefly

Plain version. Generate an image one piece at a time, each piece conditioned on everything generated so far, like writing a sentence word by word.

The chain rule of probability is exact for any ordering of the D elements of x:

$$p_\theta(x) = \prod_{i=1}^{D} p_\theta(x_i \mid x_1, \ldots, x_{i-1}).$$

Each factor is an ordinary classification or regression problem, so training is simple, stable maximum likelihood with an exact likelihood. PixelRNN and PixelCNN [25] modeled raw pixels this way with a 256-way softmax per subpixel. The drawback is sequential sampling: D network evaluations for D elements. Modern systems instead predict compressed tokens, groups of tokens or whole scales at once. ML 08 develops this family in detail.

Diffusion models, briefly

Plain version. Teach a network to remove a little noise from a noisy image. Then start from pure noise and apply the denoiser many times, slowly sculpting an image.

A diffusion model defines a fixed forward process that gradually adds Gaussian noise to data until nothing but noise remains, and learns a reverse process that undoes it step by step. Ho et al. [26] showed that training reduces, up to weighting, to predicting the noise that was added — a simple regression — and that the objective is a reweighted variational bound, which is why the taxonomy in Figure 1 lists diffusion next to the VAE. The connection to score matching (see the energy-based models section) gives a second, equivalent view. The survey by Yang et al. [27] maps the many variants. Diffusion and flow matching now dominate high-quality image generation and are the subject of ML 07.

The family map

Family	Basic training signal	Sampling mechanism	Likelihood	Main qualification
VAE	Variational bound (ELBO)	Sample latent, then decoder (one pass)	Lower bound	Bound depends on approximate inference; samples can be blurry
GAN	Adversarial discrimination	Generator pass (one pass)	None	Unstable game; mode collapse
Normalizing flow	Change-of-variables likelihood	Invertible transform (one pass)	Exact	Invertibility and same-dimension constraints
Autoregressive	Conditional likelihoods	Sequential conditional sampling	Exact	Generation dependencies can be costly
Diffusion	Denoising/score objectives	Reverse noise process (many steps)	Bound or ODE-based	Quality depends on model and sampler
Flow matching	Velocity regression	Integrate learned vector field	Needs extra computation	Integration cost
Energy-based	Energy learning (MCMC, score matching)	Often iterative sampling	Unnormalized	Normalization and mixing can be hard

Bond-Taylor et al. [3] summarize these trade-offs as a triangle that is hard to satisfy all at once: fast sampling, high sample quality, and good mode coverage. GANs are fast and sharp but can miss modes; likelihood models cover modes but were historically slower or blurrier; diffusion models reach high quality and coverage at the cost of many sampling steps. This table is a teaching synthesis, not an exclusive taxonomy. Real models combine families: a VAE compresses an image, a transformer models its latent distribution, and a diffusion objective trains that transformer.

Evaluating generative models

Plain version. A generator can be good in different ways: it can assign high probability to real data, make individually convincing samples, cover all the variety in the data, or follow its conditioning. No single number captures all of these, so use several measures and know what each misses.

Theis et al. [5] made this point sharply: average log-likelihood, sample quality and older proxies such as Parzen-window estimates can be **largely independent** of one another in high dimensions. A model can have excellent likelihood and poor samples, or the reverse. Evaluate against the purpose you have in mind.

Likelihood

For likelihood-based models, held-out NLL (or bits per dimension) is principled and comparable when dequantization and preprocessing match. It cannot be computed for GANs. And a high likelihood need not mean reliable out-of-distribution detection: Nalisnick et al. [28] found that

flows, VAEs and PixelCNNs trained on CIFAR-10 can assign **higher** likelihood to the unrelated SVHN digit images, because likelihood is dominated by low-level statistics such as smoothness rather than semantic content.

Inception Score

Salimans et al. [12] proposed the **Inception Score (IS)**. Feed each generated image x to a pre-trained ImageNet classifier (Inception-v3) to get a label distribution $p(y \mid x)$, and let $p(y) = \mathbb{E}_x[p(y \mid x)]$ be the average over samples. Then

$$\mathrm{IS} = \exp\left(\mathbb{E}_{x \sim p_g} D_{\mathrm{KL}}(p(y \mid x) \,\|\, p(y))\right).$$

IS is high when each image is confidently classified ($p(y \mid x)$ peaked) and the set covers many classes ($p(y)$ spread out). It never looks at real data, so it cannot tell whether samples resemble **your** dataset, and it is meaningful only for ImageNet-like content.

Fréchet Inception Distance

The **Fréchet Inception Distance (FID)** of Heusel et al. [29] compares real and generated images in a feature space. Extract Inception-v3 features for both sets, fit a Gaussian to each — means μ_r, μ_g and covariances Σ_r, Σ_g — and compute the Fréchet distance between the Gaussians:

$$\mathrm{FID} = \|\mu_r - \mu_g\|_2^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right).$$

Lower is better. FID uses real data and is sensitive to both quality and diversity, which made it the standard metric. But it is not an image-by-image truth measure: it depends on the feature extractor, image resizing, and especially the **number of samples** (it is biased for finite sets, so compare FIDs only with equal sample counts), and it folds quality and coverage into one number.

Precision and recall

To separate the two, Sajjadi et al. [30] defined **precision and recall for distributions**: precision asks what fraction of generated samples fall where real data lives (quality), recall asks what fraction of the real distribution the generator covers (diversity). Kynkäänniemi et al. [31] gave a simple, robust estimator that we implement below. Approximate each set's manifold by a union of balls around its points, with each ball's radius equal to the distance to the point's k-th nearest neighbor in the same set. Then

- **precision** = fraction of generated samples inside some real sample's ball;

- **recall** = fraction of real samples inside some generated sample's ball.

On images, the points are deep features rather than pixels. On our 2-D ring we can use coordinates directly:

```
import numpy as np

rng = np.random.default_rng(0)
angles = np.arange(8) * 2 * np.pi / 8
centers = 2 * np.stack([np.cos(angles), np.sin(angles)], 1)

def ring(n, modes):
    idx = rng.choice(modes, n)
    return centers[idx] + .05 * rng.standard_normal((n, 2))

def knn_radius(a, k):
    d = np.linalg.norm(a[:, None] - a[None], axis=-1)
    return np.sort(d, axis=1)[:, k]          # column 0 is the point itself

def coverage(query, ref, k=3):
    """Fraction of query points inside some reference point's k-NN ball."""
    r = knn_radius(ref, k)
    d = np.linalg.norm(query[:, None] - ref[None], axis=-1)
    return np.mean((d <= r[None]).any(axis=1))

real = ring(1000, range(8))
fakes = {
    "all 8 modes": ring(1000, range(8)),
    "only 2 modes": ring(1000, [0, 4]),
    "8 modes, blurry": ring(1000, range(8)) + .4 * rng.standard_normal((1000, 2)),
}
for name, fake in fakes.items():
    precision = coverage(fake, real)   # are fake samples on the real manifold?
    recall = coverage(real, fake)      # is the real manifold covered by fakes?
    print(f"{name:16s} precision={precision:.2f} recall={recall:.2f}")
```

```
all 8 modes      precision=0.97 recall=0.96
only 2 modes     precision=0.97 recall=0.26
8 modes, blurry  precision=0.07 recall=0.99
```

The mode-collapsed generator keeps high precision — each of its samples is perfect — but its recall drops to about 2/8. The blurry generator is the mirror image: it covers every mode but most samples are off the data. A single scalar such as FID would rank both as "worse" without saying **why**. Figure 6 shows the three cases.

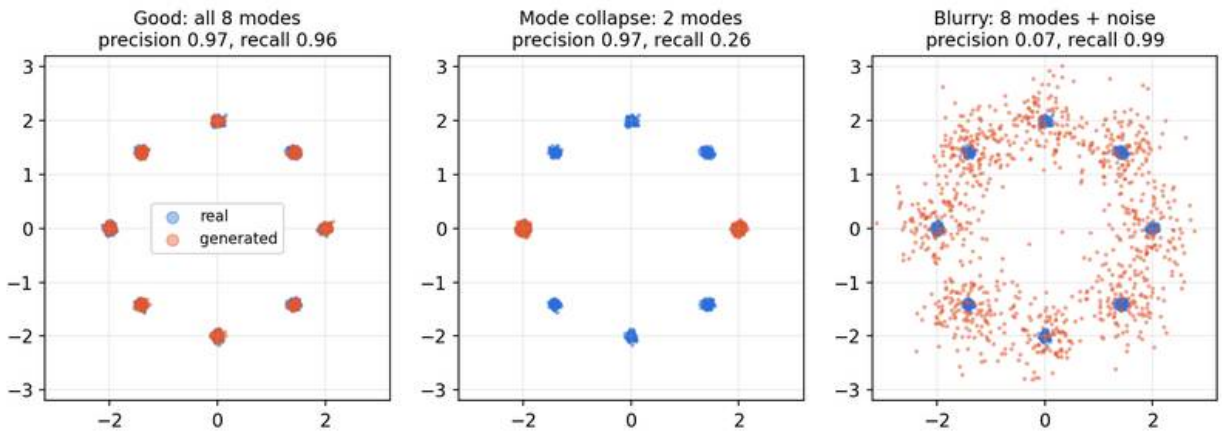


Figure 6 — k-nearest-neighbor precision and recall on a toy ring: mode collapse lowers recall, blur lowers precision.

Known limitations of current metrics

Evaluation is itself an active research area. Stein et al. [32] ran large human studies and found that no existing automatic metric strongly correlates with human judgments of realism; in particular, FID computed on Inception features under-rates the perceptual realism of diffusion models. They recommend richer self-supervised feature spaces (DINOv2 ViT-L/14) for metrics like FID and precision/recall. Jayasumana et al. [33] argue that FID's normality assumption and Inception features are poor fits for modern text-to-image models and propose **CMMD**, a maximum mean discrepancy computed on CLIP embeddings, which needs no Gaussian assumption. Practical advice: report several complementary metrics with their exact protocol (feature extractor, sample count, resolution, reference set), show uncurated samples, and run a human study when the claim is about perceived quality.

For conditional tasks, add a measure of **alignment** (does the image match the text or the low-resolution input?). For restoration, a beautiful generated texture may be unsupported by the observation; the distinction between plausible samples and measured evidence is central in DLCV 04.

Modern view

Surveys worth reading. Four reviews cover this chapter's ground from different angles.

- **Bond-Taylor et al. [3]** (IEEE TPAMI 2022) is the best single comparative review of VAEs, GANs, normalizing flows, energy-based and autoregressive models. It organizes each family by its core technique and its known pathologies, and compares them along training speed,

sampling speed, sample quality, diversity and likelihood, concluding that each family pays for its strengths with a weakness somewhere else, and that hybrids are the natural way forward.

- **Papamakarios et al. [4]** (JMLR 2021) is a unified treatment of normalizing flows: their expressive power, the design space of coupling, autoregressive, residual and continuous flows, and their uses for density estimation, inference and simulation.
- **Kingma and Welling [9]** (Foundations and Trends in ML 2019) is the authoritative tutorial-review of VAEs by their inventors: the derivation, the reparameterization estimator, tighter bounds, more flexible posteriors (including flows), hierarchical models and the β-weighted variants.
- **Yang et al. [27]** (ACM Computing Surveys 2023) surveys diffusion models — sampling acceleration, likelihood improvements, structured data, and the links to VAEs, GANs, flows and EBMs — and is the bridge to the next chapter.

The textbooks of Bishop and Bishop [1] and Prince [2] give coherent derivations of all the families, with Prince's notebooks providing runnable companions.

Where the field is (2023–2026). Diffusion and flow-matching models, usually operating in a learned latent space, now produce most state-of-the-art image and video generation. That makes the **autoencoder** a load-bearing component again: VQ-VAE [34] showed how to learn discrete latent codes that a separate prior can model, and latent diffusion [35] showed that running the generative model inside a compressed autoencoder latent space cuts cost dramatically while keeping detail. A tokenizer must balance compression, reconstruction and how easy its latent space is to model; evaluate its reconstruction first, because no downstream prior can restore detail the encoder discarded.

GANs have not disappeared. GigaGAN [36] scaled a GAN to text-to-image synthesis and emphasized what GANs still do best: single-pass sampling fast enough for interactive use, high-resolution upsampling, and a smooth, editable latent space. Huang et al. [37] revisited the training objective itself, deriving a regularized relativistic GAN loss with local convergence guarantees and showing that a minimalist modern baseline (R3GAN) without the traditional bag of tricks can surpass StyleGAN2 on standard benchmarks. Adversarial losses also live on inside other systems, for example to sharpen tokenizer reconstructions.

On evaluation, the 2023–2024 critiques of FID [32], [33] have moved the field toward richer feature extractors, separate precision/recall-style measures and human studies, but there is still no accepted single metric.

Open problems. (1) Reliable evaluation that agrees with humans and separates quality, diversity and memorization. (2) Detecting when a model reproduces training images rather than generalizing. (3) Calibrated likelihoods that are actually useful for out-of-distribution detection [28]. (4) The speed–quality–coverage trade-off: getting diffusion-level quality with GAN-level speed without losing modes. (5) Learning latent spaces that are simultaneously compressive, faithful and easy to model.

Key takeaways

- Generation requires a distribution or a defined sampling process. Maximum likelihood equals minimizing $D_{\mathrm{KL}}(p_{\mathrm{data}} \| p_\theta)$, which encourages covering every mode.
- Likelihood-based models (autoregressive, flows, VAEs, EBMs, diffusion's bound) write down or bound $\log p(x)$; implicit models (GANs) only provide a sampler.
- An ordinary autoencoder supplies reconstruction but no prior over codes, so decoding random codes need not give realistic samples.
- The ELBO equals $\log p_\theta(x)$ minus $D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x))$; reparameterization makes it trainable by backpropagation.
- With an optimal discriminator, the original GAN minimizes $2\,\mathrm{JSD} - \log 4$; the non-saturating loss, Wasserstein critics, gradient penalties, spectral normalization and R1 make the game trainable, but mode collapse must still be measured.
- Flows compute exact likelihoods via the change-of-variables formula, made cheap by triangular Jacobians in coupling layers.
- Likelihood, IS, FID and precision/recall measure different things; report several, with their exact protocol.

Exercises

1. **KL sanity check.** Substitute $\mu = 0$ and $\sigma^2 = 1$ into the Gaussian KL. Then compute the KL for $\mu = 1, \sigma^2 = 1$ in one dimension.

   Hint

   For $\mu = 0, \sigma^2 = 1$ every term is zero, because posterior and prior are the same distribution. For $\mu = 1$, $\sigma^2 = 1$ you get $\frac{1}{2}(1 + 1 - 0 - 1) = 0.5$ nats: shifting the mean by one standard deviation costs half a nat.

2. **Loss scaling.** Average the VAE's reconstruction across pixels rather than summing. What happens if KL is unchanged?

   Hint

   Reconstruction becomes 64 times smaller relative to KL. This is equivalent to a β-VAE with $\beta = 64$ on the summed loss: it effectively changes regularization (often toward posterior collapse), rather than merely changing reported units.

3. **Optimal discriminator.** At a point x where $p_{\mathrm{data}}(x) = 0.3$ and $p_g(x) = 0.1$, what is $D^*(x)$? What is D^* everywhere when $p_g = p_{\mathrm{data}}$, and what is the value of the game then?

Hint

$D^*(x) = 0.3/(0.3 + 0.1) = 0.75$. When the distributions match, $D^* = 1/2$ everywhere and $V = \log \frac{1}{2} + \log \frac{1}{2} = -\log 4$, consistent with JSD $= 0$.

4. **Affine flow.** For $x = Az + b$ with $z \sim \mathcal{N}(0, I)$, derive the density correction and explain why singular A invalidates it. Check your answer against the known Gaussian $\mathcal{N}(b, AA^\top)$.

Hint

Subtract $\log|\det A|$. Since $\det(AA^\top)^{1/2} = |\det A|$, this matches the normalizing constant of $\mathcal{N}(b, AA^\top)$. A singular map is not invertible and collapses volume, so the same-dimensional change-of-variables formula does not apply.

5. **Spot collapse.** Propose different checks for VAE posterior collapse and GAN mode collapse. Then modify the GAN code to use WGAN-GP (critic without sigmoid, gradient penalty with $\lambda = 10$, several critic steps per generator step) and compare the per-mode counts.

Hint

For a VAE, inspect the KL per latent dimension and whether decoded outputs change when you vary z. For a GAN, count samples per mode (as in our code), compute recall, and look for repeated samples at fixed conditions. They are different failures. For WGAN-GP, sample $\hat{x} = \alpha x_{\text{real}} + (1 - \alpha)x_{\text{fake}}$ with $\alpha \sim U(0, 1)$, call `torch.autograd.grad` with `create_graph=True`, and lower the learning rate if training oscillates. Compare across several seeds before concluding anything.

6. **Metric disagreement.** Using the precision/recall code, construct a fake set that has roughly the same precision and recall as the good generator but places half its samples in only four modes. What does this tell you about the limits of the metric?

Hint

Unequal mode weights can still give high recall, because recall only asks whether each real point is near **some** fake point, not whether mode frequencies match. Precision/recall measure support overlap, not density; pair them with a distribution-level measure such as FID or a per-mode histogram.

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ML 07 • Diffusion Models and Flow Matching

Derive noise, clean-image and velocity prediction, compare pixel and latent generation with JiT and representation autoencoders, and understand pretrained image/video diffusion and flow pipelines.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-07-diffusion-flow-matching/

What you'll learn

- How a known noise process turns "generate an image" into an ordinary supervised regression problem.
- How the simplified noise-prediction loss of DDPM falls out of a variational bound, step by step.
- How noise, clean-image and velocity predictions convert algebraically, why their losses and finite networks differ, and how score estimation and the SDE view connect them.
- How DDPM, DDIM and fast ODE solvers sample, and why the number of steps is a design choice.
- How classifier guidance and classifier-free guidance steer a sampler, and what they cost.
- How pixel and latent diffusion, DiT, JiT, representation autoencoders, flow matching and few-step generation fit together, and how pretrained image/video pipelines are evaluated.

The big picture

A diffusion model learns to undo controlled corruption. During training we take a real example, add a known amount of Gaussian noise, and ask a network to tell us what noise we added. Because we added the noise ourselves, the correct answer is always available: no labels, no adversary, just regression. During generation we start from pure noise and apply the learned "denoise a little" step many times until an example appears.

> **Explain like I'm five**
>
> Imagine slowly covering a drawing with pencil scribbles. You keep copies of the drawing at many scribble levels: a few lines, lots of lines, so many that the drawing is gone. A model practices one small job: look at a scribbled page, and guess which scribbles were added. After lots of practice you hand it a page that is *only* scribbles. It erases a little, looks again, erases a little more, and a brand-new drawing slowly appears — one that was never in the pile.

Why does this matter? Since 2021, diffusion-style models have produced many of the state-of-the-art results in image, video and audio synthesis, and the same machinery is used for inpainting, super-resolution, image editing, 3-D generation and scientific data. Diffusion also replaced two pain points of earlier families covered in ML 06: GANs need a fragile two-player game, and VAEs

tend to produce blurry samples. Diffusion training is a stable regression, and its samples are sharp because generation is spread over many small, easy steps.

The price is sampling cost: a classic sampler calls the network hundreds of times per image. Much of the research since 2021 — DDIM, fast ODE solvers, distillation, consistency models, flow matching, rectified flow — is about keeping the quality while cutting the number of network calls.

The chapter follows the organization of the major surveys. Yang et al. [1] group the field into efficient sampling, improved likelihood and data with special structure; Croitoru et al. [27] group it by formulation (DDPM, noise-conditioned score networks, SDEs); Lipman et al.'s flow-matching guide [2] and the monograph by Lai et al. [29] present diffusion and flows as one family of learned transport. Prince's textbook [14] is a gentle companion for the derivations. Throughout, we use **one explicit time convention per section** and say when it changes — mixing conventions is the most common source of sign errors in this area.

Forward diffusion has a closed form

Plain version. At each step, keep most of the image and add a little independent Gaussian noise. After enough steps, nothing of the image is left — only noise with a known distribution.

The idea goes back to Sohl-Dickstein et al. [15], who proposed destroying structure with a slow forward diffusion and learning to reverse it; DDPM [3] made it practical. Let $x_0 \in \mathbb{R}^d$ be a data example (an image flattened into d numbers, scaled to roughly $[-1, 1]$). Pick a **noise schedule** $\beta_1, \dots, \beta_T$ with $0 < \beta_t < 1$ and write $\alpha_t = 1 - \beta_t$. The forward (noising) process is a Markov chain

$$q(x_t \mid x_{t-1}) = \mathcal{N}\big(\sqrt{\alpha_t}\, x_{t-1},\ \beta_t I\big), \qquad t = 1, \dots, T,$$

where $\mathcal{N}(\mu, \Sigma)$ is a Gaussian with mean μ and covariance Σ, and I is the $d \times d$ identity. The factor $\sqrt{\alpha_t}$ shrinks the signal slightly so that the total variance stays bounded instead of growing; this is why this family is called **variance preserving (VP)**.

Deriving the one-shot formula

Write one step as $x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\alpha_t}\,\epsilon_t$ with $\epsilon_t \sim \mathcal{N}(0, I)$. Substituting the previous step,

$$\begin{aligned} x_t &= \sqrt{\alpha_t}\left(\sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1-\alpha_{t-1}}\,\epsilon_{t-1}\right) + \sqrt{1-\alpha_t}\,\epsilon_t \\ &= \sqrt{\alpha_t \alpha_{t-1}}\, x_{t-2} + \underbrace{\sqrt{\alpha_t(1-\alpha_{t-1})}\,\epsilon_{t-1} + \sqrt{1-\alpha_t}\,\epsilon_t}_{\text{sum of independent Gaussians}}. \end{aligned}$$

The two noise terms are independent zero-mean Gaussians, so their sum is Gaussian with variance $\alpha_t(1-\alpha_{t-1}) + (1-\alpha_t) = 1 - \alpha_t\alpha_{t-1}$. Repeating the argument down to x_0 gives, with $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$,

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar\alpha_t}\,x_0,\ (1-\bar\alpha_t)I\big) \quad\Longleftrightarrow\quad x_t = \sqrt{\bar\alpha_t}\,x_0 + \sqrt{1-\bar\alpha_t}\,\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

Here $\bar\alpha_t$ is the fraction of signal variance that survives to step t, and ϵ is a single standard Gaussian vector. This formula is what makes training cheap: we can jump straight to any noise level without simulating the earlier steps.

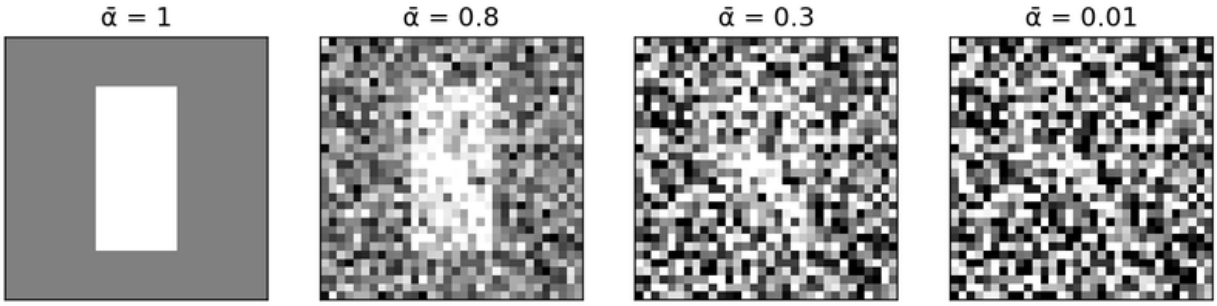


Figure 1 — The closed-form forward process at several signal levels. A shared noise realization makes the effect visible; training draws fresh noise for every example and every step.

Noise schedules and the signal-to-noise ratio

The **signal-to-noise ratio** at step t is

$$\mathrm{SNR}(t) = \frac{\bar\alpha_t}{1-\bar\alpha_t},$$

the ratio of signal variance to noise variance (for unit-variance data). It decreases monotonically from very large at $t = 1$ to nearly zero at $t = T$. Two requirements constrain the schedule: $\bar\alpha_T$ must be close to 0, so that x_T is practically a standard Gaussian (the sampler starts from $\mathcal{N}(0, I)$); and the steps in between should spend the training budget on noise levels where the network has something useful to learn.

DDPM [3] used $T = 1000$ and a **linear schedule** β_t from 10^{-4} to 0.02. Nichol and Dhariwal [19] observed that this destroys information quickly: for a large part of the chain the image is already nearly pure noise. Their **cosine schedule** defines $\bar\alpha_t = f(t)/f(0)$ with $f(t) = \cos^2\!\left(\frac{t/T+s}{1+s}\cdot\frac{\pi}{2}\right)$ and a small offset $s = 0.008$, and clips each β_t at 0.999. Figure 2 compares the two. Karras et al. [20] later argued that the schedule, the network's input/output scaling and the distribution of training noise levels are separate design choices that are best tuned independently.

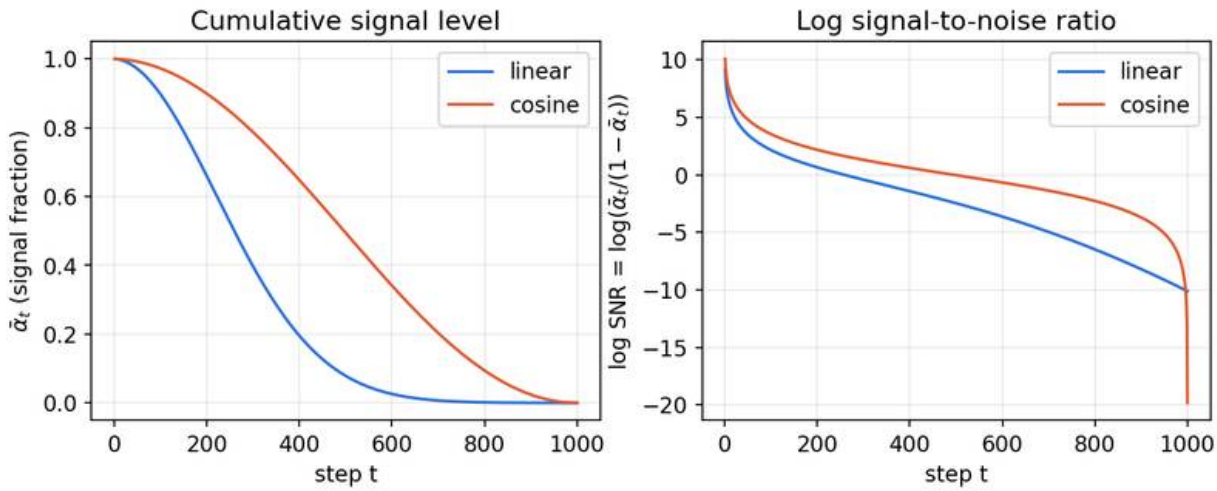


Figure 2 — Linear versus cosine schedule for $T = 1000$. The cosine schedule keeps more signal for longer and spreads the log-SNR more evenly over the steps.

The reverse process and the variational bound

Plain version. To generate, we need the reverse step: "given a noisy example, what did the slightly less noisy example look like?" We cannot compute this directly for real data, but we *can* compute it if we also know the clean example — and that turns out to give a simple training loss.

The model and the bound

The generative model is a reverse Markov chain that starts at $p(x_T) = \mathcal{N}(0, I)$ and uses learned Gaussian transitions

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(\mu_\theta(x_t, t),\ \sigma_t^2 I\big)\,,$$

where μ_θ is a neural network (with parameters θ) and σ_t^2 is a variance we will choose. This is a latent-variable model with latents $x_{1:T}$, exactly like a deep VAE whose encoder q is fixed rather than learned (see ML 06). The same Jensen-inequality argument gives an evidence lower bound. Following [3], [15], the negative bound can be rearranged into a sum of KL divergences:

$$-\log p_\theta(x_0) \;\le\; \mathbb{E}_q\Big[\underbrace{D_{\mathrm{KL}}(q(x_T \mid x_0)\,\|\,p(x_T))}_{L_T} + \sum_{t=2}^{T} \underbrace{D_{\mathrm{KL}}(q(x_{t-1} \mid x_t, x_0)\,\|\,p_\theta(x_{t-1} \mid x_t))}_{L_{t-1}} \underbrace{-\log p_\theta(x_0 \mid x_1)}_{L_0}\Big].$$

D_{KL} is the Kullback–Leibler divergence. L_T has no learnable parameters (it is small when $\bar{\alpha}_T \approx 0$); L_0 is a decoder term for the last step; and each L_{t-1} compares the model's reverse step with the **forward posterior** $q(x_{t-1} \mid x_t, x_0)$ — the reverse step that would be exact if the clean example x_0 were known.

The forward posterior is Gaussian

By Bayes' rule, $q(x_{t-1} \mid x_t, x_0) \propto q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)$. Both factors are Gaussian in x_{t-1}, so their product is Gaussian. Collecting the quadratic and linear terms in x_{t-1} (completing the square) gives

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I\big),$$

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\, x_0 + \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\, x_t, \qquad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,\beta_t.$$

The posterior mean is a weighted average of the clean example and the current noisy one; the posterior variance $\tilde{\beta}_t$ is slightly smaller than β_t. You do not have to trust the algebra: the snippet below samples the forward chain and fits a regression of x_{t-1} on x_t.

```
import numpy as np

rng = np.random.default_rng(0)
beta = np.linspace(1e-4, 0.02, 1000)
alpha = 1 - beta
abar = np.cumprod(alpha)
t, x0, n = 300, 1.3, 400_000          # check q(x_{t-1} | x_t, x_0) at step t

# Sample the forward chain: x_{t-1} from q(x_{t-1}|x_0), then one more step.
x_prev = np.sqrt(abar[t-1])*x0 + np.sqrt(1-abar[t-1])*rng.normal(size=n)
x_t = np.sqrt(alpha[t])*x_prev + np.sqrt(beta[t])*rng.normal(size=n)

# Empirical regression of x_{t-1} on x_t (the posterior mean is linear in x_t).
slope, intercept = np.polyfit(x_t, x_prev, 1)
resid_var = np.var(x_prev - (slope*x_t + intercept))

# Closed-form posterior from the derivation.
c_x0 = np.sqrt(abar[t-1])*beta[t] / (1-abar[t])
c_xt = np.sqrt(alpha[t])*(1-abar[t-1]) / (1-abar[t])
var = beta[t]*(1-abar[t-1]) / (1-abar[t])
print(f"slope     {slope:.4f}  vs  {c_xt:.4f}")
print(f"intercept {intercept:.4f}  vs  {c_x0*x0:.4f}")
print(f"variance  {resid_var:.6f}  vs  {var:.6f}")
```

The output matches to three or four decimals (slope 0.9934 vs 0.9930, variance 0.006050 vs 0.006052); the small gaps are Monte Carlo error.

From the bound to the simplified loss

Now use the forward formula backwards: $x_0 = (x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon)/\sqrt{\bar{\alpha}_t}$. Substituting into $\tilde{\mu}_t$ and simplifying with $\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1}$,

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\epsilon\right).$$

At sampling time the network sees x_t but not ϵ. So DDPM parameterizes the model mean in exactly the same form, with the unknown noise replaced by a prediction $\epsilon_\theta(x_t, t)$:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right).$$

The KL between two Gaussians with the same fixed covariance $\sigma_t^2 I$ is a scaled squared distance between their means, so

$$L_{t-1} = \frac{1}{2\sigma_t^2} \|\tilde{\mu}_t - \mu_\theta\|^2 + C = \frac{\beta_t^2}{2\sigma_t^2 \, \alpha_t \, (1-\bar{\alpha}_t)} \|\epsilon - \epsilon_\theta(x_t, t)\|^2 + C,$$

where C does not depend on θ. Every term of the bound is therefore a **weighted noise-prediction error**. Ho et al. [3] found that dropping the weight, and sampling t uniformly, trains better:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0,\, t\sim \mathcal{U}\{1,\dots,T\},\, \epsilon\sim\mathcal{N}(0,I)} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon,\ t) \right\|^2.$$

Read it as an algorithm: pick a training example, pick a random step, pick random noise, build x_t in one line, and regress the noise. The time t is fed to the network (usually through a sinusoidal embedding) because the same pixel values mean different things at different noise levels.

Two points are easy to get wrong. First, $\mathcal{L}_{\text{simple}}$ is *not* the likelihood bound: removing the weights relatively down-weights the low-noise steps, which helps sample quality but hurts the reported log-likelihood. Improved DDPM [19] recovered good likelihoods by learning the variances σ_t^2 (as an interpolation between β_t and $\tilde{\beta}_t$) with a hybrid loss. Second, the same network also yields a **clean estimate**,

$$\hat{x}_0(x_t, t) = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}},$$

which every modern sampler uses internally. When $\bar{\alpha}_t$ is tiny, the division amplifies prediction errors, so $\hat{x}_0$ at very high noise is a blurry average, not a sample.

Noise, clean data, velocity and the score

Plain version. Predicting the noise, predicting the clean image and predicting "which direction makes this example more typical" are three ways to express the same knowledge. They differ in scaling, and scaling changes how the training loss weights different noise levels.

Three parameterizations

Because $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$ is a fixed linear relation, a network can output any one of

- the noise ϵ (DDPM [3]),

- the clean example x_0,
- the **velocity** $v = \sqrt{\bar{\alpha}_t}\,\epsilon - \sqrt{1-\bar{\alpha}_t}\,x_0$, introduced by Salimans and Ho [22] for distillation, which stays well scaled at both ends of the schedule.

Given x_t and t, each converts to the others exactly. However, an unweighted MSE on ϵ, on x_0 or on v weights the noise levels differently, so the choice affects training. Karras et al. [20] made this explicit with input/output "preconditioning" coefficients that keep the network's inputs and targets at unit variance for every noise level. Note that this diffusion "velocity" is a reparameterization of signal and noise; it is related to, but not identical with, the transport velocity of flow matching later in this chapter.

Exact conversion does not make finite networks equivalent

For the variance-preserving path, write $a = \sqrt{\bar{\alpha}_t}$ and $b = \sqrt{1-\bar{\alpha}_t}$. Thus $a^2 + b^2 = 1$ and $x_t = ax_0 + b\epsilon$. The diffusion velocity $v_{\mathrm{VP}} = a\epsilon - bx_0$ completes a rotation:

$$x_0 = ax_t - bv_{\mathrm{VP}}, \qquad \epsilon = bx_t + av_{\mathrm{VP}},$$
$$\hat{x}_0^{(\epsilon)} = (x_t - b\hat{\epsilon})/a, \quad \hat{\epsilon}^{(x)} = (x_t - a\hat{x}_0)/b.$$

Here a, b are known schedule coefficients; v_{VP} is the target used in [22]; hats denote estimates; and superscripts identify the network's direct output. The last two conversions require nonzero divisors. Their instability near an endpoint is a numerical and statistical issue, even though the algebra is exact in the interior.

A noise error δ_ϵ becomes a clean-image error $-(b/a)\delta_\epsilon$. Therefore, for corresponding predictions at a fixed noise level,

$$\|\hat{x}_0 - x_0\|^2 = \frac{b^2}{a^2}\|\hat{\epsilon} - \epsilon\|^2 = \frac{1}{\mathrm{SNR}(t)}\|\hat{\epsilon} - \epsilon\|^2, \qquad \|\hat{x}_0^{(v)} - x_0\|^2 = b^2\|\hat{v}_{\mathrm{VP}} - v_{\mathrm{VP}}\|^2.$$

$\mathrm{SNR}(t) = a^2/b^2$ is the signal-to-noise ratio; the second identity follows from the rotation above. Equal unweighted losses consequently emphasize different times. Changing the target without recording time sampling and loss weighting confounds an experiment [20], [22].

There is also a capacity issue. If a network directly outputs a clean image, its final representation can discard nuisance noise. A network directly outputting noise may need to preserve more coordinates of the noisy input, or use a skip connection that supplies those coordinates. Converting one network's output afterward does not make its architecture identical to a network trained to output another target. This distinction becomes especially relevant for large pixel patches and high-dimensional latents; the JiT discussion below supplies recent evidence [42]. It does not imply that noise prediction generally fails: the earlier DDPM examples and dense image denoisers are counterexamples.

```
import numpy as np

rng = np.random.default_rng(7)
```

```
x0 = rng.normal(size=32)
eps = rng.normal(size=32)
a, b = np.sqrt(.2), np.sqrt(.8)
xt = a*x0 + b*eps
v_vp = a*eps - b*x0
assert np.allclose(a*xt - b*v_vp, x0)
assert np.allclose(b*xt + a*v_vp, eps)

# A deliberately imperfect noise estimate: compare equivalent losses.
eps_hat = eps + .03*rng.normal(size=32)
x_hat = (xt - b*eps_hat)/a
noise_mse = np.mean((eps_hat - eps)**2)
clean_mse = np.mean((x_hat - x0)**2)
assert np.allclose(clean_mse/noise_mse, b*b/(a*a))
print("Clean/noise MSE ratio:", round(clean_mse/noise_mse, 3))

# A straight flow uses a different path and a different velocity.
t = .35
zt = (1-t)*eps + t*x0
u = x0 - eps
x_from_flow = zt + (1-t)*u
u_from_clean = (x0 - zt)/(1-t)
assert np.allclose(x_from_flow, x0)
assert np.allclose(u_from_clean, u)
print("VP velocity equals straight-flow target:", np.allclose(v_vp, u))
```

The example prints a loss ratio of 4.0 and `False` for the two velocity targets. Both use the letter v in parts of the literature; a checkpoint's scheduler must identify the actual path, normalization and time direction.

> **Interactive demonstration** (**online**): open the online edition of this tutorial to use this demonstration.Open demonstration

What is a score, and why does denoising estimate it?

The **score** of a density p is $\nabla_x \log p(x)$: a vector field that points toward higher probability. If you know the score, you can sample with **Langevin dynamics**: repeatedly step a little along the score and add a little noise. Song and Ermon [17] trained "noise-conditional score networks" at many noise levels and sampled with annealed Langevin dynamics — a parallel route to diffusion that turned out to be the same idea.

The bridge is **denoising score matching** (Vincent [16]). For Gaussian corruption, the best possible noise predictor (in mean squared error) is a conditional expectation, $\epsilon^*(x_t, t) = \mathbb{E}[\epsilon \mid x_t]$, and **Tweedie's formula** [40] links it to the score of the noisy marginal p_t:

$$\nabla_{x_t} \log p_t(x_t) = -\frac{\mathbb{E}[\epsilon \mid x_t]}{\sqrt{1-\bar{\alpha}_t}} \quad \Longrightarrow \quad s_\theta(x_t, t) \approx -\frac{\epsilon_\theta(x_t, t)}{\sqrt{1-\bar{\alpha}_t}},$$

with s_θ the learned score. The same formula in "clean" units reads $\mathbb{E}[x_0 \mid x_t] = (x_t + (1 - \bar{\alpha}_t)\nabla \log p_t(x_t))/\sqrt{\bar{\alpha}_t}$. Two warnings: this is the score of the *noisy marginal distribution* at level t, not the gradient for any single training image; and it holds for the optimal predictor — a trained network approximates it.

You can verify Tweedie's formula where everything is known: one-dimensional Gaussian data, for which the best noise predictor is linear in x_t.

```
import numpy as np

rng = np.random.default_rng(0)
s, abar, n = 0.7, 0.4, 500_000     # data x0 ~ N(0, s^2); one noise level
x0 = s*rng.normal(size=n)
eps = rng.normal(size=n)
xt = np.sqrt(abar)*x0 + np.sqrt(1-abar)*eps

# "Train" the best linear noise predictor eps_hat = w * x_t by least squares.
w = (xt @ eps) / (xt @ xt)
score_from_eps = -w / np.sqrt(1-abar)       # s(x_t) = -eps_hat / sqrt(1-abar)

# Exact: x_t ~ N(0, abar s^2 + 1 - abar), so grad log p_t(x) = -x / var.
exact = -1 / (abar*s**2 + 1 - abar)
print(f"score slope from noise predictor: {score_from_eps:.4f}")
print(f"exact score slope:                {exact:.4f}")
```

It prints -1.2560 versus -1.2563: a regression on noise recovers the score.

The SDE view

Song et al. [4] took the step count to infinity. With continuous time $t \in [0, 1]$ and a rate $\beta(t)$, the VP forward process becomes the **stochastic differential equation (SDE)**

$$dx = -\tfrac{1}{2}\beta(t)\, x\, dt + \sqrt{\beta(t)}\, dw,$$

where w is a standard Wiener process (Brownian motion). A classical result says that this process run backwards in time is again an SDE, and that the only unknown in it is the score:

$$dx = \Big[-\tfrac{1}{2}\beta(t)\, x - \beta(t)\, \nabla_x \log p_t(x) \Big] dt + \sqrt{\beta(t)}\, d\bar{w},$$

with dt negative and $\bar{w}$ a reverse-time Wiener process. Moreover, there is a deterministic **probability-flow ODE** whose solutions have the same marginal distributions p_t at every time:

$$\frac{dx}{dt} = -\tfrac{1}{2}\beta(t)\Big[x + \nabla_x \log p_t(x) \Big].$$

This unifies the field. DDPM's ancestral sampler is a discretization of the reverse SDE; NCSN [17] corresponds to a "variance exploding" SDE; and the ODE turns a diffusion model into a continuous normalizing flow, so exact log-likelihoods can be computed with the instantaneous change-of-variables formula [4]. The ODE also explains why deterministic samplers such as DDIM work, and why general-purpose ODE solvers can be applied.

Sampling is a reverse process

Plain version. A loss evaluated at one random noise level is not a generator. To generate, you must chain many reverse steps with a consistent schedule, starting from pure noise — and you can choose how many steps and how much fresh randomness to inject.

DDPM ancestral sampling

DDPM samples by following the learned reverse chain:

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z, \qquad z \sim \mathcal{N}(0, I), \qquad \sigma_t^2 \in \{\beta_t,\ \tilde{\beta}_t\},$$

with no noise added at the final step. Ho et al. [3] report similar results for $\sigma_t^2 = \beta_t$ and $\sigma_t^2 = \tilde{\beta}_t$; learned variances [19] help most when using fewer steps.

Code: DDPM on a two-dimensional distribution

This complete CPU example (about 10 seconds) uses two Gaussian clusters centered at $x = \pm 1.5$ with standard deviation 0.25. It contains every operation of an image model — schedule, random-time training, noise prediction, reverse mean — without an image-scale budget. With 100 steps and this schedule, almost no signal survives to $t = T$.

```
import torch
from torch import nn
from torch.nn import functional as F

torch.manual_seed(0)
torch.set_num_threads(1)
T = 100
beta = torch.linspace(.001, .15, T)
alpha = 1-beta
abar = alpha.cumprod(0)
previous = torch.cat([torch.ones(1), abar[:-1]])
post_var = beta * (1-previous) / (1-abar)
net = nn.Sequential(nn.Linear(3, 64), nn.SiLU(),
                    nn.Linear(64, 64), nn.SiLU(), nn.Linear(64, 2))
opt = torch.optim.Adam(net.parameters(), lr=.002)
def data(n):
    side = torch.randint(2, (n, 1))*2-1
```

```
    return torch.cat([side*1.5, torch.zeros(n, 1)], 1) + .25*torch.randn(n, 2)
for step in range(3000):
    x0 = data(128)
    t = torch.randint(T, (len(x0),))
    eps = torch.randn_like(x0)
    a = abar[t, None]
    xt = a.sqrt()*x0 + (1-a).sqrt()*eps
    predicted = net(torch.cat([xt, t[:, None].float()/(T-1)], 1))
    loss = F.mse_loss(predicted, eps)
    opt.zero_grad(); loss.backward(); opt.step()
net.eval()
with torch.no_grad():
    x = torch.randn(512, 2)
    for t in reversed(range(T)):
        time = torch.full((len(x), 1), t/(T-1))
        eps = net(torch.cat([x, time], 1))
        mean = (x-beta[t]/(1-abar[t]).sqrt()*eps)/alpha[t].sqrt()
        x = mean + post_var[t].sqrt()*torch.randn_like(x) if t else mean
    assert torch.isfinite(x).all()
    print("terminal signal:", round(abar[-1].item(), 6))
    left = x[x[:, 0] < 0]
    print("fraction in left cluster:", round((x[:, 0] < 0).float().mean().item(), 3))
    print("left cluster: mean x", round(left[:, 0].mean().item(), 2),
          "std", round(left[:, 0].std().item(), 2), "(data: -1.5 and 0.25)")
```

On our machine it prints a terminal signal of 0.000349, a left-cluster fraction of 0.441, and a left-cluster mean and standard deviation of -1.47 and 0.25. The cluster *shapes* are right; the 44/56 split shows the small network has not learned the mode weights exactly. This is why a single summary number is never enough: always plot samples, and check mode coverage, within-mode spread and outliers against fresh data.

DDIM: same training, different sampler

Song, Meng and Ermon [5] noticed that $\mathcal{L}_{\text{simple}}$ only depends on the marginals $q(x_t \mid x_0)$, not on the Markov chain that produced them. Many non-Markovian forward processes share those marginals, and each gives a different reverse sampler — all usable with the *same* trained ϵ_θ. The

DDIM update, for any decreasing subsequence of steps $\tau_1 > \tau_2 > \cdots$, is

$$x_{\tau_{i+1}} = \sqrt{\bar\alpha_{\tau_{i+1}}}\,\hat{x}_0 + \sqrt{1-\bar\alpha_{\tau_{i+1}}-\sigma^2}\,\epsilon_\theta(x_{\tau_i},\tau_i) + \sigma z, \qquad \sigma = \eta\sqrt{\frac{1-\bar\alpha_{\tau_{i+1}}}{1-\bar\alpha_{\tau_i}}\Big(1-\frac{\bar\alpha_{\tau_i}}{\bar\alpha_{\tau_{i+1}}}\Big)},$$

where $\hat{x}_0$ is the clean estimate at x_{τ_i} and $\eta \geq 0$ controls stochasticity. Read it as: "estimate the clean example, then re-noise it to the next, lower noise level using the predicted noise direction." With $\eta = 1$ and consecutive steps, this recovers DDPM's posterior variance; with $\eta = 0$ the sampler

is **deterministic** and is a discretization of the probability-flow ODE. Determinism has practical consequences: the same starting noise always gives the same output, so you can interpolate in

noise space or invert a real image to its noise (DDIM inversion), which is the basis of many editing methods.

The following standalone example repeats the same small training setup so it can run independently. When experimenting in one session, reuse the already trained `net`, `abar` and `T` instead:

```
import torch
from torch import nn
from torch.nn import functional as F

torch.manual_seed(0)
torch.set_num_threads(1)
T = 100
beta = torch.linspace(.001, .15, T)
alpha = 1-beta
abar = alpha.cumprod(0)
previous = torch.cat([torch.ones(1), abar[:-1]])
post_var = beta * (1-previous) / (1-abar)
net = nn.Sequential(nn.Linear(3, 64), nn.SiLU(),
                    nn.Linear(64, 64), nn.SiLU(), nn.Linear(64, 2))
opt = torch.optim.Adam(net.parameters(), lr=.002)
def data(n):
    side = torch.randint(2, (n, 1))*2-1
    return torch.cat([side*1.5, torch.zeros(n, 1)], 1) + .25*torch.randn(n, 2)
for step in range(3000):
    x0 = data(128)
    t = torch.randint(T, (len(x0),))
    eps = torch.randn_like(x0)
    a = abar[t, None]
    xt = a.sqrt()*x0 + (1-a).sqrt()*eps
    predicted = net(torch.cat([xt, t[:, None].float()/(T-1)], 1))
    loss = F.mse_loss(predicted, eps)
    opt.zero_grad(); loss.backward(); opt.step()
net.eval()

def ddim_sample(n, steps, eta=0.0, seed=1):
    g = torch.Generator().manual_seed(seed)
    taus = torch.linspace(T-1, 0, steps).round().long()
    x = torch.randn(n, 2, generator=g)
    with torch.no_grad():
        for i, t in enumerate(taus):
            time = torch.full((n, 1), t.item()/(T-1))
            eps = net(torch.cat([x, time], 1))
            a_t = abar[t]
            a_prev = abar[taus[i+1]] if i+1 < steps else torch.tensor(1.)
            x0_hat = (x - (1-a_t).sqrt()*eps) / a_t.sqrt()
            sigma = eta*((1-a_prev)/(1-a_t)*(1-a_t/a_prev)).sqrt()
            x = (a_prev.sqrt()*x0_hat + (1-a_prev-sigma**2).sqrt()*eps
                 + sigma*torch.randn(n, 2, generator=g))
    return x

for steps in [5, 20, 100]:
    s = ddim_sample(2000, steps)
    left = s[s[:, 0] < 0]
```

```
    print(f"{steps:3d} steps: left fraction {(s[:, 0] < 0).float().mean():.3f}, "
          f"left-cluster mean x {left[:, 0].mean():.2f}, std {left[:, 0].std():.2f}")
```

The output shows the typical pattern. With 5 steps the left cluster is centered at −1.11 with standard deviation 0.39 — the samples are pulled toward the middle and smeared, because large steps follow a curved trajectory with straight lines. With 20 steps it is −1.37 and 0.30; with 100 steps −1.42 and 0.24, close to the data. No retraining was needed to change the step count.

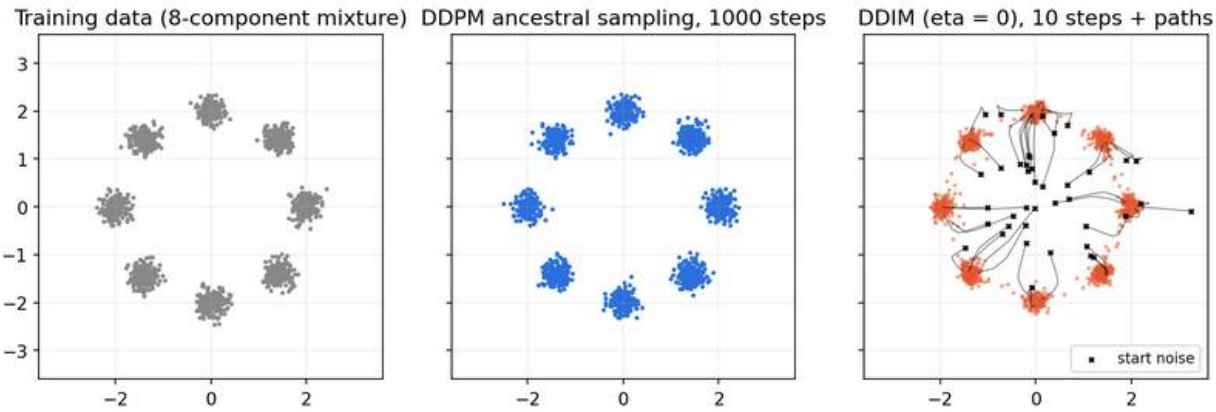


Figure 3 — Samplers driven by the exact denoiser of an 8-component mixture, so only the sampling rule differs. DDPM with 1000 steps reproduces the data; deterministic DDIM with 10 steps lands near the modes but with slightly wider, smeared clusters. Black lines are DDIM trajectories from starting noise (×).

Faster solvers

Once sampling is "solve an ODE," numerical analysis applies. Karras et al. (EDM) [20] used a second-order Heun solver with a carefully chosen spacing of noise levels and showed that sampler, schedule, preconditioning and training noise distribution can be tuned separately. DPM-Solver [21] exploits the semi-linear structure of the diffusion ODE — the linear part can be integrated exactly — to reach good quality in around 10 network evaluations. Stochastic samplers add a little fresh noise at each step; that costs more steps but can correct accumulated errors.

Whatever the solver, report the **number of function evaluations** (**NFE**): the number of network calls, which, rather than "steps," is the true sampling cost.

Steering generation: conditioning and guidance

Plain version. To generate "a cat" instead of "anything," give the network the condition. Guidance then turns a dial: follow the condition more strictly, at the cost of variety.

Conditioning

A conditional model learns $\epsilon_\theta(x_t, t, c)$ for a condition c. Class labels enter through embeddings added to the time embedding; text enters through **cross-attention** from image features to text-encoder tokens [7]; images (for super-resolution, inpainting, editing) are concatenated to the input channels or fed through extra encoders.

Classifier guidance

Bayes' rule for the noisy marginals gives $\nabla \log p_t(x_t \mid y) = \nabla \log p_t(x_t) + \nabla \log p_t(y \mid x_t)$. Dhariwal and Nichol [18] trained a separate classifier $p_\phi(y \mid x_t)$ on *noisy* images and added its gradient, scaled by w, to an unconditional model. In noise units,

$$\hat{\epsilon} = \epsilon_\theta(x_t, t) - w\sqrt{1-\bar{\alpha}_t}\,\nabla_{x_t} \log p_\phi(y \mid x_t).$$

A scale $w > 1$ samples roughly from $p_t(x_t)\,p_t(y \mid x_t)^w$, a sharpened distribution that trades diversity for fidelity. Together with architecture improvements, this was the work that showed diffusion models beating GANs on ImageNet [35] sample quality [18]. Its drawback is the extra classifier, which must be trained on noisy inputs for every noise level.

Classifier-free guidance (CFG)

Ho and Salimans [6] removed the classifier. During training, the condition is replaced by a "null" token for a random fraction of examples, so one network learns both $\epsilon_\theta(x_t, t, c)$ and $\epsilon_\theta(x_t, t, \emptyset)$. Bayes' rule says the classifier gradient equals the difference of the conditional and unconditional scores, so the implicit-classifier version of the formula above is

$$\epsilon_{\text{guided}} = \epsilon_\theta(x_t, t, \emptyset) + w(\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \emptyset)).$$

In this convention $w = 0$ is unconditional, $w = 1$ is the ordinary conditional model, and $w > 1$ extrapolates past it. (Some papers write the same thing as $(1 + w')\epsilon_c - w'\epsilon_u$, so their "scale" is $w - 1$ in our notation; always check the convention before comparing numbers.)

The snippet below uses the exact noise predictor of a four-blob mixture, so any odd behaviour is due to guidance itself, not to a badly trained network. Class 1 is the two blobs with $x > 0$.

```
import numpy as np

# Data: four Gaussian blobs at the corners of a square; class c is the sign of x.
MU = np.array([[-1.5, -1.5], [-1.5, 1.5], [1.5, -1.5], [1.5, 1.5]])
SIG = 0.2
CLASS = np.array([0, 0, 1, 1])

def eps_exact(x, abar, comps):
    """Exact E[eps | x_t] for a Gaussian mixture restricted to `comps`."""
    m = np.sqrt(abar)*MU[comps]
    var = abar*SIG**2 + 1 - abar
```

```
    logw = -((x[:, None] - m[None])**2).sum(-1) / (2*var)
    w = np.exp(logw - logw.max(1, keepdims=True))
    w /= w.sum(1, keepdims=True)
    score = -(w[..., None]*(x[:, None] - m[None])).sum(1) / var
    return -np.sqrt(1-abar)*score

abar = np.cumprod(1 - np.linspace(1e-4, 0.02, 1000))
taus = np.linspace(999, 0, 50).round().astype(int)
everything, class1 = np.arange(4), np.where(CLASS == 1)[0]
print("data: mean distance to its mode", round(SIG*np.sqrt(np.pi/2), 3))
for w in [0, 1, 3, 8]:
    x = np.random.default_rng(0).normal(size=(4000, 2))
    for i, t in enumerate(taus):                    # deterministic DDIM
        e_u, e_c = eps_exact(x, abar[t], everything), eps_exact(x, abar[t], class1)
        e = e_u + w*(e_c - e_u)                     # classifier-free guidance
        x0 = (x - np.sqrt(1-abar[t])*e) / np.sqrt(abar[t])
        a_prev = abar[taus[i+1]] if i+1 < len(taus) else 1.0
        x = np.sqrt(a_prev)*x0 + np.sqrt(1-a_prev)*e
    dist = np.linalg.norm(x[:, None] - MU[None], axis=-1).min(1)
    print(f"w={w}: class-1 fraction {(x[:, 0] > 0).mean():.3f}, "
          f"mean distance to nearest mode {dist.mean():.3f}")
```

The data's mean distance to its mode is 0.251. With $w = 0$ about half the samples (0.490) are class 1; with $w = 1$ all of them are, at distance 0.225 (50 DDIM steps slightly under-disperse). At $w = 3$ the distance is 0.259 and at $w = 8$ it grows to 0.446: strong guidance does not just "make it more class 1" — it pushes samples *past* the data, a toy version of the over-saturated colours and distorted shapes seen with very high guidance in image models. Figure 4 shows the same effect on the 8-mode mixture.

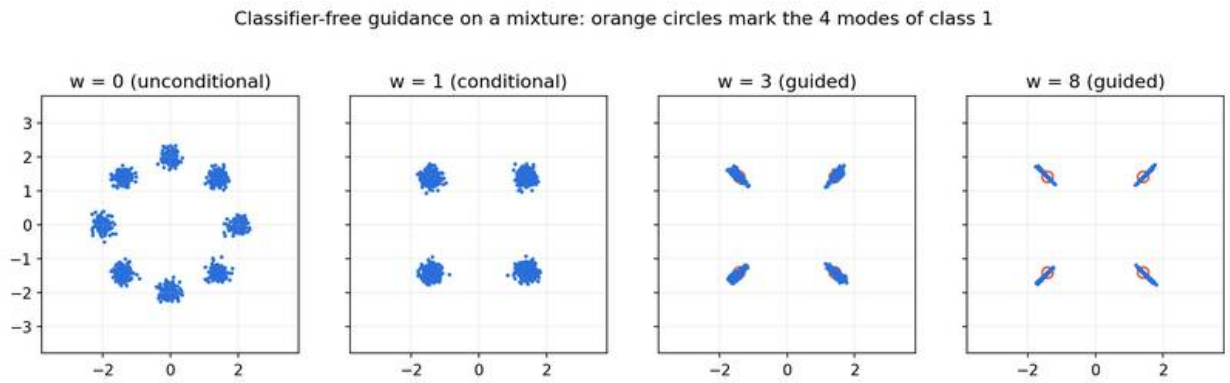


Figure 4 — Exact classifier-free guidance on an 8-mode mixture whose class 1 has 4 modes (orange circles). $w = 1$ already restricts samples to class 1; larger w shrinks clusters along some directions and stretches them along others — the samples no longer follow any data distribution.

Guidance has a compute cost too: each step needs a conditional *and* an unconditional prediction, so a 30-step guided sampler costs 60 network evaluations unless the two are fused. And w is a sampling knob, not a calibrated measure of how well the image matches the prompt.

Latent diffusion and diffusion transformers

Plain version. Pixels are expensive. Compress the image into a smaller grid of features with an autoencoder, run diffusion in that compressed space, then decode. Separately, the denoising network itself can be a transformer instead of a convolutional U-Net [37].

Latent diffusion

Rombach et al. [7] train an autoencoder first: an encoder $z = \mathcal{E}(x)$ maps an $H \times W \times 3$ image to an $h \times w \times c$ latent with a spatial downsampling factor $f = H/h$, and a decoder $\mathcal{D}$ maps back. The autoencoder is trained with reconstruction, perceptual and adversarial losses plus a light regularizer (a small KL term toward a Gaussian, or vector quantization) so that the latent space stays well behaved. Diffusion is then trained on z exactly as before, and generation ends with $\hat{x} = \mathcal{D}(z_0)$. With $f = 8$, a 512×512 image becomes a 64×64 latent: 64 times fewer spatial positions for every network call.

The trade-off is a reconstruction bottleneck: information discarded by the encoder cannot be faithfully recovered from that code, although the decoder may synthesize plausible detail. The latent's scale must also match what the diffusion model was trained on; implementations rescale latents to roughly unit variance. Text conditioning in [7] uses cross-attention layers in the U-Net, the design behind the Stable Diffusion family.

Diffusion transformers

DiT (Peebles and Xie [8]) replaces the U-Net with a Vision Transformer [41] (see ML 04) over latent patches. A latent of size $h \times w$ is cut into $p \times p$ patches, giving hw/p^2 tokens; smaller p means more tokens, more compute and better quality. The timestep and class label condition every block through **adaptive layer norm**: they predict the scale and shift of each LayerNorm, and in the "adaLN-Zero" variant also a gating factor initialized to zero, so each residual block starts as the identity. The paper's main finding is a scaling trend: across model sizes and patch sizes, more transformer compute (GFLOPs) gave consistently lower FID [8].

It helps to keep three choices separate. The **architecture** (U-Net or transformer) is the function approximator. The **objective** (noise, x_0, v, or flow velocity) is what it regresses. The **sampler** (DDPM, DDIM, ODE solver) is how predictions become an image. Stable Diffusion 3 [9] is an example of a new combination: a latent transformer with separate weights for the image and text token streams that attend jointly ("MM-DiT"), trained with a rectified-flow objective and a timestep distribution biased toward intermediate noise levels.

For video, the latent and the attention must cover time as well as space, so compression and attention patterns become the main engineering problem. Melnik et al.'s survey [31] reviews architectures, temporal-consistency mechanisms and evaluation for video diffusion. Per-frame quality

is not enough there: object permanence, motion plausibility and long-duration consistency need their own evaluation.

Pixel space, latent space and representation autoencoders

Plain version. The generator needs a place to do its repeated updates. That place can be the image itself, a compact reconstruction code, or features learned by a visual representation model. A latent stage buys a particular computational and statistical trade-off; it is not a prerequisite for generative modeling.

Why pixels become latents and then pixels again

Training images arrive as pixels, while the output must be viewable as pixels. Between these endpoints, an encoder can remove redundancy and produce a grid that is cheaper to update repeatedly [7]. Train the encoder and decoder on reconstruction first, then fit a generative distribution to encoded training images. These are two different tasks: reconstruction preserves a given example; generation learns which encoded examples are plausible.

The training and sampling routes are

$$\begin{aligned} \text{training: } & x \xrightarrow{\mathcal{E}} z_{\text{data}} \xrightarrow{\text{corrupt}} z_t \xrightarrow{\text{prediction loss}} \theta, \\ \text{generation: } & z_{\text{noise}} \xrightarrow{\text{learned sampler}} \hat{z} \xrightarrow{\mathcal{D}} \hat{x}, \\ \text{reconstruction check: } & x \xrightarrow{\mathcal{E}} \mathcal{E}(x) \xrightarrow{\mathcal{D}} \tilde{x}. \end{aligned}$$

Here x is a training image; $\mathcal{E}$ and $\mathcal{D}$ are the encoder and decoder; z_{data} is an encoded image; z_t is its corrupted state; θ parameterizes the generative predictor; $\hat{z}$ is a generated code; $\hat{x}$ is the generated image; and $\tilde{x}$ is a reconstruction. Pure text-to-image sampling does **not** encode an input image first. Editing and image-to-image generation can encode an actual input image and begin from a partially corrupted code.

The main saving occurs because the denoiser or flow predictor runs repeatedly, while encoding training data and decoding a sample occur much less often. A spatial factor f reduces grid positions by f^2, but the number of latent channels also matters. For an image with three channels and a latent with c channels, the scalar-count reduction is $3f^2/c$. It is incorrect to describe the grid reduction as the reduction in all floating-point values or total runtime.

For a transformer, token count and token width must also be distinguished. Full attention stores a matrix with a number of entries proportional to N^2, where N is the token count; feature projections depend on token width as well [41]. A patch size can reduce N in either pixels or latents. None of these counts alone predicts wall-clock speed because kernels, memory traffic, the decoder and conditioning contribute.

```
# Synthetic shape bookkeeping, not a benchmark or a specific checkpoint.
H = W = 512
channels, spatial_factor, patch = 4, 8, 2
pixel_values = H*W*3
h = H//spatial_factor
w = W//spatial_factor
latent_values = h*w*channels
pixel_tokens = (H//patch)*(W//patch)
latent_tokens = (h//patch)*(w//patch)
print("Spatial positions reduction:", (H*W)//(h*w))
print("Scalar values reduction:", pixel_values//latent_values)
print("Dense attention entries reduction:", pixel_tokens**2//latent_tokens**2)
assert pixel_values//latent_values == 48
assert pixel_tokens//latent_tokens == 64
assert pixel_tokens**2//latent_tokens**2 == 4096
```

This constructed example yields 64 times fewer spatial positions, 48 times fewer scalar values and 4096 times fewer attention entries **at the same numerical patch size in each coordinate space**. Changing patch sizes changes the comparison. A pixel transformer with much larger patches can have the same token count as a latent transformer.

The cost is information loss. If two images have the same code, a deterministic decoder cannot recover their different exact details. A decoder may invent plausible texture, but that does not restore information about the original image. The reconstruction check therefore deserves its own evaluation before generative training. Inspect fine lines, lettering and temporal changes, rather than relying only on average image error.

The following deliberately simple encoder averages blocks and the decoder repeats those averages. It demonstrates a bottleneck without pretending to reproduce a learned VAE or RAE.

```
import numpy as np

y, x = np.mgrid[:64, :64]
checker = ((x+y) % 2).astype(float)
smooth = x/63
image = .6*smooth + .4*checker
f = 4
latent = image.reshape(64//f, f, 64//f, f).mean(axis=(1, 3))
reconstruction = np.repeat(np.repeat(latent, f, axis=0), f, axis=1)
print("Synthetic reconstruction MSE:", round(np.mean((image-reconstruction)**2), 6))
assert np.mean((image-reconstruction)**2) > .04
# Recovering the exact latent does not recover the missing checkerboard.
assert np.allclose(reconstruction[0, :f], reconstruction[0, 0])
```

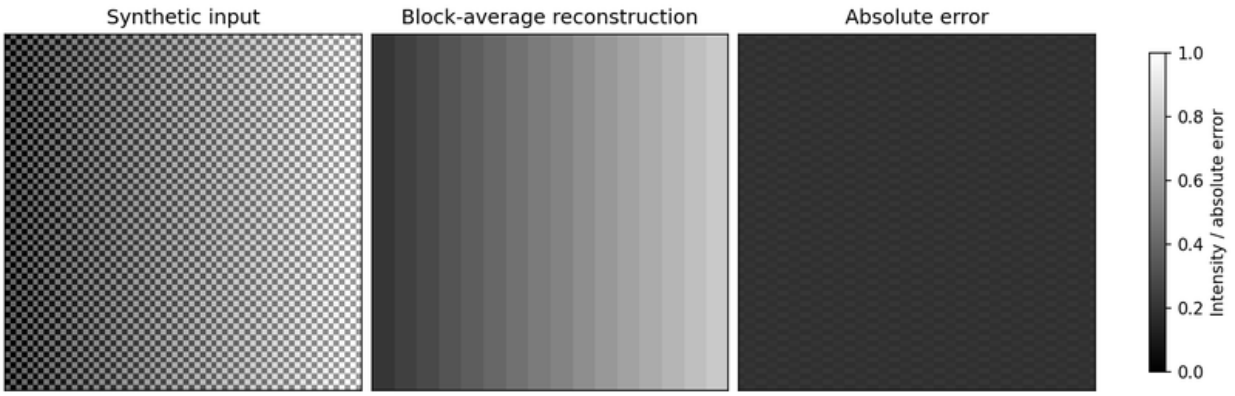


Figure 5 — A latent bottleneck can remove detail before the generative model is trained. This block-average toy is an explanation of information loss, not an evaluation of a learned autoencoder.

For editing, repeatedly encoding and decoding a result can accumulate distortion. Keep intermediate edits in the compatible latent space when the pipeline permits it, and inspect the eventual pixel output. A good unconditional generator may still be a poor reconstruction system. The claim that an autoencoder sets a ceiling concerns faithfulness to encoded information; it does not mean that a generative decoder can never synthesize new fine texture.

Reconstruction codes versus semantic representation codes

A VAE encoder is trained with a reconstruction objective and a distributional regularizer; its code is designed to support decoding and sampling [7]. A pretrained recognition encoder instead learns useful visual features. These goals can align, but recognition invariance may discard details needed for faithful reconstruction. The decoder and the retained feature structure determine whether the representation is a useful generative space.

Zheng, Ma, Tong and Xie's **Representation Autoencoders** (**RAEs**) [43] pair a frozen pretrained representation encoder with a trained decoder. Their ICLR 2026 study uses semantic visual features as the space for generative modeling. It examines high-dimensional token features, dimension-dependent noise scheduling, decoder robustness and a shallow wide diffusion head. This changes the representation being generated, rather than simply adding a feature-alignment loss to a conventional latent predictor.

Do not equate "latent" with "smaller in every dimension." A spatially sparse feature grid can have many channels per token. An RAE is also not automatically a variational model: a frozen feature extractor plus a decoder does not by itself define a Gaussian posterior, a KL regularizer or a likelihood bound. Keep reconstruction, representation quality and generative quality as separate measurements.

Tong et al. [44] extend this approach to freeform text-to-image generation in 2026. Their study compares RAE and VAE spaces at larger scales and examines decoder data composition. It finds that dimension-aware noise scheduling remains important while some small-scale architectural

and decoder modifications offer little benefit at the tested larger scales. This is a reason to re-test recipes when scale and data change, rather than treating an ImageNet result as a universal design rule.

For a new application, inspect whether the chosen encoder preserves the content that matters: fine text for documents, small defects for inspection, or frame-to-frame differences for video. Evaluate the decoder on held-out real features and on generated features. A decoder trained only on exact training codes can be sensitive to small off-distribution errors in a sampled code. Normalize features using the training recipe, and record the encoder version with the generative checkpoint.

JiT: direct clean-image prediction on pixels

Li and He's **Just image Transformers (JiT)** [42], published at CVPR 2026 after a 2025 preprint, revisit large-patch transformers operating directly on pixels. The network predicts the clean image; sampling derives a flow velocity from that prediction. Their controlled experiments support a manifold-based explanation: a bottleneck can retain structured clean-image information while making direct prediction of high-dimensional noise or velocity difficult. The reported setup uses no pretrained latent tokenizer or auxiliary representation loss. This is evidence for a particular architecture and training regime, not a proof that pixel-space models always outperform latent generators.

The crucial distinction is **direct network output versus the quantity used in the loss or solver**. For the straight path defined below, use $y_t = (1-t)\epsilon + tx$, with noise at $t = 0$ and clean data at $t = 1$. A network that directly predicts $\hat{x}_\theta(y_t, t)$ can supply

$$\hat{u}_\theta(y_t, t) = \frac{\hat{x}_\theta(y_t, t) - y_t}{1-t}, \qquad \|\hat{u}_\theta - (x-\epsilon)\|^2 = \frac{\|\hat{x}_\theta - x\|^2}{(1-t)^2}.$$

Here x is the clean image; ϵ is the noise draw; y_t is their interpolation; $\hat{x}_\theta$ is the direct prediction; and $\hat{u}_\theta$ is the corresponding transport velocity. The identity holds for $t < 1$ before numerical clipping. An x-output network can thus train with a velocity-space loss. Calling it "clean prediction" does not mean its loss is an unweighted clean-image MSE. JiT's released implementation makes this distinction explicit [42].

Near $t = 1$, division by $1 - t$ needs an endpoint convention and numerical safeguards. For the interior algebra, a fixed clean-image error is amplified increasingly as the endpoint approaches. During sampling, use the checkpoint's time distribution, noise scaling and solver recipe. Changing the direct output label alone does not reproduce the method.

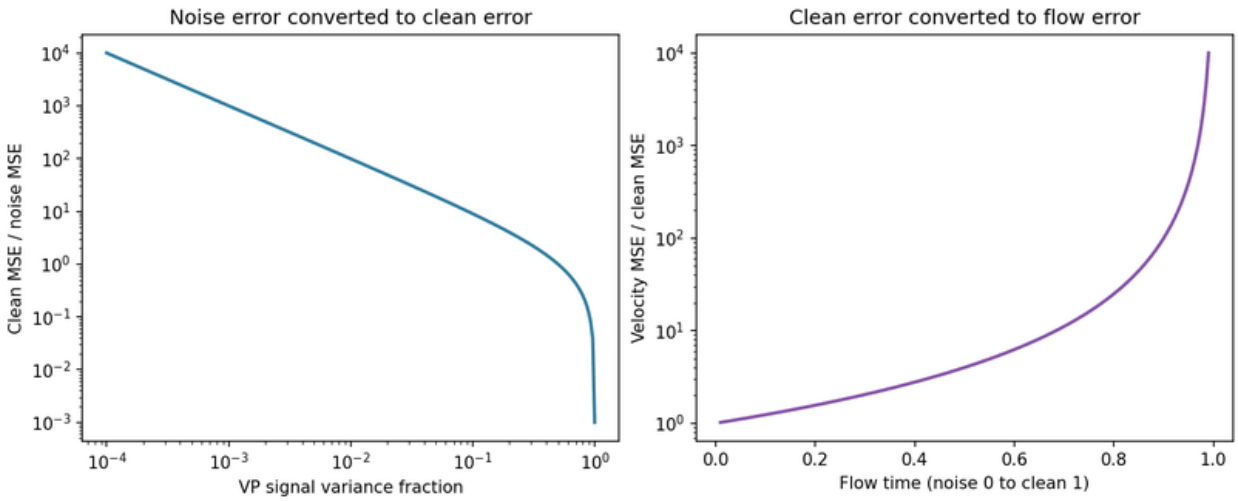


Figure 6 — Equivalent predictions use different units. The left curve shows noise-error amplification at low signal; the right shows velocity-error amplification near the clean endpoint. The curves are algebraic identities, not model performance measurements.

A useful comparison fixes training data, compute, resolution, token count and the sampler, then changes the representation or target one at a time. Compare a pixel transformer, a compact VAE-latent transformer and a representation-latent transformer on reconstruction needs, generated quality and total cost. Record encoder/decoder cost and any representation pretraining. A model that uses a pretrained visual encoder has inherited training even when its generative stage starts from scratch.

Flow matching: learn a transport velocity

Plain version. Forget noise schedules for a moment. Draw a noise point and a data point, connect them by a straight line, and train a network to predict the direction of travel at every point along such lines. To generate, drop a fresh noise point and follow the predicted directions.

Time convention change. In this section time runs from **noise at** $t = 0$ **to data at** $t = 1$, the opposite of the diffusion sections above (where large t meant more noise).

The conditional flow-matching objective

A continuous normalizing flow moves points with an ODE $dx_t/dt = v(x_t, t)$. If the velocity field v is right, it transports the noise distribution $p_0 = \mathcal{N}(0, I)$ to the data distribution p_1. Lipman et al. [10] (and, concurrently, Liu et al. [11] and Albergo and Vanden-Eijnden's stochastic interpolants [34]) showed how to train v without ever simulating the ODE.

Pair noise $z \sim \mathcal{N}(0, I)$ with a data example $x \sim p_1$ and define the straight **conditional path**

$$x_t = (1-t)\,z + t\,x, \qquad \frac{dx_t}{dt} = u_t(x_t \mid z, x) = x - z.$$

Every pair moves at constant speed in a straight line. The marginal velocity we actually want is the average of these conditional velocities over all pairs that pass through the point:

$$u_t(y) = \mathbb{E}[\,x - z \mid x_t = y\,].$$

We cannot compute this average, but we do not need to. Compare the "ideal" loss with the one we can compute:

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,x_t}\|v_\theta(x_t,t) - u_t(x_t)\|^2, \qquad \mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{t,z,x}\|v_\theta(x_t,t) - (x-z)\|^2,$$

with $t \sim \mathcal{U}[0,1]$. Expand both squares. The $\|v_\theta\|^2$ terms are identical. The cross terms are equal because, by the tower rule,

$\mathbb{E}[v_\theta(x_t,t)^\top (x-z)] = \mathbb{E}[v_\theta(x_t,t)^\top\, \mathbb{E}[x-z \mid x_t]] = \mathbb{E}[v_\theta^\top u_t(x_t)]$. The remaining terms do not involve θ. Therefore

$$\nabla_\theta \mathcal{L}_{\mathrm{CFM}} = \nabla_\theta \mathcal{L}_{\mathrm{FM}},$$

so regressing the per-pair velocity $x-z$ trains the marginal velocity field. This is the same trick as in diffusion: a target that is noisy for any single example has the right conditional mean. Generation solves $dx/dt = v_\theta(x,t)$ from $t=0$ to $t=1$ with any ODE solver.

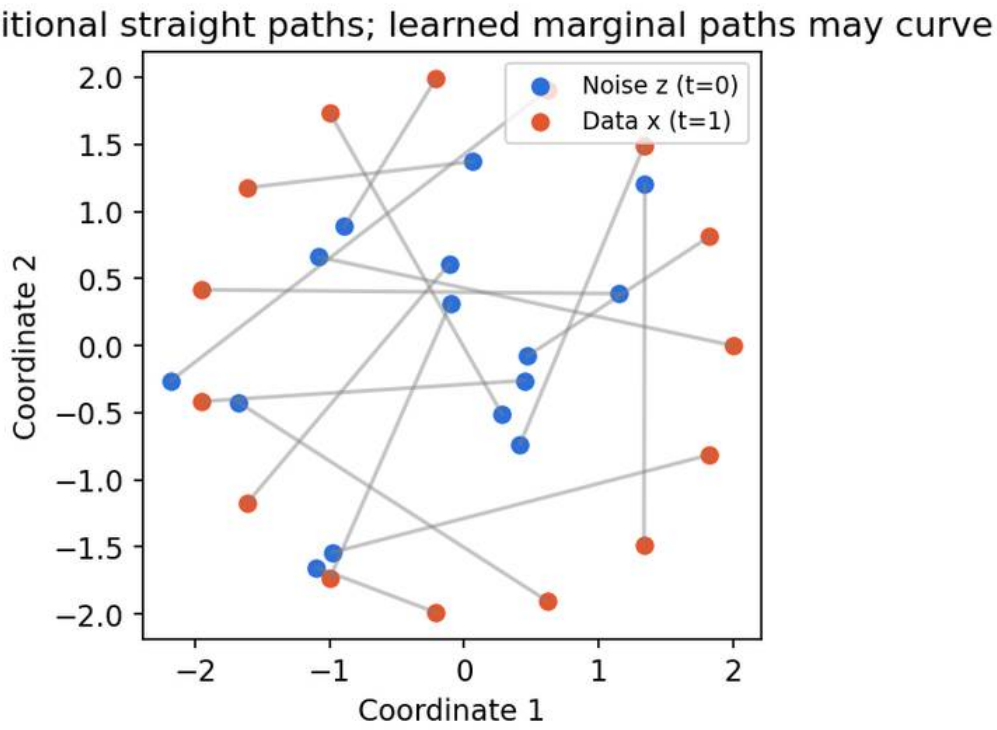


Figure 7 — Conditional training paths are straight. Paths of different pairs cross, so the marginal velocity at a crossing point is an average, and the learned trajectories can curve.

How flow matching relates to diffusion

The linear path is a Gaussian path: given x, $x_t \sim \mathcal{N}(t\,x,\ (1-t)^2 I)$. Writing $\hat{z} = \mathbb{E}[z \mid x_t]$ (a noise prediction) and using $x = (x_t - (1-t)z)/t$,

$$u_t(x_t) = \frac{x_t - \hat{z}}{t}, \qquad \nabla \log p_t(x_t) = -\frac{\hat{z}}{1-t}, \qquad \text{so} \qquad u_t(x_t) = \frac{x_t + (1-t)\,\nabla \log p_t(x_t)}{t}.$$

So a flow-matching velocity, a noise predictor and a score are again interchangeable for this path; what differs is the path (straight lines versus the VP curve), the weighting of noise levels in the loss, and the time convention. Diffusion's probability-flow ODE is itself a flow along a particular Gaussian path. Lipman et al.'s guide [2] and the stochastic-interpolant framework [34] treat both as special cases of one construction. The practical advice is the same as for parameterizations: the targets are related by derivation, not interchangeable by renaming.

Code: flow matching on two clusters

This standalone example learns the same two-cluster distribution as the DDPM snippet with linear conditional paths, then samples with Euler steps of size $1/N$.

```
import torch
from torch import nn
from torch.nn import functional as F

torch.manual_seed(0)
torch.set_num_threads(1)
net = nn.Sequential(nn.Linear(3, 64), nn.SiLU(),
                    nn.Linear(64, 64), nn.SiLU(), nn.Linear(64, 2))
opt = torch.optim.Adam(net.parameters(), lr=.002)
for step in range(3000):
    z = torch.randn(128, 2)
    side = torch.randint(2, (128, 1))*2-1
    x = torch.cat([1.5*side, torch.zeros(128, 1)], 1) + .25*torch.randn(128, 2)
    t = torch.rand(128, 1)
    xt = (1-t)*z+t*x
    loss = F.mse_loss(net(torch.cat([xt, t], 1)), x-z)
    opt.zero_grad(); loss.backward(); opt.step()
net.eval()
def euler_sample(n, steps, seed=1):
    x = torch.randn(n, 2, generator=torch.Generator().manual_seed(seed))
    with torch.no_grad():
        for k in range(steps):
            t = torch.full((n, 1), k/steps)
            x = x + net(torch.cat([x, t], 1))/steps
    return x

for steps in [1, 4, 50]:
    s = euler_sample(2000, steps)
    assert torch.isfinite(s).all()
    left = s[s[:, 0] < 0]
```

```
    print(f"{steps:2d} Euler steps: left fraction {(s[:, 0] < 0).float().mean():.3f}, "
          f"left-cluster mean x {left[:, 0].mean():.2f}, std {left[:, 0].std():.2f}")
```

It prints, for 1, 4 and 50 steps, left-cluster means of -0.18, -1.20 and -1.45 with standard deviations 0.12, 0.42 and 0.29. One Euler step lands almost at the overall data mean: at $t = 0$ the marginal velocity is $\mathbb{E}[x] - z$, so a single full step sends *every* noise point to $\mathbb{E}[x]$. Straight conditional paths do not make the learned marginal flow straight. Figure 8 shows the same effect with the exact velocity of an 8-mode mixture.

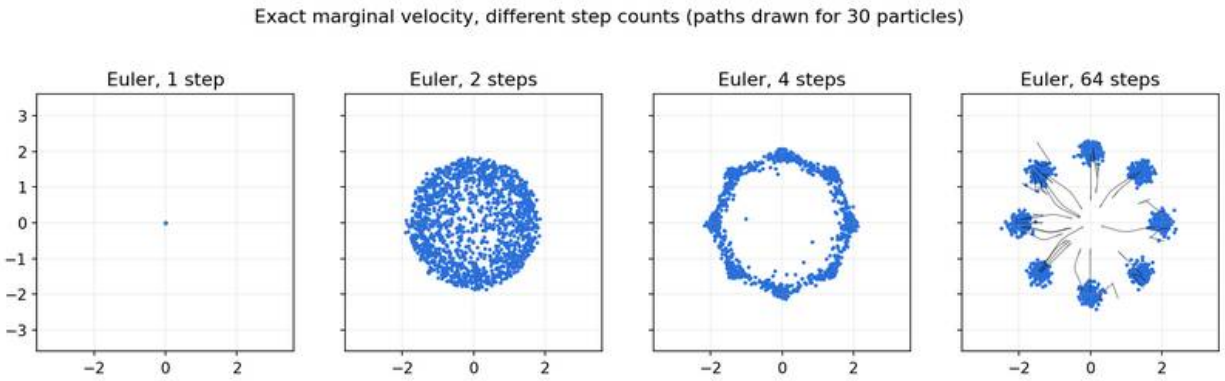


Figure 8 — Even with the exact marginal velocity, Euler sampling needs many steps: one step collapses every sample to the data mean, and the 64-step trajectories (black) bend toward the modes late in the path.

Rectified flow and reflow

Liu, Gong and Liu [11] use the same straight interpolation ("rectified flow") and add **reflow**: after training, generate pairs $(z, \hat{x})$ by solving the learned ODE from each z, then retrain on these *coupled* pairs instead of independent ones. Because the new pairs are already connected by the flow, their straight lines cross less, the learned trajectories become straighter, and fewer Euler steps are needed; a straight enough flow can then be distilled into a one-step model. Stable Diffusion 3 [9] showed that rectified-flow training of a large latent transformer works at text-to-image scale.

Pretrained image and video generators as complete pipelines

Plain version. A pretrained generator is a bundle of compatible components. Loading the denoiser weights alone does not establish the right representation, text conditioning, prediction target or sampling schedule. The image/video boundary also changes what a reconstruction must preserve.

Read the components before running the checkpoint

A practical pipeline combines a text encoder, an optional image/video encoder, a time-conditioned predictor, a sampler and an optional decoder. A separate refinement model may follow the first generator. Each component expects particular tensor shapes and scales. A scheduler declaring `v_prediction` may mean VP velocity; a flow scheduler may expect a derivative along a straight interpolation. Check the training path rather than guessing from a configuration string.

The following verified examples illustrate architectural differences. They are fixed research baselines, rather than a ranking of the latest product releases. The references include their papers or technical reports and official release documentation.

Baseline	Generation representation and predictor	What to inspect in the pretrained pipeline
SDXL [45]	Reconstruction latents with a convolutional denoiser	Text conditioning, latent scaling, optional base/refiner composition and compatible sampling configuration
Stable Diffusion 3 [9]	Latent rectified flow with a joint text–image transformer	Flow time convention, text streams, time sampling and autoencoder compatibility
FLUX.1 [46]	Latent rectified-flow transformer	Exact release variant, guidance behavior, text conditioning, autoencoder and release terms
JiT [42]	Class-conditioned clean-image prediction directly on pixels	Image normalization, patch/ resolution configuration, derived velocity and sampling recipe; this is not a freeform text-to-image checkpoint
HunyuanVideo [47]	Compressed video latents with a flow-matching transformer	Video autoencoder, image/video joint training, temporal dimensions and text-conditioning components
Wan2.1 [48]	Video diffusion transformer with a spatiotemporal autoencoder	Text-to-video versus image-to-video variant, causal video compression, temporal length and decode caching

SDXL [45] documents a larger latent image-generation architecture and optional refinement stage. Stable Diffusion 3 [9] illustrates how the same broad latent pipeline can instead use rectified flow and joint text/image attention. FLUX.1's official model card [46] describes a released rectified-flow transformer; its `.1-dev` variant is guidance-distilled. These facts do not make the checkpoints interchangeable. Ordinary two-pass CFG and a distilled guidance input can have different meanings and costs.

For image-to-image generation, encoding an input and adding partial noise trades input preservation against the ability to change it. The actual noise level is determined by the schedule, not merely by a UI label such as "strength." To compare edits, keep the same input and record the initial time, seed, prompt, guidance and reconstruction baseline. A reconstruction artifact can otherwise be mistaken for an edit requested by the text.

Video adds temporal compression and a consistency problem

Let a video have F frames, height H and width W. A hypothetical encoder compressing time by f_t and each spatial axis by f_s gives a rough latent grid of $(F/f_t) \times (H/f_s) \times (W/f_s)$ positions. Exact shapes depend on padding and special handling of the first frame. After patching, dense attention across all positions grows quadratically with the remaining token count; separately reducing time and space can therefore change cost substantially.

HunyuanVideo's technical report [47] combines a 3D video autoencoder with a flow-matching transformer and staged image/video training. Wan's report [48] explains a causal video VAE with special first-frame handling, spatial and temporal compression, and feature caching during chunked encoding/decoding. Causality in an autoencoder does not by itself imply that the entire video generator samples autoregressively; the generative attention and sampling procedure must be examined separately.

A frame can look excellent while a clip changes an object's identity, reverses motion unexpectedly or forgets an occluded object. Test reconstruction of held-out clips before generation, then evaluate appearance, motion, prompt correspondence and consistency across frames. Melnik et al. [31] provide the survey map for these video-specific issues. A per-image metric alone cannot diagnose temporal behavior.

A pretrained video generator also does not automatically become a reliable world model. Predicting plausible clips differs from predicting the consequences of a specified action. The world-model chapter develops this distinction through state, actions, rollouts and evaluation. Use the actual conditioning interface and evaluate the task the system claims to solve.

A reproducible first experiment

Start by recording a pipeline manifest: checkpoint revision, encoder/decoder revisions, text encoder, representation shape and scaling, prediction target, time direction, solver, evaluation count, guidance, dtype and seed. Then perform a reconstruction check if an autoencoder is present and generate a fixed small prompt/seed set. Compare one change at a time. Include decoder latency and peak memory when reporting the cost of a latent method.

This chapter's CPU examples test the underlying identities and failure modes without downloading weights. Full-resolution pretrained generation belongs in a separately provisioned experiment using the official release instructions [42], [45]–[48]. Treat an actual measured run, a paper's published result and a toy calculation as three different kinds of evidence.

Few-step and one-step generation

Plain version. If a sampler needs 50 network calls, train a second model that jumps along the trajectory in a few calls — or train one from scratch that is designed to jump.

The main families are:

- **Progressive distillation** [22]: a student learns to match two deterministic teacher steps with one step; repeating the procedure halves the step count each round.
- **Consistency models** [12]: learn a function $f_\theta(x_t, t)$ that maps *any* point on a probability-flow ODE trajectory to that trajectory's starting (clean) point. It must satisfy the boundary condition $f_\theta(x, \epsilon) = x$ at the smallest time ϵ, enforced by writing $f_\theta(x,t) = c_{\text{skip}}(t)\, x + c_{\text{out}}(t)\, F_\theta(x,t)$ with $c_{\text{skip}}(\epsilon) = 1$ and $c_{\text{out}}(\epsilon) = 0$. Training asks outputs at adjacent points of a trajectory to agree. In *consistency distillation*, adjacent points come from one ODE-solver step of a pretrained diffusion model; in *consistency training*, from the same data example noised at two levels, with no teacher. Sampling takes one step, or a few steps of "jump to clean, re-noise, jump again." Improved training recipes [32] simplified continuous-time consistency models and scaled them to 1.5 billion parameters on ImageNet 512 × 512 with two-step sampling.
- **Adversarial distillation**: ADD [33] combines score distillation from a large pretrained diffusion teacher with an adversarial loss so that a student produces sharp images in one to four steps.
- **Average-velocity models**: MeanFlow [13] learns the *average* velocity $u(x_t, r, t) = \frac{1}{t-r}\int_r^t v(x_s, s)\, ds$ over an interval instead of the instantaneous velocity v. Differentiating $(t-r)\,u = \int_r^t v\, ds$ with respect to t gives the identity $u = v - (t-r)\frac{d}{dt}u$, which supplies a training target without a teacher. One network call over the whole interval then generates a sample; the paper reports an FID of 3.43 with one evaluation on ImageNet 256 × 256, trained from scratch [13].

A small step count is a learned capability, not a free consequence of a nicer equation. When comparing, count NFE including guidance evaluations, the decoder's cost in latent models, resolution, quality *and* diversity — and remember that a one-step network can be larger and slower per call than a multi-step one.

Evaluating generative models

Plain version. "Does it look good?" hides at least three questions: are individual samples realistic (fidelity), does the set of samples cover the variety of the data (diversity), and does the sample match the condition (alignment)? No single number answers all three.

Fréchet Inception Distance

FID (Heusel et al. [23]) embeds real and generated images with an Inception-v3 network [36], fits a Gaussian to each set of features, and computes the Fréchet (2-Wasserstein) distance between the two Gaussians:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \operatorname{tr}\Big(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\Big),$$

where μ_r, Σ_r and μ_g, Σ_g are the feature mean and covariance of the real and generated sets and tr is the matrix trace. Lower is better. The snippet applies the formula to two-dimensional "features."

```
import numpy as np
from scipy import linalg

def frechet_distance(a, b):
    """Frechet distance between Gaussians fitted to feature sets a and b."""
    mu1, mu2 = a.mean(0), b.mean(0)
    c1, c2 = np.cov(a, rowvar=False), np.cov(b, rowvar=False)
    covmean = linalg.sqrtm(c1 @ c2).real
    return float(((mu1 - mu2)**2).sum() + np.trace(c1 + c2 - 2*covmean))

rng = np.random.default_rng(0)
centers = np.array([[-1.5, 0.], [1.5, 0.]])
real = centers[rng.integers(2, size=5000)] + .25*rng.normal(size=(5000, 2))
good = centers[rng.integers(2, size=5000)] + .25*rng.normal(size=(5000, 2))
one_mode = centers[[0]] + .25*rng.normal(size=(5000, 2))      # mode collapse
blurry = .6*rng.normal(size=(5000, 2))                         # mass in the gap
print("good     :", round(frechet_distance(real, good), 3))
print("one mode :", round(frechet_distance(real, one_mode), 3))
print("blurry   :", round(frechet_distance(real, blurry), 3))
small = [frechet_distance(real[:50], good[i*50:(i+1)*50]) for i in range(20)]
print("good, 50 samples: mean", round(np.mean(small), 3), "std", round(np.std(small), 3))
```

It prints about 0.001 for a correct sampler, 3.9 for a sampler that only produces the left mode, and 0.976 for a "blurry" sampler that puts its mass between the modes. The last line shows a practical trap: with 50 samples, even a *perfect* sampler scores 0.077 ± 0.073, because FID is biased upward for small sample sizes. Compare FIDs only with the same number of samples, the same reference set and the same feature extractor and preprocessing.

Beyond FID

- **Precision and recall** (Kynkäänniemi et al. [24]) separate fidelity from diversity: estimate the real and generated "manifolds" with k-nearest-neighbour balls in feature space; precision is the fraction of generated samples inside the real manifold, and recall the fraction of real samples inside the generated one. Our "one mode" sampler would have high precision and low recall.
- **Feature extractor matters.** Stein et al. [25] ran a large human study and found that no common metric correlated strongly with human judgments; in particular, the perceptual realism of diffusion models was not reflected in FID. They recommend features from DINOv2-ViT-L/14 [39] instead of Inception, and show that current metrics cannot separate memorization from underfitting or mode shrinkage.
- **CMMD** (Jayasumana et al. [26]) replaces Inception features with CLIP [38] embeddings and the Gaussian assumption with a maximum mean discrepancy using a Gaussian RBF kernel, giving an unbiased estimator that is more sample-efficient.

- **Likelihood.** Through the probability-flow ODE, diffusion models can report exact log-likelihoods (in bits per dimension) [4], [19]. Likelihood measures something different from visual quality and the two can disagree.
- **Conditional models** need alignment checks (does the image show what the prompt says?), usually by human rating or by vision–language models, and memorization checks against the training set.
- **Cost.** Report NFE, guidance evaluations, resolution and wall-clock time on stated hardware.

The evaluation chapter ML 10 returns to sample-size effects and honest comparisons.

Modern view

Surveys and reference texts. The literature is large; these reviews are good maps.

- Yang et al. [1] (*ACM Computing Surveys*, 2023) organize diffusion research into three improvement directions — efficient sampling, improved likelihood estimation, and data with special structure (manifolds, discrete data, invariances) — and also cover combinations with VAEs, GANs and flows and applications across domains.
- Croitoru et al. [27] (*IEEE TPAMI*, 2023) focus on vision. They identify three generic formulations — DDPMs, noise-conditioned score networks and SDEs — and categorize vision applications such as generation, super-resolution, inpainting, editing, translation and segmentation.
- Cao et al. [28] (*IEEE TKDE*, 2024) take a methods angle: fundamental formulation, algorithmic enhancements (sampling acceleration, objective design, conditioning) and applications from images to biology and healthcare.
- Lipman et al. [2] (2024) is a guide-with-code to flow matching that covers Gaussian and general probability paths, the link to diffusion and extensions to discrete and Riemannian data.
- Lai, Song, Kim, Mitsufuji and Ermon [29] (2025) is a book-length monograph presenting the variational, score-based and flow-based views as three routes to the same time-dependent velocity field, followed by guidance, fast solvers and flow-map (few-step) models.
- Shen et al. [30] (*TMLR*, 2025) survey efficiency at the algorithm level (training, sampling, compression), system level (hardware and serving) and framework level.
- Melnik et al. [31] (*TMLR*, 2024) survey video diffusion: architectures, temporal-consistency mechanisms, datasets and evaluation.

State of the art, 2023–2026. Several trends are visible across these reviews and the primary papers in this chapter. (1) *Transformers in latent space*: DiT-style backbones [8] replaced U-Nets in many large systems, and joint text–image transformers trained with rectified-flow objectives [9] became a standard text-to-image recipe. (2) *Flow matching as the default objective*: straight-path flow matching [10], [11] is now used alongside or instead of noise prediction, with diffusion understood as one choice of probability path [2], [29]. (3) *Fewer evaluations*: distillation [22], [33], consistency models [12], [32] and average-velocity models [13] bring sampling down to one to four network

calls with quality approaching multi-step samplers. (4) *Better evaluation*: the community is moving from Inception-based FID toward richer feature spaces, human studies and alignment metrics [25], [26]. (5) *Beyond images*: video, 3-D, audio and scientific data reuse the same machinery, with video's temporal consistency and cost the main open challenges [31].

Representation and prediction, 2025–2026. JiT [42] tests direct clean-image prediction on pixels; RAE [43] tests generation in frozen semantic feature spaces; its text-to-image follow-up [44] examines larger-scale behavior. These studies separate where the model operates from what the network directly predicts. The pretrained examples above [45]–[48] show why component compatibility and temporal reconstruction matter in actual image/video pipelines. Compare these choices under matched compute and data, including representation pretraining.

Open problems. Sampling cost at high resolution and long video duration; principled guidance that does not trade away diversity or produce over-saturation; evaluation metrics that agree with people and detect memorization; understanding generalization (why models produce new samples rather than copies of training data); controllability and editing with guarantees; and data provenance, consent and misuse, which are social rather than purely technical questions. The next chapter, ML 08, covers the main alternative route: autoregressive generation over pixels and tokens, and its hybrids with diffusion.

Key takeaways

- The forward process has a closed form $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$, so training reduces to "pick a random noise level and regress the noise."
- Every term of the variational bound is a weighted noise-prediction error; $\mathcal{L}_{\text{simple}}$ drops the weights, which improves samples but is not the likelihood.
- Noise, clean-image, VP velocity, score and flow velocity convert under a specified path; time direction, loss weighting, endpoint stability and network capacity matter. JiT directly outputs clean pixels while deriving a velocity for training and sampling.
- Sampling is solving a reverse SDE or ODE: DDPM, DDIM and higher-order solvers trade steps for accuracy with the same trained network; count NFE, not "steps."
- Guidance trades diversity for adherence and doubles the per-step cost for CFG; very large weights push samples off the data distribution.
- Pixels, reconstruction latents and RAE features are different generative spaces. Latent compression saves repeated updates but can lose information; pretrained image/video pipelines require compatible encoders, predictors, guidance and samplers.
- Conditional flow matching trains the marginal velocity by regressing per-pair straight-line velocities, but straight conditional paths do not make the learned flow straight — few-step generation needs reflow, distillation, consistency or average-velocity training.
- No single metric measures quality: FID depends on sample size and feature extractor; pair it with precision/recall, human or alignment studies and cost.

Exercises

1. **Recover the clean estimate.** Starting from $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$, derive $\hat{x}_0$ from a noise prediction. If ϵ_θ has an error δ, how large is the error in $\hat{x}_0$ at $\bar{\alpha}_t = 0.25$ and at $\bar{\alpha}_t = 10^{-4}$? Then derive the straight-flow velocity from a clean prediction and explain why JiT can output x while using a velocity loss.

   Hint

   $\hat{x}_0 = (x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta)/\sqrt{\bar{\alpha}_t}$, so the error is $\sqrt{1-\bar{\alpha}_t}\,\delta/\sqrt{\bar{\alpha}_t}$: about $1.7\,\delta$ at $\bar{\alpha}_t = 0.25$ and about $100\,\delta$ at 10^{-4}. At high noise, $\hat{x}_0$ is an average over many plausible images, not one image.

2. **Posterior mean in two forms.** Substitute $x_0 = (x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon)/\sqrt{\bar{\alpha}_t}$ into $\tilde{\mu}_t$ and show that it equals $\frac{1}{\sqrt{\alpha_t}}(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon)$. Then change `t` in the posterior snippet to 50 and 900 and check that the formula still matches.

   Hint

   Use $\bar{\alpha}_t = \alpha_t\bar{\alpha}_{t-1}$ and $\beta_t = 1 - \alpha_t$. The coefficient of x_t becomes $\frac{\beta_t + \alpha_t(1-\bar{\alpha}_{t-1})}{\sqrt{\alpha_t}(1-\bar{\alpha}_t)} = \frac{1}{\sqrt{\alpha_t}}$ because the numerator equals $1 - \bar{\alpha}_t$.

3. **Count compute.** A latent text-to-image model samples with 30 DDIM steps and classifier-free guidance using separate conditional and unconditional passes, then runs the decoder once. How many denoiser evaluations are needed? How does the count change for a 4-step distilled model that has guidance "baked in"? For the synthetic shape example, change the latent channels to 16 and the pixel patch size to 16. Recompute scalar counts and attention entries. Which savings remain, and which depend on patching?

   Hint

   60 denoiser evaluations plus one decoder pass; batching both CFG branches reduces wall-clock time but not the work. A guidance-distilled 4-step model needs 4 evaluations plus the decoder. Compare quality and diversity, not just the counts.

4. **Time convention.** Diffusion sections use "large t = noisy"; the flow-matching section uses "$t = 0$ noise, $t = 1$ data." Reparameterize the flow with $s = 1 - t$. What is dx/ds, and what goes wrong if you only relabel the time input?

   Hint

   $dx/ds = -v(x, 1-s)$. If you feed s to a network trained on t without flipping the sign and the time input, you integrate in the wrong direction and move samples away from the data.

5. **Guidance sweep.** In the CFG snippet, add $w \in \{0.5, 1.5, 5\}$ and also record the fraction of class-1 samples that land closer to the wrong-class modes. Plot mean distance to the nearest mode against w. Where does guidance stop helping?

Hint

Already at $w = 1$ the exact conditional model puts every sample in class 1, so larger w cannot improve class accuracy here — it only distorts the clusters. With a learned, imperfect conditional model, moderate $w > 1$ can help adherence, which is why it is used in practice.

6. **Sampler ablation.** Hold the trained flow-matching network fixed. Compare Euler with 1, 2, 5, 20 and 100 steps against a second-order Heun step (predict with Euler, then average the velocities at both ends). Use the same starting noise and measure cluster means and spreads. Does the 2-step Heun sampler beat 4-step Euler at the same number of network calls?

Hint

Heun uses two network calls per step. Plot samples, not only numbers. One Euler step of a model trained for instantaneous velocity does not test a separately trained one-step model such as a consistency or MeanFlow model.

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ML 08 • Autoregressive Vision: Pixels, Tokens and Scales

Factor an image distribution into conditionals: PixelCNN, VQ-VAE and VQGAN tokenizers, GPT-style token transformers, next-scale prediction, masked generative models, continuous tokens and how they compare with diffusion.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-08-autoregressive-vision/

What you'll learn

- How the probability chain rule turns "model a whole image" into a sequence of "predict the next piece" problems, and how to measure the result in bits per dimension.
- Why training can run in parallel even though sampling is sequential, and how causal masks and masked convolutions keep the answer hidden from its own predictor.
- How PixelRNN and PixelCNN model raw pixels, and why their sequence length eventually pushed the field toward learned tokens.
- How VQ-VAE and VQGAN tokenizers turn an image into a short grid of integers, including the codebook and commitment losses and the straight-through gradient.
- How GPT-style token transformers (Image GPT, Parti, LlamaGen), next-scale prediction (VAR), masked generative models (MaskGIT, Muse) and continuous-token models (MAR) change the unit of prediction.
- How sampling controls (temperature, top-k, top-p, classifier-free guidance, KV caching) work, and how autoregressive generators compare with diffusion models.

The big picture

An **autoregressive** (**AR**) generator builds a sample piece by piece. At each step it looks at everything it has produced so far, predicts a probability distribution for the next piece, draws from it, and appends the result. The "piece" can be a single pixel value, an integer token from a learned codebook, a vector of continuous features, or an entire map at the next resolution.

> **Explain like I'm five**
>
> Imagine drawing a comic one panel at a time. Each new panel must fit the panels already drawn. When you practice, you copy real comics, so you always know what the earlier panels look like. When you invent a new comic, you have to live with your own earlier panels, even the slightly wobbly ones. An autoregressive image model works the same way: practice on real images, then create new ones one piece at a time.

Why does this matter? Autoregression is the same recipe that powers large language models: next-token prediction with a transformer. Bringing images into that recipe lets one architecture

read and write text and pictures, reuse the engineering of language-model serving, and measure progress with an exact likelihood. The price is that images are not naturally sequences. Someone has to decide **what a token is** and **in which order tokens are produced**. Most of this chapter is about those two decisions.

Xiong et al.'s survey [1] organizes visual AR models into three families by their prediction unit: **pixel-based**, **token-based** and **scale-based**. We follow that map, then add masked (non-left-to-right) and continuous-token models, which blur the boundary with diffusion. The previous chapters on generative models and diffusion and flow matching provide the VAE and diffusion background used here; Prince [9] covers the general probabilistic material at textbook level.

The chain rule needs no independence assumption

Plain version. Any joint distribution can be written as a product of "next element given the earlier ones" distributions. That is a fact of probability, not a modeling assumption.

For an ordered sequence $x = (x_1, \dots, x_N)$,

$$p_\theta(x) = \prod_{i=1}^{N} p_\theta(x_i \mid x_{<i}), \qquad \mathcal{L}_{\mathrm{NLL}}(\theta) = -\sum_{i=1}^{N} \log p_\theta(x_i \mid x_{<i}).$$

Here x_i is the i-th element in the chosen order, $x_{<i} = (x_1, \dots, x_{i-1})$ is everything before it, θ are the network parameters, and $\mathcal{L}_{\mathrm{NLL}}$ is the negative log-likelihood of one training example (averaged over the dataset in practice).

The chain rule is exact for **any** order. Approximation enters only through the model: a finite network, a limited context window, or a restricted form for each conditional. For discrete values, each conditional is usually a softmax over the possible values; binary pixels can use a Bernoulli distribution; continuous features need a density or another conditional generator (we meet one in the MAR section).

An order is a computational choice, not a claim about physics. **Raster order** (left to right, top to bottom) is convenient but arbitrary for a two-dimensional image: the pixel below you is just as related as the pixel to your right. Different orders lead to different conditionals, and a finite model may find some orders easier to learn than others. Figure 1 shows three prediction units used in this chapter.

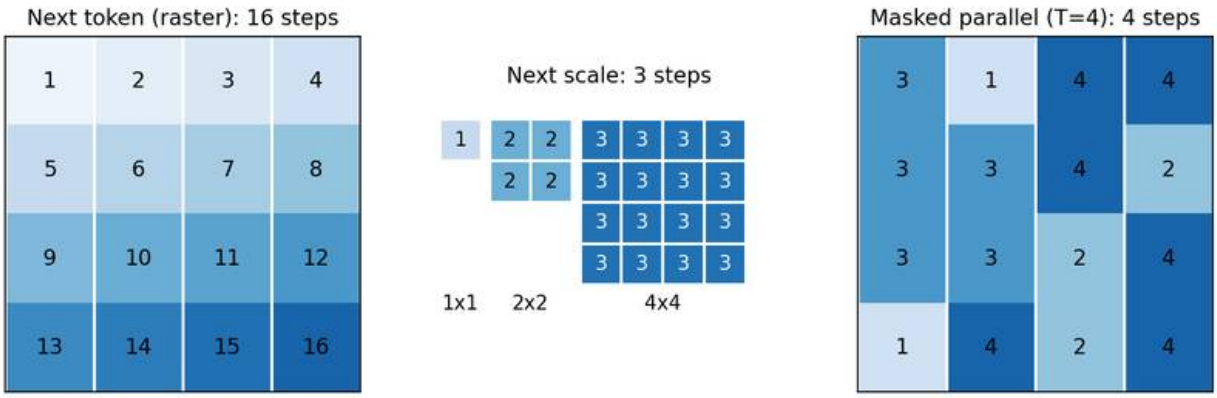


Figure 1 — The same 16 final tokens produced by three generation schemes. Numbers give the sequential step that produces each token: raster autoregression needs 16 steps, next-scale prediction 3 (one per map), and masked parallel decoding with $T = 4$ needs 4. Diagram only; no model is trained.

Measuring an AR model: bits per dimension

Because the chain rule gives an exact $\log p_\theta(x)$, AR models are usually compared by **negative log-likelihood in bits per dimension (bpd)**:

$$\text{bpd}(x) = \frac{-\log_2 p_\theta(x)}{D},$$

where D is the number of discrete values in the image (for example $D = 3 \cdot 32 \cdot 32$ for a 32×32 RGB image) and $\log_2$ converts nats to bits. A model that assigned every 8-bit value a uniform probability would score exactly 8 bpd; anything lower means the model has learned structure. Bpd is measured on held-out images, so it detects overfitting as well as underfitting.

Two cautions. First, bpd and visual quality can disagree: a model can spend most of its likelihood on imperceptible noise while still making odd global mistakes. Second, for a **token** model the likelihood is over codebook indices, not pixels. A lossy tokenizer throws information away, so a token-level likelihood is not an exact pixel-space likelihood and cannot be compared directly with pixel-model bpd.

Teacher forcing, causal masks and exposure bias

Plain version. During training the model always sees the real history; during sampling it sees its own history. A mask makes sure training never lets a position peek at its own answer.

With **teacher forcing**, every conditional in the training loss is computed using the ground-truth

previous elements. Since the whole training image is available, all N conditionals can be evaluated in a single forward pass, as long as position i is prevented from reading positions $\geq i$. A causal transformer enforces this with an attention mask; a PixelCNN enforces it with masked convolution kernels.

Two input conventions are common, and the mask must match the convention:

- In a **shifted-input** decoder, feed `[BOS, x1, x2, ...]` and predict `[x1, x2, x3, ...]`. A lower-triangular mask that *includes* the diagonal is correct, because the input at position i is the *previous* element.
- In a model that receives **unshifted** values, the dependency must be **strictly** earlier. Including the diagonal would let the network copy the value it is supposed to predict.

At inference, earlier sampled mistakes enter the context of later steps. This training/inference mismatch is often called **exposure bias**. It does not make maximum likelihood wrong (the factorization is still exact), but it matters in practice for imperfect models and long sequences: a small early error in the layout of an image can push later steps into contexts that never occurred in training.

Pixel models: PixelRNN and PixelCNN

Plain version. Treat each pixel's intensity as a 256-way classification problem whose input is all pixels above and to the left.

PixelRNN and PixelCNN were introduced together by van den Oord, Kalchbrenner and Kavukcuoglu [2]. Both model each 8-bit value with a **256-way softmax** rather than a Gaussian, so the distribution can be multimodal (a pixel can be "either black or white, but not grey"). Colour is handled by a fixed order inside each pixel, red then green then blue, so that green may depend on red and blue on both. PixelRNN uses two-dimensional LSTM layers (Row LSTM and Diagonal BiLSTM) whose dependency range is potentially unbounded; PixelCNN replaces them with stacked masked convolutions, which have a bounded receptive field but train much faster because all positions are computed in parallel.

Masked convolutions

A masked convolution multiplies its kernel by a binary mask that zeros the weights at "future" positions. In [2], **mask A** is used only in the first layer: it excludes the centre pixel, so the first layer never sees the value it predicts (and, for colour, only sees channels already predicted). **Mask B** is used in all later layers: it allows the centre position, because after the first layer the centre feature already summarizes only earlier pixels. Formally, for a $k \times k$ kernel with centre (c, c), $c = \lfloor k/2 \rfloor$,

$$M_{uv} = \begin{cases} 1 & u < c, \\ 1 & u = c \text{ and } v < c \text{ (A) or } v \le c \text{ (B)}, \\ 0 & \text{otherwise}, \end{cases}$$

where u indexes kernel rows and v kernel columns. The code below builds both masks, stacks five layers, and uses a gradient to find exactly which input pixels can influence the output at the centre of a 9×9 image.

```
import torch
from torch import nn
from torch.nn import functional as F

class MaskedConv2d(nn.Conv2d):
    """PixelCNN-style masked convolution for one channel (raster order)."""
    def __init__(self, cin, cout, k, mask_type):
        super().__init__(cin, cout, k, padding=k//2, bias=False)
        mask = torch.ones(k, k)
        c = k//2
        mask[c, c + (mask_type == "B"):] = 0   # centre (type A) and pixels to its right
        mask[c+1:] = 0                         # all rows below
        self.register_buffer("mask", mask)
    def forward(self, x):
        return F.conv2d(x, self.weight*self.mask, padding=self.padding)

torch.manual_seed(0)
layers = [MaskedConv2d(1, 8, 3, "A"), nn.Tanh()]
for _ in range(4):                             # 5 layers: reach of 5 pixels
    layers += [MaskedConv2d(8, 8, 3, "B"), nn.Tanh()]
net = nn.Sequential(*layers, MaskedConv2d(8, 1, 1, "B"))
print(net[0].mask.int())
x = torch.rand(1, 1, 9, 9, requires_grad=True)
net(x)[0, 0, 4, 4].backward()                  # logit for pixel (row 4, col 4)
dep = (x.grad[0, 0].abs() > 0).int()
print(dep)
later = torch.zeros(9, 9, dtype=torch.bool)
later[4, 4:] = True; later[5:] = True
assert dep[later].sum() == 0                   # never reads itself or later pixels
print("earlier pixels never seen (blind spot):", int(((dep == 0) & ~later).sum()))
```

The assertion passes: no pixel at or after the target influences its logit, so the model is a valid autoregressive factorization. But the printout reveals something else. Six pixels that come *earlier* in raster order, in a staircase above and to the right of the target, have zero gradient. This is the

blind spot of stacked masked convolutions (Figure 2): each layer's receptive field grows up and to the left, but the region up and to the right is never reached. The model is still correct, just less informed than it could be.

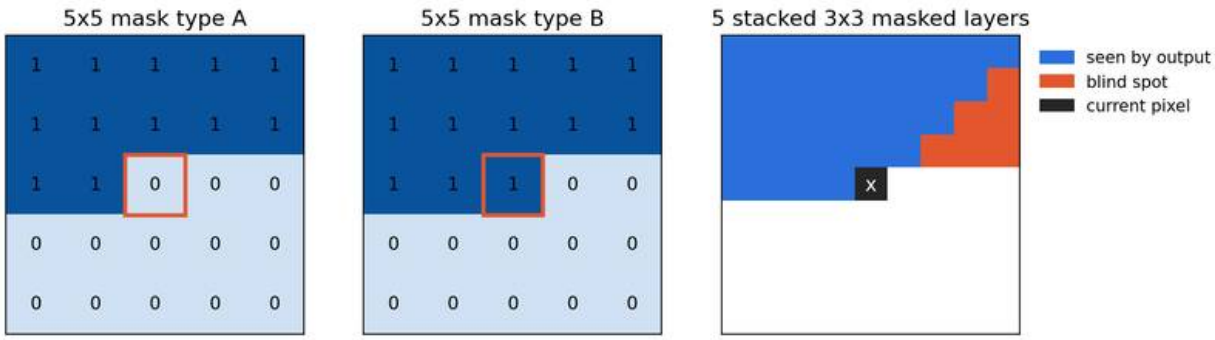


Figure 2 — Left and centre: mask A excludes the centre weight; mask B keeps it. Right: input pixels that influence the prediction at the dark cell after five stacked 3×3 masked layers, measured with a gradient. Orange cells come earlier in raster order but are never seen: the blind spot.

The Gated PixelCNN of van den Oord et al. [3] removed the blind spot by splitting the network into two stacks: a **vertical stack** that sees all rows above the current one without restriction and a **horizontal stack** that sees the current row up to the current pixel, combined after each layer. It also replaced ReLU with a gated unit $y = \tanh(W_f * x) \odot \sigma(W_g * x)$, where $*$ is convolution, σ the logistic sigmoid and $\odot$ elementwise multiplication, and showed how to condition the model on class labels or embeddings. With these changes PixelCNN matched PixelRNN's likelihood on ImageNet [23] at much lower cost [3].

Code: a minimal autoregressive model

Masked convolutions are one way to enforce causality; a transformer is not required either. The following complete CPU experiment generates 4×4 binary images whose rows all repeat the same four random bits. A single masked linear layer, whose strictly lower-triangular mask excludes the current pixel and all later ones, learns the conditionals directly.

```
import torch
from torch import nn
from torch.nn import functional as F

torch.manual_seed(0)
torch.set_num_threads(1)
class BinaryAR(nn.Module):
    def __init__(self, n=16):
        super().__init__()
        self.weight = nn.Parameter(torch.zeros(n, n))
        self.bias = nn.Parameter(torch.zeros(n))
        self.register_buffer("mask", torch.tril(torch.ones(n, n), diagonal=-1))
    def forward(self, x):
        return F.linear(x, self.weight*self.mask, self.bias)

model = BinaryAR()
opt = torch.optim.Adam(model.parameters(), lr=.04)
```

```
for step in range(600):
    first_row = torch.randint(2, (128, 4)).float()
    x = first_row.repeat(1, 4)
    loss = F.binary_cross_entropy_with_logits(model(x), x)
    opt.zero_grad(); loss.backward(); opt.step()

with torch.no_grad():
    a = torch.rand(2, 16)
    b = a.clone(); b[:, 8:] = torch.rand(2, 8)
    assert torch.allclose(model(a)[:, :9], model(b)[:, :9])
    samples = torch.zeros(8, 16)
    for i in range(16):
        p = model(samples)[:, i].sigmoid()
        samples[:, i] = torch.bernoulli(p)
    rows = samples.reshape(-1, 4, 4)
    print("one generated image:\n", rows[0].int())
    print("repeat agreement:", (rows[:, 1:] == rows[:, :1]).float().mean().item())
```

The assertion checks causality: changing positions 8 onward cannot alter the logits for positions 0 through 8, because each of those outputs reads only earlier inputs. Training then drives the agreement between later rows and the first row to 1.0, while the first row stays random (the model must not memorize one particular row).

The true distribution has 16 equally likely images, so its entropy is $\log_2 16 = 4$ bits per image, or 0.25 bits per pixel. Later rows contain no new uncertainty. A model that treated all 16 pixels as independent fair coins would need 16 bits per image. The gap between 16 and 4 bits is exactly the structure an autoregressive model captures and an independent-pixel model cannot.

Why pixel models stopped scaling

A 256×256 RGB image has $3 \cdot 256 \cdot 256 = 196{,}608$ values, so pixel-level sampling needs about two hundred thousand sequential network evaluations. Most of the likelihood is also spent on fine, locally predictable detail, while the global layout that people notice is a small fraction of the bits. Image GPT [11] showed that a plain sequence transformer trained to predict pixels autoregressively on low-resolution images learns features that transfer well under linear probing, but the pixel route stays expensive at higher resolutions. The next step was to learn a shorter, more meaningful sequence.

Discrete visual tokens: VQ-VAE and VQGAN

Plain version. Train a separate network that compresses an image into a small grid of "words" from a learned dictionary, then do autoregression over the words instead of the pixels.

VQ-VAE

VQ-VAE [4] has an encoder that produces a feature vector $z_e(x) \in \mathbb{R}^d$ at each position of a downsampled grid, a **codebook** $\{e_1, \dots, e_K\} \subset \mathbb{R}^d$ of K learned vectors, and a decoder. Each feature is replaced by its nearest code:

$$k^* = \arg\min_k \|z_e(x) - e_k\|_2^2, \qquad z_q(x) = e_{k^*}.$$

The image is now represented by the integer grid of k^* values. Training minimizes

$$\mathcal{L}_{\text{VQ}} = \underbrace{\|x - D(z_q)\|^2}_{\text{reconstruction}} + \underbrace{\|\text{sg}[z_e] - e_{k^*}\|^2}_{\text{codebook}} + \beta \underbrace{\|z_e - \text{sg}[e_{k^*}]\|^2}_{\text{commitment}},$$

where D is the decoder, $\text{sg}[\cdot]$ is the **stop-gradient** operator (identity in the forward pass, zero gradient in the backward pass), and β is a small weight on the commitment term (the snippet below uses 0.25). The reconstruction term is written as squared error here; any reconstruction likelihood can take its place.

The three terms do three jobs. The **codebook** term moves the chosen code toward the encoder output, like a k-means update. The **commitment** term pulls the encoder toward its code so that encoder outputs do not drift far from the codebook. The `argmin` has no useful gradient, so the reconstruction gradient reaches the encoder through the **straight-through estimator**: in the forward pass use z_q, in the backward pass copy the gradient from z_q to z_e as if quantization were the identity. In code this is the one-liner `z_q = z_e + (e - z_e).detach()`.

The toy below trains a 2-D "encoder", codebook and decoder on four clusters, with eight codes available:

```
import torch
from torch import nn
from torch.nn import functional as F

def quantize(z_e, codebook, beta=0.25):
    """z_e: (n, d) encoder outputs; codebook: (K, d). Returns z_q with straight-through
     ↪  gradient."""
    dist = torch.cdist(z_e, codebook)            # (n, K) Euclidean distances
    idx = dist.argmin(1)                         # nearest code index per vector
    e = codebook[idx]
    codebook_loss = F.mse_loss(e, z_e.detach())  # moves codes toward encoder outputs
    commit_loss = F.mse_loss(z_e, e.detach())    # keeps the encoder near its code
    z_q = z_e + (e - z_e).detach()               # forward: e; backward: identity
    return z_q, idx, codebook_loss + beta*commit_loss

torch.manual_seed(0)
torch.set_num_threads(1)
centres = torch.tensor([[2., 0.], [-2., 0.], [0., 2.], [0., -2.]])
encoder = nn.Linear(2, 2)                        # toy "encoder" and "decoder"
decoder = nn.Linear(2, 2)
```

```
codebook = nn.Parameter(torch.randn(8, 2)*.1)    # K = 8 codes, 4 are enough
opt = torch.optim.Adam([*encoder.parameters(), *decoder.parameters(), codebook], lr=.03)
for step in range(800):
    x = centres[torch.randint(4, (256,))] + .1*torch.randn(256, 2)
    z_q, idx, vq_loss = quantize(encoder(x), codebook)
    loss = F.mse_loss(decoder(z_q), x) + vq_loss
    opt.zero_grad(); loss.backward(); opt.step()

with torch.no_grad():
    x = centres.repeat_interleave(100, 0) + .1*torch.randn(400, 2)
    z_q, idx, _ = quantize(encoder(x), codebook)
    print("reconstruction MSE:", round(F.mse_loss(decoder(z_q), x).item(), 4))
    print("codes used:", idx.unique().numel(), "of", len(codebook))
    print("codes per cluster:", [idx[i*100:(i+1)*100].unique().tolist() for i in range(4)])
z = torch.randn(3, 2, requires_grad=True)
z_q, _, _ = quantize(z, codebook.detach())
z_q.sum().backward()
print("straight-through gradient:", z.grad[0].tolist())
```

It prints a reconstruction MSE of about 0.0096, uses **4 of the 8 codes** (one per cluster), and shows a straight-through gradient of `[1.0, 1.0]`. Two lessons hide in those numbers. The reconstruction error is almost exactly the within-cluster noise variance $0.1^2 = 0.01$: quantization has kept *which cluster* and discarded *where inside the cluster*. And half the codebook is **dead**: codes that are never selected receive no codebook gradient and never move (Figure 3). In real tokenizers, low codebook usage wastes capacity and is a standard failure to monitor.

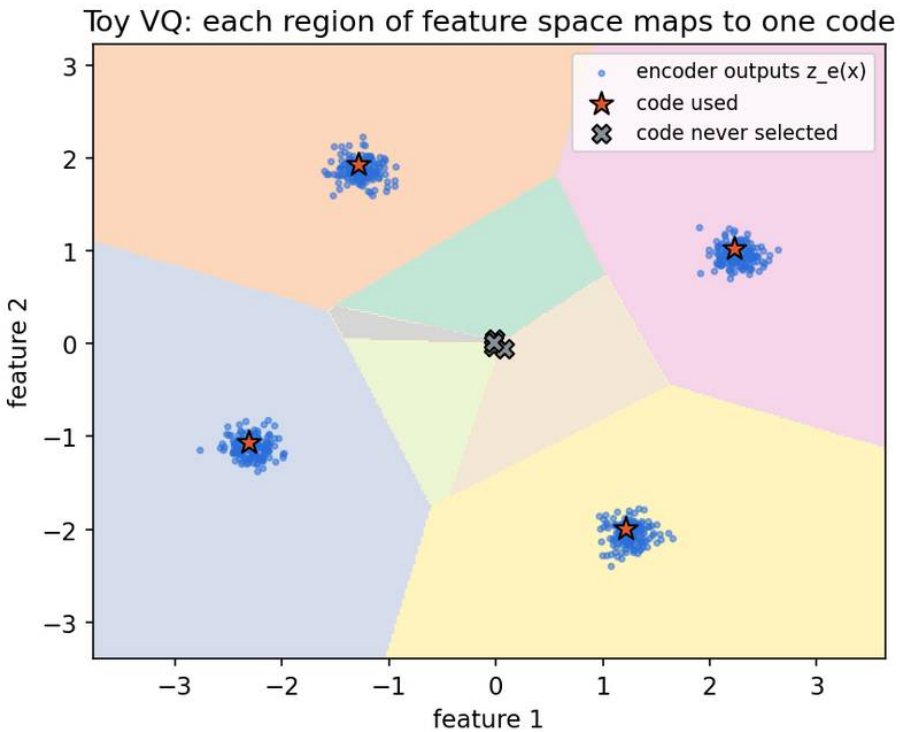


Figure 3 — Toy vector quantization. Each coloured region is the set of features mapped to one code. Four codes (stars) are used, one per cluster; four (crosses) were never selected and stayed near their initialization.

Two stages: a tokenizer, then a prior

A VQ-VAE alone is an autoencoder; it does not by itself say which token grids are likely. Generation therefore has **two stages** (Figure 4). Stage 1 trains the tokenizer. Stage 2 freezes it, converts every training image into its index grid, and trains an autoregressive model $p_\psi(k_1, \ldots, k_M)$ over the M indices. To sample, draw an index grid from the prior and decode it. In [4] the prior was a PixelCNN over codes; VQ-VAE-2 [10] used a hierarchy of two code maps (a coarse top level and a finer bottom level) with PixelCNN-style priors and produced diverse, high-fidelity ImageNet samples.

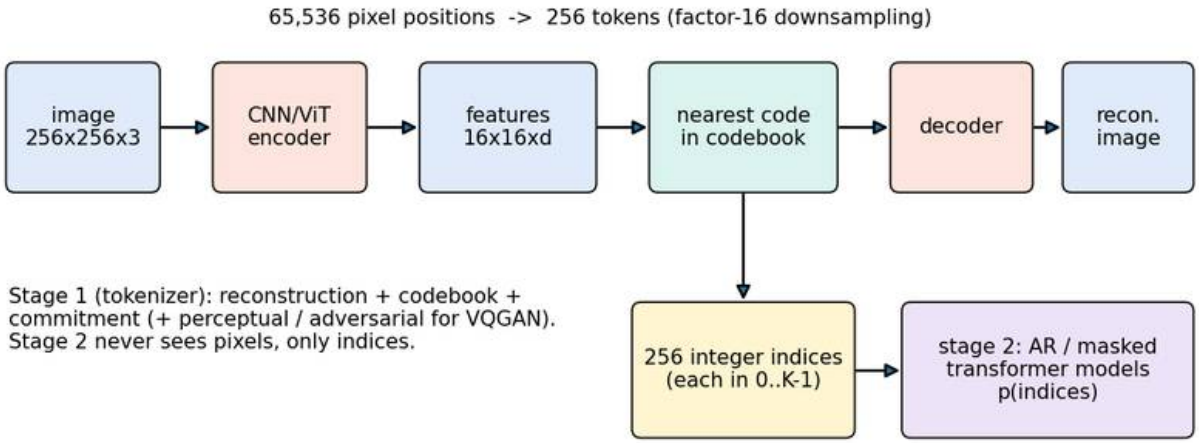


Figure 4 — Two-stage generation. The tokenizer (top row) is trained for reconstruction; the prior (bottom right) only ever sees integer indices. With factor-16 downsampling, a 256×256 image becomes 256 tokens instead of 65,536 pixel positions.

The arithmetic explains the appeal. If a tokenizer downsamples a 256×256 image by 16 in each direction, the prior sees $16 \times 16 = 256$ tokens rather than 65,536 pixel positions (or 196,608 colour values). Self-attention cost grows with the square of sequence length, so this is the difference between feasible and infeasible.

VQGAN: perceptual tokens

Esser, Rombach and Ommer's VQGAN [5] kept the two-stage design but trained the tokenizer with a **perceptual loss** and a **patch-based adversarial loss** in addition to reconstruction, and modelled the token grid with a transformer. The adversarial and perceptual terms let the decoder produce sharp, realistic texture from very few tokens, because the tokens no longer need to encode

every pixel exactly; they need to encode enough for the decoder to produce *a* plausible version. This is why VQGAN-style tokenizers became the default front end for token-based generation.

Note the division of labour: the **tokenizer** uses adversarial training, but the **prior** is an ordinary autoregressive model trained by maximum likelihood.

Codebook problems and alternatives

Real tokenizers face three recurring problems: dead codes (as in the toy), a gap between what the tokenizer reconstructs and what the prior can model, and loss of small details such as text and thin lines. Several designs respond to them:

- **ViT-VQGAN** [12] uses a Vision Transformer encoder/decoder, looks codes up in a low-dimensional **factorized** space and ℓ_2**-normalizes** codes and features, which improved codebook usage and reconstruction.
- **RQ-VAE** [15] uses **residual quantization**: each position is represented by a short stack of codes, each one quantizing what the previous ones missed. This allows a coarser spatial grid (shorter sequence) at similar fidelity.
- **FSQ** [16] (finite scalar quantization) projects to a handful of dimensions and rounds each to a few fixed levels. The implicit codebook is the product of the levels; there is no learned codebook to collapse and no commitment loss.

Compression always creates a bottleneck. Measure codebook usage, reconstruction quality and the generator's quality **separately**: a generator cannot produce detail its tokenizer cannot represent.

Autoregressive transformers over tokens

Plain version. Once an image is a sequence of integers, generating it is the same problem as generating text: a decoder-only transformer predicts the next token.

From pixels to tokens

Image GPT [11] applied a GPT-style transformer directly to sequences of low-resolution pixels.

Its main finding was about **representation learning**: features from a model trained only to predict the next pixel transfer well when a linear classifier is trained on top. Working on tokens instead of pixels shortens sequences enough to generate at higher resolution. The VQGAN transformer [5] is the classic example: a GPT-like model over VQGAN indices in raster order, which let a transformer synthesize high-resolution images that would be far too long as pixel sequences.

Text-to-image as translation: Parti

Parti [13] treats text-to-image generation like machine translation. A transformer **encoder** reads the text prompt; a transformer **decoder** autoregressively generates ViT-VQGAN image tokens [12] conditioned on it. The report scales this sequence-to-sequence model up to 20 billion parameters and introduces PartiPrompts, a benchmark of more than 1,600 English prompts spanning many categories of content.

Language-model architecture, unchanged: LlamaGen

LlamaGen [14] asks whether a vanilla language-model architecture, with no image-specific inductive biases, can generate images well when scaled. It uses a Llama-style [24] decoder (RMSNorm pre-normalization, SwiGLU feed-forward layers, 2-D rotary position embeddings) over the tokens of a factor-16 VQ tokenizer with $K = 16{,}384$ codes of dimension 8. The class label or projected text embedding is supplied as a **prefix** token, and generation proceeds in raster order. The paper releases class-conditional models from 111M to 3.1B parameters and reports that standard LLM serving software (vLLM) speeds up inference. Its broader message is architectural reuse: an image generator can share almost everything with a text model except the tokenizer.

Classifier-free guidance for autoregressive models

Classifier-free guidance (CFG) [18] was introduced for diffusion models, but it transfers directly to AR logits. During training, the condition c is randomly replaced by a learned **null** embedding so that one network learns both conditional and unconditional predictions. At each sampling step, compute both sets of logits and combine them:

$$\ell_g = \ell_u + s\,(\ell_c - \ell_u),$$

where ℓ_c are the next-token logits with the condition, ℓ_u the logits with the null condition, and s the guidance scale; $s = 1$ recovers the ordinary conditional model. This is the form LlamaGen uses [14]. Larger s usually improves prompt or class adherence and sample sharpness at the cost of diversity. It also doubles the per-step compute unless the two passes are batched, so step counts alone understate cost.

Next-scale prediction (VAR)

Plain version. Instead of predicting the next token in a fixed reading order, predict the next, sharper version of the whole image, conditioned on all coarser versions.

Raster order forces a strange dependency: the top-left of an image is generated without any knowledge of the bottom-right. VAR (Visual AutoRegressive modeling) [6] argues that this one-directional order conflicts with the two-directional correlations of image tokens and with spatial

locality, and replaces it with a **coarse-to-fine** order over token maps $r_1, \dots, r_K$ of increasing resolution:

$$p(r_1, \dots, r_K) = \prod_{k=1}^{K} p(r_k \mid r_1, \dots, r_{k-1}),$$

where r_k is an $h_k \times w_k$ grid of token indices, r_1 is 1×1 and r_K has the full latent resolution. All tokens of r_k are predicted **in parallel** given the coarser maps, so the number of sequential steps equals the number of scales, not the number of tokens.

The tokenizer is part of the method. VAR's multi-scale VQ encoder works **residually**: starting from the encoder feature map f, for $k = 1, \dots, K$ it downsamples f to $h_k \times w_k$, quantizes it to get r_k (with one codebook shared across scales), upsamples the looked-up codes back to full resolution, passes them through a small convolution ϕ_k, and subtracts the result from f before moving to the next scale [6]. Each scale therefore encodes what the coarser scales have not explained. Simply resizing ordinary token maps would not reproduce this.

During training, a **block-wise causal** attention mask lets every token of r_k attend to all tokens of $r_{\le k}$ but none of the finer scales; at inference a KV cache stores the coarser scales. VAR uses CFG as well. On class-conditional ImageNet 256×256 the paper reports that VAR improves its AR baseline's FID from 18.65 to 1.73, with faster inference, and that its models follow power-law scaling trends similar to those of language models [6]. VAR received a NeurIPS 2024 Best Paper award. As always, those numbers apply to the paper's data, resolution and training protocol.

Parallel prediction within a scale is itself an assumption: tokens in r_k are sampled independently given $r_{<k}$. The masked-model section shows what that assumption can cost on a toy problem.

Masked generative models (MaskGIT and Muse)

Plain version. Start with a grid of blanks. At each round, guess every blank at once, keep only the guesses you are most confident about, and blank out the rest again.

MaskGIT [8] uses the same VQ tokens as an AR model, but a **bidirectional** transformer that sees every position. Training is a masked-token objective: choose a mask ratio, replace that fraction of tokens with a special `[MASK]` token, and minimize cross-entropy on the masked positions,

$$\mathcal{L}_{\text{mask}} = -\mathbb{E}\Big[\sum_{i \in \mathcal{M}} \log p_\theta(y_i \mid y_{\bar{\mathcal{M}}}) \Big],$$

where y is the token grid, $\mathcal{M}$ is the set of masked positions and $y_{\bar{\mathcal{M}}}$ the unmasked tokens. The mask ratio is drawn as $\gamma(r)$ with $r \sim \mathcal{U}(0, 1)$ and a **mask schedule** γ that decreases from 1 to 0. Generation runs for T iterations. At iteration t the model predicts all masked tokens in parallel and samples them; each sample's probability serves as its **confidence**. The number of tokens that

remain masked after the iteration is

$$n_t = \lceil \gamma(t/T)\, N \rceil,$$

with N the total number of tokens, so the n_t least confident new tokens are re-masked and everything else is fixed. MaskGIT compares several schedules and selects the **cosine** schedule $\gamma(r) = \cos(\pi r/2)$, which fixes few tokens early (when the model knows little) and many late; it generates 256×256 images in 8 iterations with a 16×16 token grid and a 1,024-entry codebook [8]. Muse [17] scales the same idea to text-to-image generation, conditioning the masked-token transformer on text embeddings from a pre-trained large language model.

MaskGIT is **not** ordinary left-to-right chain-rule autoregression: the order is chosen on the fly by confidence, and several tokens are sampled in the same step. Those simultaneous tokens are drawn **independently** given the current context. The toy below shows when that matters. It trains a small bidirectional network on the repeated-row 4×4 images from earlier and decodes with the cosine schedule.

```
import math
import torch
from torch import nn
from torch.nn import functional as F

torch.manual_seed(0)
torch.set_num_threads(1)
N, MASK = 16, 2                                  # 16 binary tokens; state 2 = [MASK]
gamma = lambda r: math.cos(math.pi*r/2)          # cosine schedule: fraction still masked
net = nn.Sequential(nn.Flatten(), nn.Linear(3*N, 128), nn.ReLU(), nn.Linear(128, N))

def data(n):                                     # 4x4 images whose rows repeat
    return torch.randint(2, (n, 4)).repeat(1, 4)

def logits(tokens):                              # sees every position: bidirectional
    return net(F.one_hot(tokens, 3).float())

opt = torch.optim.Adam(net.parameters(), lr=3e-3)
for step in range(1500):
    x = data(256)
    ratio = torch.tensor([gamma(r) for r in torch.rand(256).tolist()])
    n_mask = (ratio*N).ceil().clamp(min=1)
    masked = torch.rand(256, N).argsort(1) < n_mask[:, None]   # random positions
    inp = torch.where(masked, MASK, x)
    loss = F.binary_cross_entropy_with_logits(logits(inp)[masked], x[masked].float())
    opt.zero_grad(); loss.backward(); opt.step()

@torch.no_grad()
def generate(n, T=4):
    tokens = torch.full((n, N), MASK)
    for t in range(1, T+1):
        p = logits(tokens).sigmoid()
        sample = torch.bernoulli(p).long()
```

```
        conf = torch.where(sample == 1, p, 1-p)              # prob. of the drawn value
        still = tokens == MASK
        conf = torch.where(still, conf, torch.inf)           # never re-mask fixed tokens
        proposal = torch.where(still, sample, tokens)
        n_keep_masked = math.floor(gamma(t/T)*N)             # 0 after the last step
        order = conf.argsort(1)                              # least confident first
        remask = torch.zeros_like(still)
        remask.scatter_(1, order[:, :n_keep_masked], True)
        tokens = torch.where(remask, MASK, proposal)
    return tokens

for T in (1, 2, 4, 8, 16):
    rows = generate(1000, T).reshape(-1, 4, 4)
    agree = (rows[:, 1:] == rows[:, :1]).float().mean().item()
    valid = (rows == rows[:, :1]).all(2).all(1).float().mean().item()
    print(f"T={T:2d}  repeat agreement={agree:.3f}  valid images={valid:.3f}")
```

(The code uses `floor` rather than the paper's ceiling so that the count reaches exactly zero at $t = T$ in this tiny example.) The output is:

```
T= 1  repeat agreement=0.511  valid images=0.000
T= 2  repeat agreement=0.812  valid images=0.100
T= 4  repeat agreement=0.908  valid images=0.360
T= 8  repeat agreement=0.998  valid images=0.982
T=16  repeat agreement=1.000  valid images=0.999
```

With $T = 1$ every token is sampled independently from the all-mask context, so rows almost never agree. With more iterations, the first few tokens are fixed early and later ones can condition on them; by $T = 8$ nearly every sample is valid. On natural images the dependence between tokens decoded in the same round is weaker, which is why a handful of iterations works, but the trade-off between **fewer steps** and **more independence assumptions** is real (Figure 5).

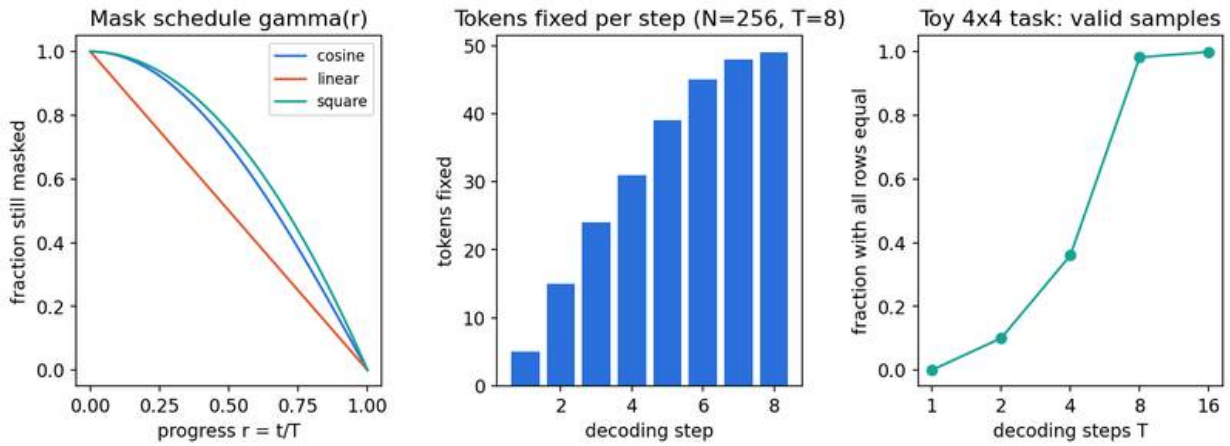


Figure 5 — Left: candidate mask schedules $\gamma(r)$. Centre: with the cosine schedule, 256 tokens and $T = 8$, few tokens are fixed early and many late. Right: on the toy repeated-row task, the fraction of valid samples grows with the number of decoding iterations because fewer tokens are sampled independently in the same step.

Continuous tokens and diffusion conditionals (MAR)

Plain version. Autoregression decides *what comes next*; a tiny diffusion model can decide *how to sample* it, so the tokens no longer need to be integers.

Vector quantization is lossy and awkward to train. Li et al. [7] observe that autoregression needs a way to model $p(x_i \mid x_{<i})$, not necessarily a categorical distribution. They let the AR (or masked AR) transformer output a conditioning vector z_i for each continuous-valued token x_i, and train a small MLP denoiser ϵ_θ with a **diffusion loss**:

$$\mathcal{L}(z_i, x_i) = \mathbb{E}_{\epsilon,t}\|\epsilon - \epsilon_\theta(x_{i,t} \mid t, z_i)\|^2, \qquad x_{i,t} = \sqrt{\bar{\alpha}_t}\, x_i + \sqrt{1-\bar{\alpha}_t}\, \epsilon,$$

where $\epsilon \sim \mathcal{N}(0, I)$, t is a diffusion time step and $\bar{\alpha}_t$ the noise schedule from ML 07. The gradient also flows into the transformer through z_i. To sample a token, run the reverse diffusion process of the small denoiser conditioned on z_i; a temperature scales the injected noise. The paper's main variant, **MAR** (masked autoregressive), uses a random order and predicts a set of tokens per step, combining the outer structure of MaskGIT with an inner continuous sampler [7].

Consequently, "AR versus diffusion" is not an either/or choice. A system can decide the order and grouping of tokens autoregressively and sample each group by diffusion. Likewise, a language decoder can autoregressively generate *text about* an image without generating any pixels; the unifying idea is factorization, not a particular output type.

Sampling, temperature and caching

Plain version. The trained model gives a distribution at every step; the sampler decides how adventurous to be when drawing from it, and the cache avoids recomputing the past.

For logits ℓ_k over K possible values, **temperature** T rescales the distribution:

$$p_T(k) = \frac{\exp(\ell_k / T)}{\sum_{j=1}^{K} \exp(\ell_j / T)}.$$

Lower T concentrates probability on the top values; larger T spreads it (Figure 6). **Top-k** sampling keeps only the k highest logits; **top-p** (nucleus) sampling keeps the smallest set of values whose probability mass reaches p. Both renormalize what remains.

```
import torch

def filtered_probs(logits, temperature=1.0, top_k=None, top_p=None):
    """Return the (renormalized) distribution actually sampled from."""
    logits = logits / temperature
    if top_k is not None:                                  # keep the k largest logits
        kth = logits.topk(top_k).values[..., -1, None]
        logits = logits.masked_fill(logits < kth, -torch.inf)
    if top_p is not None:                                  # smallest set with mass >= p
        sorted_logits, order = logits.sort(descending=True)
        cum = sorted_logits.softmax(-1).cumsum(-1)
        drop = cum - sorted_logits.softmax(-1) >= top_p  # mass before this token already >=
         ↪  p
        sorted_logits = sorted_logits.masked_fill(drop, -torch.inf)
        logits = torch.empty_like(logits).scatter(-1, order, sorted_logits)
    return logits.softmax(-1)

logits = torch.tensor([2.0, 1.0, 0.0, -1.0])
for name, kw in [("T=1", {}), ("T=0.5", {"temperature": .5}), ("T=2", {"temperature": 2.}),
                 ("top-k=2", {"top_k": 2}), ("top-p=0.8", {"top_p": .8})]:
    p = filtered_probs(logits, **kw)
    entropy = -(p[p > 0]*p[p > 0].log2()).sum().item()
    print(f"{name:10s}", [round(v, 3) for v in p.tolist()], f"entropy={entropy:.2f} bits")
```

The model's own distribution ($T = 1$) has 1.37 bits of entropy; $T = 0.5$ lowers it to 0.66 bits and $T = 2$ raises it to 1.80 bits. Top-$k = 2$ and top-$p = 0.8$ both keep the two most likely values here (the first value has mass 0.644, which is below 0.8, so the second is needed). Every one of these changes the distribution you sample from, so a sampler's visual quality should not be confused with the model's likelihood.

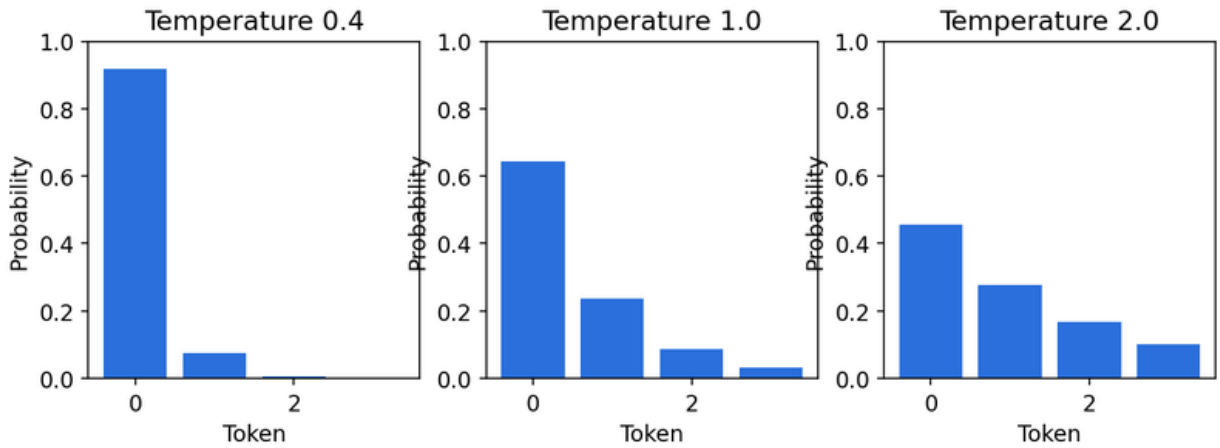


Figure 6 — Temperature changes diversity at each conditional. It cannot recover information that a tokenizer has already lost.

The KV cache

In a causal transformer, the keys and values of earlier tokens do not change when a new token is appended. A **KV cache** stores them, so each new step computes the query, key and value of the newest token only and attends to the cache:

```
import torch
from torch.nn import functional as F

torch.manual_seed(0)
d, n = 8, 6
Wq, Wk, Wv = (torch.randn(d, d) for _ in range(3))
x = torch.randn(n, d)                                  # embeddings of n tokens

# Full causal attention over the whole sequence (what training computes).
q, k, v = x@Wq, x@Wk, x@Wv
mask = torch.ones(n, n).tril().bool()
scores = (q@k.T/d**.5).masked_fill(~mask, -torch.inf)
full = scores.softmax(-1)@v

# Incremental decoding: each step computes q, k, v for the newest token only.
K_cache, V_cache, outputs = [], [], []
for i in range(n):
    K_cache.append(x[i]@Wk); V_cache.append(x[i]@Wv)
    K, V = torch.stack(K_cache), torch.stack(V_cache)
    att = F.softmax((x[i]@Wq)@K.T/d**.5, -1)        # attends to tokens 0..i
    outputs.append(att@V)
assert torch.allclose(full, torch.stack(outputs), atol=1e-5)
print("cached decoding matches full causal attention")
print("dot products per step:", [i+1 for i in range(n)], "total:", n*(n+1)//2)
```

The assertion confirms that cached decoding gives exactly the training-time causal outputs (for a single layer here; real models cache every layer and head). The cache removes recomputation but not the sequential dependency, and it grows linearly with sequence length. Step i still attends to i cached keys, so total attention work over a sequence of length n is $n(n+1)/2$, quadratic in n. This is why shorter sequences (better tokenizers) and fewer sequential steps (next-scale or masked decoding) matter so much.

For video, tokens can be ordered over space and time or grouped into chunks. Caching, compression and temporal ordering then dominate cost, and object consistency and long-duration drift need their own evaluation.

Autoregressive versus diffusion

Plain version. Both families turn one hard generation problem into many easier ones. AR splits the *image* into pieces; diffusion splits the *noise level* into steps.

Aspect	Autoregressive (token / scale)	Masked generative	Diffusion / flow (ML 07)
What is generated per step	One token (or one scale)	A confidence-chosen set of tokens	A refinement of the whole sample
Sequential steps	Tokens or scales	Fixed, typically few (T)	Solver steps (NFE)
Typical representation	Discrete codes (or continuous with a diffusion head [7])	Discrete codes	Continuous pixels or latents
Training objective	Exact NLL (cross-entropy)	Masked cross-entropy	Denoising / velocity regression (ELBO-related)
Likelihood	Exact over tokens	Not a single exact chain-rule likelihood	Bound or ODE-based estimate
Natural strengths	Shares architecture and serving with LLMs; mixes with text	Few steps; easy inpainting and editing	Strong image quality; flexible guidance
Typical weaknesses	Long sequential decoding; tokenizer bottleneck; error accumulation	Independence within a step	Many network evaluations unless distilled

Three points keep this table honest. First, the families mix: MAR [7] puts a diffusion model inside an AR loop, and latent diffusion also depends on a learned tokenizer-like autoencoder. Second, claims that one family "beats" the other (as in LlamaGen's title [14] or VAR's comparison with DiT [6]) hold for a stated benchmark, tokenizer, model size and guidance setting, not universally. Third, cost should be compared in network evaluations, guidance passes, sequence length and actual latency, not in "steps" alone.

Modern view

The field has moved from raw pixels to learned representations, from fixed raster order to flexible prediction units, and from separate image generators to models that read and write both text and images. Several surveys map this territory:

- **Xiong et al. [1]** (TMLR) is the most direct survey of visual autoregression. Its taxonomy by representation (pixel-, token- and scale-based models) is the one used in this chapter, and it covers applications in image, video, 3D and multimodal generation as well as embodied and medical settings.
- **Bond-Taylor et al. [19]** (TPAMI 2022) compare VAEs, GANs, normalizing flows, energy-based and autoregressive models side by side, including the trade-offs in run time, sample diversity and architectural constraints. It places AR models in the wider generative-model landscape: exact likelihood and stable training, paid for with slow sequential sampling.
- **Li et al. [20]** (TPAMI 2026) survey discrete tokenization for multimodal LLMs, organizing vector-quantization variants and their use in classical, single-modality LLM and multimodal LLM systems, and discussing codebook collapse and unstable gradient estimation.

- **Jia et al. [22]** survey discrete tokenizers across generation, comprehension, recommendation and retrieval, dissecting tokenizers into their sub-modules and mechanisms before organizing methods by application, and listing limitations and future directions.
- **Zhao et al. [21]** survey unified multimodal understanding-and-generation models, grouping them into diffusion-based, autoregressive-based and hybrid AR–diffusion designs, which is where most 2025–2026 activity sits.

State of the art (2023–2026). Three lines stand out in the verified literature. (i) *Scaling plain AR transformers over better tokenizers*: LlamaGen [14] shows that a language-model architecture with almost no image-specific design is competitive on class-conditional ImageNet when the tokenizer and training are good. (ii) *Changing the prediction unit*: VAR's next-scale prediction [6] and masked decoding in MaskGIT and Muse [8], [17] reduce the number of sequential steps; MAR [7] removes vector quantization by using a diffusion head per token. (iii) *Unified multimodal models*: the same next-token machinery is used to generate text and image tokens in one model, with tokenizers designed to serve both understanding and generation [20], [21]. Our interpretation is that choosing the **representation and prediction unit** now matters at least as much as choosing the backbone.

Open problems. The surveys repeatedly point to the following:

- *Tokenizer quality.* Reconstruction caps generation quality; small text, faces and fine geometry are the usual casualties. Better codebook usage and semantic structure without losing detail remain active topics [20], [22].
- *Discrete versus continuous tokens.* Discrete tokens fit LLM machinery; continuous tokens reconstruct better but need a continuous conditional model such as a diffusion head [7].
- *Speed.* Sequential decoding is latency-bound. Next-scale, masked and parallel decoding trade steps for independence assumptions.
- *Evaluation.* Token likelihood, FID-type metrics, prompt adherence and human preference measure different things; a lossy tokenizer's token likelihood is not a pixel likelihood.
- *Video and long contexts.* Token counts explode over time, and temporal consistency, object permanence and drift need dedicated evaluation [1].

Key takeaways

- Autoregression is a factorization, not a synonym for "transformer": the chain rule is exact for any order, and the model's capacity and context decide how good each conditional is.
- Training is parallel under teacher forcing; sampling is sequential. Shifted inputs and causal masks (attention masks or PixelCNN mask A/B) must be designed together.
- Pixel models are exact but long; learned tokenizers (VQ-VAE, VQGAN, RQ-VAE, FSQ) shorten the sequence at the price of a reconstruction bottleneck that the generator cannot exceed.
- Token transformers (Image GPT, VQGAN transformer, Parti, LlamaGen) reuse language-model machinery, including classifier-free guidance on logits and KV caching.

- Next-scale (VAR), masked (MaskGIT, Muse) and continuous-token (MAR) models change the **unit** of prediction; fewer sequential steps usually mean more conditional-independence assumptions.
- Temperature, top-k and top-p reshape the sampling distribution; they trade diversity for fidelity but cannot add information lost by the tokenizer.
- AR and diffusion are complementary decompositions and are increasingly combined; compare them on matched data, tokenizers, guidance and compute.

Exercises

1. **Leak the answer.** Change the `BinaryAR` mask to include the diagonal (`diagonal=0`). Explain why the training loss can fall almost to zero while sampling breaks.

   Hint

   Each output can now read its own target and simply copy it, so the training loss collapses. At sampling time the current position is still 0 (not yet drawn) when its logit is computed, so the learned "copy" rule produces nonsense. The causality assertion in the snippet would also fail.

2. **Bits per dimension.** For the repeated-row 4×4 binary data, compute the bits per pixel of (a) the true distribution, (b) a model that treats every pixel as an independent fair coin, and (c) a model that learns the first row perfectly but treats later rows as independent fair coins.

   Hint

   (a) 4 bits / 16 pixels = 0.25. (b) 16 / 16 = 1.0. (c) 4 + 12 = 16 bits per image, also 1.0 bits per pixel: learning the first row's marginal does not help, because its bits are already uniform. All gains come from modelling the dependence of later rows on the first.

3. **Tokenizer budget.** A 512×512 image is tokenized with factor-8 or factor-16 downsampling. Compute the sequence length in each case and the ratio of full self-attention costs. What might you lose with factor 16?

   Hint

   $64 \times 64 = 4096$ versus $32 \times 32 = 1024$ tokens. Attention cost scales with length squared, so the ratio is 16. With factor 16 each token must describe a 16×16 patch, so small text and thin structures are more likely to be lost unless the codebook or decoder compensates.

4. **Blind spot.** Modify the masked-convolution snippet to use 7 layers and an 11×11 input with the target at the centre. How many earlier pixels are never seen? Then sketch how a vertical stack plus a horizontal stack removes them.

Hint

The blind spot is the staircase above and to the right of the target that the up-left-growing receptive field never reaches; counting zero-gradient cells that are earlier in raster order gives 10. A vertical stack sees full rows above (no left/right restriction) and is fed into the horizontal stack, which handles the current row only, so every earlier pixel becomes reachable.

5. **Masked decoding steps.** Using $n_t = \lceil \gamma(t/T)N \rceil$ with the cosine schedule, $N = 256$ and $T = 8$, compute how many tokens are fixed at each step. Why does the schedule fix few tokens first?

Hint

Remaining masked counts are 252, 237, 213, 182, 143, 98, 50 and 0, so 4, 15, 24, 31, 39, 45, 48 and 50 tokens are fixed per step (the figure uses floor, which shifts counts by at most one). In exact arithmetic $\gamma(1) = 0$; in floating point $\cos(\pi/2)$ is a tiny positive number, so a literal ceiling leaves one token masked, another reason the snippet uses floor. Early on the model has little context, so its parallel guesses are the least reliable and the most likely to conflict; fixing only a few lets later steps condition on them.

6. **Classify the methods.** For PixelCNN, VQGAN transformer, Parti, LlamaGen, VAR, MaskGIT and MAR, give the prediction unit, the representation (pixels, discrete tokens, continuous tokens, multiscale maps), the conditional distribution type, and the number of sequential steps for a 16×16 token grid.

Hint

PixelCNN: pixels in raster order, 256-way softmax, one step per value. VQGAN transformer, Parti, LlamaGen: discrete tokens in raster order, softmax over the codebook, 256 steps. VAR: multiscale discrete maps, softmax per token with parallel tokens inside a scale, one step per scale. MaskGIT: discrete tokens, bidirectional masked prediction, T steps (8 in the paper). MAR: continuous tokens in random-order groups, a per-token diffusion head, as many steps as groups (each with its own inner diffusion steps).

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ML 09 • Foundation Models and Vision–Language Learning

What makes a model a foundation model: scaling laws, pre-training objectives, CLIP and SigLIP, DINOv2 features, promptable segmentation, multimodal language models, adaptation with linear probes and LoRA, and how to evaluate the risks.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-09-foundation-models/

What you'll learn

- What makes a model a foundation model, why the term describes training and reuse rather than an architecture, and why it need not generate text.
- What scaling laws say (a power law with a floor, compute-optimal allocation) and what they do not say.
- The main pre-training objectives — supervised, contrastive image–text, masked image modeling, self-distillation and generative — and the kind of features each tends to produce.
- How CLIP, SigLIP, DINOv2/DINOv3 and the Segment Anything family are used as reusable vision components.
- How multimodal large language models connect a vision encoder to a language model, and how LLaVA-style instruction tuning works.
- How to adapt a pretrained model — linear probe, fine-tuning, LP-FT, adapters, prompt tuning, LoRA, prompting — and how to evaluate the result honestly, including bias, hallucination and test-set contamination.

The big picture

For most of the history of computer vision, each task had its own model: one network for classifying flowers, another for detecting cars, another for segmenting tumors, each trained from scratch or from an ImageNet [43] classifier. A **foundation model** is trained once, on broad data and at large scale, and then **adapted** to many downstream tasks [1]. The word describes how the model is trained and reused, not a new layer type. A DINOv2 [15] or DINOv3 [16] image encoder, a CLIP-style image–text model [3], a promptable segmenter such as SAM [5], a multimodal language model and a broadly pretrained image generator can all play this role.

> **Explain like I'm five**
>
> Instead of teaching a new student from the beginning for every job, start with someone who has already looked at millions of pictures and read millions of captions. They already know what edges,

wheels, faces and dogs look like. For a new job you only need to give them a few examples, a short instruction, or a small new tool — not a whole new education.

Why does this matter? Three reasons. First, **labels are expensive** and pretraining moves most of the learning to data that needs no hand labels (captions, the images themselves). Second, **one backbone serves many tasks**, which concentrates engineering effort. Third, it changes the **interface**: instead of retraining, you can often specify a task with a text prompt, a click or a few examples. Bommasani et al. call the resulting concentration of methods *homogenization*, and warn that it also concentrates failure: a flaw in one shared backbone is inherited by every system built on it [1].

This chapter builds on ML 05 ·Representation Learning (what makes a feature reusable), ML 04 ·Vision Transformers (the usual backbone) and the generative chapters ML 07 and ML 08. The textbook treatment of transfer learning and representation in Bishop and Bishop [14] is a good companion. We follow the architectural map of the vision foundation-model survey by Awais et al. [39] and the multimodal survey by Caffagni et al. [2], with representative reports verified for this October 2026 edition. Examples illustrate families; they do not rank commercial products or claim that one checkpoint is best everywhere.

What makes a model a "foundation" model

Plain version. A foundation model is a large model pretrained on broad data with a generic objective, so that its internal representations or its interface can be reused for tasks it was never specifically trained on.

Bommasani et al. [1] coined the term in 2021 and emphasized two properties:

- **Emergence.** Capabilities arise from scale and the training objective rather than being designed explicitly. For example, CLIP was trained only to match images with captions, yet it can classify images into categories named in text [3]; DINO was trained without any labels, yet its attention maps outline objects [24].
- **Homogenization.** Many applications converge on a small number of pretrained backbones and recipes.

Be careful with "emergence". In this chapter it means only *a capability that was not a training target*. It does not mean an unpredictable or mysterious jump; whether a capability appears suddenly often depends on the metric used to measure it.

Formally, let f_θ be a model pretrained on an upstream dataset $\mathcal{D}_{\text{up}}$ with objective $\mathcal{L}_{\text{pre}}$:

$$\theta^\star = \arg\min_\theta \; \mathbb{E}_{x \sim \mathcal{D}_{\text{up}}}[\mathcal{L}_{\text{pre}}(f_\theta; x)].$$

A downstream task k supplies a (usually small) dataset $\mathcal{D}_k$ and a task loss ℓ_k. Adaptation chooses a small set of new or changed parameters ϕ_k and solves

$$\phi_k^\star = \arg\min_{\phi_k} \sum_{(x,y)\in\mathcal{D}_k} \ell_k(h_{\phi_k}(f_{\theta^\star}, x), y),$$

where h_{ϕ_k} is whatever we put on top of or inside the pretrained model: a linear head, an adapter, a LoRA update, a prompt, or nothing at all (pure prompting, $\phi_k = \emptyset$). The whole chapter is about the two lines above: what $\mathcal{L}_{\text{pre}}$ and $\mathcal{D}_{\text{up}}$ should be, and how small ϕ_k can be.

A foundation model does **not** have to produce text. A frozen image encoder whose features feed a hundred different classifiers is a foundation model. Conversely, a large model trained for one narrow task is not one, however big it is.

Scaling laws

Plain version. When you train bigger models on more data with more compute, the test loss usually falls smoothly and predictably — roughly as a straight line on a log–log plot — until it approaches a floor. That predictability is what lets labs plan large training runs from small ones.

Kaplan et al. [17] trained families of Transformer language models and reported that the loss scales as a power law in model size, dataset size and compute, with some trends spanning more than seven orders of magnitude, while width and depth mattered little within the tested range. A power law in model size N reads

$$L(N) = \left(\frac{N_c}{N}\right)^{\alpha_N},$$

where L is held-out cross-entropy loss, N the number of non-embedding parameters, N_c a fitted constant with the same units as N, and $\alpha_N > 0$ the fitted exponent. Taking logs gives $\log L = \alpha_N \log N_c - \alpha_N \log N$: a straight line with slope $-\alpha_N$.

Real curves bend because some loss can never be removed (the data itself is noisy or ambiguous). Hoffmann et al. [18] therefore fitted a form with an irreducible term:

$$L(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta},$$

where D is the number of training tokens, E the irreducible loss, and A, B, α, β fitted constants. For dense Transformers a common approximation of training compute is $C \approx 6ND$ floating-point operations [17] (roughly two for the forward and four for the backward pass, per parameter per token). Fixing C and minimizing L over N (with $D = C/6N$) gives a **compute-optimal** model size. Hoffmann et al. trained over 400 language models and concluded that, for compute-optimal training, model size and training tokens should be scaled equally — every doubling of model size should come with a doubling of tokens [18]. That differed from Kaplan et al.'s earlier recommendation of very large models trained on relatively modest data [17]; the disagreement is a useful

reminder that fitted exponents depend on the experimental protocol (learning-rate schedules, which runs are included, what is counted as compute).

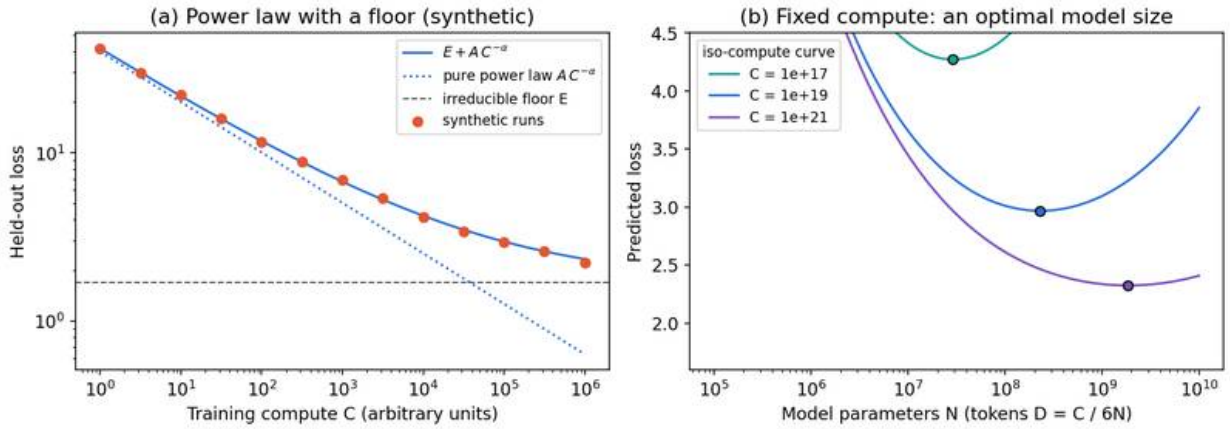


Figure 1 — (a) A power law with an irreducible floor E: a pure power law (dotted) keeps falling and over-promises at large scale. (b) With compute fixed, too small a model under-fits and too large a model sees too few tokens, so each budget has an optimal size. All constants are synthetic.

Vision has its own scaling studies. Zhai et al. [19] characterized the relationship between error rate, data and compute for Vision Transformers [21] trained on large labeled datasets, scaling to two-billion-parameter models. Cherti et al. [20] trained CLIP-style models on the public LAION dataset [34] and found power-law scaling for zero-shot classification, retrieval, linear probing and fine-tuning — but also that OpenAI's CLIP and OpenCLIP, with the same architectures and similar recipes, scale differently because their **training distributions** differ. Data is not a scalar.

Code: fit a scaling law and extrapolate

The snippet fits $L(C) = E + A\,C^{-\alpha}$ to the cheaper of 13 synthetic training runs and then predicts the expensive ones. Everything is synthetic; the point is the procedure.

```
import numpy as np
from scipy.optimize import curve_fit

rng = np.random.default_rng(0)
# Synthetic "runs": loss = E + A * C^(-alpha), with 2% multiplicative noise.
E_true, A_true, alpha_true = 1.7, 40.0, 0.3
compute = np.logspace(0, 6, 13)            # arbitrary compute units
loss = (E_true + A_true * compute**-alpha_true) * np.exp(.02 * rng.normal(size=13))

def law(c, E, A, alpha):
    return E + A * c**-alpha

# Fit only the cheaper runs, then extrapolate to the larger ones.
small = compute <= 1e4
```

```
(E, A, alpha), _ = curve_fit(law, compute[small], loss[small], p0=(1, 10, .5))
print(f"fit: E={E:.2f}  A={A:.1f}  alpha={alpha:.3f}")
for c, l in zip(compute[~small], loss[~small]):
    print(f"C={c:9.0f}  observed {l:.3f}  predicted {law(c, E, A, alpha):.3f}")

# Naive straight line in log-log space ignores the floor E.
slope = np.polyfit(np.log(compute[small]), np.log(loss[small]), 1)[0]
print(f"pure power-law slope without floor: {slope:.3f}")
```

Output: the fit recovers $E \approx 1.73$, $A \approx 40.0$, $\alpha \approx 0.300$; the extrapolated losses are within about 7% of the noisy observations; and a straight log–log line without the floor gives a slope of about -0.25, a wrong exponent. Two lessons transfer to real practice: always include the floor, and treat extrapolation over many orders of magnitude as a forecast with error bars.

What scaling laws do not tell you

- They predict **average loss** on the pretraining distribution, not accuracy on your task. Downstream metrics may improve smoothly, saturate, or move in steps.
- They hold **for a fixed recipe and data distribution** [20]. Better data filtering or a different objective changes the constants.
- They say nothing about **inference cost**. A smaller model trained on more data can be cheaper to deploy at the same loss, which is why many deployed models are trained beyond the compute-optimal point.
- Data is finite. If the high-quality data runs out, the B/D^{β} term stops shrinking however much compute you add.

Pre-training objectives: where does the target come from?

Plain version. Every pretraining method invents a prediction task. The difference is where the "answer" comes from: a human label, a caption, a hidden part of the same image, another network, or the next token.

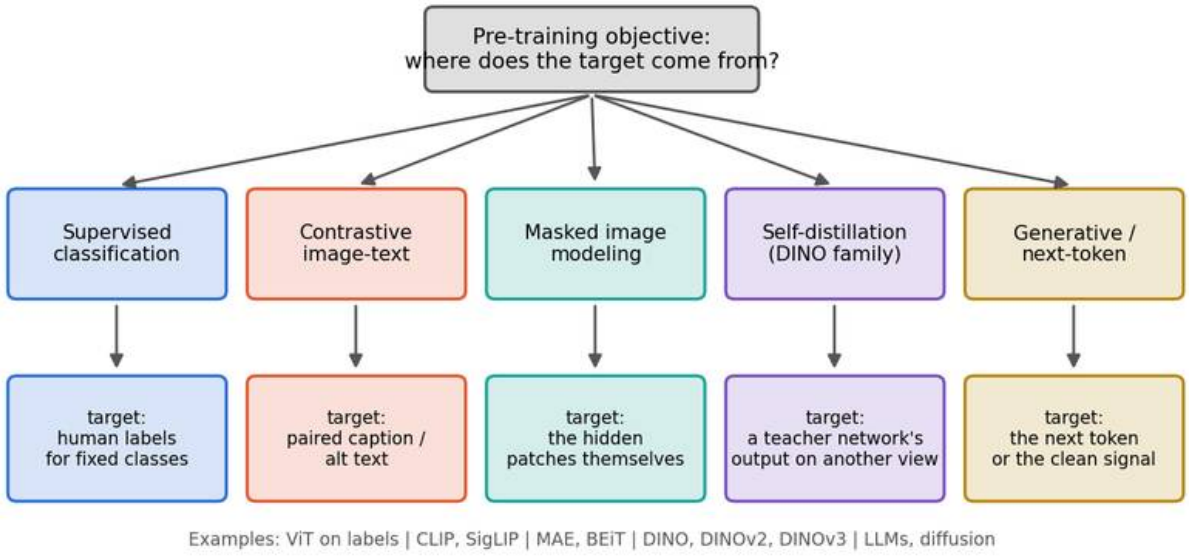


Figure 2 — Five families of pre-training objectives, organized by the source of the training target.

Objective	Target comes from	Representative models	Typical strength
Supervised classification	Human labels	ViT on large labeled sets [19], [21]	Strong global semantics for the label space
Contrastive image–text	Paired captions / alt text	CLIP [3], ALIGN [25], SigLIP [4]	Text-defined recognition, retrieval
Masked image modeling	Hidden patches of the same image	BEiT [23], MAE [22]	Strong after fine-tuning; local detail
Self-distillation	A teacher network's output on another view	DINO [24], DINOv2 [15], DINOv3 [16]	Frozen features, dense correspondence
Generative / next-token	Next token or clean signal	LLMs, diffusion (ML 07), AR (ML 08)	Generation; language interface

Supervised pre-training

Train a classifier with cross-entropy on a large labeled dataset, then drop the final layer. This was the ImageNet recipe for a decade and still works at scale [19], [21]. Its weakness is that the features are shaped by a fixed label set: distinctions that the labels do not require may be discarded.

Contrastive image–text learning

Captions come for free with web images, so CLIP [3] and ALIGN [25] train an image encoder and a text encoder together so that matching pairs score higher than mismatched ones within a batch. ALIGN showed that a large, noisy alt-text corpus with minimal filtering can still give strong representations [25]. The loss is detailed in the next section.

Masked image modeling

Hide most of an image and predict what is missing. BEiT [23] masks patches and predicts discrete visual tokens for them, borrowing the masked-language-modeling idea from BERT [44]. MAE [22] predicts the raw pixels directly and observes that masking a high proportion, e.g. 75% of patches, creates a nontrivial task; its encoder sees only the visible patches and a lightweight decoder reconstructs the rest, which makes training efficient. With $\mathcal{M}$ the set of masked patch indices, x_p the (normalized) pixels of patch p and $\hat{x}_p$ the reconstruction,

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{p \in \mathcal{M}} \|\hat{x}_p - x_p\|_2^2.$$

The loss is computed only on masked patches: predicting the visible ones would be trivial copying. Masked-modeling features are often excellent after fine-tuning but weaker as frozen features for linear probing than contrastive or self-distilled ones.

Self-distillation: the DINO family

DINO [24] — described by its authors as self-distillation with no labels — trains a **student** network g_s to match a **teacher** network g_t on different augmented views (crops) of the same image. Both output K-dimensional scores; probabilities are softmaxes with temperatures τ_s and τ_t:

$$P_s(x') = \text{softmax}\left(\frac{g_s(x')}{\tau_s}\right), \qquad P_t(x) = \text{softmax}\left(\frac{g_t(x) - c}{\tau_t}\right), \qquad \mathcal{L} = -\sum_{k=1}^{K} P_t(x)_k \log P_s(x')_k.$$

Here x and x' are two views, c is a running mean of teacher outputs (**centering**), and a small τ_t makes the teacher's distribution sharp (**sharpening**). Centering stops one dimension from dominating; sharpening stops the output from becoming uniform; together they avoid **collapse**, the trivial solution where every image maps to the same output. The teacher receives no gradients. Its weights are an exponential moving average (the momentum encoder) of the student's:

$$\theta_t \leftarrow \lambda\, \theta_t + (1 - \lambda)\, \theta_s, \qquad 0 < \lambda < 1.$$

DINO also uses multi-crop training (several small local crops for the student, global crops for the teacher), which forces local-to-global correspondence. Its authors report that the resulting ViT features contain explicit information about object segmentation that does not emerge as clearly with supervised ViTs or convnets, and that they work well as k-nearest-neighbor classifiers [24].

DINOv2 [15] scales this recipe and, crucially, curates the data: an automatic pipeline builds a dedicated, diverse, curated image dataset instead of using uncurated web images, a 1-billion-parameter ViT is trained, and it is distilled into smaller models. The authors report that these frozen features surpass OpenCLIP on most of their image-level and pixel-level benchmarks [15].

DINOv3 [16] scales data and model further and introduces **Gram anchoring**, a technique aimed at a known problem: dense (per-patch) feature maps degrading during long training schedules. It also adds post-hoc strategies for flexibility in resolution, model size and alignment with text.

Why dense features matter. A classifier needs one vector per image. Segmentation, depth estimation, correspondence and tracking need a good vector **per patch**. Self-distilled encoders are popular frozen backbones for these dense tasks precisely because their patch features stay semantically meaningful without fine-tuning.

Generative pre-training

Predicting the next token (language models, autoregressive image models in ML 08) or denoising (diffusion in ML 07) is also a pretraining objective. For language it produces the most flexible interface — instructions in natural language — which is why multimodal models attach vision to a pretrained language model rather than the other way round.

Image and text in a shared space: CLIP and SigLIP

Plain version. Train an image encoder and a text encoder so that corresponding pictures and captions point in similar directions. Then any sentence becomes a "class".

CLIP [3] normalizes image embeddings v_i and text embeddings t_j to unit length and forms scores $s_{ij} = v_i^\top t_j / \tau$ with a learned temperature τ. For a batch of B paired examples,

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2}\left[\frac{1}{B}\sum_i -\log \frac{\exp(s_{ii})}{\sum_j \exp(s_{ij})} + \frac{1}{B}\sum_j -\log \frac{\exp(s_{jj})}{\sum_i \exp(s_{ij})}\right].$$

Each image should select its matching caption among the B captions (first term, a row-wise softmax), and each caption its matching image (second term, column-wise). All other pairs in the batch act as negatives. Training uses paired text supervision, usually from existing captions or alt text; "zero-shot" downstream use does not mean the model was trained without supervision.

At inference, encode candidate descriptions such as "a photo of a dog" and compare an image with them. The text embeddings replace the class weights of a learned classifier:

$$\hat{y} = \arg\max_k v(x)^\top t(\text{prompt}_k).$$

The candidate set and wording affect the result; averaging the embeddings of several templates ("a photo of a k", "a close-up of a k") is a common way to reduce sensitivity. Zero-shot means no task-specific parameter training under the stated protocol; it does not guarantee that the classes or even the test images were absent from pretraining (see *Risks and evaluation*).

Code: paired loss and retrieval mechanics

These **synthetic embeddings** make the computation transparent. They are not outputs from a pretrained model, and their matches do not demonstrate real-world zero-shot accuracy.

```
import torch
from torch.nn import functional as F

images = F.normalize(torch.tensor([[1., .1, 0.], [.1, 1., 0.],
                                   [0., .1, 1.]]), dim=-1)
texts = F.normalize(torch.eye(3), dim=-1)
scores = images @ texts.T / .1
labels = torch.arange(3)
loss = (F.cross_entropy(scores, labels) +
        F.cross_entropy(scores.T, labels))/2
print("matching text:", scores.argmax(1).tolist())
print("paired loss:", round(loss.item(), 4))
ambiguous = F.normalize(torch.tensor([[.5, .5, .2]]), dim=-1)
print("cosines:", (ambiguous @ texts.T).tolist())
```

It prints `matching text: [0, 1, 2]`, a paired loss of about `0.0002`, and cosines of about `0.68 , 0.68, 0.27` for the ambiguous image: it is equally close to two descriptions, and no softmax can tell you which is right.

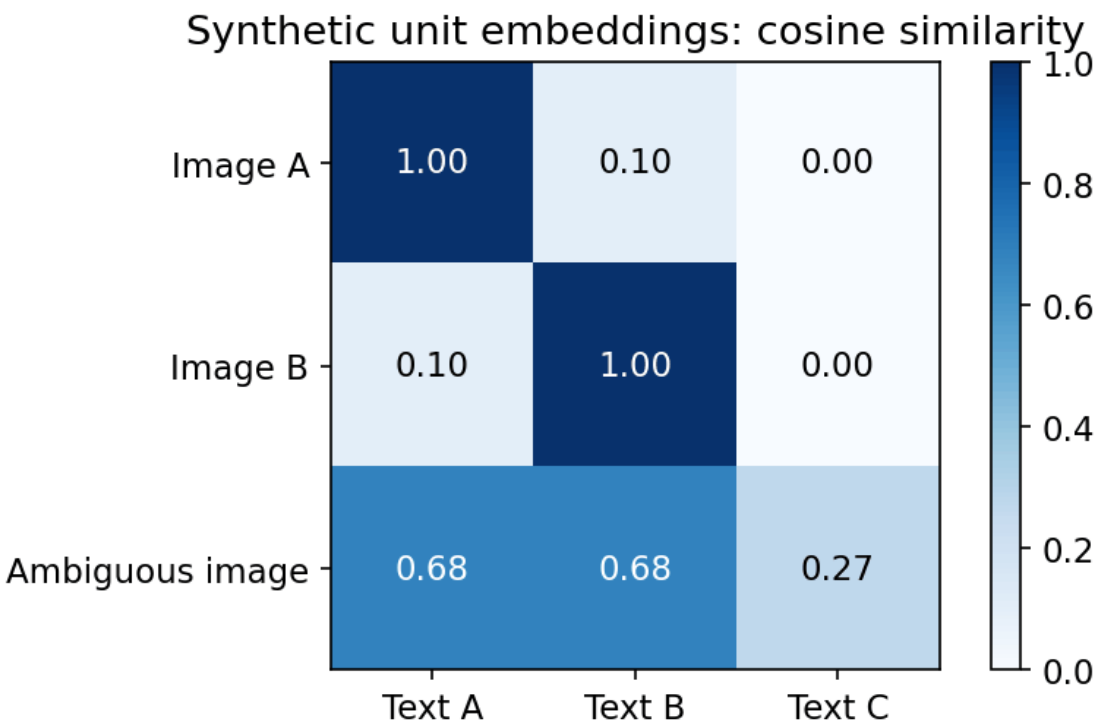


Figure 3 — Cosine similarity ranks candidates. It is not a calibrated probability that a concept is present.

SigLIP: a sigmoid instead of a softmax

SigLIP [4] replaces the batch-normalized softmax with an independent binary decision for every image–text pair. With labels $y_{ij} = +1$ for matching pairs and -1 otherwise, a learned scale a and bias b,

$$\mathcal{L}_{\text{SigLIP}} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{j=1}^{B}\log\sigma(y_{ij}(a\,v_i^\top t_j + b)),$$

where $\sigma(u) = 1/(1+e^{-u})$, so each term equals $\log(1+\exp(-y_{ij}(a\,v_i^\top t_j + b)))$. Because no term needs a sum over the whole batch, the loss does not require a global normalization across devices, which simplifies large-batch training. Since there are B positives and $B^2 - B$ negatives, the bias b is initialized so that the initial predictions are strongly negative.

Shared embeddings support retrieval and open-vocabulary recognition, but fine-grained counting, negation ("a dog **without** a collar"), spatial relations and rare domain terms are known weak points. Evaluate those requirements directly rather than inferring them from a strong overall classification score.

Promptable segmentation: the Segment Anything family

Plain version. A prompt tells the model which region or concept you want, and the output is a mask.

SAM [5] combines a heavy image encoder (run once per image), a light prompt encoder and a light mask decoder, so many prompts can be answered quickly on the same image. Spatial prompts — points, boxes or rough masks — select objects without a fixed class vocabulary. Because a single click is ambiguous (a click on a shirt could mean the shirt or the whole person), SAM predicts several candidate masks with confidence scores. It was trained on SA-1B, over 1 billion masks on 11 million licensed and privacy-respecting images, collected with a model-in-the-loop data engine [5].

SAM 2 [6] extends promptable segmentation to video with a streaming memory: the model stores features of past frames and of the prompted object, and propagates the mask through time; corrective clicks on later frames update the memory.

SAM 3 [7], released in 2025 with a report revised in March 2026, introduces **promptable concept segmentation**: given a short noun phrase (e.g. "yellow school bus"), image exemplars, or both, it returns masks and identities for **all** matching instances. Its image-level detector and memory-based video tracker share one backbone, and a **presence head** decouples recognizing whether the

concept occurs at all from localizing its instances [7].

These interfaces supply different information: a spatial point says *where*, a noun phrase says *what*, an exemplar says *what it looks like*. Report prompt creation cost and whether prompts come from people, ground truth or an automatic detector. A "zero-shot segmentation" score obtained with ground-truth boxes as prompts hides substantial annotation information.

Multimodal large language models

Plain version. Turn visual features into "words" a language model can read, then teach it to follow visual instructions.

A common design has

$$\text{image/video} \xrightarrow{\text{vision encoder}} h \xrightarrow{\text{connector}} z \xrightarrow{\text{language decoder with text}} \text{answer}.$$

Here h is a grid of visual features (for a ViT, one vector per patch), z is a sequence of vectors in the language model's token-embedding space, and the decoder treats z like the embeddings of ordinary words.

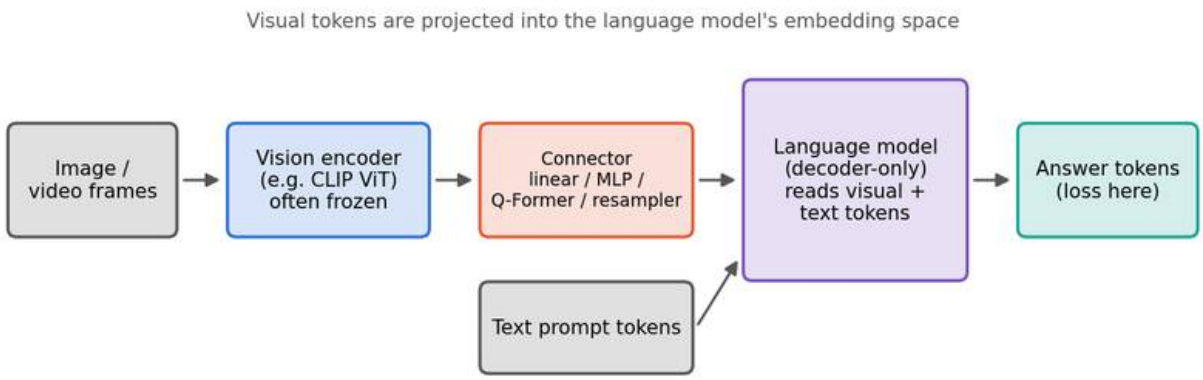


Figure 4 — The typical multimodal LLM pipeline. Visual features are converted into tokens in the language model's embedding space.

Connectors

- **Linear or MLP projection.** LLaVA [8] uses a single projection from CLIP features to the language model's embedding space; LLaVA-1.5 [28] reports that a simple MLP connector (with a CLIP ViT-L at 336-pixel resolution and academic VQA data) is surprisingly powerful and data-efficient. Every patch becomes one visual token.

- **Learned queries.** BLIP-2 [26] inserts a lightweight Querying Transformer (Q-Former) between a frozen image encoder and a frozen language model. A fixed set of learned query vectors cross-attends to the image features, so the language model always sees a small, fixed number of visual tokens. BLIP-2 trains the Q-Former in two stages: representation learning against the frozen image encoder, then generative learning against the frozen language model.
- **Resampling and cross-attention.** Flamingo [27] bridges frozen vision and language models and handles arbitrarily interleaved sequences of images and text, which enabled few-shot learning from examples written directly in the prompt.

The trade-off is token count versus detail: one token per patch keeps fine detail but makes long sequences (expensive for many images or video); a few learned queries are cheap but can lose small text.

Visual instruction tuning

LLaVA [8] used a language-only GPT-4 [45] to generate multimodal instruction-following data from image captions and boxes, then trained in two stages: first align the connector (feature alignment), then fine-tune connector and language model on the instructions. The loss is ordinary next-token cross-entropy, applied to the **answer** tokens only:

$$\mathcal{L} = -\sum_{i \in \mathcal{A}} \log p_\theta(w_i \mid z,\, w_{<i}),$$

where w_i are text tokens, $\mathcal{A}$ is the set of positions belonging to the assistant's answer, and z are the visual tokens. Instruction and prompt tokens are visible as context but masked out of the loss.

Qwen3-VL's 2025 report [9] illustrates the current direction: a family of dense and mixture-of-experts models with natively interleaved image, video and text contexts of up to 256K tokens. It is a representative report, not evidence that every task benefits from longer contexts or more generated reasoning.

What the diagram hides

Reading a document may require preserving small text through tiling or dynamic resolution. Video questions need timestamps and temporal order. These details change token counts and preprocessing; a generic "vision encoder + LLM" diagram is only the starting point. A fluent answer can also contain objects that are not in the image (**object hallucination**) or wrong counts. Distinguish object grounding, OCR, spatial reasoning, factual answering and stylistic instruction following — they need different evidence and metrics (see *Risks and evaluation*). For a survey of the whole design space, see Yin et al. [40] and Caffagni et al. [2].

Adapting a foundation model

Plain version. Use the least expensive adaptation that solves the target task, and check it on data that looks like deployment.

Strategy	Trainable components	Useful starting situation
Prompting / in-context examples	Nothing	The model's interface already fits the task
Frozen features + linear head (linear probe)	Small supervised readout	Few labels, good feature transfer
Prompt tuning / visual prompt tuning	A few learned input tokens	Many tasks share one frozen backbone
Adapters / LoRA	Small inserted or low-rank modules	Domain adaptation under memory limits
Partial fine-tuning	Selected blocks plus head	Some adaptation without full optimizer cost
Full fine-tuning (optionally LP-FT)	All model parameters	Enough data and compute; transfer alone insufficient

Linear probe

Freeze the encoder f, compute features once, and fit a linear classifier $\hat{y} = \text{softmax}(Wf(x) + b)$.

It is cheap, hard to overfit, and is also the standard **measurement** of representation quality: if a linear readout works, the information is linearly accessible in the features.

Code: why pretrained features help most with few labels

This toy experiment simulates the whole story. Observations are 300-dimensional and mix 8 **semantic** factors (which determine the class) with 8 large **nuisance** factors (think lighting and pose) plus noise. "Pretraining" sees 50 upstream classes with plenty of labels and learns an 8-dimensional projection (linear discriminant analysis, standing in for a deep encoder). Downstream, there are four **new** classes and only a few labels per class.

```
import numpy as np
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn.linear_model import LogisticRegression

rng = np.random.default_rng(0)
D, k_sem, k_nui = 300, 8, 8
sem_map = rng.normal(size=(k_sem, D))   # semantic factors -> pixels
nui_map = rng.normal(size=(k_nui, D))   # nuisance factors (light, pose...)

def sample(centers, n):
    y = rng.integers(len(centers), size=n)
```

```
    z = centers[y] + 0.7 * rng.normal(size=(n, k_sem))
    u = 4.0 * rng.normal(size=(n, k_nui))          # large nuisance variation
    x = z @ sem_map + u @ nui_map + rng.normal(size=(n, D))
    return x, y

# 1) "Pre-training": many upstream classes with plenty of labels.
up_centers = rng.normal(size=(50, k_sem))
x_up, y_up = sample(up_centers, 10_000)
encoder = LinearDiscriminantAnalysis(n_components=k_sem).fit(x_up, y_up)

# 2) Downstream: four NEW classes, few labels, linear probe only.
new_centers = rng.normal(size=(4, k_sem))
x_te, y_te = sample(new_centers, 2_000)
f_te = encoder.transform(x_te)          # frozen features, computed once
for shots in [1, 5, 20, 100]:
    raw, feat = [], []
    for repeat in range(10):
        x_tr, y_tr = sample(new_centers, 4 * shots)
        raw.append(LogisticRegression(max_iter=3000)
                   .fit(x_tr, y_tr).score(x_te, y_te))
        feat.append(LogisticRegression(max_iter=3000)
                    .fit(encoder.transform(x_tr), y_tr).score(f_te, y_te))
    print(f"{shots:>3} labels/class | raw pixels {np.mean(raw):.3f}"
          f" | frozen features {np.mean(feat):.3f}")
```

On our machine it prints accuracies of roughly 0.29 vs 0.71 with one label per class, 0.62 vs 0.95 with five, 0.96 vs 0.99 with twenty and 0.98 vs 0.99 with a hundred (raw inputs vs frozen features; chance is 0.25). The pretrained projection already discards the nuisance directions, so a handful of labels suffice; with enough labels, the raw-input probe learns to ignore the nuisance itself and the gap closes. The upstream classes are different from the downstream ones — the features transfer because both share the same semantic factors. If the downstream task depended on a factor that pretraining threw away (here, the nuisance), frozen features would fail no matter how many labels you add. That is the failure mode to test for in practice.

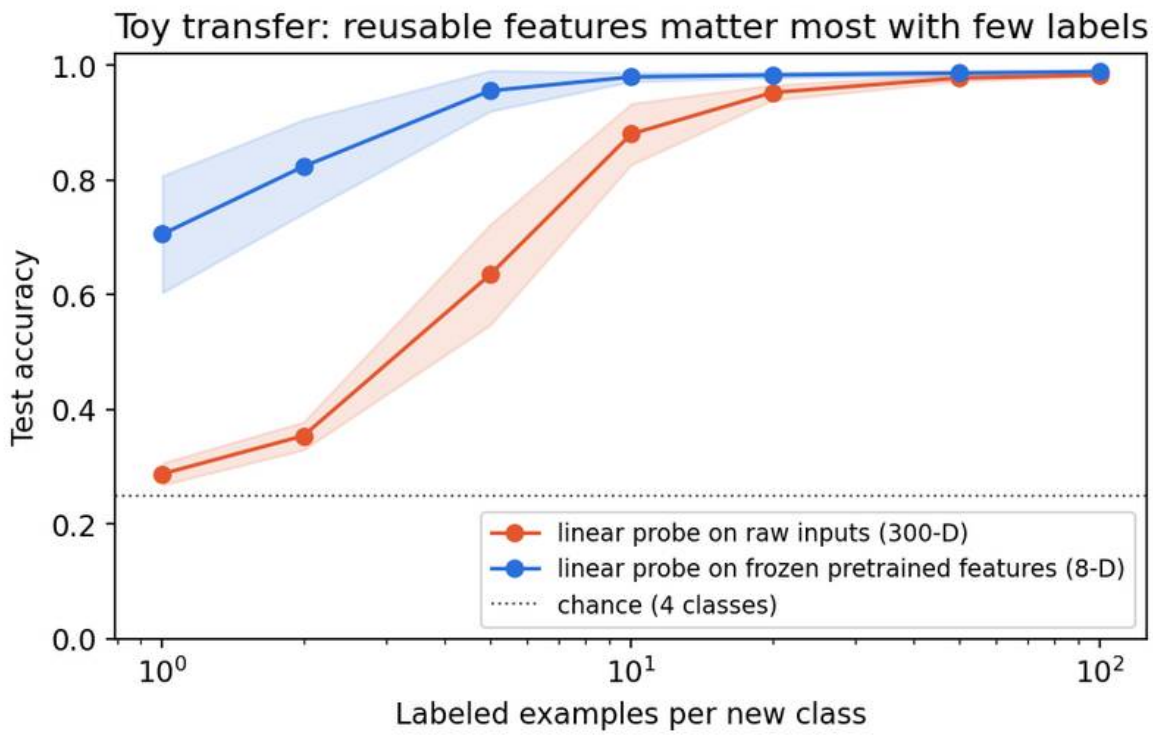


Figure 5 — The toy transfer experiment over more label budgets (mean ± one standard deviation over 10 draws). Reusable features matter most when labels are scarce.

Fine-tuning, LP-FT and robust fine-tuning

Full fine-tuning updates every weight and usually gives the best in-distribution accuracy when data suffices. But Kumar et al. [32] showed that it can distort good pretrained features: across their benchmarks, fine-tuning gave on average higher in-distribution but lower out-of-distribution accuracy than linear probing. Their explanation is that a randomly initialized head sends large, noisy gradients into the backbone early in training. **LP-FT** — first fit a linear probe, then fine-tune everything starting from that head — reduced this distortion and outperformed both alternatives in their experiments [32]. **WiSE-FT** [33] takes another route: linearly interpolate the weights of the zero-shot and fine-tuned models, $\theta_\gamma = (1-\gamma)\theta_{\text{zero-shot}} + \gamma\,\theta_{\text{ft}}$, which improved robustness under distribution shift for CLIP-style models while keeping strong target accuracy. Both are cheap to try and worth knowing before declaring that fine-tuning "hurt robustness".

Parameter-efficient fine-tuning (PEFT)

Adapters [29] insert small bottleneck MLPs inside each Transformer block and train only them, adding a few parameters per task while the original network stays fixed. **Prompt tuning** learns a few extra input tokens; **Visual Prompt Tuning (VPT)** [30] does this for ViTs, prepending learnable tokens to the patch sequence of a frozen backbone. Xu et al. [42] review the PEFT landscape with a taxonomy of additive, partial, reparameterized, hybrid and unified fine-tuning, plus experiments on parameter and memory efficiency.

LoRA [10] reparameterizes an update to a frozen weight $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ as

$$W = W_0 + \frac{\alpha}{r} BA, \qquad A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad B \in \mathbb{R}^{d_{\text{out}} \times r},$$

with rank $r \ll \min(d_{\text{in}}, d_{\text{out}})$ and a scaling constant α. Freeze W_0 and learn A, B. Trainable weights drop from $d_{\text{out}} d_{\text{in}}$ to $r(d_{\text{in}} + d_{\text{out}})$. Initialize $B = 0$ so the initial update is zero, and A randomly so learning can start. After training, BA can be merged into W_0, so inference costs nothing extra. QLoRA [31] stores the frozen base weights in 4-bit precision and backpropagates through them into LoRA adapters, cutting memory enough to fine-tune very large models on a single GPU.

Code: a low-rank update with a frozen base

This standalone example trains one adapter step on synthetic input. It shows the parameterization and initial behavior; it does not reproduce a full pretrained-model fine-tuning pipeline.

```
import torch
from torch import nn
from torch.nn import functional as F

torch.manual_seed(0)
class LoRALinear(nn.Module):
    def __init__(self, din, dout, rank=4, alpha=4):
        super().__init__()
        self.base = nn.Linear(din, dout, bias=False)
        self.base.weight.requires_grad_(False)
        self.A = nn.Parameter(torch.randn(rank, din) * .02)
        self.B = nn.Parameter(torch.zeros(dout, rank))
        self.scale = alpha/rank
    def forward(self, x):
        return self.base(x) + self.scale * F.linear(F.linear(x, self.A), self.B)

layer = LoRALinear(32, 16)
x = torch.randn(8, 32)
assert torch.allclose(layer(x), layer.base(x))
opt = torch.optim.Adam([layer.A, layer.B], lr=.01)
loss = F.mse_loss(layer(x), torch.randn(8, 16))
opt.zero_grad(); loss.backward(); opt.step()
```

```
assert layer.base.weight.grad is None
print("trainable:", sum(p.numel() for p in layer.parameters() if p.requires_grad))
```

It prints `trainable: 192`, which is $r(d_{\text{in}} + d_{\text{out}}) = 4 \times (32 + 16)$, against 512 for the full matrix. The first assertion checks that the zero-initialized B leaves the model unchanged at the start; the second, that the frozen base receives no gradient.

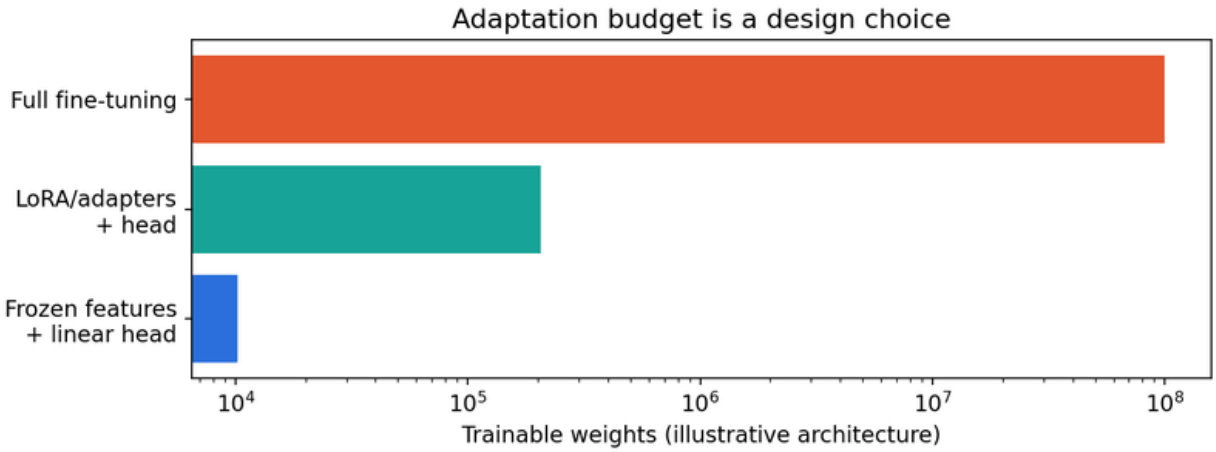


Figure 6 — Analytical trainable-weight budgets for an illustrative architecture (feature size 1024, 10 classes, rank 8 on 12 matrices, 100M parameters in total). Activation memory, frozen weight storage and optimizer state must be counted separately.

LoRA is a parameterization, not a guarantee against overfitting or forgetting. Adapters still require domain validation. Frozen BatchNorm statistics, image normalization, output-label mapping and checkpoint-specific preprocessing also need deliberate treatment.

Prompting and in-context learning

The cheapest adaptation changes no weights: write a better text prompt for CLIP, give SAM a box, or put a few solved examples into a multimodal model's context, as Flamingo did [27]. The vision–language survey by Zhang et al. [41] compares prompting, feature adaptation and fine-tuning across vision tasks. Prompting is fast to iterate but brittle: small wording changes can move results, so fix and report the prompt protocol.

Video, geometry and world models

Plain version. Beyond recognizing a frame, a model may predict change over time, infer three-dimensional structure, or support action planning.

V-JEPA 2 [11] studies self-supervised video models that predict in feature space rather than pixel space, with an action-conditioned extension used for planning. VGGT [12] predicts camera parameters, depth, point maps and tracks from one or many views in a single feed-forward pass. These illustrate two routes beyond classification: temporal state modeling and geometric inference.

A generated video that looks convincing need not obey physical constraints. A scene reconstruction can be useful without generating photorealistic pixels. Measure multi-view consistency, geometry, action-conditioned outcomes and temporal behavior according to the claimed capability. DLCV 07 ·World Models develops partial observability, learned dynamics and planning in more detail.

Risks and evaluation

Plain version. A model trained on a large slice of the internet inherits that slice's biases and gaps, may confidently describe things that are not there, and may already have seen your "test" set. Evaluation has to be designed around these facts.

Data provenance, bias and privacy

Web-scale image–text datasets are too large to inspect by hand. LAION-5B [34] made such a dataset public (5.85 billion CLIP-filtered pairs) and thereby enabled open replication such as OpenCLIP [20]; audits of its predecessor LAION-400M by Birhane et al. [35] documented explicit and stereotyping content in image–text pairs. Downstream systems inherit these associations. Licenses and consent also matter: training data, model weights and generated outputs may each carry different terms, and personal data (faces, names, medical images) raises privacy obligations that do not disappear because data was public. SAM's SA-1B is an example of a dataset whose description emphasizes licensed and privacy-respecting images [5]; record such properties for every model you deploy.

Hallucination

Multimodal LLMs can mention objects that are absent from the image. POPE [37] measures object hallucination with a polling approach: it asks yes/no questions about the presence of specific objects (including frequent and co-occurring distractors), which makes the measurement less dependent on the model's generation style than scoring free-form captions. A low POPE error does not cover counting, attributes or relations; test those separately.

Contamination and "zero-shot"

If benchmark images (or near-duplicates) are in the pretraining set, a "zero-shot" score partly measures memory. Mayilvahanan et al. [36] asked whether CLIP's generalization mainly stems

from high train–test similarity; after pruning LAION to remove images too similar to common out-of-distribution benchmarks and retraining, performance dropped on some benchmarks but remained strong overall. The lesson is two-sided: similarity explains part of the reported performance, not all of it — so measure it rather than assume either way. Practical checks: deduplicate the test set against available training data, report results on data collected after the model's data cutoff, and state the full zero-shot protocol (class names, templates, candidate set).

Benchmarks

Aggregate multimodal benchmarks such as MMMU [38] — 11.5K college-level questions from exams, quizzes and textbooks across six disciplines, with charts, diagrams, maps, tables and other image types — test knowledge and reasoning, but a single number mixes perception, OCR, domain knowledge and answer formatting. For a deployment, build a small, task-specific evaluation set with known failure cases, report variance over seeds or prompt wordings, and compare against a simple baseline (a linear probe on frozen features is often surprisingly strong). ML 10 · Evaluation and Deployment covers calibration, splits and deployment in depth.

Modern view

Surveys worth reading.

- **Bommasani et al. [1]** — the report that named foundation models. It covers capabilities (language, vision, robotics, reasoning), applications (law, health, education), technology (modeling, training, adaptation, evaluation, data) and societal impact. Its key takeaways — emergence, homogenization, and inherited failure — still frame the field.
- **Awais et al. [39]** — a computer-vision survey organizing foundation models by architecture design, training objective (contrastive versus generative), datasets, fine-tuning mechanisms and prompting, and listing open challenges in evaluation, real-world applicability, adversarial robustness and interpretability.
- **Zhang et al. [41]** — a TPAMI survey of vision–language models for vision tasks: network architectures, pre-training objectives, downstream tasks, datasets, transfer learning and knowledge distillation, with benchmarking of reviewed methods.
- **Yin et al. [40]** and **Caffagni et al. [2]** — surveys of multimodal large language models: architecture (encoder, connector, LLM), training strategy and data (pre-training, instruction tuning, alignment), evaluation, and hallucination.
- **Han et al. [13]** — a survey broadening the map to multimodal *generative* models, organized by generative category and technique.
- **Xu et al. [42]** — a critical review of parameter-efficient fine-tuning that sorts methods into additive, partial, reparameterized, hybrid and unified families and compares representative

ones experimentally.

State of the art, 2023–2026. Four trends stand out. (1) **Better frozen features**: DINOv2 [15] and DINOv3 [16] made self-supervised encoders strong enough that many dense tasks start from frozen patch features. (2) **Concept-level prompting**: segmentation moved from "click an object" (SAM [5]) to video memory (SAM 2 [6]) to "segment every instance of this phrase" (SAM 3 [7]). (3) **Long, interleaved multimodal contexts**: models such as Qwen3-VL [9] read many images, video and text together. (4) **Beyond recognition**: video prediction for planning (V-JEPA 2 [11]) and feed-forward geometry (VGGT [12]) treat foundation models as world and scene estimators.

Open problems.

- Grounded reliability: reducing hallucination and making models say "not visible" when appropriate [37].
- Honest evaluation under contamination and benchmark saturation [36].
- Data: curation as a scaling axis [15], [20], provenance, licensing and consent [34], [35].
- Efficient adaptation and inference: token budgets for high-resolution images and video; PEFT that preserves robustness [32], [33].
- Dense, 3D and temporal understanding that matches the semantic strength of image-level models.

For any new "foundation" model, record its input/output interface, training objective, data sources, supported adaptation, checkpoint license, preprocessing and task-level evaluation. This avoids treating model size or an open-ended chat interface as proof of general competence.

Key takeaways

- "Foundation" describes broad pretraining plus reuse, not a particular architecture; such models need not generate text.
- Scaling laws describe smooth, power-law-like loss improvements with a floor; compute-optimal training balances model size and data; the constants depend on data and recipe.
- The pretraining objective decides what the features are good at: contrastive for text-defined recognition, masked modeling for fine-tuning, self-distillation for frozen dense features, generative for flexible interfaces.
- CLIP-style alignment enables retrieval and text-defined classification; cosine similarity is not calibrated confidence.
- Spatial, conceptual and language prompts provide different kinds of information; report who made the prompts.
- Multimodal LLMs = vision encoder + connector + language model, trained with answer-only next-token loss; the connector sets a detail-versus-token-count trade-off.
- Start adaptation cheap (prompt, linear probe), then PEFT (LoRA, adapters), then full fine-

tuning; LP-FT and weight interpolation help preserve robustness.
- Evaluate for bias, hallucination and contamination explicitly; one aggregate benchmark number is not evidence of general competence.

Exercises

1. **Candidate dependence.** Add a fourth description to the synthetic CLIP example. Can the softmax "confidence" for an image change without changing the image?

   Hint

   Yes: the denominator of the softmax changes. Similarity to a fixed description is unchanged, but the normalized probability depends on the candidate set. Try adding a text vector close to the ambiguous image's embedding.

2. **Compute-optimal allocation.** Using the parametric law $L(N, D) = E + A/N^{\alpha} + B/D^{\beta}$ and $C = 6ND$, show that the optimal model size grows as $N^{\star} \propto C^{\beta/(\alpha+\beta)}$. What does $\alpha = \beta$ imply?

   Hint

   Substitute $D = C/(6N)$, differentiate with respect to N and set to zero: $\alpha A N^{-\alpha-1} = \beta B (6N/C)^{\beta}/N$. Solving gives $N^{\alpha+\beta} \propto C^{\beta}$. With $\alpha = \beta$, $N^{\star} \propto C^{1/2}$ and therefore $D^{\star} \propto C^{1/2}$ too: scale parameters and tokens equally, as Hoffmann et al. concluded.

3. **Break the transfer.** In the linear-probe experiment, make the downstream class depend on the *nuisance* factors instead of the semantic ones. What happens to the frozen-feature probe as labels increase, and what does that say about choosing a foundation model?

   Hint

   Draw class centers for `u` instead of `z`. The frozen 8-D features were trained to discard the nuisance directions, so they stay near chance however many labels you add, while the raw-input probe improves. A backbone is only useful if its pretraining kept the information your task needs — check with a linear probe before investing in adaptation.

4. **LoRA budget.** A ViT-B block has four 768×768 attention projections and two MLP matrices of sizes 3072×768 and 768×3072. Count trainable weights for full fine-tuning of these six matrices and for rank-8 LoRA on all of them.

   Hint

   Full: $4{\cdot}768^2 + 2{\cdot}3072{\cdot}768 = 2{,}359{,}296 + 4{,}718{,}592 = 7{,}077{,}888$. LoRA: $8{\cdot}[4{\cdot}(768{+}768) + 2{\cdot}(3072{+}768)] = 8 \cdot (6144 + 7680) = 110{,}592$, about 1.6% of the full count.

5. **Prompt fairness.** Design a SAM evaluation that separates mask quality from prompt quality.

Hint

Fix the prompt protocol and count human interactions. Evaluate automatic proposals separately from masks produced with ideal ground-truth prompts, and report both. For SAM 3, compare noun-phrase prompts with exemplar prompts on the same images.

6. **Contamination audit.** You are given a new image–text model that reports high zero-shot accuracy on a public benchmark. List three checks you would run before trusting the number.

Hint

(a) Near-duplicate search of test images against any available training data (e.g. by embedding similarity); (b) evaluation on a freshly collected set from after the data cutoff; (c) sensitivity to the prompt templates and the candidate class list; optionally compare with a linear probe and a simple baseline.

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ML 10 • Evaluation, Efficiency and Deployment

Split data without leakage, cross-validate, attach confidence intervals, check calibration and robustness under shift, measure FLOPs, memory and latency honestly, then export, compress, monitor and reproduce a model.

Online edition: https://blog.shyandram.dev/en/learn/ml/ml-10-evaluation-deployment/

What you'll learn

- How to build a baseline and a data split whose result actually supports the claim you want to make, and how leakage silently breaks that.
- How K-fold, grouped, time-ordered and nested cross-validation work, and when each one is the right tool.
- How to put an uncertainty on a test score: standard errors, the paired bootstrap and McNemar's test.
- How calibration differs from accuracy, how to measure it (Brier score, NLL, ECE) and how temperature scaling fixes part of it.
- What distribution shift is (covariate, label, concept), how robustness benchmarks probe it, and how to monitor it after deployment.
- How to count FLOPs, memory and latency correctly, and what export, quantization, pruning, distillation and inference runtimes such as ONNX Runtime and TensorRT really buy you.
- How to make an experiment reproducible and document it.

The big picture

Training a model is the middle of a project, not the end. Before anyone relies on it, you need to answer four questions. Does it work on the cases it will really see? How sure are we of that number? Does it know when it is likely to be wrong? Does it fit the device, latency and budget it has to run on? After it ships, a fifth question appears: is it still working?

> **Explain like I'm five**
>
> A student who memorized last year's exam gets full marks on last year's exam. That tells you nothing. A fair test uses new questions (a clean split). One test can be lucky, so you want to know how much the score would wobble on another day (a confidence interval). A good student also says "I'm not sure" on the questions they might get wrong (calibration). If the real exam is in a noisy hall, you check that too (robustness). If it is a timed exam, being right slowly is not enough (latency). And once the student starts a job, someone still checks their work now and then (monitoring).

This chapter closes the ML track. It gathers the practical methodology from Goodfellow et al. [1, ch. 11] and the statistical-learning view of James et al. [2], and adds the engineering steps that

turn a trained network into a dependable service. The task-specific metrics of vision (top-1, mAP, PSNR, FID and so on) are covered in DLCV 02 and DLCV 03; here we focus on what all tasks share.

Start with a meaningful baseline

Plain version. Before spending more compute, find out how well a simple method does on exactly the same split. The baseline tells you what the complicated model actually buys.

Good baselines for image tasks include: logistic regression on raw pixels or hand-designed features; a nearest-neighbor or SVM classifier; and a small linear head on frozen features from a pretrained model (see ML 09). Logistic regression learns a linear boundary; a support vector machine (SVM) maximizes a regularized margin; nearest neighbors compares examples directly; tree ensembles partition the feature space. The classical pattern-classification chapter, IP 13, develops these methods.

For reference, the binary linear SVM minimizes

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\max(0,\, 1 - y_i(w^\top x_i + b)), \qquad y_i \in \{-1, +1\},$$

where w and b are the weights and bias, x_i is the i-th feature vector, and $C > 0$ trades margin width against training errors. The term $\max(0, 1 - y_i f(x_i))$ is the **hinge loss**: it is zero for points on the correct side with margin at least 1 and grows linearly otherwise.

Code: baselines with preprocessing inside the pipeline

This example uses only bundled data. Model choice uses a validation split; the chosen model is evaluated on the test split once. Scaling and PCA are fitted inside each training pipeline, never on the full dataset.

```
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, balanced_accuracy_score

X, y = load_digits(return_X_y=True)
Xa, Xt, ya, yt = train_test_split(X, y, test_size=.2, stratify=y, random_state=0)
Xtr, Xv, ytr, yv = train_test_split(Xa, ya, test_size=.25, stratify=ya, random_state=1)
models = {      # listed from simplest to most complex
    "linear": make_pipeline(StandardScaler(),
                            LogisticRegression(max_iter=2000)),
```

```
    "PCA + RBF SVM": make_pipeline(StandardScaler(), PCA(n_components=30),
                                   SVC(C=10, gamma="scale")),
}
scores = {}
for name, model in models.items():
    model.fit(Xtr, ytr)
    scores[name] = accuracy_score(yv, model.predict(Xv))
best = max(scores, key=scores.get)   # tie -> first listed, i.e. the simpler model
pred = models[best].predict(Xt)
print("validation:", scores)
print("selected:", best)
print("test accuracy:", accuracy_score(yt, pred))
print("test balanced accuracy:", balanced_accuracy_score(yt, pred))
```

On this split both candidates score exactly 0.9639 on validation (347 of 360 images). The rule "on a tie, prefer the simpler model" was fixed in advance, so the linear model is selected, and it scores 0.9639 accuracy and 0.9636 balanced accuracy on test. Two lessons hide in this tiny run. First, a tie-break rule is part of the protocol and should be written down before you look. Second, with 360 validation images a single image is worth 0.28 percentage points, so many apparent "wins" in small experiments are noise; the section on confidence intervals makes this precise.

Data splits and leakage

Plain version. The test set must be new in the same way that deployment data will be new. If test images share a patient, a video, a scene or a time period with training images, the test score measures memory, not generalization.

Choose the split unit from the claim

Images from one patient, one video, one production batch or one camera are correlated. Randomly splitting adjacent video frames puts almost the same picture in train and test. Ask what "new" means in deployment and split by that unit: by patient if the model will see new patients, by video if it will see new videos, by site if it will be installed in new hospitals, by time if it must predict the future.



Figure 1 — Choose the split unit according to the generalization claim. Group separation prevents near-duplicate context from crossing the split.

Stratification keeps class proportions similar across splits; it does not replace grouping. Deduplicate before splitting, and keep exact or near duplicates together. When the deployment population drifts over time, a time-based split (train on the past, test on the most recent period) is the honest choice.

The many faces of leakage

Leakage is any path by which information that would not be available at prediction time reaches the model or the model-selection process. Kapoor and Narayanan found leakage errors affecting 329 papers across 17 scientific fields that use ML, and propose a taxonomy of leakage types, among them missing train/test separation, preprocessing on the full dataset, duplicates, temporal leakage and illegitimate features [4]. The scikit-learn documentation lists the most common coding mistakes [3]. A practical checklist:

Leak	Example	Fix
Preprocessing on all data	Fit `StandardScaler`, PCA or a vocabulary on train + test	Put preprocessing inside a `Pipeline` fitted on the training fold only
Correlated units	Frames of one video, slices of one scan, crops of one photo	Split by group
Duplicates	The same image scraped twice	Hash or embed and deduplicate before splitting
Temporal	Using next week's data to predict this week	Time-ordered splits
Target proxy	A feature recorded *after* the outcome (e.g. "treatment given")	Audit each feature's timestamp
Test reuse	Tuning hyperparameters while watching the test score	Validation for decisions, test once

The last row is subtle. Every time you look at the test score and change something, the test set becomes a little more like a validation set. Public benchmarks suffer from this at community scale.

Cross-validation

Plain version. Instead of one train/validation split, rotate: split the data into K parts, train K times, each time holding out a different part, and average. You use every example for evaluation once and you see how much the score varies.

K-fold and its variants

Partition the indices into K disjoint folds $F_1, \dots, F_K$. For each k, train $\hat{f}^{(-k)}$ on all folds except F_k and compute its error on F_k. The cross-validation estimate is

$$\widehat{\mathrm{Err}}_{\mathrm{CV}} = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{|F_k|} \sum_{i \in F_k} \ell(y_i, \hat{f}^{(-k)}(x_i)),$$

where ℓ is the per-example loss (for example 0–1 error) and $|F_k|$ is the fold size. Common choices are $K = 5$ or $K = 10$ [2]. The variants differ only in how folds are formed:

- **Stratified K-fold** keeps class proportions equal across folds; useful when classes are imbalanced.
- **Group K-fold** assigns each group (patient, video, site) to exactly one fold, so no group appears in both training and held-out data.
- **Time-series split** always trains on earlier data and validates on later data.
- **Leave-one-group-out** is group K-fold with one group per fold; useful with few groups.

The fold scores are not independent (training sets overlap), so the standard deviation across folds understates the true uncertainty. Treat it as a rough indication of stability, not as a formal confidence interval.

Code: what random K-fold does to grouped data

We simulate 40 patients with 25 images each. Every patient's images share a "fingerprint" (patient-specific appearance), and the label is fixed per patient. The real class signal is weak.

```
import numpy as np
from sklearn.model_selection import KFold, GroupKFold, cross_val_score
from sklearn.ensemble import RandomForestClassifier
```

```
rng = np.random.default_rng(0)
n_groups, per_group = 40, 25                    # e.g. 40 patients, 25 images each
groups = np.repeat(np.arange(n_groups), per_group)
y = np.repeat(rng.integers(0, 2, n_groups), per_group)   # label fixed per patient
identity = rng.normal(size=(n_groups, 20))[groups]       # patient "fingerprint"
signal = 0.3 * y[:, None] * np.ones((1, 20))             # weak real class signal
X = identity + signal + 0.3 * rng.normal(size=(len(y), 20))

model = RandomForestClassifier(n_estimators=200, random_state=0)
random_cv = cross_val_score(model, X, y, cv=KFold(5, shuffle=True, random_state=0))
group_cv = cross_val_score(model, X, y, cv=GroupKFold(5), groups=groups)
print(f"random K-fold accuracy: {random_cv.mean():.3f} +/- {random_cv.std():.3f}")
print(f"group  K-fold accuracy: {group_cv.mean():.3f} +/- {group_cv.std():.3f}")
```

Output:

```
random K-fold accuracy: 0.999 +/- 0.002
group  K-fold accuracy: 0.657 +/- 0.158
```

The random split reports near-perfect accuracy because the forest recognizes each patient's fingerprint and copies that patient's label. Group K-fold, which matches the claim "works on new patients", reports 0.657 with large variation across folds. Neither number is wrong as arithmetic; only one answers the question.

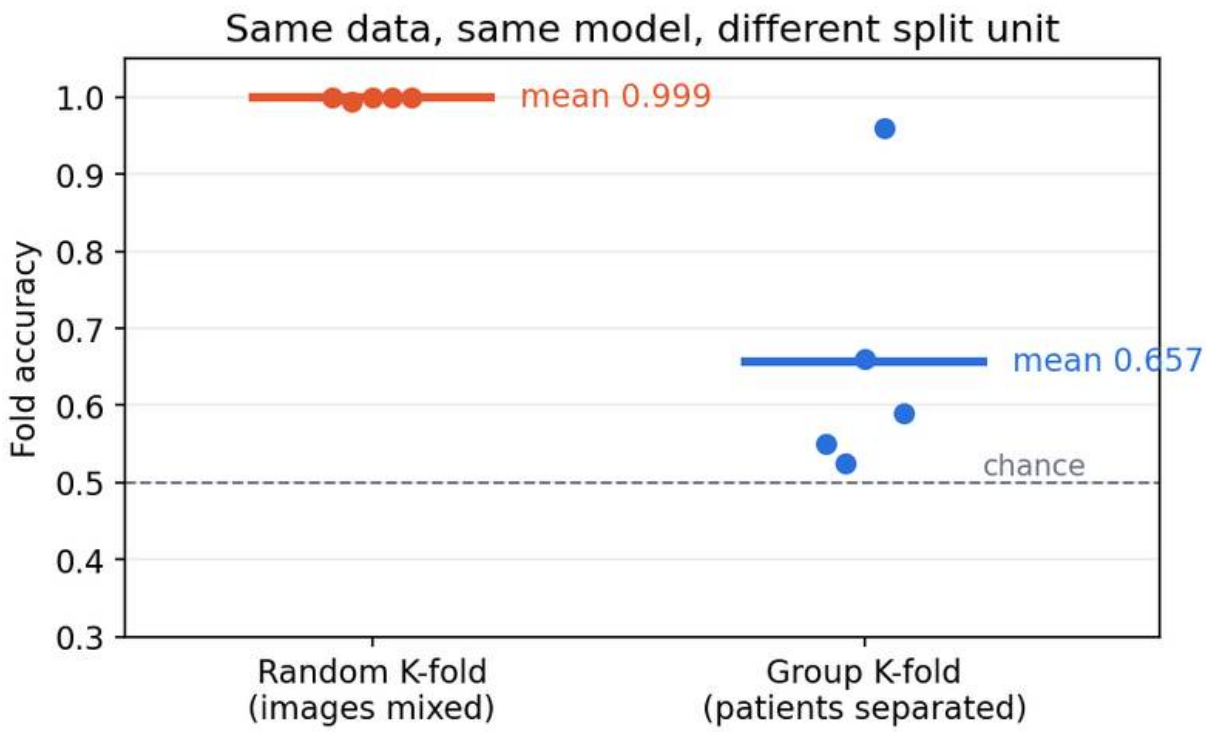


Figure 2 — Output of the code above, one dot per fold. The split unit, not the model, causes the 34-point gap.

Nested cross-validation

If you use cross-validation both to **choose** hyperparameters and to **report** performance, the reported number is optimistically biased: you picked the configuration that happened to look best on those folds. Varma and Simon demonstrated this with "null" data in which the two classes were statistically identical: after tuning an SVM by cross-validation, the CV error estimate fell below 30% on 38% of the simulated datasets, while the tuned classifiers were no better than chance on independent test data [5]; Cawley and Talbot showed that over-fitting the model-selection criterion can be as large as the differences between learning algorithms [6].

Nested CV fixes this with two loops. The outer loop holds out a fold for evaluation. Inside each outer training set, an inner CV selects hyperparameters. The outer fold is touched only to score the configuration the inner loop chose. In scikit-learn this is simply a `GridSearchCV` placed inside `cross_val_score`. The result estimates the performance of the whole *procedure* (including tuning), which is what you will run when you retrain.

For deep networks a full nested CV is often too expensive. The usual compromise is a single fixed train/validation/test split, a stated tuning budget for every method compared, and a test set that is opened once.

Metrics: a recap

Plain version. Decide what an error costs before choosing a number. One scalar rarely tells the whole story.

Task	Useful measurements	Common trap
Classification	Per-class precision/recall, macro-F1, balanced accuracy, AUROC/AUPRC	Accuracy hides rare failures; PR curves depend on prevalence
Detection/segmentation	AP, IoU/Dice, boundary errors	Comparing different IoU thresholds or protocols
Retrieval	Recall@k, mAP under a defined relevance set	Gallery duplicates inflate matches
Restoration	PSNR/SSIM, perceptual metrics, consistency with the observation	Plausible texture may not be true detail
Generation	FID/KID, coverage, conditional alignment, human study	Feature realism does not establish prompt fidelity
Video/geometry	Temporal consistency, task accuracy, geometric error	Good individual frames can hide inconsistent structure

The definitions and pitfalls of classification metrics (confusion matrix, top-k, ROC/PR curves) are worked through in DLCV 02, and those of detection (IoU, AP, COCO mAP) in DLCV 03. For

generative models, FID compares Gaussian fits (means and covariances) of feature distributions; KID instead uses a kernel maximum-mean-discrepancy estimate that is unbiased, which makes it better behaved with small sample counts [7]. Neither is a universal measure of human preference. Always state the feature extractor, reference set, sample count, resolution and sampling settings.

Two classical facts are worth remembering. With 1% positives, an always-negative classifier gets 99% accuracy and zero recall, so accuracy alone is meaningless for rare events. And precision depends on prevalence: the same detector has lower precision in a population where the event is rarer, while its recall and false-positive rate do not change.

Statistical significance and confidence intervals

Plain version. A test score is computed from a finite sample, so it would come out a little different on another sample. Report a range, not just a point, and compare two models on the *same* examples.

Standard error of an accuracy

If the n test examples are independent and the model's true accuracy is p, the number of correct predictions is binomial, and the observed accuracy $\hat{p}$ has standard error

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \qquad 95\%\ \text{interval} \approx \hat{p} \pm 1.96\,\mathrm{SE}.$$

Here $\hat{p}$ is the fraction correct and 1.96 is the 97.5th percentile of the standard normal distribution. For $\hat{p} = 0.972$ and $n = 540$, $\mathrm{SE} \approx 0.0071$, so the interval is about ± 1.4 percentage points. This normal approximation is poor when $\hat{p}$ is very close to 0 or 1 or n is small; Wilson or exact binomial intervals are better there. If test examples are grouped (several images per patient), the effective sample size is closer to the number of groups, and the interval must be computed by resampling groups.

The bootstrap

The **bootstrap** estimates the sampling distribution of any statistic by resampling the observed data with replacement [9]. Draw B resamples of size n, recompute the statistic on each, and take the 2.5th and 97.5th percentiles as an approximate 95% interval (the *percentile* interval). It works for metrics with no simple formula, such as macro-F1, AUROC or mAP.

To compare two models, use a **paired** bootstrap: resample example indices once, and evaluate both models on the same resample. Because the models usually fail on many of the same hard examples, pairing removes that shared variance and gives a much tighter interval for the difference than two separate intervals would.

McNemar's test

For two classifiers evaluated on the same test set, only the examples where they disagree carry information about which is better. Let n_{01} count examples model A gets right and B gets wrong, and n_{10} the reverse. Under the null hypothesis that both models have the same error rate, each discordant example is equally likely to fall either way, so $n_{10} \sim \text{Binomial}(n_{01} + n_{10}, \frac{1}{2})$. The exact McNemar test computes a two-sided binomial p-value from that. Dietterich compared five tests for comparing learning algorithms and found that, for algorithms that can be run only once (typical for expensive models), McNemar's test was the only one with acceptable type I error; when ten runs are affordable he recommended the 5×2 cross-validated t test [8].

Code: paired bootstrap and McNemar

```
import numpy as np
from scipy.stats import binomtest
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC

X, y = load_digits(return_X_y=True)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=0)
a = make_pipeline(StandardScaler(), LogisticRegression(max_iter=2000)).fit(Xtr, ytr)
b = make_pipeline(StandardScaler(), SVC(C=10, gamma="scale")).fit(Xtr, ytr)
ca, cb = a.predict(Xte) == yte, b.predict(Xte) == yte    # per-example correctness

# 1) Percentile bootstrap: resample test examples with replacement.
rng = np.random.default_rng(0)
idx = rng.integers(0, len(yte), size=(10_000, len(yte)))
diff = cb[idx].mean(axis=1) - ca[idx].mean(axis=1)       # paired: same resample
lo, hi = np.percentile(diff, [2.5, 97.5])
print(f"accuracy A={ca.mean():.4f}  B={cb.mean():.4f}  n={len(yte)}")
print(f"B - A = {cb.mean() - ca.mean():+.4f}, 95% bootstrap CI [{lo:+.4f}, {hi:+.4f}]")

# 2) Exact McNemar test: only the discordant examples carry information.
n01 = int(np.sum(ca & ~cb))      # A right, B wrong
n10 = int(np.sum(~ca & cb))      # A wrong, B right
p = binomtest(n10, n01 + n10, 0.5).pvalue
print(f"discordant pairs: A-only={n01}, B-only={n10}, exact McNemar p={p:.4f}")
```

Output:

```
accuracy A=0.9722  B=0.9852  n=540
B - A = +0.0130, 95% bootstrap CI [-0.0019, +0.0278]
discordant pairs: A-only=5, B-only=12, exact McNemar p=0.1435
```

The SVM is 1.3 points better, but the interval includes zero and McNemar's p-value is 0.14. Only 17 of 540 images separate the two models, and 12 versus 5 is not unusual under a fair coin. With this test set, the honest conclusion is "B is probably at least as good; the evidence that it is better is weak."

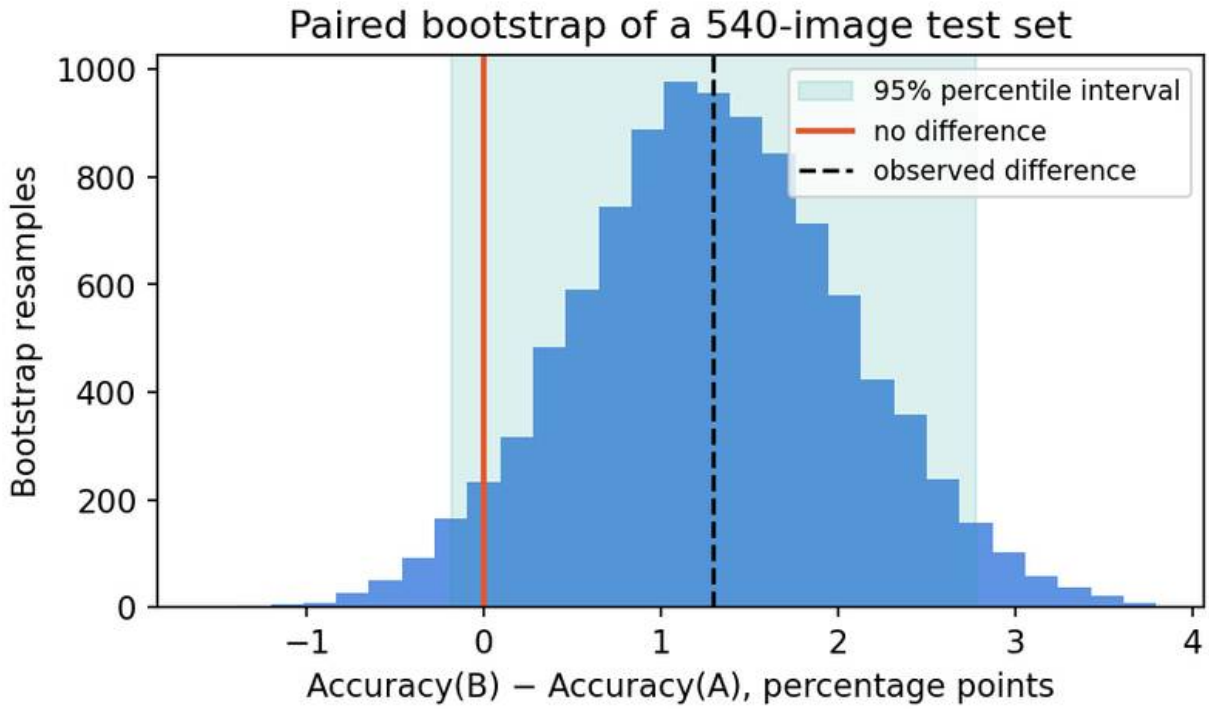


Figure 3 — The paired bootstrap distribution of Accuracy(B) − Accuracy(A) from the code above. The 95% interval crosses zero.

Other sources of variance

The test sample is not the only source of randomness. For a neural network, the random seed changes initialization, data order and augmentation; the data split itself is random; and some GPU kernels are nondeterministic. Bouthillier et al. modeled the whole benchmarking process and found that variance from data sampling, parameter initialization and hyperparameter choice markedly affects conclusions; they recommend randomizing as many of these sources as possible across repeated runs [10]. Practical rules:

- Run each configuration with several seeds (3–5 is common) and report mean and spread.
- Compare methods with paired tests on the same examples and splits.
- If you compare many models or many datasets, some will "win" by chance; adjust for multiple comparisons or confirm on a fresh set.
- Report effect sizes with intervals. A statistically significant difference can be practically irrelevant, and a large practical difference can be statistically unproven on a small set.

Calibration: knowing when you might be wrong

Plain version. A model's confidence should match how often it is right. Among all predictions made with 80% confidence, about 80% should be correct. Accuracy and calibration are separate properties.

Measuring probability quality

Let p_{ik} be the predicted probability of class k for example i, and $y_{ik} \in \{0, 1\}$ the one-hot label. Two **proper scoring rules** (scores minimized in expectation by the true probabilities) are:

$$\mathrm{NLL} = -\frac{1}{n}\sum_{i=1}^{n} \log p_{i,y_i}, \qquad \mathrm{Brier} = \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K} (p_{ik} - y_{ik})^2,$$

where y_i is the true class of example i and K is the number of classes. Some libraries divide the Brier score by K; say which convention you use.

The **expected calibration error** (ECE) groups predictions into M bins $B_1, \dots, B_M$ by their top-class confidence and compares average confidence with accuracy in each bin:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|,$$

where $\mathrm{acc}(B_m)$ is the fraction correct in bin m and $\mathrm{conf}(B_m)$ the mean confidence there. A **reliability diagram** plots acc against conf per bin; a perfectly calibrated model lies on the diagonal. ECE is not a proper scoring rule and depends on the bin count and the bin scheme, so report it alongside NLL or Brier and a reliability diagram.

Temperature scaling

Guo et al. observed that modern deep networks, which are much larger than those of earlier decades, are often overconfident, and that a single scalar fix works surprisingly well [11]. **Temperature scaling** divides the logits ℓ by $T > 0$ before the softmax:

$$\hat{p}_k = \frac{\exp(\ell_k / T)}{\sum_j \exp(\ell_j / T)}, \qquad T^\star = \arg\min_{T>0} \mathrm{NLL}_{\mathrm{cal}}(T).$$

$T > 1$ softens overconfident predictions; $T < 1$ sharpens underconfident ones. Because dividing all logits by the same positive number never changes which is largest, the predicted class (and so accuracy) is unchanged. T must be fitted on a held-out calibration set, never on the test set. Later work by Minderer et al. found that the most recent image classifiers, notably non-convolutional

ones such as Vision Transformers, are among the best calibrated, and that older trends such as calibration decaying under distribution shift are less pronounced for them [12].

Code: temperature scaling and what shift does to it

```
import numpy as np
from scipy.optimize import minimize_scalar
from scipy.special import log_softmax, softmax
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier

X, y = load_digits(return_X_y=True)
X = X / 16.0
Xa, Xte, ya, yte = train_test_split(X, y, test_size=0.4, stratify=y, random_state=0)
Xtr, Xcal, ytr, ycal = train_test_split(Xa, ya, test_size=0.3, stratify=ya, random_state=0)
net = MLPClassifier(hidden_layer_sizes=(128,), alpha=1e-6, max_iter=300,
                    random_state=0).fit(Xtr, ytr)

def logits(X):                        # log-probabilities = logits up to a constant
    return np.log(np.clip(net.predict_proba(X), 1e-12, 1.0))

def ece(prob, y, bins=10):
    conf, pred = prob.max(1), prob.argmax(1)
    edges = np.linspace(0, 1, bins + 1)
    total = 0.0
    for lo, hi in zip(edges[:-1], edges[1:]):
        m = (conf > lo) & (conf <= hi)
        if m.any():
            total += m.mean() * abs((pred[m] == y[m]).mean() - conf[m].mean())
    return total

def nll(T, z, y):
    return -log_softmax(z / T, axis=1)[np.arange(len(y)), y].mean()

T = minimize_scalar(nll, bounds=(0.05, 20), method="bounded",
                    args=(logits(Xcal), ycal)).x
print(f"fitted temperature T = {T:.2f}")
noise = np.random.default_rng(0).normal(size=Xte.shape)
for sigma in [0.0, 0.3, 0.6]:         # Gaussian pixel noise = a simple covariate shift
    Xs = np.clip(Xte + sigma * noise, 0, 1)
    z = logits(Xs)
    p_raw, p_ts = softmax(z, axis=1), softmax(z / T, axis=1)
    print(f"noise {sigma:.1f}: acc={(p_raw.argmax(1) == yte).mean():.3f}  "
          f"ECE raw={ece(p_raw, yte):.3f}  ECE scaled={ece(p_ts, yte):.3f}  "
          f"NLL raw={nll(1, z, yte):.3f}  NLL scaled={nll(T, z, yte):.3f}")
```

Output (scikit-learn may print a convergence warning, which is harmless here):

```
fitted temperature T = 1.19
noise 0.0: acc=0.972  ECE raw=0.007  ECE scaled=0.015  NLL raw=0.108  NLL scaled=0.110
```

```
noise 0.3: acc=0.794  ECE raw=0.088  ECE scaled=0.063  NLL raw=0.693  NLL scaled=0.639
noise 0.6: acc=0.435  ECE raw=0.380  ECE scaled=0.347  NLL raw=2.998  NLL scaled=2.598
```

Three observations. On clean data this small network is already well calibrated, and fitting T on 324 calibration images slightly *worsens* ECE: a fix fitted on a small set can add noise. Under pixel noise the model becomes badly overconfident: at $\sigma = 0.6$ it is right 43.5% of the time while its confidence suggests far more. Temperature scaling fitted on clean data helps only a little under shift. This matches the large study by Ovadia et al., who found that the calibration of many uncertainty methods deteriorates as shift increases [16].

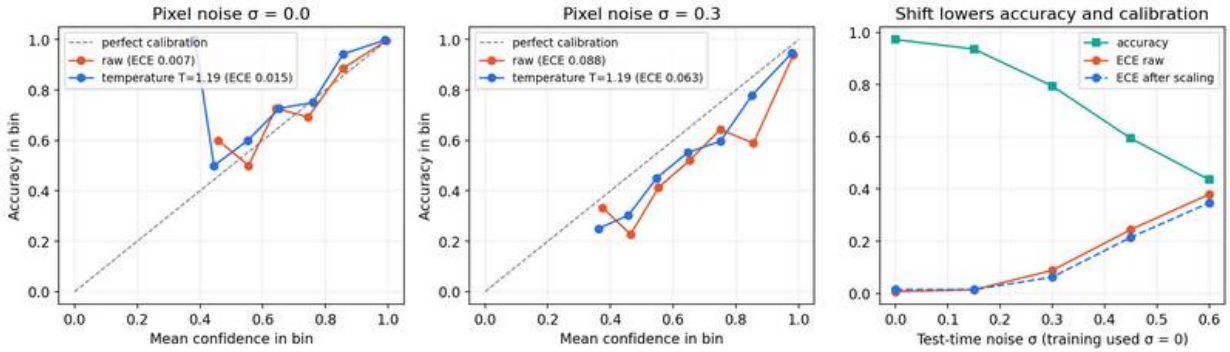


Figure 4 — Reliability diagrams and a shift sweep for the network above (same noise pattern, scaled by σ). Points below the diagonal mean overconfidence.

Robustness and distribution shift

Plain version. The world after deployment is rarely identical to the training set. Cameras change, lighting changes, populations change. A robust model degrades gracefully; a brittle one fails silently.

Three kinds of shift

Write the joint distribution as $p(x, y) = p(y \mid x)\,p(x) = p(x \mid y)\,p(y)$, with x the input and y the label. Shifts are named by what changes between training (p_{tr}) and deployment (p_{te}):

- **Covariate shift:** $p_{\text{te}}(x) \neq p_{\text{tr}}(x)$ while $p(y \mid x)$ is unchanged. New camera, new lighting, new scanner.
- **Label (prior) shift:** $p_{\text{te}}(y) \neq p_{\text{tr}}(y)$ while $p(x \mid y)$ is unchanged. A disease becomes more common during an outbreak.
- **Concept shift:** $p(y \mid x)$ itself changes. The definition of "spam" evolves.

Under pure covariate shift, the expected test loss can be rewritten as a weighted training loss, $\mathbb{E}_{\text{te}}[\ell] = \mathbb{E}_{\text{tr}}[w(x)\,\ell]$ with importance weight $w(x) = p_{\text{te}}(x)/p_{\text{tr}}(x)$. These weights are hard to estimate for images, but the identity explains why shift matters most where test inputs fall in regions with little training data. Under label shift, Lipton et al. showed that a fixed black-box classifier's confusion matrix on held-out training data, combined with its prediction frequencies on unlabeled test data, is enough to estimate the new class priors and re-weight the model [17].

Benchmarks that probe shift

- **ImageNet-C** applies 15 corruption types (noise, blur, weather, digital) at several severities to ImageNet validation images, and **ImageNet-P** adds perturbation sequences; together they measure robustness to common corruptions rather than adversarial attacks [13]. Our pixel-noise sweep is a miniature version.
- **ImageNetV2** re-created the ImageNet test set by following the original collection process as closely as possible. Accuracy dropped by 11–14% across many models, yet gains on the original test set translated into even larger gains on the new one, so the drop reflects slightly harder images rather than over-fitting to the reused test set [14]. Even a careful re-collection is a shift.
- **WILDS** gathers ten datasets with naturally occurring shifts, including tumor tissue from different hospitals, camera traps at new locations and satellite images from new countries, and reports a large gap between in-distribution and out-of-distribution performance that existing mitigation methods did not close [15].

The practical lesson: always add at least one *shifted* evaluation set built from the change you expect in deployment (new site, new device, later time), and report in-distribution and shifted results side by side.

Efficiency: FLOPs, memory and latency

Plain version. "Fast" and "small" each have several meanings. A model with fewer operations is not automatically quicker, and a model with fewer parameters does not automatically use less memory. Measure on the target device.

Counting operations

One **multiply–accumulate** (MAC) is a multiplication plus an addition, so 1 MAC $\approx$ 2 floating-point operations (FLOPs). For a fully connected layer with d_{in} inputs and d_{out} outputs, and a convolution with C_{in} input channels, C_{out} output channels, a $k \times k$ kernel and an $H' \times W'$ output:

$$\text{FLOPs}_{\text{linear}} \approx 2\, d_{\text{in}} d_{\text{out}}, \qquad \text{FLOPs}_{\text{conv}} \approx 2\, C_{\text{in}} k^2 C_{\text{out}} H' W'$$

per example (bias and activation costs are small). Libraries are inconsistent about whether they report MACs or FLOPs, so state which.

Why FLOPs do not equal speed

Dehghani et al. showed that common cost indicators (parameter count, FLOPs, throughput, latency) can disagree with each other and argued that reporting only one of them gives a misleading picture of efficiency [18]. The main reasons:

- **Memory traffic.** A layer whose weights must be read from memory for very little arithmetic is limited by memory bandwidth, not compute. A linear layer applied to a single example reads $4d_{\text{in}}d_{\text{out}}$ bytes of fp32 weights to perform $2d_{\text{in}}d_{\text{out}}$ FLOPs: 0.5 FLOPs per byte, far too little to keep a modern accelerator busy. Batching raises this ratio (called *arithmetic intensity*).
- **Parallelism.** Depthwise convolutions and many small layers have few FLOPs but use the hardware poorly.
- **Kernel support.** An operation without an optimized kernel on the target runtime may be slow regardless of its FLOP count.

Memory accounting

At inference, memory holds weights (parameters × bytes per value), activations of the layers currently alive, workspace for kernels, and for autoregressive Transformers a **KV cache** that grows with context length and batch size. During training add gradients and optimizer states: Adam keeps two extra values per parameter, so an fp32 model with Adam needs roughly 4× its weight memory before counting activations.

A latency protocol

Wall-clock latency depends on hardware, runtime, batch size, input resolution, numeric precision and thread settings, and the first calls are slower because of one-off initialization. A sound protocol:

1. Fix and report device, runtime version, precision, batch size, input size and thread count.
2. Run warm-up iterations and discard them.
3. Time many runs and report percentiles (p50 and p90 or p99), not just the mean: users experience the tail.
4. On GPUs, synchronize the device before stopping the timer, because kernels are launched asynchronously.
5. Measure end to end (pre-processing, transfers, decoding) as well as the network alone.

A **Pareto frontier** contains the candidates for which no other candidate is at least as good on every axis (accuracy, latency, memory) and strictly better on one. Presenting the frontier makes trade-offs visible without inventing a single "best model".

Deployment: export, compression and runtimes

Plain version. To ship a model, you freeze it into a portable file, often make it smaller or cheaper, and run it with an inference engine designed for the target hardware. Each step can change the outputs, so each step needs re-evaluation.

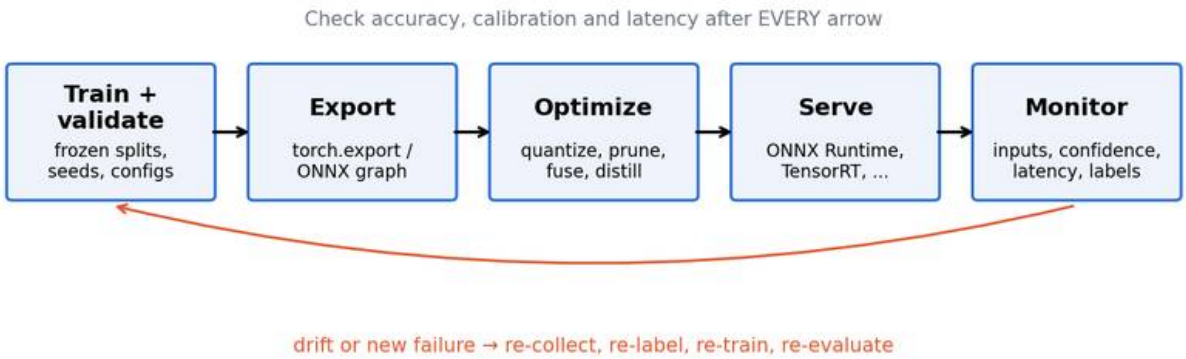


Figure 5 — A deployment loop. Accuracy, calibration and latency should be re-checked after every step, because export, compression and runtimes all change numerics.

Export and inference runtimes

Training code is written in a flexible, Python-driven framework. Deployment prefers a static description of the computation:

- `torch.export` traces a PyTorch module ahead of time into an `ExportedProgram`, a graph of tensor operations that can be serialized and executed with different inputs without the original Python class [19].
- **ONNX** (Open Neural Network Exchange) is an open standard file format for ML models, so a model trained in one framework can be run by tools from others [20].
- **ONNX Runtime** is a cross-platform inference engine for ONNX models. It partitions the graph into subgraphs and hands each to a hardware-specific *execution provider* (for example CUDA, TensorRT, OpenVINO or CoreML) [21].
- **NVIDIA TensorRT** is an SDK that compiles models (from frameworks or ONNX) into optimized engines for NVIDIA GPUs, using layer fusion and reduced precisions such as FP16, BF16, FP8 and INT8 [22].

These runtimes gain speed by fusing operations, choosing fast kernels for the exact shapes and lowering precision. The same model can therefore give slightly different outputs on different runtimes. Compare exported and original outputs on a validation batch and re-measure accuracy after conversion.

Quantization

Quantization stores and computes with fewer bits. In the common **affine (asymmetric) uniform** scheme, a real value x is mapped to an integer q and back:

$$q = \mathrm{clip}\Big(\mathrm{round}\big(\tfrac{x}{s}\big) + z,\ 0,\ 2^b - 1\Big), \qquad \hat{x} = s\,(q - z),$$

where b is the bit width, $s > 0$ the **scale** (the step between representable values), z the integer **zero point** chosen so that the real value 0 is represented exactly, and $\hat{x}$ the dequantized approximation. For a range $[x_{\min}, x_{\max}]$ that includes 0, $s = (x_{\max} - x_{\min})/(2^b - 1)$ and $z = \mathrm{round}(-x_{\min}/s)$. The *symmetric* variant fixes $z = 0$ and uses a signed range. The integer-arithmetic formulation of Jacob et al. showed how to run whole networks with 8-bit integer matrix multiplication and how to train for it [23].

Error comes from two sources: **rounding** inside the range (at most $s/2$ per value) and **clipping** outside it. Both depend on the **granularity** of the scale: one scale per tensor, per output channel, or per small group of weights. Nagel et al. give a practical guide covering these choices, *post-training quantization* (PTQ: calibrate ranges on a small representative dataset after training) and *quantization-aware training* (QAT: simulate rounding during training so the weights adapt) [24]. Gholami et al. survey the wider design space [25].

```
import numpy as np

def quantize(x, bits=8, axis=None):
    """Asymmetric (affine) uniform quantization, per tensor or per channel."""
    qmin, qmax = 0, 2**bits - 1
    lo = x.min(axis=axis, keepdims=True) if axis is not None else x.min()
    hi = x.max(axis=axis, keepdims=True) if axis is not None else x.max()
    lo, hi = np.minimum(lo, 0), np.maximum(hi, 0)      # keep 0 exactly representable
    s = (hi - lo) / (qmax - qmin)                       # scale
    z = np.round(qmin - lo / s)                         # zero point (integer)
    q = np.clip(np.round(x / s) + z, qmin, qmax)        # stored integers
    return s * (q - z)                                  # dequantized values

rng = np.random.default_rng(0)
# 64 output channels whose weight magnitudes differ by up to 100x
W = rng.normal(size=(64, 256)) * np.logspace(-2, 0, 64)[:, None]
for bits in [8, 4]:
    e_t = np.abs(quantize(W, bits) - W).mean()
    e_c = np.abs(quantize(W, bits, axis=1) - W).mean()
```

```
    print(f"{bits}-bit mean |error|: per-tensor {e_t:.5f}  per-channel {e_c:.5f}")
print("zero maps back to:", quantize(np.array([-1.0, 0.0, 3.0]))[1])
```

Output:

```
8-bit mean |error|: per-tensor 0.00632  per-channel 0.00124
4-bit mean |error|: per-tensor 0.06718  per-channel 0.02112
zero maps back to: 0.0
```

When channels have very different ranges, one shared scale wastes most of the grid on the largest channel and rounds small channels coarsely. Per-channel scales cut the error about fivefold at 8 bits here. In large language models, a few outlier activation channels cause the same problem at a larger scale, which is what methods such as GPTQ (layer-wise error-compensating weight rounding) and AWQ (protecting salient weight channels identified from activations) address for 3–4-bit weight-only quantization [42], [43].

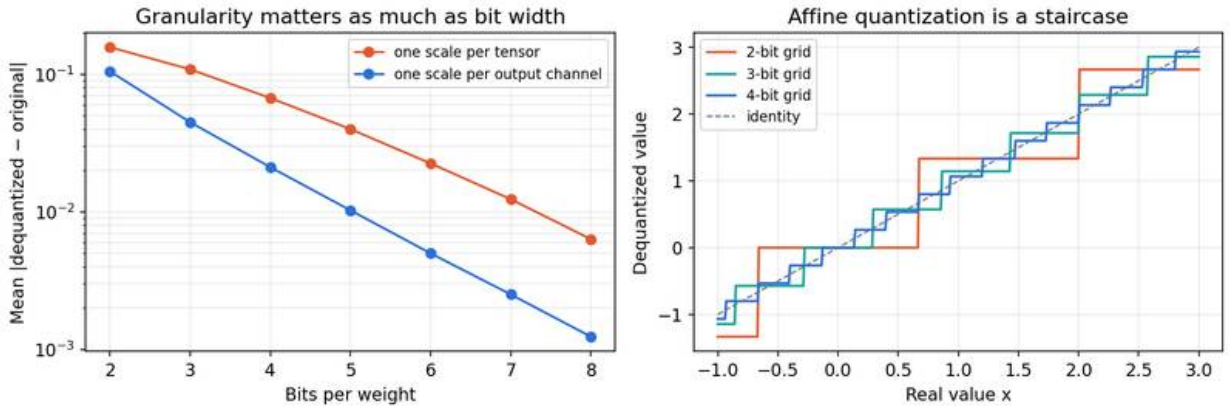


Figure 6 — Quantization error falls roughly by half for each extra bit, and the choice of scale granularity shifts the whole curve.

Pruning

Pruning removes weights, channels, attention heads or blocks that contribute little. *Unstructured* pruning zeros individual weights; it can remove a large fraction of parameters, as in Deep Compression, which combined pruning, weight sharing via quantization, and Huffman coding to shrink networks substantially [26]. But scattered zeros save time only if the runtime has sparse kernels for that pattern. *Structured* pruning removes whole channels or heads, producing smaller dense tensors that any runtime can exploit. Blalock et al. reviewed 81 pruning papers and found that inconsistent baselines and metrics made it hard to tell which methods actually work, and proposed a standardized benchmark [27]. Hoefler et al. survey sparsity in training and inference in depth [28].

Knowledge distillation

Distillation trains a small *student* to imitate a large *teacher*. Hinton et al. proposed matching the teacher's softened output distribution [29]. With temperature T, logits ℓ^t (teacher) and ℓ^s (student), softened probabilities $p^{(T)} = \mathrm{softmax}(\ell/T)$, and weight $\lambda \in [0, 1]$:

$$\mathcal{L} = (1 - \lambda)\, \mathcal{L}_{\mathrm{CE}}(y,\, p^{s,(1)}) + \lambda\, T^2\, D_{\mathrm{KL}}(p^{t,(T)} \,\|\, p^{s,(T)}).$$

The softened teacher distribution carries "dark knowledge" about which wrong classes are similar. The gradients of the soft term scale as $1/T^2$, so multiplying by T^2 keeps the two terms balanced when T changes [29]. Gou et al. organize later work by *what* is transferred (responses, intermediate features, relations between examples) and *how* (offline, online, self-distillation) [30]. A student inherits the teacher's mistakes and biases as well as its skill.

Code: weight-only int8, latency and export

This sketch stores the weights of a small MLP as int8 with one scale per output channel and measures size and latency on the CPU. It uses only core PyTorch.

```
import io, time
import numpy as np
import torch
from torch import nn

torch.manual_seed(0)
model = nn.Sequential(nn.Linear(512, 2048), nn.ReLU(),
                      nn.Linear(2048, 2048), nn.ReLU(),
                      nn.Linear(2048, 10)).eval()
x = torch.randn(32, 512)

class Int8WeightLinear(nn.Module):
    """Weight-only int8 storage, symmetric per output channel."""
    def __init__(self, lin):
        super().__init__()
        w = lin.weight.detach()
        scale = w.abs().amax(dim=1, keepdim=True) / 127
        self.register_buffer("q", torch.round(w / scale).to(torch.int8))
        self.register_buffer("scale", scale)
        self.register_buffer("bias", lin.bias.detach().clone())
    def forward(self, x):                       # dequantize, then fp32 matmul
        return nn.functional.linear(x, self.q.float() * self.scale, self.bias)

int8 = nn.Sequential(*[Int8WeightLinear(m) if isinstance(m, nn.Linear) else m
                       for m in model]).eval()

def size_mb(m):
    buf = io.BytesIO()
    torch.save(m.state_dict(), buf)
    return buf.getbuffer().nbytes / 1e6
```

```
def latency_ms(m, x, warmup=10, runs=100):
    with torch.inference_mode():
        for _ in range(warmup):                # first calls pay one-off costs
            m(x)
        times = []
        for _ in range(runs):
            t0 = time.perf_counter()
            m(x)
            times.append((time.perf_counter() - t0) * 1e3)
    return np.percentile(times, [50, 90]).round(2)

with torch.inference_mode():
    gap = (model(x) - int8(x)).abs().max().item()
    agree = (model(x).argmax(1) == int8(x).argmax(1)).float().mean().item()
print(f"fp32 file: {size_mb(model):.1f} MB  p50/p90 latency {latency_ms(model, x)} ms")
print(f"int8 file: {size_mb(int8):.1f} MB  p50/p90 latency {latency_ms(int8, x)} ms")
print(f"max |logit difference| = {gap:.4f}, top-1 agreement = {agree:.3f}")

# Ahead-of-time graph capture: a portable program, independent of this Python class.
exported = torch.export.export(model, (x,))
ops = [n.target for n in exported.graph.nodes if n.op == "call_function"]
print(type(exported).__name__, "->", [str(o).split(".")[1] for o in ops])
```

One CPU run (four threads) printed:

```
fp32 file: 21.1 MB  p50/p90 latency [2.74 4.49] ms
int8 file: 5.3 MB  p50/p90 latency [ 9.23 11.06] ms
max |logit difference| = 0.0018, top-1 agreement = 1.000
ExportedProgram -> ['linear', 'relu', 'linear', 'relu', 'linear']
```

Your latencies will differ; the pattern is the point. The file is four times smaller and the outputs barely change. Yet the int8 version is *slower*, because it converts the weights back to fp32 on every call and then runs the same fp32 matrix multiplication. Lower precision becomes faster only when the runtime has integer (or low-precision) kernels for that operation, or when the workload is limited by memory bandwidth so that reading fewer bytes dominates. This is exactly the gap between "compressed" and "accelerated" that runtimes such as TensorRT, ONNX Runtime and PyTorch's own `torchao` library [46] are built to close. The FLOP count of this MLP is $2 \times (512 \cdot 2048 + 2048 \cdot 2048 + 2048 \cdot 10) \approx 10.5$ MFLOPs per example, identical for both versions.

Monitoring after deployment

Plain version. A model can get worse without any code changing, because the data changes. Watch the inputs and outputs, collect fresh labels when possible, and decide in advance what triggers an investigation.

Sculley et al. describe how ML systems accumulate hidden "technical debt": entangled features, undeclared consumers of predictions, feedback loops where the model's outputs influence its future training data, and dependence on external data that changes without notice [33]. Paleyes et al. collected published deployment case studies and found problems at every stage of the workflow [39]. Useful signals, from cheapest to most informative:

- **System health:** latency percentiles, error rates, memory, throughput.
- **Input statistics:** image brightness, resolution, class of device, missing fields; compared with a reference window from training time.
- **Output statistics:** distribution of predicted classes and of confidence. These need no labels.
- **Delayed labels:** a small, regularly labelled sample of live traffic gives the only direct estimate of accuracy.

To compare a live window with the reference, a two-sample test is a simple starting point. The

Kolmogorov–Smirnov statistic for one-dimensional samples with empirical cumulative distribution functions F_{ref} and F_{live} is

$$D = \sup_t |F_{\text{ref}}(t) - F_{\text{live}}(t)|,$$

the largest vertical gap between the two curves. Rabanser et al. compared shift detectors and found that a two-sample test applied after reducing the inputs with a pre-trained classifier (for example to its softmax outputs) performed best across the shifts they studied [32].

```
import numpy as np
from scipy.stats import ks_2samp
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

X, y = load_digits(return_X_y=True)
X = X / 16.0
Xtr, Xlive, ytr, ylive = train_test_split(X, y, test_size=0.5, random_state=0)
Xtr, Xref, ytr = Xtr[:600], Xtr[600:], ytr[:600]   # reference window from training time
clf = LogisticRegression(max_iter=2000).fit(Xtr, ytr)
ref_conf = clf.predict_proba(Xref).max(1)     # label-free signal: top-class confidence

rng = np.random.default_rng(0)
for name, noise in [("week 1 (clean)", 0.0), ("week 2 (new camera)", 0.25)]:
    batch = rng.choice(len(Xlive), 200, replace=False)
    Xb = np.clip(Xlive[batch] + noise * rng.normal(size=Xlive[batch].shape), 0, 1)
    conf = clf.predict_proba(Xb).max(1)
    stat, p = ks_2samp(ref_conf, conf)
    acc = (clf.predict(Xb) == ylive[batch]).mean()   # usually NOT available live
    print(f"{name}: mean confidence {conf.mean():.3f}, KS={stat:.3f}, p={p:.2g}, "
          f"(hidden) accuracy {acc:.3f}")
```

Output:

```
week 1 (clean): mean confidence 0.846, KS=0.089, p=0.28, (hidden) accuracy 0.950
week 2 (new camera): mean confidence 0.757, KS=0.277, p=1.3e-08, (hidden) accuracy 0.885
```

The confidence distribution flags week 2 without a single label, and the hidden accuracy confirms that something real happened. A detector says *that* the data changed, not *whether* accuracy fell: harmless shifts also trigger it, and with large windows even tiny changes become "significant". Use it to trigger investigation and labelling, not automatic retraining.

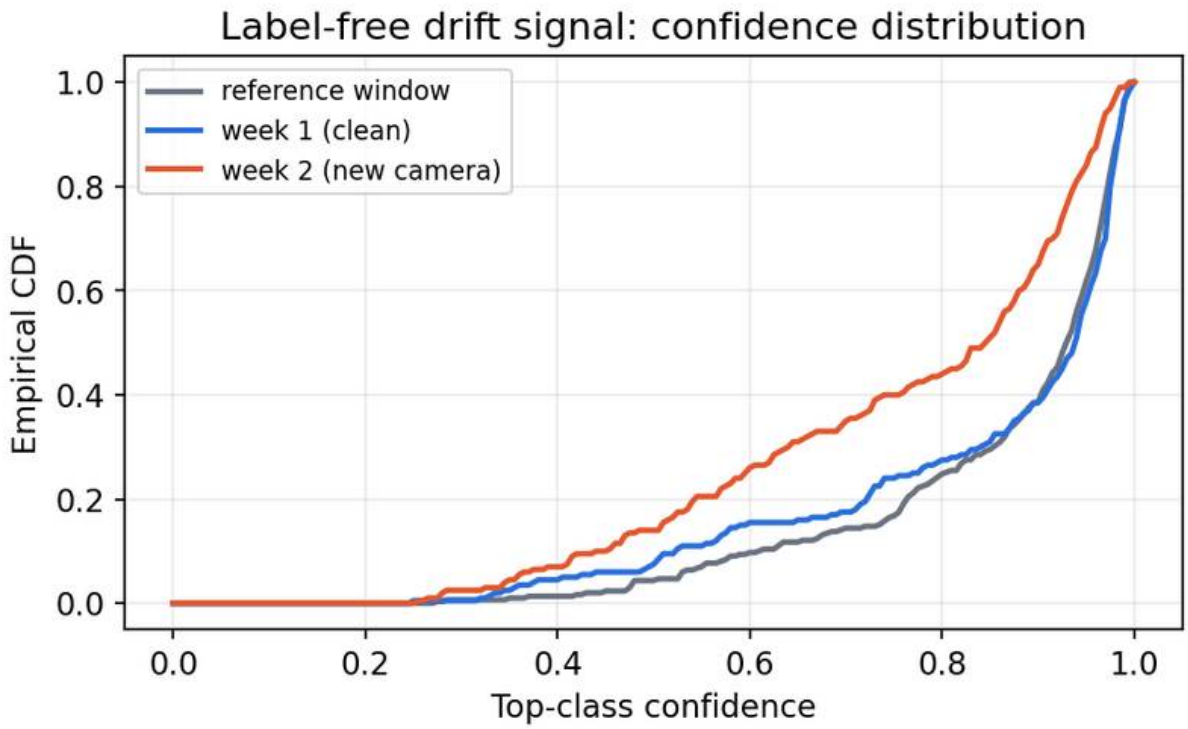


Figure 7 — The Kolmogorov–Smirnov statistic is the largest vertical gap between the reference curve and a live window's curve.

Reproducibility and documentation

Plain version. Someone else (including you in six months) should be able to rerun your experiment and get the same numbers, and a reader should be able to tell what the model is for and where it should not be used.

Making runs repeatable

Fix and record random seeds for every library, but know their limits: the PyTorch documentation states that completely reproducible results are not guaranteed across releases, commits or platforms, and recommends seeding, `torch.use_deterministic_algorithms(True)` and disabling cuDNN benchmarking when bit-exact repetition matters [34]. Beyond seeds, record:

- the dataset version, split IDs (the actual lists of example IDs) and deduplication rule;
- preprocessing and augmentation code, pretrained checkpoint and its source;
- the full configuration (optimizer, schedule, stopping rule), software environment and hardware;
- every validation decision made, and the single final test evaluation.

The NeurIPS 2019 reproducibility program, reported by Pineau et al., introduced a code-submission policy, a community reproducibility challenge and a checklist covering exactly these items [37].

Documenting models and data

Model cards are short documents that accompany a trained model and report its intended use, the evaluation data, performance broken down by relevant groups and conditions, and known limitations [35]. **Datasheets for datasets** ask dataset creators to document motivation, composition, collection process, preprocessing, recommended uses and maintenance [36]. Both make the evaluation decisions in this chapter visible to the people who will rely on them.

A capstone experiment

Choose one task with a modest, legally usable dataset. Define the deployment population and the split unit. Compare three candidates: a classical baseline, frozen pretrained features with a small head, and one fine-tuned network. For each, report the test metric with a bootstrap interval, a paired comparison against the best baseline, a reliability diagram, results on one shifted set, and latency percentiles on the target device. Write a one-page model card. Open the final test set once; if it reveals a failure and you revise the model, treat the old result as development evidence and obtain a fresh evaluation for the revised claim.

Modern view

The engineering of evaluation and deployment is now a research field in its own right. Several surveys and reviews map it.

Leakage and reproducibility. Kapoor and Narayanan's review of ML-based science documents leakage in 329 papers across 17 fields and proposes "model info sheets" that make authors argue

why their train/test separation is valid [4]. Combined with the variance study of Bouthillier et al. [10] and the NeurIPS reproducibility program [37], the consensus is that most over-optimistic results come from protocol errors, not from exotic statistics.

Uncertainty and calibration. Gawlikowski et al. survey uncertainty estimation in deep networks: they separate reducible model (epistemic) uncertainty from irreducible data (aleatoric) uncertainty, organize methods into deterministic networks, Bayesian networks, ensembles and test-time augmentation, and review calibration methods and their practical limits [38]. Ovadia et al. add the empirical warning that post-hoc calibration falls short under shift while methods that marginalize over models, such as ensembles, hold up better [16].

Deployment in practice. Paleyes et al. survey published case studies, map the challenges onto the stages of the deployment workflow and show that practitioners hit problems at every stage [39], extending the technical-debt view of Sculley et al. [33].

Compression. Cheng et al. give the classic overview of pruning, low-rank factorization, compact convolutions and distillation for CNNs [31]; Gholami et al. [25] and Nagel et al. [24] focus on quantization; Hoefler et al. on sparsity [28]; Gou et al. on distillation [30]. For large language models, Zhu et al. survey quantization, pruning, distillation and low-rank methods specific to models with billions of parameters [41]. A visible 2023–2026 trend is that weight-only quantization to 4 bits (GPTQ [42], AWQ [43]) and low-precision formats such as FP8 have moved from papers into production toolkits: TensorRT lists FP8, INT8, FP4 and INT4 among its supported precisions [22], and PyTorch's quantization tooling is consolidating in the `torchao` library [46].

Holistic evaluation. For foundation models, HELM evaluates many models on many scenarios and on several metrics at once (accuracy, calibration, robustness, fairness, bias, toxicity and efficiency) and argues that a single leaderboard number hides these trade-offs [40]. The same multi-metric habit applies to vision models.

Scaling. Scaling laws fit empirical relationships among model size, data, compute and loss. Hoffmann et al. found that, for a fixed compute budget, language models should scale parameters and training tokens roughly equally [44]; Zhai et al. studied how ViT accuracy scales with model size, data and compute [45]. Fitted constants from one domain should not be copied into another; the transferable lesson is to report the compute and data behind every result.

Open problems. Evaluation sets leaking into web-scale pretraining data (contamination) is hard to rule out; calibration under unknown shift remains unsolved; latency claims rarely transfer between hardware generations; and evaluation of generative and interactive systems (see DLCV 07) still lacks agreed protocols.

Key takeaways

- A baseline and a clean split decide whether a result means anything. Split by the unit that will be new in deployment, and fit all preprocessing inside the training fold.

- Cross-validation gives a more stable estimate on small data; use group or time-ordered folds when examples are correlated, and nested CV when tuning and reporting from the same data.
- Every test score needs an uncertainty. Compare models on the same examples with a paired bootstrap or McNemar's test.
- Accuracy, calibration and robustness are different properties. Temperature scaling fixes in-distribution overconfidence but not shift-induced overconfidence.
- FLOPs, parameters, memory and latency can disagree. Measure latency on the target hardware with warm-up and percentiles.
- Quantization, pruning and distillation save resources only when the runtime exploits them; re-evaluate accuracy after every export or compression step.
- Monitor inputs, outputs and a trickle of fresh labels after deployment, and document models and data so others can reproduce and judge them.

Exercises

1. **Leakage audit.** A colleague fits PCA on all 1,797 digits, then splits into train and test and reports test accuracy. Explain the leak, and why it exists even though PCA never sees labels. How would you fix the code?

   Hint

   The principal directions are computed using the test images, so the representation is tuned to the test distribution. Put `PCA` inside a `Pipeline` so it is fitted only on each training fold. The bias is often small for PCA, but for supervised preprocessing (feature selection using labels) it can be large.

2. **Rare positives.** A dataset has 1% positive images. What accuracy, positive recall and balanced accuracy does an always-negative classifier achieve? Which metric would you report to a doctor, and why?

   Hint

   Accuracy 99%, recall 0, balanced accuracy 0.5 (mean of recall 0 on positives and 1 on negatives). Report recall (sensitivity) and specificity or precision at a clinically chosen operating point, plus their intervals, because the cost of a missed positive is what matters.

3. **How big must the test set be?** You expect accuracy around 90% and want a 95% confidence interval of about ± 1 percentage point. Using the normal approximation, roughly how many independent test examples do you need? What changes if the images come from 50 patients?

   Hint

   Solve $1.96\sqrt{0.9 \cdot 0.1/n} = 0.01$, giving $n \approx 3{,}460$. With 50 patients the examples are not independent; the effective sample size is closer to 50 than to the number of images, so the interval will be much wider. Use a bootstrap that resamples patients.

4. **Temperature by hand.** A three-class model outputs logits $(4, 1, 0)$. Compute the top-class probability with $T = 1$ and $T = 2$. Does the predicted class change? When would a fitted T be less than 1?

Hint

$T = 1$: $e^4/(e^4 + e^1 + e^0) \approx 0.936$. $T = 2$: $e^2/(e^2 + e^{0.5} + 1) \approx 0.736$. The argmax is unchanged. $T < 1$ is fitted when the model is underconfident, for example after heavy regularization or label smoothing.

5. **Compression claim.** A pruned model has half the parameters of the original but identical latency on your GPU. Give three possible explanations and one experiment to tell them apart.

Hint

Unstructured zeros are still multiplied by dense kernels; the remaining cost is dominated by activations, memory traffic or unchanged layers; or the batch is so small that kernel-launch overhead dominates. Profile per-layer time, and compare with a structured-pruned model whose tensors are actually smaller.

6. **Design a monitor.** Your defect detector runs on a factory line; labels arrive two weeks late from manual inspection. Design a monitoring plan: what you log, which statistics you compare with which reference, what threshold triggers an alert, and what happens after an alert.

Hint

Log latency, image statistics (brightness, blur), predicted-defect rate and confidence histograms daily; compare with a reference window from validation using KS or a population-stability measure; alert on sustained change rather than single days. After an alert, label a sample immediately, compute accuracy with an interval, and only then decide whether to recalibrate, fine-tune or roll back.

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Part II

Deep Learning for Computer Vision

DLCV 01 • Deep Learning for Image Processing and Computer Vision

How deep learning reorganizes image processing, image analysis and computer vision: from the classical pattern-recognition pipeline to CNNs, Vision Transformers, self-supervised backbones and promptable foundation models.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-01-deep-learning-for-image-processing/

What you'll learn

- How the classical three levels of image work (image processing, image analysis, computer vision) map onto the two families that deep learning actually uses: **low-level** and **high-level** vision tasks.
- Why the textbook pattern-recognition pipeline (sensor → features → classifier → evaluation) was replaced by **end-to-end learning**, and how it quietly came back as "pretrained backbone + light head".
- How a convolutional layer is a bank of **learned spatial filters**, and how kernel size, stride, padding and pooling set the **receptive field** and the feature hierarchy.
- The design reasons behind the landmark CNNs (LeNet-5, AlexNet, VGG, GoogLeNet, ResNet) and the encoder–decoder (FCN, U-Net) used for dense prediction.
- How the **Vision Transformer** works, why it needs large-scale pretraining, and how Swin, ConvNeXt and self-supervised backbones (DINO, MAE, DINOv2/v3) answered it.
- A map of today's tasks and representative models: classification, detection, segmentation (the SAM family), restoration and enhancement, generation (diffusion, latent diffusion, DiT) and vision–language models (CLIP, LLaVA).

The big picture

This tutorial is the entry point of the deep-learning track. It does not go deep into any one task. Instead it gives you the map: what kinds of image problems exist, which network designs solve them, and why those designs look the way they do. The later tutorials then zoom in on image classification, object detection, image restoration, low-light enhancement and object tracking.

Explain like I'm five

Think of a photo lab with two kinds of workers. The **repair crew** takes a scratched, blurry or dark photo and hands back a cleaner photo: picture in, picture out. The **detective** looks at a photo and tells you *what* is in it: "a cat, sitting on a sofa, on the left." Old-style computer vision gave both workers a fixed toolbox designed by experts. Deep learning lets them build their own tools by looking at millions of examples. Today there is even a "super-worker" who learned from billions of pictures and can be told what to do with a few words.

Why it matters: almost every modern camera pipeline, medical-imaging product, driver-assistance system and photo app is built from the pieces in this tutorial. Knowing the map lets you pick a sensible starting point for a new problem, read papers critically, and understand which classical ideas from the Digital Image Processing series survive inside the networks (many do).

About this tutorial

This tutorial is expanded from the author's earlier post, written in Traditional Chinese, "深度學習的數位影像處理介紹" (An introduction to digital image processing with deep learning) [1]. It keeps the post's structure and viewpoints and adds the details, math and code.

Related chapters. DLCV 01A turns the ideas here into a practical training workflow: data pipelines, learning-rate schedules, regularization, debugging and fair experiments. DLCV 01B collects the building blocks (residual and attention blocks, normalization, upsampling, skips) and shows how classification and restoration networks compose them differently.

Three levels of image tasks

Plain version. "Image processing" in the broad sense covers three kinds of work. Some operations turn an image into a better image. Some pull out a particular part or property of the image. Some try to *understand* the image the way a person would.

Precise version. Following Gonzalez and Woods [2], we can sort image work into three levels by what goes in and what comes out.

Level	Input → output	Typical operations	Classical examples
Image processing (narrow sense; low level)	image → image	primitive, pixel- or neighborhood-wise steps	histogram equalization, filtering, Fourier transform (IP 03, IP 04)
Image analysis (mid level)	image → attributes or highlighted regions	segmentation, feature extraction, description, using prior knowledge	skin-color regions in a face image, vessel regions in an X-ray (IP 10, IP 12)
Computer vision (high level; understanding)	image → meaning	recognition, interpretation, decision	classification, "is there a pedestrian?" (IP 13)

The boundaries are soft. Real systems mix all three. A vessel-segmentation tool, for instance, first denoises (low level), then extracts a vessel map (mid level), and may then flag a stenosis (high level). The levels describe *kinds of computation*, not separate departments.

How deep learning redraws the map

In deep-learning papers you will almost never see the word "mid-level". The field splits tasks into just two families, according to the **shape of the output** [1]:

- **Low-level vision**: the output is an image of (roughly) the same size as the input, and the target is a pixel-accurate signal. Denoising, deblurring, super-resolution, dehazing, low-light enhancement, image-to-image translation.
- **High-level vision**: the output is a *semantic* description. A label, boxes with labels, or a label per pixel.

The old "image analysis" level is absorbed into one of the two. Segmentation produces a per-pixel map, so it *looks* like an image-to-image task. But its values are class labels that require recognition, so it is grouped with high-level vision. Edge or vessel enhancement that outputs a cleaner intensity image is grouped with low-level vision.

The reason for the split is practical. The two families differ in the loss (pixel regression versus classification), the architecture (full-resolution encoder–decoders versus downsampling backbones), the training data (pairs of degraded and clean images versus human labels) and the metrics (fidelity versus accuracy). Figure 1 shows the reorganization.

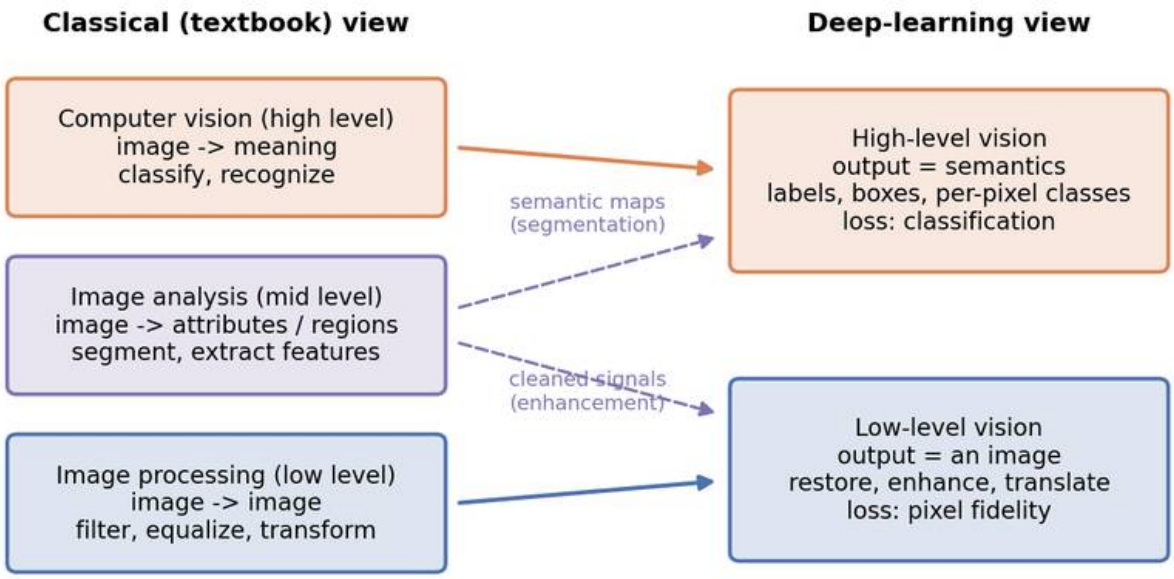


Figure 1 — The classical three levels (left) and the two families used in deep learning (right). Mid-level image analysis is split between them according to whether the output is a cleaned-up signal or a semantic map.

From the pattern-recognition pipeline to end-to-end learning

Plain version. Before deep learning, a recognition system was built like an assembly line: a camera, then a hand-designed step that measures useful properties, then a classifier that decides. Deep learning fuses the middle steps into one network that learns its own measurements.

Precise version. The classical pipeline [3] has four stages:

1. **Sensor.** Capture the pattern (camera, scanner, microscope).
2. **Feature generation and selection.** Compute a vector $\mathbf{x} = \phi(I) \in \mathbb{R}^d$ from image I, using a hand-designed map ϕ (moments, texture statistics, edge histograms, keypoint descriptors), then keep the most informative entries.
3. **Classifier design.** Learn a decision rule $\hat{y} = g(\mathbf{x}; \theta)$ with parameters θ, for example a Bayes classifier, a nearest-neighbor rule or a support-vector machine.
4. **System evaluation.** Measure the error rate on held-out data and go back to step 2 or 3 if it is too high.

Only θ is learned. The feature map ϕ is fixed by an engineer, so the whole system is only as good as ϕ. The chapter on feature extraction (IP 12) shows how much ingenuity went into those maps.

End-to-end learning makes the feature map learnable too. A deep network computes

$$\hat{y} = g(\phi(I; \mathbf{w}); \theta),$$

where $\phi(\cdot\,; \mathbf{w})$ is a stack of layers with weights $\mathbf{w}$, $g(\cdot\,; \theta)$ is a small head, and both $\mathbf{w}$ and θ are fitted jointly by minimizing a loss $\mathcal{L}(\hat{y}, y)$ over labeled pairs (I, y) with gradient descent and back-propagation. The input is raw pixels; nothing is hand-designed except the architecture and the loss.

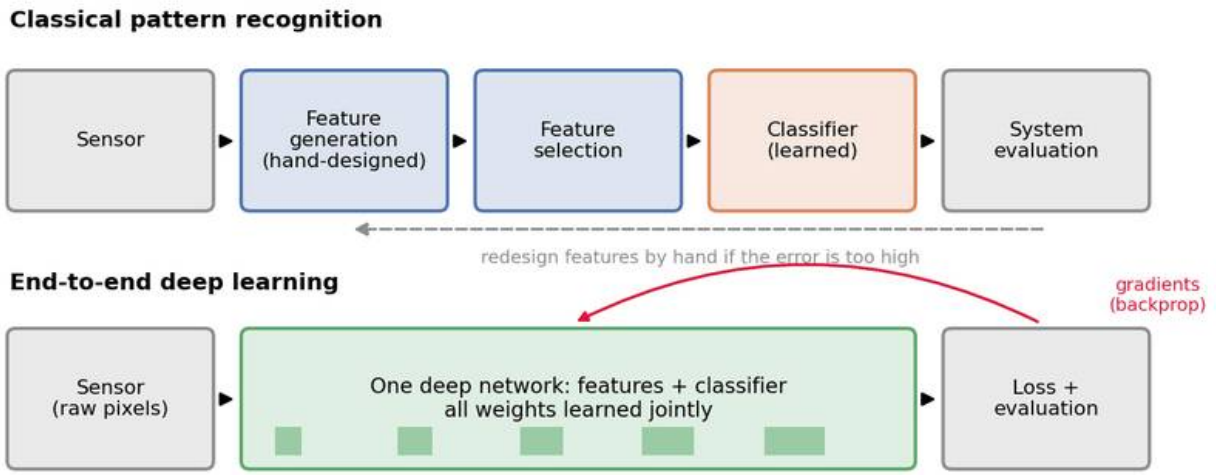


Figure 2 — Classical pattern recognition (top) versus end-to-end learning (bottom). In the deep network the functional split into 'features' and 'classifier' still exists, but there is no clean boundary between them.

The functional split still exists inside a network: early layers behave like feature extractors and the last layers like a classifier. But in practice you usually cannot point at a layer and say "here the features end" [1]. Two consequences follow.

- **Data replaces design.** End-to-end systems need many labeled examples; the hand-designed ϕ encoded prior knowledge that the network must now learn from data.
- **The split came back at a larger scale.** Since about 2021 the standard recipe is a **pretrained backbone** (trained once, often without labels, on a huge dataset) plus a **light task head** trained on your small dataset. That is "feature extractor + classifier" again, except the feature extractor is learned rather than designed. We return to this in the section on self-supervised backbones.

Convolutional neural networks

Plain version. A convolutional layer slides a small window over the image and computes a weighted sum at every position, exactly like the spatial filters of IP 03. The difference is that a CNN *learns* the weights of its filters, uses many filters at once, and stacks dozens of such layers with nonlinearities in between.

A convolutional layer is a bank of learned filters

Precise version. Let the input feature map be $X \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ (channels, height, width). A convolutional layer with C_{out} filters of size $k \times k$ computes

$$Y_o(x,y) = b_o + \sum_{c=1}^{C_{\text{in}}} \sum_{i=-r}^{r} \sum_{j=-r}^{r} W_{o,c}(i,j)\, X_c(x+i,\, y+j), \qquad o = 1, \dots, C_{\text{out}},$$

where $r = (k-1)/2$ for odd k, $W_{o,c}(i,j)$ is the weight of filter o on input channel c at offset (i, j), and b_o is a bias. Strictly this is a **correlation** (no kernel flip), as explained in the convolution tutorial; since the weights are learned, the flip makes no difference. An elementwise nonlinearity such as $\text{ReLU}(z) = \max(0, z)$ follows. Without it, a stack of convolutions would collapse into one big linear filter [1].

Two properties make this layer the right tool for images:

- **Local connectivity.** Each output depends only on a $k \times k$ neighborhood, because nearby pixels are strongly related and distant ones much less so.
- **Weight sharing.** The same filter is used at every position. This makes the layer **translation-equivariant** (shift the input, and the output shifts the same way) and slashes the number of parameters.

The saving is huge. A fully connected layer from a $224 \times 224 \times 3$ image to 64 units needs $224 \cdot 224 \cdot 3 \cdot 64 + 64 \approx 9.6$ million parameters. A 3×3 convolution from 3 to 64 channels needs $3 \cdot 3 \cdot 3 \cdot 64 + 64 = 1{,}792$, and it produces 64 *full-resolution* feature maps instead of 64 numbers.

The code below sets the two filters of a PyTorch convolution by hand to Sobel kernels, which turns the layer into an edge detector. Training would instead set these numbers by gradient descent.

```
import torch, torch.nn as nn, torch.nn.functional as F
from skimage import data, img_as_float

x = torch.tensor(img_as_float(data.camera()), dtype=torch.float32)[None, None]  # (1, 1,
↪  512, 512)
conv = nn.Conv2d(1, 2, kernel_size=3, padding=1, bias=False)
sobel_x = torch.tensor([[-1., 0., 1.], [-2., 0., 2.], [-1., 0., 1.]])
with torch.no_grad():
    conv.weight[0, 0] = sobel_x        # channel 0: responds to vertical edges
    conv.weight[1, 0] = sobel_x.T      # channel 1: responds to horizontal edges
    y = F.relu(conv(x))                # (1, 2, 512, 512): two feature maps
```

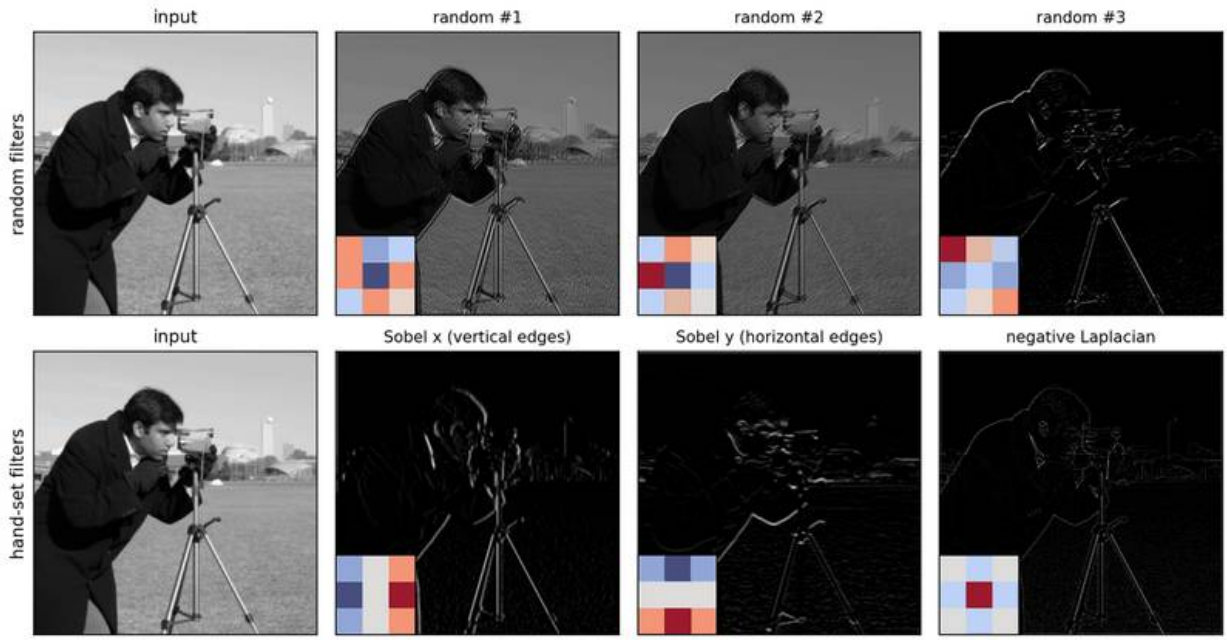


Figure 3 — Feature maps (after ReLU) of 3×3 convolutions on the camera image. Top: random initial weights, which give noisy mixtures of smoothing and edges. Bottom: hand-set Sobel and Laplacian weights. Training moves the random filters toward useful ones.

What do trained first-layer filters look like? We can train a tiny CNN on the 8×8 handwritten digits that ship with scikit-learn [36] in a few seconds on a CPU.

```
import torch, torch.nn as nn
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split

torch.manual_seed(0)
X, y = load_digits(return_X_y=True)                    # 1,797 images of 8x8 pixels
X = torch.tensor(X / 16.0, dtype=torch.float32).reshape(-1, 1, 8, 8)
Xtr, Xte, ytr, yte = train_test_split(X, torch.tensor(y), test_size=0.25, random_state=0)

net = nn.Sequential(
    nn.Conv2d(1, 8, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),     # 8x8 -> 4x4
    nn.Conv2d(8, 16, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),    # 4x4 -> 2x2
    nn.Flatten(), nn.Linear(16 * 2 * 2, 10),
)
opt = torch.optim.Adam(net.parameters(), lr=1e-2)
for epoch in range(30):
    for i in range(0, len(Xtr), 64):
        loss = nn.functional.cross_entropy(net(Xtr[i:i + 64]), ytr[i:i + 64])
        opt.zero_grad(); loss.backward(); opt.step()
print((net(Xte).argmax(1) == yte).float().mean())       # about 0.97
filters = net[0].weight.detach()[:, 0]                   # (8, 3, 3) learned kernels
```

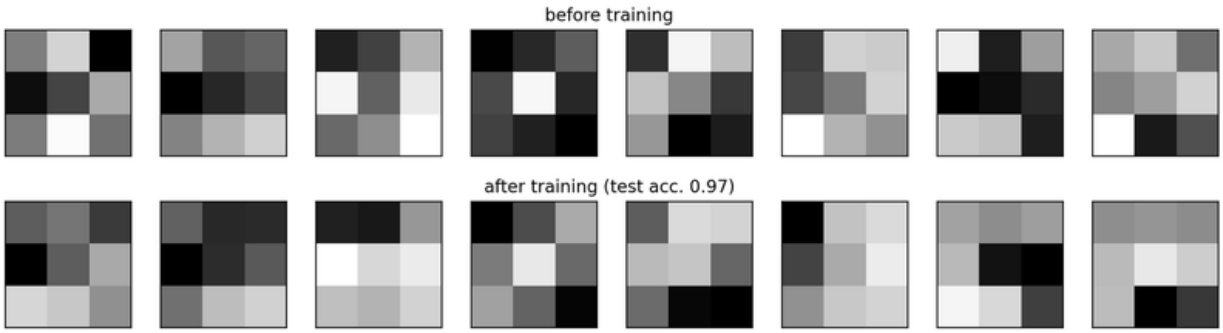


Figure 4 — The eight first-layer 3×3 filters of the tiny digit CNN, before (top) and after (bottom) training. Several drift toward oriented light-to-dark transitions, the derivative-like patterns a hand designer would choose; nobody told the network to do so. (Tiny 8×8 inputs and 3×3 kernels keep the effect modest; large networks trained on natural images show it much more clearly.)

Stride, padding and output size

Plain version. Padding adds a border so the window can sit on edge pixels. Stride makes the window jump more than one pixel at a time, which shrinks the output.

Precise version. For an input of width W, kernel size k, padding p (pixels added on each side) and stride s, the output width is

$$W_{\text{out}} = \left\lfloor \frac{W + 2p - k}{s} \right\rfloor + 1.$$

With $k = 3$, $p = 1$, $s = 1$ the size is preserved ("same" padding); with $s = 2$ it is halved. Zero padding is the default; reflect or replicate padding, the same boundary choices discussed in IP 03, are preferred in low-level networks because zero borders create dark artifacts at image edges.

Pooling

Pooling summarizes each small window by one number, usually the maximum (max pooling) or the mean (average pooling), typically with a 2×2 window and stride 2. It halves resolution, makes the representation tolerant to small shifts, and cheaply enlarges the receptive field. Many modern networks replace it with strided convolutions, which learn how to downsample. At the very end of a classification network, **global average pooling** turns a $C \times H \times W$ map into a C-vector for the classifier head.

Receptive field

Plain version. The receptive field of a neuron is the patch of the input image that can influence it. One 3×3 layer sees 3×3 pixels. Stack two, and each output sees 5×5. Downsampling makes it grow much faster.

Precise version. Number the layers $\ell = 1, \dots, L$ with kernel size k_ℓ and stride s_ℓ. Let j_ℓ be the **jump**, the distance in input pixels between two neighboring units of layer ℓ, and r_ℓ the receptive-field size, with $r_0 = 1$, $j_0 = 1$ for the input. Then

$$r_\ell = r_{\ell-1} + (k_\ell - 1)\, j_{\ell-1}, \qquad j_\ell = j_{\ell-1}\, s_\ell.$$

Each layer adds $k_\ell - 1$ units of its input, and each input unit is worth $j_{\ell-1}$ pixels. With stride 1 everywhere, L layers of 3×3 give $r_L = 2L + 1$: linear growth. Every stride-2 step doubles the jump, so later layers grow the field twice as fast, giving roughly exponential growth in depth.

```
def receptive_field(layers):
    r, j = 1, 1                       # receptive field and jump, measured in input pixels
    for k, s in layers:               # (kernel size, stride) of each layer
        r, j = r + (k - 1) * j, j * s
    return r

print(receptive_field([(3, 1)] * 3))                                                  # 7
print(receptive_field([(3, 1), (3, 1), (2, 2), (3, 1), (3, 1), (2, 2), (3, 1)])) # 24
```

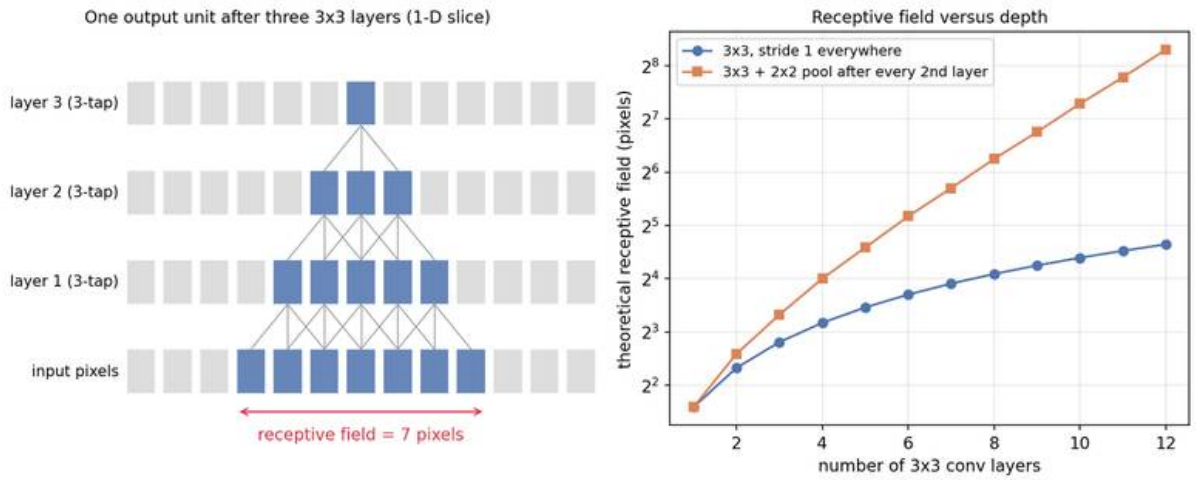


Figure 5 — Left: three stacked 3-tap layers give one output a 7-pixel receptive field. Right: theoretical receptive field versus depth, for stride 1 everywhere and for 2× downsampling after every second layer.

Interactive demonstration (online): open the online edition of this tutorial to use this demonstration.Open demonstration

Two refinements matter in practice. First, the formula gives the **theoretical** receptive field. Luo et al. [12] showed that the **effective** receptive field, measured by how much each input pixel actually affects the output gradient, is roughly Gaussian and occupies only a fraction of the theoretical one. Pixels near the center dominate. Second, the receptive field decides what a network *can* use: a denoiser whose receptive field is smaller than the noise correlation length cannot remove that noise, and a classifier whose last layer sees only part of an object cannot recognize it from context.

Feature hierarchies

Because the receptive field grows with depth while resolution shrinks, a CNN builds a **feature hierarchy**: early layers respond to edges and color blobs (Figure 4), middle layers to textures and parts, late layers to object-level patterns over large regions. A typical backbone is organized in **stages**, each ending with a 2× downsampling and doubling the channel count, so features go from "high-resolution, few channels, local" to "low-resolution, many channels, semantic". High-level tasks read the late stages. Dense prediction and low-level tasks need the early, high-resolution stages too, which is why encoder–decoders exist (below).

Landmark CNN architectures

Plain version. Between 2012 and 2016 a series of networks, each deeper than the last, won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) [6]. Each one introduced a design idea that is still in use.

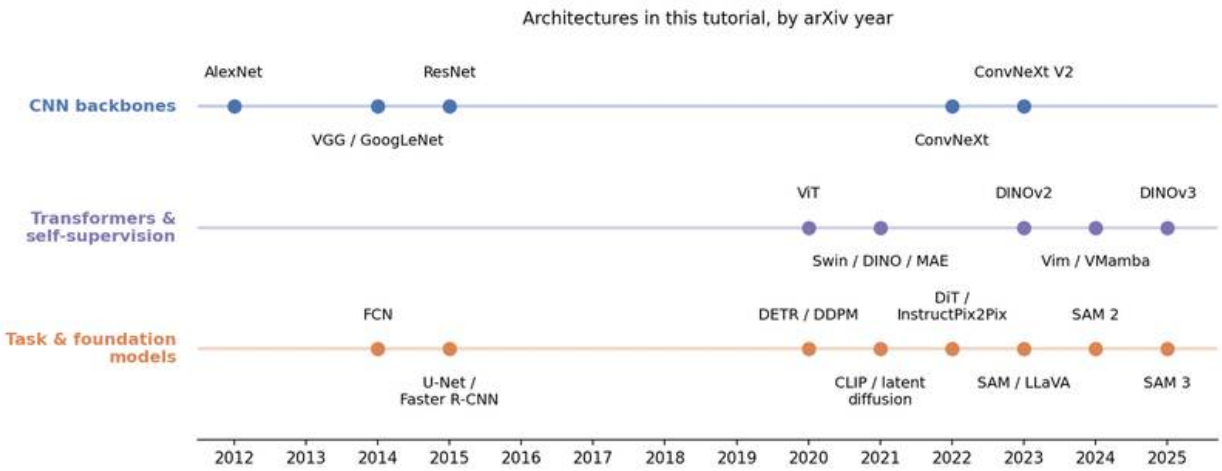


Figure 6 — A timeline of the architectures discussed in this tutorial, by arXiv year. LeNet-5 (1998) and the earlier convolutional networks of the late 1980s sit to the left of this axis.

LeNet-5 and where CNNs really began

The post's original heading said "CNN (1998~)". That year is when LeCun, Bottou, Bengio and Haffner published LeNet-5 [5], not when CNNs started: convolutional networks trained by back-propagation were already reading handwritten ZIP-code digits in 1989 [4].

LeNet-5 is a textbook pattern-recognition system in network form [1]. Layers C1 to C5 are the

feature extractor: convolutions (C1, C3, C5) alternating with subsampling layers (S2, S4) that turn a 32×32 digit image into a 120-dimensional feature vector. F6 and the output layer are the

classifier. Everything, features included, is trained by gradient descent on the classification loss. This was the end-to-end idea, already in place, but it needed 15 more years of data and compute to dominate.

AlexNet (2012)

AlexNet [7] has five convolutional layers and three fully connected layers (eight learned layers, about 60 million parameters). Its ingredients were not individually new, but the combination was: ReLU activations, which train much faster than saturating sigmoids; an efficient GPU implementation; heavy data augmentation; and dropout in the fully connected layers. It won ILSVRC 2012 by a wide margin, which is why 2012 is usually given as the start of the deep-learning era in vision.

VGG (2014)

Simonyan and Zisserman [8] asked a single question: what happens if you use only 3×3 convolutions and make the network deeper? Their best configurations had 16 and 19 weight layers. The reason 3×3 is enough is the receptive-field formula above: two stacked 3×3 layers see 5×5, three see 7×7. For C channels in and out, three 3×3 layers cost $3 \cdot 9C^2 = 27C^2$ weights versus $49C^2$ for one 7×7 layer, and they insert two extra nonlinearities. VGG's uniform design made it the default pretrained feature extractor for years.

GoogLeNet / Inception (2014)

GoogLeNet [9], 22 layers deep, won ILSVRC 2014. Its **Inception module** runs 1×1, 3×3 and 5×5 convolutions and a pooling branch in parallel and concatenates their outputs, so each block sees several scales at once. To keep this affordable, cheap 1×1 convolutions first reduce the number of channels. A 1×1 convolution mixes channels at each pixel without looking at neighbors; it is a per-pixel fully connected layer, and it remains a basic building block everywhere.

ResNet (2015) and why residual connections work

Deeper should be better, but in practice it was not: He et al. [10] observed that a plain 56-layer network had *higher training error* than a 20-layer one. That is not overfitting. It is an optimization failure: the deeper net could in principle copy the shallow one and set the extra layers to the identity, but gradient descent does not find that solution.

The fix is the **residual block** (Figure 7). Instead of asking a stack of layers to learn a mapping $H(\mathbf{x})$ directly, let it learn the residual $F(\mathbf{x}) = H(\mathbf{x}) - \mathbf{x}$ and add the input back:

$$\mathbf{y} = \sigma(\mathbf{x} + F(\mathbf{x}; W)),$$

where $\mathbf{x}$ is the block input, F is typically conv–BN–ReLU–conv–BN with weights W (BN is batch normalization), and σ is ReLU. When channel counts or resolution change, the shortcut uses a 1×1 convolution instead of the identity.

Optional: Batch normalization (in the Supplementary notes). What the BN layer computes, and why it behaves differently in training and at test time.

Why it works:

- **Identity is easy.** If extra depth is not useful, the block only has to drive F toward zero, which is far easier than making a stack of nonlinear layers imitate the identity. Initializing the last BN scale of F at zero makes each block start as the identity.
- **Gradients have a highway.** Ignoring the final nonlinearity, $\partial \mathbf{y}/\partial \mathbf{x} = I + \partial F/\partial \mathbf{x}$. The identity term carries the gradient from the loss to early layers without passing through every weight matrix, so it does not vanish with depth.
- **An ensemble view.** Unrolling L blocks gives a sum over many paths of different lengths, so a deep ResNet behaves partly like a collection of shallower networks.

```
class ResidualBlock(nn.Module):
    """Basic block: y = ReLU(x + F(x)), with F = conv-BN-ReLU-conv-BN."""
    def __init__(self, c):
        super().__init__()
        self.f = nn.Sequential(
            nn.Conv2d(c, c, 3, padding=1, bias=False), nn.BatchNorm2d(c), nn.ReLU(),
            nn.Conv2d(c, c, 3, padding=1, bias=False), nn.BatchNorm2d(c),
        )
    def forward(self, x):
        return torch.relu(x + self.f(x))

blk = ResidualBlock(16)
nn.init.zeros_(blk.f[4].weight)                  # zero the last BN scale, so F(x) = 0
x = torch.randn(2, 16, 32, 32)
print(torch.allclose(blk(x), torch.relu(x)))     # True: the block starts as the identity
 ↪  (then ReLU)
```

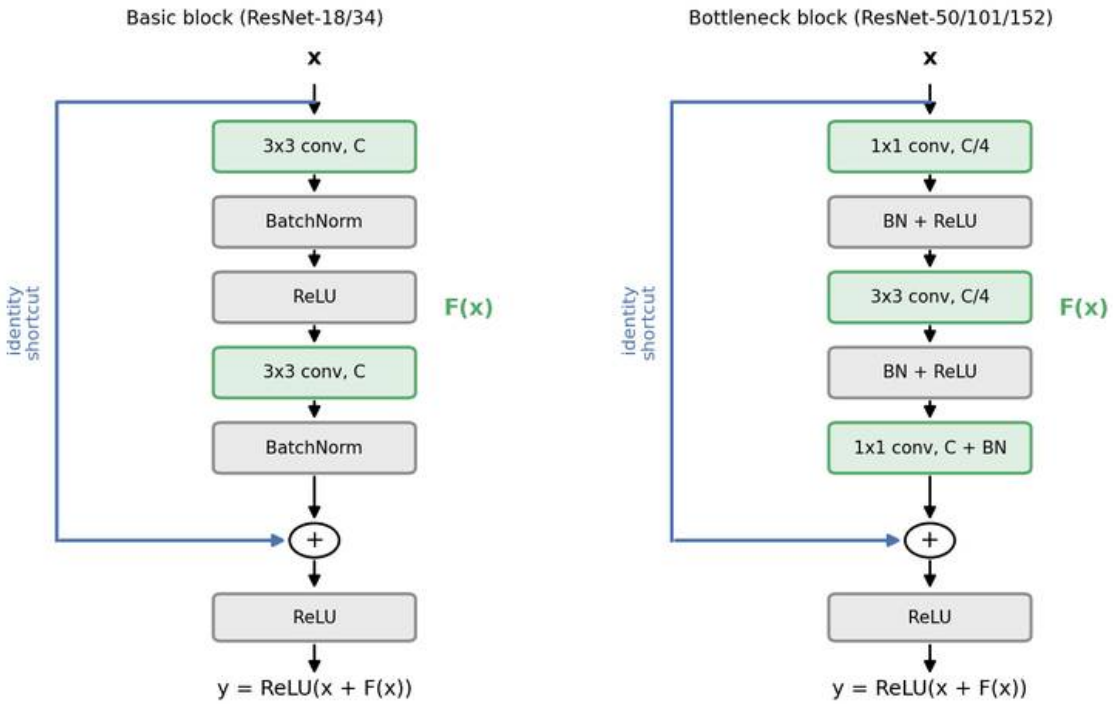


Figure 7 — Left: the basic residual block. Right: the bottleneck block used in ResNet-50/101/152, where 1×1 convolutions shrink and restore the channel count around a cheap 3×3 convolution.

On ImageNet the paper trained ResNets with **18, 34, 50, 101 and 152 layers**; the deeper three use the bottleneck block. The famous **1202-layer** network appears only in the paper's CIFAR-10 experiments [10], [11] (32×32 images), where it trained fine but overfit and tested worse than the 110-layer version. The ResNet family won ILSVRC 2015, and the residual connection is now in almost every deep network, Transformers included.

Encoder–decoders for dense prediction

Plain version. A classification backbone throws away resolution to gain meaning. For segmentation or restoration we need an answer at every pixel, so we add a second half that brings the resolution back.

Precise version. Long et al.'s **fully convolutional network** (**FCN**) [13] replaced the fully connected layers of a classifier with convolutions, so the network outputs a coarse class-score map for an input of any size, then upsampled it with learned transposed convolutions and fused it with scores from earlier, finer layers. **U-Net** [14] made this symmetric: a contracting **encoder** (conv blocks and downsampling) and an expanding **decoder** (upsampling and conv blocks), with **skip connections** that concatenate each encoder stage's feature map to the decoder stage of

the same resolution. The encoder supplies *what*; the skips supply *where*. U-Net was designed for biomedical images with few labels and strong augmentation, and the same shape now underlies medical segmentation, almost every restoration network covered in DLCV 04, and the denoiser inside many diffusion models.

The Vision Transformer

Plain version. The Transformer [15] was built for sentences: it treats the input as a sequence of tokens and lets every token look at every other token. The Vision Transformer (ViT) [16] cuts an image into small square patches, treats each patch as a "word", and feeds the sequence to an unmodified Transformer encoder.

Patchify, embed, attend

Precise version. For an image $I \in \mathbb{R}^{C\times H\times W}$ and patch size P:

1. **Patchify.** Split I into $N = HW/P^2$ non-overlapping patches and flatten each into a vector $\mathbf{p}_i \in \mathbb{R}^{CP^2}$. For a 224×224 RGB image and $P = 16$, $N = 196$ and each vector has 768 entries.
2. **Embed.** Map each patch linearly to a D-dimensional token, $\mathbf{z}_i = E\,\mathbf{p}_i$ with $E \in \mathbb{R}^{D\times CP^2}$.

   Prepend a learnable class token $\mathbf{z}_{\text{cls}}$ and add learnable **position embeddings** $\mathbf{e}_i$, because attention by itself has no notion of where a token came from:
   $Z^{(0)} = [\mathbf{z}_{\text{cls}}; \mathbf{z}_1; \dots; \mathbf{z}_N] + [\mathbf{e}_0; \dots; \mathbf{e}_N]$.
3. **Encode.** Apply L Transformer encoder blocks, each a multi-head self-attention (MSA) sublayer and an MLP sublayer, both with layer normalization (LN) and residual connections:

$$Z' = Z^{(\ell-1)} + \mathrm{MSA}(\mathrm{LN}(Z^{(\ell-1)})),$$
$$Z^{(\ell)} = Z' + \mathrm{MLP}(\mathrm{LN}(Z')).$$

4. **Head.** Feed the final class token (or the average of all tokens) to an MLP head, which plays the role of the pattern-recognition classifier [1].

A single attention head computes, for token matrix $Z \in \mathbb{R}^{(N+1)\times D}$,

$$\mathrm{Attn}(Z) = \mathrm{softmax}\!\left(\frac{QK^{\top}}{\sqrt{d}}\right)V, \qquad Q = ZW_Q,\; K = ZW_K,\; V = ZW_V,$$

where $W_Q, W_K, W_V \in \mathbb{R}^{D\times d}$ are learned projections and d is the head dimension. Row i of the softmax matrix says how much token i draws from every other token. Compared with a convolution, the "filter" is computed from the content and covers the whole image from the first layer on. The cost is quadratic in the number of tokens, $O(N^2 d)$, which is why patches rather than pixels are used.

```
img = torch.randn(1, 3, 224, 224)
P, D = 16, 768
patches = img.unfold(2, P, P).unfold(3, P, P)                    # (1, 3, 14, 14, 16, 16)
patches = patches.permute(0, 2, 3, 1, 4, 5).reshape(1, -1, 3 * P * P)
print(patches.shape)                                              # (1, 196, 768)
embed = nn.Conv2d(3, D, kernel_size=P, stride=P)                  # patchify + linear embed
 ↪  in one op
tokens = embed(img).flatten(2).transpose(1, 2)                    # (1, 196, 768)
```

The last two lines show a useful identity: patch embedding *is* a convolution with kernel size and stride both equal to P.

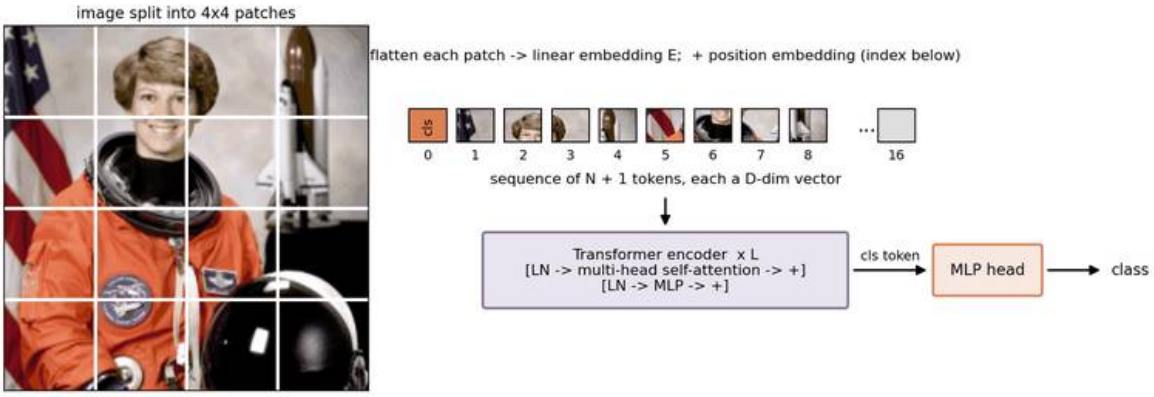


Figure 8 — ViT in one picture: patchify, linearly embed, add position embeddings and a class token, run a Transformer encoder, classify from the class token. (A 4×4 grid is drawn for legibility; ViT-B/16 on 224×224 inputs uses 14×14 patches.)

Inductive bias and the need for scale

The original post said that ViT is "harder to converge". The more precise statement [1], [16] is about **inductive bias**, the assumptions built into an architecture before it sees data. A CNN assumes locality and weight sharing, so it already "knows" that nearby pixels matter together and that a pattern means the same thing anywhere in the image. ViT assumes almost nothing: apart from the patch split, all spatial structure, even the 2-D layout of the position embeddings, must be learned.

That is a trade-off, not a defect. With little data, the CNN's assumptions are a valuable head start, and ViT trained from scratch on ImageNet-1k alone underperforms comparable ResNets. With enough data, the assumptions become a limitation, and ViT pretrained on very large datasets matched or beat the best CNNs when fine-tuned [16]. The practical lesson is that ViTs are used **pretrained**, either on huge labeled sets or, increasingly, with self-supervision.

Backbones after ViT

Plain version. After ViT, research went three ways: make Transformers more image-friendly (Swin), show that CNNs can catch up when modernized (ConvNeXt), and pretrain without labels so that one backbone serves every task (DINO, MAE, DINOv2/v3).

Swin Transformer: locality returns

Swin [17] computes attention only inside non-overlapping local windows (for example 7×7 tokens), which makes the cost linear in image size. In alternate blocks the window grid is **shifted** by half a window, so information crosses window borders. Between stages, neighboring tokens are merged to halve resolution. The result is a hierarchical, multi-scale feature pyramid shaped like a CNN backbone, which is exactly what detection and segmentation heads expect. Swin put locality and hierarchy, the CNN's inductive biases, back into a Transformer.

ConvNeXt: the CNN counterpoint

Liu et al. [18] started from a ResNet-50 and modernized it one step at a time, using the training recipe and design choices of Transformers: a "patchify" stem (a 4×4 convolution with stride 4), large 7×7 depthwise convolutions, an inverted-bottleneck MLP, fewer activation and normalization layers, and layer normalization. The resulting pure CNN, **ConvNeXt**, matched Swin in accuracy and scaling on classification, detection and segmentation. The lesson: much of ViT's advantage came from training recipes and macro design, not from attention itself. **ConvNeXt V2** [18] then added masked-autoencoder pretraining adapted to convolutions (the fully convolutional masked autoencoder, FCMAE) and a global response normalization layer. Today large foundation models mostly use ViT backbones, but convolutions remain common in low-level vision and on edge devices [1].

Self-supervised pretraining: DINO versus MAE

Self-supervised learning creates a training signal from the images themselves, so no labels are needed. Two recipes dominate, and they are often confused.

DINO: self-distillation [19]. A student network g_s and a teacher g_t with the same architecture each output a probability vector over K prototype dimensions. Several random crops of an image are made: two large "global" views and several small "local" views. The teacher sees only global views, the student sees all of them, and the student is trained to match the teacher's output:

$$\min_{\theta_s} \sum_{x \in \{x_1^g, x_2^g\}} \sum_{\substack{x' \in V \\ x' \neq x}} H(P_t(x),\, P_s(x')), \qquad H(a, b) = -a^\top \log b,$$

where V is the set of all views, P_t and P_s are the softmax outputs of teacher and student, and θ_s are the student's weights. The teacher is not trained by gradients. Its weights are an exponential moving average of the student's, $\theta_t \leftarrow \lambda\theta_t + (1-\lambda)\theta_s$ with λ close to 1, and its outputs are centered and sharpened so that training does not collapse to a constant output. Matching "small crop" to "whole image" forces the network to recognize objects from parts. A remarkable emergent property is that the attention maps of a DINO ViT segment the main objects without ever seeing a mask.

Optional: Knowledge distillation (in the Supplementary notes). The soft-target loss that DINO and "distilled into smaller models" build on.

MAE: masked prediction [20]. The masked autoencoder hides a large random subset of patches (75 % works best), encodes only the visible ones with a ViT, and asks a light decoder to reconstruct the missing pixels:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2^2,$$

where $\mathcal{M}$ is the set of masked patch indices, $\mathbf{p}_i$ the (normalized) pixels of patch i and $\hat{\mathbf{p}}_i$ the reconstruction. This "hide and predict" game is the image analogue of masked-word prediction in language models; DINO, by contrast, is a teacher–student distillation method [1]. MAE is cheap because the encoder processes only a quarter of the tokens, and it fine-tunes very well.

DINO-style features are better **frozen**, used as-is with a linear head.

DINOv2 and DINOv3: frozen general-purpose features

DINOv2 [21] scaled DINO-style self-distillation to a 1-billion-parameter ViT trained on a large, automatically **curated** image collection, then distilled it into smaller models. Its frozen features work well for both image-level tasks (classification, retrieval) and pixel-level tasks (segmentation, depth estimation) with only a light head. **DINOv3** [21] scaled data and model size further and introduced **Gram anchoring**, a regularizer that keeps dense patch features from degrading during very long training, so the same frozen backbone gives high-quality dense features.

This closes the loop with the pattern-recognition pipeline. "Feature extractor + classifier" has returned as **pretrained frozen backbone + light head**. The feature extractor is now learned once, at very large scale, without labels.

State-space backbones

Vision Mamba (Vim) and **VMamba** [22] replace attention with selective state-space models, which scan the token sequence with cost linear in its length. Images are not sequences, so these

models scan the patch grid along more than one path (Vim is bidirectional; VMamba's 2-D selective scan traverses several routes). They are attractive for high-resolution inputs but are still mostly a research direction rather than a default backbone.

High-level vision tasks

Plain version. High-level tasks differ in how *where* the answer must be: one label for the whole image, a box per object, or a label for every pixel.

Task	Output	Typical loss	Typical metric	Representative models
Classification	one label (or a probability vector)	cross-entropy	top-1 / top-5 accuracy	ResNet, ViT, ConvNeXt [10], [16], [18]
Object detection	a set of (box, class, score)	classification + box regression, or set matching	average precision (AP) over IoU thresholds	Faster R-CNN, DETR [23]
Semantic segmentation	a class per pixel	per-pixel cross-entropy, Dice (an overlap loss, below)	mean IoU	FCN, U-Net [13], [14]
Instance segmentation	a mask and class per object	detection + mask losses	mask AP	promptable: SAM family [24]

Classification

Assign one label to the whole image. The network is a backbone plus global pooling plus a linear layer, trained with cross-entropy $\mathcal{L} = -\log \hat{p}_y$, where $\hat{p}_y$ is the predicted probability of the true class y. Classification is also the **pretraining task** that produced most backbones before self-supervision. DLCV 02 covers it in depth.

Object detection

Find every object, its class and its location as an axis-aligned box (a region of interest, ROI, given by its corner coordinates or center, width and height) [1]. Two design lineages illustrate the field [23]. **Faster R-CNN** is two-stage: a region proposal network suggests candidate boxes from shared CNN features, and a second head classifies and refines each one; duplicates are removed by non-maximum suppression. **DETR** treats detection as **set prediction**: a Transformer decoder emits a fixed set of predictions, and a bipartite matching between predictions and ground-truth objects defines the loss, which removes hand-designed anchors and non-maximum suppression. Box quality is measured by intersection over union,

$$\mathrm{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|},$$

where A and B are the predicted and true regions. The closely related Dice coefficient $2|A \cap B|/(|A| + |B|)$ is used, in a differentiable soft form, as a segmentation loss. Average precision summarizes the precision–recall curve at one or more IoU thresholds. Details are in DLCV 03.

Semantic and instance segmentation

Here the region of interest is the object's exact set of pixels, not a box [1].

- **Semantic segmentation** labels each pixel with a class. Two overlapping people become one "person" region.
- **Instance segmentation** also separates objects of the same class. The two people become "person 1" and "person 2".

Semantic segmentation is the natural job of the encoder–decoders above [13], [14]. Instance segmentation adds a detection-like component that proposes one mask per object.

Promptable segmentation: SAM, SAM 2, SAM 3

Segmentation now has **promptable foundation models** [24]. **SAM** (Segment Anything) takes an image and a prompt (points, a box, or a rough mask) and returns a mask for the indicated object. A heavy ViT image encoder runs once per image, and a light prompt encoder and mask decoder run per prompt, so interaction is fast. It was trained on SA-1B, a dataset of more than 1 billion masks on 11 million images, built with a model-in-the-loop "data engine". **SAM 2** extends this to video with a streaming memory that carries the object from frame to frame, trained on a new video dataset (SA-V). **SAM 3** adds **promptable concept segmentation**: given a short noun phrase such as "yellow school bus", image exemplars, or both, it finds, segments and (in video) tracks *all* matching instances, rather than the one object a click points at. With SAM 3 the line between detection, instance segmentation and tracking becomes thin; tracking is covered in DLCV 06.

Low-level vision tasks

Plain version. Low-level tasks take an image and return an improved or transformed image of the same scene. The output is judged pixel by pixel, or by how natural it looks.

The general formulation is an inverse problem. A degraded observation $\mathbf{y}$ is modeled as

$$\mathbf{y} = \mathcal{D}(\mathbf{x}) + \mathbf{n},$$

where $\mathbf{x}$ is the clean image, $\mathcal{D}$ a degradation operator (blur, downsampling, haze, low exposure) and $\mathbf{n}$ noise, the same model as in IP 05. A network f_θ is trained on pairs to minimize, for example, $\|f_\theta(\mathbf{y}) - \mathbf{x}\|_1$. Because real paired data is scarce, pairs are usually **synthesized** by applying a modeled degradation to clean images, and the gap between synthetic and real degradations is the field's central difficulty.

- **Image restoration** recovers the clean image: denoising, deblurring, super-resolution (small to large), dehazing (hazy to clear), deraining. Architectures are full-resolution encoder–decoders (U-Net-like) or residual networks without downsampling, now often with window attention. See DLCV 04.
- **Image enhancement** makes content easier to see without a single "true" target: low-light enhancement (bringing out detail in dark images, roughly a smart brightening; turning night into day would be translation instead) and underwater enhancement. See DLCV 05.
- **Image alignment** (registration) estimates the geometric transform or dense flow that maps one image onto another, a prerequisite for burst fusion, HDR and medical image comparison.
- **Image-to-image translation** maps an image from one domain or style to another: day to night, summer to winter, sketch to photo. Face-swapping "deepfakes" are a (misused) application of the same idea. Since diffusion models, translation and editing are often done by a generative model conditioned on the input image, as discussed next.

Generation and vision–language models

Plain version. Two newer families do not fit the old map well: models that **create** images from text, and models that **talk** about images. Both rely on large-scale pretraining and on connecting images with language.

Diffusion models: from noise to image

A **denoising diffusion probabilistic model (DDPM)** [27] defines a forward process that gradually adds Gaussian noise to an image $\mathbf{x}_0$ over T steps. In closed form,

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I),$$

where $t \in \{1, \dots, T\}$ is the step and $\bar{\alpha}_t \in (0,1)$ is a fixed schedule that decreases from nearly 1 to nearly 0. A network $\boldsymbol{\epsilon}_\theta$ is trained to predict the noise that was added,

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0, t, \boldsymbol{\epsilon}} \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|_2^2,$$

and generation runs the process backwards, starting from pure noise and denoising step by step.

In other words, a generator is a **learned denoiser applied many times**, which ties generation directly to the restoration problems above.

- **Latent diffusion** [28] runs the diffusion in the compressed latent space of a pretrained autoencoder rather than on pixels, which cuts the cost enough for high-resolution text-to-image synthesis; text conditioning enters through cross-attention. It is the basis of Stable Diffusion.
- **DiT** (Diffusion Transformer) [28] replaces the U-Net denoiser with a Transformer on latent patches and shows that quality improves steadily as the model's compute grows.
- **InstructPix2Pix** [29] edits a given image according to a written instruction ("make it winter"), so classic image-to-image translation becomes a special case of conditional generation.

Vision–language models

Visual question answering (VQA), answering a free-form question about an image, once required a pipeline of image features, captioning and text question answering [1]. Two steps made it a single model.

CLIP [25] trains an image encoder f and a text encoder g on 400 million image–text pairs with a **contrastive** objective. In a batch of B pairs, with normalized embeddings $\mathbf{u}_i = f(I_i)$ and $\mathbf{v}_j = g(T_j)$ and temperature τ, the image-to-text loss is

$$\mathcal{L}_{I\to T} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(\mathbf{u}_i^\top \mathbf{v}_i/\tau)}{\sum_{j=1}^{B} \exp(\mathbf{u}_i^\top \mathbf{v}_j/\tau)},$$

and the total loss averages it with the symmetric text-to-image term. Matching pairs are pulled together and all other pairs in the batch pushed apart. Because the label space is now "any sentence", CLIP can classify zero-shot: embed the prompts "a photo of a cat", "a photo of a dog", ... and pick the one closest to the image.

LLaVA [26] connects a CLIP vision encoder to a large language model (LLM) through a projection layer that maps visual tokens into the LLM's word-embedding space, then fine-tunes on instruction-following conversations about images. Most current multimodal LLMs keep this **vision encoder + projector + LLM** pattern, so VQA, captioning and many recognition tasks are now handled by prompting one model.

Modern view

The trend across this tutorial is a move from "one task, one model" to "one foundation model + prompt or light head". The surveys below are a good map of how the field got there.

Surveys. Six reviews map the ground covered here. Li et al. [30] lay out the CNN design space component by component. Khan et al. [31] organize vision Transformer work by task and stress that Transformers need large-scale pretraining or architectural priors to be data-efficient. Jing and Tian [32] and Balestriero et al. [33] cover self-supervised learning, from early pretext tasks to today's contrastive, self-distillation and masked-modeling families. Awais et al. [34] and Zhang et al. [35] survey vision foundation models and vision–language models, including prompt types, transfer methods and open challenges such as evaluation, bias and robustness.

Optional: Guide to the Modern view surveys (in the Paper guides). What each survey covers, how it organizes the field and what it concludes.

State of the art, 2024–2026. Without quoting benchmark numbers, the landscape looks like this:

- **Backbones.** Large ViTs pretrained with self-supervision or image–text data are the default for high-level tasks, with frozen DINOv2/DINOv3 features [21] a strong baseline for dense tasks. Modern CNNs (ConvNeXt V2 [18]) remain competitive and are preferred where latency, memory or high resolution dominate. State-space backbones [22] are an active alternative.
- **Segmentation and tracking** have converged on promptable models: SAM 2 for video objects and SAM 3 for open-vocabulary concepts [24].
- **Generation** has moved to latent diffusion with Transformer denoisers [28], and editing to instruction-conditioned diffusion [29].
- **Understanding** is increasingly done by multimodal LLMs in the LLaVA pattern [26].
- **Low-level vision** still relies heavily on task-specific encoder–decoders trained on synthetic degradations, now increasingly with generative (diffusion) priors; see DLCV 04 and DLCV 05.

Open problems. (1) *Data and evaluation*: benchmarks saturate and may overlap with web-scale pretraining data, so it is hard to tell generalization from memorization [34]. (2) *Real-world degradations*: models trained on synthetic noise or blur still fail on real camera pipelines. (3) *Fidelity versus hallucination*: generative priors produce sharp but sometimes invented detail, which is unacceptable in medical or forensic use. (4) *Efficiency*: attention is quadratic in tokens, and foundation models are hard to run on devices. (5) *Dense features at scale*: DINOv3's Gram anchoring [21] exists because dense feature quality can degrade as training scales. (6) *Bias, robustness and interpretability* of models trained on uncurated web data [34].

Key takeaways

- Deep learning reduces the three classical levels to two: **low-level** (image → image, judged by fidelity) and **high-level** (image → semantics, judged by accuracy). Mid-level analysis goes to whichever its output resembles.

- **End-to-end learning** replaced hand-designed features with learned ones. The split returned at scale as **pretrained backbone + light head**.
- A convolutional layer is a bank of **learned spatial filters** with local connectivity and weight sharing. Stride, padding and pooling set the output size; the **receptive field** grows as $r_\ell = r_{\ell-1} + (k_\ell - 1)j_{\ell-1}$, and the effective field is smaller than the theoretical one.
- LeNet-5 (1998) was not the first CNN; backprop-trained convolutional networks date to 1989. ResNet was trained at **18/34/50/101/152** layers on ImageNet; **1202** layers appears only on CIFAR-10. Residual connections make the identity easy and give gradients a direct path.
- Encoder–decoders with skip connections (**FCN, U-Net**) are the standard shape for dense prediction and restoration.
- **ViT** has little inductive bias, so it needs large-scale pretraining. Swin added locality back; ConvNeXt showed a modernized CNN matches it.
- **DINO** is teacher–student self-distillation; **MAE** is masked prediction. DINOv2/v3 provide frozen, general-purpose features.
- Tasks increasingly converge on promptable or language-connected **foundation models**: SAM 1/2/3 for segmentation, latent diffusion and DiT for generation, CLIP and LLaVA for vision–language.

Exercises

1. **Receptive field by hand.** A network has, in order: conv 7×7 stride 2; max-pool 3×3 stride 2; then four conv 3×3 stride 1. Compute the theoretical receptive field and the jump after each layer, and check your answer with the `receptive_field` function.

   Hint

   Start with $r = 1, j = 1$. After the first conv: $r = 7, j = 2$. After pooling: $r = 7 + 2 \cdot 2 = 11, j = 4$. Each following 3×3 layer adds $2 \cdot 4 = 8$, so $r = 19, 27, 35, 43$ with $j = 4$.

2. **Parameter budget.** For $C = 256$ input and output channels, compare the weights (ignore biases) of one 5×5 convolution, two stacked 3×3 convolutions, and a bottleneck (1×1 to 64 channels, 3×3 at 64, 1×1 back to 256). Which have the same receptive field?

   Hint

   $25C^2 \approx 1.64$M; $2 \cdot 9C^2 \approx 1.18$M; bottleneck $256 \cdot 64 + 9 \cdot 64^2 + 64 \cdot 256 \approx 0.07$M. The first two both see 5×5; the bottleneck sees 3×3 but is far cheaper, which is why ResNet-50 and deeper use it.

3. **Why the identity is hard without a shortcut.** Inside a network, a block's input usually comes from a previous ReLU, so it is non-negative. Show that for such inputs a residual block with $F \equiv 0$ outputs exactly $\mathbf{y} = \mathbf{x}$. Then explain what a plain block (conv–BN–ReLU–conv–BN–ReLU) would have to learn to output $\mathbf{y} = \mathbf{x}$. What does this imply for very deep plain networks?

   Hint

   For $\mathbf{x} \geq 0$, $\mathrm{ReLU}(\mathbf{x} + 0) = \mathbf{x}$. A plain block must instead learn weights that exactly undo two convolutions and two normalizations, a precise and fragile solution that gradient descent rarely finds. The residual block only needs $F = 0$, which weight decay and zero-initialization already push it toward. Deep plain networks therefore struggle to "do nothing" in extra layers, which matches the degradation the ResNet authors observed.

4. **ViT token count and cost.** For a 512×512 input, compute the number of tokens for patch sizes 32, 16 and 8, and the relative cost of one global attention layer ($\propto N^2$). How does Swin's windowed attention with 7×7 windows change the scaling?

   Hint

   $N = (512/P)^2$: 256, 1024, 4096 tokens; costs scale $1 : 16 : 256$. Windowed attention costs $N \cdot 49$ per layer (each token attends to 49 others), so it grows linearly with N.

5. **Learned versus hand-set filters.** Modify the digit CNN so that its first layer is frozen to four hand-set kernels (two Sobels, a Laplacian and a box blur) plus four random kernels. Train only the rest. Compare test accuracy with the fully learned network and inspect what happens to the random kernels if you unfreeze them.

   Hint

   Set `conv.weight.requires_grad_(False)` after assigning the weights, and pass only the remaining parameters to the optimizer. On this easy dataset both versions reach similar accuracy; the point is that hand-set derivative filters are a reasonable first layer, and that unfrozen random kernels tend to drift toward oriented, derivative-like patterns (compare Figure 4).

6. **Choose a recipe.** You have 2,000 labeled microscopy images and need (a) a per-image quality label, (b) per-pixel cell masks, (c) denoised images. For each, propose a backbone, a head and a training strategy using this tutorial's map, and say which would benefit most from a frozen self-supervised backbone.

   Hint

   (a) Frozen DINOv2/v3 or fine-tuned ConvNeXt + linear head. (b) U-Net, possibly with a pretrained encoder, or SAM prompted by points and then fine-tuned. (c) A full-resolution restoration network trained on synthetic noisy/clean pairs that match the microscope's noise. Frozen semantic features help (a) and (b) most; (c) depends on pixel-accurate statistics more than on semantics.

Supplementary notes

These notes are optional background that the main text does not depend on.

Batch normalization: standardize inside the network

Imagine grading every exam on a curve: whatever the difficulty, scores are shifted and stretched so the class average is 0 and the typical spread is 1. **Batch normalization** (BN) does this to every channel of a layer's output, using the images in the current mini-batch as "the class", and then lets the network re-stretch the result with two learned numbers [37].

Take one channel k of a convolutional layer. Let B be the set of all values of that channel over the m images of the mini-batch and over all $H \times W$ positions, so $|B| = mHW$ (every position of a feature map is thus normalized in the same way [37]). Then

$$\mu_B = \frac{1}{|B|}\sum_{x\in B} x, \qquad \sigma_B^2 = \frac{1}{|B|}\sum_{x\in B}(x-\mu_B)^2, \qquad y = \gamma\,\frac{x-\mu_B}{\sqrt{\sigma_B^2+\epsilon}} + \beta,$$

where μ_B and σ_B^2 are the mean and variance of the channel in this mini-batch, ϵ is a tiny constant that avoids division by zero, and γ and β are a learned scale and shift for the channel. If the network wants the original values back, it can learn $\gamma = \sqrt{\sigma_B^2+\epsilon}$ and $\beta = \mu_B$, so no capacity is lost. In the residual block above, BN sits between the convolution and the ReLU; setting the last γ to zero (with β at its initial value 0) makes the branch output $F(\mathbf{x}) = 0$, which is the zero-initialization trick used in the code of that section.

Training versus inference. At test time one image should not depend on whichever other images happen to share its batch. So during training we also keep running averages, $\mu_{\text{run}} \leftarrow (1-\rho)\,\mu_{\text{run}} + \rho\,\mu_B$ (the same for the variance), with momentum ρ (0.1 in the code below). At inference these frozen values replace μ_B and σ_B^2, so BN becomes a fixed per-channel linear map, which can even be folded into the preceding convolution. The original paper averages the statistics over training mini-batches [37]; the exponential moving average is the usual implementation.

```
import numpy as np

rng = np.random.default_rng(0)
C, eps, momentum = 3, 1e-5, 0.1
gamma, beta = np.ones(C), np.zeros(C)             # learned scale and shift (one pair per
 ↪  channel)
run_mu, run_var = np.zeros(C), np.ones(C)         # running statistics used at inference

def batch(n=32):
    # feature maps (N, C, H, W) whose channels have very different statistics
    scale = np.array([1.0, 5.0, 0.2]).reshape(1, C, 1, 1)
    offset = np.array([0.0, 10.0, -3.0]).reshape(1, C, 1, 1)
    return rng.normal(size=(n, C, 8, 8)) * scale + offset
```

```
def bn_train(x):
    global run_mu, run_var
    mu, var = x.mean(axis=(0, 2, 3)), x.var(axis=(0, 2, 3))     # statistics of this
     ↪ mini-batch
    run_mu = (1 - momentum) * run_mu + momentum * mu
    run_var = (1 - momentum) * run_var + momentum * var
    xhat = (x - mu.reshape(1, C, 1, 1)) / np.sqrt(var.reshape(1, C, 1, 1) + eps)
    return gamma.reshape(1, C, 1, 1) * xhat + beta.reshape(1, C, 1, 1)

def bn_eval(x):
    xhat = (x - run_mu.reshape(1, C, 1, 1)) / np.sqrt(run_var.reshape(1, C, 1, 1) + eps)
    return gamma.reshape(1, C, 1, 1) * xhat + beta.reshape(1, C, 1, 1)

x = batch()
y = bn_train(x)
print("input  mean per channel:", x.mean(axis=(0, 2, 3)).round(2))
print("output mean per channel:", y.mean(axis=(0, 2, 3)).round(3))
print("output std  per channel:", y.std(axis=(0, 2, 3)).round(3))

for _ in range(100):                                # many training batches update the
 ↪ running statistics
    bn_train(batch())
print("running mean:", run_mu.round(2), " running var:", run_var.round(3))

x_new = batch()
print("eval-mode output mean:", bn_eval(x_new).mean(axis=(0, 2, 3)).round(2),
      " std:", bn_eval(x_new).std(axis=(0, 2, 3)).round(2))

one = x_new[:1]                                     # a single image
print("single image, eval mode, depends on batch? ",
      np.allclose(bn_eval(one), bn_eval(x_new)[:1]))
print("single image, batch statistics of itself: max |train - eval| =",
      np.abs(bn_train(one) - bn_eval(one)).max().round(2))
```

Run with the seed shown, this prints:

```
input  mean per channel: [-0.02  9.89 -2.99]
output mean per channel: [-0.  0. -0.]
output std  per channel: [1. 1. 1.]
running mean: [ 0.01 10.   -3.  ]  running var: [ 1.011 24.831  0.04 ]
eval-mode output mean: [ 0.02 -0.    0.02]  std: [1.   0.99 1.  ]
single image, eval mode, depends on batch?  True
single image, batch statistics of itself: max |train - eval| = 0.34
```

The three channels start with very different means and spreads and leave BN with mean 0 and standard deviation 1. After 101 training batches the running statistics are close to the values the data was generated with (means 0, 10, -3; variances 1, 25, 0.04). In eval mode an image gets the same output whether it is alone or inside a batch (the `True` line), whereas normalizing one image by its own statistics would give a different result (the last line).

Common mistakes. Forgetting to switch the model to inference mode (`model.eval()` in PyTorch), so the layer keeps using batch statistics; training with very small batches, where μ_B and σ_B^2 are noisy and differ from the running values; and evaluating on data whose statistics differ from the training data, where the stored running values no longer fit. Transformers such as ViT use **layer normalization** instead, which computes the statistics over the features of one token and therefore behaves the same in training and inference.

Why does it help? The original explanation was "reducing internal covariate shift", the drift of each layer's input distribution during training [37]. Santurkar et al. [38] tested this and reported that distributional stability of layer inputs has little to do with BN's success, and that BN instead makes the optimization landscape smoother, giving more predictive gradients. They note that the exact reasons for BN's effectiveness were still poorly understood. See ML 02 for BN next to LayerNorm and RMSNorm.

Knowledge distillation: learning from a teacher's soft answers

A student who only sees the answer key learns "this photo is a tabby cat". A student who also hears the teacher say "mostly tabby cat, a little tiger cat, almost certainly not a car" learns which classes resemble each other. **Knowledge distillation** trains a (usually smaller) student network to imitate the full output distribution of a (usually larger) teacher [39].

Let the teacher produce logits $\mathbf{v}$ and the student logits $\mathbf{z}$ for one input. A softmax with temperature T turns logits into probabilities,

$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}, \qquad p_i = \frac{\exp(v_i/T)}{\sum_j \exp(v_j/T)},$$

where q_i and p_i are the student's and teacher's probability for class i and $T = 1$ is the ordinary softmax; a larger T flattens the distribution, so the small probabilities of wrong classes become visible. Hinton et al. train the student with a weighted sum of two cross-entropies [39]:

$$\mathcal{L} = \alpha\, T^2 \left(-\sum_i p_i \log q_i \right) + (1-\alpha) \left(-\log q_y^{(1)} \right),$$

where y is the true class, $q_y^{(1)}$ is the student's probability of y at $T = 1$, and $\alpha \in [0, 1]$ weights the soft term against the hard-label term (the paper found a considerably lower weight on the hard term to work best). The gradient of the soft term with respect to the student logit z_i is $(q_i - p_i)/T$. Its size shrinks as T grows, which is why the paper multiplies it by T^2 [39]. After training, the student is run at $T = 1$.

```
import numpy as np

def softmax(z, T=1.0):
    z = np.asarray(z, dtype=float) / T
```

```
    e = np.exp(z - z.max())
    return e / e.sum()

teacher = np.array([9.0, 6.0, 2.0, -3.0])      # logits for: tabby cat, tiger cat, dog, car
for T in (1, 4):
    print(f"T={T}: teacher probabilities", softmax(teacher, T).round(3))

def soft_ce(zs, zt, T):
    return -(softmax(zt, T) * np.log(softmax(zs, T))).sum()   # cross-entropy with soft
     ↪  targets

def kd_loss(zs, zt, y, T, alpha):
    hard = -np.log(softmax(zs, 1.0)[y])                       # ordinary cross-entropy at T =
     ↪  1
    return alpha * T**2 * soft_ce(zs, zt, T) + (1 - alpha) * hard

student = np.array([5.0, 4.0, 3.0, 0.0])
print("loss T=4, alpha=0.9, true class 0:", round(kd_loss(student, teacher, 0, 4, 0.9), 3))

# the gradient of the soft term with respect to the student logits is (q - p) / T
T = 4.0
analytic = (softmax(student, T) - softmax(teacher, T)) / T
numeric = np.array([(soft_ce(student + 1e-6 * np.eye(4)[i], teacher, T)
                     - soft_ce(student - 1e-6 * np.eye(4)[i], teacher, T)) / 2e-6 for i in
                      ↪  range(4)])
print("max |analytic - numeric| gradient:", np.abs(analytic - numeric).max())

for T in (1, 4, 16, 64):                          # why the T^2 factor: gradient size shrinks
 ↪  like 1/T^2
    g = (softmax(student, T) - softmax(teacher, T)) / T
    print(f"T={T}: gradient norm {np.linalg.norm(g):.4f}, times T^2: {T**2 *
     ↪  np.linalg.norm(g):.4f}")
```

Output:

```
T=1: teacher probabilities [0.952 0.047 0.001 0.   ]
T=4: teacher probabilities [0.59  0.279 0.102 0.029]
loss T=4, alpha=0.9, true class 0: 16.462
max |analytic - numeric| gradient: 1.0457410631881103e-10
T=1: gradient norm 0.3608, times T^2: 0.3608
T=4: gradient norm 0.0653, times T^2: 1.0441
T=16: gradient norm 0.0051, times T^2: 1.2975
T=64: gradient norm 0.0003, times T^2: 1.3377
```

At $T = 1$ the teacher is almost certain (0.952 for the first class) and the second class is barely visible; at $T = 4$ the cat-versus-dog-versus-car ordering shows up. The analytic gradient matches a finite-difference check. The last four lines show that the soft-term gradient norm falls steeply with T, while the product with T^2 levels off (about 1.34 at $T = 64$): the $1/T^2$ scaling is an approximation that holds at high temperature.

Link to this tutorial. DINO's loss $H(a, b) = -a^\top \log b$ is the same soft-target cross-entropy, with two differences: the teacher is not a bigger pretrained network but a moving average of the student, and no labels are used. Its "sharpening" of the teacher is a softmax with a low temperature. Compressing a large model into smaller ones, as the text says is done for DINOv2, is the classic teacher-to-student setting. In the paper's preliminary MNIST experiment, a large regularized network made 67 test errors, a smaller unregularized one made 146, and the smaller one trained to match the large one's soft targets at $T = 20$ made 74 [39]. This is one dataset and one setting, so treat it as an illustration, not a general rate. Common mistakes: using different temperatures for teacher and student in the soft term, forgetting the T^2 factor when mixing with the hard loss, and evaluating the student at $T > 1$.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Guide to the Modern view surveys

This guide expands the short survey summary in the Modern view: for each review, what it covers, how it organizes the field and what it concludes.

CNN architectures. Li et al. [30] review CNNs from their history through the convolution operation itself, classic and advanced architectures (the LeNet–ResNet line, lightweight and attention-augmented networks), 1-D, 2-D and multi-dimensional convolutions, and applications. They also cover the improvements made to each component (layer design, activation and loss functions, regularization, optimization, fast computation), run experiments to derive rules of thumb for choosing functions and hyperparameters, and close with open issues and promising directions. It is the best single place to see the CNN design space laid out component by component.

Vision Transformers. Khan et al. [31] (ACM Computing Surveys) organize Transformer work by task: the foundations (self-attention, large-scale pretraining, bidirectional encoding), then recognition (classification, detection, segmentation, action recognition), generative modeling, multimodal tasks (VQA, visual reasoning, grounding), video, low-level vision (super-resolution, enhancement, colorization) and 3-D point clouds. Their central contrast matches this tutorial's: Transformers model long-range dependencies with minimal inductive bias, so they need large-scale pretraining or architectural priors (windows, hierarchies) to be data-efficient.

Self-supervised visual learning. Jing and Tian [32] (TPAMI) survey the pre-2020 generation of self-supervised methods, organized by pretext task (generation-based, context-based, free-semantic-label-based and cross-modal), along with the standard protocol of evaluating features by transfer to downstream tasks. Balestriero et al. [33] cover the modern generation in a practical "cookbook" and group methods into four families: deep metric learning (contrastive), self-

distillation (the DINO family), canonical-correlation-style methods, and masked image modeling (MAE). Their emphasis on the many hidden "knobs" (augmentations, projector heads, collapse prevention) explains why reproducing self-supervised results is hard.

Foundation models for vision. Awais et al. [34] (TPAMI) categorize vision foundation models by modality pairing (vision with text, audio, depth), training objective (contrastive versus generative) and **prompt type**: textual, visual (points, boxes, masks, as in SAM) and heterogeneous. Their open challenges are evaluation and benchmarking, real-world understanding and contextual reasoning, bias, adversarial robustness and interpretability. Zhang et al. [35] (TPAMI) focus on vision–language models for visual recognition: architectures and pretraining objectives, transfer methods (prompt tuning, adapters) and knowledge distillation into detectors and segmenters. They argue that web-scale image–text data reduces the dependence on crowd-sourced labels and lets one model serve many tasks.

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DLCV 01A • Training Deep Vision Models in Practice

A practical workflow for training vision networks: data pipelines and augmentation, initialization, learning-rate finding and schedules, batch size, regularization, mixed precision, fine-tuning, debugging and fair experiments.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-01a-training-practice/

What you'll learn

- A short, ordered checklist for any new training run, and where each step's theory lives on this site.
- How to build a data pipeline that does not leak: normalization statistics, class imbalance, and augmentation chosen per task, including geometric consistency for boxes, masks and keypoints.
- How to find a learning rate with a range test, and how warmup, step decay, cosine and one-cycle schedules use it; how the learning rate should follow the batch size, and when gradient accumulation is exactly equivalent to a bigger batch.
- Which regularizers to reach for in practice (decoupled weight decay, dropout, stochastic depth, early stopping, an EMA of the weights) and what mixed precision changes.
- How to fine-tune a pretrained backbone: freezing, discriminative learning rates, and linear probing followed by fine-tuning.
- How to debug a run (overfit one batch, read curves, watch gradient norms, chase NaNs) and how to run a fair experiment with baselines, ablations and several seeds.

The big picture

You already know how convolutional and transformer networks for images are built (DLCV 01). This chapter is about the other half of the job: getting one to train well, quickly, and knowing whether a result is real.

> **Explain like I'm five**
>
> Training a network is like cooking a new dish in a big kitchen. First you taste one spoonful to check that the recipe works at all (overfit one batch). You check how hot the oven can go before things burn (the learning-rate range test). You heat up slowly instead of putting the pan straight on full flame (warmup), and you turn the heat down at the end so nothing burns (the decay). And before you tell anyone your new recipe is better, you cook it several times, because one good dinner can be luck (several seeds).

Why spend a chapter on this? Because the recipe often matters as much as the architecture. He et al. combined training refinements, such as learning-rate warmup and cosine decay, with small

changes to the network and raised the ImageNet top-1 accuracy of ResNet-50 from 75.3% to 79.29% [1]; DLCV 02 describes a similar result. And training failures are often **silent**: a network with a bug in its data pipeline usually still trains, just worse, so nothing crashes to warn you [2]. The habits in this chapter are how practitioners catch those bugs early.

The theory behind most tools is already covered elsewhere on this site, and this chapter links to it instead of repeating it:

Topic	Covered in depth in
Floating point, gradient checks, a debugging order for NaN values	ML 01C
SGD, momentum, Adam, AdamW, initialization, normalization layers, gradient clipping, basic schedules	ML 02
Baselines, splits and leakage, significance tests, reproducibility, latency and deployment	ML 10
How batch normalization works; knowledge distillation	DLCV 01 notes
Classification recipes: augmentation, loss and optimizer choices, transfer learning, calibration	DLCV 02

What this chapter adds is the **workflow**: the order in which to do things, the quick tests that catch mistakes, and the defaults that work for most vision models.

A training checklist

Plain version. Do the cheap checks first and change one thing at a time. Most wasted weeks come from training a large model for days before checking that a small one can learn at all.

The list below is the whole chapter in one place. Each item links to the section that explains it.

1. **Fix the evaluation first.** Define the split unit, the metric and a simple baseline before training anything (data pipeline; ML 10).
2. **Look at the data.** Plot a batch *after* the full input pipeline (decoding, resizing, augmentation, normalization), with its labels drawn on top. This catches most pipeline bugs in a minute.
3. **Start from known-good defaults.** A standard architecture, He initialization, a pretrained backbone if one exists, AdamW or SGD with momentum (initialization).
4. **Overfit one batch.** The training loss on a single fixed batch must go to almost zero (debugging).
5. **Find the learning rate** with a short range test, then pick a schedule with warmup and a decay (learning rate).
6. **Set the batch size** from memory, and scale the learning rate with it (batch size).

7. **Train the full model, log everything** (losses, metrics, learning rate, gradient norms), and keep the best checkpoint by validation score.
8. **Regularize only when you see overfitting**, one tool at a time (regularization).
9. **Speed up** with mixed precision once the run is correct (mixed precision).
10. **Compare fairly:** same budget, same data, several seeds, one change per ablation; open the test set once at the end (fair experiments).

The running example

To show each step with real numbers, the snippets train a small multilayer perceptron (MLP) written in NumPy: 64 inputs, one hidden layer of 32 ReLU units and 10 outputs, trained with SGD and momentum on scikit-learn's 8×8 handwritten digits [3]. It is not a deep vision model, but every training decision in this chapter applies to it, and each run takes well under a second on a CPU. All numbers called "our toy runs" below come from this code; the figure script `scripts/figures/dl_01a_training_practice.py` reproduces them.

The code below defines it once: data loading with a stratified split, normalization with training-split statistics, a two-layer network with He initialization, backpropagation, and a training loop that accepts any learning-rate schedule. The loop also keeps an exponential moving average of the weights, which the EMA section uses. Later snippets reuse these functions, so run this block first.

```
import numpy as np
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split

X, y = load_digits(return_X_y=True)                    # 1,797 images of 8x8 pixels, 10
 ↪  classes
X_tmp, X_te, y_tmp, y_te = train_test_split(X, y, test_size=0.3, stratify=y, random_state=0)
X_tr, X_va, y_tr, y_va = train_test_split(X_tmp, y_tmp, test_size=0.25, stratify=y_tmp,
 ↪  random_state=0)
mu, sd = X_tr.mean(0), X_tr.std(0) + 1e-8              # statistics of the TRAINING split
 ↪  only
X_tr, X_va, X_te = [(a - mu) / sd for a in (X_tr, X_va, X_te)]

def init(seed, sizes=(64, 32, 10)):                    # He initialization for ReLU layers
    rng = np.random.default_rng(seed)
    return [[rng.normal(0, np.sqrt(2 / m), (m, n)), np.zeros(n)]
            for m, n in zip(sizes[:-1], sizes[1:])]

def forward(P, X):
    h = [X]
    for i, (W, b) in enumerate(P):
        z = h[-1] @ W + b
        h.append(np.maximum(z, 0) if i < len(P) - 1 else z)   # ReLU; the last layer gives
         ↪  logits
```

```
    return h

def loss_and_grads(P, X, y):
    h = forward(P, X)
    z = h[-1] - h[-1].max(1, keepdims=True)             # stable log-softmax
    logp = z - np.log(np.exp(z).sum(1, keepdims=True))
    loss = -logp[np.arange(len(y)), y].mean()           # cross-entropy
    d = np.exp(logp)                                    # softmax probabilities
    d[np.arange(len(y)), y] -= 1
    d /= len(y)                                         # dLoss/dlogits
    grads = []
    for i in reversed(range(len(P))):
        grads.insert(0, [h[i].T @ d, d.sum(0)])
        if i:
            d = (d @ P[i][0].T) * (h[i] > 0)            # back through the ReLU
    return loss, grads

def evaluate(P, X, y):
    z = forward(P, X)[-1]
    z = z - z.max(1, keepdims=True)
    logp = z - np.log(np.exp(z).sum(1, keepdims=True))
    return -logp[np.arange(len(y)), y].mean(), (z.argmax(1) == y).mean()

def train(schedule, peak, epochs=15, bs=16, momentum=0.9, seed=0, ema_decay=0.99):
    P = init(seed)
    V = [[np.zeros_like(W), np.zeros_like(b)] for W, b in P]   # momentum buffers
    E = [[W.copy(), b.copy()] for W, b in P]                    # EMA copy of the weights
    rng = np.random.default_rng(seed + 1000)            # data order depends on the seed too
    T, t, log = epochs * int(np.ceil(len(y_tr) / bs)), 0, []
    for _ in range(epochs):
        order = rng.permutation(len(y_tr))
        for i in range(0, len(y_tr), bs):
            idx = order[i:i + bs]
            _, G = loss_and_grads(P, X_tr[idx], y_tr[idx])
            lr = schedule(t, T, peak)
            t += 1
            for (W, b), (gW, gb), v, e in zip(P, G, V, E):   # SGD with momentum, in place
                v[0] = momentum * v[0] + gW
                v[1] = momentum * v[1] + gb
                W -= lr * v[0]
                b -= lr * v[1]
                e[0] += (1 - ema_decay) * (W - e[0])    # e <- decay * e + (1 - decay) * w
                e[1] += (1 - ema_decay) * (b - e[1])
        log.append(evaluate(P, X_tr, y_tr) + evaluate(P, X_va, y_va))
    return P, np.array(log), E   # log columns: train loss, train acc, val loss, val acc
```

The split is 942 training, 315 validation and 540 test images; the test images are not touched in this chapter.

The data pipeline

Plain version. The data pipeline turns files on disk into the tensors the network sees. It decides what the model can learn and what the evaluation can prove, so it deserves the first and most careful look.

Splits, leakage and the test set

The test set must be new in the same way deployment data will be new. Split by the unit that will be new (patient, video, camera, site or time period), deduplicate before splitting, fit every preprocessing step on the training split only, and use the validation set for all decisions so that the test set is opened once. ML 10 covers the forms leakage takes and how to measure uncertainty; this chapter assumes those rules.

Normalization statistics

Networks train best when each input channel has a mean near zero and a spread near one. The standard step is

$$x'_c = \frac{x_c - \mu_c}{\sigma_c},$$

where x_c is a pixel value in channel c (for example red, green or blue), μ_c and σ_c are the mean and standard deviation of channel c, and x'_c is the value the network sees. Two practical rules follow.

- **Training from scratch:** compute μ_c and σ_c on the training split only, store them with the model, and apply the same stored values at validation, test and deployment time. The running example does exactly this, per pixel rather than per channel.
- **Fine-tuning a pretrained model:** use the statistics the model was *pretrained* with, not those of your new dataset. The first layer's filters were learned for inputs on that scale.

The most common bug here is a mismatch: images scaled to $[0, 1]$ in training but raw 0–255 values in deployment, or RGB in one place and BGR in another. Both still produce predictions, just bad ones.

Class imbalance

Plain version. If one class has nine hundred examples and another twenty, the network can score well by mostly ignoring the rare class. You can rebalance what the network sees, how much each mistake costs, or where the classifier starts.

Buda et al. compared remedies for convolutional networks on benchmarks of increasing difficulty [4]. In their experiments oversampling the rare classes was the dominant method in almost all scenarios, it worked best when it removed the imbalance completely, and, unlike for some classical models, it did not cause the networks to overfit; they also recommend thresholding to compensate for the class priors when overall accuracy is what matters [4]. The three levers look like this:

```
import numpy as np

counts = np.array([900, 80, 20])                          # training images per class
labels = np.repeat([0, 1, 2], counts)

w = counts.sum() / (len(counts) * counts)                 # option 1: weight the loss
print("loss weight per class:", np.round(w, 2))

p = 1.0 / counts[labels]                                  # option 2: oversample rare classes
p /= p.sum()
rng = np.random.default_rng(0)
drawn = labels[rng.choice(len(labels), size=3000, p=p)]
print("class share in 3,000 sampled examples:", np.round(np.bincount(drawn) / 3000, 3))

prior = counts / counts.sum()                             # option 3: start the output bias at
 ↪  the prior
b0 = np.log(prior)
print("initial softmax output:", np.round(np.exp(b0) / np.exp(b0).sum(), 3))
```

```
loss weight per class: [ 0.37  4.17 16.67]
class share in 3,000 sampled examples: [0.33  0.338 0.332]
initial softmax output: [0.9  0.08 0.02]
```

Option 1 multiplies each example's loss by $w_k = N/(K\,n_k)$, where N is the number of training examples, K the number of classes and n_k the count of class k. Option 2 draws each example with probability proportional to $1/n_k$, so every class appears equally often. Option 3 sets the output biases to $b_k = \log(n_k/N)$, so the untrained network already predicts the class frequencies; Lin et al. use the same idea in dense detection, starting every location at a foreground probability of about 0.01 [5] (see the focal loss in DLCV 03). Whatever you choose, report per-class metrics (DLCV 02).

Augmentation as a regularizer

Plain version. Augmentation shows the network a slightly different version of each image every time, so it cannot memorize pixels and must learn what stays the same. The golden rule: a transformation is allowed only if the correct label of the new image is known.

That rule decides the augmentation for each task:

Task	Usually safe	Needs care or is wrong
Classification of photos	random crops and rescaling, horizontal flips, color jitter, mild blur	vertical flips for upright scenes; flips for text, digits and left/right-sensitive classes
Detection	flips, crops, scaling, color jitter, applied to the image **and** the boxes	crops that cut objects: clip the box and drop it if too little is left
Segmentation	the same geometry applied to the image **and** the mask	interpolating the mask: resize labels with nearest neighbor, never bilinear
Keypoints and pose	geometry applied to the image **and** the points	a horizontal flip must also **swap** left and right keypoint labels (DLCV 06A)
Restoration and low-level vision	flips, 90° rotations, crops applied identically to input and target	color or blur augmentation that changes the target itself
Medical and remote sensing	rotations where orientation carries no meaning	intensity changes that alter diagnostic values

For dense tasks the key word is **consistency**: every geometric change applied to the pixels must be applied to the labels in the same coordinates. The snippet flips and crops a photo of coins together with a segmentation mask and a bounding box for one coin, then checks that the transformed box equals the box of the transformed mask:

```
import numpy as np
from scipy import ndimage
from skimage import data, filters, measure, segmentation

img = data.coins()                                          # 303 x 384 grayscale photo
fg = segmentation.clear_border(img > filters.threshold_otsu(img))
lab = measure.label(ndimage.binary_fill_holes(fg))
coin = max(measure.regionprops(lab), key=lambda r: r.area)
mask = lab == coin.label                                    # segmentation target for one coin
box = tuple(int(v) for v in coin.bbox)                      # detection target (y0, x0, y1, x1)
H, W = img.shape

def hflip(img, mask, box):
    y0, x0, y1, x1 = box
    return img[:, ::-1], mask[:, ::-1], (y0, W - x1, y1, W - x0)

def crop(img, mask, box, top, left, h, w):
    y0, x0, y1, x1 = box                                    # shift, then clip to the crop
    new = (max(y0 - top, 0), max(x0 - left, 0), min(y1 - top, h), min(x1 - left, w))
    return img[top:top + h, left:left + w], mask[top:top + h, left:left + w], new

i2, m2, b2 = crop(*hflip(img, mask, box), top=90, left=0, h=180, w=240)
r, c = np.nonzero(m2)
print("original box:               ", box)
print("box after flip + crop:      ", b2)
print("box of the transformed mask:", (int(r.min()), int(c.min()), int(r.max()) + 1,
 ↪  int(c.max()) + 1))
```

```
original box:                (156, 315, 218, 380)
```

```
box after flip + crop:       (66, 4, 128, 69)
box of the transformed mask: (66, 4, 128, 69)
```

Under a horizontal flip of an image of width W, a box with columns x_0 to x_1 moves to columns $W - x_1$ to $W - x_0$: the edges swap roles. Flipping the image but not its labels gives the network contradictory targets (Figure 1, right). Karpathy lists exactly this kind of mistake, flipping images without flipping their labels, among the bugs that make a network train silently worse instead of failing [2]. In practice, use a library that transforms images together with their masks and boxes, such as Albumentations [6], and still verify it once with a check like the one above.

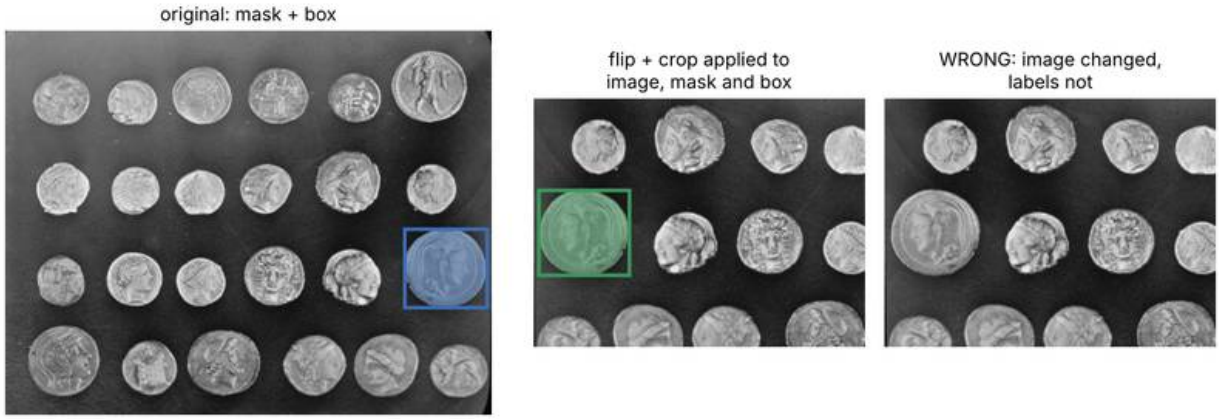


Figure 1 — Geometric augmentation for dense tasks. A flip and crop must move the mask and the box with the pixels (middle). Transforming only the image (right) leaves annotations that no longer match any object; here the old box falls outside the crop entirely.

How strong? Augmentation strength is a hyperparameter: tune it on the validation set, and expect a strongly augmented model to show a *higher* training loss than validation loss. Shorten and Khoshgoftaar survey the families of image augmentation [7]; DLCV 02 shows the mixing augmentations of modern classification recipes.

Initialization and normalization in practice

Plain version. A good start keeps the signal at a sensible size through every layer, so the first steps of training are spent learning instead of recovering from a bad scale. For standard architectures, the defaults below are well tested; use them unless you have a reason not to.

- **Weights.** Use He initialization for layers followed by ReLU-like activations: draw weights with variance $2/n_{\text{in}}$, where n_{in} is the number of inputs to a unit, which keeps the activation variance roughly constant across layers [8]. Biases start at zero. ML 02 derives this.

- **Residual blocks.** Initialize the scale γ of the last batch-normalization layer in each residual branch to zero. Each block then starts as the identity, the signal travels through the shortcuts, and early training is easier; Goyal et al. found that this helped all their models and especially large-batch training [9], and He et al. include it in their bag of tricks [1].
- **Output layer.** Initialize the classifier bias at the log class prior when classes are imbalanced (option 3 above).
- **Pretrained weights.** When a pretrained backbone exists for your input type, start from it. A new head is the only part initialized from scratch.

Normalization layers in practice. Batch normalization (BN) [10] standardizes each channel with the mean and variance of the current mini-batch, then keeps running averages for inference (DLCV 01 note). Two habits avoid most BN problems:

1. **Switch modes.** Training mode uses batch statistics and updates the running averages; evaluation mode uses the stored averages. Forgetting to switch to evaluation mode at test time is a classic bug.
2. **Watch the per-device batch size.** BN statistics are computed over the examples on one device. With only one or two images per device, as is common in detection and segmentation, the statistics become noisy. Group Normalization, which normalizes groups of channels within each example, does not depend on the batch size and stays accurate where BN degrades; Wu and He also describe the common detection practice of *freezing* a pretrained backbone's BN, using its stored statistics as a fixed linear layer [11].

Transformers use a normalization computed within each example, which does not depend on the batch; ML 02 compares the normalization layers.

Choosing the learning rate

Plain version. The learning rate is the size of each step downhill. Too small and training crawls; too large and the loss jumps around or explodes. The best rate also changes during training: larger steps help early, smaller steps help at the end.

Bengio wrote that the learning rate is often the single most important hyperparameter [12]. The procedure here: measure the usable range once, then let a schedule move through it.

The learning-rate range test

Plain version. Run training once while raising the learning rate a little at every step, and watch the loss. Where the loss stops falling and starts to climb, the rate has become too large.

Smith introduced this **learning-rate range test** together with cyclical learning rates: train for a short time while increasing the rate between a low and a high value, and read off the range in which the network still learns [13]. In his later guidance, the rate where the loss is lowest (or the accuracy highest) is the largest usable value, a suitable maximum for a schedule that ramps up and down, while a constant learning rate must be smaller or the network will not begin to converge; for a single-cycle schedule he suggests a minimum 10 to 20 times below the maximum [14].

Smith reads the curve from the validation loss or accuracy; our version uses the training loss of each batch, which costs nothing extra. It increases the rate exponentially from 10^{-4} to 10 over at most 200 steps, so that every decade gets equal time, and smooths the noisy batch loss with an exponential moving average:

$$\bar{\ell}_t = \beta\,\bar{\ell}_{t-1} + (1-\beta)\,\ell_t, \qquad \hat{\ell}_t = \frac{\bar{\ell}_t}{1-\beta^{t+1}},$$

where ℓ_t is the loss of the batch at step t, $\bar{\ell}_t$ the running average (starting from $\bar{\ell}_{-1} = 0$), $\beta = 0.9$ the smoothing factor, and $\hat{\ell}_t$ the bias-corrected average that is plotted; the correction removes the pull toward the zero starting value. The test stops when the smoothed loss exceeds four times its minimum. The code below reuses `init` and `loss_and_grads` from the running example.

```
def lr_range_test(lo=1e-4, hi=10.0, steps=200, bs=16, momentum=0.9, seed=0):
    P = init(seed)
    V = [[np.zeros_like(W), np.zeros_like(b)] for W, b in P]
    rng = np.random.default_rng(seed)
    lrs, smooth, avg = [], [], 0.0
    for t in range(steps):
        lr = lo * (hi / lo) ** (t / (steps - 1))        # exponential sweep
        idx = rng.choice(len(y_tr), bs, replace=False)
        loss, G = loss_and_grads(P, X_tr[idx], y_tr[idx])
        avg = 0.9 * avg + 0.1 * loss                    # smooth the noisy batch loss
        lrs.append(lr)
        smooth.append(avg / (1 - 0.9 ** (t + 1)))       # bias-corrected average
        if not np.isfinite(loss) or smooth[-1] > 4 * min(smooth):
            break                                       # the loss has blown up: stop
        for (W, b), (gW, gb), v in zip(P, G, V):
            v[0] = momentum * v[0] + gW
            v[1] = momentum * v[1] + gb
            W -= lr * v[0]
            b -= lr * v[1]
    return np.array(lrs), np.array(smooth)

lrs, smooth = lr_range_test()
best = lrs[smooth.argmin()]
print(f"lowest smoothed loss {smooth.min():.3f} at lr {best:.3f}; stopped at lr
 ↪  {lrs[-1]:.3f}")
print(f"suggested peak for a one-cycle or warmup schedule: about {best:.2f}")
print(f"suggested constant learning rate: about {best / 10:.3f}")
```

```
lowest smoothed loss 0.270 at lr 0.138; stopped at lr 0.392
```

```
suggested peak for a one-cycle or warmup schedule: about 0.14
suggested constant learning rate: about 0.014
```

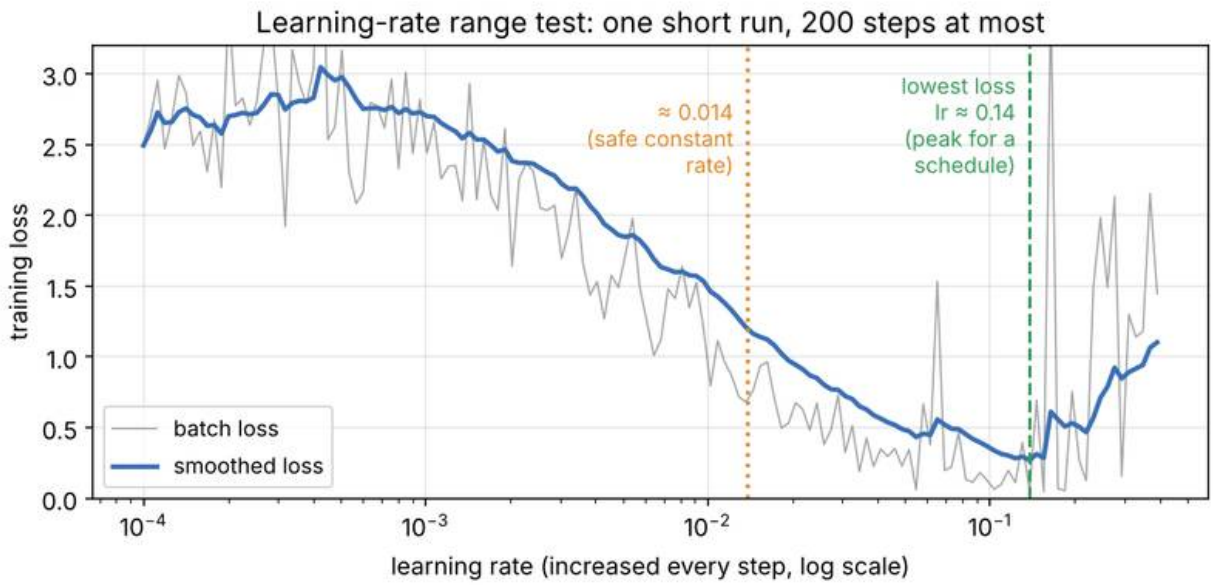


Figure 2 — A learning-rate range test on the toy MLP (our run). The loss falls once the rate is large enough to make progress, bottoms out near 0.14 and then climbs; the test stopped at 0.39.

Read Figure 2 from left to right: too small to change anything, then the useful range where the loss falls steeply, then steps that are too large. The code takes a tenth of the minimum as the constant rate; that factor is our rule of thumb, not a value from [14]. A range test costs a few hundred steps, so rerun it whenever you change the model, batch size, optimizer or weight decay. Treat the result as a range, not an exact value: the curve is noisy and depends on the seed.

Warmup

Plain version. Start with a small learning rate and raise it to the target over the first few hundred or thousand steps. At the start, the weights are random and the optimizer's running statistics are empty, so a full-size step can throw the model into a bad region it never leaves.

With a linear warmup over W steps, the rate at step $t < W$ is

$$\eta_t = \eta_{\text{peak}} \frac{t+1}{W},$$

where η_{peak} is the target rate reached at the end of warmup. Goyal et al. showed why the *shape* matters: in large-batch ImageNet training, a constant low rate for the first epochs followed by a jump to the target made the training error spike, while a gradual ramp avoided that spike [9]. Warmup matters most for large batches, large learning rates, adaptive optimizers and Transformers, and

it is cheap insurance in fine-tuning. Goyal et al. report that even a constant low-rate warmup was particularly helpful when prototyping detection and segmentation models that fine-tune pre-trained layers together with new ones [9]. Our toy warms up over the first 5% of the steps.

Schedules: step, cosine and one-cycle

Plain version. After warmup, the rate should fall. How it falls matters less than *that* it falls: large steps explore, small steps at the end settle into a good region.

ML 02 introduces step decay, cosine annealing and warmup; this section adds one-cycle and compares all of them on one problem. Write t for the current step, T for the total number of steps and η_{peak} for the largest rate. The four schedules in Figure 3 are:

- **Constant:** $\eta_t = \eta_{\text{peak}}$. Simple, but the final steps are as large as the first ones.
- **Step decay:** multiply the rate by a factor γ (often 0.1) at fixed milestones, for example at 50% and 75% of training. The classic choice for CNNs trained with SGD.
- **Warmup + cosine:** after W warmup steps, $\eta_t = \frac{1}{2}\eta_{\text{peak}}(1 + \cos(\pi q))$ with progress $q = (t - W)/(T - W)$ running from 0 to 1, so the rate glides from η_{peak} to zero. The cosine shape comes from SGDR, which also restarted it periodically [15]; He et al. adopted the plain cosine decay for ImageNet classification, where it improved accuracy over their step-decay baseline [1].
- **One-cycle:** a single long rise from a low rate to η_{peak}, then a fall to a value far below the starting rate. Smith and Topin used this "1cycle" policy with a large maximum rate to train some networks much faster than with standard schedules, which they called super-convergence [16]. Our version follows the defaults of PyTorch's `OneCycleLR`: a cosine-shaped rise over the first 30% of steps from $\eta_{\text{peak}}/25$, then a cosine fall to 10^4 times below the start [17]. That implementation also lowers the momentum while the rate rises, as Smith recommends [14]; our toy keeps momentum fixed.

```
import numpy as np

def constant(t, T, peak):
    return peak

def step_decay(t, T, peak, milestones=(0.5, 0.75), factor=0.1):
    return peak * factor ** sum(t >= m * T for m in milestones)

def warmup_cosine(t, T, peak, warmup=0.05):
    W = max(1, int(warmup * T))
    if t < W:
        return peak * (t + 1) / W                       # linear warmup
    q = (t - W) / max(1, T - W)                         # progress after warmup, 0 -> 1
    return 0.5 * peak * (1 + np.cos(np.pi * q))         # cosine decay to 0

def one_cycle(t, T, peak, pct=0.3, div=25, final_div=1e4):
```

```
    up, lo = max(1, int(pct * T)), peak / div
    if t < up:                                          # rise from peak/div to peak
        return lo + (peak - lo) * 0.5 * (1 - np.cos(np.pi * t / up))
    q = (t - up) / max(1, T - up)                       # fall far below the start
    end = lo / final_div
    return end + (peak - end) * 0.5 * (1 + np.cos(np.pi * q))

T = 900
for f in (constant, step_decay, warmup_cosine, one_cycle):
    print(f"{f.__name__:13s}", [f"{f(t, T, 0.1):.2g}" for t in (0, 20, 270, 450, 700, 899)])
```

```
constant      ['0.1', '0.1', '0.1', '0.1', '0.1', '0.1']
step_decay    ['0.1', '0.1', '0.1', '0.01', '0.001', '0.001']
warmup_cosine ['0.0022', '0.047', '0.084', '0.054', '0.013', '3.4e-07']
one_cycle     ['0.004', '0.0053', '0.1', '0.081', '0.023', '1e-06']
```

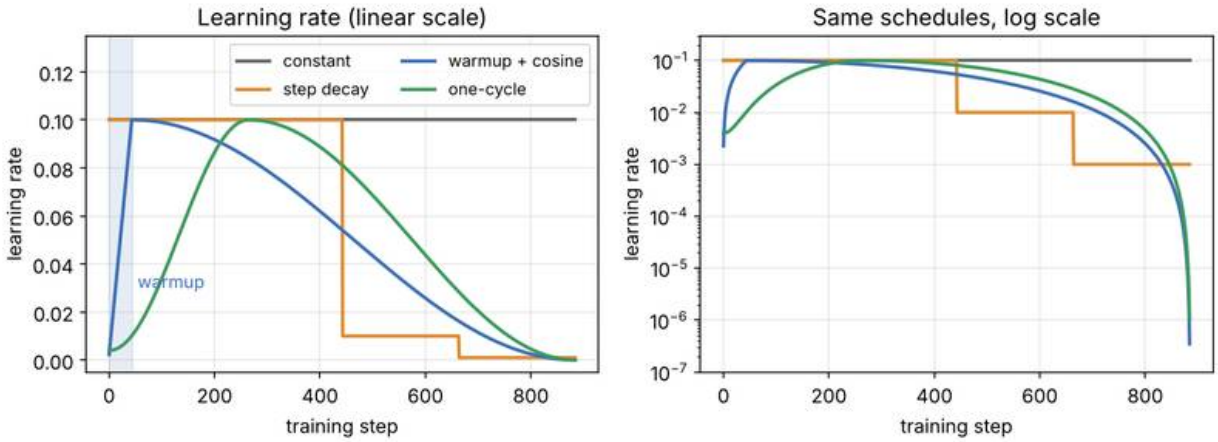


Figure 3 — The four schedules used in this chapter with a peak of 0.1 over 885 steps (15 epochs of the toy). The log-scale view (right) shows how far cosine and one-cycle fall at the end.

Optional: Cosine annealing with warm restarts (in the Supplementary notes). The SGDR formula with restart periods, and when restarts are useful.

What the schedules do on the toy

The loop below trains the running example once with each schedule at a peak of 0.1, and once more with a constant rate of 0.01. It uses `train` from the running example and the schedule functions just above.

```
for f in (constant, step_decay, warmup_cosine, one_cycle):
    _, log, _ = train(f, peak=0.1)
    print(f"{f.__name__:13s} peak 0.1   final train loss {log[-1, 0]:8.3f}   val acc
     ↪  {log[-1, 3]:.3f}")
```

```
_, log, _ = train(constant, peak=0.01)
print(f"{'constant':13s} peak 0.01  final train loss {log[-1, 0]:8.3f}   val acc {log[-1,
 ↪  3]:.3f}")
```

```
constant      peak 0.1   final train loss  995.468   val acc 0.917
step_decay    peak 0.1   final train loss    0.000   val acc 0.940
warmup_cosine peak 0.1   final train loss    0.000   val acc 0.971
one_cycle     peak 0.1   final train loss    0.000   val acc 0.956
constant      peak 0.01  final train loss    0.021   val acc 0.959
```

Figure 4 shows the runs epoch by epoch. Three lessons come out of this toy, and they match the advice above:

1. **A peak near the edge needs a schedule.** At a constant 0.1, close to the range-test minimum, the loss explodes and validation accuracy swings between 81% and 92%. The same peak is fine inside a schedule that ramps up and decays (warmup + cosine reached 97.1%, one-cycle 95.6% in this run).
2. **Decay rescues, but late.** Step decay spends half of training at the unstable rate; once the rate drops, during epoch 8, the loss recovers and validation accuracy reaches 94.0%.
3. **A constant rate can work if it is small.** At 0.01, a tenth of the peak as the range test suggested, the constant schedule trains smoothly (95.9%), just less far in the same budget: its training loss is still 0.021.

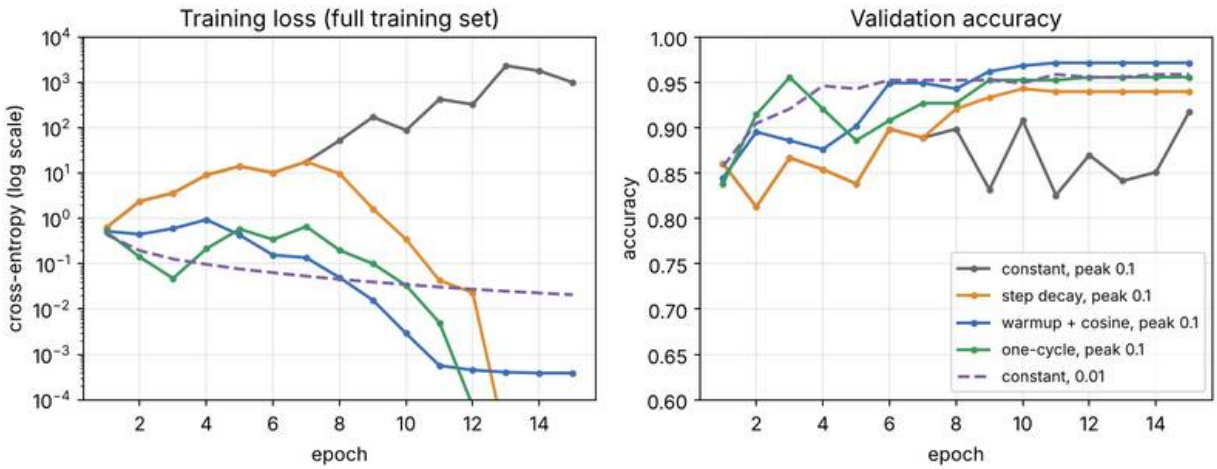


Figure 4 — The schedules of Figure 3 on the toy MLP, seed 0 (our runs). With a peak of 0.1, the constant rate diverges, step decay recovers after its first drop, and warmup + cosine and one-cycle train stably. The dashed line is a constant rate of 0.01.

Do not read the ranking between cosine and one-cycle into this figure: it is one seed, and the seed study below shows that their order flips from seed to seed. The robust lesson is the first one.

Which schedule to pick. Warmup followed by cosine decay is a safe default; step decay is fine when reproducing an older paper; one-cycle suits short budgets; a constant rate with a short final decay helps when the training length is not known in advance (Modern view). Schedules are defined in *steps*, so recompute T when the epochs, batch size or dataset size change.

Batch size and gradient accumulation

Plain version. A bigger batch gives a less noisy gradient, so each step can be bigger. If you double the batch, you can usually double the learning rate, up to a point. (ML 02 defines batches, iterations and epochs.)

Goyal et al. trained ResNet-50 [18] on ImageNet [19] with mini-batches of up to 8,192 images without losing accuracy, using two rules [9]:

- **Linear scaling rule:** when the mini-batch size is multiplied by k, multiply the learning rate by k and keep the other hyperparameters, such as weight decay, unchanged. He et al. state the same rule in the form $\eta = 0.1 \times B/256$ for a ResNet-50 trained with SGD at batch size B [1].
- **Gradual warmup:** ramp the rate from the small-batch value to the scaled value over the first 5 epochs.

The rule is an approximation. It assumes the gradient changes little over the k small steps that one big step replaces, which fails early in training (hence warmup) and beyond some batch size; in their ImageNet experiments that limit was around 8,000 images [9]. For adaptive optimizers the relationship is less simple, so re-run the range test after a large change in batch size.

Optional: Accurate, Large Minibatch SGD (Goyal et al.) (in the Paper guides). Where the scaling rule comes from, how to read their warmup comparison, and the implementation pitfalls they list.

Gradient accumulation. When the batch you want does not fit in memory, run several smaller *micro-batches*, add up their gradients, and take one optimizer step. With a loss that is a mean over examples and k micro-batches of equal size,

$$\nabla_\theta L_{\text{batch}} = \frac{1}{k} \sum_{j=1}^{k} \nabla_\theta L_j,$$

where L_{batch} is the mean loss over all $k \cdot m$ examples, L_j the mean loss over micro-batch j of m examples, and θ the parameters. The snippet checks this on the toy (it reuses `init` and `loss_and_grads` from the running example):

```
P = init(0)
xb, yb = X_tr[:32], y_tr[:32]
_, full = loss_and_grads(P, xb, yb)                    # one batch of 32

acc = [[np.zeros_like(W), np.zeros_like(b)] for W, b in P]
```

```
k = 4                                                  # 4 micro-batches of 8
for j in range(k):
    _, G = loss_and_grads(P, xb[8 * j:8 * j + 8], yb[8 * j:8 * j + 8])
    for a, (gW, gb) in zip(acc, G):
        a[0] += gW / k                                 # divide by k: mean of means
        a[1] += gb / k
diff = max(np.abs(a[0] - g[0]).max() for a, g in zip(acc, full))
print(f"largest difference, accumulated vs full batch: {diff:.1e}")
```

```
largest difference, accumulated vs full batch: 1.1e-16
```

The difference is floating-point rounding: the update is the same. Two things break the equivalence in real models. Batch normalization computes its statistics per micro-batch, so accumulation does not give BN a larger batch. And a loss averaged over a variable number of items per image (valid pixels in segmentation, boxes in detection, tokens in a caption) needs the accumulated sum divided by the *total* count, not by k.

Optional: Gradient accumulation when the loss is not a plain mean (in the Supplementary notes). The correct weighting for per-pixel and per-box losses, and which counters (schedule, weight decay, EMA) must advance per optimizer step.

Regularization in practice

Plain version. Regularizers make memorizing harder so that the network learns patterns that generalize. They all pull in the same direction, so add them one at a time and only when the validation curve shows overfitting.

Smith and Topin stress a point that is easy to miss: large learning rates regularize too, so a high peak needs less of the other kinds, and all forms (large learning rates, small batch sizes, weight decay, dropout) must be balanced together [16]. When you change the schedule, re-tune weight decay and augmentation, and vice versa.

Weight decay. Weight decay shrinks every weight a little at each step. With plain SGD this equals adding an L2 penalty to the loss; with adaptive optimizers such as Adam [20] it does not, because the penalty's gradient gets rescaled per parameter. Loshchilov and Hutter's AdamW decouples the decay from the gradient step [21]; ML 02 derives the difference. Two practical rules:

- In PyTorch, `torch.optim.AdamW` applies decoupled decay, while the `weight_decay` argument of `torch.optim.Adam` adds an L2 term to the gradient unless `decoupled_weight_decay=True` is set [22]. Use decoupled decay with Transformers and modern CNN recipes.
- Do not decay biases and the scale and shift of normalization layers. He et al. list this "no bias decay" among their refinements [1]; DLCV 02 shows the two parameter groups in code.

Dropout and stochastic depth. Dropout zeroes random activations during training [23]; stochastic depth skips whole residual blocks at random, with a skip probability that grows with depth [24]. Dropout is common in classifier heads and Transformer blocks, stochastic depth in deep residual networks and vision Transformers. Both change behavior between training and evaluation mode.

Early stopping and checkpoint selection. Evaluate on the validation set at regular intervals, keep the best checkpoint, and stop when the score has not improved for a while (the "patience"). In Prechelt's study on multilayer perceptrons, slower stopping criteria gave small improvements in generalization at a much higher training cost [25]. The Tuning Playbook recommends saving checkpoints regularly and selecting the best one retrospectively [26].

EMA of the weights

Plain version. Keep a slowly moving average of the weights next to the weights themselves, and evaluate the average. The last few noisy steps cancel out, much as averaging several photos of the same scene removes noise.

DLCV 02 lists the EMA among the parts of a modern classification recipe; this section is about choosing its window. After every optimizer step, update

$$\bar{\theta} \leftarrow \mu\,\bar{\theta} + (1-\mu)\,\theta,$$

where θ are the current weights, $\bar{\theta}$ their exponential moving average (EMA), and $\mu \in [0, 1)$ the decay. The average effectively covers about the last $1/(1-\mu)$ steps: about 100 steps for $\mu = 0.99$, about 1,000 for $\mu = 0.999$. Averaging the iterates of stochastic gradient descent is an old idea in optimization [27], and stochastic weight averaging (SWA) brought an equal-weight version to deep learning [28]. Morales-Brotons et al. studied the EMA of weights systematically: EMA models generalized better than the last iterate and needed less learning-rate decay, and batch normalization limited how long the averaging window could be unless its statistics were recomputed [29].

The window must be short compared with the run. Our toy makes this concrete (it reuses `train`, `constant` and `evaluate` from above):

```
for decay in (0.99, 0.999):
    raw, avg = [], []
    for s in range(5):
        P, log, E = train(constant, peak=0.05, seed=s, ema_decay=decay)
        raw.append(evaluate(P, X_va, y_va)[1])
        avg.append(evaluate(E, X_va, y_va)[1])
    print(f"decay {decay}: window ~{1 / (1 - decay):.0f} steps   "
          f"val acc raw {np.mean(raw):.3f}   EMA {np.mean(avg):.3f}")
print("steps in the whole run:", 15 * int(np.ceil(len(y_tr) / 16)))
```

```
decay 0.99: window ~100 steps   val acc raw 0.963   EMA 0.963
decay 0.999: window ~1000 steps   val acc raw 0.963   EMA 0.945
steps in the whole run: 885
```

On this small, easy problem the EMA brings no gain at $\mu = 0.99$, and at $\mu = 0.999$ it is *worse*: its window of about 1,000 steps is longer than the whole 885-step run, so the average still carries weight from the poor early iterates. EMA pays off in long runs with noisy updates, which is where it is used in practice.

Optional: EMA of weights: choosing the decay (in the Supplementary notes). The effective window, a bias correction for the start, batch-normalization statistics and how to evaluate the average.

Mixed precision and loss scaling

Plain version. Doing most arithmetic with 16-bit numbers halves memory traffic and uses fast hardware units, but 16 bits cannot hold everything. Keep a few sensitive quantities in 32 bits, and scale the loss so that tiny gradients do not round to zero.

Micikevicius et al. made float16 training work with two techniques: a float32 master copy of the weights that accumulates the updates, and loss scaling, which multiplies the loss by a factor S before backpropagation and divides the gradients by S before the update [30]. Both problems are easy to reproduce:

```
import numpy as np

g = np.float32(2e-8)                                  # a small, real gradient value
print("float16 gradient:          ", np.float16(g))     # underflows to zero
S = np.float32(1024)                                  # loss scale
g16 = np.float16(g * S)                               # scaled: representable in float16
print("scaled, then unscaled:     ", np.float32(g16) / S)
print("float16 weight 1 + 1e-4:   ", np.float16(1) + np.float16(1e-4))   # update lost
print("float32 weight 1 + 1e-4:   ", np.float32(1) + np.float32(1e-4))
```

```
float16 gradient:           0.0
scaled, then unscaled:      2.0023435e-08
float16 weight 1 + 1e-4:    1.0
float32 weight 1 + 1e-4:    1.0001
```

The first line is **underflow**: 2×10^{-8} is below the smallest positive float16 value, so it becomes zero, and scaling by 1,024 moves it back into range. The third line is a **lost update**: near 1.0, float16 numbers are spaced too far apart to register a change of 10^{-4}, which is why the weights themselves stay in float32.

In PyTorch, automatic mixed precision combines `torch.autocast`, which runs each operation in a suitable precision, with `torch.amp.GradScaler`, which performs dynamic loss scaling; its bfloat16 example on the CPU uses autocast alone [31]. bfloat16 keeps the 8 exponent bits of float32, and therefore its range, so it rarely needs loss scaling (ML 01C). Make the run correct in float32 first, then switch on mixed precision and check that the loss curve barely changes; if NaNs appear only in mixed precision, look for operations that overflow in 16 bits. ML 01C explains the formats in detail.

Transfer learning and fine-tuning strategies

Plain version. A pretrained network already knows edges, textures and object parts. Fine-tuning teaches it your task without destroying that knowledge: change the new parts quickly and the old parts gently.

DLCV 02 defines linear probing and full fine-tuning; this section is about choosing between them and making fine-tuning gentle. Yosinski et al. measured how general the features of each layer are: early layers learn features that transfer across datasets, later layers are more task-specific, and transferability falls as the source and target tasks drift apart, though transferred features beat random ones even for distant tasks [32]. That is the reason behind the strategies below, ordered from cheapest to most flexible:

Strategy	What trains	When it fits
Linear probe	only a new linear head on frozen features	very little data; features already good for the task; also to *measure* feature quality
Partial fine-tuning	the head and the last stages	moderate data; target close to the pretraining domain
Full fine-tuning	everything, with a small learning rate for the backbone	enough data or a large domain gap
Linear probe, then fine-tune (LP-FT)	first the head alone, then everything	when robustness to distribution shift matters
Parameter-efficient fine-tuning	small added modules, such as low-rank updates (LoRA [33])	very large models; many tasks sharing one backbone

Three tools make full fine-tuning gentler:

- **Discriminative learning rates.** Give each layer group its own rate, smaller for earlier layers. Howard and Ruder introduced this as *discriminative fine-tuning*, together with *gradual unfreezing* (train the last layer first, then unfreeze one more layer group per epoch) [34]. In vision recipes the backbone often gets a rate 10 to 100 times smaller than the head (DLCV 02).
- **LP-FT.** Kumar et al. showed that full fine-tuning with a randomly initialized head can *distort* good pretrained features: while the random head is being fitted, the backbone changes for training-like inputs but much less for different-looking ones. Training the linear head first,

then fine-tuning everything (LP-FT), kept the in-distribution gain of fine-tuning and the out-of-distribution robustness of linear probing in their experiments [35].

- **Warmup and BN.** A short warmup protects the pretrained weights from the first large gradients of the new head, and with small per-device batches the backbone's BN layers are often frozen (above).

Optional: Discriminative learning rates and gradual unfreezing (in the Supplementary notes). How to set per-layer rates, the layer-wise decay rule, and how to unfreeze in stages.

Optional: Fine-Tuning can Distort Pretrained Features (Kumar et al.) (in the Paper guides). The argument for LP-FT and how to read its in-distribution and out-of-distribution results.

Debugging a training run

Plain version. Most training bugs do not crash; they make the model a little worse. Debugging means running cheap tests that a correct pipeline must pass, in an order that isolates the cause (Figure 5).

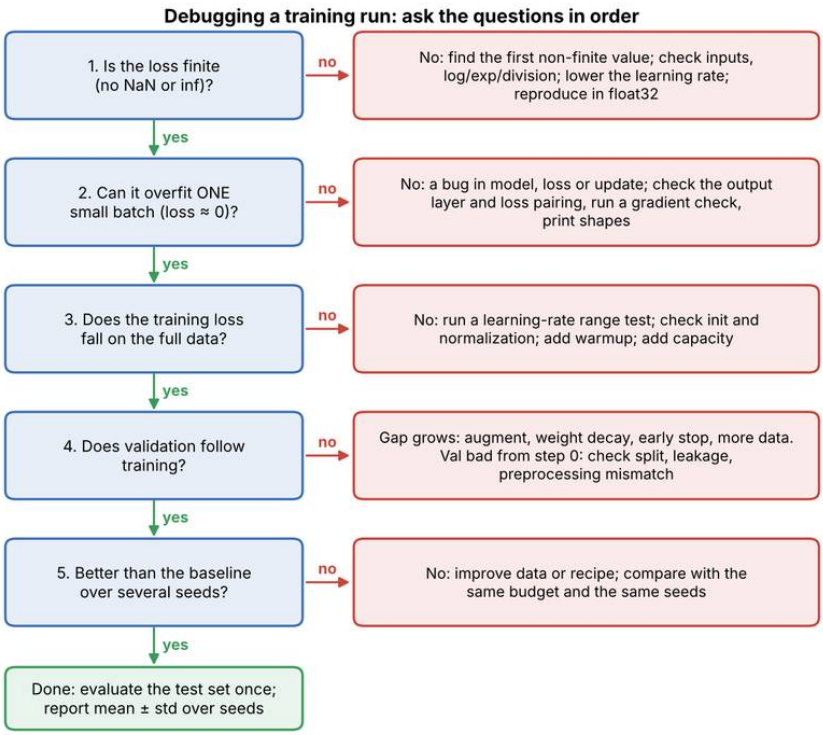


Figure 5 — A debugging order. Each question assumes the ones above it are answered yes, so a failure points to a small set of causes.

Overfit one batch first

Plain version. Take one small batch and train on it over and over. A correct model and training loop can memorize a handful of images, so the loss must fall to nearly zero. If it cannot, something is broken, and you have found that out in seconds instead of days.

Karpathy's recipe for training networks puts this test early: overfit a single small batch, and before that, check that the loss at initialization is what it should be [2]. For K balanced classes and a well-initialized final layer, the first predictions are close to uniform, so the cross-entropy starts near $\ln K$ ($\ln 10 \approx 2.30$ for our digits). The snippet checks that, then runs the one-batch test on 16 digits twice: once correctly, and once with a bug that is easy to make, applying softmax in the model *and* inside the loss. It reuses `init` and `forward` from the running example.

```
def overfit_one_batch(double_softmax=False, steps=300, lr=0.1, seed=0):
    P = init(seed)
    xb, yb = X_tr[:16], y_tr[:16]                        # ONE fixed batch of 16 images
    for step in range(steps):
        h = forward(P, xb)
        z = h[-1]
        if double_softmax:                               # BUG: the "logits" are already
         ↪  probabilities
            z = np.exp(z - z.max(1, keepdims=True))
            z = z / z.sum(1, keepdims=True)
        s = z - z.max(1, keepdims=True)
        p = np.exp(s) / np.exp(s).sum(1, keepdims=True)
        loss = -np.log(p[np.arange(16), yb]).mean()
        d = p.copy()
        d[np.arange(16), yb] -= 1
        d /= 16
        if double_softmax:                               # chain rule through the extra
         ↪  softmax
            d = z * (d - (d * z).sum(1, keepdims=True))
        for i in reversed(range(len(P))):
            gW, gb = h[i].T @ d, d.sum(0)
            if i:
                d = (d @ P[i][0].T) * (h[i] > 0)
            P[i][0] -= lr * gW
            P[i][1] -= lr * gb
    return loss

print(f"loss at init:          {evaluate(init(0), X_tr, y_tr)[0]:.4f}   (ln 10 =
 ↪  {np.log(10):.4f})")
print(f"correct loss:          {overfit_one_batch():.4f}")
print(f"double-softmax bug:    {overfit_one_batch(double_softmax=True):.4f}")
print(f"floor of the bug:      {-np.log(np.e / (np.e + 9)):.4f}")
```

```
loss at init:          2.8204   (ln 10 = 2.3026)
correct loss:          0.0036
double-softmax bug:    1.4661
floor of the bug:      1.4612
```

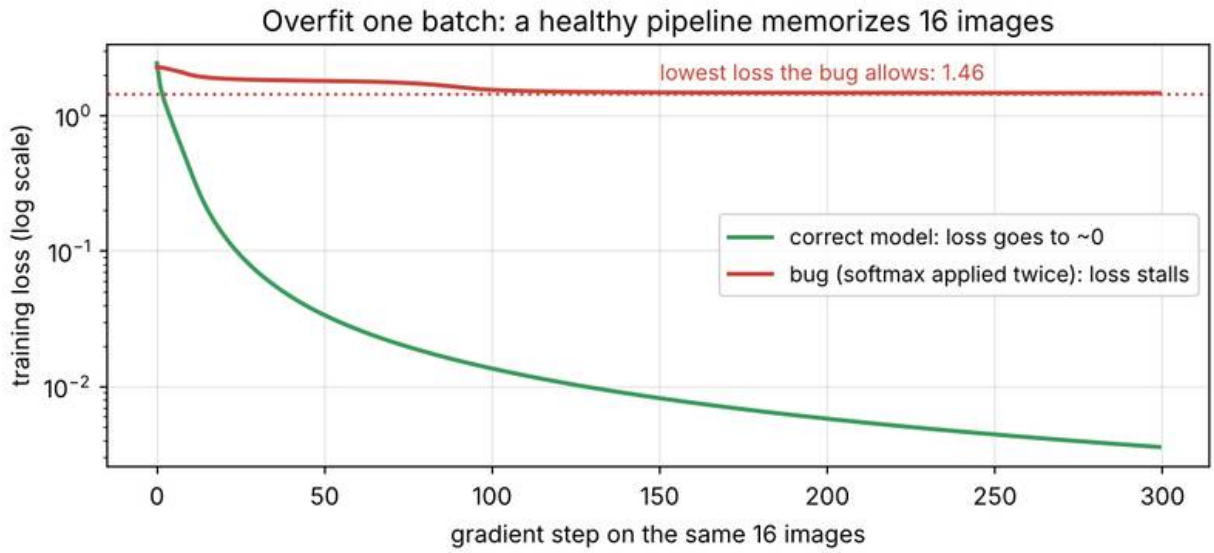


Figure 6 — Overfitting one batch (our run). The correct pipeline memorizes 16 images; the double-softmax bug cannot go below 1.46, however long it trains.

The initial loss is somewhat above $\ln 10$ because the He-initialized output layer produces logits of moderate size, so the first predictions are not exactly uniform; that is harmless here. A value far above $\ln K$ means the network starts out confidently wrong, and scaling the last layer's initial weights down fixes it. The bug has a hard floor. After the first softmax, the "logits" passed to the loss lie between 0 and 1, so the best the second softmax can do is give the true class e^1 against e^0 for each of the nine others: a probability of $e/(e+9) \approx 0.23$ and a loss of 1.46. On the full dataset this bug does not crash and still produces a usable classifier, so it can go unnoticed; the one-batch test exposes it in seconds.

A passing test proves less than it seems. Zhang et al. showed that standard image classifiers can fit random labels, and even random noise images, perfectly [36], so memorizing one batch says that the model, the loss and the update can learn, not that the labels are correct. Mislabeled or misaligned data must be caught by looking at it (checklist item 2).

Reading loss curves

Plain version. A loss curve is the main record of a run: its shape tells you which part of the pipeline to suspect. Plot training and validation curves together, on a log scale for the loss.

ML 02 explains the four basic shapes (flat, overfitting, converged, healthy). A few more patterns are typical of vision training:

- **Stuck exactly at $\ln K$.** The network outputs uniform probabilities: the gradient is not reaching it (a detached tensor, frozen parameters, a learning rate of zero) or the labels are constant.

- **Loss spikes that recover.** The learning rate is near the edge of stability, as in Figure 4, or a few batches contain corrupted images or labels. Log the indices of the batches with the largest loss and look at them.
- **Validation better than training.** Normal with strong augmentation or dropout, because the training loss is measured on harder, augmented inputs.
- **Validation accuracy flat while validation loss rises.** The network becomes overconfident on the examples it gets wrong; the ranking is fine, but the probabilities are not (calibration in DLCV 02).
- **A drop at every milestone.** Expected with step decay. If the curve is still falling steeply at the end of a cosine schedule, the budget is too short.
- **Good validation, bad deployment.** Suspect a preprocessing mismatch, or *shortcut learning*: the network relies on a cue that the validation set shares with the training set but deployment data lacks [37] (DLCV 02).

Gradient norms, dead units and NaNs

Plain version. Log a few health numbers every few hundred steps: the size of each layer's gradient, the fraction of units that never activate, and whether anything has become NaN. When training goes wrong, these numbers usually go wrong first.

The snippet computes two of them after three epochs at three constant learning rates. It reuses `train`, `forward`, `loss_and_grads` and `constant` from above.

```
def health(P, X, y):
    _, G = loss_and_grads(P, X, y)
    h = forward(P, X)[1]                                # hidden ReLU activations
    dead = (h.max(0) == 0).mean()                       # units that are zero for every input
    norms = [np.linalg.norm(gW) for gW, _ in G]
    return norms, dead

for peak in (0.01, 0.1, 0.3):
    P, log, _ = train(constant, peak=peak, epochs=3)
    norms, dead = health(P, X_tr, y_tr)
    print(f"lr {peak:<5} train loss {log[-1, 0]:9.3g}  grad norms "
          f"{norms[0]:9.3g} {norms[1]:9.3g}  dead units {dead:.0%}")
```

```
lr 0.01  train loss     0.127  grad norms    0.0664    0.0496  dead units 0%
lr 0.1   train loss      3.61  grad norms      1.44      2.11  dead units 0%
lr 0.3   train loss   2.1e+08  grad norms  1.02e+04  1.31e+04  dead units 0%
```

The gradient norms grow by more than five orders of magnitude between the healthy and the diverging run, long before the loss becomes NaN, so logging them is cheap insurance. Gradient

clipping (ML 02) is a sensible guard, but it hides a too-large learning rate rather than fixing it. No units die here, because this toy fails by explosion. A **dead ReLU** outputs zero for every input, so it gets no gradient and never recovers; many dead units, typically after a huge step or with strongly negative biases, mean lost capacity.

When the loss becomes NaN or infinite, find the *first* operation that produced a non-finite value rather than clipping everything: check inputs, then the forward pass, the loss, the gradients and the update size, and reproduce in float32 before blaming mixed precision. ML 01C gives this order in detail, and its section on checking derivatives shows how to test a hand-written backward pass.

Reproducibility and fair experiments

Plain version. A result is only useful if it can be repeated and if the comparison behind it is fair. Fix the seeds, log what you did, and run each setting several times, because random chance alone can move a score by more than the improvement you hope to show.

Seeds, determinism and logging

Seed every random number generator: Python's, NumPy's, the framework's and the data-loader workers'. Even then, runs can differ: the PyTorch documentation states that completely reproducible results are not guaranteed across releases, commits or platforms, or between CPU and GPU. Within one setup it recommends `torch.use_deterministic_algorithms(True)`, disabling cuDNN benchmarking, and seeding data-loader workers through `worker_init_fn` and a `generator`, and it warns that deterministic operations are often slower [38]. Log enough to rerun the experiment: code version, configuration, data version and split identifiers, seeds, environment and hardware, and the curves of loss, metrics, learning rate and gradient norm (ML 10).

How much do seeds matter?

The seed changes the initialization and the order of the data, and therefore the result. The snippet trains two configurations on the toy, A (warmup + cosine) and B (one-cycle), each with five seeds, and compares them pair by pair (it reuses `train` and the schedule functions):

```
seeds = range(5)
A = [train(warmup_cosine, peak=0.1, seed=s)[1][-1, 3] for s in seeds]
B = [train(one_cycle, peak=0.1, seed=s)[1][-1, 3] for s in seeds]
print("A (warmup + cosine):", np.round(A, 3))
print("B (one-cycle):      ", np.round(B, 3))
d = np.array(B) - np.array(A)                          # paired: same seed, same data order
```

```
print(f"A mean {np.mean(A):.3f} +- {np.std(A, ddof=1):.3f}   B mean {np.mean(B):.3f} +-
 ↪  {np.std(B, ddof=1):.3f}")
print(f"paired difference B - A: mean {d.mean():+.3f}, range {d.min():+.3f} to
 ↪  {d.max():+.3f}")
```

```
A (warmup + cosine): [0.971 0.962 0.956 0.956 0.959]
B (one-cycle):       [0.956 0.952 0.952 0.978 0.956]
A mean 0.961 +- 0.007   B mean 0.959 +- 0.011
paired difference B - A: mean -0.002, range -0.016 to +0.022
```

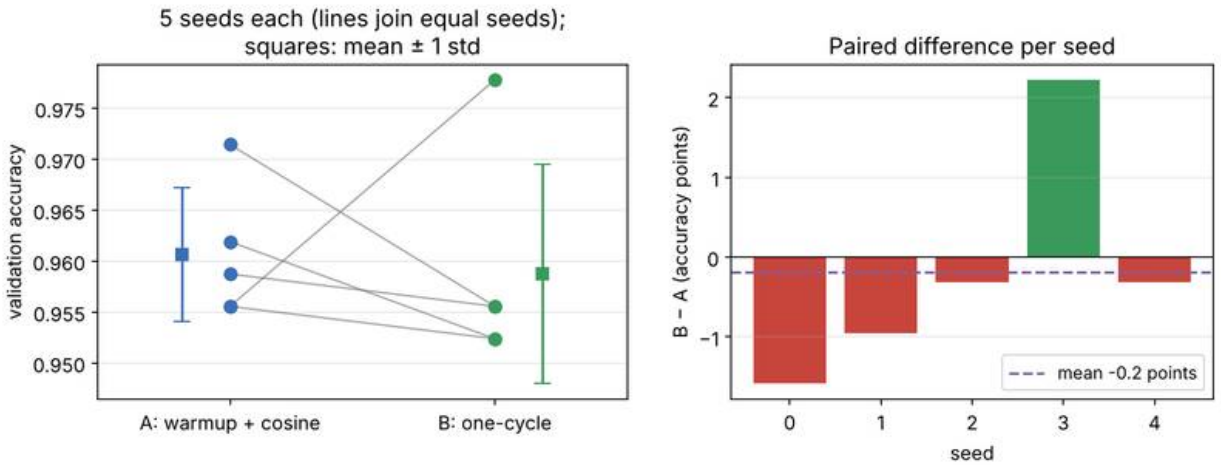


Figure 7 — Seed variance on the toy (our runs, validation set of 315 images). A single seed could support either conclusion: seed 3 says B wins by 2.2 points, seed 0 says A wins by 1.6. Over five seeds the two are indistinguishable.

With only seed 3 you would report that one-cycle wins by 2.2 points; with seed 0, that it loses by 1.6. Over five seeds the mean difference is 0.2 points with a spread far larger than that: no detectable difference on this toy (and one validation image is worth about 0.3 points here). Report the mean and standard deviation over seeds, compare configurations **paired** on the same seeds and splits, and use the confidence intervals and paired tests of ML 10 before claiming an improvement. Published large-scale results do this too: Goyal et al. report every ImageNet error as a mean and standard deviation over five trials [9].

Baselines, ablations and budgets

A fair experiment changes one thing and holds everything else fixed:

- **Baselines.** Always include a simple baseline and the previous best recipe, trained by you in the same pipeline. A new method compared with a baseline from an old paper mixes the effect of the method with the effect of the recipe (DLCV 02).

- **Equal tuning budget.** Give every method the same number of trials for its hyperparameters. A method tuned for a week against a baseline run once wins by tuning, not by design.
- **Equal training budget.** Compare at the same number of steps or the same compute; a longer schedule alone can explain a gain.
- **Ablations.** Remove one component at a time from the full method, with several seeds each, and report the drop. An ablation that changes two things at once cannot tell you which one mattered.
- **One test evaluation.** Make every decision on validation data; evaluate the final choice on the test set once.

The Deep Learning Tuning Playbook organizes this into an incremental strategy: decide on the goal of each round of experiments, separate the hyperparameters being studied from those that must be re-tuned for each setting ("nuisance" hyperparameters) and those held fixed, and adopt a change only when the evidence is strong [26].

Modern view

Plain version. The core habits of this chapter are stable. Recent work changes how training methods are compared, how much of the schedule must be fixed in advance, and how large pretrained models are adapted.

What the surveys and guides say

Practical training knowledge lives more in reviews and guides than in single papers. Bengio's chapter in *Neural Networks: Tricks of the Trade* collects recommendations for gradient-based training, from the learning rate and mini-batch size to early stopping and hyperparameter search [12]. The Deep Learning Tuning Playbook covers choosing the architecture, optimizer, batch size and initial configuration, an incremental, scientific tuning process, how long to train, and pipeline details such as checkpoint selection and experiment tracking [26]. Shorten and Khoshgoftaar organize image augmentation into geometric and color transformations, kernel filters, mixing images, random erasing, feature-space and adversarial augmentation, generative and style-transfer augmentation, and meta-learned policies, and also discuss test-time augmentation [7]. Xin et al. divide parameter-efficient fine-tuning of vision models into addition-based, partial-based, unified-based and multi-task tuning, and propose a unified benchmark [39].

Comparing training methods fairly is itself hard. Dahl et al. argue that the community cannot reliably identify training-algorithm improvements, because it is unclear when training is complete, results are sensitive to workload details, and methods that need tuning are hard to compare fairly; they introduced the AlgoPerf benchmark, which measures time to reach a target result on fixed hardware across several workloads [40].

The state of the art, 2023–2026

- **Benchmarked training algorithms.** The first AlgoPerf competition drew 18 submissions from 10 teams. In the ruleset where submitters provide a search space, the winner used non-diagonal preconditioning and beat popular methods such as Adam even on wall-clock time; in the ruleset with no tuning allowed, the winner was based on Schedule-Free AdamW [41].
- **Fewer schedules to tune.** Defazio et al.'s Schedule-Free approach removes the need to know the stopping step in advance by unifying scheduling and iterate averaging, without new hyperparameters beyond those of momentum optimizers [42]. A simpler alternative is a constant learning rate followed by a short "cooldown" decay, named warmup-stable-decay (WSD) in MiniCPM [43]. Hägele et al. found that this scales as predictably as a cosine schedule, that a cooldown over a modest fraction of training is enough, and that weight averaging improves results along the trajectory at no extra training cost; their experiments are on language models, so verify the transfer to vision workloads yourself [44].
- **Weight averaging everywhere.** Beyond the EMA study above [29], averaging is also used in fine-tuning: WiSE-FT interpolates between the zero-shot and fine-tuned weights of a pretrained model and improves robustness under distribution shift [45], and model soups average several fine-tuned models to improve accuracy without extra inference cost [46].
- **Fine-tuning foundation models.** With very large pretrained backbones, full fine-tuning can be replaced by linear probing, LP-FT [35], or parameter-efficient methods such as LoRA [33], [39]. The practical questions shift from "which architecture" to "which pretrained model, and how much of it to adapt".

Open problems

1. **Hyperparameters that transfer.** Best learning rates, warmup lengths and weight decays still change with model size, batch size and dataset; the range test and the linear scaling rule are approximations, not laws.
2. **Fair, affordable comparison.** Tuning budgets differ between papers, and seeds are often too few to separate methods. Benchmarks such as AlgoPerf address this for training algorithms [40], but no comparable standard exists for most vision tasks.
3. **Determinism at scale.** Bit-exact reproducibility is not guaranteed across hardware and software versions [38], which makes small regressions hard to diagnose in large training runs.
4. **Augmentation for dense and specialized tasks.** How to choose augmentation automatically for detection, segmentation, medical and remote-sensing images, where geometry and intensity carry label information, is still an open question.
5. **Fine-tuning without forgetting.** Adapting a foundation model to a narrow task while keeping

its robustness out of distribution is still an active research question [35], [45].

Key takeaways

- Do cheap checks first: look at a batch after the full pipeline, check the initial loss (about $\ln K$), and overfit one batch before training the full model.
- Keep data honest: split by the unit that will be new, compute normalization statistics on the training split (or use the pretrained model's), and transform boxes, masks and keypoints together with the pixels.
- Find the learning rate with a short range test; use the minimum as the peak of a schedule with warmup and decay, and a smaller value (a tenth is a simple rule of thumb) for a constant rate.
- Warmup protects the first steps; the decay at the end matters more than its exact shape. Warmup + cosine is a safe default; scale the rate linearly with the batch size, with warmup, and recompute schedules in steps.
- Gradient accumulation equals a larger batch only for mean losses over equal-size micro-batches and without batch normalization; weight per-pixel and per-box losses by their total counts.
- Regularizers interact (a large learning rate is one): add them one at a time, use decoupled weight decay without decay on biases and normalization parameters, and keep the best checkpoint by validation.
- Fine-tune gently: smaller rates for earlier layers, a trained head before full fine-tuning when robustness matters, warmup, and care with BN on small batches.
- One seed proves little: report mean and spread over several seeds, compare paired runs with equal budgets, ablate one change at a time, and open the test set once.

Exercises

1. **Range test, then schedule.** Rerun the range test with batch sizes 8, 16 and 64 (change `bs`). How does the learning rate of the minimum move? Then train with `warmup_cosine` at each batch size, using a peak taken from its own range test, and compare with the linear scaling rule.

   Hint

   In our runs (seeds 0 and 1) the minimum sat at about 0.15–0.16 for batch 8, 0.13–0.14 for batch 16 and 0.35 for batch 64: it moves up with the batch size, but not in exact proportion, and a short test is noisy. Remember that T in steps shrinks as the batch grows at a fixed number of epochs, so the larger batches also get fewer updates.

2. **A one-batch test that passes for the wrong reason.** Shuffle `y_tr[:16]` (so labels no longer match images) and run `overfit_one_batch`. What happens to the loss, and what does that tell you about what the test can and cannot detect? Which check from the checklist would have caught the shuffled labels?

Hint

The loss still goes to nearly zero (0.0056 in our run with the labels permuted by `np.random.default_rng(0)`), because a network can memorize any 16 labels, as Zhang et al. showed for random labels at scale [36]. The test checks the model, the loss and the update, not the data. Looking at a batch with its labels drawn on it (item 2) catches misaligned labels; so does a validation accuracy near chance.

3. **Flip a keypoint.** Extend the `hflip` function so it also transforms a list of keypoints given as `(x, y, name)` with names such as `left_eye` and `right_eye`. Write a check that fails if a flipped `left_eye` is still called `left_eye`.

Hint

The coordinates follow $x \to W - 1 - x$ for pixel centers (or $W - x$ for continuous coordinates, matching the box convention used here), y unchanged. Then swap every left/right name pair with a lookup table. Applying the flip twice must return the original keypoints exactly, which is a good test.

4. **Accumulation with ignored pixels.** A segmentation loss averages the per-pixel cross-entropy over pixels whose label is not "ignore". Micro-batch 1 has 1,000 valid pixels with mean loss 0.5; micro-batch 2 has 3,000 valid pixels with mean loss 0.9. What is the mean loss of the combined batch, and what does averaging the two micro-batch means give instead?

Hint

The combined mean is $(1000 \cdot 0.5 + 3000 \cdot 0.9)/4000 = 0.8$. Averaging the means gives $(0.5 + 0.9)/2 = 0.7$, which underweights the micro-batch with more valid pixels. The same weighting applies to the gradients: accumulate sums and divide by the total count.

5. **How many seeds?** Using `train`, run configurations A and B with 10 seeds each instead of 5. Does the mean difference change sign? Estimate the standard error of the paired mean difference, $s_d/\sqrt{n}$, where s_d is the standard deviation of the per-seed differences and n the number of seeds, and decide whether a 0.5-point improvement could be detected with 10 seeds.

Hint

In our run with seeds 0–9 the sign does change: the mean difference B − A becomes about +0.6 points, against −0.2 points with five seeds. With $s_d \approx 1.3$ points, the standard error is $1.3/\sqrt{10} \approx 0.4$ points, so a 0.5-point improvement is only about one standard error away from zero: not convincing. Larger validation sets and more seeds both help; ML 10 shows paired bootstrap intervals.

6. **Fine-tuning plan.** You have 2,000 labeled X-ray images, a backbone pretrained on natural photos and one GPU that fits 8 images per batch. Write a training plan: normalization statistics, augmentations (which ones are unsafe?), how to handle batch normalization, the fine-tuning strategy, the learning-rate procedure and how you will decide that a change helped.

Hint

Use the pretrained normalization statistics; avoid intensity augmentations that change diagnostic values, and be careful with flips if laterality matters; freeze the backbone BN statistics (or use group normalization in new layers) because 8 images per batch give noisy statistics; start with a linear probe, then LP-FT with smaller rates for early layers and a short warmup; run a range test for the head; split by patient; compare variants over several seeds with paired intervals.

Supplementary notes

These notes are optional: each takes one point from the main text and treats it in more detail. Nothing in the main text, code or exercises depends on them.

Cosine annealing with warm restarts

This note gives the full SGDR schedule, of which the main text uses only the single cosine decay.

Loshchilov and Hutter's SGDR (stochastic gradient descent with warm restarts) runs several cosine decays one after another [15]. Within the i-th run, the rate is

$$\eta_t = \eta^i_{\min} + \tfrac{1}{2}\left(\eta^i_{\max} - \eta^i_{\min}\right)\left(1 + \cos\left(\pi \frac{T_{\text{cur}}}{T_i}\right)\right),$$

where $\eta^i_{\max}$ and $\eta^i_{\min}$ are the upper and lower rates of that run, T_{cur} is the number of epochs since the last restart (updated at every batch, so it takes fractional values), and T_i is the length of run i in epochs. When $T_{\text{cur}} = 0$ the rate is $\eta^i_{\max}$; when $T_{\text{cur}} = T_i$ it reaches $\eta^i_{\min}$, and the next run restarts at the top. The lengths grow geometrically, $T_{i+1} = T_{\text{mult}} \cdot T_i$, starting from T_0; the paper studies settings such as $T_0 = 1, T_{\text{mult}} = 2$ and $T_0 = 10, T_{\text{mult}} = 2$ [15].

Steps. (1) Choose T_0 and T_{mult}. (2) Track T_{cur} in fractional epochs. (3) At $T_{\text{cur}} = T_i$, set $T_{\text{cur}} = 0$ and $T_{i+1} = T_{\text{mult}} T_i$.

Why restart? The paper's motivation is *anytime* performance: at the end of every run the rate is small, so the model is in a good state at many points of training, not only at the end. The paper also builds an ensemble from the snapshots taken just before each restart [15].

Pitfalls. The total length should end exactly at the end of a run; with $T_0 = 1$ and $T_{\text{mult}} = 2$ that means $1+2+4+\ldots$ epochs, for example 31 or 63. A run that is cut in the middle of a cycle ends at a high rate and gives a poor final model. The jump back to $\eta_{\max}$ at each restart briefly increases the loss, which is expected; if you combine restarts with a warmup, decide explicitly whether each restart repeats it.

Gradient accumulation when the loss is not a plain mean

This note gives the correct weighting when micro-batches contribute different numbers of items to the loss, and lists what else must follow the optimizer step instead of the micro-batch.

Suppose micro-batch j contains n_j loss items (valid pixels, matched boxes or tokens) with summed loss $S_j = \sum_{i=1}^{n_j} \ell_{ij}$, where ℓ_{ij} is the loss of item i in micro-batch j. The loss of the full batch is the mean over all items,

$$L = \frac{\sum_{j=1}^{k} S_j}{\sum_{j=1}^{k} n_j}, \qquad \nabla_\theta L = \frac{\sum_{j=1}^{k} \nabla_\theta S_j}{N}, \quad N = \sum_{j=1}^{k} n_j,$$

where k is the number of micro-batches and N the total item count. Averaging the micro-batch means, $\frac{1}{k}\sum_j S_j/n_j$, gives each micro-batch the same weight whatever its size, which is a different objective whenever the n_j differ (Exercise 4).

Steps. (1) For each micro-batch, compute the *summed* loss S_j and the count n_j. (2) Backpropagate S_j and add the gradients. (3) Before the optimizer step, divide the accumulated gradient by N. If N is only known after all micro-batches (it usually is), keep the gradients unscaled until then; in data-parallel training, N must be summed across devices too.

Counters that follow the optimizer step. The learning-rate schedule, weight decay, gradient clipping and the EMA update should all advance once per *optimizer* step, not once per micro-batch. Clipping must be applied to the accumulated gradient, after the division by N; clipping each micro-batch separately clips a different quantity.

What accumulation cannot change. Batch normalization still sees only the micro-batch; so do in-batch negatives in contrastive losses, where the number of negatives per example is set by the micro-batch size. Dropout masks differ per micro-batch, which is harmless. With mixed precision and a loss scaler, the PyTorch documentation specifies that the gradients stay scaled and the scale factor stays constant during one accumulation cycle, and that the inf/NaN check, step skipping and scale update happen once per effective batch [47].

EMA of weights: choosing the decay

This note explains how to choose the EMA decay μ, why the average is biased at the start, and how batch normalization affects it.

Effective window. After n updates, the EMA is a weighted sum of past weights, $\bar\theta_n = (1-\mu)\sum_{s=1}^{n} \mu^{n-s}\theta_s + \mu^n \bar\theta_0$, where θ_s are the weights after step s and $\bar\theta_0$ is the initial average. The weight of step s falls by a factor μ per step, so the mean age of the averaged weights is $\mu/(1-\mu)$ steps, about $1/(1-\mu)$ for μ near 1. Choose μ so that this window is a small fraction of the run: with $\mu = 0.999$ and a run of 885 steps, as in the main text, the average is dominated by early weights.

Bias at the start. If $\bar{\theta}_0$ is the initial (random) weights, the term $\mu^n \bar{\theta}_0$ dies out slowly: after $n = 1/(1-\mu)$ steps its weight is still about $e^{-1} \approx 0.37$. Two fixes follow from the formula. Either start the EMA only after warmup, from the current weights, or start from $\bar{\theta}_0 = 0$ and divide by the total weight, $\hat{\theta}_n = \bar{\theta}_n/(1-\mu^n)$, the same bias correction Adam applies to its moment estimates; the weights $(1-\mu)\mu^{n-s}/(1-\mu^n)$ then sum to one at every step.

Batch normalization. The EMA averages the learnable weights, but BN layers also store running statistics computed for the *current* weights. Morales-Brotons et al. found that BN is the limiting factor when choosing the decay: without recomputing the statistics, long averaging windows fail, while recomputing them allows larger windows [29]. SWA, which averages checkpoints with equal weight, recomputes BN statistics with an extra pass over the training data after training [28]. To recompute, reset the running statistics, run the averaged model in training mode without gradient updates over the training data (or a large sample of it), then switch to evaluation mode.

Evaluation. Evaluate both the raw and the averaged weights at every validation step and keep the better one; it costs one extra forward pass over the validation set. EMA models are also often used as "teacher" models in larger training pipelines [29].

Discriminative learning rates and gradual unfreezing

This note shows how to set a separate learning rate for each layer group and how to unfreeze a pretrained network in stages.

Discriminative learning rates. Split the parameters into groups $\theta^1, \dots, \theta^L$ from the input ($l = 1$) to the head ($l = L$), and give group l its own rate η^l. The update is $\theta^l_t = \theta^l_{t-1} - \eta^l \nabla_{\theta^l} J$, where J is the training objective. Howard and Ruder chose the head's rate η^L by fine-tuning only the last layer and then set $\eta^{l-1} = \eta^l/2.6$ for each lower layer [34]; the factor 2.6 is an empirical choice from their language-model experiments, not a universal constant. The general pattern is a geometric decay,

$$\eta^l = \eta^L \, \alpha^{L-l}, \qquad 0 < \alpha \leq 1,$$

where α is the per-group decay factor: $\alpha = 1$ is ordinary fine-tuning, and small α approaches freezing the early layers. In practice, use groups such as the stem, each backbone stage and the head, rather than every layer.

Gradual unfreezing. Instead of training everything from the start, unfreeze the last layer group first and train for one epoch, then unfreeze the next lower group and train all unfrozen groups for another epoch, and continue until the whole network trains [34]. The motivation is the generality gradient measured by Yosinski et al.: later layers are the most task-specific and should adapt first [32].

Edge cases. (1) Frozen layers that contain BN should normally be in evaluation mode too, otherwise their running statistics keep changing even though their weights do not. (2) Keep frozen

parameters out of the optimizer, or give them no gradient, so that no update of any kind, including weight decay, touches them. (3) When a group is unfrozen, its optimizer state (momentum, or Adam's moment estimates) starts from zero; a short warmup for that group avoids a sudden jump. (4) A schedule defined as a multiplier of each group's base rate keeps the ratios η^l/η^L fixed during warmup and decay.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Accurate, Large Minibatch SGD (Goyal et al.)

The problem. Training on many GPUs at once requires a large total batch, because each GPU needs enough work to be efficient. But larger batches had been reported to hurt accuracy. Goyal et al. asked whether this loss was an optimization problem that could be fixed [9].

The idea. Two rules: scale the learning rate linearly with the batch size, and ramp it up gradually over the first epochs. The scaling rule follows from a simple comparison. k SGD steps with small batches and rate η move the weights by η times the sum of k small-batch mean gradients, each taken at a slightly different point; one step with a k-times larger batch moves them by $\hat{\eta}$ times the mean of the same gradients taken at the starting point. If the gradients barely change over the k steps, the two updates match when $\hat{\eta} = k\eta$. The paper names the two situations where that assumption fails: early in training, when the network changes quickly, which the warmup addresses, and beyond some batch size, where accuracy degrades.

What it showed. ResNet-50 on ImageNet with a batch of 8,192 images across 256 GPUs, trained in one hour, matched the accuracy of the batch-256 baseline [9].

How to read its table. The key table compares warmup strategies at batch 8k with the baseline, each as mean and standard deviation of the top-1 validation error over 5 trials: baseline 23.60% $\pm$ 0.12, no warmup 24.84% $\pm$ 0.37, constant warmup 25.88% $\pm$ 0.56, gradual warmup 23.74% $\pm$ 0.09 [9]. Read it in three steps. First, the spread of each row: differences smaller than about two standard deviations are not meaningful. Second, the surprise: a constant low rate for warmup is *worse* than no warmup, because the switch to the full rate causes a spike from which training never fully recovers. Third, the authors attribute the no-warmup gap to higher *training* error, not to worse generalization, which is what makes it an optimization problem.

Implementation pitfalls it lists. Scaling the cross-entropy loss is not equivalent to scaling the learning rate when weight decay is present; momentum needs a correction when the learning rate changes if the implementation folds the rate into the momentum buffer; BN statistics are

computed per GPU over 32 images, and this per-worker batch is part of the definition of the loss; the zero-γ initialization of the last BN in each residual block helps, especially at large batches.

Limits. The experiments are ImageNet classification with SGD and momentum, plus transfer of the pretrained models to detection and instance segmentation; the rule is an approximation whose limit (about 8k images in their setting) depends on the task and model, and it says nothing direct about adaptive optimizers.

Why it matters here. It is the source of two defaults in this chapter, linear scaling with warmup and zero-γ residual initialization, and an example of reporting variance over several trials.

Fine-Tuning can Distort Pretrained Features (Kumar et al.)

The problem. Given a good pretrained model, should you train only a new linear head (linear probing) or update all weights (fine-tuning)? Fine-tuning is known to be more accurate on data like the training data. Kumar et al. asked what happens on shifted data [35].

The idea. When fine-tuning starts from a fixed or random head, the head and the feature extractor are optimized together. The features of the in-distribution training inputs are updated to suit the head, while the features of out-of-distribution inputs change less, so the final head fits the *distorted* in-distribution features and works poorly on the others. The authors prove this tradeoff for overparameterized two-layer linear networks and show that, in that setting, early stopping does not fix it: fine-tuning never passes through parameters that do better out of distribution than linear probing. Their remedy is to train the head first (linear probing), then fine-tune everything starting from that head (LP-FT).

What it showed. On ten distribution-shift benchmarks, fine-tuning was on average 2% more accurate in distribution but 7% less accurate out of distribution than linear probing, and LP-FT beat full fine-tuning by about 1% in distribution and 10% out of distribution [35].

How to read its results. Look at two columns for every method: in-distribution (ID) and out-of-distribution (OOD) accuracy. A method that wins ID and loses OOD is fitting features specific to the training distribution. The gaps depend on how good the pretrained features are and how large the shift is: the paper's claim is that fine-tuning can underperform when the features are good and the shift is large, not that it always does.

Limits. The analysis assumes good pretrained features; with poor features or small shifts, plain fine-tuning can be the better choice. LP-FT adds a stage whose length and learning rate need tuning.

Why it matters here. It explains why "train the head first" is a sensible default when a robust model matters, and why the head's initialization is part of a fine-tuning recipe, not a detail.

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DLCV 01B • Building Blocks and How to Compose Them: Classification and Restoration Networks

A catalogue of the blocks deep vision networks are built from, the rules for composing them into classifiers and into image-restoration networks, side-by-side comparisons of matched models, and how to explore a design space without fooling yourself.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-01b-building-blocks/

What you'll learn

- The vocabulary of a network's anatomy (stem, stages, blocks, downsampling layers, neck, head) and a catalogue of the blocks vision networks are built from, each in one plain line with a precise version where it matters.
- How the same blocks are composed differently for **classification** (shrink the resolution, widen the features, pool, classify) and for **image restoration** (keep the resolution or come back to it, learn a residual, keep the intensity range).
- Why some blocks that help classifiers hurt restorers: batch normalization, transposed convolutions and early downsampling, each shown with a small NumPy experiment.
- How to read three matched pairs (ResNet versus EDSR and NAFNet, Swin versus SwinIR, ViT versus Restormer) as lists of changes with a reason for each.
- How to explore a design space honestly: reading ablations, changing one thing at a time, and comparing populations of models as RegNet does.
- The composition mistakes that most often cost accuracy, memory or a week of debugging.

The big picture

A vision network looks complicated in a paper figure, but it is built from a small number of block types, repeated and connected. Two networks that use the same blocks can behave very differently because they are **composed** differently: where the resolution changes, where the skips go, where normalization sits, how the output is formed. This chapter is about those composition decisions.

> **Explain like I'm five**
>
> Think of a toolbox: saws, brushes, glue. A furniture maker uses them to turn a big log into one small chair leg: cut away most of the wood, keep only what matters. A picture restorer uses the same tools very differently: the painting must come back the same size, every brushstroke in place, just cleaner. A classifier is the furniture maker: it throws away pixels until only the answer ("cat") is left. A restoration network is the restorer: it must give back every pixel. Same tools, opposite plans.

Why it matters. Practical work is rarely inventing a block. It is choosing a backbone, adapting it to a new task, or deciding which of a paper's ten changes actually helped. A classifier backbone used unchanged for denoising throws away detail, and a block that helped in one paper may not help in your design space.

Where the basics are. The blocks themselves are taught elsewhere on this site, and this chapter links there instead of repeating them: convolution, receptive field and residual blocks in ML 03 and DLCV 01; the Transformer block, Swin and efficient attention in ML 04; the classification backbones in DLCV 02; encoder–decoders and ASPP in DLCV 03B; restoration backbones and upsampling in DLCV 04 and DLCV 04A. All code here uses NumPy and SciPy only, runs on a CPU in seconds, and uses synthetic data or `skimage.data`.

Anatomy of a network

Plain version. Almost every vision network has three parts: a **stem** that turns pixels into features, a **body** of repeated blocks, and a **head** that turns features into the answer. What differs between tasks is mostly what happens to the image size along the way.

Precise version. Write a feature map as a tensor of shape $C \times H \times W$, where C is the number of channels (the **width**) and $H \times W$ the spatial size. The common parts are:

- **Stem**: the first one or two layers. ResNet uses a 7×7 convolution with stride 2 and a 3×3 max-pool with stride 2, so the body starts at $1/4$ of the input resolution [1]. ConvNeXt replaces this with a "patchify" 4×4 convolution with stride 4 [2]; ViT cuts the image into 16×16 patches [3]. Restoration networks use a single 3×3 convolution at full resolution, which SwinIR calls **shallow feature extraction** [4].
- **Stage**: a run of blocks that share one resolution and one width. A **block** is the repeated unit (a residual block, a Transformer block).
- **Downsampling layer** (transition): halves H and W, usually while doubling C. It can be a strided convolution inside the first block of a stage (ResNet), a separate 2×2 convolution with stride 2 (ConvNeXt, NAFNet), patch merging (Swin), or pixel unshuffle (Restormer).
- **Neck**: an optional module that mixes features from several stages, such as the Feature Pyramid Network (FPN) used by detectors [5].
- **Head**: global average pooling and a linear layer for classification; a final convolution, and for super-resolution an upsampler, for restoration.

The **output stride** of a stage is the ratio between the input size and the stage's feature-map size. ResNet's four stages have output strides 4, 8, 16 and 32 [1]; so do Swin's [6]. Networks that

keep one resolution and width throughout, like ViT, are **isotropic**; those with stages at decreasing resolution are **hierarchical**.

ResNet also states the most copied rule of macro design: when the feature map is halved, the number of filters is doubled "so as to preserve the time complexity per layer" [1]. A 3×3 convolution with C input and output channels on an $H \times W$ map costs $9HWC^2$ multiply-accumulates (MACs); after halving the map and doubling the width it costs $9 \cdot \frac{H}{2} \cdot \frac{W}{2} \cdot (2C)^2 = 9HWC^2$, the same. Figure 1 shows the three macro-architectures this chapter compares.

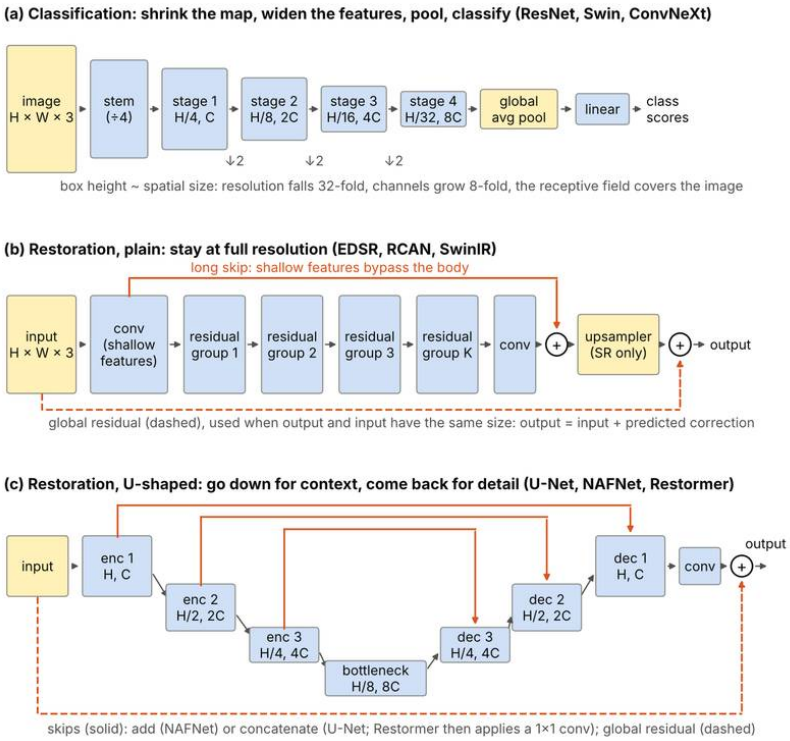


Figure 1 — Three macro-architectures. (a) A classifier is a funnel: resolution falls, width grows, and global pooling removes the spatial grid. (b) A plain restorer keeps full resolution and adds long and global skips. (c) A U-shaped restorer goes down for context and comes back for detail.

The block catalogue

Plain version. A block has two jobs: mix information across **space** (neighboring pixels or tokens) and mix information across **channels** (features at one position). Every block in the catalogue does both, in a different proportion and at a different cost. Around them sit three helpers: skips, normalization and nonlinearities.

The MetaFormer paper makes this split explicit: a Transformer block is a **token mixer** (attention) followed by a **channel mixer** (the MLP), each in a residual branch with normalization, and replacing attention by a simple spatial pooling operator still gave competitive accuracy on ImageNet [7], which suggests that the general structure carries much of the performance [8]. Figure 2 shows the blocks of this chapter in that common format.

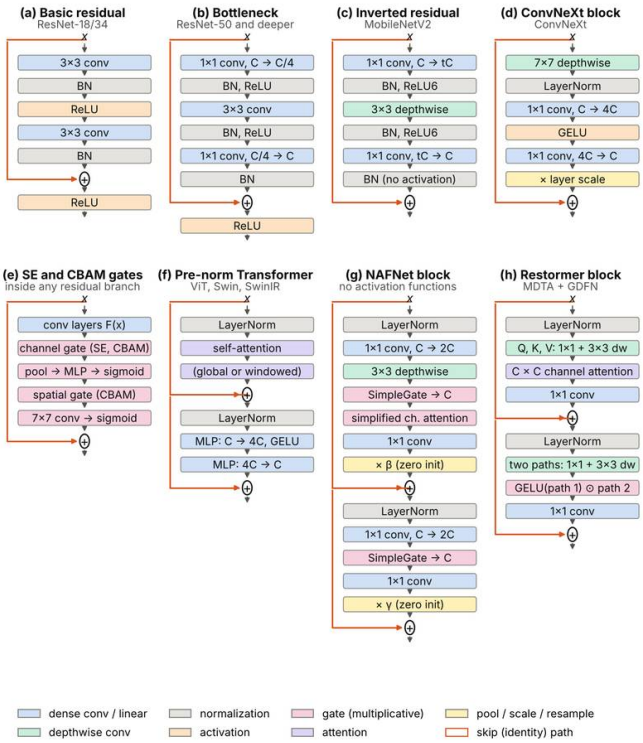


Figure 2 — The block catalogue, drawn in one format. Colors mark the kind of operation; orange arrows are identity skips. (a)–(d) are classification blocks, (e) are attention add-ons, (f) is the Transformer block, (g)–(h) are restoration blocks.

The table gives each block one plain line and says where its basics are taught.

Block	One plain line	Used in	Basics
Residual (basic)	Learn a correction $F(x)$ and add it to the input	ResNet-18/34, EDSR	DLCV 01
Bottleneck	Squeeze channels with 1×1, filter, expand back	ResNet-50 and deeper	DLCV 02
Depthwise separable	Filter each channel alone, then mix channels with 1×1	MobileNet [9]	DLCV 02
Inverted residual	Expand, filter depthwise, project back linearly; skip between thin ends	MobileNetV2 [10]	ML 03

Block	One plain line	Used in	Basics
ConvNeXt block	Large depthwise kernel, then a Transformer-style MLP	ConvNeXt	DLCV 02
SE / CBAM	Reweight channels (and positions) by a gate computed from pooled statistics	SENet, RCAN, NAFNet	DLCV 02 note
Self-attention	Every token looks at every other token	ViT	ML 04
Window attention	Attention inside local windows that shift between layers	Swin, SwinIR	ML 04
Channel (transposed) attention	Attention between channels, so cost grows linearly with pixels	Restormer	this chapter
Gated block	Multiply two feature halves instead of applying an activation	NAFNet, Restormer's GDFN	this chapter

Residual connections and residual scaling

Plain version. A residual block adds its input back to its output, so the block only has to learn a correction, and the identity path carries signal and gradient through any depth. The open questions are where to put normalization and activations relative to the addition, and how large the correction should be at the start.

Precise version. The general form is

$$y = x + \alpha F(x),$$

where x is the block input, F the residual branch (convolutions, attention, normalization), y the output and α a scale on the branch. Three placements of the nonlinearity are common:

- **Post-activation** (original ResNet): $y = \mathrm{ReLU}(x + F(x))$, with BN inside F [1].
- **Pre-activation**: BN and ReLU move inside the branch, before each convolution, so the shortcut is a pure identity; He et al. found this easier to train at very large depth [11]. The pre-norm Transformer block, $y = x + \mathrm{MHA}(\mathrm{LN}(x))$, is the same idea (ML 04).
- **No activation after the addition**: EDSR, MobileNetV2, ConvNeXt, NAFNet and Restormer all end the block with the addition.

Several designs control α explicitly. EDSR found training numerically unstable once the number of feature maps grew large, and fixed it with a constant residual scaling of 0.1 in its final model of 32 blocks with 256 channels [12]. ConvNeXt uses Layer Scale [13], a learnable per-channel α, initialized to 10^{-6} [2]. NAFNet's code multiplies each branch by learnable per-channel parameters initialized to zero, so every block starts as the identity [14]; zero-initializing the last BN scale of a

ResNet branch does the same (DLCV 01A). The shared reason: a deep stack of identity blocks is easy to optimize, and each block can then grow its correction.

Skips also come at two scales. **Short skips** wrap one block; **long skips** wrap a group of blocks or the whole body. RCAN's "residual in residual" structure uses both: residual groups with long skip connections, each containing blocks with short skip connections [15].

Bottlenecks, inverted residuals and depthwise separable convolutions

Plain version. Spatial filtering and channel mixing need not happen in one expensive layer. Bottlenecks run the expensive 3 × 3 on fewer channels; depthwise convolutions filter each channel alone and leave mixing to cheap 1 × 1 layers. These blocks change cost by large factors, so it pays to count.

Precise version. A dense $k \times k$ convolution from a to b channels has k^2ab weights and, at stride 1, costs k^2ab MACs per output pixel. A depthwise $k \times k$ convolution on a channels has k^2a of each.

The ResNet bottleneck reduces C to $C/4$, filters, and expands back [1]. The MobileNetV2 **inverted residual** does the opposite: it expands C to tC with a 1 × 1 convolution (expansion factor $t = 6$ in the paper's main experiments), filters with a 3 × 3 depthwise convolution, and projects back with a **linear** 1 × 1 convolution, without an activation, because the authors argue that a ReLU in the narrow layer destroys information [10]. The paper gives its cost as $h \cdot w \cdot d' \cdot t\,(d' + k^2 + d'')$ multiply-adds for an $h \times w$ input with d' input and d'' output channels [10]. The snippet counts weights and MACs per output pixel for the blocks of Figure 2 at $C = 64$, on a 128 × 128 map, and checks the inverted residual against that formula.

```
# Weights (no biases or norm parameters) and MACs per output pixel of common blocks,
# all with C input and C output channels at stride 1. A dense k x k conv from a to b
# channels has k*k*a*b weights and costs k*k*a*b MACs per pixel; a depthwise one k*k*a.
def blocks(C, n_tokens):
    g = int(2.66 * C)                                   # Restormer GDFN hidden width
    return {                                            # name: (weights, MACs per pixel)
        'basic residual (2 x 3x3)':     (18 * C * C, 18 * C * C),
        'bottleneck (1x1, 3x3, 1x1)':   (2 * C * C // 4 + 9 * (C // 4) ** 2,) * 2,
        'depthwise separable 3x3':      (9 * C + C * C,) * 2,
        'inverted residual, t = 6':     (12 * C * C + 54 * C,) * 2,
        'ConvNeXt (7x7 dw, 4x MLP)':    (49 * C + 8 * C * C,) * 2,
        'NAFNet block':                 (7 * C * C + 18 * C, 6 * C * C + 18 * C),  # SCA
         ↪ acts on a pooled vector
        'SE add-on, r = 16':            (2 * C * C // 16, C),                      # one
         ↪ multiply per pixel
        'ViT block, global attention':  (12 * C * C, 12 * C * C + 2 * n_tokens * C),
        'Swin block, 8 x 8 windows':    (12 * C * C, 12 * C * C + 2 * 64 * C),
        'Restormer block (MDTA+GDFN)':  (4 * C * C + 27 * C + 3 * g * C + 18 * g,
                                         4 * C * C + 27 * C + 2 * C * C // 4 + 3 * g * C +
                                          ↪ 18 * g),  # 4 heads
```

```
    }

C, side = 64, 128
for name, (p, m) in blocks(C, side * side).items():
    print(f"{name:30s} {p:7d} weights  {m:8d} MACs/pixel")

# Check the inverted residual against MobileNetV2's formula h*w*d'*t*(d' + k^2 + d'') per
 ↪  pixel.
t, k = 6, 3
assert blocks(C, 1)['inverted residual, t = 6'][1] == C * t * (C + k * k + C)
```

```
basic residual (2 x 3x3)          73728 weights     73728 MACs/pixel
bottleneck (1x1, 3x3, 1x1)         4352 weights      4352 MACs/pixel
depthwise separable 3x3            4672 weights      4672 MACs/pixel
inverted residual, t = 6          52608 weights     52608 MACs/pixel
ConvNeXt (7x7 dw, 4x MLP)         35904 weights     35904 MACs/pixel
NAFNet block                      29824 weights     25728 MACs/pixel
SE add-on, r = 16                   512 weights        64 MACs/pixel
ViT block, global attention       49152 weights   2146304 MACs/pixel
Swin block, 8 x 8 windows         49152 weights     57344 MACs/pixel
Restormer block (MDTA+GDFN)       53812 weights     55860 MACs/pixel
```

Four lessons come out of these numbers.

1. **At equal width, the blocks differ by more than 10 times in cost.** The bottleneck and the separable block are about 16 to 17 times cheaper than the basic block. Real networks therefore never compare blocks at equal width: MobileNetV2 uses an inverted residual on thin tensors, ResNet-50 uses bottlenecks on wide ones. Compare blocks at equal cost, not at equal C.
2. **Global attention over pixels is out of reach for restoration.** With every pixel of a 128×128 map as a token, the ViT block costs about 37 times more per pixel than a Swin block with 8×8 windows, and the gap grows with image size. This is why restoration Transformers use windows (SwinIR) or attention across channels (Restormer); see ML 04 for the general cost argument.
3. **Attention add-ons are cheap in MACs.** An SE gate adds $C^2/8$ weights and one multiply per channel and pixel; its cost is memory traffic, not arithmetic.
4. **MACs are not latency.** Depthwise layers do few operations per byte moved, so they can run slower than their MAC count suggests; ML 03 and the DLCV 02 efficiency section discuss this. Measure on the target hardware.

Whether a cheaper block is also a better one depends on the design space. RegNet's population study found that its best models used neither a bottleneck nor an inverted bottleneck, and that depthwise convolution did even worse than the inverted bottleneck in its setting [16], while ConvNeXt found the inverted bottleneck slightly better within its own sequence of changes [2]. Both can be true: the comparison depends on the other choices held fixed.

Attention inside a CNN: SE and CBAM

Plain version. A gate looks at a summary of the whole feature map and decides which channels (or positions) to turn up or down for this particular image. It is a cheap way to give a convolutional network some content-dependent, image-wide behavior.

Precise version. Squeeze-and-excitation (SE) computes a channel gate $s = \sigma(W_2\,\delta(W_1\,\bar{u}))$ from the vector $\bar{u}$ of channel means, where W_1 and W_2 are the weights of a two-layer MLP that shrinks the dimension by a ratio r and restores it, δ is a ReLU and σ the sigmoid, and multiplies each channel u_c of the feature map by s_c; the formula and code are in the DLCV 02 note [17]. CBAM extends it in two ways [18]: its channel gate pools each channel by both average and maximum and passes both vectors through one shared MLP, and a second, **spatial** gate pools across channels and applies a 7×7 convolution, so the block can also turn whole regions up or down. CBAM's ablations found the sequential channel-then-spatial order better than running the two gates in parallel [18].

Optional: CBAM's channel and spatial gates (in the Supplementary notes). The two formulas with every symbol defined, and how they relate to SE and to NAFNet's simplified channel attention.

Where it goes. Both SE and CBAM sit on the residual branch, before the addition: SE applies the gate to the non-identity branch "before summation with the identity branch" [17], and CBAM applies its module to the convolution outputs of each ResNet block [18]. RCAN brought channel attention into super-resolution [15]; NAFNet later simplified it to $X \cdot W\,\mathrm{pool}(X)$, a single linear map of the pooled vector with no ReLU or sigmoid, and lost nothing in its ablation [19]. The Modern view's attention survey places SE in the "channel attention" family and CBAM in "channel and spatial" [20].

Token mixers: self-, window- and channel attention

Plain version. Attention mixes information between tokens with weights computed from the content, instead of fixed weights as in a convolution. The design question is **which tokens talk to which**: all of them (global), those in the same local window, or, in Restormer's twist, the channels instead of the pixels.

Precise version. Global multi-head self-attention over N tokens of width C costs about $2N^2C$ MACs for the scores and the weighted sum, on top of $4NC^2$ for the projections; window attention restricts it to $M \times M$ windows, giving $2M^2NC$ (ML 04) [6]. Restormer's **multi-Dconv head transposed attention** (MDTA) instead computes attention between channels [21]:

$$\hat{X} = W_p\,\mathrm{Attention}(\hat{Q},\hat{K},\hat{V}) + X, \qquad \mathrm{Attention}(\hat{Q},\hat{K},\hat{V}) = \hat{V}\cdot\mathrm{Softmax}(\hat{K}\cdot\hat{Q}/\alpha),$$

where X is the input feature map and $\hat{X}$ the output, $Q = W_d^Q W_p^Q Y$ (and likewise K, V) is computed from the layer-normalized input Y by a 1×1 convolution W_p^Q followed by a 3×3 depthwise convolution W_d^Q, the hats denote reshaped matrices $\hat{Q} \in \mathbb{R}^{HW \times C}$, $\hat{K} \in \mathbb{R}^{C \times HW}$, $\hat{V} \in \mathbb{R}^{HW \times C}$, α is a learnable temperature, and W_p without a superscript is the output 1×1 convolution. The attention map $\hat{K}\hat{Q}$ has size $C \times C$ instead of $HW \times HW$, so the cost grows linearly with the number of pixels, and every output pixel still depends on the whole image through the channel statistics. The depthwise convolutions put local context back in. As in ordinary multi-head attention, the channels are split into heads with separate maps.

Restormer's feed-forward part (GDFN) is covered with the gates below. The trade-off between the mixers returns in the restoration sections: window attention mixes space strongly but locally, channel attention reaches globally at low cost, and a comparative study argued that Restormer's spatial mapping is weaker than SwinIR's for super-resolution [22].

Normalization: BN, LN, GN, and when to leave it out

Plain version. A normalization layer rescales features to a standard size and then lets the network learn its preferred scale. The layers differ in **which numbers share statistics**. In classification the choice mostly affects training stability; in restoration the wrong choice can change the brightness and contrast of the output.

Precise version. All the common layers compute $y = \gamma(x-\mu)/\sqrt{\sigma^2 + \epsilon} + \beta$ over some set of values (ML 02 gives the template). For a mini-batch of shape $N \times C \times H \times W$:

Layer	Statistics computed over	Depends on the batch?
BatchNorm (BN) [23]	N, H, W for each channel	yes; running averages at test time
LayerNorm (LN) [24]	the features of one sample (for images, implementations differ: all of C, H, W, or C at each position)	no
InstanceNorm (IN) [25]	H, W for each sample and channel	no
GroupNorm (GN) [26]	H, W and a group of channels, for each sample	no; $G = 1$ gives LN, $G = C$ gives IN

BN is the default in classification CNNs and works well with large batches. Its known failure is small batches: GN's authors report a ResNet-50 on ImageNet with 10.6% lower error than its BN counterpart at a batch size of 2 [26]. What BN computes, and why training and inference differ, is in the DLCV 01 note.

Why restorers dropped BN. EDSR removed BN from its residual blocks, arguing that because BN normalizes the features it removes "range flexibility" from the network; the change improved accuracy and cut training memory by about 40% compared with SRResNet, the residual SR network of the SRGAN paper [27], [12]. ESRGAN added that BN tends to introduce artifacts when training and test statistics differ, especially in deeper networks trained with a GAN loss, and removed

it as well [28]. This is not a law: DnCNN combined residual learning with BN for Gaussian denoising and reported that the two help each other [29]. The snippet shows the mechanism with a fixed, untrained residual branch $r(x)$ (convolution, normalization, ReLU, 1×1 convolution) and the global residual $y = x + r(x)$. The same mid-gray image A is restored inside two different batches.

```
import numpy as np
from scipy.ndimage import gaussian_filter, correlate

rng = np.random.default_rng(0)

def image(level, size=32):
    """A smooth synthetic image with mean brightness `level` (values roughly in [0, 1])."""
    t = gaussian_filter(rng.standard_normal((size, size)), 2)
    return level + 0.15 * t / t.std()

C = 4
W1 = rng.normal(0.1, 0.2, (C, 3, 3))           # 3x3 conv, 1 -> C channels (fixed random
 ↪  weights)
w2 = rng.normal(0.0, 0.05, C)                  # 1x1 conv, C -> 1 channel

def features(batch):                           # (N, H, W) -> (N, C, H, W)
    return np.stack([[correlate(x, W1[c], mode='nearest') for c in range(C)] for x in batch])

def branch(batch, norm, running=None):
    """Residual branch: conv -> normalize -> ReLU -> 1x1 conv."""
    f = features(batch)
    if norm == 'bn-train':                     # per channel, over the whole batch
        mu, var = f.mean((0, 2, 3), keepdims=True), f.var((0, 2, 3), keepdims=True)
    elif norm == 'bn-eval':                    # per channel, stored training statistics
        mu, var = running
    else:                                      # 'instance': per sample and channel
        mu, var = f.mean((2, 3), keepdims=True), f.var((2, 3), keepdims=True)
    h = np.maximum((f - mu) / np.sqrt(var + 1e-5), 0)
    return np.tensordot(w2, h, axes=(0, 1))    # (N, H, W)

train = np.stack([image(rng.uniform(0.3, 0.7)) for _ in range(64)])
f = features(train)
running = (f.mean((0, 2, 3), keepdims=True), f.var((0, 2, 3), keepdims=True))

A = image(0.5)
dark = np.stack([image(0.15) for _ in range(7)])
bright = np.stack([image(0.85) for _ in range(7)])
for norm in ['bn-train', 'bn-eval', 'instance']:
    for name, others in [('dark', dark), ('bright', bright)]:
        batch = np.concatenate([A[None], others])
        out = A + branch(batch, norm, running)[0]          # global residual: y = x + r(x)
        print(f"{norm:8s}  A with 7 {name:6s} images: output mean {out.mean():.3f}")

# Per-sample normalization forgets each image's level and contrast:
B = 0.5 * A + 0.3
r_A, r_B = branch(A[None], 'instance')[0], branch(B[None], 'instance')[0]
```

```
print("instance-norm branch, A vs 0.5*A + 0.3: max |r_A - r_B| =", np.abs(r_A -
 ↪  r_B).max().round(4))
print("outputs with the global residual: means", (A + r_A).mean().round(3), (B +
 ↪  r_B).mean().round(3))
```

```
bn-train  A with 7 dark   images: output mean 0.629
bn-train  A with 7 bright images: output mean 0.482
bn-eval   A with 7 dark   images: output mean 0.509
bn-eval   A with 7 bright images: output mean 0.509
instance  A with 7 dark   images: output mean 0.521
instance  A with 7 bright images: output mean 0.521
instance-norm branch, A vs 0.5*A + 0.3: max |r_A - r_B| = 0.0005
outputs with the global residual: means 0.521 0.575
```

In training mode, BN gives the same image (input mean 0.492) an output brightness of 0.629 or 0.482 depending only on its batch companions (Figure 3). In evaluation mode the batch no longer matters, but the stored statistics come from other images, so the network behaves differently at test time from how it was trained: the mismatch EDSR and ESRGAN describe. Per-sample normalization is batch-independent but blind in the other direction: the branch gives the same output for A and for $0.5A + 0.3$ (up to ϵ), so only the identity path carries the image's own level and contrast. That is why restorers that do normalize (NAFNet, Restormer and SwinIR use LayerNorm) keep it **inside** residual branches with a global residual [19], [21], [4]. The weights are random and untrained; training can partly compensate, but the batch dependence is structural.

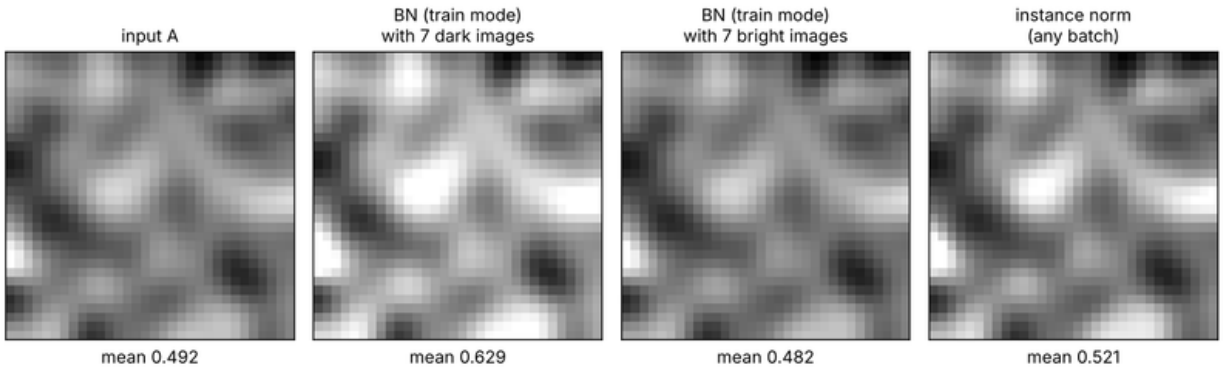


Figure 3 — The same image through the same untrained residual branch. With batch normalization in training mode its output brightness depends on the other images in the batch; with per-sample normalization it does not.

NAFNet's ablation shows the other side: adding LayerNorm to its plain block let it train with a 10 times larger learning rate and raised PSNR [30] by 0.44 dB on the SIDD denoising benchmark [31] and by 3.39 dB on the GoPro deblurring benchmark [32], [19]. Normalization helps restorers too, as long as it does not take away the information the output needs.

Activations and gates

Plain version. A nonlinearity is what lets stacked layers do more than one linear map. The classic choice applies a fixed function to every number (ReLU, GELU). A gate instead **multiplies** two learned features, so one feature decides how much of the other passes. Modern restoration blocks use gates and, in NAFNet's case, nothing else.

Precise version. ReLU, GELU and the SwiGLU feed-forward block [33] are defined in ML 02. The gated linear unit (GLU) of Dauphin et al. computes $f(X) \odot \sigma(g(X))$ with two linear maps f, g and a sigmoid σ [34]. NAFNet's **SimpleGate** drops the sigmoid and the second map [19]:

$$\text{SimpleGate}(X) = X_1 \odot X_2, \qquad X = [X_1; X_2],$$

where X is a feature map with $2C$ channels, X_1 and X_2 are its first and second C channels, and $\odot$ is element-wise multiplication. The product of two linear functions of the input is quadratic, so the block is nonlinear without any activation function; the authors note that GELU itself, $x\,\Phi(x)$ with Φ the standard normal CDF [35], is a special case of a gate. Restormer's GDFN uses $\phi(W_d^1 W_p^1 \,\text{LN}(X)) \odot W_d^2 W_p^2 \,\text{LN}(X)$ with ϕ the GELU, two paths of 1×1 and 3×3 depthwise convolutions, and a channel expansion $\gamma = 2.66$ [21]. In NAFNet's ablation, replacing GELU by SimpleGate gave +0.08 dB on SIDD and +0.41 dB on GoPro [19].

Upsampling: transposed convolution, pixel shuffle and resize-convolution

Plain version. Upsampling makes a feature map larger. There are three common ways: let each input pixel "paint" a learned patch into the output (transposed convolution), compute several values per input pixel and rearrange them (pixel shuffle), or enlarge first and convolve afterwards (resize-convolution). The first two can easily paint a regular checkerboard pattern; the third cannot.

Precise version. The three methods and the exact pixel-shuffle index rule are in DLCV 04 and DLCV 04A [36]. What matters for composition is a counting argument. With stride s, the output pixels fall into s^2 **phases** (their row and column positions modulo s). In a transposed convolution, the output at a given phase receives only some of the kernel's taps. Its value on a flat input equals the sum of those taps, the **phase sum**. If the phase sums differ, a flat input comes out as a periodic pattern. One cause, which Odena et al. call **uneven overlap**, occurs when the kernel size is not divisible by the stride: the phases then receive different numbers of taps [37]. Even with divisible sizes, random weights give unequal phase sums, so a freshly initialized layer still makes a pattern; Aitken et al. list these two causes and strided operations in the loss as the three sources of checkerboard artifacts [38]. A sub-pixel convolution has the same freedom, since each phase has its own filter. A resize-convolution (nearest-neighbor upsampling, then an ordinary

convolution) applies the same full kernel at every output pixel, so all phases see the same sum. The snippet upsamples a flat 16 × 16 input by 2.

```
import numpy as np
from scipy.ndimage import correlate

rng = np.random.default_rng(0)

def transposed_conv(x, w, s):
    """Each input pixel 'paints' a scaled copy of the k x k kernel w; copies are s pixels
     ↪  apart."""
    k, (H, W) = w.shape[0], x.shape
    out = np.zeros((s * (H - 1) + k, s * (W - 1) + k))
    for i in range(H):
        for j in range(W):
            out[s * i:s * i + k, s * j:s * j + k] += x[i, j] * w
    return out

def resize_conv(x, w, s):
    """Nearest-neighbour upsampling by s, then an ordinary 'same' convolution with w."""
    return correlate(np.kron(x, np.ones((s, s))), w, mode='nearest')

def subpixel_conv(x, ws, s):
    """s*s ordinary convolutions whose outputs are interleaved (pixel shuffle)."""
    out = np.zeros((s * x.shape[0], s * x.shape[1]))
    for n, w in enumerate(ws):
        out[n // s::s, n % s::s] = correlate(x, w, mode='nearest')
    return out

x = np.ones((16, 16))                          # flat input: a good upsampler returns a flat
 ↪  image
cases = {
    'transposed k=3, s=2, uniform w': transposed_conv(x, np.ones((3, 3)), 2)[4:-4, 4:-4],
    'transposed k=4, s=2, uniform w': transposed_conv(x, np.ones((4, 4)), 2)[4:-4, 4:-4],
    'transposed k=4, s=2, random w ': transposed_conv(x, rng.uniform(0.5, 1.5, (4, 4)),
     ↪  2)[4:-4, 4:-4],
    'sub-pixel, 4 random 3x3 kernels': subpixel_conv(x, rng.uniform(0.5, 1.5, (4, 3, 3)),
     ↪  2)[4:-4, 4:-4],
    'resize + conv, random 3x3 w   ': resize_conv(x, rng.uniform(0.5, 1.5, (3, 3)), 2)[4:-4,
     ↪  4:-4],
}
for name, y in cases.items():
    phases = [y[a::2, b::2].mean() for a in (0, 1) for b in (0, 1)]     # the four 2x2
     ↪  sub-pixel phases
    print(f"{name}: phase means {np.round(phases, 2)}  relative spread {np.ptp(y) /
     ↪  y.mean():.2f}")
```

```
transposed k=3, s=2, uniform w: phase means [4. 2. 2. 1.]  relative spread 1.30
transposed k=4, s=2, uniform w: phase means [4. 4. 4. 4.]  relative spread 0.00
transposed k=4, s=2, random w : phase means [4.04 3.22 5.01 3.85]  relative spread 0.44
sub-pixel, 4 random 3x3 kernels: phase means [ 8.71 10.13  9.49  8.91]  relative spread 0.15
resize + conv, random 3x3 w   : phase means [7.86 7.86 7.86 7.86]  relative spread 0.00
```

With $k = 3$ and $s = 2$, one phase receives 4 taps, two receive 2 and one receives 1, even with uniform weights: the $4:2:2:1$ pattern is built into the shapes. With $k = 4$ every phase receives 4 taps, so uniform weights give a flat output, but random weights do not. Only resize-convolution is flat for any weights. Figure 4 shows the same four upsamplers on a flat input and on a photograph. Training can push the phase sums together, and Odena et al. report their best results with nearest-neighbor resize-convolution [37]; for pixel shuffle, initializing all phases from one kernel removes the pattern at the start [38], as noted in DLCV 04A.

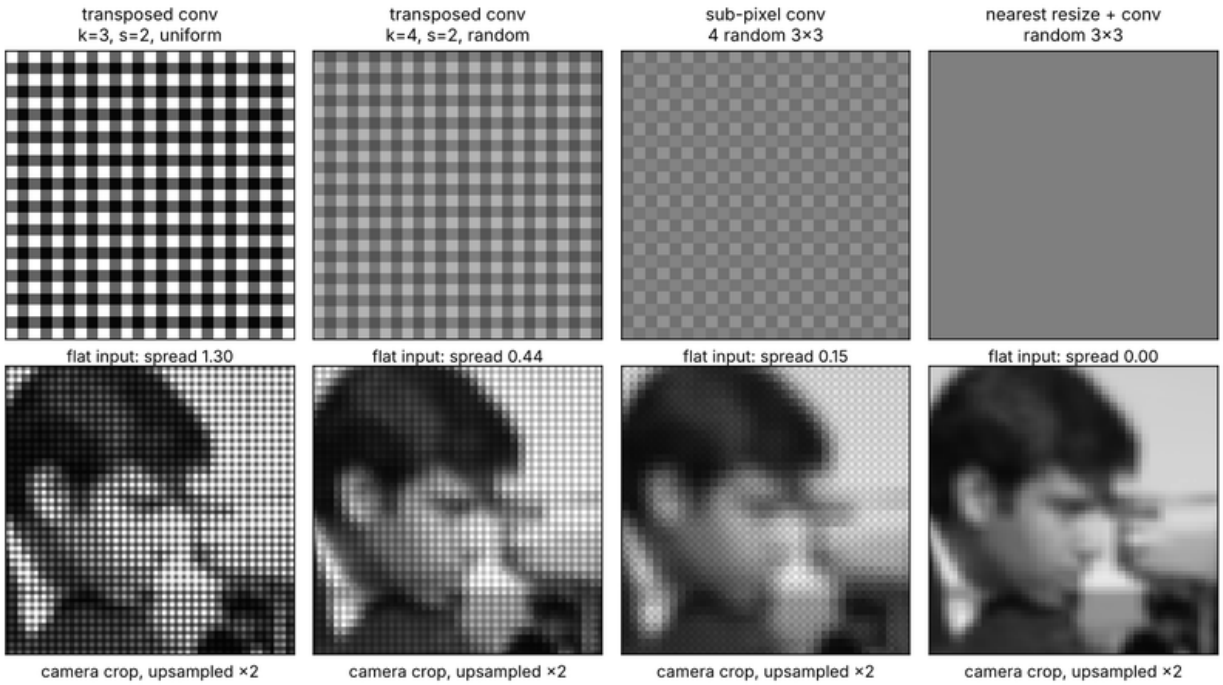


Figure 4 — Checkerboard artifacts from untrained upsamplers. Top: a flat input (relative spread printed below). Bottom: a camera crop upsampled by 2 with the same kernels, rescaled to unit average gain. Only resize-convolution is free of the pattern by construction.

Optional: Phase sums of a transposed convolution in 1-D (in the Supplementary notes). How many taps each output phase receives for any kernel size and stride, and why divisibility is necessary but not sufficient.

Downsampling has matching choices: strided convolution (ResNet), pooling, a separate 2×2 convolution with stride 2 (ConvNeXt, NAFNet [14]), patch merging (Swin) and pixel unshuffle (Restormer [21]). ConvNeXt reports that its separate 2×2 downsampling layers made training diverge until it added normalization layers wherever the spatial resolution changes [2].

Skip connections: add or concatenate

Plain version. A skip connection brings an earlier feature map to a later layer. It can be **added** (the two maps must have the same shape, and the result keeps the width) or **concatenated** (the maps

are stacked along channels, and the next layer decides how to combine them). Concatenation is more flexible and more expensive.

Precise version. Concatenation followed by a 1×1 convolution contains addition as a special case: with encoder map e and decoder map d, each with C channels, the 1×1 convolution with weight matrix $[I \;\; I]$ (two $C \times C$ identity blocks side by side) maps $[e; d]$ to $e + d$. So concatenation can learn anything addition can, and more, at the cost of a wider layer and of storing both maps. The snippet checks the identity and counts the cost of the first convolution after each merge in a 4-level U-Net decoder with widths 32 to 256, on a 256×256 input.

```
import numpy as np

rng = np.random.default_rng(0)

# 1) Concatenation followed by a 1x1 convolution contains addition as a special case.
C, H, W = 8, 5, 5
enc, dec = rng.standard_normal((C, H, W)), rng.standard_normal((C, H, W))
cat = np.concatenate([enc, dec])                        # (2C, H, W)
W_fuse = np.hstack([np.eye(C), np.eye(C)])              # a 1x1 conv is a (C_out x C_in)
 ↪  matrix
print("concat + 1x1 conv [I I] == add:", np.allclose(np.tensordot(W_fuse, cat, 1), enc +
 ↪  dec))

# 2) What each choice costs in the decoder of a 4-level U-Net on a 256 x 256 input.
widths, size = [32, 64, 128, 256], 256
totals = {'add': [0, 0], 'concat': [0, 0], 'concat + 1x1': [0, 0]}
for level, c in enumerate(widths):
    hw = (size // 2 ** level) ** 2                      # pixels at this level
    first = {'add': 9 * c * c,                          # first 3x3 conv after the merge
             'concat': 9 * 2 * c * c,
             'concat + 1x1': 2 * c * c + 9 * c * c}     # 1x1 halves channels, then 3x3
    for k, p in first.items():
        totals[k][0] += p
        totals[k][1] += p * hw
for k, (p, m) in totals.items():
    print(f"{k:13s} merge-layer weights {p:7d}   MACs {m / 1e9:.2f} G")
```

```
concat + 1x1 conv [I I] == add: True
add           merge-layer weights  783360   MACs 2.42 G
concat        merge-layer weights 1566720   MACs 4.83 G
concat + 1x1  merge-layer weights  957440   MACs 2.95 G
```

Concatenating doubles the merge layer's weights and MACs; a 1×1 fusion first, as Restormer does, recovers most of the saving. The choice differs across well-known designs:

- **U-Net** concatenates the (cropped) encoder map with the decoder map after a 2×2 up-convolution that halves the channels [39].

- **FPN** adds: the coarser map is upsampled by 2 with nearest-neighbor interpolation and added to a 1×1-projected backbone map, and a 3×3 convolution on each merged map reduces the aliasing of the upsampling [5] (DLCV 03).
- **Restormer** concatenates and then halves the channels with a 1×1 convolution at all levels except the top [21].
- **NAFNet**'s code adds the encoder map to the upsampled decoder map [14].
- **DenseNet** concatenates the outputs of all earlier layers in a block [40] (DLCV 02 note).

Multi-scale modules: FPN, ASPP and the U-Net

Plain version. Objects and degradations come at many sizes. Multi-scale modules let one network look at several scales at once: by building a pyramid of feature maps (FPN), by probing one map with filters of several sizes (ASPP), or by going down and back up (U-Net).

These are composition patterns more than blocks, and each is taught where it is used: FPN in DLCV 03 [5]; atrous spatial pyramid pooling (ASPP), parallel atrous convolutions at several rates, in DLCV 03B [41]; the U-Net in DLCV 01 and the IP 10 paper guide [39]. One rule connects them: **context comes from going coarse, detail comes from a path that stayed fine**. FPN's lateral connections, U-Net's skips and the low-level skip of the segmentation decoders in DLCV 03B are all that second path.

Composing for classification

Plain version. A classifier must answer one question about the whole image. So it shrinks the image quickly, spends its computation on small, wide feature maps where each position already sees a large region, and finally throws away the spatial grid altogether.

The principles, with the reason for each:

1. **Downsample early with a stem.** The first layers run on the most pixels, so the stem cuts the resolution by 4 before the body starts (ResNet's 7×7 stride-2 convolution plus max-pool [1]; ConvNeXt's 4×4 stride-4 patchify layer, which cost nothing in accuracy in its roadmap [2]).
2. **Halve the resolution and double the width at each stage.** This keeps the cost per layer roughly constant (see the anatomy section) and grows the receptive field geometrically.
3. **Put most blocks in the third stage.** ResNet-50 uses (3, 4, 6, 3) blocks per stage; ConvNeXt changed this to (3, 3, 9, 3), following Swin's 1:1:3:1 ratio, and gained accuracy [2]. The third stage, at output stride 16, has a large receptive field and still enough positions.
4. **Pool globally, then classify linearly.** Global average pooling makes the output invariant to

where the object is and lets the network accept any input size. The ViT paper found that a class token and global average pooling work similarly well, but need different learning rates [3].

5. **Put expensive global mixing where tokens are few.** Global attention is affordable in late stages; hybrid designs put convolutions early and attention late (ML 04). Cheap gates like SE can go in every block.

6. **Batch normalization is fine** when batches are large; use GN or LN when they are not.

How fast does the receptive field grow? The recurrence of ML 03 covers strided stacks; the snippet also handles upsampling by walking back from one output unit and tracking the interval of inputs that reach it. It compares ResNet-50 with two restorers on a 256 × 256 input: a plain full-resolution body of the EDSR-baseline shape (16 residual blocks of two 3 × 3 convolutions, 64 channels, plus head and tail) and a 4-level U-Net with widths 32 to 512.

```
def analyze(layers, H=256):
    """Exact 1-D receptive field of one output unit, plus the multiply-accumulates (MACs)
    of a square H x H input. A layer is ('conv', k, stride, c_in, c_out) or ('up', f),
    where 'up' is nearest-neighbour upsampling by f (the convolution after it pays the
     ↪  cost)."""
    macs, size = 0, H
    for layer in layers:                                # forward: track the map size and cost
        if layer[0] == 'conv':
            _, k, s, cin, cout = layer
            size //= s
            macs += size * size * k * k * cin * cout
        else:
            size *= layer[1]
    rf = 0
    for p0 in range(16):                                # a few output positions (alignment
     ↪  matters)
        lo = hi = p0
        for layer in reversed(layers):                  # backward: which inputs reach unit p0?
            if layer[0] == 'conv':
                _, k, s, _, _ = layer
                pad = (k - 1) // 2
                lo, hi = s * lo - pad, s * hi - pad + k - 1
            else:
                lo, hi = lo // layer[1], hi // layer[1]
        rf = max(rf, hi - lo + 1)
    return rf, macs, size

# Classification: ResNet-50 (stride on the 3x3 conv of each first block, max-pool as a k=3,
 ↪  s=2 layer).
resnet = [('conv', 7, 2, 3, 64), ('conv', 3, 2, 64, 0)]        # stem conv, then max-pool
 ↪  (no MACs)
cin = 64
for width, blocks, stride in [(64, 3, 1), (128, 4, 2), (256, 6, 2), (512, 3, 2)]:
    for b in range(blocks):
        s = stride if b == 0 else 1
        resnet += [('conv', 1, 1, cin, width), ('conv', 3, s, width, width),
```

```
                ('conv', 1, 1, width, 4 * width)]
        if b == 0:
            resnet += [('conv', 1, 1, cin, 4 * width)]          # projection shortcut (cost
             ↪ only)
        cin = 4 * width

# Restoration A: plain full-resolution body, 64 channels, 32 convs of 3x3
 ↪ (EDSR-baseline-like).
plain = [('conv', 3, 1, 3, 64)] + [('conv', 3, 1, 64, 64)] * 32 + [('conv', 3, 1, 64, 3)]

# Restoration B: 4-level U-Net, 2 convs per level, widths double as resolution halves.
w = [32, 64, 128, 256, 512]
unet = [('conv', 3, 1, 3, w[0])]
for i in range(4):
    unet += [('conv', 3, 1, w[i], w[i])] * 2 + [('conv', 2, 2, w[i], w[i + 1])]
unet += [('conv', 3, 1, w[4], w[4])] * 2
for i in reversed(range(4)):
    unet += [('up', 2), ('conv', 1, 1, w[i + 1], w[i]),
             ('conv', 3, 1, 2 * w[i], w[i]), ('conv', 3, 1, w[i], w[i])]   # concat skip
              ↪ doubles c_in
unet += [('conv', 3, 1, w[0], 3)]

for name, net in [('ResNet-50', resnet), ('plain 64ch x 34', plain), ('U-Net 4 levels',
 ↪ unet)]:
    rf, macs, size = analyze(net)
    print(f"{name:16s} RF {rf:4d} px   output {size:3d}x{size:<3d}   {macs / 1e9:6.2f}
     ↪ GMACs")
```

```
ResNet-50        RF  427 px   output   8x8       5.34 GMACs
plain 64ch x 34  RF   69 px   output 256x256    77.54 GMACs
U-Net 4 levels   RF  206 px   output 256x256    14.47 GMACs
```

The U-Net's skips are left out because its deepest path sets the receptive field (a brute-force check over all paths gives the same 206). Figure 5 plots the growth. ResNet-50 reaches 427 pixels for 5.3 GMACs but returns an 8×8 map; the plain restorer grows by 2 pixels per layer, 69 pixels for 77.5 GMACs; the U-Net reaches 206 pixels at full output resolution for less than a fifth of that compute. These are theoretical receptive fields; the effective one, measured by gradients, is smaller and concentrated in the center [42] (DLCV 01).

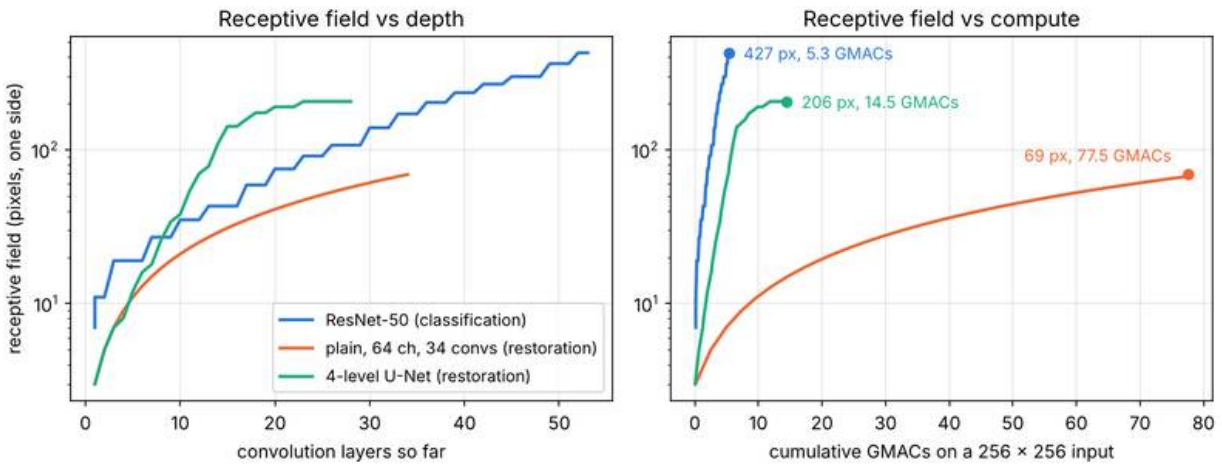


Figure 5 — Receptive field (one side, theoretical) as a function of depth and of compute, for the three networks of the snippet. Downsampling buys receptive field cheaply; a full-resolution body pays for every pixel of it.

Optional: Receptive fields through upsampling and skips (in the Supplementary notes). The backward interval method step by step, why alignment matters after upsampling, and why the deepest path decides.

Composing for restoration

Plain version. A restoration network returns an image of the same size (or larger, for super-resolution) that is almost the input. So it keeps the resolution, or comes back to it with skips; it predicts only the correction; it avoids layers that change the intensity range; and it buys context without paying full resolution for every layer.

The principles:

1. **Keep the resolution, or come back to it.** The comparative study behind X-Restormer sorts restoration networks into three architectures: U-shaped encoder–decoders (such as Restormer and NAFNet), plain residual-in-residual networks that keep the original size (RCAN, SwinIR) and multi-stage progressive networks, which process the image in several sub-networks one after another [22]. It describes the plain design as favorable for reconstructing high-frequency detail but expensive, and the U-shaped one as handling large inputs with a receptive field that grows quickly through downsampling [22]. Figure 1(b) and (c) show the two main shapes.

2. **Learn the residual.** The output is the input plus a predicted correction, $\hat{x} = y + R(y)$ for degraded input y and network R. VDSR predicted the residual between the high-resolution image and its interpolated input [43]; DnCNN predicts the noise [29]; SwinIR uses $I_{\text{HQ}} = H(I_{\text{LQ}}) + I_{\text{LQ}}$ for tasks without upsampling [4]; Restormer outputs a residual image R and returns $\hat{I} = I + R$ [21]; NAFNet's code adds the input to the last convolution's output [14]. The correction is small and centered on zero, so no capacity is spent copying the image.

3. **Add a long skip for low frequencies.** SwinIR sends its shallow features directly to the reconstruction module, $H_{\text{REC}}(F_0 + F_{\text{DF}})$, because shallow features mainly contain low frequencies and the deep features can then focus on the lost high frequencies [4]. EDSR and RCAN use long skips around the body or around groups [12], [15].

4. **Preserve the intensity range.** No BN in the body (EDSR, ESRGAN), normalization only inside residual branches (NAFNet, Restormer, SwinIR), as shown in the normalization section [12], [28].

5. **Match the receptive field to the degradation, and pay for it wisely.** The X-Restormer study argues that super-resolution depends mostly on local information and strong spatial mapping, that U-shaped downsampling and upsampling help denoising, and that motion deblurring and dehazing need large-range or global information [22]. The receptive-field snippet shows the price: a full-resolution body grows its field by a few pixels per layer at full cost, downsampling grows it fast and cheaply, and channel attention or state-space scans reach the whole image at linear cost [21], [44].

6. **Design the output layer for the target.** The last layer is a plain convolution (SwinIR's reconstruction for denoising, Restormer's final convolution, NAFNet's ending layer [4], [21], [14]); a ReLU or sigmoid there would clip the correction. For super-resolution, the upsampler goes at the end so the body runs at low resolution (DLCV 04A).

7. **Plan for full-size test images.** Restorers train on crops and test on whole images. A U-Net needs sizes divisible by 2^{levels}, which NAFNet's code handles by padding [14]. Global statistics, such as channel attention's pooling, cover a crop in training but the whole image at test time; NAFNet notes that this degrades performance and adopts a remedy from earlier work [19], and Restormer trains on progressively larger patches [21].

Optional: A Comparative Study of Image Restoration Networks for General Backbone Network Design (X-Restormer) (in the Paper guides). Which backbone wins on which degradation, and why.

Side by side: what changed and why

Plain version. The fastest way to see composition at work is to put a classification model next to the restoration model derived from it and list the differences. Each difference has a reason from the two lists above.

ResNet versus EDSR and NAFNet

Component	ResNet [1]	EDSR [12]	NAFNet [19], [14]	Why it changed
Stem	7×7 stride 2 + max-pool	one 3×3 conv	one 3×3 conv	keep every pixel
Resolution	4 stages, output stride 32	constant (low-resolution input size)	U-shape, 2×2 stride-2 conv down, 1×1 conv + pixel shuffle up	detail versus context
Block	conv-BN-ReLU-conv-BN, add, ReLU	conv-ReLU-conv, ×0.1 (large model), add	LN, 1×1, 3×3 dw, SimpleGate, SCA, 1×1, $\times\beta$, add; then a gated MLP, $\times\gamma$, add	range preservation; cheap gating
Normalization	BN	none	LayerNorm inside the branches	see the normalization section
Skips	short only	short + long skip around the body	short + add-skips between levels + global residual	low frequencies bypass the body
Head	global average pool + FC	upsampler (conv + pixel shuffle) + conv	3×3 conv + input	output is an image
Loss	cross-entropy	L1	PSNR loss in the ablations	regression targets

EDSR's own table lists the step from SRResNet: 16 blocks of 64 filters for both baseline models, BN removed, L1 instead of L2, and for the full model 32 blocks of 256 filters with residual scaling 0.1, growing from 1.5M to 43M parameters [12].

Swin versus SwinIR

Component	Swin [6]	SwinIR [4]	Why it changed
Input	4×4 patch partition and linear embedding	3×3 conv (shallow features) at full resolution	one token per pixel
Hierarchy	patch merging between 4 stages	none: 6 residual Swin Transformer blocks (RSTB) at one resolution	restoration needs every pixel
Window	$M = 7$	$M = 8$ (7 for JPEG artifact removal)	JPEG [45] codes 8×8 blocks; the authors observed a drop with 8
Block group	Swin layers	6 Swin layers + a 3×3 conv + a residual connection per RSTB	the conv adds translation equivariance; the residual helps aggregation
Head	global average pool + linear	conv (+ pixel shuffle for SR) on $F_0 + F_{\mathrm{DF}}$	low frequencies bypass the body
Width	96 channels and up	180 channels, 6 heads	balance accuracy and size

SwinIR's ablation reports that the residual connection inside each RSTB improves PSNR [30] by 0.16 dB, and that a 1×1 convolution there brings little improvement, possibly because it cannot extract local neighboring information as a 3×3 convolution does [4]. The JPEG window size is a good example of a composition choice that interacts with the data: the authors suggest the drop with $M = 8$ may come from JPEG's 8×8 block partitions.

ViT versus Restormer

Component	ViT [3]	Restormer [21]	Why it changed
Tokens	16 × 16 patches	every pixel, after a conv embedding	no information is thrown away
Token mixer	global self-attention over patches, $N \times N$ map	MDTA: channel attention, $C \times C$ map, with depthwise convs	linear cost in pixels, global reach
Channel mixer	MLP, 4× expansion, GELU	GDFN: gated, depthwise convs, expansion 2.66	gating and local context
Shape	isotropic, one resolution	4-level U-shape, pixel-unshuffle down, pixel-shuffle up, concat + 1 × 1 skips, refinement stage	multi-scale at manageable cost
Head	class token + MLP	conv to a residual image, added to the input	output is an image
Training	fixed resolution	progressive patch sizes, starting at 128 × 128	learn from larger regions

Restormer's configuration uses [4, 6, 6, 8] blocks, [1, 2, 4, 8] heads and [48, 96, 192, 384] channels from level 1 to level 4, plus 4 blocks in the refinement stage [21]. Both columns use LayerNorm in pre-norm residual branches: the Transformer block itself survived; what changed is the token mixer and the macro-architecture around it.

Exploring a design space

Plain version. A design question ("does block X help?") has no answer in general; it has an answer for a given set of other choices, data and budget. So change one thing at a time, keep the budget fixed, look at more than one task or seed, and when you can, compare populations of models rather than single winners.

Reading an ablation

NAFNet's paper is a model of a readable ablation. Starting from a plain block of convolution, ReLU and shortcut in a U-Net, under a fixed budget of 16 GMACs, the authors add one change at a time and report PSNR on SIDD [31] (real-camera denoising) and GoPro [32] (motion deblurring) [19]:

Step	SIDD (dB)	GoPro (dB)
PlainNet	39.29	28.51
+ LayerNorm (and a 10× larger learning rate)	39.73	31.90
ReLU → GELU	39.71	32.11

Step	SIDD (dB)	GoPro (dB)
+ channel attention	39.85	32.35
GELU → SimpleGate	39.93	32.76
CA → simplified CA	39.96	32.85

What to check when you read such a table:

- **Is the budget fixed?** Here it is (16 GMACs), so a gain is not just "more compute".
- **Are the steps cumulative?** Yes: each row includes the previous changes, so the table measures each change **given** the earlier ones, not in isolation. A different order could give different increments.
- **Do the tasks agree?** Not always: GELU is neutral on SIDD and helps on GoPro. A change that matters for one degradation may not matter for another.
- **Is the training recipe confounded with the block?** The LayerNorm row also changes the learning rate, because LayerNorm made the larger rate stable. The authors say so explicitly; many papers do not.
- **How large is the noise?** A single run per row gives no error bar. Differences of a few hundredths of a dB should be read as "no loss", which is how the paper reads the last row.

Optional: Simple Baselines for Image Restoration (NAFNet) (in the Paper guides). The problem the paper poses, how it builds and then simplifies its block, and what its claims do and do not mean.

ConvNeXt's roadmap reads the same way for classification: a ResNet-50 went from 76.1% to 78.8% ImageNet top-1 with a modern training recipe alone, and then to 82.0% through a sequence of architectural changes, the largest single one adding less than the recipe [2]. That is why architectures trained with different recipes cannot be compared (DLCV 01A).

Optional: The ConvNeXt roadmap step by step (in the Supplementary notes). Every step with its accuracy, and how the compute changed along the way.

Comparing populations: design spaces

RegNet replaces "which single model is best?" with "which **design space** produces good models more often?" [16]. A design space is a set of models defined by a few parameters (number of blocks per stage, widths, bottleneck ratio, group width). The tool is the **error empirical distribution function**:

$$F(e) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[e_i < e],$$

where e_i is the error of the i-th of n models sampled from the space and trained, $\mathbf{1}[\cdot]$ is 1 when its argument is true and 0 otherwise, and $F(e)$ is the fraction of models with error below e. A space whose curve lies higher is better as a whole, a far more robust comparison than two hand-tuned models. RegNet sampled $n = 500$ models at 360 to 400 million FLOPs, trained each for 10 epochs, and kept constraints that did not hurt the curve (shared bottleneck ratio and group width) or improved it (widths and depths increasing across stages) [16]. The best models had about 20 blocks across compute regimes, used no bottleneck, and benefited from SE [16].

Optional: Designing Network Design Spaces (RegNet) (in the Paper guides). How the study refines a design space step by step and how to read its distribution plots.

Optional: RegNet's quantized linear widths (in the Supplementary notes). The three equations that turn a line into stage widths and depths, with a tested snippet.

A practical protocol

1. Fix the budget (MACs, parameters, memory or measured latency) and the training recipe before comparing blocks.
2. Change one thing at a time, and keep a log of every run, including the ones that did not help.
3. Evaluate on more than one task, dataset or seed; report the spread.
4. Confirm at the target model size; a block that helps a tiny model may not help a large one.
5. For architecture searches, think in terms of the three dimensions of the NAS survey: the search space, the search strategy and how performance is estimated [46]. Cheap estimates such as short training (RegNet's 10 epochs) are fine for ranking populations but should be confirmed with full training.

Common composition mistakes

Plain version. Most architecture bugs are not wrong blocks; they are right blocks in the wrong place. The list gives the symptom, the cause and the fix.

- **BN in a restoration body.** Symptom: batch-dependent brightness or color shifts, or test-time artifacts. Fix: remove BN, or normalize per sample inside residual branches with a global residual.
- **Small per-device batches with BN** (detection, segmentation, high resolution). Fix: GroupNorm, or freeze the pretrained BN statistics (DLCV 01A).
- **Transposed convolution with a kernel size not divisible by the stride.** Symptom: checkerboard texture. Fix: resize-convolution, or pixel shuffle with a matched initialization.
- **No global residual in a restorer.** The network must learn to copy the image before it can correct it. Fix: predict the correction.

- **A ReLU or sigmoid at the output.** Corrections cannot be negative, or highlights saturate. Fix: a linear last layer; clip only for display and evaluation.
- **A receptive field smaller than the degradation.** Symptom: large blur or haze remains. Fix: downsampling levels, dilation, or global mixing; compute the receptive field first.
- **Downsampling too early in a restorer**, or a classification backbone used unchanged: fine detail is lost. Fix: a full-resolution path (skips, a refinement stage).
- **Global attention at full resolution** runs out of memory. Use windows, channel attention or downsampling.
- **Input sizes not divisible by 2^{levels} in a U-Net**, or odd sizes with concatenated skips. Symptom: shape errors or misaligned skips. Fix: pad and crop, as NAFNet's code does [14].
- **Counting parameters as speed.** Depthwise and concatenation-heavy designs can be slow despite few parameters or MACs. Measure latency and memory.
- **Comparing designs trained with different recipes or budgets**: the recipe alone can be worth more than any single block [2].
- **Ignoring interactions with the data**, such as SwinIR's window size and JPEG's blocks [4].

Modern view

What the surveys say

Khan et al. [47] review CNN architectures and sort their innovations into seven categories: spatial exploitation, depth, multi-path, width, feature-map exploitation, channel boosting and attention. The taxonomy is useful here because most categories are composition choices (depth, width, multi-path connections) rather than new layer types.

Guo et al. [20] survey attention in computer vision and classify it by what it selects: channel attention (what to attend, e.g. SE), spatial attention (where, including self-attention), temporal attention (when), branch attention (which branch), and the two hybrids channel-and-spatial (e.g. CBAM) and spatial-and-temporal. Their framing of attention as "dynamic weight adjustment" based on the input matches the gate view of this chapter.

Khan et al. [48] review Transformers in vision, from the basics of self-attention and large-scale pretraining to applications in recognition, detection, segmentation, generation, video and low-level vision, compare the advantages and limitations of popular designs, and end with open research directions.

Elsken, Metzen and Hutter [46] organize neural architecture search along three dimensions: the

search space, the search strategy and the performance-estimation strategy. Their survey is the natural next step after the manual design-space exploration of this chapter.

Chen et al. [22] is a comparative study rather than a survey: it benchmarks five restoration backbones, among them SwinIR, Restormer and NAFNet, on five tasks, classifies them by architecture (U-shaped, plain residual-in-residual, multi-stage) and core operator (convolution, spatial self-attention, transposed self-attention), and finds that networks strong on one task often fall short on another.

The state of the art, 2023–2026

- **Convolutions and attention converge on one block shape.** ConvNeXt V2 adds a Global Response Normalization layer and masked-autoencoder pretraining to ConvNeXt [49]; the MetaFormer view treats the token mixer as one interchangeable part [8].
- **Pretraining matters as much as the block.** Battle of the Backbones compared many pretrained backbones across tasks in more than 1,500 training runs and found that convolutional networks pretrained with supervision on large datasets still performed best on most of the tasks it considered, while self-supervised backbones were competitive in matched comparisons [50].
- **Restoration mixes operators.** HAT combines channel attention with window self-attention and adds overlapping cross-attention between windows [51]; X-Restormer replaces half of Restormer's channel-attention blocks with spatial attention blocks to improve its spatial mapping [22]; MambaIR and MambaIRv2 replace attention with state-space scans that reach the whole image at linear cost, adding local enhancement and channel attention, and in v2 non-causal "attentive" modeling [44], [52].

Open problems

1. **General restoration backbones.** No single composition is best on all degradations; the comparative study frames task generality as the open target [22].
2. **Fair comparisons.** Architecture, recipe and pretraining are entangled, and population-level studies like RegNet's need hundreds of training runs [16], which is costly for high-resolution restoration models.
3. **Receptive field in practice.** Theoretical fields are easy to compute; how much context a trained restorer actually uses is harder to measure, and attribution analyses suggest it can be less than the design allows [51], [42].
4. **Cost models.** MACs, memory traffic and latency disagree, especially for depthwise convolutions, gating and attention; design rules still depend on hardware.
5. **Train–test shift inside the architecture.** Global statistics (pooling, normalization, attention)

behave differently on crops and full images, and the remedies in use change the training or test procedure rather than the architecture [19], [21].

Key takeaways

- A network is a stem, stages of repeated blocks, downsampling layers and a head; most design decisions are about where the resolution changes and where the skips go, not about inventing blocks.
- Every block mixes space and channels at some cost; count weights and MACs per pixel, compare blocks at equal cost, and measure latency separately.
- Classifiers shrink the resolution early, double the width as they halve the map, put most blocks in the third stage and pool globally; their receptive field grows geometrically and cheaply.
- Restorers keep or recover full resolution, learn a residual, send low frequencies through long skips, avoid BN in the body, and end with a linear layer.
- Batch normalization makes a restored image depend on its batch; per-sample normalization removes that dependence but also hides the image's own level, so it belongs inside residual branches.
- Transposed convolutions and untrained sub-pixel layers make checkerboards when their phase sums differ; resize-convolution cannot.
- Concatenation plus a 1×1 convolution contains addition; it costs about twice as much at the merge.
- Read ablations for a fixed budget, cumulative steps, agreement across tasks and confounded recipes; when possible, compare populations of models, not single winners.

Exercises

1. **Halve and double.** A stage runs 3×3 convolutions with 128 channels on a 56×56 map. (a) How many MACs does one such layer cost? (b) The next stage halves the map and doubles the width; what does one layer cost now? (c) What happens to the cost if the width is tripled instead?

   Hint

   (a) $9 \cdot 56^2 \cdot 128^2 = 462{,}422{,}016$ MACs. (b) $9 \cdot 28^2 \cdot 256^2$ is the same number. (c) With $3 \cdot 128 = 384$ channels the cost is $9 \cdot 28^2 \cdot 384^2$, that is $9/4$ of (a): width enters squared, area linearly.

2. **Block budget.** Using the `blocks` function, find the expansion factor t at which an inverted residual with C input and output channels costs the same as a basic residual block, for $C = 64$. Then find the width C' at which an inverted residual with $t = 6$ costs the same as a basic block with $C = 64$. What does this say about comparing blocks "at equal width"?

Hint

Set $t(2C^2 + 9C) = 18C^2$; for $C = 64$ this gives $t = 18 \cdot 64/(2 \cdot 64 + 9) \approx 8.4$. For the second question solve $12C'^2 + 54C' = 73{,}728$, which gives $C' \approx 76$. The inverted residual is not "cheap" at equal width; it is cheap because MobileNets run it on thin tensors.

3. **BN with a batch of one.** In the normalization snippet, restore A alone (a batch of size 1) with `'bn-train'`. Compare the output mean with the `'instance'` result and explain the relation between BN at batch size 1 and instance normalization. Then add 32 random mid-gray images to the batch and see how close training mode comes to evaluation mode.

Hint

With $N = 1$, BN's statistics over (N, H, W) are exactly the per-sample, per-channel statistics, so the result equals instance normalization (0.521). With many images from the training distribution, the batch statistics approach the running statistics and the output approaches the evaluation-mode value.

4. **Make a transposed convolution flat.** For $k = 4$, $s = 2$, write the condition on the 16 kernel weights under which a flat input gives a flat output (away from the borders). Construct such a kernel from a random 2×2 kernel, check it with the upsampling snippet, and relate it to resize-convolution.

Hint

The four phase sums $\sum_{m,n} w[a+2m, b+2n]$ for $a, b \in \{0, 1\}$ must be equal. One construction is `w = np.kron(v, np.ones((2, 2)))` for any 2×2 kernel `v`: every phase then receives each entry of `v` exactly once, so all phase sums equal the sum of `v`. Resize-convolution has the same property for every kernel, which is why it is flat.

5. **Receptive field of a design.** Modify the U-Net in the receptive-field snippet to three levels instead of four, and to four levels with three convolutions per level. Which change buys more receptive field per GMAC? Then replace the 32 middle layers of the plain network by 16 layers with dilation 2 (extend `analyze` with a dilation entry). What is the new receptive field, and what problem from DLCV 03B does this design risk?

Hint

A dilated layer with kernel k and dilation d spans $d(k-1)+1$ inputs, so replace `k - 1` by `d * (k - 1)` in the backward step. Removing a level removes the coarsest, cheapest and most receptive-field-rich part of the U-Net; adding convolutions per level adds cost at every resolution. Stacking layers with the same dilation leaves holes in the receptive field (gridding).

Supplementary notes

These notes are optional: each takes one point from the main text and treats it in more detail. Nothing in the main text, code or exercises depends on them.

CBAM's channel and spatial gates

This note writes out the two gates of CBAM and compares them with the other channel gates of the chapter.

For a feature map $F \in \mathbb{R}^{C\times H\times W}$, CBAM computes [18]

$$M_c(F) = \sigma(\mathrm{MLP}(\mathrm{AvgPool}(F)) + \mathrm{MLP}(\mathrm{MaxPool}(F))), \qquad M_s(F') = \sigma(f^{7\times 7}([\mathrm{AvgPool}_c(F'); \mathrm{MaxPool}_c(F')])),$$

where $M_c(F) \in \mathbb{R}^{C\times 1\times 1}$ is the channel gate, AvgPool and MaxPool reduce each channel to one number, the MLP has one hidden layer of C/r units with a ReLU (reduction ratio r) and is shared between the two pooled vectors, σ is the sigmoid, $F' = M_c(F) \odot F$ is the channel-refined map,

$\mathrm{AvgPool}_c$ and $\mathrm{MaxPool}_c$ pool **along the channel axis** to give two $H \times W$ maps, $[\cdot;\cdot]$ concatenates them, $f^{7\times 7}$ is a 7×7 convolution with one output channel, $M_s(F') \in \mathbb{R}^{1\times H\times W}$ is the spatial gate, and the output is $M_s(F') \odot F'$. The product $\odot$ broadcasts each gate over the axes it lacks: the channel gate over all positions, the spatial gate over all channels.

Relation to the other gates. SE is the channel gate with average pooling only [17]. RCAN's channel attention is the SE form applied inside each residual block of a super-resolution network [15]. NAFNet's simplified channel attention keeps the pooling and a single linear map but drops the MLP's ReLU and the sigmoid, $\mathrm{SCA}(X) = X * W\,\mathrm{pool}(X)$, where $*$ is channel-wise multiplication [19]. All of them are multiplicative gates computed from global statistics, which is why NAFNet reads channel attention as a special case of a gated linear unit [19].

Pitfalls. The pooled statistics are computed over the whole map, so they differ between a training crop and a full test image; this is one form of the train–test shift discussed in the restoration principles. The spatial gate's 7×7 kernel sees only a local window of the pooled maps, so it is a local, not a global, spatial attention.

Phase sums of a transposed convolution in 1-D

This note derives how many kernel taps each output position of a transposed convolution receives, which is the arithmetic behind the checkerboard snippet.

Setup. A 1-D transposed convolution with stride s and kernel $w_0, \dots, w_{k-1}$ maps input x_i to

$$y_n = \sum_i x_i\, w_{n-si}, \qquad 0 \le n - si < k,$$

where n is the output position and the sum runs over the inputs i whose kernel copy covers n. Write $n = sm + p$ with phase $p \in \{0, \dots, s-1\}$. The kernel index $n - si = p + s(m - i)$ takes the values $p, p + s, p + 2s, \dots$ that are less than k.

Taps per phase. Phase p therefore receives the taps $w_p, w_{p+s}, w_{p+2s}, \dots$, and their number is

$$T_p = \left\lceil \frac{k - p}{s} \right\rceil,$$

for $0 \le p < s$, away from the borders. For $k = 3, s = 2$: $T_0 = 2, T_1 = 1$. In 2-D the counts multiply, giving the $4 : 2 : 2 : 1$ pattern of the snippet. All T_p are equal exactly when s divides k; this is Odena et al.'s condition for even overlap [37].

Phase sums. On a flat input $x_i = 1$, the output at phase p is the phase sum $S_p = \sum_{t \ge 0,\ p+st<k} w_{p+st}$. A flat output requires $S_0 = S_1 = \cdots = S_{s-1}$. Equal tap counts make this possible but do not enforce it: random weights break it, which is the "random initialization" source of artifacts described by Aitken et al. [38].

Resize-convolution. Nearest-neighbor upsampling by s followed by a convolution with kernel v of length r is itself a transposed convolution with stride s whose kernel is v convolved with a box of s ones. Each of its phase sums is $\sum_t v_t$, the same for every phase, so a flat input always gives a flat output, whatever v is. That is the precise sense in which resize-convolution is "biased against" checkerboards.

Receptive fields through upsampling and skips

This note explains the backward interval method used by the receptive-field snippet, and when it is exact.

The question. For one output unit at position p of the last layer, which input positions can change its value? For a chain of layers, answer it backward: start with the interval $[p, p]$ at the output and map it through each layer, from the last to the first, to the interval of positions in that layer's input that feed it.

Convolution. A layer with kernel size k, stride s and padding $q = \lfloor (k-1)/2 \rfloor$ computes output unit u from inputs $su - q, \dots, su - q + k - 1$. An interval $[\ell, h]$ of outputs therefore depends on the inputs

$$[\, s\ell - q, \;\; sh - q + k - 1 \,],$$

where ℓ and h are the first and last output positions and the new endpoints are the first input of unit ℓ and the last input of unit h. Positions outside the image stand for padding; they do not limit the interval, so the result is the theoretical receptive field away from borders.

Nearest-neighbor upsampling by f. Output unit u copies input unit $\lfloor u/f \rfloor$, so $[\ell, h]$ maps to $[\lfloor \ell/f \rfloor, \lfloor h/f \rfloor]$. Because of the floor, the size of the result depends on where ℓ falls relative to the blocks

of f copies: two neighboring output units can have receptive fields that differ by up to one input unit of that layer, which can be many pixels deep in a network. That is why the snippet takes the maximum over 16 consecutive output positions, a full period of a 4-level U-Net.

Skips. At a merge (addition or concatenation) the input interval is the union of the intervals through the two branches. In a U-Net, the branch through the deeper level passes through the same encoder output and then through more layers; with nearest-neighbor upsampling and a stride-2 downsampling whose window contains its own position, the deeper branch's interval contains the skip branch's interval. So the union equals the deepest path, and the snippet can list that path alone. The brute-force check mentioned in the main text built the U-Net in 1-D with positive weights, pushed a unit impulse from every nearby input position through it, and recorded which ones reached a given output; it found the same 206 pixels.

Edge cases. Dilated convolutions make the dependence set a lattice with holes, not an interval, so the interval method reports the span but not the gaps (see the gridding note in DLCV 03B). Attention layers with global reach make the receptive field the whole input after one layer; window attention lets each token see its own $M \times M$ window, so one layer adds at most $M - 1$ tokens of reach, and the shifted windows of the next layer carry it across window borders. Araujo, Norris and Sim give closed-form expressions for receptive field size, stride and offset in general networks, including how to handle multiple paths [53].

The ConvNeXt roadmap step by step

This note lists the steps of ConvNeXt's "modernization" of ResNet-50 with the ImageNet top-1 accuracy the paper reports after each one [2].

Step	Change	Top-1 (%)
0	ResNet-50 with its original training	76.1
1	modern training recipe (300 epochs, AdamW, heavy augmentation)	78.8
2	stage compute ratio (3, 3, 9, 3)	79.4
3	patchify stem, 4 × 4 stride 4	79.5
4	depthwise (grouped) convolution, width raised to 96	80.5
5	inverted bottleneck	80.6
6	depthwise convolution moved up	79.9
7	kernel size 7 × 7	80.6
8	ReLU replaced by GELU	80.6
9	fewer activations	81.3
10	fewer normalization layers	81.4
11	LayerNorm instead of BatchNorm	81.5
12	separate 2 × 2 stride-2 downsampling layers, with normalization where resolution changes	82.0

How to read it. The steps are cumulative, so each number is conditional on everything above

it. Step 6 lowers accuracy on purpose: it prepares step 7, and the two must be judged together. Steps 8 and 11 change accuracy little by themselves; the paper adopts them anyway, so the final design contains choices that were neutral when they were made. Several steps also change the compute: step 4 raises it, and steps 5 and 6 lower it, so the paper tracks FLOPs next to accuracy. The roadmap is a single path through a large design space, which the authors acknowledge for the stage ratio: a better distribution of computation "is likely to exist" [2].

RegNet's quantized linear widths

This note writes out how RegNet turns three numbers into a network's stage widths and depths.

RegNet first describes per-block widths with a line [16]:

$$u_j = w_0 + w_a \cdot j, \qquad 0 \le j < d,$$

where d is the network depth in blocks, j the block index, $w_0 > 0$ the initial width, $w_a > 0$ the slope, and u_j the continuous width of block j. To get a few stages of constant width, it introduces a multiplier $w_m > 0$ and writes each u_j as a power of it:

$$u_j = w_0 \cdot w_m^{s_j}, \qquad w_j = w_0 \cdot w_m^{\lfloor s_j \rceil},$$

where s_j is a real exponent, $\lfloor s_j \rceil$ rounds it to the nearest integer, and w_j is the quantized width. Blocks with the same rounded exponent form one stage: stage i has width $w_0 \cdot w_m^i$ and $d_i = \sum_j \mathbf{1}[\lfloor s_j \rceil = i]$ blocks. A ratio $w_m = 2$ gives the classic doubling of width from stage to stage; RegNet found that good models had w_m near 2.5 [16]. When it studies four-stage networks, the paper ignores parameter combinations that give a different number of stages [16]; the snippet shows only the three equations, with illustrative values that are not one of the paper's models.

```
import numpy as np

def regnet_stages(d, w0, wa, wm):
    u = w0 + wa * np.arange(d)                    # continuous widths, one per block
    s = np.log(u / w0) / np.log(wm)               # exponents with u = w0 * wm**s
    q = np.round(s).astype(int)                   # quantized exponents
    widths = w0 * wm ** q
    return [(int(w0 * wm ** i), int((q == i).sum())) for i in np.unique(q)], widths

stages, widths = regnet_stages(d=16, w0=48, wa=36, wm=2.5)
print("stage (width, blocks):", stages)
print("block widths:", widths.astype(int).tolist())
```

```
stage (width, blocks): [(48, 1), (120, 3), (300, 8), (750, 4)]
block widths: [48, 120, 120, 120, 300, 300, 300, 300, 300, 300, 300, 300, 750, 750, 750, 750]
```

Two things to notice. Depth per stage comes out of the line, not from a separate choice: a steeper slope moves blocks into later, wider stages. And the first stage can end up with a single block, which is one of the parameter combinations a practical generator would filter or adjust.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Simple Baselines for Image Restoration (NAFNet)

Chen, Chu, Zhang and Sun, ECCV 2022 [19].

The problem. State-of-the-art restorers had grown complex in two ways the authors call inter-block complexity (multi-stage pipelines, multi-scale fusion) and intra-block complexity (elaborate attention and feed-forward designs). The question is whether that complexity is necessary.

The idea. Fix a simple single-stage U-Net and build the block from scratch, adding one common component at a time under a fixed compute budget: LayerNorm, GELU, channel attention. This gives a "baseline". Then simplify: observe that GELU and channel attention can both be read as gates, replace GELU by SimpleGate (multiply two channel halves) and channel attention by simplified channel attention (a linear map of the pooled vector). The resulting network has no nonlinear activation functions at all, hence its name, Nonlinear Activation Free Network.

What it showed. The ablation table reproduced in this chapter: every step is reported on two tasks, and the final simplifications cost nothing. In the abstract, the full models exceed the previous best results on GoPro and SIDD with a fraction of their computational cost (8.4% on GoPro, less than half on SIDD).

How to read it. The ablation is cumulative, so each increment is conditional on the steps before it; the LayerNorm step changes the learning rate as well; the two tasks do not always agree. The claim "activations are not necessary" is about pointwise activation functions; the network is still nonlinear through its multiplicative gates.

Why it matters here. It is the clearest example of the composition view: the macro-architecture is held fixed and deliberately plain, and almost all of the result comes from the arrangement of a few ordinary parts.

Designing Network Design Spaces (RegNet)

Radosavovic, Kosaraju, Girshick, He and Dollár, CVPR 2020 [16].

The problem. Manual design and neural architecture search both produce single models, which say little about why they work and may not transfer to other settings.

The idea. Study design **spaces**: sample hundreds of models from a parameterized space, train each briefly, and compare spaces by their error distribution functions. Refine the space step by

step by testing simple constraints (shared bottleneck ratio, shared group width, increasing widths and depths), keeping those that keep or improve the distribution while shrinking the space. The final space, RegNet, is described by a quantized linear rule for widths.

What it showed. Simple, regular models do as well as or better than the unconstrained space. Several findings go against common practice: a depth of about 20 blocks is good across compute regimes, the best models use no bottleneck, and inverted bottlenecks and depthwise convolutions do worse in this space; SE helps (RegNetY). The paper also compares RegNet models with popular existing networks under comparable training settings.

How to read it. The error EDF plots are the core: look at the whole curve, not just its best point. The findings are conditional on the space (ResNet-like blocks, ImageNet, the compute range and the short training schedule used for sampling); the authors check that the trends hold for longer training and other regimes, but conclusions about blocks such as depthwise convolution should not be transferred to very different spaces without testing.

Why it matters here. It turns "explore the design space" into a method with a statistic, and it is the main reason this chapter says "compare populations, not single winners".

A Comparative Study of Image Restoration Networks for General Backbone Network Design (X-Restormer)

Chen, Li, Pu, Liu, Zhou, Qiao and Dong, ECCV 2024 [22].

The problem. Restoration backbones are usually designed and evaluated on one or two tasks, and a network that is excellent on one task can be mediocre on another.

The idea. Benchmark five representative backbones, among them SwinIR, Restormer and NAFNet, on five tasks (super-resolution, denoising, deblurring, deraining, dehazing) under common settings, and explain the differences through the characteristics of each degradation and each network: macro-architecture (U-shaped, plain residual-in-residual, multi-stage) and core operator (convolution, spatial self-attention, transposed self-attention).

What it showed. The authors report that SwinIR is strong on super-resolution but falls short on deblurring and dehazing, that Restormer is strong on dehazing and deraining but weaker on super-resolution, and that Restormer ranks in the top two on all five tasks. They attribute Restormer's weakness to weaker spatial mapping and add spatial attention to half of its blocks, giving X-Restormer, without adding parameters.

How to read it. Check that every network is trained with the same data and schedule for each task; note that the U-shaped networks need the low-resolution input upsampled first for super-resolution, which changes their cost and behavior on that task; and treat the functional explanations (local versus global information needs) as hypotheses supported by the benchmark, not as measured properties.

Why it matters here. It is the restoration counterpart of the side-by-side tables: it links each composition choice to the kind of degradation it serves.

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DLCV 02 • Deep Image Classification

How deep networks decide what an image shows: losses and metrics, datasets, the CNN and transformer architectures and why they were designed that way, modern training recipes, pre-training, robustness, calibration and interpretability.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-02-image-classification/

What you'll learn

- How image classification is posed precisely: single-label, multi-label and fine-grained tasks; logits, softmax and cross-entropy; top-1/top-5 accuracy; the confusion matrix and the metrics built on it (precision, recall, F1, MCC, macro and micro averages, ROC-AUC, average precision, mAP); and what it means for a classifier to be calibrated.
- What the standard training datasets (MNIST, CIFAR, ImageNet, iNaturalist) are for, which evaluation benchmarks test what (ImageNet-ReaL, ImageNetV2, robustness sets, fine-grained and transfer suites, long-tailed and zero-shot evaluation), and how to run an honest evaluation protocol.
- The design reasoning behind each major backbone, from AlexNet, VGG, Inception and ResNet to MobileNet, EfficientNet, RegNet, ConvNeXt, ViT, DeiT and Swin, with parameter and cost arithmetic you can check yourself.
- Which parts of a modern training recipe matter (augmentation, Mixup/CutMix, label smoothing, AdamW, warmup + cosine, stochastic depth, EMA) and why recipes alone moved ResNet-50 by several points.
- How pre-training changed the job: linear probes and fine-tuning, contrastive and masked self-supervision, DINOv2-style frozen features and CLIP zero-shot classification.
- How to evaluate honestly: distribution shift, adversarial examples, shortcut learning, temperature scaling, efficiency metrics and Grad-CAM.

The big picture

Image classification asks one question of a whole image: *which of these labels fits?* It is the oldest deep-learning benchmark task and still the place where most backbone networks are born. A detector, a segmenter or a restoration network usually starts from a backbone that was first trained, or at least evaluated, as an image classifier.

> **Explain like I'm five**
>
> Think of a child learning animals from a picture book. At first the child only sees blobs. With many pages, the child learns to spot ears, then faces, then "that's a cat". A deep classifier does the same in layers: early layers notice edges and colors, middle layers notice parts like eyes and wheels, and the

last layer votes for a label. Training is the parent saying "no, that's a dog" thousands of times. The rest of this tutorial is about building a better picture book (data), a better brain (architecture), a better way of teaching (training recipe), and better exams (evaluation), because a child who only ever saw cats on sofas may decide that "sofa" means "cat".

This tutorial assumes you know what a convolution, a CNN layer and backpropagation are. The classical side of classification (Bayes classifiers, perceptrons, a small CNN written in NumPy, softmax and cross-entropy derived by hand) is in IP 13. The broad history of deep learning for images is in DLCV 01. Here we go deep on classification itself. For a refresher on activation functions and optimizers, see ML 02 ·Neural Networks. A good single survey of the CNN side of this story is Li et al. [1].

Related chapters. DLCV 02A turns classification into comparison: face recognition, embeddings and margin-based losses for open-set problems. DLCV 01B compares how classification and restoration networks compose the same building blocks.

The classification problem

Plain version. The network looks at an image and outputs one score per class. We turn scores into probabilities, compare them with the true label, and nudge the network so the right class gets more probability next time.

Single-label, multi-label and fine-grained

Let $\mathbf{x} \in \mathbb{R}^{H\times W\times 3}$ be an RGB image and $\mathcal{Y} = \{1, \dots, K\}$ a set of K classes. A network f_θ with parameters θ maps the image to a vector of **logits** $\mathbf{z} = f_\theta(\mathbf{x}) \in \mathbb{R}^K$, one unnormalized score per class.

- **Single-label (multi-class)** classification assumes exactly one correct class $y \in \mathcal{Y}$. The logits are turned into a probability vector with the softmax,

$$p_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, \qquad k = 1, \dots, K,$$

where p_k is the model's estimate of $P(y = k \mid \mathbf{x})$. Training minimizes the **cross-entropy**, the negative log-probability of the true class, averaged over a mini-batch:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B}\sum_{i=1}^{B} \log p_{y_i}(\mathbf{x}_i),$$

where B is the batch size and y_i the label of image $\mathbf{x}_i$. The gradient with respect to the logits is $\partial \mathcal{L}/\partial \mathbf{z} = \mathbf{p} - \mathbf{e}_y$ (per image), with $\mathbf{e}_y$ the one-hot vector of the true class; DIP Chapter 13 derives it.

- **Multi-label** classification allows any subset of labels (a street photo can contain "car", "person" and "bicycle"). Each class gets an independent sigmoid $\sigma(z_k) = 1/(1+e^{-z_k})$ and a binary cross-entropy, $-\sum_k [t_k \log \sigma(z_k) + (1 - t_k) \log(1 - \sigma(z_k))]$, where $t_k \in \{0, 1\}$ says whether class k is present. Metrics change too: mean average precision over classes replaces accuracy.
- **Fine-grained** classification separates visually similar subordinate classes, such as 200 bird species or 1,000 aircraft variants. The loss is the same as single-label, but the signal lives in small parts (a beak shape, a wing bar), classes are often long-tailed, and label noise from non-expert annotators is common. iNaturalist [6] is the standard large fine-grained benchmark.

Metrics: top-1, top-5 and beyond

Top-1 accuracy is the fraction of images whose highest-scoring class is the true one. **Top-k accuracy** counts an image as correct if the true class is among the k highest scores. The ImageNet challenge reported top-5 error because many images contain several objects, and some classes are nearly indistinguishable [2]. Top-5 hides real mistakes, so modern papers mostly report top-1. For imbalanced data, report **balanced accuracy** (the mean of per-class recalls) or per-class results, because a classifier can score high accuracy by ignoring rare classes.

```
import torch
import torch.nn.functional as F

logits = torch.tensor([[2.0, 1.0, 0.1, -1.0],    # 2 images, 4 classes
                       [0.5, 2.5, 2.4, 0.0]])
y = torch.tensor([0, 2])

p = logits.softmax(dim=1)
ce = F.cross_entropy(logits, y)                       # mean of -log p[i, y_i]
ce_ls = F.cross_entropy(logits, y, label_smoothing=0.1)
top1 = (logits.argmax(1) == y).float().mean()
top2 = (logits.topk(2, dim=1).indices == y[:, None]).any(1).float().mean()
print(p.round(decimals=3))
print(f"CE {ce:.3f}  CE+LS {ce_ls:.3f}  top-1 {top1:.2f}  top-2 {top2:.2f}")

# multi-label: one independent sigmoid per class, binary cross-entropy
targets = torch.tensor([[1., 0., 1., 0.], [0., 1., 1., 0.]])
print(f"BCE {F.binary_cross_entropy_with_logits(logits, targets):.3f}")
```

The second image is a near-tie between classes 1 and 2. Top-1 counts it as wrong; top-2 counts it as right. That one example is the whole argument for and against top-5.

The confusion matrix and the metrics built on it

Plain version. Accuracy collapses all mistakes into one number. A confusion matrix keeps them apart: it counts, for every true class, what the model said. Almost every other classification metric is a ratio of cells in that table, and each ratio answers a different question.

The binary case: TP, FP, FN, TN

Take one class as **positive** (say "sick") and the rest as **negative**. Each prediction lands in one of four cells:

	Predicted positive	Predicted negative
Actually positive	true positive (TP)	false negative (FN)
Actually negative	false positive (FP)	true negative (TN)

> **Explain like I'm five**
>
> A smoke alarm is a classifier. When there is a fire and it rings, that is a true positive. When you burn toast and it rings, that is a false positive: annoying, but nobody is hurt. When there is a fire and it stays quiet, that is a false negative: the dangerous one. When nothing happens and it stays quiet, that is a true negative. A good alarm is not just "right most of the time" (it is quiet almost every day, so that is easy). It must ring for fires and stay quiet for toast. Different metrics weigh those two goals differently.

With $N = \mathrm{TP} + \mathrm{FP} + \mathrm{FN} + \mathrm{TN}$ test examples, the standard ratios are

$$\text{accuracy} = \frac{\mathrm{TP} + \mathrm{TN}}{N}, \quad \text{precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad \text{recall (TPR)} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}},$$

$$\text{specificity (TNR)} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}, \quad \mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} = 1 - \mathrm{TNR}.$$

- **Precision** answers "when the model says positive, how often is it right?" It matters when a false alarm is costly: a spam filter that hides real mail, an automatic product recall.
- **Recall**, also called **sensitivity** or the **true positive rate (TPR)**, answers "of all real positives, how many did we catch?" It matters when a miss is costly: cancer screening, a defect detector on a production line.
- **Specificity**, the **true negative rate (TNR)**, is recall for the negative class. The **false positive rate (FPR)** is its complement, the fraction of negatives wrongly flagged.

Precision and recall pull against each other: lowering the decision threshold flags more items, which raises recall and usually lowers precision. Single numbers that combine them are:

$$F_1 = \frac{2\,\mathrm{P}\,\mathrm{R}}{\mathrm{P}+\mathrm{R}} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}, \qquad F_\beta = \frac{(1+\beta^2)\,\mathrm{P}\,\mathrm{R}}{\beta^2\,\mathrm{P}+\mathrm{R}},$$

where P is precision, R recall and $\beta > 0$ says how many times more recall matters than precision (F_2 favours recall, $F_{0.5}$ favours precision). F_1 is the harmonic mean, so it is high only when *both* parts are high. Note what F_1 ignores: true negatives never appear in it.

Two further summaries use all four cells:

$$\text{balanced accuracy} = \frac{\mathrm{TPR}+\mathrm{TNR}}{2}, \qquad \mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN}-\mathrm{FP}\cdot\mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}.$$

Balanced accuracy is the mean of the per-class recalls, so a classifier that ignores one class scores 0.5 however rare that class is. The **Matthews correlation coefficient** (MCC) is the Pearson correlation between the true and predicted binary labels: $+1$ is perfect, 0 is no better than chance, -1 is perfectly inverted. Chicco and Jurman [57] argue that it should be preferred over accuracy and F_1 for binary evaluation because it is high only when the classifier does well in all four cells. Sokolova and Lapalme [58] give a systematic comparison of these measures and of how each one reacts to changes in the confusion matrix.

Why accuracy misleads at 99:1. Suppose 10 of 1,000 scans are sick. A "model" that always answers *healthy* scores 99.0% accuracy. Model B flags 30 scans and catches 8 of the 10 sick ones; it scores *lower* accuracy (97.6%) yet is the only useful one:

```
import numpy as np
from sklearn.metrics import (accuracy_score, balanced_accuracy_score, confusion_matrix,
                             f1_score, matthews_corrcoef, precision_score, recall_score)

# 1,000 scans, 10 sick (label 1). Model A says "healthy" to everyone.
# Model B flags 30 scans and catches 8 of the 10 sick ones.
y = np.r_[np.ones(10), np.zeros(990)].astype(int)
pred_a = np.zeros(1000, dtype=int)
pred_b = np.r_[np.ones(8), np.zeros(2), np.ones(22), np.zeros(968)].astype(int)

for name, p in [("all-negative", pred_a), ("model B", pred_b)]:
    tn, fp, fn, tp = confusion_matrix(y, p).ravel()
    print(f"{name:12s} TP={tp} FP={fp} FN={fn} TN={tn}  "
          f"acc={accuracy_score(y, p):.3f} bal-acc={balanced_accuracy_score(y, p):.3f} "
          f"P={precision_score(y, p, zero_division=0):.3f} R={recall_score(y, p):.3f} "
          f"F1={f1_score(y, p, zero_division=0):.3f} MCC={matthews_corrcoef(y, p):.3f}")
# all-negative TP=0 FP=0 FN=10 TN=990  acc=0.990 bal-acc=0.500 P=0.000 R=0.000 F1=0.000
 ↪  MCC=0.000
# model B      TP=8 FP=22 FN=2 TN=968  acc=0.976 bal-acc=0.889 P=0.267 R=0.800 F1=0.400
 ↪  MCC=0.454
```

Balanced accuracy, F_1 and MCC all rank model B first; accuracy ranks it last. Rules of thumb: use accuracy only when classes are roughly balanced and all errors cost the same; report precision and recall (or F_β with a β chosen from the real costs) when one kind of error matters more; use balanced accuracy or MCC as a single imbalance-aware summary.

The multi-class confusion matrix

With K classes the confusion matrix $C \in \mathbb{N}^{K\times K}$ has entry C_{ij} = number of test images whose **true** class is i (row) and whose **predicted** class is j (column); this is the convention of scikit-learn [62]. The diagonal holds the correct predictions. Dividing each row by its sum gives the **row-normalized** matrix, whose diagonal is the per-class recall and whose off-diagonal entries show *which* classes are confused. Figure 1 shows one for a logistic regression trained on only 200 of scikit-learn's 8×8 digits: most mistakes are $3 \to 8$, $4 \to 1$ and $7 \to 9$, pairs that share strokes.

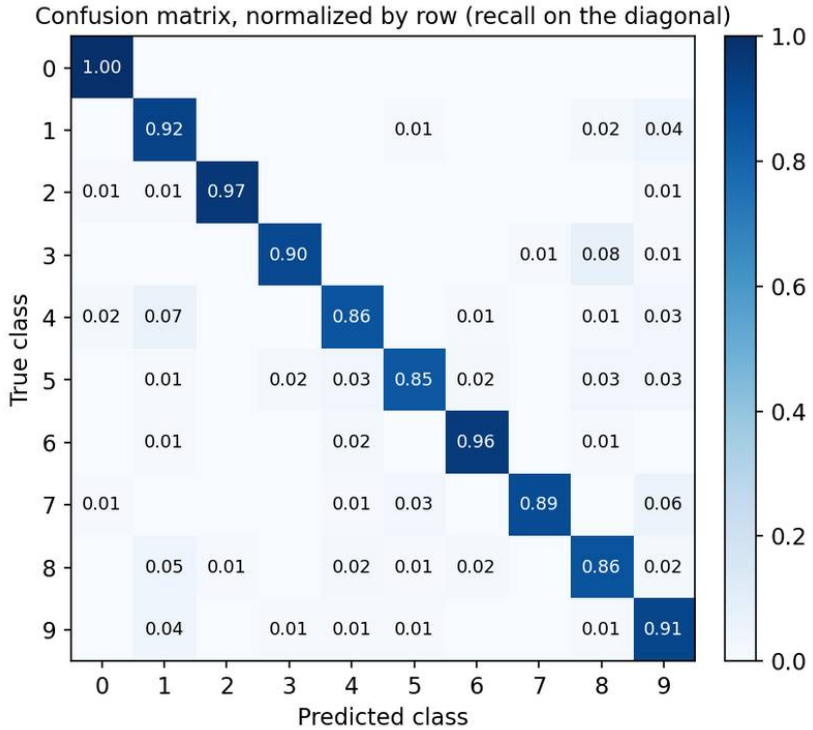


Figure 1 — Row-normalized confusion matrix of a logistic regression trained on 200 digits and tested on 899 (accuracy 91.1%). Each row sums to 1; the diagonal is the recall of each class, and the bright off-diagonal cells name the confusions worth fixing.

Treating each class k in turn as "positive" and all others as "negative" gives per-class counts: $\mathrm{TP}_k = C_{kk}$, $\mathrm{FP}_k = \sum_{i\neq k} C_{ik}$ (the rest of column k), $\mathrm{FN}_k = \sum_{j\neq k} C_{kj}$ (the rest of row k). Hence per-class precision $\mathrm{P}_k = C_{kk} / \sum_i C_{ik}$ and recall $\mathrm{R}_k = C_{kk} / \sum_j C_{kj}$. There are three common ways to turn K per-class numbers into one:

- **Macro average**: the plain mean over classes, $\mathrm{F}_1^{\text{macro}} = \frac{1}{K}\sum_k \mathrm{F}_{1,k}$. Every class counts equally, so rare classes weigh as much as common ones.
- **Micro average**: sum TP_k, FP_k, FN_k over classes first, then compute the metric once. In single-

label multi-class problems every error is one FP (for the predicted class) and one FN (for the true class), so micro precision = micro recall = micro F_1 = accuracy.

- **Weighted average**: the mean of per-class values weighted by each class's **support** n_k (number of true examples). It sits close to micro and hides rare-class failures almost as well.

On an imbalanced test set the averages can differ a lot. If a model is good on a common class and poor on rare ones, micro averaging (dominated by the common class) gives a high score and macro averaging a much lower one; that gap is the signal. Report macro when every class matters (rare diseases, rare species); report micro/accuracy when every *image* matters equally.

Optional: A worked macro-versus-micro example (in the Supplementary notes). A small three-class confusion matrix worked through by hand.

Cohen's kappa [59] is accuracy corrected for chance: it compares the observed agreement between predictions and labels with the agreement a classifier would reach by guessing each class at its observed rate. It is 1 for perfect agreement and 0 at chance level, and it is common when a model is compared with human annotators.

Optional: Cohen's kappa: agreement beyond chance (in the Supplementary notes). The formula and its value for the digit classifier below.

```
from sklearn.datasets import load_digits
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (average_precision_score, classification_report,
                             confusion_matrix, precision_recall_fscore_support,
                              ↪  roc_auc_score)
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import label_binarize

X, y = load_digits(return_X_y=True)
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.5, stratify=y, random_state=0)
clf = LogisticRegression(max_iter=2000).fit(X_tr[:200], y_tr[:200])   # only 200 labels
pred, prob = clf.predict(X_te), clf.predict_proba(X_te)

cm = confusion_matrix(y_te, pred)                     # rows = true, columns = predicted
cm_norm = confusion_matrix(y_te, pred, normalize="true")   # each row sums to 1 (Figure 1)
print(classification_report(y_te, pred, digits=3))    # per-class P, R, F1, support +
 ↪  averages
for avg in ["macro", "micro", "weighted"]:
    p, r, f, _ = precision_recall_fscore_support(y_te, pred, average=avg)
    print(f"{avg:8s} P={p:.3f} R={r:.3f} F1={f:.3f}")
print(f"ROC-AUC (one-vs-rest, macro) {roc_auc_score(y_te, prob, multi_class='ovr'):.4f}")
Y = label_binarize(y_te, classes=range(10))           # one binary column per class
print(f"mean AP over classes {average_precision_score(Y, prob, average='macro'):.4f}")
# macro    P=0.914 R=0.911 F1=0.912
# micro    P=0.911 R=0.911 F1=0.911
# weighted P=0.915 R=0.911 F1=0.912
# ROC-AUC (one-vs-rest, macro) 0.9958
# mean AP over classes 0.9742
```

Here macro and micro agree because the digit classes are balanced; on an imbalanced test set they would not.

Threshold-free evaluation: ROC and precision–recall curves

Every metric above needs a decision rule (argmax, or "positive if the score exceeds t"). A scoring classifier can be evaluated at *all* thresholds at once. Sweep t from high to low and record a point per threshold:

- The **ROC curve** plots TPR against FPR. Its area, **ROC-AUC**, equals the probability that a randomly chosen positive gets a higher score than a randomly chosen negative; 0.5 is chance, 1.0 is perfect separation. Fawcett [54] is the standard introduction.
- The **precision–recall (PR) curve** plots precision against recall. Its summary, **average precision** (AP), is the precision averaged over the recall levels at which a new positive is retrieved,

$$\mathrm{AP} = \sum_{n} (\mathrm{R}_n - \mathrm{R}_{n-1})\, \mathrm{P}_n,$$

where P_n and R_n are precision and recall at the n-th threshold, ordered by decreasing score (this is scikit-learn's definition [62]; benchmarks such as PASCAL VOC [61] have used interpolated variants, so check which one a paper reports).

Why PR under heavy imbalance. FPR divides by the number of negatives. When negatives outnumber positives 99 to 1, even thousands of false alarms barely move FPR, so the ROC curve looks excellent while precision is poor. Precision divides by the number of *flagged* items, so false alarms hit it directly, and the random baseline of AP equals the positive rate rather than 0.5. Davis and Goadrich [55] proved that a curve dominates in ROC space if and only if it dominates in PR space, and showed that optimizing ROC-AUC does not optimize AP; Saito and Rehmsmeier [56] showed with simulations and a literature review that PR plots are more informative than ROC plots on imbalanced data. Figure 2 makes this concrete: two classifiers with the same ROC-AUC (0.95) on a task with 1% positives differ clearly in AP (0.45 against 0.36).

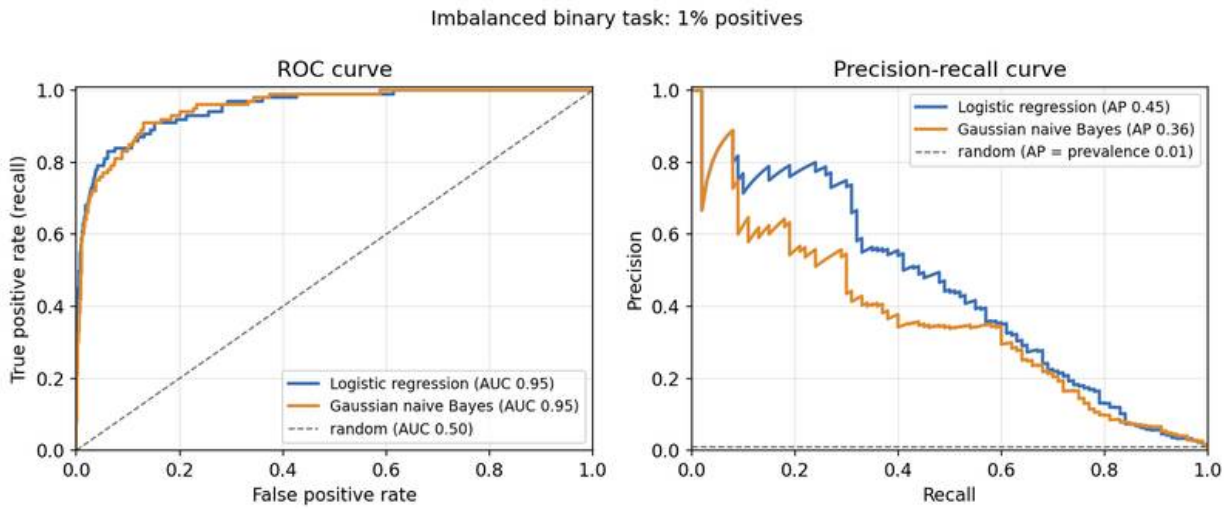


Figure 2 — ROC (left) and precision–recall (right) curves for two classifiers on a synthetic task with 1% positives (100 of 10,000 test points). Both have ROC-AUC 0.95, yet logistic regression keeps much higher precision at every recall level (AP 0.45 against 0.36). An all-negative classifier would score 99% accuracy here.

For $K > 2$ classes, both curves are computed **one-vs-rest**: class k is positive, all others negative, using the score p_k. The K areas are then macro-averaged (`roc_auc_score(..., multi_class="ovr")`), or micro-averaged by pooling all (k, image) pairs.

Multi-label metrics

In multi-label classification each label is its own yes/no problem, so the standard summary is the **average precision of each class**, averaged over classes into **mAP**, as in the PASCAL VOC classification task [61]. Thresholded metrics either compute a binary metric per label and average it over labels (macro or micro, as above), or score each image's predicted label set and average over images.

Optional: Multi-label metrics in detail (in the Supplementary notes). Example-based F1, Hamming loss and a small worked example in code.

Calibration: are the probabilities honest?

Accuracy only checks the argmax. **Calibration** checks whether the confidence means what it says: among all predictions made with confidence 0.8, about 80% should be right. Formally, a classifier is calibrated if $P(\hat{y} = y \mid \hat{p} = c) = c$ for every confidence c, where $\hat{y}$ is the predicted class and $\hat{p} = \max_k p_k$ its probability. The usual summary is the **expected calibration error**: split predictions into M confidence bins B_m and compute

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \, |\,\mathrm{acc}(B_m) - \mathrm{conf}(B_m)|,$$

where n is the number of test images, $\mathrm{acc}(B_m)$ the accuracy inside bin m and $\mathrm{conf}(B_m)$ the mean confidence there. Guo et al. [47] found that modern deep networks, trained long with cross-entropy, tend to be **overconfident**, and that a single scalar fix works remarkably well. We return to it in the robustness section.

Datasets and why they matter

Plain version. A benchmark is a frozen exam. It lets everyone compare methods fairly, but once a whole field studies for the same exam, scores start to measure the exam as much as the skill. It helps to keep two roles apart: datasets used to *train* a model and datasets used only to *evaluate* it.

Training datasets

Dataset	Images	Classes	Resolution	Role
MNIST [3]	60k train / 10k test	10 digits	28×28 gray	sanity check, teaching
CIFAR-10 / CIFAR-100 [4]	50k train / 10k test	10 / 100 (20 superclasses)	32×32 RGB	small-scale architecture and recipe studies
ImageNet-1k (ILSVRC 2012) [2]	about 1.28M train / 50k val	1,000	variable, usually trained at 224×224	the standard backbone benchmark
ImageNet-21k [5]	roughly 13–14M (versions differ)	about 21k	variable	larger-scale supervised pre-training
iNaturalist [6]	hundreds of thousands, by year	thousands of species	variable	fine-grained, long-tailed

MNIST is solved and now only catches bugs. CIFAR is small enough to run many controlled experiments, which is why regularization papers love it. ImageNet-1k became the reference after the 2012 challenge [2], [10]: architectures are usually reported as "top-1 on ImageNet-1k val at 224×224", and almost every pre-trained backbone in a model zoo has an ImageNet number attached. ImageNet-21k is the larger hierarchy from which ImageNet-1k was drawn; Ridnik et al. [5] made pre-training on it practical with a cleaned, semantic-hierarchy-aware recipe.

Evaluation benchmarks

Plain version. Training data teaches; evaluation data examines. A model that only ever sat one exam may have learned that exam. So modern papers grade a backbone on a *battery* of test sets,

each built to probe one thing: the original distribution, a fresh sample of it, corrupted or stylized images, unusual viewpoints, fine-grained categories, small labelled budgets or rare classes.

The evaluation sets you will meet most often fall into five groups:

- **In distribution.** The ImageNet-1k validation set [2] is the de facto test set; **ImageNet-ReaL** [8] re-labels the same images with multiple labels per image.
- **A fresh sample. ImageNetV2** [7] repeats the original collection process to build new test images.
- **Distribution shift. ImageNet-C** [42] adds synthetic corruptions, **ImageNet-R** [43] contains renditions such as art and cartoons, **ImageNet-A** [44] collects natural photos that fool a ResNet-50, **ImageNet-Sketch** [63] contains sketches, and **ObjectNet** [64] shows objects at unusual rotations, backgrounds and viewpoints.
- **Long tail and fine grain. ImageNet-LT** [73] trains on a long-tailed subset and tests on balanced data; iNaturalist [6], CUB-200-2011 [65] and Stanford Cars [66] test fine-grained recognition.
- **Transfer.** Suites such as the 12 datasets of Kornblith et al. [33] and **VTAB** [72] measure how well frozen or fine-tuned features adapt to small, specialized tasks.

Two practical warnings. Several fine-grained benchmarks report **mean per-class accuracy** instead of plain accuracy, so do not mix the two in one comparison. And image–text models such as CLIP [40] are often evaluated **zero-shot**, without any training on the target data: prompt wording then becomes a hyperparameter that must not be tuned on the test set, and web-scale pre-training data may already contain the test images.

Optional: Evaluation benchmarks in detail (in the Supplementary notes). The size, purpose and usual metric of each benchmark, the two transfer suites, and the zero-shot results reported for CLIP.

How reliable are the numbers?

Three lines of work changed how we read ImageNet accuracy. Rebuilding the test set by the original procedure (ImageNetV2 [7]) lowered every model's accuracy but largely kept their ranking. Re-labelling studies [8], [9] showed that single labels and label errors can change which model looks best. And every dataset has its own bias in viewpoint and background, so test on independently collected data whenever possible.

Optional: How reliable are ImageNet numbers? (in the Supplementary notes). The size of the ImageNetV2 drop and what the re-labelling studies found.

Read a benchmark number as a claim about a benchmark

"81.0% top-1" means: on these 50,000 particular validation images, with this preprocessing and this label set. Report the resolution, the number of crops, whether extra data or distillation was used, and the variance across seeds when differences are under a point.

Evaluation protocol

A benchmark number is only as honest as the protocol behind it.

- **Three splits, three jobs.** Fit parameters on the **training** set. Choose hyperparameters, early stopping, temperature, thresholds and prompts on a **validation** set (or by cross-validation inside the training data). Touch the **test** set once, at the end. When a benchmark only publishes a validation split, as with ImageNet-1k, hold out part of the training set for tuning.
- **Never tune on the test set.** Every decision made after looking at test results (an extra augmentation, another epoch, a different checkpoint) fits the test set a little. Repeated over a field's worth of papers, this is **test-set reuse**, or adaptive overfitting to a benchmark. ImageNetV2 [7] found that ranking was largely preserved on a fresh sample, which is reassuring, but the same study also shows that absolute numbers do not transfer and that a fresh test set is the only direct check.
- **Report variation.** Train with several random seeds (at least three, more for small datasets) and report mean $\pm$ standard deviation, or a confidence interval. With 10,000 test images, ImageNetV2 [7] quotes 95% intervals of about $\pm 1\%$; a 0.3-point gain on such a set is not a result.
- **Report the whole recipe.** Resolution, number of crops, extra data, distillation, which variant of a metric (top-1 or mean per-class accuracy, which AP definition) and which version of a label set (original or ReaL).

CNN architectures and the reasons behind them

Plain version. Each famous CNN fixed one specific problem of its predecessors: training too slow, too many parameters, too hard to optimize when deep, too expensive on a phone. Knowing the problem each one solved is more useful than memorizing their layer tables.

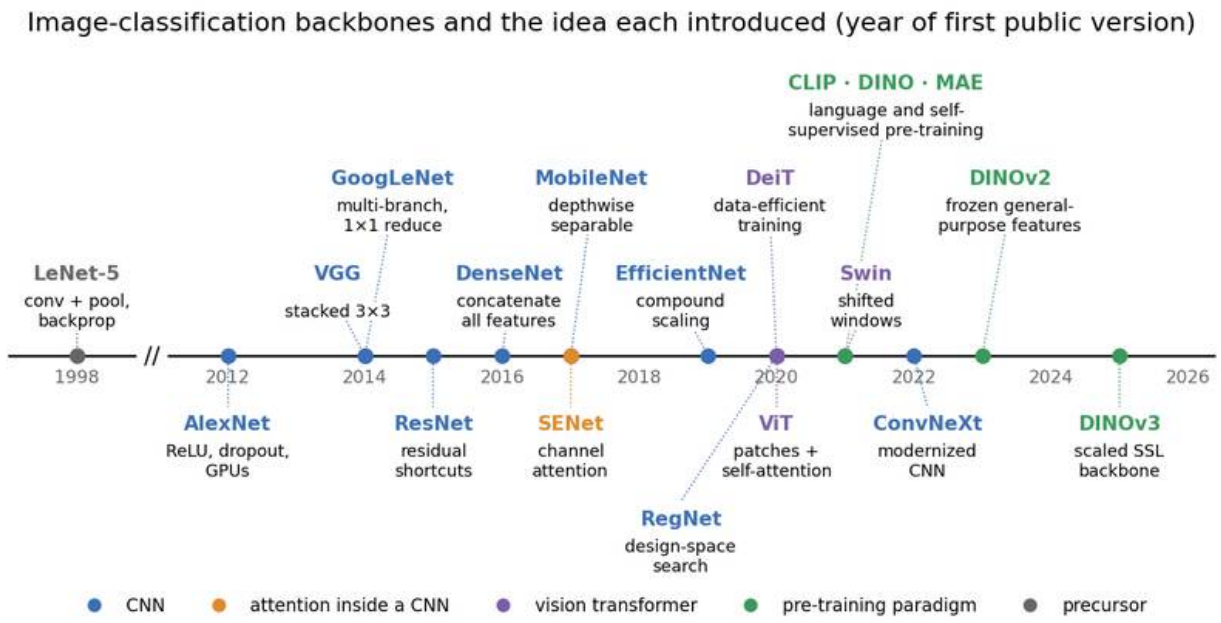


Figure 3 — Image-classification backbones and the main idea each introduced. Dates are the year of the first public version (usually arXiv). Colors separate pure CNNs, attention inside CNNs, vision transformers and pre-training paradigms.

AlexNet: depth becomes trainable

Before 2012, the best ImageNet systems used hand-crafted features (SIFT, Fisher vectors) with linear classifiers. AlexNet [10] was a five-convolution, three-fully-connected network trained end to end on ImageNet. Three ingredients made it work at that scale. **ReLU** activations, $\max(0, z)$, do not saturate for positive inputs, so gradients do not vanish the way they do with tanh, and the paper reports several times faster training. **Dropout** in the large fully connected layers fought overfitting; those layers held most of the parameters. **GPUs** made it feasible: the model was split across two GPUs. Heavy data augmentation (random crops and flips, color perturbation) was the fourth, often forgotten, ingredient. The design lesson: with enough data and compute, learned features beat engineered ones.

VGG: small kernels, stacked

VGG [11] asked a cleaner question: what happens if we only use 3×3 convolutions and go deeper (16–19 weight layers)? Two stacked 3×3 layers see a 5×5 region; three see 7×7. For C input and output channels,

$$\underbrace{2 \cdot 3^2 C^2}_{\text{two } 3\times 3} = 18C^2 \;<\; \underbrace{5^2 C^2}_{\text{one } 5\times 5} = 25C^2,$$

ignoring biases. So the stack has fewer parameters for the same receptive field, and it inserts an extra nonlinearity between the two layers. VGG's uniform design made it a favourite feature extractor for years. Its weakness is cost: the fully connected head and the wide early layers make it heavy.

GoogLeNet / Inception: width, branches and 1×1 bottlenecks

GoogLeNet [12] attacked cost instead. Each **Inception module** runs 1×1, 3×3, 5×5 convolutions and pooling in parallel and concatenates the outputs, so the next layer can choose the right scale.

Running big kernels on many channels would be expensive, so 1×1 **convolutions** first reduce the channel count. A 1×1 convolution is a per-pixel linear map across channels: it mixes channels without looking at neighbours, and it is the cheapest way to change width. GoogLeNet also replaced the large fully connected head with **global average pooling**. The follow-up Inception-v3 paper [13] factorized large kernels into stacks or into $1 \times n$ and $n \times 1$ pairs, and introduced **label smoothing**, which we meet again in the training section.

ResNet: learning residuals

The problem. Stacking more plain layers eventually made networks *worse*, even on the training set. This "degradation" is an optimization problem, not overfitting: a deeper network could in principle copy a shallower one and set the extra layers to identity, yet SGD did not find that solution.

The idea. He et al. [14] let a block learn a residual $\mathcal{F}$ and add its input back:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}; \{W_i\}) + \mathbf{x},$$

where $\mathbf{x}$ is the block input, $\mathcal{F}$ is two or three convolution layers with weights W_i, and $\mathbf{y}$ the output. If the best mapping is close to identity, the block only has to push $\mathcal{F}$ toward zero, which is easy. When the shapes differ (more channels, stride 2), the shortcut becomes a 1×1 projection.

Why it trains. Because every block adds its input back, the output of a deep stack equals the input of any earlier block plus a sum of residuals. The gradient of the loss therefore has a direct path through the shortcuts to every block, so it cannot vanish through a long product of small factors. A follow-up paper [15] also found that keeping the shortcut a *pure* identity (moving BN and ReLU into the residual branch, "pre-activation") makes even deeper networks easier to train.

Optional: Why residual networks train: the unrolled gradient (in the Supplementary notes). The two-line derivation behind this argument.

Optional: Batch normalization (a supplementary note in DLCV 01). What the BN layer used throughout this page computes, and why training and inference use different statistics.

The bottleneck. For deeper ResNets (50 layers and up) each block becomes 1×1 (reduce) $\to 3\times3$ $\to 1\times1$ (expand), shown in Figure 4(c). At 256 channels, a basic block of two 3×3 convolutions has $2\cdot9\cdot256^2 \approx 1.18$M weights; a bottleneck that reduces to 64 channels has $256\cdot64+9\cdot64^2+64\cdot256 \approx 70$k. The expensive 3×3 runs on a thin tensor, which is what makes 50–152 layers affordable.

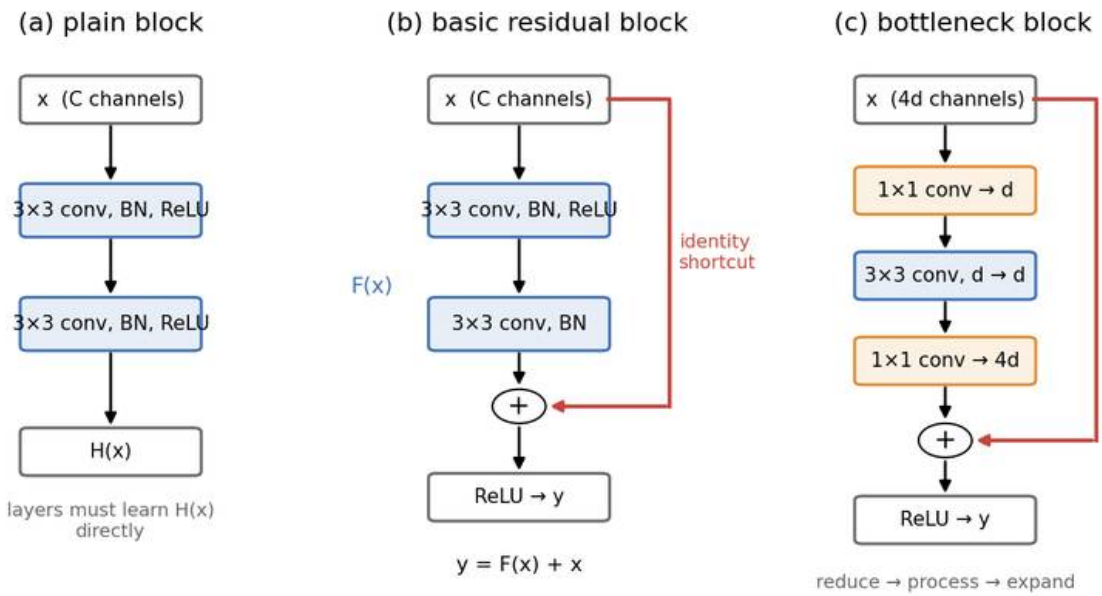


Figure 4 — (a) A plain block must learn the full mapping $H(x)$. (b) A residual block learns $F(x) = H(x) - x$ and adds the input back through an identity shortcut. (c) The bottleneck block runs the 3×3 convolution on a reduced width d and expands back to $4d$.

The basic block used in this tutorial's experiments, written in PyTorch:

```
import torch.nn as nn
import torch.nn.functional as F

class BasicBlock(nn.Module):
    def __init__(self, cin, cout, stride=1):
        super().__init__()
        self.conv1 = nn.Conv2d(cin, cout, 3, stride, 1, bias=False)
        self.bn1 = nn.BatchNorm2d(cout)
        self.conv2 = nn.Conv2d(cout, cout, 3, 1, 1, bias=False)
        self.bn2 = nn.BatchNorm2d(cout)
        self.short = nn.Identity()
        if stride != 1 or cin != cout:   # projection shortcut when shapes change
            self.short = nn.Sequential(nn.Conv2d(cin, cout, 1, stride, bias=False),
             ↪  nn.BatchNorm2d(cout))

    def forward(self, x):
        out = F.relu(self.bn1(self.conv1(x)))
        out = self.bn2(self.conv2(out))
```

```
        return F.relu(out + self.short(x))   # y = F(x) + x
```

Two later CNN designs refine these ideas. **DenseNet** [16] gives each layer in a block the concatenated outputs of *all* earlier layers, which reuses features with few parameters but costs memory.

RegNet [19] studied whole populations of ResNet-like networks and found that good ones follow simple rules for how width grows from stage to stage.

Optional: Two more CNN designs: DenseNet and RegNet (in the Supplementary notes). How each design works and the lesson it teaches.

MobileNet: factorize the convolution

MobileNet [17] targets phones. A standard $K \times K$ convolution from M to N channels on an $H \times W$ map costs HWK^2MN multiply–adds. A **depthwise separable** convolution splits it into a **depthwise** $K \times K$ convolution that filters each channel separately (HWK^2M) and a **pointwise** 1×1 convolution that mixes channels ($HWMN$). The cost ratio is

$$\frac{HWK^2M + HWMN}{HWK^2MN} = \frac{1}{N} + \frac{1}{K^2},$$

so with $K = 3$ and many output channels the separable version is roughly 8–9 times cheaper. The bet is that spatial filtering and channel mixing do not need to be learned jointly. That bet held up so well that depthwise convolutions reappear in EfficientNet and ConvNeXt.

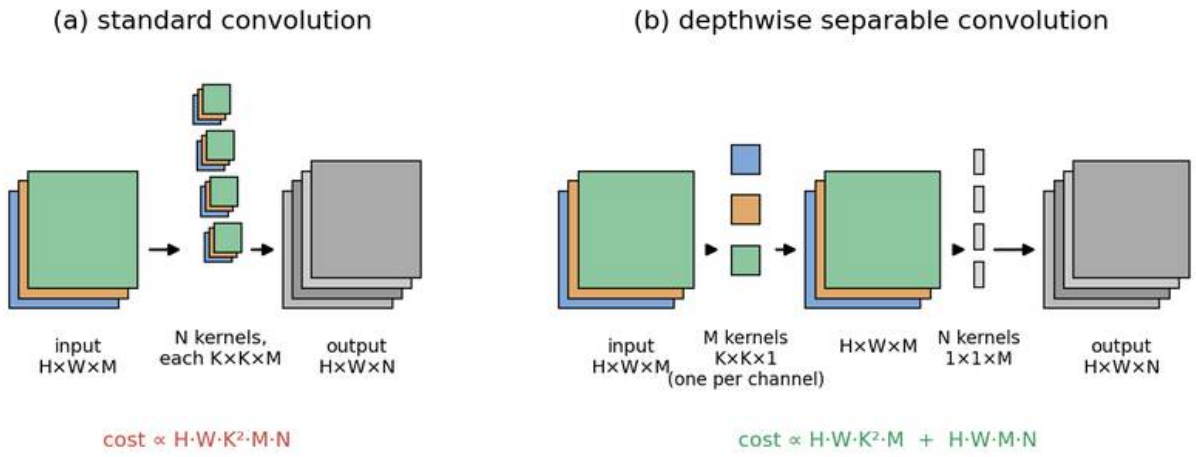


Figure 5 — Standard convolution filters and mixes channels in one step. Depthwise separable convolution filters each channel alone (depthwise) and then mixes channels with 1×1 kernels (pointwise).

```
import torch.nn as nn
```

```
def n_params(m):
    return sum(p.numel() for p in m.parameters())

M, N, K = 128, 128, 3
standard = nn.Conv2d(M, N, K, padding=1, bias=False)
separable = nn.Sequential(nn.Conv2d(M, M, K, padding=1, groups=M, bias=False),   # depthwise
                          nn.Conv2d(M, N, 1, bias=False))                        #
                           ↪ pointwise
print("standard:", n_params(standard), " separable:", n_params(separable),
      f" ratio {n_params(separable) / n_params(standard):.3f} vs 1/N + 1/K^2 = {1/N +
       ↪ 1/K**2:.3f}")
# standard: 147456  separable: 17536  ratio 0.119 vs 1/N + 1/K^2 = 0.119
```

EfficientNet: scale depth, width and resolution together

Given a good small network, how should we make it bigger? Deeper, wider, or higher input resolution? Tan and Le [18] argued these interact: a higher resolution image needs more depth (larger receptive field) and more width (finer patterns). **Compound scaling** grows all three with one coefficient ϕ:

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi, \qquad \text{s.t. } \alpha\beta^2\gamma^2 \approx 2, \;\; \alpha, \beta, \gamma \geq 1,$$

where d, w, r multiply depth, width and resolution. FLOPs scale roughly as $d\,w^2 r^2$, so each unit increase of ϕ roughly doubles the cost. The constants are found by a small grid search on the base network (itself found by architecture search, using inverted-bottleneck blocks with depthwise convolutions and squeeze-and-excitation). The idea, "scale in a balanced way", outlived the specific constants.

ConvNeXt: a CNN modernized step by step

After vision transformers took over the leaderboards, Liu et al. [20] asked how much of their advantage came from attention and how much from everything else. They started from a ResNet-50 trained with a transformer-style recipe and changed one thing at a time: stage ratios like Swin's, a "patchify" stem (a 4×4 stride-4 convolution), depthwise convolutions with larger 7×7 kernels, an inverted bottleneck, fewer activations and normalizations, GELU and LayerNorm. The result, ConvNeXt, matched Swin-style transformers of similar cost on ImageNet and downstream tasks. The lesson: architecture families matter less than the many small design and training choices that come with them.

Attention and transformers

Plain version. A convolution only looks at a small neighbourhood with fixed weights. Attention lets each position decide, based on content, which other positions to listen to. Vision transformers

apply this to image patches.

Attention first entered CNNs in a light form. The **squeeze-and-excitation** (**SE**) block [21] summarizes each channel of a feature map by its global average, and a tiny network turns these summaries into weights between 0 and 1 that rescale the channels. The network can then emphasise, say, "fur" channels when the whole image looks like an animal. SE blocks are part of EfficientNet and many mobile networks.

Optional: Squeeze-and-excitation: attention inside a CNN (in the Supplementary notes). The SE formula and a few lines of PyTorch.

ViT: an image as a sequence of patches

The Vision Transformer [22] removes convolutions almost entirely. An image $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ is cut into $N = HW/P^2$ non-overlapping $P\times P$ patches. Each patch is flattened and linearly projected to a D-dimensional **token**. A learnable **class token** is prepended and learnable **position embeddings** are added:

$$\mathbf{z}_0 = [\,\mathbf{x}_{\text{class}};\ \mathbf{x}_p^1\mathbf{E};\ \dots;\ \mathbf{x}_p^N\mathbf{E}\,] + \mathbf{E}_{\text{pos}},$$

where $\mathbf{x}_p^i \in \mathbb{R}^{P^2C}$ is the i-th flattened patch, $\mathbf{E} \in \mathbb{R}^{P^2C\times D}$ the patch embedding and $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1)\times D}$ the position embeddings. A stack of L standard transformer encoder layers (multi-head self-attention and MLP, each with LayerNorm and a residual connection) processes the sequence, and a small head reads the final class token to produce logits. Self-attention for one head is

$$\text{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\Big(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\Big)\mathbf{V},$$

with $\mathbf{Q} = \mathbf{Z}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{Z}\mathbf{W}_K$, $\mathbf{V} = \mathbf{Z}\mathbf{W}_V$ linear projections of the token matrix $\mathbf{Z}$, and d the key dimension. The attention matrix is $(N+1)\times(N+1)$: every patch can attend to every other patch from the first layer, so the receptive field is global immediately.

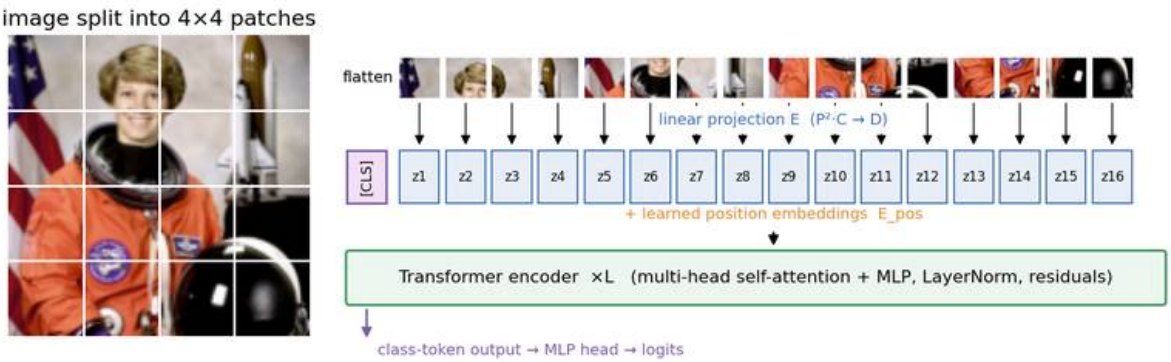


Figure 6 — The ViT input pipeline. Patches become tokens through one shared linear projection (equivalently, a $P\times P$ convolution with stride P). A class token and learned position embeddings are added, and the

class token's final state is classified.

```
import torch
import torch.nn as nn

B, C, H, W, P, D = 2, 3, 32, 32, 8, 64
img = torch.randn(B, C, H, W)

# 1) patchify: (B, C, H, W) -> (B, N, P*P*C) with N = (H/P)*(W/P)
patches = img.unfold(2, P, P).unfold(3, P, P)                 # (B, C, H/P, W/P, P, P)
patches = patches.permute(0, 2, 3, 1, 4, 5).reshape(B, -1, C * P * P)
N = patches.shape[1]

# 2) linear patch embedding + class token + learned position embedding
embed = nn.Linear(C * P * P, D)          # same as nn.Conv2d(C, D, kernel_size=P, stride=P)
cls = nn.Parameter(torch.zeros(1, 1, D))
pos = nn.Parameter(torch.randn(1, N + 1, D) * 0.02)
z = torch.cat([cls.expand(B, -1, -1), embed(patches)], dim=1) + pos   # (B, N+1, D)

# 3) one head of scaled dot-product self-attention
Wq, Wk, Wv = (nn.Linear(D, D, bias=False) for _ in range(3))
q, k, v = Wq(z), Wk(z), Wv(z)
A = (q @ k.transpose(1, 2) / D ** 0.5).softmax(dim=-1)        # (B, N+1, N+1), rows sum to 1
out = A @ v
print("tokens:", N + 1, " attention:", tuple(A.shape), " class-token output:", tuple(out[:,
 ↪  0].shape))
```

Why ViT needs data. A CNN has built-in assumptions, called **inductive biases**: locality (a pixel mostly relates to its neighbours) and translation equivariance (the same filter everywhere). ViT has almost none; apart from patch extraction and position embeddings, it must learn spatial structure from data. The ViT paper [22] reports that, trained on ImageNet-1k alone, ViTs fell behind comparable ResNets, but pre-trained on much larger datasets (ImageNet-21k or the 300M-image JFT) they matched or beat them. Weak priors cost data at small scale and stop limiting the model at large scale. The same paper also tried **hybrid** models, feeding the feature map of a ResNet stem into the transformer instead of raw patches; hybrids helped at small compute budgets and the gap closed as models grew.

DeiT: training ViTs on ImageNet alone

DeiT [23] showed that the "ViT needs huge data" conclusion was partly a statement about recipes. With strong augmentation (RandAugment, Mixup, CutMix, random erasing), stochastic depth, repeated augmentation and AdamW, a ViT-B trained only on ImageNet-1k reached 83.1% top-1.

DeiT also added a **distillation token**: a second learnable token, like the class token, whose output is trained to match a teacher network's predictions (they found a convnet teacher worked best), while the class token is trained on the true label. This is a transformer-specific form of knowledge distillation [51]. At test time the two heads' outputs are combined.

Optional: Knowledge distillation (a supplementary note in DLCV 01). The distillation loss and the role of the temperature.

Swin: windows, shifts and a hierarchy

Global attention costs grow quadratically with the number of tokens, which is a problem for high-resolution inputs and for dense tasks. Swin [24] computes attention only inside non-overlapping $M \times M$ windows, and shifts the window grid by half a window in alternating layers so information crosses window borders. Token count is reduced between stages by merging 2×2 neighbouring patches, giving a CNN-like pyramid with strides 4, 8, 16, 32. For an $h \times w$ token map with C channels, the paper gives

$$\Omega(\text{MSA}) = 4hwC^2 + 2(hw)^2C, \qquad \Omega(\text{W-MSA}) = 4hwC^2 + 2M^2hwC,$$

so windowed attention is linear in the number of tokens hw for a fixed window size M. Swin brought back two CNN priors, locality and hierarchy, and became a general-purpose backbone for detection and segmentation as well as classification. The broad landscape of vision transformers is surveyed by Khan et al. [25].

Training recipes

Plain version. The same network can gain or lose several points of accuracy depending on how it is trained. A modern recipe is a bundle of small tricks that each make memorizing harder or optimization smoother.

Data augmentation

- **Random resized crop** samples a crop covering a random fraction of the image area and a random aspect ratio, then resizes it to the training resolution. It teaches scale and position invariance and is the default for ImageNet training. **Horizontal flips** are added when they are label-preserving (they are for most photos, not for text or digits).
- **Mixup** [26] blends two images and their labels: $\tilde{\mathbf{x}} = \lambda \mathbf{x}_i + (1-\lambda)\mathbf{x}_j$ and $\tilde{\mathbf{y}} = \lambda \mathbf{y}_i + (1-\lambda)\mathbf{y}_j$, with $\lambda \sim \text{Beta}(\alpha, \alpha)$ and $\mathbf{y}$ one-hot. It encourages linear behaviour between training examples.
- **CutMix** [27] pastes a random rectangle from image j into image i and mixes the labels in proportion to the pasted area. Unlike Mixup, every pixel stays natural, and the network must recognize objects from partial views.
- **RandAugment** [28] replaces learned augmentation policies with two numbers: apply N random operations (rotate, shear, posterize, contrast, ...) at a shared magnitude M. Its paper re-

ports that this tiny search space matches or beats earlier automated policies, and that the right strength depends on model and dataset size.

```
import numpy as np
import torch
import torch.nn.functional as F

def mixup(x, y, num_classes, alpha=0.2):
    lam = float(np.random.beta(alpha, alpha))
    idx = torch.randperm(x.size(0))
    y1 = F.one_hot(y, num_classes).float()
    return lam * x + (1 - lam) * x[idx], lam * y1 + (1 - lam) * y1[idx]

def cutmix(x, y, num_classes, alpha=1.0):
    lam = float(np.random.beta(alpha, alpha))
    idx = torch.randperm(x.size(0))
    H, W = x.shape[-2:]
    h, w = int(H * np.sqrt(1 - lam)), int(W * np.sqrt(1 - lam))
    cy, cx = np.random.randint(H), np.random.randint(W)
    y0, y1_, x0, x1_ = max(cy - h // 2, 0), min(cy + h // 2, H), max(cx - w // 2, 0), min(cx
     ↪  + w // 2, W)
    x = x.clone()
    x[..., y0:y1_, x0:x1_] = x[idx][..., y0:y1_, x0:x1_]
    lam = 1 - (y1_ - y0) * (x1_ - x0) / (H * W)            # actual area kept
    y1 = F.one_hot(y, num_classes).float()
    return x, lam * y1 + (1 - lam) * y1[idx]

def soft_ce(logits, soft_targets):
    return -(soft_targets * logits.log_softmax(1)).sum(1).mean()
```

Label smoothing

Cross-entropy with one-hot targets keeps pushing the true logit up forever; the loss only reaches zero at infinite margin. **Label smoothing** [13] replaces the target with

$$\mathbf{q} = (1 - \varepsilon)\,\mathbf{e}_y + \frac{\varepsilon}{K}\mathbf{1},$$

where ε (often 0.1) is the smoothing strength, $\mathbf{e}_y$ the one-hot vector and $\mathbf{1}$ the all-ones vector. The optimum is now a finite logit gap, which regularizes the network and usually reduces overconfidence. In PyTorch it is one argument: `F.cross_entropy(logits, y, label_smoothing=0.1)`.

Optimizer, schedule and regularizers

- **AdamW** [29] decouples weight decay from the adaptive gradient step: instead of adding $\lambda\theta$ to the gradient (which Adam then rescales per parameter), it shrinks the weights directly, $\theta \leftarrow \theta - \eta(\hat{\mathbf{m}}/(\sqrt{\hat{\mathbf{v}}} + \epsilon) + \lambda\theta)$, where η is the learning rate, $\hat{\mathbf{m}}, \hat{\mathbf{v}}$ Adam's moment estimates and λ the decay. It is the default for transformers and for modern CNN recipes. Biases and normalization parameters are usually excluded from decay.
- **Warmup + cosine schedule.** Large learning rates are unstable in the first steps, when normalization statistics and Adam's moment estimates are still poor, so the learning rate rises linearly for a few epochs. It then follows a half cosine down to near zero, $\eta_t = \frac{1}{2}\eta_{\max}(1 + \cos(\pi t/T))$ with t the step after warmup and T the remaining steps, an idea popularized by SGDR [30].
- **Stochastic depth** [31] randomly skips whole residual branches during training (the block becomes identity), with a skip probability that grows with depth. It shortens the effective depth during training and acts like an ensemble of shallower networks.
- **EMA of weights** keeps an exponential moving average $\bar{\theta} \leftarrow \mu\bar{\theta} + (1 - \mu)\theta$ (with μ close to 1, e.g. 0.999) and evaluates $\bar{\theta}$. It smooths out the noise of the last SGD steps.

```
import math
import torch

def warmup_cosine(step, total, warmup):
    if step < warmup:
        return (step + 1) / warmup                      # linear warmup
    p = (step - warmup) / max(1, total - warmup)
    return 0.5 * (1 + math.cos(math.pi * p))            # cosine decay to 0

model = torch.nn.Linear(10, 3)
decay = [p for n, p in model.named_parameters() if p.ndim > 1]      # weights
no_decay = [p for n, p in model.named_parameters() if p.ndim <= 1]  # biases, norm scales
opt = torch.optim.AdamW([{"params": decay, "weight_decay": 0.05},
                         {"params": no_decay, "weight_decay": 0.0}], lr=1e-3)
total = 1000
sched = torch.optim.lr_scheduler.LambdaLR(opt, lambda s: warmup_cosine(s, total, warmup=50))
ema = torch.optim.swa_utils.AveragedModel(model,
 ↪  multi_avg_fn=torch.optim.swa_utils.get_ema_multi_avg_fn(0.999))
for step in range(total):
    loss = model(torch.randn(16, 10)).pow(2).mean()
    opt.zero_grad(); loss.backward(); opt.step(); sched.step(); ema.update_parameters(model)
```

A small experiment

To see the pieces working together, we trained a 78k-parameter ResNet ("TinyResNet": a stem, three basic blocks with 16, 32 and 64 channels, global average pooling) on scikit-learn's 8×8 digits, upsampled to 32 × 32 so that the strided stages have room to work. To make overfitting visible

we used only **100 training images**, 500 images for calibration and 1,197 for testing, with AdamW (weight decay 0.05), batch size 32, 40 epochs and two epochs of warmup before a cosine decay. One run used no augmentation. The other used random rotation ($\pm 15°$), scaling and shifts (a random-resized-crop-like affine; no flips, since a flipped digit is a different symbol) together with label smoothing 0.1. All code is in `scripts/figures/dl_02_image_classification.py` and runs on a CPU in about a minute.

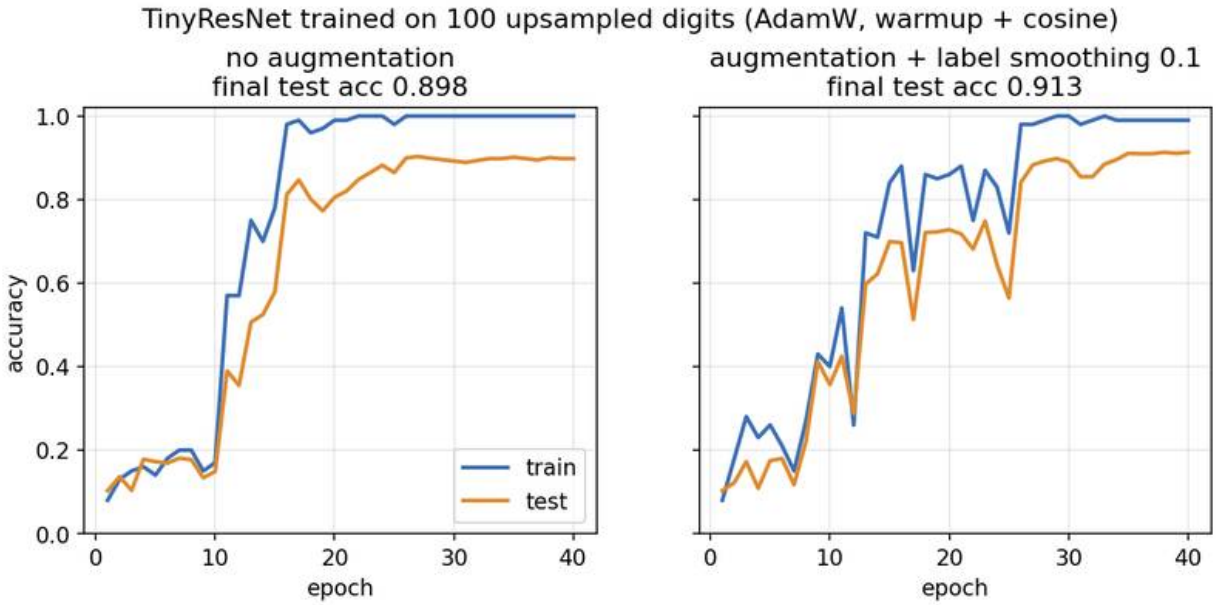


Figure 7 — Training curves on 100 training digits. Without augmentation (left) the network memorizes the training set and reaches 89.8% test accuracy. With augmentation and label smoothing (right) it reaches 91.3%. The gap is modest on such an easy task; on natural images the effect of the recipe is much larger.

The no-augmentation model reaches 100% training accuracy after about twenty epochs: with 100 images, memorization is easy. The augmented model sees a slightly different version of each digit every epoch, learns more slowly, and generalizes a little better (91.3% against 89.8%). Do not read too much into 1.5 points on one seed; the point is the mechanism.

"ResNet strikes back"

The most striking evidence for recipes comes from Wightman, Touvron and Jégou [32]. They kept the 2015 ResNet-50 architecture fixed and retrained it with a modern recipe: random resized crops and flips, RandAugment, Mixup and CutMix, the LAMB optimizer with a cosine schedule, a binary cross-entropy loss and longer training. The vanilla ResNet-50 reached **80.4% top-1** on ImageNet at 224×224 without extra data or distillation, against 76.1% for the standard torchvision weights they compare with. Two consequences follow. First, many "new architecture beats ResNet-50" claims compared a new model with a modern recipe against an old model with an

old recipe. Second, when you compare architectures, train them with the same, tuned recipe, as ConvNeXt [20] and RegNet [19] did.

Transfer learning and fine-tuning

Plain version. Training a backbone from scratch needs a lot of data. Instead, start from a network already trained on a big dataset and adapt it to your task. You can either keep its features frozen and train only a new last layer, or gently retrain everything.

Let g_ϕ be a pre-trained backbone that maps an image to a feature vector $\mathbf{h} = g_\phi(\mathbf{x}) \in \mathbb{R}^D$, and $\mathbf{W} \in \mathbb{R}^{K \times D}$ a new linear head.

- **Linear probe.** Freeze ϕ and train only $\mathbf{W}$ (a logistic regression on fixed features). It is cheap, needs few labels, and is the standard way to *measure* how good a representation is.
- **Full fine-tuning.** Update ϕ and $\mathbf{W}$ together, usually with a learning rate for the backbone 10–100 times smaller than for the head, a short warmup, and sometimes layer-wise learning-rate decay (smaller rates for earlier layers). It adapts the features to the new domain and usually wins when there is enough labelled data.
- In between lie partial fine-tuning (only the last stages), and parameter-efficient methods that add small trainable modules to a frozen backbone.

Kornblith et al. [33] measured how ImageNet accuracy predicts transfer performance. For fine-tuning, better ImageNet models generally transferred better, with a strong correlation. For fixed features, the correlation was weaker and depended on regularization choices: some settings that improved ImageNet accuracy (for example, label smoothing and dropout) produced *worse* frozen features for other tasks. A good classifier and a good feature extractor are related, but not the same thing.

```
import torch
import torch.nn as nn

backbone = nn.Sequential(nn.Conv2d(3, 32, 3, 2, 1), nn.ReLU(), nn.Conv2d(32, 64, 3, 2, 1),
 ↪  nn.ReLU(),
                          nn.AdaptiveAvgPool2d(1), nn.Flatten())          # stand-in for a
                           ↪  pre-trained model
head = nn.Linear(64, 5)

# linear probe: freeze the backbone, train only the head
for p in backbone.parameters():
    p.requires_grad = False
probe_opt = torch.optim.AdamW(head.parameters(), lr=1e-3)

# full fine-tuning: unfreeze, use a smaller learning rate for the backbone than for the head
for p in backbone.parameters():
    p.requires_grad = True
```

```
ft_opt = torch.optim.AdamW([{"params": backbone.parameters(), "lr": 1e-5},
                            {"params": head.parameters(), "lr": 1e-3}], weight_decay=0.05)
```

Self-supervised and foundation-model classifiers

Plain version. Labels are expensive; images are not. Self-supervised learning invents a task whose answer is in the image itself (are these two crops from the same photo? what was under this mask?), learns features by solving it, and only then uses a few labels.

Contrastive learning: SimCLR and MoCo

Contrastive methods pull together the embeddings of two augmented **views** of the same image and push apart views of different images. In SimCLR [34], each view must pick out its partner among all the other views in the batch, a classification task whose "wrong answers" are the other images; strong augmentation (especially cropping plus color distortion) and large batches turned out to be essential. MoCo [35] gets many negatives without huge batches by keeping a queue of embeddings produced by a slowly updated copy of the encoder.

Optional: Contrastive learning in detail: SimCLR and MoCo (in the Supplementary notes). The InfoNCE loss with every symbol defined, and how MoCo's momentum encoder works.

DINO and DINOv2: self-distillation

DINO [36] drops explicit negatives. A student network sees several crops (including small local ones) and is trained to match the output distribution of a teacher that sees global crops. The teacher is an EMA of the student; centering and sharpening of the teacher outputs prevent collapse to a constant. With ViTs, DINO features showed striking properties: the class token's attention maps segment objects without any segmentation labels, and a simple k-NN classifier on frozen features works well. DINOv2 [38] scaled the approach with an automatically curated dataset and a large ViT distilled into smaller ones, producing general-purpose **frozen features** that work across classification, segmentation and depth without fine-tuning. DINOv3 [39] continued the scaling and introduced "Gram anchoring" to keep dense patch features from degrading during long training.

MAE: masked image modeling

MAE [37] borrows the masked-language-model idea. It masks a large fraction of patches (75% in the paper), feeds only the visible patches to a ViT encoder, and asks a light decoder to reconstruct the missing pixels. Because the encoder skips masked tokens, pre-training is cheap. MAE features are less linearly separable than contrastive ones (linear probes are weaker), but they **fine-tune**

very well. This is a recurring pattern: different pre-training objectives favour different adaptation methods.

CLIP: zero-shot classification with language

CLIP [40] trains an image encoder g and a text encoder h on hundreds of millions of image–caption pairs with a symmetric contrastive loss, so that matching pairs have high cosine similarity.

A classifier is then built **without any training images**: write a prompt per class, such as `a photo of a dog.`, embed it, and predict

$$\hat{y} = \arg\max_k \frac{g(\mathbf{x})^\top h(t_k)}{\|g(\mathbf{x})\| \, \|h(t_k)\|},$$

where t_k is the prompt for class k. The text embeddings act as the weights of a linear classifier, computed from words. The paper reports that zero-shot CLIP matched the accuracy of a ResNet-50 trained on ImageNet's 1.28M labelled images, and that averaging several prompt templates ("a photo of a ...", "a blurry photo of a ...") helps. Zero-shot accuracy depends strongly on prompt wording, on whether the class names are visually meaningful, and on how well the web data covers the domain.

Robustness and evaluation pitfalls

Plain version. A classifier that scores 90% on its test set may fail badly when the photos come from a different camera, when someone adds invisible noise on purpose, or when the "easy clue" it learned (backgrounds, watermarks) is missing. And even when it is right, its confidence may be wrong.

Distribution shift: ImageNet-C, -R and -A

Hendrycks and colleagues built three widely used shifted test sets for ImageNet classifiers.

ImageNet-C [42] applies 15 algorithmic corruptions (noise, blur, weather, digital artefacts) at five severities and summarizes the error relative to a baseline (mean corruption error).

ImageNet-R [43] collects renditions (art, cartoons, sculptures, sketches, toys) of 200 ImageNet classes and tests whether a model understands shape and concept rather than photographic texture. **ImageNet-A** [44] contains natural, unmodified photos that were filtered to fool a ResNet-50; standard models do very poorly on it. These sets reveal different weaknesses, and improving one does not guarantee improving the others. Large-scale pre-training (as in CLIP and DINOv2) tends to improve robustness to such natural shifts much more than architecture changes at fixed data.

Adversarial examples: FGSM

Goodfellow et al. [45] explained adversarial examples by the **linearity** of networks in high dimensions: a tiny change to every input dimension, aligned with the weights, adds up to a large change in the output. The **fast gradient sign method** perturbs an input in one step:

$$\mathbf{x}_{\text{adv}} = \text{clip}\left(\mathbf{x} + \varepsilon \, \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(f_\theta(\mathbf{x}), y))\right),$$

where ε bounds the change per pixel (ℓ_∞ norm), $\mathcal{L}$ is the loss and clip keeps pixels in the valid range.

```
import torch.nn.functional as F

def fgsm(model, x, y, eps):
    x = x.clone().requires_grad_(True)
    F.cross_entropy(model(x), y).backward()
    return (x + eps * x.grad.sign()).clamp(-1, 1).detach()
```

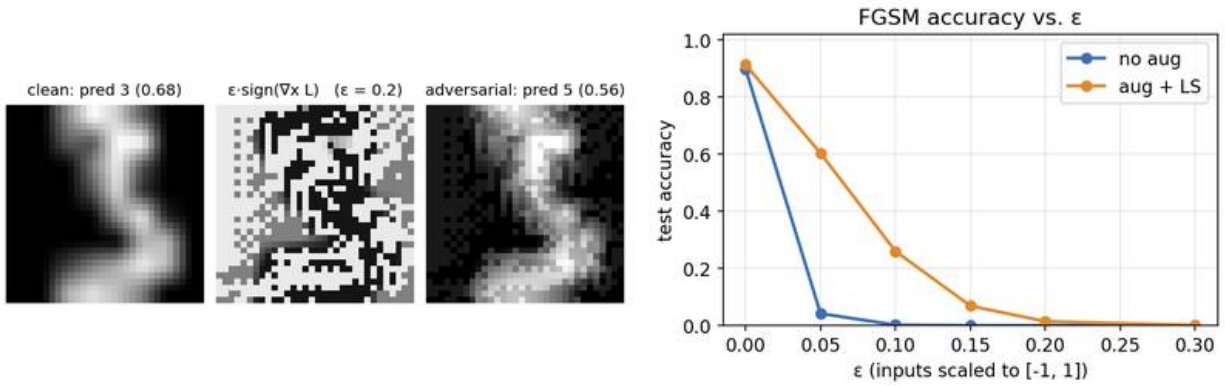


Figure 8 — FGSM on the TinyResNet models (inputs in $[-1, 1]$). At $\varepsilon = 0.05$ test accuracy falls from 89.8% to 4.2% for the model trained without augmentation and from 91.3% to 60.3% for the model trained with augmentation and label smoothing; at $\varepsilon = 0.2$ both are near zero.

Two cautions about Figure 8. First, the augmented model looks more robust at small ε, but label smoothing is known to flatten the loss surface around training points, which can weaken one-step gradient attacks without giving real robustness; stronger iterative attacks are the proper test. Second, $\varepsilon = 0.05$ on a $[-1, 1]$ scale is visible on an 8×8 digit; on natural images, imperceptible perturbations already fool standard ImageNet models.

Shortcut learning

Geirhos et al. [46] collect many failures under one name: **shortcut learning**. A network minimizes the training loss with whatever cue is easiest, which is often not the cue we intended: background

instead of animal, hospital tag instead of pathology, texture instead of shape. Shortcuts look fine on an i.i.d. test set, because the same shortcut exists there. They fail under distribution shift. The prescription is to test out of distribution on purpose, design tests that break the suspected shortcut, and use interpretability tools such as Grad-CAM (below) to check what the model attends to.

Calibration and temperature scaling

Temperature scaling [47] divides all logits by one scalar $T > 0$ before the softmax, $\mathbf{p} = \text{softmax}(\mathbf{z}/T)$. T is fitted by minimizing the negative log-likelihood on a held-out **calibration** split (never the test set). $T > 1$ softens overconfident predictions; $T < 1$ sharpens underconfident ones. Because dividing by T does not change the argmax, accuracy is unchanged.

```
import torch
import torch.nn.functional as F

def fit_temperature(logits, y, iters=300):
    """Learn one scalar T > 0 on a held-out calibration split."""
    log_t = torch.zeros(1, requires_grad=True)
    opt = torch.optim.Adam([log_t], lr=0.05)
    for _ in range(iters):
        loss = F.cross_entropy(logits / log_t.exp(), y)
        opt.zero_grad(); loss.backward(); opt.step()
    return log_t.exp().item()

def ece(probs, y, n_bins=10):
    conf, pred = probs.max(1)
    correct = (pred == y).float()
    edges = torch.linspace(0, 1, n_bins + 1)
    total = 0.0
    for lo, hi in zip(edges[:-1], edges[1:]):
        m = (conf > lo) & (conf <= hi)
        if m.any():
            total += m.float().mean() * (correct[m].mean() - conf[m].mean()).abs()
    return float(total)

# toy check: an over-confident "model" = true logits multiplied by 3
torch.manual_seed(0)
true_logits = torch.randn(4000, 10)
y = torch.distributions.Categorical(logits=true_logits).sample()
over = 3 * true_logits
T = fit_temperature(over[:2000], y[:2000])
print(f"T = {T:.2f}  ECE {ece(over[2000:].softmax(1), y[2000:]):.3f} -> "
      f"{ece((over[2000:] / T).softmax(1), y[2000:]):.3f}")
# T = 2.90  ECE 0.345 -> 0.018
```

In this toy check the logits are three times too large, and the fitted temperature (2.90) recovers that

factor. On our real TinyResNet (the model trained without augmentation), the miscalibration went the *other* way.

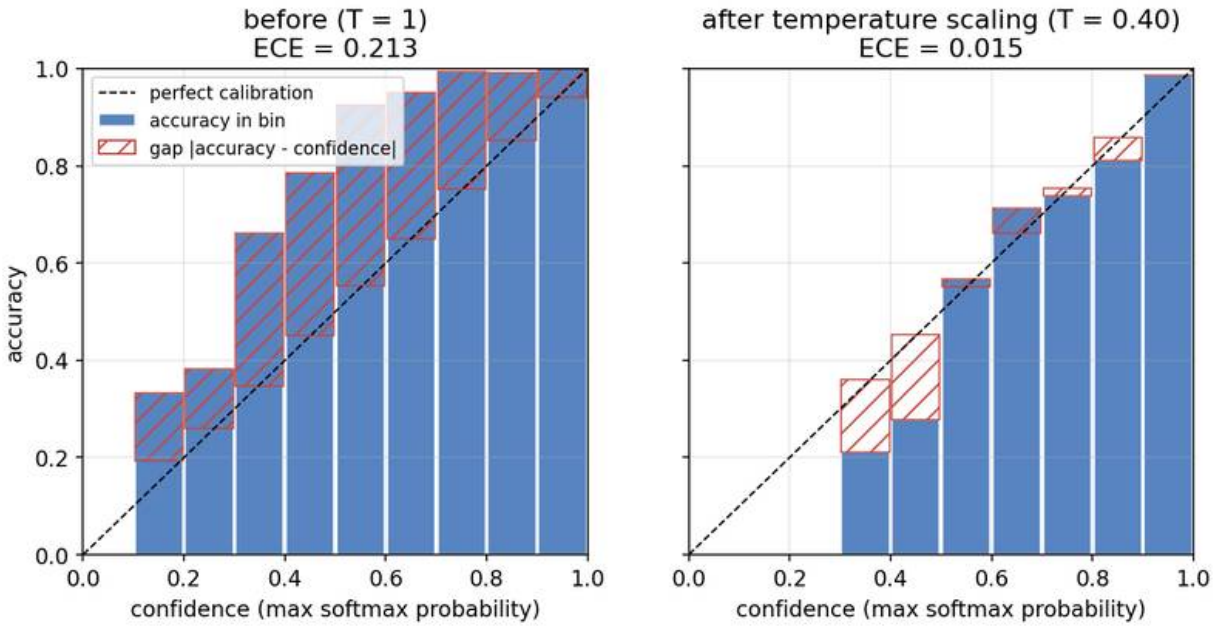


Figure 9 — Reliability diagrams for the TinyResNet trained without augmentation, on 1,197 test digits. Before scaling (left), accuracy exceeds confidence in most bins: the model is underconfident (ECE 0.213). A temperature $T = 0.40$ fitted on 500 separate calibration images brings ECE to 0.015 and test NLL from 0.515 to 0.298, without changing any prediction.

Why underconfident? Our tiny model was trained briefly with strong weight decay, which keeps logits small. Guo et al. [47] observed overconfidence in large, long-trained modern networks, and identified depth, width, batch normalization and *reduced* weight decay as contributing factors. Miscalibration has no fixed sign; measure it. Also note that calibration is itself fragile under distribution shift: a temperature fitted on in-distribution data does not guarantee calibrated confidence on ImageNet-C-style shifts.

Efficiency

Plain version. A model must fit the device and the time budget. Parameter count, compute, and actual speed are three different measurements, and only the last one is what users feel.

- **Parameters** determine model size on disk and in memory. A float32 parameter takes 4 bytes.
- **FLOPs** (or multiply–accumulates, MACs; one MAC is usually counted as two FLOPs) measure arithmetic per image. For a convolution with $K \times K$ kernels, C_{in} input and C_{out} output channels on an $H' \times W'$ output map, MACs $= H'W'K^2C_{\text{in}}C_{\text{out}}$. Compute grows with the square of

resolution.

- **Latency and throughput** are measured on target hardware. Depthwise convolutions have few FLOPs but low arithmetic intensity, so they often run slower on GPUs than their FLOP count suggests; DenseNet's concatenations cost memory bandwidth. Always benchmark.
- **Quantization** stores weights and activations as 8-bit integers instead of 32-bit floats. Jacob et al. [50] described integer-only inference with quantization-aware training, which simulates rounding during training so the network learns to tolerate it. Post-training quantization with a small calibration set is the cheap alternative.
- **Distillation** [51] trains a small student to match a large teacher's soft predictions, as DeiT did with its distillation token.

Interpretability: CAM and Grad-CAM

Plain version. A heat map over the image shows which regions pushed the score for a class up. It is a sanity check, not a proof of reasoning.

Class activation mapping (CAM) [48] works for networks that end with global average pooling and one linear layer. If A^k is the k-th feature map of the last convolutional layer and w_k^c the weight connecting it to class c, then the class score is $\sum_k w_k^c \cdot \text{mean}(A^k)$, and the map $\sum_k w_k^c A^k$ shows where that score comes from. **Grad-CAM** [49] generalizes this to any differentiable architecture by replacing the weights with spatially averaged gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}, \qquad L_{\text{Grad-CAM}}^c = \text{ReLU}\Big(\sum_k \alpha_k^c A^k \Big),$$

where y^c is the logit of class c, A_{ij}^k the activation of channel k at position (i, j), and Z the number of positions. The ReLU keeps only regions with a positive influence on the class. The coarse map (8×8 in our network) is upsampled to the image size.

```
import torch.nn.functional as F

def grad_cam(model, x, target=None):
    model.eval()
    feats = model.features(x)                          # (1, K, h, w) last conv stage
    feats.retain_grad()
    logits = model.fc(feats.mean(dim=(2, 3)))
    c = logits.argmax(1).item() if target is None else target
    logits[0, c].backward()
    alpha = feats.grad.mean(dim=(2, 3), keepdim=True)  # channel importance weights
    cam = F.relu((alpha * feats).sum(1, keepdim=True))
    cam = F.interpolate(cam, size=x.shape[-2:], mode="bilinear", align_corners=False)
    return (cam / (cam.max() + 1e-8))[0, 0].detach(), c
```

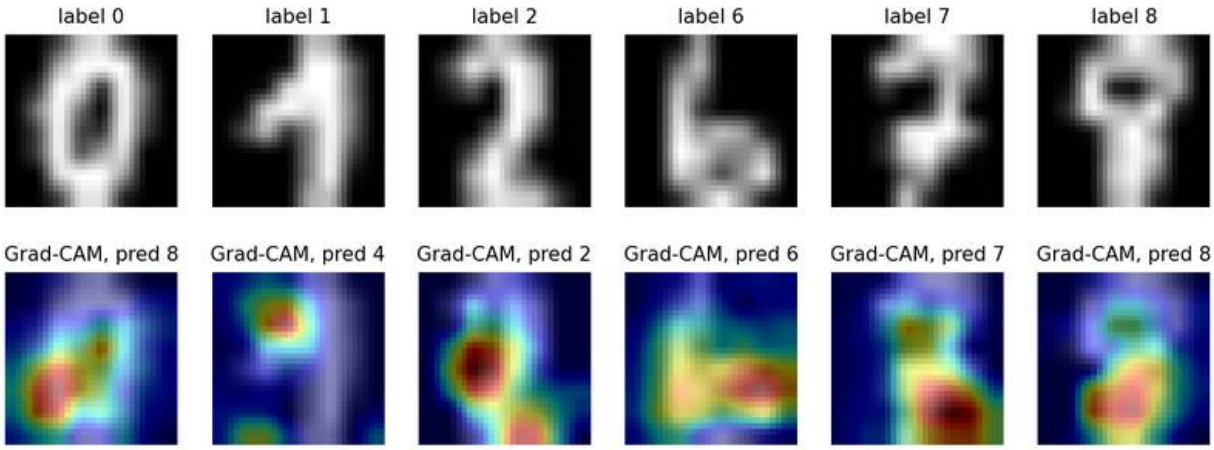


Figure 10 — Grad-CAM for the predicted class on the first test example of six digits, using the augmented TinyResNet. Two are misclassified: the 0 is called an 8, with heat on the lower-left part of the loop, and the 1 is called a 4, with heat on the hook at its upper left. The maps point to where the misleading evidence was found.

Explanations are most useful exactly when the model is wrong, as in Figure 10: they show which strokes looked like the wrong class. But treat saliency maps with care. They are coarse (limited by the last feature map's resolution), and they explain one class score for one input, not the model's general reasoning.

Modern view

Surveys worth reading. Li et al. [1] review CNNs from basic building blocks to advanced models, with experiments that compare activation functions, losses and optimizers. Khan et al. [25] survey vision transformers and stress their reliance on large-scale pre-training and the cost of quadratic attention. Gui et al. [41] divide self-supervised learning into context-based, contrastive, generative and hybrid methods. On benchmark reliability, the ImageNetV2 [7], re-labelling [8] and label-error [9] studies together argue for multiple test sets, multi-label evaluation and reporting variance.

Optional: Guide to the Modern view surveys (in the Paper guides). What each survey covers and how it organizes the field.

The 2023–2026 picture. Five trends define current practice.

1. **Pre-train once, adapt cheaply.** Most new classifiers are not trained from scratch. They start from a large backbone trained with self-supervision (DINOv2 [38], DINOv3 [39]), image–text contrast (CLIP [40] and open reproductions), or masked modeling (MAE [37]), and adapt it with a linear probe, a k-NN rule, or a short fine-tune.

2. **Zero-shot and open-vocabulary classification.** Image–text models turn the label set into text. The class list can change at test time without retraining, which blurs the line between classification and retrieval.
3. **Scaling laws for vision.** Zhai et al. [52] scaled ViTs to billions of parameters and characterized how error falls with compute, data and model size, including the observation that larger models are more sample-efficient in few-shot transfer. Cherti et al. [53] fitted reproducible power laws for open CLIP training on public data, and found that the scaling behaviour depends on the training data distribution and the downstream task.
4. **Architectures converged.** ConvNeXt [20] and Swin [24] show that convolutions and attention, given the same recipe and scale, land in a similar place. The choice is now driven by hardware, resolution and downstream tasks (dense prediction favours hierarchical designs) more than by top-1 accuracy.
5. **Evaluation moved beyond ImageNet-1k top-1.** Robustness suites (ImageNet-C/R/A [42]–[44] and ImageNetV2 [7]), calibration, and transfer to many downstream datasets are now standard parts of a backbone paper.

Open problems. Robustness to natural distribution shift still lags clean accuracy, and gains from robust training methods are often specific to the shift they target. Shortcut learning [46] is hard to detect without counterfactual tests. Calibration under shift is unsolved. Long-tailed and fine-grained recognition remain hard because the rare classes have the fewest examples. Benchmark saturation and label noise make small accuracy differences on ImageNet-1k uninformative. And the dominant foundation models are trained on web-scale data whose composition, licensing and biases are hard to audit, which matters directly for what a "zero-shot" classifier will and will not recognize.

Key takeaways

- A classifier outputs logits; softmax + cross-entropy trains single-label tasks, independent sigmoids + binary cross-entropy train multi-label tasks. Report top-1 accuracy and, where it matters, calibration (ECE) and per-class results.
- Each landmark CNN fixed one bottleneck: ReLU/GPUs (AlexNet), small stacked kernels (VGG), 1×1 reductions and multi-branch modules (Inception), identity shortcuts for optimization (ResNet), depthwise separable convolutions for cost (MobileNet), balanced scaling (EfficientNet).
- ViTs trade convolutional inductive biases for flexibility; they need more data or a stronger recipe (DeiT), and windows plus hierarchy (Swin) make them practical backbones. ConvNeXt shows that the recipe explains much of the gap.
- Training recipes matter as much as architecture: augmentation (random resized crop, Mixup, CutMix, RandAugment), label smoothing, AdamW, warmup + cosine, stochastic depth and EMA lifted an unchanged ResNet-50 to 80.4% top-1.

- Pre-trained backbones are the default starting point. Linear probes measure representations; fine-tuning adapts them. CLIP turns class names into classifier weights for zero-shot recognition.
- Accuracy hides *which* errors a model makes. Read the confusion matrix; under class imbalance report precision/recall, balanced accuracy or MCC, and prefer the precision–recall curve and AP over ROC-AUC. State whether an average is macro (every class equal) or micro (every image equal; equals accuracy for single-label tasks).
- Separate training data from evaluation data: judge a backbone on a battery of test sets (ImageNet val and ReaL, ImageNetV2, ImageNet-C/R/A/Sketch, ObjectNet, fine-grained and transfer suites), tune only on validation data, and report mean $\pm$ std over seeds.
- Benchmark accuracy is fragile: test on shifted data, beware shortcuts and adversarial inputs, fit a temperature on held-out data, and inspect Grad-CAM maps, especially on errors.

Exercises

1. **Receptive fields and parameters.** Show that three stacked 3×3 convolutions (stride 1) have a 7×7 receptive field. With C input and output channels and no biases, compare their parameter count with a single 7×7 convolution.

   Hint

   Each 3×3 layer adds 2 pixels to the receptive field: $3 \to 5 \to 7$. Parameters: $3 \cdot 9C^2 = 27C^2$ versus $49C^2$, about 45% fewer, plus two extra nonlinearities.

2. **Bottleneck arithmetic.** A ResNet bottleneck has $4d$ input channels and reduces to d. Count the weights (no biases) of the 1×1–3×3–1×1 branch as a function of d, and compare with a basic block of two 3×3 convolutions at width $4d$. What is the ratio?

   Hint

   Bottleneck: $4d \cdot d + 9d^2 + d \cdot 4d = 17d^2$. Basic block at width $4d$: $2 \cdot 9 \cdot 16d^2 = 288d^2$. The ratio is $17/288 \approx 0.059$, about 17 times fewer weights.

3. **Label smoothing optimum.** For one example with K classes and smoothing ε, assume all wrong-class logits are equal. Show that the cross-entropy with smoothed targets is minimized when the logit gap between the true class and each wrong class is $\log\left((1 - \varepsilon + \varepsilon/K) \,/\, (\varepsilon/K)\right)$. Evaluate it for $K = 1000$, $\varepsilon = 0.1$.

   Hint

   The minimum of cross-entropy over $\mathbf{p}$ is at $\mathbf{p} = \mathbf{q}$. So $p_y = 1 - \varepsilon + \varepsilon/K$ and every other $p_k = \varepsilon/K$, and the logit gap is the log of their ratio: $\log(0.9001/0.0001) \approx 9.1$. Without smoothing the optimal gap is infinite.

4. **ViT token count.** A ViT with 16×16 patches receives a 224×224 image. How many tokens enter the encoder? If the image is fine-tuned at 384 × 384, how many tokens are there, by what factor does the attention-matrix size grow, and what must be done with the position embeddings?

   Hint

   $(224/16)^2 = 196$ patches plus one class token: 197. At 384: $24^2 + 1 = 577$. The attention matrix grows by $(577/197)^2 \approx 8.6$. The learned position embeddings form a 14×14 grid; they are interpolated (2-D) to 24 × 24, which is what the ViT paper does when fine-tuning at higher resolution.

5. **Temperature does not change accuracy.** Prove that for any $T > 0$, $\arg\max_k \mathrm{softmax}(\mathbf{z}/T)_k = \arg\max_k z_k$. Then, using `scripts/figures/dl_02_image_classification.py`, fit the temperature on the *test* set instead of the calibration split and compare the reported ECE. Why is that number optimistic?

   Hint

   Division by a positive constant and the exponential are both strictly increasing, and the softmax denominator is shared. Fitting T on the test set uses test labels to tune a parameter, so the reported ECE is no longer an honest estimate of performance on new data.

6. **Build a shortcut.** Modify the digit experiment so that in the training set every "3" has a bright pixel in the top-left corner, but the test set does not. Train the TinyResNet, report test accuracy on the 3s, and look at Grad-CAM maps for a few training 3s. What do you expect, and how would you detect this problem if you did not know the shortcut was there?

   Hint

   The network can classify training 3s from the corner pixel alone, so test accuracy on 3s drops while training accuracy stays perfect, and Grad-CAM tends to highlight the corner. Without knowing the shortcut, look for heat maps that fall on background, evaluate on independently collected data, and run counterfactual tests (erase or move suspicious regions and see whether predictions change).

7. **Macro versus micro F1.** A three-class model is tested on 100 images. Rows are true classes, columns predictions: class A (90 images) → 88 A, 1 B, 1 C; class B (5 images) → 3 A, 2 B, 0 C; class C (5 images) → 4 A, 0 B, 1 C. Compute per-class precision, recall and F_1, then macro F_1 and micro F_1. Which number would you put in a report about detecting the rare classes B and C, and why?

   Hint

   Column sums are 95, 3 and 2. Precision: $88/95 = 0.926$, $2/3 = 0.667$, $1/2 = 0.500$. Recall: $88/90 = 0.978$, $2/5 = 0.400$, $1/5 = 0.200$. Using $F_1 = 2\mathrm{TP}/(2\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$: $176/185 = 0.951$, $4/8 = 0.500$, $2/7 = 0.286$. Macro $F_1 = (0.951 + 0.500 + 0.286)/3 = 0.579$. Micro F_1 equals accuracy, $91/100 = 0.910$. Micro is dominated by class A; macro exposes that B and C are mostly missed, so macro (plus per-class recall) is the honest number for rare classes. You can check the result with `precision_recall_fscore_support(..., average="macro")`.

Supplementary notes

These notes are optional background. Nothing in the main text or the exercises depends on them.

A worked macro-versus-micro example

This note works through the macro, micro and weighted averages of the main text on a small, imbalanced three-class example.

A worked example. 100 test images from three classes with supports 80 (A), 15 (B) and 5 (C):

true predicted	A	B	C
A (80)	76	3	1
B (15)	4	10	1
C (5)	3	0	2

Column sums are 83, 13 and 4, so precision is $76/83 = 0.916$, $10/13 = 0.769$ and $2/4 = 0.500$; recall is $76/80 = 0.950$, $10/15 = 0.667$ and $2/5 = 0.400$; per-class F_1 is 0.933, 0.714 and 0.444. Then

$$\mathrm{F}_1^{\text{macro}} = \tfrac{1}{3}(0.933 + 0.714 + 0.444) = 0.697, \qquad \mathrm{F}_1^{\text{micro}} = \text{accuracy} = \tfrac{76+10+2}{100} = 0.880,$$

and the support-weighted F_1 is $0.80 \cdot 0.933 + 0.15 \cdot 0.714 + 0.05 \cdot 0.444 = 0.875$. The 18-point gap between micro and macro is the signal: the model is good on the common class and poor on the rare ones. Report macro when every class matters (rare diseases, rare species); report micro/accuracy when every *image* matters equally.

Cohen's kappa: agreement beyond chance

This note defines the chance-corrected agreement measure mentioned in the main text.

Cohen's kappa [59] corrects observed agreement for agreement expected by chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \qquad p_o = \frac{1}{N}\sum_k C_{kk}, \qquad p_e = \frac{1}{N^2}\sum_k \Big(\sum_j C_{kj}\Big)\Big(\sum_i C_{ik}\Big),$$

where p_o is the accuracy and p_e the accuracy a classifier would get by predicting each class at its observed rate independently of the input. It was designed for agreement between two human raters and is still common when comparing a model with annotators.

For the digit classifier of the main text, $p_o = 0.911$, $p_e = 0.100$ and $\kappa = 0.901$ (scikit-learn's `cohen_kappa_score` [62] gives the same value). With ten balanced classes, agreement by chance is small, so kappa stays close to accuracy.

Multi-label metrics in detail

This note gives the definitions and a small example for the multi-label metrics summarized in the main text.

In multi-label classification each image has a binary target vector $\mathbf{t} \in \{0,1\}^K$ and the model a score vector. Each label is its own binary problem, so the natural ranking metric is the **per-class AP** and its mean over classes, **mAP**, which became standard with the PASCAL VOC classification task [61]. Thresholded metrics come in two families, reviewed by Zhang and Zhou [60]:

- **Label-based** metrics compute a binary metric per label and macro- or micro-average over labels, exactly as in the multi-class case.
- **Example-based** metrics compute a score per *image* and average over images. Example-based F_1 is $\frac{1}{N}\sum_i \frac{2|\hat{Y}_i \cap Y_i|}{|\hat{Y}_i|+|Y_i|}$, where Y_i and $\hat{Y}_i$ are the true and predicted label sets of image i.
- **Hamming loss** is the fraction of wrong (image, label) bits, $\frac{1}{NK}\sum_i |\hat{Y}_i \,\triangle\, Y_i|$, where $\triangle$ is the symmetric difference. It is easy to read but, with many rare labels, an empty prediction already looks good.

```
import numpy as np
from sklearn.metrics import average_precision_score, f1_score, hamming_loss

# 4 street photos, 3 labels: car, person, bicycle
T = np.array([[1, 1, 0], [1, 0, 0], [0, 1, 1], [1, 1, 1]])            # ground truth
S = np.array([[.9, .6, .2], [.8, .4, .3], [.3, .7, .4], [.6, .2, .7]])  # sigmoid scores
P = (S >= 0.5).astype(int)                                            # threshold 0.5

print("per-class AP:", average_precision_score(T, S, average=None).round(3))
print(f"mAP {average_precision_score(T, S, average='macro'):.3f}")
print(f"example-based F1 {f1_score(T, P, average='samples'):.3f}  "
      f"macro F1 {f1_score(T, P, average='macro'):.3f}  micro F1 {f1_score(T, P,
      ↪ average='micro'):.3f}")
print(f"Hamming loss {hamming_loss(T, P):.3f}")
# per-class AP: [1.    0.917 1.   ]
# mAP 0.972
# example-based F1 0.867  macro F1 0.822  micro F1 0.857
# Hamming loss 0.167
```

Evaluation benchmarks in detail

This note gives the full tables behind the short list of evaluation benchmarks in the main text.

The table lists the evaluation sets you will meet most often. Sizes are those stated by the papers or official pages cited.

Benchmark	What it measures	Size / classes	Usually reported
ImageNet-1k val [2]	in-distribution accuracy; the de facto test set, since ILSVRC test labels stay on the evaluation server	50,000 images, 1,000 classes	top-1 (top-5)
ImageNet-ReaL [8]	the same validation images with re-annotated, multi-label "reassessed" labels	46,837 images with 57,553 labels (3,163 images left without a label are discarded)	ReaL accuracy: top-1 counted correct if it is in the label set
ImageNetV2 [7]	generalization to a new sample drawn by repeating the original collection process	10,000 images per variant (10 per class); three sampling variants, MatchedFrequency is the usual one	top-1
ImageNet-C [42]	robustness to synthetic corruptions	validation images × 15 corruptions × 5 severities	mean corruption error
ImageNet-R [43]	renditions: art, cartoons, sculpture, toys and more	30,000 images, 200 classes	top-1 over the 200 classes
ImageNet-A [44]	natural photos filtered to fool a ResNet-50	7,500 images, 200 classes	top-1 over the 200 classes
ImageNet-Sketch [63]	domain shift to black-and-white sketches	50,000 images, 1,000 classes (50 per class)	top-1
ObjectNet [64]	bias-controlled photos with varied rotation, background and viewpoint; test set only	50,000 images, 313 classes (113 overlap ImageNet)	top-1 on the overlapping classes
ImageNet-LT [73]	long-tailed training, balanced testing	115.8k training images, 1,000 classes, 5 to 1,280 per class; balanced test set	top-1, often split by class frequency
iNaturalist [6]	naturally long-tailed, fine-grained species	see the training-dataset table in the main text	top-1

Fine-grained and transfer benchmarks. A backbone is also judged by how well its features transfer to small, specialized datasets, either with a linear probe on frozen features or with fine-tuning (see the transfer-learning section of the main text).

Dataset	Domain	Size / classes	Usually reported
CUB-200-2011 [65]	bird species, with part and attribute annotations	11,788 images, 200 classes	top-1
Stanford Cars [66]	car make, model and year	196 classes; 8,144 train / 8,041 test	top-1
FGVC-Aircraft [67]	aircraft variants in a three-level hierarchy	10,000 images, 100 variants	mean per-class accuracy
Oxford-IIIT Pets [68]	cat and dog breeds	37 classes, about 200 images each	mean per-class accuracy
Oxford Flowers-102 [69]	flower species	102 classes, 40 to 258 images each; 2,040 train / 6,149 test	mean per-class accuracy

Dataset	Domain	Size / classes	Usually reported
Food-101 [70]	dishes; training labels deliberately left noisy	101,000 images, 101 classes (750 train / 250 test per class)	top-1
Caltech-101 [71]	generic objects	101 categories plus background, about 40 to 800 images each	mean per-class accuracy

Train/test sizes for Cars and Flowers follow the splits listed by Kornblith et al. [33]. Note that several of these benchmarks report **mean per-class accuracy** (macro-averaged recall) instead of plain accuracy, because their test sets are not balanced; mixing the two in one comparison is a common error.

Two **suites** bundle such datasets so that one number summarizes transfer:

- Kornblith et al. [33] used **12 datasets**: Food-101, CIFAR-10, CIFAR-100, Birdsnap, SUN397, Stanford Cars, FGVC-Aircraft, PASCAL VOC 2007 classification, Describable Textures, Oxford-IIIT Pets, Caltech-101 and Oxford Flowers-102. They report top-1 accuracy, mean per-class accuracy, or 11-point mAP (VOC 2007) depending on the dataset, under linear probing, fine-tuning and training from scratch.
- **VTAB** [72] contains 19 tasks in three groups: 7 *natural* (standard camera images), 4 *specialized* (satellite and medical imagery) and 8 *structured* (scene geometry, such as counting or distance, mostly from simulated environments). In **VTAB-1k** each task provides only 1,000 labelled training examples (800 for training and 200 for validation), so it tests how well a representation adapts from few labels to tasks unlike its pre-training data.

Zero-shot evaluation. Image–text models such as CLIP [40] are evaluated without any training on the target dataset: the class names are turned into text prompts, and the image is assigned to the most similar prompt (see the CLIP section of the main text). The CLIP paper benchmarked over 30 existing datasets this way, reported that zero-shot CLIP beat a linear probe on ResNet-50 features on 16 of 27 datasets, and found that an ensemble of 80 prompt templates improved ImageNet accuracy by 3.5 points over a single default prompt. Two cautions follow. Prompts and class-name wording are hyperparameters, so tuning them on the test set leaks test information exactly as tuning a learning rate on it would. And because the pre-training data are web-scale and mostly unpublished, "zero-shot" does not guarantee that the test images, or near-duplicates, were never seen.

How reliable are ImageNet numbers?

Three lines of work changed how we read ImageNet accuracy.

- **Reproduction.** Recht et al. [7] rebuilt test sets for CIFAR-10 and ImageNet by repeating the

original collection procedure as closely as possible (ImageNetV2). Every model lost accuracy: 3–15% on CIFAR-10 and 11–14% on ImageNet. But the ranking was largely preserved, and gains on the old test set translated into gains on the new one. Their reading is that the drop comes from a subtle shift in difficulty between the old and new samples, not from the community overfitting to the test set by reusing it.

- **Labels.** Beyer et al. [8] re-annotated the ImageNet validation set with multiple labels per image and found that recent "progress" partly reflects fitting the idiosyncrasies of the original single-label annotations. Northcutt et al. [9] estimated that label errors are pervasive across common test sets and showed that they can change which model looks best.
- **Bias.** Every dataset has a "signature" in viewpoint, background and photographer style. DIP Chapter 13 reviews the dataset-bias studies; the practical lesson is to test on independently collected data whenever possible.

Why residual networks train: the unrolled gradient

This note gives the short derivation behind the main text's explanation of why residual networks train well.

Why it trains. Unroll L blocks of the pre-activation form $\mathbf{x}_{l+1} = \mathbf{x}_l + \mathcal{F}(\mathbf{x}_l)$, as analysed in the follow-up paper on identity mappings [15]:

$$\mathbf{x}_L = \mathbf{x}_l + \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i), \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{x}_l} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}_L}\Big(\mathbf{I} + \frac{\partial}{\partial \mathbf{x}_l} \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i)\Big),$$

where $\mathcal{L}$ is the loss and $\mathbf{I}$ the identity matrix. The identity term carries the gradient from the top straight down to any block, so it cannot vanish through a product of many small Jacobians. That paper also found that keeping the shortcut a *pure* identity (moving BN and ReLU inside the residual branch, "pre-activation") trains even deeper networks more easily.

Two more CNN designs: DenseNet and RegNet

This note describes the two CNN designs that the main text summarizes in two sentences.

DenseNet: reuse everything. DenseNet [16] pushes feature reuse further: inside a dense block,

layer ℓ receives the **concatenation** of all earlier feature maps, $\mathbf{x}_\ell = H_\ell([\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{\ell-1}])$, where $[\cdot]$ is channel concatenation and H_ℓ is BN–ReLU–conv. Each layer adds only a few channels (the "growth rate"), so the network is parameter-efficient, and every layer has a short path to the loss. The cost is memory: concatenated activations grow with depth, which made DenseNets slower in practice than their parameter count suggests. A useful lesson for later: parameters, FLOPs and wall-clock time are three different things.

RegNet: design spaces instead of single designs. Radosavovic et al. [19] studied populations

of networks rather than single winners. Starting from a broad space of ResNet-like networks, they repeatedly constrained it (for example, shared bottleneck ratios, widths that increase across stages) and checked whether the *distribution* of errors improved. The surviving space, RegNet, describes stage widths with a simple quantized linear rule. The lesson is methodological: good architectures follow simple regularities, and comparing distributions of models is more reliable than comparing one tuned model against another.

Squeeze-and-excitation: attention inside a CNN

This note gives the formula and code for the squeeze-and-excitation block summarized in the main text.

The SE block [21] adds cheap, content-dependent **channel** weights to any CNN. It "squeezes" each channel to one number by global average pooling, passes the vector through a small two-layer MLP, and multiplies each channel by a sigmoid gate:

$$\mathbf{s} = \sigma(W_2\,\delta(W_1\,\bar{\mathbf{u}})), \qquad \tilde{\mathbf{u}}_c = s_c\,\mathbf{u}_c,$$

where $\mathbf{u}_c$ is channel c of the feature map, $\bar{\mathbf{u}}$ the vector of channel means, δ a ReLU, W_1 reduces the dimension by a ratio r and W_2 restores it. The network can now emphasise "fur" channels when the global context says "animal". SE blocks are part of EfficientNet and many mobile networks.

```
import torch
import torch.nn as nn

class SE(nn.Module):
    """Squeeze-and-excitation: reweight channels using global context."""
    def __init__(self, c, r=16):
        super().__init__()
        self.fc = nn.Sequential(nn.Linear(c, c // r), nn.ReLU(), nn.Linear(c // r, c),
        ↪  nn.Sigmoid())

    def forward(self, x):                       # x: (B, C, H, W)
        s = x.mean(dim=(2, 3))                  # squeeze: (B, C)
        w = self.fc(s)[:, :, None, None]        # excitation: (B, C, 1, 1) in (0, 1)
        return x * w

print(SE(64)(torch.randn(2, 64, 8, 8)).shape)
```

Contrastive learning in detail: SimCLR and MoCo

This note gives the loss behind the contrastive methods summarized in the main text.

Contrastive methods pull together the embeddings of two augmented **views** of the same image and push apart views of different images. SimCLR [34] takes a batch of B images, makes two random views of each, and minimizes the InfoNCE (NT-Xent) loss for each positive pair (i, j):

$$\ell_{i,j} = -\log \frac{\exp(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k=1}^{2B} \mathbb{1}_{[k \neq i]} \exp(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)},$$

where $\mathbf{z}$ are projected embeddings, sim is cosine similarity, τ a temperature and the other $2B - 2$ views in the batch act as negatives. It is a $2B - 1$-way softmax classification: "which of these is my partner?". SimCLR found that strong augmentation (especially crop plus color distortion), a non-linear projection head and large batches were essential. MoCo [35] avoids the need for huge batches by keeping a **queue** of negatives encoded by a slowly updated **momentum encoder**, whose weights are an EMA of the main encoder.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Guide to the Modern view surveys

This guide expands the short survey summary in the Modern view: what each review covers and how it organizes the field.

Surveys worth reading. Li et al. [1] review CNNs from their history and basic building blocks through classic and advanced models, highlighting the key ideas behind each, and run experiments that compare activation functions, loss functions and optimizers to derive rules of thumb; they close with applications of 1-D, 2-D and multi-dimensional convolution and a list of open issues. Khan et al. [25] survey vision transformers across tasks. For classification they group them into uniform-scale models (ViT, DeiT), multi-scale hierarchical models (such as Swin), hybrids with convolutions, and self-supervised transformers (such as DINO), and they stress the reliance on large-scale pre-training and the cost of quadratic attention. Gui et al. [41] survey self-supervised learning and divide it into context-based methods (pretext tasks such as rotation prediction or jigsaw puzzles), contrastive methods (with negative pairs, self-distillation as in DINO, or feature decorrelation), generative methods (masked image modeling as in MAE), and hybrids of contrastive and generative objectives, before covering combinations with semi-supervised and multi-modal learning. On the reliability of benchmarks, the ImageNetV2 study [7], the re-labelling study of Beyer et al. [8] and the label-error analysis of Northcutt et al. [9] together argue for multiple test sets, multi-label evaluation and reporting variance.

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DLCV 02A • Face Recognition and Metric Learning

How deep networks recognize people they never saw in training: verification and identification in closed and open sets, detection and landmark alignment, normalized embeddings, contrastive and triplet losses, margin-based softmax losses (SphereFace, CosFace, ArcFace), TAR at a fixed FAR, ROC and DET curves, CMC and FPIR, datasets and their withdrawal, demographic differentials, privacy and presentation attacks.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-02a-face-metric-learning/

What you'll learn

- Why face recognition is an **open-set** problem, and how the four tasks differ: verification (1:1), closed-set identification (1:N), open-set identification, and the classification you met in DLCV 02.
- The pipeline of a modern system: detect the face, find five landmarks, align the crop with a similarity transform, map it to a unit-length **embedding**, and compare embeddings with cosine similarity.
- How embeddings are trained: pair-based losses (contrastive, triplet with hard-negative mining) and the margin-based softmax losses **SphereFace**, **CosFace** and **ArcFace**, with every symbol defined and their geometry on the hypersphere, plus a tested NumPy experiment.
- How recognition is measured: genuine and impostor scores, TAR at a fixed FAR, ROC and DET curves, EER, rank-1 and CMC for closed-set search, and FNIR/FPIR for open-set search, all with code.
- What the datasets are (LFW, IJB-B/C, MegaFace, MS-Celeb-1M, WebFace260M, synthetic sets), why some were withdrawn, and what the evidence says about demographic differentials, privacy and law, presentation attacks and other failure modes.

The big picture

A classifier picks one of a fixed list of answers. A face recognition system cannot work that way: the people it will meet next year are not in its training set. Instead it learns a way to **measure how alike two faces are**, and then decides with a threshold.

> **Explain like I'm five**
>
> Imagine a guard at a school gate who has never met most of the visitors. The guard cannot memorize everyone in the world. Instead, each visitor shows a photo card, and the guard asks one question:

> "Does the face in front of me look enough like the face on this card?" Over many years the guard has learned which differences do not matter (a new haircut, glasses, a shadow) and which do (the shape of the eyes and nose). Face recognition models learn the same skill: turn every face into a short list of numbers, so that two photos of the same person give similar lists and photos of different people give different ones.

This idea, **metric learning**, is much wider than faces. The same recipe (learn an embedding, compare by distance, decide by a threshold) powers person re-identification (DLCV 06), image retrieval, speaker verification and the contrastive self-supervised methods of ML 05. Faces are where the recipe was pushed hardest, because the scale is extreme (millions of identities, error rates of one in a million) and the stakes are high (phone unlocking, border control, police searches).

Face recognition is also a technology with real social costs: errors fall unevenly on different groups, training data was often collected without consent, and some uses are restricted by law. This chapter treats those topics as part of the engineering, factually and with verified sources.

Two surveys frame the field: Wang and Deng's review of deep face recognition [1] and NIST's report on demographic effects in commercial algorithms [2]. We return to both in the Modern view.

Related chapters. The softmax, cross-entropy, ROC and calibration basics are in DLCV 02. Re-identification, the triplet loss for people and the CMC metric are in DLCV 06. Contrastive learning and InfoNCE are in ML 05, and image–text embeddings (CLIP) in ML 09. The classical view of nearest-neighbor matching and of open-set rejection is in PR 04 and PR 10.

Recognition as comparison: the open-set problem

Plain version. A face system answers one of two questions. *Verification*: "Are these two photos the same person?" *Identification*: "Who, among the people I know, is this?" In both cases the system compares a new photo with stored ones; it never needs to have seen the person during training.

Verification, identification, closed and open set

Let x be a face image and $f_\theta(x) \in \mathbb{R}^d$ its embedding, produced by a network with parameters θ.

A comparison of two images x_a, x_b produces a **similarity score**

$$s(x_a, x_b) = \cos \angle(f_\theta(x_a), f_\theta(x_b)) = \frac{f_\theta(x_a)^\top f_\theta(x_b)}{\|f_\theta(x_a)\| \, \|f_\theta(x_b)\|} \in [-1, 1],$$

where $\|\cdot\|$ is the Euclidean norm. People who have stored reference images are **enrolled**; their stored embeddings (one or several per person) form the **gallery**, and a new image to be recognized is a **probe**. A fixed **threshold** t turns a score into a decision.

Task	Question	Output	Typical use
Verification (1:1)	Is probe x the claimed person c?	accept if $s(x, g_c) \geq t$, else reject	phone unlock, passport gate
Closed-set identification (1:N)	Which of the N enrolled people is x?	$\arg\max_n s(x, g_n)$	benchmarks; a known, closed group
Open-set identification (1:N)	Is x enrolled, and if so, who?	best match if its score $\geq t$, else "not enrolled"	watch lists, photo search
Classification (DLCV 02)	Which training class is x?	one of the classes seen in training	not used for faces at test time

Here g_n is the gallery embedding of person n and N is the gallery size. Figure 1 draws the three recognition tasks. The important line is the last one: the classes of a face classifier are the *training* identities, and at test time almost every probe belongs to someone else. NIST's description of how deployed systems work makes the same point: a face recognition algorithm has no built-in notion of a particular person; it extracts a feature vector from people it has usually never seen and compares two vectors to produce a score [2].

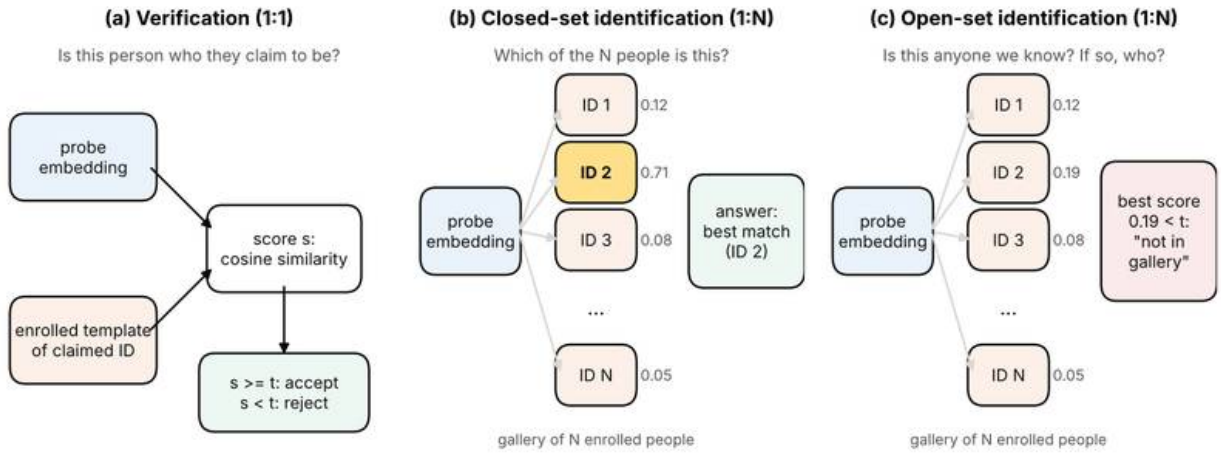


Figure 1 — (a) Verification compares a probe with one claimed identity. (b) Closed-set identification always returns the best match. (c) Open-set identification also needs a threshold, because the person may not be enrolled at all.

Closed versus open set is a property of the evaluation, not of the network. A closed-set identification test guarantees that every probe has a mate in the gallery, so "pick the best match" is always right in principle. An open-set test includes probes of people who are not enrolled, and then a threshold is unavoidable. Most real searches are open-set. The same distinction appears for generic classifiers as open-set recognition in PR 10; for faces it is the normal case, not an edge case.

Why not just train a classifier?

Plain version. A classifier's output layer has one unit per training person. New people have no unit. What survives is the layer *before* the output: a general description of a face. The whole craft of face recognition is to train that layer well.

A standard classifier trained with softmax and cross-entropy on C identities already learns a useful penultimate layer. DeepFace [6], one of the first deep systems, trained a network on an identity-labeled dataset of more than 4,000 people and showed that the learned representation generalizes to faces in unconstrained photos; the output layer itself is not used for new people. FaceNet [7] instead trained the embedding directly with a triplet loss. The two lines (classification-style training and pair-based training) then merged into the margin-based softmax losses of 2017–2019 that we study below.

What makes a good embedding for open-set use? Two properties, stated by SphereFace as the ideal criterion: the **largest distance within a class** should be smaller than the **smallest distance between classes** [12]. A plain classifier only needs each training sample to fall on the correct side of a decision boundary; it does not care whether the classes are compact. Margins are how we ask for compactness.

The face recognition pipeline

Plain version. Before any comparison, every photo goes through the same steps: find the face, find a few landmarks (eyes, nose, mouth corners), rotate and scale the face so that these landmarks land at fixed positions, then run the embedding network.

1. **Detection** finds a box around each face.
2. **Landmark localization** finds a few keypoints, usually five: the two eye centers, the nose tip and the two mouth corners.
3. **Alignment** warps the image so that the landmarks match a fixed template, and crops a small, fixed-size image (for example 112×112 pixels).
4. **Embedding** maps the crop to a vector, usually L_2-normalized to unit length.
5. **Comparison** computes cosine similarities against one template (verification) or a gallery (identification) and applies a threshold.

Detection and landmarks

Face detection is object detection specialized to one class, and the general ideas of DLCV 03 apply. The classical milestone was the Viola–Jones cascade, summarized in DLCV 03's classical

detectors note. Two deep detectors are common front ends in the face-recognition literature, and both predict the five landmarks together with the box:

- **MTCNN** [4] is a cascade of three convolutional networks that predict face boxes and landmark positions from coarse to fine, trained with an online hard-sample mining strategy.
- **RetinaFace** [5] is a single-stage detector with multi-task heads; its authors manually annotated five landmarks on the WIDER FACE dataset and showed that this extra supervision helps detect hard faces.

These are not historical details: CosFace aligned its faces with MTCNN landmarks [13], and the ArcFace journal version with RetinaFace landmarks [16]. Detector and landmark errors propagate into everything that follows, so a recognition result is always a result *of the whole pipeline*.

Alignment by a similarity transform

Plain version. We want every face to look straight at the camera, at the same size and position. A similarity transform (rotate, scale uniformly, shift) is just flexible enough to do this without distorting the face's shape.

A **similarity transform** maps a point $\mathbf{p} \in \mathbb{R}^2$ to

$$T(\mathbf{p}) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \mathbf{p} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}, \qquad a = \sigma \cos\phi,\ b = \sigma \sin\phi,$$

where $\sigma > 0$ is the scale, ϕ the rotation angle and (t_x, t_y) the translation. It has four parameters (a, b, t_x, t_y), and the equations are *linear* in them. Given detected landmarks $\mathbf{p}_1, \dots, \mathbf{p}_5$ and template positions $\mathbf{q}_1, \dots, \mathbf{q}_5$, we choose the parameters that minimize $\sum_i \|T(\mathbf{p}_i) - \mathbf{q}_i\|^2$: ten equations, four unknowns, one least-squares solve. Both CosFace and SphereFace describe their crops as obtained "by similarity transformation" from detected landmarks [12], [13].

```
import numpy as np

# A toy 5-point template (left eye, right eye, nose tip, left and right mouth
# corners) in a 112 x 112 crop. The coordinates are our own illustration.
template = np.array([[38., 46], [74, 46], [56, 66], [42, 86], [70, 86]])

def fit_similarity(src, dst):
    """Least-squares a, b, tx, ty with dst ~ [[a, -b], [b, a]] @ src + t."""
    n = len(src)
    A = np.zeros((2 * n, 4))
    A[0::2] = np.c_[src[:, 0], -src[:, 1], np.ones(n), np.zeros(n)]
    A[1::2] = np.c_[src[:, 1],  src[:, 0], np.zeros(n), np.ones(n)]
    a, b, tx, ty = np.linalg.lstsq(A, dst.reshape(-1), rcond=None)[0]
    return np.array([[a, -b, tx], [b, a, ty]])

# "Detected" landmarks: the template rotated by 20 degrees, scaled by 1.8,
```

```
# shifted into a large photo, plus 1-pixel detector noise.
rng = np.random.default_rng(0)
ang, sc = np.radians(20), 1.8
R = sc * np.array([[np.cos(ang), -np.sin(ang)], [np.sin(ang), np.cos(ang)]])
detected = template @ R.T + [310, 205] + rng.normal(scale=1.0, size=(5, 2))

M = fit_similarity(detected, template)          # photo -> crop
warped = detected @ M[:, :2].T + M[:, 2]
scale = np.hypot(M[0, 0], M[1, 0])
print(f"scale {scale:.3f} (expect {1 / sc:.3f}), "
      f"rotation {np.degrees(np.arctan2(M[1, 0], M[0, 0])):.1f} deg (expect -20.0)")
print(f"RMS landmark error after alignment: {np.sqrt(((warped - template) **
 ↪  2).sum(1).mean()):.2f} px")
```

```
scale 0.559 (expect 0.556), rotation -19.8 deg (expect -20.0)
RMS landmark error after alignment: 0.56 px
```

The fit recovers the scale and rotation almost exactly, and the remaining error (0.56 pixels) is what the 1-pixel landmark noise leaves after averaging over five points. The matrix `M` is then used to warp the whole image into the crop.

Optional: Aligning a face: the least-squares similarity transform (in the Supplementary notes). The normal equations, why a full affine or projective warp is avoided, and the edge cases (reflections, profile faces, wrong landmarks).

The embedding and the template

The embedding network is an ordinary image backbone from DLCV 02 (a ResNet, or a vision transformer) whose last layer outputs a d-dimensional vector instead of class scores. FaceNet used $d = 128$ [7]; the ArcFace experiments use $d = 512$ with 112×112 crops [16]. When a person is enrolled with several images or a video, the embeddings are combined into one **template**; ArcFace simply uses the feature center (mean) of all images in the template [16]. Since the network's output is normalized before comparison, it is natural to renormalize the mean as well.

Embeddings on the hypersphere

Plain version. If we scale every embedding to length 1, all of them live on the surface of a sphere. Then "how alike are two faces" becomes "how small is the angle between their arrows". Length, which often reflects image quality rather than identity, no longer interferes.

Why normalize, and why cosine

For unit vectors $\mathbf{u}, \mathbf{v}$ (that is, $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$),

$$\|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\,\mathbf{u}^\top\mathbf{v} = 2 - 2\cos\theta,$$

where θ is the angle between them. So on the sphere, squared Euclidean distance and cosine similarity are the same ranking, and a threshold on one is a threshold on the other. Normalization removes one degree of freedom that a network can otherwise misuse: with an unnormalized softmax, a network can lower its loss on easy samples simply by making their features longer, without making the classes more compact. NormFace analyzed this and showed that training directly on normalized features needs an extra scale factor to converge [11].

High dimension adds a useful fact: **random directions are almost orthogonal**. If $\mathbf{u}$ and $\mathbf{v}$ are independent and uniformly distributed on the unit sphere in $\mathbb{R}^d$, then $\mathbb{E}[\cos\theta] = 0$ and the standard deviation of $\cos\theta$ is $1/\sqrt{d}$. Impostor scores therefore cluster tightly around 0, and the threshold only has to beat the tail of that narrow distribution.

```
import numpy as np

rng = np.random.default_rng(0)
e = rng.normal(size=(2000, 512))
e /= np.linalg.norm(e, axis=1, keepdims=True)       # 2,000 random unit vectors
a, b = e[0], e[1]
cos = a @ b
print(f"cos = {cos:.4f}   |a-b|^2 = {np.sum((a - b) ** 2):.4f}   2-2cos = {2 - 2 * cos:.4f}")

S = e @ e.T
off = S[np.triu_indices(len(e), 1)]                 # all distinct pairs
print(f"random pairs: mean cos {off.mean():.4f}, std {off.std():.4f}, "
      f"1/sqrt(512) = {1 / np.sqrt(512):.4f}, max {off.max():.3f}")
```

```
cos = 0.0472   |a-b|^2 = 1.9057   2-2cos = 1.9057
random pairs: mean cos -0.0000, std 0.0442, 1/sqrt(512) = 0.0442, max 0.214
```

Among almost two million random pairs in 512 dimensions, the largest cosine is only 0.214. A trained network does not produce random embeddings, but the picture holds: different people sit at angles near 90°, and the job of training is to pull each person's images into a small cap around one direction.

Pair-based losses: contrastive and triplet

Plain version. The most direct way to train an embedding is to show the network pairs or triples of images and say "these two should be close, those two should be far". The difficulty is that almost all such examples are already easy, so the network learns little from them unless we search for the hard ones.

Contrastive loss

The contrastive loss, introduced for face verification with a Siamese network by Chopra, Hadsell and LeCun [8] and developed by Hadsell, Chopra and LeCun [9], looks at one pair at a time:

$$\mathcal{L}_{\text{con}}(x_a, x_b) = y\, D^2 + (1-y)\,[\,m - D\,]_+^2, \qquad D = \|f_\theta(x_a) - f_\theta(x_b)\|,$$

where $y = 1$ if the pair shows the same person and $y = 0$ otherwise, $m > 0$ is a margin, and $[z]_+ = \max(z, 0)$. Same-person pairs are pulled together; different-person pairs are pushed apart only until they are m apart. Up to constant factors and the label convention, this squared-hinge form is the one in [9]; [8] introduced the underlying idea of training a Siamese network so that a learned distance separates same-person from different-person pairs.

Triplet loss and hard-negative mining

The **triplet loss** of FaceNet [7] compares distances instead of thresholding them. Its definition, and its use in re-identification, are in DLCV 06 and ML 05; in FaceNet's form, with squared distances on unit-length embeddings,

$$\mathcal{L}_{\text{tri}}(a, p, n) = [\,\|f(a) - f(p)\|^2 - \|f(a) - f(n)\|^2 + \alpha\,]_+,$$

where a is an anchor image, p a positive (same person), n a negative (another person), $f = f_\theta$ the embedding, and $\alpha > 0$ the margin. What this chapter adds is the **mining** question. For an anchor–positive pair with squared distance d_{ap}, a negative at squared distance d_{an} is

- **easy** if $d_{an} \geq d_{ap} + \alpha$ (zero loss, zero gradient),
- **semi-hard** if $d_{ap} \leq d_{an} < d_{ap} + \alpha$ (farther than the positive, but inside the margin; this is FaceNet's definition [7]),
- **hard** if $d_{an} < d_{ap}$ (closer than the positive).

FaceNet selected semi-hard negatives online within large mini-batches, because the very hardest negatives early in training led to bad local minima [7]. A later and widely used alternative, **batch-hard** mining, takes for every anchor in a batch of P identities × K images the *farthest* positive and the *closest* negative [10].

```
import numpy as np

rng = np.random.default_rng(0)
P, Kimg, alpha = 8, 4, 0.2                  # 8 people x 4 images, margin
centres = rng.normal(size=(P, 64))
y = np.repeat(np.arange(P), Kimg)
e = centres[y] + 1.5 * rng.normal(size=(len(y), 64))
```

```
e /= np.linalg.norm(e, axis=1, keepdims=True)
d2 = ((e[:, None] - e[None]) ** 2).sum(-1)  # squared distances

easy = semi = hard = 0
for a in range(len(y)):
    for p in np.flatnonzero((y == y[a]) & (np.arange(len(y)) != a)):
        dn = d2[a, y != y[a]]               # distances to all negatives
        hard += (dn < d2[a, p]).sum()
        semi += ((dn >= d2[a, p]) & (dn < d2[a, p] + alpha)).sum()
        easy += (dn >= d2[a, p] + alpha).sum()
tot = easy + semi + hard
print(f"{tot} triplets: easy {easy / tot:.1%}, semi-hard {semi / tot:.1%}, hard {hard /
 ↪  tot:.1%}")

# batch-hard: for each anchor, farthest positive and closest negative
same = y[:, None] == y[None]
hp = np.where(same, d2, -np.inf).max(1)
hn = np.where(~same, d2, np.inf).min(1)
loss = np.maximum(hp - hn + alpha, 0)
print(f"batch-hard loss {loss.mean():.3f}; anchors with non-zero loss: {(loss > 0).sum()} of
 ↪  {len(y)}")
```

```
2688 triplets: easy 92.8%, semi-hard 5.4%, hard 1.9%
batch-hard loss 0.213; anchors with non-zero loss: 28 of 32
```

In this synthetic batch, 92.8% of all 2,688 valid triplets are easy and contribute nothing. Averaging over all triplets would dilute the few useful ones; batch-hard mining finds a non-zero loss for 28 of the 32 anchors. This is the core weakness of pair-based training at face scale: the number of possible triplets grows with the cube of the dataset size, almost all of them are easy, and the result depends on the sampling strategy and the batch composition.

Margin-based softmax losses

Plain version. Instead of comparing images with each other, give every training identity a learned "anchor direction" on the sphere, and train each image to point at its own identity's direction *by a clear margin*: not just closer than to the other directions, but closer by a fixed angle. One loss term per image then compares it with all identities at once.

From softmax to normalized softmax

Start from the softmax cross-entropy of DLCV 02. For a feature $\mathbf{x}_i \in \mathbb{R}^d$ of image i with identity label $y_i \in \{1, \dots, C\}$, a last layer with weight columns $W_j \in \mathbb{R}^d$ and biases b_j gives

$$\mathcal{L}_{\text{softmax}} = -\log \frac{e^{W_{y_i}^\top \mathbf{x}_i + b_{y_i}}}{\sum_{j=1}^{C} e^{W_j^\top \mathbf{x}_i + b_j}}.$$

Set $b_j = 0$ and write the logit as $W_j^\top \mathbf{x}_i = \|W_j\| \, \|\mathbf{x}_i\| \cos\theta_j$, where θ_j is the angle between $\mathbf{x}_i$ and W_j. Now normalize both: $\|W_j\| = 1$, and $\|\mathbf{x}_i\|$ fixed to a constant s. Then the logits are $s\cos\theta_j$ and the prediction depends on angles only [11], [13], [16]:

$$\mathcal{L}_{\text{norm}} = -\log \frac{e^{s\cos\theta_{y_i}}}{\sum_{j=1}^{C} e^{s\cos\theta_j}}.$$

The **scale** $s > 0$ is necessary because cosines lie in $[-1, 1]$: without it, even a perfect angular arrangement would leave the softmax far from confident, so the loss could not become small. NormFace analyzed this failure to converge [11], and CosFace derived a lower bound for s that grows with the number of classes [13]; both CosFace and ArcFace use $s = 64$ [13], [16].

Optional: How large must the scale s be? (in the Supplementary notes). CosFace's lower bound with every symbol defined, a worked example, and what goes wrong when s is too small or too large.

Each column W_j is now a unit vector: a **class center**, or **proxy**, for identity j on the sphere. This is the key structural change compared with pair-based losses. Every image is compared with all C class centers in every step, so there is nothing to mine, and the comparison is global rather than limited to one batch. The class centers exist only for training; at test time the last layer is discarded and only the embedding network is kept [16].

Three ways to add a margin

The normalized softmax is satisfied when $\theta_{y_i} < \theta_j$ for all $j \neq y_i$: the feature is closer in angle to its own center than to any other. That puts the decision boundary exactly halfway between two centers, so samples near the boundary are correct but barely. A **margin** makes the target logit harder to achieve by modifying it from $s\cos\theta_{y_i}$ to $s\,\psi(\theta_{y_i})$ with $\psi(\theta) < \cos\theta$, while the other logits stay $s\cos\theta_j$:

$$\mathcal{L}_{\text{margin}} = -\log \frac{e^{s\,\psi(\theta_{y_i})}}{e^{s\,\psi(\theta_{y_i})} + \sum_{j\neq y_i} e^{s\cos\theta_j}}.$$

The three classic losses differ only in ψ:

Loss	Target logit $\psi(\theta)$	Margin type	Value in the paper
SphereFace (A-Softmax) [12]	$\cos(m\theta)$, extended to stay monotone (below)	multiplicative angular	$m = 4$
CosFace (LMCL) [13]; also AM-Softmax [14]	$\cos\theta - m$	additive cosine	$m = 0.35$, $s = 64$
ArcFace [15], [16]	$\cos(\theta + m)$	additive angular	$m = 0.5$, $s = 64$

Here m is the margin (an integer for SphereFace, a cosine offset for CosFace and an angle in radians for ArcFace), and $\theta = \theta_{y_i}$ is the angle to the sample's own class center; the last column gives the values reported in the papers. Because $\cos(m\theta)$ is not monotone on $[0, \pi]$, SphereFace uses $\psi(\theta) =$

$(-1)^k \cos(m\theta) - 2k$ for $\theta \in [k\pi/m, (k+1)\pi/m]$, $k \in \{0, \dots, m-1\}$ [12]. The original SphereFace normalized only the weights and needed help from an ordinary softmax early in training [16]. AM-Softmax [14], published at about the same time as CosFace, uses the same additive cosine margin.

```
import numpy as np

def sphereface_psi(theta, m=4):
    """SphereFace's monotone extension of cos(m*theta) to [0, pi]."""
    k = np.floor(theta * m / np.pi)
    return (-1) ** k * np.cos(m * theta) - 2 * k

print(" angle  softmax  SphereFace  CosFace  ArcFace")
for deg in [10, 30, 45, 60, 90]:
    t = np.radians(deg)
    print(f"{deg:5d}   {np.cos(t):6.3f}   {sphereface_psi(t):8.3f}   "
          f"{np.cos(t) - 0.35:6.3f}   {np.cos(t + 0.5):6.3f}")
```

```
 angle  softmax  SphereFace  CosFace  ArcFace
   10    0.985      0.766    0.635    0.781
   30    0.866     -0.500    0.516    0.520
   45    0.707     -1.000    0.357    0.282
   60    0.500     -1.500    0.150    0.024
   90    0.000     -3.000   -0.350   -0.479
```

The table shows the target logit (divided by s) at five angles. Every margin lowers the logit of the correct class, so the loss stays large until the angle is small. The penalties grow differently: CosFace subtracts the same 0.35 at every angle; ArcFace's penalty is small near $0°$ (0.204 at $10°$) and larger at wide angles; SphereFace with $m = 4$ is very strict at wide angles.

The geometry on the circle

Plain version. Picture two class centers on a circle. Without a margin, the circle is split exactly halfway between them. With a margin, each class claims less than half, and a strip in between belongs to nobody: a sample there still has a loss. Training pushes samples out of that strip, toward their own center.

For two classes with centers W_1, W_2 an angle φ apart, and a feature on the arc between them, a sample of class 1 is "satisfied" when $\psi(\theta_1) > \cos\theta_2$ with $\theta_2 = \varphi - \theta_1$. Solving gives the class-1 boundary:

- normalized softmax: $\theta_1 < \varphi/2$ (no gap);
- SphereFace: $m\,\theta_1 < \theta_2$, so $\theta_1 < \varphi/(m+1)$; the gap between the two classes' boundaries is $\varphi(m-1)/(m+1)$, which **grows with** φ;

- CosFace: $\cos\theta_1 - m > \cos(\varphi - \theta_1)$; the gap is a fixed amount in *cosine*, so its size in angle depends on φ;
- ArcFace: $\theta_1 + m < \theta_2$, so $\theta_1 < (\varphi - m)/2$; the gap is exactly m radians for every φ.

ArcFace's authors call this a margin equal to the **geodesic distance** on the hypersphere, which is where the name comes from [15], [16]. Figure 2 draws all four cases for $\varphi = 80°$; SphereFace is shown with $m = 2$ so that its zone fits on the same picture.

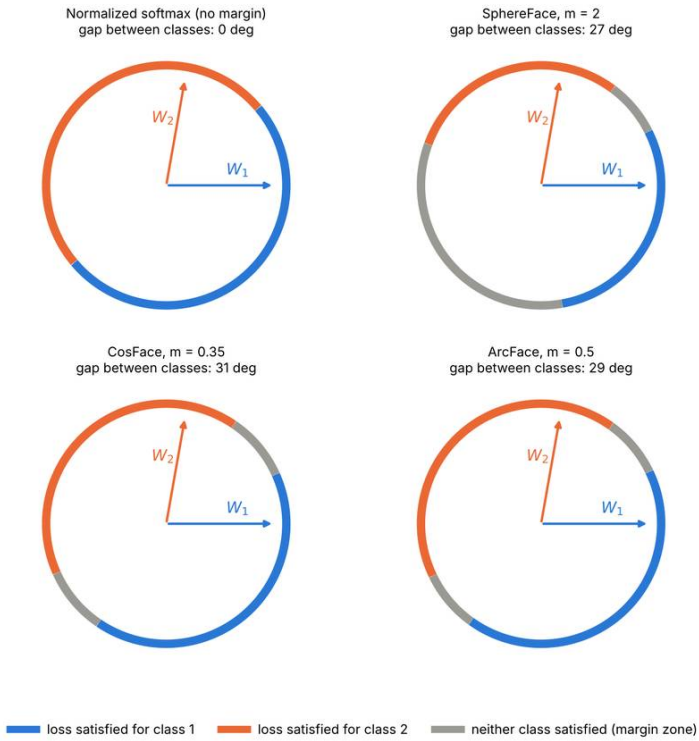


Figure 2 — Where each loss is satisfied for two classes whose centers are 80° apart. Gray arcs are the margin zones, where a sample of either class still has a loss. The gap between the centers is 0° without a margin, 27° for SphereFace ($m = 2$), 31° for CosFace ($m = 0.35$) and 29° ($= 0.5$ rad) for ArcFace ($m = 0.5$).

One practical detail follows from the formula: $\cos(\theta + m)$ decreases only while $\theta + m \leq \pi$. For a sample whose angle to its own center is larger than $\pi - m$ (a badly misclassified or mislabeled sample), the ArcFace target logit would start to *increase* again as the sample moves away. An implementation has to handle this range explicitly, for example by switching to a monotone penalty there. In our toy code below, all angles stay well below that range.

Optional: Margin losses side by side: one formula and its edge cases (in the Supplementary notes). The combined margin $\cos(m_1\theta + m_2) - m_3$, the derivation of each boundary, the gradient of the ArcFace logit, and why easy samples stop contributing.

A toy experiment: softmax versus ArcFace

Plain version. We train the same small network on synthetic "faces" twice: once with the ordinary softmax and once with ArcFace. Then we look at how tight the training clusters are and, more importantly, how well the network separates *people it never saw*.

The data mimic the face problem. Each synthetic identity is a random point in a 4-dimensional "identity" subspace of a 16-dimensional input; every image adds a large random "nuisance" component (standing in for pose and lighting) in the other 12 dimensions and small noise everywhere. The network is one hidden layer of 64 ReLU units followed by a linear map to a $d = 4$ embedding. We implement the forward pass and the gradients by hand, including the chain rule through the normalization and through $\cos(\theta + m)$, and use plain gradient descent. We train on 100 identities and test verification on 50 identities that never appear in training. TAR at FAR $= 1\%$ (defined in the next section) is the fraction of same-person test pairs accepted when the threshold lets through 1% of different-person pairs.

```
import numpy as np

rng = np.random.default_rng(0)
D, K = 16, 4                                        # input dim, identity dims
U = np.linalg.qr(rng.normal(size=(D, D)))[0]      # random orthonormal basis

def sample(ids, n_per):
    """Identity signal in 4 directions, large 'pose/light' nuisance in 12."""
    y = np.repeat(np.arange(len(ids)), n_per)
    nuis = 1.5 * rng.normal(size=(len(y), D - K))
    x = ids[y] @ U[:, :K].T + nuis @ U[:, K:].T + 0.3 * rng.normal(size=(len(y), D))
    return x, y

unit = lambda v, ax: v / np.linalg.norm(v, axis=ax, keepdims=True)

def train(x, y, C, d=4, margin=None, s=16.0, steps=800, lr=0.2, H=64):
    """One-hidden-layer embedding net + class centres W, plain gradient descent.
    margin=None: ordinary softmax on raw features; otherwise ArcFace logits."""
    r = np.random.default_rng(1)
    W1 = r.normal(size=(D, H)) / 4; b1 = np.zeros(H)
    W2 = r.normal(size=(H, d)) / 8; W = r.normal(size=(d, C))
    Y = np.eye(C)[y]
    for _ in range(steps):
        h = np.maximum(x @ W1 + b1, 0); z = h @ W2          # embeddings
        if margin is None:
            L = z @ W
        else:
            f, w = unit(z, 1), unit(W, 0)
            th = np.arccos(np.clip(f @ w, -1 + 1e-7, 1 - 1e-7))   # angles to centres
            L = s * np.where(Y > 0, np.cos(th + margin), np.cos(th))
        p = np.exp(L - L.max(1, keepdims=True)); p /= p.sum(1, keepdims=True)
        g = (p - Y) / len(y)                                  # dLoss/dlogits
        if margin is None:
            gz, gW = g @ W.T, z.T @ g
```

```
        else:                                                  # chain rule through
         ↪ cos(theta+m)
            gc = g * s * np.where(Y > 0, np.sin(th + margin) / np.sin(th), 1.0)
            gf, gw = gc @ w.T, f.T @ gc
            gz = (gf - (gf * f).sum(1, keepdims=True) * f) / np.linalg.norm(z, axis=1,
             ↪ keepdims=True)
            gW = (gw - (gw * w).sum(0, keepdims=True) * w) / np.linalg.norm(W, axis=0,
             ↪ keepdims=True)
        gh = (gz @ W2.T) * (h > 0)
        W1 -= lr * x.T @ gh; b1 -= lr * gh.sum(0); W2 -= lr * h.T @ gz; W -= lr * gW
    return lambda xx: unit(np.maximum(xx @ W1 + b1, 0) @ W2, 1)

def intra_angle(e, y):            # mean angle (degrees) to own class mean direction
    cen = unit(np.stack([e[y == j].mean(0) for j in np.unique(y)]), 1)
    return np.degrees(np.arccos(np.clip((e * cen[y]).sum(1), -1, 1))).mean()

def tar_at_far(e, y, far=0.01):   # verification on all pairs of unseen people
    i, j = np.triu_indices(len(y), 1)
    sc, same = (e[i] * e[j]).sum(1), y[i] == y[j]
    thr = np.quantile(sc[~same], 1 - far)
    return (sc[same] > thr).mean()

ids = rng.normal(size=(150, K))
xtr, ytr = sample(ids[:100], 10)          # 100 training identities
xte, yte = sample(ids[100:], 10)          # 50 identities never seen in training
for name, m in [("softmax", None), ("norm-softmax", 0.0), ("ArcFace m=0.5", 0.5)]:
    emb = train(xtr, ytr, 100, margin=m)
    print(f"{name:14s} train intra {intra_angle(emb(xtr), ytr):5.1f} deg | "
          f"unseen intra {intra_angle(emb(xte), yte):5.1f} deg | "
          f"unseen TAR@FAR=1% {tar_at_far(emb(xte), yte):.3f}")
```

```
softmax        train intra  27.1 deg | unseen intra  43.9 deg | unseen TAR@FAR=1% 0.082
norm-softmax   train intra  16.1 deg | unseen intra  30.1 deg | unseen TAR@FAR=1% 0.098
ArcFace m=0.5  train intra  11.6 deg | unseen intra  24.4 deg | unseen TAR@FAR=1% 0.248
```

The script runs in about ten seconds on a CPU. In this toy run the mean angle of a training image to its class center drops from 27.1° (softmax) to 16.1° (normalization alone) to 11.6° (ArcFace), and the clusters of *unseen* people tighten too (43.9° to 24.4°). The verification rate on unseen people at FAR = 1% roughly triples, from 0.082 to 0.248. Two cautions. First, the absolute numbers are low because a 4-dimensional embedding of noisy data is a hard problem; only the comparison matters. Second, these are single runs; with two other random seeds the ordering softmax < normalized softmax < ArcFace stayed the same in our runs, but the exact values changed.

Figure 3 shows the same effect in a 2-D embedding with six identities, where it can be seen directly: softmax spreads each class along a ray (it can lower its loss by lengthening features), while ArcFace packs each class into a short arc.

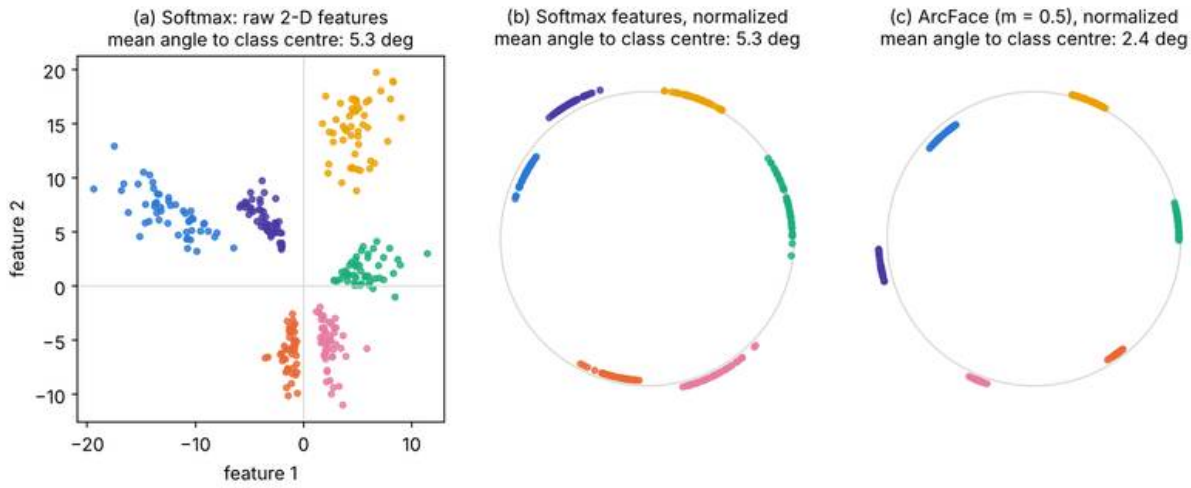


Figure 3 — Six synthetic identities embedded in 2-D by the toy network. (a) Softmax features spread radially. (b) Normalized, the softmax classes cover wide arcs (mean angle to the class center 5.3°). (c) ArcFace with $m = 0.5$ gives tighter arcs (2.4°). Numbers are from this toy run.

Margins tighten clusters of the training identities directly. Whether that transfers to new people depends on how many and how varied the training identities are; with very few training identities, a network can separate them perfectly and still learn nothing general. That is one reason why face recognition training sets contain hundreds of thousands to millions of identities (see the datasets section).

Class centers, sub-centers and adaptive margins

Plain version. Once each identity has a center, several refinements follow naturally. They all keep the margin softmax and change one thing about the centers or the margin.

Center loss [17], which predates the angular losses, pulls each feature toward a learned center of its class. Sub-center ArcFace [18], [16] gives each identity several centers, so that noisy or mislabeled images can gather away from the clean ones. MagFace [19] and AdaFace [20] tie the margin to image quality, using the feature magnitude, so that very poor images do not dominate training. Partial FC [21] cuts the cost of millions of centers by updating only a sampled subset of them in each step. The one thing to remember: the class centers are a training device, and these methods tune how strictly and how cheaply images are pulled toward them.

Optional: Beyond one center per class: sub-centers, quality-aware margins and Partial FC (in the Supplementary notes). How each variant changes the loss, the problem it targets, and what to watch for when you use it.

Evaluating verification

Plain version. Compare many pairs of photos of the same person (**genuine** pairs) and many pairs of different people (**impostor** pairs). A good system gives genuine pairs high scores and impostor pairs low scores. Any threshold then trades two errors: letting impostors in, and turning genuine users away.

Genuine and impostor scores, FAR and TAR

For a threshold t on the similarity score s,

$$\mathrm{FAR}(t) = P(s \geq t \mid \text{impostor}), \qquad \mathrm{FRR}(t) = P(s < t \mid \text{genuine}), \qquad \mathrm{TAR}(t) = 1 - \mathrm{FRR}(t).$$

FAR is the **false accept rate**, FRR the **false reject rate** and TAR the **true accept rate**. NIST's evaluations use the equivalent names **false match rate** (FMR) and **false non-match rate** (FNMR) for one-to-one comparisons [2], [3]. These are exactly the false-positive rate and the false-negative rate of DLCV 02; the "positive" class is "same person".

The two distributions are very unequal in size. With P people and k images each, there are $P\binom{k}{2}$ genuine pairs but about $\binom{Pk}{2} - P\binom{k}{2}$ impostor pairs, which grows with the square of P. Because impostor pairs are so many, operating points are stated as **TAR at a fixed FAR**: "TAR@FAR = 10^{-4}" is the fraction of genuine pairs accepted when the threshold is set so that one impostor pair in ten thousand is accepted. Benchmarks such as IJB-B and IJB-C report TAR@FAR = 10^{-4} [16], and NIST's ongoing one-to-one evaluation ranks algorithms by FNMR at FMR = 10^{-6} on one of its datasets [3].

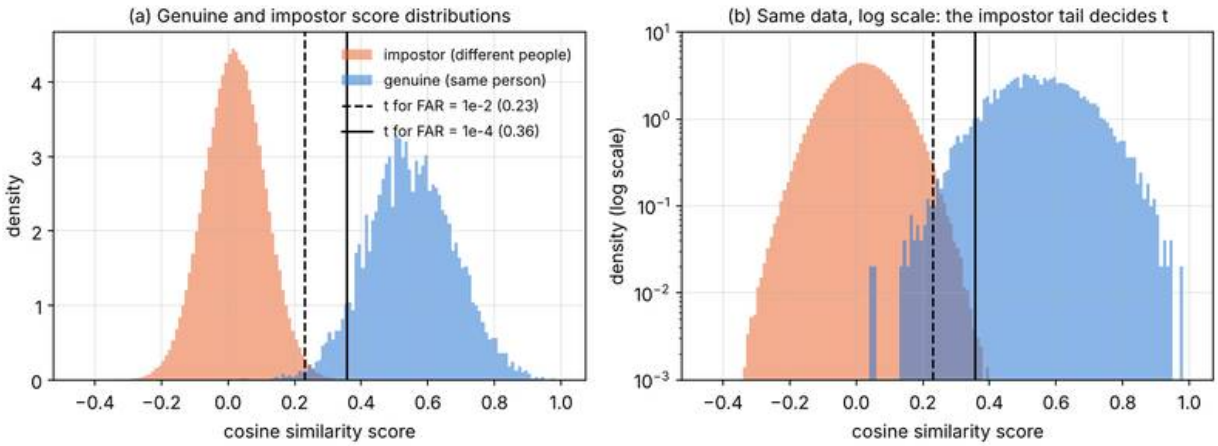


Figure 4 — Synthetic genuine (blue) and impostor (orange) score distributions. (a) Linear scale. (b) Log scale: the threshold for a small FAR is set by the far tail of the impostor distribution, which is invisible on a

linear axis.

```
import numpy as np
from sklearn.metrics import roc_curve

rng = np.random.default_rng(0)
genuine = np.clip(rng.normal(0.55, 0.13, 5_000), -1, 1)       # same-person cosines
impostor = np.clip(rng.normal(0.02, 0.09, 500_000), -1, 1)    # different-person cosines
y = np.r_[np.ones(genuine.size), np.zeros(impostor.size)]
fpr, tpr, thr = roc_curve(y, np.r_[genuine, impostor])

for far in [1e-2, 1e-3, 1e-4]:
    i = np.searchsorted(fpr, far, side="right") - 1          # last point with FPR <= far
    print(f"FAR <= {far:.0e}: threshold {thr[i]:.3f}, TAR {tpr[i]:.3f}, "
          f"false accepts {int(round(fpr[i] * impostor.size))}")
i = np.argmin(np.abs(fpr - (1 - tpr)))
print(f"EER ~ {(fpr[i] + 1 - tpr[i]) / 2:.4f} at threshold {thr[i]:.3f}")
```

```
FAR <= 1e-02: threshold 0.231, TAR 0.994, false accepts 4825
FAR <= 1e-03: threshold 0.300, TAR 0.975, false accepts 495
FAR <= 1e-04: threshold 0.356, TAR 0.936, false accepts 49
EER ~ 0.0072 at threshold 0.241
```

The **equal error rate** (EER) is the error at the threshold where FAR equals FRR; it is a convenient single number but describes an operating point that a secure system rarely uses. Notice how the threshold moves: lowering FAR from 10^{-2} to 10^{-4} raises the threshold from 0.231 to 0.356 and costs 5.8 points of TAR. Notice also how few errors the last line is based on: 49 false accepts. A FAR of 10^{-6} cannot be measured at all with 500,000 impostor pairs; you need millions of them, and they must be reasonably independent.

Optional: Measuring very small error rates (in the Supplementary notes). The "rule of three", binomial confidence intervals, and why impostor pairs built from the same images are not independent.

ROC and DET curves

The ROC curve of DLCV 02 plots TAR against FAR. For face verification it is always drawn with a **logarithmic FAR axis**, because the interesting region is FAR between 10^{-6} and 10^{-2}, which a linear axis squeezes into the left edge. The **detection error trade-off (DET) curve** [22] plots both error rates, FRR against FAR, with both axes transformed by the inverse of the standard normal cumulative distribution function Φ^{-1} (the "normal deviate" scale). If genuine and impostor scores are both Gaussian, the DET curve is a straight line, and lower-left is better. Martin et al. introduced it for speaker recognition and explained why it tends to be close to linear [22]; it is now standard for biometric systems.

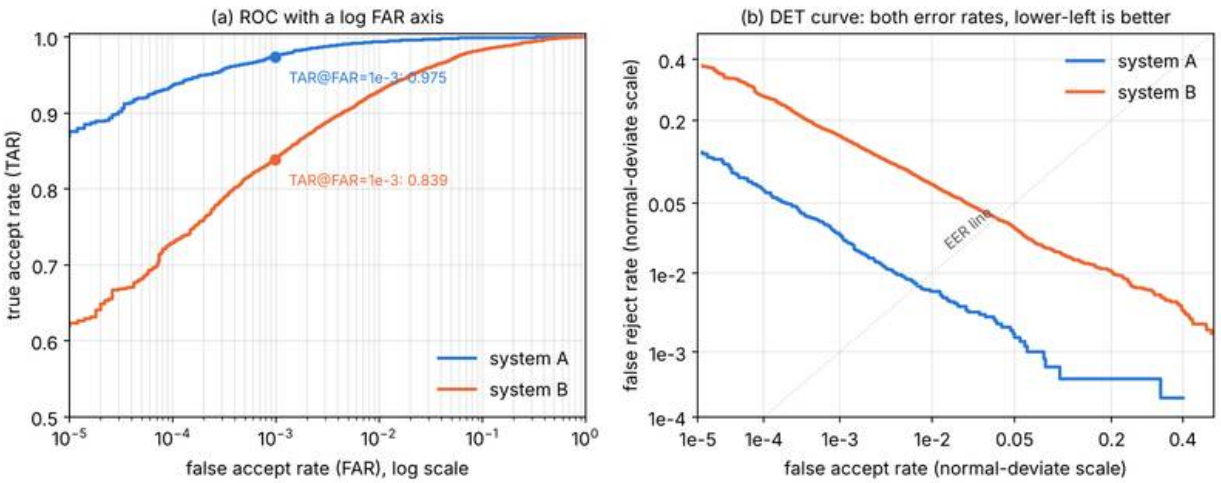


Figure 5 — (a) ROC with a log FAR axis; dots mark TAR at FAR = 10^{-3}. (b) DET curves of the same two synthetic systems: both error rates on normal-deviate axes. The EER is where a curve crosses the dotted diagonal.

Choosing the threshold

The threshold is part of the system, not of the network, and it must be chosen on data that represent deployment. Three rules follow from the formulas above. (1) Fix the FAR you can tolerate from the application (a phone unlock and a forensic search have very different costs), then choose t as the corresponding quantile of impostor scores on a **held-out** set; tuning t on the test set inflates results. (2) Report TAR and FAR *together* at that threshold, never one alone. (3) Check that the impostor scores of your deployment look like those of your calibration set: a new camera, a new population or a new image quality changes the impostor distribution and therefore the real FAR at a fixed t. We return to the last point in the fairness section.

Evaluating identification

Plain version. In a search, the probe is compared with every enrolled person. In a closed-set test we ask: was the right person ranked first (or in the top k)? In an open-set test we also ask: when the person is not enrolled at all, did the system stay silent?

Closed set: rank-1 and the CMC curve

For closed-set identification, sort the gallery by similarity to the probe. **Rank-1 accuracy** is the fraction of probes whose mate is ranked first, and the **cumulative match characteristic** CMC(k)

is the fraction whose mate is within the top k. The formula and code are in DLCV 06; a face gallery usually holds one template per person, so mAP is rarely used.

The crucial effect is **gallery size**. Each extra enrolled person is one more chance for an impostor to outscore the true mate. MegaFace was built to measure this: it added up to one million distractor photos to the gallery and reported identification and verification as the number of distractors grew from 10 to 1M [24].

Open set: FNIR and FPIR

In an open-set search, the system returns a candidate only if its score reaches a threshold t. Two error rates describe it [2]:

$$\mathrm{FNIR}(t) = \frac{\#\{\text{mated searches whose mate is not returned with score} \geq t\}}{\#\{\text{mated searches}\}}, \qquad \mathrm{FPIR}(t) = \frac{\#\{\text{non-mated searches returning any candidate with score} \geq t\}}{\#\{\text{non-mated searches}\}}.$$

A **mated** search is one whose probe person is enrolled; a **non-mated** search is one whose probe person is not. FNIR (false negative identification rate) and FPIR (false positive identification rate) play the roles of FRR and FAR for one-to-many search; in this simple version a candidate is returned only at rank 1. If every one of the N gallery comparisons of a non-mated probe were an independent trial with false match rate $\mathrm{FMR}(t)$, then

$$\mathrm{FPIR}(t) \approx 1 - (1 - \mathrm{FMR}(t))^{N} \approx N \cdot \mathrm{FMR}(t) \quad \text{for small } N \cdot \mathrm{FMR}(t),$$

so at a fixed threshold FPIR grows roughly in proportion to the gallery size. This is why one-to-many search needs far lower one-to-one false match rates than verification.

```
import numpy as np

rng = np.random.default_rng(0)
dim, noise = 128, 1.1
unit = lambda v: v / np.linalg.norm(v, axis=-1, keepdims=True)

def identify(n_gallery, n_probe=300, thr=0.36):
    people = rng.normal(size=(n_gallery + n_probe, dim))
    gallery = unit(people[:n_gallery] + noise * rng.normal(size=(n_gallery, dim)))
    mated = rng.choice(n_gallery, n_probe, replace=False)            # enrolled people
    p_in = unit(people[mated] + noise * rng.normal(size=(n_probe, dim)))
    p_out = unit(people[n_gallery:] + noise * rng.normal(size=(n_probe, dim)))  # not
     ↪  enrolled
    S_in, S_out = p_in @ gallery.T, p_out @ gallery.T
    rank1 = (S_in.argmax(1) == mated).mean()
    # open set: a search "returns" the top candidate only if its score >= thr
    fnir = 1 - ((S_in.argmax(1) == mated) & (S_in.max(1) >= thr)).mean()
    fpir = (S_out.max(1) >= thr).mean()
    return rank1, fnir, fpir

for n in [300, 3_000, 30_000]:
```

```
    r1, fnir, fpir = identify(n)
    print(f"gallery {n:6d}: rank-1 {r1:.3f} | threshold 0.36: FNIR {fnir:.3f}, FPIR
     ↪  {fpir:.3f}")
```

```
gallery    300: rank-1 0.990 | threshold 0.36: FNIR 0.090, FPIR 0.010
gallery   3000: rank-1 0.953 | threshold 0.36: FNIR 0.113, FPIR 0.043
gallery  30000: rank-1 0.903 | threshold 0.36: FNIR 0.117, FPIR 0.330
```

With the same synthetic embedding quality, growing the gallery from 300 to 30,000 people lowers rank-1 from 0.990 to 0.903, and at a fixed threshold the share of non-enrolled probes that trigger a false hit rises from 1% to 33%. Figure 6 shows the CMC curves and the FPIR-versus-threshold curves of a similar run. An identification result is meaningless without the gallery size, and an open-set system must be re-thresholded when the gallery grows.

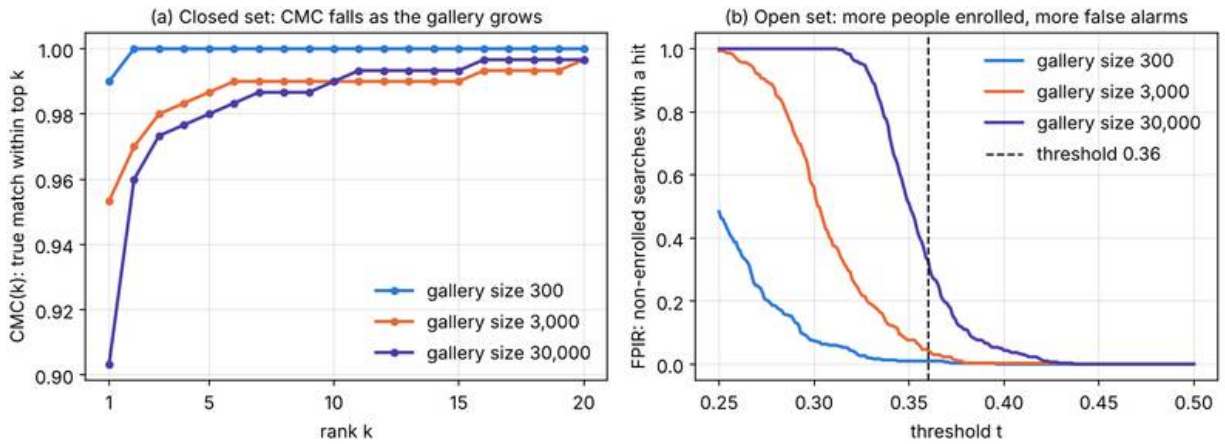


Figure 6 — A synthetic search experiment. (a) Closed-set CMC drops as the gallery grows. (b) Open-set FPIR at each threshold: at a fixed threshold, more enrolled people means more false alarms for non-enrolled probes.

Datasets and benchmarks

Plain version. Training sets for faces are huge lists of web photos grouped by person; test sets define pairs or searches with a fixed protocol. Several of the best-known sets were collected from the web without the consent of the people in them, and some have been withdrawn.

Dataset	Role	What it contains (as described by its source)	Status
LFW [23]	verification test	unconstrained news photos; about 13K photos of 5K people [24]	saturated; a disclaimer added in 2019 warns against concluding commercial suitability from it [30]
IJB-B [27]	verification, identification, clustering	1,845 subjects; 21,798 still images and 55,026 frames from 7,011 videos, with human-labeled boxes and eye/nose locations	standard test set; media under a Creative Commons license [27]
IJB-C [28]	as IJB-B, larger	3,531 subjects; 31.3K still images and 117.5K frames from 11,779 videos [16]	standard test set [16]
MegaFace [24], [25]	identification at scale	1M distractor photos of more than 690K individuals; a later training set of 4.7M photos of 672,057 identities	**decommissioned**; data no longer distributed [25]
MS-Celeb-1M [26]	training	web images for a benchmark of recognizing one million celebrities linked to a knowledge base	**taken down in April 2019** [30]
WebFace260M / WebFace42M [29]	training	4M identities / 260M faces (noisy) and a cleaned 2M identities / 42M faces	described by its authors as the largest public training set [29]
DigiFace-1M [31]	training	synthetic faces rendered with a computer-graphics pipeline	synthetic, not photographs of real people

LFW (Labeled Faces in the Wild) [23] defined unconstrained face verification and its standard protocols. Its accuracy is now saturated: state-of-the-art models exceed 99.8% on it [31], so it no longer separates good systems from very good ones. **IJB-B** and **IJB-C** (IARPA Janus Benchmarks) [27], [28] replaced it as the standard academic test: they include video frames, large pose variation and protocols at operationally relevant points such as TAR@FAR $= 10^{-4}$, and they evaluate *templates* (sets of images) rather than single images. **MegaFace** [24] made gallery size the central question. **NIST's evaluations** [2], [3] use large operational datasets that are not public and test submitted algorithms directly, so they are the reference for deployment-scale error rates.

Withdrawn datasets. MS-Celeb-1M was taken down by Microsoft in April 2019; Peng, Mathur and Narayanan note that the removal followed critical reports, that many of the "celebrities" were fairly ordinary people, that the images had been aggregated without consent, and that Microsoft gave as its reason that the research challenge was over [30]. MegaFace's own site states that the challenge has concluded and that the data are no longer distributed [25]. The same study found that retracted data often keep circulating through copies, cleaned derivatives and pretrained models, and that licenses did little to restrict such use [30]; the DukeMTMC case in DLCV 06 is similar.

In practice: do not use withdrawn datasets or their derivatives, record the provenance and license of every training set, and expect reviewers to ask.

Synthetic training data is one response. DigiFace-1M renders faces with a computer-graphics pipeline and motivates this with bias, label noise and the lack of consent in web-crawled data [31]; DCFace generates identities and styles with a diffusion model [32]. Boutros et al. survey this direction and its open challenges [33].

Fairness and demographic differentials

Plain version. A system can have the same overall accuracy as another and still make many more errors for some groups of people than for others. Because a deployed system usually uses *one* threshold for everybody, small differences in score distributions between groups can turn into large differences in error rates.

What the measurements show

The most extensive public measurement is NIST's 2019 report on demographic effects [2]. It processed 18.27 million images of 8.49 million people from four U.S. government datasets through 189 mostly commercial algorithms from 99 developers. Its main findings, in the report's own framing:

- **False positives** differed far more than false negatives: across demographic groups, false positive rates often varied by factors of 10 to beyond 100, while false negatives were more algorithm-specific and often varied by factors below 3.
- The pattern **depended on the algorithm and the data**. For example, with high-quality application photos the highest false positive rates were in West and East African and East Asian people for many algorithms, but with a number of algorithms developed in China the East Asian effect was reversed. False positives were also higher for women than for men and elevated for the elderly and children.
- **Image quality** mattered for false negatives: results differed between well-controlled mugshots and lower-quality border-crossing photos.
- Some highly accurate one-to-many algorithms showed **undetectable** false positive differentials, so differentials are not inevitable.
- The report recommends that every demographic claim state which error (false positive or false negative) is meant, at a fixed threshold, for each group, because many earlier reports did not [2].

Academic studies add complementary evidence. Wang et al. built the Racial Faces in-the-Wild (RFW) test set and measured racial bias in commercial APIs and research algorithms [34]. The

widely cited Gender Shades study [35] measured error rates of up to 34.7% for darker-skinned women versus at most 0.8% for lighter-skinned men; note that it evaluated commercial **gender classification**, not face recognition, a distinction NIST explicitly asks reporters to make [2]. Drozdowski et al. survey bias estimation and mitigation across biometrics [36].

A toy: one threshold, two groups

The mechanism is simple enough to simulate. Assume two groups A and B with identical genuine score distributions, but B's impostor scores are slightly higher: different members of B look a little more alike to the network. In this toy we simply assume the shift; in real systems its causes have to be investigated. Set one global threshold for FMR $= 10^{-3}$ on the pooled impostors.

```
import numpy as np

rng = np.random.default_rng(0)
n_imp, n_gen = 400_000, 4_000
# Same genuine distribution; group B's impostor scores sit slightly higher.
imp = {"A": rng.normal(0.00, 0.09, n_imp), "B": rng.normal(0.06, 0.09, n_imp)}
gen = {"A": rng.normal(0.55, 0.13, n_gen), "B": rng.normal(0.55, 0.13, n_gen)}

pooled = np.r_[imp["A"], imp["B"]]
t_global = np.quantile(pooled, 1 - 1e-3)          # FMR = 0.1% on the pooled data
print(f"global threshold {t_global:.3f}")
for g in "AB":
    print(f"  group {g}: FMR {(imp[g] >= t_global).mean():.5f}, "
          f"FNMR {(gen[g] < t_global).mean():.4f}")
for g in "AB":
    t_g = np.quantile(imp[g], 1 - 1e-3)             # per-group threshold, FMR 0.1% each
    print(f"group {g} own threshold {t_g:.3f}: FNMR {(gen[g] < t_g).mean():.4f}")
```

```
global threshold 0.322
  group A: FMR 0.00018, FNMR 0.0403
  group B: FMR 0.00182, FNMR 0.0400
group A own threshold 0.279: FNMR 0.0210
group B own threshold 0.337: FNMR 0.0517
```

A shift of 0.06 in the mean impostor score, two-thirds of one standard deviation, makes group B's false match rate 10 times group A's at the same threshold (0.182% against 0.018%), although the two groups' false non-match rates are equal. Choosing a separate threshold per group equalizes FMR, but then group B's FNMR rises to 5.2% while A's falls to 2.1%: one error rate or the other differs. Which trade-off is acceptable is a policy and legal question, not a purely technical one, and group-specific thresholds raise their own concerns. The technical lessons are firmer: measure both error rates per group at the operating threshold, on data like the deployment data, and remember that the overall FAR can hide a tenfold difference (Figure 7).

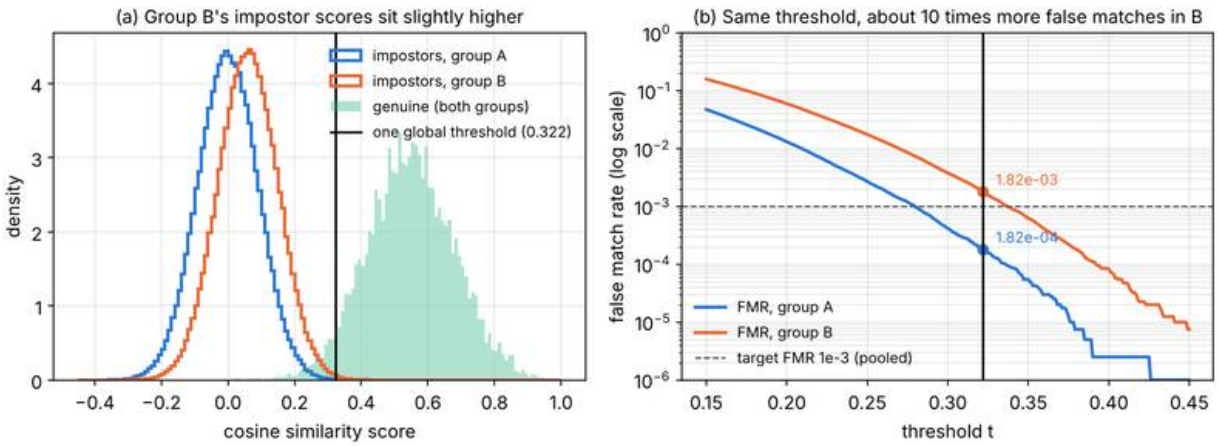


Figure 7 — A synthetic two-group example. (a) Group B's impostor distribution is shifted by 0.06. (b) At one global threshold, group B's false match rate is about 10 times group A's, although the pooled rate meets the 10^{-3} target.

Presentation attacks

Plain version. A recognition network answers "who is this?", not "is this a real, live face?". A printed photo, a phone screen showing a video, or a mask can be presented to the camera. Detecting such attacks is a separate task.

A **presentation attack** is an attempt to fool the sensor with an artifact; **presentation attack detection** (PAD), also called face anti-spoofing, decides whether the input is *bona fide* (a live person) or an attack. Yu et al. survey deep-learning methods for it [37]: beyond binary "live versus attack" labels, many methods use pixel-wise auxiliary supervision such as pseudo depth maps; the field now emphasizes **domain generalization** (attacks, cameras and lighting not seen in training) and **open-set** attacks of new types; and multi-modal sensors (depth, infrared) or special sensors (light field, flash) help. PAD runs alongside recognition, and the two error types are evaluated separately; a strong recognizer with no PAD is easy to fool with a good photo of the enrolled person. Digital attacks such as morphed passport photos and deepfakes are related but distinct threats.

Privacy, consent and the law

Plain version. A face embedding is a biometric identifier: it can link a person's photos across the internet and cameras, and a person cannot change their face the way they change a password.

Many legal systems therefore treat it as sensitive data. This section describes the context; it is not legal advice.

Two points from European law illustrate how regulators view face recognition (rules differ between jurisdictions, and they change):

- Under the EU's General Data Protection Regulation, "biometric data for the purpose of uniquely identifying a natural person" is one of the categories of **sensitive** personal data, which are subject to specific processing conditions (GDPR Article 9) [38].
- The EU AI Act (Regulation (EU) 2024/1689) lists among its **prohibited practices** the untargeted scraping of the internet or CCTV material to create or expand facial recognition databases, and real-time remote biometric identification for law enforcement purposes in publicly accessible spaces; the prohibitions apply from 2 February 2025 [39]. The Regulation itself defines the exact scope and conditions.

For engineers, the practical consequences are concrete. Know where every training image came from and under what terms; the withdrawn datasets above show that "publicly visible" is not the same as "available for any use". Treat stored templates as sensitive data: limit who can query a gallery, log searches, and delete templates that are no longer needed. Prefer designs that keep templates on the user's device for verification when the application allows it. Document the intended use and the measured error rates per group, as recommended by NIST [2]. Synthetic training data [31], [33] reduces, but does not remove, these concerns, because the deployed system still processes real faces.

Failure modes and practical advice

Plain version. Most failures come from a mismatch between the conditions in training and testing: a face turned away, half covered, much older, blurred, or from a population the system rarely saw. A threshold tuned in one setting silently changes meaning in another.

- **Pose.** Large yaw (profile views) breaks five-point alignment and removes half the face. IJB-B was built with human-labeled face boxes precisely because datasets filtered by a commodity face detector under-represent such poses and occlusions [27], and the ArcFace authors cite pose and age gaps as conditions where plain softmax features degrade [16].
- **Occlusion.** Masks, sunglasses, hands and hair hide parts of the face; detectors may still fire but landmarks drift, and the embedding moves toward the impostor region.
- **Age.** Templates enrolled years earlier match less well; MegaFace evaluated performance with respect to age [24], and NIST found elevated false positives for the elderly and children [2].
- **Image quality.** Low resolution, motion blur, compression and poor lighting raise false non-matches; NIST linked several false-negative differentials to image quality rather than to demographics [2]. Quality-aware methods (MagFace, AdaFace [19], [20]) and quality checks at

capture time help.

- **Threshold drift.** A threshold chosen on one dataset gives a different FAR on another camera, population or gallery size (Figures 6 and 7). Recalibrate on deployment-like impostor pairs, and monitor score distributions after deployment, as in ML 10.
- **Label noise and identity overlap.** Web-collected training sets contain mislabeled images and the same person under two names; sub-center ArcFace [18] and dataset cleaning [29] target this. Overlap between training identities and test identities inflates open-set results, so check it before reporting.
- **Evaluation shortcuts.** Reporting only accuracy on LFW, only rank-1 on a small closed gallery, or only one error rate hides exactly the failures above.

Modern view

What the surveys say

Wang and Deng's survey [1] is the standard map of deep face recognition. It sorts the losses into **Euclidean-distance-based** (contrastive, triplet, center loss), **angular/cosine-margin-based** (SphereFace, CosFace, ArcFace) and **softmax variations** (feature and weight normalization), and it traces how Euclidean-distance-based losses dominated the early years and how angular/cosine-margin losses and feature and weight normalization took over from 2017. Drozdowski et al. [36] survey demographic bias across biometrics: how bias is estimated, how it might be mitigated, and the social and technical questions that remain. Yu et al. [37] survey deep face anti-spoofing, including methods designed for domain generalization and open-set attacks. Boutros et al. [33] survey synthetic data for face recognition, organize its use cases and discuss the challenges that remain.

Optional: Guide to the Modern view surveys (in the Paper guides). What each survey covers, how it is organized and how to use it as a map.

The state of the art, 2023–2026

- **Margin-based softmax losses on very large datasets remain the standard recipe.** The main changes since ArcFace are adaptive margins tied to image quality (MagFace [19], AdaFace [20]), robustness to label noise (sub-center ArcFace [18]) and efficient training over millions of identities (Partial FC [21], applied with both CNN and ViT backbones).
- **Benchmarks moved to low-FAR, template-based and operational evaluation.** LFW is saturated [31]; IJB-B/C at TAR@FAR = 10^{-4} became the academic norm [16], and NIST's FRTE

(the evaluation known as FRVT until 2023 [3]) tests submitted algorithms on large operational data at false match rates as low as 10^{-6}.

- **Synthetic and privacy-aware training data** is an active direction, with graphics-rendered (DigiFace-1M [31]) and diffusion-generated (DCFace [32]) datasets, motivated by consent, bias and label-noise problems of web data [33].
- **Fairness is measured and reported**, not assumed: NIST's demographic analysis [2] and dedicated test sets such as RFW [34] made per-group error rates part of evaluation.

Open problems

1. **Generalization to unseen conditions**: extreme pose, occlusion, very low resolution and long age gaps.
2. **Demographic differentials**: measuring both error types per group on deployment-like data, and reducing differentials without simply moving errors from one group to another.
3. **Training data without consent problems** that still reaches the accuracy of web-scale real data [33].
4. **Presentation and digital attacks** of types not seen in training [37].
5. **Calibrated, quality-aware scores**, so that a threshold keeps its meaning across cameras and populations.
6. **Governance**: dataset stewardship after release [30], and legal limits that differ between jurisdictions [38], [39].

Key takeaways

- Face recognition is open-set: test identities are not training classes. The network learns an embedding; a threshold on cosine similarity makes the decision.
- Verification (1:1), closed-set identification (1:N) and open-set identification are different tasks with different metrics; always state which one and, for search, the gallery size.
- The pipeline is detect → five landmarks → similarity-transform alignment → normalized embedding → cosine comparison; errors early in the pipeline propagate.
- Pair-based losses (contrastive, triplet) need hard-negative mining because almost all pairs are easy; margin-based softmax losses compare each image with all class centers and need no mining.
- SphereFace, CosFace and ArcFace change only the target logit: $\cos(m\theta)$, $\cos\theta - m$ and $\cos(\theta + m)$ with scale s; ArcFace's margin is a constant geodesic gap of m radians on the hypersphere.
- Report TAR at a stated FAR (with the number of impostor pairs), ROC on a log axis or DET curves; for search, rank-1/CMC (closed set) and FNIR/FPIR (open set). FPIR grows with gallery size at a fixed threshold.

- Datasets carry history: LFW is saturated, IJB-B/C and NIST FRTE are the references, and MS-Celeb-1M and MegaFace are no longer distributed.
- One global threshold can give one group ten times the false match rate of another; measure both error types per group, and treat face templates as sensitive biometric data.

Exercises

1. **Open or closed?** For each system, say whether it performs verification, closed-set identification or open-set identification, and which error matters most: (a) unlocking a phone, (b) sorting a family's photo album into its five members, (c) searching a watch list of 10,000 people at a stadium entrance.

   Hint

   (a) Verification; a false accept is a security failure, a false reject is an inconvenience (the user can retry). (b) Closed-set identification only if every photo shows one of the five; real albums contain guests, so it is open-set in practice. (c) Open-set identification; most people passing are not on the list, so FPIR at the chosen threshold, multiplied by the number of people passing, gives the number of false alarms.

2. **The ArcFace boundary.** Two class centers are $\varphi = 60°$ apart. For a feature on the arc between them, find the angle from W_1 at which class 1's loss stops being satisfied for normalized softmax, ArcFace with $m = 0.5$ rad and SphereFace with $m = 4$. Which margin is largest here, and what happens to SphereFace's margin when $\varphi = 120°$?

   Hint

   Softmax: 30°. ArcFace: $(\varphi - m)/2 = (60° - 28.6°)/2 \approx 15.7°$, so the gap is 28.6°. SphereFace: $\varphi/(m+1) = 12°$, gap $\varphi(m-1)/(m+1) = 36°$. At $\varphi = 120°$ SphereFace's gap doubles to 72°, while ArcFace's stays 28.6°.

3. **CosFace in the toy.** Change the toy experiment so that the target logit is $s(\cos\theta - m)$ with $m = 0.35$ (CosFace). What is $\partial(\text{target logit})/\partial\cos\theta$ now? Run it and compare the three printed numbers with ArcFace's.

   Hint

   Replace `np.cos(th + margin)` with `np.cos(th) - margin` in the logits, and the gradient factor `np.sin(th + margin) / np.sin(th)` with 1 (the derivative of $\cos\theta - m$ with respect to $\cos\theta$ is 1, so the whole factor is just s). Expect a result close to ArcFace's; the exact numbers depend on the run.

4. **How many impostor pairs?** You want to claim FAR $= 10^{-5}$. (a) With 300,000 impostor pairs and zero false accepts observed, what is the approximate 95% upper bound on the true FAR?

(b) How many impostor pairs do you need for the bound to reach 10^{-5} with zero errors? (c) Why might your pairs be worth fewer "independent trials" than their count?

Hint

(a) Rule of three: about $3/300{,}000 = 10^{-5}$. (b) About 300,000, but only if no errors occur; with a few errors you need more. (c) Pairs that reuse the same images or the same people are correlated: one unusual image can produce many false accepts at once.

5. **Gallery growth.** In the identification code, a non-mated probe at threshold 0.36 triggers a false hit with probability 0.010 for a gallery of 300. Use FPIR $\approx 1 - (1 - \text{FMR})^N$ to estimate the per-comparison FMR, then predict FPIR for galleries of 3,000 and 30,000. Compare with the printed values and explain any difference.

Hint

$1 - (1 - \text{FMR})^{300} = 0.010$ gives FMR $\approx 3.4 \times 10^{-5}$, predicting about 0.096 for 3,000 and 0.64 for 30,000. The printed 0.043 and 0.330 are lower. With only 300 non-mated probes, an FPIR of 0.010 is just 3 false hits, so the FMR estimate is very uncertain; estimate FMR directly from many impostor comparisons instead.

6. **Fair thresholds.** In the two-group code, find the single threshold that equalizes the two groups' FNMR, and one that makes FMR equal. Can one threshold do both? What would you report to a decision maker?

Hint

Because the genuine distributions are identical, *any* single threshold gives equal FNMR (up to sampling noise: 0.0403 against 0.0400 in the printed run); no single threshold gives equal FMR, because the impostor distributions differ. Report both error rates for both groups at the proposed threshold, and the alternatives (group thresholds, a better model), and leave the choice of trade-off to the people accountable for it.

Supplementary notes

These notes are optional: each takes one point from the main text and treats it in more detail. Nothing in the main text, code or exercises depends on them.

Aligning a face: the least-squares similarity transform

This note derives the alignment fit used in the main text and lists what can go wrong with it.

The problem. Given n detected landmarks $\mathbf{p}_i = (u_i, v_i)$ and template positions $\mathbf{q}_i = (u'_i, v'_i)$, find a, b, t_x, t_y minimizing

$$E(a, b, t_x, t_y) = \sum_{i=1}^{n} \left[(au_i - bv_i + t_x - u'_i)^2 + (bu_i + av_i + t_y - v'_i)^2 \right].$$

Each landmark contributes two rows to a linear system $A\boldsymbol{\beta} = \mathbf{r}$ with $\boldsymbol{\beta} = (a, b, t_x, t_y)^\top$: the row $(u_i, -v_i, 1, 0)$ with right side u'_i, and the row $(v_i, u_i, 0, 1)$ with right side v'_i. The least-squares solution satisfies the normal equations $A^\top A\boldsymbol{\beta} = A^\top \mathbf{r}$, which `np.linalg.lstsq` solves stably.

A closed form. Center both point sets: $\bar{\mathbf{p}} = \frac{1}{n}\sum_i \mathbf{p}_i$, $\tilde{\mathbf{p}}_i = \mathbf{p}_i - \bar{\mathbf{p}}$, and likewise $\tilde{\mathbf{q}}_i$. Then the optimal a and b are

$$a = \frac{\sum_i \tilde{\mathbf{p}}_i^\top \tilde{\mathbf{q}}_i}{\sum_i \|\tilde{\mathbf{p}}_i\|^2}, \qquad b = \frac{\sum_i (\tilde{u}_i \tilde{v}'_i - \tilde{v}_i \tilde{u}'_i)}{\sum_i \|\tilde{\mathbf{p}}_i\|^2},$$

where $(\tilde{u}_i, \tilde{v}_i)$ and $(\tilde{u}'_i, \tilde{v}'_i)$ are the coordinates of $\tilde{\mathbf{p}}_i$ and $\tilde{\mathbf{q}}_i$, and the translation is $\mathbf{t} = \bar{\mathbf{q}} - \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \bar{\mathbf{p}}$. Scale and angle follow as $\sigma = \sqrt{a^2 + b^2}$ and $\phi = \operatorname{atan2}(b, a)$. The numerator of b is a sum of 2-D cross products, so $b = 0$ exactly when the best rotation is zero.

Why not a more flexible warp? DeepFace went the other way and used explicit 3D face modeling with a piecewise affine warp [6], but most later systems settled on the simple 2-D similarity transform. A full affine transform (six parameters) can also shear and stretch the face differently in x and y; a projective one (eight parameters) can do more. With only five noisy landmarks, these extra degrees of freedom mostly fit the noise and distort the face's proportions, which are part of what identifies a person. The similarity transform preserves angles and ratios of distances.

Edge cases.

- *Reflections.* The parameterization $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ cannot mirror the image, which is what we want; a mirrored face is a different image (though horizontal flipping is sometimes used as test-time augmentation).
- *Degenerate configurations.* If all landmarks are (nearly) the same point, $\sum_i \|\tilde{\mathbf{p}}_i\|^2 \approx 0$ and the scale is undefined; this happens for tiny faces or failed detections and should be rejected before alignment.
- *Profile faces.* For large yaw, one eye and mouth corner are hidden, and the detector's estimates for them are guesses. A five-point frontal template then pulls the face into an unnatural crop. Systems either use pose-specific templates or accept that alignment quality falls with pose.
- *One bad landmark.* Least squares is sensitive to an outlier: one landmark that is 20 pixels off moves the whole fit. Checking the residual (as the code prints) and rejecting fits with a large residual is a simple safeguard.

How large must the scale s be?

This note explains why normalized softmax needs the scale s and how CosFace bounds it from below.

Why a scale at all. With unit features and unit weights, every logit is a cosine in $[-1, 1]$. Even in the best case, the correct class has logit 1 and every other class has logit at least -1, so the softmax probability of the correct class is at most $e^1/(e^1 + (C-1)e^{-1})$. For $C = 10{,}000$ classes this is about 7×10^{-4}: the network cannot make the loss small however well it arranges the angles, and training stalls. NormFace identified this problem and showed that a scale parameter after normalization is necessary [11].

CosFace's lower bound. Suppose the learned features of each class lie around their class center, and let P_W be the desired expected minimum posterior probability of a class center (the probability the softmax assigns to class j for a feature exactly at its center W_j). CosFace shows that this requires [13]

$$s \geq \frac{C-1}{C} \ln \frac{(C-1)\,P_W}{1-P_W},$$

where C is the number of classes and ln the natural logarithm. The bound grows with C (logarithmically) and with the desired confidence P_W.

A worked example. For $C = 10{,}000$ and $P_W = 0.9$: $\frac{9{,}999}{10{,}000} \ln \frac{9{,}999 \times 0.9}{0.1} = \ln(89{,}991) \times 0.9999 \approx 11.4$. For $C = 100{,}000$ and $P_W = 0.99$ the bound is about 16.1. The value $s = 64$ used by CosFace and ArcFace [13], [16] is far above these bounds.

```
import numpy as np

def s_lower_bound(C, P):
    return (C - 1) / C * np.log((C - 1) * P / (1 - P))

for C, P in [(10_000, 0.9), (100_000, 0.99)]:
    print(f"C = {C:7,d}, P_W = {P}: s >= {s_lower_bound(C, P):.1f}")
```

```
C =  10,000, P_W = 0.9: s >= 11.4
C = 100,000, P_W = 0.99: s >= 16.1
```

Too small or too large. If s is below the bound, the loss cannot get close to zero, and even well-separated samples keep receiving gradients that push them around. If s is very large, the softmax becomes nearly a hard maximum: the probabilities saturate at 0 or 1 early, most samples stop contributing gradients, and training can become unstable. In practice s is a hyperparameter fixed for a given number of classes; the scale and the margin interact, because $s \cdot m$ sets how much the target logit is lowered.

Margin losses side by side: one formula and its edge cases

This note writes the three margins as one formula, derives their two-class boundaries, and gives the gradient used in the toy code.

One formula. The ArcFace journal paper writes SphereFace, ArcFace and CosFace as special cases of a **combined margin** target logit [16]

$$\psi(\theta) = \cos(m_1\theta + m_2) - m_3,$$

where m_1 is SphereFace's multiplicative angular margin, m_2 ArcFace's additive angular margin and m_3 CosFace's additive cosine margin. $(m_1, m_2, m_3) = (1, 0, 0)$ is normalized softmax; $(1, 0.5, 0)$ is ArcFace; $(1, 0, 0.35)$ is CosFace; and the paper also tries combinations of the three [16].

Two-class boundaries. Let the centers W_1 and W_2 be an angle φ apart and consider a feature on the arc between them, at angle θ_1 from W_1 and $\theta_2 = \varphi - \theta_1$ from W_2. Class 1 is satisfied when $\psi(\theta_1) > \cos\theta_2$.

- *ArcFace:* $\cos(\theta_1 + m) > \cos(\varphi - \theta_1)$. Both arguments lie in $[0, \pi]$ (when $\theta_1 + m \le \pi$), where cosine is decreasing, so the condition is $\theta_1 + m < \varphi - \theta_1$, that is $\theta_1 < (\varphi - m)/2$. By symmetry class 2 is satisfied when $\theta_1 > (\varphi + m)/2$, so the zone where neither is satisfied has width exactly m, independent of φ.
- *SphereFace* (on the first branch of ψ): $m\theta_1 < \varphi - \theta_1$, so $\theta_1 < \varphi/(m+1)$; the gap is $\varphi - 2\varphi/(m+1) = \varphi(m-1)/(m+1)$. Close centers get a small margin and distant centers a large one.
- *CosFace:* $\cos\theta_1 - \cos(\varphi - \theta_1) > m$. Using $\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$, this is $2\sin\frac{\varphi}{2}\sin(\frac{\varphi}{2} - \theta_1) > m$, so $\theta_1 < \frac{\varphi}{2} - \arcsin(\frac{m}{2\sin(\varphi/2)})$. The angular gap is $2\arcsin(\frac{m}{2\sin(\varphi/2)})$: larger for close centers, and undefined (no satisfiable region on the arc) if $2\sin(\varphi/2) < m$. For $\varphi = 80°$ and $m = 0.35$ it is about 31°, as in Figure 2.

The gradient of the ArcFace logit. Training uses the derivative of the target logit with respect to the cosine $c = \cos\theta$, because the cosine is what the network computes. With $\theta = \arccos c$,

$$\frac{\partial}{\partial c}\, s\cos(\theta + m) = s\,\sin(\theta + m)\,\frac{1}{\sin\theta},$$

since $\frac{d}{d\theta}\cos(\theta + m) = -\sin(\theta + m)$ and $\frac{d\theta}{dc} = -1/\sin\theta$. This is the factor `np.sin(th + margin) / np.sin(th)` in the toy code. It is larger than 1 whenever $\sin(\theta + m) > \sin\theta$, that is for $\theta < (\pi - m)/2$, so the margin *amplifies* the gradient on samples that are already close to their center, until the softmax probability saturates. Two edge cases follow. Near $\theta = 0$ the factor divides by $\sin\theta \approx 0$, so implementations clip the cosine away from ± 1 (the toy code clips at $1 - 10^{-7}$). For $\theta + m > \pi$ the factor becomes negative and the logit is no longer monotone, as noted in the main text; such samples need a fallback penalty.

Why easy samples stop contributing. The gradient with respect to the logits is $p_j - [j = y_i]$, where p_j is the softmax probability. When $s\,\psi(\theta_{y_i})$ is far above every $s\cos\theta_j$, $p_{y_i} \to 1$ and all these

differences go to zero. The margin moves the point where this happens: a sample must be inside its class's satisfied region by a clear amount before its gradient vanishes, which is exactly how margins create compact clusters.

Beyond one center per class: sub-centers, quality-aware margins and Partial FC

This note describes, at the level of the papers' own summaries, how four extensions change the basic margin loss and what each one is for.

Sub-center ArcFace [18], [16]. Each identity j gets K unit vectors $W_{j,1}, \dots, W_{j,K}$ instead of one. For a feature $\mathbf{x}$, the similarity to class j is the largest of its K cosines, $\cos\theta_j = \max_k \cos\angle(\mathbf{x}, W_{j,k})$, and the usual ArcFace loss is applied to these class-level cosines. A sample therefore only has to be near *any* of its class's sub-centers. In web-collected data, most clean images of a person gather around one dominant sub-center, while mislabeled or very unusual images form small non-dominant ones; the authors use this to automatically clean noisy training data [16]. Pitfall: with large K and little noise, the sub-centers of one person can drift apart and the class is no longer compact, which is why the journal version describes dropping non-dominant sub-centers after cleaning [16].

MagFace [19]. Instead of fixing the feature length, MagFace lets the magnitude $\|\mathbf{x}\|$ vary within a range, makes the angular margin depend on it and adds a regularizer on it, such that the magnitude increases monotonically with how likely the face is to be recognized. The magnitude then doubles as a **face-quality score**, which can be used to reject poor images before matching. The paper also describes an adaptive within-class structure: easy samples are pulled toward the class center and hard samples pushed away, which reduces overfitting to noisy, low-quality images [19].

AdaFace [20]. AdaFace starts from the observation that a margin loss emphasizes hard (misclassified) samples, but a hard sample may be hard because it is unrecognizable (very low quality) rather than informative. It makes the margin a function of image quality, approximated by the feature norm, so that the emphasis on hard samples is strong for good-quality images and weak for very poor ones [20]. Its experiments target low-quality benchmarks such as IJB-S and TinyFace as well as IJB-B and IJB-C [20].

Partial FC [21]. The last layer of a margin softmax holds one center per identity; with millions of identities it dominates memory and computation. Partial FC keeps all centers but, in each iteration, computes the loss only over the positive centers of the batch and a random subset of negative centers, and updates only those. This cuts computation and also reduces two side problems of huge web datasets: conflicts between classes that are actually the same person, and rare updates of tail classes [21]. Pitfall: the sampled subset changes the softmax denominator, so the sampling rate is a hyperparameter that affects results, not just speed.

Measuring very small error rates

This note explains how many comparisons are needed to support a claimed FAR, and why counting pairs overstates the evidence.

The rule of three. Suppose each of n impostor comparisons independently gives a false accept with probability p, and you observe zero false accepts. The probability of that outcome is $(1-p)^n$. The largest p that is still consistent with zero errors at the 95% level solves $(1-p)^n = 0.05$, so $n\ln(1-p) = \ln 0.05 \approx -3.0$, and for small p, $\ln(1-p) \approx -p$, giving

$$p_{95} \approx \frac{3}{n}.$$

So zero false accepts in 30,000 impostor pairs supports only "FAR below about 10^{-4}", and a claim of 10^{-6} needs at least three million impostor comparisons with no errors.

With some errors. If you observe $k > 0$ false accepts in n trials, the estimate is $\hat{p} = k/n$, and an exact (Clopper–Pearson) interval can be computed from the beta distribution, for example with `scipy.stats.beta.ppf`. The relative uncertainty is roughly $1/\sqrt{k}$: with $k = 49$ false accepts (as in the main text's FAR = 10^{-4} line) the estimate is good to about $\pm 30\%$ at the 95% level; with $k = 4$ it is good to about a factor of two.

Pairs are not independent. Impostor pairs are usually formed by comparing every image with every other image. If one person's image is unusual (blurred, half-occluded, or simply similar to many others), it takes part in thousands of pairs and can produce many false accepts at once. The effective number of independent trials is then closer to the number of *people* or *images* than to the number of pairs. Resampling by person (a bootstrap over identities, as in the bootstrap section of ML 10) gives more honest intervals than the binomial formula.

Before deep learning: eigenfaces and Fisherfaces

This note connects the chapter to the classical subspace methods for faces.

Eigenfaces. Turk and Pentland represented a face image as a long vector of pixel values and applied principal component analysis to a training set of aligned faces [40]. The leading principal components, viewed as images, are the "eigenfaces"; a face is described by its coordinates in the subspace they span, and recognition is nearest-neighbor matching in that subspace. The method is PCA exactly as in PR 05. Its weakness is the one PR 05 points out for PCA in general: the directions of largest variance in face images are often due to lighting and pose, not identity.

Fisherfaces. Belhumeur, Hespanha and Kriegman replaced PCA by Fisher's linear discriminant analysis, which finds directions that maximize between-class scatter relative to within-class scatter [41] (PR 05). Because the within-class scatter matrix of high-dimensional pixel vectors is singular when there are few images per person, they first reduce dimension with PCA, then apply LDA in the reduced space [41].

The link to this chapter. Both methods learn a linear embedding and compare faces by distance, which is the same structure as a deep face system. Fisherfaces already optimize a ratio of between-class to within-class spread, the same goal that margins pursue; deep networks replace the linear map by a learned nonlinear one and the scatter ratio by a margin softmax.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

ArcFace (Deng et al., CVPR 2019; TPAMI 2022)

The problem. By 2018 several losses added margins to the softmax (SphereFace, CosFace), with different geometric meanings and training difficulties. The authors wanted a margin with a clear geometric interpretation that is also easy to train at scale [15].

The idea. Normalize features and class weights, rescale by s, and add an angular margin m to the angle between a feature and its own class center: the target logit becomes $s\cos(\theta+m)$. Because the margin is added to the angle, it equals a geodesic distance on the hypersphere [15]. The journal version [16] extends the work with sub-center ArcFace for noisy web data, a combined-margin analysis that places SphereFace, CosFace and ArcFace in one formula, and further experiments (including inverting the model to synthesize faces).

What it showed, and how to read the tables. The experiments compare losses with the backbone and training data held fixed, written as [training set, network, loss] [16]. Read the tables in that spirit: compare rows that differ only in the loss; a change of training set (for example a cleaned versus a raw web set) moves results by more than many loss changes. The main test sets are LFW and similar pair-based sets, MegaFace, and IJB-B/IJB-C; for IJB the reported number is TAR at FAR $= 10^{-4}$ on template pairs, where a template's embedding is the mean of its images' embeddings [16]. The experimental setup (five RetinaFace landmarks, 112 × 112 crops, 512-D embeddings, $s = 64$, $m = 0.5$) is documented in the paper [16] and has become a common default.

Limits. The gains on LFW are tiny because LFW is saturated; the informative comparisons are at low FAR on IJB-B/C and on large galleries. Several training sets used in this line of work have since been withdrawn or changed [30], which makes exact reproduction difficult. The margin assumes labels are correct; with heavy label noise the plain loss suffers, which is what sub-centers address.

Why it matters here. ArcFace is the reference form of the margin-based softmax in this chapter, and its geometric argument (constant geodesic margin) is the one Figure 2 illustrates.

NIST FRVT Part 3: Demographic Effects (Grother, Ngan and Hanaoka, 2019)

The problem. Public claims about bias in face recognition were often based on small studies, on different tasks (such as gender classification), or on a single error type. NIST set out to quantify demographic differentials of contemporary algorithms on large operational datasets [2].

The idea. Run many submitted one-to-one and one-to-many algorithms on four large U.S. government image collections with sex, age and race or country-of-birth metadata, and report false positive and false negative rates per group at fixed thresholds [2].

What it showed. The executive summary's main result is that false positive differentials were much larger than false negative ones, often by factors of 10 to beyond 100 across groups, and that the size and even the direction of effects depended on the algorithm and the dataset; image quality explained several false negative differences; and some accurate one-to-many algorithms showed undetectable false positive differentials [2].

How to read it. Always check three things for any number you quote: which error (FMR or FNMR, FPIR or FNIR), at which threshold, and on which dataset. The report gives results per algorithm in extensive annexes, so "face recognition is biased by X" is not a statement the report supports; "algorithm A on dataset D at threshold t has false match rates that differ by X between groups" is. The datasets are not public, so results cannot be reproduced outside NIST, and the algorithms were prototypes submitted at the time, not necessarily deployed products [2].

Limits. Country of birth is used as a proxy for race in three of the four datasets [2]; the report measures differentials but does not decide which differences are acceptable; and algorithms have changed since 2019, so current per-algorithm results should be taken from NIST's ongoing evaluations [3].

Why it matters here. It is the basis for the fairness section's main claims and for the advice to report both error types per group at the operating threshold.

Guide to the Modern view surveys

This guide expands the short survey summary in the Modern view: what each review covers, how it is organized and how to use it.

Deep face recognition: A survey (Wang and Deng, *Neurocomputing*, 2021) [1]. Coverage: architectures, loss functions, face processing, datasets and protocols, and application scenes. Organization: losses in three families (Euclidean-distance-based, angular/cosine-margin-based, softmax variations); face processing in two ("one-to-many augmentation", which generates many images with pose variation from one image, and "many-to-one normalization", which recovers a canonical frontal view); scenes such as cross-factor, heterogeneous, multiple-media and industrial recognition. Use it as a map when you meet an unfamiliar loss or benchmark: find its family,

then read the original paper.

Demographic bias in biometrics: A survey on an emerging challenge (Drozdowski et al., *IEEE Transactions on Technology and Society*, 2020) [36]. Coverage: algorithmic bias in biometrics generally, the literature on estimating and mitigating it, and the related technical and social issues. Use it to see how fairness is measured in different studies, which makes it easier to compare their claims.

Deep learning for face anti-spoofing: A survey (Yu et al., *IEEE TPAMI*, 2023) [37]. Coverage: deep PAD with binary and pixel-wise supervision, domain generalization and open-set protocols, and multi-modal or specialized sensors. Use it to understand what the PAD benchmarks test and which protocols are hardest.

Synthetic data for face recognition: Current state and future prospects (Boutros et al., *Image and Vision Computing*, 2023) [33]. Coverage: a taxonomy of use cases of synthetic face data, recent recognition models trained on it, and the remaining challenges. Use it to place DigiFace-1M, DCFace and later synthetic datasets.

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DLCV 03 • Object Detection: From R-CNN to DETR

How deep networks find and label every object in an image: box regression, IoU, anchors, NMS and feature pyramids; two-stage, one-stage, anchor-free and transformer detectors; open-vocabulary models; and how detection is measured.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-03-object-detection/

What you'll learn

- What a detector outputs (boxes, class labels, confidences), and how detection differs from classification, semantic segmentation and instance segmentation.
- The building blocks every modern detector shares, in math and in code: box parameterization and regression targets, IoU and its GIoU/DIoU/CIoU variants, anchors and label assignment, non-maximum suppression, feature pyramids and the focal loss.
- How the four big families work and why each was invented: two-stage (R-CNN to Cascade R-CNN), one-stage (YOLO, SSD, RetinaNet), anchor-free (CornerNet, CenterNet, FCOS) and set prediction with transformers (DETR to RT-DETR).
- How open-vocabulary and grounding detectors (OWL-ViT, Grounding DINO, YOLO-World) and promptable foundation models change what "the class list" means.
- How detection is evaluated (precision–recall, AP, VOC and COCO protocols, AP50/AP75/APs/m/l), which datasets matter, and what limits speed in deployment.

The big picture

A classifier answers one question about a whole image: *what is this?* A detector answers a harder one: *what is where?* It must find every object of interest, draw a tight rectangle (a **bounding box**) around each, name it, and say how sure it is. The number of objects is not known in advance: there may be zero, one or three hundred.

> **Explain like I'm five**
>
> Imagine a "Where's Waldo?" page. A **classifier** only tells you "this page has Waldo on it." A **detector** puts a sticker on Waldo, another sticker on every dog and every balloon, and writes on each sticker what it is and how sure it is ("dog, pretty sure"). The tricky parts are the ones you would also find tricky: tiny things far away, things hiding behind other things, and not putting two stickers on the same dog.

Detection is the entry point of most "real" vision systems. Self-driving cars detect pedestrians and vehicles; phones detect faces before focusing; factories detect defects; trackers (see the tracking

chapter linked at the end) mostly *link detections over time*. The field also produced many ideas that now appear everywhere in deep learning: multi-task losses, feature pyramids, the focal loss, and set prediction with bipartite matching.

This tutorial assumes you know CNNs and image classification (DLCV 02) and the classical feature ideas of IP 12. Two surveys frame the whole story: Zou et al., *Object Detection in 20 Years* [1], and Liu et al.'s IJCV survey of deep generic detection [2]. We return to both in the Modern view.

Related chapters. DLCV 03A follows the YOLO family and real-time DETRs in depth, and DLCV 03B moves from boxes to pixel masks (semantic, instance and panoptic segmentation).

The detection problem

Plain version. The input is an image. The output is a short list of (box, class, score) triples. The training data is a list of human-drawn boxes with class names.

Precise version. Given an image $I \in \mathbb{R}^{H\times W\times 3}$, a detector outputs a set

$$\hat{Y} = \{(\hat{b}_i, \hat{c}_i, \hat{s}_i)\}_{i=1}^{N},$$

where $\hat{b}_i \in \mathbb{R}^4$ is a box, $\hat{c}_i \in \{1, \dots, K\}$ is one of K classes, $\hat{s}_i \in [0, 1]$ is a confidence score, and N varies from image to image. The ground truth is a set $Y = \{(b_j, c_j)\}_{j=1}^{M}$ of M annotated objects.

Two box formats are common: corners (x_1, y_1, x_2, y_2) and center–size (c_x, c_y, w, h). Boxes are **axis-aligned**; rotated boxes are a separate variant used for aerial images and text.

Three properties make detection harder than classification:

1. **Variable-size output.** The network must decide *how many* things to output. Every architecture in this tutorial is, at heart, a different answer to this problem.
2. **Localization and recognition at once.** The model must be invariant to *what* changes about an object's appearance but sensitive to *where* it is, a tension that shapes the design of every head.
3. **Extreme imbalance.** A typical image has a handful of objects but tens of thousands of candidate positions and sizes, nearly all background.

Detection versus its neighbours

Figure 1 compares the four classic recognition tasks on one synthetic scene. **Classification** gives one label per image. **Detection** gives a box and a label per object. **Semantic segmentation** labels every pixel with a class, so two touching balls merge into one blob. **Instance segmentation** gives

each object its own pixel mask, which is detection plus a mask per box; Mask R-CNN [16] is the standard way to get it, as we will see.

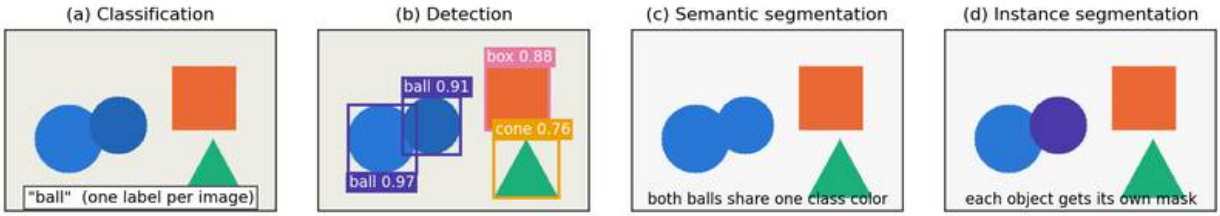


Figure 1 — The same scene seen by four tasks: (a) classification, (b) detection, (c) semantic segmentation, (d) instance segmentation.

Classical detection: sliding windows and hand-made features

Plain version. Before deep learning, detection was "classification in a loop": cut out a window at every position and size, ask a classifier whether it contains the object, and keep the windows that say yes.

Sliding windows. For a window of fixed size $w \times h$ slid with stride s over an image pyramid with scale factor λ, the number of windows is roughly

$$N_{\text{win}} \approx \sum_{k \geq 0} \frac{(H\lambda^{-k} - h)(W\lambda^{-k} - w)}{s^2},$$

where k indexes the pyramid levels. For a 640×480 image, a 64×128 window, $s = 8$ and $\lambda = 1.2$, that is on the order of 10^4 windows for one aspect ratio, all of which must be classified. This is the template matching of IP 13 scaled up, and it has the same weaknesses: one template shape, many scales, and a lot of wasted work on background.

Three classical detectors show how far hand-made features went. **Viola–Jones** (2001) [5] made face detection real-time: simple rectangle features are computed in constant time, and a cascade of ever stricter classifiers throws away most background windows after a few cheap tests. **HOG + linear SVM** (2005) [6] described each window by histograms of gradient orientations, which made pedestrian detection practical. **Deformable part models** (DPM) [7] added a few movable part filters around a coarse whole-object filter, and led the PASCAL VOC benchmark [3] until 2012.

Optional: Classical detectors in detail: Viola–Jones, HOG and DPM (in the Supplementary notes). The integral image, the HOG descriptor and the DPM scoring formula.

Why this era ended. Classical detectors were *slow* (an exhaustive scan over positions, scales and aspect ratios, each scored by a hand-designed pipeline) and *brittle* (one hand-designed feature

type, a few rigid templates, and poor transfer to new categories, viewpoints and clutter). They also had no way to share features across classes. Deep detectors kept the useful ideas, such as scanning in a feature pyramid, cascades, parts and hard-negative mining, and replaced the hand-made features with learned ones.

Core building blocks

Every detector in the rest of this tutorial is assembled from the same parts. We define each one carefully, with code you can run.

Box parameterization and regression targets

Plain version. A network is bad at outputting absolute pixel coordinates, but good at outputting small corrections. So we start from a **reference box** (a proposal, an anchor or a point) and predict how to nudge it.

Given a reference box $P = (P_x, P_y, P_w, P_h)$ in center–size form and its matched ground-truth box $G = (G_x, G_y, G_w, G_h)$, the standard targets introduced with R-CNN [8] are

$$t_x = \frac{G_x - P_x}{P_w}, \qquad t_y = \frac{G_y - P_y}{P_h}, \qquad t_w = \log \frac{G_w}{P_w}, \qquad t_h = \log \frac{G_h}{P_h}.$$

Dividing the shift by the reference size makes the target **scale-invariant** (a 10-pixel error matters more for a 20-pixel box than for a 400-pixel box). The **log** for width and height makes scaling symmetric (doubling and halving are $+\log 2$ and $-\log 2$) and keeps the decoded size positive: $\hat{G}_w = P_w \exp(\hat{t}_w)$. In practice the targets are often divided by fixed standard deviations, such as $(0.1, 0.1, 0.2, 0.2)$, so that all four have similar magnitude.

```
import numpy as np

def to_cxcywh(b):
    return np.stack([(b[:, 0] + b[:, 2]) / 2, (b[:, 1] + b[:, 3]) / 2,
                     b[:, 2] - b[:, 0], b[:, 3] - b[:, 1]], axis=1)

def to_xyxy(c):
    return np.stack([c[:, 0] - c[:, 2] / 2, c[:, 1] - c[:, 3] / 2,
                     c[:, 0] + c[:, 2] / 2, c[:, 1] + c[:, 3] / 2], axis=1)

def encode(anchors, gt, std=(0.1, 0.1, 0.2, 0.2)):
    """Regression targets (tx, ty, tw, th) of gt relative to anchors (both xyxy)."""
    P, G = to_cxcywh(anchors), to_cxcywh(gt)
    t = np.stack([(G[:, 0] - P[:, 0]) / P[:, 2],
                  (G[:, 1] - P[:, 1]) / P[:, 3],
                  np.log(G[:, 2] / P[:, 2]),
                  np.log(G[:, 3] / P[:, 3])], axis=1)
```

```
    return t / np.array(std)          # normalize so all four have similar scale

def decode(anchors, t, std=(0.1, 0.1, 0.2, 0.2)):
    P = to_cxcywh(anchors)
    t = t * np.array(std)
    c = np.stack([P[:, 0] + t[:, 0] * P[:, 2],
                  P[:, 1] + t[:, 1] * P[:, 3],
                  P[:, 2] * np.exp(t[:, 2]),
                  P[:, 3] * np.exp(t[:, 3])], axis=1)
    return to_xyxy(c)

anchors = np.array([[0, 0, 32, 32], [100, 100, 228, 164]], float)
gt      = np.array([[4, 2, 40, 30], [90, 110, 250, 170]], float)
t = encode(anchors, gt)
print(np.round(t, 3))
print(np.allclose(decode(anchors, t), gt))   # round trip
```

```
[[ 1.875  0.     0.589 -0.668]
 [ 0.469  1.25   1.116 -0.323]]
True
```

These targets are usually trained with the **smooth L1** (Huber) loss of Fast R-CNN [9],

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5\,x^2 & \text{if } |x| < 1, \\ |x| - 0.5 & \text{otherwise,} \end{cases}$$

applied to each coordinate difference $x = \hat{t} - t$. It behaves like L2 for small errors (smooth gradients) and like L1 for large ones (robust to outliers early in training). Its weakness is that the four coordinates are treated as independent numbers, while the evaluation metric, IoU, treats the box as a whole. That mismatch motivated IoU-based losses.

IoU and its variants

Plain version. IoU asks: of all the area covered by either box, what fraction is covered by both? It is 1 for a perfect match and 0 for boxes that do not touch.

For two boxes A and B,

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|},$$

where $|\cdot|$ is area. IoU is scale-invariant, and it is the yardstick for evaluation (a detection counts as correct if its IoU with an unmatched ground-truth box exceeds a threshold), for label assignment and for NMS. Figure 2 shows how demanding the thresholds are: shifting a square by only about 18% of its side along both axes already drops IoU to 0.5.

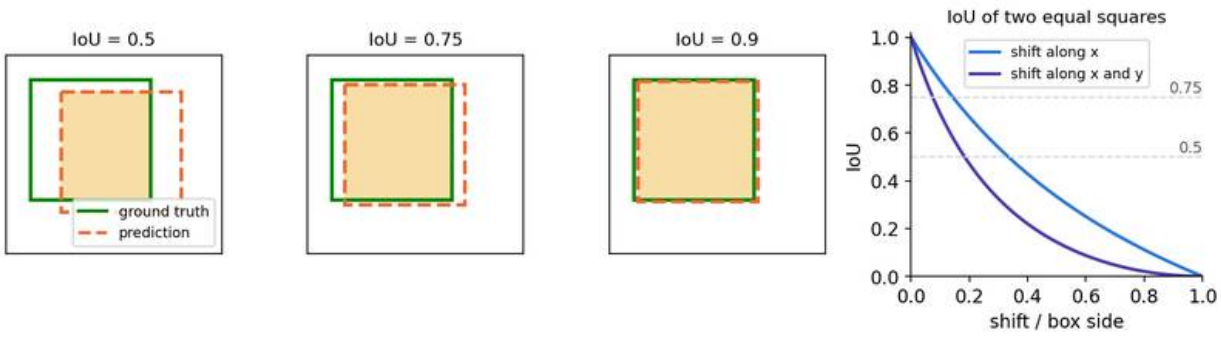


Figure 2 — What IoU thresholds look like. Left: a prediction at IoU 0.5, 0.75 and 0.9 with the ground truth. Right: IoU of two equal squares as one is shifted along x, or along x and y.

As a loss, $1 - \text{IoU}$ has a flaw: when two boxes do not overlap, IoU is 0 no matter how far apart they are, so the gradient is zero. Three fixes are widely used:

- **GIoU** [11] adds a penalty based on the smallest box C that encloses both: $\text{GIoU} = \text{IoU} - \frac{|C \setminus (A \cup B)|}{|C|}$, which lies in $(-1, 1]$ and keeps decreasing as non-overlapping boxes drift apart.
- **DIoU** [12] penalizes the distance between centers directly: $\text{DIoU} = \text{IoU} - \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2}$, where ρ is the Euclidean distance between the box centers $\mathbf{b}$ and $\mathbf{b}^{gt}$, and c is the diagonal length of C. It converges faster than GIoU because it pulls the centers together even when one box contains the other, where the GIoU penalty vanishes.
- **CIoU** [12] adds an aspect-ratio term: $\text{CIoU} = \text{DIoU} - \alpha v$, with $v = \frac{4}{\pi^2}\left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w}{h}\right)^2$ measuring aspect-ratio mismatch and $\alpha = \frac{v}{(1-\text{IoU})+v}$ a trade-off weight.

The corresponding losses are $1 - \text{GIoU}$, $1 - \text{DIoU}$ and $1 - \text{CIoU}$.

```
import numpy as np

def iou_family(a, b, eps=1e-9):
    """a, b: (N, 4) boxes as (x1, y1, x2, y2). Returns IoU, GIoU, DIoU, CIoU."""
    ix1, iy1 = np.maximum(a[:, 0], b[:, 0]), np.maximum(a[:, 1], b[:, 1])
    ix2, iy2 = np.minimum(a[:, 2], b[:, 2]), np.minimum(a[:, 3], b[:, 3])
    inter = np.clip(ix2 - ix1, 0, None) * np.clip(iy2 - iy1, 0, None)
    wa, ha = a[:, 2] - a[:, 0], a[:, 3] - a[:, 1]
    wb, hb = b[:, 2] - b[:, 0], b[:, 3] - b[:, 1]
    union = wa * ha + wb * hb - inter
    iou = inter / (union + eps)
    # smallest enclosing box C
    cx1, cy1 = np.minimum(a[:, 0], b[:, 0]), np.minimum(a[:, 1], b[:, 1])
    cx2, cy2 = np.maximum(a[:, 2], b[:, 2]), np.maximum(a[:, 3], b[:, 3])
    area_c = (cx2 - cx1) * (cy2 - cy1)
    giou = iou - (area_c - union) / (area_c + eps)
    # center distance over enclosing-box diagonal
    rho2 = ((a[:, 0] + a[:, 2]) - (b[:, 0] + b[:, 2])) ** 2 / 4 \
```

```
        + ((a[:, 1] + a[:, 3]) - (b[:, 1] + b[:, 3])) ** 2 / 4
    c2 = (cx2 - cx1) ** 2 + (cy2 - cy1) ** 2
    diou = iou - rho2 / (c2 + eps)
    v = 4 / np.pi**2 * (np.arctan(wb / hb) - np.arctan(wa / ha)) ** 2
    alpha = v / (1 - iou + v + eps)
    ciou = diou - alpha * v
    return iou, giou, diou, ciou

gt = np.array([[10, 10, 50, 50]] * 3, float)
pred = np.array([[20, 20, 60, 60],    # overlapping, shifted
                 [60, 10, 100, 50],   # disjoint, nearby
                 [200, 10, 240, 50]], float)  # disjoint, far away
for name, v in zip(["IoU", "GIoU", "DIoU", "CIoU"], iou_family(pred, gt)):
    print(f"{name:5s}", np.round(v, 3))
```

```
IoU   [0.391 0.    0.   ]
GIoU  [ 0.311 -0.111 -0.652]
DIoU  [ 0.351 -0.258 -0.662]
CIoU  [ 0.351 -0.258 -0.662]
```

The two disjoint predictions both have IoU 0, but GIoU and DIoU correctly say that the far one is worse. CIoU equals DIoU here because all boxes are squares, so there is no aspect-ratio mismatch.

Anchors and label assignment

Plain version. Instead of searching freely, a dense detector places a fixed set of reference boxes, **anchors**, at every cell of a feature map, and each anchor asks: "is there an object roughly like me here, and how should I move to fit it?"

At a feature map with stride s (each cell covers $s \times s$ input pixels), anchors are centered at $((i+\frac{1}{2})s, (j+\frac{1}{2})s)$ with a few scales σ and aspect ratios $r = h/w$, giving widths $w = \sigma/\sqrt{r}$ and heights $h = \sigma\sqrt{r}$ so that the area stays σ^2. Faster R-CNN uses 3 scales × 3 ratios = 9 anchors per location [10] (Figure 3, left).

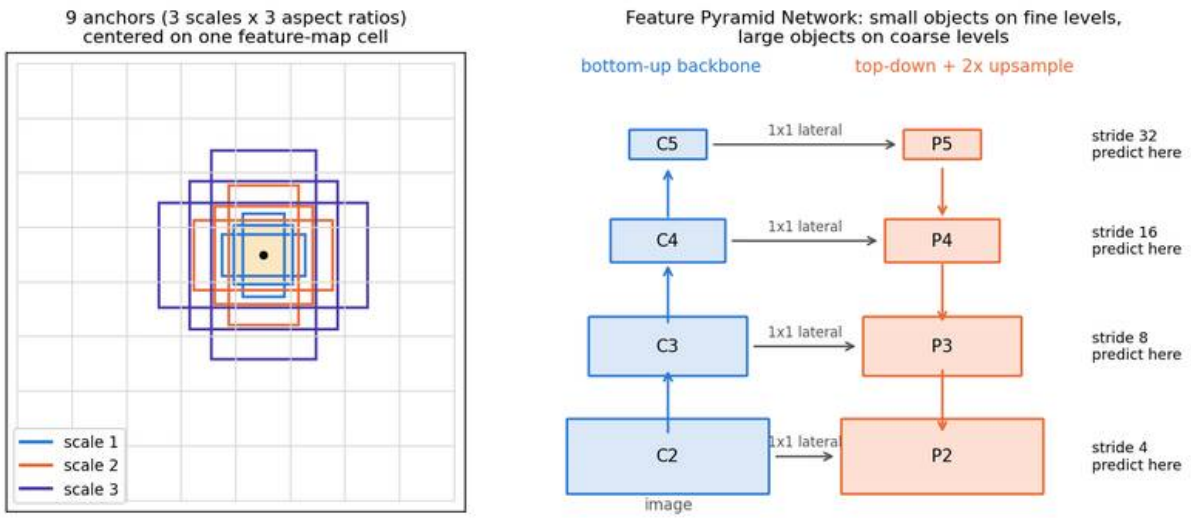


Figure 3 — Left: nine anchors (3 scales × 3 aspect ratios) at one feature-map cell. Right: a Feature Pyramid Network; each level predicts objects of a matching size range.

Label assignment decides which anchors are trained as which object. The classic rule is IoU-based: in the region proposal network of Faster R-CNN an anchor is *positive* if its IoU with some ground-truth box is at least 0.7 (or if it is the best anchor for that box), *negative* if its IoU with every box is below 0.3, and ignored otherwise [10]; RetinaNet uses 0.5 and 0.4 [15]. Each positive anchor regresses toward its matched box with the targets above. Assignment turned out to matter as much as architecture: anchor-free detectors (FCOS [29]) assign by *location* and object size instead, and DETR [30] replaces all fixed rules by a learned one-to-one matching.

Anchors have real costs: they introduce hyperparameters (scales, ratios, thresholds) that must be tuned per dataset, and they multiply the number of predictions. YOLO9000 [19] chose anchor shapes by running k-means on the training boxes, using $1 - \text{IoU}$ as the distance.

Optional: Anchor shapes by k-means (in the Supplementary notes). The clustering loop, why $1 - IoU$ replaces Euclidean distance, and a small run.

Non-maximum suppression (NMS) and Soft-NMS

Plain version. A dense detector fires many times on the same object (neighboring anchors all see it). NMS keeps the most confident box and deletes the others that overlap it too much.

Greedy NMS, per class: sort the boxes by score; repeatedly take the highest-scoring remaining box M, keep it, and remove every box b_i with $\text{IoU}(M, b_i) > N_t$ (typically $N_t = 0.5$ to 0.7). The weakness is crowded scenes: two real people standing close may overlap by more than N_t, and the lower-scoring one is deleted, losing recall.

Soft-NMS [13] replaces deletion with a score decay. With the Gaussian variant,

$$s_i \leftarrow s_i \exp\!\left(-\frac{\mathrm{IoU}(M, b_i)^2}{\sigma}\right),$$

where s_i is the score of box b_i and σ controls how fast overlapping boxes are penalized. A heavy overlap still pushes the score down a lot, but a genuine neighbor survives with a reduced score instead of disappearing.

```
import numpy as np

def iou_one_to_many(box, boxes):
    x1 = np.maximum(box[0], boxes[:, 0]); y1 = np.maximum(box[1], boxes[:, 1])
    x2 = np.minimum(box[2], boxes[:, 2]); y2 = np.minimum(box[3], boxes[:, 3])
    inter = np.clip(x2 - x1, 0, None) * np.clip(y2 - y1, 0, None)
    area = lambda b: (b[..., 2] - b[..., 0]) * (b[..., 3] - b[..., 1])
    return inter / (area(box) + area(boxes) - inter)

def nms(boxes, scores, thr=0.5):
    order = np.argsort(-scores)
    keep = []
    while order.size:
        i = order[0]
        keep.append(i)
        ious = iou_one_to_many(boxes[i], boxes[order[1:]])
        order = order[1:][ious <= thr]          # drop everything that overlaps too much
    return np.array(keep)

def soft_nms(boxes, scores, sigma=0.5, score_thr=0.001):
    scores = scores.copy()
    idx = np.arange(len(scores))
    keep = []
    while idx.size:
        j = np.argmax(scores[idx]); i = idx[j]
        keep.append(i)
        idx = np.delete(idx, j)
        ious = iou_one_to_many(boxes[i], boxes[idx])
        scores[idx] *= np.exp(-ious**2 / sigma)  # decay instead of delete
        idx = idx[scores[idx] > score_thr]
    return np.array(keep), scores

# two people standing close together + duplicates
boxes = np.array([[10, 10, 60, 110], [14, 12, 62, 112], [8, 6, 58, 104],
                  [40, 15, 90, 115], [43, 18, 92, 118]], float)
scores = np.array([0.95, 0.90, 0.80, 0.85, 0.70])
print("greedy NMS keeps:", nms(boxes, scores, 0.5))
keep, s = soft_nms(boxes, scores)
print("Soft-NMS order:", keep, "rescored:", np.round(s[keep], 3))
```

```
greedy NMS keeps: [0 3]
Soft-NMS order: [0 3 2 4 1] rescored: [0.95  0.761 0.183 0.145 0.058]
```

Greedy NMS keeps one box per person. Soft-NMS keeps everything but ranks the duplicates far

below the two real detections; a final score threshold or the AP computation then treats them as low-confidence guesses. Figure 4 shows the same effect on a synthetic scene.

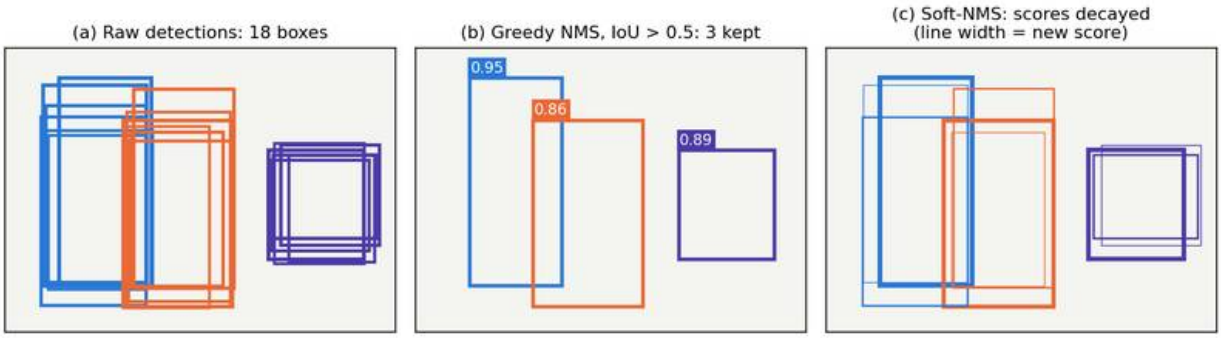


Figure 4 — NMS on synthetic detections. (a) 18 raw boxes around three objects. (b) Greedy NMS at IoU 0.5 keeps one box per object. (c) Soft-NMS keeps all boxes but decays the scores of overlapping ones (thinner line = lower score).

NMS has three further problems: it has a hand-tuned threshold; it is a sequential, data-dependent step that is awkward to run on accelerators; and it is not part of training, so the network is never taught to avoid duplicates. Removing it is one of the main motivations behind DETR and YOLOv10 [24].

Feature pyramids (FPN)

Plain version. Small objects need high-resolution feature maps; large objects need features that have seen a lot of context. A feature pyramid gives you both, at every scale.

A CNN backbone naturally produces a pyramid $\{C_2, C_3, C_4, C_5\}$ with strides $4, 8, 16, 32$, but its shallow levels are semantically weak. The **Feature Pyramid Network** [14] adds a top-down path: starting from the coarsest level, it upsamples by 2 and adds a 1×1-projected lateral connection from the backbone,

$$P_l = \mathrm{Conv}_{3\times3}\big(\mathrm{Conv}_{1\times1}(C_l) + \mathrm{Up}_{2\times}(P_{l+1})\big),$$

so every output level P_l has the same channel width and strong semantics. In a two-stage detector with FPN, a region of interest of size $w \times h$ is pooled from level $l = \lfloor l_0 + \log_2(\sqrt{wh}/224) \rfloor$ (with $l_0 = 4$ in [14]), so a region twice as large moves one level up; dense detectors use a similar size-to-level rule. Later necks (extra bottom-up paths, weighted fusion) refine the same idea; nearly every detector after 2017 uses one (Figure 3, right).

Focal loss and class imbalance

Plain version. If 99.9% of your candidates are background, a normal loss is dominated by millions of easy "nothing here" examples. The focal loss turns down the volume on examples the model

already gets right.

Let $p \in [0,1]$ be the predicted probability of the foreground class and $y \in \{0,1\}$ the label. Define $p_t = p$ if $y = 1$ and $p_t = 1 - p$ otherwise. The **focal loss** [15] is

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t),$$

where $\gamma \geq 0$ is the focusing parameter and α_t a class-balancing weight (α for foreground, $1-\alpha$ for background). With $\gamma = 0$ this is weighted cross-entropy. With the paper's default $\gamma = 2, \alpha = 0.25$, an easy negative with $p_t = 0.99$ is down-weighted by $(0.01)^2 = 10^{-4}$ relative to cross-entropy, while hard examples keep almost their full weight. The loss is normalized by the number of positive anchors. Two-stage detectors avoid the problem differently (the proposal stage removes most background, then a fixed foreground:background sampling ratio), and SSD [26] uses **hard negative mining** (keeping only the highest-loss negatives at about 3:1). The focal loss made a *dense* one-stage detector as accurate as two-stage ones for the first time.

Two-stage detectors: the R-CNN family

Plain version. First propose a few thousand places where *something* might be; then look closely at each one and decide *what* it is and exactly where its edges are.

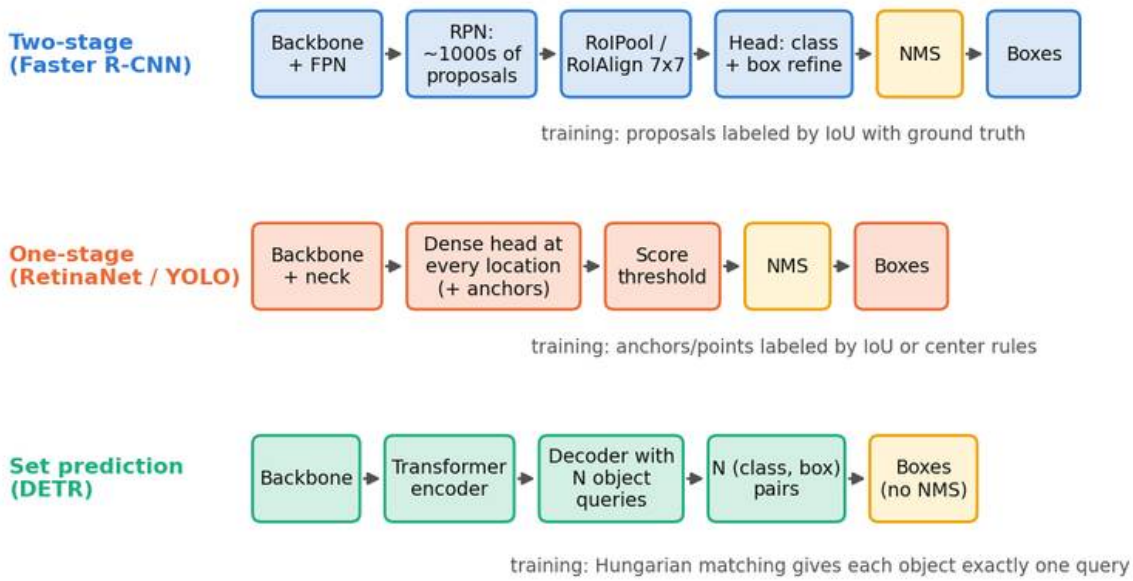


Figure 5 — Three detector pipelines. NMS (yellow) is a post-processing step for two-stage and one-stage detectors; DETR-style set prediction removes it by learning one-to-one assignments.

R-CNN (2014) [8] ran a bottom-up grouping algorithm to get about 2,000 class-agnostic region proposals per image, warped each one to a fixed size, passed it through a classification-pretrained CNN, scored the features with per-class linear SVMs and refined the box with the regression targets above. It improved mean AP on PASCAL VOC 2012 by more than 30% relative to the previous best result, and it established the recipe "pretrain on classification, fine-tune on detection" that is still standard. It was very slow, because the CNN ran separately on every proposal.

Fast R-CNN (2015) [9] runs the CNN *once* on the whole image and then crops each proposal from the shared feature map with **RoI pooling**: the proposal's projection onto the feature map is divided into a fixed $H \times W$ grid (e.g. 7×7) and each bin is max-pooled, so every proposal yields the same-size feature regardless of its shape. A single network is trained with a multi-task loss,

$$L = L_{\text{cls}}(p, u) + \lambda\,[u \geq 1]\,L_{\text{loc}}(t^u, v),$$

where p is the softmax over $K + 1$ classes, u the true class (0 for background), $[u \geq 1]$ is 1 for foreground proposals only, t^u the predicted box offsets for class u, v the target offsets and L_{loc} the smooth L1 loss.

Faster R-CNN (2015) [10] made the proposals themselves learned. A small **Region Proposal Network** (RPN) slides over the shared feature map and, for each of the 9 anchors per location, predicts an objectness score and box offsets. The top proposals (after NMS) feed the Fast R-CNN head. Because both stages share the backbone, proposals become almost free, and the whole detector is one network trained end to end.

Mask R-CNN (2017) [16] adds a small fully convolutional branch that predicts a binary mask per class inside each RoI, giving instance segmentation. Its key detail is **RoIAlign**: RoI pooling rounds the proposal boundaries and bin edges to the feature grid, which shifts features by up to half a stride. RoIAlign keeps real-valued coordinates and samples each bin with bilinear interpolation. This sounds minor, but pixel-accurate masks need it, and box accuracy at high IoU improves too.

Optional: RoI pooling versus RoIAlign (in the Supplementary notes). The bilinear sampling formula and a worked example of the rounding error.

Cascade R-CNN (2018) [17] addresses a subtle problem. A head trained with positives at IoU ≥ 0.5 learns to produce boxes that are "good at 0.5" but not at 0.75. Simply raising the threshold leaves too few positives and overfits. Cascade R-CNN chains several heads trained at increasing IoU thresholds (0.5, 0.6, 0.7), each refining the boxes of the previous one, so each stage receives a training distribution of higher quality. It was a large gain on the strict COCO metrics.

Two-stage detectors are accurate, flexible (the RoI head can predict masks, keypoints or attributes) and still the default for many instance-segmentation tasks, but the per-RoI head and NMS cost time.

One-stage detectors: YOLO, SSD and RetinaNet

Plain version. Skip the proposal step. Make one dense prediction at every location of the feature map in a single forward pass, then clean it up with NMS.

YOLO v1 (2016) [18] divided the image into an $S \times S$ grid ($S = 7$). Each cell predicted $B = 2$ boxes, each with a confidence, plus one set of C class probabilities, so the output was a single $S \times S \times (5B + C)$ tensor ($7 \times 7 \times 30$ for PASCAL VOC's 20 classes). The cell containing an object's center was responsible for it. It ran in real time and, because it saw the whole image, made fewer background errors than Fast R-CNN, but it localized less precisely and struggled with small objects and groups, since each cell could only predict a fixed number of boxes and one class.

The **YOLO family** then grew through many versions [19]–[25]. The core idea stayed the same, one dense pass over a grid followed by NMS, while each version improved one part: anchors and predictions at several scales, stronger backbones and training tricks, better label assignment, attention, and finally training that no longer needs NMS (YOLOv10 [24]).

Optional: The YOLO family, version by version (in the Supplementary notes). A table of the versions with a verifiable paper and the main idea of each.

SSD (2016) [26] attached small convolutional predictors to *several* feature maps of different resolutions, each with its own set of default boxes (anchors) of several aspect ratios, which handled scale far better than YOLO v1. It trained with hard negative mining and heavy data augmentation.

RetinaNet (2017) [15] combined an FPN backbone, dense anchors (9 per location) and two small subnetworks for classification and box regression, trained with the focal loss. The paper's analysis showed that class imbalance, not the architecture, had been the main reason one-stage detectors lagged behind two-stage ones.

Anchor-free detectors: points instead of boxes

Plain version. Anchors are guesses about what objects look like. Anchor-free detectors drop them and predict boxes from *points*: corners, centers, or every pixel inside the object.

CornerNet (2018) [27] detects an object as a pair of keypoints: a heatmap for top-left corners, a heatmap for bottom-right corners, and an **embedding** per corner. Corners from the same object are trained to have similar embeddings, so pairing is done by embedding distance. Because a corner often lies outside the object, the paper introduced **corner pooling**, which looks along rows and columns to collect evidence for a corner.

CenterNet, "Objects as Points" (2019) [28] represents each object by its center. The network

outputs a heatmap $\hat{Y} \in [0,1]^{\frac{W}{R}\times\frac{H}{R}\times K}$ (output stride R, one channel per class), plus a size (w,h) and a sub-pixel offset at each location. Ground-truth centers are splatted with a Gaussian, and the heatmap is trained with a pixel-wise focal loss. At inference, peaks are simply local maxima, found with a 3×3 max-pooling comparison, so **no IoU-based NMS** is needed. The same design extends to 3D boxes and human pose by regressing more quantities at the center.

FCOS (2019) [29] treats detection like segmentation. Every feature-map location that falls inside a ground-truth box is a positive sample and regresses the four distances to the box sides,

$$l^* = x - x_1, \qquad t^* = y - y_1, \qquad r^* = x_2 - x, \qquad b^* = y_2 - y,$$

where (x,y) is the location in image coordinates and (x_1,y_1,x_2,y_2) the box. Each FPN level only handles objects whose largest regression distance falls in its own range, which resolves most ambiguity when boxes overlap. Locations far from the object's center produce poor boxes, so FCOS adds a **centerness** branch,

$$\text{centerness} = \sqrt{\frac{\min(l^*,r^*)}{\max(l^*,r^*)} \cdot \frac{\min(t^*,b^*)}{\max(t^*,b^*)}},$$

that is 1 at the center and decays toward the edges; it multiplies the class score at test time so off-center boxes rank lower before NMS. FCOS removed every anchor hyperparameter and matched or beat anchor-based RetinaNet.

Set prediction with transformers: DETR and its descendants

Plain version. Instead of predicting thousands of candidates and deleting duplicates, predict a small *set* of answers directly, and train so that each real object is claimed by exactly one prediction.

DETR (2020) [30] uses a CNN backbone, a transformer encoder over the feature map, and a transformer decoder that takes N learned **object queries** (100 in the paper) and outputs N (class, box) predictions in parallel. There are no anchors, no proposals and no NMS. Since N exceeds the number of objects, a special "no object" class $\emptyset$ pads the ground truth to size N.

The key is training with **bipartite matching**. Let $y = \{y_j\}$ be the padded ground truth and $\hat{y} = \{\hat{y}_i\}_{i=1}^{N}$ the predictions. Find the permutation $\hat{\sigma}$ of N elements with lowest total matching cost,

$$\hat{\sigma} = \arg\min_{\sigma\in\mathfrak{S}_N} \sum_{j=1}^{N} \mathcal{L}_{\text{match}}(y_j, \hat{y}_{\sigma(j)}), \qquad \mathcal{L}_{\text{match}} = -\mathbb{1}_{\{c_j\neq\emptyset\}}\, \hat{p}_{\sigma(j)}(c_j) + \mathbb{1}_{\{c_j\neq\emptyset\}}\, \mathcal{L}_{\text{box}}(b_j, \hat{b}_{\sigma(j)}),$$

where $\mathfrak{S}_N$ is the set of permutations, c_j and b_j are the class and box of target j, $\hat{p}_{\sigma(j)}(c_j)$ is the predicted probability of class c_j for the matched query, and

$\mathcal{L}_{\text{box}} = \lambda_{L1}\|b_j - \hat{b}_{\sigma(j)}\|_1 + \lambda_{\text{iou}}(1 - \text{GIoU}(b_j, \hat{b}_{\sigma(j)}))$. This assignment problem is solved exactly by

the **Hungarian algorithm** [31] in polynomial time. The training loss (the **set loss**) then uses the matched pairs:

$$\mathcal{L}_{\text{Hungarian}} = \sum_{j=1}^{N} \Big[-\log \hat{p}_{\hat{\sigma}(j)}(c_j) + \mathbb{1}_{\{c_j \neq \emptyset\}}\, \mathcal{L}_{\text{box}}(b_j, \hat{b}_{\hat{\sigma}(j)}) \Big],$$

with the log-probability term down-weighted by a factor of 10 for $c_j = \emptyset$ to balance the many "no object" slots. Because each object is matched to exactly one query, a duplicate prediction is *penalized* as a false "no object", and the decoder's self-attention learns to suppress duplicates, which is why NMS is unnecessary.

The matching step is short with SciPy. The cost below combines the three terms with DETR's weights; `linear_sum_assignment` accepts the rectangular $N \times M$ matrix directly (DETR computes the L1 term on normalized (c_x, c_y, w, h); corner format is used here for brevity).

```
import numpy as np
from scipy.optimize import linear_sum_assignment

def giou_matrix(a, b):
    x1 = np.maximum(a[:, None, 0], b[None, :, 0]); y1 = np.maximum(a[:, None, 1], b[None, :,
     ↪  1])
    x2 = np.minimum(a[:, None, 2], b[None, :, 2]); y2 = np.minimum(a[:, None, 3], b[None, :,
     ↪  3])
    inter = np.clip(x2 - x1, 0, None) * np.clip(y2 - y1, 0, None)
    area = lambda b: (b[:, 2] - b[:, 0]) * (b[:, 3] - b[:, 1])
    union = area(a)[:, None] + area(b)[None, :] - inter
    cx1 = np.minimum(a[:, None, 0], b[None, :, 0]); cy1 = np.minimum(a[:, None, 1], b[None,
     ↪  :, 1])
    cx2 = np.maximum(a[:, None, 2], b[None, :, 2]); cy2 = np.maximum(a[:, None, 3], b[None,
     ↪  :, 3])
    c = (cx2 - cx1) * (cy2 - cy1)
    return inter / union - (c - union) / c

rng = np.random.default_rng(1)
N, C = 6, 3                                      # 6 queries, 3 real classes (+ "no object")
logits = rng.normal(size=(N, C + 1))
prob = np.exp(logits) / np.exp(logits).sum(1, keepdims=True)
pred_boxes = np.array([[.10, .10, .40, .50], [.55, .20, .90, .60], [.12, .08, .42, .48],
                       [.30, .60, .60, .95], [.00, .00, .20, .20], [.50, .50, .70, .70]])
gt_boxes  = np.array([[.10, .10, .40, .50], [.32, .62, .62, .93], [.56, .22, .88, .62]])
gt_labels = np.array([0, 2, 1])

w_cls, w_l1, w_giou = 1.0, 5.0, 2.0              # weights used by DETR
cost = (-w_cls * prob[:, gt_labels]                          # (N, M) class term
        + w_l1 * np.abs(pred_boxes[:, None] - gt_boxes[None]).sum(-1)
        - w_giou * giou_matrix(pred_boxes, gt_boxes))
rows, cols = linear_sum_assignment(cost)         # works on rectangular N x M matrices
print("query -> gt:", dict(zip(rows.tolist(), cols.tolist())))
print("unmatched queries (trained as 'no object'):", sorted(set(range(N)) -
 ↪  set(rows.tolist())))
```

```
query -> gt: {0: 0, 1: 2, 3: 1}
unmatched queries (trained as 'no object'): [2, 4, 5]
```

Query 2 is a near-duplicate of query 0, yet it is left unmatched and will be trained toward "no object". That is the mechanism that replaces NMS.

The price. DETR needed very long training schedules (hundreds of epochs, compared with roughly a dozen for Faster R-CNN) and was weak on small objects, because global attention over a high-resolution feature map is expensive, so it used only a single coarse level. The descendants fixed these problems one by one.

Three descendants show the direction. **Deformable DETR** [32] lets each query look at only a few sampled points across several feature levels, which makes multi-scale features affordable and shortens training a lot. **DINO** [33] (not the self-supervised method of the same name) speeds up convergence further with denoising training and better query initialization. **RT-DETR** [34] makes the design fast enough for real-time use, still without NMS.

Optional: DETR descendants in detail (in the Supplementary notes). How each one works and the results its paper reports.

Today the DETR line and the YOLO line are converging: YOLOv10 borrows one-to-one assignment to drop NMS, and real-time DETRs borrow efficient convolutional necks.

Open-vocabulary and grounding detectors

Plain version. A closed-set detector can only find the 80 or 365 classes it was trained on. An open-vocabulary detector takes the class list *as text at test time*, so you can ask for "red fire hydrant" or "forklift" without retraining.

The trick is to replace the last classification layer, a fixed matrix of K class weight vectors, with **text embeddings**. For a region feature $\mathbf{z}_i$ and the text embedding $\mathbf{e}_k$ of a prompt such as "a photo of a forklift", the class score is a scaled cosine similarity,

$$s_{ik} = \tau \cdot \frac{\mathbf{z}_i^\top \mathbf{e}_k}{\|\mathbf{z}_i\| \, \|\mathbf{e}_k\|},$$

where τ is a learned temperature. Any phrase you can embed becomes a class. The vision side must learn to align regions with language, which requires image–text data far larger than any box-annotated dataset.

- **OWL-ViT (2022)** [35] starts from a transformer image encoder and a text encoder pretrained contrastively on image–text pairs, removes the final pooling, and attaches light box and class

heads to every output token. The class head is the text embedding above. It supports zero-shot (text query) and one-shot (image-exemplar query) detection, and the paper shows consistent gains with model scale.

- **Grounding DINO (2024)** [36] fuses language deeply into a DINO detector: a feature enhancer with cross-attention between image and text, language-guided query selection, and a cross-modality decoder. It is pretrained on detection, grounding (phrase-to-region) and caption data, and reports 52.5 AP on COCO zero-shot transfer (with no COCO training data) and a 26.1 mean AP on a benchmark collection of diverse real-world detection datasets.
- **YOLO-World (2024)** [37] brings open vocabulary to real-time detectors with a vision–language path aggregation network (RepVL-PAN) and region–text contrastive pretraining. Its *prompt-then-detect* mode encodes the user's vocabulary once offline, so inference costs about the same as a closed-set YOLO. It reports 35.4 AP on LVIS at 52.0 FPS on a V100 GPU.

Promptable foundation models go one step further. SAM 3 [38] (2025) accepts a short noun phrase or image exemplars as a *concept prompt* and returns masks with identities for **all** matching instances in images and videos; the authors built a data engine producing a dataset with 4M unique concept labels and report doubling the accuracy of existing systems on their promptable concept segmentation benchmark. In such systems, detection becomes one output of a general "find this concept" model rather than a separately trained network.

Small objects and dense scenes

Plain version. Tiny objects have very few pixels, and stride-32 features may squeeze them into less than one cell. Crowds are hard because objects hide each other and NMS confuses neighbors with duplicates.

Small objects. COCO defines "small" as area below 32^2 pixels [40]. After a stride-16 backbone, such an object covers at most about 2×2 feature cells, and its appearance is mostly lost in downsampling. IoU also becomes harsh: a 2-pixel error on a 10×10 box can drop IoU below 0.7. Common remedies are high-resolution feature levels (P_2 with stride 4), feature pyramids, image tiling or slicing at inference, larger input resolutions, copy-paste and scale augmentation, and assignment rules that do not starve small boxes of positive samples. The survey and benchmark of Cheng et al. [39] reviews these strategies and introduces two large small-object benchmarks: SODA-D (24,828 traffic images, 278,433 instances, 9 categories) and SODA-A (2,513 high-resolution aerial images, 872,069 instances, 9 classes).

Dense and occluded scenes. In crowds, true objects overlap heavily, so a fixed NMS threshold trades recall against duplicates. Soft-NMS [13] helps; set-prediction detectors avoid the threshold altogether by learning which predictions are duplicates. Occlusion also makes box annotation ambiguous (full extent or visible part?), and evaluation then depends on the protocol. These issues return in tracking, where objects disappear and reappear.

Evaluating detectors

Plain version. Sort all detections by confidence. Walk down the list: each detection is either a hit (it matches an unclaimed real object well enough) or a false alarm. Track how precision and recall change; the area under that curve is the Average Precision.

Precision, recall and AP

For one class and one IoU threshold θ, sort all detections in the test set by descending score. A detection is a **true positive** (TP) if its IoU with a not-yet-matched ground-truth box of the same class is at least θ; otherwise it is a **false positive** (FP). Each ground-truth box can be matched only once, which is how duplicates are punished. After the top k detections,

$$\text{precision}(k) = \frac{\text{TP}(k)}{k}, \qquad \text{recall}(k) = \frac{\text{TP}(k)}{M},$$

where M is the number of ground-truth objects of that class. To make the curve monotone, precision is **interpolated** with its upper envelope, $p_{\text{interp}}(r) = \max_{r' \geq r} p(r')$. Then:

- **VOC 11-point AP** (PASCAL VOC up to 2009 [3]): $\text{AP} = \frac{1}{11} \sum_{r \in \{0, 0.1, \ldots, 1\}} p_{\text{interp}}(r)$.
- **VOC all-point AP** (VOC 2010 onwards [41]): the exact area under the interpolated curve, $\text{AP} = \sum_n (r_{n+1} - r_n)\, p_{\text{interp}}(r_{n+1})$ over the distinct recall values.
- **COCO AP** [40]: the mean of p_{interp} at 101 recall points $0, 0.01, \ldots, 1$.

Figure 6 shows a PR curve, its envelope and the AP area.

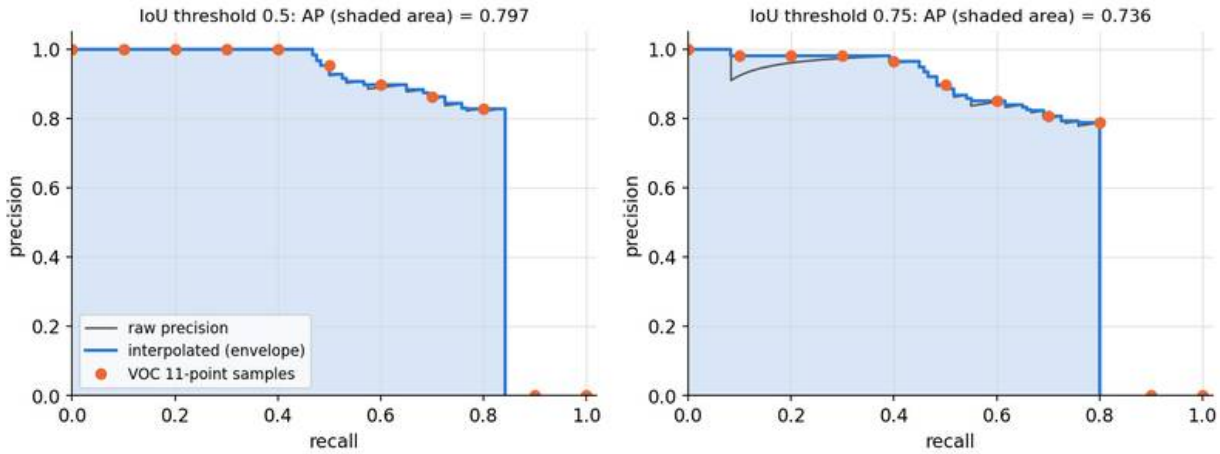


Figure 6 — Precision–recall on synthetic detections at two IoU thresholds. The shaded area under the

interpolated envelope is the all-point AP; orange dots are the VOC 11-point samples. Precision drops to 0 where recall ends, because some objects were never detected.

The COCO protocol

COCO's primary metric, written **AP** or **mAP@[.5:.95]**, averages AP over 10 IoU thresholds $\theta \in \{0.50, 0.55, \dots, 0.95\}$ and over all 80 categories [40]:

$$\mathrm{AP}_{\mathrm{COCO}} = \frac{1}{10\,K} \sum_{\theta} \sum_{k=1}^{K} \mathrm{AP}_k(\theta).$$

It rewards precise localization, not just finding the object. Companion metrics are AP_{50} (the VOC-style metric), AP_{75} (strict), and AP_S, AP_M, AP_L for small (area $< 32^2$), medium (32^2 to 96^2) and large ($> 96^2$) objects. At most 100 detections per image are scored. Average recall (AR) at 1, 10 and 100 detections per image is also reported. Note that COCO calls its category-averaged metric "AP"; "mAP" in papers usually means the same thing.

The code below computes everything from scratch on synthetic predictions (jittered ground truth, about 15% misses and some random false positives).

```
import numpy as np

def box_iou(a, b):
    """Pairwise IoU between (N,4) and (M,4) xyxy boxes -> (N, M)."""
    x1 = np.maximum(a[:, None, 0], b[None, :, 0]); y1 = np.maximum(a[:, None, 1], b[None, :,
     ↪  1])
    x2 = np.minimum(a[:, None, 2], b[None, :, 2]); y2 = np.minimum(a[:, None, 3], b[None, :,
     ↪  3])
    inter = np.clip(x2 - x1, 0, None) * np.clip(y2 - y1, 0, None)
    area = lambda b: (b[:, 2] - b[:, 0]) * (b[:, 3] - b[:, 1])
    return inter / (area(a)[:, None] + area(b)[None, :] - inter)

def match(dets, gts, thr):
    """dets: list over images of (boxes, scores); gts: list of gt boxes.
    Returns scores, TP flags (sorted by score), and number of GT."""
    recs = []
    for img, ((db, ds), gb) in enumerate(zip(dets, gts)):
        for k in range(len(ds)):
            recs.append((ds[k], img, k))
    recs.sort(key=lambda r: -r[0])                      # global ranking by confidence
    used = [np.zeros(len(g), bool) for g in gts]
    tp = np.zeros(len(recs))
    for n, (s, img, k) in enumerate(recs):
        gb = gts[img]
        if len(gb) == 0:
            continue
        ious = box_iou(dets[img][0][k:k + 1], gb)[0]
        ious[used[img]] = -1                            # each GT can be matched once
        j = ious.argmax()
        if ious[j] >= thr:
```

```
            tp[n] = 1; used[img][j] = True
    return tp, sum(len(g) for g in gts)

def pr_curve(tp, n_gt):
    ctp, cfp = np.cumsum(tp), np.cumsum(1 - tp)
    return ctp / n_gt, ctp / (ctp + cfp)                # recall, precision

def ap(rec, prec, mode="all"):
    r = np.concatenate([[0], rec, [1]]); p = np.concatenate([[0], prec, [0]])
    p = np.maximum.accumulate(p[::-1])[::-1]            # monotone envelope
    if mode == "all":                                   # VOC2010+: exact area
        i = np.where(r[1:] != r[:-1])[0]
        return np.sum((r[i + 1] - r[i]) * p[i + 1])
    pts = np.linspace(0, 1, 11 if mode == "voc11" else 101)
    return np.mean([p[r >= t].max() if np.any(r >= t) else 0 for t in pts])

# synthetic data: 50 images, 1-4 objects each; detector = jittered GT + false positives
rng = np.random.default_rng(0)
gts, dets = [], []
for _ in range(50):
    n = rng.integers(1, 5)
    xy = rng.uniform(0, 200, (n, 2)); wh = rng.uniform(20, 80, (n, 2))
    g = np.hstack([xy, xy + wh]); gts.append(g)
    keep = rng.random(n) > 0.15                         # miss ~15% of objects
    jit = rng.normal(0, 0.06, (n, 4)) * np.hstack([wh, wh])
    db = (g + jit)[keep]; ds = rng.uniform(0.4, 1.0, keep.sum())
    m = rng.integers(0, 3)                              # 0-2 false positives
    fxy = rng.uniform(0, 200, (m, 2)); fwh = rng.uniform(20, 80, (m, 2))
    db = np.vstack([db, np.hstack([fxy, fxy + fwh])]); ds = np.concatenate([ds,
     ↪  rng.uniform(0, 0.7, m)])
    dets.append((db, ds))

for thr in (0.5, 0.75):
    tp, n_gt = match(dets, gts, thr)
    rec, prec = pr_curve(tp, n_gt)
    print(f"IoU {thr}: VOC11 {ap(rec, prec, 'voc11'):.3f}  all-point {ap(rec, prec):.3f}"
          f"  COCO-101 {ap(rec, prec, 'coco'):.3f}")
aps = [ap(*pr_curve(*match(dets, gts, t)), "coco") for t in np.arange(0.5, 0.96, 0.05)]
print("AP@[.5:.95] =", round(float(np.mean(aps)), 3))
```

```
IoU 0.5: VOC11 0.776  all-point 0.797  COCO-101 0.797
IoU 0.75: VOC11 0.749  all-point 0.736  COCO-101 0.738
AP@[.5:.95] = 0.532
```

Three lessons are visible. The three interpolation rules give similar but *not identical* numbers, so AP values computed with different protocols must never be compared. AP@[.5:.95] is much lower than AP50 because jittered boxes fail the strict thresholds. And this is a single class; real mAP averages over classes, so rare classes count as much as frequent ones.

What AP does not tell you

AP summarizes ranking quality over the whole test set, so it does not choose an operating threshold, and it does not separate *types* of errors: localization errors, confusion with similar classes, duplicates and background false positives all look the same. When diagnosing a model, break errors down by type and by object size, and look at the images.

Datasets and benchmarks

Plain version. Detection progress has been driven by a few public datasets; each one changed what "good" meant.

Dataset	Classes	Scale (from the papers)	Annotations	Why it matters
PASCAL VOC [3]	20	Small by today's standards	Boxes (+ some masks)	The 2005–2012 benchmark; AP50
MS COCO [4]	91 types labeled; 80 used for detection	328k images, 2.5M labeled instances	Boxes + instance masks	Cluttered everyday scenes, many small objects; AP@[.5:.95]
Objects365 [42]	365	Large-scale, high-quality boxes	Boxes	Common pretraining set for strong detectors
Open Images V4 [43]	600 boxable classes	9.2M images; 15.4M boxes	Boxes, image labels, relationships	Very large; 19.8k concepts at image level
LVIS [44]	over 1000	~2M masks in 164k images	Instance masks	Long-tailed vocabulary; rare-class evaluation

Two trends stand out. Category counts grew (20, then 80, then hundreds and over a thousand), and with them came **long-tail** problems: in LVIS many categories have only a handful of training examples, so a loss or sampler that ignores frequency fails on rare classes. And the biggest models are now pretrained on Objects365 or on image–text data before fine-tuning on COCO, which makes "trained on COCO only" and "with extra data" separate leaderboards.

Speed, accuracy and deployment

Plain version. In practice you pick a point on a curve: more accuracy usually means more computation. What counts is the latency on *your* hardware, end to end.

- **FLOPs are not latency.** FLOPs count arithmetic, but runtime also depends on memory traffic, operator support and parallelism. Depth-wise convolutions and attention can be cheap in FLOPs but slow on some accelerators.

- **Measure end to end.** Report latency including preprocessing, the network *and* post-processing. NMS cost grows with the number of candidate boxes (worst case quadratic in their number), is data-dependent, and often runs on the CPU. RT-DETR's paper [34] analyses exactly this and is one reason NMS-free designs such as RT-DETR and YOLOv10 [24] are attractive for deployment.
- **Input resolution** is the strongest knob: halving the side length cuts compute by about 4 times but hurts small objects most.
- **Quantization** (e.g. 8-bit integer weights and activations) and **re-parameterization** (training a multi-branch block, then folding it into a single convolution, as used by YOLOv7 [22]) reduce latency with small accuracy loss. Box-regression outputs and the score head are often the most sensitive to quantization, so calibrate with real data and check AP75, not just AP50.
- **Batch size 1** is the usual edge setting; throughput numbers measured with large batches do not transfer.

A tiny detector you can train

To see box regression and an IoU loss work end to end, the PyTorch script below trains a small CNN to localize one rectangle or ellipse in noisy 32×32 images. It is a single-object detector: the network directly regresses (c_x, c_y, w, h) in $[0, 1]$ through a sigmoid, trained with L1 plus a GIoU loss, as DETR does for its matched queries.

```
import time, torch, torch.nn as nn
torch.manual_seed(0)
S = 32
torch.set_num_threads(4)

def make_batch(n):
    """One white rectangle or ellipse per 32x32 noisy image; target = (cx, cy, w, h) in [0,
     ↪  1]."""
    img = 0.1 * torch.randn(n, 1, S, S)
    wh = torch.rand(n, 2) * 0.4 + 0.15
    c = wh / 2 + torch.rand(n, 2) * (1 - wh)
    yy, xx = torch.meshgrid(torch.arange(S), torch.arange(S), indexing="ij")
    xx, yy = (xx[None] + 0.5) / S, (yy[None] + 0.5) / S
    dx = (xx - c[:, 0, None, None]) / (wh[:, 0, None, None] / 2)
    dy = (yy - c[:, 1, None, None]) / (wh[:, 1, None, None] / 2)
    rect = (dx.abs() <= 1) & (dy.abs() <= 1)
    ell = dx**2 + dy**2 <= 1
    shape = torch.where((torch.rand(n) < 0.5)[:, None, None], rect, ell)
    return img + shape[:, None].float(), torch.cat([c, wh], 1)

def xyxy(b):
    return torch.cat([b[:, :2] - b[:, 2:] / 2, b[:, :2] + b[:, 2:] / 2], 1)

def giou(p, g):
    p, g = xyxy(p), xyxy(g)
```

```
    lt, rb = torch.max(p[:, :2], g[:, :2]), torch.min(p[:, 2:], g[:, 2:])
    inter = (rb - lt).clamp(min=0).prod(1)
    area = lambda b: (b[:, 2:] - b[:, :2]).prod(1)
    union = area(p) + area(g) - inter
    c = (torch.max(p[:, 2:], g[:, 2:]) - torch.min(p[:, :2], g[:, :2])).prod(1)
    return inter / union, inter / union - (c - union) / c

net = nn.Sequential(
    nn.Conv2d(1, 16, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),   # 16
    nn.Conv2d(16, 32, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),  # 8
    nn.Conv2d(32, 32, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),  # 4
    nn.Flatten(), nn.Linear(32 * 4 * 4, 128), nn.ReLU(),
    nn.Linear(128, 4), nn.Sigmoid())                               # (cx, cy, w, h) in [0, 1]
opt = torch.optim.Adam(net.parameters(), 2e-3)

t0 = time.time()
for step in range(500):
    x, y = make_batch(64)
    pred = net(x)
    iou, g = giou(pred, y)
    loss = (pred - y).abs().sum(1).mean() + (1 - g).mean()        # L1 + GIoU loss
    opt.zero_grad(); loss.backward(); opt.step()

with torch.no_grad():
    x, y = make_batch(1000)
    iou, _ = giou(net(x), y)
print(f"train time {time.time() - t0:.1f}s | test mean IoU {iou.mean():.3f}"
      f" | IoU>=0.5: {(iou >= 0.5).float().mean():.1%} | IoU>=0.75: {(iou >=
      ↪ 0.75).float().mean():.1%}")
```

```
train time 21.1s | test mean IoU 0.885 | IoU>=0.5: 100.0% | IoU>=0.75: 97.8%
```

(Training time depends on your CPU.) Because there is exactly one object, there is no assignment and no NMS. Exercise 6 asks you to extend this to several objects, which is where every hard problem in this tutorial reappears.

Modern view

Figure 7 places the methods of this tutorial on a timeline, using the year of each cited publication.

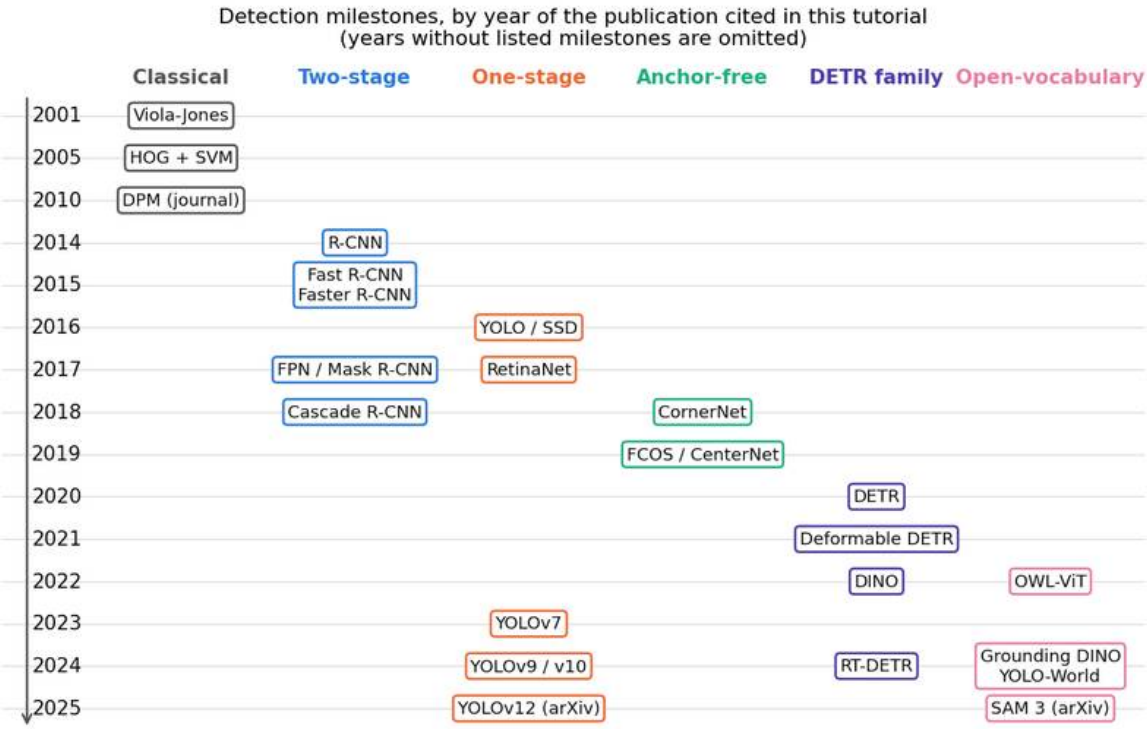


Figure 7 — Milestones by year of the publication cited in this tutorial, grouped by family. Years without a listed milestone are omitted.

What the surveys say

Five surveys frame the field. Zou et al. [1] give the broadest history, from Viola–Jones to deep detectors, and show that most "new" ideas have older roots: multi-scale scanning became feature pyramids, and hard-negative mining became the focal loss. Liu et al. [2] are the best single reference for the CNN era and stress that training strategies, data and evaluation mattered as much as architectures. Shehzadi et al. [45] review recent DETR improvements, Zhu and Chen [46] organize open-vocabulary methods and warn that vocabulary overlap between pretraining and test data complicates evaluation, and Cheng et al. [39] show that small objects remain far harder than large ones.

Optional: Guide to the Modern view surveys (in the Paper guides). What each survey covers, how it organizes the field and what it concludes.

The state of the art, 2023–2026

- **Real-time closed-set detection** is now a contest between the YOLO line (YOLOv9 [23], YOLOv10 [24], YOLOv12 [25]) and real-time DETRs (RT-DETR [34]). The two lines borrow

from each other: one-to-one assignment and NMS-free heads in YOLO, convolutional necks and hybrid encoders in DETRs. Attention is now affordable at real-time budgets.

- **High-accuracy closed-set detection** is dominated by DETR-style models such as DINO [33] with large backbones and large-scale pretraining (Objects365 [42] and beyond). Gains on COCO have become small, and the benchmark is close to saturated for the largest models, which is pushing attention to harder benchmarks: LVIS [44], open vocabulary and cross-domain evaluation.
- **Open-vocabulary and grounding detection** (OWL-ViT [35], Grounding DINO [36], YOLO-World [37]) is moving into practice: in many applications you now *prompt* a detector rather than train one, and then optionally fine-tune it.
- **Promptable foundation models** (SAM 3 [38]) unify detection, segmentation and tracking under concept prompts.

Open problems

1. **Localization quality versus confidence.** Classification score and box quality are trained separately and are often poorly correlated, so the "best" box can be suppressed by a more confident but worse one.
2. **Small, crowded and occluded objects** remain far from solved [39].
3. **Long-tail and open-world recognition.** Rare classes and truly unseen objects, and knowing *when* the model does not know.
4. **Evaluation.** COCO AP hides error types, is saturated at the top, and open-vocabulary evaluation depends on prompts and on what the pretraining data already contained [46].
5. **Efficiency on real hardware**, including NMS-free designs, quantization-friendly heads and fair latency reporting.

Key takeaways

- Detection outputs a variable-size set of (box, class, score) triples; every architecture is a different answer to "how many, and which one is responsible for which object?"
- Boxes are regressed as scale-normalized offsets from a reference $(t_x, t_y$, log-scale $t_w, t_h)$; IoU-based losses (GIoU, DIoU, CIoU) align training with the IoU-based metric.
- Anchors and assignment rules decide which predictions learn which object; NMS removes duplicates afterwards, and Soft-NMS softens its failure in crowds.
- FPN handles scale; the focal loss handles the foreground–background imbalance that made dense detectors weak.
- Two-stage (R-CNN, Fast, Faster, Mask, Cascade) refine proposals; one-stage (YOLO, SSD, RetinaNet) predict densely in one pass; anchor-free (CornerNet, CenterNet, FCOS) predict from

points; DETR predicts a set with Hungarian matching and needs no NMS.
- Open-vocabulary detectors turn the class list into text prompts, and promptable foundation models merge detection, segmentation and tracking.
- Measure with the right protocol: VOC 11-point, VOC all-point and COCO 101-point AP differ; COCO AP averages IoU 0.50–0.95, and APs/m/l expose scale weaknesses.

Exercises

1. **Encode by hand.** An anchor has center $(100, 100)$ and size 64×32. A ground-truth box has center $(110, 96)$ and size 80×40. Compute (t_x, t_y, t_w, t_h) without normalization, then decode them back.

   Hint

   $t_x = 10/64 = 0.156$, $t_y = -4/32 = -0.125$, $t_w = \log(80/64) = 0.223$, $t_h = \log(40/32) = 0.223$. Decoding: $c_x = 100 + 0.156 \cdot 64 = 110$, $w = 64e^{0.223} = 80$, and similarly for y and h.

2. **IoU versus shift.** Show that two equal $a \times a$ squares shifted by d along one axis have IoU $= (a - d)/(a + d)$. What shift gives IoU 0.5 and 0.75? Repeat for an equal shift along both axes (Figure 2, right).

   Hint

   Intersection $a(a - d)$, union $2a^2 - a(a - d) = a(a + d)$. IoU 0.5 at $d = a/3$; IoU 0.75 at $d = a/7$. For a diagonal shift, IoU $= (a - d)^2/(2a^2 - (a - d)^2)$, so IoU 0.5 at $d = a(1 - \sqrt{2/3}) \approx 0.18a$.

3. **When greedy NMS fails.** Modify the NMS example so that two different people overlap with IoU 0.6, and run greedy NMS at thresholds 0.5 and 0.7, and Soft-NMS. Which recovers both people, and what new problem does a high threshold create?

   Hint

   At 0.5 the lower-scored person is deleted. At 0.7 both survive, but duplicates of the same person with IoU between 0.5 and 0.7 also survive, adding false positives. Soft-NMS keeps the second person with a reduced score ($e^{-0.36/0.5} \approx 0.49$ times its original score), which still counts toward recall in AP.

4. **Focal loss numbers.** With $\gamma = 2$ and $\alpha = 0.25$, compute the focal loss and the plain cross-entropy for a background anchor with $p = 0.01$ (easy) and $p = 0.6$ (hard). By what factor does each change?

   Hint

   For background, $p_t = 1 - p$ and $\alpha_t = 0.75$. Easy: CE $= -\log 0.99 = 0.01005$; FL $= 0.75 \cdot 0.01^2 \cdot 0.01005 \approx 7.5 \times 10^{-7}$, a factor of about 1.3×10^4 smaller. Hard: CE $= -\log 0.4 = 0.916$; FL $= 0.75 \cdot 0.6^2 \cdot 0.916 \approx 0.247$, only about 3.7 times smaller.

5. **Protocols disagree.** Using the AP code, find (by changing the random seed or the jitter) a case where VOC 11-point AP is *higher* than all-point AP, and one where it is lower. Explain why neither is always larger.

Hint

The 11-point rule samples the envelope at fixed recalls, so it overestimates when the envelope is high just to the right of a sample point and drops right after it, and underestimates in the opposite case. The output above already shows both: at IoU 0.5 the 11-point value is lower, at 0.75 it is higher.

6. **From one object to many.** Extend the tiny PyTorch detector to images with up to three shapes. Option A: predict an 8×8 grid of (objectness, box) like YOLO v1, assigning each object to the cell containing its center. Option B: predict 5 (class, box) slots and train with Hungarian matching like DETR. Compare the failure modes.

Hint

Option A fails when two centers fall in the same cell and needs NMS to remove duplicates from neighboring cells. Option B needs no NMS, but trains more slowly, and early in training the matching is unstable (the same object may be assigned to different slots from step to step). Use `scipy.optimize.linear_sum_assignment` on a detached cost matrix.

Detections are the raw material of tracking. The next step, linking boxes across video frames and keeping identities, is covered in DLCV 06 ·Object Tracking.

Supplementary notes

These notes are optional background: details of methods that the main text summarizes or only mentions. Nothing in the main text or the exercises depends on them.

Classical detectors in detail: Viola–Jones, HOG and DPM

This note gives the details behind the three classical detectors summarized in the main text: how each one computes its features and scores a window.

Viola–Jones (2001) [5] made real-time face detection possible with three tricks. *Haar-like features* (differences of sums over adjacent rectangles) can be evaluated in constant time using an **integral image** $S(x,y) = \sum_{x' \le x, y' \le y} I(x',y')$, so any rectangle sum needs four lookups. *AdaBoost* selects a small set of informative features out of a huge pool. An *attentional cascade* of increasingly complex classifiers rejects most background windows after only a few feature evaluations, which is the first big idea for beating the imbalance problem: spend computation only where it might matter.

HOG + linear SVM (2005) [6] replaced raw intensities with **histograms of oriented gradients**, close relatives of the gradient-orientation descriptors in IP 12. The detection window is divided into small cells; each cell stores a histogram of gradient orientations weighted by magnitude; overlapping blocks of cells are contrast-normalized; the concatenated vector is scored by a linear SVM. HOG is robust to small shifts and illumination changes, and it made pedestrian detection practical.

Deformable part models (DPM) [7] added structure. An object is a coarse *root* HOG filter plus several higher-resolution *part* filters that may shift relative to the root, paying a quadratic deformation cost:

$$\text{score}(p_0) = F_0 \cdot \phi(p_0) + \sum_{i=1}^{n} \max_{p_i} \left[F_i \cdot \phi(p_i) - d_i \cdot \psi(p_i - p_0) \right] + \beta,$$

where p_0 is the root position, p_i the position of part i, F_i the learned filters, $\phi(\cdot)$ the HOG features at a position, ψ the displacement features (dx, dy, dx^2, dy^2), d_i learned deformation weights and β a bias. Part positions are latent variables, trained with a *latent SVM*. DPM and its variants dominated the PASCAL VOC benchmark [3] until 2012.

Anchor shapes by k-means: clustering with an IoU distance

Plain version. Anchor shapes are a guess about which boxes the data contains. Instead of guessing, let the training boxes vote: group boxes of similar shape and use each group's typical shape as one anchor.

Reduce every training box to its width and height (w, h), as if all boxes shared a corner, because an anchor only needs a shape. For a box $b = (w, h)$ and a centroid $c = (w_c, h_c)$ the distance used by YOLO9000 [19] is

$$d(b, c) = 1 - \text{IoU}(b, c), \qquad \text{IoU}(b, c) = \frac{\min(w, w_c)\ \min(h, h_c)}{wh + w_c h_c - \min(w, w_c)\ \min(h, h_c)},$$

where the numerator is the overlap area of two boxes that share a corner and the denominator is their union. The algorithm is the usual k-means loop:

1. choose k training boxes as the initial centroids;
2. assign each box to the centroid with the smallest d, which is the one with the highest IoU;
3. replace each centroid by the mean (w, h) of its members (this note's choice; the paper states the distance, not the update rule);
4. repeat steps 2–3 until the assignments stop changing.

Each pass costs $O(Nk)$ IoU evaluations for N boxes. The choice of d matters because a 10-pixel error is large for a 20-pixel box and negligible for a 200-pixel box. Squared Euclidean distance lets big boxes dominate the clustering, while $1 - \text{IoU}$ is blind to absolute size. The paper picks k by plotting the mean IoU between each box and its nearest centroid against k, and settles on $k = 5$ as

a compromise between recall and model complexity [19]. On VOC 2007 it reports a mean IoU of 61.0 for 5 IoU-clustered priors, 58.7 for 5 priors clustered with the sum-of-squares distance, 60.9 for 9 hand-picked anchor boxes from Faster R-CNN [10], and 67.2 for 9 IoU-clustered priors (its Table 1).

```
import numpy as np

def wh_iou(wh, cent):
    """IoU of (w, h) pairs when boxes share a corner: (N, 2) x (K, 2) -> (N, K)."""
    inter = np.minimum(wh[:, None, 0], cent[None, :, 0]) * np.minimum(wh[:, None, 1],
     ↪  cent[None, :, 1])
    union = wh[:, :1] * wh[:, 1:] + (cent[:, 0] * cent[:, 1])[None, :] - inter
    return inter / union

def kmeans(wh, k, dist, iters=50, seed=0):
    cent = wh[np.random.default_rng(seed).choice(len(wh), k, replace=False)].copy()
    for _ in range(iters):
        if dist == "iou":   # distance 1 - IoU: nearest centroid = highest IoU
            assign = wh_iou(wh, cent).argmax(axis=1)
        else:               # ordinary squared Euclidean distance on (w, h)
            assign = ((wh[:, None] - cent[None]) ** 2).sum(-1).argmin(axis=1)
        for c in range(k):
            if np.any(assign == c):
                cent[c] = wh[assign == c].mean(axis=0)
    return cent

# synthetic "training boxes": 1000 small and 100 large, aspect ratios spread around 1
rng = np.random.default_rng(0)
area = np.concatenate([rng.lognormal(np.log(20**2), 0.5, 1000),
 ↪  rng.lognormal(np.log(200**2), 0.8, 100)])
ratio = np.exp(rng.normal(0, 0.6, area.size))            # h / w
wh = np.stack([np.sqrt(area / ratio), np.sqrt(area * ratio)], axis=1)

for dist in ["euclid", "iou"]:
    cent = kmeans(wh, 4, dist)
    n_small = int((cent.prod(axis=1) < 50**2).sum())
    print(f"{dist:6s} mean IoU to nearest centroid {wh_iou(wh,
     ↪  cent).max(axis=1).mean():.3f}, "
          f"centroids for small boxes (area < 50^2): {n_small} of 4")
```

```
euclid mean IoU to nearest centroid 0.633, centroids for small boxes (area < 50^2): 2 of 4
iou    mean IoU to nearest centroid 0.668, centroids for small boxes (area < 50^2): 3 of 4
```

Most of these synthetic boxes are small, and the IoU distance spends three of four anchors on them instead of two, which raises the mean IoU. The size of the gain depends on the data and the initialization; this is one seeded run on synthetic boxes, not a benchmark. Two cautions: the clusters are only a local optimum, so different seeds give different anchors, and they describe the training set only, so they have to be recomputed for a new dataset.

RoI pooling versus RoIAlign: sampling without rounding

Plain version. RoI pooling cuts a region out of the feature map with scissors that can only follow the grid lines, so the cut snaps to the nearest line. RoIAlign cuts exactly where the box is and reads the values between grid lines by blending the neighbouring cells.

Let the feature map F have stride s (one cell covers $s \times s$ image pixels), and let an RoI be given in image pixels. A pixel coordinate x becomes the feature coordinate x/s. Here cell j covers $[j, j+1)$ and its value sits at its center $j + \frac{1}{2}$.

RoI pooling [9] rounds the RoI to whole cells, splits it into an $H \times W$ grid of bins, rounds the bin edges to whole cells too, and takes the maximum inside each bin. Every rounding moves a boundary by up to half a cell, which is $s/2$ image pixels, and the error is different for every RoI [16].

RoIAlign [16] keeps every coordinate real-valued. Each bin gets a regular grid of sample points (the paper uses four regularly spaced points per bin; the snippet below uses two along x), and each point is read by bilinear interpolation. For a point (x, y) in feature coordinates, set $u = x - \frac{1}{2}$, $v = y - \frac{1}{2}$, $j = \lfloor u \rfloor$, $i = \lfloor v \rfloor$, $a = u - j$ and $b = v - i$. Then

$$F(x, y) = (1-b)\big[(1-a)F_{i,j} + aF_{i,j+1}\big] + b\big[(1-a)F_{i+1,j} + aF_{i+1,j+1}\big],$$

where $F_{i,j}$ is the feature value of the cell in row i and column j, a and b are the fractional offsets of the point from cell center (j, i), and the four terms weight the four surrounding cells. The bin's output is the maximum or the average of its sample points; the paper uses the average. The paper reports that results are not sensitive to the exact sample positions or their number, as long as nothing is rounded [16]. Gradients reach the four neighbours of every sample point, so the layer trains like any other.

The snippet uses a feature map whose value equals the x coordinate of each cell center, so a correct pooling should return the x coordinate of the middle of each bin. Both methods use averages here, to isolate position errors from the effect of taking a maximum. The RoI keeps its width of 64 pixels and moves right by 4 pixels per row.

```
import numpy as np

STRIDE = 16
F = np.tile(np.arange(8) + 0.5, (8, 1))   # F[i, j] = x-coordinate of the centre of cell j
 ↪  (in cells)

def roi_pool_avg(F, x1, x2, bins=2):
    """Quantized pooling along x (average). Returns the mean cell-centre read by each bin."""
    s, e = int(round(x1 / STRIDE)), int(round(x2 / STRIDE))     # quantize the RoI
    edges = s + (e - s) * np.arange(bins + 1) / bins
    out = []
    for a, b in zip(edges[:-1], edges[1:]):
        lo, hi = int(np.floor(a)), int(np.ceil(b))              # quantize the bin edges
        out.append(F[0, lo:hi].mean())
    return np.array(out)
```

```
def bilinear(F, x, y=0.0):
    """Sample F at a continuous point (x, y); cell j has its centre at x = j + 0.5."""
    u, v = x - 0.5, y - 0.5
    j0, i0 = int(np.floor(u)), int(np.floor(v))
    du, dv = u - j0, v - i0
    j0c, j1c = np.clip([j0, j0 + 1], 0, F.shape[1] - 1)
    i0c, i1c = np.clip([i0, i0 + 1], 0, F.shape[0] - 1)
    return ((1 - dv) * ((1 - du) * F[i0c, j0c] + du * F[i0c, j1c])
            + dv * ((1 - du) * F[i1c, j0c] + du * F[i1c, j1c]))

def roi_align_avg(F, x1, x2, bins=2, n=2):
    """RoIAlign along x: n sample points per bin on a regular grid, no rounding, then
     ↪  average."""
    a, b = x1 / STRIDE, x2 / STRIDE
    w = (b - a) / bins
    return np.array([np.mean([bilinear(F, a + k * w + (t + 0.5) * w / n) for t in range(n)])
                     for k in range(bins)])

for x1, x2 in [(22, 86), (26, 90), (30, 94)]:                    # same width, moved right
 ↪  by 4 px each step
    true_c = (np.arange(2) + 0.5) / 2 * (x2 - x1) / STRIDE + x1 / STRIDE   # true bin
     ↪  centres (cells)
    print(f"RoI x = [{x1}, {x2}] px   true bin centres {np.round(true_c, 3)}  "
          f"RoIPool {np.round(roi_pool_avg(F, x1, x2), 3)}  RoIAlign
           ↪  {np.round(roi_align_avg(F, x1, x2), 3)}")
```

```
RoI x = [22, 86] px   true bin centres [2.375 4.375]  RoIPool [2. 4.]  RoIAlign [2.375 4.375]
RoI x = [26, 90] px   true bin centres [2.625 4.625]  RoIPool [3. 5.]  RoIAlign [2.625 4.625]
RoI x = [30, 94] px   true bin centres [2.875 4.875]  RoIPool [3. 5.]  RoIAlign [2.875 4.875]
```

RoIAlign follows the box exactly, because bilinear interpolation reproduces a linear ramp. RoI pooling is off by 0.375, 0.375 and 0.125 cells, which is 6, 6 and 2 pixels at stride 16. Moving the RoI from the first row to the second by 4 pixels makes its output jump by a whole cell (16 pixels), and moving it again by 4 pixels changes nothing. A real feature map is not a ramp, but this staircase response is what RoI pooling does to the features.

How to read the paper's evidence. Look first at Table 2 (c) and (d) of [16]. With a ResNet-50-C5 backbone, whose stride is 32, the rows RoIPool and RoIAlign give mask AP 23.6 and 30.9 (+7.3), and AP75 21.6 and 32.1 (+10.5); box AP75 rises from 26.9 to 36.4. The paper also tests RoIWarp, which interpolates bilinearly but still rounds the RoI, and finds it on a par with RoI pooling. So the gain comes from removing the rounding, not from the interpolation. When you read such an ablation, check that the compared rows share the backbone, the data split and the schedule (the table caption states training on trainval35k and testing on minival), and note that single numbers are reported; I found no repeated runs or variances in the text.

Common mistake. Forgetting the half-cell offset. If a cell value is treated as sitting at its corner while the RoI arithmetic assumes its center (or the reverse), every sample is read half a cell away,

which is as large as the rounding error RoIAlign was designed to remove.

The YOLO family, version by version

This note lists the YOLO versions summarized in the main text, one row per version, with the main idea each paper introduced.

After YOLO v1 [18], the **YOLO family** evolved quickly. The table lists only versions with a paper we could verify; many widely used releases are maintained as software packages without papers and are not covered here.

Version	Year	Main ideas (from the paper)
YOLO [18]	2016	Single-pass grid regression; real time
YOLO9000 / v2 [19]	2016 (arXiv)	Anchors chosen by k-means with an IoU distance; batch norm; higher-resolution training
YOLOv3 [20]	2018 (tech report)	Deeper backbone; predictions at three scales; independent logistic class scores
YOLOv4 [21]	2020 (arXiv)	Systematic study of training tricks ("bag of freebies") and cheap architectural add-ons ("bag of specials")
YOLOv7 [22]	2023 (CVPR)	Trainable bag-of-freebies: re-parameterized modules, improved label assignment, efficient layer aggregation
YOLOv9 [23]	2024 (ECCV)	Programmable gradient information (PGI) to keep information through deep layers; GELAN backbone
YOLOv10 [24]	2024 (NeurIPS)	NMS-free training with consistent dual (one-to-many and one-to-one) assignments; efficiency-driven design
YOLOv12 [25]	2025 (arXiv)	Attention-centric design that keeps real-time speed

A recurring theme is visible: the core "dense grid + NMS" idea stayed, while label assignment, necks, training recipes and, most recently, attention and NMS-free heads changed.

DETR descendants in detail

This note describes the three DETR descendants summarized in the main text, with the results their papers report.

- **Deformable DETR (2021)** [32] replaces dense attention with **deformable attention**: each query attends to only a few learned sampling points around a reference point, across multiple

feature levels. This makes multi-scale features affordable and, according to the paper, reaches better accuracy than DETR (especially on small objects) with 10 times fewer training epochs.

- **DINO (2022)** [33] (not to be confused with the self-supervised method of the same name) combines *contrastive denoising training* (noised copies of ground-truth boxes are fed as extra queries and must be reconstructed, while slightly more noised ones must be rejected), *mixed query selection* (initializing anchor positions from encoder features) and a *look-forward-twice* box update. It reports 49.4 AP after 12 epochs and 51.3 AP after 24 epochs on COCO with a standard 50-layer CNN backbone, and 63.2 AP on COCO val2017 with a large backbone and Objects365 pretraining.
- **RT-DETR (2024)** [34] targets real-time use. An *efficient hybrid encoder* applies attention only to the coarsest level and fuses scales with convolutions, and *IoU-aware query selection* picks better initial queries. The paper reports 53.1 / 54.3 AP on COCO at 108 / 74 FPS on a T4 GPU for its smaller and larger variants, faster and more accurate than YOLO models of comparable size at the time, without NMS.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Guide to the Modern view surveys

This guide expands the short survey summary in the Modern view: for each review, what it covers, how it organizes the field and what it concludes.

Object Detection in 20 Years (Zou et al., *Proceedings of the IEEE*, 2023) [1] is the broadest historical review. It covers milestone detectors from the traditional era (Viola–Jones, HOG, DPM) to deep two-stage and one-stage methods, detection datasets and metrics, fundamental building blocks (multi-scale detection, context, hard-negative mining, NMS, box regression), speed-up techniques, and recent state-of-the-art methods. Its main message is that most "new" ideas have older roots: multi-scale scanning became feature pyramids, cascades became cascaded heads, hard-negative mining became the focal loss, and NMS has been rethought several times.

Deep Learning for Generic Object Detection: A Survey (Liu et al., *IJCV*, 2020) [2] reviews over 300 contributions in depth and organizes them by detection framework, object feature representation, object proposal generation, context modeling, training strategies and evaluation metrics. It is the best single reference for the pre-transformer CNN era, and it stresses that progress came as much from training strategies, data and evaluation as from architectures.

Object Detection with Transformers: A Review (Shehzadi et al., *Sensors*, 2025) [45] analyses 21 recent improvements to DETR, grouped by changes to the backbone, query design and attention mechanism, and compares them on common benchmarks. Its taxonomy tracks the fixes described

in the main text: multi-scale and sparse attention for small objects and speed, and better query initialization and denoising for convergence.

A Survey on Open-Vocabulary Detection and Segmentation: Past, Present, and Future (Zhu and Chen, *IEEE TPAMI*, 2024) [46] organizes open-vocabulary methods by how they use weak supervision into six families: visual–semantic space mapping, novel visual feature synthesis, region-aware training, pseudo-labeling, knowledge distillation and transfer learning. It covers 2D, 3D and video tasks, and it highlights how evaluation of "novel" classes is complicated by vocabulary overlap between pretraining and test data.

Towards Large-Scale Small Object Detection (Cheng et al., *IEEE TPAMI*, 2023) [39] focuses on the most stubborn failure mode, reviews the remedies by category and contributes the SODA benchmarks described in the main text. It shows that small-object accuracy remains far below that of larger objects even for strong detectors.

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DLCV 03A • Real-Time Detection: The YOLO Family and RT-DETR

How detectors fast enough for live video are built, trained and measured: latency budgets, the backbone–neck–head design, the YOLO line from YOLO v1 to YOLO26, real-time DETRs, and the move from NMS to one-to-one label assignment.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-03a-real-time-detection/

What you'll learn

- What "real-time" means in numbers: latency versus throughput, frames per second, and which parts of the pipeline (pre-processing, the network, NMS) a speed figure does or does not include.
- The anatomy shared by modern one-stage detectors: a backbone, a neck (FPN and PAN), and a decoupled head that predicts boxes on several grids, with or without anchors.
- The YOLO lineage from YOLO v1 (2015) to YOLO26 (2026): which versions have peer-reviewed papers, which are software releases, and the handful of ideas that actually changed the field.
- How real-time DETRs (RT-DETR and its successors) made transformer detectors fast, and why the YOLO and DETR lines now meet at one idea: **one-to-one label assignment**, which removes NMS.
- The training recipe that every modern real-time detector shares (mosaic and mixup, weight averaging, IoU and distribution losses), and how to measure, export and choose a model without being fooled by speed numbers.

The big picture

A detector is "real-time" when it keeps up with its camera. That sounds like a property of the network, but it is really a property of the whole system: the image has to be resized, the network has to run, duplicate boxes have to be removed, and all of that has to finish before the next frame arrives. This chapter is about the family of detectors designed around that deadline.

> **Explain like I'm five**
>
> Imagine a guard at a gate who must write down every car that passes. If cars come every two seconds, the guard has two seconds per car, no matter how good the handwriting is. A slow guard who writes beautifully is useless, because cars pile up. Real-time detectors are guards trained to be quick: they look at the whole scene once, write a short list, and cross out the lines that say the same car twice. The newest guards are trained so well that they never write a car twice, so they can skip the crossing-out step altogether.

Why this matters: most detectors that run in products (cameras, robots, phones, cars, factory lines) are real-time detectors, and most come from two lines of work. The **YOLO** line ("You Only Look Once") started in 2015 and has more than a dozen versions from several teams. The **real-time DETR** line started in 2023 and brought transformer detectors to the same speed. Both now converge on one idea: train the network so that each object gets exactly one prediction, so that no step is needed to remove duplicates.

This chapter assumes DLCV 03. Its Core building blocks section defines box regression targets, IoU and its variants (GIoU, DIoU, CIoU), anchors and label assignment, non-maximum suppression (NMS), feature pyramids and the focal loss; its sections on one-stage detectors, anchor-free detectors and DETR introduce the families we build on. We do not repeat that material; we use it.

What "real-time" means

Plain version. A detector is real-time when the time it needs per frame is shorter than the time between frames. Everything else in this section is about measuring that time honestly.

Latency, throughput and frames per second

Latency is the time from receiving one frame to having its final list of boxes. **Throughput** is how many frames are finished per second when many frames are processed together. They are different numbers, and real-time systems care about latency.

For a single frame, the end-to-end latency is a sum of stages:

$$T_{\text{e2e}} = T_{\text{pre}} + T_{\text{net}} + T_{\text{post}},$$

where T_{pre} is pre-processing (decoding, resizing and normalizing the image, copying it to the accelerator), T_{net} is the forward pass of the network, and T_{post} is post-processing (decoding boxes, the confidence threshold, NMS, copying results back). A camera running at f frames per second leaves a budget of $1/f$ seconds per frame: about 33.3 ms at 30 FPS and 16.7 ms at 60 FPS. The system keeps up only if $T_{\text{e2e}} \le 1/f$; in practice it must also leave time for whatever uses the boxes (tracking, control, display).

Throughput with batches of size B is

$$\text{throughput} = \frac{B}{T_{\text{batch}}(B)},$$

where $T_{\text{batch}}(B)$ is the time to process one batch of B frames. Accelerators work best on large batches, so throughput grows with B, but every frame waits for its whole batch, so latency grows too. A live camera delivers one frame at a time, which is why real-time papers usually report

latency at **batch size 1**; the YOLOX report, for example, measures in FP16 with batch 1 on one GPU [10]. Converting latency to FPS as $1000/T$ (with T in milliseconds) is only meaningful at batch 1.

What is counted, and what is not

Two papers can both say "5 ms" and mean different things. A speed number depends on:

- **Hardware and software stack.** GPU or CPU model, the inference engine (for example, TensorRT on NVIDIA GPUs [45] or a runtime that executes ONNX models [44]), its version, and the numeric precision (FP32, FP16 or INT8).
- **Input size.** Most COCO results for real-time detectors use 640 × 640 inputs. Compute in the convolutional layers grows roughly with the number of pixels, so doubling the side length roughly quadruples it.
- **Batch size**, as above.
- **Which stages are timed.** Many YOLO papers historically reported only the forward pass. The YOLOX report states that its latencies are "measured without post-processing" [10]. The RT-DETR paper points out that YOLO detectors typically report model speed and exclude NMS time, and builds an **end-to-end speed benchmark** that adds the NMS step back in [23].

The rule for this chapter: **never compare two speed numbers unless they were measured on the same hardware, at the same precision, input size and batch size, over the same stages.** A table that mixes numbers from different papers violates this rule more often than not.

Why NMS time is not a constant

Plain version. The network's forward pass takes about the same time for every image. NMS does not: its cost depends on how many boxes survive the confidence threshold, and that depends on the image.

Greedy NMS (defined in DLCV 03) takes the highest-scoring box, computes its IoU with every remaining box, deletes those above the IoU threshold, and repeats. If n boxes enter and few are deleted, the number of IoU computations approaches

$$\sum_{k=1}^{n-1} k = \frac{n(n-1)}{2},$$

where k counts the boxes still remaining after each kept box. If most boxes are duplicates of a few objects, they are deleted early and the work is much smaller. So NMS time depends on

the number of candidate boxes, on the confidence threshold that decides how many enter, and on the scene. The RT-DETR paper measured this directly: it counted how many boxes survive confidence thresholds from 0.001 to 0.25, and found that with a confidence threshold of 0.001 and an IoU threshold of 0.7, YOLOv8 reached its best AP but its NMS time was also at a higher level [23]. It also found that the anchor-based detectors it tested produced about three times as many boxes as the anchor-free ones, and correspondingly needed more NMS time [23].

The snippet times a NumPy greedy NMS on synthetic candidates. Each scene has either 5 or 500 objects, and the candidates are jittered copies of them; counts of IoU computations are deterministic, while the times depend on your machine.

```
import time
import numpy as np

def nms_count(boxes, scores, thr=0.5):
    """Greedy NMS that also counts how many box-pair IoUs it computes."""
    order = np.argsort(-scores)
    keep, n_iou = [], 0
    while order.size:
        i, rest = order[0], order[1:]
        keep.append(i)
        n_iou += rest.size
        x1 = np.maximum(boxes[i, 0], boxes[rest, 0]); y1 = np.maximum(boxes[i, 1],
         ↪  boxes[rest, 1])
        x2 = np.minimum(boxes[i, 2], boxes[rest, 2]); y2 = np.minimum(boxes[i, 3],
         ↪  boxes[rest, 3])
        inter = np.clip(x2 - x1, 0, None) * np.clip(y2 - y1, 0, None)
        area = lambda b: (b[..., 2] - b[..., 0]) * (b[..., 3] - b[..., 1])
        iou = inter / (area(boxes[i]) + area(boxes[rest]) - inter)
        order = rest[iou <= thr]
    return len(keep), n_iou

def random_boxes(n, n_objects, rng, size=640):
    """n candidate boxes jittered around n_objects true objects."""
    centres = rng.uniform(40, size - 40, (n_objects, 2))
    owner = rng.integers(0, n_objects, n)
    c = centres[owner] + rng.normal(0, 6, (n, 2))
    wh = rng.uniform(30, 60, (n, 2))
    return np.hstack([c - wh / 2, c + wh / 2]), rng.random(n)

rng = np.random.default_rng(0)
for n_objects in (5, 500):                        # few objects vs a crowded scene
    for n in (100, 1000, 8400):
        b, s = random_boxes(n, n_objects, rng)
        t0 = time.perf_counter()
        kept, n_iou = nms_count(b, s)
        ms = (time.perf_counter() - t0) * 1e3
        print(f"objects {n_objects:3d} | boxes {n:4d} | kept {kept:4d} | IoUs {n_iou:9,d} |
         ↪  {ms:6.1f} ms")
```

```
objects   5 | boxes  100 | kept   21 | IoUs       770 |    0.8 ms
objects   5 | boxes 1000 | kept   43 | IoUs     7,809 |    1.4 ms
```

```
objects   5 | boxes 8400 | kept   86 | IoUs    90,699 |    6.8 ms
objects 500 | boxes  100 | kept   94 | IoUs     4,621 |    2.2 ms
objects 500 | boxes 1000 | kept  513 | IoUs   209,490 |   22.1 ms
objects 500 | boxes 8400 | kept 1259 | IoUs 3,039,292 |  154.7 ms
```

(One run on one CPU core; your times will differ, the IoU counts will not.) With the same 8,400 candidates, the crowded scene needs about 34 times as many IoU computations as the sparse one. Real implementations run on the GPU and cap the number of boxes, so absolute times are much smaller, but the shape of the effect is the same: post-processing time grows with the number of candidates and with crowding. That is the work an NMS-free detector avoids. Figure 1 repeats the experiment over a range of candidate counts.

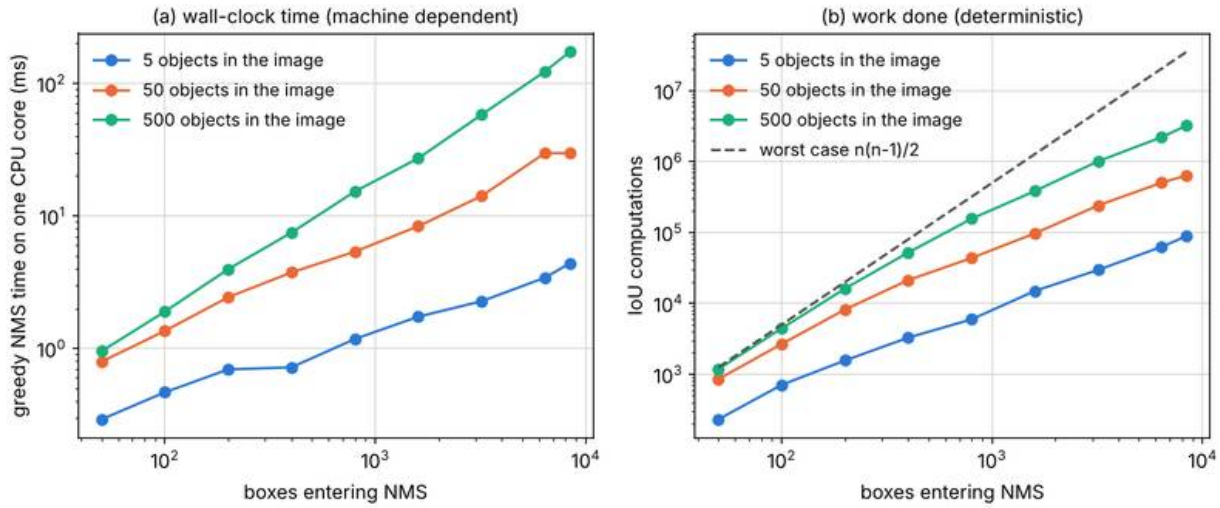


Figure 1 — Cost of greedy NMS on synthetic candidates (toy run in NumPy). (a) Wall-clock time on one CPU core, which depends on the machine. (b) Number of IoU computations, which is deterministic; the dashed line is the worst case $n(n-1)/2$.

Anatomy of a modern one-stage detector

Plain version. Almost every real-time detector has three parts. A **backbone** turns the image into feature maps at several resolutions. A **neck** mixes information between those resolutions. A **head** looks at every cell of every map and predicts "is there an object here, of which class, and where are its edges?"

Backbone, neck and head

The **backbone** is a convolutional network (or, increasingly, a network with some attention layers) that downsamples the image in stages. The last three stages, usually called C_3, C_4, C_5, have **strides** 8, 16 and 32: one cell of C_5 summarizes a 32×32 pixel patch. Deep stages see more context but have coarse resolution; shallow stages are sharp but semantically weak.

The **neck** fixes that mismatch. A Feature Pyramid Network (FPN) [30] adds a top-down path, so that coarse, semantic features are upsampled and added to finer levels. A Path Aggregation Network (PAN) [31] then adds a bottom-up path, so that precise, high-resolution information flows back up to the coarse levels. The outputs P_3, P_4, P_5 keep the strides 8, 16 and 32. YOLOv4 lists PAN among its path-aggregation choices [8], and most later YOLO versions use an FPN-plus-PAN neck in some form.

The **head** runs a few small convolutions on every pyramid level. Early YOLO heads were *coupled*: one set of convolutions produced the class scores and the box together. YOLOX showed that a **decoupled head**, with separate branches for classification and regression, converges faster and improves accuracy [10]; nearly every later version uses one. Figure 2 summarizes the design.

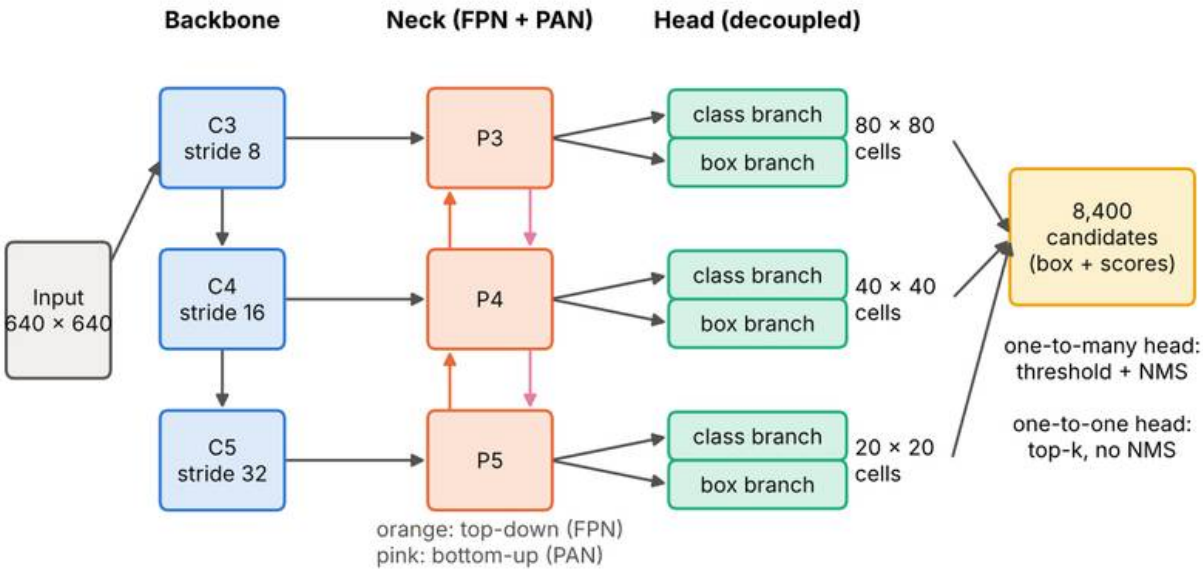


Figure 2 — The backbone–neck–head design shared by modern one-stage detectors, for a 640 × 640 input. Each cell of each grid makes one prediction (anchor-free case), giving 8,400 candidates per image.

For a 640×640 input and strides 8, 16 and 32, the three grids have $80^2 + 40^2 + 20^2 = 8{,}400$ cells. An anchor-free head makes one prediction per cell; an anchor-based head with A anchors per cell makes $8{,}400\,A$. Each prediction is a vector of $4 + K$ numbers (box and K class scores), plus an objectness score in some designs. That output tensor is what post-processing has to turn into a short list of boxes.

Anchor-based and anchor-free prediction

Plain version. An anchor-based head says "start from this template box and adjust it". An anchor-free head says "from this cell centre, the object's left edge is this far, its top edge is that far", and so on.

Anchor-based (YOLOv2 to YOLOv5, YOLOv7). YOLOv2 replaced hand-picked anchors with shapes found by k-means clustering of the training boxes (it chose $k = 5$) [6]. For a cell with top-left corner (c_x, c_y) (in grid units) and an anchor of width p_w and height p_h, the network outputs $(t_x, t_y, t_w, t_h, t_o)$ and the box is decoded as

$$b_x = \sigma(t_x) + c_x, \qquad b_y = \sigma(t_y) + c_y, \qquad b_w = p_w\, e^{t_w}, \qquad b_h = p_h\, e^{t_h},$$

where σ is the logistic sigmoid, (b_x, b_y) is the box centre and (b_w, b_h) its size, all in grid units (multiply by the stride for pixels), and $\sigma(t_o)$ is the predicted objectness, trained toward $\Pr(\text{object}) \cdot \text{IoU}$ [6]. The sigmoid keeps the centre inside its own cell, which the YOLOv2 paper introduced to make early training stable [6]. YOLOv3 kept this scheme, predicted 3 boxes per cell at 3 scales, and replaced the softmax over classes by independent logistic classifiers [7].

Anchor-free (YOLOX, YOLOv6, YOLOv8 and later). YOLOX reduced the predictions per location from 3 to 1 and predicted box offsets directly, without anchors [10]. The common form today predicts, at a cell centre (x, y) on a level with stride s, four non-negative distances to the box sides, in units of the stride:

$$x_1 = x - s\,\hat{l}, \qquad y_1 = y - s\,\hat{t}, \qquad x_2 = x + s\,\hat{r}, \qquad y_2 = y + s\,\hat{b},$$

where $(\hat{l}, \hat{t}, \hat{r}, \hat{b})$ are the predicted left, top, right and bottom distances and (x_1, y_1, x_2, y_2) the decoded box corners. This is the same parameterization as the anchor-free detectors of DLCV 03. The Ultralytics documentation describes YOLOv8 as using an "anchor-free split" head [13]. Dropping anchors removes anchor hyperparameters, makes the head simpler, and produces fewer candidates, which also means less NMS work [23].

Decoding the output tensor

The snippet builds the 8,400 cell centres, plants one confident prediction in random background noise, and decodes it with the anchor-free formula.

```
import numpy as np

def make_grid(img_size=640, strides=(8, 16, 32)):
    """Centre (x, y) in pixels and stride of every prediction, level by level."""
    centres, st = [], []
    for s in strides:
        n = img_size // s
        ys, xs = np.mgrid[0:n, 0:n]
        centres.append(np.stack([(xs.ravel() + 0.5) * s, (ys.ravel() + 0.5) * s], axis=1))
```

```
        st.append(np.full(n * n, s))
    return np.concatenate(centres).astype(float), np.concatenate(st).astype(float)

def decode(raw, centres, strides, conf=0.25):
    """raw: (N, 4 + K) = distances (l, t, r, b) in stride units, then K class logits."""
    ltrb = raw[:, :4] * strides[:, None]                    # back to pixels
    boxes = np.concatenate([centres - ltrb[:, :2], centres + ltrb[:, 2:]], axis=1)
    scores = 1 / (1 + np.exp(-raw[:, 4:]))                  # one sigmoid per class
    cls, best = scores.argmax(axis=1), scores.max(axis=1)
    keep = best > conf                                      # confidence threshold
    return boxes[keep], cls[keep], best[keep]

centres, strides = make_grid()
print("predictions per 640x640 image:", len(centres))

rng = np.random.default_rng(0)
K = 3                                                       # three classes
raw = np.concatenate([rng.uniform(0, 4, (len(centres), 4)),  # random distances
                      rng.normal(-6, 1, (len(centres), K))], axis=1)  # background logits
# plant one confident object (class 1) at the stride-16 cell centred on (296, 200)
i = np.flatnonzero((strides == 16) & (centres[:, 0] == 296) & (centres[:, 1] == 200))[0]
raw[i, :4] = [5.0, 3.0, 4.0, 2.5]
raw[i, 4 + 1] = 4.0
boxes, cls, score = decode(raw, centres, strides)
print("kept:", len(boxes), "| box:", boxes[0], "| class:", cls[0], "| score:",
 ↪  round(float(score[0]), 3))
```

```
predictions per 640x640 image: 8400
kept: 1 | box: [216. 152. 360. 240.] | class: 1 | score: 0.982
```

The planted distances $(5, 3, 4, 2.5)$ at stride 16 become $(80, 48, 64, 40)$ pixels, giving the box from $(216, 152)$ to $(360, 240)$. In a trained detector many neighbouring cells would fire on the same object; what happens to those duplicates is the subject of the label-assignment section below.

The YOLO lineage

Plain version. "YOLO" is not one model but a name used by many teams over a decade. The useful way to read the history is to ignore the version numbers and follow a few ideas: grids, then anchors, then multi-scale heads, then better training recipes, then anchor-free heads with learned assignment, and most recently attention and NMS-free heads.

A timeline

Figure 3 places the versions on a timeline by the year they first appeared in public, grouped by who released them, and marks whether each has a peer-reviewed paper. Three facts are easy

to miss. First, **version numbers do not form one lineage**: after YOLOv3, its original authors stopped, and later numbers were taken by different groups, sometimes within weeks of each other. Second, several of the most widely used versions (YOLOv5, YOLOv8, YOLO11) were released as **software without a paper**; their official documentation says so [9], [13], [16]. Third, the name is unregulated: in August 2026 a preprint titled "YOLOv14" was posted to arXiv and withdrawn by its authors about two weeks later [52], so a new number by itself says nothing about quality.

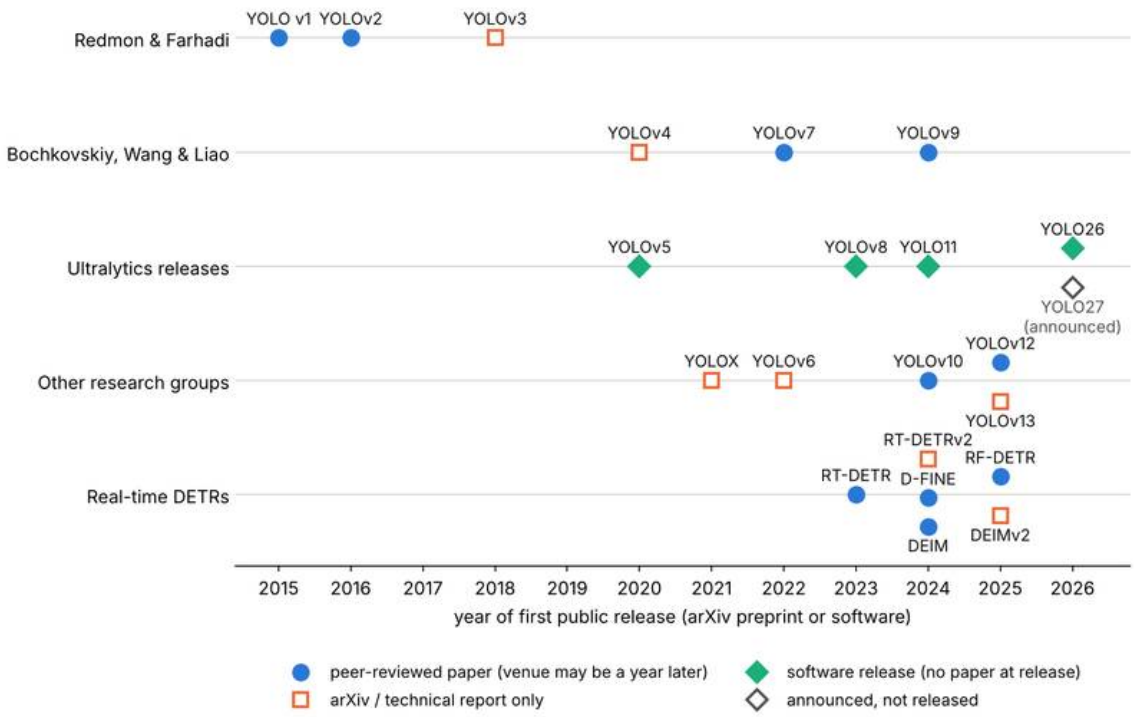


Figure 3 — The YOLO family and real-time DETRs by year of first public release. Filled circles have a peer-reviewed paper (often published a year after the preprint), open squares are arXiv preprints or technical reports only, filled diamonds are software releases without a paper at release time, and the open diamond is announced but not released.

The table lists each version's primary source and the idea worth remembering. "Paper" means the version was introduced by a paper; the venue column says whether that paper was peer reviewed.

Version	First public	Primary source	Peer reviewed?	The idea to remember
YOLO v1 [5]	2015	Paper (CVPR 2016)	Yes	One network predicts all boxes from an $S \times S$ grid in one pass

Version	First public	Primary source	Peer reviewed?	The idea to remember
YOLOv2 [6]	2016	Paper (CVPR 2017)	Yes	Anchors from k-means "dimension clusters"; sigmoid-bounded centres
YOLOv3 [7]	2018	Tech report (arXiv)	No	Predictions at 3 scales; independent logistic class scores
YOLOv4 [8]	2020	arXiv	No	A systematic "bag of freebies" (training tricks) and "bag of specials" (cheap modules); mosaic augmentation
YOLOv5 [9]	2020	Software (Ultralytics)	No paper	A widely used PyTorch implementation and training pipeline
YOLOX [10]	2021	arXiv	No	Anchor-free, decoupled head, SimOTA dynamic label assignment
YOLOv6 [11]	2022	Tech report (arXiv)	No	Re-parameterizable backbone and neck; task-aligned assignment; quantization-aware design
YOLOv7 [12]	2022	Paper (CVPR 2023)	Yes	E-ELAN aggregation; planned re-parameterization; lead-guided assignment
YOLOv8 [13]	2023	Software (Ultralytics)	No paper	Anchor-free split head in a multi-task package
YOLOv9 [14]	2024	Paper (ECCV 2024)	Yes	Programmable gradient information (PGI); GELAN
YOLOv10 [15]	2024	Paper (NeurIPS 2024)	Yes	NMS-free training with consistent dual assignments
YOLO11 [16]	2024	Software (Ultralytics)	No paper	Refined backbone and neck; same package and tasks
YOLOv12 [17]	2025	Paper (NeurIPS 2025)	Yes	Attention-centric design (area attention) at real-time speed
YOLOv13 [18]	2025	arXiv	No	Hypergraph-based correlation module (HyperACE)

Version	First public	Primary source	Peer reviewed?	The idea to remember
YOLO26 [19]	2026	Software (Jan 2026); arXiv paper (Jun 2026)	No	NMS-free by default; DFL removed; new optimizer and assignment for small objects
YOLO27 [20]	—	Announced only	—	Not released as of October 2026; no weights or code

Optional: The YOLO family, version by version (in the Supplementary notes). Two or three paragraphs per version: what changed, what the paper or documentation says about it, and its limits.

Five ideas to remember

You do not need to remember sixteen versions. Five ideas carry most of the progress.

1. **One pass over a grid.** YOLO v1 made detection a single regression from the image to a fixed-size tensor: a 7×7 grid with 2 boxes per cell and 20 class scores, a $7 \times 7 \times 30$ tensor for PASCAL VOC [5]. Every later version keeps this "look once, predict densely" structure.
2. **Several scales, joined by a neck.** From YOLOv3 on, predictions come from several strides [7], and from YOLOv4 on the levels are fused by FPN/PAN-style necks [8]. Small objects are found on the stride-8 grid, large ones on the stride-32 grid.
3. **The training recipe is half the model.** YOLOv4's main contribution was to test many training-time tricks ("bag of freebies", which cost nothing at inference) and cheap modules ("bag of specials") systematically [8]. Later versions differ as much in augmentation, assignment and losses as in architecture.
4. **Learned label assignment.** Deciding which predictions are trained as positives for which object is a design choice, not a detail. YOLOX's SimOTA picks a dynamic number of lowest-cost predictions per object [10]; YOLOv6 and YOLOv8-era models use task-aligned assignment, which ranks predictions by a mix of class score and IoU [11], [34].
5. **Removing NMS.** YOLOv10 trains a second, one-to-one head alongside the usual one-to-many head and uses only the one-to-one head at inference, so no NMS is needed [15]. YOLO26 makes this the default deployment path [19]. This is the same idea DETR introduced (see below), now inside a YOLO.

Two further threads matter for deployment: **re-parameterization** (train a multi-branch block, then fold it into a single convolution for inference; RepVGG [32], used in YOLOv6 [11] and YOLOv7 [12]) and **attention at real-time cost** (YOLOv12's area attention [17], YOLOv13's hypergraph module [18]).

Optional: Folding a re-parameterized block into one convolution (in the Supplementary notes). The batch-norm folding formula and a NumPy check that three branches become one convolution with the same output.

Real-time DETRs

Plain version. DETR showed that a transformer can output the final set of boxes directly, with no anchors and no NMS, but it was slow to train and slow to run. Real-time DETRs keep the "direct set" idea and redesign the expensive parts.

Why the original DETR was slow

DETR [21] (introduced in DLCV 03) had two separate speed problems.

- **Slow training.** With one-to-one Hungarian matching [40], each object supervises exactly one of the object queries, so each image provides few positive training signals, and the queries start from learned embeddings that carry no information about the image. DETR therefore needed very long training schedules.
- **Expensive encoder.** Self-attention over all feature-map positions costs time quadratic in the number of positions. DETR used only one coarse feature map to stay affordable, which hurt small objects. Deformable DETR [22] made multi-scale features affordable by letting each query attend to a few sampled points, but RT-DETR's analysis found that the multi-scale encoder then became the computational bottleneck, because concatenating all levels makes the token sequence very long [23].

RT-DETR

RT-DETR (Zhao et al., CVPR 2024; the arXiv title is "DETRs beat YOLOs on real-time object detection") [23] was the first DETR designed for real-time use. It feeds the last three backbone stages $\{S_3, S_4, S_5\}$ into an encoder and changes three things:

1. **Efficient hybrid encoder.** Attention is applied only *within* the coarsest level (a module the paper calls AIFI, attention-based intra-scale feature interaction), where tokens are few and semantics are rich. The levels are then fused across scales with convolutions (CCFF, CNN-based cross-scale feature fusion), as in a YOLO neck. The paper argues that attention over all levels together is redundant, since high-level features are computed from low-level ones [23].
2. **Uncertainty-minimal query selection.** Instead of learned queries that ignore the image, a fixed number of encoder features are selected as initial object queries. The paper defines a feature's uncertainty as the disagreement between its predicted localization quality and its classification score, $U(\hat{X}) = \|P(\hat{X}) - C(\hat{X})\|$, where $\hat{X}$ is an encoder feature and P and C are its localization

and classification predictions, and adds it to the classification loss so that selected queries are both confident and well localized [23].

3. **Adjustable decoder depth.** Because the decoder refines boxes layer by layer and every layer has a prediction head, layers can be removed at inference to trade accuracy for speed **without retraining**; the paper observes that removing a few final decoder layers costs little accuracy [23].

Like DETR, RT-DETR is trained with one-to-one matching and needs **no NMS**. The paper reports 53.1% and 54.3% AP on COCO at 108 and 74 FPS on an NVIDIA T4 GPU for its ResNet-50 and ResNet-101 versions [23]; those FPS values come from its own end-to-end benchmark (TensorRT, FP16), which also times the NMS of the YOLO models it compares against.

Optional: Reading the RT-DETR paper (in the Paper guides). Its NMS analysis, its end-to-end speed benchmark, and what to check in its comparison tables.

After RT-DETR

Real-time DETRs moved quickly between 2024 and 2026. The successors below are the ones whose primary sources we verified; each keeps one-to-one matching at inference.

Model	Source	Main idea
RT-DETRv2 [24]	arXiv, 2024	"Bag of freebies" for RT-DETR: different numbers of sampling points per scale; an optional discrete sampling operator that is easier to deploy; dynamic augmentation and scale-adaptive hyperparameters
D-FINE [25]	ICLR 2025	Boxes as probability distributions refined layer by layer (fine-grained distribution refinement), plus self-distillation from the last decoder layer to earlier ones
DEIM [26]	CVPR 2025	Faster convergence: more positives per image through "dense one-to-one" matching (more objects per image via mosaic and mixup) and a matchability-aware loss
RF-DETR [27]	ICLR 2026	A pretrained vision-transformer backbone (DINOv2 [47]) and weight-sharing neural architecture search that finds many accuracy–latency trade-offs without retraining

Model	Source	Main idea
DEIMv2 [28]	arXiv, 2025	DEIM with DINOv3 [48] features adapted to multi-scale detection for the larger models, and pruned convolutional backbones for the smallest ones

Optional: Real-time DETRs after RT-DETR, in detail (in the Supplementary notes). What each successor changes, how distribution refinement decodes a box, and what dense one-to-one matching does to the number of positives.

Label assignment and the NMS-free trend

Plain version. During training, every object must be "given" to some predictions, which are then taught to fire on it. If an object is given to many predictions (one-to-many), training is rich and fast, but at test time several predictions fire on the same object and NMS must clean up. If each object is given to exactly one prediction (one-to-one), duplicates are trained away and NMS is unnecessary, but each object provides less training signal. Modern detectors use both: one-to-many to train, one-to-one to predict.

One-to-many versus one-to-one

Let $\mathcal{P}$ be the set of predictions (one per cell for an anchor-free YOLO, one per query for a DETR) and $\mathcal{G}$ the ground-truth objects. Both styles start from a **matching score** between a prediction i and an object j. The task-aligned score introduced by TOOD [34], in the form written in the YOLOv10 paper [15], is

$$m_{ij} = s_{ij} \cdot p_{ij}^{\alpha} \cdot \mathrm{IoU}(\hat{b}_i, b_j)^{\beta},$$

where p_{ij} is the predicted score of prediction i for the class of object j, $\hat{b}_i$ and b_j are the predicted and true boxes, $s_{ij} \in \{0, 1\}$ is a **spatial prior** that is 1 only if the prediction's anchor point lies inside the object, and $\alpha, \beta \geq 0$ balance classification against localization. The YOLOv10 paper quotes $\alpha = 0.5$ and $\beta = 6$ as YOLOv8's defaults [15]. Because m uses the network's own current predictions, the assignment changes during training; this is what "dynamic" or "prediction-aware" assignment means.

- **One-to-many (o2m).** For each object j, the top-k predictions by m_{ij} become positives. TOOD uses the top 13 [34]; YOLOX's SimOTA uses a cost instead of a score and chooses k separately for each object [10], following OTA's optimal-transport view of assignment [33].

- **One-to-one (o2o).** Each object gets exactly one prediction, and each prediction at most one object. Solving this optimally is a bipartite matching problem, solved by the Hungarian algorithm [40] as in DETR [21]:

$$\hat{\pi} = \arg\max_{\pi} \sum_{j \in \mathcal{G}} m_{\pi(j)\, j},$$

where π ranges over injective maps from objects to predictions (no prediction is used twice). DETR-style detectors write the same problem as a minimum-cost matching with cost $-m$ or a combination of class, L1 and GIoU terms.

Why does one-to-one remove NMS? In one-to-many training, the second- and third-best predictions for an object are *taught to fire* on it, so at test time they do, and NMS is needed. In one-to-one training, every prediction other than the matched one is a negative, so the network is *taught not to produce duplicates*. Why not use one-to-one only? Because it provides far fewer positives, which slows convergence and weakens the features, the same problem that made DETR slow.

Dual assignment, hybrid matching and dense one-to-one

The solutions from both lines are variations of "use one-to-many for training signal, one-to-one for the output":

- **YOLOv10: consistent dual assignments** [15]. A second head with the same structure as the usual one is trained with one-to-one assignment, while the original head keeps one-to-many assignment; both share the backbone and neck, which therefore receive the rich one-to-many signal. At inference only the one-to-one head is used. The paper picks the one-to-one match by "top one selection", which it reports performs as well as Hungarian matching with less training time, and sets the same α and β for both heads ("consistent matching metric") so that the one-to-one head's best prediction is also highly ranked by the one-to-many head. In its ablation on the small model, measured end to end on a T4 GPU with TensorRT FP16 at batch 1 including NMS where it is used, the one-to-many head alone reached 44.9 AP at 7.07 ms, the one-to-one head alone 43.4 AP at 2.44 ms, and the dual-trained one-to-one head 44.3 AP at 2.44 ms [15]. The latency difference is NMS.
- **H-DETR: hybrid matching** [29]. A DETR is given an extra group of queries trained with one-to-many matching during training only, to add positives; inference uses only the one-to-one queries.
- **DEIM: dense one-to-one** [26]. Keep one-to-one matching, but increase the number of objects per training image (with mosaic and mixup) so that one-to-one matching produces more positives.
- **YOLO26: progressive loss** [19]. The total loss is a weighted sum of the one-to-many and one-to-one losses, $\mathcal{L} = \alpha(t)\, \mathcal{L}_{\text{o2m}} + (1 - \alpha(t))\, \mathcal{L}_{\text{o2o}}$, where $\alpha(t)$ decreases linearly over the training epochs t, so supervision shifts gradually toward the head that is used at inference.

Optional: Reading the YOLOv10 paper (in the Paper guides). The supervision-gap argument behind the consistent matching metric, and how to read its ablation tables.

A toy comparison

The snippet places an 8×8 grid of anchor points on a 100×100 image with three objects, gives every point a noisy predicted box and a score, computes m_{ij}, and compares top-4 one-to-many assignment with Hungarian one-to-one assignment.

```
import numpy as np
from scipy.optimize import linear_sum_assignment

def iou_matrix(a, b):
    x1 = np.maximum(a[:, None, 0], b[None, :, 0]); y1 = np.maximum(a[:, None, 1], b[None, :,
     ↪  1])
    x2 = np.minimum(a[:, None, 2], b[None, :, 2]); y2 = np.minimum(a[:, None, 3], b[None, :,
     ↪  3])
    inter = np.clip(x2 - x1, 0, None) * np.clip(y2 - y1, 0, None)
    area = lambda b: (b[:, 2] - b[:, 0]) * (b[:, 3] - b[:, 1])
    return inter / (area(a)[:, None] + area(b)[None, :] - inter)

rng = np.random.default_rng(3)
gt = np.array([[20, 20, 60, 70], [55, 30, 95, 80], [10, 80, 40, 100]], float)
# 8 x 8 grid of anchor points (stride 12.5 on a 100 x 100 image), each with a predicted box
ys, xs = np.mgrid[0:8, 0:8]
pts = np.stack([(xs.ravel() + 0.5) * 12.5, (ys.ravel() + 0.5) * 12.5], axis=1)
inside = ((pts[:, None, 0] > gt[None, :, 0]) & (pts[:, None, 0] < gt[None, :, 2]) &
          (pts[:, None, 1] > gt[None, :, 1]) & (pts[:, None, 1] < gt[None, :, 3]))      #
           ↪  spatial prior s
near = inside.argmax(axis=1)                        # the object each point "looks at"
pred = gt[near] + rng.normal(0, 5, (len(pts), 4))  # noisy box predictions
p = np.where(inside.any(axis=1), rng.uniform(0.3, 0.9, len(pts)), rng.uniform(0, 0.2,
 ↪  len(pts)))

alpha, beta = 0.5, 6.0                              # the defaults quoted in the YOLOv10 paper
m = inside * p[:, None] ** alpha * iou_matrix(pred, gt) ** beta   # (points, objects)

# one-to-many: the top-k points by m for every object become positives
k = 4
o2m = {}
for j in range(len(gt)):
    best = np.argsort(-m[:, j])[:k]
    o2m[j] = best[m[best, j] > 0].tolist()
# one-to-one: Hungarian matching on cost -m (each object gets exactly one point)
rows, cols = linear_sum_assignment(-m)
o2o = {int(c): [int(r)] for r, c in zip(rows, cols)}
print("one-to-many positives:", o2m)
print("one-to-one positives: ", dict(sorted(o2o.items())))
print("positives per image: o2m", sum(map(len, o2m.values())), "| o2o", len(o2o))
top1 = m.argmax(axis=0)                             # YOLOv10's cheaper top-1 choice
print("top-1 equals Hungarian here:", all(top1[j] == o2o[j][0] for j in o2o))
```

```
one-to-many positives: {0: [26, 34, 42, 43], 1: [37, 46, 38, 30], 2: [58, 50, 49, 57]}
one-to-one positives:  {0: [26], 1: [37], 2: [58]}
positives per image: o2m 12 | o2o 3
top-1 equals Hungarian here: True
```

One-to-many gives four times as many positives, and Figure 4 shows their cost: four overlapping boxes per object, all taught to fire. One-to-one keeps the single best point per object, which is also the top-ranked point under one-to-many, the property YOLOv10's consistent matching metric is designed to encourage. Here the cheap per-object top-1 choice agrees with Hungarian matching because no two objects compete for the same point; when they do, top-1 can assign one point to two objects and Hungarian matching cannot.

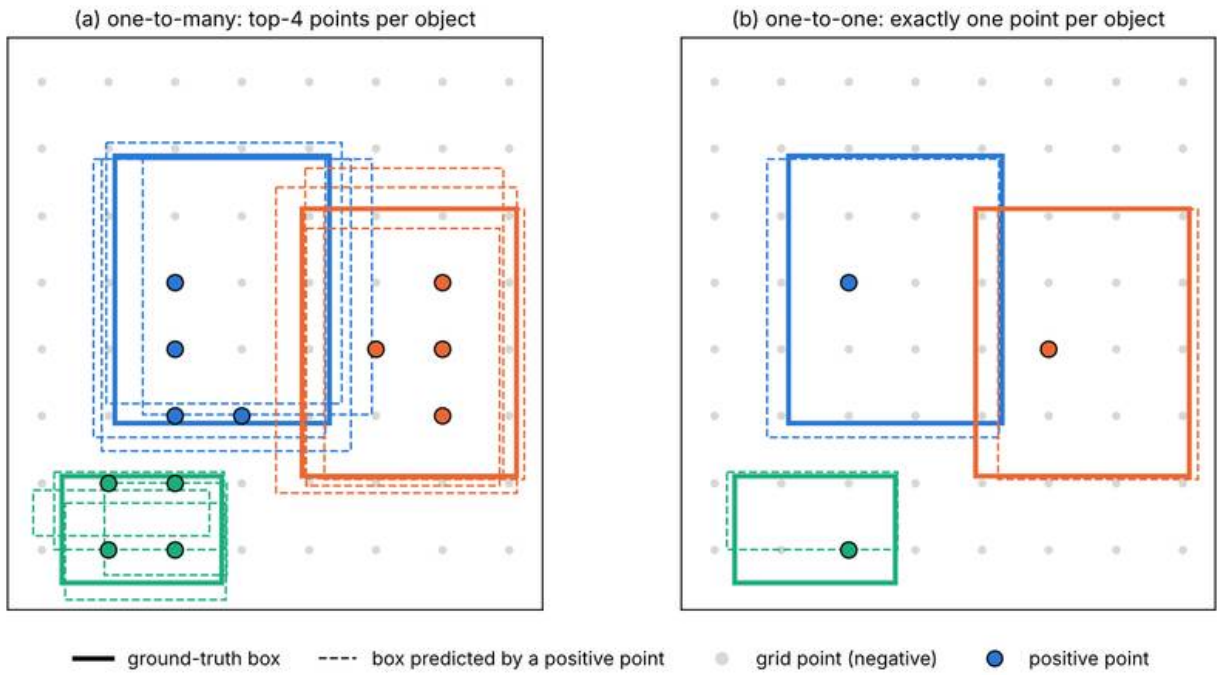


Figure 4 — Label assignment on a toy scene. (a) One-to-many: the top-4 points per object by the task-aligned score are positives, so four overlapping predictions per object are trained to fire. (b) One-to-one (Hungarian): exactly one positive per object; every other point is trained as background.

Training recipe essentials

Plain version. Real-time detectors are small, so they depend more than large models on how they are trained. The recipe has three parts: augmentations that make every image harder and more varied, a smoothed copy of the weights for evaluation, and losses that match the evaluation metric.

Data augmentation: mosaic and mixup

Mosaic, introduced in YOLOv4, combines 4 training images into one, so each training image shows objects in 4 different contexts and at reduced scale [8]. **Mixup** forms a convex combination of two training examples,

$$\tilde{x} = \lambda\, x_a + (1-\lambda)\, x_b, \qquad \lambda \sim \mathrm{Beta}(a, a),$$

where x_a, x_b are two images, $\lambda \in [0,1]$ the mixing weight and $a > 0$ a hyperparameter [38]; it was designed for classification and later adapted to detection, as the YOLOX report notes [10]. YOLOX turns both augmentations off for the last 15 epochs, so that training ends on images that look like test images, and found that with this strong augmentation ImageNet pre-training was no longer beneficial [10]. DEIM uses mosaic and mixup on purpose to put more objects in each image, which gives one-to-one matching more positives [26].

Weight averaging (EMA)

The YOLOX baseline adds "EMA weights updating" to its training [10]. The model used for evaluation is an exponential moving average of the training weights,

$$\bar{\theta}_t = d\,\bar{\theta}_{t-1} + (1-d)\,\theta_t,$$

where θ_t are the weights after training step t, $\bar{\theta}_t$ the averaged weights, and d a decay close to 1. The average smooths out the noise of the last optimization steps, at no inference cost: only $\bar{\theta}$ is exported.

Loss components

A modern real-time detector's loss has a classification term, one or more box terms, and, for NMS-free models, a one-to-one counterpart of each.

- **Classification with quality-aware targets.** Instead of training the class score toward 1 for positives, several losses train it toward the positive's IoU, so that the score also ranks localization quality. The quality focal loss of GFL [35] and the varifocal loss of VarifocalNet [36] are the two standard forms. The varifocal loss is

$$\mathrm{VFL}(p,q) = \begin{cases} -q\,(q\log p + (1-q)\log(1-p)) & q > 0, \\ -\alpha\, p^{\gamma}\log(1-p) & q = 0, \end{cases}$$

where p is the predicted score, q the target (the IoU between the predicted box and its ground truth for a positive, 0 for a negative), and α, γ down-weight easy negatives as in the focal loss [39]. YOLOv6 selected VFL as its classification loss [11].

- **IoU-type box losses.** CIoU (defined in DLCV 03 and introduced with DIoU [37]) is part of YOLOv4's bag of freebies [8]; YOLOv6 chose among SIoU and GIoU losses [11].
- **Distribution focal loss (DFL)** [35]. Instead of predicting one number per box side, the head predicts a discrete probability distribution over $n+1$ values $y_0 < y_1 < \ldots < y_n$ (spaced Δ apart; GFL uses $\Delta = 1$ in stride units) and decodes the side as its expected value,

$$\hat{y} = \sum_{i=0}^{n} S_i\, y_i, \qquad \mathrm{DFL}(S_i, S_{i+1}) = -\big((y_{i+1} - y)\log S_i + (y - y_i)\log S_{i+1}\big),$$

where S_i is the softmax probability of value y_i, y is the true distance and $y_i \le y \le y_{i+1}$ are its two neighbouring values. DFL pushes probability onto the two values that bracket the target; its minimum is at $S_i = (y_{i+1}-y)/(y_{i+1}-y_i)$ and $S_{i+1} = (y-y_i)/(y_{i+1}-y_i)$, where the expected value equals y exactly [35]. A broad distribution signals an uncertain edge, for example an occluded one. The GFL paper found results insensitive to n (it tested 12 to 18) and suggests a small Δ such as 1 [35]; the YOLO26 paper describes DFL as adopted in Ultralytics models since YOLOv8, with 16 bins per side [19].

```
import numpy as np

K, stride = 16, 8                          # 16 bins per box side, a stride-8 level
bins = np.arange(K)                        # support y_0 = 0, ..., y_15 = 15 (in stride units)

def softmax(z):
    e = np.exp(z - z.max())
    return e / e.sum()

def dfl(logits, y):
    """Distribution focal loss for one box side with continuous target y (stride units)."""
    S = softmax(logits)
    i = int(np.floor(y))                   # neighbours y_i <= y <= y_i + 1
    return -((i + 1 - y) * np.log(S[i]) + (y - i) * np.log(S[i + 1]))

y = 5.3                                    # true distance: 5.3 cells = 42.4 pixels
flat = np.zeros(K)                         # untrained head: uniform distribution
sharp = np.full(K, -10.0); sharp[5], sharp[6] = np.log(0.7), np.log(0.3)   # the ideal shape
for name, z in [("uniform", flat), ("ideal", sharp)]:
    S = softmax(z)
    print(f"{name:7s} expected distance {S @ bins:5.2f} cells = {S @ bins * stride:6.1f} px
     ↪  | DFL {dfl(z, y):.3f}")
print("largest distance this head can output:", (K - 1) * stride, "px")
```

```
uniform expected distance  7.50 cells =   60.0 px | DFL 2.773
ideal   expected distance  5.30 cells =   42.4 px | DFL 0.611
largest distance this head can output: 120 px
```

Two lessons. The ideal distribution decodes to exactly 5.3 cells, but its loss is not zero (0.611 is the entropy of the split 0.7 / 0.3), so DFL values are not "errors in pixels". And the support is finite:

with 16 bins a side can be at most 15 strides long. The YOLO26 paper gives this range limit and the extra head parameters as its reasons for removing DFL [19]. Figure 5 shows the three cases.

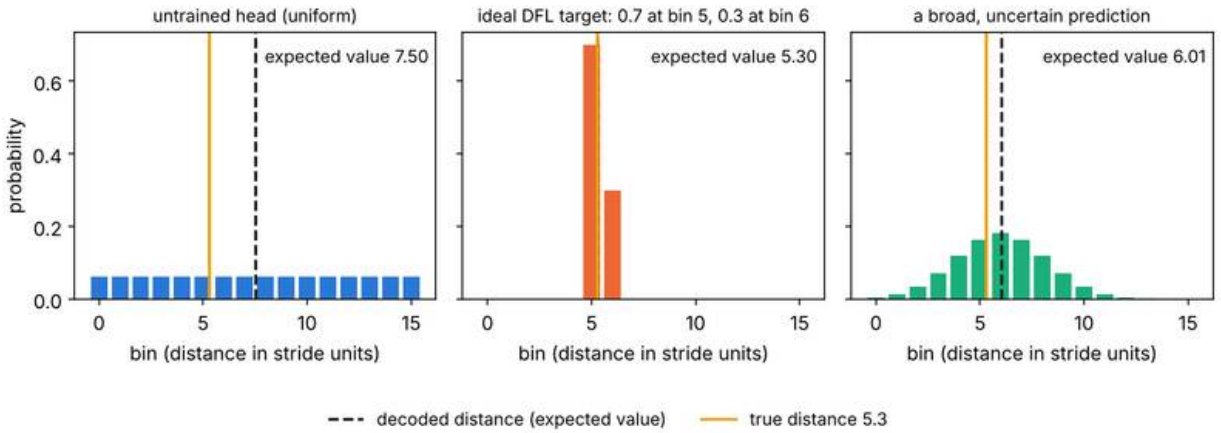


Figure 5 — How a distribution-based box side is decoded. The dashed line is the expected value (the predicted distance); the yellow line is the true distance 5.3. DFL trains the head toward the middle panel; a broad distribution, as on the right, expresses uncertainty about the edge.

Evaluation and deployment

Plain version. Accuracy is measured the same way for every detector; speed is not. Most mistakes in comparing real-time detectors come from speed numbers measured under different conditions.

Accuracy: COCO mAP

Real-time detectors are compared on COCO [41] with the COCO protocol [42] described in DLCV 03: AP averaged over 10 IoU thresholds from 0.50 to 0.95 and over the 80 categories, written AP or mAP@[.5:.95], with at most 100 detections per image. Three cautions apply specifically to this family:

- **Split.** Most papers report on val2017; some also report test-dev. Do not mix them.
- **Extra data.** Some results use pre-training on Objects365 [43] (RT-DETR reports both [23]); such numbers belong in a separate column.
- **Confidence threshold.** AP is computed from a ranked list, so evaluation uses a very low confidence threshold (the RT-DETR analysis cites 0.001 for YOLOv8's best AP [23]). A deployed system uses a much higher one. The speed of an NMS-based detector at the evaluation threshold is therefore not its speed in deployment, and vice versa.

Speed: a measurement checklist

When you measure, or read, a latency number, check:

1. **Hardware, engine and precision** are stated (for example, T4 GPU, TensorRT, FP16), and the same for every model being compared.
2. **Batch size 1** for latency; throughput numbers with large batches do not transfer to a live camera.
3. **End to end**: pre-processing, the network and post-processing including NMS, or a clear statement of what is excluded. YOLOv10 and RT-DETR time NMS with the TensorRT efficient-NMS plugin and average over the COCO val images, because NMS time depends on the image [15], [23].
4. **Warm-up and averaging**: the first iterations of any engine are slow (memory allocation, kernel selection), so discard them and average over many images.
5. **Input size and the confidence threshold** used during timing.

Export, reduced precision and re-parameterization

Training frameworks are not deployment engines. A trained detector is usually **exported** to an exchange format such as ONNX [44] and compiled by an inference engine for the target device, for example TensorRT for NVIDIA GPUs [45]. Three things change on the way:

- **Precision.** FP16 halves memory traffic with little accuracy change on modern GPUs. INT8 quantization stores weights and activations as 8-bit integers with a scale per tensor or channel, using integer arithmetic at inference [46]. It needs calibration data, and the box and score outputs are the most sensitive parts, so check AP at strict IoU thresholds after quantizing, not only AP50. YOLOv6 reports that re-parameterized models are harder to quantize and changed its training to fix this [11].
- **Graph simplification.** Re-parameterized blocks are folded into single convolutions, and batch normalization is folded into the preceding convolution (see the Supplementary notes).
- **Unsupported operators.** Every operator must exist in the target engine. RT-DETRv2 added a discrete sampling operator for this reason [24], and YOLO26 cites simpler export as one reason for removing DFL [19]. NMS-free models export more simply, because the output already is the final list.

Failure modes and how to choose a model

Plain version. A real-time detector fails where any detector fails (small, crowded, unusual objects), and a few places of its own: low resolution, quantization, and speed numbers that do not hold on your device.

Failure modes to test for.

- **Small objects.** At 640 × 640 input, an object of 16 × 16 pixels covers 2 × 2 cells of the finest (stride 8) grid. YOLO26 addresses one version of this problem explicitly: with center-based assignment, a tiny box may contain no cell centre at all and get no positive sample, so its small-target-aware assignment enlarges tiny boxes for candidate selection only [19].
- **Crowds.** NMS-based models delete true neighbours at strict IoU thresholds or keep duplicates at loose ones (DLCV 03 discusses Soft-NMS); one-to-one models avoid the threshold but must have learned to separate neighbours.
- **Domain shift.** COCO-trained weights know everyday photographs; aerial, medical, industrial and night images need fine-tuning. RF-DETR was designed for fine-tuning on new target datasets [27].
- **Large objects at high resolution** for distribution-based heads, because of the finite range discussed above [19].
- **Deployment drift.** Re-measure accuracy after INT8 export, a new input size or a new confidence threshold; the paper's number does not cover them.

How to choose.

1. Fix the **latency budget** on the *target device*, end to end, at batch 1.
2. Take two or three candidate families (for example, one recent YOLO and one real-time DETR) and, for each, the largest size that fits the budget **on your device**.
3. Evaluate them on **your data** (after fine-tuning), with AP and with the error types that matter for the application (misses of small objects, false alarms).
4. Prefer NMS-free models when post-processing is a large part of the budget, when the scene is crowded, or when the deployment engine handles NMS poorly.
5. Check the **license and maintenance** of the code. The Ultralytics releases, for example, are distributed under the AGPL-3.0 license according to their citation entries [13], [16], which matters for commercial products.

If the class list is open-ended ("find any forklift"), a closed-set detector is the wrong tool; see the open-vocabulary detectors of DLCV 03.

Modern view

What the surveys say

A Comprehensive Review of YOLO Architectures in Computer Vision: From YOLOv1 to YOLOv8 and YOLO-NAS (Terven, Córdova-Esparza and Romero-González, *Machine Learning and Knowledge Extraction*, 2023) [1] walks through 16 versions in order, starting from the metrics and post-processing, then the architecture and training tricks of each. Its discussion organizes the history by four axes: anchors (absent in YOLO v1, added in YOLOv2, removed again from YOLOX on), framework (Darknet, then PyTorch), backbone (Darknet, then cross-stage partial connections, re-parameterization and architecture search), and the constant speed–accuracy trade-off. It expects COCO to be replaced by harder benchmarks and YOLO models to expand into tracking and multimodal tasks.

YOLO Evolution: A Comprehensive Benchmark and Architectural Review of YOLOv12, YOLO11, and Their Previous Versions (Jegham, Koh, Abdelatti and Hendawi, arXiv, 2024, revised 2025) [2] is an experimental comparison rather than a history: it evaluates versions from YOLOv3 to YOLOv12 on datasets chosen for varying object sizes, aspect ratios and small single-class objects, and reports precision, recall, mAP, processing time, GFLOPs and model size. Its takeaway is that no version wins everywhere: in its experiments YOLO11 gave the best overall balance, YOLOv10 was fast but weaker on overlapping objects, and YOLOv12's added complexity did not pay off. Treat these as one group's results on its datasets, not general rankings.

Ultralytics YOLO Evolution: An Overview of YOLO27, YOLO26, YOLO11, YOLOv8, and YOLOv5 (Sapkota and Karkee, arXiv, 2025, revised 2026) [3] reviews only the Ultralytics releases, covering architecture, benchmarking, deployment (export and quantization) and applications, and lists open challenges: dense scenes, combining CNNs with transformers, open-vocabulary perception, domain generalization and hardware-aware optimization. It is a third-party review, not an Ultralytics publication.

Object Detection with Transformers: A Review (Shehzadi et al., *Sensors*, 2025) [4] covers the DETR side: improvements to DETR grouped by backbone, query design and attention mechanism, which is the context for the real-time DETRs above.

The state of the art, 2023–2026

- **NMS-free is becoming the default.** YOLOv10 [15] introduced it to YOLOs in 2024; YOLO26 [19] ships it as the default deployment path; every real-time DETR [23]–[28] is NMS-free by

construction. The announced YOLO27 describes query-based decoding without NMS for its larger models [20].

- **Attention is affordable.** YOLOv12 [17] and YOLOv13 [18] put attention-like modules inside YOLOs; RT-DETR [23] and its successors restrict attention to where it is cheap.
- **Pretrained vision transformers as backbones.** RF-DETR [27] and DEIMv2 [28] start from self-supervised backbones (DINOv2 [47], DINOv3 [48]), which mainly helps transfer to new datasets.
- **Training efficiency.** DEIM [26] and YOLO26 [19] (with a new optimizer and progressive loss) target shorter training rather than only higher AP.

Open problems

1. **Fair speed comparison.** There is no shared, maintained end-to-end benchmark across devices; most tables still mix conditions.
2. **Small and crowded objects** remain the main accuracy gap at real-time input sizes.
3. **Quantization-friendly heads.** Score and box outputs lose accuracy at INT8, and attention layers complicate low-precision deployment.
4. **Peer review and reproducibility.** Several widely deployed versions have no peer-reviewed paper, and version numbers are claimed freely; claims need independent checks.
5. **Beyond closed sets.** Bringing open-vocabulary detection to the same latency as closed-set YOLOs without losing accuracy.

Key takeaways

- "Real-time" is a latency budget for the whole pipeline at batch size 1 on a stated device; a speed number without hardware, precision, input size and timed stages is not comparable.
- NMS cost depends on the number of candidates and on the scene, which is why end-to-end benchmarks include it and why NMS-free detectors are attractive.
- Modern one-stage detectors share a backbone, an FPN/PAN neck and a decoupled head on stride 8/16/32 grids; anchor-free heads predict four distances per cell.
- The YOLO name covers many teams; YOLO v1, v2, v7, v9, v10 and v12 have peer-reviewed papers, while YOLOv5, YOLOv8 and YOLO11 are software releases, and YOLO26's paper appeared on arXiv after its release.
- The ideas that matter are single-pass grid prediction, multi-scale necks, systematic training recipes, learned label assignment, re-parameterization and, now, NMS-free one-to-one heads.
- One-to-many assignment gives rich training signal but duplicates; one-to-one assignment removes duplicates but gives few positives; dual heads, hybrid matching and dense one-to-one combine the two.

- RT-DETR made DETRs real-time with a hybrid encoder, uncertainty-minimal query selection and adjustable decoder depth; its successors improve box representation, convergence and backbones.

Exercises

1. **Budget arithmetic.** A camera runs at 25 FPS. Pre-processing takes 3 ms, the network 18 ms, and NMS between 1 ms (empty street) and 9 ms (crowd). Does the system keep up in both cases, and how much time is left for tracking? Which single change would help most in crowds?

   Hint

   The budget is 40 ms. The total is 22 ms in the empty street and 30 ms in the crowd, leaving 18 ms and 10 ms. The NMS term is the only one that depends on the scene, so an NMS-free head (or a higher confidence threshold, at a cost in recall) helps most in crowds.

2. **Count the candidates.** For a 1280×1280 input with strides 8, 16, 32 and 64, how many predictions does an anchor-free head make? How many does an anchor-based head with 3 anchors per cell make at strides 8, 16 and 32 for a 640×640 input? Use the NMS snippet to compare the cost of the two candidate counts in a crowded scene.

   Hint

   $160^2 + 80^2 + 40^2 + 20^2 = 34{,}000$. The anchor-based head makes $3 \times 8{,}400 = 25{,}200$. With the snippet's crowded scene, both are far beyond 8,400 candidates, and the IoU count grows faster than linearly, so a confidence threshold before NMS matters even more.

3. **When top-1 is not a matching.** Modify the toy assignment so that two objects overlap strongly and the same grid point has the highest score for both. Compare per-object top-1 with Hungarian matching. What goes wrong with top-1, and how often would you expect this in real images?

   Hint

   Top-1 assigns the same point to both objects, so that point gets two conflicting targets and one object effectively has no positive. Hungarian matching gives the second object its second-best point. It happens when object centres are closer than about one stride, which is common for small objects in crowds and rare otherwise.

4. **The range of DFL.** With 16 bins and $\Delta = 1$, what is the largest box a stride-32 cell can describe if the cell centre is at the box centre? What happens to a 1200-pixel-wide object in a 1280×1280 image? Name two ways to extend the range.

Hint

Each side can reach at most $15 \times 32 = 480$ pixels, so the widest box is 960 pixels. A 1200-pixel-wide object is truncated (the predicted box is too small). You can use more bins, larger Δ (coarser bins), a coarser extra level (stride 64), or a head that regresses the distance directly, as YOLO26 does.

5. **Two speed tables.** Paper A reports 4.1 ms on an A100 GPU, FP16, batch 32, forward pass only. Paper B reports 6.3 ms on a T4 GPU, FP16, batch 1, end to end with NMS. List every reason why you cannot conclude that A is faster, and design the experiment that would let you compare them.

Hint

Different GPUs, different batch sizes (A's per-image time at batch 32 hides latency), different timed stages (A excludes pre- and post-processing). Run both exported models on one device with one engine and precision, at batch 1, at the same input size, timing end to end over the same images after warm-up, with the confidence threshold you will deploy.

6. **Design a training schedule.** You train an NMS-free detector with a one-to-many and a one-to-one head. Write a schedule for the loss weight $\alpha(t)$ of the one-to-many loss and for mosaic augmentation over 300 epochs, and justify each choice from the sections above.

Hint

For example: α decreases linearly from a large value to a small one, as in YOLO26's progressive loss, so the backbone first learns from many positives and the inference head gets most of the weight at the end; mosaic and mixup stay on until the last 15 epochs, as in YOLOX, so the final epochs match the test distribution. Keep the one-to-one head's matching score consistent with the one-to-many one, as in YOLOv10.

Detections from a real-time detector are the input to tracking, covered in DLCV 06 ·Object Tracking.

Supplementary notes

These notes are optional background: version-by-version details and one derivation that the main text summarizes. Nothing in the main text or the exercises depends on them.

The YOLO family, version by version

Plain version. Each version below changed one or two things. The notes say what, according to the version's own paper or documentation, and what to keep in mind.

YOLO v1 [5]. The image is divided into an $S \times S$ grid ($S = 7$); each cell predicts $B = 2$ boxes with confidences and one set of C class probabilities, an $S \times S \times (5B + C)$ tensor. The cell containing an object's centre is responsible for it. Because each cell predicts only two boxes and one class, nearby small objects compete for the same cell, the main weakness that later versions address.

YOLOv2 / YOLO9000 [6]. Adds batch normalization [49], anchors with shapes from k-means on the training boxes using a $1 - \text{IoU}$ distance (5 clusters), sigmoid-bounded centre offsets (the equation in the main text), a passthrough layer that brings finer features to the output, and multi-scale training (the input size changes during training so one model works at several resolutions).

YOLOv3 [7]. A deeper residual backbone the report calls Darknet-53, predictions at 3 scales with 3 anchors each, and independent logistic classifiers with binary cross-entropy instead of a softmax, so that a box can carry overlapping labels. It was written as a short technical report.

YOLOv4 [8]. A systematic study: a CSPDarknet53 backbone (cross-stage partial connections [50]), a PAN neck [31], and a long list of tested tricks. The "bag of freebies" (training-only changes) includes mosaic augmentation and the CIoU loss [37]; the "bag of specials" (cheap inference-time modules) includes receptive-field and attention modules. The paper also stresses training on a single conventional GPU.

YOLOv5 [9]. Released by Ultralytics in June 2020 as a PyTorch code base, with five model sizes from nano to extra large. The documentation states that no paper was published "due to the rapidly evolving nature of the models". The original YOLOv5 is anchor-based (the RT-DETR paper uses it as its anchor-based example [23]); the current Ultralytics package loads an anchor-free variant called YOLOv5u.

YOLOX [10]. Returns to anchor-free prediction (one prediction per location instead of three), adds the decoupled head, strong augmentation (mosaic and mixup, switched off for the last 15 epochs) and SimOTA. SimOTA computes a cost $c_{ij} = L_{ij}^{\text{cls}} + \lambda L_{ij}^{\text{reg}}$ for each object–prediction pair, where L^{cls} and L^{reg} are the classification and regression losses and λ a weight, and makes the k lowest-cost predictions inside a centre region positive, with k chosen per object; it simplifies OTA [33], which solves the assignment as optimal transport. YOLOX also tried an NMS-free one-to-one variant and found it slightly less accurate and slower, so left it optional, two years before YOLOv10.

YOLOv6 [11]. An industrial design report: a re-parameterizable backbone (EfficientRep) and neck (Rep-PAN) built from RepVGG-style blocks [32], an efficient decoupled head, task-aligned assignment [34], VFL for classification [36], self-distillation, and a quantization scheme for re-parameterized models.

YOLOv7 [12]. "Trainable bag of freebies": E-ELAN (an extended efficient layer aggregation block), a scaling rule for concatenation-based models, *planned* re-parameterized convolution (the paper shows that some branch combinations hurt when re-parameterized and plans where to use them), and coarse-to-fine lead-guided label assignment: the predictions of the final (lead) head

and the ground truth generate soft labels for both the lead head and an auxiliary training head, with coarser labels (more positive cells) for the auxiliary head to favour recall.

YOLOv8 [13]. Released by Ultralytics on 10 January 2023 in the `ultralytics` Python package. The documentation lists an anchor-free split head and improved backbone and neck, and states that no paper was published. The YOLO26 paper describes DFL as part of the Ultralytics head since YOLOv8 [19].

YOLOv9 [14]. Targets the *information bottleneck*: as data passes through many layers, information needed for the loss can be lost, and gradients become unreliable. Programmable gradient information (PGI) adds an auxiliary reversible branch during training that supplies complete input information to the loss, and is removed at inference, so it costs nothing at test time. GELAN (generalized ELAN) is the accompanying backbone design.

YOLOv10 [15]. Consistent dual assignments for NMS-free training (main text and Paper guides), plus an efficiency-driven design (a lighter classification head, decoupled spatial and channel downsampling, a rank-guided block allocation) and an accuracy-driven one (large-kernel convolutions and a partial self-attention module).

YOLO11 [16]. Released by Ultralytics on 27 September 2024. The documentation describes improved backbone and neck design and refined training, across detection, segmentation, classification, pose and oriented boxes; no paper.

YOLOv12 [17]. Argues that attention had been too slow for real-time detectors because of its quadratic cost and memory access, and makes it fast with *area attention*: the feature map is split into l equal areas (default $l = 4$) and attention runs within each, which reduces cost while keeping a large receptive field. R-ELAN (a residual aggregation block) stabilizes training of the larger models. Peer reviewed at NeurIPS 2025.

YOLOv13 [18]. Models higher-order correlations among positions with a hypergraph: HyperACE treats pixels of multi-scale features as vertices and learns the hyperedges, and FullPAD distributes the enhanced features to the backbone, neck and head; depthwise-separable blocks keep the model light. arXiv only at the time of writing.

YOLO26 [19]. Released by Ultralytics in January 2026, with an arXiv paper in June 2026. The dual-head design gives NMS-free inference by default (an NMS path remains optional). DFL is removed: the paper gives two reasons, the head size (it reports that removing DFL reduces YOLO11n from 2.6M to 2.3M parameters) and the finite range of $(K - 1)$ strides per side with $K = 16$ bins. Training adds MuSGD, a hybrid of SGD and the Muon optimizer [51]; progressive loss (main text); and STAL, which enlarges boxes smaller than the finest stride to a reference size *for candidate selection only*, so that tiny objects always get positive samples.

YOLO27 [20]. Announced on the Ultralytics documentation, which states that the models are in final development, not yet available, with no launch date. The announced design uses CNN heads predicting on fewer scales for the smallest models and query-based decoding without NMS

for the larger ones. Treat it as an announcement until weights and a paper exist.

Real-time DETRs after RT-DETR, in detail

Plain version. The successors attack three remaining weaknesses: boxes predicted as four plain numbers, slow convergence from few positives, and backbones that transfer poorly to new data.

RT-DETRv2 [24] keeps the RT-DETR architecture and adds training and deployment improvements: a different number of deformable-attention sampling points for each feature scale, so the decoder extracts multi-scale features selectively; an optional discrete sampling operator that replaces the `grid_sample` operator, which removes a deployment obstacle; and dynamic data augmentation and hyperparameters adapted to the model scale.

D-FINE [25] redefines box regression in a DETR as **fine-grained distribution refinement** (**FDR**): each decoder layer predicts probability distributions over offsets for the four edges and refines the previous layer's distributions in a residual way, so boxes are corrected step by step, with uncertainty, rather than regressed as fixed coordinates. **Global optimal localization self-distillation** (**GO-LSD**) then distills the refined distributions of the final layer into the earlier layers. This connects the DETR line to the distribution idea of DFL in the main text.

DEIM [26] addresses the shortage of positives under one-to-one matching without adding extra decoders. **Dense one-to-one** matching keeps the one-to-one rule but increases the number of targets per image, which mosaic and mixup achieve easily. The **matchability-aware loss** (**MAL**) then adjusts the classification penalty by the quality of each match, so that low-IoU matches with high confidence are handled better. The paper applies DEIM to RT-DETRv2 and D-FINE and reports that, with half the training epochs, it slightly exceeds their accuracy.

RF-DETR [27] starts from a DETR with a DINOv2-pretrained vision-transformer backbone [47] and uses **weight-sharing neural architecture search**: one network is fine-tuned on the target dataset while varying parameters such as input resolution and patch size during training, and thousands of sub-configurations with different accuracy–latency trade-offs can then be evaluated without retraining. Its focus is transfer to arbitrary target datasets rather than COCO alone.

DEIMv2 [28] extends DEIM with DINOv3 features [48]: for the larger models it uses DINOv3-pretrained or distilled backbones and a Spatial Tuning Adapter that converts the single-scale transformer output into the multi-scale features a detector needs; for the smallest models it uses depth- and width-pruned convolutional backbones.

Folding a re-parameterized block into one convolution

Plain version. A block with three parallel branches (a 3×3 convolution, a 1×1 convolution and a shortcut) trains well but runs slowly. Because every branch is linear until the activation, the three can be added into a single 3×3 convolution after training, and the network computes exactly the same function, faster.

At inference, batch normalization [49] is an affine map per channel. For a bias-free convolution with kernel W followed by batch normalization with learned scale γ, shift β, and running mean μ and variance σ^2,

$$\mathrm{BN}(W * x) = \gamma\,\frac{W * x - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta = W' * x + b', \qquad W' = \frac{\gamma}{\sqrt{\sigma^2 + \epsilon}}\,W, \quad b' = \beta - \frac{\gamma\,\mu}{\sqrt{\sigma^2 + \epsilon}},$$

where $*$ is convolution, ϵ a small constant, and all operations are per output channel. A 1×1 kernel is a 3×3 kernel that is zero except at its centre, and the identity is a 1×1 kernel equal to the identity matrix across channels. Once each branch is written as a 3×3 kernel with a bias, the sum of branches is one convolution with the summed kernel and summed bias. This is the structural re-parameterization of RepVGG [32].

```
import numpy as np

def conv(x, w, b):
    """'Same' convolution (cross-correlation). x: (C, H, W), w: (O, C, k, k), b: (O,)."""
    k = w.shape[-1]; p = k // 2
    xp = np.pad(x, ((0, 0), (p, p), (p, p)))
    H, W = x.shape[1:]
    out = sum(np.einsum("oc,chw->ohw", w[:, :, i, j], xp[:, i:i + H, j:j + W])
              for i in range(k) for j in range(k))
    return out + b[:, None, None]

def fold_bn(w, gamma, beta, mu, var, eps=1e-5):
    """A bias-free conv followed by inference-mode batch norm -> one conv with a bias."""
    scale = gamma / np.sqrt(var + eps)
    return w * scale[:, None, None, None], beta - mu * scale

rng = np.random.default_rng(0)
C = 4                                                  # channels in = channels out
x = rng.normal(size=(C, 10, 10))
bn = lambda: (rng.uniform(0.5, 1.5, C), rng.normal(size=C), rng.normal(size=C),
 ↪  rng.uniform(0.5, 2, C))
w3, w1 = rng.normal(size=(C, C, 3, 3)), rng.normal(size=(C, C, 1, 1))
bn3, bn1, bn_id = bn(), bn(), bn()

def bn_apply(z, g, b, mu, var, eps=1e-5):
    return g[:, None, None] * (z - mu[:, None, None]) / np.sqrt(var[:, None, None] + eps) +
     ↪  b[:, None, None]

# training-time block: 3x3 branch + 1x1 branch + identity branch, each with its own batch
 ↪  norm
```

```
y_train = (bn_apply(conv(x, w3, np.zeros(C)), *bn3) + bn_apply(conv(x, w1, np.zeros(C)),
 ↪  *bn1)
           + bn_apply(x, *bn_id))

# deployment: express every branch as a 3x3 kernel, fold its batch norm, then add the kernels
w1_as3 = np.pad(w1, ((0, 0), (0, 0), (1, 1), (1, 1)))  # 1x1 kernel in the centre of a 3x3
 ↪  one
w_id = np.zeros((C, C, 3, 3)); w_id[np.arange(C), np.arange(C), 1, 1] = 1.0
folded = [fold_bn(w, *p) for w, p in [(w3, bn3), (w1_as3, bn1), (w_id, bn_id)]]
w_dep, b_dep = sum(f[0] for f in folded), sum(f[1] for f in folded)
y_dep = conv(x, w_dep, b_dep)
print("same output:", np.allclose(y_train, y_dep))
print("branches at training: 3 | convolutions at inference: 1")
```

```
same output: True
branches at training: 3 | convolutions at inference: 1
```

The folding is exact only for inference-mode batch normalization (fixed running statistics) and only before the nonlinearity. The YOLOv7 paper's "planned" re-parameterization exists because some combinations, such as an identity branch inside a block that already has a residual connection, hurt accuracy [12], and YOLOv6 reports that folded models can be harder to quantize [11].

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Reading the YOLOv10 paper

The paper. A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han and G. Ding, "YOLOv10: Real-time end-to-end object detection," NeurIPS 2024 [15].

The problem it poses. YOLOs are trained with one-to-many assignment and therefore need NMS, which adds latency and a threshold to tune. DETR-style one-to-one assignment removes NMS but gives weak supervision. Separately, the authors argue that the YOLO architecture had never been designed as a whole for efficiency and accuracy.

The idea. Two heads with the same structure share the backbone and neck. One is trained one-to-many, the other one-to-one; only the one-to-one head is used at inference. The key analysis is the **supervision gap**: if the two heads used different matching parameters, the prediction chosen by the one-to-one head need not be among the best of the one-to-many head, so the two heads would

pull the shared features in different directions. The appendix derives that the gap is smallest when the one-to-one head's α and β are proportional to the one-to-many head's ($\alpha_{o2o} = r\,\alpha_{o2m}$, $\beta_{o2o} = r\,\beta_{o2m}$), and the paper takes $r = 1$. Figure 2(b) of the paper supports this by counting how often the one-to-one match falls in the one-to-many top-1, top-5 and top-10.

What it showed. Table 3 isolates the assignment on the small model: one-to-many only, 44.9 AP at 7.07 ms; one-to-one only, 43.4 AP at 2.44 ms; both heads, 44.3 AP at 2.44 ms. Table 4 shows that the consistent setting is as good as or better than the alternatives tried. Table 2 adds the efficiency and accuracy design changes step by step.

What to check when you read it.

- **What the latency includes.** End-to-end latency is measured on a T4 GPU with TensorRT FP16, at batch 1, averaged over COCO val images, with the TensorRT efficient-NMS plugin for models that need NMS and I/O excluded (appendix). The paper also reports a forward-only latency for comparisons with YOLOs trained the usual way; do not mix the two columns.
- **The price of NMS-free.** In Table 3 the one-to-one head is 0.6 AP below the one-to-many head with NMS. The gain is latency and simplicity, not accuracy.
- **Single runs.** The ablations report single numbers; differences of a few tenths of AP may be within run-to-run variation.
- **Baselines.** Comparisons with other detectors use their official weights and the paper's own benchmark, which is the right way to compare speed, but the training recipes differ between models.

Why it matters here. It is the clearest demonstration that the YOLO and DETR lines differ in assignment more than in architecture, and it set the pattern (train with both, infer with one-to-one) that YOLO26 adopted.

Reading the RT-DETR paper

The paper. Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu and J. Chen, "DETRs beat YOLOs on real-time object detection," CVPR 2024 [23].

The problem it poses. DETRs need no NMS but were too slow for real time; YOLOs are fast but their reported speeds usually leave NMS out. The paper asks both "how much does NMS really cost?" and "can a DETR be fast?".

The idea. First, measure. Section 3 analyses NMS: the number of boxes that survive the confidence threshold, the NMS execution time under different thresholds (using the TensorRT efficient-NMS kernel on a T4 GPU in FP16), and the resulting AP. It then defines an end-to-end benchmark on COCO val2017 that appends NMS to YOLO detectors with the thresholds each uses for its

reported accuracy. Second, build. The hybrid encoder (AIFI + CCFF), uncertainty-minimal query selection and adjustable decoder depth described in the main text.

What it showed. Under its benchmark, anchor-free YOLOs were faster end to end than anchor-based ones at equal accuracy, because they produce fewer boxes. RT-DETR with ResNet-50 and ResNet-101 backbones reports 53.1% and 54.3% AP at 108 and 74 FPS on a T4 GPU, and the ablations compare encoder variants and query-selection schemes.

What to check when you read it.

- **Which NMS time is reported.** The NMS analysis times only the main NMS kernel of a multi-kernel plugin; the end-to-end benchmark excludes I/O and memory copies. Both are stated in the text.
- **Which numbers use extra data.** Results with Objects365 pre-training are reported separately; keep them apart from COCO-only results.
- **Version differences.** The first arXiv version called the query selection "IoU-aware"; the final version calls it "uncertainty-minimal" and defines it with the uncertainty U in the main text. Cite the final version.
- **What "beats" means.** The comparison is at matched accuracy and on one GPU type and engine. The ranking can change on CPUs or edge accelerators, where attention and deformable sampling may be slower relative to convolutions.

Why it matters here. It made end-to-end speed (with NMS) the expected way to report real-time detectors, and its successors (RT-DETRv2, D-FINE, DEIM, RF-DETR, DEIMv2) all build on its design.

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DLCV 03B • Image Segmentation: Semantic, Instance and Panoptic

How deep networks label every pixel: the three segmentation tasks, fully convolutional and encoder–decoder networks, atrous convolution and multi-scale context, losses and metrics (mIoU, mask AP, panoptic quality), mask classification with MaskFormer and Mask2Former, and promptable and open-vocabulary segmentation.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-03b-image-segmentation/

What you'll learn

- The three segmentation tasks, defined precisely: semantic (a class per pixel), instance (a mask per object) and panoptic (a class and an object identity for every pixel), and the "stuff" versus "things" distinction behind them.
- How a classification network becomes a per-pixel predictor: fully convolutional networks, output stride, upsampling of scores, encoder–decoders with skip connections, atrous (dilated) convolution and multi-scale context modules (ASPP, pyramid pooling).
- Which losses train segmenters (per-pixel cross-entropy, Dice, focal) and why class imbalance makes the choice matter, with a NumPy experiment.
- How each task is measured, in code: pixel accuracy and mean IoU, mask AP, and panoptic quality $\mathrm{PQ} = \mathrm{SQ} \times \mathrm{RQ}$.
- How mask classification (MaskFormer, Mask2Former) solves all three tasks with one architecture, and where promptable (SAM family) and open-vocabulary models fit in.

The big picture

A detector draws a box around each object (DLCV 03). A box is a rough answer: it contains background pixels, and it says nothing about the object's exact outline. **Segmentation** gives the exact answer. It decides, for every pixel, what that pixel belongs to.

> **Explain like I'm five**
>
> Give a child a colouring book page of a street and some crayons. **Semantic segmentation** is colouring by kind: every car blue, every person orange, the road grey, the sky light blue. Two cars that touch become one blue blob. **Instance segmentation** is putting a numbered sticker on each car and each person and colouring exactly inside that object's outline; the road and sky are left blank. **Panoptic segmentation** does both: every bit of the page gets a colour, and each car and person also gets its own number.

Why it matters. A self-driving car needs to know which pixels are drivable road, not just that "a road is somewhere in this box". A radiologist measuring a tumour needs its exact area. A photo editor that removes a person needs that person's outline. Robots that grasp objects, satellite images of fields and forests, and medical scans all need per-pixel answers.

This chapter is the deep-learning sequel of two classical chapters. IP 10 and IP 11 segment an image with thresholds, region growing, graph cuts, watersheds and active contours, using no training data; this chapter learns the segmentation from labeled examples. DLCV 03 introduced boxes, IoU, anchors, feature pyramids, the focal loss and Mask R-CNN; we reuse those ideas and link back instead of repeating them. The promptable Segment Anything models are introduced in DLCV 01 and ML 09; here we only place them among the other tasks. Three surveys frame the field: Minaee et al. for the CNN era [1], Li et al. for transformer-based segmentation [2], and Zhou et al. for segmentation with foundation models [3]. We return to them in the Modern view.

Three tasks, one label map

Plain version. All three tasks write labels onto the pixel grid. They differ in *which* labels: a class name only (semantic), an object identity for countable objects only (instance), or both for every pixel (panoptic).

Precise version. Let $\Omega = \{1, \dots, H\} \times \{1, \dots, W\}$ be the set of pixel positions of an $H \times W$ image and let $\{1, \dots, K\}$ be the class labels.

- **Semantic segmentation** predicts a label map $y : \Omega \to \{1, \dots, K\}$, one class $y(\mathbf{x})$ per pixel $\mathbf{x}$. It does not distinguish two objects of the same class.
- **Instance segmentation** predicts a set $\{(m_j, c_j, s_j)\}_{j=1}^{N}$, where $m_j \in \{0,1\}^{H \times W}$ is a binary mask, c_j its class, $s_j \in [0,1]$ a confidence score and N varies per image. Masks may overlap, and only countable object classes are segmented. It is detection with a mask instead of a box.
- **Panoptic segmentation** [4] maps each pixel i to a pair (l_i, z_i), where l_i is the class and z_i an instance id. Pixels with the same pair form one **segment**. Segments cannot overlap, since each pixel carries one pair, and no confidence scores are required.

The panoptic paper names the two kinds of classes after an older distinction in vision. **Things** are countable objects with a well-defined shape, such as people, cars and animals. **Stuff** is amorphous regions of similar texture or material, such as sky, road and grass [4]. For a stuff class the instance id is ignored: all the sky in an image is one segment. Which classes are stuff and which are things is a design choice of each dataset. Pixels that are ambiguous or outside every class get a special **void** label and are not evaluated.

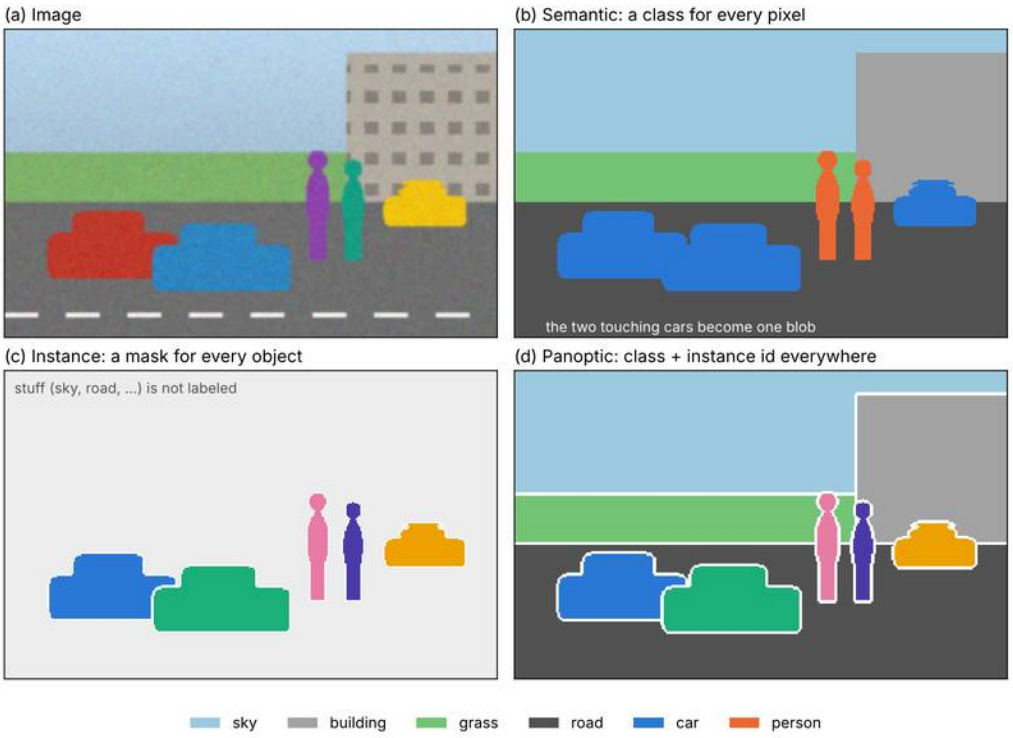


Figure 1 — The three tasks on one synthetic scene. (b) Semantic: the two touching cars become one region. (c) Instance: one mask per object, stuff is not labeled. (d) Panoptic: every pixel gets a class, and every thing also gets its own identity.

	Semantic	Instance	Panoptic
Labels	class per pixel	mask + class + score per object	(class, instance id) per pixel
Classes covered	stuff and things	things only	stuff and things
Overlaps allowed	no	yes	no
Confidence scores	no	yes	no
Standard metric	mean IoU	mask AP	panoptic quality (PQ)

The table explains why the three tasks long had separate methods: their outputs have different shapes, and each metric rewards something different. Most of this chapter follows that history: per-pixel classifiers for semantic segmentation first, then detector-based instance segmentation, then the panoptic task that forced the two to meet, and finally one architecture that handles all three.

From a classifier to a per-pixel classifier

Plain version. Take an image classifier, remove the part that throws away "where", and it outputs a small grid of class scores instead of one score vector. Enlarge that grid back to the image size,

and you have a segmenter.

Fully convolutional networks

Long, Shelhamer and Darrell's **fully convolutional network** (**FCN**) [5] replaced the fully connected layers of a classification network with convolutions. A convolution works on an input of any size, so the network now maps an $H \times W$ image to a score map

$$Z \in \mathbb{R}^{K \times \frac{H}{s} \times \frac{W}{s}},$$

where K is the number of classes and s is the **output stride**, the ratio between the input resolution and the score-map resolution. A classification backbone pools and strides repeatedly; for the networks FCN started from, $s = 32$, so a 512×512 image yields a 16×16 grid of class scores. Each score vector $Z[:, u, v]$ summarizes a large window of the image, which is good for recognizing *what* is there, but one cell covers 32×32 pixels, which is bad for saying exactly *where* a boundary runs.

FCN trains the whole network end to end with a per-pixel cross-entropy loss after upsampling Z to the input size, starting from weights pretrained for classification. To recover detail, it fuses predictions from finer layers: the stride-32 scores are upsampled by 2 and added to a score map computed from the stride-16 layer, and again for stride 8. The paper calls the three versions FCN-32s, FCN-16s and FCN-8s, and shows that each finer fusion gives more detailed outlines [5]. Its summary of the idea is a good one-liner for the whole field: combine "what" from a deep, coarse layer with "where" from a shallow, fine layer. DLCV 01 introduces the same idea in its section on encoder–decoders.

Upsample the scores, not the labels

Plain version. If you decide the class at low resolution and then enlarge the decision, every boundary becomes a staircase of $s \times s$ blocks. If you first enlarge the *scores* smoothly and then decide per pixel, the boundaries fall between grid points.

Bilinear interpolation computes the score at a fine position from the four nearest coarse cells. For a fine pixel that sits at fractional coarse coordinates $(u + a, v + b)$ with u, v integers and $a, b \in [0, 1)$,

$$\tilde{Z}_k = (1-a)(1-b)\, Z_k[u, v] + a(1-b)\, Z_k[u+1, v] + (1-a)\, b\, Z_k[u, v+1] + a\, b\, Z_k[u+1, v+1],$$

where Z_k is the score map of class k and $\tilde{Z}_k$ the interpolated score. The predicted label is $\hat{y} = \arg\max_k \tilde{Z}_k$. FCN implements upsampling as a strided transposed convolution initialized to bilinear interpolation, and lets the network learn it [5]; transposed convolution is explained in ML 03. Most later models simply use fixed bilinear upsampling for the last step.

The experiment below isolates the effect of output stride. It builds an **ideal** coarse score map, the

fraction of each class inside each $s \times s$ cell, an idealized output for a stride-s head, and compares the two orders of "enlarge" and "decide".

```
import numpy as np
from scipy import ndimage as ndi

# Synthetic ground truth: 0 background, 1 large disk, 2 thin bar, 3 small square
H = W = 256
yy, xx = np.mgrid[0:H, 0:W]
gt = np.zeros((H, W), int)
gt[(yy - 120) ** 2 + (xx - 90) ** 2 < 60 ** 2] = 1
gt[40:216, 178:184] = 2                       # 6 pixels wide
gt[196:216, 60:80] = 3                        # 20 x 20 pixels
K = 4

def iou_per_class(gt, pred, K):
    return np.array([((gt == k) & (pred == k)).sum() / ((gt == k) | (pred == k)).sum()
                     for k in range(K)])

onehot = np.stack([(gt == k).astype(float) for k in range(K)])
print("stride | nearest labels: mIoU (per class) | bilinear scores: mIoU (per class)")
for s in [4, 8, 16, 32]:
    # ideal stride-s scores: the class fractions inside each s x s cell
    coarse = onehot.reshape(K, H // s, s, W // s, s).mean(axis=(2, 4))
    # (a) decide at low resolution, then enlarge the label map
    lab_nn = np.kron(coarse.argmax(0), np.ones((s, s), int))
    # (b) enlarge the scores with bilinear interpolation, then decide per pixel
    up = np.stack([ndi.zoom(c, s, order=1, mode="nearest", grid_mode=True) for c in coarse])
    lab_bi = up.argmax(0)
    a, b = iou_per_class(gt, lab_nn, K), iou_per_class(gt, lab_bi, K)
    print(f"{s:6d} | {a.mean():.3f} {np.round(a, 2)} | {b.mean():.3f} {np.round(b, 2)}")
```

```
stride | nearest labels: mIoU (per class) | bilinear scores: mIoU (per class)
     4 | 0.906 [0.99 0.97 0.67 1.  ] | 0.994 [1.   1.   0.99 0.99]
     8 | 0.825 [0.98 0.93 0.75 0.64] | 0.883 [0.99 0.99 0.7  0.86]
    16 | 0.565 [0.95 0.89 0.   0.41] | 0.578 [0.97 0.97 0.   0.37]
    32 | 0.421 [0.92 0.76 0.   0.  ] | 0.473 [0.96 0.93 0.   0.  ]
```

Three lessons come out of this toy run (the numbers are from the code above, not from any paper). Bilinear upsampling of scores is almost free and always at least as good for the large disk. But no upsampling can bring back what the coarse grid never saw: at stride 16 the 6-pixel bar is a minority in every cell and vanishes, and at stride 32 the 20-pixel square vanishes too. Thin and small structures need a small output stride or a decoder that brings fine features back. And mean IoU punishes those losses heavily, because the bar and the square count as much as the background (Figure 2).

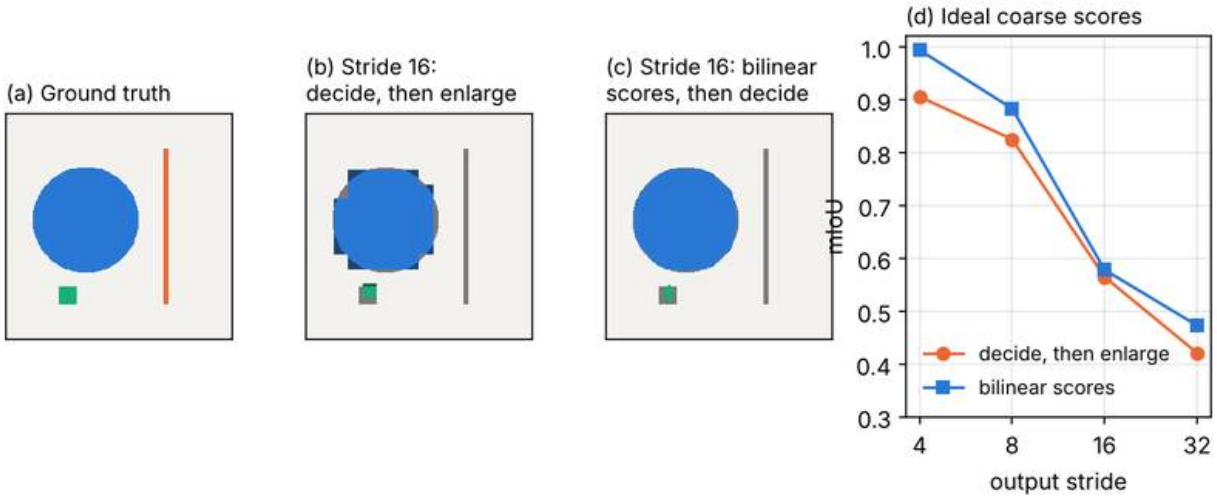


Figure 2 — Output stride limits what a segmenter can recover, even with ideal coarse scores. (b, c) Stride 16; dark overlay marks wrong pixels. (d) mIoU of the toy map for both upsampling orders.

Encoder–decoders and skip connections

Plain version. Instead of one big jump from coarse to fine, climb back up the resolution ladder one step at a time, and at each step borrow the detailed feature map that the encoder had at the same size.

U-Net [6] made this symmetric: a contracting **encoder** of convolutions and downsampling, an expanding **decoder** of upsampling and convolutions, and at each level a skip connection that concatenates the encoder's feature map to the decoder's. The encoder supplies *what*, the skips supply *where*. The paper's design choices (unpadded convolutions, the overlap-tile strategy, a weighted loss for touching cells and strong elastic augmentation) are covered in the U-Net reading guide of IP 10.

Optional: U-Net: a reading guide to the paper (in the Paper guides of IP 10). What the paper claims, the output-size arithmetic of valid convolutions, and how to read its tables.

The opposite design keeps the decoder light. **DeepLabv3+** [7] runs the encoder to output stride 16, upsamples its features by 4, concatenates low-level features from the backbone at stride 4 (after a 1×1 convolution that reduces their channels), refines with a few 3×3 convolutions and upsamples by 4 again. The paper motivates the decoder by noting that a plain 16-fold bilinear upsampling, a "naive decoder", may not recover object details. Figure 3 compares the three shapes: FCN's few fused score maps, U-Net's full ladder, and DeepLab's context module plus a single skip.

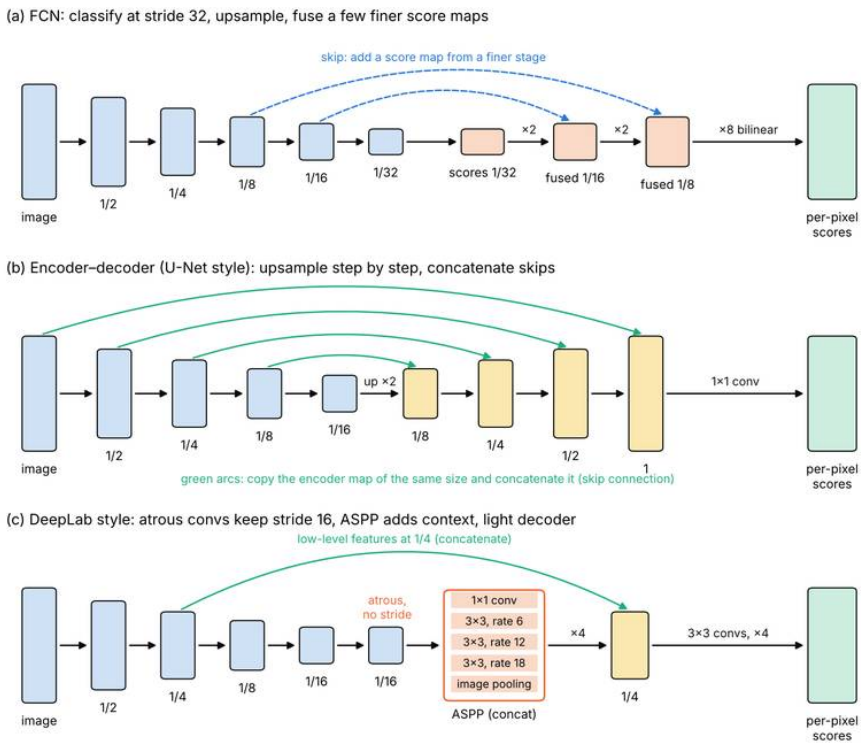


Figure 3 — Three ways to get from coarse features back to pixels. (a) FCN fuses score maps from a few finer stages. (b) A U-Net-style encoder–decoder upsamples step by step and concatenates same-size encoder maps. (c) DeepLabv3+ keeps stride 16 with atrous convolutions, adds multi-scale context with ASPP, and uses one low-level skip.

Context without losing resolution

Plain version. To label a pixel you need two things that pull in opposite directions: a wide view (is this grey patch a road or a wall?) and fine resolution (where exactly does the road end?). Atrous convolution and pooling pyramids give the wide view without shrinking the map.

Atrous (dilated) convolution

Downsampling is the usual way a CNN grows its receptive field, the region of the input that one output can see (DLCV 01 and ML 03 derive it). An **atrous** convolution ("à trous", with holes) spreads the taps of a small kernel apart instead. In 1-D, DeepLab [8] defines the output y of an atrous convolution of input x with a filter w of length K as

$$y[i] = \sum_{k=1}^{K} x[i + r \cdot k]\, w[k],$$

where i indexes output positions and r is the **rate**: the input is sampled every r positions. Rate $r = 1$ is ordinary convolution. A $k \times k$ kernel with rate r covers $k_e = k + (k-1)(r-1)$ pixels per side with the same k^2 weights and the same cost [8]. Yu and Koltun [9] called the same operation a dilated convolution and showed that stacking dilations that double from layer to layer grows the receptive field exponentially without losing resolution.

The DeepLab recipe is to set the stride of the last downsampling layer to 1 and replace the convolutions that follow by atrous convolutions with rate 2, so that they still see the same image region with the same pretrained weights; repeating this for one more stage doubles the density again [8]. Typical models output stride 8 or 16 instead of 32 [10]. The price is compute and memory: stride 8 has four times as many positions as stride 16 in the last stages.

```
import numpy as np

def atrous_1d(x, w, r):
    """y[i] = sum_k x[i + r*k] w[k] (DeepLab's Eq. 1), taps centred, zero padding."""
    K = len(w); c = K // 2
    xp = np.pad(x, r * c)
    return np.array([sum(xp[i + r * (k)] * w[k] for k in range(K)) for i in range(len(x))])

x = np.zeros(21); x[10] = 1.0                  # an impulse in the middle
w = np.array([1.0, 1.0, 1.0])
for r in [1, 2, 4]:
    print(f"rate {r}: taps at offsets {[(k - 1) * r for k in range(3)]},",
          "response", atrous_1d(x, w, r).astype(int))

def receptive_field(kernels, rates, strides):
    """Receptive field (in input pixels) of a stack of conv layers."""
    rf, jump = 1, 1                             # jump = distance between neighbouring
    ↪ outputs, in input pixels
    for k, r, s in zip(kernels, rates, strides):
        rf += (r * (k - 1)) * jump
        jump *= s
    return rf, jump

print("three 3x3, rates 1,1,1:", receptive_field([3, 3, 3], [1, 1, 1], [1, 1, 1]))
print("three 3x3, rates 1,2,4:", receptive_field([3, 3, 3], [1, 2, 4], [1, 1, 1]))
print("three 3x3, stride 2   :", receptive_field([3, 3, 3], [1, 1, 1], [2, 2, 2]))
```

```
rate 1: taps at offsets [-1, 0, 1], response [0 0 0 0 0 0 0 0 0 1 1 1 0 0 0 0 0 0 0 0 0]
rate 2: taps at offsets [-2, 0, 2], response [0 0 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 0]
rate 4: taps at offsets [-4, 0, 4], response [0 0 0 0 0 0 1 0 0 0 1 0 0 0 1 0 0 0 0 0 0]
three 3x3, rates 1,1,1: (7, 1)
three 3x3, rates 1,2,4: (15, 1)
three 3x3, stride 2   : (15, 8)
```

The last two lines are the whole argument for atrous convolution. Three layers with rates 1, 2, 4 see 15 input pixels, exactly as many as three stride-2 layers, but they keep one output per input pixel (jump 1) instead of one per 8 pixels (Figure 4).

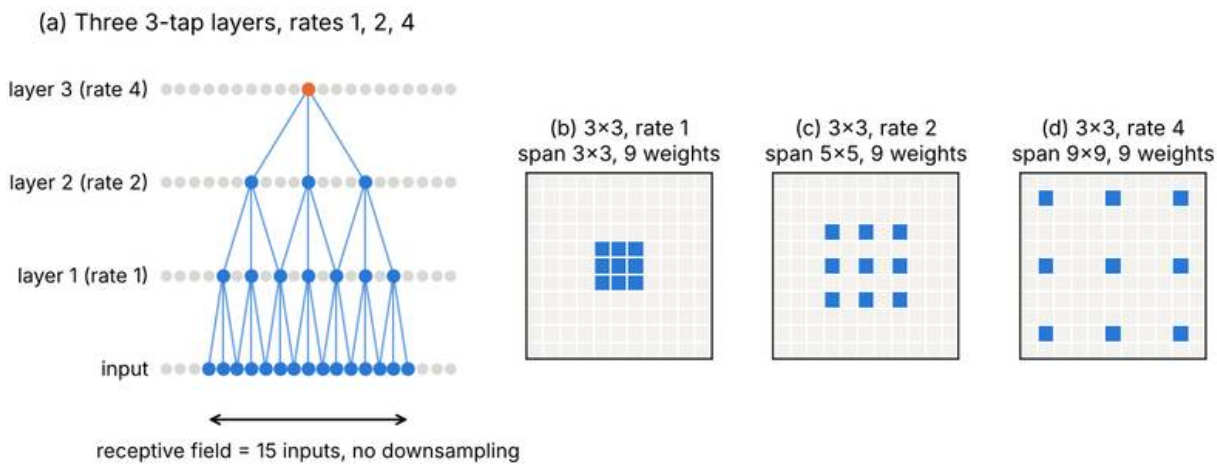


Figure 4 — Atrous convolution. (a) Stacking rates 1, 2 and 4 covers 15 inputs at full resolution. (b–d) A 3×3 kernel at rates 1, 2 and 4 keeps 9 weights but spans 3×3, 5×5 and 9×9 pixels.

Dilation has a catch. A single dilated layer only reads a sparse lattice of inputs, and stacking layers whose rates share a common factor leaves holes that no path ever reaches, a checkerboard artifact that Wang et al. call **gridding** [11]. The fix is to choose rates such as 1, 2, 5 that fill each other's gaps.

Optional: Gridding in stacked dilated convolutions (in the Supplementary notes). Which rate sequences leave holes, the hybrid-dilation rule, and a check in code.

Multi-scale context: ASPP and pyramid pooling

Plain version. Objects come at many sizes, and what a pixel is often depends on what surrounds it. So look at each position through several windows of different sizes at once, including one that sees the whole image, and let the network combine the answers.

Atrous spatial pyramid pooling (ASPP) [8] applies several atrous convolutions with different rates in parallel to the same feature map and fuses their outputs, which probes the image at several effective scales. DeepLabv3 [10] uses one 1×1 convolution and three 3×3 convolutions with rates 6, 12 and 18 at output stride 16 (rates doubled at stride 8), plus an **image-level feature**: global average pooling of the whole map, a 1×1 convolution, and bilinear upsampling back. The reason for the image-level branch is a boundary effect the paper points out: as the rate approaches the feature-map size, more and more of the 3×3 taps fall outside the map, and the filter degenerates to a 1×1 filter that only uses its centre weight. The snippet counts this on a 65×65 map.

```
import numpy as np
```

```
n = 65                                              # feature map is n x n
for r in [1, 6, 12, 18, 24, 36, 64]:
    pos = np.arange(n)
    inside_1d = np.array([[(0 <= p + r * d < n) for d in (-1, 0, 1)] for p in pos]).sum(1)
    mean_taps = np.mean(np.outer(inside_1d, inside_1d))   # taps inside the map, averaged
     ↪  over positions
    print(f"rate {r:2d}: mean valid taps {mean_taps:.2f} of 9")
```

```
rate  1: mean valid taps 8.82 of 9
rate  6: mean valid taps 7.93 of 9
rate 12: mean valid taps 6.92 of 9
rate 18: mean valid taps 5.98 of 9
rate 24: mean valid taps 5.11 of 9
rate 36: mean valid taps 3.58 of 9
rate 64: mean valid taps 1.06 of 9
```

At rate 64 a "3×3" filter is, on average, a single tap. Global pooling is the cheap and reliable way to get image-wide context.

The **Pyramid Scene Parsing Network** (**PSPNet**) [12] reaches the same goal with pooling instead of dilation. Its pyramid pooling module pools the final feature map into 1×1, 2×2, 3×3 and 6×6 grids, reduces each with a 1×1 convolution, upsamples all of them back and concatenates them with the original features. The paper motivates it with errors of an FCN baseline on scene parsing, such as a "mismatched relationship": a boat on a river labeled as a car, although a car is seldom over a river [12].

Early DeepLab versions also sharpened boundaries after the network with a fully connected conditional random field, which encourages pixels of similar colour and position to share a label [8]. DeepLabv3 improves on the earlier versions without it [10].

Optional: Dense CRF post-processing (in the Supplementary notes). The energy, its two Gaussian kernels, and why mean-field inference makes it fast.

Attention as context

A transformer gives every token a view of every other token in each layer, so "context" no longer needs a special module (ML 04 explains the Vision Transformer, ViT [13]). The catch is resolution: plain ViT has one coarse token grid, and global attention over many tokens is costly. **SegFormer** [14] is a widely used answer for semantic segmentation. Its encoder is hierarchical, producing multi-scale features like a CNN, uses a reduced attention that shortens the key and value sequences, and needs no positional encoding, which avoids interpolating position codes when the test resolution differs from the training one. Its decoder is only a few MLP layers that fuse the multi-scale features. Hierarchical vision transformers are covered in ML 04.

Losses for dense prediction

Plain version. A segmenter is trained by comparing its per-pixel probabilities with the true label map. Averaging a classification loss over pixels is the default, but when one class covers 99% of the image, that average mostly measures the easy class.

Per-pixel cross-entropy. With $p_{\mathbf{x}}(k)$ the softmax probability of class k at pixel $\mathbf{x}$ and $y(\mathbf{x})$ the true class,

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{|\Omega'|} \sum_{\mathbf{x} \in \Omega'} w_{y(\mathbf{x})} \log p_{\mathbf{x}}(y(\mathbf{x})),$$

where Ω' is the set of labeled (non-void) pixels and w_k an optional weight per class, for example inversely proportional to the class frequency. Void pixels are simply left out of the sum; most frameworks implement this with an "ignore index".

Dice loss. Milletari et al. [15] proposed to maximize the Dice coefficient between a soft prediction and the binary ground truth directly, as an alternative to re-weighting samples, because the anatomy of interest often occupies only a small part of a medical volume and training then tends to predict background everywhere. For foreground probabilities p_i and labels $g_i \in \{0, 1\}$ over N pixels, their form is

$$D = \frac{2 \sum_{i=1}^{N} p_i g_i}{\sum_{i=1}^{N} p_i^2 + \sum_{i=1}^{N} g_i^2}, \qquad \mathcal{L}_{\mathrm{Dice}} = 1 - D.$$

A common variant uses $\sum p_i + \sum g_i$ in the denominator; with binary p and g both equal the overlap measure $2|A \cap B| / (|A| + |B|)$, a close relative of IoU. For several classes, the Dice loss is computed per class and averaged. Because D is a ratio over the whole image, a small object counts as much as a large background.

Focal loss. The focal loss [16] multiplies the cross-entropy of each pixel by $(1 - p_t)^{\gamma}$, where p_t is the probability of the true class, so that easy pixels contribute little; it is derived in DLCV 03. Combinations are common: MaskFormer, discussed below, trains each binary mask with a weighted sum of a focal loss and a Dice loss [17].

The experiment compares the losses on a 64×64 image whose foreground is a disk of 49 pixels, for three candidate predictions. "IoU@0.5" thresholds the probabilities at 0.5 and computes the IoU that an evaluation would report.

```
import numpy as np

H = W = 64
yy, xx = np.mgrid[0:H, 0:W]
r2 = (yy - 32) ** 2 + (xx - 32) ** 2
g = (r2 <= 4 ** 2).astype(float)                # small foreground disk
print(f"foreground pixels: {int(g.sum())} of {g.size} ({100 * g.mean():.1f}%)")
```

```
def bce(p, g):
    return -np.mean(g * np.log(p) + (1 - g) * np.log(1 - p))

def weighted_bce(p, g):                          # weight each class by inverse frequency
    wf, wb = 0.5 / g.mean(), 0.5 / (1 - g.mean())
    return -np.mean(wf * g * np.log(p) + wb * (1 - g) * np.log(1 - p))

def focal(p, g, gamma=2.0):
    pt = np.where(g == 1, p, 1 - p)
    return -np.mean((1 - pt) ** gamma * np.log(pt))

def soft_dice_loss(p, g, eps=1e-6):
    return 1 - (2 * (p * g).sum() + eps) / (p.sum() + g.sum() + eps)

def iou_at_half(p, g):                           # the metric we actually report
    m = p > 0.5
    return (m & (g == 1)).sum() / (m | (g == 1)).sum()

preds = {
    "all background (p=0.05)":  np.full((H, W), 0.05),
    "right shape, timid (0.4)": np.where(g == 1, 0.4, 0.05),
    "bloated, confident (0.9)": np.where(r2 <= 8 ** 2, 0.9, 0.05),
}
print(f"{'prediction':26s} {'BCE':>6s} {'wBCE':>6s} {'focal':>6s} {'Dice':>6s}
 ↪  {'IoU@0.5':>8s}")
for name, p in preds.items():
    print(f"{name:26s} {bce(p, g):6.3f} {weighted_bce(p, g):6.3f} "
          f"{focal(p, g):6.3f} {soft_dice_loss(p, g):6.3f} {iou_at_half(p, g):8.3f}")

# Where does the BCE gradient come from? dL/dz = p - g per pixel (z = logit)
p = preds["all background (p=0.05)"]
grad = np.abs(p - g)
print(f"share of |BCE gradient| from background pixels: {grad[g == 0].sum() /
 ↪  grad.sum():.2f}")
```

```
foreground pixels: 49 of 4096 (1.2%)
prediction                   BCE   wBCE  focal   Dice  IoU@0.5
all background (p=0.05)    0.087  1.524  0.032  0.981    0.000
right shape, timid (0.4)   0.062  0.484  0.004  0.855    0.000
bloated, confident (0.9)   0.133  0.119  0.068  0.791    0.249
share of |BCE gradient| from background pixels: 0.81
```

Read the table row by row (all numbers are from this toy run). Plain binary cross-entropy (BCE) prefers "all background" to the bloated prediction, the only one that finds the object at the 0.5 threshold, because the bloated mask is wrong on about 150 background pixels and those outnumber the object. The unweighted focal loss, without the class weight α, makes the same choice. Class-weighted BCE and Dice both rank the bloated prediction best and "all background" worst, in the same order as the IoU. The last line shows why plain BCE behaves this way: 81% of its gradient comes from background pixels that are already almost right. Note also how harshly soft Dice treats the timid prediction: the 4,047 background pixels at $p = 0.05$ add about 202 to the

denominator, more than four times the object's area, so a soft prediction must be confident on the background too.

In practice, cross-entropy (often with class weights or the focal modulation) is the default for natural-scene semantic segmentation, Dice or a Dice + cross-entropy sum is common for medical images with small structures, and mask-based models train each mask with Dice + focal. Losses that also penalize boundary or topology errors exist; three built from active-contour energies are summarized in IP 11's Supplementary notes.

Optional: The gradient of the soft Dice loss (in the Supplementary notes). The derivative in closed form, why every background pixel gets the same push, and edge cases such as empty masks.

Measuring semantic segmentation

Plain version. Count, over the whole test set, how many pixels of each true class were given each predicted class. Every common score is a different summary of that table. The standard one, mean IoU, asks for each class "of all pixels that are this class or were called this class, what fraction got it right?", and then averages over classes.

Let n_{ij} be the number of pixels of true class i predicted as class j, accumulated over all test images, and $t_i = \sum_j n_{ij}$ the number of pixels of class i. FCN [5] defines four metrics, with K classes:

$$\text{pixel accuracy} = \frac{\sum_i n_{ii}}{\sum_i t_i}, \qquad \text{mean accuracy} = \frac{1}{K}\sum_i \frac{n_{ii}}{t_i},$$

$$\text{mean IoU} = \frac{1}{K}\sum_i \frac{n_{ii}}{t_i + \sum_j n_{ji} - n_{ii}}, \qquad \text{frequency-weighted IoU} = \frac{1}{\sum_k t_k}\sum_i t_i \frac{n_{ii}}{t_i + \sum_j n_{ji} - n_{ii}}.$$

In the IoU of class i, the numerator n_{ii} counts true positives and the denominator counts the union of true and predicted pixels: t_i (true positives plus misses) plus $\sum_j n_{ji} - n_{ii}$ (false alarms). **Mean IoU (mIoU)** is the standard metric of PASCAL VOC, Cityscapes and ADE20K.

```
import numpy as np

def confusion(gt, pred, K, ignore=255):
    """K x K matrix: rows = true class, columns = predicted class; 'ignore' pixels are
     ↪  skipped."""
    keep = gt != ignore
    return np.bincount(K * gt[keep] + pred[keep], minlength=K * K).reshape(K, K)

def seg_metrics(C):
    tp = np.diag(C).astype(float)
    t = C.sum(1)                    # pixels of each true class
    p = C.sum(0)                    # pixels predicted as each class
    iou = tp / (t + p - tp)
    return {"pixel acc": tp.sum() / C.sum(), "mean acc": np.mean(tp / t),
            "mIoU": iou.mean(), "fwIoU": (t * iou).sum() / t.sum(), "IoU": np.round(iou, 3)}
```

```
# Toy: road (0) covers most of the image, car (1) is medium, pole (2) is thin
H, W = 120, 160
gt = np.zeros((H, W), int)
gt[60:100, 40:90] = 1
gt[10:110, 130:133] = 2
pred = gt.copy()
pred[60:100, 86:90] = 0                          # car slightly too short
pred[10:110, 130:133] = 0                        # the pole is missed entirely
m = seg_metrics(confusion(gt, pred, 3))
for k, v in m.items():
    print(f"{k:9s}: {v:.3f}" if np.isscalar(v) else f"{k:9s}: {v}")
```

```
pixel acc: 0.976
mean acc : 0.640
mIoU     : 0.631
fwIoU    : 0.953
IoU      : [0.974 0.92  0.   ]
```

Missing the pole entirely costs 1.6% of pixel accuracy and almost nothing in frequency-weighted IoU, but a third of the mean IoU. That is why mIoU, which weighs every class equally, is the headline number. Four practical details matter when you compare papers:

- The confusion matrix is accumulated over the **whole dataset**, then the IoUs are computed. Averaging per-image IoUs gives a different number.
- Void pixels are excluded through the ignore label, as in `confusion` above.
- mIoU still favours large objects *within* a class: one large car outweighs ten small ones.

  Cityscapes therefore also reports an instance-level **iIoU**, in which each pixel's contribution is weighted by the ratio of the class's average instance size to the size of its ground-truth instance [18].
- Region overlap is insensitive to boundary quality on large objects, where a few pixels of boundary error hardly change the IoU. **Boundary IoU** [19] measures overlap only in a thin band along the contours.

Optional: Boundary IoU in detail (in the Supplementary notes). Its definition, the band width, and a check in code that it scores the same boundary error equally on small and large objects.

Instance segmentation

Plain version. Instance segmentation is detection with an outline. The dominant approach finds each object first, as a detector would, and then segments the inside of its box.

Detect, then segment: Mask R-CNN

Mask R-CNN [20] adds a small fully convolutional mask branch to the two-stage detector Faster R-CNN [21], which DLCV 03 explains together with RoIAlign. For each region of interest, the branch predicts K binary masks of size $m \times m$, one per class (28×28 in the paper's variant built on a feature pyramid network, FPN [22]), and the loss uses a per-pixel sigmoid and the average binary cross-entropy on the mask of the ground-truth class only. Mask and class prediction are therefore **decoupled**: the box branch chooses the class, and masks of different classes do not compete, unlike the per-pixel softmax of an FCN; the paper's ablation finds that the per-class sigmoid gives large gains over a softmax [20]. At test time the $m \times m$ mask of the predicted class is resized to the box and binarized at 0.5.

Optional: RoI pooling versus RoIAlign (in the Supplementary notes of DLCV 03). Why rounding the box to the feature grid shifts masks, and the bilinear sampling that fixes it.

The fixed 28×28 grid is also Mask R-CNN's weakness: a large object's outline is drawn from a few hundred cells and comes out oversmoothed. **PointRend** [23] treats mask prediction like rendering in computer graphics: it starts from a coarse mask, repeatedly upsamples it bilinearly, and recomputes predictions only at the most uncertain points of the denser grid (for a binary mask, those with probabilities closest to 0.5), which lie mostly along the boundary.

Other routes exist. **Bottom-up** methods first predict per-pixel quantities, such as an embedding vector or a distance to the object centre, and then group pixels into instances; a watershed on a predicted distance map, with markers found by the network, is the learned version of the marker-controlled watershed of IP 10. And **query-based** methods predict a set of masks directly, as described in the mask-classification section below.

Mask AP

Instance segmentation is scored exactly like detection, with the COCO average precision of DLCV 03, except that the IoU between a prediction and a ground-truth object is computed on masks instead of boxes [20]. The metric needs a confidence score per mask, because AP ranks predictions, and it allows overlapping masks: a detector may output several overlapping hypotheses and is only penalized for those that do not match. This is why instance outputs cannot be used as a panoptic map without further processing.

Panoptic segmentation

Plain version. The panoptic task asks for one coherent picture: every pixel labeled, stuff and things alike, no overlaps, and a metric that treats a sky region and a car the same way.

The task and its metric

Kirillov et al. [4] defined the task (the format is the one given in "Three tasks, one label map") together with a metric, **panoptic quality (PQ)**. PQ is computed in two steps.

1. Matching. A predicted segment p and a ground-truth segment g of the same class match only if their IoU is strictly greater than 0.5. Because segments of a panoptic map cannot overlap, this match is **unique**: if p_1 and p_2 are disjoint predictions, then $\mathrm{IoU}(p_1, g) + \mathrm{IoU}(p_2, g) \le 1$, since each $|p_i \cap g|/|p_i \cup g| \le |p_i \cap g|/|g|$ and the two intersections share no pixel. So at most one prediction can exceed 0.5 for each g, and vice versa [4]. No optimal assignment step is needed.

2. Scoring. For each class, the matches split the segments into true positives TP (matched pairs), false positives FP (unmatched predictions) and false negatives FN (unmatched ground-truth segments). Then

$$\mathrm{PQ} = \frac{\sum_{(p,g)\in \mathrm{TP}} \mathrm{IoU}(p,g)}{|\mathrm{TP}| + \tfrac{1}{2}|\mathrm{FP}| + \tfrac{1}{2}|\mathrm{FN}|} = \underbrace{\frac{\sum_{(p,g)\in \mathrm{TP}} \mathrm{IoU}(p,g)}{|\mathrm{TP}|}}_{\text{segmentation quality (SQ)}} \times \underbrace{\frac{|\mathrm{TP}|}{|\mathrm{TP}| + \tfrac{1}{2}|\mathrm{FP}| + \tfrac{1}{2}|\mathrm{FN}|}}_{\text{recognition quality (RQ)}},$$

where the sums run over matched pairs and $|\cdot|$ counts elements of a set. **SQ** is the average IoU of matched segments, and **RQ** is the familiar F1 score of detection. PQ is computed per class and averaged over classes, so a rare class counts as much as a frequent one; every segment counts equally regardless of its area. $\mathrm{PQ}^{\mathrm{St}}$ and $\mathrm{PQ}^{\mathrm{Th}}$ average over stuff and thing classes only. Void pixels are removed from predictions before matching, and unmatched predictions that consist mostly of void (or of a "crowd" region of the same class) are not counted as false positives [4]. The two factors are not independent, since SQ is measured only on the matched segments, so always report all three.

The code implements this matching and scoring on the toy scene of Figure 5 (void handling reduced to its simplest form).

```
import numpy as np

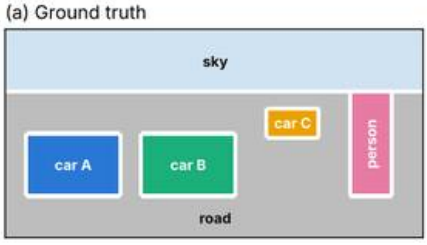
def panoptic_quality(gt_seg, gt_cls, pr_seg, pr_cls):
    """gt_seg/pr_seg: integer segment-id maps (0 = void); *_cls: dict segment id -> class.
    Matches segments of the same class with IoU > 0.5 and returns per-class (PQ, SQ, RQ, TP,
     ↪  FP, FN)."""
    out = {}
    for c in sorted(set(gt_cls.values()) | set(pr_cls.values())):
        g_ids = [i for i, k in gt_cls.items() if k == c]
        p_ids = [i for i, k in pr_cls.items() if k == c]
        ious, matched_p = [], set()
        for gi in g_ids:
            G = gt_seg == gi
            for pj in p_ids:
                P = (pr_seg == pj) & (gt_seg != 0)          # ignore predicted pixels on void
```

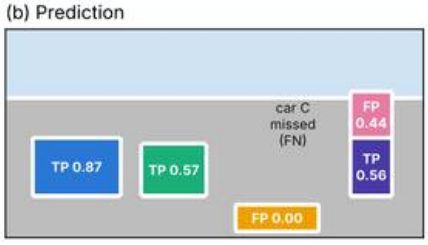
```
                iou = (G & P).sum() / (G | P).sum()
                if iou > 0.5:                                  # unique by the matching theorem
                    ious.append(iou); matched_p.add(pj)
        tp = len(ious); fn = len(g_ids) - tp; fp = len(p_ids) - len(matched_p)
        sq = np.mean(ious) if tp else 0.0
        rq = tp / (tp + 0.5 * fp + 0.5 * fn)
        out[c] = (sq * rq, sq, rq, tp, fp, fn)
    return out

H, W = 100, 200
SKY, ROAD, CAR, PERSON = 0, 1, 2, 3
def scene(boxes, sky_rows):
    seg = np.full((H, W), 2); seg[:sky_rows] = 1          # segment 1 = sky, 2 = road
    cls = {1: SKY, 2: ROAD}
    for sid, (c, r0, r1, c0, c1) in enumerate(boxes, start=3):
        seg[r0:r1, c0:c1] = sid; cls[sid] = c
    return seg, cls

gt_seg, gt_cls = scene([(CAR, 50, 80, 10, 55), (CAR, 50, 80, 65, 110),
                        (CAR, 38, 52, 125, 150), (PERSON, 30, 80, 165, 185)], sky_rows=30)
pr_seg, pr_cls = scene([(CAR, 52, 80, 13, 55),             # car A, slightly off
                        (CAR, 55, 80, 65, 96),             # car B, too small
                        (CAR, 84, 97, 110, 150),           # a car that is not there
                        (PERSON, 30, 52, 165, 185),        # the person, split in two
                        (PERSON, 52, 80, 165, 185)], sky_rows=33)   # car C is missed
names = ["sky", "road", "car", "person"]
res = panoptic_quality(gt_seg, gt_cls, pr_seg, pr_cls)
for c, (pq, sq, rq, tp, fp, fn) in res.items():
    print(f"{names[c]:6s} PQ {pq:.3f} = SQ {sq:.3f} x RQ {rq:.3f}  (TP {tp}, FP {fp}, FN
    ↪  {fn})")
pqs = np.array([v[0] for v in res.values()])
print(f"PQ {pqs.mean():.3f} | PQ_stuff {pqs[:2].mean():.3f} | PQ_things
↪  {pqs[2:].mean():.3f}")
```

```
sky    PQ 0.917 = SQ 0.917 x RQ 1.000  (TP 1, FP 0, FN 0)
road   PQ 0.805 = SQ 0.805 x RQ 1.000  (TP 1, FP 0, FN 0)
car    PQ 0.482 = SQ 0.723 x RQ 0.667  (TP 2, FP 1, FN 1)
person PQ 0.373 = SQ 0.560 x RQ 0.667  (TP 1, FP 1, FN 0)
PQ 0.644 | PQ_stuff 0.861 | PQ_things 0.428
```





(c) Panoptic quality

class	SQ	RQ	PQ
sky	0.917	1.000	0.917
road	0.805	1.000	0.805
car	0.723	0.667	0.482
person	0.560	0.667	0.373

PQ (mean of 4 classes) = 0.644

labels in (b): IoU with the best ground-truth segment of that class

Figure 5 — Panoptic quality on a toy scene. Each predicted segment is labeled with its best IoU against ground truth of its class; only IoU above 0.5 counts as a match.

The toy shows how PQ reads errors (numbers from this run). The split person still gets a match, because the lower piece covers 56% of it, but the upper piece becomes a false positive: splitting an object costs recognition quality, not just segmentation quality. Car B, at IoU 0.57, counts as fully recognized, but its low IoU pulls the car's SQ down. A missed car and an invented car each cost half a segment in the denominator.

Early panoptic systems: merge two outputs

Plain version. The first panoptic systems simply ran a semantic segmenter and an instance segmenter and pasted the results together.

The panoptic paper's own baseline [4] does this with a non-maximum-suppression-like procedure. It sorts the instance masks by confidence, drops low-scoring ones, and walks down the list: each mask loses the pixels already claimed by earlier masks and is kept only if enough of it remains. The surviving thing segments are then combined with the semantic map, and any pixel claimed by both a thing and a stuff class goes to the thing. **Panoptic FPN** [24] then put both tasks into one network: a Mask R-CNN with a feature pyramid network (FPN [22], see DLCV 03) plus a light semantic branch that upsamples and sums all pyramid levels at 1/4 scale. The semantic branch predicts the stuff classes plus a single "other" class for all thing pixels, so the two branches do not fight over objects.

Optional: Panoptic Segmentation: a reading guide (in the Paper guides). How the paper defines the task and metric, what its human-consistency study shows, and how to read its tables.

Mask classification: one model for all three tasks

Plain version. Instead of asking "what class is this pixel?", ask "which regions are there, and what is each region?". Predict a fixed number of candidate masks and one class (or "nothing") for each. A semantic map, a set of instances and a panoptic map are then just different ways of reading the same list.

MaskFormer

Cheng, Schwing and Kirillov [17] call the FCN formulation **per-pixel classification** and the Mask R-CNN and DETR formulation **mask classification**. Their observation is that mask classification is general enough for semantic segmentation too. **MaskFormer** predicts a set of N probability–mask pairs $\{(p_i, m_i)\}_{i=1}^{N}$, where p_i is a distribution over K classes plus a "no object" label $\emptyset$, and $m_i \in [0,1]^{H \times W}$ is a soft mask. Three modules produce them:

1. A backbone and a **pixel decoder** produce per-pixel embeddings $\mathcal{E}_{\text{pixel}} \in \mathbb{R}^{C_E \times H \times W}$, where C_E is the embedding dimension.
2. A **transformer decoder** turns N learnable queries into N per-segment embeddings by attending to the image features, as in DETR [25].
3. A linear classifier gives p_i, and a small MLP gives a mask embedding $\mathcal{E}_{\text{mask}}[:, i] \in \mathbb{R}^{C_E}$. The mask is a dot product followed by a sigmoid,

$$m_i[h, w] = \sigma(\mathcal{E}_{\text{mask}}[:, i]^\top \, \mathcal{E}_{\text{pixel}}[:, h, w]),$$

where σ is the logistic sigmoid and (h, w) a pixel position.

Training matches predictions to ground-truth segments one to one, as DETR does for boxes (DLCV 03), but with a matching cost built from class probabilities and mask losses instead of boxes. Matched predictions get a cross-entropy class loss and a focal + Dice mask loss; unmatched ones are trained to predict $\emptyset$ [17].

At inference, two rules read the same output in different ways [17]. **Semantic inference** marginalizes over queries, $\hat{y}[h, w] = \arg\max_c \sum_i p_i(c)\, m_i[h, w]$, with c ranging over the K real classes.

Panoptic inference assigns each pixel to one query, $\arg\max_{i:\, c_i \neq \emptyset} p_i(c_i)\, m_i[h, w]$, where c_i is the most likely class of query i, so two queries of the same class give two instances. The snippet runs both rules on a tiny 6×10 example.

```
import numpy as np

rng = np.random.default_rng(0)
# Ground-truth layout of a tiny 6 x 10 image: sky on top, road below, two cars on the road
region = np.zeros((6, 10), int)               # 0 sky, 1 road, 2 car A, 3 car B
region[2:] = 1
region[3:5, 1:4] = 2
region[3:5, 6:9] = 3

# Pixel decoder output: a C_E = 4 dimensional embedding per pixel (here: noisy one-hot of
 ↪  the region)
E_pixel = 6 * np.eye(4)[region].transpose(2, 0, 1) - 3 + 0.5 * rng.standard_normal((4, 6,
 ↪  10))

# Transformer decoder output for N = 5 queries: a mask embedding and class probabilities
# over K = 3 classes (sky, road, car) plus "no object" (last column)
E_mask = np.array([[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 1, 0], [0, 0, 0, 1], [.2, .2, .2,
 ↪  .2]], float)
p = np.array([[.90, .05, .03, .02],          # query 0: sky
              [.05, .85, .05, .05],          # query 1: road
              [.02, .08, .85, .05],          # query 2: a car
              [.03, .07, .75, .15],          # query 3: another car
              [.05, .05, .05, .85]])         # query 4: nothing

m = 1 / (1 + np.exp(-np.einsum("nc,chw->nhw", E_mask, E_pixel)))   # N masks, sigmoid(dot
 ↪  product)
```

```
# Semantic inference: sum over queries of p_i(c) * m_i, then argmax over the K real classes
sem = np.einsum("nk,nhw->khw", p[:, :3], m).argmax(0)
# Panoptic inference: each pixel goes to the query maximizing p_i(c_i) * m_i (no-object
 ↪  queries dropped)
c_i = p.argmax(1)
keep = c_i != 3
score = p[np.arange(5), c_i][:, None, None] * m
seg = np.where(keep[:, None, None], score, -1).argmax(0)
print("semantic (0 sky, 1 road, 2 car):\n", sem)
print("panoptic segment = query index:\n", seg)
print("query classes:", c_i, "-> car segments:", [int(i) for i in np.unique(seg) if c_i[i]
 ↪  == 2])
```

```
semantic (0 sky, 1 road, 2 car):
 [[0 0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0 0]
 [1 1 1 1 1 1 1 1 1 1]
 [1 2 2 2 1 1 2 2 2 1]
 [1 2 2 2 1 1 2 2 2 1]
 [1 1 1 1 1 1 1 1 1 1]]
panoptic segment = query index:
 [[0 0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0 0]
 [1 1 1 1 1 1 1 1 1 1]
 [1 2 2 2 1 1 3 3 3 1]
 [1 2 2 2 1 1 3 3 3 1]
 [1 1 1 1 1 1 1 1 1 1]]
query classes: [0 1 2 2 3] -> car segments: [2, 3]
```

The semantic map merges both cars into class 2; the panoptic map keeps them apart as segments 2 and 3, from exactly the same network output. The paper reports that MaskFormer is on par with per-pixel classification on Cityscapes (19 classes) and better on datasets with larger vocabularies [17].

Mask2Former and its successors

Mask2Former [26] kept MaskFormer's meta-architecture (backbone, pixel decoder, transformer decoder) and changed the decoder so that it trains faster and segments instances well. Its key component is **masked attention**: each query attends only to the pixels inside the mask it predicted at the previous layer. In layer l, with query features $X_l \in \mathbb{R}^{N\times C}$,

$$X_l = \mathrm{softmax}(\mathcal{M}_{l-1} + Q_l K_l^{\top})V_l + X_{l-1}, \qquad \mathcal{M}_{l-1}(x,y) = \begin{cases} 0 & \text{if } M_{l-1}(x,y) = 1, \\ -\infty & \text{otherwise,} \end{cases}$$

where Q_l is a linear projection of X_{l-1}, K_l and V_l are projections of the image features, and M_{l-1} is the previous layer's mask prediction, resized and binarized at 0.5. Adding $-\infty$ before the softmax

gives zero attention weight outside the mask. The authors hypothesize that local features suffice to update a query, and that the slow convergence of earlier models came from cross-attention having to learn to focus on the object. Two further changes are a round-robin use of the pixel decoder's 1/32, 1/16 and 1/8 feature maps in successive decoder layers, for small objects, and computing the mask loss on sampled points instead of full masks, which the paper reports reduces training memory three-fold [26].

Optional: Mask2Former: a reading guide (in the Paper guides). What "universal" means in the paper, its ablations, how to read its comparison tables and the limitations it states.

Two later directions build on this design. **OneFormer** [27] trains a *single* model for all three tasks: it derives semantic and instance labels from panoptic annotations, samples one task per training image, and conditions the queries on a "task token", so that one set of weights serves every task.

EoMT [28] removes the task-specific parts altogether. The authors argue that a large plain ViT with extensive pretraining can learn what the adapter, pixel decoder and transformer decoder supplied; they concatenate learnable queries to the patch tokens before the last ViT blocks, which then process both jointly, and report accuracy similar to the more complex models at much higher speed (up to four times faster with ViT-L).

Promptable and open-vocabulary segmentation

Plain version. The models above segment a fixed list of classes they were trained on. Two newer families relax that list. Promptable models segment *whatever you point at*; open-vocabulary models segment *whatever you name*.

Promptable segmentation. The Segment Anything Model (SAM) [29] returns a mask for a point, box or rough-mask prompt, and for an ambiguous point (shirt or person?) it returns several valid masks with confidence scores; it was trained on SA-1B, a dataset of over 1 billion masks on 11 million images built with a model-in-the-loop data engine. SAM 2 [30] extends this to video with a streaming memory, and SAM 3 [31] adds **promptable concept segmentation**: given a short noun phrase or image exemplars, it returns masks and identities for *all* matching instances. These models are described in DLCV 01 and ML 09, and SAM 2 as a tracker in DLCV 06. In terms of this chapter's three tasks: SAM's masks carry **no class**, so on their own they are neither semantic nor panoptic output; a box from a detector plus SAM gives instance masks; and SAM 3's concept prompts make it an open-vocabulary instance segmenter. When you read "zero-shot" results with such models, check where the prompts came from: prompts derived from ground-truth boxes give the model a lot of information.

Open-vocabulary segmentation. The key ingredient is a vision–language model such as CLIP [32], whose text encoder maps any class name to an embedding in the same space as image features (ML 09 covers it). Replace a segmenter's fixed classifier weights with text embeddings of the class names and the vocabulary becomes a list you type at test time. **LSeg** [33] does exactly

this: a transformer image encoder produces per-pixel embeddings, a text encoder embeds the label names, and training aligns each pixel's embedding with the text embedding of its class, so that unseen labels can be used at test time without retraining. **CAT-Seg** [34] instead builds a **cost volume** of cosine similarities between CLIP image and text embeddings and aggregates it, which the authors find lets them fine-tune CLIP's encoders without overfitting to the training classes, a problem that made most earlier methods freeze CLIP. Zhu and Chen's survey [35] warns that evaluating "unseen" classes is difficult because the image–text pretraining data may already contain them.

Datasets and benchmarks

Plain version. Pixel labels are expensive, so a few public datasets carry the field. Each one fixes a class list, a stuff/thing split and an evaluation protocol.

Dataset	Content	Scale and classes (from the papers)	Tasks
PASCAL VOC 2012 [36]	everyday objects	20 object classes + background (as described in [8])	semantic
Cityscapes [18]	street scenes from 50 cities	5,000 finely and 20,000 coarsely annotated images; 30 classes, 19 used for evaluation; 8 with instance labels [4]	semantic, instance, panoptic
ADE20K [37]	diverse scenes, objects and parts	20,210 / 2,000 / 3,000 train / val / test images; benchmark of the top 150 classes (SceneParse150); 100 things + 50 stuff for panoptic [4]	semantic, panoptic
COCO [38] + COCO-Stuff [39]	everyday scenes	instance masks for 80 thing classes [24]; COCO-Stuff adds pixel labels for 91 stuff classes to all 164K images of COCO 2017; 80 things + 53 stuff in COCO panoptic [17]	instance, semantic, panoptic
Mapillary Vistas [40]	street-level images worldwide	25k images; 28 stuff + 37 thing classes [4]	semantic, instance, panoptic

Three facts about these datasets shape every result. First, **annotation is costly and noisy**: the panoptic paper's human-consistency study, in which two annotations of the same images are scored against each other, gives PQ values well below 100 on all three datasets it examines [4].

Second, **class lists are small** compared with the vocabulary of the world, which motivates the

open-vocabulary models above. Third, **domains differ**: Cityscapes was recorded in 50 cities, primarily in Germany, and deliberately not in adverse weather [18], so a model trained on it meets different road markings, weather and cameras elsewhere. Bird's-eye-view map segmentation for driving, which predicts labels on a ground-plane grid instead of the image, is covered in DLCV 09.

Failure modes and practical advice

Plain version. Segmenters fail in predictable places: where structures are thin or small, along boundaries, on rare classes, and on images unlike the training set.

- **Thin and small structures** (poles, wires, distant pedestrians) disappear at large output strides, as the first experiment showed. Use output stride 8 or a decoder with high-resolution skips, and evaluate with per-class IoU, not only mIoU.
- **Boundaries** come out smooth or blocky. Bilinear score upsampling, decoders, point-based refinement such as PointRend [23] and boundary-aware metrics [19] address this.
- **Class imbalance** biases the network toward frequent classes. Weighted cross-entropy, focal or Dice terms, and sampling images that contain rare classes help (see the loss experiment).
- **Context errors**: a region labeled with a class that makes no sense in its surroundings (the boat labeled as a car in [12]). Global context modules and attention reduce them.
- **Fragmented masks and holes**: per-pixel classifiers have no notion of shape. Morphological clean-up of predicted masks (opening, closing, hole filling; see IP 09) is a cheap and common post-processing step.
- **High-resolution inputs** exceed GPU memory. Run on overlapping tiles and blend the scores, the same overlap-tile idea as U-Net's.
- **Domain shift**: performance drops on new cities, weather, sensors or medical scanners. Measure it with a held-out domain, not only a held-out split.

Modern view

What the surveys say

Minaee et al. [1] review the CNN era of semantic and instance segmentation in ten families: fully convolutional networks, convolutional models with graphical models (such as CRFs), encoder–decoders, multi-scale and pyramid networks, R-CNN-based models for instances, dilated convolutions and the DeepLab family, recurrent networks, attention models, generative and adversar-

ial models, and convolutional models combined with active contours. They also review datasets and reported results. Most families appear in this chapter's sections, with FCN, U-Net, PSPNet, DeepLab and Mask R-CNN as examples.

Li et al. [2] organize transformer-based segmentation around one **meta-architecture** with a feature extractor, object queries and a transformer decoder, and classify methods by which component they modify: better backbones, cross-attention designs in the decoder, ways of optimizing the queries, and so on. They also cover video and point-cloud segmentation, foundation-model tuning, efficiency and medical segmentation, and they re-evaluate many methods on common benchmarks.

Zhou et al. [3] review more than 300 methods of the foundation-model era, splitting the field into **generic** segmentation (semantic, instance, panoptic) and **promptable** segmentation (interactive, referring and few-shot). A central theme is that segmentation knowledge *emerges* in models trained without masks, such as CLIP, text-to-image diffusion models and self-supervised ViTs, which training-free methods exploit. **Zhu and Chen** [35] focus on open-vocabulary detection and segmentation and on how hard fair evaluation of "novel" classes is.

The state of the art, 2023–2026

- **Universal query-based architectures** are the default for closed-vocabulary benchmarks: Mask2Former [26] and its descendants such as OneFormer [27] handle semantic, instance and panoptic segmentation with one design, and specialized per-pixel models remain common where speed or simplicity matters (SegFormer [14], DeepLab-style models [7]).
- **Simpler models on larger pretraining**: EoMT [28] argues that compute is better spent on scaling a plain ViT encoder than on adding decoders and adapters.
- **Promptable foundation models** (SAM [29], SAM 2 [30], SAM 3 [31]) turned class-agnostic masks, video propagation and concept prompts into reusable components, and their data engines changed how segmentation datasets are built.
- **Open-vocabulary segmentation** builds on vision–language models (LSeg [33], CAT-Seg [34]); in many applications a model is now prompted with class names instead of retrained.

Open problems

1. **Boundaries, small objects and thin structures.** Mask2Former's authors state that it still struggles with small objects [26], and region metrics hide boundary errors on large objects [19].
2. **One model, trained once.** Universal architectures are still often trained separately per task and dataset; Mask2Former trained only on panoptic labels is slightly worse on the instance

and semantic metrics than when trained on each task [26].

3. **Open-vocabulary evaluation** depends on prompts and on what the pretraining data contained [35].
4. **Data.** Pixel labels are expensive; the surveys point to scalable data engines, synthetic image–mask pairs from generative models and weak or no supervision as directions, each with its own domain gap [3], [2].
5. **Efficiency.** Foundation-model segmenters are heavy; distillation, compression and parameter-efficient tuning are open engineering problems [3].
6. **Understanding emergence.** Why models trained without masks learn to segment is not well explained [3].

Key takeaways

- Semantic segmentation labels every pixel with a class; instance segmentation outputs a scored, possibly overlapping mask per thing; panoptic segmentation gives every pixel a class and an instance id, with stuff and things treated uniformly and no overlaps.
- A fully convolutional network outputs a score map at output stride s; upsample scores (bilinearly), not labels, and remember that nothing smaller than about half a cell survives a large stride.
- Detail comes back through decoders with skip connections (U-Net, DeepLabv3+); context comes from atrous convolution, ASPP with image pooling, pyramid pooling or attention.
- Per-pixel cross-entropy is dominated by the frequent class; class weights, the focal loss or Dice-type losses counter imbalance, and they rank predictions differently.
- mIoU (from a dataset-wide confusion matrix) weighs classes equally; mask AP scores ranked, overlapping instance masks; PQ = SQ × RQ scores non-overlapping segments matched at IoU above 0.5, which is unique.
- Mask classification (MaskFormer, Mask2Former) predicts N mask–class pairs and reads them as semantic, instance or panoptic output; masked attention made it competitive on all three tasks.
- Promptable models (SAM family) segment what you point at but give no class; open-vocabulary models (LSeg, CAT-Seg) segment what you name, with evaluation pitfalls of their own.

Exercises

1. **Output stride by hand.** A road scene is 1024 × 2048 pixels and a pole is 6 pixels wide. With ideal coarse scores (class fractions per cell) and argmax decisions, at which output strides can the pole survive at all, and why does its position relative to the grid matter? Check your answer by moving the bar in the first snippet by one or two pixels.

Hint

The pole wins a cell only if it covers more than half of the cell's width, so $s < 12$; with $s = 8$ it survives only in cells where at least 5 of its 6 columns fall inside one 8-pixel cell. At $s = 4$ it always covers at least one full cell. Shifting the bar changes which cells it dominates, which is why the bar's stride-4 IoU in the nearest-label column of the snippet is 0.67, not 1 (a tie at exactly half a cell goes to the background, the first class).

2. **Atrous arithmetic.** (a) What is the effective size of a 3×3 kernel at rate 12? (b) A backbone stage often starts with stride 2 followed by 3×3 convolutions. If you remove the stride and use rate 2 instead, what happens to the output stride, the receptive field of the following layers, and the cost of those layers?

Hint

(a) $k_e = 3 + 2 \cdot 11 = 25$. (b) The output stride halves (for example 32 to 16), every later layer covers the same image region as before (the rate compensates for the denser grid), and the later layers now run on four times as many positions, so their cost grows about four-fold.

3. **Losses with an empty mask.** Modify the loss snippet so that the image contains *no* foreground. What does each loss return for the "all background (p=0.05)" prediction, and what happens to the soft Dice loss without the `eps` term? How do implementations usually avoid the problem?

Hint

With $g = 0$ everywhere, the Dice numerator is 0 (plus `eps`) and the denominator is $\sum p_i = 204.8$, so the loss is almost 1 however good the prediction is; for $p = 0$ everywhere it is $0/0$ without `eps`, and with `eps` it becomes $1 - \text{eps}/\text{eps} = 0$. `iou_at_half` also divides 0 by 0 when neither mask has a pixel. Common fixes: add a smoothing constant to numerator and denominator, compute Dice over the whole batch instead of per image, or combine Dice with cross-entropy.

4. **PQ by hand.** In the PQ snippet, replace the two person pieces by a single prediction rows 32 to 80. Predict the new person SQ, RQ and PQ before running the code. Then make car B larger so that its IoU drops below 0.5 by over-segmenting; how do the car's TP, FP and FN counts change?

Hint

The single person prediction has IoU $48 \cdot 20/(50 \cdot 20) = 0.96$, so SQ = 0.96, RQ = 1 and PQ = 0.96. If car B's IoU falls to 0.5 or below, that pair is no longer a match: car TP drops from 2 to 1, and FP and FN each rise by one.

5. **Semantic from panoptic, and back.** Using the mask-classification snippet, (a) derive the semantic map from the panoptic segment map and check it equals `sem`; (b) make query 3 predict "road" instead of "car" with probability 0.6 and explain both outputs. (c) Why can a semantic map not be turned back into a panoptic map in general?

Hint

(a) Map each segment id to its class, `c_i[seg]`. (b) Query 3's mask still covers car B, so in the panoptic map car B stays segment 3, but that segment is now labeled road; because a stuff class has a single segment per image, the final output merges it with the road. In the semantic map car B becomes road too. Only one car is left in both. (c) A semantic map has no instance ids: touching objects of one class form one region, and splitting it needs information the map does not contain.

Supplementary notes

These notes are optional: each takes one point from the main text and treats it in more detail. Nothing in the main text, code or exercises depends on them.

Gridding in stacked dilated convolutions

This note explains which sequences of dilation rates leave holes in the receptive field, and a simple rule for choosing rates that do not.

The problem. Take one output of a stack of n dilated convolutions with kernel size k and rates $r_1, \dots, r_n$, and ask which input positions influence it. In 1-D, the set of reachable offsets is the sum set $\{\sum_l r_l d_l : d_l \in \{-c, \dots, c\}\}$ with $c = (k-1)/2$. If every r_l is a multiple of some integer $q > 1$, every reachable offset is a multiple of q as well: the output sees one input in q along each axis, a regular lattice with holes. Neighbouring outputs then read disjoint lattices, and the output map shows a checkerboard pattern. Wang et al. [11] call this **gridding** and note that it gets worse at high rates, where a sparse kernel no longer covers any local neighbourhood.

The rule. Their hybrid dilated convolution (HDC) uses different rates in consecutive layers, arranged in a sawtooth pattern, chosen so that the final receptive field covers a square region without holes. Define the maximum distance between two nonzero taps seen through layers i to n recursively as

$$M_i = \max\left[M_{i+1} - 2r_i,\ M_{i+1} - 2(M_{i+1} - r_i),\ r_i\right], \qquad M_n = r_n,$$

where r_i is the rate of layer i. The design goal is $M_2 \le k$. For $k = 3$, rates $(1, 2, 5)$ give $M_2 = 2$ and work, while $(1, 2, 9)$ give $M_2 = 5$ and do not [11]. They also state that the rates within a group should not have a common factor such as 2, 4, 8.

A check in code. Counting the paths from each input offset to one output makes the holes visible.

```
import numpy as np

def coverage(rates, k=3):
    """How many paths connect each input offset to one output of a stack of 1-D dilated
     ↪  convs."""
    reach = np.array([1])
    for r in rates:
        taps = np.zeros(r * (k - 1) + 1, int); taps[::r] = 1      # a k-tap kernel with rate
         ↪  r
        reach = np.convolve(reach, taps)
    return reach

for rates in [(2, 2, 2), (1, 2, 4), (1, 2, 5), (1, 2, 9)]:
    c = coverage(rates)
    print(f"rates {rates}: span {len(c)}, holes {int((c == 0).sum())}, paths {c}")
```

```
rates (2, 2, 2): span 13, holes 6, paths [1 0 3 0 6 0 7 0 6 0 3 0 1]
rates (1, 2, 4): span 15, holes 0, paths [1 1 2 1 3 2 3 1 3 2 3 1 2 1 1]
rates (1, 2, 5): span 17, holes 0, paths [1 1 2 1 2 2 2 2 1 2 2 2 2 1 2 1 1]
rates (1, 2, 9): span 25, holes 4, paths [1 1 2 1 2 1 1 0 0 1 1 2 1 2 1 1 0 0 1 1 2 1 2 1 1]
```

Equal rates of 2 leave every second input unseen; a jump from 2 to 9 leaves gaps between the copies of the inner pattern. The path counts also show that coverage is uneven even without holes: some offsets are reached by three paths, others by one.

Edge cases. ASPP is not affected in the same way, because its branches run in parallel on the same map and are combined, not stacked. And a backbone whose earlier, undilated layers already mix neighbouring pixels softens gridding in practice; the problem is strongest when several highly dilated layers follow each other.

Dense CRF post-processing

This note describes the fully connected conditional random field that early DeepLab versions applied to the network's output to sharpen boundaries.

The model. Treat every pixel i as a node with a label x_i and minimize the energy [8], [41]

$$E(\mathbf{x}) = \sum_i \theta_i(x_i) + \sum_{i<j} \theta_{ij}(x_i, x_j),$$

where the **unary** term $\theta_i(x_i) = -\log P(x_i)$ uses the network's probability $P(x_i)$ at pixel i, and the **pairwise** term connects *every* pair of pixels:

$$\theta_{ij}(x_i, x_j) = \mu(x_i, x_j) \left[w_1 \exp\Big(-\frac{\|\mathbf{p}_i - \mathbf{p}_j\|^2}{2\sigma_\alpha^2} - \frac{\|\mathbf{I}_i - \mathbf{I}_j\|^2}{2\sigma_\beta^2} \Big) + w_2 \exp\Big(-\frac{\|\mathbf{p}_i - \mathbf{p}_j\|^2}{2\sigma_\gamma^2} \Big) \right].$$

Here $\mu(x_i, x_j) = 1$ if $x_i \neq x_j$ and 0 otherwise (a Potts penalty, so only disagreeing pairs pay), $\mathbf{p}_i$ is the position and $\mathbf{I}_i$ the RGB colour of pixel i, w_1, w_2 are weights, and $\sigma_\alpha, \sigma_\beta, \sigma_\gamma$ set the scales of the kernels. The first, **bilateral** kernel says that pixels close in both position and colour should share a label; the second only enforces spatial smoothness.

Why it is tractable. A fully connected graph over a megapixel image has about 5×10^{11} pixel pairs. Krähenbühl and Koltun [41] showed that under a mean-field approximation, in which the label distribution factorizes over pixels, each update of all pixels is a Gaussian filtering in a joint position–colour feature space; approximate high-dimensional filtering then makes each update linear, instead of quadratic, in the number of pixels.

What it does and does not do. It snaps boundaries to colour edges, the same intuition as the bilateral filter of IP 03, and can fill small holes. It cannot recover an object the network missed, because the unary term dominates far from boundaries, and it adds parameters to tune on a validation set. DeepLabv3 reports its improvements without this post-processing [10].

The gradient of the soft Dice loss

This note derives the derivative of the soft Dice coefficient and explains its unusual behaviour.

The derivative. Use the sum form $D = 2I/S$ with $I = \sum_i p_i g_i$ and $S = \sum_i p_i + \sum_i g_i$, where $p_i \in [0, 1]$ are predicted probabilities and $g_i \in \{0, 1\}$ the labels. By the quotient rule, for pixel j,

$$\frac{\partial D}{\partial p_j} = \frac{2g_j S - 2I}{S^2} = \frac{2\,(g_j S - I)}{S^2},$$

because $\partial I/\partial p_j = g_j$ and $\partial S/\partial p_j = 1$. (Milletari et al. [15] give the analogous expression for their squared denominator.)

What it says.

- Every **background** pixel ($g_j = 0$) gets the same gradient $-2I/S^2$, whatever its current probability: Dice pushes all background probabilities down equally hard, unlike cross-entropy, whose push $p_j - g_j$ is proportional to each pixel's error.
- Every **foreground** pixel gets the same positive gradient $2(S - I)/S^2$.
- The size of the push depends on global sums. When the predicted and true masks are both small, S is small and the gradients are large; this is how Dice "listens" to small objects, and also why it can be unstable early in training.

The code checks the formula against finite differences.

```
import numpy as np
```

```
rng = np.random.default_rng(1)
g = (rng.random(200) < 0.05).astype(float)       # 200 pixels, about 5% foreground
p = rng.uniform(0.05, 0.6, 200)                  # some soft prediction

def dice(p, g):
    return 2 * (p * g).sum() / (p.sum() + g.sum())

I, S = (p * g).sum(), p.sum() + g.sum()
grad = 2 * (g * S - I) / S ** 2                  # analytic dD/dp_j
eps = 1e-6
num = np.array([(dice(p + eps * np.eye(200)[j], g) - dice(p - eps * np.eye(200)[j], g)) / (2
 ↪  * eps)
                for j in range(200)])
print("max |analytic - numeric|:", f"{np.abs(grad - num).max():.1e}")
print("distinct gradient values on background pixels:", np.unique(grad[g ==
 ↪  0].round(12)).size)
print(f"foreground gradient {grad[g == 1][0]:+.5f}, background gradient {grad[g ==
 ↪  0][0]:+.5f}")
```

```
max |analytic - numeric|: 9.0e-12
distinct gradient values on background pixels: 1
foreground gradient +0.02793, background gradient -0.00099
```

Edge cases. If the ground truth is empty ($g = 0$), then $I = 0$ and $D = 0$ for any nonzero prediction, so the loss is 1 even for a nearly perfect prediction; if both are zero the ratio is undefined. Implementations add a smoothing constant to the numerator and denominator, compute the sums over the whole batch rather than per image, or add cross-entropy so that the gradient never depends on the Dice ratio alone. With these probabilities the gradient chain goes through a sigmoid or softmax, which multiplies each pixel's push by $p_j(1 - p_j)$, so saturated pixels still learn slowly.

Boundary IoU in detail

This note defines Boundary IoU [19] and shows why it complements mask IoU.

The problem with mask IoU. For a large object, almost all of its pixels are far from the boundary, so a prediction can get the outline wrong by several pixels and still have a mask IoU above 0.9. The Boundary IoU paper shows this sensitivity gap across object sizes and proposes a measure focused on the contour.

Definition. For a ground-truth mask G and a prediction P, let G_d be the set of pixels within distance d of the contour of G, and P_d likewise for P. Boundary IoU is

$$\text{Boundary IoU}(G, P) = \frac{|(G_d \cap G) \cap (P_d \cap P)|}{|(G_d \cap G) \cup (P_d \cap P)|},$$

the IoU of the two **inner boundary bands**. The paper sets d to a fixed fraction of the image diagonal (2% in its analysis), motivated by an observation that two annotators' contours of the same

object rarely differ by more than about 1% of the image diagonal, regardless of object size. Replacing mask IoU by Boundary IoU in the AP and PQ protocols gives the paper's Boundary AP and Boundary PQ.

A check in code. Predict each disk with a boundary that is 2 pixels too far out, once for a small and once for a large disk.

```
import numpy as np
from scipy import ndimage as ndi

def boundary_band(mask, d):
    """Pixels of the mask that lie within distance d of its contour."""
    return mask & (ndi.distance_transform_edt(mask) <= d)

def mask_iou(a, b):
    return (a & b).sum() / (a | b).sum()

def boundary_iou(g, p, d):
    return mask_iou(boundary_band(g, d), boundary_band(p, d))

n = 256
yy, xx = np.mgrid[0:n, 0:n]
d = 0.02 * np.hypot(n, n)                        # 2% of the image diagonal, about 7 pixels
for r in [10, 100]:
    g = (yy - 128) ** 2 + (xx - 128) ** 2 <= r ** 2
    p = (yy - 128) ** 2 + (xx - 128) ** 2 <= (r + 2) ** 2      # boundary 2 pixels too far
     ↪ out
    print(f"radius {r:3d}: mask IoU {mask_iou(g, p):.3f}, boundary IoU {boundary_iou(g, p,
     ↪ d):.3f}")
```

```
radius  10: mask IoU 0.719, boundary IoU 0.596
radius 100: mask IoU 0.961, boundary IoU 0.562
```

Mask IoU calls the large disk nearly perfect and the small one poor, for the same 2-pixel error. Boundary IoU scores both errors similarly, because it looks only at a band of the same width in both cases.

Edge cases. When both masks are so small or thin that every pixel lies within d of its contour, the bands are the whole masks and Boundary IoU equals mask IoU, so small objects are not over-penalized. The band is computed per object, so the measure is meant for instance and panoptic segments, not for a whole semantic map at once.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Panoptic Segmentation: a reading guide

Kirillov, He, Girshick, Rother and Dollár, *Panoptic Segmentation*, CVPR 2019 [4]. This guide summarizes what the paper proposes and how to read its results.

The problem. Semantic segmentation (stuff, per-pixel labels, mIoU) and instance segmentation (things, scored masks, AP) had grown into separate tasks with separate datasets, methods and metrics. The authors argue that the split may hide something important, and that earlier joint "scene parsing" work faded partly for lack of an agreed task definition and metric.

The idea. The paper contributes a task and a metric, not a network. The task: every pixel gets a (class, instance id) pair, stuff ids are ignored, segments cannot overlap, and no confidence scores are needed. The metric: PQ, with a matching step whose IoU threshold of 0.5 makes the match unique (Theorem 1 in the paper, proved in a few lines) and a scoring step that factors into SQ and RQ. The desiderata are stated explicitly: completeness (stuff and things treated alike), interpretability and simplicity.

What it shows.

- *Human consistency* (its Tables 1 and 2). Doubly annotated images (30 from Cityscapes, 64 from ADE20K, 46 from Vistas) are scored against each other. Human PQ is far from perfect, stuff and things are similarly hard for humans, and for small objects RQ drops sharply while SQ stays reasonable: annotators miss small objects more than they mis-draw them.
- *The threshold* (Figures 6 and 7). With an optimal bipartite matching and no threshold, fewer than 16% of matched pairs have IoU below 0.5, and lowering the threshold to 0.25 changes PQ much less than raising it to 0.75, which supports 0.5 as the default.
- *Machine baselines* (Tables 3–6). Outputs of leading semantic and instance systems, obtained from the challenge organizers, are merged with the heuristic described in the main text. AP and PQ track each other for instance methods, and mIoU and PQ for semantic methods, with one telling exception: on ADE20K a semantic method that produces many small spurious patches loses much more PQ than mIoU, because PQ counts segments and mIoU counts pixels.
- *Humans versus machines* (Table 6). Machines are close to humans in SQ but far behind in RQ: at the time, the main gap was recognition, not segmentation.

How to read the tables. The human study uses small image sets, so the authors report bootstrapped error ranges; check them before reading small differences. The human-versus-machine comparison uses different test images and averages over slightly different class sets, which the paper itself flags as approximate. The paper also warns that human PQ values are not comparable across datasets, because class counts, coverage and scene complexity differ. The machine results are merges of competition entries, not a single trained model, so they describe the state of separate systems rather than a method.

Limits. PQ gives every segment equal weight regardless of area, so a tiny missed object costs as much as a missed building; this is by design, but it changes which errors dominate. The 0.5

threshold ignores how good a match is beyond "more than half", which SQ only partly recovers. Boundary quality is not measured separately, a gap Boundary PQ [19] later addressed.

Why it matters here. PQ is the metric of the panoptic section and the motivation for the universal models that follow; the paper's observation that recognition, not segmentation, was the bottleneck is a useful lens for reading later results.

Mask2Former: a reading guide

Cheng, Misra, Schwing, Kirillov and Girdhar, *Masked-attention Mask Transformer for Universal Image Segmentation*, CVPR 2022 [26]. This guide explains what the paper claims and what to look for in its tables.

The problem. "Universal" mask-classification models such as MaskFormer could address every segmentation task with the same architecture, but they lagged behind the best specialized model on instance segmentation and were expensive to train: the paper notes that MaskFormer needed 300 epochs on COCO and fit only one image on a 32 GB GPU. Researchers therefore kept building specialized architectures per task.

The idea. Keep MaskFormer's meta-architecture and fix the transformer decoder:

- **masked attention**, which restricts each query's cross-attention to the foreground of its own previous mask prediction (the equation in the main text);
- **multi-scale features in round robin**, feeding the pixel decoder's 1/32, 1/16 and 1/8 maps to successive decoder layers, repeated L times;
- **optimization changes**: cross-attention before self-attention, learnable query features that are supervised directly (the authors say they behave like a region proposal network), and no dropout;
- **point-sampled mask loss**: the loss is computed on $K = 12544$ (112×112) sampled points, with uniform sampling for the matching cost and importance sampling for the final loss, following PointRend; the paper reports training memory falling from 18 GB to 6 GB per image.

What it shows. The same architecture, trained separately per task and dataset, reaches or exceeds the specialized state of the art of the time on COCO, ADE20K, Cityscapes and Mapillary Vistas, for panoptic, instance and semantic segmentation (the abstract quotes 57.8 PQ on COCO panoptic, 50.1 AP on COCO instance and 57.7 mIoU on ADE20K). In the ablations (its Table 4), masked attention gives the largest improvement, more for instance and panoptic than for semantic segmentation, and the high-resolution multi-scale strategy matters too.

How to read the tables.

- *Check the training schedule column.* In the COCO panoptic table (its Table 1), DETR is trained for 500+25 epochs, MaskFormer for 300 and Mask2Former for 50; part of the paper's message is convergence speed, so compare at equal epochs where the paper provides them.
- *Check backbone and pretraining.* Rows marked with a dagger use backbones pretrained on a larger classification dataset than the others, and the strongest rows use 200 queries instead of 100. Compare rows with the same backbone before crediting the decoder.
- *Separate task-specific and panoptic-trained numbers.* The table also reports AP^{Th}_{pan} and $mIoU_{pan}$, which come from the same model trained only with panoptic annotations; they are not the same as the numbers of models trained for instance or semantic segmentation.
- *Look at cost columns.* Parameters, FLOPs and frames per second are reported next to accuracy; the largest models are much slower.

Limits stated by the authors. A universal *architecture* is not yet a universal *model*: trained only on panoptic labels, Mask2Former is slightly worse on instance and semantic metrics than the same model trained with task-specific annotations (their Table 7), so it still needs per-task training. It also still struggles with small objects and does not fully exploit multi-scale features.

Why it matters here. Mask2Former is the reference design behind the "mask classification" section and the baseline in most later tables, including OneFormer's single-model training and EoMT's encoder-only simplification.

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DLCV 04 • Deep Image Restoration: Low-Level Vision

One degradation model, many tasks: how deep networks undo blur, noise, low resolution, haze, rain, snow and darkness, from SRCNN and DnCNN to Restormer, all-in-one models and diffusion priors.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-04-image-restoration/

What you'll learn

- What "low-level vision" means, and why super-resolution, denoising, deblurring, dehazing, deraining, desnowing and low-light enhancement are really **one problem**: invert a degradation $y = \mathcal{D}(x) + n$.
- How deep restoration is set up in practice: paired data, **synthetic degradation pipelines**, and the gap to real-world (blind) degradations. You will build a pipeline and train a tiny residual denoiser in PyTorch.
- The milestones and current families for each task, from SRCNN, DnCNN, DehazeNet and the GoPro deblurring benchmark to SwinIR, Restormer, NAFNet, MambaIR and diffusion-based SR (SUPIR, OSEDiff).
- How **all-in-one** models (AirNet, TransWeather, PromptIR, AdaIR, DA-CLIP, InstructIR) handle unknown or mixed degradations with one network.
- Which losses and metrics to use (full-reference, no-reference and user studies), why PSNR and "looks real" disagree (the **perception–distortion trade-off**), and how NTIRE challenges track the field.

The big picture

Low-level vision is about the pixels themselves. Before a system can recognize a face, read a sign or detect a car, the image has to be clear enough to look at. Restoration models are the part of the pipeline that takes a damaged photo (too small, noisy, blurred, foggy, rainy, dark) and returns a better estimate of the clean scene.

> **Explain like I'm five**
>
> Imagine someone hands you a photo that went through the washing machine: it is smudged, speckled and faded. You have seen thousands of clean photos in your life, so you can guess what this one *used to* look like: faces have two eyes, edges of buildings are straight, the sky is smooth. A restoration network does the same thing. It has looked at millions of "before and after" pairs, so it learned what clean pictures look like and how washing-machine damage changes them, and it uses both to undo the damage.

Why it matters: every phone camera runs denoising, demosaicking and super-resolution before you ever see a picture. Satellite, microscope, medical and surveillance images are restored before they are measured. Video streaming services upscale low-bitrate frames. Self-driving cars need their detectors to work in rain and fog. And from a research point of view, restoration is where many architectural ideas were tested first: residual learning, channel attention, efficient transformers and diffusion priors all have well-known restoration variants.

About this tutorial

This tutorial is expanded from an earlier post, Low-Level Vision Task — Image Restoration 簡介 (written in Traditional Chinese) [1]. It keeps the post's order and viewpoints, folds in its 2026 updates and corrections, and adds the math, code and survey literature.

We assume you know CNNs and backpropagation (DLCV 01) and the classical restoration chapter (IP 05), which covers noise models, Wiener filtering and the degradation model in detail. We link to it rather than repeat it.

Related chapters. Three chapters go deeper into parts of this one: DLCV 04A (single-image super-resolution), DLCV 04B (video super-resolution) and DLCV 04C (generative restoration with GAN and diffusion priors). DLCV 05B covers image fusion. DLCV 01B explains why restoration networks are built from blocks differently than classifiers.

What is low-level vision?

Plain version. High-level vision asks "what is in the picture?" (classification, detection). Low-level vision asks "what does the picture actually look like, pixel by pixel?" One is about *understanding*, the other about *perceiving*.

The analogy in the earlier post [1] comes from how people interact with the world: we first *sense* (light hits the retina and we see edges, colors and contrast) and then *interpret* (that is a cup). Both are necessary. If the sensing is poor (the cup is blurry), the interpretation suffers. Low-level vision tasks exist to make the sensing good again: when we say an object looks "blurry", our perception of it is not sharp, and a deblurring task is created to make it "clear".

Precise version. A low-level vision task maps an image to an image of (usually) the same content:

$$f_\theta : \mathbb{R}^{H\times W\times C} \to \mathbb{R}^{H'\times W'\times C'},$$

where the output is a pixel-wise estimate rather than a label. H, W, C are the input height, width and channels, and H', W', C' those of the output ($H' = sH$ for super-resolution by a factor s, otherwise usually equal). This has three consequences that shape the whole field:

1. **Dense, high-resolution outputs.** Every pixel matters, so architectures cannot downsample aggressively and must keep fine detail, which makes them expensive at high resolution.

2. **Supervision is cheap to synthesize.** Take a clean image, damage it with a known process, and you have an input–target pair. This is a huge advantage over labeled data for recognition, and also the source of the field's main weakness (synthetic damage is not real damage).
3. **"Correct" is ambiguous.** Many clean images are consistent with one damaged image. Which one is "best" depends on whether you value fidelity to the truth or natural appearance, as we will see in the evaluation section.

The family of tasks is broad: deblurring, super-resolution, denoising, inpainting, color calibration, infrared–visible fusion and more. Papers with Code used to be the standard place to browse these task lists and leaderboards, but it has shut down and its address now redirects to Hugging Face's trending-papers page, so its per-task leaderboards are no longer available. A reliable way to follow progress today is the yearly **NTIRE** challenge reports (New Trends in Image Restoration and Enhancement, a CVPR workshop), which we discuss near the end [2], [3].

One problem, many faces: the degradation model

Plain version. Every task in this tutorial can be written as "the camera saw a clean scene x, something damaged it, and we got y." Only the damage changes. If you can write the damage down, you can name the task.

The unified view

$$y = \mathcal{D}(x) + n,$$

where x is the unknown clean image, $\mathcal{D}$ is a (possibly nonlinear, possibly unknown) degradation operator, n is noise, and y is what we observe. Restoration means estimating x from y. Here are the specific forms used in the post [1] and in the rest of this tutorial. Figure 1 shows each one applied to the same photo.

Blur (the textbook model). As in IP 05,

$$g(u,v) = (h * f)(u,v) + n(u,v),$$

where f is the sharp image, h the point spread function (blur kernel), $*$ 2-D convolution, n additive noise and g the observed image, all indexed by pixel coordinates (u,v). Denoising is the special case $h = \delta$ (a unit impulse, so no blur).

Super-resolution. The low-resolution image is a blurred, subsampled and noisy version of the high-resolution one:

$$y = (x * k)\downarrow_s + n,$$

where k is a blur (anti-aliasing) kernel, $\downarrow_s$ keeps every s-th pixel in each direction, and s is the scale factor ($2, 3, 4, \ldots$). The classic "bicubic" benchmark setting replaces all of this with MATLAB-style bicubic downsampling and no noise.

Compression. $y = \mathrm{JPEG}_q(x)$, where q is the quality factor. JPEG quantizes 8×8 DCT coefficients (see IP 08), producing blocking and ringing. It is not additive and not linear, so it is usually just added to the pipeline as a black box.

Haze (the atmospheric scattering model).

$$I(p) = J(p)\, t(p) + A\,(1 - t(p)), \qquad t(p) = e^{-\beta d(p)},$$

where p is a pixel, I the hazy image, J the haze-free scene radiance, A the global atmospheric light (the color of the haze), $t \in (0, 1]$ the transmission, β the scattering coefficient and d the scene depth. The first term is the attenuated scene; the second is "airlight" scattered toward the camera. Far objects (large d) get small t and fade into A.

Rain. For rain streaks the common model is additive,

$$I = B + R,$$

where B is the rain-free background and R the rain layer. Raindrops stuck on a lens or windshield behave differently: they block and refract the background, so a popular model is $I = (1 - M) \odot B + R$, where M is a binary raindrop mask, $\odot$ the element-wise product and R the light passing through the drops [39].

Low light (Retinex). Retinex theory [5] separates an image into reflectance and illumination:

$$S = R \cdot L,$$

where S is the observed image, R the reflectance (the intrinsic color and texture of surfaces, what we want), L the illumination and $\cdot$ the element-wise product. A dark photo has small L; enhancement estimates L and brightens it, while noise that was hidden in the dark gets amplified.



Figure 1 — One clean image x pushed through the degradation models of this section. Each restoration task tries to undo one (or several) of these.

Although none of these models is exact, they all have the same shape: a clean image goes in, an operator plus noise comes out. That is the post's central point [1]: seen through the degradation model, these are all **the same kind of problem**, and the same tools (and often the same networks) apply to all of them. Figure 2 organizes them by where the damage comes from.

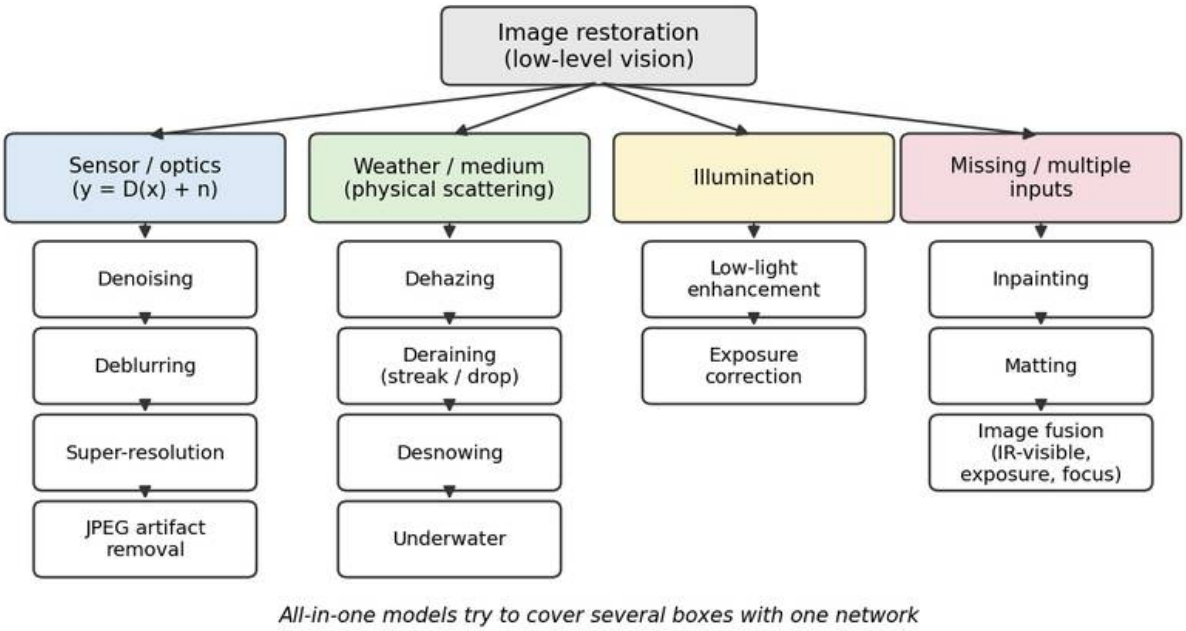


Figure 2 — A task taxonomy of low-level vision by the source of degradation. All-in-one models try to cover several boxes with a single network.

Why the problem is hard: ill-posedness

For almost every $\mathcal{D}$ above, many different x produce the same y. Downsampling by 4 throws away 15 of every 16 pixels; blur suppresses high frequencies to near zero; haze with $t \approx 0$ erases distant objects. The inverse problem is **ill-posed**, and we need a **prior** on what clean images look like. In the Bayesian (MAP) form,

$$\hat{x} = \arg\min_{x} \; \tfrac{1}{2}\|y - \mathcal{D}(x)\|_2^2 + \lambda\, \Phi(x),$$

where the first term (data fidelity) says "$\hat{x}$ must explain the observation", Φ is a regularizer encoding the prior (small Φ for natural-looking images), and $\lambda > 0$ balances the two. The whole history of the field is a history of better priors Φ.

Before deep learning: the era of hand-crafted priors

Before deep learning, each task had its own carefully reasoned prior, and success depended on having enough physical or statistical knowledge of the problem. Denoising used local smoothness, non-local self-similarity and sparsity; deblurring used Wiener and constrained least-squares filters plus heavy-tailed gradient priors; dehazing used the dark channel prior [25]; super-resolution used interpolation and example-based dictionaries; low-light enhancement

used histogram equalization and Retinex decompositions. IP 05 covers the core of this toolbox (noise models, mean/median/adaptive filters, inverse, Wiener and CLS filtering) and ends with a short history of the move to learned denoisers. The view in [1] is that once deep learning surpassed what these models could do, the decisive factor stopped being a strong theoretical prior for each task and became a strong **learning mechanism and feature extractor**. The degradation model did not disappear: it moved from the solver into the **data generator**.

The deep learning formulation

Plain version. Make lots of (damaged, clean) pairs, and train a network to map damaged to clean. Most of the hard work is in making the damage realistic.

Supervised training on pairs

Given a dataset of clean images $\{x_i\}$ and a degradation sampler, we build pairs (y_i, x_i) with $y_i = \mathcal{D}_i(x_i) + n_i$, and fit

$$\theta^\star = \arg\min_\theta \frac{1}{N} \sum_{i=1}^{N} \ell(f_\theta(y_i),\, x_i),$$

where f_θ is the restoration network with parameters θ, N the number of training pairs and ℓ a loss (L1, L2, perceptual, adversarial, ...; see the losses section). Training is done on random crops ("patches", e.g. 64×64 to 256×256) with flips and rotations, because convolutional and windowed-attention networks are translation-equivariant and full-resolution images do not fit in GPU memory.

A useful fact explains a lot of what follows: with the L2 loss, the optimal network outputs the **posterior mean** $f^\star(y) = \mathbb{E}[x \mid y]$. When many sharp images are consistent with y, their average is blurry. This is why MSE-trained super-resolution looks smooth, and why GANs and diffusion models were brought in.

Synthetic degradation pipelines

Pairs come from three sources. (1) **Synthetic**: degrade clean images with a known model (bicubic downsampling, Gaussian noise, Gaussian blur, JPEG). Cheap and unlimited, but limited by how realistic the model is. (2) **Captured**: photograph the same scene twice, e.g. a short and a long exposure, or a burst averaged into a clean reference (as in the SIDD real-noise dataset [22]), or averaging high-speed video frames to create real motion blur (GoPro [44]). Realistic but expensive

and specific to one camera. (3) **Unpaired / self-supervised**: learn from degraded images alone, using GANs, cycle consistency or noise statistics. Most strong models today train on synthetic pairs from a rich degradation pipeline and evaluate on both synthetic and captured data.

A typical "classic" pipeline is blur → downsample → noise → JPEG, mirroring the order in which a camera and an image-sharing site would damage a photo:

```
import numpy as np, cv2, torch, torch.nn.functional as F
from skimage import data, img_as_float32

def gaussian_kernel(sigma, size=21):
    ax = torch.arange(size) - size // 2
    g = torch.exp(-ax**2 / (2 * sigma**2))
    k = torch.outer(g, g)
    return k / k.sum()

def degrade(x, scale=4, sigma_blur=1.6, sigma_noise=0.03, jpeg_q=40, rng=None):
    """x: float tensor (C,H,W) in [0,1]. Returns the low-quality image y = D(x) + n."""
    rng = rng or np.random.default_rng(0)
    C = x.shape[0]
    k = gaussian_kernel(sigma_blur).to(x)[None, None].repeat(C, 1, 1, 1)
    y = F.conv2d(F.pad(x[None], (10,) * 4, mode="reflect"), k, groups=C)   # 1) blur  h * x
    y = y[..., ::scale, ::scale]                                           # 2) downsample
     ↪  (.)↓s
    y = y + sigma_noise * torch.from_numpy(rng.standard_normal(y.shape).astype(np.float32))
     ↪  # 3) noise n
    y = y.clamp(0, 1)[0]
    img = (y.permute(1, 2, 0).numpy() * 255).round().astype(np.uint8)      # 4) JPEG round
     ↪  trip
    ok, buf = cv2.imencode(".jpg", img[..., ::-1], [cv2.IMWRITE_JPEG_QUALITY, jpeg_q])
    img = cv2.imdecode(buf, cv2.IMREAD_COLOR)[..., ::-1]
    return torch.from_numpy(img.copy()).permute(2, 0, 1).float() / 255

x = torch.from_numpy(img_as_float32(data.astronaut())).permute(2, 0, 1)    # (3, 512, 512)
y = degrade(x)
print(x.shape, "->", y.shape)        # torch.Size([3, 512, 512]) -> torch.Size([3, 128, 128])
```

In training, every parameter (kernel width and shape, noise level and type, JPEG quality, even the order) is **sampled at random per image**, so the network sees a distribution of degradations instead of a single one.

Real-world and blind degradations

A model trained on one fixed $\mathcal{D}$ (say, bicubic ×4) breaks on real photos, whose degradation is unknown, spatially varying and a mix of several effects. This is the **blind** (or real-world) restoration problem. Two broad strategies exist:

- **Make the synthetic distribution wide enough to cover reality.** Real-ESRGAN [4] applies the classic pipeline *twice* in sequence ("high-order" degradation, with randomized blur kernels, resizing, noise and JPEG in each round) and adds sinc filters to simulate ringing and overshoot. Trained on purely synthetic data, it generalizes much better to real low-quality images than bicubic-trained models.
- **Bring in a strong generative prior.** Pretrained text-to-image diffusion models already know what natural images look like. DiffBIR [16], SUPIR [17] and OSEDiff [18] condition such a model on the degraded input, so that the prior "fills in" plausible detail. This is the main recent trend in real-world SR and is discussed below.

A third, complementary strategy is to **estimate the degradation** (noise level, kernel, degradation type) and feed it to the network, as FFDNet's noise-level map [21] and the all-in-one models do. In blind super-resolution the main unknown is the blur kernel k in $y = (x * k) \downarrow_s + n$, and a whole family of methods gives the kernel to the network, estimates and corrects it in a loop, or estimates it from the test image alone [87]–[91]. The result to remember: explicit kernel methods are accurate and interpretable when the degradation really is "blur, downsample, noise", but a wrong kernel propagates into the output. That is why most real-world models, such as BSRGAN [92], Real-ESRGAN [4] and the diffusion-prior methods, learn the degradation **implicitly** from a randomized pipeline instead.

Optional: Blind super-resolution: conditioning on and estimating the kernel (in the Supplementary notes). How SRMD, USRNet, IKC, DAN and KernelGAN handle the kernel, and when the explicit route beats the implicit one.

Optional: Plug-and-play priors and half-quadratic splitting (in the Supplementary notes). How a solver can split the data term from the prior so that any denoiser acts as the prior, the idea behind deep unfolding networks such as USRNet.

Super-resolution

Plain version. Make a small image bigger *and* sharper, i.e. predict the missing high-resolution pixels rather than just stretching the existing ones.

This section is an overview; DLCV 04A ·Image Super-Resolution treats super-resolution in depth: degradation models, classical and learned methods, perceptual and diffusion SR, real-world SR and how to evaluate it.

Problem setup

Single-image super-resolution (SISR) estimates $x \in \mathbb{R}^{sH \times sW \times C}$ from $y \in \mathbb{R}^{H \times W \times C}$ under $y = (x * k) \downarrow_s + n$. The traditional answer is **interpolation**: nearest-neighbor, bilinear and bicubic.

Bicubic, which fits a cubic polynomial through a 4×4 neighborhood, is the best known and is still used everywhere, both as a baseline and as the standard way to create benchmark low-resolution inputs. Interpolation only redistributes existing information; it cannot recreate the high frequencies that $\downarrow_s$ removed.

How a network gets bigger: three upsampling strategies

Because the output is larger than the input, every SR network must decide *where* the resolution increases.

a. Pre-upsampling (interpolate first). SRCNN [6] bicubically enlarges the input to the target size, then applies three convolution layers (patch extraction, non-linear mapping, reconstruction). Simple and intuitive, but every layer runs at high resolution, which is slow, and the network must clean up interpolation artifacts.

b. Transposed convolution (deconvolution). FSRCNN [7] works at low resolution and enlarges only at the end with a learned transposed convolution, giving a large speed-up over SRCNN. Transposed convolutions with a stride that does not divide the kernel size overlap unevenly and tend to produce **checkerboard artifacts** [9], so they are now rare as the final SR upsampler, although they remain common elsewhere in neural networks (for example in decoders of segmentation and generative models).

c. Sub-pixel convolution (pixel shuffle). ESPCN [8] computes Cs^2 channels at low resolution and rearranges them into an image s times larger:

$$\mathrm{PS}(T)_{c,\,sh+i,\,sw+j} = T_{cs^2+si+j,\,h,\,w}, \qquad 0 \le i,j < s,$$

where T is the low-resolution feature tensor with Cs^2 channels, (h, w) a low-resolution position, (i, j) the sub-pixel offset and c the output channel. All learning happens at low resolution, and the rearrangement itself has no parameters. It is the dominant choice in modern SR networks (`torch.nn.PixelShuffle`).

Milestones and current families

Plain version. SR networks first got deeper, then learned to draw realistic texture, and most recently borrowed transformers and large pretrained generative models.

The main steps: **SRCNN** [6] was the first end-to-end CNN for SR; **EDSR** [10] and **RCAN** [11] made SR networks very deep (EDSR without batch normalization, RCAN with channel attention); **SRGAN** [12] and **ESRGAN** [13] added adversarial and perceptual losses, giving sharp, realistic texture instead of the blurry MSE average. Today's families are transformers such as **SwinIR** [14] and **HAT** [15], state-space models such as MambaIR [49], and **diffusion priors**

(DiffBIR [16], SUPIR [17] and the one-step OSEDiff [18]), which produce far more convincing textures on real photos but can also invent detail that was not in the scene. Models are usually trained on **DIV2K** [19] and compared by PSNR and SSIM on the luminance channel; real-world SR, which has no ground truth, relies on perceptual and no-reference metrics (see Evaluation). Once outputs looked much better than interpolation (around 2018), how to *judge* SR became a central question (see the perception–distortion section).

Optional: Super-resolution milestones, families and benchmarks (in the Supplementary notes). What each milestone changed, how the diffusion-prior methods differ, and the benchmark conventions.

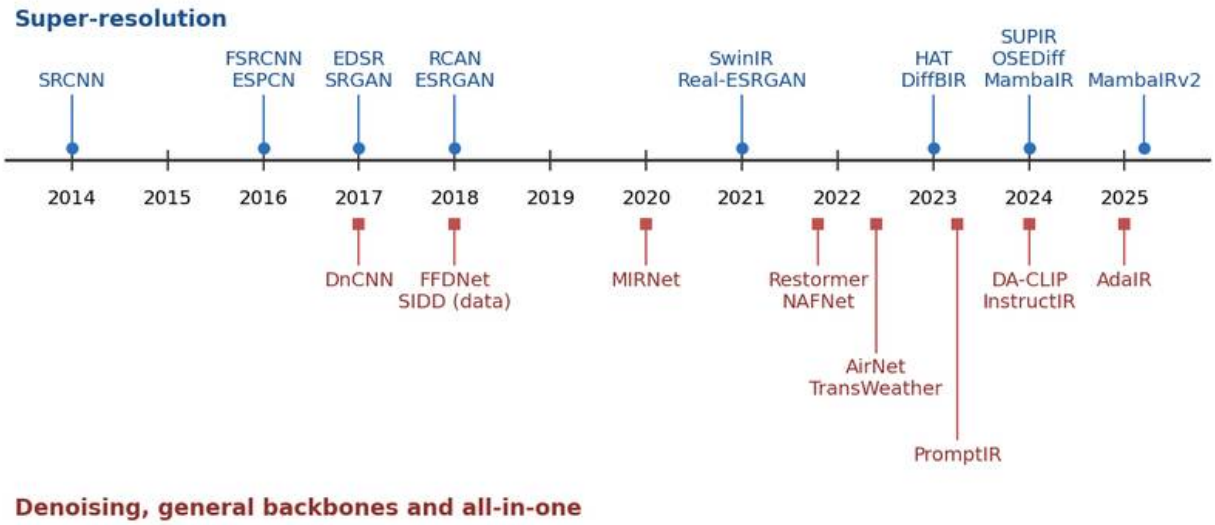


Figure 3 — Milestones in super-resolution (top) and in denoising, general backbones and all-in-one restoration (bottom), placed by publication year (positions within a year are only for legibility).

Denoising

Plain version. Remove the random grain from a photo without wiping out the real texture. Noise is the simplest degradation, so denoising became the testbed where most deep restoration ideas appeared first.

Problem setup

$y = x + n$. In the synthetic setting $n \sim \mathcal{N}(0, \sigma^2 I)$ (additive white Gaussian noise, AWGN, with standard deviation σ, usually quoted on a 0–255 scale such as $\sigma = 25$). Real camera noise is different: it is signal-dependent (photon shot noise grows with brightness), spatially correlated after demosaicking, and reshaped by the camera's processing pipeline. The Poisson–Gaussian model of such noise, written out for dark images, is in DLCV 05.

Milestones

DnCNN [20] made two choices that became standard. It uses **residual learning**: instead of predicting x, the network $\mathcal{R}_\theta$ predicts the noise, and the output is

$$\hat{x} = y - \mathcal{R}_\theta(y),$$

where $\mathcal{R}_\theta(y) \approx n$. The residual (noise) is a "simpler" function to learn than the clean image, which made deep plain CNNs with batch normalization train well. A single DnCNN can also be trained for a range of noise levels (blind Gaussian denoising), and the same network handles SR and JPEG deblocking.

FFDNet [21] feeds a **noise-level map** M (same size as the image, value σ at each pixel) to the network along with a downsampled version of the input. Changing M at test time trades noise removal against detail preservation, and a non-uniform M handles spatially varying noise. It is fast enough to run on a CPU.

Real noise. The **SIDD** dataset [22] provides real smartphone noisy images with clean references obtained by careful capture and processing of image bursts. Models trained only on AWGN do poorly on it, which pushed the field toward realistic noise synthesis and captured training data.

General backbones. MIRNet [30], **Restormer** [23] and **NAFNet** [24] are general restoration architectures that reported state-of-the-art results on SIDD as well as on deblurring and other tasks. They are covered in the backbones section.

Code: a tiny residual denoiser

The snippet trains a five-layer DnCNN-style network on random 32×32 patches cut from four `skimage` images, with fresh Gaussian noise every step, then tests on an image it has never seen. On a laptop CPU it takes well under a minute (we measured about one minute on a heavily loaded machine).

```
import time, numpy as np, torch, torch.nn as nn
from skimage import data, img_as_float32
from skimage.color import rgb2gray

torch.manual_seed(0); torch.set_num_threads(2)
rng = np.random.default_rng(0)
sigma = 25 / 255                                    # noise level (on a [0,1] scale)
train_imgs = [img_as_float32(im) for im in (data.camera(), data.coins(), data.moon(),
 ↪  data.page())]
test_img = img_as_float32(rgb2gray(data.astronaut()))   # held out

def random_patches(n=32, p=32):
    batch = []
    for _ in range(n):
```

```
        im = train_imgs[rng.integers(len(train_imgs))]
        i, j = rng.integers(0, im.shape[0] - p), rng.integers(0, im.shape[1] - p)
        batch.append(im[i:i + p, j:j + p])
    return torch.from_numpy(np.stack(batch))[:, None]   # (n, 1, p, p)

class TinyDnCNN(nn.Module):
    """conv-ReLU, (depth-2) x conv-BN-ReLU, conv. Predicts the NOISE, not the image."""
    def __init__(self, depth=5, ch=32):
        super().__init__()
        layers = [nn.Conv2d(1, ch, 3, padding=1), nn.ReLU(inplace=True)]
        for _ in range(depth - 2):
            layers += [nn.Conv2d(ch, ch, 3, padding=1), nn.BatchNorm2d(ch),
             ↪  nn.ReLU(inplace=True)]
        layers += [nn.Conv2d(ch, 1, 3, padding=1)]
        self.body = nn.Sequential(*layers)
    def forward(self, y):
        return y - self.body(y)                         # x_hat = y - R(y): residual learning

net = TinyDnCNN()
opt = torch.optim.Adam(net.parameters(), lr=2e-3)
t0 = time.time()
for step in range(300):
    x = random_patches()
    y = x + sigma * torch.randn_like(x)                 # synthetic pairs, made on the fly
    loss = nn.functional.l1_loss(net(y), x)
    opt.zero_grad(); loss.backward(); opt.step()
print(f"trained in {time.time() - t0:.0f} s, last L1 = {loss.item():.4f}")

net.eval()
with torch.no_grad():
    x = torch.from_numpy(test_img)[None, None]
    y = x + sigma * torch.randn_like(x)
    x_hat = net(y).clamp(0, 1)
mse = lambda a, b: torch.mean((a - b) ** 2).item()
print(f"PSNR noisy {10*np.log10(1/mse(y.clamp(0,1), x)):.2f} dB -> denoised
 ↪  {10*np.log10(1/mse(x_hat, x)):.2f} dB")
# e.g. PSNR noisy 20.74 dB -> denoised 28.64 dB
```

Even this toy model, trained for 300 small steps on four images, gains about 8 dB on an unseen photo and beats a hand-tuned Gaussian filter (Figure 4). Drag the slider to compare the noisy input with the tiny network's output.



Noisy ($\sigma = 25$)



Tiny residual CNN

Slider comparing a grainy grayscale astronaut face with the same crop after denoising by a small CNN, which is clean but slightly smoothed.

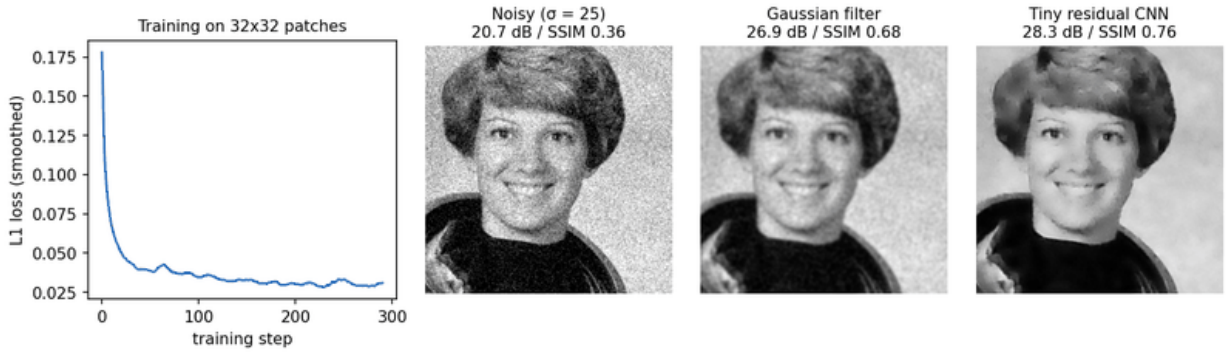


Figure 4 — The tiny residual denoiser. PSNR and SSIM are computed on the full 512 × 512 test image. The CNN removes more noise than a Gaussian filter while keeping edges sharper, but it also flattens fine texture, a typical MSE/L1 behavior. Values vary slightly between runs and machines.

Dehazing

Plain version. Remove the gray-white veil of fog or smog, restoring contrast and color, especially for distant objects. In the post's words it is an extension of denoising and deblurring, with a physical model attached.

The classical recipe and the dark channel prior

Classical methods estimate the unknowns of $I = Jt + A(1 - t)$ (the transmission map t and the airlight A), then invert the model. The hard part is finding a reliable relationship that pins those unknowns down. The most famous one is He et al.'s **dark channel prior** [25]. In haze-free outdoor images, most local patches contain some pixel that is very dark in at least one color channel (shadows, dark or colorful objects):

$$J^{\text{dark}}(p) = \min_{c \in \{r,g,b\}} \Big(\min_{q \in \Omega(p)} J^c(q) \Big) \approx 0,$$

where $\Omega(p)$ is a small patch around p and J^c is color channel c. Applying the same min–min operation to the haze model and using $J^{\text{dark}} \approx 0$ gives a transmission estimate

$$\tilde{t}(p) = 1 - \omega \min_c \Big(\min_{q \in \Omega(p)} \frac{I^c(q)}{A^c} \Big),$$

where $\omega < 1$ (0.95 in the paper) keeps a little haze for depth perception. Then $\hat{J} = (I - A)/\max(\tilde{t}, t_0) + A$, with a small t_0 to avoid dividing by zero. The prior fails on large bright regions similar

to the airlight (white walls, sky), and the coarse patch-based t must be refined to follow edges.

Deep dehazing: two schools

The post [1] divides deep dehazing into two schools, and the split still describes the field well.

1. Physics-guided ("apply the formula"). Learn the unknown parameters, then invert the model. **DehazeNet** [26] uses a CNN (with Maxout units and a bounded ReLU) to regress the transmission t from hazy patches. **AOD-Net** [27] simplifies the scattering model further by folding t and A into a single function K:

$$J(p) = K(p)\, I(p) - K(p) + b, \qquad K(p) = \frac{\frac{1}{t(p)}(I(p) - A) + (A - b)}{I(p) - 1},$$

where b is a constant bias. A light-weight CNN predicts K, so the whole network is end-to-end, and it can be attached in front of a detector and trained jointly.

2. Strong general architectures ("big learner"). Skip the physics and learn $I \mapsto J$ directly with a powerful network. **GridDehazeNet** [28] uses an attention-based multi-scale grid; **FFA-Net** [29] stacks channel- and pixel-attention blocks with adaptive feature fusion. These designs transfer well to other degradations, which is exactly the route taken by general restoration networks such as MIRNet [30]. Transformer dehazers followed: **DehazeFormer** [31] adapts the Swin Transformer to dehazing with a modified normalization layer, activation and spatial aggregation.

Datasets

- **RESIDE** [32]: a large benchmark with synthetic indoor and outdoor hazy images (generated from depth with the scattering model) plus real hazy images for task-driven evaluation. Its SOTS test set is the most quoted dehazing benchmark.
- **O-HAZE** [33] and **Dense-Haze** [34]: real hazy/haze-free pairs captured with professional haze machines, outdoor and with dense haze respectively. They are small but expose the synthetic-to-real gap.
- **KITTI** [35] is *not* a dehazing dataset. It is an autonomous-driving benchmark (stereo, optical flow, odometry, 3-D detection, tracking) recorded around Karlsruhe. Some dehazing papers synthesize fog on its images (it has depth from LiDAR and stereo) or use it to measure how dehazing affects downstream detection.

Desnowing

Plain version. Remove falling snow. It looks like a cousin of deraining, but it also behaves like dehazing.

Snow is harder than it looks. Snowflakes vary hugely in size and transparency, from opaque white blobs to semi-transparent streaks, and heavy snow also creates a **veiling effect** much like haze. So a desnowing model has to handle both the flakes and the fog-like veil. **DesnowNet** [36] introduced the **Snow100K** dataset (synthetic snow over real images, plus real snowy images) and a network that estimates both the translucency of snow and the residual it leaves. **JSTASR** [37] explicitly models snow size and transparency and adds a veiling-effect removal step. Desnowing is now usually studied inside "adverse weather" all-in-one models such as TransWeather [52].

Deraining

Plain version. Remove rain from photos. "Rain" means two very different things: thin streaks falling through the air, and drops sitting on the lens or windshield.

Rain streaks versus raindrops

- **Rain streaks** are thin, bright, oriented lines overlaid on the scene, modeled as $I = B + R$. Distant streaks accumulate into a haze-like veil. Methods learn to separate the two layers. An early influential deep method jointly detects and removes rain [38] and comes with widely used synthetic datasets of light and heavy rain.
- **Raindrops** on a glass surface act as tiny lenses: they hide and refract the background, are often out of focus, and have no consistent orientation. The attentive GAN of Qian et al. [39] uses an attention map (where are the drops?) to guide both the generator and the discriminator, and introduced a paired raindrop dataset captured through glass with and without water drops.

Toward real scenes

Most deraining datasets are synthetic, and models trained on synthetic streaks often fail on real rain. The **NTIRE 2025 Day and Night Raindrop Removal for Dual-Focused Images** challenge [40] reflects the shift to real data: it uses the real-captured **Raindrop Clarity** dataset, covering daytime and nighttime scenes and both raindrop-focused and background-focused shots (14,139 training, 240 validation and 731 test images). The focus setting matters: when the camera focuses on the drops, the background is blurred too, so raindrop removal becomes partly deblurring.

Low-light enhancement (brief)

Plain version. Make a dark photo look properly exposed, with visible details and natural colors, without blowing up the noise.

Training compares dark images with normally exposed references of the same scene, for example the **LOL** dataset of paired low/normal-light images introduced with **RetinexNet** [41]. Classical methods rely on contrast adjustment (histogram equalization) and Retinex-based models (and, less often, an inverted atmospheric scattering model). Deep methods follow two lines: Retinex-inspired networks that decompose $S = R \cdot L$ and adjust L (RetinexNet [41]), and general well-designed networks. **Retinexformer** [42] (ICCV 2023), which its authors describe as the first Transformer-based low-light enhancement method, combines both: a one-stage Retinex formulation with illumination-guided attention. Low-light enhancement has enough of its own (zero-reference methods, curves, noise, color, video) to deserve a separate chapter: see DLCV 05.

Image fusion (brief)

Plain version. Combine several images of the same scene, each good at something different, into one image that has the best of each.

Three common settings appear in the post [1]: **infrared and visible** fusion (thermal targets from IR, textures and color from the visible camera), **multi-exposure** fusion (detail from under- and over-exposed shots, an HDR-like result) and **multi-focus** fusion (an all-in-focus image from shots focused at different depths). Unlike the other tasks, there is usually no ground-truth fused image, so deep fusion relies on unsupervised losses that keep gradients and intensities from the sources, or on GANs. The survey by Zhang et al. [43] organizes deep fusion methods by these scenarios and by learning strategy.

Deblurring

Plain version. Recover a sharp image from one smeared by camera shake, object motion or defocus.

Problem setup

Deblurring is the direct instance of $g = h * f + n$. If the kernel h is known it is **non-blind** deblurring (the setting of the classical Wiener and CLS filters in DIP 5, and of learned non-blind solvers); if h

is unknown it is **blind** deblurring. Real motion blur in dynamic scenes is worse: different objects move differently and depth varies, so there is no single global h.

Blind deblurring: estimating the kernel

Plain version. If you knew exactly how the camera shook, undoing the blur would be a well-studied filtering problem. In blind deblurring you do not know the shake, so you must guess the shake *and* the sharp photo together. The trick is to guess them in turns: first sharpen with your current guess of the shake, then use the sharpest edges you got to improve the guess of the shake, and repeat.

The joint problem. For a spatially uniform blur, blind deconvolution solves

$$(\hat{f}, \hat{h}) = \arg\min_{f,h} \|h * f - g\|_2^2 + \lambda_f \, \phi(f) + \lambda_h \, \psi(h) \quad \text{subject to} \quad h \geq 0, \;\; \textstyle\sum_{u,v} h(u,v) = 1,$$

where g is the blurred image, f the unknown sharp image, h the unknown kernel (a small window, e.g. 15×15 to 75×75 pixels for camera shake), ϕ an image prior (typically a sparse gradient penalty, because natural images have mostly flat regions and a few strong edges), ψ a kernel prior, and $\lambda_f, \lambda_h > 0$ their weights. The constraints say that a blur kernel only redistributes light: it is non-negative and preserves total brightness. Even so, the kernel can only be recovered up to a small shift, because moving f one way and h the other leaves $h * f$ unchanged.

Why it is badly ill-posed. Far more unknowns (f and h) than equations, and many pairs explain g exactly. Worse, the most obvious pair is wrong: $h = \delta$ (no blur) and $f = g$ fits the data perfectly. Levin et al. [74] showed that with the usual sparse gradient priors, the blurry image often has a *lower* prior cost than the sharp one, because blur shrinks gradients and ϕ rewards small gradients. So naive joint MAP estimation over (f, h) tends to return the no-blur solution. Their analysis points to a better-posed target: estimate only the kernel, averaging over all the sharp images that could have produced g. A kernel has few unknowns compared with the many pixel observations, so it is much better determined than the full image; once $\hat{h}$ is known, a non-blind deconvolution recovers f.

Classical remedies. The methods that worked in practice either approximate that averaging over images (Fergus et al. [75], with variational Bayes), **predict and select sharp edges** so that the kernel is estimated only from reliable gradients (Cho and Lee [77] sharpen the current estimate with a shock filter, as the code below does; Xu and Jia [78] keep only informative edges), or change the prior so that the sharp image wins (Krishnan et al. [79], Pan et al. [80]). All of them share the loop below.

The standard recipe. Almost all classical blind deblurring methods follow the same loop. Starting from a delta kernel at the coarsest level of an image pyramid, alternate

$$f^{(t+1)} = \arg\min_{f} \|h^{(t)} * f - g\|_2^2 + \lambda_f \, \phi(f),$$

$$h^{(t+1)} = \arg\min_{h \geq 0,\ \sum h = 1} \|h * \nabla \tilde{f}^{(t+1)} - \nabla g\|_2^2 + \lambda_h \|h\|_2^2,$$

where t is the iteration, ∇ the image gradient, and $\tilde{f}$ the latent image after edge prediction or selection (sharpened, with weak gradients set to zero). The kernel step works in the gradient domain because convolution commutes with differentiation, $\nabla(h * f) = h * \nabla f$, and gradients are where the blur information sits; with a quadratic penalty it has a closed form in the Fourier domain, followed by projection onto the constraints. After a few iterations the kernel is upsampled to the next pyramid level (**coarse-to-fine**), which avoids poor local minima because a large blur is only a few pixels wide at coarse scales. Finally, with $\hat{h}$ fixed, a **non-blind deconvolution** produces the output image: Wiener or constrained least squares filtering as in IP 05, or a hyper-Laplacian or learned non-blind solver.

Deep learning and the kernel. Networks entered blind deblurring in three ways: they **predict the kernel** (or a deconvolution filter) and then deconvolve classically [81]–[83]; they **fit a kernel and an image to one photo** with untrained generator networks and no training data, as SelfDeblur [85] does; or they **skip the kernel** and map blurred to sharp images end to end, like the multi-scale network of Nah et al. [44], DeblurGAN [86], Restormer and NAFNet.

Optional: Classical and learned blind-deblurring methods (in the Supplementary notes). The kernel-only MAP formula and what each method named above does.

Optional: Deep Image Prior: the network as the prior (in the Paper guides). A reading guide to the paper behind SelfDeblur's image generator.

Which is preferable? Kernel-based methods suit **spatially uniform** blur and give an **interpretable** kernel that can be checked and reused, but they are slower and break when the blur varies across the image. Kernel-free networks handle **non-uniform** blur from object motion and depth and are the current state of the art on dynamic-scene benchmarks such as GoPro, but they generalize only as far as their training blur and give no explicit kernel. The survey of Zhang et al. [69] summarizes this shift (the full comparison is in the same note).

Code: a toy alternating blind deconvolution. The snippet blurs a 256 × 256 `camera` image with a curved 15 × 15 camera-shake kernel, then alternates a closed-form image step (a CLS-like filter with a quadratic gradient penalty) and a kernel step (least squares on the strongest 2% of shock-filtered gradients, projected onto $h \geq 0$, $\sum h = 1$). It runs in about a second on a CPU.

```
import numpy as np
from numpy.fft import fft2, ifft2
from scipy.ndimage import convolve, gaussian_filter, laplace
```

```
from skimage import data, img_as_float, transform

rng = np.random.default_rng(0)
f_true = transform.resize(img_as_float(data.camera()), (256, 256), anti_aliasing=True)
K = 15                                              # kernel support K x K
h_true = np.zeros((K, K))
for t in np.linspace(0, 1, 40):                     # a curved "camera-shake" path
    h_true[round(3 + 8 * t), round(2 + 10 * t + 2 * np.sin(3 * t))] += 1
h_true = gaussian_filter(h_true, 0.5); h_true /= h_true.sum()
g = convolve(f_true, h_true, mode="wrap") + 0.005 * rng.standard_normal(f_true.shape)

def otf(h, shape):                                  # centered kernel -> transfer function
    p = np.zeros(shape); p[:h.shape[0], :h.shape[1]] = h
    return fft2(np.roll(p, (-(h.shape[0] // 2), -(h.shape[1] // 2)), (0, 1)))

G = fft2(g)
Dx, Dy = otf(np.array([[0, -1, 1]]), g.shape), otf(np.array([[0], [-1], [1]]), g.shape)

def image_step(h, lam):        # f = argmin ||h*f - g||^2 + lam ||grad f||^2  (closed form,
 ↪  CLS-like)
    H = otf(h, g.shape)
    return np.real(ifft2(np.conj(H) * G / (np.abs(H)**2 + lam * (np.abs(Dx)**2 +
     ↪  np.abs(Dy)**2))))

def shock(f, n=5, dt=0.5):     # edge prediction: a shock filter turns soft edges into steps
    f = gaussian_filter(f, 0.7)
    for _ in range(n):
        gy, gx = np.gradient(f)
        f = f - dt * np.sign(laplace(f)) * np.hypot(gx, gy)
    return f

def kernel_step(f, keep=0.02, beta=1e-3):   # least squares in the gradient domain, strong
 ↪  edges only
    num = den = 0
    for D in (Dx, Dy):
        fx = np.real(ifft2(D * fft2(f)))
        fx[np.abs(fx) < np.quantile(np.abs(fx), 1 - keep)] = 0
        Fx = fft2(fx)
        num, den = num + np.conj(Fx) * D * G, den + np.abs(Fx)**2
    h = np.roll(np.real(ifft2(num / (den + beta))), (K // 2, K // 2), (0, 1))[:K, :K]
    h = np.maximum(h, 0)                            # project onto h >= 0 ...
    return h / h.sum()                              # ... and sum(h) = 1

def best_shift(fn):            # blind deconvolution recovers h (and f) only up to a small
 ↪  translation
    return min(fn(dy, dx) for dy in range(-3, 4) for dx in range(-3, 4))
kernel_error = lambda h: best_shift(lambda dy, dx: np.linalg.norm(np.roll(h, (dy, dx), (0,
 ↪  1)) - h_true)) / np.linalg.norm(h_true)
psnr = lambda f: -best_shift(lambda dy, dx: 10 * np.log10(np.mean((np.clip(np.roll(f, (dy,
 ↪  dx), (0, 1)), 0, 1) - f_true)[8:-8, 8:-8] ** 2)))

h = np.zeros((K, K)); h[K // 2, K // 2] = 1
h = gaussian_filter(h, 1.0); h /= h.sum()           # start from an almost-delta kernel
 ↪  ("no blur")
```

```
print(f"kernel error at start: {kernel_error(h):.2f}")
for _ in range(30):                                  # alternate: image update, then kernel
 ↪  update
    h = kernel_step(shock(image_step(h, lam=0.02)))
print(f"kernel error after 30 alternations: {kernel_error(h):.2f}")
print(f"PSNR  blurry {psnr(g):.2f} dB | estimated kernel {psnr(image_step(h, 1e-2)):.2f} dB"
      f" | true kernel {psnr(image_step(h_true, 1e-2)):.2f} dB")
# kernel error at start: 1.19
# kernel error after 30 alternations: 0.57
# PSNR  blurry 22.59 dB | estimated kernel 27.11 dB | true kernel 29.67 dB
```

The relative kernel error roughly halves, and the final non-blind step with the estimated kernel recovers most of the gap between the blurry input and the true-kernel result. Be honest about what this toy is: the blur is synthetic, uniform and circular (so the FFT model is exact), the noise is low, and the parameters (`lam`, `keep`, the number of shock iterations) were tuned on this one image. There is no image pyramid, so larger kernels would fail, and with real photos you would also need boundary handling, saturated-pixel handling and robust losses. Both errors are measured after the best small shift, because of the translation ambiguity explained above.

Data and milestones

Nah et al. [44] (CVPR 2017) addressed this with the **GoPro** dataset: they recorded sharp video with a high-speed camera and averaged consecutive frames to produce realistically blurred images, keeping the middle frame as the sharp target. Their network is **multi-scale**: it restores a coarse version first, then refines at finer scales, like the coarse-to-fine schemes of classical blind deblurring, but without estimating a kernel at all. GoPro is still the most common deblurring benchmark.

Today the most common baselines are the general backbones **Restormer** [23] and **NAFNet** [24]. NAFNet's striking finding is that the usual nonlinear activations (ReLU, GELU, Sigmoid) are not needed: they can be replaced by a simple element-wise multiplication of two feature halves ("SimpleGate") or removed, giving a "nonlinear activation free" network that is both simpler and stronger on GoPro and SIDD. The deblurring survey by Zhang et al. [69] covers the wider landscape (blind and non-blind, video, defocus, losses).

Inpainting (brief)

Plain version. Fill in missing or damaged regions (scratches, removed objects, holes) with plausible content.

Inpainting is the restoration task closest to image **generation**: inside a large hole there is nothing to restore, only something to invent that matches the surroundings. The degradation is a mask,

$y = m \odot x$, where m is 0 in the hole and 1 elsewhere. **Masked autoencoders** (MAE) [45] can be seen as inpainting used as a pre-training task: hide 75% of image patches and train a ViT to reconstruct them, which turns out to teach strong visual representations. For practical inpainting,

LaMa [46] uses fast Fourier convolutions with an image-wide receptive field and training on large masks to fill big holes at high resolution; today, diffusion-based inpainting is the default for large, semantic edits.

Matting (brief)

Plain version. Cut out the foreground (a person, their hair) with soft, partially transparent edges so it can be pasted on a new background.

Matting assumes each pixel is a blend of foreground and background,

$$I = \alpha F + (1 - \alpha)B,$$

where $\alpha \in [0, 1]$ is the opacity (alpha matte), F the foreground color and B the background color. With seven unknowns per pixel (α and two RGB colors) and three observations, the problem is badly underdetermined, so classical methods ask the user for a **trimap** (definite foreground, definite background, unknown band). It can be seen as a soft variant of segmentation. **Background**

Matting [47] replaces the trimap with an extra photo of the background without the subject, and trains networks first on synthetic composites and then on real images with an adversarial loss that judges how natural the new composites look.

General backbones

Plain version. Across all these tasks, a handful of network designs keep winning. They are general "image-to-image engines" that you can train on any of the degradations above.

Backbone	Core idea	Why it suits restoration
U-Net [48]	Encoder–decoder with skip connections at every scale	Large receptive field from the downsampling path, fine detail from the skips
SwinIR [14]	Residual groups of shifted-window self-attention blocks at full resolution	Content-adaptive weights; strong for SR, denoising, JPEG
Restormer [23]	U-Net of transformer blocks; attention computed across **channels** (transposed attention) plus gated feed-forward	Cost linear in the number of pixels, so it scales to large images

Backbone	Core idea	Why it suits restoration
NAFNet [24]	U-Net of simple conv blocks with LayerNorm, SimpleGate and simplified channel attention; no nonlinear activations	Very strong accuracy per FLOP; the default "simple baseline"
MambaIR / MambaIRv2 [49], [50]	Selective state-space (Mamba) scans with local enhancement and channel attention; v2 adds "attentive" non-causal scanning	Global receptive field at linear cost

Design reasons that recur. (1) **Global residual**: predict $x - y$ (or x minus an upsampled y) so the network learns only the missing part. (2) **No batch normalization** in most SR/restoration nets since EDSR [10]: BN statistics differ between training patches and full test images and hurt color/range fidelity. (3) **Large receptive field** at manageable cost: multi-scale U-Nets, windowed or channel attention, state-space scans. Restoration needs context (to tell a texture from noise) *and* full-resolution detail. (4) **Attention or gating** for content-adaptive processing: channel attention (RCAN), gating (NAFNet, Restormer). The post's "Extension" list predicted new architectures from ViT, flow, GAN and diffusion; by 2026 the state-space model joined CNNs and transformers as a third backbone option [49], [50].

All-in-one and universal restoration

Plain version. Instead of training one model for rain, another for haze and another for noise, train *one* model that figures out what is wrong with the image and fixes it.

Motivation

The post's "Extension" section lists all-in-one restoration as one of the hottest topics of 2023, starting around 2021. The motivation is practical: in a deployed system we do not know in advance which degradation (or mixture) a given image has, and weather can change or combine at any moment. A bank of task-specific models needs a reliable classifier in front and wastes memory; a single model that adapts is more convenient. The difficulty is **task interference**: different degradations need different, sometimes opposite operations (denoising smooths, deblurring sharpens), so naive multi-task training underperforms specialists.

Representative methods

- **AirNet** [51] (CVPR 2022): a contrastive degradation encoder learns, without labels, *how* the image is degraded and guides the restorer; its noise + rain + haze setting became the standard three-task benchmark.
- **TransWeather** [52] (CVPR 2022): one transformer for rain, fog and snow, with learnable weather-type queries.
- **PromptIR** [53] (NeurIPS 2023): lightweight prompt blocks inject degradation-specific information into a Restormer-style decoder.
- **AdaIR** [54] (ICLR 2025): different degradations affect different frequency bands, so it adapts low- and high-frequency features separately.
- **DA-CLIP** [55] (ICLR 2024): an adapted CLIP model supplies a degradation embedding that guides the restorer.
- **InstructIR** [56] (ECCV 2024): a natural-language instruction ("remove the rain") conditions the network.

Optional: All-in-one methods in more detail (in the Supplementary notes). How each of the six models represents the degradation and injects it into the network.

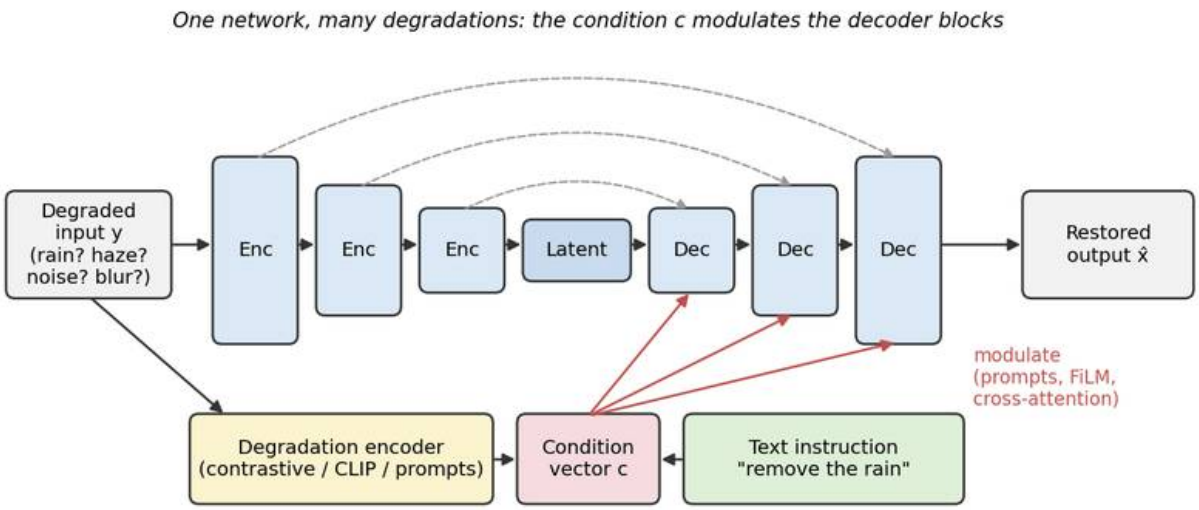


Figure 5 — The common template behind all-in-one restoration: an encoder–decoder backbone plus a degradation-aware condition c, inferred from the image (AirNet, PromptIR, AdaIR, DA-CLIP) or given as text (InstructIR), that modulates the decoder.

The post's original motivation is still the core; what changed is *who decides* the degradation type. It moved from fixed task labels to learned prompts, to vision–language embeddings and finally to human instructions, often backed by large pretrained models. The all-in-one survey [72] discusses this in detail.

Loss functions

Plain version. The loss decides what "good" means. Pixel losses reward being close to the truth on average; perceptual and adversarial losses reward looking real; frequency losses reward getting fine detail.

Pixel losses. With $\hat{x} = f_\theta(y)$,

$$\ell_2 = \frac{1}{P}\sum_p (\hat{x}_p - x_p)^2, \qquad \ell_1 = \frac{1}{P}\sum_p |\hat{x}_p - x_p|, \qquad \ell_\varepsilon = \frac{1}{P}\sum_p \sqrt{(\hat{x}_p - x_p)^2 + \varepsilon^2},$$

where P is the number of pixel values and ε a small constant that makes the last one a smooth, differentiable version of L1. L2 directly optimizes PSNR but penalizes large errors heavily and converges to the posterior mean. Zhao et al. [57] showed that the loss alone changes results noticeably with the architecture fixed, and that L1 and perceptually motivated losses (including combinations with SSIM) often beat L2 even on L2-related metrics. L1 is the default in most modern SR and restoration papers.

Perceptual (feature) loss. Compare images in the feature space of a pretrained classification network ϕ (usually VGG) [58]:

$$\ell_{\text{feat}} = \frac{1}{C_j H_j W_j}\|\phi_j(\hat{x}) - \phi_j(x)\|_2^2,$$

where ϕ_j is the feature map of layer j with C_j channels and size $H_j \times W_j$. It tolerates small misalignments and rewards matching edges and textures rather than exact values.

Adversarial loss. A discriminator D_ψ learns to tell restored from real images and the restorer learns to fool it, e.g. $\ell_{\text{adv}} = -\log D_\psi(\hat{x})$. This pushes outputs onto the manifold of natural images (SRGAN [12], ESRGAN [13], Real-ESRGAN [4]) at the cost of possible hallucinated texture. In practice GAN-based restorers use a weighted sum such as $\ell = \ell_1 + \lambda_p \ell_{\text{feat}} + \lambda_a \ell_{\text{adv}}$ with weights λ_p, λ_a.

Frequency losses. Networks tend to fit low frequencies first and under-fit fine detail. Adding a loss on the Fourier transform $\mathcal{F}$, such as $\|\mathcal{F}(\hat{x}) - \mathcal{F}(x)\|_1$ on complex values or on amplitude, directly penalizes missing high frequencies. The **focal frequency loss** [59] goes further and adaptively up-weights the frequency components that are currently hardest to reconstruct. Frequency terms are now common in deblurring and all-in-one models.

Evaluation metrics

Plain version. "Is the restored image close to the truth?" and "Does it look like a real photo?" are different questions, answered by different metrics, and you cannot maximize both at once.

Full-reference metrics

PSNR (peak signal-to-noise ratio):

$$\mathrm{PSNR}(x, \hat{x}) = 10 \log_{10} \frac{\mathrm{MAX}^2}{\mathrm{MSE}(x, \hat{x})},$$

where MAX is the largest possible pixel value (1 or 255) and MSE is the mean squared error. It is measured in dB; +1 dB is a 21% reduction in MSE.

SSIM (structural similarity) [60] compares local means, variances and covariance:

$$\mathrm{SSIM}(a, b) = \frac{(2\mu_a\mu_b + C_1)(2\sigma_{ab} + C_2)}{(\mu_a^2 + \mu_b^2 + C_1)(\sigma_a^2 + \sigma_b^2 + C_2)},$$

computed over local windows a, b of the two images and averaged, where μ are local means, σ^2 local variances, σ_{ab} the local covariance and C_1, C_2 small constants for stability. It correlates better with perceived structure than PSNR but still prefers smooth results.

LPIPS [61] measures distance between deep-network features (calibrated on human similarity judgments). Lower is better, and it rewards realistic textures that PSNR penalizes.

```
import numpy as np
from scipy import ndimage as ndi
from skimage import data, img_as_float32
from skimage.metrics import peak_signal_noise_ratio, structural_similarity

x = img_as_float32(data.camera())
y = np.clip(x + np.random.default_rng(0).normal(0, 0.1, x.shape), 0, 1).astype(np.float32)
x_hat = ndi.gaussian_filter(y, 1.0)

def psnr(ref, est, peak=1.0):
    mse = np.mean((ref.astype(np.float64) - est) ** 2)
    return 10 * np.log10(peak ** 2 / mse)

print(f"PSNR  ours {psnr(x, x_hat):.2f} dB | skimage {peak_signal_noise_ratio(x, x_hat,
 ↪  data_range=1):.2f} dB")
print(f"SSIM  {structural_similarity(x, x_hat, data_range=1):.3f}")
# PSNR  ours 27.20 dB | skimage 27.20 dB
# SSIM  0.639
```

Always report the conventions (RGB or Y channel, border crop, data range, 8-bit rounding): differences in these alone can shift published PSNR by tenths of a dB.

No-reference metrics

Plain version. A no-reference (or *blind*) metric looks at a single image and guesses how good a person would say it is, without a "correct answer" to compare against. It is like a teacher grading

an essay without an answer key: the teacher relies on what good essays usually look like. Restoration needs such metrics because the photos we most want to fix (real blurry, noisy, rainy, dark photos) have no clean version anywhere.

Full-reference, reduced-reference and no-reference

Image quality assessment (IQA) methods are grouped by how much they know about the pristine image x:

- **Full-reference (FR)**: the whole x is available, and the metric compares $\hat{x}$ with it pixel by pixel or feature by feature (PSNR, SSIM [60], LPIPS [61]). This is what synthetic benchmarks use.
- **Reduced-reference (RR)**: only a few summary features of x are available (for example statistics sent alongside a compressed video), and the metric compares the same features computed on $\hat{x}$.
- **No-reference (NR, blind)**: only $\hat{x}$ is available. The metric must carry its own model of what "good" looks like.

Restoration needs NR metrics for three reasons. (1) **Real test images have no ground truth**: real-world SR, deraining and dehazing are evaluated on photos where nobody has the clean version.

(2) **The reference itself can be imperfect or only one of many valid answers**: a GAN or diffusion output can look excellent yet have low PSNR because its texture differs from the reference's (the perception–distortion trade-off below). (3) **NR scores can steer systems at run time**, for example to choose among outputs or parameters when no reference exists. The price: an NR metric is a *model of human opinion*, and every model has blind spots.

How NR metrics are validated: correlation with human opinion

Plain version. We trust an NR metric only if it ranks images the way people do. So researchers collect many human ratings on many images and check how well the metric's scores agree with them.

Mean opinion score. In a subjective study, each image i is rated by several observers, and the average rating is its **mean opinion score**, $\mathrm{MOS}_i = \frac{1}{R}\sum_{r=1}^{R} s_{ir}$, where s_{ir} is the score of rater r and R the number of raters. Some datasets publish a **differential** score (DMOS), based on the difference between the ratings of the distorted image and its reference; for DMOS, lower usually means better.

SRCC (Spearman rank-order correlation) measures whether the metric puts the images in the

same *order* as people:

$$\mathrm{SRCC} = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n\,(n^2-1)},$$

where n is the number of test images and d_i the difference between the rank of image i by predicted score and its rank by MOS (this short form assumes no tied ranks; with ties one computes the Pearson correlation of the ranks). SRCC is unchanged by any monotonic transformation of the scores, so it measures **prediction monotonicity**.

PLCC (Pearson linear correlation) measures how *linearly* the scores agree:

$$\mathrm{PLCC} = \frac{\sum_{i=1}^{n}(q_i-\bar{q})(m_i-\bar{m})}{\sqrt{\sum_{i=1}^{n}(q_i-\bar{q})^2}\,\sqrt{\sum_{i=1}^{n}(m_i-\bar{m})^2}},$$

where q_i is the metric's score for image i, m_i its MOS, and $\bar{q}$, $\bar{m}$ their means. Because metric scales are arbitrary, it is common practice to first pass q_i through a fitted monotonic (usually logistic) function and compute PLCC on the mapped scores, so PLCC measures **prediction accuracy** after calibration. Both lie in $[-1, 1]$. For "lower is better" metrics such as NIQE the raw correlation with MOS is negative, and papers report its absolute value. A metric is typically trained on part of a dataset and tested on the rest, and its cross-dataset correlation is the more honest test of generalization.

Human-rated datasets. NR metrics are trained and tested on image-quality datasets with human scores. What matters for restoration is *where the distortions come from*: synthetic damage (LIVE [104], TID2013 [105], KADID-10k [106]), real camera photos (KonIQ-10k [107], SPAQ [108]) or the outputs of restoration algorithms themselves (PIPAL [109]). Only datasets of the last kind contain the images a restoration paper actually needs to score.

Optional: Human-rated IQA datasets (in the Supplementary notes). What each dataset contains, in one table.

Family 1: natural-scene-statistics (NSS) models

Plain version. Clean photos share hidden statistical habits. If you remove local brightness and local contrast from a photo, the leftovers follow a bell-shaped histogram of a typical shape, and neighboring leftovers are only weakly related. Blur, noise and compression change that shape in characteristic ways. NSS metrics measure the shape and ask how unusual it is.

MSCN coefficients. For a grayscale image I, the **mean-subtracted contrast-normalized** (MSCN) coefficients are [94]

$$\hat{I}(i,j) = \frac{I(i,j)-\mu(i,j)}{\sigma(i,j)+C}, \qquad \mu(i,j) = \sum_{k,l} w_{k,l}\, I(i+k, j+l), \qquad \sigma(i,j) = \sqrt{\sum_{k,l} w_{k,l}\,\big(I(i+k,j+l)-\mu(i,j)\big)^2},$$

where (i, j) is a pixel position, w a small circularly symmetric Gaussian window whose weights sum to 1, μ and σ the local weighted mean and standard deviation, and C a small constant (1 for 8-bit images) that prevents division by zero in flat regions. This is a divisive normalization, loosely modeled on contrast gain control in early human vision.

The generalized Gaussian fit. The histogram of $\hat{I}$ is modeled by a zero-mean generalized Gaussian distribution (GGD),

$$f(t; \alpha, \sigma^2) = \frac{\alpha}{2\beta\,\Gamma(1/\alpha)} \exp\!\big(-(|t|/\beta)^{\alpha}\big), \qquad \beta = \sigma\sqrt{\frac{\Gamma(1/\alpha)}{\Gamma(3/\alpha)}},$$

where t is an MSCN value, $\alpha > 0$ the **shape** ($\alpha = 2$ is a Gaussian, $\alpha = 1$ a Laplacian; smaller α means a peakier histogram with heavier tails, larger α a flatter top), σ^2 the variance, β the corresponding scale and Γ the gamma function. A simple way to estimate α is **moment matching**: the ratio $\mathbb{E}[t^2]/(\mathbb{E}|t|)^2 = \Gamma(1/\alpha)\,\Gamma(3/\alpha)/\Gamma(2/\alpha)^2$ depends only on α, so one computes the left side from the data and searches for the α that matches it.

BRISQUE [94] (Blind/Referenceless Image Spatial Quality Evaluator) computes the GGD shape and variance of the MSCN coefficients, plus parameters of *asymmetric* GGDs fitted to products of neighboring MSCN coefficients in four orientations (horizontal, vertical and the two diagonals), at two image scales. A support-vector regressor trained on human scores of synthetically distorted images maps these features to a quality score. Lower is better. It is **opinion-aware**: it needs human scores for training.

NIQE [62] (Natural Image Quality Evaluator) is **opinion-unaware**: it never sees human scores or distorted images. It computes similar NSS features on sharp patches, fits a multivariate Gaussian (MVG) with mean ν_1 and covariance Σ_1 to features from a corpus of pristine images, fits another MVG (ν_2, Σ_2) to the test image's patches, and reports

$$D(\nu_1, \nu_2, \Sigma_1, \Sigma_2) = \sqrt{(\nu_1 - \nu_2)^{\top}\Big(\frac{\Sigma_1 + \Sigma_2}{2}\Big)^{-1}(\nu_1 - \nu_2)},$$

a Mahalanobis-like distance between the two Gaussians ($^{\top}$ is the transpose). Lower is better; zero means "statistically like the pristine corpus".

Two relatives appear often in restoration papers. **PIQE** [96] applies MSCN statistics block by block, needs no opinion scores, and also gives a map of where the image looks distorted. The **Perceptual Index** (PI) of the 2018 PIRM super-resolution challenge [63] combines NIQE with a score learned by Ma et al. [95] from human ratings of super-resolved images; lower PI means better perceptual quality, and many perceptual SR papers still report it.

Optional: PIQE and the Perceptual Index (in the Supplementary notes). How PIQE pools block distortions, the PI formula, and how the PIRM challenge used it.

Code: MSCN statistics in practice. The snippet computes MSCN coefficients, the GGD shape by moment matching, the MSCN variance and the correlation between horizontal neighbors (a simplified version of BRISQUE's pairwise features) on `camera` and on blurred, noisy and over-sharpened versions:

```
import numpy as np
from scipy.ndimage import gaussian_filter
from scipy.special import gamma
from skimage import data, img_as_float

def mscn(img, sigma=7/6, C=1.0):
    """Mean-subtracted contrast-normalized coefficients of a grayscale image in [0, 255]."""
    mu = gaussian_filter(img, sigma)
    sd = np.sqrt(np.abs(gaussian_filter(img * img, sigma) - mu * mu))
    return (img - mu) / (sd + C)

def ggd_shape(x):
    """Moment-matching estimate of the generalized Gaussian shape alpha (2 = Gaussian)."""
    alphas = np.arange(0.2, 10.0, 0.001)
    r_theory = gamma(1 / alphas) * gamma(3 / alphas) / gamma(2 / alphas) ** 2
    r_data = np.mean(x ** 2) / np.mean(np.abs(x)) ** 2
    return alphas[np.argmin((r_theory - r_data) ** 2)], np.mean(x ** 2)

rng = np.random.default_rng(0)
clean = img_as_float(data.camera()) * 255
versions = {
    "clean": clean,
    "blur (sigma=2)": gaussian_filter(clean, 2),
    "noise (sigma=20)": np.clip(clean + rng.normal(0, 20, clean.shape), 0, 255),
    "over-sharpened": np.clip(clean + 2.0 * (clean - gaussian_filter(clean, 2)), 0, 255),
}
for name, img in versions.items():
    m = mscn(img)
    alpha, var = ggd_shape(m)
    rho = np.mean(m[:, :-1] * m[:, 1:]) / var           # correlation of horizontal neighbors
    print(f"{name:17s} GGD shape {alpha:4.2f} | MSCN variance {var:4.2f} | neighbor corr.
     ↪  {rho:+.2f}")
# clean             GGD shape 1.56 | MSCN variance 0.28 | neighbor corr. +0.12
# blur (sigma=2)    GGD shape 0.87 | MSCN variance 0.04 | neighbor corr. +0.89
# noise (sigma=20)  GGD shape 3.12 | MSCN variance 0.68 | neighbor corr. -0.10
# over-sharpened    GGD shape 2.35 | MSCN variance 0.43 | neighbor corr. +0.09
```

Each distortion pushes the features in its own direction (Figure 6). **Blur** removes local contrast, so most MSCN values collapse toward zero: tiny variance, a very peaky histogram (small α) and strongly correlated neighbors. **Noise** adds independent energy: larger variance, a flat-topped histogram (α above 2) and neighbor correlation near zero (slightly negative, because subtracting the local mean couples neighbors). **Over-sharpening** moves the features *in the same direction as noise*. That is the core weakness of NSS metrics for restoration: they cannot tell restored texture

from invented high-frequency energy. Note also that the absolute numbers depend on content and on C (the clean `camera` image is not a unit Gaussian), which is why BRISQUE needs a trained regressor and NIQE a pristine reference corpus rather than fixed thresholds.

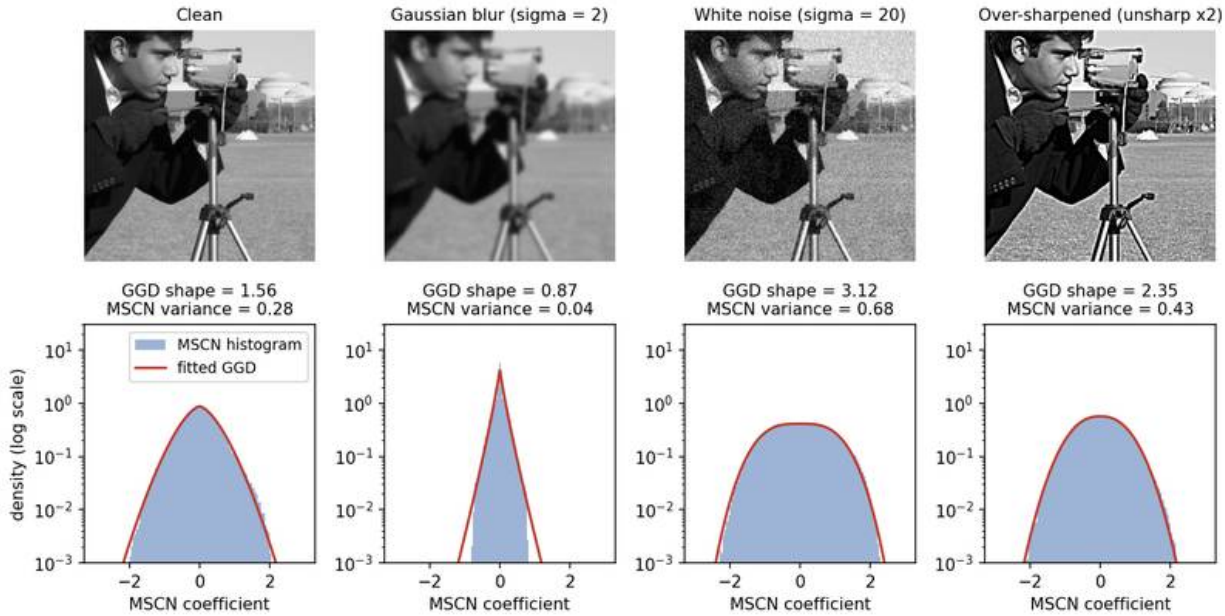


Figure 6 — MSCN statistics, the basis of BRISQUE, NIQE and PIQE. Blur makes the histogram narrow and peaky; noise and over-sharpening both make it wider and flatter, so an NSS metric sees them as similar changes.

Family 2: learned deep models

Plain version. Instead of hand-made statistics, train a neural network to predict human ratings directly. Deep features capture content, semantics and complex artifacts far better, but the model only knows the kinds of images and opinions it was trained on.

Five widely used examples: **NIMA** [97] predicts the whole distribution of human ratings rather than one number; **HyperIQA** [98] adapts its quality rule to the image content, since the same blur is judged differently on a sky and on a face; **MUSIQ** [99] reads the image at its native resolution and aspect ratio, so no resizing changes the artifacts it judges; **MANIQA** [100] is a transformer that won the no-reference track of the NTIRE 2022 perceptual IQA challenge, whose data include restoration and GAN outputs; and **TOPIQ** [101] lets high-level semantics decide which low-level distortions matter.

All of these are **opinion-aware** and output "higher is better" scores, and each depends on its training dataset: a model trained on KonIQ-10k and the same architecture trained on PIPAL can

rank restoration methods differently.

Optional: How the learned NR assessors work (in the Supplementary notes). The architecture and training target behind each of the five models.

Family 3: vision-language and large multimodal models

Plain version. Models trained on huge collections of images and text already "know" what a good or bad photo looks like. You can ask them, either by comparing the image with text prompts or by literally asking a chatbot-style model to rate it.

CLIP-IQA [102] compares the CLIP embedding of the image with those of an opposite prompt pair such as "Good photo." and "Bad photo." and turns the two similarities into a score between 0 and 1 (higher is better). The basic version needs no opinion scores at all; other prompt pairs ("Sharp photo." / "Blurry photo.") probe single attributes, and the wording of the prompts changes the scores. **Q-Align** [103] fine-tunes a large multimodal model to answer the way human raters do, with a rating word from bad to excellent, and turns its probabilities for the five words into a score. Such models are strong across datasets but heavy, and they inherit the biases of their base model.

Optional: CLIP-IQA and Q-Align scores (in the Supplementary notes). The two formulas that turn similarities and word probabilities into a quality score.

Distribution-level metrics

Some metrics score a *set* of outputs rather than one image. **FID** [64] compares the distribution of Inception features of the restored set with that of a set of real images; it was designed for generative models and is used for generative restoration such as diffusion SR, but it needs many images and says nothing about any single one. **IS** (Inception Score), mentioned in the post, measures the confidence and diversity of class predictions; it suits class-conditional generation rather than restoration and is rarely used there today.

User studies and paired comparisons

Plain version. In the end, people decide what looks good. The two standard ways to ask them are to rate images on a scale or to choose the better of two. Such a study is also the yardstick every NR metric above is validated against, and it is the strongest evidence a new method can bring.

Mean opinion score (MOS). Each of N raters gives an image a score s_n on a fixed scale (often 1 to 5), and the MOS is their average, as defined for metric validation above. ITU-R BT.500 [111]

standardizes such tests: viewing conditions, rating scales, single- and double-stimulus and comparison methods, and screening of unreliable raters.

Paired comparison. Absolute scales drift between raters; choosing the better of two images is easier and more consistent. The **Bradley–Terry** model [112] turns many such choices into one strength $\pi_i > 0$ per method i, fitted so that method i is preferred over method j with probability $\pi_i/(\pi_i + \pi_j)$. Sorting methods by their fitted strengths gives one consistent ranking, although each rater only ever compared two images at a time.

Optional: Fitting the Bradley–Terry model (in the Supplementary notes). The maximum-likelihood update and a runnable three-method example.

Good practice: hide method names, randomize the order and the left/right position, show the input for context, use enough images *and* raters, report confidence intervals or significance, and state the display and viewing conditions. A study with a handful of raters and hand-picked images is a demonstration, not evidence.

Pitfalls

- **Metrics can be gamed.** A restorer that adds grain or sharpens edges can improve NSS scores, as the code above shows, without becoming more faithful. Any differentiable metric (and any learned IQA network) used as a training loss or for model selection can be optimized directly, and the network may learn to please the metric rather than people. If a method uses CLIP or an IQA network in its loss, evaluating it with the same model is not independent.
- **Domain mismatch.** Most opinion-aware metrics were trained on synthetic distortions (LIVE, TID2013, KADID-10k) or on camera photos (KonIQ-10k, SPAQ). Restoration outputs contain different artifacts: hallucinated textures, over-smoothing, GAN patterns. This mismatch was the explicit motivation for PIPAL [109], whose authors found that existing IQA methods could not fairly evaluate GAN-based restoration. For low light, the problems are exposure and color, which most NSS features normalize away; see the low-light-specific discussion in DLCV 05.
- **Metrics disagree.** NSS, learned and vision-language metrics often rank the same methods differently, and all can disagree with people. Scores also depend on the implementation, checkpoint, color space, input size and whether the image is resized, so numbers copied from different papers are often not comparable.
- **"Higher NR quality" is not "more correct".** An NR metric cannot know whether a sharp face or line of text matches the real scene. A hallucinated license plate can score perfectly.

Recommended practice

1. When a reference exists, lead with FR metrics (PSNR, SSIM, LPIPS) and add NR metrics as a complement; place your method on the perception–distortion plane (next subsection).

2. On real images without references, report **several NR metrics from different families** (for example NIQE or PI, one or two learned models such as MUSIQ, MANIQA or TOPIQ, and a vision-language one such as CLIP-IQA or Q-Align), and say which way each one points.
3. Use one shared implementation and state it, with versions and checkpoints. The open-source

   IQA-PyTorch toolbox (`pyiqa`) [110] reimplements most metrics in this section (NIQE, BRISQUE, PIQE, the Ma score and PI, NIMA, HyperIQA, MUSIQ, MANIQA, TOPIQ, CLIP-IQA, Q-Align, and FR metrics) behind one interface.
4. Do not tune on the judge: keep at least one metric that played no role in training or model selection.
5. Back important claims with a **user study** (mean opinion scores or paired comparisons, blind and randomized); the user-study subsection above describes MOS and Bradley–Terry paired-comparison protocols. When metrics disagree with each other or with people, say so.

Summary table

The four families at a glance:

Family	Examples	Needs human scores?	Better	Typical blind spot
NSS	NIQE, BRISQUE, PIQE, PI	NIQE and PIQE no; BRISQUE and PI (through Ma) yes	Lower	Rewards added grain and sharpening; normalizes away exposure and color
Learned deep	NIMA, HyperIQA, MUSIQ, MANIQA, TOPIQ	Yes	Higher	Depends on its training data; restoration artifacts differ from the training distortions
Vision-language	CLIP-IQA, Q-Align	CLIP-IQA (zero-shot) no; Q-Align yes	Higher	Prompt wording; biases of the base model; heavy
Distribution-level	FID	No (pretrained Inception features)	Lower	Needs many images; says nothing about a single one

Optional: Metric-by-metric summary table (in the Supplementary notes). Family, training requirement, direction and main sensitivity of all thirteen metrics.

The perception–distortion trade-off

Blau and Michaeli [65] formalized the tension. Let $\Delta(x, \hat{x})$ be any distortion measure (MSE, $1 - $ SSIM, ...) and $d(p_X, p_{\hat{X}})$ a divergence between the distribution of real images and that of restored images (perceptual quality: small d means outputs are statistically indistinguishable from real images). The **perception–distortion function** is

$$P(D) = \min_{p_{\hat{X}|Y}} d(p_X, p_{\hat{X}}) \quad \text{subject to} \quad \mathbb{E}[\Delta(X, \hat{X})] \le D,$$

where the minimum is over all (possibly random) restoration methods. They proved that $P(D)$ is non-increasing and (for divergences convex in their second argument) convex: there is an unattainable region near the origin. A method cannot be both minimal-distortion and perfectly realistic. In particular, the MMSE estimator $\mathbb{E}[x \mid y]$ has the lowest distortion but a blurry, unrealistic output distribution; to look real you must accept higher MSE. GANs (and now diffusion models) are principled ways to move along the bound.

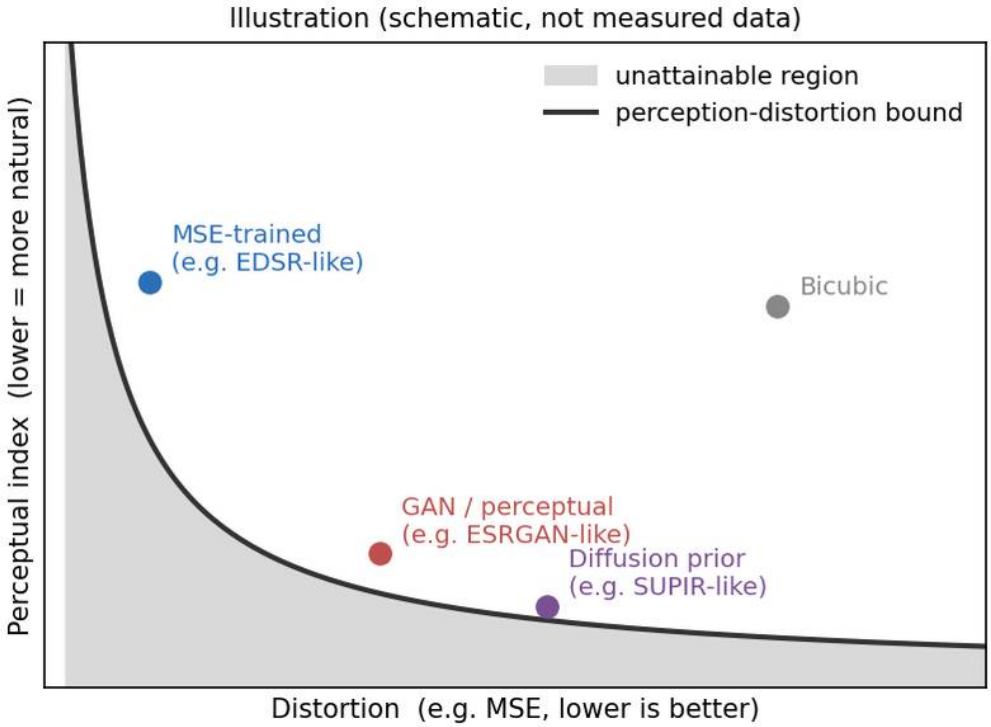


Figure 7 — Schematic of the perception–distortion trade-off [65] (not measured data). The bound is convex and decreasing; methods trade fidelity (left) against realism (bottom).

This is the update in the post's metric section: since diffusion models can generate plausible but unfaithful detail, the gap between fidelity (PSNR/SSIM) and perceptual quality has become central. The **NTIRE 2025** ×4 SR challenge therefore ran two tracks: a restoration track ranked by PSNR and a perceptual track ranked by a perceptual score [3], whereas the 2024 edition ranked by PSNR [2]. Practical advice: report both a distortion and a perceptual metric, and say where on the trade-off your method aims.

NTIRE challenges

Plain version. Each year a CVPR workshop runs public competitions on fixed datasets and publishes a report describing the winning ideas: the easiest way to see what currently works.

The NTIRE challenge reports are a good way to follow restoration now that per-task leaderboards are gone. Each report describes the task, data, evaluation protocol and the methods of top teams. Examples cited in this tutorial:

- **NTIRE 2017 SR** [19]: introduced DIV2K; EDSR [10] won.
- **NTIRE 2024** ×4 **SR** [2]: bicubic ×4 on DIV2K, ranked by PSNR.
- **NTIRE 2025** ×4 **SR** [3]: separate restoration (PSNR) and perceptual tracks.
- **NTIRE 2025 raindrop removal** [40]: real day and night raindrop images with dual focus.
- **NTIRE 2024 and 2025 low-light enhancement**: the two editions of the low-light challenge are summarized in DLCV 05.

Two patterns appear across recent reports: winning entries are usually ensembles or large hybrids of the backbones above (HAT-, Restormer-, NAFNet- or Mamba-style, sometimes with diffusion priors) trained with heavy data and compute, and challenges increasingly use real captured data and perceptual tracks.

Modern view

The post closes with an extension list (medical images, multimodal, unsupervised, new architectures, all-in-one, low-level plus high-level, real-time, underwater). Most of these are now active areas. Here we summarize the survey literature task by task, then the 2023–2026 state of the art and the open problems.

Survey papers

- **Super-resolution** (Wang, Chen and Hoi [66]): describes SR models as choices of upsampling position and operator, network design, losses and training, and flagged unsupervised, real-world SR as the main open direction.
- **Denoising** (Tian et al. [67]; Elad, Kawar and Vaksman [68]): the realism of the noise model is often the bottleneck; denoisers are now building blocks for other inverse problems and for diffusion models, and the frontier moved toward sampling from the posterior.
- **Deblurring** (Zhang et al. [69]): dynamic-scene deblurring moved from kernel estimation to end-to-end multi-scale networks trained on captured pairs such as GoPro.

- **Dehazing** (Gui et al. [70]) and **deraining** (Yang et al. [71]): physics-guided and physics-free designs coexist, and realistic degradation models and real-world evaluation matter more than further architecture tweaks on synthetic data.
- **All-in-one** (Jiang et al. [72]): a taxonomy by architecture, learning paradigm and core innovation; inconsistent task sets complicate comparisons, and generalization to unseen and mixed degradations is open.
- **Diffusion restoration** (Li et al. [73]): the best perceptual quality, but costly and with fidelity and hallucination concerns; few-step distillation and better evaluation are the key directions.

Optional: Survey papers in detail (in the Paper guides). The full title, venue, taxonomy and takeaway of each survey.

State of the art, 2023–2026

1. **All-in-one and blind restoration became the mainstream setting**, not a niche topic. Methods moved from learned degradation representations (AirNet) to prompts (PromptIR), frequency-aware adaptation (AdaIR) and language guidance (DA-CLIP, InstructIR).
2. **Pretrained diffusion models as generative priors** transformed real-world SR and blind restoration (DiffBIR, SUPIR), and one-step variants (OSEDiff) cut the inference cost from tens or hundreds of steps to one.
3. **State-space models** (MambaIR, MambaIRv2) joined CNNs and transformers as backbone options with global receptive fields at linear cost.
4. **Evaluation caught up with generation**: challenges separate fidelity from perceptual tracks [3], and real captured datasets (SIDD, O-HAZE, Dense-Haze, Raindrop Clarity) are standard alongside synthetic benchmarks.

Open problems

- **Real-world generalization.** Synthetic pipelines still miss real camera processing, spatially varying blur and mixed weather; models degrade on unseen combinations.
- **Fidelity versus hallucination.** Generative priors can invent text, faces or textures. For medical, forensic and scientific use this is unacceptable, and we lack good metrics for "faithful but sharp".
- **Evaluation.** PSNR/SSIM reward blur; no-reference metrics can be gamed; human studies are costly. Agreement between metrics and human preference for generative restoration is still weak.

- **Efficiency.** High-resolution, real-time (and video) restoration on phones and cameras remains a strong constraint, for transformers and diffusion models especially.
- **Task interference and scaling in all-in-one models**, and how to add a new degradation without retraining.
- **Low-level plus high-level.** Restoration that helps downstream detection or segmentation, rather than human viewers, is a different objective (task-driven evaluation, as in RESIDE [32]).

A perspective from the original post

The post ends with two observations that still hold. First, because each task is simple to state, the race for state of the art is largely won by special architectures, which means a lot of compute and complicated experiments; but these tasks are very practical and easy to apply, so there is much room in **applications** and in **combining tasks** (exactly where all-in-one restoration went). Second, compared with high-level tasks the learning curve is gentle: a beginner can easily picture what the model is doing (a damaged image goes in, a cleaner one comes out), which makes low-level vision a good door into deep learning research.

Key takeaways

- Low-level vision is about *perceiving* pixels well; almost every task is an instance of $y = \mathcal{D}(x) + n$ with a different $\mathcal{D}$ (blur, downsampling, JPEG, scattering, rain layer, Retinex illumination, mask).
- Deep learning moved the degradation model from the solver into the **data generator**: realistic, randomized synthetic pipelines (and captured pairs) are as important as the network.
- Upsampling in SR evolved from bicubic pre-upsampling (SRCNN) through transposed convolution (FSRCNN) to sub-pixel convolution (ESPCN), which is the standard today.
- A few general backbones dominate: U-Net-style CNNs (NAFNet), transformers (SwinIR, Restormer, HAT) and state-space models (MambaIR); key design habits are global residuals, no batch norm, large receptive fields and gating/attention.
- All-in-one models (AirNet, TransWeather, PromptIR, AdaIR, DA-CLIP, InstructIR) handle unknown degradations by conditioning one network on a learned, prompted or text-given degradation description.
- L2/L1 losses give faithful but smooth results; perceptual, adversarial and diffusion-based methods give realistic but less faithful ones. The **perception–distortion trade-off** says you cannot have both at the limit, so report both kinds of metric.
- Blind deblurring and blind SR must estimate the **kernel** as well as the image. Naive joint MAP prefers the no-blur solution [74]; working methods marginalize over the image, use edge prediction or sharpness-favoring priors, alternate image and kernel updates coarse-to-fine and

finish with non-blind deconvolution. Deep methods either estimate or condition on the kernel (SelfDeblur, kernel-aware blind-SR networks) or skip it entirely (GoPro-trained networks, BSRGAN, Real-ESRGAN).
- KITTI is a driving benchmark, not a dehazing dataset; for dehazing use RESIDE plus real pairs such as O-HAZE and Dense-Haze.
- No-reference metrics are models of human opinion, validated by SRCC/PLCC against mean opinion scores. NSS metrics (BRISQUE, NIQE, PIQE, PI) measure deviations of MSCN statistics; learned (NIMA, HyperIQA, MUSIQ, MANIQA, TOPIQ) and vision-language (CLIP-IQA, Q-Align) models learn from ratings. All can be gamed by sharpening or noise and suffer from the gap between their training distortions and restoration artifacts, so report several, together with FR metrics and a user study.

Exercises

1. **Degradation order.** Using the `degrade` function, compare (a) blur → downsample → noise → JPEG with (b) noise → blur → downsample → JPEG at the same parameters. Visually and with the noise standard deviation measured on a flat patch, explain why the order matters for what the network has to learn.

   Hint

   In (b), the blur and downsampling filter the noise: its standard deviation drops and it becomes spatially correlated (smooth blotches rather than per-pixel grain). In (a), the noise stays white at the low resolution. A network trained only on (a) will under-remove correlated noise at test time. This is one reason Real-ESRGAN randomizes and repeats the stages.

2. **Pixel shuffle by hand.** A feature tensor has shape $(Cs^2, H, W) = (4, 2, 2)$ with $s = 2$, $C = 1$, and channel k is filled with the constant k. Write the 4×4 output of pixel shuffle using the formula in the SR section, then check with `torch.nn.PixelShuffle(2)`.

   Hint

   Output pixel $(2h + i, 2w + j)$ takes channel $2i + j$. So every 2×2 block is $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$, and the 4×4 output repeats that block four times. Each output block gathers one value from each of the four channels at the same low-resolution position.

3. **Residual or direct?** Modify `TinyDnCNN.forward` to return `self.body(y)` (predict the clean image directly) and train both versions with the same seed and steps. Compare PSNR and the loss curves. Then repeat with $\sigma = 5/255$. When does residual learning help most?

   Hint

With few training steps, the residual version usually converges faster and reaches a higher PSNR, because at initialization it already outputs roughly y (good PSNR) and only needs to learn a small correction. At low noise the direct version struggles most, since it must learn a near-identity mapping through several ReLU layers.

4. **Dark channel by numbers.** A hazy pixel has $I = (0.75, 0.78, 0.80)$, the darkest value of I^c/A^c over its patch and channels is 0.70, and $A = (0.9, 0.9, 0.9)$. With $\omega = 0.95$ and $t_0 = 0.1$, compute $\tilde{t}$ and the recovered $\hat{J}$ for this pixel.

   Hint

   $\tilde{t} = 1 - 0.95 \times 0.70 = 0.335$. Then $\hat{J}^c = (I^c - 0.9)/0.335 + 0.9$, giving about $(0.452, 0.542, 0.601)$. The output is darker and more saturated than the input, as expected when the veil is removed.

5. **Perception versus distortion.** Take the tiny denoiser's output and add a small amount of fresh Gaussian noise ($\sigma = 3/255$) to it. Compute PSNR and SSIM before and after. The image may look *more* natural (less plastic) although both metrics drop. Relate this to the perception–distortion function $P(D)$.

   Hint

   The MMSE-like output lies at low distortion but far from the natural-image distribution (too smooth). Adding grain moves the output distribution closer to real photos (lower d) at the price of higher MSE: a move along, or toward, the convex bound, not past it.

6. **Design an all-in-one experiment.** You must build a single model for noise ($\sigma \in [0, 50]$), rain streaks and haze. Describe the training data mix, how you would condition the network (no labels at test time), the baselines you would compare against, and how you would detect task interference.

   Hint

   Mix datasets per task with balanced sampling; add a degradation encoder (contrastive as in AirNet, or prompts as in PromptIR) trained jointly. Compare with three single-task models of the same backbone and size, and with a version without conditioning. Interference shows up as the all-in-one model falling behind the single-task model on one task while matching it on others; also test on mixed degradations (rain + haze) unseen in training.

7. **Kernel estimation by hand.** In the blind-deconvolution snippet, (a) remove the `shock(...)` call and set `keep=1.0` (use all gradients) and compare the kernel error and final PSNR; (b) start from a 3×3 box kernel instead of the near-delta; (c) replace `h_true` by a straight 25-pixel motion path in a 31×31 window. Explain each result with the discussion of Levin et al. [74] and the coarse-to-fine strategy.

Hint

(a) With all gradients, the many weak gradients of the smoothed latent image dominate the least-squares fit, and the estimate no longer improves (in our run the kernel error grew instead of shrinking and the final PSNR stayed at the blurry input's level). Selecting a few strong, well-predicted edges is what makes the kernel step informative, which is the practical answer to the "no-blur" bias of naive joint MAP. (b) The loop is a non-convex alternation, so the start matters; a different start can converge to a different local minimum. (c) A large kernel at full resolution is hard to find from a delta start: in our run the error dropped much less and most of the true-kernel PSNR was not recovered. Estimating the kernel on a downsampled image first (where it is only a few pixels long) and upsampling the result is what the image pyramid is for.

8. **Validating an NR metric.** Five restored images have mean opinion scores MOS $= (4.1, 3.2, 2.5, 1.8, 3.9)$ and NIQE scores $(3.4, 4.2, 5.1, 6.3, 3.0)$. (a) Compute SRCC by hand and interpret its sign. (b) Would SRCC change if you reported $\log(\text{NIQE})$ instead? Would PLCC? (c) Run the MSCN snippet on an over-sharpened version of a *denoised* image and explain why an NSS metric might rate it better than the plain denoised output.

Hint

(a) Ranking from smallest to largest, the MOS ranks of images 1–5 are $(5, 3, 2, 1, 4)$ and the NIQE ranks are $(2, 3, 4, 5, 1)$. The rank differences are $(3, 0, -2, -4, 3)$, so $\sum d_i^2 = 38$ and SRCC $= 1 - 6 \cdot 38/(5 \cdot 24) = -0.9$. The sign is negative because lower NIQE means better quality; one reports $|\text{SRCC}| = 0.9$. Only images 1 and 5 are swapped. (b) SRCC depends only on ranks, so any monotonic transform such as log leaves it unchanged; PLCC changes, which is why PLCC is usually computed after a fitted monotonic mapping. (c) Denoising leaves a smooth image whose MSCN statistics resemble mild blur (low variance, peaky histogram). Sharpening restores high-frequency energy and pushes the statistics back toward those of natural images, whether or not the added detail is real.

Supplementary notes

These notes are optional background that the main text does not depend on.

Blind super-resolution: conditioning on and estimating the kernel

Plain version. In blind super-resolution the main unknown is the blur kernel k in $y = (x * k) \downarrow_s + n$. A method can hand the kernel to the network, estimate it and correct it in a loop, estimate it from the test image alone, or skip it and learn the degradation implicitly. The three explicit families are:

- **Condition the network on the kernel.** SRMD [87] gives a single SR network the degradation as extra input: the blur kernel, reduced to a low-dimensional vector, and the noise level are

stretched into maps of the same size as the low-resolution image and concatenated with it. One network then handles many kernels and noise levels, provided someone supplies them.

USRNet [91] goes further with **deep unfolding**: it unrolls a half-quadratic-splitting MAP solver into a network whose data module uses the given kernel, scale and noise level in closed form and whose prior module is a learned denoiser, so one model serves arbitrary known kernels.

- **Estimate the kernel, then correct it.** IKC [88] observes that a wrong kernel causes characteristic artifacts (too large gives over-sharpening, too small gives over-smoothing), and uses them to iteratively correct an initial kernel estimate while a kernel-conditioned SR network (SFTMD) restores the image. DAN [90] unfolds the alternation into one end-to-end network: an *estimator* predicts the kernel from both the low-resolution input and the current SR result, a *restorer* super-resolves with the current kernel, and the two alternate, the same idea as the classical blind-deblurring loop of the main text.
- **Estimate the kernel from the image alone.** KernelGAN [89] trains a small GAN on the single test image at test time: its generator learns to downscale the image so that the discriminator cannot tell the patches of the downscaled image from those of the original. Because patches recur across scales in natural images, the learned downscaler is an estimate of the image's own SR kernel, which is then given to a non-blind SR method.

Explicit-kernel methods are interpretable and accurate when the degradation really is "blur, downsample, noise" with a roughly uniform kernel. They are fragile when it is not (JPEG, camera processing, spatially varying blur), and errors in the estimated kernel propagate into the result. That motivated the **implicit** route: BSRGAN [92] trains on a practical degradation model with randomly shuffled blur (isotropic and anisotropic Gaussian kernels), downsampling, noise and JPEG steps, and Real-ESRGAN [4] uses its high-order pipeline; neither estimates a kernel at test time. Diffusion-prior methods [16]–[18] are implicit as well. The survey by Liu et al. [93] organizes blind SR methods by how they model the degradation and which data (external datasets or the single test image) they use, and compares them on synthetic and real images.

Plug-and-play priors and half-quadratic splitting

Plain version. A deblurring method has two jobs: keep the result consistent with the blurry photo, and make it look like a natural image. Half-quadratic splitting (HQS) gives each job to its own step and lets the two steps take turns. The "natural image" step is exactly what a denoiser does, so you can plug in any denoiser, from total variation to a CNN, without rewriting the rest.

Precise version. Let $y = Hx + n$ be the observation, where H is the matrix form of the known degradation (here a convolution with a blur kernel h; for super-resolution it also contains sub-sampling). The MAP problem of the ill-posedness section is

$$\min_x \ \tfrac{1}{2}\|y - Hx\|_2^2 + \lambda\, \Phi(x),$$

where Φ is the regularizer (negative log-prior) and $\lambda > 0$ its weight. HQS [113] copies x into an auxiliary variable z and ties the copies with a quadratic penalty:

$$\min_{x,z} \ \tfrac{1}{2}\|y - Hx\|_2^2 + \lambda\, \Phi(z) + \tfrac{\mu}{2}\|x - z\|_2^2,$$

where $\mu > 0$ is the penalty weight. Alternating over the two variables gives

$$x^{t+1} = (H^\top H + \mu I)^{-1}(H^\top y + \mu z^t),$$
$$z^{t+1} = \arg\min_z \ \tfrac{1}{2}\|z - x^{t+1}\|_2^2 + \tfrac{\lambda}{\mu}\, \Phi(z),$$

where t is the iteration and I the identity matrix. The first step involves only the data. For a convolution, H is diagonal in the Fourier domain, so it is a pointwise division: $\hat{x} = (\overline{\hat{h}}\,\hat{y} + \mu\,\hat{z})/(|\hat{h}|^2 + \mu)$, with hats for 2-D Fourier transforms and the bar for the complex conjugate. The second step involves only the prior, and it is the MAP estimate of a clean image from x^{t+1} corrupted by Gaussian noise of standard deviation $\sigma_t = \sqrt{\lambda/\mu}$ [115]. That is why a denoiser can play the prior: write $z^{t+1} = \mathcal{D}_{\sigma_t}(x^{t+1})$ with $\mathcal{D}_\sigma$ any Gaussian denoiser at noise level σ, and Φ never has to be written down. This is the idea of plug-and-play priors [114] (introduced there with ADMM, another splitting scheme), and [115] plugged learned CNN denoisers into HQS; the set of denoisers at different σ plays the role of Φ. Because $\sigma_t = \sqrt{\lambda/\mu}$, a growing μ means a shrinking denoiser strength: start with strong smoothing and let the data term take over. USRNet [91] unrolls this loop into a network whose data module is the closed-form step and whose prior module is a learned denoiser. The survey of Elad et al. [68] discusses the wider use of denoisers as building blocks for inverse problems.

Code. The snippet below blurs `camera` with a known Gaussian kernel, adds noise, and compares a one-step Tikhonov filter (the $z = 0$ case of the data step) with ten HQS iterations that plug in a total variation denoiser [116] or a non-local means denoiser [117] from scikit-image. It runs in about two seconds.

```
import numpy as np
from numpy.fft import fft2, ifft2
from skimage import data, img_as_float, transform
from skimage.restoration import denoise_tv_chambolle, denoise_nl_means

rng = np.random.default_rng(0)
x = transform.resize(img_as_float(data.camera()), (256, 256), anti_aliasing=True)
ax = np.arange(15) - 7
h = np.exp(-(ax[:, None] ** 2 + ax[None, :] ** 2) / (2 * 1.5 ** 2)); h /= h.sum()   # known
 ↪  Gaussian blur
P = np.zeros(x.shape); P[:15, :15] = h
H = fft2(np.roll(P, (-7, -7), (0, 1)))                                               #
 ↪  transfer function of h
y = np.real(ifft2(H * fft2(x))) + 0.01 * rng.standard_normal(x.shape)                # y = h
 ↪  * x + n
Y = fft2(y)
psnr = lambda a: 10 * np.log10(1 / np.mean((np.clip(a, 0, 1) - x) ** 2))
```

```
def data_step(z, mu):          # x = argmin 1/2||y - h*x||^2 + mu/2||x - z||^2  (closed form
 ↪  in Fourier domain)
    return np.real(ifft2((np.conj(H) * Y + mu * fft2(z)) / (np.abs(H) ** 2 + mu)))

def pnp_hqs(denoise, lam=1e-5, sigmas=np.geomspace(0.05, 0.02, 10)):
    z = y.copy()
    for s in sigmas:                       # denoiser noise level s = sqrt(lam / mu),
     ↪  decreasing
        mu = lam / s ** 2                  # so mu grows from iteration to iteration
        z = denoise(data_step(z, mu), s)   # prior step: any Gaussian denoiser
    return z

tv  = lambda a, s: denoise_tv_chambolle(a, weight=s)
nlm = lambda a, s: denoise_nl_means(a, patch_size=5, patch_distance=6, h=0.8 * s, sigma=s,
 ↪  fast_mode=True)

best = max((psnr(data_step(np.zeros_like(y), mu)), mu) for mu in (1e-3, 3e-3, 1e-2, 3e-2))
 ↪  # Tikhonov: z = 0
print(f"blurred+noisy {psnr(y):.2f} dB | Tikhonov (best of 4 mu) {best[0]:.2f} dB")
print(f"plug-and-play HQS with TV denoiser  {psnr(pnp_hqs(tv)):.2f} dB")
print(f"plug-and-play HQS with NLM denoiser {psnr(pnp_hqs(nlm)):.2f} dB")
# blurred+noisy 25.79 dB | Tikhonov (best of 4 mu) 28.34 dB
# plug-and-play HQS with TV denoiser  30.17 dB
# plug-and-play HQS with NLM denoiser 30.45 dB
```

Both plugged-in denoisers beat the one-step filter by more than 1.8 dB, with the data step unchanged. Read this as a demonstration, not a benchmark: it is one image, the kernel is exactly known, and `lam` and the schedule were chosen by trying a few values on this image (the same values are used for both denoisers; the 0.28 dB gap between them is small next to the effect of changing `lam`, below). The method is sensitive to `lam`: with the TV denoiser, `lam=1e-4` gives 28.55 dB and `lam=1e-6` gives 25.17 dB.

When it fails. The degradation H must be known (a wrong kernel is propagated by the data step); the denoiser is only asked to work at the noise levels σ_t of the schedule, so a learned denoiser needs to cover them; and for a denoiser that does not correspond to an explicit Φ (such as non-local means) convergence of the loop is not guaranteed in general, so the iteration count and schedule are simply fixed. In [115], for example, the denoiser level decays exponentially from 49 to a value between 1 and 15 (on a 0–255 scale) over 30 iterations.

Super-resolution milestones, families and benchmarks

Plain version. The main text summarizes how super-resolution evolved in one paragraph. This note gives what each milestone changed, how the current families work, and the benchmark conventions.

Milestones.

- **SRCNN** (2014/2016) [6]: the first end-to-end CNN for SR, three layers on a bicubic-upsampled input.
- **FSRCNN** (2016) [7] and **ESPCN** (2016) [8]: post-upsampling for real-time speed.
- **EDSR** (2017) [10]: a deep residual network with batch normalization removed (BN hurt the range flexibility needed for restoration), residual scaling for stable training, and a multi-scale variant. It won the NTIRE 2017 SR challenge.
- **RCAN** (2018) [11]: very deep "residual-in-residual" structure with **channel attention**, which re-weights feature channels using global statistics, so the network can focus on high-frequency information.
- **SRGAN** (2017) [12] and **ESRGAN** (2018) [13]: adversarial and perceptual (VGG feature) losses produce sharp, realistic textures instead of the blurry MSE average. ESRGAN improved the generator (Residual-in-Residual Dense Blocks without BN), used a relativistic discriminator and computed the perceptual loss on features before activation.

By around 2018 these models already produced results that looked much better than interpolation to human observers, and the field started to question how to *judge* SR at all (see the perception–distortion section).

Current families.

- **Transformers.** SwinIR [14] (2021) uses residual Swin Transformer blocks (self-attention inside shifted local windows) for SR, denoising and JPEG artifact removal. HAT [15] (CVPR 2023) combines channel attention with window self-attention and an overlapping cross-attention module to "activate more pixels" for each output. Both are standard baselines.
- **State-space models.** MambaIR [49] and MambaIRv2 [50] replace attention with selective state-space (Mamba) scans that give a global receptive field at linear cost; see the backbones section.
- **Real-world (blind) SR.** Real-ESRGAN [4] and its high-order degradation pipeline, described in the main text.
- **Diffusion priors.** DiffBIR [16] removes degradations first and then regenerates detail with a pretrained latent diffusion model; SUPIR [17] scales this up with a large diffusion backbone, a 20-million-image training set with text descriptions and negative-quality prompts; OSEDiff [18] distills the process into a **single** diffusion step that starts from the low-quality image itself, using variational score distillation to keep outputs realistic. These methods produce far more convincing textures on real photos, but they can also invent detail that was not in the scene.

Datasets and metrics. The standard training set is **DIV2K** [19] (2K-resolution images introduced for NTIRE 2017). Evaluation uses small classic test sets of natural, urban and comic images, reporting PSNR and SSIM on the luminance (Y) channel, usually after cropping s border pixels. Real-world SR is evaluated with perceptual and no-reference metrics (LPIPS, NIQE and others;

see Evaluation).

Classical and learned blind-deblurring methods

Plain version. The main text summarizes how blind deblurring escapes the "no-blur" trap. This note gives the formula behind estimating the kernel alone and a short description of each method the main text names.

The joint problem in detail. In the blind-deconvolution objective of the main text, $(\hat{f}, \hat{h}) = \arg\min_{f,h} \|h * f - g\|_2^2 + \lambda_f \phi(f) + \lambda_h \psi(h)$ subject to $h \geq 0$ and $\sum_{u,v} h(u,v) = 1$, where g is the blurred image, f the unknown sharp image, h the unknown kernel (supported on a small window, e.g. 15×15 to 75×75 pixels for camera shake), ϕ an image prior (typically a sparse gradient penalty $\sum_p |\nabla f(p)|^\alpha$ with $0 < \alpha \leq 1$, because natural images have mostly flat regions and a few strong edges), ψ a kernel prior (e.g. $\|h\|_2^2$ or $\|h\|_1$, favoring compact, sparse motion paths), and λ_f, $\lambda_h > 0$ their weights. The constraints say a blur kernel only redistributes light: it is non-negative and preserves total brightness. They also remove one ambiguity, because $(c\,h, f/c)$ explains g as well as (h, f) for any $c > 0$. A second ambiguity cannot be removed: shifting f one way and h the other leaves $h * f$ unchanged, so the kernel is only recovered up to a translation.

Estimating the kernel alone. Levin et al. [74] showed that naive joint MAP over the image and the kernel tends to return the no-blur solution. Their analysis points to a better-posed target: estimate only the kernel, integrating out the image,

$$\hat{h} = \arg\max_h \; p(h \mid g) = \arg\max_h \; p(h) \int p(g \mid f, h)\, p(f)\, df,$$

where $p(g \mid f, h)$ is the Gaussian likelihood given by the data term, $p(f)$ and $p(h)$ the priors behind ϕ and ψ, and the integral averages over all sharp images. A kernel has few unknowns compared with the many pixel observations, so its posterior is much better determined than that of the full image. Once $\hat{h}$ is known, a non-blind deconvolution recovers f.

Classical remedies. The methods that worked in practice either approximate that marginalization or change the prior so the sharp solution wins:

- **Variational Bayes.** Fergus et al. [75] fit a heavy-tailed mixture-of-Gaussians model of image gradients and approximate the kernel posterior with variational Bayes, estimating the kernel coarse-to-fine; the image is then recovered by a standard non-blind method.
- **Edge prediction and selection.** Shan et al. [76] alternate kernel and image estimation with a natural-image gradient prior and a careful noise model to suppress ringing. Cho and Lee [77] made the alternation fast: they *predict* sharp edges from the current latent image with bilateral and shock filtering, and estimate the kernel in the gradient domain with FFTs. Xu and Jia [78] observed that thin edges narrower than the kernel can mislead estimation, so they select only informative edges in a first phase and refine the kernel's support in a second phase.

- **Priors that prefer sharp images.** Krishnan et al. [79] replace $\|\nabla f\|_1$ by the normalized sparsity $\|\nabla f\|_1 / \|\nabla f\|_2$, which is scale-invariant and *increases* with blur, so it no longer favors the blurry solution. Pan et al. [80] use the dark channel (the same quantity as in He et al.'s dehazing prior [25]): blur averages dark pixels with bright neighbors, so the dark channel becomes less sparse, and an L_0 penalty on it favors sharp images.

Deep learning and the kernel. Networks entered blind deblurring in three ways.

- **Predict the kernel (or a filter), then deconvolve.** Sun et al. [81] trained a CNN to classify the local motion kernel of each patch among a set of linear motions, smoothed the resulting non-uniform motion field with a Markov random field, then deconvolved. Schuler et al. [82] built a deep network with stages that mimic the classical loop (feature extraction, kernel estimation, image estimation) and trained it end-to-end on synthetic blur. Chakrabarti [83] trained a network to predict, for each patch, the Fourier coefficients of a deconvolution filter, used the result to estimate one global kernel, and finished with standard non-blind deconvolution.
- **Self-supervised, per-image estimation.** Deep Image Prior [84] showed that the structure of an untrained generator network is itself a useful image prior. SelfDeblur [85] uses two generators, an encoder–decoder for the image and a fully connected network with a softmax output for the kernel (the softmax enforces $h \geq 0$ and $\sum h = 1$), and fits both to the single blurred image. No training data is needed, but each image takes an optimization run.
- **Kernel-free, end-to-end.** The multi-scale network of Nah et al. [44], DeblurGAN [86] (a conditional GAN with a content loss) and the general backbones Restormer and NAFNet map blurred to sharp images directly, never forming h.

Kernel-based or kernel-free? Kernel-based methods fit **spatially uniform** blur (camera shake of a distant, static scene), give an **interpretable** intermediate result that can be checked and reused (for example to deblur other frames from the same exposure), and generalize to new kernels as long as the physics holds; they are slower and break when the blur varies across the image or the image has few edges. Kernel-free networks handle **non-uniform** blur from object motion and depth, are fast at test time, and are the current state of the art on dynamic-scene benchmarks such as GoPro, but they generalize only as far as their training blur distribution and give no explicit kernel. The survey of Zhang et al. [69] summarizes this shift.

All-in-one methods in more detail

Plain version. All six models of the main text condition one restoration network on a description of the degradation. They differ in where that description comes from and how it enters the network.

- **AirNet** [51] (CVPR 2022), the "airnet" of the original post: a contrastive-learning degradation encoder extracts a representation of *how* the image is degraded, without labels, and that representation guides the restoration network. It handles unknown corruption types with one model; its noise + rain + haze setting became the standard "three-task" benchmark.
- **TransWeather** [52] (CVPR 2022): a single transformer encoder–decoder for rain, fog and snow, with learnable **weather-type queries** in the decoder.
- **PromptIR** [53] (NeurIPS 2023): a Restormer-style network with lightweight **prompt blocks**. Learnable prompt components are combined with weights predicted from the input features and injected into the decoder, encoding degradation-specific information. Tested on denoising, deraining and dehazing.
- **AdaIR** [54] (ICLR 2025): observes that different degradations affect **different frequency subbands**, then mines low- and high-frequency features from the input and modulates them adaptively. Covers five degradations: noise, haze, rain, motion blur and low light.
- **DA-CLIP** [55] (ICLR 2024): adapts the vision–language model CLIP with a trained controller so that its image encoder yields both a clean-content embedding and a degradation embedding from a corrupted image; these guide a restoration network through cross-attention.
- **InstructIR** [56] (ECCV 2024): the user writes a natural-language instruction ("remove the rain", "make it brighter"), a text encoder embeds it, and the embedding modulates the restoration network.

Human-rated IQA datasets

Plain version. No-reference metrics are trained and validated on image-quality datasets in which people have rated many images. The datasets differ mainly in where the distortions come from, and that decides whether a metric trained on them can judge restoration outputs.

Dataset	Distortions	What it contains
LIVE (Release 2) [104]	Synthetic	Images with five distortion types (JPEG, JPEG2000, Gaussian blur, white noise, JPEG2000 bit errors), with DMOS
TID2013 [105]	Synthetic	25 reference images × 24 distortion types × 5 levels = 3,000 distorted images; scores from 971 experiments with observers in five countries
KADID-10k [106]	Synthetic	81 pristine images × 25 distortion types × 5 levels = 10,125 images, 30 crowdsourced ratings each
KonIQ-10k [107]	Authentic (in the wild)	10,073 photos with about 1.2 million ratings from 1,459 crowd workers

Dataset	Distortions	What it contains
SPAQ [108]	Authentic (smartphones)	11,125 photos from 66 smartphones, with quality, attribute (brightness, colorfulness, contrast, noisiness, sharpness) and scene labels plus EXIF data
PIPAL [109]	Restoration outputs	Outputs of image restoration algorithms, including GAN-based ones, with over 1.13 million human judgments converted to scores by an Elo system

Synthetic datasets control the distortion exactly but contain only "textbook" damage. Authentic datasets contain real camera problems but no restoration outputs. Only datasets such as PIPAL contain the kind of images a restoration paper actually needs to score.

PIQE and the Perceptual Index

Plain version. Both are no-reference metrics built on the natural-scene statistics of the main text: PIQE looks for distorted blocks, and PI adds a score learned from human ratings of super-resolved images.

PIQE [96] (Perception-based Image Quality Evaluator) also uses MSCN coefficients but works **block by block**: it divides the image into 16×16 blocks, flags blocks that show blocking artifacts or noise, and pools the distortion of the flagged blocks into one score. It needs no training on opinion scores, and it gives a local map of where the image looks distorted. In the common MATLAB implementation the score lies between 0 and 100, lower is better.

Ma et al.'s score and the Perceptual Index. Ma et al. [95] ran human studies on super-resolved images and learned a no-reference metric for SR from those scores, using statistical features in spatial and frequency domains and a two-stage regression; their score ("Ma", higher is better) lies roughly between 0 and 10. The 2018 PIRM challenge on perceptual SR [63] combined it with NIQE into the **Perceptual Index**,

$$\mathrm{PI} = \tfrac{1}{2}((10 - \mathrm{Ma}) + \mathrm{NIQE}),$$

so that lower PI means better perceptual quality. PIRM ranked methods by PI within three regions of the perception–distortion plane, defined by RMSE at most 11.5, 12.5 and 16, so that realism was compared at a fixed distortion budget. PI is still reported in many perceptual SR papers.

How the learned NR assessors work

Plain version. Each of these models is a neural network trained to predict human quality ratings. They differ in what they predict, how they look at the image and which data they learned from.

- **NIMA** [97] predicts the whole *distribution* of human ratings with a CNN, not a single number. With p_k the predicted probability of score k (on a 1–10 scale), the reported score is the mean $\hat{s} = \sum_k k\, p_k$, and the spread of the distribution says how much people would disagree. Checkpoints trained on aesthetic ratings judge "is this a nice photo", not "is this a faithful restoration".
- **HyperIQA** [98] targets authentic, in-the-wild distortions. It splits assessment into three stages:

  a backbone first *understands the content*, a **hyper network** then generates the weights of the quality-prediction network from that content (the "perception rule" adapts to what is in the picture, since the same blur is judged differently on a sky and on a face), and the generated network predicts the score.

- **MUSIQ** [99] is a multi-scale transformer that takes the image at its **native resolution and aspect**

  ratio plus downscaled versions, using hash-based 2-D spatial and scale embeddings, so no resizing or cropping changes the very artifacts being judged.

- **MANIQA** [100] combines vision-transformer features with transposed (channel) attention blocks and scale Swin-transformer blocks, and predicts the score with a dual-branch, patch-weighted head. It won the no-reference track of the NTIRE 2022 perceptual IQA challenge, whose data include outputs of restoration and GAN-based algorithms, so it is closer to the restoration setting than metrics trained on camera distortions.
- **TOPIQ** [101] follows a **top-down** idea: high-level semantic features decide where to look, and a cross-scale attention mechanism uses them to weight lower-level, distortion-sensitive features (its network is called CFANet, with a ResNet-50 backbone). The same design works for FR and NR assessment and is light compared with transformer-based IQA models.

CLIP-IQA and Q-Align scores

Plain version. Both metrics ask a model trained on images and text how good a photo looks. The formulas below show how its answer becomes a number.

CLIP-IQA [102] compares the CLIP embedding of the image with the embeddings of an antonym prompt pair such as "Good photo." and "Bad photo.". With $\mathbf{x}$ the image embedding, $\mathbf{t}_1, \mathbf{t}_2$ the two text embeddings and $s_i = \mathbf{x}^\top \mathbf{t}_i / (\|\mathbf{x}\|\, \|\mathbf{t}_i\|)$ their cosine similarities, the score is

$$\text{CLIP-IQA} = \frac{e^{s_1}}{e^{s_1} + e^{s_2}} \in [0, 1],$$

higher is better. The basic version is **zero-shot** (no opinion scores at all), and other prompt pairs ("Noisy photo." / "Clean photo.", "Sharp photo." / "Blurry photo.") probe single attributes. The wording of the prompts changes the scores.

Q-Align [103] fine-tunes a large multimodal model to answer the way human raters do, with a **text rating level** (bad, poor, fair, good, excellent) instead of a number. At test time, with X_{l_i} the model's logit for level l_i, the score is the expected level under a softmax restricted to the five words,

$$S = \sum_{i=1}^{5} i \cdot \frac{e^{X_{l_i}}}{\sum_{j=1}^{5} e^{X_{l_j}}},$$

where $i = 1, \dots, 5$ indexes the levels from bad to excellent. Such models are strong across datasets but heavy, and they inherit the biases of their base model.

Fitting the Bradley–Terry model

Plain version. The Bradley–Terry model of the main text gives each method one strength. This note shows how the strengths are fitted from a table of pairwise wins.

Each method i has a strength $\pi_i > 0$, and

$$P(i \text{ preferred over } j) = \frac{\pi_i}{\pi_i + \pi_j},$$

where i and j index two methods. The strengths are fitted by maximum likelihood to the observed wins and are defined only up to a common scale. With W_i the total number of wins of method i and n_{ij} the number of comparisons between i and j, a simple iteration $\pi_i \leftarrow W_i / \sum_{j \neq i} n_{ij}/(\pi_i + \pi_j)$ converges to the maximum-likelihood solution:

```
import numpy as np

# wins[i, j] = number of raters who preferred method i over method j (20 raters per pair)
wins = np.array([[0, 14, 17],
                 [6,  0, 12],
                 [3,  8,  0]], dtype=float)

def bradley_terry(wins, iters=200):
    n = wins.shape[0]
    games = wins + wins.T                                  # comparisons made for each pair
    pi = np.ones(n)
    for _ in range(iters):
        denom = (games / (pi[:, None] + pi[None, :])).sum(axis=1)
        pi = wins.sum(axis=1) / denom                      # the update above
        pi /= pi.sum()                                     # fix the arbitrary scale
    return pi
```

```
pi = bradley_terry(wins)
print("strengths:", pi.round(3))
print("P(method 0 preferred over method 1) =", round(pi[0] / (pi[0] + pi[1]), 3), "|
 ↪  observed", 14 / 20)
# strengths: [0.629 0.233 0.137]
# P(method 0 preferred over method 1) = 0.729 | observed 0.7
```

The fitted probability (0.729) differs slightly from the observed share (0.7) because one strength per method has to explain all three pairs at once.

Metric-by-metric summary table

Plain version. This table lists every no-reference metric of the main text, plus FID, with its family, what it needs for training, which direction is better and what it mainly reacts to.

Metric	Family	Needs training?	Better	Mainly sensitive to
NIQE [62]	NSS	Pristine images only (opinion-unaware)	Lower	Deviations of local statistics: blur, noise, compression
BRISQUE [94]	NSS + regression	Yes, human scores on synthetic distortions	Lower	The same statistics, mapped to human scores
PIQE [96]	NSS, block-wise	No opinion scores	Lower	Blocking and noise in local blocks; gives a distortion map
Ma [95]	Handcrafted features + regression	Yes, human scores on SR outputs	Higher	SR artifacts, texture and sharpness
PI [63]	NSS + learned	Through its Ma part	Lower	Perceptual quality of SR; rewards texture, including amplified noise
NIMA [97]	Learned CNN	Yes, rating distributions	Higher	Technical or aesthetic quality, depending on checkpoint
HyperIQA [98]	Learned CNN + hyper network	Yes	Higher	Authentic distortions, judged in the context of content
MUSIQ [99]	Learned transformer	Yes	Higher	Multi-scale quality at native resolution
MANIQA [100]	Learned transformer	Yes	Higher	Restoration and GAN artifacts (trained on such data)
TOPIQ [101]	Learned CNN, top-down attention	Yes	Higher	Distortions in semantically important regions
CLIP-IQA [102]	Vision-language	No (zero-shot version)	Higher	Overall "good photo" impression; depends on prompts

Metric	Family	Needs training?	Better	Mainly sensitive to
Q-Align [103]	Large multimodal model	Yes, ratings as text levels	Higher	Human-like overall quality judgments
FID [64]	Distribution-level	Pretrained Inception features	Lower	Distribution gap between a set of outputs and real images

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Deep Image Prior: the network as the prior

Plain version. Restoration networks are usually credited to what they learned from millions of images. Deep Image Prior (DIP) [84], [118] asks how much comes from the *shape* of a convolutional network alone. Its recipe uses one damaged photo and a network with random weights: fit the network to the photo, and stop before it has copied everything. Smooth and repeated structure is fitted first and noise last, so the early result is a cleaned picture, like a student who copies the main shapes of a drawing before the speckles.

The problem. Inverse problems need a prior, and the options in this tutorial are a hand-written Φ (total variation, sparse gradients) or a prior learned from a dataset. The paper asks whether a prior can come from the architecture of an untrained generator, with no data at all.

The idea. Write the image as the output of a network, $x = f_\theta(z)$, where z is a fixed random input tensor (of the image's spatial size) and θ the network weights. For a data term $E(x; y)$ that compares x with the observation y, such as $\|x - y\|^2$ for denoising, $\|d(x) - y\|^2$ for super-resolution (with d a downsampling operator) or $\|(x - y) \odot m\|^2$ for inpainting (with m a binary mask of the known pixels), solve

$$\theta^\star = \arg\min_\theta \; E(f_\theta(z); y), \qquad x^\star = f_{\theta^\star}(z),$$

with gradient descent from a random θ_0, and stop after a task-dependent number of steps. There is no explicit regularizer: the prior is the set of images the network reaches easily from its random start, plus the stopping time. The paper's learning curves (its Fig. 4) show the mechanism: a natural image is fitted far faster than the same image after its pixels are scrambled, and faster than white noise, so the parametrization has "high impedance to noise and low impedance to signal".

What it showed. All numbers below are from the journal version [118]. For denoising nine colour images at $\sigma = 25$, the paper reports 29.22 dB after 1800 steps, 30.43 dB when the restored images

from the last iterations are averaged (exponential sliding window), and 31.00 dB when two runs are also averaged; two non-learned reference denoisers get 31.42 dB and 30.26 dB. For ×4 super-resolution on the five images of its Table 2 the average PSNR is 29.89 dB, against 28.44 dB for bicubic interpolation and 31.58 to 32.10 dB for two trained networks. Its Fig. 6 shows a JPEG-compressed image restored after 100, 600, 2400 and 50,000 steps: artifacts disappear around 2400 steps and the network overfits the input by 50,000.

How to read it. Start with Fig. 6 (the effect of stopping time) and Fig. 3 (why early stopping helps), then Tables 1 and 2. Check: (a) the stopping time is a hyperparameter set per task (1800 steps for denoising, 2000 for ×4 super-resolution, 4000 for ×8, 6000 for the text-removal inpainting, 5000 for large holes, learning rates from 0.1 to 0.001 across experiments), so ask how it was chosen when no ground truth exists; (b) which numbers use the averaging trick, which costs extra runs; (c) that the comparison for super-resolution is against *non-trained* methods and is described by the authors as approaching, not matching, trained networks; (d) the PSNR convention (the paper states center crops; check channel and border handling before comparing to other tables); (e) the headline is one optimization run per image, and the gain from averaging two runs (30.43 to 31.00 dB) shows that run-to-run variation is not negligible.

Limits. The authors state that the method is probably too slow for most practical uses (several minutes per image) and that a feed-forward network trained for one task would do better and faster. It must be stopped at the right moment; run long enough it fits the noise. The architecture matters a great deal: in the paper's inpainting comparison, adding U-Net-style skip connections hurts, and for the denoising benchmark in the paper different architectures give very different scores. It cannot supply semantic content: the authors do not expect it to work for large holes that need object knowledge such as faces.

Why it matters here. It supports two points of this tutorial: the structure of a convolutional network is itself an image prior, and per-image optimization can replace training data, as in SelfDeblur [85]. It also marks the boundary with the diffusion priors of the main text [16]–[18], which bring a learned notion of what objects look like that an untrained network cannot.

Survey papers in detail

Plain version. The Modern view summarizes these surveys in one line each. Here are their full titles, venues, taxonomies and takeaways.

- **Wang, Chen and Hoi, "Deep Learning for Image Super-resolution: A Survey" (TPAMI 2021)** [66]. Organizes SR into supervised, unsupervised and domain-specific SR; within supervised SR it dissects models by upsampling position (pre-, post-, progressive, iterative up-and-down), upsampling operator (interpolation, transposed convolution, sub-pixel), network design (residual, recursive, dense, attention, multi-path), losses and training strategies. Takeaway: most SR progress can be described as choices along these axes, and unsupervised/real-world SR was flagged as the main open direction, which the field then pursued.

- **Tian et al., "Deep Learning on Image Denoising: An Overview" (Neural Networks 2020)** [67]. Groups CNN denoisers into additive white noise, real noise, blind denoising, and hybrid noise with blur or low resolution, and compares them on public benchmarks. Takeaway: the realism of the noise model, not the network, is often the bottleneck.
- **Elad, Kawar and Vaksman, "Image Denoising: The Deep Learning Revolution and Beyond" (SIAM J. Imaging Sciences 2023)** [68]. Traces denoising from classical priors to deep networks, then argues that denoisers are now building blocks for other inverse problems (plug-and-play, regularization by denoising) and for diffusion-based image generation, and that restoration should embrace multiple valid solutions (posterior sampling) instead of one answer. Takeaway: the field's frontier moved from "better MMSE denoisers" to "sampling from the posterior".
- **Zhang et al., "Deep Image Deblurring: A Survey" (IJCV 2022)** [69]. Covers causes of blur, datasets and metrics, and a taxonomy of CNN deblurring by architecture, loss and application (faces, text, stereo), for both blind and non-blind settings. Takeaway: dynamic-scene deblurring moved from kernel estimation to end-to-end, multi-scale networks trained on captured pairs such as GoPro.
- **Gui et al., "A Comprehensive Survey and Taxonomy on Single Image Dehazing Based on Deep Learning" (ACM Computing Surveys 2023)** [70]. Splits methods into supervised, semi-supervised and unsupervised, and within each by whether and how they use the scattering model; also reviews datasets, network modules, losses and metrics, with experiments on baselines. Takeaway: physics-guided and physics-free designs coexist, and the synthetic-to-real gap is the main open problem.
- **Yang et al., "Single Image Deraining: From Model-Based to Data-Driven and Beyond" (TPAMI 2021)** [71]. Reviews rain appearance models, model-based methods (priors and optimization) and data-driven methods (architectures, constraints, losses, datasets). Takeaway: realistic rain models and real-world evaluation matter more than further architecture tweaks on synthetic data.
- **Jiang et al., "A Survey on All-in-One Image Restoration: Taxonomy, Evaluation and Future Trends" (TPAMI, 2025)** [72]. Organizes all-in-one methods by architectural design, learning paradigm and core innovation (degradation representations, prompts, mixture-of-experts, language guidance), and consolidates datasets and evaluation protocols. Takeaway: comparisons are complicated by inconsistent task sets, and generalization to unseen and mixed degradations remains open.
- **Li et al., "Diffusion Models for Image Restoration and Enhancement: A Comprehensive Survey" (IJCV 2025)** [73]. Categorizes diffusion-based restoration by learning paradigm (su-

pervised versus zero-shot use of a pretrained prior), conditioning strategy, framework design and modeling strategy, with focus on SR, deblurring, inpainting and blind/real-world restoration. Takeaway: diffusion gives the best perceptual quality but at high compute cost and with fidelity/hallucination concerns; distillation to few steps and better evaluation are key directions.

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DLCV 04A • Image Super-Resolution

Why super-resolution is ill-posed, how degradation models, sub-pixel convolution, residual, GAN, transformer and diffusion networks attack it, and how to evaluate the results honestly.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-04a-super-resolution/

What you'll learn

- Why single-image super-resolution (SR) is **ill-posed**: one low-resolution image is consistent with many high-resolution images, and a network trained with a pixel loss returns their blurry average.
- How the **degradation model** (bicubic only, or blur, downsampling, noise and JPEG) decides what a model can do, why bicubic-trained models fail on real photos, and how aliasing looks in the frequency domain.
- The classical toolbox (interpolation kernels, example-based and self-similarity methods, sparse coding, learned linear regressors), and a linear regressor you can train on a CPU in about a second.
- How CNN, GAN, transformer and diffusion SR networks are built: pre- and post-upsampling, the exact index rule of **sub-pixel convolution** (**pixel shuffle**), residual and attention networks, perceptual losses, and the **perception–distortion trade-off**.
- Blind and real-world SR, arbitrary-scale, video and reference-based SR, and efficient SR, in brief.
- How SR is evaluated in practice (Y-channel PSNR and SSIM with border shaving, the standard test sets, NTIRE challenges), and the typical failure modes: hallucinated text and faces, texture artifacts, color shifts and over-smoothing.

The big picture

Super-resolution makes a small image bigger *and* sharper. Stretching the pixels is easy; the hard part is filling in the detail that the small image never recorded. A modern SR model fills it in from what it learned about natural images, and that is both its strength and its risk.

> **Explain like I'm five**
>
> Imagine a friend describes a picture to you over the phone: "a woman with short brown hair, smiling, in front of a light wall." You can draw a good picture from that, but you cannot draw *her* exact freckles, because the description did not include them. Many different drawings fit the same description. A careful artist draws soft, safe shapes where they are unsure (the result looks blurry). A bold artist

draws crisp freckles anyway (the result looks real, but the freckles may be in the wrong place). Super-resolution networks are artists of one kind or the other, and this chapter is about how they are trained, how to tell them apart, and how to check their work.

Why it matters: phones zoom digitally, video and games are often rendered small and upscaled, satellites and microscopes are limited by their optics and sensors, and old photos are restored for archives. SR is also where many low-level vision ideas were first tested: residual learning, channel attention, perceptual losses, window attention and diffusion priors.

This chapter extends the short SR section of DLCV 04, which places SR among the other restoration tasks and defines the generic losses and metrics. We do not repeat those definitions; we link to them and add what is specific to SR. You should know basic sampling and interpolation (IP 02) and the Fourier view of aliasing (IP 04). All code below uses NumPy, SciPy, scikit-image, scikit-learn and OpenCV only, runs on a CPU in seconds, and uses images from `skimage.data`.

The problem: one input, many answers

Plain version. A low-resolution (LR) image has fewer numbers than the high-resolution (HR) image we want. Many HR images shrink to the same LR image. A model has to choose one of them, and the choice is a matter of what it believes natural images look like.

Setup and notation

Single-image super-resolution (SISR) estimates an HR image $x \in \mathbb{R}^{sH \times sW \times C}$ from one LR image $y \in \mathbb{R}^{H \times W \times C}$, where s is the **scale factor** (2, 3, 4, ...), $H \times W$ the LR size and C the number of color channels. The standard forward model, already introduced in DLCV 04, is

$$y = (x * k) \downarrow_s + n,$$

where k is a blur kernel (the camera's optics and pixel integration, or a deliberate anti-aliasing filter), $*$ is 2-D convolution applied to each channel, $\downarrow_s$ keeps one sample out of every $s \times s$ block, and n is noise. Multi-frame SR, the older form of the problem, combines several shifted LR images of the same scene and is reviewed by Park et al. [1]; this chapter is about the single-image case, with a short section on video.

Why the problem is ill-posed

Count the numbers. For a scale factor s, each LR pixel summarizes s^2 HR pixels, so the HR image has s^2 times more unknowns than there are measurements. At $s = 4$, a perfect, noise-free LR image fixes only one linear combination out of every 16 HR values; the other 15 directions are invisible. Mathematically, the degradation operator $\mathcal{D}(x) = (x * k) \downarrow_s$ has a large **null space**: any HR pattern z with $\mathcal{D}(z) = 0$ can be added to a solution without changing y. Fine textures, sharp

edge positions and small text live largely in that null space. Noise makes it worse, because it also corrupts the information we *do* measure.

So SR needs a **prior** $p(x)$, a model of which HR images are plausible. With Bayes' rule, every HR candidate gets a posterior probability

$$p(x \mid y) \propto p(y \mid x)\, p(x),$$

where $p(y \mid x)$ is the likelihood (how well $\mathcal{D}(x)$ explains y given the noise) and $p(x)$ the prior. Classical methods write the prior by hand; deep SR learns it from (LR, HR) pairs.

The posterior-mean blur

Most SR networks are trained by minimizing a pixel loss over pairs. With the squared error, the best possible network outputs the **posterior mean**

$$\hat{x}_{\mathrm{MMSE}}(y) = \mathbb{E}[x \mid y] = \int x\, p(x \mid y)\, dx,$$

the minimum mean-squared-error (MMSE) estimate. (With the L1 loss the optimum is the per-pixel posterior median, which behaves similarly.) When the posterior contains many sharp candidates that differ in the exact position of an edge or the phase of a texture, their average is a smooth compromise that is *not itself* a plausible image. This is the blur you see in MSE-trained SR, and no amount of training data removes it: it is the correct answer to the question the loss asks.

A one-dimensional toy makes this concrete. An HR signal of 64 samples is a single step edge at an unknown position. We average blocks of 4 samples and add noise with standard deviation 0.2, then compute the exact posterior over the 32 possible edge positions, its mean, and one random sample from it.

```
import numpy as np

rng = np.random.default_rng(0)
n, s, sigma = 64, 4, 0.2                # HR length, scale factor, LR noise level
cand = np.arange(16, 48)                # possible edge positions t (the prior: uniform)
X = (np.arange(n)[None, :] >= cand[:, None]).astype(float)    # every HR step edge x_t
DX = X.reshape(len(cand), n // s, s).mean(axis=2)              # their LR versions D(x_t)

err_mmse, err_samp, jump_mmse, jump_samp = [], [], [], []
for _ in range(2000):
    t = rng.integers(len(cand))
    y = DX[t] + sigma * rng.standard_normal(n // s)               # observed LR signal
    logp = -((y - DX) ** 2).sum(axis=1) / (2 * sigma**2)          # log posterior over t
    post = np.exp(logp - logp.max()); post /= post.sum()
    x_mmse = post @ X                                             # posterior mean E[x | y]
    x_samp = X[rng.choice(len(cand), p=post)]                     # one posterior sample
    err_mmse.append(np.mean((x_mmse - X[t]) ** 2)); err_samp.append(np.mean((x_samp - X[t])
    ↪  ** 2))
    jump_mmse.append(np.diff(x_mmse).max()); jump_samp.append(np.diff(x_samp).max())
```

```
print(f"MSE   posterior mean {np.mean(err_mmse):.4f} | posterior sample
 ↪  {np.mean(err_samp):.4f} "
      f"| ratio {np.mean(err_samp) / np.mean(err_mmse):.2f}")
print(f"largest one-pixel jump (a real edge has 1.00): mean {np.mean(jump_mmse):.2f} |
 ↪  sample {np.mean(jump_samp):.2f}")
```

```
MSE   posterior mean 0.0072 | posterior sample 0.0143 | ratio 1.97
largest one-pixel jump (a real edge has 1.00): mean 0.48 | sample 1.00
```

The posterior mean has half the error of a posterior sample, but its edges are ramps: on average its largest one-pixel jump is 0.48, while every real edge (and every sample) jumps by 1.00. The sample looks exactly like a real edge, only sometimes in the wrong place. The ratio of about 2 is not a coincidence. Blau and Michaeli proved that an estimator whose outputs have exactly the distribution of real images (for example, a posterior sampler) never needs more than twice the MSE of the MMSE estimator, which is 3 dB of PSNR [2]. Figure 1 shows one example and the whole range between the two estimators.

Optional: Why a posterior sample costs twice the MMSE (in the Supplementary notes). It derives the factor of 2 in a few lines.

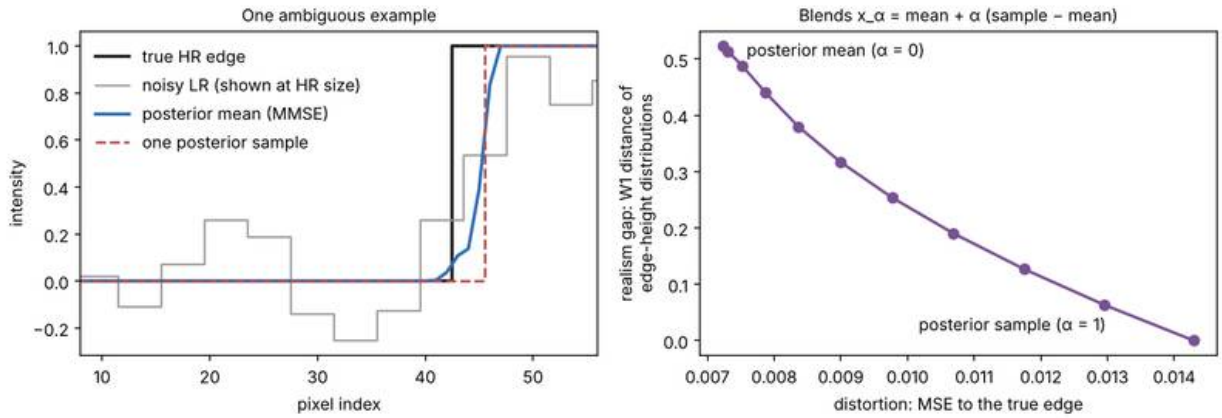


Figure 1 — The posterior-mean blur on a 1-D step-edge toy (×4, noise $\sigma = 0.2$). Left: one ambiguous example. Right: blending the posterior mean with a posterior sample trades distortion (MSE) for realism (distance between the distribution of edge heights in the outputs and in real edges). No blend is best on both axes.

This toy contains the whole story of the chapter. MSE-trained CNNs approximate the posterior mean. GANs and diffusion models try to output samples instead. Evaluation has to decide which of the two you wanted.

Degradation models

Plain version. To train an SR network you need pairs, and you make them by shrinking HR images. *How* you shrink them is a modeling choice, and the network learns to undo exactly that shrinking and nothing else.

The bicubic benchmark setting

Most SR benchmarks create LR images with MATLAB's `imresize` at its default settings, which use bicubic interpolation and, when shrinking, an anti-aliasing filter that is on by default [3], [4]. There is no added noise, no camera blur and no compression. This "bicubic" setting is clean and reproducible, which made years of results comparable, but real cameras do not shrink images this way.

Blur, downsample, noise, compression

A more realistic model follows the physics of a camera and the path of a photo through the internet:

$$y = C_q\big(Q(A_s(x * k) + n)\big), \qquad A_s(z)[u,v] = \frac{1}{s^2}\sum_{a=0}^{s-1}\sum_{b=0}^{s-1} z[su+a,\ sv+b],$$

where k is the optical blur (often modeled as a Gaussian), A_s averages each $s \times s$ block (the light collected by one larger sensor pixel) and keeps one value per block, $[u, v]$ indexes LR pixels, n is sensor noise, Q rounds to 8 bits and C_q is JPEG compression at quality q. The order matters: noise added after downsampling stays fine-grained in the LR image, while noise added before the blur would be smoothed out.

The snippet implements the first three steps plus quantization (JPEG is left out to keep it short), then upsamples with four standard interpolators and reports PSNR in the way SR papers do: on the luminance (Y) channel, with a border of s pixels removed. The details of that protocol are explained in the evaluation section.

```
import numpy as np, cv2
from scipy import ndimage as ndi
from skimage import data, img_as_float64

def degrade(hr, s=4, blur=1.2, noise=0.01, rng=None):
    """hr: (H, W, 3) in [0, 1], H and W divisible by s. Returns the LR image y."""
    rng = rng or np.random.default_rng(0)
    x = ndi.gaussian_filter(hr, sigma=(blur, blur, 0))          # 1) lens blur
    H, W, C = x.shape
    x = x.reshape(H // s, s, W // s, s, C).mean(axis=(1, 3))      # 2) s x s pixel
     ↪  integration + decimation
    x = x + noise * rng.standard_normal(x.shape)                  # 3) sensor noise
```

```
    return np.round(np.clip(x, 0, 1) * 255) / 255                # 4) 8-bit quantization

def y_channel(rgb):
    """BT.601 luma on the [16, 235] scale (the MATLAB rgb2ycbcr convention)."""
    return 16 + rgb @ np.array([65.481, 128.553, 24.966])

def psnr(a, b, shave, peak):
    if shave:
        a, b = a[shave:-shave, shave:-shave], b[shave:-shave, shave:-shave]
    return 10 * np.log10(peak**2 / np.mean((a - b) ** 2))

hr = img_as_float64(data.astronaut())                            # (512, 512, 3)
lr = degrade(hr)                                                 # (128, 128, 3)
print("LR shape:", lr.shape)
for name, flag in [("nearest", cv2.INTER_NEAREST), ("bilinear", cv2.INTER_LINEAR),
                   ("bicubic", cv2.INTER_CUBIC), ("lanczos", cv2.INTER_LANCZOS4)]:
    sr = cv2.resize(lr, (512, 512), interpolation=flag)
    sr = np.clip(sr, 0, 1)
    y_hr, y_sr = y_channel(hr), y_channel(sr)
    print(f"{name:8s}  Y, shave 4: {psnr(y_hr, y_sr, 4, 255):.2f} dB | "
          f"Y, no shave: {psnr(y_hr, y_sr, 0, 255):.2f} dB | RGB, shave 4: {psnr(hr, sr, 4,
          ↪  1):.2f} dB")
```

```
LR shape: (128, 128, 3)
nearest   Y, shave 4: 24.62 dB | Y, no shave: 24.66 dB | RGB, shave 4: 23.06 dB
bilinear  Y, shave 4: 25.35 dB | Y, no shave: 25.40 dB | RGB, shave 4: 23.83 dB
bicubic   Y, shave 4: 26.36 dB | Y, no shave: 26.40 dB | RGB, shave 4: 24.81 dB
lanczos   Y, shave 4: 26.50 dB | Y, no shave: 26.54 dB | RGB, shave 4: 24.94 dB
```

The ranking is the same under every protocol, but Y and RGB differ by about 1.5 dB, so only compare numbers measured the same way. Figure 2 shows the stages of the pipeline on a crop, with JPEG added at the end.



Figure 2 — A blur → downsample → noise → JPEG degradation at ×4. Each stage removes or corrupts information that an SR model would have to bring back. The LR panels are enlarged with nearest-neighbor interpolation only for display.

Why bicubic-trained models fail on real photos

Plain version. A network trained on one kind of shrinking learns the exact "signature" of that shrinking. Give it an image shrunk differently and it applies the wrong correction.

A model trained on bicubic pairs has learned to invert one specific, mild kernel with no noise. A real phone image has a wider, camera-specific blur, demosaicking and denoising traces, sensor noise and compression. Two things then go wrong. If the real blur is **wider** than the training blur, the model under-corrects and the result stays soft. If the real image is **sharper** or noisier than expected, the model over-corrects: it sharpens noise and JPEG blocks and draws halos around edges. Efrat et al. showed for classical methods that an accurate blur model matters more than a sophisticated image prior, and that the default blur models of SR algorithms can differ from real camera blur, which typically leads to over-smoothed results [5]. Cai et al. made the same point for deep networks by capturing real LR–HR pairs with two DSLR cameras at different focal lengths (the RealSR dataset): models trained on their real pairs did better on real images than models trained on simulated bicubic data [6]. We will reproduce the mismatch effect with a tiny model in the next section (Figure 6). Real-world SR, which tries to cover the whole space of camera degradations, has its own section below.

Aliasing and the frequency view

Plain version. Shrinking an image can make fine stripes turn into fake coarse stripes. Once that happens, the small image cannot tell you whether the coarse stripes were real or fake, and an SR model has to guess.

Sampling at the LR grid can only represent frequencies up to the LR **Nyquist limit**, $1/(2s)$ cycles per HR pixel (0.125 at $s = 4$). In one dimension, if $X(f)$ is the spectrum of the HR signal with f in cycles per HR sample, keeping every s-th sample gives the LR spectrum

$$Y(f') = \frac{1}{s} \sum_{m=0}^{s-1} X\left(\frac{f' + m}{s}\right), \qquad -\tfrac{1}{2} \le f' < \tfrac{1}{2},$$

where f' is in cycles per LR sample and X is periodic with period 1. Each LR frequency f' is the sum of s HR frequencies $(f' + m)/s$ that "fold" onto it: this folding is **aliasing**, explained in detail in IP 04. A blur before sampling (the k in the model) attenuates the HR frequencies above $1/(2s)$, so fewer of them fold; that is why cameras and `imresize` blur first.

For SR this has two consequences. First, content above the LR Nyquist limit is either removed by the blur (and must be invented) or aliased into the LR image (and must be *unfolded*, which is only possible with a prior that says which of the folded frequencies is more plausible). Second, even below the Nyquist limit the blur has weakened the HR frequencies, so a large part of what an SR model does in practice is **deblurring**. The right panel of Figure 3 shows both: bicubic

interpolation falls below the HR spectrum already before 0.125 cycles per pixel, and a learned linear model (introduced below) recovers much of that band but adds little above it.

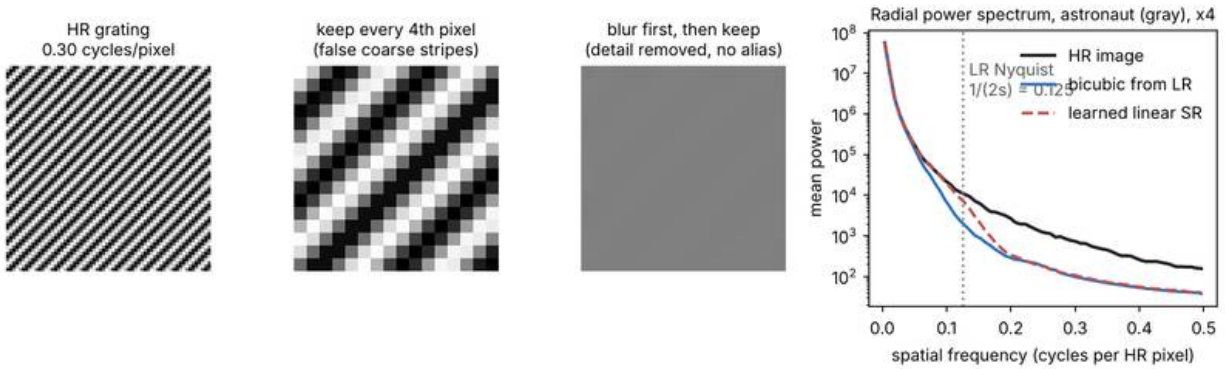


Figure 3 — Left: a grating at 0.30 cycles per pixel decimated by 4 without a prefilter turns into false coarse stripes (each axis component of about 0.21 cycles per pixel becomes about −0.15 cycles per LR pixel); blurring first removes it instead. Right: radial power spectra at ×4. Interpolation cannot create energy above the LR Nyquist limit (dotted line), and the blur also weakens frequencies below it.

Classical super-resolution

Plain version. Before deep learning, SR methods either used a fixed formula to fill in pixels (interpolation), copied detail from similar image patches (example-based methods), or learned a simple mapping from small to large patches. Each of these ideas survives inside modern networks.

Interpolation as a fixed linear filter

Interpolation places the LR samples on the HR grid and fills the gaps with a weighted sum of nearby samples. In one dimension,

$$\hat{x}(t) = \sum_{u} y[u]\, \phi\left(\frac{t + \frac{1}{2}}{s} - \frac{1}{2} - u \right),$$

where t is an HR pixel index, u an LR pixel index, ϕ the **interpolation kernel** and the half-pixel terms align the centers of LR and HR pixels (the convention OpenCV uses). In 2-D the kernel is applied separably along rows and columns. IP 02 derives nearest-neighbor, bilinear and bicubic interpolation; here we compare them as filters.

- **Nearest neighbor**: ϕ is a box of width 1. Blocky, but it never invents new values.
- **Linear**: ϕ is a triangle of width 2. Continuous, but soft.

- **Cubic convolution** (Keys) [7]: a piecewise cubic of width 4 with a free parameter a; Keys showed that $a = -\frac{1}{2}$ gives the best approximation order within that family. Its small negative lobes sharpen edges slightly and can cause mild ringing. Libraries do not all use the same a: we resized a single bright pixel with OpenCV's `INTER_CUBIC` and its output matched the Keys kernel with $a = -0.75$ exactly, not $a = -0.5$ (Exercise 1).
- **Lanczos**: a sinc windowed by a wider sinc, $\phi(t) = \mathrm{sinc}(t)\,\mathrm{sinc}(t/a)$ for $|t| < a$ and 0 otherwise, with $\mathrm{sinc}(t) = \sin(\pi t)/(\pi t)$ and a the number of lobes kept on each side [8]. OpenCV's `INTER_LANCZOS4` uses $a = 4$.

The ideal reconstructor for band-limited signals is the sinc itself, whose frequency response is 1 up to the Nyquist limit and 0 above. Figure 4 compares the kernels and their frequency responses. Wider kernels approximate the ideal better, which is why Lanczos scored highest in the snippet above (26.50 dB versus 26.36 dB for bicubic), but every interpolator is **linear and shift-invariant**: it cannot create frequencies above the LR Nyquist limit, and it treats an edge and a flat region with the same weights.

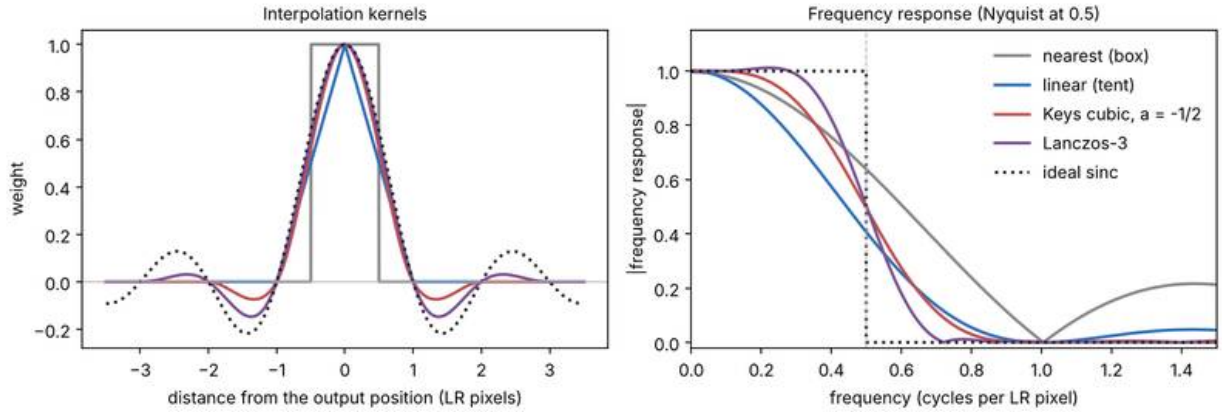


Figure 4 — Interpolation kernels (left) and the magnitude of their frequency responses (right). A good interpolator passes frequencies below the LR Nyquist limit (0.5 cycles per LR pixel) and blocks the spectral copies above it.



Figure 5 — ×4 interpolation of the degraded astronaut image (blur $\sigma = 1.2$, noise $\sigma = 0.01$). Interpolation redistributes the information in the LR image but cannot restore what the blur and decimation removed.

Example-based SR, self-similarity and sparse coding

Plain version. If a small blurry patch looks like the blurry version of a sharp patch you have seen before, paste in the sharp version. The examples can come from other images or from the same image at a coarser scale.

Freeman et al. built such pairs from training images [9], Glasner et al. showed that small patches recur within and across scales of the same image, so the input itself supplies examples [10], and Yang et al. coded LR and HR patches with paired dictionaries that share one sparse code [11]. SR-CNN later read this sparse-coding pipeline as a three-layer network (extract patches, map them, reconstruct), which is where deep SR started [12].

Optional: Example-based SR and sparse coding in detail (in the Supplementary notes). It gives the sparse-coding objective and the internal-learning idea behind ZSSR.

Learned linear regressors

Plain version. Instead of searching for similar patches at test time, fit one simple formula per kind of patch in advance: "for a patch that looks like this, add this correction." Then SR is just a few matrix multiplications.

Anchored neighborhood regression (ANR) replaces the sparse coding step with a precomputed ridge regressor for each dictionary atom: a LR patch is assigned to its most correlated atom and multiplied by that atom's matrix [13]. A+ fits each regressor on training samples near the atom instead of on neighboring atoms, which improved quality at the same speed [14]. RAISR hashes each patch by the angle, strength and coherence of its local gradient and applies a filter learned by least squares for that hash bucket [15]. The common structure is

$$\hat{r} = W_{c(p)}^{\top}\tilde{p}, \qquad W_c = \arg\min_{W} \sum_{i \in c} \|W^{\top}\tilde{p}_i - r_i\|_2^2 + \lambda \|W\|_F^2 = (P_c^{\top}P_c + \lambda I)^{-1} P_c^{\top} R_c,$$

where p is an LR patch, $\tilde{p}$ its feature vector, $c(p)$ the class (atom or hash bucket) it belongs to, r the HR detail to predict, $(\tilde{p}_i, r_i)$ the training pairs of class c stacked as the rows of P_c and R_c, λ the ridge weight and $\|\cdot\|_F$ the Frobenius norm.

The snippet below implements the simplest version, with a single class. For every LR pixel it takes the surrounding 7×7 LR patch (mean removed, plus a constant 1 for the bias) and predicts the 4×4 block of HR **residual**, the difference between the HR image and its bicubic upsampling. Predicting s^2 values per LR pixel and then rearranging them into an $s \times s$ block is exactly what a sub-pixel convolution does, as we will see in the CNN section: this model is a one-layer, linear ESPCN. We train it twice on eleven `skimage` images, once for "sharp" LR inputs (block averaging only) and once for "blurry" LR inputs (Gaussian blur with $\sigma = 1.2$ before averaging), and test both on the astronaut image, which is not in the training set.

```
import numpy as np, cv2
from numpy.lib.stride_tricks import sliding_window_view
```

```
from scipy import ndimage as ndi
from skimage import data, img_as_float64
from skimage.color import rgb2gray

s, p = 4, 7                                  # scale factor, LR patch size

def degrade(x, blur):                        # blur (optional) -> s x s average
    x = ndi.gaussian_filter(x, blur) if blur else x
    H, W = x.shape[0] // s * s, x.shape[1] // s * s
    return x[:H, :W].reshape(H // s, s, W // s, s).mean(axis=(1, 3))

def bicubic(lr):
    return cv2.resize(lr, None, fx=s, fy=s, interpolation=cv2.INTER_CUBIC)

def features(lr):                            # one p x p patch per LR pixel, mean
↪   removed, plus a bias
    f = sliding_window_view(np.pad(lr, p // 2, mode="reflect"), (p, p)).reshape(lr.size, p *
    ↪   p)
    return np.c_[f - f.mean(axis=1, keepdims=True), np.ones(lr.size)]

def to_blocks(r):                            # HR residual -> one s*s vector per LR pixel
↪   (space-to-depth)
    h, w = r.shape[0] // s, r.shape[1] // s
    return r.reshape(h, s, w, s).transpose(0, 2, 1, 3).reshape(h * w, s * s)

def from_blocks(b, h, w):                    # inverse: depth-to-space (pixel shuffle)
    return b.reshape(h, w, s, s).transpose(0, 2, 1, 3).reshape(h * s, w * s)

def train(images, blur, lam=1e-3):
    A, B = [], []
    for im in images:
        im = im[: im.shape[0] // s * s, : im.shape[1] // s * s]
        lr = degrade(im, blur)
        A.append(features(lr)); B.append(to_blocks(im - bicubic(lr)))
    A, B = np.concatenate(A), np.concatenate(B)
    return np.linalg.solve(A.T @ A + lam * np.eye(A.shape[1]), A.T @ B)   # ridge: (p*p+1,
    ↪   s*s)

def super_resolve(lr, Wt):
    return bicubic(lr) + from_blocks(features(lr) @ Wt, *lr.shape)

def psnr(a, b):
    a, b = a[s:-s, s:-s], b[s:-s, s:-s]          # shave s border pixels
    return 10 * np.log10(1.0 / np.mean((a - np.clip(b, 0, 1)) ** 2))

train_imgs = [img_as_float64(im) for im in (data.camera(), data.moon(), data.coins(),
↪   data.page(),
              data.text(), data.brick(), data.grass(), data.gravel(),
              rgb2gray(data.coffee()), rgb2gray(data.chelsea()), rgb2gray(data.rocket()))]
models = {"trained sharp": train(train_imgs, 0.0), "trained blurry": train(train_imgs, 1.2)}
test = img_as_float64(rgb2gray(data.astronaut()))           # never seen in training
print("weights per model:", models["trained sharp"].shape)
for name, blur in [("sharp", 0.0), ("blurry", 1.2)]:
    lr = degrade(test, blur)
```

```
    row = "  ".join(f"{m} {psnr(test, super_resolve(lr, W)):.2f}" for m, W in models.items())
    print(f"test {name:6s} | bicubic {psnr(test, bicubic(lr)):.2f} | {row}  (dB)")
```

```
weights per model: (50, 16)
test sharp  | bicubic 25.61 | trained sharp 26.05  trained blurry 25.43  (dB)
test blurry | bicubic 24.98 | trained sharp 25.66  trained blurry 26.26  (dB)
```

Each model has only $50 \times 16 = 800$ weights and is fitted in closed form in about a second. With the matching degradation it gains 0.44 dB (sharp inputs) and 1.28 dB (blurry inputs) over bicubic on an unseen image. The off-diagonal entries reproduce the failure described earlier. The blurry-trained model applied to sharp inputs **over-sharpens** and ends up below bicubic (25.43 dB versus 25.61 dB); the sharp-trained model applied to blurry inputs under-corrects (25.66 dB versus 26.26 dB for the matched model). Figure 6 shows the crops and the 16 learned filters, one per sub-pixel offset. The slider compares bicubic with the matched linear model on the blurry input.



Bicubic



Linear SR (*matched*)

Slider comparing a blurry bicubic ×4 upsampling of the astronaut's face with the sharper output of the learned linear regressor on the same crop.

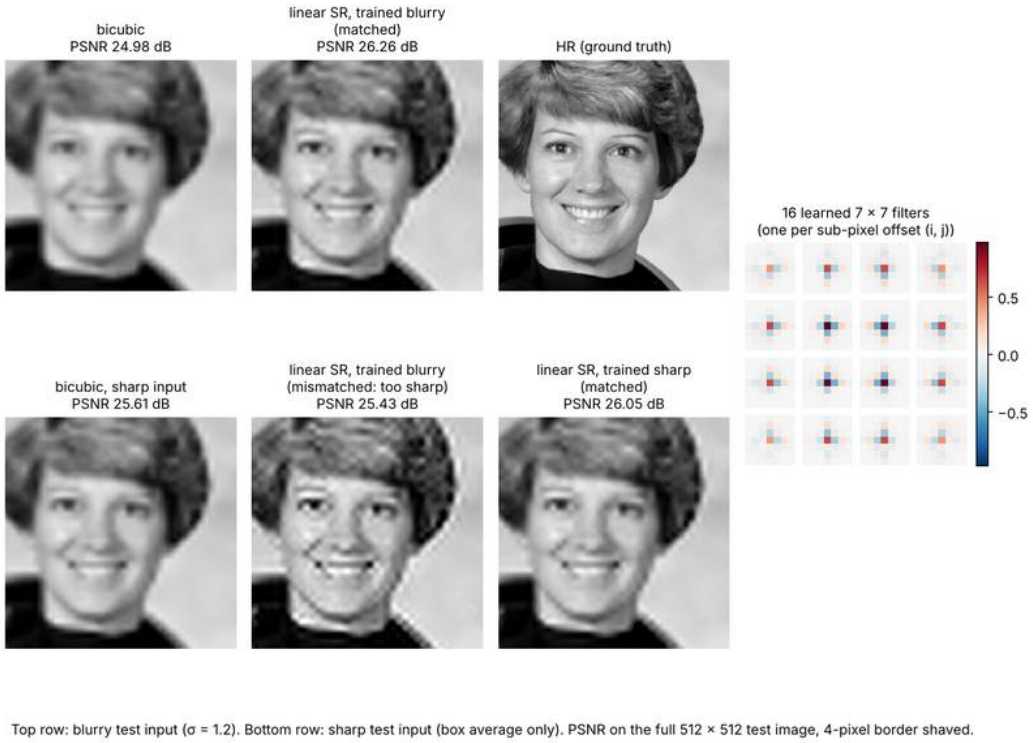


Figure 6 — A learned linear regressor (one 7 × 7 filter per sub-pixel offset, predicting the residual over bicubic). Matched training and test degradations help; a mismatch (bottom middle) over-sharpens and scores below bicubic. PSNR on the full grayscale test image with a 4-pixel border shaved.

CNN super-resolution

Plain version. A CNN is a much richer version of the regressor above: many stacked, non-linear filters that look at a large neighborhood. The main design questions are where to increase the resolution, how to make deep networks train, and how to spend the computation.

Pre-upsampling: SRCNN and VDSR

SRCNN, the first end-to-end CNN for SR, upsamples the input with bicubic interpolation first and then applies three convolution layers that its authors interpret as patch extraction, non-linear mapping and reconstruction, the same three steps as sparse-coding SR [12]. VDSR made the network much deeper (20 weight layers of 3 × 3 filters) and made that depth trainable with two choices: it predicts only the **residual** between the HR image and the interpolated input, and it uses a very high learning rate kept stable by adjustable gradient clipping [16]. The residual idea is the same as in our linear model: the bicubic image already contains the low frequencies, so the network only has to add the missing detail, which is a small, zero-centered signal.

Pre-upsampling is simple and handles any scale with one network (just change the interpolation factor), but every layer runs at HR size, so a layer costs s^2 times more than it would at LR size, and the network must also clean up interpolation artifacts. Figure 7 contrasts it with post-upsampling.

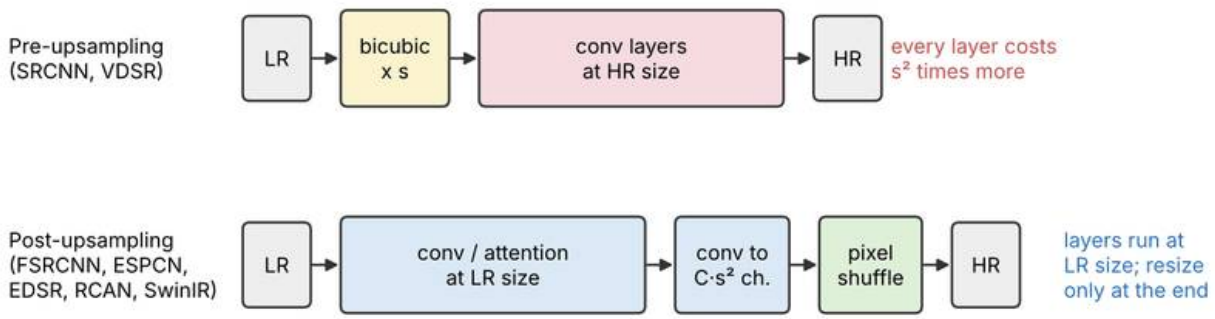


Figure 7 — Pre-upsampling (SRCNN, VDSR) versus post-upsampling with sub-pixel convolution (ESPCN and most later networks). FSRCNN also upsamples at the end, but with a transposed convolution.

Post-upsampling and sub-pixel convolution

FSRCNN moved the upsampling to the end with a learned transposed (fractionally strided) convolution, shrank the feature dimension before the mapping layers, and reported a speed-up of more than 40 times over SRCNN [17]. ESPCN proposed the **sub-pixel convolution layer**: the last convolution produces Cs^2 channels at LR resolution, and a fixed rearrangement, now called **pixel shuffle** (or depth-to-space), turns them into C channels at s times the resolution [18]. All learned layers run at LR size, and the rearrangement has no parameters. EDSR, RCAN and SwinIR all use it in their upsampling modules [19], [20], [21].

The precise version is an index rule. Let T be the LR feature tensor of shape (Cs^2, H, W) and $X = \mathrm{PS}(T)$ the output of shape (C, sH, sW). Then

$$X[c,\ sh+i,\ sw+j] = T[c\,s^2 + s\,i + j,\ h,\ w], \qquad 0 \le c < C,\ \ 0 \le h < H,\ \ 0 \le w < W,\ \ 0 \le i, j < s,$$

where c is the output channel, (h, w) an LR position and (i, j) the **sub-pixel offset** inside the $s \times s$ output block that belongs to that LR position. In words: the s^2 channels that belong to output channel c are read in row-major order of (i, j) and spread over one $s \times s$ block. This is the ordering of PyTorch's `PixelShuffle`, whose implementation reshapes the channel axis into (C, s, s) and permutes it next to the height and width axes [22]. The inverse operation, **space-to-depth** (pixel unshuffle), reads each $s \times s$ block back into s^2 channels; some efficient networks use it at the input to work at a lower resolution. Figure 8 draws the rule for $s = 2$.

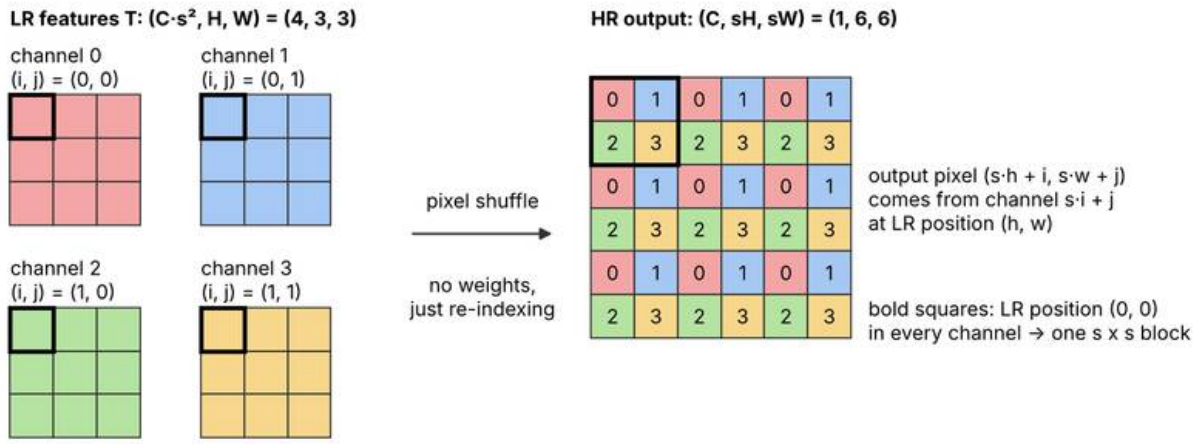


Figure 8 — Pixel shuffle for $s = 2$ and $C = 1$. The four values at one LR position, one from each channel, become one 2×2 output block.

The NumPy version below writes pixel shuffle and its inverse as one reshape, one transpose and one reshape each, checks the index rule element by element, and checks that the two operations are exact inverses.

```
import numpy as np

def pixel_shuffle(t, s):
    """Depth-to-space: (C*s*s, H, W) -> (C, H*s, W*s)."""
    Cs2, H, W = t.shape
    C = Cs2 // (s * s)
    return t.reshape(C, s, s, H, W).transpose(0, 3, 1, 4, 2).reshape(C, H * s, W * s)

def pixel_unshuffle(x, s):
    """Space-to-depth: (C, H*s, W*s) -> (C*s*s, H, W)."""
    C, Hs, Ws = x.shape
    H, W = Hs // s, Ws // s
    return x.reshape(C, H, s, W, s).transpose(0, 2, 4, 1, 3).reshape(C * s * s, H, W)

rng = np.random.default_rng(0)
s, C, H, W = 3, 2, 4, 5
t = rng.standard_normal((C * s * s, H, W))
x = pixel_shuffle(t, s)
print(x.shape)                                                  # (2, 12, 15)

# The index rule: output (c, s*h + i, s*w + j) comes from channel c*s*s + s*i + j at (h, w).
ok = all(x[c, s*h + i, s*w + j] == t[c*s*s + s*i + j, h, w]
         for c in range(C) for h in range(H) for w in range(W)
         for i in range(s) for j in range(s))
print("index rule holds:", ok)
print("unshuffle(shuffle(t)) == t:", np.array_equal(pixel_unshuffle(x, s), t))
print("shuffle(unshuffle(x)) == x:", np.array_equal(pixel_shuffle(pixel_unshuffle(x, s), s),
 ↪  x))
```

```
(2, 12, 15)
index rule holds: True
unshuffle(shuffle(t)) == t: True
shuffle(unshuffle(x)) == x: True
```

Pixel shuffle is a permutation of array entries, so it is exactly invertible and its gradient is the inverse permutation. A sub-pixel convolution is therefore s^2 ordinary convolutions, one per sub-pixel offset, with interleaved outputs; our linear regressor is the case with one layer. If the s^2 filters start from unrelated random values, the first outputs show a checkerboard pattern; Aitken et al. initialize them to act like one convolution followed by nearest-neighbor upsampling, which removes it [23].

Deep residual and attention networks

Plain version. Deeper networks see more context and model more complex detail, but they are harder to train. The fixes were the same as in image classification (residual connections), plus a few SR-specific ones.

EDSR adapted residual networks to SR and removed the batch normalization layers, which its authors found unnecessary for SR and costly in GPU memory; to train a wide model stably (32 residual blocks with 256 feature channels), it scales each residual branch by 0.1 before adding it [19]. EDSR also trained with the L1 loss instead of L2, and its multi-scale variant MDSR shares one body across ×2, ×3 and ×4 with scale-specific heads and tails. It won the NTIRE 2017 SR challenge [19], [24]. RCAN went much deeper (over 400 layers) with a **residual-in-residual** structure, long skip connections around groups of residual blocks with short skips inside, so that low-frequency information can bypass the body, and with **channel attention**, which rescales each feature channel using a weight computed from global statistics of that channel [20]. These networks are trained on DIV2K (see the benchmarks section) with random crops, flips and rotations; the general recipe and the loss functions are covered in DLCV 04.

Perceptual and adversarial SR

Plain version. If the pixel loss always returns the safe, blurry average, change the loss: reward outputs that look like real photos, even if their details are not in exactly the right place.

Perceptual losses and GANs

Johnson, Alahi and Fei-Fei trained SR networks with a **perceptual loss**, the distance between features of a pretrained classification network (VGG [25]) for the output and for the target, and

obtained sharper edges and textures than with a pixel loss at a lower PSNR [26]. SRGAN added an **adversarial loss**: a discriminator learns to tell SR outputs from real HR images, and the SR network is trained to fool it, which "pushes the solution to the natural image manifold" in the authors' description, combined with a VGG content loss [27]. ESRGAN improved each part (dense residual blocks without batch normalization, a relativistic discriminator that judges whether real images look *more* realistic than generated ones, and a perceptual loss on VGG features before the activation) and won the PIRM 2018 perceptual SR challenge in its region [28], [29]. A typical generator loss is

$$\mathcal{L} = \lambda_1 \, \|\hat{x} - x\|_1 + \lambda_p \, \|\phi(\hat{x}) - \phi(x)\|_2^2 + \lambda_a \, \mathcal{L}_{\text{adv}}(\hat{x}),$$

where $\hat{x}$ is the SR output, x the HR target, ϕ a fixed layer of a pretrained network, $\mathcal{L}_{\text{adv}}$ the adversarial term computed with the discriminator, and $\lambda_1, \lambda_p, \lambda_a \geq 0$ weights chosen by hand. GANs are explained in ML 06.

The perception–distortion trade-off

Blau and Michaeli defined **distortion** as the expected dissimilarity between $\hat{x}$ and x (MSE, for example) and **perceptual quality** as the distance between the distribution of outputs and the distribution of real images, and proved that the two cannot both be minimized: the best perceptual quality achievable at a given distortion gets worse as the allowed distortion shrinks (the bound is non-increasing, and convex for common divergences), and the MMSE estimator sits at its low-distortion, low-realism end [2]. The formal statement is in DLCV 04. Our toy in Figure 1 traces such a curve: moving from the posterior mean toward a posterior sample raises MSE from 0.0072 to 0.0143 and closes the realism gap. Two practical consequences follow for SR:

- A method should state *where* on the trade-off it aims. Comparing a GAN to an MSE network by PSNR alone, or by a perceptual metric alone, says nothing about which is better.
- Benchmarks must measure both axes. The PIRM 2018 challenge did this by splitting the perception–distortion plane into three regions by RMSE (at most 11.5, 12.5 and 16) and ranking methods within each region by a perceptual index built from two no-reference metrics, NIQE [30] and the SR-specific metric of Ma et al. [31], [29].

Transformer SR

Plain version. Transformers let each output pixel draw on information from far away in the image, chosen by content rather than by distance. This can help on repeated structures such as windows, tiles and text.

SwinIR applies residual Swin Transformer [32] blocks, self-attention inside local windows that shift between layers so that information crosses window borders, between a shallow convolutional feature extractor and a sub-pixel reconstruction module [21]. Window attention scales linearly

with image size, which matters for dense outputs. HAT used an attribution analysis to argue that SwinIR-style models use only a limited spatial range of the input, and "activates more pixels" by combining channel attention with window self-attention and adding an overlapping cross-attention module between neighboring windows; it also pre-trains on a large dataset for the same SR task before fine-tuning [33]. DAT alternates spatial and channel self-attention in consecutive blocks and adds modules that let the two interact [34]. State-space models such as MambaIR replace attention with selective scans that reach the whole image at linear cost [35]. The general ideas of attention and window attention are in ML 04.

These networks are usually trained with an L1 loss and evaluated by PSNR, so they are still **posterior-mean estimators**: they reduce the error of the average, they do not escape it. Their sharper outputs come from a better use of context, not from sampling.

Diffusion-based SR

Plain version. A diffusion model learns to turn noise into realistic images step by step. Conditioned on the LR image, it generates an HR image that fits the input, and different random seeds give different plausible answers: it is a posterior sampler, not an averager.

SR3 is the representative first method: a U-Net [36] trained to remove noise at many levels, conditioned on the upsampled LR image, generates the HR image by starting from pure Gaussian noise and refining it over many steps [37]. In a human study on ×8 face SR, its outputs fooled raters close to half the time, a rate the GAN baselines did not reach [37]. For real photos, current methods start from a large **pretrained text-to-image diffusion model** and add a way to condition it on the LR image. StableSR, for example, trains a time-aware encoder while keeping the pretrained model fixed, and adds a feature-wrapping module whose scalar weight lets the user trade realism against fidelity at inference time [38]. The diffusion mechanics are in ML 07.

For SR, three facts matter most. Diffusion outputs are samples, so they sit at the realistic, higher-distortion end of the perception–distortion trade-off and score lower PSNR than regression networks. Their details are generated, so they can be wrong (see Failure modes). And they are slow: early methods needed hundreds of network evaluations per image, and recent work cuts this to a handful of steps (ResShift, 15 steps [39]) or a single step (OSEDiff [40]). Moser et al. list compute, poor comparability across papers and color shifts among the open problems of diffusion SR [41]; StableSR adds an explicit color-correction step for that reason [38].

Blind and real-world SR

Plain version. In the real world nobody tells the model how the image was shrunk. Either the model learns to handle a very wide range of damage, or it first estimates the damage and then undoes it.

Degradation synthesis pipelines

The **implicit** route trains one network on a degradation distribution wide enough to cover real photos. BSRGAN's practical degradation model applies blur (two convolutions with isotropic and anisotropic Gaussian kernels), downsampling (randomly chosen nearest, bilinear or bicubic), Gaussian noise, JPEG compression and processed camera sensor noise in a **randomly shuffled order** [42]. Real-ESRGAN instead applies the classical blur–resize–noise–JPEG chain **twice** ("high-order" degradation, second order in practice), adds sinc filters to simulate the ringing and overshoot artifacts of real images, and uses a U-Net discriminator [43]. Both are trained purely on synthetic pairs and tested on real images. The `degrade` function of the degradation snippet above is the first-order version of these pipelines.

Kernel estimation

The **explicit** route keeps the classical model $y = (x * k) \downarrow_s + n$ and estimates the kernel k and noise level, or gives them to the network as extra inputs. It works well when the estimate is right, and our mismatch experiment shows what happens when it is wrong: over-sharpening or over-smoothing. Liu et al. organize blind SR along two axes, explicit versus implicit modeling of the degradation, and external datasets versus the single input image [44]; DLCV 04 describes the main explicit methods.

Optional: Three kernel-aware methods (in the Supplementary notes). It summarizes SRMD, IKC and KernelGAN in a few lines each.

Real captured pairs

The third route captures real data. RealSR pairs images of the same scene taken with two DSLR cameras at different focal lengths and registers them, so the "degradation" is that of a real lens and sensor [6]. DRealSR extends the idea to five DSLR cameras to cover more diverse degradation processes [45]. Captured pairs are the most realistic data, but they are expensive, small, specific to the cameras used, and need careful alignment between the two zoom levels.

Arbitrary-scale SR

Plain version. Most SR networks are trained for one fixed factor such as ×4. Arbitrary-scale networks treat the image as a continuous function, so you can ask for any output size, including non-integer factors like ×2.7.

Meta-SR replaced the fixed upsampling layer with a **meta-upscale module** that predicts the weights of the upscaling filters from the scale factor and the relative position of each output pixel

[46]. LIIF went further and learned a **local implicit image function** [47]:

$$\hat{I}(x_q) = f_\theta(z^*,\ x_q - v^*),$$

where x_q is a continuous query coordinate in the image plane, z^* the latent feature vector (from a CNN encoder applied to the LR image) whose grid position v^* is nearest to x_q, and f_θ a small MLP that outputs the RGB value at x_q. Because x_q is continuous, the same network renders any output size; its authors show outputs up to ×30, far beyond the training scales [47]. The price is one MLP evaluation per output pixel, and quality at scales far outside training is limited by what the encoder saw.

Optional: LIIF's three refinements (in the Supplementary notes). It explains feature unfolding, local ensemble and cell decoding.

Video and reference-based SR (brief)

Plain version. Extra images of the same scene contain extra information. Video SR uses neighboring frames; reference-based SR uses a separate high-resolution photo of similar content.

Neighboring video frames show the scene at slightly different sub-pixel positions, the original multi-frame SR situation [1], so detail aliased in one frame may be visible in another; the difficulty is aligning moving content. BasicVSR is a widely used baseline built from bidirectional recurrent propagation and optical-flow alignment [48]. Reference-based SR transfers textures from an HR reference image by matching features, for example with attention in TTSR [49]; it is very faithful when the reference matches and must degrade gracefully when it does not.

Efficiency

Plain version. Many SR uses (phones, video calls, game rendering) need results in milliseconds on small chips. Efficient SR asks how small and fast a network can be while staying above a quality threshold.

Post-upsampling with pixel shuffle is the first efficiency rule: almost all computation happens at LR size. Beyond that, efficient SR networks reuse features cheaply. IMDN, for example, splits features step by step into a "distilled" part that is kept and a part that is processed further, then fuses them with a contrast-aware channel attention [50]. Parameters, FLOPs and runtime measure different costs, so the NTIRE efficient SR challenges score all three. In 2024 and 2025 the task was ×4 SR, and entries had to reach a PSNR of at least 26.90 dB on the challenge's validation set and 26.99 dB on its test set while minimizing runtime, parameters and FLOPs [51], [52]. Diffusion models are far from these budgets, which is why one-step distillation is an active topic.

Evaluating SR

Plain version. SR results are judged in two ways: how close the output is to the true HR image (fidelity) and how natural it looks (perceptual quality). For fidelity, the community uses a fixed recipe whose small details change the numbers, so you must follow it exactly. The generic metrics, no-reference metrics and user studies are explained in DLCV 04; this section covers what is specific to SR.

Y-channel PSNR and SSIM with border shaving

The classical protocol, inherited from MATLAB code of early papers and still used by most papers and challenges, is:

1. Convert both the HR image and the SR output to YCbCr and keep the luma channel

$$Y = 16 + 65.481\,R + 128.553\,G + 24.966\,B,$$

where $R, G, B \in [0, 1]$ are the (gamma-encoded) color values and Y lies in $[16, 235]$. These are the BT.601 coefficients as implemented by MATLAB's `rgb2ycbcr` (and by `skimage.color.rgb2ycbcr`) [53], [54]. 2. Remove a border of pixels from both images before comparing ("shaving"), so that border effects of padding do not dominate. The width varies: EDSR reports benchmark results on Y with a border equal to the scale factor [19], and the NTIRE 2024 and 2025 ×4 challenges exclude a 4-pixel border and compute on Y [55], [56]. 3. Compute PSNR with peak value 255 and SSIM [57] on what remains. Definitions are in DLCV 04.

Why luma? The eye is more sensitive to luminance detail than to color detail, and many early methods super-resolved only Y. Wang et al. note that the choice of color space and channel for training or evaluation alone can change results greatly, by up to 4 dB in one reported case [4]. One part of that difference is pure arithmetic: the coefficients above sum to 219, so $Y = 16 + 219\,\ell$ with $\ell \in [0, 1]$, and computing PSNR with peak 255 on Y adds a constant $20 \log_{10}(255/219) \approx 1.32$ dB compared with computing it on ℓ with peak 1. Our first snippet showed a similar gap between Y and RGB (about 1.5 dB). Rounding to 8 bits, clipping and the resizer used to make the LR images also shift results slightly, so fair comparisons run every method through one evaluation script.

The standard test sets

SR papers report results on a small set of public test sets, each with a different character [4]:

Test set	Images	Content	Source
Set5	5	baby, bird, butterfly, head, woman	[58]
Set14	14	people, animals, flowers, comic, slides	[59]

Test set	Images	Content	Source
BSD100 (B100)	100	natural scenes, the test images of the Berkeley segmentation dataset	[60]
Urban100	100	buildings and city scenes with regular structures	[61]
Manga109	109	Japanese comic volumes (line art and screen tones)	[62], [63]
DIV2K validation	100	diverse 2K-resolution photos	[64]

Set5 and Set14 are tiny, so a difference of a few hundredths of a dB on them means little on its own. All of these sets are used in the bicubic setting; for real-world SR, papers add captured sets such as RealSR and DRealSR [6], [45] and no-reference metrics.

Why PSNR and perceptual rankings disagree

Three reasons, all visible in this chapter. First, the posterior-mean argument: PSNR is maximized by the average of all plausible answers, which is blurry, so any method that commits to one sharp answer loses PSNR even when it looks better (Figure 1). Second, misalignment: a texture drawn with the right statistics but shifted by half a period has a large pixel error. Third, the metrics were built for different things: no-reference and learned perceptual metrics such as LPIPS [65] reward natural-looking textures and can be fooled by over-sharpening, while PSNR rewards smoothness. Challenges therefore report both. The NTIRE 2025 ×4 challenge ran a restoration track ranked by PSNR and a perceptual track ranked by one combined score of six full-reference and no-reference quality metrics, including LPIPS and NIQE [56], [30]. The generic advice from DLCV 04 applies: report a fidelity metric, a perceptual metric and, for generative SR, a user study.

Datasets, benchmarks and NTIRE challenges

Plain version. A few datasets and a yearly competition define most of what "state of the art" means in SR. Knowing what they measure tells you what a leaderboard number does and does not mean.

- **DIV2K** [64] contains 1,000 diverse 2K-resolution images, split 800/100/100 for training, validation and testing [56]. It was introduced for the NTIRE 2017 challenge and is the standard training set. It is often combined with **Flickr2K**, which the NTIRE 2025 report describes as 2,650 2K images [56]; recent challenges add much larger collections.
- **NTIRE 2017** [24] was the first SR challenge of the series. Track 1 used standard bicubic downscaling; Track 2 used unknown downscaling operators that could be learned from the provided LR–HR training pairs; each had ×2, ×3 and ×4 competitions.

- **NTIRE 2020 real-world SR** [66] provided no paired data at all: one set of source images and a set of unpaired high-quality targets, with a track of synthetic processing artifacts and a track of real smartphone images, ranked by a human study of perceptual quality.
- **NTIRE 2024 and 2025 ×4 SR** [55], [56] returned to bicubic ×4 on DIV2K; the 2025 edition added the perceptual track described above.
- **NTIRE efficient SR** [51], [52] ranks ×4 models by runtime, parameters and FLOPs above a fixed PSNR.

Failure modes

Plain version. An SR model can fail by being too cautious (blurry) or too bold (invented detail). The bold failures are more dangerous, because the result looks convincing.

- **Over-smoothing.** MSE- and L1-trained models return the posterior mean: waxy skin, flat grass, soft hair. This is expected behavior, not a bug (Figure 1).
- **Hallucinated text and faces.** At large scales a generative model can turn illegible text into legible but *wrong* characters, and turn a few pixels of a face into a specific, sharp face that is not the person who was photographed. For forensic, medical or scientific images this is unacceptable. Ren et al. argue that such hallucinations are not captured well by fidelity metrics or by no-reference quality metrics, and propose scoring them with a multimodal large language model [67].
- **Texture artifacts.** GAN-based models can add repetitive, noise-like textures or sharpen JPEG blocks into structure; ESRGAN was designed partly to reduce such "unpleasant artifacts" of SRGAN [28]. Our mismatched linear model shows the same effect in miniature: halos and exaggerated edges (Figure 6).
- **Color shifts.** Diffusion-based SR can shift global colors away from the input [41], [38]; a cheap check is to downsample the output and compare its colors with the LR input.
- **Degradation mismatch.** A model trained on the wrong degradation under- or over-corrects (the Real-world section and Figure 6). Testing on several degradations, not only bicubic, exposes this.
- **Tiling seams.** Large images are often processed in patches; non-overlapping patches can leave visible borders, as noted for patch-based processing by the SwinIR authors [21].

A useful sanity check for any SR output is **consistency with the input**: degrade the output with the assumed model and compare it with y. A large difference means the model changed content that the LR image did measure.

Modern view

Survey papers

Plain version. Four surveys map the field: one for deep SR in general, one for blind SR, one for real-world SR and one for diffusion-based SR. Read them in that order.

Wang, Chen and Hoi organize deep SR by upsampling framework, upsampling operator, network design, losses and learning strategies [4]. Liu et al. classify blind SR by how the degradation is modeled (explicitly or implicitly) and which data are used (external datasets or the single input image) [44]. Chen et al. group real-world SR into degradation modeling, captured image pairs, domain translation and self-learning [68]. Moser et al. cover the theory of diffusion models for SR and its open problems, such as compute and color shifts [41]. Together they point to the open problems discussed in this chapter: realistic degradations, faithful generation and evaluation.

Optional: Paper guides (in the Paper guides). Each of the four surveys has a short reading guide there.

State of the art, 2023–2026

1. **Fidelity-oriented SR** uses attention and state-space backbones (HAT, DAT, MambaIR [33], [34], [35]) trained on large data, and is benchmarked each year by the NTIRE ×4 restoration track [55], [56].
2. **Real-world SR moved to pretrained diffusion priors** (for example StableSR [38]), and the main engineering effort is now speed: from hundreds of steps to 15 (ResShift [39]) and to one step (OSEDiff [40]).
3. **Evaluation is catching up with generation**: separate fidelity and perceptual tracks [56], and first attempts to measure hallucination directly [67].
4. **Efficiency** has its own benchmark with fixed quality thresholds [51], [52].

Open problems

- **Faithful generative SR.** How to get diffusion-level realism without inventing content, and how to show a user which details are measured and which are guessed.
- **Evaluation.** PSNR rewards blur, no-reference metrics can be gamed, and hallucination metrics are new. There is no agreed protocol for real images without ground truth.
- **Real degradations.** Synthetic pipelines still miss camera processing pipelines and spatially varying blur; captured datasets are small and camera-specific.

- **Cost.** One-step diffusion is still far heavier than the networks of the efficient SR challenges, and video adds the problem of temporal consistency.
- **Arbitrary scales and extreme scales.** Behavior far outside the training scales, and at factors where almost everything is generated, is poorly understood.

Key takeaways

- SR is ill-posed: at ×4 each LR pixel constrains one combination of 16 HR values, so a prior decides the rest. Pixel losses give the **posterior mean**, which is blurry by construction; a posterior sample looks real and costs at most twice the MSE.
- The **degradation model** defines the task. Bicubic benchmarks are clean but unrealistic; a mismatched blur makes a model under- or over-sharpen, as the tiny linear regressor showed.
- Aliasing folds frequencies above $1/(2s)$ into the LR image; blur removes them. Much of practical SR is deblurring below the LR Nyquist limit plus prior-driven guessing above it.
- Modern networks upsample at the end with **pixel shuffle**: $X[c, sh + i, sw + j] = T[cs^2 + si + j, h, w]$, an exact permutation with an exact inverse (space-to-depth).
- Residual learning, no batch normalization, residual scaling and attention (EDSR, RCAN, SwinIR, HAT) push fidelity; perceptual, adversarial and diffusion methods (SRGAN, ESRGAN, SR3, StableSR) push realism. The **perception–distortion trade-off** says you must choose.
- Evaluate on Y with a shaved border, state the protocol, use the standard test sets, and add perceptual metrics and a user study for generative models.
- Watch for hallucinated text and faces, texture artifacts, color shifts, over-smoothing and tiling seams; check outputs for consistency with the LR input.

Exercises

1. **Which cubic does OpenCV use?** Make a 1-D signal of 16 zeros with a single 1 in the middle, upsample it by 4 with `cv2.resize(..., interpolation=cv2.INTER_CUBIC)`, and compare the result with the Keys kernel evaluated at the half-pixel-aligned positions $(t + \frac{1}{2})/4 - \frac{1}{2} - 8$ for $a = -0.5$ and $a = -0.75$. Then plot both kernels' frequency responses and compare their values at the LR Nyquist frequency.

   Hint

   The output of a resize of a unit impulse is the kernel itself, sampled at the output positions. In our run the maximum absolute difference was 0.0 for $a = -0.75$ and about 0.037 for $a = -0.5$. A more negative a gives deeper negative lobes, a response that stays near 1 a little longer and then overshoots slightly: a sharper but more ringing interpolator. Repeat with `INTER_LANCZOS4` and the Lanczos kernel with

$a = 4$; the match is close but not exact (we measured a maximum difference of about 0.0016), because the library normalizes the weights.

2. **Protocol arithmetic.** (a) Show that Y-PSNR with peak 255 equals the PSNR of $\ell = (Y-16)/219$ with peak 1 plus $20\log_{10}(255/219)$. (b) In the first snippet, change the shave from 4 to 0 and to 16 pixels and explain the size of the change. (c) Why is shaving necessary for a network with zero padding but nearly irrelevant for `cv2.resize`?

Hint

(a) $Y_{\text{hr}} - Y_{\text{sr}} = 219(\ell_{\text{hr}} - \ell_{\text{sr}})$, so the MSE scales by 219^2 and PSNR becomes $10\log_{10}(255^2/(219^2\,\text{MSE}_\ell)) = \text{PSNR}_\ell + 20\log_{10}(255/219) \approx \text{PSNR}_\ell + 1.32$ dB. (b) Shaving changes the result by only a few hundredths of a dB here (24.66 versus 24.62 dB for nearest neighbor), because OpenCV extrapolates borders reasonably and the image edges are smooth. (c) Zero padding makes a convolutional network see black outside the image, so the outputs near the border are systematically worse; shaving removes a region whose quality depends on the padding choice rather than on the method.

3. **A channel-ordering bug.** Suppose one library orders the Cs^2 channels as $k = cs^2 + si + j$ (our formula) and another as $k = (si + j)C + c$. You load weights trained with the first ordering into the second. For $C = 3$ and $s = 2$, which of the 12 channels end up in the right place? What does the output look like?

Hint

Solve $4c + q = 3q + c$ with $q = 2i + j$: $3c = 2q$, so only $(c, q) = (0, 0)$ and $(2, 3)$ coincide, 2 channels out of 12 (we checked this by enumerating all 12). The other values are moved to the wrong color channel and the wrong sub-pixel offset, so the output shows a fine color mosaic in every 2×2 block even though the coarse image is recognizable. For $C = 1$ the two orderings agree, which is why such bugs often appear only when moving from grayscale to color models.

4. **Make the linear model better.** Starting from the regressor snippet, (a) change the patch size p to 3, 5, 9 and 11; (b) split LR pixels into classes by the angle of the local gradient (as RAISR does) and fit one regressor per class; (c) add noise with $\sigma = 0.01$ to the LR training images but not to the test images, and the other way round. Report PSNR for each change and explain.

Hint

(a) On the blurry test input we measured 25.94 dB for $p = 3$, 26.18 dB for $p = 5$ and 26.26 dB for $p = 7, 9$, 11: beyond 7×7 a linear model gains nothing. (b) Four angle classes (computed from the gradient of the slightly smoothed LR image) gave 26.33 dB and eight gave 26.32 dB, a small gain: classes let the filter follow edge direction, which is what makes RAISR and A+ better than one global filter. (c) Training on noisy inputs teaches the model to smooth, so it under-sharpens clean inputs; training on clean inputs teaches it to amplify fine detail, so it amplifies noise. This is the noise version of the blur mismatch in Figure 6.

5. **Posterior mean by hand.** Two HR pixels $a, b \in \{0, 1\}$ are independent and equally likely to be 0 or 1. The LR pixel is $y = (a+b)/2$ with no noise. (a) For each value of y, give the posterior mean of (a, b) and a posterior sample. (b) Compute the expected per-pixel MSE of both estimators. (c) Which estimator produces outputs that look like the HR data?

Hint

(a) $y = 0$ and $y = 1$ determine (a, b) exactly. For $y = 0.5$ (probability $1/2$) the posterior is $(0, 1)$ or $(1, 0)$ with equal probability, so the mean is $(0.5, 0.5)$ and a sample is one of the two. (b) Only $y = 0.5$ contributes. The mean has squared error 0.25 per pixel there, so its MSE is $0.5 \times 0.25 = 0.125$. A sample is right half the time and wrong by 1 on both pixels otherwise, so its MSE is $0.5{\times}0.5{\times}1 = 0.25$: exactly twice as large. (c) Only the sample: real pixels are 0 or 1, and the value 0.5 never occurs in the HR data.

Supplementary notes

These notes are optional background. Nothing in the main text or the exercises depends on them.

Example-based SR and sparse coding in detail

Plain version. Classical example-based methods keep a library of (small blurry patch, sharp patch) pairs. Sparse coding compresses that library into two matched dictionaries, so each blurry patch is described by a few "atoms", and the same few atoms in the sharp dictionary draw the sharp patch.

Freeman, Jones and Pasztor chose, for each LR patch of a new image, an HR training patch that matched it and was compatible with its neighbors [9]. Glasner, Bagon and Irani observed that patches recur within and across scales of one image, so the LR image and its downscaled copies supply LR–HR examples without external data [10]; Huang, Singh and Ahuja widened that internal search with perspective and affine transformations, which helps on urban scenes [61]. Zero-shot SR (ZSSR) brought the internal idea to deep learning: it trains a small image-specific CNN at test time on examples extracted only from the input image [69].

Yang et al. learned an LR dictionary D_ℓ and an HR dictionary D_h whose columns are paired atoms [11]. For an LR patch feature p_ℓ they solve

$$\min_{\alpha} \|D_\ell\, \alpha - p_\ell\|_2^2 + \lambda \|\alpha\|_1, \qquad \hat{p}_h = D_h\, \alpha,$$

where α is the sparse code, $\|\alpha\|_1$ encourages few non-zero coefficients, $\lambda > 0$ sets the sparsity and $\hat{p}_h$ is the HR patch estimate; overlapping patches are averaged. Zeyde, Elad and Protter simplified and sped up this pipeline [59], and anchored regression (ANR, A+) replaced the sparse-coding step by precomputed ridge regressors [13], [14], which is the model of the main text's snippet.

Three kernel-aware methods

Plain version. These methods either tell the network what the blur was, or work out the blur from the image itself.

- **SRMD** gives one network the blur kernel (reduced to a short vector) and the noise level, stretched into maps of the LR image's size and concatenated with it, so one model serves many degradations if someone supplies them [70].
- **IKC** starts from a kernel guess and corrects it in several rounds, using the artifacts a wrong kernel produces (too wide gives over-sharpening, too narrow gives over-smoothing) [71].
- **KernelGAN** trains a small GAN on the single test image: its generator learns to downscale the image so that the patches of the result cannot be told apart from those of the original, and the learned downscaler is the kernel estimate [72].

LIIF's three refinements

Plain version. LIIF looks up the nearest stored feature for each query point. Three small additions make that lookup richer, smooth, and aware of how big the output pixel is.

In the LIIF formula of the main text, **feature unfolding** concatenates the 3×3 neighborhood of each latent code into one vector, so each code describes a small region. **Local ensemble** evaluates the MLP with the four latent codes around x_q and blends the four predictions with area weights, so the output does not jump when the nearest code z^* changes. **Cell decoding** also gives the MLP the height and width of the output pixel being rendered, so it knows whether it is predicting a fine or a coarse sample [47].

Why a posterior sample costs twice the MMSE

Plain version. The posterior mean sits exactly in the middle of all plausible answers. The true image is one random point in that cloud, and a posterior sample is another, independent random point in the same cloud. Two random points are, on average, twice as far apart (in squared distance) as one random point is from the center.

Precise version. Let X be the HR image, Y the LR observation, $\mu(Y) = \mathbb{E}[X \mid Y]$ the posterior mean, and $\hat{X}$ a sample drawn from $p(x \mid Y)$ independently of X given Y. Then, conditioned on Y,

$$\begin{aligned}\mathbb{E}[\|X - \hat{X}\|^2 \mid Y] &= \mathbb{E}[\|(X - \mu) - (\hat{X} - \mu)\|^2 \mid Y] \\ &= \mathbb{E}[\|X - \mu\|^2 \mid Y] + \mathbb{E}[\|\hat{X} - \mu\|^2 \mid Y] - 2\,\mathbb{E}[(X - \mu)^\top (\hat{X} - \mu) \mid Y] \\ &= 2\,\mathbb{E}[\|X - \mu\|^2 \mid Y],\end{aligned}$$

where $\|\cdot\|$ is the Euclidean norm over all pixels. The second line expands the square. In the third, the two squared terms are equal because X and $\hat{X}$ have the same conditional distribution, and the cross term vanishes because, given Y, the two are independent and each has mean μ. Taking the expectation over Y gives $\mathbb{E}\|X - \hat{X}\|^2 = 2\,\mathbb{E}\|X - \mu(Y)\|^2$: the sampler's MSE is exactly twice the minimum MSE, a 3 dB PSNR gap. Its output distribution equals the distribution of real images, so its perceptual quality is perfect. Blau and Michaeli use this construction to show that perfect perceptual quality never costs more than a factor of 2 in MSE [2].

The same algebra gives the curve in Figure 1. For the blend $X_\alpha = \mu + \alpha(\hat{X} - \mu)$ with $0 \le \alpha \le 1$,

$$\mathbb{E}\|X - X_\alpha\|^2 = (1 + \alpha^2)\,\mathbb{E}\|X - \mu\|^2,$$

where the cross term vanishes for the same reason. The distortion grows smoothly from the MMSE value at $\alpha = 0$ to twice that value at $\alpha = 1$, while the blend's outputs move from the blurry posterior mean toward realistic samples. In our toy, the measured MSE went from 0.0072 to 0.0143.

```
import numpy as np

rng = np.random.default_rng(1)
mu = rng.standard_normal((100000, 1))                  # posterior means for many
 ↪  observations y
X = mu + rng.standard_normal(mu.shape)                 # the truth: one draw from p(x | y) =
 ↪  N(mu, 1)
X_hat = mu + rng.standard_normal(mu.shape)             # an independent posterior sample
for a in (0.0, 0.5, 1.0):
    print(f"alpha {a:.1f}: MSE {np.mean((X - (mu + a * (X_hat - mu))) ** 2):.3f}  (theory {1
     ↪  + a * a:.3f})")
```

```
alpha 0.0: MSE 1.002  (theory 1.000)
alpha 0.5: MSE 1.250  (theory 1.250)
alpha 1.0: MSE 1.996  (theory 2.000)
```

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Wang, Chen and Hoi: Deep Learning for Image Super-Resolution (TPAMI 2021)

Problem. By 2019 there were many deep SR papers, each combining choices about upsampling, architecture, loss and training, which made it hard to see which choice mattered [4].

Idea. Split the field into supervised SR, unsupervised SR and domain-specific SR (faces, hyperspectral images, video and others), and break supervised models into independent axes: where

the upsampling happens (pre-, post-, progressive, iterative up-and-down), which operator does it, how the network is designed (residual, recursive, dense, attention and others), which losses and which training strategies are used.

What it showed. Most published models are a combination of choices along these axes; the survey also tabulates the standard datasets (the counts in this chapter's test-set table come from it) and notes that the color space used for training or evaluation can change results by up to 4 dB.

What to check when you read it. Its comparisons predate transformers and diffusion models. Read the axes as a vocabulary for describing any new model, and read the future-directions section (network design, learning strategies, evaluation metrics, unsupervised SR, real-world scenarios) as the list of problems the field then worked on.

Liu et al.: Blind Image Super-Resolution, a Survey and Beyond (TPAMI 2022)

Problem. Models trained for one known degradation fail on images with an unknown one, and blind SR methods used incompatible assumptions [44].

Idea. Classify methods on two axes: explicit modeling (the classical kernel-and-noise model, estimated or given) versus implicit modeling (a degradation learned from data distributions), and external training data versus the single input image (internal statistics, as in KernelGAN and ZSSR).

What it showed. The taxonomy separates methods that look similar but assume different things, and its comparison covers both synthetic and real images, together with the datasets and earlier challenges on blind SR.

What to check when you read it. For each method, ask which degradations it can represent at all. A method that is excellent on Gaussian kernels may have no way to model JPEG or camera processing.

Chen et al.: Real-World Single Image Super-Resolution, a Brief Review (Information Fusion 2022)

Problem. Results on synthetic data overestimate how well methods super-resolve real images [68].

Idea. Review the datasets and metrics for real-world SR and group methods into four families: degradation-modeling-based, image-pairs-based (captured data such as RealSR), domain-translation-based (learning the mapping between real and synthetic domains) and self-learning-based (using the test image itself).

What it showed. A compact map of the real-world setting with comparisons on benchmark data.

What to check when you read it. Look at which metrics each comparison uses. On real images without ground truth, only no-reference metrics are available, and different ones can rank methods differently.

Moser et al.: Diffusion Models, Image Super-Resolution, and Everything (TNNLS 2025)

Problem. Diffusion models changed SR quickly, and the literature is large and hard to enter [41].

Idea. Explain the theory of diffusion models first, then organize SR methods by input domain (pixel, latent, wavelet), conditioning, guidance, corruption spaces and zero-shot use of pretrained models, with a section on domain-specific applications.

What it showed. Diffusion models give the most realistic SR, but the survey lists concrete open issues: high compute, poor comparability because papers use different data and metrics, limited explainability and color shifts.

What to check when you read it. When you compare two diffusion SR papers, check the number of sampling steps, the test data and the metrics before comparing their numbers; the survey's comparability section explains why these differ.

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DLCV 04B • Video Super-Resolution: Using Time to Recover Detail

How video super-resolution uses several misaligned low-resolution frames to recover detail one frame cannot hold: the video degradation model, optical flow and warping, flow, deformable and attention alignment, sliding-window, recurrent and transformer designs, real-world and diffusion VSR, and how results are measured.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-04b-video-super-resolution/

What you'll learn

- Why video super-resolution (VSR) is more than running an image model on every frame: neighboring frames see the scene at slightly different positions, so together they hold detail that no single frame holds, and the output must also be stable over time.
- The video degradation model (motion, blur, downsampling, noise and compression per frame), and a tested multi-frame reconstruction that beats single-frame interpolation by several dB once the motion is known.
- Optical flow and backward warping with bilinear sampling, and the three ways networks align frames: explicit flow warping, deformable convolution and attention.
- The main temporal designs, with representative methods: sliding-window (VESPCN, DUF, EDVR, TDAN), recurrent (FRVSR, RBPN), bidirectional recurrent (BasicVSR, IconVSR), grid propagation (BasicVSR++) and transformers (VRT, RVRT).
- Real-world and compressed video (COMISR, RealBasicVSR) and diffusion-based VSR (Upscale-A-Video and later models), and how VSR is evaluated: Y-channel PSNR/SSIM per frame, BI and BD settings, temporal-consistency measures, Vid4, REDS4, Vimeo-90K-T and UDM10.

The big picture

A video is a stack of photos taken a fraction of a second apart. Each photo is small and a bit blurry, but the camera or the objects move a little between shots. Video super-resolution turns such a clip into a sharper, larger clip by combining what all the nearby frames saw.

> **Explain like I'm five**
>
> Imagine looking at a painting through a window screen. The wires hide some of the painting, so one look gives you only bits of it. Now move your head a tiny bit and look again: the wires hide *different* bits. After a few small head movements you have seen almost the whole painting, as long as you remember how far you moved each time. A VSR model does exactly this: it works out how far everything moved between frames (that is the "remember how far you moved" part) and fills in the

hidden bits from the other frames. If it forgets the movement, the bits land in the wrong places and the picture gets smeared.

Why it matters: most of the pixels people look at are video. Streams drop to low resolution when bandwidth is short, old films and home videos were recorded at low resolution, and phones, surveillance cameras, satellites and microscopes record small, noisy image sequences. A good VSR model makes these sharper without the frame-to-frame shimmering of a frame-by-frame image model. VSR is also where ideas now common in video models were developed: flow-based feature alignment, deformable alignment, long-range recurrent propagation and video transformers.

This chapter builds on DLCV 04, which covers the degradation model, training on synthetic pairs, losses and the generic image metrics; we link to it instead of repeating it. Single-image super-resolution, the setting of one frame in and one frame out, is covered in DLCV 04A. Here we focus on what time adds.

Why video is not just many images

Plain version. If you upscale each frame on its own, you throw away two things: the extra information in the other frames, and the requirement that the output should not change from frame to frame unless the scene changes.

Sub-pixel motion gives new information

Suppose a camera samples a scene on a coarse grid of pixels. In one frame, details that fall between the sampling points are lost (or worse, folded into wrong, low-frequency patterns, which is called **aliasing**). If the scene shifts by a quarter of a pixel in the next frame, the same grid now samples the scene at different places. Four frames shifted by 0, $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ of a pixel together sample the scene four times more densely than any one of them (Figure 1, in one dimension; in two dimensions, a factor s needs s^2 distinct shifts). This is the basic reason multi-frame super-resolution can beat single-image super-resolution: the extra frames are new measurements, not copies [1].

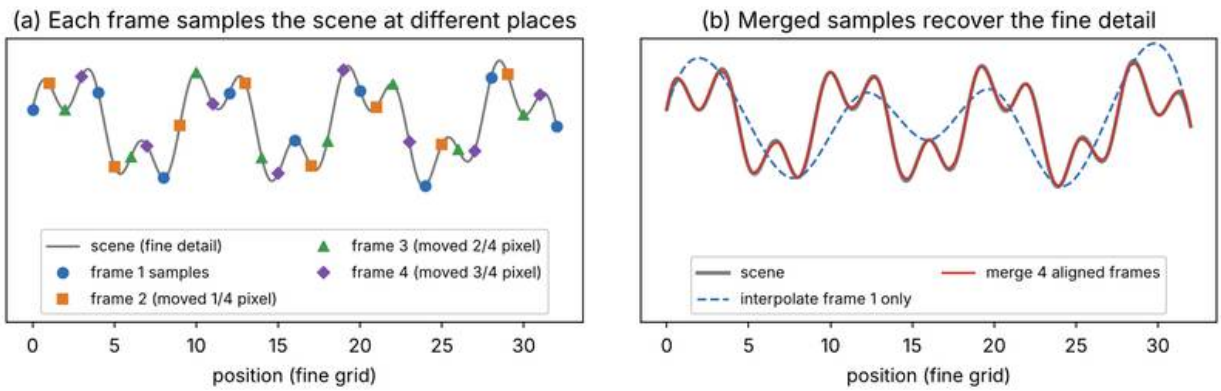


Figure 1 — One-dimensional illustration. (a) Four frames sample the same scene at positions shifted by quarter-pixel steps. (b) Interpolating one frame misses the fine detail; merging the four aligned sample sets recovers it. Synthetic signal.

Two conditions must hold. First, the motion must be **sub-pixel**: a shift by a whole number of low-resolution pixels samples the same places again and adds only noise averaging. Second, we must **know the motion** accurately enough to put each sample in the right place. Real motion is not a global shift: objects move differently, rotate, deform, appear and disappear. Estimating and compensating that motion is the core problem of VSR, and most of this chapter is about it.

Temporal redundancy

Consecutive frames are also highly redundant: most of the scene is seen again. Independent noise averages out over repeated views, and a model can reuse work done for earlier frames. Video codecs exploit the same redundancy by predicting each frame from the previous one with motion vectors (see the motion-compensated prediction in IP 08).

Temporal consistency and flicker

A single-image model, applied frame by frame, makes small independent decisions in every frame. Where it invents texture (grass, hair, water, noise-like detail), the invented texture differs from frame to frame, and the video **flickers** or "boils". Such temporal artifacts are visible even when each frame looks fine on its own. VESPCN [2] and FRVSR [3] both list temporal consistency as a main motivation for processing frames jointly instead of independently. So a VSR model has two goals: each frame should be faithful and sharp, and consecutive frames should change only as the scene changes.

The video degradation model

Plain version. Each low-resolution frame is the high-resolution frame at that moment, blurred, shrunk, made noisy and usually compressed. The frames differ because the scene moved.

One equation per frame

Extending the super-resolution model of DLCV 04 to a clip $\{x_t\}_{t=1}^{T}$ of T high-resolution (HR) frames gives

$$y_t = C((x_t * k) \downarrow_s + n_t), \qquad t = 1, \dots, T,$$

where y_t is the low-resolution (LR) frame at time t, k a blur kernel (camera optics, sensor integration, anti-aliasing filter), $*$ 2-D convolution, $\downarrow_s$ keeps every s-th pixel in each direction (s is the scale factor, usually 4 in benchmarks), n_t noise that is new in every frame, and C lossy video compression (the identity in clean benchmarks). Motion enters because each x_t is a moved version of its neighbors. Near frame t we can write

$$x_{t+j} \approx \mathcal{W}_{t \to t+j}(x_t),$$

where $\mathcal{W}_{t \to t+j}$ is the motion (warping) from frame t to frame $t + j$, j is a small frame offset, and $\approx$ hides what motion cannot explain: occluded regions, objects entering the frame, lighting changes. Combining the two equations,

$$y_{t+j} \approx C((\mathcal{W}_{t \to t+j}(x_t) * k) \downarrow_s + n_{t+j}),$$

so every neighboring LR frame is a *different* measurement of the same unknown x_t. VSR estimates x_t from $y_{t-K}, \dots, y_{t+K}$ (a window of $2K + 1$ frames) or from the whole clip.

Multi-frame reconstruction as an inverse problem

If the motions, blur and scale were all known and there were no compression, x_t could be found by least squares:

$$\hat{x}_t = \arg\min_{x} \frac{1}{2K+1} \sum_{j=-K}^{K} \| DB\, \mathcal{W}_{t \to t+j}(x) - y_{t+j} \|_2^2,$$

where B is blurring with k, D is the downsampling $\downarrow_s$, and the sum runs over the frames in the window. With one frame this problem is badly underdetermined ($s^2 = 16$ unknown HR pixels per LR pixel for $s = 4$). With many frames at different sub-pixel positions it becomes much better determined. Gradient descent on this objective repeatedly simulates the LR frames from the current estimate, compares them with the observed frames and spreads the error back onto the HR grid; this is the classical **iterative back-projection** idea introduced by Irani and Peleg [4], [1], which also inspired the RBPN architecture discussed later [5].

Optional: Classical multi-frame super-resolution (in the Supplementary notes). Fusing registered samples on the fine grid, robust (median) fusion, and when the multi-frame problem is well posed.

A toy experiment: what alignment buys

The snippet builds a 16-frame LR video from one `skimage` photo by random sub-pixel camera shifts, Gaussian blur, ×4 subsampling and noise. It then compares four reconstructions of the first frame: cubic interpolation of that frame alone; the average of all upsampled frames *without* alignment; the least-squares solver above using only the first frame; and the same solver using all

16 frames with the true shifts and with shifts **estimated from the LR frames** by sub-pixel phase correlation [6]. It runs in about ten seconds on a laptop CPU.

```
import numpy as np
from scipy import ndimage as ndi
from skimage import data, color
from skimage.registration import phase_cross_correlation

rng = np.random.default_rng(0)
s, T, blur, noise = 4, 16, 1.2, 0.01                        # scale, frames, blur sigma (HR
↪  px), noise std
hr = color.rgb2gray(data.astronaut())[64:320, 128:384]      # 256 x 256 ground-truth frame,
↪  values in [0, 1]
shifts = rng.uniform(0, s, size=(T, 2)); shifts[0] = 0      # sub-pixel camera motion of each
↪  frame (HR px)

def degrade(x, d):                                          # y_t = D B W_t x : shift, blur,
↪  keep every s-th pixel
    return ndi.gaussian_filter(ndi.shift(x, d, order=3, mode="reflect"), blur)[::s, ::s]
def degrade_T(r, d):                                        # adjoint: zero-fill, blur, shift
↪  back
    z = np.zeros((r.shape[0] * s, r.shape[1] * s)); z[::s, ::s] = r
    return ndi.shift(ndi.gaussian_filter(z, blur), -d, order=3, mode="reflect")
def upsample(y):                                            # cubic-spline interpolation on
↪  the HR grid
    u, v = np.mgrid[0:y.shape[0] * s, 0:y.shape[1] * s] / s
    return ndi.map_coordinates(y, [u, v], order=3, mode="nearest")
def psnr(x):
    c = (slice(8, -8), slice(8, -8))                        # skip a border, as benchmarks do
    return 10 * np.log10(1 / np.mean((np.clip(x, 0, 1)[c] - hr[c]) ** 2))

lr = np.stack([degrade(hr, d) + noise * rng.standard_normal((64, 64)) for d in shifts])
print("LR video:", lr.shape)

def reconstruct(frames, ds, iters=20):
    """Multi-frame least squares by gradient descent (iterative back-projection)."""
    x = np.mean([ndi.shift(upsample(f), -d, order=3, mode="reflect") for f, d in zip(frames,
    ↪  ds)], 0)
    for _ in range(iters):
        grad = np.mean([degrade_T(degrade(x, d) - f, d) for f, d in zip(frames, ds)], 0)
        x -= s * s * grad
```

```
    return x

# shifts estimated from the LR frames themselves (sub-pixel phase correlation), in HR pixels
est = np.array([-s * phase_cross_correlation(lr[0], f, upsample_factor=100)[0] for f in lr])
print(f"mean shift error: {np.abs(est - shifts).mean():.2f} HR px")

print(f"single frame, cubic interpolation {psnr(upsample(lr[0])):.2f} dB")
print(f"average of frames, no alignment   {psnr(np.mean([upsample(f) for f in lr], 0)):.2f}
 ↪  dB")
print(f"single frame, same solver         {psnr(reconstruct(lr[:1], shifts[:1])):.2f} dB")
print(f"{T} frames, true shifts            {psnr(reconstruct(lr, shifts)):.2f} dB")
print(f"{T} frames, estimated shifts       {psnr(reconstruct(lr, est)):.2f} dB")
# LR video: (16, 64, 64)
# mean shift error: 0.13 HR px
# single frame, cubic interpolation 25.50 dB
# average of frames, no alignment   20.99 dB
# single frame, same solver         25.81 dB
# 16 frames, true shifts            30.30 dB
# 16 frames, estimated shifts       29.33 dB
```

Three lessons from this toy run (all numbers are from our run, not from any paper). First, **averaging without alignment is worse than using one frame** (20.99 against 25.50 dB): misaligned frames smear the image, exactly the "forgot how far you moved" failure of the analogy. Second, the gain comes from the **frames, not the solver**: the same least-squares solver on one frame barely beats interpolation (25.81 dB), while 16 frames give 30.30 dB. Third, the motion can be **estimated from the LR frames themselves** with an average error of about an eighth of an HR pixel, which keeps most of the gain (29.33 dB). Figure 2 shows the crops and how PSNR grows with the number of frames.

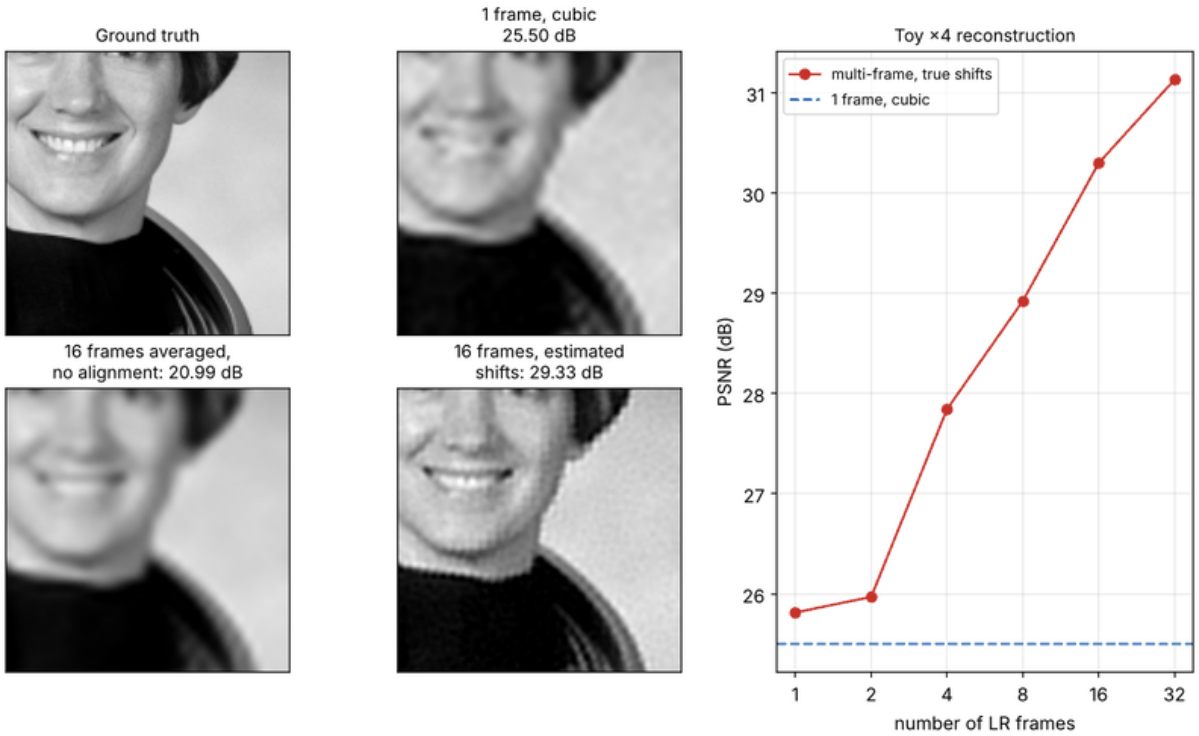


Figure 2 — The toy multi-frame experiment. Left: crops of the ground truth and of three reconstructions (PSNR on the 256 × 256 frame without an 8-pixel border). Right: PSNR of the multi-frame solver with true shifts as more LR frames are used. Values from our run of the figure script.

Drag the slider to compare single-frame interpolation with the 16-frame reconstruction on an enlarged crop.



1 frame, cubic



16 frames, estimated shifts

Slider comparing a blurry enlarged crop of the astronaut's smile from one frame with a sharper crop from sixteen aligned frames, where the teeth and the edge of the hair are clearer.

Real videos break the toy's assumptions (global shifts, known blur, no occlusion or compression). Deep VSR keeps the structure (estimate motion, align, fuse, reconstruct) and learns each part from data. Figure 3 shows the pipeline that almost every method follows in some form; BasicVSR [7] names its four components **propagation, alignment, aggregation and upsampling**.

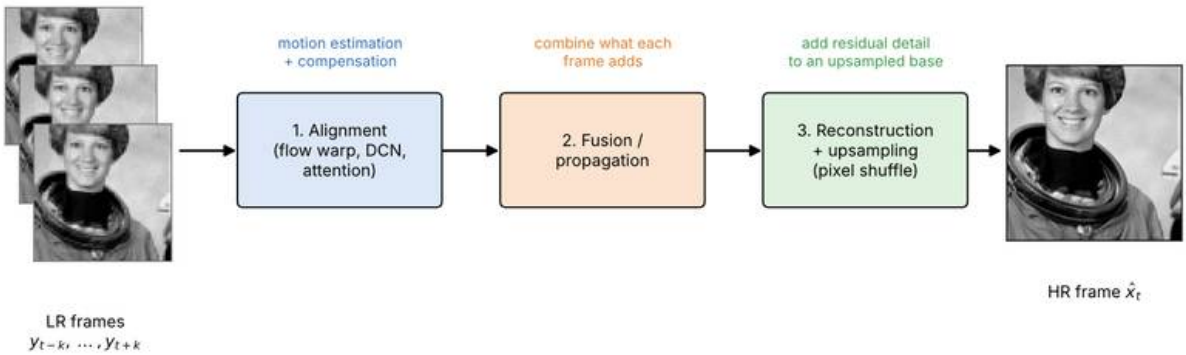


Figure 3 — The generic VSR pipeline. Neighboring LR frames (or features) are aligned to the current frame, fused or propagated over time, and reconstructed at the higher resolution, usually as a residual added to an upsampled base image.

Motion: optical flow and warping

Plain version. Optical flow is a map that says, for every pixel, where that bit of the scene went in the next frame. Warping uses that map to move one frame onto another, so that the same object sits at the same pixel in both.

Optical flow

The classical starting point is **brightness constancy**: a point keeps its brightness as it moves,

$$I(u + \delta u,\, v + \delta v,\, t + 1) = I(u, v, t),$$

where $I(u, v, t)$ is the intensity at column u, row v and time t, and $(\delta u, \delta v)$ is the displacement of that point between frames t and $t + 1$. The field of displacements over all pixels is the **optical flow** $F(p) = (\delta u(p), \delta v(p))$ for each pixel p. One equation per pixel cannot determine two unknowns

(the "aperture problem": along a straight edge only the motion across the edge is visible), so every method adds an assumption that neighboring pixels move similarly. Horn and Schunck [8] add a global smoothness penalty; Farnebäck's method [9], available in OpenCV, fits a quadratic polynomial to each neighborhood and estimates the displacement from how the polynomial moves.

Deep flow estimators learn this mapping from data. SPyNet [10] estimates large motions coarse to fine: at each level of an image pyramid it warps one image by the current flow and a small network predicts the remaining, small update; it is compact and is the flow module of BasicVSR [7]. RAFT [11] builds correlation volumes between all pairs of pixels and refines a flow field iteratively with a recurrent unit. BasicVSR, for example, starts from a pretrained SPyNet and fine-tunes it together with the rest of the model [7].

Optional: Brightness constancy and the classical flow solvers (in the Supplementary notes). The linearized flow equation, the aperture problem, and what global (Horn–Schunck), local-window and polynomial-expansion (Farnebäck) solvers add.

Backward warping with bilinear sampling

To align frame $t+1$ to frame t, we use the flow from t to $t+1$ and **pull** values: for every pixel p of the reference frame, look up where that point is in the other frame and read the value there,

$$\hat{I}_{t+1\to t}(p) = I_{t+1}(p + F_{t\to t+1}(p)),$$

where $\hat{I}_{t+1\to t}$ is frame $t+1$ warped into the coordinates of frame t and $F_{t\to t+1}(p)$ is the displacement of pixel p. This is **backward warping**: every output pixel gets exactly one value, with no holes. (Forward warping, pushing each source pixel to its new place, leaves holes and collisions.) The position $q = p + F(p)$ is usually not on the pixel grid, so the value is interpolated **bilinearly** from the four surrounding pixels:

$$I(q) = (1-a)(1-b)\,I_{00} + a(1-b)\,I_{10} + (1-a)\,b\,I_{01} + a\,b\,I_{11},$$

where $q = (u_0 + a,\, v_0 + b)$ with integer (u_0, v_0) and fractions $a, b \in [0, 1)$, and I_{ij} is the value at $(u_0 + i,\, v_0 + j)$. Bilinear sampling is differentiable with respect to both the image values and the flow, so a network can learn the flow end to end through the warp; this "bilinear sampler" is the core of spatial transformer networks [12] and of every flow-based VSR model.

The snippet below implements backward warping by hand, checks it against `cv2.remap`, builds two frames in which the background moves by $(1.5, -2)$ pixels and a textured square moves by $(6, 4)$ pixels, and warps the second frame onto the first with the true flow and with Farnebäck flow from OpenCV.

```
import numpy as np, cv2
from scipy import ndimage as ndi
from skimage import data, transform

def bilinear_warp(img, flow):
```

```
    """Backward warping: out(p) = img(p + flow(p)), bilinear sampling. flow[..., 0] = dx,
     ↪  flow[..., 1] = dy."""
    H, W = img.shape
    yy, xx = np.mgrid[0:H, 0:W].astype(np.float64)
    x = np.clip(xx + flow[..., 0], 0, W - 1); y = np.clip(yy + flow[..., 1], 0, H - 1)
    x0 = np.minimum(np.floor(x).astype(int), W - 2); y0 =
     ↪  np.minimum(np.floor(y).astype(int), H - 2)
    ax, ay = x - x0, y - y0                                  # fractional parts = bilinear
     ↪  weights
    return ((1 - ay) * ((1 - ax) * img[y0, x0] + ax * img[y0, x0 + 1])
            + ay * ((1 - ax) * img[y0 + 1, x0] + ax * img[y0 + 1, x0 + 1]))

# two frames: the background moves by (dx, dy) = (1.5, -2.0) px, a textured square moves by
 ↪  (6, 4) px
bg = transform.resize(data.camera() / 255.0, (256, 256), anti_aliasing=True)
obj = transform.resize(data.coins() / 255.0, (60, 60), anti_aliasing=True)
def frame(t):
    f = ndi.shift(bg, (-2.0 * t, 1.5 * t), order=3, mode="reflect")   # ndi.shift takes
     ↪  (rows, cols)
    r, c = 90 + 4 * t, 100 + 6 * t
    f[r:r + 60, c:c + 60] = obj
    return f
I0, I1 = frame(0).astype(np.float32), frame(1).astype(np.float32)

# true backward flow from frame 0 to frame 1: where is pixel p of frame 0 found in frame 1?
flow = np.zeros((256, 256, 2), np.float32); flow[..., 0], flow[..., 1] = 1.5, -2.0
flow[90:150, 100:160] = (6.0, 4.0)
mine = bilinear_warp(I1, flow)
H, W = I0.shape; yy, xx = np.mgrid[0:H, 0:W].astype(np.float32)
ocv = cv2.remap(I1, xx + flow[..., 0], yy + flow[..., 1], cv2.INTER_LINEAR,
 ↪  borderMode=cv2.BORDER_REPLICATE)
print(f"max |ours - cv2.remap| = {np.abs(mine - ocv)[5:-5, 5:-5].max():.4f}")

u8 = lambda a: (np.clip(a, 0, 1) * 255).round().astype(np.uint8)        # Farneback expects
 ↪  8-bit images
est = cv2.calcOpticalFlowFarneback(u8(I0), u8(I1), None, pyr_scale=0.5, levels=4, winsize=21,
                                   iterations=5, poly_n=7, poly_sigma=1.5, flags=0)
m0 = np.zeros((H, W), np.float32); m0[90:150, 100:160] = 1          # square in frame 0
m1 = np.zeros((H, W), np.float32); m1[94:154, 106:166] = 1          # square in frame 1
occl = (bilinear_warp(m1, flow) > 0) & (m0 == 0)                  # background of frame 0
 ↪  hidden in frame 1
inner = (slice(8, -8), slice(8, -8))
for name, img in [("no alignment", I1), ("warped, true flow", mine), ("warped, Farneback",
 ↪  bilinear_warp(I1, est))]:
    e = np.abs(img - I0)
    print(f"{name:18s} mean |error| {e[inner].mean():.4f}   on occluded pixels
     ↪  {e[occl].mean():.4f}")
print("Farneback median flow (dx, dy): square", np.median(est[100:140, 110:150], axis=(0,
 ↪  1)).round(2),
      " background", np.median(est[200:240, 20:60], axis=(0, 1)).round(2))
# max |ours - cv2.remap| = 0.0000
# no alignment       mean |error| 0.0562   on occluded pixels 0.2367
# warped, true flow  mean |error| 0.0089   on occluded pixels 0.2617
# warped, Farneback  mean |error| 0.0133   on occluded pixels 0.1733
```

```
# Farneback median flow (dx, dy): square [6. 4.]  background [ 1.42 -1.88]
```

Warping with the true flow cuts the mean difference between the frames by a factor of about six, and Farnebäck's estimate gets most of the way there (our run). But look at the **occluded pixels**: the strip of background that the square covers in frame $t + 1$. There, even the *true* flow gives a large error, because the content simply is not in frame $t + 1$; it is hidden behind the square. No flow can fix that. A VSR model must learn to trust the warped frame where it is consistent and to ignore it where it is not (Figure 4).

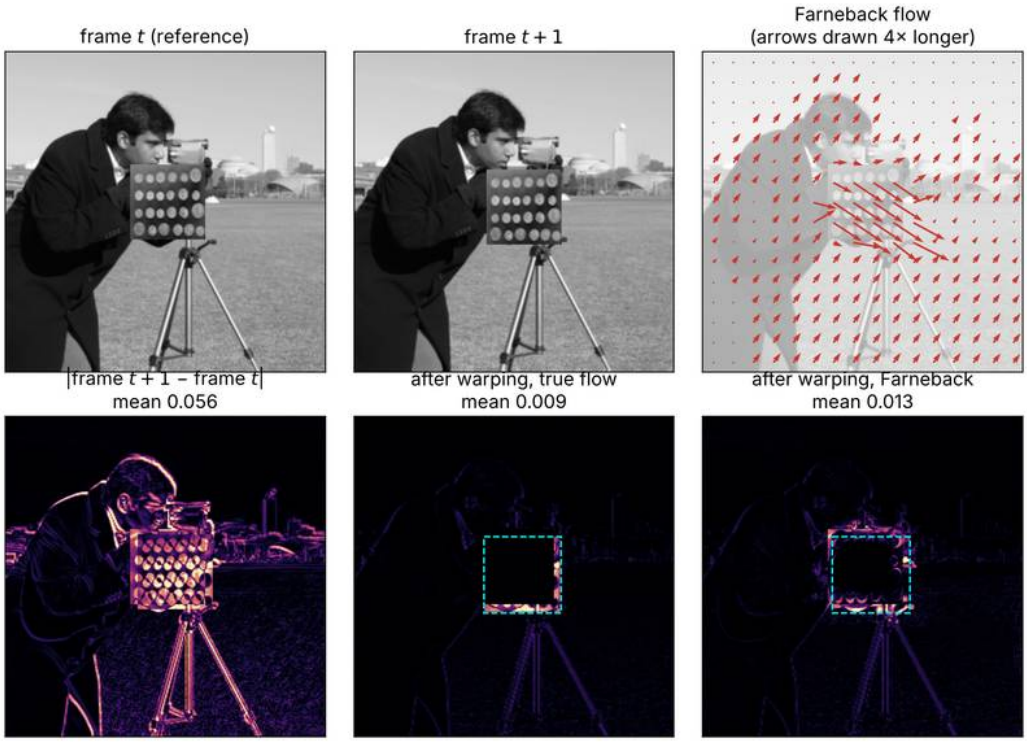


Figure 4 — Backward warping. Warping frame t+1 onto frame t removes most of the difference, but not in the occluded strip (dashed box) where background visible in frame t is hidden in frame t+1. In our run the Farneback errors concentrate at the motion boundary. Mean absolute errors from our run.

Alignment strategies

Plain version. Before frames can be combined, the model must line them up. There are three families: move the frames with a flow map (explicit), let the convolution itself learn where to look (deformable), or let every position search for related content (attention).

Explicit alignment: flow-based warping

The direct approach estimates a flow from each neighbor to the reference frame and warps the neighbor with it.

- **VESPCN** [2] (CVPR 2017) trains a fast multi-resolution spatial-transformer motion-compensation module end to end with a spatio-temporal version of the sub-pixel convolution (pixel shuffle) network ESPCN [13].
- **TOFlow** [14] (IJCV 2019) trains the flow network jointly with the restoration network, so the flow is **task-oriented**: it moves whatever makes the output best, even where that differs from physical motion. The paper also introduced Vimeo-90K.
- **Image versus feature warping.** Early methods warp the LR *images*. Inaccurate flow then produces blurry or wrong warped images, and the loss of detail cannot be undone later. Later methods compute flow on images but warp *feature maps*, so subsequent layers can correct small misalignments; the BasicVSR ablation found feature warping better than image warping [7].

Flow-based alignment is interpretable and works well for moderate motion. Its weaknesses are the cost of the flow network, errors at occlusions and motion boundaries, and the fact that one flow vector per pixel can point to only one place.

Implicit alignment: deformable convolution

A normal 3×3 convolution reads its input on a fixed grid around each pixel. A **deformable convolution** [15] adds a learned offset to each of the sampling positions:

$$y(p) = \sum_{i=1}^{K} w_i \, x(p + p_i + \Delta p_i(p)),$$

where $y(p)$ is the output at pixel p, K the number of kernel taps (9 for 3×3), w_i the weight of tap i, p_i its fixed grid position (for example $(-1, -1)$), $\Delta p_i(p)$ a learned, real-valued offset predicted by a small network for every pixel and tap, and $x(\cdot)$ is read with bilinear sampling. DCNv2 [16] also multiplies each tap by a learned modulation weight in $[0, 1]$, which lets the layer down-weight unreliable positions.

For alignment, the offsets are predicted from the features of the reference frame and a neighbor, and the deformable convolution is applied to the neighbor's features. **TDAN** [17] aligns each supporting frame to the reference this way at the feature level without computing optical flow. **EDVR** [18] applies deformable alignment in a **Pyramid, Cascading and Deformable (PCD)** module, coarse to fine over a feature pyramid to handle large motions, and fuses the aligned features with **temporal and spatial attention (TSA)**; it won all four tracks of the NTIRE 2019 video restoration

challenges.

Chan et al. [19] showed that a deformable convolution can be decomposed into several spatial warps (one per kernel tap, each with its own offset field) followed by a convolution. So deformable alignment is a generalization of flow warping with several flow fields instead of one; this **offset diversity** is why it aligns better. The same work, and BasicVSR++ [20], report that deformable alignment is unstable to train: offsets can grow very large ("offset overflow"). BasicVSR++ therefore predicts the deformable offsets as a **residual on top of optical flow** (flow-guided deformable alignment), combining the stability of flow with the diversity of deformable sampling.

Optional: Deformable convolution as warping plus convolution (in the Supplementary notes). The decomposition in two lines, a tested example, and why several offset fields help at occlusions.

Alignment without an explicit motion step

Some models skip motion estimation altogether and let the network absorb motion. **DUF** [21] predicts, for every output pixel, a dynamic upsampling filter and a residual from a spatio-temporal neighborhood of the input frames, so motion is handled implicitly by the content-dependent filters. **PFNL** [22] fuses frames progressively and uses a non-local operation, which compares every position with every other, instead of motion estimation and compensation. 3-D convolutions over (time, height, width) are a third option. These methods avoid flow errors, but their effective search range is bounded by the receptive field, so large motions are harder.

Attention and transformers

Attention lets each position compare itself with many candidate positions in other frames and take a weighted average of the best matches, which is a soft, learned version of motion search.

VRT [23] divides the video into short clips and uses **temporal mutual self-attention (TMSA)**: mutual attention between two frames performs joint motion estimation, alignment and fusion, while self-attention extracts features; the clip partition shifts in alternate layers so information crosses clip borders, and a **parallel warping** module adds flow-based warping of neighboring features. VRT is applied to video SR, deblurring, denoising, frame interpolation and space-time SR. **RVRT** [24] uses **guided deformable attention**: starting from optical flow, it predicts several relevant locations in the previous clip and aggregates their features with attention weights.

Architectures by temporal design

Plain version. Once frames can be aligned, the question is which frames each output gets to see, and how information travels along the video. There are four main answers, shown in Figure 5.

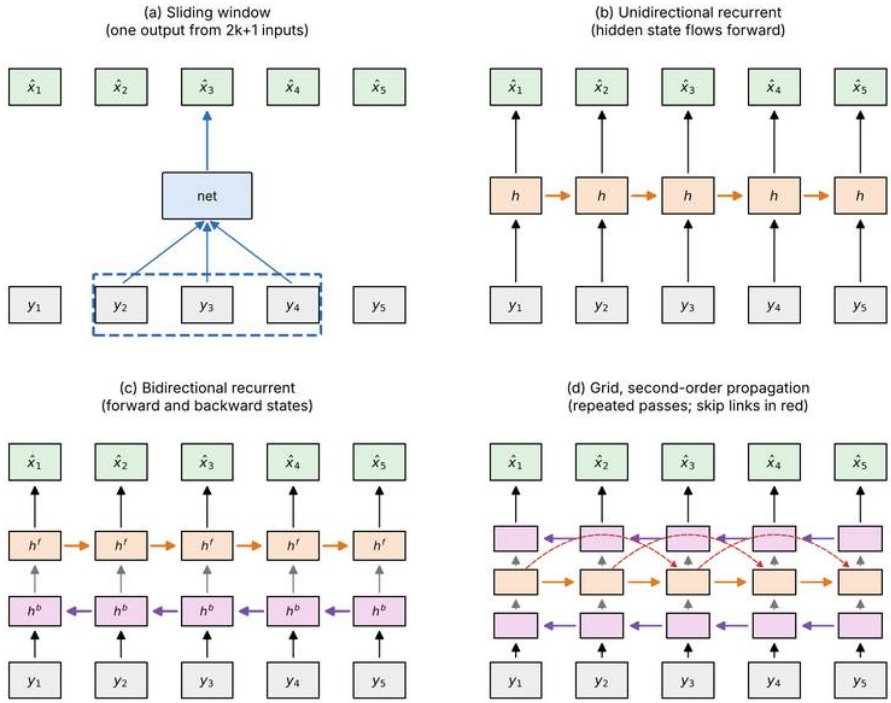


Figure 5 — Temporal designs. (a) Sliding window: each output is computed from a fixed window of 2K + 1 frames. (b) Unidirectional recurrent: a hidden state carries information forward. (c) Bidirectional recurrent: separate forward and backward states; each output combines both at its time step. (d) Grid propagation with second-order connections (BasicVSR++): repeated backward and forward passes, with links that also skip one frame.

Sliding window

A **sliding-window** model takes $2K + 1$ consecutive LR frames and produces one HR frame, the center one, then moves the window by one frame. VESPCN, DUF, TDAN and EDVR are sliding-window models. It is simple and parallel, and errors cannot accumulate over time. FRVSR [3] points out two weaknesses: each input frame is processed and warped $2K + 1$ times, and each output is estimated independently, which limits temporal consistency. A fixed window also cannot use information from farther away; the BasicVSR authors measured that cutting the test sequences into shorter independent segments lowers PSNR, most strongly at segment ends [7].

Recurrent

A **recurrent** model keeps a hidden state h_t that summarizes the past and updates it frame by frame:

$$h_t = F(y_t,\ \mathcal{W}(h_{t-1}, F_{t\to t-1})), \qquad \hat{x}_t = U(h_t),$$

where F is a network that combines the current LR frame y_t with the previous state warped to frame t by the flow $F_{t\to t-1}$ (the warp $\mathcal{W}$ is the backward warping of the previous section), and U is the reconstruction and upsampling head. Because the state is aligned before reuse, information can travel over many frames at a fixed cost per frame. (For recurrent networks in general, see ML 03a.) **FRVSR** [3] feeds the *previous HR output*, warped by flow and rearranged to LR size, into the network for the next frame, which encourages temporally consistent results and needs only one warp per step. **RBPN** [5] is a hybrid: within a window it integrates the neighboring frames one at a time with a recurrent encoder–decoder inspired by back-projection, feeding the estimated motion as an input instead of warping the frames.

A unidirectional recurrent model sees only the past. Early frames receive little information, and errors in the state (from a bad flow, an occlusion or a scene cut) are carried into later frames: **error accumulation**.

Bidirectional recurrent

BasicVSR [7] runs two recurrent branches, one forward and one backward in time:

$$h^b_t = F_b(y_t, y_{t+1}, h^b_{t+1}), \qquad h^f_t = F_f(y_t, y_{t-1}, h^f_{t-1}), \qquad \hat{x}_t = U(h^f_t, h^b_t),$$

where h^b_t and h^f_t are the backward and forward features at frame t, F_b and F_f the two branches (each estimates flow with SPyNet between its two input frames, warps the propagated feature with it and refines it with residual blocks), and U concatenates both features and upsamples them with pixel shuffle. Every output now sees the whole clip, past and future. The paper's ablations found a clear PSNR drop for a unidirectional variant at early frames and a larger drop without alignment, which supports its recipe: bidirectional propagation plus flow-based feature alignment is enough to be a strong baseline. Its extension **IconVSR** adds an *information-refill* module that re-extracts features from sparse keyframes to limit error accumulation at occlusions and boundaries, and *coupled propagation* that feeds the backward branch's output into the forward branch.

The price is latency: the backward branch needs the *future* frames, so the whole clip (or a long chunk) must be available before the first output. That suits offline processing and streaming with a buffer, not live video.

Optional: BasicVSR and IconVSR (in the Paper guides). What the four-component analysis showed, the ablations to look at and the limits of the setup.

Grid and second-order propagation

BasicVSR++ [20] makes propagation richer in two ways. **Grid propagation** passes features backward and forward several times in alternation, so each frame's features are refined repeatedly with information from the whole sequence. **Second-order connections** let the state at frame t use frames $t-1$ *and* $t-2$ (a second-order Markov chain instead of first-order), so information can bypass one bad frame, which helps at occlusions and fine structures. With the flow-guided deformable alignment described above, BasicVSR++ reports a 0.82 dB PSNR gain over BasicVSR at a similar parameter count, and it won three tracks and placed second in a fourth in the NTIRE 2021 video super-resolution and compressed video enhancement challenges [20].

Transformers and recurrent–transformer hybrids

VRT [23] processes all frames **in parallel** with long-range temporal attention at several scales, which is powerful but heavy. RVRT [24] describes the trade-off directly: parallel models fuse temporal information well but have a large model size and high memory use, while recurrent models share parameters across frames and are compact but are weaker at long-range modeling and harder to parallelize. RVRT processes neighboring frames as a clip in parallel *inside* a globally recurrent framework, using the inferred previous clip to estimate the next, with guided deformable attention for clip-to-clip alignment.

The table summarizes the representative methods of this chapter.

Method	Venue	Temporal design	Alignment
VESPCN [2]	CVPR 2017	sliding window	flow (spatial transformer), image level
FRVSR [3]	CVPR 2018	recurrent (previous HR output)	flow, image level
DUF [21]	CVPR 2018	sliding window	none (dynamic filters)
TOFlow [14]	IJCV 2019	sliding window	task-oriented flow
EDVR [18]	CVPRW 2019	sliding window	deformable (PCD) + attention fusion
RBPN [5]	CVPR 2019	recurrent over a window	motion as input, back-projection
PFNL [22]	ICCV 2019	sliding window (progressive fusion)	none (non-local)
TDAN [17]	CVPR 2020	sliding window	deformable, feature level
BasicVSR / IconVSR [7]	CVPR 2021	bidirectional recurrent	flow, feature level
BasicVSR++ [20]	CVPR 2022	grid, second-order	flow-guided deformable
RVRT [24]	NeurIPS 2022	clip-recurrent transformer	guided deformable attention
VRT [23]	TIP 2024	parallel transformer	mutual attention + parallel warping

Real-world video and compression

Plain version. Benchmark LR videos are made by one clean, known recipe. Real videos are noisy, blurred in unknown ways and, above all, compressed. A model trained on the clean recipe often sharpens the compression artifacts instead of the scene.

Compression is the main real-world degradation

Most video is stored and streamed with block-based codecs such as H.264/AVC [25], which predict each block from earlier frames with motion vectors and quantize the residual. The resulting artifacts (blocking, ringing, smeared texture, mosquito noise around edges) differ from frame to frame: frames coded without reference to others (intra frames) look different from predicted frames, and the quality changes with the bitrate. These artifacts are not independent noise and they move with the content, so a VSR model can mistake them for real detail and propagate them.

COMISR [26] designs VSR for compressed input, with bidirectional recurrent warping, detail-preserving flow estimation and a Laplacian enhancement module that take compression properties such as the position of intra frames into account, and evaluates over a range of compression rates.

Real-world VSR: RealBasicVSR

RealBasicVSR [27] (CVPR 2022) found that long-term propagation, which helps for mild degradations, can **exaggerate severe artifacts**: noise and compression artifacts are propagated and amplified along with the useful information. Its fix is an image **pre-cleaning** module applied to each frame before a BasicVSR network. It also introduced **VideoLQ**, a set of real low-quality videos without ground truth, evaluated with no-reference metrics and visual comparison.

Optional: RealBasicVSR (in the Paper guides). The three trade-offs it identifies and how to read its experiments.

The degradation pipelines are the video versions of the blind pipelines in DLCV 04: random blur, resizing, noise and compression, with the extra choice of applying the same or different parameters to all frames of a clip.

Diffusion-based VSR

Pretrained text-to-image diffusion models produce very realistic textures for real-world image SR (see DLCV 04). In video, their randomness is a problem: independent sampling per frame produces different textures in every frame, which is flicker at its worst. Recent methods therefore add

temporal machinery to a pretrained generative model: temporal layers plus flow-guided propagation of latents across the sequence (Upscale-A-Video [28], CVPR 2024), a motion-guided loss that steers the sampling path (MGLD-VSR [29], ECCV 2024), a text-to-*video* model that already models motion (STAR [30], ICCV 2025), a diffusion transformer for videos of arbitrary length and resolution (SeedVR [31], CVPR 2025), and one-step distillation to cut the cost (DLoRAL [32], NeurIPS 2025).

Optional: Diffusion-based VSR models (in the Supplementary notes). What each of these five models adds to a pretrained diffusion model to keep the video consistent.

These models give the most convincing textures on real videos, but they inherit the image-level risk of **hallucinated detail** (plausible content that was never in the scene) and add a video-level risk: detail that is plausible in each frame but not consistent across frames. They are also far more expensive than CNN or transformer VSR.

Training a VSR model

Plain version. Training is the same as for image restoration, except that each training example is a short clip, not a single image.

Pairs are made by degrading HR clips with the model above: the bicubic (BI) or blur-downsampling (BD) recipe for benchmarks (see Evaluation), or a randomized real-world pipeline. Each step samples a short clip and random crops at the same location in all frames; flips and rotations are applied identically to all frames, and the frame order can be reversed as temporal augmentation (BasicVSR reverses Vimeo-90K's 7-frame sequences this way [7]). Recurrent models are trained on longer clips (BasicVSR's REDS setting uses 15-frame inputs) so that propagation is exercised; RealBasicVSR's finding that longer sequences beat larger batches points the same way [27]. In BasicVSR the pretrained flow network and feature extractor are frozen for the first 5,000 iterations, and the flow network is then trained with a smaller learning rate than the rest [7].

The loss is usually an L1-type pixel loss on every output frame (BasicVSR uses the smooth version $\rho(z) = \sqrt{z^2 + \varepsilon^2}$ with a small constant ε, where z is the difference between an output and a ground-truth pixel [7]), with perceptual and adversarial terms for real-world models; DLCV 04 covers these. Some methods add a **temporal loss** that penalizes differences between consecutive outputs after motion compensation, such as the short-term warping loss of Lai et al. [33], which compares each output frame with the previous output warped by optical flow, masked where the warp is unreliable.

Evaluation

Plain version. Score every frame against the ground truth, average the scores, and say exactly how the LR frames were made. Then check the video in motion, because per-frame scores cannot see flicker.

Per-frame fidelity: Y-channel PSNR and SSIM

Most VSR papers report PSNR and SSIM [34] (defined in DLCV 04) on the **luminance (Y) channel**, computed for every frame and averaged over the frames of a clip and then over clips. The Y channel follows ITU-R BT.601 [35]: luma $E_Y = 0.299E_R + 0.587E_G + 0.114E_B$ from gamma-corrected RGB in $[0, 1]$, quantized on the 8-bit "studio" scale as $Y = 219E_Y + 16$. The convention is not universal: in the BasicVSR and BasicVSR++ comparison tables, all results are computed on Y except REDS4, which is computed on RGB [7], [20]. Mixing conventions changes numbers by several dB, as the snippet shows.

```
import numpy as np
from skimage import data, color

def rgb_to_y(rgb):
    """BT.601 luma on the 8-bit studio scale: Y = 16 + 219 * (0.299 R + 0.587 G + 0.114 B),
     ↪  RGB in [0, 1]."""
    return 16 + 65.481 * rgb[..., 0] + 128.553 * rgb[..., 1] + 24.966 * rgb[..., 2]

def psnr(a, b, peak=255.0):
    return 10 * np.log10(peak ** 2 / np.mean((a.astype(np.float64) - b) ** 2))

rng = np.random.default_rng(0)
gt = np.stack([np.roll(data.astronaut(), 2 * t, axis=1) for t in range(5)]) / 255.0   # 5
 ↪  ground-truth frames
out = np.clip(gt + rng.normal(0, 0.02, gt.shape) * np.array([1.0, 0.3, 1.5]), 0, 1)  #
 ↪  unequal error per channel

y_scores = [psnr(rgb_to_y(o), rgb_to_y(g)) for o, g in zip(out, gt)]
rgb_scores = [psnr(255 * o, 255 * g) for o, g in zip(out, gt)]
print("matches skimage rgb2ycbcr:", np.allclose(rgb_to_y(gt[0]), color.rgb2ycbcr(gt[0])[...,
 ↪  0]))
print(f"Y-channel PSNR, mean of frames: {np.mean(y_scores):.2f} dB")
print(f"RGB PSNR,       mean of frames: {np.mean(rgb_scores):.2f} dB")
# matches skimage rgb2ycbcr: True
# Y-channel PSNR, mean of frames: 43.74 dB
# RGB PSNR,       mean of frames: 33.94 dB
```

The same output scores about 10 dB higher on Y than on RGB here (our run), partly because Y is a weighted average of the three channels, so independent channel errors partly cancel, and partly

because Y spans only 219 of the 255 levels while the peak in the formula stays 255. Always state the channel, the border crop and whether the frames were rounded to 8 bits.

Degradation settings: BI and BD

Two LR recipes dominate the clean benchmarks, and results under them are not comparable:

- **BI (bicubic):** downsample each HR frame by ×4 with MATLAB's `imresize`, which blurs with a cubic filter and subsamples. The Vimeo-90K SR benchmark was built this way [14].
- **BD (blur-downsampling):** blur each HR frame with a Gaussian of standard deviation $\sigma = 1.6$, then keep every fourth pixel, as specified in BasicVSR [7].

BD includes a stronger, known blur, so a model must deblur as well as upsample. The BasicVSR tables report REDS4 under BI only, UDM10 under BD only, and Vimeo-90K-T and Vid4 under both [7]. When you read a comparison table, check the setting of each column first.

Temporal consistency

Per-frame PSNR is blind to flicker: errors that change randomly from frame to frame can give the same average PSNR as stable errors. Three kinds of measures are used:

- **Warping error** [33]: warp output frame $t+1$ onto frame t with optical flow and measure the masked difference,

$$E_{\text{warp}}(V_t, V_{t+1}) = \frac{1}{\sum_i M_t^{(i)}} \sum_{i=1}^{N} M_t^{(i)} \|V_t^{(i)} - \hat{V}_{t+1}^{(i)}\|_2^2,$$

where $V_t^{(i)}$ is pixel i of output frame t, $\hat{V}_{t+1}$ is frame $t+1$ warped to frame t, N the number of pixels and $M_t^{(i)} \in \{0,1\}$ a non-occlusion mask; the video score averages over all consecutive pairs. Its weakness: a blurry, detail-free video is very stable and scores well.

- **tOF and tLP** (TecoGAN [36]) compare the temporal behavior of the output with that of the ground truth instead of demanding zero change:

$$\text{tOF} = \|\text{OF}(b_{t-1}, b_t) - \text{OF}(g_{t-1}, g_t)\|_1, \qquad \text{tLP} = |\text{LP}(b_{t-1}, b_t) - \text{LP}(g_{t-1}, g_t)|,$$

where b_t and g_t are the output and ground-truth frames, OF is optical flow estimated with Farnebäck's method and LP is the LPIPS perceptual distance [37]. The TecoGAN authors introduced them because a plain warped-difference measure "can be easily deceived" by blurry results such as bicubic interpolation.

- **Temporal profiles**: one row of pixels followed through time and stacked into an image; speckle and streaks reveal flicker. VESPCN used them for qualitative comparison [2].

The snippet builds a toy scene with known motion and four "outputs": independent per-frame invented texture, the same outputs smoothed recursively along the motion, the same smoothing with the motion ignored, and a blurry output. Flicker is measured in the spirit of tOF: the temporal change of the output after motion compensation, minus that of the ground truth.

```
import numpy as np
from scipy import ndimage as ndi
from skimage import data, transform

rng = np.random.default_rng(0)
T, v = 12, (1, 2)                                        # 12 frames, the scene moves 1 px
 ↪  down and 2 px right per frame
base = transform.resize(data.camera() / 255.0, (256, 256), anti_aliasing=True)
gt = np.stack([np.roll(base, (v[0] * t, v[1] * t), axis=(0, 1)) for t in range(T)])
warp_prev = lambda f: np.roll(f, v, axis=(0, 1))          # exact motion compensation for
 ↪  this toy scene

def fake_detail():                                        # zero-mean "hallucinated
 ↪  texture", new every frame
    n = ndi.gaussian_filter(rng.standard_normal(base.shape), 0.8)
    return 0.03 * n / n.std()

per_frame = gt + np.stack([fake_detail() for _ in range(T)])  # each frame restored
 ↪  independently
smoothed, unaligned, blurry = per_frame.copy(), per_frame.copy(),
 ↪  np.stack([ndi.gaussian_filter(g, 1.5) for g in gt])
for t in range(1, T):
    smoothed[t] = 0.3 * per_frame[t] + 0.7 * warp_prev(smoothed[t - 1])   # recursive filter
     ↪  along the motion
    unaligned[t] = 0.3 * per_frame[t] + 0.7 * unaligned[t - 1]           # same filter,
     ↪  motion ignored

c = (slice(None), slice(16, -16), slice(16, -16))          # ignore borders (np.roll wraps
 ↪  around)
def psnr(out):
    return np.mean([10 * np.log10(1 / np.mean((o - g) ** 2)) for o, g in zip(out[c], gt[c])])
def flicker(out):
    """Mean |temporal change of output - temporal change of ground truth| after motion
     ↪  compensation."""
    d_out = out[1:] - np.stack([warp_prev(f) for f in out[:-1]])
    d_gt = gt[1:] - np.stack([warp_prev(f) for f in gt[:-1]])
    return np.abs(d_out - d_gt)[c].mean()

for name, out in [("per-frame", per_frame), ("aligned smoothing", smoothed),
                  ("unaligned smoothing", unaligned), ("blurry output", blurry)]:
    print(f"{name:20s} PSNR {psnr(out):5.2f} dB   flicker {flicker(out):.4f}")
# per-frame            PSNR 30.50 dB   flicker 0.0338
# aligned smoothing    PSNR 36.40 dB   flicker 0.0082
# unaligned smoothing  PSNR 21.74 dB   flicker 0.0223
```

```
# blurry output        PSNR 25.99 dB   flicker 0.0000
```

Read the four lines together (all from our toy run). Propagating information **along the motion** reduces flicker and, in this toy, also improves PSNR, because the invented texture is zero-mean noise that averages out; real hallucinated texture is not, so do not expect PSNR gains of this size.

Propagating **without alignment** ghosts the moving scene: PSNR collapses and flicker stays high.

And the **blurry output** has zero flicker but poor PSNR, which is the TecoGAN warning in miniature: always report a temporal measure *together with* a fidelity or perceptual measure. Figure 6 shows the same effects as motion-compensated temporal profiles.

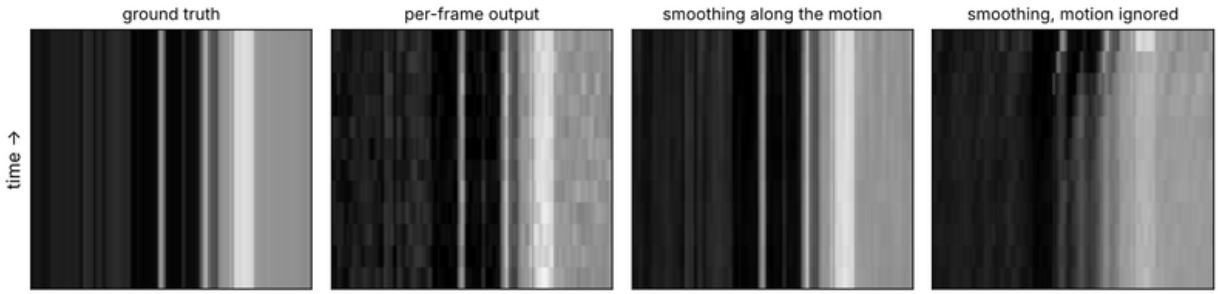


Figure 6 — Motion-compensated temporal profiles of the toy scene: one scene row followed over 12 frames (time runs downward). Flicker appears as row-to-row speckle; unaligned smoothing smears the structure.

Datasets and test sets

Dataset	What it is	Typical use
Vimeo-90K [14]	89,800 video clips (720p or higher) from Vimeo; the SR benchmark uses 7-frame sequences at 448 × 256, with ×4 LR inputs of 112 × 64 made by MATLAB `imresize`	training; test split **Vimeo-90K-T**
REDS [38]	300 sequences of 100 frames at 720 × 1280 (240 training, 30 validation, 30 test), captured for the NTIRE 2019 video challenges, with large and complex motion	training; **REDS4** test set
REDS4	four REDS training clips (000, 011, 015, 020) held out for testing, following EDVR [7]	test under BI
Vid4 [39]	four short sequences from Liu and Sun's Bayesian adaptive VSR work	test under BI and BD
UDM10	ten test sequences released with PFNL [22]	test under BD

Dataset	What it is	Typical use
VideoLQ [27]	real-world low-quality videos without ground truth	real-world evaluation (no-reference metrics, visual)

The Liu et al. survey notes that test videos evolved from Vid4 toward REDS as methods aimed at more complex motion and scene changes [40]; BasicVSR describes REDS4 as containing large and complicated motions [7]. Because Vimeo-90K-T sequences have only seven frames, long-range propagation has little to work with there: BasicVSR reports slightly lower scores on Vimeo-90K-T than sliding-window methods such as EDVR and attributes this to the short sequences [7].

Challenges

Challenges at the NTIRE (CVPR) and AIM (ECCV) workshops created datasets and track the state of the art. **NTIRE 2019** introduced REDS [38]; its video SR challenge [41] had a bicubic track and a track with realistic motion blur, and EDVR won all four video restoration tracks that year [18]. **NTIRE 2021** video SR [42] had a restoration-quality track and a lower-frame-rate track, which turns the problem into space-time SR (raising resolution and frame rate together); BasicVSR++ reports three first places and one second place across that challenge and the compressed video enhancement challenge [20]. **Efficiency challenges** judge quality per unit of compute: the Mobile AI & AIM 2022 report [43] asked for real-time ×4 VSR on a phone's neural processing unit, trained on REDS and judged on runtime and power, and the AIM 2024 challenge on efficient VSR for compressed content [44] targets streaming with ×4 upscaling from 540p to 4K and ×3 from 360p to 1080p.

Efficiency and streaming

Plain version. A video model must run on every frame, often in real time, and often before the future frames exist. Those constraints shape the architecture as much as accuracy does.

- **Cost per frame.** A sliding window processes each input frame $2K + 1$ times unless features are cached; a recurrent model processes each frame once and carries a state, so its cost per frame does not grow with the length of the video [3].
- **Latency and causality.** Bidirectional and grid propagation need future frames, so the first output waits for the end of the chunk. Live video (calls, game streaming) allows only causal designs or short look-ahead buffers, at some cost in quality (the unidirectional ablation of BasicVSR [7]). Long videos are processed in chunks, and quality dips at chunk ends, as BasicVSR's segment experiment shows.
- **Memory.** Parallel transformers such as VRT attend across many frames at several scales; their model size and memory use are the drawback RVRT addresses with clip-level recurrence [24].

Diffusion VSR is costlier still, which motivates one-step models [32].

- **Deployment.** The efficiency challenges set the target for phones and streaming: few parameters and operations, real-time frame rates and low power, with quality judged against interpolation baselines rather than against the largest models [43], [44].

Failure modes

Plain version. VSR fails where its main assumption fails: that neighboring frames show the same content, slightly moved.

- **Occlusion and disocclusion.** Content hidden in the neighbors (Figure 4) has no extra measurements; a model that trusts the warped features there produces ghosts or doubled edges. Modulation masks and attention weights let a model down-weight such regions [20].
- **Large and complex motion.** Fast, non-rigid motion and motion blur make alignment inaccurate; coarse-to-fine alignment (PCD in EDVR, the pyramid in SPyNet) helps but does not remove the problem. REDS was used to stress exactly this [18].
- **Error accumulation.** In recurrent models a wrong state is carried forward, especially after occlusions or scene cuts; IconVSR's keyframe refill and BasicVSR++'s second-order links are direct countermeasures [7], [20]. Long propagation can also amplify severe input artifacts, which RealBasicVSR counters with pre-cleaning [27].
- **Flicker.** Independent per-frame decisions, in single-image models or in diffusion sampling, cause temporal instability that per-frame metrics cannot see.
- **Hallucinated detail.** GAN and diffusion VSR can create plausible texture, text or faces that were never there, and in video the invented content may also drift. For surveillance, forensic, medical or scientific video, prefer fidelity-oriented models and state which kind of model produced an output.
- **Domain gap.** Models trained on BI or BD degradations do not account for compression and real noise [26], [27].

Modern view

Survey papers

- **Liu et al., "Video Super-Resolution Based on Deep Learning: A Comprehensive Survey"** (**Artificial Intelligence Review, 2022**) [40]. Reviews 37 deep VSR methods in seven sub-

categories by **how they use inter-frame information**: *with alignment* (motion estimation and compensation, deformable convolution) or *without alignment* (2-D and 3-D convolution, recurrent CNNs, non-local operations, others), and compares them on benchmarks. Takeaway: the field moved from aligned short windows toward a mix of designs and long sequences, with REDS replacing Vid4 as the demanding test set.

- **Tu et al., "Optical Flow for Video Super-Resolution: A Survey" (Artificial Intelligence Review, 2022)** [45]. Covers metrics and classical and deep flow-based VSR, with an analysis of representative deep methods. Takeaway: dense, sub-pixel flow is among the most common tools for motion compensation, so understanding flow is a large part of understanding VSR alignment.
- **Baniya et al., "A Survey of Deep Learning Video Super-Resolution" (IEEE Transactions on Emerging Topics in Computational Intelligence, 2024)** [46]. A component taxonomy (inputs, alignment, fusion, refinement and reconstruction, upsampling), plus losses, quality and efficiency evaluation, applications and trends such as online, real-world, resource-constrained and space-time VSR. Takeaway: design choices are often driven by benchmark gains rather than explained, and practical settings deserve more attention.

State of the art, 2023–2026

1. **Propagation plus alignment is the core of fidelity-oriented VSR.** Bidirectional or grid recurrent propagation with flow-guided deformable alignment (BasicVSR++) and parallel or clip-recurrent transformers (VRT, RVRT) are the reference designs of this chapter, trading accuracy against memory and latency.
2. **Real-world VSR moved to generative priors.** After CNN-based real-world models such as RealBasicVSR, diffusion-based VSR built on image diffusion models (Upscale-A-Video, MGLD-VSR), then on video diffusion models and diffusion transformers (STAR, SeedVR), with temporal consistency as the central technical problem.
3. **Cost reduction of generative VSR.** One-step diffusion (DLoRAL) follows the same path as one-step image SR.
4. **Efficiency and compression as first-class targets**, with challenge tracks for mobile hardware and compressed streaming content [43], [44].

Open problems

- **Faithful versus plausible.** Measuring and controlling hallucination in video, where invented detail may also be temporally unstable, is unsolved.

- **Temporal evaluation.** Warping error can be gamed by blur, tOF and tLP depend on flow and feature estimates, there is no agreed perceptual measure of temporal quality, and user studies are costly.
- **Real degradations and codecs.** Synthetic pipelines still miss real camera processing and codec behavior; ground truth for real videos is rarely available.
- **Long videos, online processing and scene cuts.** Combining the quality of bidirectional and global models with causal, constant-memory processing remains difficult.
- **Occlusions and large motion.** Alignment is still the weak point under fast, non-rigid motion and disocclusion.
- **Cost of generative VSR**, which is still far above that of CNN and transformer VSR for long, high-resolution video.

Key takeaways

- Video SR differs from image SR because neighboring frames, shifted by sub-pixel motion, are **new measurements** of the same scene, and because the output must be **temporally consistent**.
- The video degradation model applies blur, downsampling, noise and compression to every frame, and links frames through motion; without knowing the motion, frames cannot be combined (unaligned averaging is worse than one frame).
- Optical flow plus **backward warping with bilinear sampling** is the basic alignment tool; it fails at occlusions even with perfect flow.
- Alignment comes in three families: explicit flow warping (VESPCN, TOFlow, BasicVSR), deformable convolution (TDAN, EDVR; a generalization of warping with diverse offsets, stabilized by flow guidance in BasicVSR++) and attention (VRT, RVRT).
- Temporal designs trade accuracy, latency and cost: sliding windows are local and parallel, recurrent models are compact but accumulate errors, bidirectional and grid propagation see the whole clip but need future frames, and transformers model long range at high memory cost.
- Real-world VSR is dominated by compression artifacts; propagation can amplify them (RealBasicVSR), and diffusion-based VSR gives realistic texture at the risk of hallucination and flicker.
- Report Y-channel (or RGB) PSNR/SSIM per frame with the exact BI or BD setting, add a temporal-consistency measure, and remember that a blurry video scores well on stability alone.

Exercises

1. **Integer versus sub-pixel motion.** In the multi-frame snippet, replace the random shifts by integer shifts (`rng.integers(0, s, size=(T, 2)).astype(float)`, keeping `shifts[0] =`

`0`) and rerun. Then try all shifts equal to zero. Explain the PSNR you get in each case using the sampling argument of Figure 1.

Hint

Integer shifts in HR pixels still land the LR samples on different HR phases (an HR shift of 1 is a quarter LR pixel), so they still help; what matters is the shift *modulo s* HR pixels. With all shifts zero every frame samples the same HR positions, so only the noise averages out and the result stays close to the single-frame solver. Try shifts that are multiples of 4 HR pixels to see the "no new information" case directly.

2. **Registration error budget.** Add Gaussian noise with standard deviation 0.25, 0.5 and 1.0 HR pixels to the true shifts before calling `reconstruct`, and plot PSNR against the shift error. Compare with the 0.13-pixel error of phase correlation. What does this suggest about how accurate alignment must be for super-resolution, compared with denoising?

Hint

PSNR should fall quickly once the shift error approaches the size of the detail being recovered (a fraction of an HR pixel), because samples land in the wrong places and the reconstruction blurs or rings. Denoising by averaging tolerates larger errors, since it only needs the same content, not its exact sub-pixel position. This is why VSR needs sub-pixel accurate flow, while many video denoisers can work with coarser alignment.

3. **Forward versus backward warping.** Write a forward-warping function that pushes each pixel of frame $t + 1$ to $p - F(p)$ (rounded to the nearest pixel) and compare its output with `biline ar_warp` on the two-frame example. Count holes (pixels that receive no value) and collisions (pixels that receive several).

Hint

Holes appear where the scene expands or is disoccluded (behind the moving square), and collisions where it contracts or is occluded. Backward warping has neither, because every output pixel reads exactly one (interpolated) value, which is why networks use it; the price is that it needs the flow defined on the *reference* frame.

4. **Sliding window versus recurrent cost.** A sliding-window model with a 7-frame window runs a feature extractor on every input frame of the window and one flow network per neighbor. A bidirectional recurrent model runs one feature extractor per frame and two flow estimations per frame (one per direction). Count feature extractions and flow estimations per output frame for each design, with and without caching features between windows. Which design has lower latency for the first output of a 100-frame clip?

Hint

Without caching the window model does 7 feature extractions and 6 flow estimations per output; with caching, 1 new extraction per output but still 6 flows (each neighbor must be aligned to a new center). The bidirectional model does 1 extraction and 2 flows per frame. But its first output needs the backward pass over all 100 frames, while the window model needs only frames 1 to 4. Cost per frame favors recurrence; latency favors the window.

5. **Flicker metric pitfalls.** In the flicker snippet, replace the zero-mean fake detail by a *fixed* texture pattern that moves with the scene (generate it once and roll it with the scene). Rerun and explain why the flicker measure changes while PSNR does not change much. What does this say about what tOF/tLP-style measures can and cannot detect?

Hint

A texture that moves consistently with the scene has the same temporal change as the ground truth, so the motion-compensated flicker drops to about zero, while PSNR stays roughly as low as before because the texture is still wrong. Temporal measures detect *inconsistency*, not *incorrectness*: a consistently hallucinated texture passes them. Fidelity (or perceptual) metrics and temporal metrics answer different questions.

Supplementary notes

These notes are optional background that the main text does not depend on.

Classical multi-frame super-resolution

Plain version. Before deep learning, multi-frame super-resolution was solved as an estimation problem: register the frames, put their samples on a fine grid, and solve for the HR image that explains all of them, with a prior for what is still missing.

Precise version. With global translations d_j, a known blur B and downsampling D, the observation model is $y_j = DBS_{d_j}x + n_j$ for frames $j = 1, \dots, M$, where S_d shifts the image by d. Farsiu et al. [1] observe that for pure translations and a space-invariant blur the blur and shift operators commute, $BS_{d_j} = S_{d_j}B$, so the problem splits into two steps: (1) estimate the blurred HR image $z = Bx$ from the LR frames, by copying every registered LR sample to its position on the HR grid and combining the samples that land on the same HR pixel; (2) deblur z to get x. With an L2 data term, step 1 averages the samples at each HR pixel; Farsiu et al. instead minimize an L1 data term, $\min_x \sum_j \|DBS_{d_j}x - y_j\|_1 + \lambda\,\Phi_{\mathrm{BTV}}(x)$, where Φ_{BTV} is their edge-preserving "bilateral TV" regularizer and $\lambda > 0$ its weight. The L1 version turns step 1 into a **pixel-wise median**, which is robust to outlier frames and to errors in motion and blur estimation. Iterative back-projection, introduced by Irani and Peleg [4], is steepest descent on the data term, as in the main-text snippet.

When it is well posed. Counting unknowns, $\times s$ SR needs at least s^2 frames with distinct sub-pixel phases; random shifts need more, because some phases repeat and others are missing. The blur sets a second limit: frequencies that B suppresses to near zero cannot be recovered by any number of frames, only by a prior. That is one reason the toy gains in Figure 2 grow more slowly as frames are added.

Code. Step 1 with the shifts rounded to the HR grid shows two effects: with few frames many HR pixels receive no sample at all, and the fused image is close to the *blurred* image Bx, not to x, so step 2 is still needed.

```
import numpy as np
from scipy import ndimage as ndi
from skimage import data, color

rng = np.random.default_rng(0)
s, blur = 4, 1.2
hr = color.rgb2gray(data.astronaut())[64:320, 128:384]
def fuse(T):
    shifts = rng.uniform(0, s, size=(T, 2)); shifts[0] = 0
    acc, cnt = np.zeros_like(hr), np.zeros_like(hr)
    i, j = np.mgrid[0:64, 0:64]
    for d in shifts:
        y = ndi.gaussian_filter(ndi.shift(hr, d, order=3, mode="reflect"), blur)[::s, ::s]
        u, v = np.rint(s * i - d[0]).astype(int), np.rint(s * j - d[1]).astype(int)
        ok = (u >= 0) & (u < 256) & (v >= 0) & (v < 256)
        np.add.at(acc, (u[ok], v[ok]), y[ok]); np.add.at(cnt, (u[ok], v[ok]), 1)
    z = acc / np.maximum(cnt, 1)
    err = np.abs(z - ndi.gaussian_filter(hr, blur))[cnt > 0].mean()
    return (cnt > 0).mean(), err
for T in (4, 16, 64):
    cov, err = fuse(T)
    print(f"{T:3d} frames: {100 * cov:5.1f}% of HR pixels filled, mean |z - Bx| on filled
     ↪  pixels {err:.4f}")
#   4 frames:  18.4% of HR pixels filled, mean |z - Bx| on filled pixels 0.0052
#  16 frames:  66.9% of HR pixels filled, mean |z - Bx| on filled pixels 0.0061
#  64 frames:  97.7% of HR pixels filled, mean |z - Bx| on filled pixels 0.0030
```

With $s = 4$ there are 16 sub-pixel phases, so even 16 random shifts fill only about two thirds of the HR grid in our run, and 64 are needed to fill almost all of it. The small remaining error comes from rounding the shifts to whole HR pixels.

Limits. These methods depend on accurate registration with a simple motion model and on a known blur; local, non-rigid motion and occlusion break the commuting-operator argument, which is where learned alignment took over. A modern descendant is burst photography: Wronski et al. [47] use natural hand tremor to obtain a burst of raw frames with small offsets, then align and merge them into a full-color image without a separate demosaicing step, with robustness to local motion, occlusion and scene changes.

Brightness constancy and the classical flow solvers

Plain version. Flow solvers assume that a point keeps its brightness as it moves, and that neighbors move together. The first assumption gives one equation per pixel; the second supplies the missing equations.

Precise version. Linearizing brightness constancy $I(u+\delta u, v+\delta v, t+1) = I(u,v,t)$ with a first-order Taylor expansion gives the **optical flow constraint**

$$I_u\,\delta u + I_v\,\delta v + I_t = 0,$$

where I_u, I_v are the spatial derivatives and I_t the temporal derivative of the intensity at the pixel. It is one equation in two unknowns: only the component of motion along the gradient (I_u, I_v) is determined (the aperture problem), and the linearization is valid only for small displacements, which is why practical solvers work coarse to fine on image pyramids. Horn and Schunck [8] minimize, over the whole image,

$$\iint (I_u\,\delta u + I_v\,\delta v + I_t)^2 + \alpha^2(\|\nabla\delta u\|^2 + \|\nabla\delta v\|^2)\,du\,dv,$$

where $\alpha > 0$ weights a smoothness penalty on the flow field; the result is dense but blurred across motion boundaries. Local methods instead assume constant flow in a small window and solve the constraint for all window pixels by least squares. Farnebäck [9] approximates each neighborhood by a quadratic polynomial, $I(\mathbf{z}) \approx \mathbf{z}^\top A\mathbf{z} + \mathbf{b}^\top\mathbf{z} + c$ with $\mathbf{z}$ the local coordinates, A a symmetric matrix, $\mathbf{b}$ a vector and c a scalar fitted to the data; a pure shift by $\mathbf{d}$ keeps A and changes $\mathbf{b}$ to $\mathbf{b} - 2A\mathbf{d}$, so $\mathbf{d}$ follows from the change in $\mathbf{b}$, averaged over a neighborhood and refined over a pyramid. This is the solver used in the main-text snippet, and the flow estimator in the TecoGAN tOF metric [36].

Why learned flow replaced them in VSR. Classical solvers break under brightness change, large motion, occlusion and textureless regions, and they are not optimized for the downstream task. Learned estimators (SPyNet [10], RAFT [11]) are more accurate and can be fine-tuned end to end, and TOFlow [14] showed that the flow that helps restoration most need not be the physically correct one.

Deformable convolution as warping plus convolution

Plain version. A deformable convolution with K taps behaves like K ordinary warps, each with its own small flow field, followed by a 1×1-style weighted sum. So it can "look" in several places at once, while flow warping looks in one.

Precise version. Rewrite the deformable convolution of the main text, for a single input channel, as

$$y(p) = \sum_{i=1}^{K} w_i \, \tilde{x}_i(p), \qquad \tilde{x}_i(p) = x(p + p_i + \Delta p_i(p)),$$

where $\tilde{x}_i$ is the input backward-warped by the flow field $p_i + \Delta p_i(\cdot)$, one field per tap i. Each $\tilde{x}_i$ is a flow-warped copy of x, and y is a learned weighted sum of the K warped copies. Chan et al. [19] use this decomposition to show that flow-based alignment is the special case in which all offset fields are equal to one flow (up to the fixed tap positions), and that the extra **offset diversity** of deformable alignment is what improves results. At an occlusion, for example, some taps can sample the background on one side of the occluder while others sample it on the other side.

Code. The decomposition can be checked numerically with the `bilinear_warp` function of the main text:

```
import numpy as np
from scipy import ndimage as ndi
from skimage import data, transform

def bilinear_warp(img, flow):
    H, W = img.shape
    yy, xx = np.mgrid[0:H, 0:W].astype(np.float64)
    x = np.clip(xx + flow[..., 0], 0, W - 1); y = np.clip(yy + flow[..., 1], 0, H - 1)
    x0 = np.minimum(np.floor(x).astype(int), W - 2); y0 =
     ↪  np.minimum(np.floor(y).astype(int), H - 2)
    ax, ay = x - x0, y - y0
    return ((1 - ay) * ((1 - ax) * img[y0, x0] + ax * img[y0, x0 + 1])
            + ay * ((1 - ax) * img[y0 + 1, x0] + ax * img[y0 + 1, x0 + 1]))

rng = np.random.default_rng(0)
x = transform.resize(data.camera() / 255.0, (128, 128), anti_aliasing=True)
w = rng.normal(size=9)                                          # 3x3 kernel weights
taps = [(dx, dy) for dy in (-1, 0, 1) for dx in (-1, 0, 1)]
# (a) all offsets zero: deformable conv = ordinary (correlation-style) convolution
zero = np.zeros((128, 128, 2))
y_def = sum(wi * bilinear_warp(x, zero + np.array(t)) for wi, t in zip(w, taps))
y_conv = ndi.correlate(x, w.reshape(3, 3), mode="nearest")
print("zero offsets match ndi.correlate:", np.allclose(y_def[2:-2, 2:-2], y_conv[2:-2,
 ↪  2:-2]))
# (b) every tap shares one flow f: deformable conv = warp by f, then ordinary convolution
f = np.zeros((128, 128, 2)); f[..., 0], f[..., 1] = 3.0, -2.0     # integer flow keeps the
 ↪  check exact
y_shared = sum(wi * bilinear_warp(x, f + np.array(t)) for wi, t in zip(w, taps))
y_warp_conv = ndi.correlate(bilinear_warp(x, f), w.reshape(3, 3), mode="nearest")
print("shared offsets = warp then conv:", np.allclose(y_shared[6:-6, 6:-6],
 ↪  y_warp_conv[6:-6, 6:-6]))
# zero offsets match ndi.correlate: True
# shared offsets = warp then conv: True
```

Case (a) is an ordinary convolution and case (b) is flow warping followed by an ordinary convolution; a general deformable convolution gives each of the nine taps its own offset field.

Why training is unstable. The offsets Δp_i are unconstrained regression outputs. Early in training they can drift to very large values, pointing far outside the relevant region ("offset overflow") and producing useless features. Chan et al. add an offset-fidelity loss that keeps offsets close to optical flow [19]; BasicVSR++ instead predicts offsets as residuals on top of the flow and multiplies each tap by a sigmoid modulation mask, which also acts as an attention map over the warped copies [20].

Diffusion-based VSR models

Plain version. All diffusion VSR models start from a generator that already knows what natural images (or videos) look like, condition it on the LR video, and add something that keeps the generated detail the same from frame to frame. They differ in what they add.

Precise version.

- **Upscale-A-Video** [28] is a text-guided latent diffusion model. Locally, it inserts temporal layers into the U-Net and the VAE decoder for consistency within short sequences; globally, a training-free **flow-guided recurrent latent propagation** module propagates and fuses latents across the whole sequence. Text prompts can guide texture creation, and an adjustable noise level trades restoration against generation.
- **MGLD-VSR** [29] guides the latent sampling path of a pretrained latent diffusion model with a **motion-guided loss** computed from the temporal dynamics of the LR video, and fine-tunes a temporal module inserted in the decoder with a sequence-oriented loss.
- **STAR** [30] starts from a text-to-video diffusion model, which already models motion, and adds a local information enhancement module before the global attention blocks (to enrich local detail and suppress degradation artifacts) and a dynamic frequency loss that emphasizes different frequency components at different diffusion steps (to keep fidelity).
- **SeedVR** [31] is a diffusion transformer for generic video restoration of arbitrary length and resolution, built on shifted window attention with variable-sized windows at the spatial and temporal borders, a causal video autoencoder, and mixed image and video training.
- **DLoRAL** [32] trains a one-step model with two LoRA branches: a consistency branch, learned with a cross-frame retrieval module from degraded inputs, and a detail branch trained afterwards to add spatial detail while staying within the consistency branch's temporal representation. Both are merged into the diffusion model for single-step inference.

How to compare them. Check whether the comparison uses synthetic data with ground truth (fidelity metrics), real videos (no-reference metrics, user studies), or AI-generated videos, and whether a temporal-consistency measure is reported next to per-frame quality; generative models can trade one for the other.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

BasicVSR and IconVSR

The problem. By 2020 VSR methods had become complex and dissimilar (multi-scale deformable alignment with attention in EDVR, several projection modules in RBPN), which made them slow, hard to reimplement and hard to compare fairly. BasicVSR [7] asks which components actually matter.

The idea. Decompose VSR methods into four functions, **propagation, alignment, aggregation and upsampling**, and test the options for each with minimal designs. The paper's Table 1 classifies existing methods this way (local, unidirectional or bidirectional propagation; no alignment, flow or deformable alignment; concatenation-based aggregation; pixel-shuffle upsampling). BasicVSR itself is the simplest combination that worked best: bidirectional propagation, flow-based (SPyNet) alignment of *features*, concatenation and pixel shuffle.

What it showed. The ablations are the core of the paper, and the numbers here are the paper's own. Splitting test sequences into independent segments lowers PSNR, most at segment ends, which argues for long-range propagation (its Fig. 3). A unidirectional variant is markedly worse at early frames and, overall, about 0.5 dB worse than the bidirectional model (Fig. 4). Removing alignment costs 1.19 dB, and warping images instead of features costs 0.17 dB. The extension IconVSR adds keyframe information-refill and coupled propagation to reduce error accumulation, reporting up to 0.31 dB over BasicVSR.

How to read it. Start with Table 1 and Section 3.1, which together are a compact map of the field. In the main comparison table, check: (a) the degradation of each column (REDS4 under BI only, UDM10 under BD only, Vimeo-90K-T and Vid4 under both); (b) the channel (Y everywhere except REDS4, which is RGB); (c) the training set behind each column (REDS for REDS4, Vimeo-90K for the rest); (d) runtimes, which depend on hardware and LR size. In the ablations, note that each one modifies a variant of BasicVSR, so the drops measure the component's value inside this design, not in general.

Limits. Bidirectional propagation needs the whole sequence, so BasicVSR is an offline method. Flow-based alignment inherits flow errors under large motion and occlusion, which BasicVSR++ addresses with flow-guided deformable alignment [20]. The study is on clean BI/BD degradations; RealBasicVSR [27] shows that the same long-range propagation can amplify severe real-world artifacts.

Why it matters here. It gives the vocabulary used throughout this chapter, and it is the baseline

from which BasicVSR++, RealBasicVSR and several diffusion VSR models were built.

RealBasicVSR

The problem. Models trained on BI or BD degradations fail on real videos, and simply training BasicVSR-style models on random real-world degradations raises new trade-offs that earlier work ignored [27].

The idea. The paper identifies three trade-offs and proposes a simple fix for each. (1) *Detail synthesis versus artifact suppression*: long-term propagation spreads noise and artifacts along with detail, so an **image pre-cleaning module** (a stack of residual blocks) cleans each LR frame independently before a BasicVSR network propagates them. It is trained with an extra *cleaning loss* that compares its output with a downsampled ground truth, and at test time a *dynamic refinement* scheme applies it repeatedly until the change falls below a threshold. The paper shows that a recurrent cleaning module, or no cleaning loss, fails to remove the artifacts. (2) *Speed versus performance in training*: loading many long, degraded clips makes training slow, so a **stochastic degradation scheme** loads half of each sequence and flips it in time, and lets the degradation parameters change from frame to frame as a random walk to keep the data diverse; this reduces training time by up to 40% without loss of quality. (3) *Batch versus length*: under a fixed memory budget, longer sequences use temporal information more effectively than larger batches and make inference more stable.

What it showed. RealBasicVSR, a BasicVSR with the cleaning module, fine-tuned with perceptual and adversarial losses (with the cleaning module detached from those two losses), compares favorably in quality and efficiency with real-world image and video SR models on real videos.

The paper also releases the **VideoLQ** dataset of real low-quality videos as a common benchmark.

How to read it. Begin with the analysis of the cleaning module (what happens when propagation is applied to uncleaned inputs) and the batch-versus-length experiments, since they explain the design. For VideoLQ, check that the comparison uses no-reference metrics and visual results because there is no ground truth, and read the visual comparisons for the trade-off between sharper texture and invented artifacts.

Limits. No-reference metrics on VideoLQ are only weakly tied to human judgment, so the comparison relies partly on visual examples. The degradation model is synthetic, so codecs and camera pipelines outside it can still break the model. A cleaning module that removes too much also removes real detail.

Why it matters here. It explains why real-world VSR cannot just reuse the clean-benchmark recipe, and it is the standard non-diffusion baseline that diffusion VSR papers such as Upscale-A-Video compare with [28].

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DLCV 04C • Generative Image Restoration: Sampling a Plausible Answer

Why regression-trained restorers blur, what it means to sample from the posterior instead, and how GAN priors, conditional and zero-shot diffusion, latent-diffusion and flow-based restorers do it, with the evaluation, cost and hallucination risks that come with them.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-04c-generative-restoration/

What you'll learn

- Why a network trained with a pixel loss returns the **posterior mean** $\mathbb{E}[x \mid y]$, and why that average of many plausible images looks blurry.
- What "generative" restoration means: return a **sample** from the posterior $p(x \mid y)$, an image that is both realistic and consistent with the input, and what the **perception–distortion trade-off** says that costs.
- The main families: GAN-based restorers and pretrained GAN priors; conditional diffusion restorers versus **zero-shot posterior sampling** with a pretrained diffusion model (DPS, DDRM, DDNM, ΠGDM); latent-diffusion restorers built on text-to-image models; one-step distilled models; flow-based restorers.
- The "prior step + data-consistency step" loop behind zero-shot diffusion restoration, built from scratch on a 2-D toy with an exact score.
- How to evaluate a stochastic restorer honestly (fidelity, realism, consistency with the input, several seeds, compute), how hallucination happens, and when *not* to use generative restoration.

The big picture

DLCV 04 trained restoration networks to be close to the clean image on average. That works well for mild damage. When the damage destroys a lot of information (a 16-pixel-wide face, a heavily compressed photo, a large hole), "close on average" produces a smooth, waxy guess. Generative restoration takes a different stance: since many clean images are consistent with the input, *pick one of them*, and make sure it looks like a real photo.

> **Explain like I'm five**
>
> A friend shows you a tiny, blurry photo of a car's number plate and asks what it says. One way to answer is to imagine every plate that could look like this blur and mix them all together: you get a

gray smudge, which is never exactly wrong but is useless to read. Another way is to draw one plate that really could look like that blur: crisp letters, nice and clear. That drawing *looks* great, but it is only one possibility, and the letters may not be the real ones. Regression restorers draw the smudge; generative restorers draw the crisp plate. This chapter is about drawing crisp plates well, and about never forgetting that they might be wrong.

Why it matters. The most convincing real-world super-resolution and face or old-photo restoration of recent years comes from generative models, GANs first and diffusion models since about 2021. The same machinery is used for medical and scientific inverse problems, where an invented detail can do real harm. So you need two skills at once: how these methods work, and how to measure what they invent.

Where this chapter sits. It builds on the degradation model, losses and metrics of DLCV 04 and the generative models of ML 06; diffusion and flow matching are derived in ML 07, and here we only use them. Super-resolution, which has its own chapters (DLCV 04A for images, DLCV 04B for video), appears here only as an example. All code runs in seconds on a CPU with NumPy and SciPy, on toy problems whose exact answer is known, so every claim can be checked.

Why regression gives blur: the posterior mean

Plain version. If several clean images could have produced the same damaged input, a network trained to minimize the squared error cannot pick one: picking the wrong one costs too much. Its safest answer is their **average**, and an average of differently placed edges is a soft ramp. Blur is not a bug in the network; it is the correct answer to the question we asked.

One input, many answers

Write the measurement as $y = \mathcal{D}(x) + n$, as in DLCV 04. For super-resolution, deblurring, inpainting or heavy compression, $\mathcal{D}$ throws information away, so a whole set of clean images x explains the same y almost equally well. In the Bayesian language, the uncertainty that remains after seeing y is the **posterior distribution**

$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{p(y)},$$

where $p(x)$ is the **prior** (how likely an image is before we look at y: natural images are likely, noise is not), $p(y \mid x)$ is the **likelihood** (how well x explains the measurement; for Gaussian noise of standard deviation σ it is proportional to $\exp(-\|y - \mathcal{D}(x)\|^2 / 2\sigma^2)$), and $p(y)$ is a normalizing constant that does not depend on x. Mild damage gives a narrow posterior, and every reasonable method agrees. Severe damage gives a wide posterior, often with several separate modes ("the edge is here" or "two pixels to the right").

The math: squared error is minimized by the posterior mean

A restoration network f_θ trained on pairs (y, x) with the squared error minimizes the expected loss $\mathbb{E}\,\|f_\theta(y) - x\|^2$ over the data distribution. For each fixed y, the best output $\hat{x}$ minimizes

$$\mathbb{E}[\|\hat{x} - x\|^2 \mid y] = \|\hat{x} - \mathbb{E}[x \mid y]\|^2 + \mathbb{E}[\|x - \mathbb{E}[x \mid y]\|^2 \mid y],$$

where the expectation is over the posterior $p(x \mid y)$ and $\mathbb{E}[x \mid y]$ is the posterior mean. The second term does not depend on $\hat{x}$ (it is the irreducible uncertainty), so the minimum is at

$$\hat{x}_{\text{MMSE}}(y) = \mathbb{E}[x \mid y] = \int x\, p(x \mid y)\, dx,$$

the **minimum mean squared error (MMSE)** estimator. A large enough network trained long enough approaches it. (The L1 loss gives the per-pixel posterior *median* instead: sometimes sharper, but still chosen pixel by pixel rather than as one coherent image; DLCV 04 compares the losses.) The posterior mean is the right answer if you are scored by MSE or PSNR, and the wrong one if you want an image that looks real: averaging images with edges in different places gives an image with no sharp edge at all.

Code: a toy where the mean is blurry and every sample is sharp

We need a problem whose posterior we can compute exactly. The clean "images" are 1-D signals of 64 pixels with one step edge (0 on the left, 1 on the right) at an unknown position between pixels 8 and 55, all equally likely. The degradation blurs with a Gaussian of standard deviation 3 pixels, keeps one pixel in eight and adds Gaussian noise of standard deviation 0.2. With only 48 possible clean signals, the posterior is an exact list of 48 probabilities.

```
import numpy as np
from scipy.ndimage import gaussian_filter1d
from scipy.stats import wasserstein_distance

rng = np.random.default_rng(0)
n, factor, sigma = 64, 8, 0.2
positions = np.arange(8, 56)                                        # where the edge may be
X = (np.arange(n)[None, :] >= positions[:, None]).astype(float)  # every possible clean
 ↪  signal
A = lambda x: gaussian_filter1d(x, 3.0, axis=-1, mode="nearest")[..., ::factor]  # blur,
 ↪  keep 1 of 8
AX = A(X)                                                           # every candidate, degraded

def posterior(y):
    """Exact p(edge position | y): uniform prior, Gaussian noise of std sigma."""
    logp = -np.sum((y - AX) ** 2, axis=1) / (2 * sigma**2)
    p = np.exp(logp - logp.max())
    return p / p.sum()

truth, mean, post, prior = [], [], [], []
for _ in range(2000):
```

```
    x = X[rng.integers(len(positions))]
    y = A(x) + sigma * rng.standard_normal(n // factor)
    p = posterior(y)
    truth.append(x)
    mean.append(p @ X)                                          # posterior mean (MMSE)
    post.append(X[rng.choice(len(positions), p=p)])             # a sample from p(x | y)
    prior.append(X[rng.integers(len(positions))])               # a sample that ignores y
truth = np.array(truth)

for name, est in [("posterior mean", mean), ("posterior sample", post), ("prior sample",
 ↪  prior)]:
    est = np.array(est)
    mse = np.mean((est - truth) ** 2)
    w1 = wasserstein_distance(est.ravel(), truth.ravel())       # pixel values like real
     ↪  ones?
    jump = np.abs(np.diff(est, axis=1)).max(axis=1).mean()      # sharpest step (real: 1.0)
    print(f"{name:16s} MSE {mse:.4f} | W1 {w1:.4f} | largest step {jump:.2f}")
# posterior mean   MSE 0.0169 | W1 0.0243 | largest step 0.22
# posterior sample MSE 0.0333 | W1 0.0010 | largest step 1.00
# prior sample     MSE 0.2485 | W1 0.0076 | largest step 1.00
```

Three estimators are compared over 2,000 random test signals. The columns are the MSE to the clean signal; a crude "realism" score, the 1-D Wasserstein distance W1 between the distribution of output pixel values and that of real pixel values (real signals contain only 0s and 1s, so in-between gray values are unrealistic); and the height of the sharpest step in the output (1.0 for every real signal). All numbers in this chapter's code blocks come from our own runs.

- The **posterior mean** has the lowest MSE, but its sharpest step averages 0.22: it is a ramp, not an edge, and its gray values make it statistically unlike any real signal.
- A **posterior sample** is a real edge (step 1.00, W1 near zero) placed where the data allow. Its MSE is 0.0333, almost exactly **twice** the posterior mean's 0.0169. That factor of two is not a coincidence (next section).
- A **prior sample** ignores y. It is perfectly realistic and useless: its MSE is about 15 times worse. Realism alone is not restoration. (Its W1 of 0.0076 instead of 0 is sampling noise of the 2,000-signal test.)

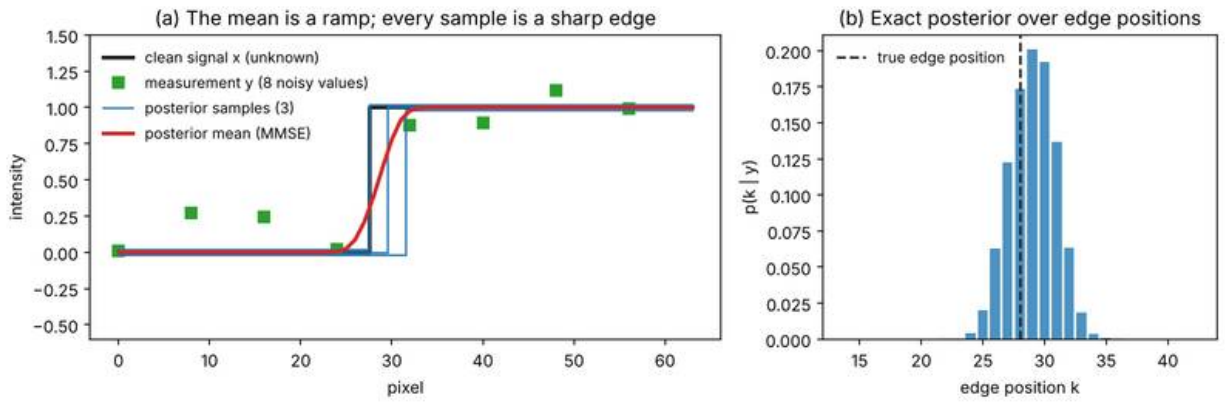


Figure 1 — The step-edge toy for one input. (a) The posterior mean (red) is a smooth ramp; each posterior sample (blue) is a sharp edge, but they disagree about where. (b) The exact posterior over the edge position: the measurement narrows the position to a few pixels but cannot pin it down.

What "generative" restoration means

Plain version. A generative restorer does not output the average of all plausible answers. It outputs *one* plausible answer, ideally drawn at random with the right probabilities, so that if you ran it many times you would see the whole range of possibilities the input allows.

Precise version. A generative (or stochastic, or posterior-sampling) restorer is a random map: given y and a random seed z, it returns $\hat{x} = G(y, z)$ whose distribution, for fixed y, is meant to approximate $p(x \mid y)$. Three ingredients appear in every method of this chapter:

1. **A prior** $p(x)$ of natural images, learned by a generative model (a GAN, a diffusion model, a flow, a codebook).
2. **A likelihood or data-consistency term** that ties the output to the input, either explicitly ($\|y - \mathcal{D}(\hat{x})\|^2$ with a known $\mathcal{D}$) or implicitly (a network that is shown y and trained on pairs).
3. **A way to produce samples**: one forward pass of a generator, or many steps of a sampler.

A convenient identity connects the three. Taking the gradient of the logarithm of Bayes' rule with respect to x removes the constant $p(y)$:

$$\nabla_x \log p(x \mid y) = \nabla_x \log p(x) + \nabla_x \log p(y \mid x),$$

where $\nabla_x \log p(x)$ is the **score** of the prior (a vector field pointing toward more likely images) and $\nabla_x \log p(y \mid x)$ is the gradient of the likelihood (pointing toward images that explain y better). Every sampler that follows "uphill directions plus noise" can therefore be split into a *prior push* and a *data push*. This split is the backbone of zero-shot diffusion restoration below.

There are two broad ways to build such a restorer:

- **Learn the posterior directly** (supervised): train a conditional generator on pairs (y, x), as conditional GANs and conditional diffusion models do. $\mathcal{D}$ never has to be written down, but a new degradation needs new training.
- **Combine a prior with a known likelihood** (zero-shot): train an *unconditional* generative model on clean images once, then steer its sampler with $\mathcal{D}$ and y at test time, as GAN inversion and DPS-type methods do. One model serves many tasks, but $\mathcal{D}$ must be known and sampling is usually slower.

The perception–distortion trade-off

Plain version. You cannot have both the lowest possible error and perfectly realistic outputs. The lowest-error answer is the blurry average; perfectly realistic answers must commit to details, and a committed guess is sometimes wrong. The good news is that the price of realism is bounded: for squared error, at most a factor of two.

Precise version. Blau and Michaeli [1] define, for any restoration method that outputs $\hat{X}$ from Y, its **distortion** $\mathbb{E}[\Delta(X, \hat{X})]$, where Δ is any per-image distortion measure such as squared error, and its **perceptual quality** as a divergence $d(p_X, p_{\hat{X}})$ between the distribution of real images p_X and the distribution of restored images $p_{\hat{X}}$ (zero means the outputs are statistically indistinguishable from real images, for example by a perfect discriminator). The **perception–distortion function** is

$$P(D) = \min_{p_{\hat{X}|Y}} d(p_X, p_{\hat{X}}) \quad \text{subject to} \quad \mathbb{E}[\Delta(X, \hat{X})] \le D,$$

where the minimum is over all, possibly random, restoration methods $p_{\hat{X}|Y}$ and D is the distortion budget. They prove that $P(D)$ is non-increasing, and convex when d is convex in its second argument [1]. So there is an **unattainable region** near the origin of the (distortion, perception) plane: lowering distortion past a point forces outputs away from the real-image distribution. They also show that this holds for *any* distortion measure, not only for MSE.

Two points on the curve matter most. Let $D_{\min}$ be the lowest possible distortion (achieved by the MMSE estimator for squared error) and $D_{\max}$ the lowest distortion achievable with perfect perceptual quality ($p_{\hat{X}} = p_X$). For squared error the paper proves (in its extended arXiv version [1])

$$D_{\max} \le 2\, D_{\min},$$

and the bound is reached by **posterior sampling**, $p_{\hat{X}|Y} = p_{X|Y}$. In PSNR terms, perfect realism never needs to cost more than about 3 dB, because $10 \log_{10} 2 \approx 3.01$. This is exactly the factor of two in our toy (0.0333 versus 0.0169). The paper also notes that the bound need not be tight: some problems allow perfect realism for less than twice the MMSE.

Optional: Why posterior sampling exactly doubles the MSE (in the Supplementary notes). A four-line proof of the factor of two.

Figure 2 places the toy's estimators on the plane. Between the posterior mean and the posterior sample we also plot simple blends $(1-\lambda)\,\hat{x}_{\mathrm{MMSE}} + \lambda\,\hat{x}_{\mathrm{sample}}$, a simple way to trade the two. Their expected MSE is $(1+\lambda^2)D_{\mathrm{min}}$, because a posterior sample's deviation from the mean is uncorrelated with the truth's deviation, so $\lambda = 1$ gives the factor of two. The blends are one achievable path; the true bound $P(D)$ may lie below it.

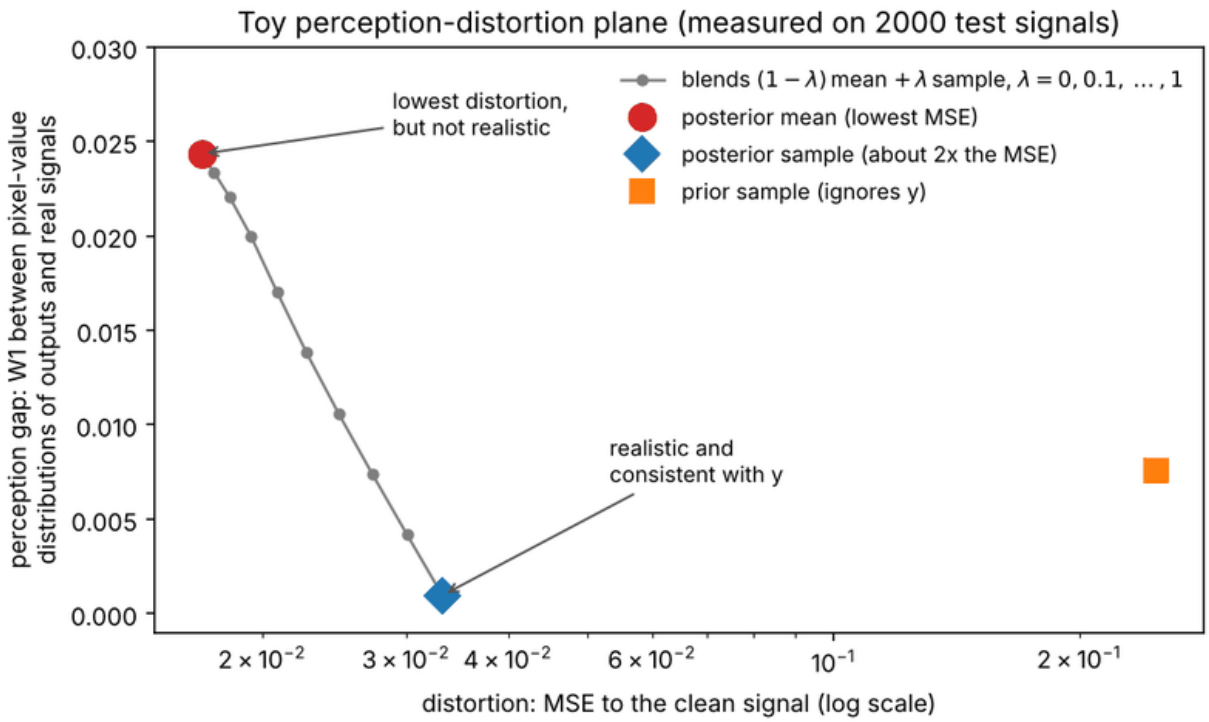


Figure 2 — The perception–distortion plane for the step-edge toy, measured on 2,000 test signals. Moving from the posterior mean to a posterior sample lowers the perception gap to near zero and doubles the MSE. A prior sample is also realistic but ignores the input, so it is far to the right.

What this means in practice:

- **A method is a point, not a score.** Report a distortion metric (PSNR, SSIM) *and* a perceptual one, and say which point your method aims for. Comparing a GAN restorer with an MSE restorer by PSNR alone is comparing two different targets.
- **"Better PSNR and better realism" is possible only for methods away from the bound.** Once a method is near the curve, it can only move along it.
- **Many generative restorers expose a knob** that slides along the curve: a GAN loss weight, a fidelity weight in CodeFormer [10], a feature-wrapping scalar in StableSR [24], a guidance

scale in DiffBIR [25]. Choosing the knob is choosing a point on the trade-off.

Optional: Blau and Michaeli, "The Perception-Distortion Tradeoff" (in the Paper guides). How to read the theorems and the super-resolution comparison.

Family 1: GAN-based restoration and GAN priors

Plain version. A GAN adds a critic that has seen many real photos and complains whenever the restored image looks fake. The restorer learns to produce outputs the critic accepts, which pushes it off the blurry average and onto realistic texture. A second idea reuses a GAN that was already trained to *generate* faces or scenes: restoration becomes "find the image this generator can make that matches the damaged input".

Adversarial and perceptual losses

The first route keeps supervised training and changes the loss (DLCV 04 has the formulas). SRGAN [2] combined an adversarial loss with a loss on pretrained VGG features, following the perceptual-loss idea of Johnson et al. [3]; ESRGAN [4] improved the generator and discriminator; Real-ESRGAN [5] trained such a model on a rich synthetic degradation pipeline for real photos (DLCV 04A covers them in detail). In this chapter's terms, the total loss

$$\mathcal{L} = \mathcal{L}_{\text{pixel}} + \lambda_p \, \mathcal{L}_{\text{feat}} + \lambda_a \, \mathcal{L}_{\text{adv}}$$

has a pixel term $\mathcal{L}_{\text{pixel}}$ (L1 or L2, pulling toward the posterior mean), a feature term $\mathcal{L}_{\text{feat}}$ (matching deep features, which tolerates small shifts), and an adversarial term $\mathcal{L}_{\text{adv}}$ (pulling the output distribution toward real images), with weights $\lambda_p, \lambda_a \geq 0$. Blau and Michaeli point out that the adversarial term approximates a divergence $d(p_X, p_{\hat{X}})$, so minimizing this loss is a Lagrangian form of the perception–distortion problem, and changing λ_a moves along the curve [1].

Two caveats. Most such networks are **deterministic**, so they give *one* plausible answer, not samples with the right probabilities. And GAN training can **mode-collapse** (ML 06), while the adversarial term rewards any texture the discriminator likes, including textures that were never in the scene.

Pretrained GANs as image priors

The second route separates "what images look like" from "what this input says". A GAN generator G trained on clean images, for example a StyleGAN [6] trained on faces, maps a latent code w to an image $G(w)$. Its outputs form a learned **image manifold**: restoration can search on that manifold for an image consistent with y.

GAN inversion by optimization. Solve

$$\hat{w} = \arg\min_{w} \|\mathcal{D}(G(w)) - y\|^2 + \lambda R(w), \qquad \hat{x} = G(\hat{w}),$$

where R keeps w in the region the generator was trained on and λ weights it. Nothing is trained per task, and $\mathcal{D}$ can be any differentiable degradation. PULSE [7] does this for face super-resolution, searching a StyleGAN's latent space for faces that downscale to the input; different random starts give different faces that all fit, a direct picture of "many answers".

Generators inside a feed-forward network. To avoid one optimization per image, GLEAN [8] and GFP-GAN [9] put a pretrained, frozen GAN inside an encoder–decoder: the encoder reads the damaged image, the generator supplies realistic features, and one forward pass gives the result. CodeFormer [10] uses a learned discrete codebook of face parts instead of a GAN latent space and has a test-time knob that trades fidelity for quality.

Optional: GAN-prior restorers in more detail (in the Supplementary notes). How PULSE, GLEAN, GFP-GAN and CodeFormer use their priors, with a 2-D inversion toy.

Strengths and limits. GAN priors give sharp, realistic results in one pass on narrow domains with good generators, such as aligned faces. But they can only produce what the generator can produce: an input unlike the training faces is pulled toward a "typical" face, which is a fidelity and fairness problem, and the PULSE authors discuss such biases explicitly [7]. A diffusion-restoration survey [44] names unstable training and fake-looking textures as the two main weaknesses of GAN-based restoration, one motivation for the diffusion priors that follow. The survey of Xia et al. [47] reviews GAN inversion and its use for restoration and editing.

Family 2: diffusion models as priors

Plain version. A diffusion model is a denoiser trained at every noise level. Starting from pure noise and denoising a little at a time turns noise into a realistic image. For restoration, we either *teach the denoiser to look at the damaged input* while it denoises (conditional diffusion), or we take a denoiser that only knows clean images and, after each small denoising step, *nudge* the current guess so that it agrees with the damaged input (zero-shot posterior sampling).

Diffusion in one page

The full derivation is in ML 07; we need only four facts, in the variance-preserving DDPM convention [11].

Forward noising. For a clean image x_0 and steps $t = 1, \dots, T$,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$

where $\bar{\alpha}_t \in (0, 1)$ decreases from about 1 (almost clean) to about 0 (pure noise) along a chosen schedule and ϵ is standard Gaussian noise.

A denoiser is a score estimator. A network trained to predict ϵ from x_t gives the score of the noisy distribution p_t: $\nabla_{x_t} \log p_t(x_t) \approx -\epsilon_\theta(x_t, t)/\sqrt{1-\bar{\alpha}_t}$ (ML 07) [12].

Tweedie's formula [13] turns the score into a clean-image estimate,

$$\hat{x}_0(x_t) = \mathbb{E}[x_0 \mid x_t] = \frac{x_t + (1-\bar{\alpha}_t)\,\nabla_{x_t} \log p_t(x_t)}{\sqrt{\bar{\alpha}_t}},$$

where $\hat{x}_0$ is the *posterior mean* of the clean image given the noisy one, so it is blurry when t is large, exactly as in the first section.

Sampling. Start from $x_T \sim \mathcal{N}(0, I)$ and repeatedly draw x_{t-1} from a Gaussian centered between x_t and $\hat{x}_0(x_t)$ with a small, known variance (ML 07). Each step costs one network call; classic samplers use hundreds. Annealed Langevin dynamics with score networks [48] and the continuous-time SDE view [12] describe the same idea in other terms.

Code: diffusion with an exact score. Image-scale diffusion needs a trained network. On a 2-D toy we can skip the network: if the "image" distribution is a mixture of Gaussians, every noisy marginal p_t is again a mixture of Gaussians, and its score has a closed form. The prior below has four blobs with weights 0.1, 0.2, 0.3, 0.4 and standard deviation 0.3.

```
import numpy as np

rng = np.random.default_rng(1)
mu = np.array([[-2.0, -2.0], [-2.0, 2.0], [2.0, -2.0], [2.0, 2.0]])  # a 2-D "image" prior:
w = np.array([0.1, 0.2, 0.3, 0.4])                                    # four Gaussian blobs
s2 = 0.3**2                                                           # each with variance s2
 ↪  * I

T = 200
beta = np.linspace(1e-4, 0.05, T)
alpha = 1 - beta
abar = np.cumprod(alpha)

def score(x, ab):
    """Exact grad log p_t(x) of the noised mixture, plus responsibilities r."""
    m, v = np.sqrt(ab) * mu, ab * s2 + (1 - ab)            # component k at time t: N(m_k, v
     ↪  I)
    d2 = ((x[:, None, :] - m[None]) ** 2).sum(-1)         # (N, K) squared distances
    logr = np.log(w) - d2 / (2 * v)
    r = np.exp(logr - logr.max(1, keepdims=True)); r /= r.sum(1, keepdims=True)
    return (r[:, :, None] * (m[None] - x[:, None, :])).sum(1) / v, r

def x0_hat(x, ab):                                         # Tweedie: E[x_0 | x_t]
    return (x + (1 - ab) * score(x, ab)[0]) / np.sqrt(ab)

def ddpm_step(x, t, x0):
    """Sample x_{t-1} from the Gaussian q(x_{t-1} | x_t, x_0 = x0)."""
    ab_prev = abar[t - 1] if t > 0 else 1.0
```

```
    mean = (np.sqrt(ab_prev) * beta[t] * x0 + np.sqrt(alpha[t]) * (1 - ab_prev) * x) / (1 -
     ↪  abar[t])
    var = beta[t] * (1 - ab_prev) / (1 - abar[t])
    return mean + np.sqrt(var) * rng.standard_normal(x.shape)

x = rng.standard_normal((4000, 2))                          # start from pure noise
for t in reversed(range(T)):
    x = ddpm_step(x, t, x0_hat(x, abar[t]))
_, r = score(x, 1.0)
print("signal left at t=T:", round(abar[-1], 5))
print("fraction per blob :", np.bincount(r.argmax(1), minlength=4) / len(x), "(target", w,
 ↪  ")")
print("spread inside blob:", round(float(np.sqrt(((x - mu[r.argmax(1)]) ** 2).mean())), 3),
 ↪  "(target 0.3)")
# signal left at t=T: 0.00612
# fraction per blob : [0.09675 0.2005  0.3075  0.39525] (target [0.1 0.2 0.3 0.4] )
# spread inside blob: 0.283 (target 0.3)
```

The sampler reproduces the blob weights to within sampling error. The spread inside each blob is 0.283 instead of 0.3: with only 200 coarse steps the reverse process is slightly too narrow (with 1,000 smaller steps we measured 0.296). This is the first lesson about **step count**: fewer steps are cheaper and less exact. Figure 3 shows the particle cloud contracting from noise to the four blobs.

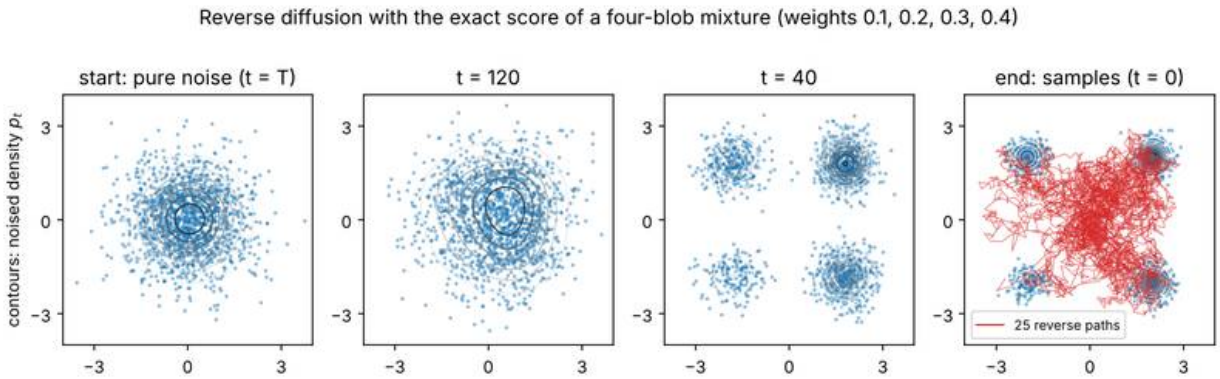


Figure 3 — Reverse diffusion with the exact score of a four-blob prior. Gray contours show the noised density at each time; blue points are 1,500 samples; red lines (right) are 25 individual reverse paths. Paths are noisy because every step injects fresh noise.

Route A: train a conditional diffusion restorer

The direct approach lets the denoiser see the input: train $\epsilon_\theta(x_t, t, y)$ on pairs, so the sampler draws from an approximation of $p(x \mid y)$. **SR3** [14] did this for super-resolution: the upsampled low-resolution image is given to the U-Net with the noisy image, and sampling refines pure noise step by step. In the paper's human study on 8× face super-resolution, raters were fooled by SR3 close to 50% of the time, against at most 34% for GAN baselines [14]. **Palette** [15] applied the recipe to

colorization, inpainting, uncropping and JPEG restoration, and showed that one multi-task model can match task-specific ones.

Conditional models need pairs and new training for each family of degradations, but they cope with noise and with operators that are hard to write down (camera processing, compression).

Starting from pure noise wastes steps re-creating what y already contains; **ResShift** [16] instead builds a chain that moves between the high- and low-resolution images by shifting their residual, and reports competitive results with only 15 sampling steps.

Route B: zero-shot posterior sampling with a pretrained model

Plain version. Train one unconditional diffusion model on clean images. To restore, run its sampler as usual, but after every step check the current clean guess against the input and correct it a little. The model contributes "looks real"; the correction contributes "agrees with what we measured".

Precise version. Apply the score identity of the previous section to the noisy variable x_t:

$$\nabla_{x_t} \log p_t(x_t \mid y) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log p_t(y \mid x_t),$$

where the first term is the pretrained model's score and the second is the **noisy likelihood**: how well the noisy state x_t explains y. The first term we have. The second is the hard part, because y depends on the *clean* image, and

$$p_t(y \mid x_t) = \int p(y \mid x_0)\, p(x_0 \mid x_t)\, dx_0$$

averages the likelihood over all clean images that could have produced x_t, which is intractable for an image-scale model. The methods differ mainly in how they approximate this term, and they all end up with the same loop (Figure 4): a **prior step** (one reverse-diffusion step with the pretrained model) and a **data-consistency step** (a gradient step, a projection or a closed-form correction toward $\mathcal{D}(\hat{x}_0) \approx y$).

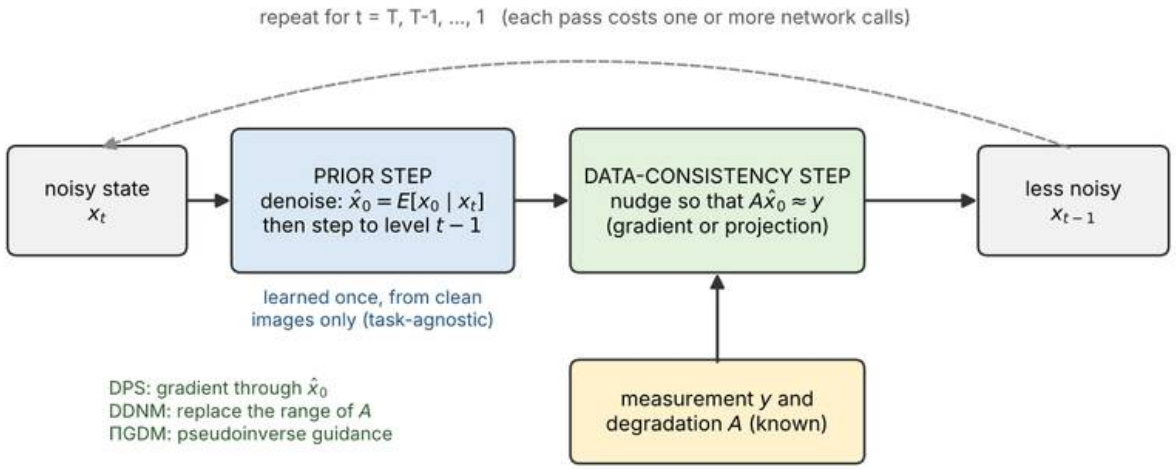


Figure 4 — The loop behind zero-shot diffusion restoration. The prior step uses a model trained only on clean images; the data step uses the known degradation. Methods differ in exactly how, and where in the step, the data correction is applied.

DPS (diffusion posterior sampling) [17] is the simplest to state. It replaces the average over $p(x_0 \mid x_t)$ by its mean, $p_t(y \mid x_t) \approx p(y \mid \hat{x}_0(x_t))$. With Gaussian noise this gives the data step

$$x_{t-1} \leftarrow x'_{t-1} - \zeta_t \nabla_{x_t} \|y - \mathcal{D}(\hat{x}_0(x_t))\|^2,$$

where x'_{t-1} is the result of the ordinary prior step and $\zeta_t > 0$ is a step size; the paper uses $\zeta_t = \zeta'/\|y - \mathcal{D}(\hat{x}_0)\|$ with a constant ζ'. The gradient passes through $\hat{x}_0$, so it needs **backpropagation through the denoiser** at every step. Because it never projects exactly onto the measurements, DPS also works with noisy and nonlinear degradations (the paper includes Poisson noise, Fourier phase retrieval and non-uniform deblurring).

The other members differ mainly in the data step:

- **DDRM** (denoising diffusion restoration models) [18] handles linear degradations in the coordinates of the degradation's singular value decomposition, treating measured and unmeasured coordinates differently; no backpropagation, few steps.
- **DDNM** (denoising diffusion null-space model) [19] copies the part of $\hat{x}_0$ that the measurement determines from the data, A^+y (with A^+ the pseudoinverse of a linear A), and keeps the model's guess only in the part the measurement cannot see.
- **ΠGDM** (pseudoinverse-guided diffusion models) [20] builds its guidance from a pseudoinverse of the degradation and an explicit uncertainty for $\hat{x}_0$; it backpropagates through the denoiser but not through the degradation, so it also covers non-differentiable measurements such as JPEG.

- **DiffPIR** [21] puts plug-and-play splitting (a data step and a denoiser step in turn, as described in DLCV 04) inside the diffusion sampler and reports good results with at most 100 network evaluations.

Optional: How the zero-shot diffusion samplers differ (in the Supplementary notes). The data step of each method, the range–null-space split and the SVD view, with a three-line check.

Code: a DPS-style sampler where the true posterior is known. We reuse the four-blob prior and observe one linear projection of the 2-D "image", $y = Ax + n$ with $A = [1, 1]/\sqrt{2}$ and noise standard deviation $\sigma_y = 0.2$. The blobs at $(-2, 2)$ and $(2, -2)$ both have $Ax \approx 0$, so a measurement near 0 is genuinely ambiguous. Prior and likelihood are Gaussian within each blob, so the exact posterior is again a four-blob mixture and we can check the sampler against it. The Jacobian of $\hat{x}_0$ is taken by finite differences, cheap in 2-D; at image scale this is the backpropagation step.

```
# Continues the previous snippet (same session): mu, w, s2, T, abar, x0_hat, ddpm_step.
rng = np.random.default_rng(3)
A = np.array([[1.0, 1.0]]) / np.sqrt(2)          # we only observe one projection of x
sig_y = 0.2
x_true = np.array([1.8, -2.1])
y = A @ x_true + sig_y * rng.standard_normal(1)

# Reference: the exact posterior of a Gaussian mixture under a linear-Gaussian measurement.
S = (A @ A.T)[0, 0] * s2 + sig_y**2               # variance of y within one blob
logpost = np.log(w) - (y - mu @ A.T)[:, 0] ** 2 / (2 * S)
post_w = np.exp(logpost - logpost.max()); post_w /= post_w.sum()
gain = s2 * A.T / S                               # Kalman gain, shape (2, 1)
post_mu = mu + (y - mu @ A.T) @ gain.T            # posterior mean inside each blob
post_cov = s2 * (np.eye(2) - gain @ A)
k = rng.choice(4, size=2000, p=post_w)
exact = post_mu[k] + rng.multivariate_normal(np.zeros(2), post_cov, size=2000)
mmse = post_w @ post_mu                           # posterior mean
least_sq = (np.linalg.pinv(A) @ y).ravel()        # minimum-norm solution of A x = y

def data_grad(x, ab, h=1e-4):
    """grad_x ||y - A x0_hat(x)||^2 through Tweedie's estimate (finite-difference
    ↪ Jacobian)."""
    J = np.stack([(x0_hat(x + h * e, ab) - x0_hat(x - h * e, ab)) / (2 * h) for e in
    ↪ np.eye(2)], -1)
    resid = y - x0_hat(x, ab) @ A.T               # (N, 1)
    return -2 * np.einsum("nij,ik,nk->nj", J, A.T, resid), np.abs(resid)

x = rng.standard_normal((2000, 2))
for t in reversed(range(T)):
    x_prior = ddpm_step(x, t, x0_hat(x, abar[t]))  # 1) prior step: one reverse-diffusion
    ↪ step
    g, res = data_grad(x, abar[t])
    x = x_prior - 0.1 / (res + 1e-8) * g          # 2) data-consistency step, DPS step size

blob = lambda pts: np.bincount(np.argmin(((pts[:, None] - mu) ** 2).sum(-1), 1),
↪ minlength=4) / len(pts)
```

```
print("least squares", least_sq.round(2), "| posterior mean", mmse.round(2), "| truth",
 ↪  x_true)
for name, pts in [("exact posterior", exact), ("DPS", x)]:
    print(f"{name:15s} blob shares {blob(pts)} | median |y - Ax| {np.median(np.abs(y - pts @
     ↪  A.T)):.3f}"
          f" | mean sq. error {np.mean(((pts - x_true) ** 2).sum(1)):.2f}")
for name, est in [("least squares", least_sq), ("posterior mean", mmse)]:
    print(f"{name:15s} sq. error {np.sum((est - x_true) ** 2):.2f}")
# least squares [0.14 0.14] | posterior mean [ 0.5 -0.3] | truth [ 1.8 -2.1]
# exact posterior blob shares [0.    0.409 0.591 0.   ] | median |y - Ax| 0.122 | mean sq.
 ↪  error 12.97
# DPS             blob shares [0.     0.4055 0.5945 0.    ] | median |y - Ax| 0.100 | mean
 ↪  sq. error 13.09
# least squares   sq. error 7.77
# posterior mean  sq. error 4.93
```

Read the output with Figure 5:

- The **least-squares** solution $A^{+}y$ and the **posterior mean** both lie *between* the two plausible blobs, where the prior is almost empty: the "blurry" answers of this toy. The posterior mean has the lowest squared error (4.93) because it hedges.
- The **DPS-style samples** fall in the two right blobs with shares 0.41 and 0.59, matching the exact posterior. Their mean squared error (13.09) is much higher, because about 40% of them land in the other blob: a confident, wrong answer. That is posterior sampling working as intended, and also the 2-D version of a hallucinated detail.
- DPS fits the measurement slightly *more* tightly than the true posterior (median residual 0.100 versus 0.122): it is an approximation, and its residual depends on the step size ζ' (Exercise 3). In high dimensions such deviations are much harder to see.

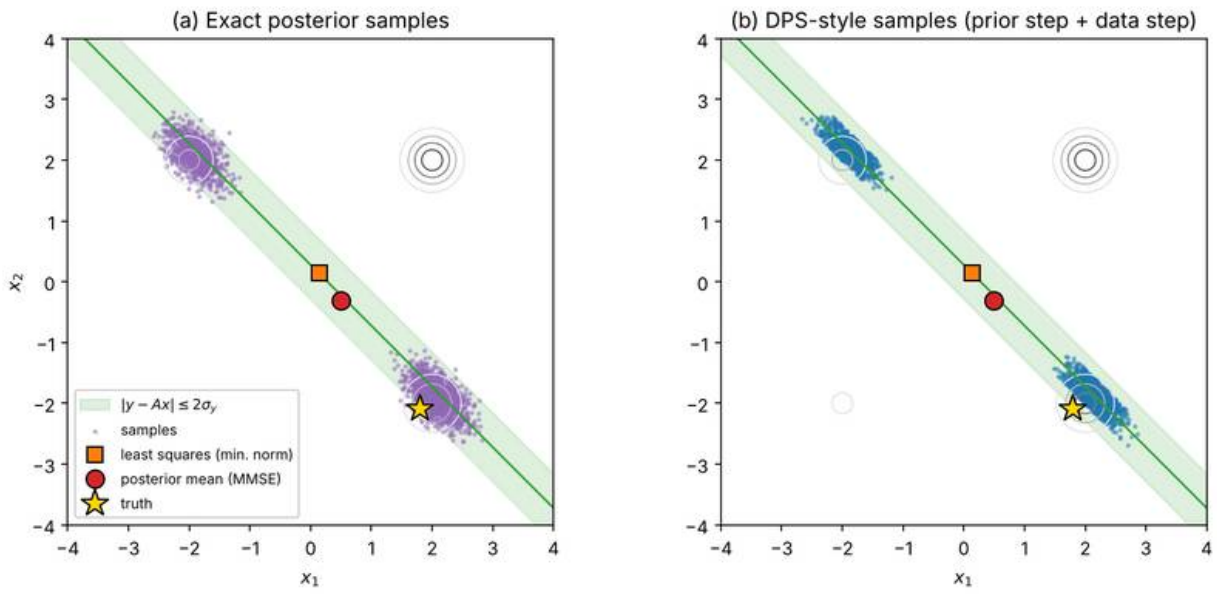


Figure 5 — A linear inverse problem with a known posterior. (a) Exact posterior samples; (b) DPS-style samples from the loop of Figure 4. Both land in the two blobs the measurement allows, while least squares and the posterior mean land in the empty space between them.

Optional: Chung et al., "Diffusion Posterior Sampling for General Noisy Inverse Problems" (in the Paper guides). What the paper approximates, what it measured and what to check in its tables.

Choosing between conditional and zero-shot

Question	Conditional (SR3, Palette, ResShift)	Zero-shot (DPS, DDRM, DDNM, ΠGDM, DiffPIR)
Training	One model per family of degradations, on pairs	One unconditional model on clean images
Degradation at test time	Implicit; can be unknown if it was in training	Must be known (or estimated) and usually differentiable
Noise and real camera pipelines	Handled if they were in training	Handled only as far as the likelihood model allows
Cost per image	Tens to hundreds of steps, no backpropagation	Often hundreds of steps; DPS-type methods also backpropagate
Good fit	One task with plenty of training pairs	Known physics, new operators, no retraining

A rough rule: if you can collect or synthesize pairs for exactly your degradation, train a conditional model (or a non-generative one). If the operator is known physics (a blur kernel, a sampling mask, a tomography projector) and retraining for every setting is impractical, zero-shot sampling is attractive.

Family 3: latent-diffusion and foundation-model restorers

Plain version. The strongest image generators are large text-to-image diffusion models trained on huge image collections. They know what grass, fabric and brick look like far better than any restoration dataset could teach. Recent real-world restorers keep such a model frozen (or almost frozen), add a small network that shows it the damaged image, and let it paint in the missing detail. The result can look stunning, and it can invent things.

Building on a pretrained text-to-image model

Latent diffusion [22] runs diffusion in the compressed latent space of an autoencoder instead of on pixels, which is what makes large text-to-image models affordable (ML 07). ControlNet [23] showed how to add a new image condition to such a model, which its authors describe as pretrained with billions of images, without damaging it: the pretrained model is locked, a trainable copy of its encoding layers receives the condition, and the two are joined by zero-initialized convolutions, so training starts from the unchanged model. Restoration methods adapt both ideas.

StableSR [24] keeps the generator fixed and injects the low-quality image through a time-aware encoder, with a user-set scalar that trades quality for fidelity. **DiffBIR** [25] first removes the degradation with an ordinary restoration network and then regenerates lost detail with a ControlNet-style adapter, again with a tunable fidelity guidance. **SUPIR** [26] scales up the prior and the data (the paper describes 20 million high-resolution images with text annotations) and lets text prompts steer the restoration.

Fewer steps: from hundreds to one

Multi-step latent restorers are slow, and their random starting noise makes outputs vary from run to run. Distilled and re-designed models reduce both: **SinSR** [27], **OSEDiff** [28] and **TSD-SR** [30] restore in one step, and **InvSR** [29] in one to five, mostly by starting from the low-quality image instead of pure noise and by distilling a many-step teacher. The general distillation families (progressive, consistency, adversarial) are described in ML 07.

Optional: Latent and one-step restorers, method by method (in the Supplementary notes). What StableSR, DiffBIR, SUPIR, SinSR, OSEDiff, InvSR and TSD-SR each add.

Two consequences. First, a one-step restorer that starts from the input is nearly deterministic: it gives *one* plausible answer, like a GAN, rather than samples with the right spread. Second, step count is not the whole cost: a one-step model may have a larger network, and latent methods also pay for the autoencoder's encoder and decoder at full resolution.

The hallucination problem

Plain version. A strong prior fills gaps with what is *typical*, not with what was *there*. Where the input says little, the output is mostly prior: it looks real, and nothing in the image tells you it was invented. Where it shows up:

- **Text and symbols.** Letters on signs, plates and documents are very specific: a small error changes the meaning. A generative restorer can produce clean, legible, wrong characters.
- **Faces and identity.** Face priors produce *a* face from the training distribution; for a specific person, a slightly different nose or eye is an identity change.
- **Textures.** Brick, fabric and foliage may be replaced by a different but equally plausible pattern.
- **Latent decoders.** The autoencoder of a latent restorer has its own reconstruction error; encode and decode a clean image to see what it alone changes, as ML 07 recommends.
- **Medical and scientific images.** A plausible structure that is not there, or a real lesion smoothed away, is a diagnostic error. Bhadra et al. [35] analyze this in tomographic reconstruction by splitting an estimate into the part determined by the measurements and a **null-space** part supplied by the prior, and define a hallucination map from it.

The null-space view explains the mechanism in one line, using the decomposition behind DDNM: for a linear A, any estimate $\hat{x}$ satisfies $\hat{x} = A^{+}A\hat{x} + (I - A^{+}A)\hat{x}$. Data consistency constrains only the first part. Everything in the second part, which is large when A discards a lot, comes from the prior, so **a perfectly data-consistent output can still be largely invented**. Consistency with y is necessary, never sufficient.

Measuring hallucination is itself a research topic. A recent study of generative super-resolution argues that hallucinated details are poorly captured both by fidelity metrics and by no-reference quality models, and proposes a "Hallucination Score" obtained by prompting a multimodal large language model, which its authors found to align closely with human judgments [36].

Family 4: flow matching and rectified flow restorers

Plain version. Flow matching learns a smooth "velocity field" that carries noise to images along nearly straight paths, so it needs few steps. For restoration, one can carry a *blurry best guess* to a realistic image, or plug a pretrained flow into the same prior-plus-data loop as diffusion.

Flow matching [31] and rectified flow [32] train a network to predict the velocity of a path between a simple distribution and the data distribution; samples come from integrating an ordinary differential equation, and straight paths need few integration steps (ML 07). Two verified restoration uses:

- **PMRF** (posterior-mean rectified flow) [33] targets the point $D_{\max}$ of the perception–distortion section directly: the estimator with the lowest MSE among those whose outputs are distributed like real images. Following a theoretical result the paper cites, it first predicts the **posterior mean** (an ordinary MMSE network) and then **transports** that blurry estimate to a realistic image with a rectified flow model that approximates the optimal transport map. This is a clean way to think about the whole chapter: first get the best average, then move it as little as possible to make it look real.
- **PnP-Flow** [34] defines a time-dependent denoiser from a pretrained flow-matching model and alternates gradient steps on the data-fidelity term, reprojections onto the learned flow path, and denoising: the loop of Figure 4 with a flow instead of a diffusion model, avoiding backpropagation through the ODE.

Evaluating generative restoration

Plain version. Ask four separate questions: is the output close to the truth (when you have it)? Does it look real? Does it agree with the input? And how much does it change when you run it again? One number cannot answer all four, and a pretty picture answers none of them.

The metric definitions are in DLCV 04. For a generative restorer, use them as follows.

1. **Fidelity, when a reference exists.** PSNR and SSIM [39] (low distortion), LPIPS [38] (perceptual distance to the reference). Expect generative methods to lose PSNR against regression methods. An exact posterior sampler loses about 3 dB against the MMSE estimator, so a much larger gap to a strong regression model means the method is far from the best trade-off, not that it is "more realistic".
2. **Realism.** No-reference metrics such as NIQE [41], MUSIQ [42] and CLIP-IQA [43] score single images; FID [40] compares the *distribution* of a set of outputs with real images, which is the quantity $d(p_X, p_{\hat{X}})$ of the trade-off but needs many images. No-reference metrics can be gamed by adding texture, so never use them alone.
3. **Consistency with the input.** Re-apply the degradation and compare: $\|y - \mathcal{D}(\hat{x})\|$ should be about the noise level. This needs no reference and catches outputs that ignore the input. It cannot catch null-space hallucinations (previous section).
4. **Stochasticity.** A sampler gives a distribution of outputs. Report it.

Reporting several samples. The snippet continues the step-edge toy: one test input, eight restorations per run, and three runs with different seeds, as three users of the same method would see.

```
# Continues the first snippet (same session): X, A, posterior, positions, sigma, n, factor.
rng = np.random.default_rng(5)
```

```
x = X[20]                                                  # a test signal: edge at pixel 28
y = A(x) + sigma * rng.standard_normal(n // factor)
p = posterior(y)
for seed in range(3):                                      # three "users" with different
 ↪  seeds
    r = np.random.default_rng(seed)
    S = X[r.choice(len(positions), size=8, p=p)]           # 8 restorations of the same
     ↪  input
    err = ((S - x) ** 2).mean(axis=1)
    print(f"seed {seed}: MSE mean {err.mean():.4f} ± {err.std():.4f} | best of 8
     ↪  {err.min():.4f}"
          f" | edges at {sorted(set(np.argmax(S, axis=1).tolist()))}")
S = X[rng.choice(len(positions), size=200, p=p)]
std = S.std(axis=0)                                        # per-pixel spread across samples
print("pixels with std > 0.1:", np.flatnonzero(std > 0.1), "| true edge at", np.argmax(x))
# seed 0: MSE mean 0.0273 ± 0.0268 | best of 8 0.0000 | edges at [23, 24, 26, 27, 28, 29]
# seed 1: MSE mean 0.0273 ± 0.0103 | best of 8 0.0156 | edges at [25, 26, 27, 29, 30]
# seed 2: MSE mean 0.0293 ± 0.0213 | best of 8 0.0000 | edges at [24, 25, 26, 27, 28]
# pixels with std > 0.1: [22 23 24 25 26 27 28 29 30] | true edge at 28
```

Three lessons, each with a reporting rule:

- **The spread is as large as the mean.** On one input, the per-sample MSE varies by about as much as its average. Report the mean and the spread over samples (and over seeds), not one sample.
- **"Best of K" is an oracle.** Picking the sample closest to the ground truth gives an MSE of zero for two of the three seeds, a number no user can obtain because users do not have the ground truth. If a paper reports best-of-K, it must say so, and it should not be compared with single-sample results. (DPS, for example, reports the best of four samples for phase retrieval, following common practice for that problem, and says so [17].)
- **Sample variation is a free uncertainty map.** The per-pixel standard deviation over 200 samples is large exactly around the edge (pixels 22 to 30) and zero elsewhere. Showing such a map next to the restored image tells a viewer where *not* to trust the details. For calibrated guarantees, conformal methods can wrap any image-to-image regressor with per-pixel intervals that contain the true value with a chosen probability [37].

Compute. Report the number of network evaluations (NFE), including guidance and any extra passes, the resolution, the hardware and wall-clock time, and for latent methods the autoencoder's cost. As a concrete anchor, the DPS paper reports about 95 seconds per 256 × 256 FFHQ image and about 600 seconds per ImageNet image, each with 1,000 network evaluations on one RTX 2080 Ti GPU [17]. Distilled one-step restorers cut the step count to one, but compare full pipelines, not step counts.

A minimal honest report: PSNR/SSIM and LPIPS on references; two or three no-reference metrics from different families, plus FID on a large enough set; the re-degradation residual; mean and

spread over samples and seeds; NFE and time; a user study for headline claims; and a gallery that includes failures, not only the best-looking results.

Failure modes and when NOT to use generative restoration

Plain version. Generative restoration is the right tool when the goal is a pleasant, plausible picture. It is the wrong tool when the picture will be used as evidence.

Typical failure modes.

- **Hallucinated specifics**: text, digits, faces, small objects, medical findings (see the hallucination section).
- **Ignoring the input**: weak data consistency, especially with strong guidance toward the prior or a prompt, gives realistic images of a slightly different scene; color shifts are a known issue of diffusion super-resolution [46].
- **Mismatch between assumed and real degradation**: zero-shot methods need the right $\mathcal{D}$. A wrong blur kernel or noise level is propagated by the data step, and the prior then makes the wrong answer look convincing.
- **Out-of-distribution inputs**: a prior trained on faces or natural photos pulls microscopy, satellite or medical images toward the wrong "typical" appearance, and faces unlike the training data toward the typical face (the bias discussed for PULSE [7]).
- **Instability across runs and sizes**: different seeds give different details, and models trained at a fixed size need special handling for larger images (StableSR's progressive aggregation sampling is one answer [24]).
- **Cost**: hundreds of network calls per image, or a very large one-step network.

When not to use it, or to use it only with explicit labeling and uncertainty:

- **Evidence and measurement**: forensic, legal, journalistic and surveillance use, license plates, documents to be read, anything where a person will act on a detail. Prefer an MMSE restorer (smooth, but it does not commit to specifics) or show a set of samples with an uncertainty map.
- **Medicine and science**: diagnosis, quantification and any downstream measurement. Use methods validated for the task, with hallucination analysis such as [35], and keep the measured data visible.
- **Mild degradations**: when the posterior is narrow (light noise, 2× super-resolution of a clean image), a regression network already gives a sharp answer, faster and with less risk.

- **Pipelines that feed another algorithm** (detection, OCR, measurement): restoration should be judged by the downstream task; invented detail can create false detections.

A short decision rule: if a wrong but realistic detail is worse than a blurry region, do not use a generative restorer without uncertainty reporting.

Modern view

Survey papers.

- **Li et al., "Diffusion Models for Image Restoration and Enhancement: A Comprehensive Survey" (IJCV 2025)** [44]. Splits diffusion restoration into **supervised** methods (grouped by how they condition the sampler) and **zero-shot** methods (projection, decomposition and posterior estimation), adds blind and real-world restoration, and compares methods on several tasks. Takeaway: the best perceptual quality, with sampling cost, model size, real-world degradations and evaluation as the main obstacles.
- **Daras et al., "A Survey on Diffusion Models for Inverse Problems" (arXiv 2024)** [45]. Covers methods that use a *pretrained* diffusion model without further training, in four families by how they treat the measurement term: **explicit approximations** of $\nabla \log p_t(y \mid x_t)$ (DPS, ΠGDM and the DDRM family among them), **variational inference**, **CSGM-type** latent optimization and **asymptotically exact** Monte Carlo methods; a second taxonomy sorts them by how consistency is enforced (gradients, projections, sampling, optimization). Takeaway: many seemingly different methods compute a measurement error and "lift" it back to image space in different ways, and the field lacks a standard benchmark of quality versus compute.
- **Moser et al., "Diffusion Models, Image Super-Resolution, and Everything: A Survey" (IEEE TNNLS 2025)** [46]. Concentrates on super-resolution: foundations, input domains (pixel or latent), conditioning, guidance, corruption spaces and zero-shot methods. Takeaway: diffusion closed much of the gap to human preference, while computational cost, comparability between papers, explainability and color shifts are open.
- **Xia et al., "GAN Inversion: A Survey" (IEEE TPAMI 2023)** [47]. Reviews how images are mapped back into a pretrained GAN's latent space and how this supports restoration and editing. Takeaway: the GAN-prior methods of Family 1 are special cases of inversion, and their limits are the generator's coverage and the inversion's cost.

Optional: Li et al., "Diffusion Models for Image Restoration and Enhancement: A Comprehensive Survey" (in the Paper guides). How to use the survey's taxonomy and comparison tables.

State of the art, 2023–2026.

1. **Pretrained text-to-image models became the default prior for real-world restoration**, connected through adapters (StableSR, DiffBIR, SUPIR), with explicit fidelity knobs because uncontrolled generation drifts from the input.
2. **Step counts fell from hundreds to one** through residual-shifting chains, distillation and inversion (ResShift, SinSR, OSEDiff, InvSR, TSD-SR). One-step models trade the posterior's spread for speed and repeatability.
3. **Zero-shot posterior sampling matured into a family** with shared structure (DPS, DDRM, DDNM, ΠGDM, DiffPIR) and was extended to flows (PnP-Flow), with surveys that unify the approximations [45].
4. **Theory re-entered practice**: the perception–distortion bound now guides method design directly, as in PMRF's "posterior mean, then transport" [33].
5. **Hallucination started to be measured**, not only mentioned, for example with dedicated scores [36].

Open problems.

- **Measuring faithfulness**: no accepted metric says "sharp *and* correct". Fidelity metrics penalize all detail; no-reference metrics reward invented detail.
- **True posterior sampling at image scale**: zero-shot methods are approximate (our toy shows a small bias even in 2-D), and few-step models give little diversity. Calibrated uncertainty for high-resolution images is largely open.
- **Unknown degradations**: zero-shot methods need $\mathcal{D}$; supervised ones need the right training distribution.
- **Cost**: high-resolution and video restoration with large priors remain expensive.
- **Responsible use**: domain shift, demographic bias of face priors and the labeling of generated content are unsolved in practice.

Key takeaways

- With a squared-error loss, the optimal restorer outputs the **posterior mean** $\mathbb{E}[x \mid y]$; when the input allows several answers, that average is blurry. Blur is the correct answer to the MSE question.
- **Generative restoration** returns a plausible sample from $p(x \mid y)$ instead: realistic and consistent

with the input, but committed to details that may be wrong.

- The **perception–distortion trade-off** [1] makes realism and low distortion incompatible beyond a bound; for MSE, perfect realism costs at most twice the minimum MSE (about 3 dB), reached by posterior sampling.
- GAN losses and **pretrained GAN priors** (PULSE, GLEAN, GFP-GAN; codebook priors such as CodeFormer) restore in one pass but only within the generator's domain.
- Diffusion restoration is either **conditional** (trained on pairs: SR3, Palette, ResShift) or **zero-shot** (a pretrained prior plus a data-consistency step: DPS, DDRM, DDNM, ΠGDM, DiffPIR), all built on the loop "prior step, then data step".
- **Latent-diffusion restorers** (StableSR, DiffBIR, SUPIR) give the most convincing real-world results; **distilled** ones (SinSR, OSEDiff, InvSR, TSD-SR) need one or a few steps. All can hallucinate, and data consistency cannot detect null-space inventions.
- Evaluate with fidelity, realism, re-degradation consistency, **several samples and seeds** (never an undisclosed best-of-K), uncertainty maps and full compute cost.
- Do not use generative restoration where a realistic wrong detail is worse than a blurry region: evidence, medicine, science, text to be read.

Exercises

1. **The factor of two, by hand.** In the step-edge toy, suppose that for one input the posterior puts probability 0.5 on an edge at pixel 30 and 0.5 on an edge at pixel 32, and the truth is at 30. Compute the MSE (per pixel, over 64 pixels) of the posterior mean and the *expected* MSE of a posterior sample. Is the ratio two? Explain why the factor of two in the theorem is an average over inputs, not a guarantee for each input.

   Hint

   The two candidates differ only at pixels 30 and 31. The mean is 0.5 there, so its MSE is $2 \times 0.25/64 \approx 0.0078$; a sample is right with probability 0.5 and wrong by 1 at two pixels otherwise, so its expected MSE is $0.5 \times 2/64 \approx 0.0156$: a ratio of two. If instead the posterior put 0.9 on the wrong position, the ratio for *this* truth would be $1.8/1.62 \approx 1.1$. The theorem averages over x and y jointly, where the truth itself is distributed like the posterior.

2. **Walking the trade-off.** Using the first snippet, compute MSE and W1 for (a) the posterior mean plus Gaussian noise of standard deviation 0.1, (b) the posterior *median* edge position (choose the position where the posterior's cumulative probability first exceeds 0.5), and (c) the posterior *mode*. Place them on Figure 2's plane. Which are near the posterior-sample point?

   Hint

Adding noise does not create edges: in our run the largest step was only 0.40, W1 grew to about 0.095 (noise creates many unrealistic in-between values) and the MSE grew to 0.0274, so (a) moves *away* from both goals. The median and the mode are single real edges, so their W1 is near zero like a sample's (0.0009 and 0.0011 in our run), and their MSE (0.0245 and 0.0259) is *below* a random sample's (0.0338) because they never pick an unlikely position. They are deterministic, though: they do not reproduce the posterior's spread, which a set-level metric could detect (all outputs for similar inputs agree).

3. **DPS step size.** In the DPS-style snippet, replace the constant 0.1 in the data step by 0.02, 0.5 and 2.0. For each, report the blob shares, the median residual $|y - Ax|$ and the mean squared error, and compare with the exact posterior. Then replace the normalized step $0.1/(\text{res} + 10^{-8})$ g by an unnormalized step $0.2\,g$. What do you conclude about how DPS's step size relates to the noise level σ_y?

Hint

With the normalized step, each data step has roughly the same length whatever the residual, so the final median residual settles near the constant itself (in our runs 0.021, 0.100, 0.496 and 1.994 for 0.02, 0.1, 0.5 and 2.0). Large values leave residuals far above the noise level and distort the blob shares (0.24 and 0.76 at 2.0, against 0.41 and 0.59 exactly); at 0.02 about 1% of the samples ended in a blob the measurement rules out, because the early corrections were too weak. The unnormalized step $0.2\,g$ gave a median residual of 0.007, far tighter than the noise level allows. Neither choice is tied to σ_y the way an exact posterior is: the step size is a tuned hyperparameter, one reason DPS is an approximation.

4. **Steps and accuracy.** In the diffusion snippet, run the sampler with $T = 50$ (and `beta = np.linspace(1e-4, 0.2, T)`) and with $T = 1000$ (and `np.linspace(1e-4, 0.01, T)`). Record the spread inside each blob. Relate the result to the cost discussion and to why distilled one-step restorers are trained rather than obtained by simply using fewer steps.

Hint

In our runs the spread was about 0.263 for 50 steps, 0.283 for 200 and 0.296 for 1,000 (target 0.3): coarse steps make the reverse process too narrow even with the *exact* score. A one-step model cannot get its accuracy by taking one huge step of the same sampler; it must be trained (distilled or built differently) to jump correctly.

5. **Null-space hallucination.** Take a 1-D signal of length 16 and A = average of consecutive pairs (an 8×16 matrix). Compute $P = A^{+}A$ and verify that P is a projection ($P^2 = P$). Construct two different signals with exactly the same measurement Ax, one smooth and one with a sharp checkerboard pattern. Which one would a generative prior of natural signals pick, and which part of each signal is "invented"?

Hint

The null space of pair-averaging contains every pattern that alternates sign within a pair, $(+a, -a)$ with any amplitude a per pair; adding it to a smooth signal leaves Ax unchanged. The within-pair

differences, $(I - P)x$, are invisible to the data and supplied entirely by the prior: a smooth-signal prior picks small ones, a texture prior may add some. This is the hallucination-map idea of [35].

6. **Design an evaluation.** You must decide whether a one-step latent-diffusion restorer can replace an MSE-trained network for restoring scanned archival documents. Write the evaluation protocol: data, metrics, number of samples and seeds, a check for hallucinated characters, compute measurements, and a decision rule.

Hint

Use real scans with references where possible (rescan at high resolution) plus synthetic degradations of clean documents. Report PSNR/SSIM/LPIPS, a no-reference metric, and, crucially, an OCR character error rate against the true text for both methods and for several seeds; also report how often two seeds disagree on a character (an uncertainty signal). Include NFE, time and memory including the autoencoder. Decide by the task: if the generative model raises character errors or changes characters between seeds, keep the MSE network for reading and use the generative one only for display, clearly labeled.

Supplementary notes

These notes are optional background that the main text does not depend on.

Why posterior sampling exactly doubles the MSE

Plain version. A posterior sample is like a second, independent draw of the truth. The distance between two independent draws is, on average, twice the distance of either draw from their common mean.

Precise version. Fix y and let $m = \mathbb{E}[x \mid y]$. If $\hat{x}$ is drawn from $p(x \mid y)$ independently of the true x, both x and $\hat{x}$ have mean m and the same covariance given y. Then

$$\mathbb{E}[\|\hat{x} - x\|^2 \mid y] = \mathbb{E}[\|\hat{x} - m\|^2 \mid y] + \mathbb{E}[\|x - m\|^2 \mid y] - 2\,\mathbb{E}[(\hat{x} - m)^\top (x - m) \mid y] = 2\,\mathbb{E}[\|x - m\|^2 \mid y],$$

where the cross term vanishes because $\hat{x} - m$ and $x - m$ are independent with zero mean given y, and the two remaining terms are equal because $\hat{x}$ and x have the same conditional distribution. Averaging over y gives $\mathrm{MSE}(\text{sample}) = 2\,\mathrm{MSE}(\text{MMSE})$. The blends of Figure 2 follow the same way: $(1 - \lambda)m + \lambda\hat{x} - x = (m - x) + \lambda(\hat{x} - m)$, and the two parts are uncorrelated, so the MSE is $(1 + \lambda^2)D_{\min}$. Blau and Michaeli give this argument in an appendix of [1]. Note what the proof needs: an *exact* posterior sample. An approximate sampler can be above or below the factor of two.

GAN-prior restorers in more detail

Plain version. All four methods ask a pretrained generator "which image that you can draw matches this input?". They differ in how they ask: by searching the generator's input space for each image (PULSE), or by training an encoder that asks in one shot (GLEAN, GFP-GAN, CodeFormer).

The methods. PULSE [7] optimizes the latent code of a pretrained StyleGAN [6] so that the generated face, downscaled, matches the low-resolution input, and restricts the search to the region where high-dimensional Gaussian latent vectors concentrate (near a sphere), so outputs stay realistic. Its paper includes a discussion of the method's limitations and biases. GLEAN [8] builds an encoder–bank–decoder network around a pretrained GAN whose weights stay fixed during training: the encoder turns the low-resolution input into latent codes and features for the "latent bank", and a decoder with multi-resolution skip connections fuses the bank's features into the output, for large upscaling factors and several categories (faces, cats, cars, buildings). GFP-GAN [9] pairs a U-Net degradation-removal module with a pretrained face GAN and connects them with channel-split spatial feature transform layers, which modulate part of the GAN's features with spatial information from the input while leaving the rest free, as a balance between realness and fidelity. CodeFormer [10] learns a discrete codebook of high-quality face parts, predicts the code sequence of a degraded face with a transformer (restoration becomes code prediction), and adds a controllable feature transformation whose weight trades fidelity against quality at test time.

A 2-D inversion toy. If a generator can produce several images that all match the input, an optimizer started from different random points finds different ones. Take a toy "generator" $G(w) = (\cos w, \sin w)$, which maps a scalar latent w to the unit circle (the "manifold of real images"), and a measurement $y = x_1 + n$ that observes only the first coordinate. Minimizing $(\cos w - y)^2$ has two solutions on the circle for $|y| < 1$, $w = \pm \arccos y$, the upper and lower points with the same x_1. The snippet runs gradient descent from 10 random starts:

```
import numpy as np

rng = np.random.default_rng(0)
y = 0.3                                          # we observe only x1 = cos(w)
w = rng.uniform(-np.pi, np.pi, size=10)          # 10 random starting latents
for _ in range(500):
    w -= 0.2 * 2 * (np.cos(w) - y) * (-np.sin(w))  # gradient of (cos w - y)^2
x = np.stack([np.cos(w), np.sin(w)], 1)
print(np.unique(x.round(3), axis=0))
# [[ 0.3   -0.954]
#  [ 0.3    0.954]]
```

Both answers explain y exactly and both lie on the manifold: this is the PULSE situation [7] in miniature. Which one an optimizer returns depends on its start, not on any probability, so plain inversion gives plausible answers but not posterior samples with the right frequencies.

How the zero-shot diffusion samplers differ

Plain version. DPS, DDRM, DDNM, ΠGDM and DiffPIR all run the loop of Figure 4. They differ in the data step: a gradient through the denoiser, a split of the image into measured and unmeasured coordinates, a projection that copies the measured part from the data, a pseudoinverse-based guidance term, or a plug-and-play data subproblem.

The methods. DDRM [18] handles linear degradations $y = Hx + n$ through the singular value decomposition of H; its paper reports competitive results with as few as 20 network evaluations and needs no backpropagation. DDNM [19] uses the range–null-space projection below; DDNM+ adds handling of noise. ΠGDM [20] approximates $p(x_0 \mid x_t)$ by a Gaussian $\mathcal{N}(\hat{x}_0, r_t^2 I)$ with a time-dependent standard deviation r_t; for linear measurements this makes $p_t(y \mid x_t)$ Gaussian too, and the guidance becomes a vector–Jacobian product through the denoiser only, which is why it extends to non-differentiable measurements given a suitable pseudoinverse. DiffPIR [21] alternates, inside the diffusion sampler, a data subproblem solved for the current clean estimate and a denoising step by the diffusion model, the plug-and-play splitting of classical restoration.

Range–null-space projection and the SVD view. A linear measurement sees only part of an image. You can split any image into the part the measurement sees and the part it cannot see; a zero-shot restorer may invent the invisible part but should copy the visible part from the data.

Precise version. For a linear operator $A \in \mathbb{R}^{m \times d}$ with Moore–Penrose pseudoinverse A^+, the matrix $P = A^+A$ is the orthogonal projector onto the row space of A, and every x splits as

$$x = \underbrace{A^+A\,x}_{\text{range part}} + \underbrace{(I - A^+A)\,x}_{\text{null-space part}}, \qquad A\,(I - A^+A)\,x = 0,$$

where I is the $d \times d$ identity. For noiseless data $y = Ax$, the range part equals A^+y. DDNM [19] therefore corrects each clean estimate as $\hat{x}_0 \leftarrow A^+y + (I - A^+A)\,\hat{x}_0$: the measured part is copied from the data, the unmeasured part is kept from the diffusion model. DDRM [18] reaches a similar split through the singular value decomposition $A = U\Sigma V^\top$ (with U, V orthogonal and Σ diagonal): in the coordinates $V^\top x$, each coordinate is either scaled by a nonzero singular value (measured, possibly with noise of known size) or multiplied by zero (unmeasured), and its sampler treats the two kinds separately. A three-line check:

```
import numpy as np

A = np.kron(np.eye(8), [[0.5, 0.5]])              # average consecutive pairs: 8 x 16
P = np.linalg.pinv(A) @ A                         # projector onto the measured part
x = np.random.default_rng(0).standard_normal(16)
print(np.allclose(P @ P, P), np.allclose(A @ (x - P @ x), 0), int(round(np.trace(P))))
# True True 8
```

The projector is idempotent, the null-space part is invisible to A, and the trace shows that 8 of the 16 dimensions are measured: the other 8 are entirely up to the prior.

Latent and one-step restorers, method by method

Plain version. The latent restorers differ in how they show the damaged image to a frozen generator and in which fidelity knob they offer; the one-step models differ in how they teach a network to jump straight to the answer.

Multi-step latent restorers. StableSR [24] keeps the pretrained text-to-image model fixed and trains a time-aware encoder that injects features of the low-quality image; a controllable feature wrapping module mixes encoder features into the autoencoder's decoder with a user-set scalar, to recover fidelity lost to the sampler's randomness; and progressive aggregation sampling handles images larger than the model's native size. DiffBIR [25] splits blind restoration into degradation removal (a restoration module gives a smooth, faithful image) and information regeneration (IRControlNet, an adapter for a latent diffusion model trained on such cleaned conditions), with a region-adaptive restoration guidance whose scale trades realness against fidelity at sampling time. SUPIR [26] combines a large generative prior and model scaling with a training set of 20 million high-resolution, high-quality images with descriptive text, text prompts to steer restoration, negative-quality prompts to push away from low quality, and a restoration-guided sampling method meant to limit fidelity problems.

One- and few-step models. ResShift [16] shortened the chain by moving between the high- and low-resolution images (15 steps in the paper). SinSR [27] derives a deterministic sampler from an accelerated diffusion super-resolution model, distills it into a one-step student and adds a consistency-preserving loss that also uses the ground truth; the paper reports up to a tenfold speed-up. OSEDiff [28] starts from the low-quality image, fine-tunes the pretrained network with small trainable layers and regularizes the one-step output with variational score distillation in latent space. InvSR [29] trains a noise predictor that places the low-quality image at an intermediate point of a pretrained diffusion trajectory, so sampling can take any number of steps from one to five. TSD-SR [30] proposes a target score distillation objective and a distribution-aware sampling module for one-step real-world super-resolution.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Blau and Michaeli, "The Perception-Distortion Tradeoff" (CVPR 2018)

The problem. By 2017, super-resolution methods trained with adversarial losses produced images that people preferred, yet they scored worse in PSNR and SSIM than MSE-trained networks. Was this a flaw of the metrics, or something unavoidable?

The idea. Define perceptual quality as a property of the *output distribution*, the divergence between the distribution of restored images and that of real images (which a perfect discriminator would detect), and study the best possible distortion for each level of perceptual quality, over all possible estimators [1].

What it showed. The perception–distortion function is monotone, and convex for convex divergences, so the trade-off is a mathematical property of the problem, not of a particular metric or network, and it holds for every distortion measure. For squared error, perfect perceptual quality costs at most twice the minimum MSE, reached by posterior sampling (in the extended arXiv version). GAN training with a distortion term and an adversarial term is a principled way to approach the bound. The paper then plots many super-resolution methods of the time on the plane, using no-reference metrics as proxies for perceptual quality.

How to read it. Start with the definitions and Theorem 2 (the trade-off), then the factor-of-two result and its proof in the appendix, which is the argument in our Supplementary notes. When reading the experimental plots, check: (a) that the "perceptual quality" axis there is a no-reference metric (a proxy), not the divergence of the theory; (b) that the bound is drawn qualitatively, since it cannot be computed for real images; (c) which color channel and evaluation protocol were used for PSNR.

Limits. The theory is about distributions and expectations; it says nothing about which *particular* details an estimator invents, so it does not measure hallucination. The bound is not computable for natural images, and the factor of two is an upper bound, not a prediction.

Why it matters here. It is the theory behind every section of this chapter: why MSE networks blur, why generative methods lose PSNR, and why a method should be described by a point on a plane rather than a single score.

Chung et al., "Diffusion Posterior Sampling for General Noisy Inverse Problems" (ICLR 2023)

The problem. Earlier diffusion solvers for inverse problems mostly enforced exact consistency with the measurement by projection, which works for noiseless linear problems but amplifies noise and does not apply to nonlinear measurements [17].

The idea. Approximate the intractable noisy likelihood $p(y \mid x_t)$ by $p(y \mid \hat{x}_0(x_t))$, with $\hat{x}_0$ from Tweedie's formula, and take a gradient step on $\|y - \mathcal{A}(\hat{x}_0)\|^2$ after each ordinary reverse step (Algorithm 1 in the paper; a Poisson version uses a weighted norm). The paper bounds the error of this approximation (a "Jensen gap") and argues that it shrinks as measurement noise grows.

What it showed. On FFHQ and ImageNet at 256 × 256 (1,000 validation images each) with Gaussian measurement noise of standard deviation 0.05, it compared DPS with DDRM, manifold-constrained gradients, plug-and-play ADMM, total-variation optimization and score-SDE solvers

on inpainting, super-resolution and Gaussian and motion deblurring, reporting FID and LPIPS; it also handled Poisson noise, Fourier phase retrieval and non-uniform deblurring. The appendix reports about 95 seconds (FFHQ) and 600 seconds (ImageNet) per image with 1,000 network evaluations on one RTX 2080 Ti.

How to read it. Read Section 3 with Figure 2 (the graphical model of y, x_0, x_t), then Algorithm 1, then the step-size choice $\zeta_i = \zeta'/\|y - \mathcal{A}(\hat{x}_0(x_i))\|$ and the per-task values of ζ' in the appendix. Check: (a) that the tables report FID and LPIPS, so they measure realism and perceptual similarity, not PSNR; (b) that phase-retrieval numbers use the best of four samples for every method; (c) the NFE and wall-clock comparison in the appendix, where DPS is far slower than DDRM; (d) how sensitive results are to ζ', which is tuned per task.

Limits. The likelihood approximation is uncontrolled for small noise and strongly multimodal $p(x_0 \mid x_t)$; samples are not guaranteed to follow the true posterior (our toy shows a small bias even in 2-D); each step needs backpropagation through the network; and the degradation must be known and differentiable.

Why it matters here. DPS is the clearest instance of the prior-step + data-step loop and the template for many later zero-shot methods, and its appendix is a model of the cost reporting this chapter asks for.

Li et al., "Diffusion Models for Image Restoration and Enhancement: A Comprehensive Survey" (IJCV 2025)

The problem. Diffusion-based restoration grew quickly, with very different conditioning schemes, training regimes and benchmarks, which made methods hard to compare and follow [44].

The idea. Organize the field by learning paradigm (supervised versus zero-shot), then by how each method conditions or constrains the sampler, then by how it addresses blind and real-world degradations, and close with experimental comparisons and open directions.

What it showed. Its sections give a usable map: supervised methods are grouped by the form of conditioning and modeling (low-quality image as condition, pre-processed references, revised diffusion processes, residual generation, latent diffusion, efficient transfer learning); zero-shot methods by projection, decomposition and posterior estimation; real-world methods by degradation simulation, kernel estimation, domain translation and degradation-invariant learning. Its experiments section collects results on super-resolution, inpainting, deblurring and real-world restoration with PSNR, SSIM, LPIPS and FID. Its discussion highlights sampling efficiency, model size and compression, degradation estimation, all-in-one restoration and language-guided restoration.

How to read it. Use Sections 3 and 4 as a catalogue: find the group your problem belongs to, then read the two or three papers it cites there. In the comparison tables, check which numbers were

rerun by the authors and which were copied from papers, the resolution and number of steps, and whether a method's training data overlaps the test sets. Read Section 6 for open problems.

Limits. Like any survey of a fast field, it lags the newest methods (for example the 2025 one-step distillation methods), and its categories overlap: one method can fit several groups. Its numbers are not a benchmark in their own right.

Why it matters here. It places every method in this chapter's families into a larger map, and its supervised/zero-shot split is the same as this chapter's Route A and Route B.

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DLCV 05 • Low-Light Image Enhancement

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- How to evaluate enhancement honestly: which of the general metrics from DLCV 04 (full-reference, no-reference, user studies) tell you something about dark images, the low-light-specific measures (lightness order error, colorfulness, noise), the **GT-mean** pitfall, and task-driven evaluation.
- Which datasets and challenges (LOL, LOL-v2, LSRW, SDSD, SID, FiveK, ExDark, NTIRE 2024/2025) are used for what, and what remains unsolved.

The big picture

A photo taken in the dark is not just a normal photo with the brightness turned down. The sensor collected very few photons, so the picture is dim, noisy, and often has a color cast. Low-light image enhancement aims to produce the image you *would* have gotten with good light: well exposed, with natural colors, clean details, and no new artifacts.

> **Explain like I'm five**
>
> Imagine you are drawing in a room with the lights almost off. Your drawing comes out faint and smudgy, and you are not sure which crayon was which color. Turning on the lights later and photocopying the drawing "brighter" won't help: the smudges get brighter too, and the wrong colors stay wrong. A good low-light enhancer is like a friend who looks at your faint drawing and *redraws* it as it would look in daylight: same picture, right colors, without the smudges.

Why it matters: phones shoot in the dark every day, cars and robots must see at night, and surveillance, astronomy-like scientific imaging and medical endoscopy all fight the same physics. Low light also hurts downstream models: a detector trained on daytime photos loses accuracy at night. That makes LLIE both a picture-quality task and a robustness task for computer vision.

About this tutorial

This tutorial is expanded from an earlier Medium post on LLIE [1], which is written in Traditional Chinese (read it here). It follows the post's outline and adds math, code, figures and a survey of the literature.

Related chapters. DLCV 05A shows how enhancement models stay small and fast enough for phones and 4K video, and DLCV 05B covers fusing several images (infrared–visible, multi-focus, multi-exposure) into one.

Goal and difficulty

Plain version. We want to map a dark image to a normally exposed one. The hard part is that brightening and repairing are different jobs: brightening changes *tone* (how light or dark each region is), while repairing recovers *details and clean colors* that the darkness and noise destroyed.

Precise version. Unlike image-to-image translation, which changes an image's *attributes* (summer to winter, photo to painting), LLIE should keep the scene and its content exactly as they are and only undo the damage caused by insufficient light. The post [1] splits that damage into two difficulties:

1. **Tone enhancement.** Lift the exposure, globally and locally, without crushing highlights or flattening contrast. Bright regions such as lamps must stay bright, dark regions must open up, and the ordering of brightness between regions should be preserved.
2. **Detail restoration.** Recover texture and color that are buried in noise and quantization, while suppressing that noise rather than amplifying it.

In deep learning, detail restoration is what a network does anyway once it is trained to output a clean image, so most method design goes into the *tone* part (which image properties or physical model to build in) and the *learning mechanism* (which supervision and losses to use), rather than into a dedicated detail module.

Why it is not simple brightening

A linear gain $y = c\,x$ with $c > 1$ fails in four ways, all visible in Figure 1:

- **Noise.** Photon arrival is random. With few photons, the signal-to-noise ratio is low, and a gain multiplies signal and noise equally. The image gets brighter, not cleaner.

- **Color cast.** Low light is often colored (tungsten, sodium street lamps), and the camera's white balance and the three color channels have different noise levels. Scaling the channels equally keeps the cast; scaling them separately may produce strange colors.
- **Saturation and over-exposure.** Pixels already near the top (lamps, windows, screens) clip at 1. A global gain big enough for the shadows destroys the highlights.
- **Quantization.** A dark 8-bit image may use only 20–40 distinct gray levels. Stretching them produces visible banding.

A model of low-light capture

It helps to have an explicit degradation model, both for understanding and for **synthesizing training data**. Work in *linear* light (values proportional to photon counts), not on gamma-encoded pixels (see IP 03 on gamma). A widely used model is the Poisson–Gaussian model of Foi et al. [7]:

$$z = \frac{1}{\eta}\,\mathcal{P}(\eta\, k\, x) + n, \qquad n \sim \mathcal{N}(0, \sigma_r^2),$$

where $x \in [0,1]$ is the clean linear image, $k \in (0,1]$ the fraction of light that reached the sensor, η a scale that converts intensity to photon counts (a larger η means more photons and less relative noise), $\mathcal{P}(\cdot)$ a Poisson random variable modeling **shot noise**, and n Gaussian **read noise** with standard deviation σ_r. The camera then applies a tone curve and quantizes. The variance of z is approximately $kx/\eta + \sigma_r^2$: it grows with the signal, so noise is *signal-dependent*, which is why "add Gaussian noise" is a poor stand-in for real low light.

```
import numpy as np
from skimage import data, img_as_float

rng = np.random.default_rng(0)
clean = img_as_float(data.coffee())            # H x W x 3, sRGB-encoded, in [0, 1]

def synth_low_light(x, k=0.05, peak=2000.0, read_std=8e-4, gamma=2.2):
    lin = k * x ** gamma                              # 1) less light, in linear units
    lin = rng.poisson(lin * peak) / peak              # 2) shot noise (Poisson)
    lin = lin + rng.normal(0, read_std, lin.shape) # 3) read noise (Gaussian)
    out = np.clip(lin, 0, 1) ** (1 / gamma)           # 4) re-encode for display
    return np.round(out * 255) / 255                  # 5) 8-bit quantization

dark = synth_low_light(clean)
print(clean.mean(), dark.mean())                      # about 0.39 -> 0.10
```

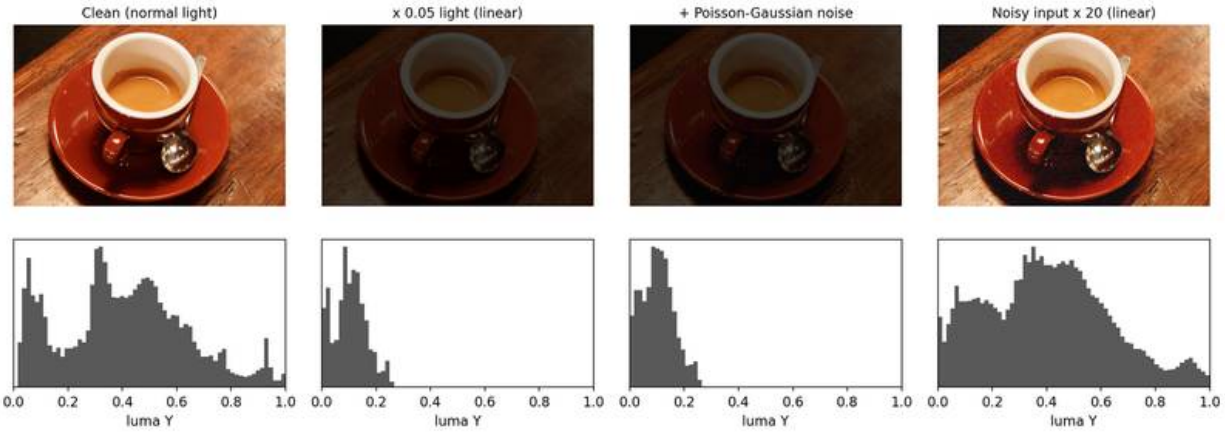


Figure 1 — Synthetic low light. Removing 95% of the light squeezes the histogram into the bottom quarter. Noise is hard to see in the dark image, but undoing the exposure with a perfect gain (right) reveals it: brightening and denoising are separate problems.

> **Interactive demonstration (online)**: open the online edition of this tutorial to use this demonstration.Open demonstration

The right panel is the most important picture in this tutorial: even the *ideal* brightness correction leaves the noise, which is why LLIE is a restoration problem (see DLCV 04 ·Image Restoration) and not only a tone-mapping problem.

Non-deep-learning LLIE and the characteristics of low-light images

Plain version. Before deep learning there were two main families: **histogram-based** methods, which redistribute gray levels, and **Retinex-based** methods, which split an image into "lighting" and "objects" and fix the lighting. A third idea, borrowed from dehazing, fits into the Retinex family, and statistical methods extend all of them.

Low-light images in color space

Many color spaces separate a "how bright" channel from "what color" channels. Different spaces call it value, intensity, luma or lightness, and each defines it differently. With $R, G, B \in [0, 1]$ (see IP 06 for the spaces themselves):

$$\begin{aligned} \text{HSV:} \quad & V = \max(R, G, B), \\ \text{HSI:} \quad & I = \tfrac{1}{3}(R + G + B), \\ \text{YCbCr (BT.601):} \quad & Y' = 0.299\,R' + 0.587\,G' + 0.114\,B', \\ \text{CIELAB:} \quad & L^* = 116\,f(Y_{\text{lin}}/Y_n) - 16. \end{aligned}$$

Here R', G', B' are gamma-encoded values; Y_{lin} is linear-light luminance and Y_n that of the reference white; $f(t) = t^{1/3}$ above a small threshold and linear below it. Some remarks:

- V **is the "bright channel".** It is the per-pixel maximum, the counterpart of the *dark channel* (a per-patch minimum) used in dehazing [14]. In a dark image, V is the channel least damaged by the darkness, which is why LIME [17] uses it to initialize the illumination.
- **The luma weights.** ITU-R BT.601 [8] defines luma as $Y' = 0.299R' + 0.587G' + 0.114B'$. That full-range form is what JPEG/JFIF uses. For 8-bit *limited-range* ("studio") video, BT.601 places black at 16 and white at 235: $Y = 16 + 219\,Y'$. With $R', G', B' \in [0, 255]$ this becomes $Y = 16 + 0.257R' + 0.504G' + 0.098B'$, because $219/255 \times (0.299, 0.587, 0.114) \approx (0.257, 0.504, 0.098)$. Both formulas describe the same luma; they differ only in range and offset.
- L^* is designed to be perceptually uniform, so equal steps look roughly equal to a person. That makes it a good channel for contrast enhancement.

The practical lesson: since brightness can be separated from color, you can **enhance only the brightness channel** and keep the chroma. Figure 2 shows that the choice of channel matters a lot.

```
R, G, B = dark[..., 0], dark[..., 1], dark[..., 2]
V = dark.max(axis=2)                          # HSV value
I = dark.mean(axis=2)                         # HSI intensity
Y = 0.299 * R + 0.587 * G + 0.114 * B         # BT.601 luma, full range (JPEG/JFIF)
Y_limited = 16 + 219 * Y                      # 8-bit limited ("studio") range, 16..235
print(np.allclose(Y_limited, 16 + (65.481 * R + 128.553 * G + 24.966 * B)))  # True
```

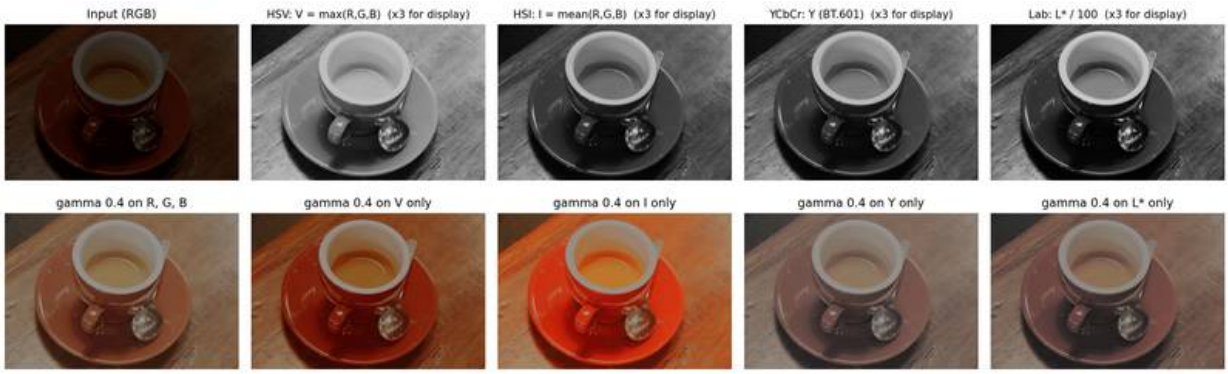


Figure 2 — The same gamma curve applied in different color spaces. Raising V or I scales all three channels by one factor, which keeps hue but strengthens saturation (very orange). Raising Y or L with fixed chroma desaturates the result. Neither is 'correct': tone and color are coupled.*

Two effects are visible. Scaling RGB by a common factor (HSV, HSI) preserves hue and keeps the ratio between channels, so saturated objects become *very* saturated. Changing only Y' or L^* while keeping Cb/Cr or a*/b* fixed adds brightness without adding color, so the image looks washed

out. This coupling of tone and color is exactly what later methods such as HVI-CIDNet [27] try to handle with a learned color space.

Histogram equalization and CLAHE

Plain version. Histogram equalization (HE) spreads the gray levels of an image so that all brightness values are used about equally. A dark image, whose values are packed near zero, gets stretched across the full range.

Precise version. For an image with L levels, normalized histogram $p_r(r_k) = n_k/(MN)$, HE uses the scaled cumulative distribution function as the mapping:

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j), \qquad k = 0, \dots, L-1,$$

where n_k is the number of pixels at level r_k and MN the image size. The slope of T is proportional to the histogram, so heavily populated gray levels get pulled apart. The derivation is in IP 03. For color images, the usual recipe is to convert to a space with a brightness channel, equalize that channel only, and convert back, for example RGB → Lab → equalize L^* → RGB.

Global HE has two problems in low light: it over-stretches large flat regions (including their noise), and one mapping cannot serve both a dark corner and a bright lamp. **Adaptive HE** computes a mapping per tile, and **CLAHE** (contrast-limited AHE) clips each tile's histogram at a limit before computing the CDF, which caps the slope of T and therefore the noise gain, then interpolates between neighboring tiles to avoid seams [9], [10].

Optional: CLAHE: equalizing locally without amplifying noise (in the Supplementary notes). The clipping step and the tile blending written out as formulas, with a small experiment on noise gain.

Gamma correction

A power law $s = r^\gamma$ with $\gamma < 1$ lifts shadows more than highlights; its slope $\gamma r^{\gamma-1}$ is large near $r = 0$. It is cheap and monotonic but global: one γ for the whole picture, no denoising, and the same desaturation-or-oversaturation choice as above depending on where it is applied.

```
from skimage import color, exposure

gamma_out = dark ** 0.4                                    # power-law curve on every
 ↪  channel
lab = color.rgb2lab(dark)
L = lab[..., 0] / 100                                      # L* scaled to [0, 1]
he = lab.copy();    he[..., 0] = 100 * exposure.equalize_hist(L)
clahe = lab.copy(); clahe[..., 0] = 100 * exposure.equalize_adapthist(L, clip_limit=0.01)
he_out, clahe_out = color.lab2rgb(he), color.lab2rgb(clahe)
```

```
for name, im in [("gamma", gamma_out), ("HE", he_out), ("CLAHE", clahe_out)]:
    print(f"{name:6s} mean {im.mean():.3f}   std {im.std():.3f}")
```



Figure 3 — Classical enhancers on the noisy input of Figure 1. All of them brighten; none of them removes noise, and the Retinex outputs show strong color noise where the input was darkest.

Retinex theory

Plain version. Retinex says that what the eye (or camera) receives is "the object's own colors" multiplied by "the light falling on it". If we can estimate the light, we can divide it out and see the object as it really is, then put back a better light.

Precise version. Land's Retinex theory [11] models an observed image S as

$$S(\mathbf{x}) = R(\mathbf{x}) \circ L(\mathbf{x}),$$

where $\mathbf{x}$ is the pixel position, $R \in [0, 1]$ is the **reflectance** (a property of the surfaces, assumed constant under any lighting), L is the **illumination** (the light falling on the scene, which varies smoothly in space except at shadow boundaries), and $\circ$ is element-wise multiplication. Note the word: L is *illumination*, not luminance. "Luminance" usually refers to a brightness channel of a color space, such as Y above.

Recovering two unknowns from one product is ill-posed, so every Retinex method adds a prior. The classic one is "illumination is smooth".

Single-scale Retinex (SSR) [12] estimates L with a Gaussian blur G_σ and works in the log domain, where the product becomes a sum:

$$\log R_c(\mathbf{x}) = \log S_c(\mathbf{x}) - \log\left(G_\sigma * S_c\right)(\mathbf{x}),$$

for each color channel c, with $*$ convolution and σ the surround scale. Small σ gives strong local contrast and halos at edges; large σ gives better tone but weaker detail. **Multi-scale Retinex (MSR)** [13] averages SSR outputs at several scales (typically three) with weights w_n:

$$\log R_c^{\mathrm{MSR}} = \sum_{n=1}^{N} w_n \left[\log S_c - \log(G_{\sigma_n} * S_c)\right].$$

Because SSR and MSR work per channel, they tend to gray out the colors. **MSRCR** (MSR with color restoration) [13] multiplies the result by a color-restoration factor based on each channel's share of $R + G + B$; other variants apply MSR to an intensity channel only and keep the input's chromaticity.

```
from scipy import ndimage as ndi

def ssr(img, sigma=40.0, eps=1e-3):
    log_s = np.log(img + eps)
    log_l = np.log(ndi.gaussian_filter(img, sigma=(sigma, sigma, 0)) + eps)  # L ~ Gaussian
     ↪  blur
    r = log_s - log_l                                          # log R = log S - log L
    lo, hi = np.percentile(r, [1, 99])                         # robust stretch for display
    return np.clip((r - lo) / (hi - lo), 0, 1)

msr = np.mean([ssr(dark, s) for s in (15, 80, 250)], axis=0)   # MSR, equal weights
 ↪  (simplified)
```

For *enhancement* we usually do not want R alone, which looks flat and unnatural. We want to brighten L and recombine: $\hat{S} = R \circ \hat{L}$ with, for example, $\hat{L} = L^\gamma$ and $\gamma < 1$. This "decompose, adjust L, recombine" pattern is the template for most Retinex-based deep networks.



Figure 4 — A LIME-style Retinex decomposition. The illumination is initialized with $\max(R, G, B)$ and smoothed; the reflectance is S / L. Brightening L with a gamma of 0.35 and multiplying back gives a natural result, while R alone looks flat and over-saturated.

LIME: Retinex as optimization

LIME [17] estimates only the illumination. It initializes it with the bright channel, $\hat{T}(\mathbf{x}) = \max_c S_c(\mathbf{x})$, then refines it by solving a structure-aware smoothing problem: keep T close to

$\hat{T}$, but make it smooth except where the image has strong structure. The enhanced image is S/T^{γ}, and because dividing amplifies noise, LIME optionally denoises afterward. LIME's images (often called the LIME set) became a standard test set, and the method is a good example of the classical recipe: a physical model, a hand-designed prior, and an optimization.

Another model: dehazing an inverted image

The Retinex model is not the only image-formation model. Dehazing uses the **atmospheric scattering model**, written out in DLCV 04: a hazy observation I is the haze-free scene radiance J attenuated by a transmission t, plus a global atmospheric light A. This model cannot be applied to a dark image directly, because a dark image does not look like a hazy one.

Dong et al. observed that an **inverted** low-light image, $1 - S$, *does* look hazy: dark regions become bright and low-contrast, like fog. They therefore invert the image, run a dehazing algorithm (in the spirit of the dark channel prior [14]), and invert the result back [15], [16]. With I_{haze} the hazy image and J_{haze} the haze-free result, the inversions are

$$I_{\text{haze}} = 1 - S_{\text{Retinex}}, \qquad J_{\text{haze}} = 1 - \hat{S}_{\text{enhanced}}.$$

From the "what is reliable" angle, dehazing assumes the *dark* channel marks haze-free content; in a low-light image the dark channel is mostly damaged pixels, so attention shifts to the *bright* channel, the region least affected by the darkness. The LIME paper [17] analyzes this inversion approach and shows that it can be rewritten in Retinex form, so the two models make essentially the same assumption, with Retinex being the simpler statement.

Statistical approaches

Many classical methods add statistics or priors on top of these models: naturalness constraints that keep the lightness order of a scene (NPE [18]), layered or two-dimensional histogram models (the method behind the DICM set [59]), and fusion of several synthetic exposures. They are hand-crafted, need no training data, and generalize in a predictable way; their weak point is noise, which they usually ignore or hand to a separate denoiser.

Deep-learning LLIE

Plain version. Deep networks learn the mapping from dark to bright from data. The question is not whether to use a network but **what training signal** to use: paired dark/bright photos, unpaired photos, no reference at all, or a generative model of what bright images look like. Physics (usually Retinex) can be built into any of these.

Deep learning is itself a large optimization, and its black-box nature lets us drop many hand-made assumptions. A common first split is **theory-based** methods (usually Retinex) versus **theory-free** methods (strong generic networks). In practice the two mix, so the split is not binary. Figure 5 groups methods mainly by training signal instead, following the surveys [2], [3].

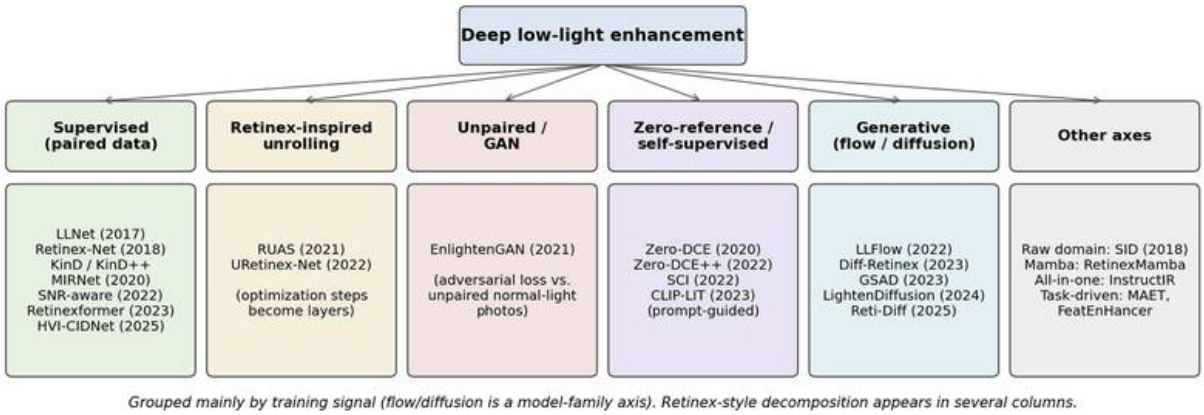


Figure 5 — A taxonomy of deep LLIE, mainly by training signal (the flow/diffusion column is a model-family axis). Retinex decomposition is an architectural choice that appears in several columns, not a column of its own.

Supervised methods (paired data)

Training on pairs (x_i, y_i) of low-light and normal-light images follows the general recipe of DLCV 04: fit f_θ with a reconstruction loss, typically $\mathcal{L} = \|f_\theta(x) - y\|_1$ plus perceptual or structural terms (see DLCV 04 for these losses). Paired data is expensive (two exposures of a static scene, or synthesis), and the network learns the *specific* brightness and color style of the reference images, which matters for evaluation later.

Representative examples: **LLNet** [19] (2017), one of the first deep LLIE models, learned brightening and denoising jointly with a denoising autoencoder; Tao et al. [20] (2017) denoised with a CNN and then enhanced contrast with a bright-channel model; **MIRNet** [24] (2020) is a general multi-resolution restoration backbone; **SNR-aware** [25] (2022) estimates a per-pixel signal-to-noise map and uses long-range attention where the local neighborhood is too noisy to trust; and **HVI-CIDNet** [27] (2025) learns a color space that decouples color from intensity, attacking the problem of Figure 2.

Optional: More deep LLIE methods (in the Supplementary notes). How each of these five networks is built.

Retinex-based networks

Retinex-Net [21] (2018) changed the goal from "estimate R" to "**both the input and the output have their own R and L**". The pipeline is: decompose the low and normal images, enhance L to $\hat{L}$, recombine with R. It also introduced the LOL dataset (below). Its decomposition network is trained only through consistency, without ground-truth R or L, using

$$\mathcal{L}_{\text{decom}} = \mathcal{L}_{\text{recon}} + \lambda_{ir}\,\mathcal{L}_{ir} + \lambda_{is}\,\mathcal{L}_{is},$$

$$\mathcal{L}_{\text{recon}} = \sum_{i\in\{\text{low,normal}\}} \sum_{j\in\{\text{low,normal}\}} \lambda_{ij} \left\| R_i \circ I_j - S_j \right\|_1,$$

$$\mathcal{L}_{ir} = \|R_{\text{low}} - R_{\text{normal}}\|_1, \qquad \mathcal{L}_{is} = \sum_i \left\| \nabla I_i \circ \exp\left(-\lambda_g \nabla R_i\right) \right\|.$$

In the paper's notation $S = R \circ I$, so I here is the illumination (L in our notation). S_j is the input image j, R_i and I_i the decomposed reflectance and illumination of image i, and ∇ the horizontal plus vertical gradient. The terms say:

- $\mathcal{L}_{\text{recon}}$: each decomposition must reproduce its own image ($\lambda_{ii} = 1$), and crossing reflectance and illumination between the pair should still roughly work ($\lambda_{ij} = 0.001$ for $i \neq j$).
- $\mathcal{L}_{ir}$: the two images show the same scene, so their reflectances should be equal; this is how the network learns what "reflectance" means.
- $\mathcal{L}_{is}$: illumination should be smooth, *except* where the reflectance has a strong edge. The weight $\exp(-\lambda_g \nabla R)$ relaxes the smoothness penalty across object boundaries. This is the learned version of LIME's structure-aware prior.

The paper uses $\lambda_{ir} = 0.001$, $\lambda_{is} = 0.1$ and $\lambda_g = 10$. The enhancement network is trained with $\|R_{\text{low}} \circ \hat{I} - S_{\text{normal}}\|_1$ plus the same smoothness term.

The pattern spread quickly. **KinD** [22] (2019) and **KinD++** [23] (2021) add a reflectance-restoration step, because the dark image's reflectance is noisy and color-shifted; **Retinexformer** [26] (2023) adds perturbation terms for noise and artifacts to the Retinex model, trains in one stage, uses the illumination estimate to guide self-attention, and became a common baseline; **RetinexMamba** [42] (2024 preprint) swaps attention for Mamba state-space blocks. "Retinex decomposition plus a new backbone" has been a stable design pattern for years. A less common route builds lightweight networks on the atmospheric scattering model instead (CPGA-Net [28], CPGA-Net+ [29]).

Optional: More deep LLIE methods (in the Supplementary notes). Retinexformer's perturbed Retinex model and the prior-guided CPGA-Net line in more detail.

Retinex-inspired unrolling

Unrolling (or unfolding) takes an iterative optimization algorithm for the Retinex problem and turns each iteration into a network stage with learnable parameters. The architecture inherits the algorithm's structure and interpretability, and training replaces hand-tuned priors.

- **RUAS** [30] (2021) unrolls optimization models for illumination estimation and noise removal, searches the architecture of the prior modules, and is trained with a cooperative *reference-free* strategy, so it also belongs in the unsupervised column. It is very small and fast.
- **URetinex-Net** [31] (2022) unfolds a Retinex decomposition optimization into a network with learned modules for initialization, the unfolded updates of R and L, and illumination adjustment.

Unpaired training: EnlightenGAN

EnlightenGAN [32] (2021) uses a generative adversarial network (GAN) so that it needs only a set of dark images and an *unrelated* set of well-lit images. A U-Net generator, guided by a self-regularized attention map derived from the input's illumination, produces the enhanced image. A **global** discriminator judges the whole image and a **local** discriminator judges random patches,

so the output looks like a well-lit photo both overall and in detail. A **self-feature-preserving** loss compares VGG features of input and output, keeping content in place since there is no ground truth to do so.

Zero-reference methods and curve estimation

Plain version. Zero-DCE does not output a new image. It outputs a *brightening curve per pixel*, applies it, and is trained with losses that judge the result without any reference photo: "is it well exposed, is the color balanced, is the curve smooth, is local contrast kept?"

The LE curve. Zero-DCE [33] (2020) defines a quadratic **light-enhancement curve**

$$\mathrm{LE}(x;\alpha) = x + \alpha\, x\,(1-x), \qquad \alpha \in [-1,1],$$

where $x \in [0,1]$ is a pixel value. The curve has three useful properties: $\mathrm{LE}(0) = 0$ and $\mathrm{LE}(1) = 1$, so the range is preserved and no clipping is needed; it is monotonic for $|\alpha| \le 1$, because $\mathrm{d}\,\mathrm{LE}/\mathrm{d}x = 1 + \alpha(1-2x) \ge 0$, so brightness order is kept; and it is differentiable, so it can be trained by backpropagation. One quadratic is not flexible enough, so it is applied **iteratively**:

$$\mathrm{LE}_n(\mathbf{x}) = \mathrm{LE}_{n-1}(\mathbf{x}) + \mathcal{A}_n(\mathbf{x})\,\mathrm{LE}_{n-1}(\mathbf{x})(1 - \mathrm{LE}_{n-1}(\mathbf{x})),$$

with LE_0 the input, $n = 1, \ldots, 8$, and $\mathcal{A}_n$ a **per-pixel, per-channel** parameter map instead of a single α. A seven-layer CNN (DCE-Net, 32 channels per layer, about 79k parameters, no downsampling) predicts $8 \times 3 = 24$ maps with a tanh output. Each pixel thus gets its own high-order curve, which is how local adjustment arises from a global-looking formula.

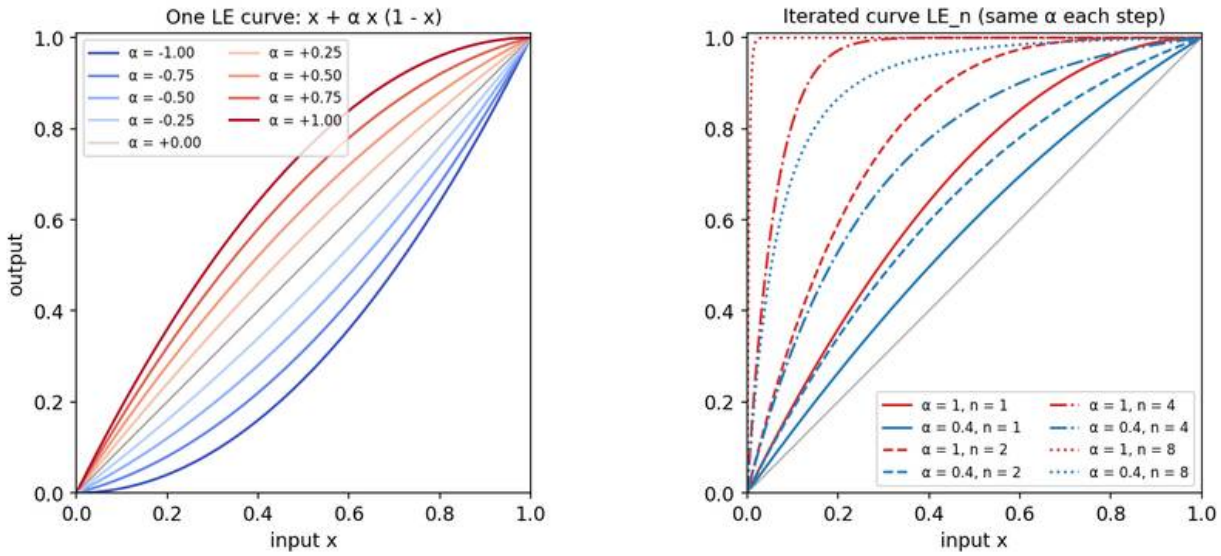


Figure 6 — The LE curve. Left: one iteration for α from −1 to 1. Right: iterating sharpens the curve; with $\alpha = 1$ and $n = 8$, a value of 0.02 is already mapped to almost 1, so tiny noisy values in the shadows are amplified enormously.

```
def le_iterate(x, alphas):
    """Zero-DCE curve: LE_n = LE_{n-1} + a_n * LE_{n-1} * (1 - LE_{n-1})."""
    y = x.copy()
    for a in alphas:              # a can be a scalar or a per-pixel map
        y = y + a * y * (1 - y)
    return y

x = np.array([0.0, 0.02, 0.1, 0.3, 0.6, 1.0])
print(le_iterate(x, [1.0] * 8).round(3))   # [0. 0.994 1. 1. 1. 1.]
print(le_iterate(x, [0.4] * 8).round(3))   # [0. 0.245 0.695 0.93 0.985 1.]
```

The first line shows the danger: for small x, each step multiplies by about $1 + \alpha$, so eight steps with $\alpha = 1$ give a gain near $2^8 = 256$. Whatever noise sits in the shadows is amplified by the same factor.

Non-reference losses. Zero-DCE is trained with four losses that need no ground truth:

$$\mathcal{L}_{spa} = \frac{1}{K} \sum_{i=1}^{K} \sum_{j \in \Omega(i)} \left(|Y_i - Y_j| - |I_i - I_j| \right)^2,$$

$$\mathcal{L}_{exp} = \frac{1}{M}\sum_{k=1}^{M}|Y_k - E|, \qquad \mathcal{L}_{col} = \sum_{(p,q)\in\varepsilon}(J^p - J^q)^2,$$

$$\mathcal{L}_{tv_A} = \frac{1}{N}\sum_{n=1}^{N}\sum_{c\in\{R,G,B\}}\left(|\nabla_x A_n^c| + |\nabla_y A_n^c|\right)^2.$$

In $\mathcal{L}_{spa}$ (**spatial consistency**), K is the number of 4×4 regions, $\Omega(i)$ the four neighbors of region i, and Y and I the mean intensities of a region in the enhanced and input images: contrast between neighboring regions should be preserved. In $\mathcal{L}_{exp}$ (**exposure control**), Y_k is the mean intensity of the k-th of M non-overlapping 16×16 regions and $E = 0.6$ a target well-exposedness level. In $\mathcal{L}_{col}$ (**color constancy**), J^p is the mean of channel p over the image and $\varepsilon = \{(R,G),(R,B),(G,B)\}$: the gray-world assumption that the average scene color is gray. In $\mathcal{L}_{tv_A}$ (**illumination smoothness**), N is the number of iterations and ∇_x, ∇_y horizontal and vertical gradients: neighboring pixels should get similar curves. The total is

$\mathcal{L}_{spa} + \mathcal{L}_{exp} + W_{col}\mathcal{L}_{col} + W_{tv_A}\mathcal{L}_{tv_A}$ with $W_{col} = 0.5$ and $W_{tv_A} = 20$ in the paper; training uses multi-exposure images from SICE [56], which provide both under- and over-exposed inputs.

Follow-ups refined the idea:

- **Zero-DCE++** [34] (2022) uses depthwise separable convolutions, predicts only **3** curve maps and reuses them across all iterations, and estimates them on a downsampled input before resizing them back. This cuts the model to about 10k parameters and makes it fast enough to run at interactive rates even on a CPU.
- **SCI** [35] (2022) learns illumination with a cascade of weight-sharing stages and a **self-calibrated** module that makes the stages' outputs converge during training, so that at test time a **single** basic block suffices. It is trained with unsupervised losses.
- **CLIP-LIT** [36] (2023) targets *backlit* images. It uses the vision-language model CLIP and learns a pair of text prompts, one for backlit and one for well-lit images, then refines the prompts and the enhancement network iteratively, so the CLIP similarity to the "well-lit" prompt acts as the training signal.

A zero-reference toy in PyTorch

To see these losses at work, the code below trains a tiny Zero-DCE-style network for about half a minute on CPU. It trains on random crops of four `skimage` images, each darkened by a random factor in linear light, and is tested on the coffee image, which it never sees during training. No ground truth is used. (The loss weights differ from the paper's because the terms are normalized differently: squared rather than absolute exposure error and mean rather than summed gradients.)

```
import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
from skimage import data, img_as_float
from skimage.transform import resize

torch.manual_seed(0); rng = np.random.default_rng(0)
darken = lambda x, k: (k * x ** 2.2) ** (1 / 2.2)          # less light, in linear units
train = [resize(img_as_float(f()), (256, 256), anti_aliasing=True)
         for f in (data.astronaut, data.chelsea, data.rocket, data.immunohistochemistry)]

def batch(n=4, size=64):                                    # random crops, random darkness
    xs = []
    for _ in range(n):
        img = train[rng.integers(len(train))]
        i, j = rng.integers(0, 256 - size, 2)
        xs.append(darken(img[i:i + size, j:j + size], rng.uniform(0.02, 0.3)))
    return torch.tensor(np.stack(xs), dtype=torch.float32).permute(0, 3, 1, 2)

class TinyDCE(nn.Module):
    def __init__(self, n_iter=8, w=16):
        super().__init__()
        self.n_iter = n_iter
        self.body = nn.Sequential(
            nn.Conv2d(3, w, 3, padding=1), nn.ReLU(), nn.Conv2d(w, w, 3, padding=1),
            ↪  nn.ReLU(),
            nn.Conv2d(w, w, 3, padding=1), nn.ReLU(), nn.Conv2d(w, 3 * n_iter, 3,
            ↪  padding=1), nn.Tanh())
    def forward(self, x):
        alphas = self.body(x).chunk(self.n_iter, dim=1)    # 8 maps of shape B x 3 x H x W
        y = x
        for a in alphas:
            y = y + a * y * (1 - y)                         # the LE curve
        return y, torch.cat(alphas, 1)

def L_exp(y, E=0.6, p=16):   # mean brightness of 16x16 patches should be near E
    return (F.avg_pool2d(y.mean(1, keepdim=True), p) - E).pow(2).mean()
def L_col(y):                # gray-world: channel means should agree
    r, g, b = y.mean((2, 3)).unbind(1)
    return ((r - g) ** 2 + (r - b) ** 2 + (g - b) ** 2).mean()
def L_tv(a):                 # curve parameters should vary smoothly in space
    return (a[..., 1:, :] - a[..., :-1, :]).pow(2).mean() + (a[..., :, 1:] - a[..., :,
     ↪  :-1]).pow(2).mean()
def L_spa(y, x, p=4):        # keep the contrast between neighbouring 4x4 regions
    Y, X = (F.avg_pool2d(t.mean(1, keepdim=True), p) for t in (y, x))
    return sum(((torch.diff(Y, dim=d) - torch.diff(X, dim=d)) ** 2).mean() for d in (2, 3))

net = TinyDCE()
opt = torch.optim.Adam(net.parameters(), lr=1e-3)
for step in range(600):                                     # ~20-40 s on a laptop CPU
    x = batch()
    y, a = net(x)
```

```
    loss = L_spa(y, x) + 10 * L_exp(y) + 5 * L_col(y) + 200 * L_tv(a)
    opt.zero_grad(); loss.backward(); opt.step()

test = darken(img_as_float(data.coffee()), 0.05)          # an image never used in training
with torch.no_grad():
    out, _ = net(torch.tensor(test, dtype=torch.float32).permute(2, 0, 1)[None])
print(f"mean brightness: {test.mean():.3f} -> {out.mean():.3f}")   # about 0.10 -> 0.50
```

Drag the slider to compare the dark input and the toy's output:



Dark input



Zero-reference toy

The dark coffee-cup image and the toy network's output, which is much brighter, with visible wood grain and a lighter, slightly greenish and less saturated color than the original photo.

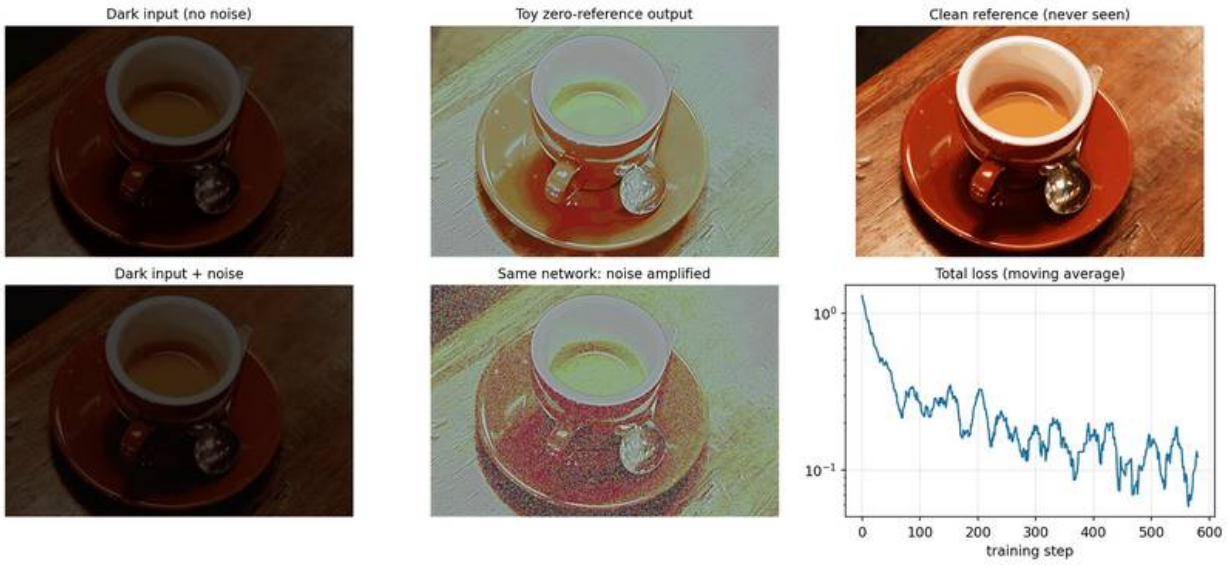


Figure 7 — What the zero-reference losses do and don't do. The exposure loss gives good brightness, but the gray-world color loss pulls a red-brown scene toward gray (top middle versus top right), and with real

noise the per-pixel curves amplify it (*bottom middle*).

The toy shows the strengths and the limits of the zero-reference idea in one picture. Without a single reference image, the network learns a sensible brightening that transfers to an unseen photo. But each loss is a *prior*, and priors can be wrong: gray-world fails for scenes that really are red or brown, a fixed exposure target E flattens deliberately dark regions, and nothing in the losses asks for denoising. Real zero-reference methods are trained on many more images, and some add explicit noise handling, but these failure modes are visible in published results too.

Normalizing flows and diffusion models

LLFlow [37] (2022) observes that one dark image has *many* acceptable bright versions, and a pixel-wise loss averages them into a blurry compromise. It learns a conditional normalizing flow: an invertible network that maps normal-light images to a Gaussian distribution, conditioned on features of the dark input. Those features are guided by priors that echo the classical methods: a histogram-equalized version of the input, a color map (close to the Retinex reflectance) and a noise map.

From 2023, **diffusion models** entered LLIE. They generate the output by iteratively denoising random noise, conditioned on the dark image, and so can synthesize plausible detail that regression networks blur out. Diff-Retinex [38] generates the Retinex reflectance and illumination with diffusion, LightenDiffusion [40] decomposes in a latent space and trains on unpaired data, Reti-Diff [41] uses latent diffusion only to produce compact priors for a restoration transformer, and GSAD [39] regularizes the diffusion trajectory with the image structure. The trade-off is cost and fidelity: many denoising steps make inference slow, and generated details may look right without being *true*.

Optional: More deep LLIE methods (in the Supplementary notes). What each diffusion model adds, and a survey taxonomy of the sub-field.

Raw-domain low light

Everything above works on processed sRGB images, where the camera's pipeline (demosaicking, white balance, denoising, tone curve, compression) has already discarded information. **Learning to See in the Dark** [44] (2018) instead feeds **raw** sensor data to a U-Net that replaces the whole pipeline. Its SID dataset has 5,094 short-exposure raw images paired with 424 distinct long-exposure references, from a Sony α7S II (Bayer sensor) and a Fujifilm X-T2 (X-Trans sensor), with exposure ratios of ×100, ×250 and ×300. Raw data keeps the linear relation between photons and values and has more bits, so extreme low light that is hopeless in sRGB becomes recoverable. The cost is that the model is tied to one sensor.

All-in-one restoration

Low-light enhancement is also one task inside all-in-one restoration models, for example **InstructIR** [43] (2024), which restores an image according to a natural-language instruction such as "make this photo brighter". The all-in-one family as a whole is described in DLCV 04.

Low light for downstream tasks

If the goal is *detection* rather than pretty pictures, the usual baseline is "enhance, then detect" (see DLCV 03 ·Object Detection). But enhancement tuned for people is not necessarily what a detector needs. **MAET** [45] (2021) models the low-light degradation physically and trains the detector to also predict the degradation parameters, so that features become aware of it. **FeatEnHancer** [46] (2023) enhances hierarchical *features* inside the network, trained directly with the downstream loss, instead of enhancing pixels.

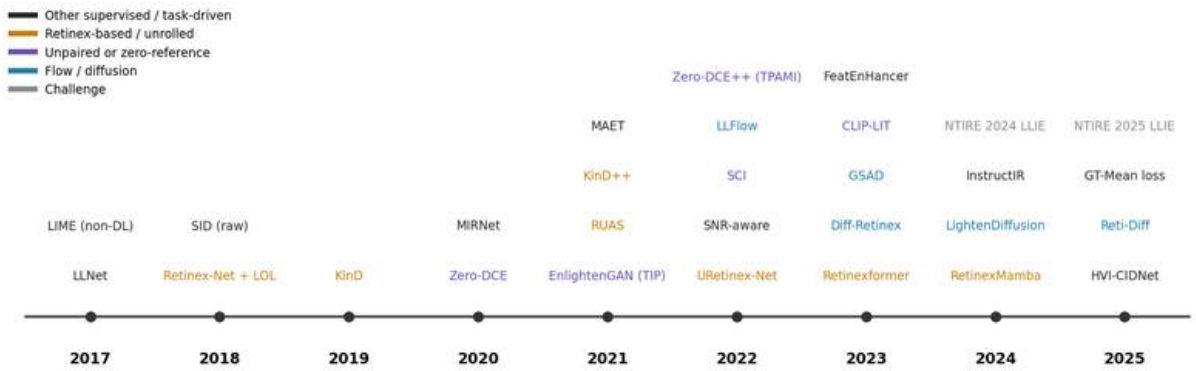


Figure 8 — A timeline of the methods in this tutorial (by publication year). Colors mark Retinex-based or unrolled designs, unpaired or zero-reference training, flow and diffusion models, and challenges.

Evaluation

Plain version. The general evaluation toolkit (full-reference and no-reference metrics, learned and vision-language assessors, user studies, and the pitfalls and reporting practice that go with them) is explained once, in DLCV 04. This section does not repeat it. It covers what is different for dark images: why no-reference scores matter so much here, how the common ones mislead on exposure and color, measures made for enhancement, the GT-mean pitfall, and task-driven evaluation.

Full-reference metrics

On paired sets, LLIE papers report PSNR, SSIM [47] and LPIPS [48] exactly as defined in DLCV 04. The one low-light-specific caution is brightness: a small global exposure difference from the reference costs a lot of PSNR, which is the subject of the GT-mean pitfall below.

No-reference metrics: why low light depends on them

Plain version. A no-reference (NR) metric looks at one image and guesses how good it is, with no "correct answer" to compare against. Low-light papers need these metrics because their most realistic test photos have no correct answer, and because even when one exists, it is only one of many acceptable brightness levels.

Precise version. Two facts push LLIE toward NR evaluation:

1. **Real test sets are unpaired.** LIME [17], MEF [58], NPE [18], DICM [59] and VV [60] are small sets of real, often non-uniformly lit photos with no ground truth. Paired sets such as LOL [21] exist, but they were captured under controlled conditions (a static scene, two exposures), and a model can do well on them without generalizing (the generalization gap reported in [2]). Results on the unpaired sets are therefore shown in almost every LLIE paper, and only NR metrics and human judgments can score them.
2. **The "correct" brightness is ambiguous.** Even with a reference, the normal-light image is one photographer's choice of exposure and color. An output that is a little brighter or warmer can look as good, yet lose several dB of PSNR (this is the reason for the GT-mean protocol below). An NR metric at least does not punish a different but valid exposure.

The price is that an NR metric is a *model of human preference*, and every model has blind spots. The general theory of no-reference image quality assessment (IQA), including how NIQE, BRISQUE, MUSIQ and CLIP-IQA are built and how NR metrics are validated against human opinion, is covered in DLCV 04. The subsections below do not repeat it; they focus on how each family behaves on *enhanced low-light* images.

Classical natural-scene-statistics metrics: NIQE, BRISQUE and PI

NIQE [49], BRISQUE [50] and the Perceptual Index (PI) [63], [64] are natural-scene-statistics (NSS) metrics: they measure how far the statistics of an image (its MSCN coefficients) are from those of clean photos, and lower is better for all three. The MSCN coefficients and the BRISQUE and NIQE constructions are given in DLCV 04, together with a runnable snippet showing how blur, noise and over-sharpening move the features; the definition of PI is in the DLCV 04 notes. Low-light papers use these metrics heavily; Zero-DCE [33], for example, reports PI on the unpaired sets alongside a user study.

What they reward, and how they mislead on exposure. All three were designed for *structural* distortions (blur, noise, compression, ringing), not for exposure or color:

- MSCN normalization divides out local mean and contrast by construction, so a global brightness change barely moves the features. An under-enhanced image and a well-exposed one can get similar scores.
- Grayscale features ignore color casts and saturation, the very errors of Figure 2.
- Noise and over-sharpening add high-frequency energy, which can make a smooth, noise-amplified output look "more natural" to these statistics than a clean but slightly soft one. HE and CLAHE outputs, with strong local contrast, are a typical case.
- The pristine corpus and training distortions contain few dark, night or backlit scenes, so "natural" means "daylight-like".

Low-light-specific measures: lightness order, colorfulness and noise

Plain version. Some measures target what goes wrong in enhancement specifically: Did the method swap which regions are brighter than others? Are the colors vivid or washed out? How much noise did brightening reveal?

Lightness Order Error (LOE). The NPE paper [18] argues that a natural enhancement should keep the *relative order* of lightness in the scene: if a lamp was brighter than a wall before, it should still be after, because that order carries the direction of the light and the sense of depth. LOE counts how often the order of two pixels flips. With $L(x) = \max_{c\in\{R,G,B\}} I_c(x)$ the lightness of the input and $L_e(x)$ that of the enhanced image,

$$\mathrm{LOE} = \frac{1}{m}\sum_{x=1}^{m} \mathrm{RD}(x), \qquad \mathrm{RD}(x) = \sum_{y=1}^{m} U(L(x), L(y)) \oplus U(L_e(x), L_e(y)),$$

where m is the number of pixels, $U(p,q) = 1$ if $p \geq q$ and 0 otherwise, and $\oplus$ is exclusive-or, so $\mathrm{RD}(x)$ is the number of pixels whose order relative to x changed. Lower is better. Three practical points:

- The cost is $O(m^2)$, so implementations evaluate LOE on a subsampled image. The raw value grows with m, so numbers are comparable only at the same subsampling. Many papers do not state theirs.
- The reference is the **input**, not a clean image. Any monotonic tone curve applied in the max channel gets LOE = 0, including doing nothing at all. And because the noise in the dark input already scrambles the order of nearby pixels, a method that *removes* noise is penalized.
- LOE says nothing about whether the result is bright enough. It is a guard against unnatural local tone mapping, not a quality score.

Colorfulness. Hasler and Süsstrunk [65] fit a simple colorfulness measure to perceptual ratings, using opponent-color axes on 8-bit RGB:

$$rg = R - G, \qquad yb = \tfrac{1}{2}(R + G) - B, \qquad M = \sqrt{\sigma_{rg}^2 + \sigma_{yb}^2} + 0.3\sqrt{\mu_{rg}^2 + \mu_{yb}^2},$$

where μ and σ are the mean and standard deviation over the image, and the weight 0.3 comes from the authors' fit. Higher means more colorful. It is useful as a descriptive statistic for desaturation (luma-only enhancement, Figure 2) or oversaturation, but it has no notion of the *correct* color, and noise in the chroma channels inflates σ_{rg} and σ_{yb}.

Noise level of the output. Since brightening reveals noise (Figure 1), it helps to report the noise of the output. Immerkær's fast estimator [66] filters the image with a mask that cancels smooth structure and keeps noise,

$$N = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}, \qquad \hat{\sigma} = \sqrt{\frac{\pi}{2}}\,\frac{1}{6\,(W-2)(H-2)} \sum_{i,j} |(I * N)(i,j)|,$$

where $W \times H$ is the image size and the sum runs over interior pixels. It assumes white Gaussian noise, so it overestimates on textured images and underestimates spatially correlated noise, which is what real camera pipelines and many networks produce. Use it to compare outputs of the *same* input, not across scenes.

The snippet below recreates the dark coffee image of Figure 1 and compares five "enhancements" plus the clean photo.

```
import numpy as np
from scipy import ndimage as ndi
from skimage import color, data, exposure, img_as_float

rng = np.random.default_rng(0)
clean = img_as_float(data.coffee())

def synth_low_light(x, k=0.05, peak=2000.0, read_std=8e-4, gamma=2.2):   # as in Figure 1
    lin = k * x ** gamma
    lin = rng.poisson(lin * peak) / peak
    lin = lin + rng.normal(0, read_std, lin.shape)
    return np.round(np.clip(lin, 0, 1) ** (1 / gamma) * 255) / 255

def loe(ref, out, step=8):
    """Lightness Order Error on a strided subsample (m = 50 x 75 = 3750 pixels here)."""
    L, Le = (im.max(axis=2)[::step, ::step].ravel() for im in (ref, out))
    U, Ue = L[:, None] >= L[None, :], Le[:, None] >= Le[None, :]
    return (U ^ Ue).sum(axis=1).mean()                  # mean over x of RD(x)

def colorfulness(im):
    """Hasler-Suesstrunk colorfulness, RGB scaled to 0..255."""
    R, G, B = (255 * im[..., c] for c in range(3))
    rg, yb = R - G, 0.5 * (R + G) - B
    return np.hypot(rg.std(), yb.std()) + 0.3 * np.hypot(rg.mean(), yb.mean())

def noise_sigma(im):
    """Immerkaer's noise estimate on BT.601 luma, in 8-bit units."""
```

```
    Y = 255 * (0.299 * im[..., 0] + 0.587 * im[..., 1] + 0.114 * im[..., 2])
    N = np.array([[1, -2, 1], [-2, 4, -2], [1, -2, 1]])
    H, W = Y.shape
    return np.sqrt(np.pi / 2) / (6 * (W - 2) * (H - 2)) * np.abs(ndi.convolve(Y, N)[1:-1,
     ↪  1:-1]).sum()

dark = synth_low_light(clean)
lab = color.rgb2lab(dark); L = lab[..., 0] / 100
he = lab.copy(); he[..., 0] = 100 * exposure.equalize_hist(L)
cl = lab.copy(); cl[..., 0] = 100 * exposure.equalize_adapthist(L, clip_limit=0.01)
outputs = {
    "dark input": dark,
    "gain x20 (linear)": np.clip(20 * dark ** 2.2, 0, 1) ** (1 / 2.2),
    "gamma 0.4": dark ** 0.4,
    "HE on L*": np.clip(color.lab2rgb(he), 0, 1),
    "CLAHE on L*": np.clip(color.lab2rgb(cl), 0, 1),
    "clean reference": clean,
}
for name, im in outputs.items():
    print(f"{name:18s} mean {im.mean():.2f}  LOE {loe(dark, im):6.1f}  "
          f"colorfulness {colorfulness(im):5.1f}  noise {noise_sigma(im):5.2f}")
# dark input         mean 0.10  LOE    0.0  colorfulness  23.1  noise  3.32
# gain x20 (linear)  mean 0.39  LOE    0.3  colorfulness  90.0  noise 12.93
# gamma 0.4          mean 0.34  LOE    0.0  colorfulness  61.6  noise 11.19
# HE on L*           mean 0.48  LOE  334.7  colorfulness  28.3  noise 16.17
# CLAHE on L*        mean 0.40  LOE  657.6  colorfulness  27.1  noise 23.06
# clean reference    mean 0.39  LOE  322.0  colorfulness  76.9  noise  4.69
```



Figure 9 — Brighter is not automatically better. HE is the brightest output but the most order-breaking after CLAHE and nearly colorless; the noisy linear gain is more 'colorful' than the clean photo; and the clean photo itself has a high LOE, because LOE is measured against the noisy input.

Read the numbers together, not one at a time:

- **Brightness alone ranks the wrong way.** HE has the highest mean (0.48) but also a large LOE, the second-highest noise and almost no color left. CLAHE's local remapping breaks lightness order the most.
- **Colorfulness is fooled by noise.** The linear gain scores 90, *higher* than the clean photo's 77, partly because amplified chroma noise raises σ_{rg} and σ_{yb}.
- **LOE penalizes restoration.** The clean photo, the ideal answer, gets LOE 322 against the noisy input, while doing nothing at all gets 0. Low LOE is necessary for a natural tone curve, not sufficient for quality.
- **Only the noise estimate separates the clean photo** (4.7) from the bright but noisy outputs (11 to 23). No single NR number captures "well exposed, natural color, and clean".

Learned no-reference assessors

Neural and vision-language assessors (NIMA, MUSIQ, CLIP-IQA, MANIQA, Q-Align and others) are described in DLCV 04 and DLCV 04; they can all be applied to LLIE outputs and share common implementations, e.g. the IQA-PyTorch toolbox [75]. Three points are specific to low light:

- **Resolution.** Avoiding resizing matters for LLIE, where noise and fine texture are exactly what resizing would hide, so an assessor that takes the image at its native resolution, such as MUSIQ [68], is a natural fit.
- **Exposure prompts and independence.** CLIP-IQA [69] can be probed with an exposure-specific prompt pair such as "Bright photo." / "Dark photo.", which is convenient for exposure, but the exact wording changes the score. The same CLIP-similarity idea is used as a *training* signal by CLIP-LIT [36], so a CLIP-based metric should not be the only judge of a CLIP-trained enhancer.
- **LLIE-specific assessors.** Kannan et al. [72] (2023) collected human opinion scores for images restored by different LLIE methods, benchmarked existing NR-IQA models on them, and proposed an *opinion-unaware* model that learns distortion-aware features of restored low-light images by self-supervised contrastive learning. Datasets of this kind are the right test bed for deciding which metric to trust on enhancement outputs.

Caveats for low light. Learned assessors are trained mostly on daylight photos. They tend to rate brighter, sharper and more saturated images higher, which favors over-enhancement, and a deliberately dark night scene can be penalized even when it is the faithful result. Their scores

depend on the checkpoint (which training set), the input size and the implementation, so report those. And when a method is tuned with a learned metric or the same backbone in its loss, that metric is no longer an independent judge.

User studies and paired comparisons

Mean opinion scores (standardized in ITU-R BT.500 [73]), Bradley–Terry paired comparison [74] and good practice for running a study are described in DLCV 04. Two low-light papers show the pattern:

- Zero-DCE [33] had 15 subjects rate results on a 1–5 scale over the 202 images of NPE, LIME, MEF, DICM and VV, with instructions to look for over- or under-exposure, color deviation, and unnatural texture or noise.
- EnlightenGAN [32] used paired comparisons: 9 subjects compared pairs of outputs from five methods on 23 images, judging visible noise, exposure artifacts and unrealistic color or texture, and a Bradley–Terry fit gave the ranking.

The criteria given to the raters (exposure, color deviation, noise, unnatural texture) are a good starting list for a new low-light study.

Choosing and reporting no-reference metrics

The general comparison of NR metrics (family, training requirement, direction, sensitivity) is the metric-by-metric summary table in the DLCV 04 notes, and its recommended practice applies unchanged. The table below keeps only what changes for low-light outputs.

Metric	Better	Caveat for low light
NIQE [49]	Lower	Nearly blind to global exposure and color; can prefer noisy, high-contrast outputs
BRISQUE [50]	Lower	Trained on daylight distortions; ignores color cast
PI [63], [64]	Lower	Designed for super-resolution; rewards detail, including amplified noise
LOE [18] (task-specific)	Lower	Depends on subsampling; 0 for identity; penalizes denoising
Colorfulness [65] (task-specific, descriptive)	Context-dependent	No notion of correct color; inflated by chroma noise
Noise estimate [66] (task-specific, descriptive)	Lower	Assumes white noise; confused by texture and correlated noise
NIMA [67]	Higher	Aesthetic models reward attractive, not faithful, results
MUSIQ [68]	Higher	Trained mostly on daylight photos; checkpoint-dependent

Metric	Better	Caveat for low light
CLIP-IQA [69]	Higher	Prompt wording matters; not independent of CLIP-trained enhancers
MANIQA [70]	Higher	Not trained on exposure errors specifically
Q-Align [71]	Higher	Heavy; biases of the base model; version-dependent
MOS / paired comparison [73], [74]	Higher MOS / higher strength	Costly; small studies are noisy; protocol must be reported

What to do on top of the general practice:

1. **On paired sets**, lead with full-reference metrics (PSNR, SSIM, LPIPS) under the standard protocol, and add NR scores only as a complement.
2. **On unpaired sets**, report *several* NR metrics from different families (one NSS, one or two learned, and LOE), state the implementation, checkpoint and image size, and show many visual examples, including failures.
3. **Expect disagreement.** NSS and learned metrics often rank methods differently, and both can disagree with people; the survey of Li et al. [2] notes the frequent mismatch between metric scores and visual quality, and Liu and Fan [5] find a further gap between image quality and downstream task performance. When metrics disagree, say so instead of picking the favorable one.

The GT-mean pitfall

Plain version. On paired benchmarks, a small difference in *overall brightness* changes PSNR a lot. Some papers rescale each output so its average brightness matches the ground truth before scoring. That is information you never have in practice.

Precise version. "GT-mean" evaluation scales the output by $\lambda = \mathbb{E}[y]/\mathbb{E}[f(x)]$ and computes the metric on $\lambda f(x)$, where $\mathbb{E}[\cdot]$ is the image mean. Because LLIE is ambiguous about the "right" brightness, this can raise PSNR substantially. The Retinexformer authors provide this mode in their code for comparability with methods such as LLFlow and KinD and recent diffusion models, while explicitly saying they do *not* recommend it, because it uses the ground truth to correct the output [26]. Liao et al. [51] (2025) study the brightness mismatch between outputs and references and propose a GT-mean *loss* for training, reporting both standard and GT-mean metrics. When reading tables: check which setting is used, never mix the two, and prefer methods that are good under the standard protocol.

Task-driven evaluation

Since LLIE is often a preprocessing step, another test is to run a detector or segmenter on the enhanced images (on ExDark [57], for example) and report its accuracy. Liu and Fan's 2025 review [5] finds a disconnect: supervised methods often produce images with high perceptual quality but only modest gains for vision tasks, while zero-shot methods, despite lower image-quality scores, give more consistent gains across tasks. The benchmark of Liu et al. [4] also evaluates face detection in low light. A complete evaluation reports fidelity, no-reference quality, task performance, and runtime.

Benchmark datasets

Plain version. Datasets differ in whether they have pairs, whether the darkness is real or synthetic, whether they are raw or sRGB, and whether they carry labels for detection.

Dataset	Type	Content and what it is for
LOL (v1) [21]	Paired, real, sRGB	500 low/normal pairs captured by changing exposure (485 train / 15 test), 400×600. The most common LLIE benchmark; small test set.
LOL-v2 [52]	Paired, real + synthetic subsets	A larger *Real* subset of captured pairs and a *Synthetic* subset of synthesized pairs. Report the two subsets separately.
LSRW [53]	Paired, real	5,650 pairs from a Nikon D7500 (3,170) and a Huawei P40 Pro (2,480); 5,600 train, 50 test. Tests generalization across devices.
SDSD [54]	Paired video	Dynamic indoor and outdoor scenes captured twice with a mechatronic system for alignment; often split into frames for image tests.
SID [44]	Paired, raw	5,094 short-exposure raw images with 424 long-exposure references (Sony and Fuji). Extreme low light, raw-to-sRGB.
MIT-Adobe FiveK [55]	Paired, retouching	5,000 raw photos, each retouched by five experts. Used for tone and color enhancement (photo retouching).
SICE [56]	Multi-exposure sequences	Under- to over-exposed captures of each scene with a reference; used to train exposure correction (e.g. Zero-DCE).
VE-LOL [4]	Paired + detection	VE-LOL-L for enhancement and VE-LOL-H with face annotations for high-level evaluation.
ExDark [57]	Unpaired, labeled	7,363 low-light images, 12 object classes, 10 lighting conditions. Task-driven evaluation (detection).

Dataset	Type	Content and what it is for
LIME [17], MEF [58], NPE [18], DICM [59], VV [60]	Unpaired test sets	Small collections of real low-light or non-uniformly lit photos for visual comparison and no-reference metrics.

Challenges: NTIRE 2024 and 2025

The NTIRE workshop at CVPR (see DLCV 04 for the series in general) ran LLIE challenges in consecutive years. The **NTIRE 2024** challenge [61] covered ultra-high-resolution (4K and beyond), non-uniform illumination, backlighting, extreme darkness and night scenes; it had 428 registered participants and 22 teams with valid submissions. The **NTIRE 2025** challenge [62] had 762 registered participants and 28 teams with valid entries. The reports summarize the winning designs and are a good snapshot of what works under a fixed, fair protocol.

Modern view

The field has moved from hand-made priors to learned models in under a decade, but the core questions in the post [1] are the same: how to model the light, how to handle noise, and how to know whether an output is good. The surveys and benchmarks below are the best entry points:

- **Li et al. [2]** (TPAMI 2022), the most cited deep-LLIE survey: a taxonomy by learning strategy, network structure, use of Retinex, data format and losses; it reports a generalization gap and a frequent mismatch between metric scores and visual quality.
- **Zheng et al. [3]**: a similar taxonomy, plus new test data that mixes under- and over-exposure and a night video set.
- **Liu et al. [4]** (IJCV 2021): the VE-LOL benchmark, which made task-driven evaluation mainstream.
- **Liu and Fan [5]** (Neurocomputing 2025): image-quality metrics and downstream task performance disagree, and zero-shot methods give the most consistent task gains.
- **Adhikarla et al. [6]** (2025): diffusion models for LLIE in six families; realism and flexibility at a high inference cost.
- **NTIRE reports [61], [62]**: the current practice of top teams under a common protocol.

Optional: Guide to the LLIE surveys (in the Paper guides). What each survey covers, how it organizes the field and which data it adds.

State of the art (2023–2026). Three directions dominate. First, **Retinex-guided backbones**: transformers (Retinexformer [26]) and state-space models (RetinexMamba [42]) that keep a light-versus-reflectance structure. Second, **generative models**: diffusion (Diff-Retinex [38], GSAD

[39], LightenDiffusion [40], Reti-Diff [41]) and flows (LLFlow [37]). Third, **better representations and protocols**: learned color spaces (HVI-CIDNet [27]), brightness-mismatch-aware training and evaluation (GT-Mean loss [51]), language-guided and all-in-one restoration (CLIP-LIT [36], InstructIR [43]), and task-aware enhancement (MAET [45], FeatEnHancer [46]).

Open problems.

- **Real noise.** Most sRGB methods train on data whose noise does not match real sensors after a camera pipeline. Better noise models [7], raw-domain training [44], or joint enhancement and denoising remain central.
- **Video consistency.** Frame-by-frame enhancement flickers. Paired video data (SDSD [54]) is scarce, and temporal consistency is still hard, especially with generative models.
- **Evaluation.** The ill-posed brightness target, GT-mean protocols, unreliable no-reference metrics for exposure, and the gap between perceptual and task metrics make comparisons fragile.
- **Efficiency on edge devices.** Phones and cars need real-time, low-power models. Curve methods (Zero-DCE++ [34], SCI [35]) and lightweight prior-based networks [28] show that tiny models can work, while diffusion models remain far too slow for this use.
- **Generalization.** Models trained on one dataset or camera often fail on another (the generalization gap reported in [2]); mixed over- and under-exposure, backlight and colored artificial light are still difficult.

Key takeaways

- LLIE is **tone enhancement plus restoration**: a perfect brightness correction still leaves noise, color cast and quantization (Figure 1).
- Brightness can be enhanced in a separate channel (V, I, Y', L^*), but the choice couples to saturation. BT.601 luma is $0.299R' + 0.587G' + 0.114B'$; $16 + 0.257R' + 0.504G' + 0.098B'$ is the same luma in 8-bit limited range.
- Retinex models an image as reflectance times **illumination**, $S = R \circ L$; SSR/MSR/LIME estimate L with smoothness priors, and the inverted-image dehazing approach rests on essentially the same assumption.
- Deep methods are best organized by **training signal**: supervised, unrolled, unpaired (EnlightenGAN), zero-reference (Zero-DCE, SCI, CLIP-LIT) and generative (LLFlow, diffusion). Retinex decomposition appears across them.
- Zero-DCE's LE curve $x + \alpha x(1-x)$ preserves range and order; its non-reference losses are priors (exposure, gray world, smoothness, contrast) that can fail on real noise and colorful scenes.
- Report full-reference, no-reference and task-driven results, **never mix GT-mean and standard**

numbers, and remember that higher PSNR does not guarantee better detection.

- No-reference metrics each see one aspect: NIQE/BRISQUE/PI see structure but are nearly blind to exposure and color, LOE sees lightness order but not brightness or noise, colorfulness is inflated by chroma noise, and learned assessors inherit daylight biases. **Brighter is not automatically better** (Figure 9): report several NR metrics from different families, plus a well-designed user study. The general NR toolbox is in DLCV 04.

Exercises

1. **Monotonic curves.** Show that $\mathrm{LE}(x;\alpha) = x + \alpha x(1-x)$ is monotonically non-decreasing on $[0, 1]$ if and only if $|\alpha| \le 1$. What goes wrong for $\alpha = 1.5$?

   Hint

   The derivative is $1 + \alpha(1-2x)$, which is linear in x, so its minimum on $[0,1]$ is at an endpoint: $1+\alpha$ at $x=0$ or $1-\alpha$ at $x=1$. Both are ≥ 0 exactly when $|\alpha| \le 1$. For $\alpha = 1.5$ the slope at $x=1$ is -0.5, so near-white pixels would get darker as the input gets brighter, reversing the brightness order and creating artifacts in highlights.

2. **Gain of the iterated curve.** For a small input x, show that eight iterations with a constant α multiply x by approximately $(1+\alpha)^8$. Use this to explain why zero-reference curve methods amplify shadow noise, and suggest one change to the losses or architecture that would reduce the problem.

   Hint

   For small x, $x(1-x) \approx x$, so each step gives $x \mapsto (1+\alpha)x$. With $\alpha = 1$ that is $2^8 = 256$. Noise with standard deviation σ becomes 256σ. Possible fixes: add a denoising branch or a noise-aware loss, limit α where the estimated SNR is low (in the spirit of SNR-aware), or penalize high-frequency energy in the output that is not present in a smoothed input.

3. **Two luma formulas.** A pixel has $(R', G', B') = (200, 100, 50)$ in 8 bits. Compute its full-range BT.601 luma and its limited-range luma. Show that they encode the same quantity.

   Hint

   Full range: $0.299 \cdot 200 + 0.587 \cdot 100 + 0.114 \cdot 50 = 124.2$. Limited range: $16 + 0.257 \cdot 200 + 0.504 \cdot 100 + 0.098 \cdot 50 \approx 122.7$. Converting back: $(122.7 - 16) \cdot 255/219 \approx 124.2$. The only differences are the scale $219/255$ and the offset 16.

4. **Retinex losses without ground truth.** In Retinex-Net, why is $\mathcal{L}_{ir} = \|R_{\text{low}} - R_{\text{normal}}\|_1$ needed? Describe a trivial decomposition that zeroes the self-reconstruction terms of $\mathcal{L}_{\text{recon}}$ and $\mathcal{L}_{is}$ but is useless, and explain how $\mathcal{L}_{ir}$ rules it out.

Hint

Set $I_i \equiv 1$ (perfectly smooth) and $R_i = S_i$. Then each image is reconstructed exactly and the smoothness loss is zero, but nothing has been decomposed. Since $S_{\text{low}} \neq S_{\text{normal}}$, this gives $R_{\text{low}} \neq R_{\text{normal}}$, so $\mathcal{L}_{ir}$ is large. Requiring equal reflectances forces the brightness difference between the pair into the illumination.

5. **GT-mean by hand.** Your model outputs an image with mean 0.40; the ground truth has mean 0.50, and otherwise the output is perfect up to that global scale. What is the GT-mean scale factor, and what is the PSNR before and after rescaling if the image is a constant? Why is the rescaled number not a fair comparison with a method evaluated normally?

Hint

$\lambda = 0.50/0.40 = 1.25$. Before: MSE $= 0.1^2 = 0.01$, PSNR $= 20$ dB. After rescaling the output equals the reference, so the error is 0 and PSNR is unbounded. A real method never has the ground-truth mean at test time, so GT-mean numbers reward the right *structure* while ignoring the hardest part of the task, choosing the brightness.

6. **Change the color prior.** In the PyTorch toy, replace the gray-world loss with a loss that keeps the *chromaticity* of the input, for example $\|y/\sum_c y_c - x/\sum_c x_c\|_1$. Train again and compare the coffee result with Figure 7. When is each prior preferable?

Hint

Add a small epsilon to the sums. Preserving input chromaticity keeps the red-brown cup red, but it also keeps any color cast of the light source (for example a sodium lamp). Gray world removes casts but desaturates scenes that are genuinely colorful. Real methods often combine a weak color prior with learned color correction from data.

7. **LOE by hand.** A three-pixel input has lightness $L = (0.1, 0.2, 0.3)$, and an enhancer outputs $L_e = (0.5, 0.4, 0.9)$. Compute $\text{RD}(x)$ for each pixel and the LOE. Then explain why, in Figure 9, the clean reference gets a *higher* LOE (322) than the gamma curve (0), and what this implies for comparing a denoising enhancer with a plain tone curve using LOE alone.

Hint

Only the pair $(1, 2)$ changes order: $0.1 < 0.2$ in the input but $0.5 > 0.4$ in the output. That flips both $U(L_1, L_2)$ and $U(L_2, L_1)$, so RD $= (1, 1, 0)$ (self-pairs never flip) and LOE $= 2/3$. A gamma curve applied per channel is monotonic, so it keeps the order of $\max(R, G, B)$ exactly and gets 0. The clean image does not share the *noise* of the dark input, and that noise has scrambled the order of many

nearby, nearly equal pixels, so the clean image disagrees with the input on those pairs. LOE therefore rewards keeping the input's noise and penalizes real restoration; it should be read together with a noise or quality measure, never alone.

Supplementary notes

These notes are optional background. Nothing in the main text or the exercises depends on them.

CLAHE: equalizing locally without amplifying noise

Plain version. Imagine correcting the contrast of a photo one small tile at a time, but with a rule: no gray level may claim more than its fair share of the tile. A big flat patch of sky then cannot be stretched into a noisy mess, because its one crowded gray level is capped. The corrected tiles are finally blended so that no tile border shows.

Precise version. Split the image into tiles, for example an 8×8 grid. For tile t, let $h_t(j)$ be the number of its pixels in gray bin $j = 0, \dots, B-1$, and N the number of pixels in the tile. Plain adaptive HE uses the cumulative histogram of each tile as its mapping, as in the global case. CLAHE first clips the histogram at a cap $c = \beta N$ and spreads the clipped pixels evenly over all bins:

$$\tilde{h}_t(j) = \min\left(h_t(j),\, c\right) + \frac{E_t}{B}, \qquad E_t = \sum_j \max\left(h_t(j) - c,\, 0\right), \qquad T_t(r) = \sum_{j=0}^{r} \frac{\tilde{h}_t(j)}{N}.$$

Here $\beta \in (0, 1]$ is the (normalized) clip limit, the largest fraction of a tile's pixels that one bin may hold; c is the resulting cap in pixels; E_t is the number of clipped pixels, which keeps the total at N; and $T_t(r)$ maps the bin index r to an output in $[0, 1]$ (multiply by $L - 1$ for gray levels). Implementations normalize the clip limit differently; the one above is the convention of `skimage.exposure.equalize_adapthist` (default $\beta = 0.01$), and a value of 1 or more means no clipping, which is plain AHE.

The reason clipping controls noise is the slope. Going from bin r to $r + 1$ raises T_t by $\tilde{h}_t(r)/N$, and the identity map raises it by $1/B$, so the local gain of the mapping relative to the identity is $B\,\tilde{h}_t(r)/N$. Since $\tilde{h}_t(r) \le c + E_t/B$,

$$\text{gain} \;\le\; B\beta + \frac{E_t}{N}.$$

Here $B\beta$ is the cap expressed as a multiple of the average bin count, and $E_t/N \le 1$ is the fraction of clipped pixels. An unclipped dark tile has almost all its pixels in a few bins, so its gain is huge; clipping bounds it by a small number.

Interpolation. Applying T_t strictly inside each tile would leave visible seams. Instead each tile owns the pixel at its center, and a pixel at fractional position $(u, v) \in [0, 1]^2$ between four neighboring tile centers is mapped by all four tiles and blended bilinearly:

$$\hat{s}(\mathbf{x}) = (1-u)(1-v)\,T_{00}(r) + u(1-v)\,T_{10}(r) + (1-u)v\,T_{01}(r) + uv\,T_{11}(r),$$

where r is the pixel's bin and $T_{00}, T_{10}, T_{01}, T_{11}$ are the mappings of the four surrounding tiles. Implementations differ at the image border (scikit-image pads by reflecting the image). The cost is one histogram pass plus one lookup pass, linear in the number of pixels, plus $O(B)$ per tile.

The snippet below builds a dark, noisy test image, prints the largest gain of one tile's mapping with and without clipping, and compares the noise left in a flat patch (the sky, where the true signal is smooth).

```
import numpy as np
from skimage import data, exposure, img_as_float

rng = np.random.default_rng(0)
img = img_as_float(data.camera())
dark = np.clip(0.25 * img ** 1.5 + rng.normal(0, 0.01, img.shape), 0, 1)   # dark + noisy

def tile_map(tile, beta, bins=256):
    """Mapping T for one tile; beta = max fraction of the tile's pixels one bin may hold."""
    h = np.bincount((tile * (bins - 1)).astype(int).ravel(), minlength=bins).astype(float)
    cap = beta * tile.size
    excess = np.clip(h - cap, 0, None).sum()          # pixels above the cap
    h = np.minimum(h, cap) + excess / bins            # clip, then spread the excess evenly
    return np.cumsum(h) / h.sum(), excess / tile.size  # CDF in [0, 1], excess fraction E/N

tile = dark[300:364, 100:164]                         # a 64 x 64 tile (4096 pixels)
for beta in (1.0, 0.01):                              # 1.0 = no clipping
    T, e = tile_map(tile, beta)
    print(f"beta={beta:<5}  max slope of T = {256 * np.diff(T).max():.2f}   E/N = {e:.2f}")

sky = (slice(0, 60), slice(300, 360))                 # flat region, noise only
for name, out in [("input", dark),
                  ("HE", exposure.equalize_hist(dark)),
                  ("AHE", exposure.equalize_adapthist(dark, clip_limit=1.0)),
                  ("CLAHE 0.01", exposure.equalize_adapthist(dark, clip_limit=0.01))]:
    print(f"{name:10s} mean {out.mean():.3f}  noise std in flat patch {out[sky].std():.3f}")
```

Printed output:

```
beta=1.0    max slope of T = 35.88   E/N = 0.00
beta=0.01   max slope of T = 3.47   E/N = 0.91
input      mean 0.103  noise std in flat patch 0.011
HE         mean 0.506  noise std in flat patch 0.066
AHE        mean 0.517  noise std in flat patch 0.274
CLAHE 0.01 mean 0.408  noise std in flat patch 0.089
```

In this tile the largest gain falls from 35.88 to 3.47; with $B = 256$, $\beta = 0.01$ and $E_t/N = 0.91$ the bound above gives $2.56 + 0.91 = 3.47$. In the flat patch the noise standard deviation is 0.011 in the input; global HE raises it to 0.066, unclipped AHE (`clip_limit=1.0`) to 0.274, and CLAHE with

$\beta = 0.01$ to 0.089. So clipping removes most of the extra noise gain of AHE but not all of it. These numbers come from one seed and one image; treat them as an illustration, not a benchmark.

When it fails. CLAHE does not denoise, it only limits how much it amplifies; very dark, noisy inputs still gain visible grain (Figure 3 and Figure 9). Tiles that are too small treat noise as texture, and tiles that are too large behave like global HE. Because each tile remaps independently, brightness order across tile borders can change, which is why LOE (introduced later in this tutorial) is high for CLAHE. For color images, run it on a lightness channel such as L^*, as in the main text, never on R, G and B separately. The method is described in [9], [10]; the clip-limit convention above is the one used by scikit-image, not necessarily the one in those papers.

More deep LLIE methods

Plain version. The main text names the representative deep LLIE methods in a sentence each. This note says how each one is built: first the supervised networks, then the Retinex-based ones, then the diffusion models.

Supervised networks.

- **LLNet** [19] (2017) was one of the first deep LLIE models: a stacked sparse denoising autoencoder trained on synthetically darkened and noised patches, learning brightening and denoising jointly.
- **CNN + bright channel prior** (Tao et al. [20], 2017) handled the two difficulties in sequence: a CNN denoises first, then contrast is enhanced with an atmospheric-scattering-style model driven by the bright channel, an early example of combining learning with the inversion idea of the main text.
- **MIRNet** [24] (2020) is a general restoration backbone that keeps a full-resolution stream alongside parallel lower-resolution streams, exchanges information between them, fuses them with selective-kernel attention, and was evaluated on enhancement as well as denoising and super-resolution.
- **SNR-aware** [25] (2022) estimates a per-pixel signal-to-noise map. Where SNR is high, local convolution is enough; where SNR is very low, the local neighborhood is unreliable, so the model uses long-range transformer attention to borrow information from elsewhere in the image.
- **HVI-CIDNet** [27] (2025) attacks the color-space problem of Figure 2: sRGB is very sensitive to color and gives color bias and brightness artifacts, and HSV fixes brightness but creates red and black noise. The authors propose the HVI space, built from polarized hue/saturation maps and a *learnable* intensity, plus a network that decouples color and intensity.

Retinex-based networks.

- **KinD** [22] (2019) splits the work into three subnetworks: decomposition, *reflectance restora-*

tion (because the dark image's reflectance is noisy and color-shifted, the "R is unchanged" assumption is relaxed), and illumination adjustment with a user-controllable brightness ratio.

KinD++ [23] (2021) refines the design to reduce visual defects.

- **Retinexformer** [26] (2023) notes that both capture and brightening corrupt the components, and rewrites the model with perturbations: $S = (R + \hat{R}) \circ (L + \hat{L})$, where $\hat{R}$ and $\hat{L}$ model noise, artifacts and color distortion. It is trained in **one stage** (instead of Retinex-Net's multi-stage training) with an L1 loss, and its **illumination-guided multi-head self-attention** uses the illumination estimate to steer attention, so regions with different exposure interact. It became a common baseline.
- **RetinexMamba** [42] (2024 preprint) keeps the Retinex-transformer structure and replaces attention with Mamba state-space blocks, which scale linearly with the number of pixels. "Retinex decomposition plus a new backbone" has been a stable design pattern for years.

A less common route builds on the atmospheric scattering model rather than plain Retinex. CPGA-Net [28] combines channel priors with gamma correction in a lightweight network that processes global and local information, and CPGA-Net+ [29] revisits the theoretical illumination model for efficiency. That line of work also reports that the features learned by different functional modules follow the physical roles of the priors, which suggests that a network guided by a model is not a completely opaque black box.

Diffusion models. Diffusion models generate the output by iteratively denoising random noise, conditioned on the dark image:

- **Diff-Retinex** [38] (2023) first decomposes the image with a Retinex transformer, then uses conditional diffusion models to generate the restored reflectance and illumination.
- **GSAD** [39] (2023) adds a global structure-aware regularization that uses the image structure to guide a curvature-based constraint on the diffusion trajectory, plus an uncertainty-guided term for hard regions.
- **LightenDiffusion** [40] (2024) performs Retinex decomposition in a latent space and trains with *unpaired* data.
- **Reti-Diff** [41] (2025) runs a latent diffusion model to produce compact Retinex priors that guide a restoration transformer, which reduces the cost of running diffusion at full resolution.

The trade-off is cost and fidelity: many denoising steps make inference slow, and generated details may look right without being *true*. The survey by Adhikarla et al. [6] organizes this sub-field into six families: intrinsic decomposition, spectral and latent, accelerated, guided, multimodal, and autonomous.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Guide to the LLIE surveys

Plain version. The Modern view lists these surveys in one line each. Here is what each covers, how it organizes the field and which data it adds.

Li et al., TPAMI 2022 [2]. The most cited deep-LLIE survey. It organizes methods by learning strategy (supervised, reinforcement, unsupervised, zero-shot and semi-supervised learning), network structure, use of the Retinex model, data format (sRGB or raw), loss functions, and training and test data. It also introduces LLIV-Phone, an image and video dataset captured with many different phone cameras under diverse illumination, and an online platform for comparing methods. Its main lessons are a generalization gap (many methods struggle on the phone data) and a frequent mismatch between metric scores and visual quality.

Zheng et al., "Low-Light Image and Video Enhancement: A Comprehensive Survey and Beyond" [3]. It reviews traditional and deep methods with a similar taxonomy and goes further on data: it proposes SICE_Grad and SICE_Mix, which mix under- and over-exposure inside one image, and Night Wenzhou, an aerial and street night video set, to test methods on cases that existing benchmarks miss. It was the source of the post's own overview figures.

Liu et al., IJCV 2021 [4]. "Benchmarking low-light image enhancement and beyond" is a benchmark more than a survey: the VE-LOL dataset with paired data and face annotations, a comparison using full-reference, no-reference and semantic metrics, and a joint enhancement-plus-detection model. It made task-driven evaluation mainstream.

Liu and Fan, Neurocomputing 2025 [5]. This review asks which enhancement actually helps downstream vision and finds the disconnect described above between image-quality metrics and task performance, with zero-shot methods giving the most consistent task gains.

Adhikarla et al., 2025 [6]. A survey of diffusion models for LLIE with a six-family taxonomy and a performance analysis. Its message is that diffusion brings realism and flexibility but at a high inference cost, so acceleration, guidance and latent-space designs are the active topics.

NTIRE reports [61], [62]. These are not surveys but show the current practice of top teams under a common protocol, with high-resolution and difficult illumination.

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DLCV 05A • Lightweight Image Enhancement: Curves, Bilateral Grids and Lookup Tables

How enhancement models stay small and fast enough for phones and 4K video: what to count, why predicting a transform beats predicting an image, and how curves, bilateral grids, learnable 3D lookup tables and tiny networks do it.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-05a-lightweight-enhancement/

What you'll learn

- What "lightweight" means for an enhancement model: parameters, multiply–accumulates (MACs) per megapixel, memory traffic, and measured latency on the target device, and why 4K inputs change the picture.
- Why enhancement is unusually well suited to tiny models: the output is mostly a smooth remapping of the input's own values, so a model can **predict a transform** instead of painting every output pixel.
- The four families of fast enhancers: curve estimation, bilateral grids and guided upsampling (HDRNet, the deep guided filter), learnable 3D lookup tables (and their successors), and very small CNN or transformer networks.
- How a 3D lookup table is applied with **trilinear interpolation**, and how a bilateral grid is fitted at low resolution and **sliced** back to full resolution, with NumPy code you can run in seconds.
- How to report efficiency fairly (what to count, how to time, what to state) and how these models fail: banding, halos, color shifts, noise and flicker.

The big picture

Most enhancement networks in DLCV 05 are judged by how good their pictures look. This chapter adds a second requirement: the model must run on a phone, inside a camera, or on every frame of a video, often at 4K resolution, within a few milliseconds and a small power budget. That requirement changes the design more than any loss function does.

> **Explain like I'm five**
>
> Imagine you must recolor a huge mural, and you are allowed only a tiny helper. A slow way is to have the helper repaint every brick. A fast way is to have the helper look at a small photo of the mural and write a short recipe card: "make the dark bricks a bit lighter, make the reds a little warmer, and do it more on the left wall than on the right." Then a simple machine follows the card brick by brick, which

is quick because it never has to think. Lightweight enhancers are recipe writers: a small network writes the card (a curve, a color table or a grid of small recipes), and a cheap rule applies it to every pixel.

Why it matters:

- **Every pixel, every frame.** A phone camera preview or a video pipeline enhances tens of frames per second. A model that is fine for one photo on a server can be far too slow, too hot or too memory-hungry for that.
- **Resolution keeps growing.** A 4K frame has 8.3 million pixels and a 12-megapixel photo 12 million. Most of the cost of a full-resolution convolutional network grows in proportion to the number of pixels, so a network that is cheap at 512 × 512 can be 30 times more expensive on a 4K frame.
- **Enhancement has structure we can exploit.** Unlike denoising or super-resolution, tone and color enhancement mostly *rearranges values that are already there*. That one observation leads to designs that cost a few hundred arithmetic operations per pixel instead of tens of thousands.

This chapter builds on DLCV 05 (Retinex, Zero-DCE curves, low-light datasets and evaluation) and on the classical tools of IP 03 (intensity transformations, the guided and bilateral filters). It does not repeat them; it explains how they become fast learned models.

What "lightweight" means

Plain version. "Lightweight" is not one number. A model can be small on disk but slow, fast on a GPU but slow on a phone, or cheap in arithmetic but limited by memory. You need to say which cost you mean, at which resolution and on which device.

Precise version. Five costs matter for an enhancement model. The general background (MACs versus FLOPs, why FLOPs are not latency, a latency protocol) is in ML 10; here is what each means for enhancement.

1. **Parameters** P: the number of learned values. They decide the size of the model file and the weight memory. Zero-DCE++ has about 10K parameters [1]; the U-Net-like design counted below has about 1.9 million.
2. **Arithmetic** per image, counted in multiply–accumulates. For a convolution layer,

$$\text{MACs} = H_{\text{out}}\, W_{\text{out}}\, k^2\, C_{\text{in}}\, C_{\text{out}},$$

where $H_{\text{out}} \times W_{\text{out}}$ is the size of the output feature map, $k \times k$ the kernel size, and C_{in}, C_{out} the numbers of input and output channels. One MAC is one multiplication and one addition, so 1

MAC $\approx$ 2 FLOPs; state which unit you use. 3. **MACs per pixel** (or per megapixel): the arithmetic divided by the number of *input* pixels HW. This is the fairest single number for enhancement,

because the same model is applied at very different resolutions. A layer that runs at full resolution contributes $k^2 C_{\text{in}} C_{\text{out}}$ MACs per pixel; a layer that runs on a fixed-size thumbnail contributes almost nothing per pixel at 4K. 4. **Memory traffic.** Every feature map has to be written to memory and read back by the next layer. A feature map with C channels at full resolution occupies $HWCb$ bytes, where b is the bytes per value (2 for fp16, 1 for int8). On phones and embedded chips, moving these bytes often costs more time and energy than the arithmetic. 5. **Latency and energy** on the target device: the measured time from input frame to output frame, and the energy per frame. These are what users feel, and they can only be measured, not computed.

The code below counts the first four costs for three designs, by pure arithmetic: (1) a seven-layer, 32-channel CNN that runs at full resolution, in the spirit of the DCE-Net of Zero-DCE [2] (the real DCE-Net has skip connections, so its count differs); (2) a U-Net-like restorer [3] with four levels and 32 to 256 channels; and (3) an HDRNet-like design [4] in which a small CNN looks at a fixed 256 × 256 copy of the image and predicts a 16 × 16 × 8 grid of color transforms, followed by a cheap per-pixel step that applies them (explained in the bilateral-grid section).

```
def conv(cin, cout, k=3, scale=1.0, fixed=None):
    """A k x k convolution on (scale x the input pixels), or on a fixed number of pixels."""
    return dict(params=k * k * cin * cout + cout, macs=k * k * cin * cout, cout=cout,
     ↪  scale=scale, fixed=fixed)

def cost(layers, H, W, per_pixel_apply=0):
    """Parameters, multiply-accumulates (MACs) and the largest feature map (fp16 bytes)."""
    px = [l["fixed"] or H * W * l["scale"] for l in layers]
    macs = sum(l["macs"] * p for l, p in zip(layers, px)) + per_pixel_apply * H * W
    peak = max(2 * l["cout"] * p for l, p in zip(layers, px))
    return sum(l["params"] for l in layers), macs, peak

# (1) a seven-layer, 32-channel CNN at full resolution (in the spirit of DCE-Net)
tiny = [conv(3, 32)] + [conv(32, 32) for _ in range(5)] + [conv(32, 24)]
# (2) a U-Net-like restorer: 4 levels, 32-64-128-256 channels, two convs per level
unet = [conv(3, 32), conv(32, 32), conv(32, 64, scale=1/4), conv(64, 64, scale=1/4),
        conv(64, 128, scale=1/16), conv(128, 128, scale=1/16),
        conv(128, 256, scale=1/64), conv(256, 256, scale=1/64),
        conv(256 + 128, 128, scale=1/16), conv(128, 128, scale=1/16),
        conv(128 + 64, 64, scale=1/4), conv(64, 64, scale=1/4),
        conv(64 + 32, 32), conv(32, 32), conv(32, 3, k=1)]
# (3) HDRNet-like: a CNN on a fixed 256 x 256 copy predicts a 16x16x8 grid of 3x4 affine maps
grid_net = [conv(3, 8, fixed=128**2), conv(8, 16, fixed=64**2), conv(16, 32, fixed=32**2),
            conv(32, 64, fixed=16**2), conv(64, 64, fixed=16**2), conv(64, 64, fixed=16**2),
            conv(64, 96, k=1, fixed=16**2)]
apply = 9 + 48 + 8 * 12 + 12   # guide (3x3 matrix + 16-piece curves), trilinear slice,
 ↪  affine

for H, W, name in [(1080, 1920, "1080p"), (2160, 3840, "4K")]:
    print(f"--- {name}: {H * W / 1e6:.2f} megapixels")
    for label, layers, extra in [("tiny full-res CNN", tiny, 0), ("U-Net-like", unet, 0),
                                 ("low-res grid + slicing", grid_net, apply)]:
        p, m, peak = cost(layers, H, W, extra)
```

```
        print(f"{label:23s} params {p:>9,d}  GMACs {m / 1e9:7.1f}  MACs/pixel {m / (H *
         ↪  W):8.0f}"
              f"  largest feature map {peak / 2**20:7.2f} MiB")
```

```
--- 1080p: 2.07 megapixels
tiny full-res CNN       params    54,072  GMACs   111.7  MACs/pixel    53856  largest
 ↪  feature map  126.56 MiB
U-Net-like              params 1,946,947  GMACs   336.4  MACs/pixel   162240  largest
 ↪  feature map  126.56 MiB
low-res grid + slicing  params   104,624  GMACs     0.4  MACs/pixel      183  largest
 ↪  feature map    0.25 MiB
--- 4K: 8.29 megapixels
tiny full-res CNN       params    54,072  GMACs   446.7  MACs/pixel    53856  largest
 ↪  feature map  506.25 MiB
U-Net-like              params 1,946,947  GMACs  1345.7  MACs/pixel   162240  largest
 ↪  feature map  506.25 MiB
low-res grid + slicing  params   104,624  GMACs     1.4  MACs/pixel      170  largest
 ↪  feature map    0.25 MiB
```

Three lessons are in these numbers (all from our own arithmetic, not from any paper):

- **Parameters mislead.** The "tiny" CNN has half the parameters of the grid design, yet it needs about 300 times more arithmetic, because every one of its layers runs at full resolution. Parameter count says almost nothing about the cost of a full-resolution network.
- **Resolution multiplies everything at full resolution.** Going from 1080p to 4K multiplies the MACs of designs (1) and (2) by exactly 4, the ratio of pixel counts. The grid design grows only through its small per-pixel step.
- **Memory traffic can be the real limit.** One 32-channel fp16 feature map at 4K is 3840×2160×32×2 bytes, about 506 MiB, and a full-resolution network writes and reads several of them per frame. The grid design's largest feature map is a quarter of a mebibyte.

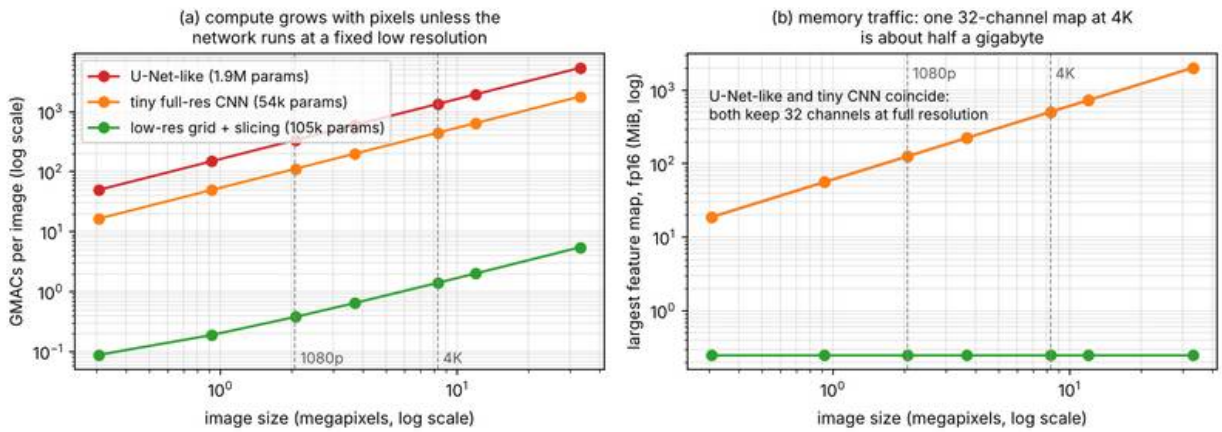


Figure 1 — Cost versus resolution for the three designs of the snippet (our arithmetic). Full-resolution networks scale with the pixel count; a network that runs on a fixed thumbnail does not, and only its cheap per-pixel step grows.

Why enhancement suits tiny models

Plain version. When you brighten or recolor a photo, the edges, textures and objects stay exactly where they were. Only the *values* change, and they change in a smooth, predictable way: similar colors in the same neighborhood get similar treatment. So a model does not have to redraw the image; it only has to describe how to remap it.

Precise version. Write the enhanced image as a pixel-wise function of the input,

$$O(\mathbf{x}) = T_{\mathbf{x}}(I(\mathbf{x})),$$

where $\mathbf{x}$ is the pixel position, $I(\mathbf{x}) \in [0,1]^3$ the input RGB value, $O(\mathbf{x})$ the output value, and $T_{\mathbf{x}} : [0,1]^3 \to [0,1]^3$ a color transform that may depend on the position. Two assumptions make T cheap to represent:

1. **Smooth in space.** $T_{\mathbf{x}}$ changes slowly with $\mathbf{x}$, except across edges of the image. Lighting, exposure and a photographer's local edits are all low-frequency.
2. **Smooth in color.** For a fixed position, $T_{\mathbf{x}}$ is a smooth function of the color, so a few parameters (a curve, a 3×4 matrix, a coarse table) describe it.

Under these assumptions T has far fewer degrees of freedom than the image. It can be **predicted at low resolution** by a small network and then **applied at full resolution** by a fixed, cheap rule. The image's fine detail is never processed by the network at all; it passes through T untouched, which is why these methods keep edges sharp even though their networks see only a thumbnail.

The families in this chapter differ only in how they represent T:

Family	Representation of T	Depends on position?	Per-pixel cost of applying it
Global curves (1D LUTs)	One curve per channel	No	3 lookups
Per-pixel curves (Zero-DCE)	Curve parameters per pixel	Yes, but predicted at full resolution or upsampled	a few MACs per iteration
Bilateral grid (HDRNet)	3×4 affine maps on a coarse (x, y, intensity) grid	Yes, edge-aware	slicing plus a matrix–vector product
3D LUT	A table over the RGB cube	No (unless extended)	one trilinear interpolation

When the assumptions fail. Denoising, deblurring, super-resolution and inpainting must create or remove fine structure, so they are *not* smooth remappings, and these tricks do not apply to them. Low-light enhancement is in between: the tone part is a smooth remapping, but the noise

part is not, which is the split between tone enhancement and detail restoration described in DLCV 05. Keep this limit in mind for every method below: a fast transform can brighten noise, but it cannot remove it.

Family 1: curves and tone mapping

Plain version. The cheapest transform is a tone curve: a rule that says "input value 0.1 becomes 0.3", stored as a small table. A network only has to choose the curve, once per image or once per pixel.

Precise version. A **global per-channel curve** (a 1D LUT) maps each channel independently, $O_c(\mathbf{x}) = T_c(I_c(\mathbf{x}))$ for $c \in \{R, G, B\}$, where each $T_c : [0,1] \to [0,1]$ is stored as a table of, say, 256 values. The gamma curve and histogram equalization of IP 03 are curves of this kind. Applying it costs one table lookup (or one interpolation) per channel per pixel, independent of how complicated the curve is.

Global curves have two limits. They cannot treat a dark corner differently from a bright window, because they ignore position. And because each channel is mapped on its own, they cannot express transforms that mix channels, such as "make skin tones warmer but leave the sky alone" or a change of saturation that depends on hue.

Per-pixel curves. Zero-DCE [2] predicts the parameters of an iterated quadratic curve for every pixel, which makes the curve local; its curve and its zero-reference losses are explained in DLCV 05. From the efficiency angle, the important design is its successor Zero-DCE++ [1]: it uses depth-wise separable convolutions, predicts only three curve-parameter maps that are reused across iterations, and estimates them on an input **downsampled by a factor of 12** (the default) before resizing the maps back to full resolution. The authors report a network of about 10K parameters running at 1000 frames per second on a GPU and 11 on a CPU for a 1200 × 900 image [1]. The step "estimate on a thumbnail, resize the *parameter maps*, apply at full resolution" is the core idea of this chapter in its simplest form: the curve parameters are smooth in space, so resizing them loses little, while resizing the *image* would lose its detail.

The lesson of curves. A curve is a transform that a small network can choose and a cheap rule can apply. The rest of the chapter makes that transform richer: local and edge-aware (bilateral grids) or able to mix colors (3D LUTs).

Family 2: bilateral grids and guided upsampling

Plain version. Imagine cutting the image into large tiles, and inside each tile, also sorting the pixels into a few brightness levels. For each (tile, brightness level) box, store one small color recipe. A pixel finds its box by *where it is* and *how bright it is*, so two neighboring pixels on opposite

sides of an edge (one dark, one bright) use different recipes, and the edge stays sharp even though the tiles are big.

Precise version. A **bilateral grid** [5] is a 3D array indexed by two coarse spatial coordinates and one coarse intensity coordinate. Chen, Paris and Durand introduced it to accelerate edge-aware filters such as the bilateral filter [6] by working in this downsampled space, building on the signal-processing view of the bilateral filter by Paris and Durand [7] that is mentioned in the IP 03 notes. For enhancement, each grid node stores a color transform. HDRNet [4] stores a 3×4 affine matrix per node:

$$A[i, j, k] \in \mathbb{R}^{3\times 4}, \qquad i = 0, \dots, G_y - 1,\ j = 0, \dots, G_x - 1,\ k = 0, \dots, G_z - 1,$$

where $G_y \times G_x$ is the number of spatial nodes and G_z the number of intensity nodes. To process a full-resolution pixel at row y and column x, three steps follow.

1. Guide. Compute a scalar guide value $g(\mathbf{x}) \in [0, 1]$ at full resolution, for example the luma, or (as in HDRNet) a learned per-pixel function of the RGB value.

2. Slice. Map the pixel to continuous grid coordinates $(u_y, u_x, u_z) = (y\, s_y,\ x\, s_x,\ g(\mathbf{x})\,(G_z - 1))$, where s_y and s_x convert pixel coordinates to grid coordinates, and interpolate the grid there:

$$\bar{A}(\mathbf{x}) = \sum_{i,j,k} \tau(u_y - i)\, \tau(u_x - j)\, \tau(u_z - k)\, A[i, j, k], \qquad \tau(t) = \max(1 - |t|, 0),$$

where τ is the "tent" (linear interpolation) weight. Only the 8 nodes around the pixel have non-zero weight, so this is a **trilinear interpolation** in (row, column, guide). HDRNet calls this operation **slicing**; it has no learned parameters and is differentiable with respect to both the grid and the guide [4].

3. Apply. Multiply the pixel's color by its interpolated matrix:

$$O(\mathbf{x}) = \bar{A}(\mathbf{x}) \begin{bmatrix} I(\mathbf{x}) \\ 1 \end{bmatrix},$$

where $I(\mathbf{x})$ is the RGB column vector and the appended 1 lets the fourth column of $\bar{A}$ act as an offset. Each output channel is thus an affine combination of the three input channels.

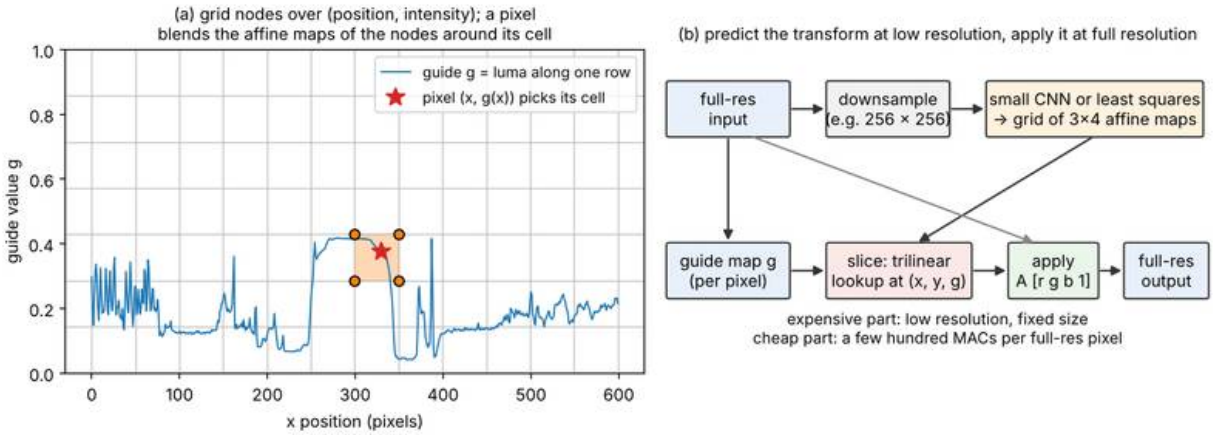


Figure 2 — A bilateral grid of affine color transforms. Left: in one image row, each pixel is located by its position and its guide value, so pixels on the two sides of an edge fall into different intensity cells. Right: the expensive part (the network or the fit) runs on a thumbnail; slicing and applying cost a fixed, small number of operations per full-resolution pixel.

Why the intensity axis matters. If the grid had only spatial axes, neighboring pixels would always get nearly the same transform, and a strong edge inside a cell would be blurred into a halo (a bright fringe on the dark side or a dark fringe on the bright side). The intensity axis separates them: a dark pixel and a bright pixel at the same position read different nodes. This is the same "only average with pixels that look alike" idea as the bilateral filter.

HDRNet. Gharbi et al. [4] train a CNN that takes a fixed 256 × 256 copy of the image, splits into a local path (convolutions at 16 × 16) and a global path (fully connected layers that summarize the whole image), fuses them, and outputs a 16 × 16 × 8 grid with 12 coefficients per node. The guide is a learned pointwise function: a 3 × 3 color matrix followed by a learned piecewise-linear curve per channel, summed. The loss is computed on the full-resolution output, so the network learns a grid that looks right *after* slicing. The authors report that it processes high-resolution images on a smartphone in milliseconds and provides a real-time viewfinder at 1080p [4]. They trained it to imitate many operators, including human retouching from MIT-Adobe FiveK [8], and note that the outputs are slightly softer than the reference and that operators which create new edges or need non-local information (sharpening, dehazing, matting) fit the local affine model poorly.

Optional: Paper guide: HDRNet (in the Paper guides). The architecture, the training data, and what to check in its tables.

Bilateral guided upsampling. The same machinery can accelerate *any* slow operator, learned or not. Chen et al. [9] run the operator on a downsampled image, fit a bilateral grid of affine transforms that maps the downsampled input to the downsampled output, and slice that grid with the full-resolution input. The code below does exactly this, in a simplified form.

The deep guided filter. Wu et al. [10] use the guided filter [11] (see the IP 03 note) as a differentiable layer for the same purpose. A network produces a low-resolution output O_l; the layer

fits the guided filter's local linear model $O_l \approx A_l \circ I_l + b_l$ in small windows of the low-resolution guide I_l, upsamples the coefficient maps A_l, b_l bilinearly to A_h, b_h, and returns $O_h = A_h \circ I_h + b_h$, where I_h is the full-resolution guide and $\circ$ is element-wise multiplication. The guide itself can be learned by two 1×1 convolutions. Compared with a bilateral grid, it has no intensity axis, so it relies on the window model rather than on intensity bins to keep edges.

Code: fit a grid at low resolution, slice at full resolution

The experiment: a dark coffee image (15% of the light) is the full-resolution input. A "retouching operator" brightens it locally, Retinex-style (darker surroundings get a stronger gamma lift, see DLCV 05), and boosts saturation. Pretend this operator is too slow to run at full resolution. We run it only on a 4× smaller copy, then compare three ways of producing the full-resolution result against what the operator would have produced at full resolution: (a) upsample the small output image; (b) fit one global tone curve per channel on the small pair; (c) fit a $9 \times 13 \times 8$ bilateral grid of affine transforms on the small pair and slice it with the full-resolution input.

The fit is weighted least squares per node: each low-resolution pixel p contributes to its 8 surrounding nodes with its trilinear weights w_p, and each node solves $\min_A \sum_p w_p \|A[I_p; 1] - O_p\|^2 + \lambda \|A - A_{\text{id}}\|^2$, where A_{id} is the identity transform and the small ridge λ keeps empty nodes at the identity.

```
import numpy as np
from scipy.ndimage import gaussian_filter
from skimage import data, img_as_float
from skimage.metrics import peak_signal_noise_ratio as psnr
from skimage.transform import resize

luma = lambda x: x @ np.array([0.299, 0.587, 0.114])

def local_brighten(x):
    """The 'expensive' operator to imitate: a Retinex-style local gamma plus a saturation
     ↪  boost."""
    Y = luma(x)
    base = gaussian_filter(Y, x.shape[1] / 40)             # smooth illumination estimate
    Y2 = Y ** (0.35 + 0.9 * base)                           # darker surroundings ->
     ↪  stronger lift
    out = x * (Y2 / np.maximum(Y, 1e-3))[..., None]         # same brightness change on R,
     ↪  G, B
    return np.clip(Y2[..., None] + 1.3 * (out - Y2[..., None]), 0, 1)   # 30% more saturation

full = (0.15 * img_as_float(data.coffee()) ** 2.2) ** (1 / 2.2)   # 400 x 600, 15% of the
↪  light
low = resize(full, (100, 150), anti_aliasing=True)          # 4x smaller in each direction
target_low = local_brighten(low)                             # the operator runs only at low
↪  resolution
reference = local_brighten(full)                             # ground truth, for scoring only

def corners(x, shape):
    """For every pixel: the 8 surrounding grid nodes (rows, cols, luma) and their trilinear
     ↪  weights."""
```

```
    H, W = x.shape[:2]
    gy, gx, gz = shape
    rows, cols = np.meshgrid(np.linspace(0, gy - 1, H), np.linspace(0, gx - 1, W),
     ↪  indexing="ij")
    u = np.stack([rows, cols, luma(x) * (gz - 1)], -1)      # continuous grid coordinates
    i0 = np.minimum(u.astype(int), np.array(shape) - 2)
    d = u - i0
    for c in np.ndindex(2, 2, 2):
        w = np.prod(np.where(c, d, 1 - d), axis=-1)
        yield tuple((i0 + c).reshape(-1, 3).T), w.ravel()

def fit_grid(x, y, shape=(9, 13, 8), lam=1e-4):
    """One 3x4 affine color transform per node, by weighted least squares with a small
     ↪  ridge."""
    X = np.concatenate([x.reshape(-1, 3), np.ones((x[..., 0].size, 1))], 1)   # N x 4: (R,
     ↪  G, B, 1)
    Y = y.reshape(-1, 3)
    XtX, XtY = np.zeros(shape + (4, 4)), np.zeros(shape + (4, 3))
    for idx, w in corners(x, shape):
        np.add.at(XtX, idx, w[:, None, None] * X[:, :, None] * X[:, None, :])
        np.add.at(XtY, idx, w[:, None, None] * X[:, :, None] * Y[:, None, :])
    eye = np.eye(4)[:, :3]                                     # empty nodes fall back to the
     ↪  identity
    return np.linalg.solve(XtX + lam * np.eye(4), XtY + lam * eye)   # (gy, gx, gz, 4, 3)

def slice_apply(A, x):
    """Slice: interpolate the grid at every full-resolution pixel (guide = luma), then
     ↪  apply."""
    X = np.concatenate([x.reshape(-1, 3), np.ones((x[..., 0].size, 1))], 1)
    Ap = sum(w[:, None, None] * A[idx] for idx, w in corners(x, A.shape[:3]))
    return np.clip(np.einsum("nk,nkc->nc", X, Ap).reshape(x.shape), 0, 1)

def fit_curves(x, y, bins=64):
    """Baseline: one global tone curve per channel (mean output for each input level)."""
    curves = []
    for c in range(3):
        k = np.minimum((x[..., c] * bins).astype(int), bins - 1).ravel()
        n = np.bincount(k, minlength=bins)
        mean = np.bincount(k, y[..., c].ravel(), bins) / np.maximum(n, 1)
        centers = (np.arange(bins) + 0.5) / bins
        curves.append((centers[n > 0], mean[n > 0]))
    return lambda z: np.stack([np.interp(z[..., c], *curves[c]) for c in range(3)], -1)

A = fit_grid(low, target_low)
results = {
    "upsample the low-res output": resize(target_low, full.shape, order=1),
    "global tone curve per channel": fit_curves(low, target_low)(full),
    "bilateral grid 9x13x8, sliced": slice_apply(A, full),
}
print(f"grid coefficients: {A.size}   mean brightness {full.mean():.2f} ->
 ↪  {reference.mean():.2f}")
for name, out in results.items():
    print(f"{name:30s} PSNR vs full-resolution operator {psnr(reference, out,
     ↪  data_range=1):5.2f} dB")
```

```
grid coefficients: 11232   mean brightness 0.16 -> 0.37
upsample the low-res output    PSNR vs full-resolution operator 28.48 dB
global tone curve per channel  PSNR vs full-resolution operator 23.02 dB
bilateral grid 9x13x8, sliced  PSNR vs full-resolution operator 36.01 dB
```

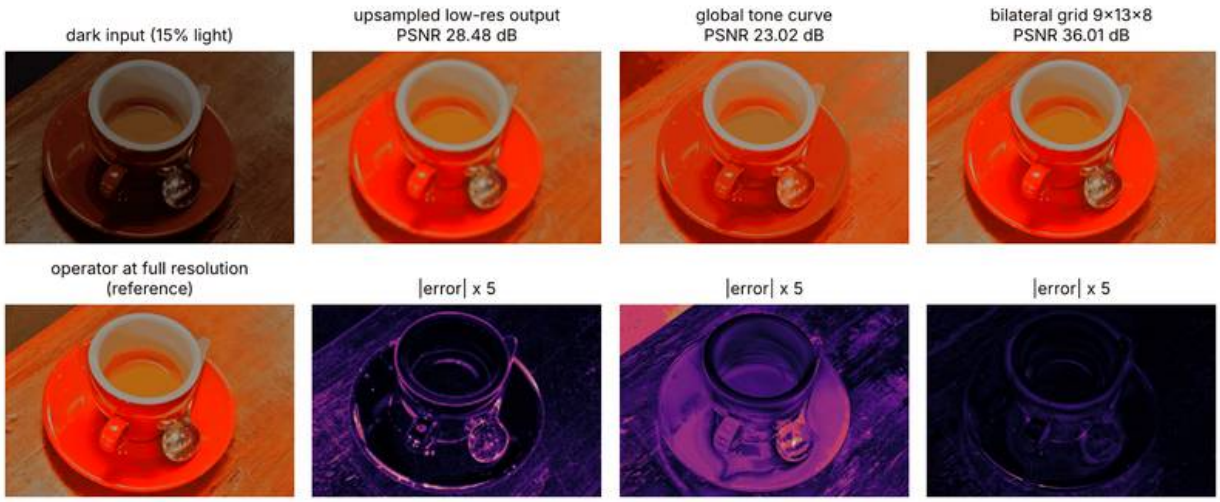


Figure 3 — Upsample the transform, not the image (our toy). Upsampling the small output loses detail at every edge; a global curve keeps detail but cannot vary across the image; a bilateral grid fitted on the small pair keeps both. Its residual errors sit along strong edges, where one cell must serve both sides.

Drag the slider to compare the dark input with the grid's output:



Dark input



Grid, fitted at 1/4 size

The dark coffee-cup image and the bilateral-grid result, which is much brighter, with vivid red cup and saucer and sharp wood grain.

What the numbers say (one image, one operator, our own run):

- **The image is the wrong thing to upsample.** The upsampled output has the right colors but has lost the fine detail, so it loses about 7.5 dB to the grid. The grid never resizes the image; the detail comes straight from the full-resolution input through $\bar{A}$.
- **A global transform is not enough.** The operator brightens dark surroundings more, so no single curve fits; the global curve is the worst of the three.
- **About 11 thousand numbers describe the whole edit.** The grid holds $9 \times 13 \times 8 \times 12 = 11{,}232$ coefficients for a 240,000-pixel image, and applying it costs a fixed amount of work per pixel.

Optional: Fitting and slicing a bilateral grid in detail (in the Supplementary notes). The normal equations per node, the role of the ridge, hard versus soft binning, and how grid size trades accuracy against halos.

Family 3: learnable 3D lookup tables

Plain version. A 3D lookup table (3D LUT) is a color cube: for a grid of input colors it stores the output color. Any pixel's color falls inside one small box of the cube, and its output is blended from the box's 8 corners. Fixed, hand-tuned 3D LUTs are widely used for color and tone adjustment in camera imaging pipelines and photo editing tools [12]; the learned version lets a small network choose the table for each photo.

Precise version. A 3D LUT with M nodes per axis is an array $\Phi \in \mathbb{R}^{M\times M\times M\times 3}$; node (i,j,k) holds the output color for the input color $(\frac{i}{M-1}, \frac{j}{M-1}, \frac{k}{M-1})$. For an input color $(r,g,b) \in [0,1]^3$, compute the continuous lattice coordinates and split them into an integer corner and a fractional offset:

$$u_r = r\,(M-1), \quad i = \min\left(\lfloor u_r \rfloor,\, M-2\right), \quad d_r = u_r - i,$$

and likewise j, d_g for green and k, d_b for blue, where $\lfloor\cdot\rfloor$ rounds down, the cap $M-2$ keeps the upper corner $i+1$ inside the table when $r=1$, and $d_r, d_g, d_b \in [0,1]$ are the offsets inside the cell.

Trilinear interpolation blends the 8 corners of the cell:

$$\Phi(r,g,b) = \sum_{a,b',c\in\{0,1\}} w_a(d_r)\, w_{b'}(d_g)\, w_c(d_b)\; \Phi[i+a,\, j+b',\, k+c], \qquad w_0(d) = 1-d,\; w_1(d) = d,$$

where a, b', c select the lower (0) or upper (1) corner along each axis. The 8 weights are non-negative and sum to 1, so the output is a weighted average of the 8 stored colors. It is the same formula as slicing a bilateral grid, with the axes being (R, G, B) instead of (row, column, intensity).

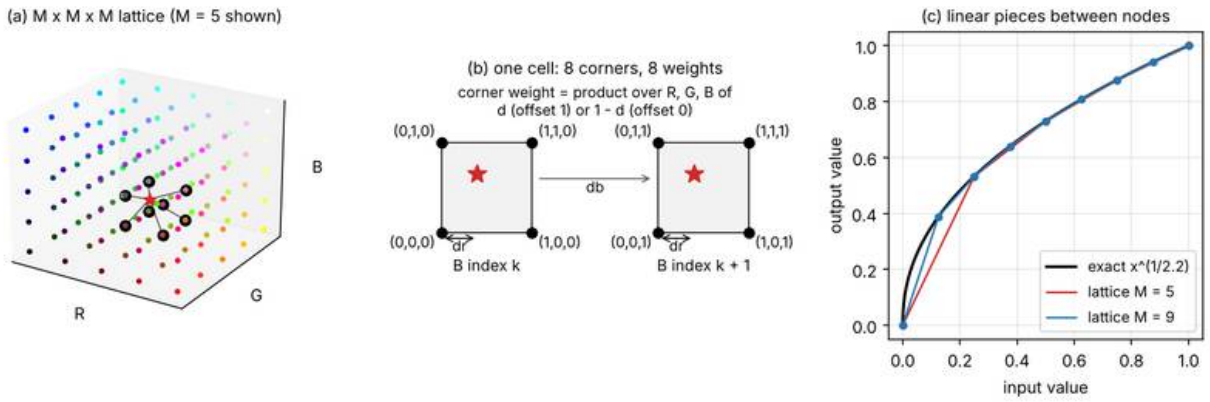


Figure 4 — A 3D LUT and trilinear interpolation. (a) The lattice; an input color is located in one cell. (b) The 8 corner weights are products of the offsets. (c) Along one axis, interpolation is piecewise linear, so a curve that bends sharply between nodes (here, a gamma curve near black) is approximated poorly unless the lattice is fine there.

Why LUTs are fast. Applying a LUT costs 8 table reads and about 24 multiply–adds per pixel, *regardless of how complicated the color transform is*. A transform built from dozens of color operations, or one that no formula describes, costs the same as the identity once it is baked into the table. The work is a few memory reads and multiply–adds per pixel with no data-dependent branching, which suits GPUs and fixed-function hardware well. The price is memory and smoothness: the table has $3M^3$ values, and it can only represent transforms that are close to linear inside each cell.

Image-adaptive 3D LUTs. Zeng et al. [12] made LUTs learnable and content-dependent. They learn N basis LUTs $\Phi_1, \dots, \Phi_N$ and a small CNN that looks at a 256×256 downsampled copy of the image and predicts weights $w_1, \dots, w_N$. The image's own LUT is the weighted sum

$$\Phi = \sum_{n=1}^{N} w_n \, \Phi_n,$$

which is applied to the full-resolution image with one trilinear interpolation per pixel. Because the interpolation is linear in the table entries, fusing the tables first and interpolating once gives the same result as interpolating each table and fusing the outputs. In the paper, $M = 33$ and $N = 3$; each LUT has $3 \cdot 33^3 = 107{,}811$ values, the whole model has fewer than 600K parameters, and it processes a 4K image in under 2 ms on one Titan RTX GPU [12]. Training adds two regularizers on the tables: a smoothness term that penalizes differences between neighboring nodes, and a monotonicity term that penalizes outputs that decrease when an input channel increases, so that brightness order is preserved. The authors also state the limits clearly: one LUT per image applies the same color transform everywhere, so local adjustments are impossible, and since each color is transformed independently, noise in a night image is amplified rather than removed [12].

The snippet below implements the trilinear LUT, checks it against direct formulas, and times it against a long color pipeline.

```
import time
import numpy as np
from skimage import data, img_as_float

img = img_as_float(data.coffee())                    # 400 x 600 x 3, RGB in [0, 1]

def build_lut(f, M=33):
    """Sample a color transform f on an M x M x M lattice: table of shape (M, M, M, 3)."""
    g = np.linspace(0, 1, M)
    r, gr, b = np.meshgrid(g, g, g, indexing="ij")
    return f(np.stack([r, gr, b], axis=-1))

def apply_lut(lut, x):
    """Trilinear interpolation of the table at every RGB value of x."""
    M = lut.shape[0]
    u = x * (M - 1)                                  # continuous lattice coordinates
    i0 = np.minimum(u.astype(int), M - 2)            # lower corner of the enclosing cell
    d = u - i0                                       # fractional offsets in [0, 1]
    out = np.zeros_like(x)
    for cr in (0, 1):
        for cg in (0, 1):
            for cb in (0, 1):
                w = ((d[..., 0] if cr else 1 - d[..., 0]) * (d[..., 1] if cg else 1 - d[...,
                 ↪  1])
                     * (d[..., 2] if cb else 1 - d[..., 2]))
                out += w[..., None] * lut[i0[..., 0] + cr, i0[..., 1] + cg, i0[..., 2] + cb]
    return out

identity = lambda x: x
gamma = lambda x: x ** (1 / 2.2)
print(f"identity, M=33: max error {np.abs(apply_lut(build_lut(identity), img) -
 ↪  img).max():.1e}")
for M in (9, 17, 33, 65):
    err = np.abs(apply_lut(build_lut(gamma, M), img) - gamma(img))
    worst = img.ravel()[err.argmax()]
    print(f"gamma 1/2.2, M={M:2d}: mean error {err.mean():.4f}  max error {err.max():.4f}
     ↪  (at input {worst:.3f})")

def grade(x, n_ops):
    """A long color pipeline: n_ops small rounds of channel mixing and a soft S-curve."""
    mix = np.array([[0.80, 0.15, 0.05], [0.10, 0.80, 0.10], [0.05, 0.15, 0.80]])
    step = np.eye(3) + (mix - np.eye(3)) / n_ops     # n_ops small steps, same total effect
    for _ in range(n_ops):
        x = np.clip(x @ step.T, 0, 1)
        x = x + (0.3 / n_ops) * np.sin(2 * np.pi * x) / (2 * np.pi)
    return x

def ms(fn, reps=5):
    fn(); t = time.perf_counter()
    for _ in range(reps): fn()
    return 1000 * (time.perf_counter() - t) / reps

for n in (1, 8, 64):
```

```
    lut = build_lut(lambda x: grade(x, n))
    err = np.abs(apply_lut(lut, img) - grade(img, n)).max()
    print(f"n_ops={n:2d}: direct {ms(lambda: grade(img, n)):6.1f} ms | apply LUT "
          f"{ms(lambda: apply_lut(lut, img)):5.1f} ms | max difference {err:.4f}")
```

```
identity, M=33: max error 3.3e-16
gamma 1/2.2, M= 9: mean error 0.0210  max error 0.1098 (at input 0.031)
gamma 1/2.2, M=17: mean error 0.0090  max error 0.0801 (at input 0.016)
gamma 1/2.2, M=33: mean error 0.0041  max error 0.0585 (at input 0.008)
gamma 1/2.2, M=65: mean error 0.0014  max error 0.0427 (at input 0.004)
n_ops= 1: direct   16.5 ms | apply LUT  78.3 ms | max difference 0.0002
n_ops= 8: direct   84.6 ms | apply LUT  83.1 ms | max difference 0.0002
n_ops=64: direct  645.1 ms | apply LUT  87.8 ms | max difference 0.0002
```

(The timings are from one run on a 4-core CPU and will differ on yours; the errors will not.) Three observations:

- **The identity LUT is exact** up to floating-point rounding, because trilinear interpolation reproduces any function that is linear in each channel.
- **Steep curves need fine lattices where they are steep.** The gamma curve has an infinite slope at black, so the worst error is always in the darkest input bin, and doubling M shrinks the *mean* error by a factor of 2 to 3 but the *worst* error much more slowly. This is exactly the problem that non-uniform lattices address (AdaInt, below), and it matters for low light, where most pixels are near black.
- **The cost is flat.** The direct pipeline's time grows with its length; the LUT's time stays about the same for 1, 8 or 64 operations, and its output matches to 0.0002. In NumPy the LUT is slower than a one-step pipeline, because eight gathers are expensive in Python; on hardware with texture interpolation the lookup is nearly free. Building the table costs $33^3 = 35{,}937$ evaluations of the pipeline, about 15% of this image's 240,000 pixels and a vanishing fraction of a 4K frame's.

Successors. Later work kept the "small CNN chooses a LUT" idea and attacked its limits; each is described in one line here and in more detail in the notes:

- **Spatial-aware 3D LUTs** (Wang et al. [13]) predict pixel-wise weight maps as well as global weights, so different regions blend the basis LUTs differently; the authors report about 4 ms for a 4K image on one V100 GPU.
- **AdaInt** (Yang et al. [14]) learns **non-uniform** node positions along each axis, so the lattice is dense where the transform bends and sparse where it is nearly linear.
- **SepLUT** (Yang et al. [15]) splits the transform into per-channel 1D LUTs followed by a smaller 3D LUT: the 1D curves spread the input colors over the cube so that more of the 3D LUT's cells are used.

- **4D LUT** (Liu et al. [16]) adds a fourth input, a learned **context** value per pixel, so two pixels with the same RGB but different content (sky versus sea) can be mapped differently; interpolation becomes quadrilinear.
- **LUTs with bilateral grids** (Kim and Cho [17]) bring spatial information into the LUT model through bilateral grids and replace a long decoder with slicing operations.

Optional: Learnable LUT variants in detail (in the Supplementary notes). How AdaInt locates a color in a non-uniform lattice, what SepLUT's 1D stage does to cell usage, and the memory of 3D and 4D tables.

Optional: Paper guide: image-adaptive 3D LUTs (in the Paper guides). The problem, the idea, what the tables show and how to read the timing claims.

Family 4: very small enhancement networks

Plain version. Some methods stay with an ordinary network that outputs the image (or an illumination map), but make that network extremely small, or arrange the training so that only a tiny part of it is needed at test time.

Precise version. Three representative designs, each verified against its paper:

- **SCI** (self-calibrated illumination, Ma et al. [18]) learns illumination with a cascade of weight-sharing stages during training. A self-calibration module makes the stages' results converge, so that at inference a **single basic block** is used, which cuts the computation drastically. It is trained with unsupervised losses. The lesson for efficiency: a training-time structure can be much larger than the inference-time network.
- **IAT** (Illumination Adaptive Transformer, Cui et al. [19]) splits a simplified camera pipeline into local and global components. A local branch, built from depthwise convolutions and kept at the input resolution, predicts a per-pixel multiply map and add map. A global branch uses a few attention queries (nine for a 3×3 color matrix, one for a gamma value) to predict the parameters of global operations, which are then applied to the locally adjusted image. The authors report about 90K parameters and an average inference time of about 0.004 s per image [19]. Its global part is a parametric transform in the sense of this chapter, chosen by attention instead of a lookup.
- **Prior-guided lightweight networks.** CPGA-Net (Weng et al. [20]) combines channel priors with gamma correction in a lightweight network for low-light enhancement, so that classical image-formation knowledge replaces some of the capacity a larger network would need.

The **NTIRE 2026 Efficient Low-Light Image Enhancement challenge** [21] shows what such designs look like under a fixed budget. Its data are phone photographs at 3024×4032, the submitted model size had to be below 1 MB, and the ranking aggregated several full-reference and

no-reference quality metrics, with fewer parameters breaking ties. The report groups the submissions into lightweight U-Net or encoder–decoder designs with depthwise separable convolutions, Retinex-style decompositions, small attention modules, color-space decoupling (for example HVI), classical priors such as histogram equalization, and structural re-parameterization or partial convolutions [21].

Efficiency techniques

Plain version. A handful of tricks recur in every fast enhancer. The most important one is the theme of this chapter: think at low resolution, and only *apply* at full resolution. The others make the low-resolution thinking itself cheaper.

Run at low resolution, upsample the transform

The total cost of a predict-then-apply design is

$$\text{cost} = C_{\text{net}}(h \times w) + c_{\text{apply}} \cdot HW,$$

where $C_{\text{net}}(h \times w)$ is the cost of the network on an $h \times w$ thumbnail, which does not depend on the full resolution $H \times W$, and c_{apply} is the per-pixel cost of slicing or lookup (165 MACs in our HDRNet-like count; for a 3D LUT, about 24 multiply–adds for the 8-corner blend plus the weights). At 4K the second term dominates, and it is small. The design choices are what to upsample (curve parameters, affine coefficients, LUT weights; never the output image) and how (bilinear for smooth maps; slicing with a guide where edges must be respected).

Separable and depthwise layers

A depthwise separable convolution replaces one $k \times k$ convolution from C_{in} to C_{out} channels by a $k \times k$ convolution per channel followed by a 1×1 convolution that mixes channels, which reduces the cost by the factor $1/C_{\text{out}} + 1/k^2$ (MobileNet [22]; derivation in DLCV 02 and ML 03). Zero-DCE++ [1] and many challenge entries [21] use it. Two cautions: depthwise layers have low arithmetic intensity, so on some hardware they run slower than their MAC count suggests, and in a very small network the 1×1 layers dominate, so the saving is smaller than the formula implies for large C. MobileNetV2's inverted residual blocks [23] are a common building block in this setting.

Structural re-parameterization

Plain version. Train a block with several parallel branches (which trains well), then merge the branches into one plain convolution for deployment (which runs fast). The merged network computes exactly the same function.

RepVGG [24] trains blocks with a 3×3 convolution, a 1×1 convolution and an identity branch, each followed by batch normalization, and sums them. At inference, every branch is linear, so each batch normalization can be folded into its kernel and bias, the 1×1 kernel and the identity can be written as 3×3 kernels with zeros around the center, and the three kernels can be added into one. Efficient enhancement networks, including entries in the NTIRE 2026 efficient challenge [21], use this to get multi-branch training with single-branch inference cost.

Optional: Structural re-parameterization step by step (in the Supplementary notes). The folding formulas and a NumPy check that the fused convolution matches the three-branch block to about 10^{-15}.

Quantization

Storing and computing with 8-bit integers instead of 32-bit floats cuts memory by 4 and usually speeds up inference on mobile hardware; integer-only inference was made practical by Jacob et al. [25], and the affine quantization scheme is explained in ML 10. Two points are specific to enhancement. First, the *output* is an image that people look at, so quantization error is directly visible, especially as banding in smooth dark gradients after strong brightening (see Failure modes). Second, the per-pixel step (slicing, LUT interpolation, curve application) is cheap but runs on every pixel, so its numeric format matters too: an 8-bit LUT output is fine for display, but intermediate coordinates need enough fractional bits for smooth interpolation. Re-measure quality after quantizing, at full resolution.

Other recurring tricks

Space-to-depth (pixel unshuffle) at the input folds each $s \times s$ block into channels so that every layer runs at $1/s^2$ of the pixels (DLCV 04A). Weight sharing across iterations (Zero-DCE++ reuses one set of curve maps; SCI shares weights across stages) keeps the parameter count low. And global information is gathered cheaply, by a few fully connected layers on a tiny map (HDRNet's global path) or a few attention queries (IAT), instead of by deep stacks of convolutions.

Measuring efficiency honestly

Plain version. Efficiency claims are easy to make and hard to compare. A fair comparison says exactly what was counted, at which resolution, on which hardware, and includes every step from input frame to output frame.

Counting. Dehghani et al. show that parameters, FLOPs, throughput and latency can disagree and that reporting only one of them gives a misleading picture [26]. For enhancement:

- **Count at the deployment resolution**, and also report MACs per megapixel so that readers can rescale. A FLOP count at 256×256 says nothing about 4K for a full-resolution network, and the opposite is true for thumbnail-based designs.

- **Include the full-resolution step.** Slicing, LUT interpolation, guided-filter upsampling and resizing are often custom operators that automatic FLOP counters skip. They are the part of a predict-then-apply design that grows with resolution, so leaving them out understates the cost exactly where it matters.
- **State MACs or FLOPs**, and whether a "parameter" includes LUT entries (a 33-node 3D LUT is about 108K values, more than the whole of Zero-DCE++ at about 10K parameters [1] or IAT at about 90K [19]).
- **Report memory**: peak activation memory at the deployment resolution, not only the parameter count.

Timing. Follow the protocol in ML 10: fix and report device, runtime, precision, batch size 1, resolution and thread count; warm up; report percentiles; synchronize the GPU before stopping the clock; and time the whole path, including color conversion, resizing and memory transfers. Challenge series that measure runtime on real phones make this concrete: the Mobile AI 2022 learned-ISP challenge, for example, evaluated runtime on a Snapdragon 8 Gen 1 GPU with TensorFlow Lite, and its report states that the solutions process Full HD photos in less than 20–50 ms on that platform [27]. A GPU number from a paper (such as the Titan RTX or V100 timings above) tells you about relative speed on that GPU, not about your phone.

Quality. Fast models must also be judged on quality, with the metrics and pitfalls of DLCV 04 and the low-light specifics of DLCV 05. Photo-enhancement papers in this chapter mostly evaluate on MIT-Adobe FiveK [8] (raw photos retouched by five experts) with PSNR, SSIM and a color difference; low-light models use the datasets in DLCV 05. Two additions matter for fast models:

evaluate at **full resolution**, because halos and banding from coarse grids or tables may vanish

when the output is downsampled for scoring; and evaluate **after quantization and export**, on the actual runtime, because that is the model users get.

What to report, as a checklist: parameters (including tables); MACs at the deployment resolution and per megapixel, including the per-pixel step; peak memory; latency percentiles with the full hardware and software description; quality at full resolution, after quantization; and a few full-resolution crops of the hardest cases.

Failure modes

Plain version. Fast enhancers fail in characteristic ways that follow from their shortcuts: tables are coarse, grids are coarse, transforms ignore noise, and each frame is processed alone.

- **Banding from quantization and coarse tables.** Strong brightening stretches the few gray levels of a dark 8-bit image apart, so a smooth gradient becomes visible steps. A coarse or 8-bit-stored

LUT adds its own steps and piecewise-linear kinks. The snippet below brightens a smooth dark ramp after quantizing it at different bit depths.

```
import numpy as np

ramp = np.linspace(0, 0.1, 4000)                # a smooth dark gradient (e.g. a night sky)
curve = lambda x: x ** (1 / 2.2)                # a strong brightening curve
for bits in (8, 10, 12):
    q = np.round(ramp * (2**bits - 1)) / (2**bits - 1)          # sensor / file bit depth
    out = np.round(curve(q) * 255)                              # 8-bit display output
    levels = np.unique(out)
    print(f"{bits:2d}-bit input: {len(np.unique(q)):4d} input levels -> {len(levels):3d}
    ↪  output levels,"
          f" largest jump {np.diff(levels).max():.0f} display levels")
```

```
 8-bit input:   27 input levels ->  27 output levels, largest jump 21 display levels
10-bit input:  103 input levels ->  67 output levels, largest jump 11 display levels
12-bit input:  411 input levels ->  84 output levels, largest jump 6 display levels
```

The brightened ramp spans about 90 display levels (from 0 to $0.1^{1/2.2} \approx 0.35$ of 255), but an 8-bit input provides only 27 of them, so the gradient is shown as 27 bands; the largest jump is the first step above black, where the curve is steepest. No enhancer can invent the missing levels without adding noise or dithering, which is one reason raw or higher-bit-depth inputs matter for low light (DLCV 05).

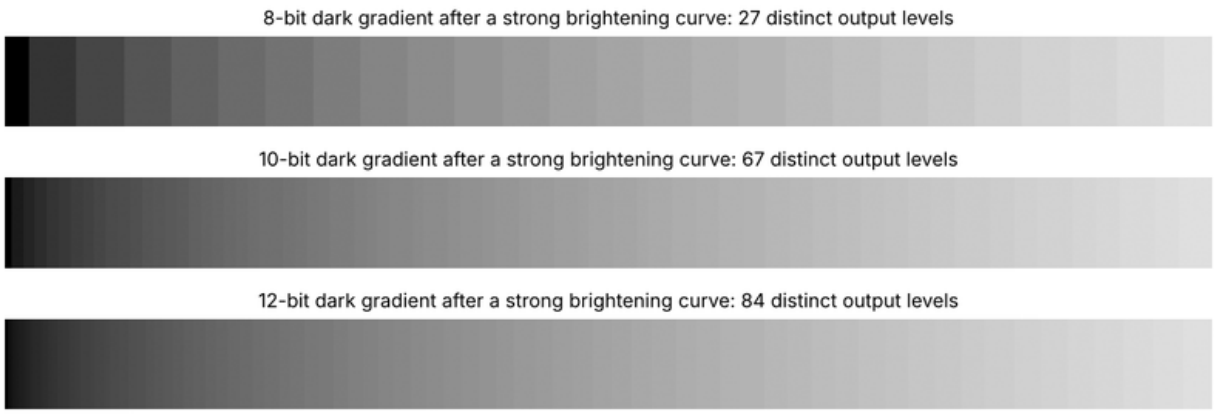


Figure 5 — Banding after strong brightening (our synthetic ramp). The same curve applied to an 8-, 10- and 12-bit version of one dark gradient; the number of distinct output levels limits how smooth the result can look.

- **Halos from coarse grids.** A grid cell spans many pixels; where an edge separates two regions that need different transforms but fall into the same intensity bin, both sides receive a blend, and a fringe appears along the edge. Figure 3's residual errors along the cup rim are this effect

in miniature. Finer intensity bins, a better (learned) guide, or a finer grid reduce it at a cost. HDRNet's authors also note slightly softer outputs from the heavy downsampling [4].

- **Color shifts.** Per-channel curves change hue whenever they treat R, G and B differently; a LUT trained on one dataset may map colors it never saw (saturated neon, skin under colored light) unpredictably, because nodes far from the training colors are constrained only by the smoothness regularizer. Gray-world-style losses push colorful scenes toward gray (see the toy in DLCV 05).
- **Amplified noise.** A transform that brightens shadows multiplies their noise by the local slope of the curve, and pointwise or local-affine transforms cannot average noise away. The 3D LUT authors report exactly this on night images [12]. A fast enhancer for real low light needs a denoiser somewhere, or a curve whose slope is limited where the signal-to-noise ratio is low.
- **Temporal flicker on video.** When each frame is processed independently, the predicted LUT weights, curve parameters or grid coefficients jump a little from frame to frame, and the whole image pulses in brightness or color. Smoothing the predicted parameters over time (for example, an exponential moving average of the LUT weights) is cheap because the parameters are few. Lai et al. [28] proposed a general post-processing network that makes per-frame processed video temporally consistent without knowing which operator was applied.

Modern view

Surveys and reports. There is no dedicated survey of lightweight enhancement that this chapter could verify, so four documents together give the overview:

- **Li et al., TPAMI 2022 [29]**, the deep low-light survey already used in DLCV 05, includes a computational-efficiency comparison of LLIE methods (runtime, trainable parameters and FLOPs, averaged over 32 images of 1200 × 900 on one GPU). Its conclusion is that curve estimation (Zero-DCE) had the shortest runtime among the compared methods because it only estimates a few curve parameters with a small network.
- **Dehghani et al., ICLR 2022 [26]** is about efficiency reporting in general: cost indicators contradict each other, and incomplete reporting leads to wrong conclusions about which model is more efficient.
- **The NTIRE 2026 efficient low-light challenge report [21]** is the most recent snapshot of what works under a hard size limit (below 1 MB) on high-resolution phone photographs.
- **The Mobile AI and AIM 2022 learned-ISP challenge report [27]** shows how runtime can be measured on a real phone GPU under a common protocol, for a closely related task (turning raw sensor data into a finished photo).

Optional: Paper guide: NTIRE 2026 efficient low-light challenge (in the Paper guides). How the challenge

was set up, how its ranking works, and what its tables can and cannot tell you.

State of the art (2023–2026). Three lines are active. First, **lookup tables keep gaining locality and context**: 4D LUTs with a learned context channel [16] and LUTs combined with bilateral grids [17] narrow the gap between global tables and local operators while keeping per-pixel cost tiny. Second, **efficient low-light networks are converging on a recipe**: small encoder–decoders with depthwise convolutions, a Retinex-style or color-space decomposition, a little attention, and re-parameterization for inference, as the NTIRE 2026 submissions show [21]. Third, **parametric ISP-style models** such as IAT [19] predict a few interpretable global parameters (color matrix, gamma) plus a light local correction, which makes them both small and easy to constrain.

Open problems.

- **Local, noise-aware and fast at once.** Transform-based models are fast because they do not touch fine structure, which is also why they cannot denoise. Combining them with a cheap denoiser without losing the speed advantage is unsolved in general.
- **Video.** Temporal consistency at a few milliseconds per 4K frame, without optical flow, remains hard; per-frame models flicker and video models are heavier.
- **Fair, comparable efficiency numbers.** Papers time on different GPUs, at different resolutions, with or without the full-resolution step; on-device measurements are rare outside challenges.
- **Quantization-aware enhancement.** Most methods are trained in floating point and evaluated there; the visible effects of 8-bit deployment (banding, color steps) are seldom reported.
- **Generalization.** Small models have little capacity to cover unusual lighting, cameras and scenes; image-adaptive tables and grids help, but behavior far from the training data is poorly understood.

Key takeaways

- "Lightweight" has several meanings: parameters, MACs per megapixel, memory traffic and measured latency. For full-resolution networks, arithmetic and memory grow with the pixel count, so a 4K frame costs four times a 1080p frame.
- Tone and color enhancement is mostly a **smooth remapping** of the input's values, so a model can **predict a transform at low resolution and apply it at full resolution**. Upsample the transform, never the image.
- The transform can be a curve (Zero-DCE++), a **bilateral grid** of affine maps sliced with a guide (HDRNet, bilateral guided upsampling), a guided-filter coefficient map (deep guided filter),

or a **3D LUT** applied by trilinear interpolation (image-adaptive 3D LUTs and successors).
- Trilinear interpolation blends 8 corners with weights that are products of the fractional offsets; it is exact for functions that are linear in each channel and weakest where the transform bends sharply, such as gamma near black.
- Applying a LUT costs the same for any transform, but a single LUT is global; spatial weights, 4D context, bilateral grids and non-uniform lattices add locality and accuracy.
- Very small networks (SCI, IAT, prior-guided networks) and techniques such as depthwise separable layers, structural re-parameterization and quantization shrink the low-resolution part further.
- Report parameters (including tables), MACs at the deployment resolution including the per-pixel step, peak memory, on-device latency percentiles and full-resolution quality after quantization; watch for banding, halos, color shifts, amplified noise and flicker.

Exercises

1. **MACs per megapixel.** A full-resolution network has five 3×3 convolutions with 16 channels in and out, plus a first layer $3 \to 16$ and a last layer $16 \to 3$. Compute its MACs per pixel and its total GMACs at 1080p and at a 12-megapixel photo. Then redesign it so that the five middle layers run after a 2×2 pixel unshuffle (64 channels at a quarter of the pixels). What happens to MACs per pixel?

   Hint

   Per pixel:

   $9 \cdot 3 \cdot 16 + 5 \cdot 9 \cdot 16 \cdot 16 + 9 \cdot 16 \cdot 3 = 432 + 11{,}520 + 432 = 12{,}384$ MACs. Multiply by 2.07×10^6 and 12×10^6 pixels: about 25.7 and 148.6 GMACs. After unshuffling, each middle layer has $9 \cdot 64 \cdot 64 = 36{,}864$ MACs per *low-resolution* pixel, which is 9,216 per input pixel, four times *more* than before (2,304). Unshuffling saves only if you also reduce the channel count; the quantity to keep fixed is (channels)^2 × (pixel fraction).

2. **Exactness of trilinear interpolation.** Show that trilinear interpolation reproduces exactly any function of the form $f(r, g, b) = \alpha + \beta r + \gamma g + \delta b + \ldots$ that is linear in each variable separately (it may contain rg, rb, gb and rgb terms), and that it does not reproduce r^2. Use this to explain why a color matrix, but not a gamma curve, is represented exactly by any $M \geq 2$.

   Hint

   Along one axis, interpolation between two nodes reproduces any linear function of that variable exactly. Trilinear interpolation is the composition of three 1D interpolations, one per axis, so it reproduces products of linear functions, i.e. every multilinear polynomial. For r^2 on $[0, 1]$ with nodes at 0 and 1, the interpolant at $r = 0.5$ is 0.5 while $r^2 = 0.25$. A color matrix with offset is linear, hence exact; a gamma curve is not multilinear, so its error depends on M (and, as in the snippet, is worst where the curve is steepest).

3. **Grid size versus halos.** In the bilateral-grid snippet, change the grid to $(5, 7, 8)$, $(17, 25, 8)$ and $(9, 13, 4)$, and add one more experiment: fit the $(9, 13, 8)$ grid on the *full-resolution* pair instead of the low-resolution one. Report PSNR for each and explain which error comes from the grid model and which from running the operator at low resolution.

Hint

In our run, $(5, 7, 8)$ gave 33.79 dB, $(9, 13, 4)$ 34.71 dB, $(9, 13, 8)$ 36.01 dB and $(17, 25, 8)$ 37.49 dB; fitting $(9, 13, 8)$ on the full-resolution pair gave 37.23 dB. Fewer spatial cells or intensity bins means each cell must serve more different content, which shows up as errors along edges (halos). The full-resolution fit is an upper bound for that grid size: the remaining gap to it comes from the operator's output at low resolution not being exactly the downsampled full-resolution output.

4. **LUT memory.** Compute the number of stored values and the memory in fp16 for: a 33-node 3D LUT; three of them as basis LUTs; a 17-node 3D LUT; and a 4D LUT with 33 nodes on each color axis and 8 on a context axis. Compare with the 10K parameters reported for Zero-DCE++. Why do LUT methods often use fewer, smaller tables in later work?

Hint

$3 \cdot 33^3 = 107{,}811$ values (about 211 KiB in fp16); three basis LUTs about 323K values; $3 \cdot 17^3 = 14{,}739$; and $3 \cdot 33^3 \cdot 8 = 862{,}488$ for the 4D LUT. Tables grow with the cube (or fourth power) of the node count, so they quickly dominate a model's size even though they cost almost nothing to apply. This is the motivation of SepLUT (a small 3D LUT after 1D LUTs) and AdaInt (put nodes only where needed).

5. **Temporal smoothing.** Simulate a video: for 200 frames, scale the coffee image by a light level k_t that rises slowly from 0.1 to 0.2. A "model" predicts one gamma per frame: the value γ_t that makes the mean of x^{γ_t} equal to 0.4, plus a random prediction error with standard deviation 0.02. Measure the mean absolute frame-to-frame change of the output's mean brightness with the raw γ_t and with an exponential moving average $\bar{\gamma}_t = \beta\bar{\gamma}_{t-1} + (1 - \beta)\gamma_t$, $\beta = 0.8$. Then derive the reduction factor and discuss the lag that smoothing introduces.

Hint

Find γ_t by 1D root finding (for example `scipy.optimize.brentq`). In our run the mean absolute change of the output mean was about 0.025 per frame without smoothing and about 0.0039 with it, a ratio of 0.15. For white prediction noise, the frame-to-frame difference of the moving average has standard deviation $(1-\beta)\sqrt{2/(1+\beta)}\,\sigma$ against $\sqrt{2}\,\sigma$ for the raw values, a ratio of $(1-\beta)/\sqrt{1+\beta} \approx 0.149$ for $\beta = 0.8$, which matches. The price is lag: the average responds to a real change of lighting with a mean delay of $\beta/(1-\beta) = 4$ frames. Real systems trade these against each other, or detect scene cuts and reset the average.

Supplementary notes

These notes are optional. Each takes one point from the main text and goes into finer detail; nothing in the main text, code or exercises depends on them.

Fitting and slicing a bilateral grid in detail

This note writes out the least-squares fit used in the bilateral-grid snippet, the choices it makes, and how they relate to bilateral guided upsampling [9].

The model. Each node $n = (i, j, k)$ of a $G_y \times G_x \times G_z$ grid holds a matrix $A_n \in \mathbb{R}^{4\times 3}$ (the transpose of the 3×4 form in the main text, which is convenient for least squares). A pixel p with input color $I_p \in \mathbb{R}^3$ has the augmented vector $\mathbf{x}_p = [I_p^\top, 1]^\top \in \mathbb{R}^4$, a target color $O_p \in \mathbb{R}^3$, and trilinear weights $w_{pn} \geq 0$ for its 8 surrounding nodes ($w_{pn} = 0$ for all other nodes, and $\sum_n w_{pn} = 1$).

The local fit. The snippet minimizes, separately for each node,

$$E_n(A_n) = \sum_p w_{pn} \left\| A_n^\top \mathbf{x}_p - O_p \right\|^2 + \lambda \left\| A_n - A_{\mathrm{id}} \right\|_F^2,$$

where $\|\cdot\|_F$ is the Frobenius norm, $A_{\mathrm{id}} = [\, I_3;\, \mathbf{0}^\top \,]$ is the identity transform (no change, zero offset) and $\lambda > 0$ is a small ridge weight. Setting the gradient to zero gives the normal equations

$$\Big(\underbrace{\textstyle\sum_p w_{pn}\, \mathbf{x}_p \mathbf{x}_p^\top}_{S_n \in \mathbb{R}^{4\times 4}} + \lambda I_4 \Big) A_n = \underbrace{\textstyle\sum_p w_{pn}\, \mathbf{x}_p O_p^\top}_{T_n \in \mathbb{R}^{4\times 3}} + \lambda A_{\mathrm{id}},$$

which is exactly what `XtX`, `XtY` and `np.linalg.solve` compute. Accumulating S_n and T_n needs one pass over the low-resolution pixels; solving needs one 4×4 system per node.

Why the ridge is needed. S_n is a weighted scatter matrix of the colors near node n. If a node sees no pixels (an intensity bin that does not occur in that region), $S_n = 0$ and the ridge alone gives $A_n = A_{\mathrm{id}}$. If it sees pixels of almost one color, S_n is close to rank 1: the fit can match the target at that color, but the slopes in other color directions are arbitrary. The ridge pulls them toward the identity. Without it, a full-resolution pixel whose color is slightly different from what the node saw can receive an extreme slope, which shows up as speckles or color blotches. Too large a λ makes every node close to the identity and the edit too weak.

Soft versus hard assignment. The snippet uses the same trilinear weights for fitting and slicing ("soft" binning), so each node's matrix is fitted to the pixels that will later read it. Assigning each pixel only to its nearest node ("hard" binning) is simpler but fits each node to a slightly different set of pixels than slicing uses, which adds error at cell boundaries.

What bilateral guided upsampling adds. Chen et al. [9] add a smoothness term on the affine matrices to the data term, which, in their words, prevents false edges and amplified noise. The

resulting energy couples neighboring nodes, so empty or poorly constrained nodes are filled from their neighbors instead of falling back to the identity. They either solve it as one global optimization (a large sparse linear system instead of many 4×4 ones) or approximate it by fitting the matrices locally and then blurring them in grid space; depending on the parameters, that fast approximation reduces to known methods such as the guided filter [11]. They report that fitting takes about 10 ms on mobile devices and that applying the curves at full resolution can be done by a simple GPU shader.

Grid size. More spatial cells follow spatial changes of the edit more closely; more intensity bins separate the two sides of an edge better. Both reduce halos, but each node then sees fewer pixels, so the fit becomes noisier and the ridge or smoothness term must do more work. Exercise 3 measures this trade-off on the toy.

Pitfalls. The guide must be computed identically at fitting and slicing time (the snippet uses luma for both); a guide in a different color space or gamma encoding shifts every pixel to the wrong intensity bin. The grid's spatial coordinates must map the image corners to the grid corners at both resolutions (the snippet uses `linspace(0, G - 1, H)`), or the edit shifts by a fraction of a cell.

Learnable LUT variants in detail

This note describes, at the level of their key mechanism, the LUT variants listed in the main text.

Spatial-aware 3D LUTs [13]. A two-head weight predictor outputs a weight vector for image-level adaptation and a weight *map* for pixel-wise fusion of several spatial-aware LUTs. Two pixels with the same color can therefore be mapped differently if they lie in different regions, at the cost of fusing tables (or outputs) per pixel instead of once per image.

AdaInt [14]. A uniform lattice places nodes at $0, \frac{1}{M-1}, \dots, 1$ on every axis. AdaInt predicts, per image and per axis, $M-1$ positive **intervals** $\Delta_1, \dots, \Delta_{M-1}$ that sum to 1 (for example with a softmax), and places node m at the cumulative sum $q_m = \sum_{l \le m} \Delta_l$ with $q_0 = 0$. To look up a value r on that axis, find the interval with $q_i \le r < q_{i+1}$ (a search over the sorted node positions) and use the offset

$$d_r = \frac{r - q_i}{q_{i+1} - q_i},$$

where q_i and q_{i+1} are the positions of the two nodes around r. The 8 corner weights are then formed exactly as in the uniform case. The paper's operator for this lookup, AiLUT-Transform, also passes gradients to the interval sizes, so the lattice positions are learned end to end. The effect is the one our gamma experiment asked for: dense nodes where the transform bends (for example near black for a brightening curve) and sparse nodes where it is nearly linear.

SepLUT [15]. The transform is split into a per-channel 1D LUT stage followed by a 3D LUT. The 1D curves are cheap and can be fine, so they handle each channel's tone curve; they also

redistribute the input colors so that they spread over more of the 3D LUT's cells. The authors argue that this better **cell utilization** lets the 3D stage be much smaller. Memory goes from $3M^3$ to $3M^3_{\text{small}} + 3K$, where K is the number of entries per 1D LUT.

4D LUT [16]. A context encoder predicts a per-pixel context value $c(\mathbf{x})$, and the table is indexed by (r, g, b, c). Interpolation becomes **quadrilinear**: 16 corners, with weights that are products of four offsets. With M_c nodes on the context axis the table has $3M^3M_c$ values, which is why such tables use few context nodes.

LUTs with bilateral grids [17]. Kim and Cho transform bilateral grids into a spatial feature domain to bring spatial information into a 3D LUT model, and use slicing operations in place of the long decoding path of earlier spatially aware designs, which reduces inference time and parameters.

A common pitfall. All these methods are trained on photo-retouching data (most often FiveK [8]), whose inputs cover the RGB cube unevenly. Nodes in rarely visited corners of the cube are set only by the smoothness and monotonicity regularizers, so they behave sensibly but are not learned from data. Check outputs on saturated and very dark colors, which are exactly those corners.

Structural re-parameterization step by step

This note shows why a RepVGG-style block [24] with three branches can be replaced at inference by a single 3×3 convolution, and checks the claim numerically.

Batch normalization at inference is affine. With learned scale γ_o and shift β_o and stored running mean μ_o and variance σ_o^2 for output channel o,

$$\mathrm{BN}(y)_o = \gamma_o \frac{y_o - \mu_o}{\sqrt{\sigma_o^2 + \epsilon}} + \beta_o = s_o\, y_o + (\beta_o - s_o \mu_o), \qquad s_o = \frac{\gamma_o}{\sqrt{\sigma_o^2 + \epsilon}},$$

where y_o is the convolution output in channel o and ϵ a small constant. So BN after a convolution with kernel W (no bias) equals a convolution with kernel $s_o W_o$ and bias $\beta_o - s_o \mu_o$.

Every branch is a 3×3 convolution. A 1×1 kernel is a 3×3 kernel that is zero except at the center. The identity map is a 3×3 kernel with a 1 at the center of the (o, o) slice and zeros elsewhere. Convolution is linear in the kernel, so the sum of three convolutions of the same input is one convolution with the summed kernel and the summed bias:

$$W_{\text{fused}} = \sum_{\text{branch } m} s^{(m)} \odot W^{(m)}, \qquad b_{\text{fused}} = \sum_{m} (\beta^{(m)} - s^{(m)} \mu^{(m)}),$$

where $W^{(m)}$ is branch m's kernel written in 3×3 form, $s^{(m)}, \beta^{(m)}, \mu^{(m)}$ are its BN quantities, and $\odot$ scales each output channel's kernel by its s_o. The ReLU comes *after* the sum, so it is not affected.

```
import numpy as np
from numpy.lib.stride_tricks import sliding_window_view

def conv3x3(x, w, b):
    """x: (C, H, W), w: (Cout, C, 3, 3), b: (Cout,). Zero padding keeps the size."""
    patches = sliding_window_view(np.pad(x, ((0, 0), (1, 1), (1, 1))), (3, 3), axis=(1, 2))
    return np.einsum("chwij,ocij->ohw", patches, w) + b[:, None, None]

def bn(y, gamma, beta, mean, var, eps=1e-5):
    """Batch normalization at inference: a fixed per-channel scale and shift."""
    s = gamma / np.sqrt(var + eps)
    return s[:, None, None] * (y - mean[:, None, None]) + beta[:, None, None]

rng = np.random.default_rng(0)
C, H, W = 8, 32, 32
x = rng.standard_normal((C, H, W))
w3, w1 = rng.standard_normal((C, C, 3, 3)) * 0.1, rng.standard_normal((C, C, 1, 1)) * 0.1
bns = [(rng.uniform(0.5, 1.5, C), rng.normal(0, 0.1, C), rng.normal(0, 0.1, C),
 ↪  rng.uniform(0.5, 2, C))
       for _ in range(3)]                                  # (gamma, beta, mean, var) per
        ↪  branch
zero_b = np.zeros(C)

# Training-time block: 3x3 conv + BN, 1x1 conv + BN, identity + BN, summed
w1_as_3 = np.pad(w1, ((0, 0), (0, 0), (1, 1), (1, 1)))     # a 1x1 kernel is a 3x3 kernel
 ↪  with zeros
w_id = np.zeros((C, C, 3, 3)); w_id[np.arange(C), np.arange(C), 1, 1] = 1   # identity as a
 ↪  3x3 kernel
branches = [w3, w1_as_3, w_id]
y_train = sum(bn(conv3x3(x, w, zero_b), *p) for w, p in zip(branches, bns))

# Inference-time block: fold each BN into its kernel, then add the three kernels
w_fused, b_fused = np.zeros_like(w3), np.zeros(C)
for w, (g, beta, mean, var) in zip(branches, bns):
    s = g / np.sqrt(var + 1e-5)
    w_fused += s[:, None, None, None] * w
    b_fused += beta - s * mean
y_infer = conv3x3(x, w_fused, b_fused)
print(f"max difference: {np.abs(y_train - y_infer).max():.1e}")
print("weights: training", w3.size + w1.size, "+ 3 BNs -> inference", w_fused.size, "+ bias")
```

```
max difference: 2.7e-15
weights: training 640 + 3 BNs -> inference 576 + bias
```

Edge cases. The identity branch exists only when the input and output have the same number of channels and the same size (stride 1). Padding must match: the 3×3 branch uses one pixel of zero padding, and the 1×1 branch must be padded so that its output aligns, which writing it as a centered 3×3 kernel does automatically. Folding is exact only with BN in inference mode (fixed running statistics). And the saving is in latency and memory traffic, not in the arithmetic of the 3×3 branch itself: the inference block does one convolution instead of three plus an addition.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

HDRNet: deep bilateral learning for real-time image enhancement

Problem. Phone cameras need complex, content-dependent photographic edits (tone mapping, a retoucher's style) at viewfinder rates on full-resolution frames, which a full-resolution CNN cannot deliver on a phone [4].

Idea. Let a CNN see only a 256×256 copy of the frame and predict a $16 \times 16 \times 8$ bilateral grid of 3×4 affine color transforms, combining a local path and a global path (fully connected layers) so that decisions can depend on the whole image. A learned pointwise guide and a parameter-free, differentiable **slicing** layer bring the transforms back to full resolution, where they are applied pixel by pixel. The loss is computed on the full-resolution output.

What it showed. The same architecture can imitate many operators, including hand-engineered pipelines, Photoshop actions and expert retouching from FiveK [8], and the authors report a real-time 1080p viewfinder on a phone. Its training is offline from input/output pairs, so it can learn edits for which no algorithm exists.

What to check when you read it. Separate the timing of the network (on the thumbnail) from that of slicing and applying (at full resolution), and note on which processor each runs. Read the limitations section: outputs are slightly softer than the reference, and operators that create edges or need non-local reasoning (sharpening, dehazing, matting) fit the local affine model poorly. Note that the global path's fully connected layers fix the input size of the thumbnail.

Why it matters here. It is the clearest statement of this chapter's principle, "predict the transform at low resolution, apply it at full resolution", and its slicing operation reappears in later LUT and grid methods [17].

Image-adaptive 3D LUTs (Zeng et al., TPAMI 2022)

Problem. Learned photo enhancers either gave unsatisfying results or needed too much computation and memory for high-resolution photos of 12 megapixels and more [12].

Idea. Learn a few basis 3D LUTs and a small CNN on a 256×256 thumbnail that predicts content-dependent weights to fuse them into one LUT per image, trained end to end with pairwise or unpaired learning, with smoothness and monotonicity regularizers on the tables.

What it showed. A model with fewer than 600K parameters that processes a 4K image in under 2 ms on one Titan RTX GPU, while scoring well on FiveK and HDR+ benchmarks by PSNR, SSIM

and a color-difference metric [12]. It started a line of work on learnable LUTs (spatial-aware LUTs, AdaInt, SepLUT, 4D LUTs).

What to check when you read it. The timing tables are on a desktop GPU with a custom interpolation kernel; check whether a comparison includes data transfer and the thumbnail resize. Check which expert's retouching is the target on FiveK and which split is used, because published numbers differ with both. Read the limitations: one LUT per image cannot make local adjustments, and noise is amplified on night images.

Why it matters here. It shows how far a purely global, per-image transform can go, and its stated limits are exactly what the successors in this chapter address.

NTIRE 2026 efficient low-light image enhancement challenge report

Problem. Efficient low-light enhancement had no common benchmark with a hard resource limit on realistic, high-resolution phone photographs [21].

Idea. A challenge with phone images at 3024 × 4032 covering dim, extremely dark, backlit, unevenly lit and night scenes, a model-size limit of 1 MB, and a ranking that aggregates each team's ranks over several full-reference and no-reference quality metrics, with fewer parameters breaking ties.

What it showed. 207 registered participants, 27 teams with valid entries and 17 with valid fact sheets, and a set of recurring design patterns: lightweight encoder–decoders with depthwise separable convolutions, Retinex-style decomposition, small attention modules, color-space decoupling, classical priors and re-parameterization [21].

What to check when you read it. The efficiency constraint is model size, not runtime or FLOPs on a fixed device, so the tables tell you which designs fit in 1 MB, not which run fastest on a phone. Rank aggregation over metrics means a team can win without being best on any single metric; look at the individual metric columns. Training hardware listed by teams is not an efficiency measurement.

Why it matters here. It is the most recent independent snapshot of efficient low-light design, and it shows how strongly the measurement protocol shapes what "efficient" means.

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DLCV 05B • Image Fusion: Infrared–Visible, Multi-Focus and Multi-Exposure

How to combine several images of one scene into a single better image: what infrared–visible, multi-focus and multi-exposure fusion ask for, the classical rules (decision maps, Laplacian pyramids, wavelets, exposure fusion), deep fusion without ground truth, and why fusion metrics disagree and can be gamed.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-05b-image-fusion/

What you'll learn

- What image fusion is, and what the three most common tasks ask for: **infrared–visible** fusion (a warm target from the thermal camera, texture from the visible one), **multi-focus** fusion (one all-in-focus image from partly sharp shots) and **multi-exposure** fusion (one well-exposed picture from a bracket of exposures, without recovering HDR radiance).
- Why plain averaging fades contrast, and the classical rules that avoid it: **focus measures and decision maps**, **Laplacian-pyramid** and **wavelet** fusion, Mertens-style **exposure fusion**, and briefly guided-filtering and sparse-representation fusion. You will run each one in NumPy and OpenCV.
- How deep fusion networks are trained when **no ground-truth fused image exists**: intensity, gradient and SSIM-based losses, and why the loss is in effect the definition of the answer.
- The main families of deep fusion models (learned decision maps, autoencoders, end-to-end unsupervised networks, GANs, task-driven, diffusion and text-guided models), what registration-robust fusion adds, and which datasets are used for each task.
- How fusion is evaluated without a reference (entropy, standard deviation, spatial frequency, mutual information, $Q^{AB/F}$, SSIM- and VIF-based scores), why these metrics disagree, and how easily they are gamed.

The big picture

Many cameras see the same scene in different ways. A thermal camera sees a person in the dark but not the texture of the wall behind them. A lens focused on the foreground blurs the background. A short exposure keeps the sky but loses the shadows. **Image fusion** takes two or more such images and builds one image that keeps the useful part of each.

Explain like I'm five

Imagine three friends each took a photo of the same playground. One photo is sharp only on the swings, one is sharp only on the slide, and one was taken with special glasses that show which children are warm. You want one picture to put on the wall. You could cut out the best piece of each photo and glue the pieces together, but you have to be careful: if you cut badly, people will see the seams, and if the photos were not taken from the same spot, the pieces will not line up. Image fusion is that cutting and gluing, done carefully, pixel by pixel.

Why it matters:

- **Night driving, surveillance and search-and-rescue** use visible and thermal cameras together. A driver or an operator looks at one screen, so the two views must be combined into one image that shows both the warm pedestrian and the road markings.
- **Photography fuses images too.** A bracket of exposures can become one picture of a bright window and a dark room; focus stacking gives macro and microscope photos that are sharp from front to back.
- **Fusion is unusual among image tasks: there is no correct answer to copy.** Nobody owns an image that is simultaneously thermal and visible. That single fact shapes the whole field: how methods are designed, how networks are trained and why evaluation is so hard.

This chapter builds on two earlier ones. DLCV 04 placed image fusion in the restoration taxonomy in one paragraph; here it gets the full treatment. IP 07 built the wavelet transform that wavelet fusion uses; this chapter adds the Laplacian pyramid, which IP 07 does not cover. For no-reference image quality assessment in general, see DLCV 04; here we focus on the metrics specific to fusion.

What image fusion is

Plain version. Fusion takes several images of the *same* scene, taken by different sensors or with different camera settings, and returns *one* image. Each input is good at something different, and the output should be good at all of it.

Precise version. Given K registered source images $I_1, \dots, I_K$ of size $H \times W$, a fusion method is a map

$$F = \Phi(I_1, \dots, I_K), \qquad F \in \mathbb{R}^{H \times W \times C},$$

where F is the fused image, C its number of channels (1 for gray, 3 for color) and Φ the fusion rule, which may be hand-designed or learned. "Registered" means that the same pixel position (i, j) shows the same point of the scene in every source. Most rules can be written as a **weighted combination** of the sources, either of the pixels themselves or of some transform of them:

$$F(i,j) = \sum_{k=1}^{K} w_k(i,j)\, I_k(i,j), \qquad \sum_{k=1}^{K} w_k(i,j) = 1, \quad w_k(i,j) \geq 0,$$

where $w_k(i,j)$ is the weight of source k at pixel (i,j). Almost everything in this chapter is a way of choosing those weights: from a focus measure, from exposure quality, per frequency band, or by a network.

This chapter is about **pixel-level** fusion [1], which outputs an image. The alternative is to feed both sources to a network and combine them only at the level of its decisions, which can serve one task well but gives up the fused image that a person can look at and that other tasks can reuse [2]. Deep fusion networks sit in between: they combine features inside the network but still output pixels.

Three tasks

The three tasks of this chapter differ in what each source contributes, in how hard alignment is, and in whether any ground truth can exist (Figure 1).

	Infrared–visible	Multi-focus	Multi-exposure
Sources	Thermal (long-wave infrared) image + visible image, two different sensors	Two or more shots of one camera, focused at different depths	Two or more shots of one camera, with different exposure times
What each source contributes	IR: warm targets, visible through darkness and smoke; visible: texture, color, fine detail	The region that is sharp in each shot	Shadows from long exposures, highlights from short ones
Typical rule	Keep the salient (warm) targets of IR and the texture of visible	**Select**, per region, the sharpest source	**Blend** with weights that favor well-exposed, detailed pixels
Alignment	Hard: two lenses, two resolutions, different appearance	Easy if camera and scene are still	Easy for static scenes; moving objects cause ghosts
Ground truth?	No	Only for synthetic data (blur a sharp image)	No (a tone-mapped HDR image is one possible reference, not *the* answer)

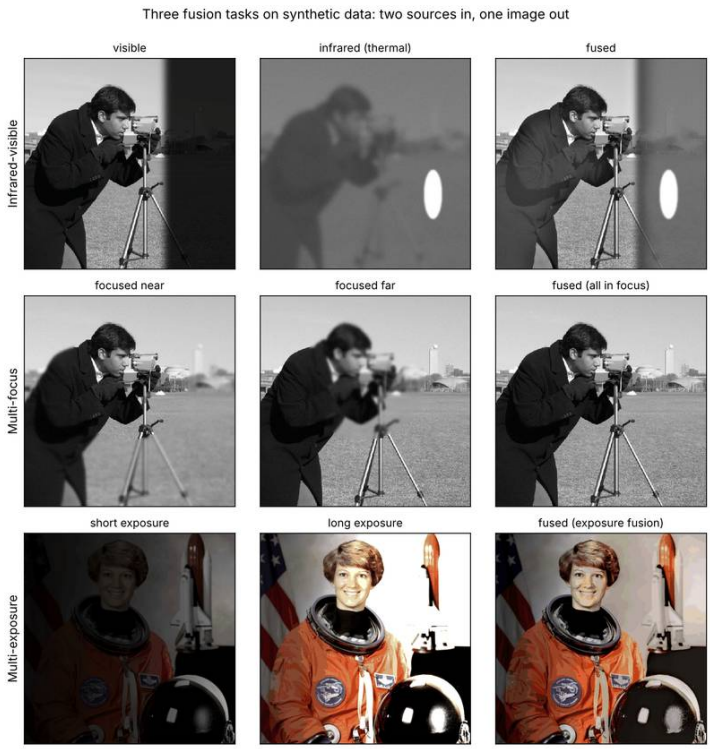


Figure 1 — The three fusion tasks on synthetic data built from scikit-image test images: infrared–visible (pyramid fusion), multi-focus (decision map) and multi-exposure (exposure fusion of the two shots). All three are produced by code in this chapter.

Infrared–visible fusion. A thermal camera measures emitted heat, so a warm target such as a person stands out from its background day and night and in bad weather, while a visible camera measures reflected light and shows texture and detail in a way that matches human vision, but little in the dark [3], [2]. The goal is an image in which warm targets stand out and the background keeps its visible detail. Because the two images look so different, "keep the best of each" is ambiguous: the fused image cannot be both as bright as the thermal target and as dark as the visible shadow at the same pixel.

Multi-focus fusion. A lens has a limited depth of field: objects at one distance are sharp, nearer and farther ones are blurred. Taking several shots with different focus settings and fusing them gives an **all-in-focus** image (also called focus stacking or extended depth of field). Here, unlike the other tasks, a natural answer exists: at each pixel, the source that is in focus is simply correct. That is why multi-focus methods often *select* rather than blend.

Multi-exposure fusion (MEF). A camera sensor records a limited range of brightness in one exposure. A bracket of exposures covers a wider range. The classical route is **high-dynamic-**

range (HDR) imaging: recover the camera response and the scene radiance from the bracket [4], then compress it to a displayable image with a tone-mapping operator. Multi-exposure fusion skips the radiance step and directly blends the 8-bit exposures into a well-exposed 8-bit image [5]. It needs no camera calibration and is fast; the price is that its output is a pleasant picture, not a physical measurement.

Registration: the step everything assumes

Plain version. Fusion compares the sources pixel by pixel, so the same pixel must show the same point in every source. If one image is shifted by a few pixels, every edge appears twice.

Registration (alignment) estimates the geometric transform between the sources and warps them onto a common grid; for moving scenes it becomes dense motion estimation, which DLCV 04B explains. For multi-focus and multi-exposure shots from one camera on a tripod, registration is almost free. For infrared–visible pairs it is a real problem, and a prerequisite of fusion [3]: the two sensors differ in viewpoint, pixel distribution and resolution, so accurately registered pairs are hard to obtain, yet most existing methods need pixel-level registered pairs and existing benchmarks consist of pixel-aligned pairs [2]. We return to methods that relax this assumption at the end of the deep-learning section.

The simplest rule: averaging, and why it fades contrast

Plain version. Averaging the sources keeps a bit of everything, but anything that appears in only one source comes out at half strength. Fusion's whole difficulty is to avoid that.

Precise version. Take two sources I_A and I_B that share a background b, and suppose a feature of amplitude a (an edge, a texture, a warm target) appears only in I_A: $I_A = b + a\,s$, $I_B = b$, where s is the feature's shape. The average is

$$F_{\text{avg}} = \tfrac{1}{2}(I_A + I_B) = b + \tfrac{a}{2}\,s,$$

so the feature's contrast is halved. With K sources and the feature in one of them, it drops to a/K. In multi-focus fusion it is worse: the blurred source adds a blurred copy of the edge, so the average shows a sharp edge with a soft halo. In infrared–visible fusion, a target that is bright in IR and dark in the visible image ends up gray (Figure 6, last column). In multi-exposure fusion, averaging an under- and an over-exposed shot gives a flat, low-contrast image.

The fix is to make the weights $w_k(i, j)$ depend on *where* each source is informative: high weight for the sharp source, the warm target, the well-exposed pixel. The rest of the classical section is about how to measure "informative" and how to change the weights without creating seams.

Multi-focus fusion: focus measures and decision maps

Plain version. Sharp regions have strong fine detail; blurred ones do not. Measure the detail around every pixel in every shot, pick the shot with the most, and copy its pixel.

Precise version. A **focus measure** $M_k(i,j) \geq 0$ scores how much high-frequency detail source k has around pixel (i,j). A simple choice, of the kind used in classical multi-focus fusion [5], is the local energy of the Laplacian,

$$M_k = G_\sigma * (\nabla^2 I_k)^2,$$

where $\nabla^2 I_k = \partial^2 I_k/\partial x^2 + \partial^2 I_k/\partial y^2$ is the Laplacian of the source (a second-derivative filter, see IP 03), the square makes it an energy, and $G_\sigma *$ denotes smoothing with a Gaussian of standard deviation σ, which averages the energy over a neighborhood. Blur removes high frequencies, so the in-focus source has the larger M_k. The **decision map** records the winner,

$$D(i,j) = \arg\max_k M_k(i,j), \qquad F(i,j) = I_{D(i,j)}(i,j),$$

where $D(i,j) \in \{1, \ldots, K\}$ is the index of the source chosen at pixel (i,j).

A raw decision map is noisy: in flat regions (sky, a plain wall) no source has detail, and the winner is decided by noise. Two post-processing steps are standard. **Cleaning** removes small islands, for example with a majority vote (median filter) or morphological operations, on the assumption that in-focus regions are large and connected. **Softening** blurs the binary map slightly near its boundaries so that the switch from one source to another does not leave a visible seam.

The snippet builds a synthetic pair from the camera test image: inside a disk shot A is sharp, outside it shot B is sharp, and defocus is modeled as a Gaussian blur. Because we made the blur ourselves, we have the all-in-focus truth and can measure PSNR, which is impossible with real multi-focus pairs.

```
import numpy as np
from scipy import ndimage as ndi
from skimage import data, img_as_float
from skimage.metrics import peak_signal_noise_ratio as psnr

rng = np.random.default_rng(0)
x = img_as_float(data.camera())                       # 512 x 512: the all-in-focus "truth"
yy, xx = np.mgrid[:512, :512]
near = (yy - 200) ** 2 + (xx - 250) ** 2 < 150 ** 2   # the part that is in focus in shot A
blurred = ndi.gaussian_filter(x, 3.0)                 # defocus modeled as a Gaussian blur
A = np.where(near, x, blurred) + 0.01 * rng.standard_normal(x.shape)  # sharp inside the disk
B = np.where(near, blurred, x) + 0.01 * rng.standard_normal(x.shape)  # sharp outside it

def focus_measure(img, s=2.0):
    """Local energy of the Laplacian: large where the image has sharp detail."""
```

```
    return ndi.gaussian_filter(ndi.laplace(img) ** 2, s)

raw = focus_measure(A) > focus_measure(B)                # decision map: True = take A
clean = ndi.median_filter(raw.astype(float), 31) > 0.5   # majority vote removes small
 ↪  islands
w = ndi.gaussian_filter(clean.astype(float), 2.0)        # soften the seam a little

results = {"average": (A + B) / 2,
           "raw decision map": np.where(raw, A, B),
           "cleaned, soft map": w * A + (1 - w) * B}
for name, F in results.items():
    print(f"{name:18s} PSNR = {psnr(x, np.clip(F, 0, 1), data_range=1):.2f} dB")
textured = ndi.gaussian_filter(np.abs(ndi.laplace(x)), 2) > 0.01   # where the choice matters
for name, m in [("raw", raw), ("cleaned", clean)]:
    print(f"{name:7s} map agrees with the true focus map on {np.mean(m == near):.1%} of all
     ↪  pixels, "
          f"{np.mean((m == near)[textured]):.1%} of textured pixels")
```

```
average            PSNR = 29.97 dB
raw decision map   PSNR = 39.22 dB
cleaned, soft map  PSNR = 39.81 dB
raw     map agrees with the true focus map on 80.9% of all pixels, 91.3% of textured pixels
cleaned map agrees with the true focus map on 91.2% of all pixels, 98.4% of textured pixels
```

The decision map wins by about 10 dB over averaging, because it copies sharp pixels instead of mixing them with blurred ones (Figure 2). Cleaning makes the map much more faithful (98.4% instead of 91.3% agreement on textured pixels), but the PSNR gain is small, 0.6 dB, because most of the raw map's mistakes are in flat regions where shots A and B look the same anyway. These numbers are from our toy, not from any paper.

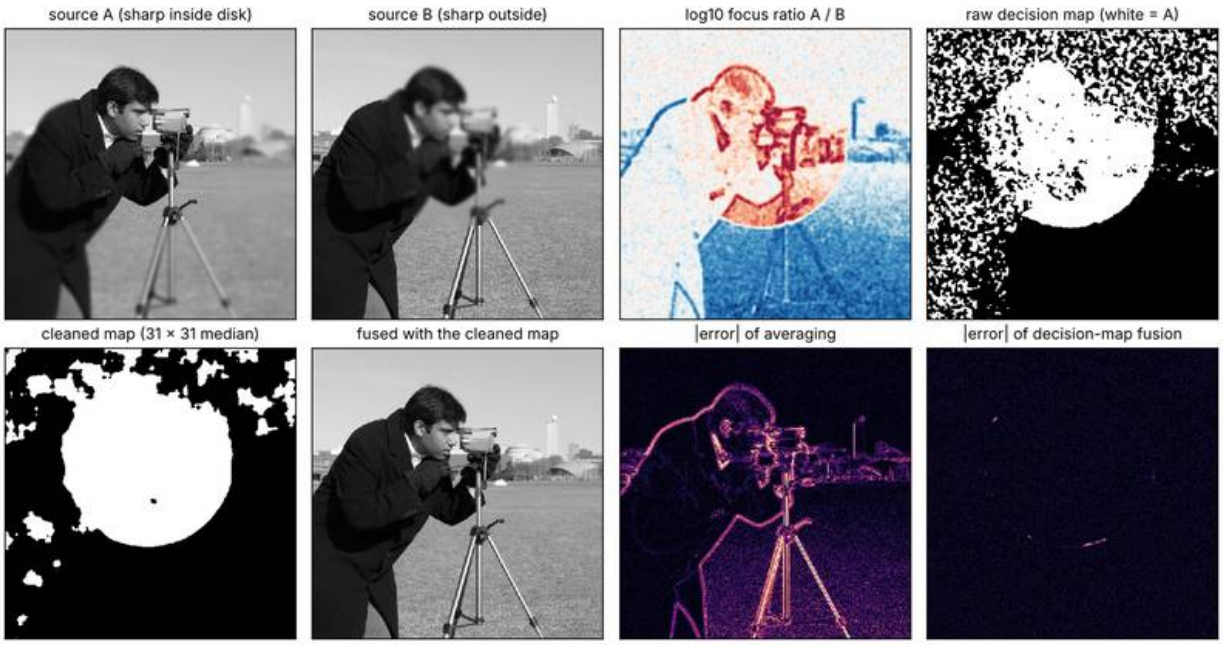


Figure 2 — Multi-focus fusion with a focus measure. The focus ratio is decided by noise in the flat sky, which the median filter mostly cleans up; there both sources are equally good, so the leftover blobs do no harm. The error maps (same color scale) show why selection beats averaging: averaging leaves an error at every edge.

What the toy hides. Real defocus is not a Gaussian blur pasted onto a mask. Near a depth edge, the blurred foreground spreads over the background in the far-focused shot, so *neither* shot is sharp there. This **defocus spread effect** is why methods that do well on simulated data or on the popular Lytro pairs often fail on real photographs with strong defocus spread [6], [7]; Figure 6 shows it on a layered version of our toy.

Optional: The defocus spread effect (in the Supplementary notes). Why selection fails near depth edges, what a layered defocus model predicts, and what methods do about it.

Multi-scale fusion: Laplacian pyramids and wavelets

Plain version. Split each source into layers, from fine detail to coarse brightness, and fuse layer by layer: keep the strongest fine detail from whichever source has it, and blend the coarse layers gently. Seams disappear because coarse layers change slowly.

The Laplacian pyramid

Precise version. The **Laplacian pyramid** of Burt and Adelson [8] represents an image as a stack of band-pass images plus a small low-pass residual. Let $G_0 = I$ and

$$G_{\ell+1} = \mathrm{REDUCE}(G_\ell), \qquad L_\ell = G_\ell - \mathrm{EXPAND}(G_{\ell+1}), \quad \ell = 0, \dots, N-1,$$

where REDUCE blurs with a small low-pass kernel and keeps every second row and column (OpenCV's `pyrDown`), EXPAND upsamples by a factor of 2 and interpolates with the same kernel (`pyrUp`), G_ℓ is the **Gaussian pyramid** level ℓ (a blurred, smaller copy of the image) and L_ℓ the **Laplacian pyramid** level ℓ, the detail that G_ℓ has and the coarser level does not. The top of the pyramid is the residual $L_N = G_N$. Level ℓ has $2^{-\ell}$ times the width and height of the image. The image is recovered exactly by the **collapse**

$$G_N = L_N, \qquad G_\ell = L_\ell + \mathrm{EXPAND}(G_{\ell+1}), \quad \ell = N-1, \dots, 0,$$

and $I = G_0$. Exactness holds for any EXPAND, because each L_ℓ stores exactly what EXPAND fails to predict. The name comes from the authors' observation that the encoding is equivalent to sampling the image with Laplacian operators of many scales [8].

Optional: Building and collapsing a Laplacian pyramid (in the Supplementary notes). The kernel, the size bookkeeping for odd image sizes, why reconstruction is exact, and how many levels to use.

Fusing two pyramids

Pyramid fusion decomposes each source, combines the pyramids level by level, and collapses the result. Classic pyramid fusion selects the most salient Laplacian coefficients of the sources [9], [5]; a simple version of the rule has two parts:

$$L_\ell^F(i,j) = \begin{cases} L_\ell^A(i,j) & \text{if } |L_\ell^A(i,j)| \geq |L_\ell^B(i,j)|, \\ L_\ell^B(i,j) & \text{otherwise,} \end{cases} \qquad L_N^F = \tfrac{1}{2}(L_N^A + L_N^B),$$

where L_ℓ^A, L_ℓ^B and L_ℓ^F are level ℓ of the pyramids of source A, source B and the fused image. The **max-abs** rule keeps, at every position and scale, the coefficient with the larger magnitude, which is the stronger local feature; the coarse residual is averaged because it carries overall brightness, where no source should simply win. Selecting per coefficient at every scale means the switch between sources is spread over the support of the pyramid's filters, which is wider at coarser levels; this is the multiresolution blending idea that Burt and Adelson introduced for joining image mosaics without visible seams [10]. Burt and Kolczynski later refined pyramid fusion to make it more robust to small shifts and noise and to handle features of similar salience but opposite contrast [9].

The snippet continues from the multi-focus example (it uses `A`, `B`, `x` and `psnr`).

```
import cv2
import pywt

def laplacian_pyramid(img, levels=5):
    G = [img.astype(np.float32)]
    for _ in range(levels):
        G.append(cv2.pyrDown(G[-1]))                    # blur, then keep every 2nd pixel
    L = [G[i] - cv2.pyrUp(G[i + 1], dstsize=G[i].shape[::-1]) for i in range(levels)]
    return L + [G[-1]]                                  # band-pass details + coarse base

def collapse(L):
    img = L[-1]
    for detail in reversed(L[:-1]):
        img = cv2.pyrUp(img, dstsize=detail.shape[::-1]) + detail
    return img

def pick_stronger(p, q):
    return np.where(np.abs(p) >= np.abs(q), p, q)       # "max-abs" selection rule

def pyramid_fuse(a, b, levels=5):
    La, Lb = laplacian_pyramid(a, levels), laplacian_pyramid(b, levels)
    fused = [pick_stronger(p, q) for p, q in zip(La[:-1], Lb[:-1])]
    fused.append((La[-1] + Lb[-1]) / 2)                 # base: average the coarse level
    return collapse(fused).astype(float)

def wavelet_fuse(a, b, wavelet="db2", levels=4):
    ca, cb = pywt.wavedec2(a, wavelet, level=levels), pywt.wavedec2(b, wavelet, level=levels)
    fused = [(ca[0] + cb[0]) / 2]                       # approximation band: average
    for da, db in zip(ca[1:], cb[1:]):                  # (H, V, D) detail bands per level
        fused.append(tuple(pick_stronger(p, q) for p, q in zip(da, db)))
    return pywt.waverec2(fused, wavelet)

print(f"pyramid reconstruction error: {np.abs(collapse(laplacian_pyramid(A)) -
 ↪  A).max():.1e}")
for name, F in [("Laplacian pyramid", pyramid_fuse(A, B)), ("wavelet (db2)", wavelet_fuse(A,
 ↪  B))]:
    print(f"{name:18s} PSNR = {psnr(x, np.clip(F, 0, 1), data_range=1):.2f} dB")
```

```
pyramid reconstruction error: 5.9e-08
Laplacian pyramid  PSNR = 37.60 dB
wavelet (db2)      PSNR = 36.43 dB
```

The reconstruction error of about 10^{-7} is single-precision rounding: the pyramid itself is lossless. Pyramid fusion beats averaging by more than 7 dB but stays about 2 dB below the decision map. The reason is visible in Figure 3: the per-coefficient choice is noisy, especially at fine levels, where the blurred source's noise sometimes wins in flat areas. Comparing smoothed coefficient energies instead of single coefficients, the "area-based" selection of Li et al. [11], closes most of the gap (Exercise 3). The general lesson: when the true answer is a selection (multi-focus), spatially coherent decisions beat per-coefficient ones.

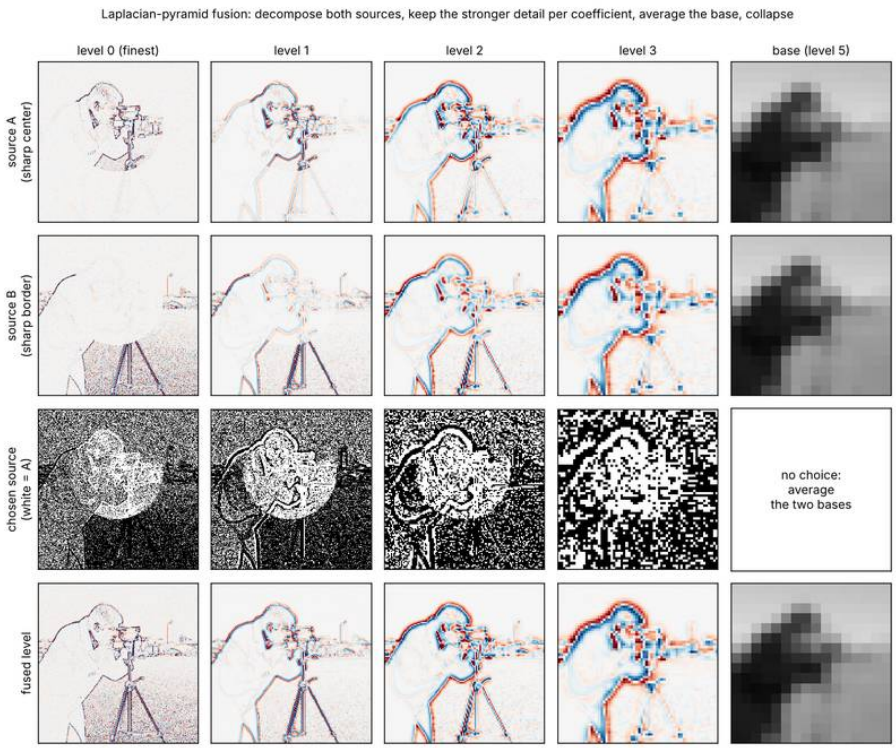


Figure 3 — Laplacian-pyramid fusion of the multi-focus pair. Each source is split into band-pass levels (red and blue are positive and negative coefficients); the max-abs rule picks the stronger coefficient per position (row 3, white = source A), the base is averaged, and collapsing the fused pyramid gives the result.

Wavelet fusion

Wavelet fusion follows the same plan with the discrete wavelet transform of IP 07 instead of the pyramid: decompose both sources, keep the stronger detail coefficient in each horizontal, vertical and diagonal band, average the approximation band, and invert. Li, Manjunath and Mitra [11] proposed it with an **area-based** maximum-selection rule (compare activity over a small window, not single coefficients) and a **consistency verification** step that makes a coefficient follow the choice of its neighbors, and argued that the wavelet transform's compactness, orthogonality and directional selectivity make it better suited than the Laplacian pyramid. In our toy, the simple per-coefficient rule gives 36.43 dB, about 1 dB below the pyramid. One known weakness is that the decimated wavelet transform is not shift-invariant: shifting the image by one pixel changes its coefficients, so a per-coefficient decision can change too. Greater shift invariance was also one of the aims of Burt and Kolczynski's refinement of pyramid fusion [9].

Exposure fusion: well-exposed images without HDR

Plain version. For each pixel of each exposure, ask three questions: is there detail here, is the color vivid, and is the pixel neither black nor white? Turn the answers into a weight, and blend the exposures with those weights, using a pyramid so that the blend leaves no seams.

Precise version. Mertens, Kautz and Van Reeth [5] compute, for exposure k at pixel (i, j), three **quality measures**:

- **contrast** $C_{ij,k}$: the absolute response of a Laplacian filter applied to the grayscale exposure (high at edges and texture);
- **saturation** $S_{ij,k}$: the standard deviation of the R, G and B values at the pixel (vivid colors have a large spread, washed-out ones a small spread);
- **well-exposedness** $E_{ij,k} = \prod_{c\in\{R,G,B\}} \exp\!\left(-\frac{(I^c_{ij,k}-0.5)^2}{2\sigma^2}\right)$: a Gaussian bump around mid-gray, applied to each color channel $I^c_{ij,k} \in [0,1]$ and multiplied, with $\sigma = 0.2$ in their implementation.

They are combined by a product with exponents and normalized over the N exposures:

$$W_{ij,k} = (C_{ij,k})^{\omega_C}\,(S_{ij,k})^{\omega_S}\,(E_{ij,k})^{\omega_E}, \qquad \hat{W}_{ij,k} = \frac{W_{ij,k}}{\sum_{k'=1}^{N} W_{ij,k'}},$$

where $\omega_C, \omega_S, \omega_E \geq 0$ set the influence of each measure (an exponent of 0 switches a measure off) and $\hat{W}_{ij,k}$ is the normalized weight, which sums to 1 over the exposures at each pixel. The product acts like an "and": a pixel gets a high weight only if it is detailed *and* colorful *and* well exposed [5].

Why a pyramid. Blending pixels directly, $R_{ij} = \sum_k \hat{W}_{ij,k} I_{ij,k}$, produces visible seams wherever the weights change quickly, because the exposures have very different absolute brightness [5]. Smoothing the weights with a Gaussian removes the seams but creates halos around edges. Exposure fusion instead blends **per pyramid level**: each Laplacian level of each exposure is weighted by the matching level of a Gaussian pyramid of its weight map,

$$L\{R\}_\ell = \sum_{k=1}^{N} G\{\hat{W}_k\}_\ell \; L\{I_k\}_\ell,$$

where $L\{\cdot\}_\ell$ is level ℓ of a Laplacian pyramid and $G\{\cdot\}_\ell$ level ℓ of a Gaussian pyramid; the result R is obtained by collapsing $L\{R\}$ [5]. Fine detail is blended with sharp weights and coarse brightness with very smooth weights, which is exactly what removes both seams and halos.

The snippet builds a synthetic scene with an 8-stop light falloff (dark on the left, bright on the right), photographs it with three exposure times through a simple camera model (gamma, clipping, 8-bit quantization), and fuses the bracket. It checks the result against OpenCV's implementation, `cv2.createMergeMertens`.

```
import numpy as np
import cv2
from skimage import data, img_as_float

# A synthetic high-dynamic-range scene: linear radiance times a light falloff of 8 stops.
rgb = img_as_float(data.astronaut()) ** 2.2                       # undo display gamma ->
 ↪  linear
light = 2.0 ** np.linspace(-6, 2, rgb.shape[1])[None, :, None]    # dark left, bright right
radiance = rgb * light

def camera(E, t):
    """Expose for time t, apply a 1/2.2 display gamma, clip and quantize to 8 bits."""
    return np.round(np.clip((t * E) ** (1 / 2.2), 0, 1) * 255) / 255

stack = [camera(radiance, t) for t in (1 / 8, 1.0, 8.0)]          # under, middle, over

def mertens_weights(imgs, sigma=0.2, eps=1e-12):
    W = []
    for I in imgs:
        gray = I.mean(axis=2).astype(np.float32)
        C = np.abs(cv2.Laplacian(gray, cv2.CV_32F))                # contrast
        S = I.std(axis=2)                                          # saturation
        E = np.prod(np.exp(-(I - 0.5) ** 2 / (2 * sigma ** 2)), axis=2)  # well-exposedness
        W.append(C * S * E + eps)
    W = np.array(W)
    return W / W.sum(axis=0)                                       # weights sum to 1 per
     ↪  pixel

def gauss_pyr(img, n):
    G = [img.astype(np.float32)]
    for _ in range(n):
        G.append(cv2.pyrDown(G[-1]))
    return G

def lap_pyr(img, n):
    G = gauss_pyr(img, n)
    return [G[i] - cv2.pyrUp(G[i + 1], dstsize=G[i].shape[1::-1]) for i in range(n)] +
     ↪  [G[-1]]

def exposure_fusion(imgs):
    n = int(np.log2(min(imgs[0].shape[:2])))                       # full depth: 9 levels for
     ↪  512 px
    fused = None
    for I, Wk in zip(imgs, mertens_weights(imgs)):
        level = [g[..., None] * l for g, l in zip(gauss_pyr(Wk, n), lap_pyr(I, n))]
        fused = level if fused is None else [f + l for f, l in zip(fused, level)]
    out = fused[-1]
    for detail in reversed(fused[:-1]):
        out = cv2.pyrUp(out, dstsize=detail.shape[1::-1]) + detail
    return np.clip(out, 0, 1)

R = exposure_fusion(stack)
naive = np.clip(np.sum(mertens_weights(stack)[..., None] * np.array(stack), axis=0), 0, 1)
```

```
ref = cv2.createMergeMertens(1.0, 1.0, 1.0).process(
    [(255 * I[..., ::-1]).round().astype(np.uint8) for I in stack])[..., ::-1]   # OpenCV
     ↪  uses BGR
for name, I in [("under", stack[0]), ("middle", stack[1]), ("over", stack[2]),
                ("per-pixel blend", naive), ("pyramid blend", R)]:
    ok = np.mean(np.all((I > 0.05) & (I < 0.95), axis=2))        # no channel near black or
     ↪  clipped
    print(f"{name:15s} well-exposed pixels: {100 * ok:4.1f}%")
print(f"mean |ours - OpenCV MergeMertens| = {np.abs(R - np.clip(ref, 0, 1)).mean():.4f}")
```

```
under           well-exposed pixels: 45.8%
middle          well-exposed pixels: 58.7%
over            well-exposed pixels: 41.2%
per-pixel blend well-exposed pixels: 72.3%
pyramid blend   well-exposed pixels: 73.8%
mean |ours - OpenCV MergeMertens| = 0.0027
```

Our implementation agrees with OpenCV's to within 0.003 on a $[0, 1]$ scale; the small difference presumably comes from implementation details such as pyramid depth, border handling and input rounding. The crude "well-exposed pixels" count rises from at most 59% for any single exposure to 74% after fusion. Note what the count misses: the per-pixel blend scores almost as well (72%) yet looks badly broken, with seams and speckle where the weights jump (Figure 4, bottom left). A single number does not see artifacts, a theme the evaluation section returns to.

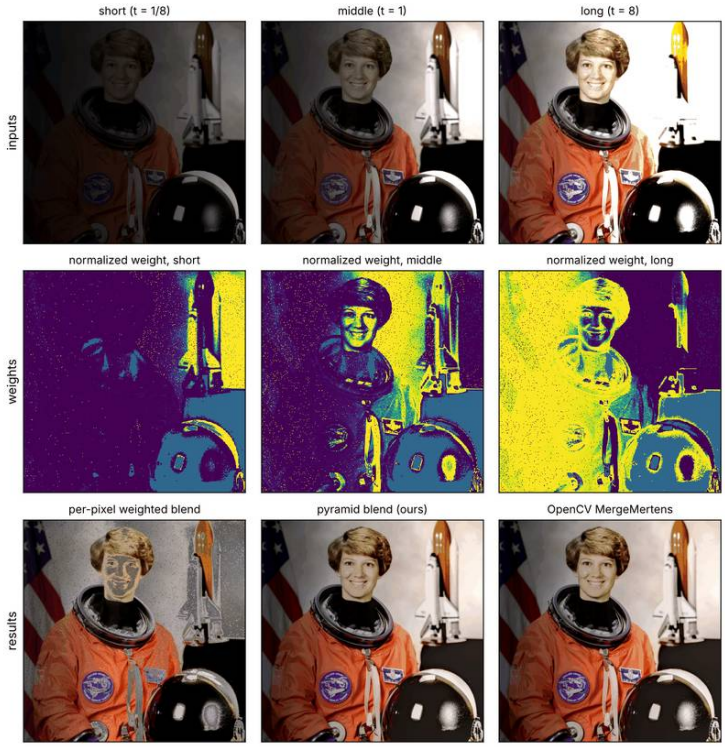


Figure 4 — Exposure fusion of a synthetic bracket. The weights pick the well-exposed, detailed parts of each exposure (in flat black areas all weights are tiny and the normalization gives each exposure one third). Blending pixels directly with these weights fails; blending pyramid levels works and matches OpenCV's implementation.

Exposure fusion is not HDR. The output keeps no radiance values, so it cannot be relit or re-exposed, and with strongly varying brightness the multiresolution blend can create a spurious low-frequency brightness change; Mertens et al. note that a taller pyramid partly solves it [5]. Exposure fusion also assumes perfectly aligned exposures, possibly after registration [5]; with camera or object motion, the result shows ghosts (Figure 6). Finally, Mertens et al. compare their blending with earlier pyramid fusion that *selects* the most salient coefficients and find the selection results too dark and with color shifts on exposure stacks [5]: for exposures, blending beats selecting, the opposite of multi-focus fusion.

Optional: Exposure fusion (Mertens, Kautz and Van Reeth) (in the Paper guides). What the paper set out to do, what its comparison figures show, and the limitations it states.

Two more classical tools: guided filtering and sparse codes

Plain version. Two later ideas improved the classical rules: make the weight maps follow object edges using an edge-aware filter, and describe image patches with a learned dictionary so that "how much detail" is measured on a richer representation.

Guided-filtering fusion (GFF). Li, Kang and Hu [12] split each source into a **base layer** (large-scale intensity variations) and a **detail layer** (small-scale detail), and fuse each layer with a weighted average whose weight maps are refined by the guided filter [13] so that they respect spatial consistency, that is, neighboring pixels with similar intensities get similar weights. The guided filter is explained in IP 03, and its learned and fast variants in DLCV 05A. The authors report state-of-the-art results at the time for multispectral, multi-focus, multimodal and multi-exposure images with one method [12]. The design shows the general recipe of the classical era: a two- or multi-scale decomposition, an activity measure, and a smoothing step that turns the activity into clean weights.

Sparse-representation (SR) fusion. Yang and Li [14] represent the sources with sparse coefficients over an overcomplete dictionary, combine the coefficients with a choose-max rule, and reconstruct the fused image from the combined coefficients and the dictionary; by changing the approximation criterion of the sparse coding, the same scheme also restores noisy sources while fusing them. Later work learned the dictionary from the source patches themselves [15]. SR methods were among the strongest classical multi-focus methods, but they are slow: in one benchmark they took more than seven minutes to fuse an image pair [7].

Optional: Sparse-representation fusion (in the Supplementary notes). The sparse coding model, the patch-by-patch fusion steps, the activity measure and where the cost comes from.

Evaluating fusion without ground truth

Plain version. Without a correct fused image to compare with, fusion is scored by rules of thumb: does the result have a lot of detail, a lot of contrast, a lot of information from the sources? Each rule captures one aspect, so they disagree, and a method can be tuned to look good on the chosen ones.

This section comes before deep learning on purpose: deep fusion networks are trained with losses built from the same ingredients as these metrics, so understanding the metrics explains the losses.

Why fusion has no single right answer

For multi-focus fusion on synthetic data, the all-in-focus image is known and PSNR or SSIM [16] against it is meaningful, as in our first snippet. For infrared–visible and multi-exposure fusion, no such image exists, and benchmark studies agree that no single metric is better than all others [17], [18], [7]. Benchmarks therefore report many metrics: VIFB implements 13 [17], and the multi-exposure benchmark MEFB and the multi-focus benchmark MFIFB 20 each [18], [7]. VIFB groups them into four families: information-theory-based, image-feature-based, structural-similarity-based and human-perception-based [17]. Its results show the consequence: one method wins the information-theory metrics, another the feature-based ones and a third the structural ones, and no method wins all [17].

Optional: VIFB: a visible and infrared image fusion benchmark (in the Paper guides). How the benchmark was built, how to read its tables, and why its best-scoring methods looked worst.

Metrics that look only at the fused image

Let F be the fused image, of size $M \times N$, with values in $[0, 1]$.

Entropy (EN) measures how spread out the histogram is:

$$\mathrm{EN} = -\sum_{l=0}^{L-1} p_l \log_2 p_l,$$

where L is the number of gray levels (histogram bins) and p_l the fraction of pixels in bin l. EN is in bits; a larger value means the gray levels are used more evenly. It is easily affected by noise, so benchmark studies use it only as an auxiliary metric [18].

Standard deviation (SD) measures global contrast, $\mathrm{SD} = \sqrt{\frac{1}{MN}\sum_{i,j}(F(i,j)-\mu)^2}$, where μ is the mean of F (some implementations omit the $\frac{1}{MN}$, which changes the scale but not the ranking).

Spatial frequency (SF) measures the overall activity of neighboring differences [19], [18]:

$$\mathrm{SF} = \sqrt{\mathrm{RF}^2 + \mathrm{CF}^2}, \qquad \mathrm{RF}^2 = \frac{1}{MN}\sum_{i,j}\left(F(i,j) - F(i,j-1)\right)^2, \quad \mathrm{CF}^2 = \frac{1}{MN}\sum_{i,j}\left(F(i,j) - F(i-1,j)\right)^2,$$

where RF (row frequency) uses horizontal differences and CF (column frequency) vertical ones; normalizations differ between implementations. Large SF means many strong edges and much texture.

These three never look at the sources. They reward "more": more gray levels, more contrast, more edges. Noise supplies all three for free.

Metrics that compare with the sources

Mutual information (MI) measures how much the fused image tells about each source [20]:

$$\mathrm{MI} = \mathrm{MI}_{A,F} + \mathrm{MI}_{B,F}, \qquad \mathrm{MI}_{X,F} = \sum_{x,f} p_{X,F}(x,f)\log_2\frac{p_{X,F}(x,f)}{p_X(x)\,p_F(f)},$$

where $p_{X,F}$ is the joint histogram of source $X \in \{A,B\}$ and F (normalized to sum to 1), and p_X, p_F its marginals. MI is large when the gray level of F predicts the gray level of each source well (see IT 01 for the definition). Hossny, Nahavandi and Creighton showed analytically that this classic MI sum is not a valid fusion performance measure on its own [21], and benchmark toolkits implement a normalized version attributed to them, which divides each term by the sum of the two entropies [18].

Optional: Mutual information as a fusion metric (in the Supplementary notes). Why copying one source scores well, how the bin count changes the value, and what normalization fixes.

Gradient-based fusion performance $Q^{AB/F}$ (Xydeas and Petrović [22]) measures how much **edge information** of the sources survives in F. At every pixel it compares the edge strength g and orientation α (from image gradients) of each source X with those of F, turns the comparison into a preservation value $Q^{XF}(i,j) = Q_g^{XF}(i,j)\,Q_\alpha^{XF}(i,j)$, the product of an edge-strength and an orientation preservation value, and averages with weights $w^X(i,j)$ that express the importance of each source [18]:

$$Q^{AB/F} = \frac{\sum_{i,j}\left(Q^{AF}(i,j)\,w^A(i,j) + Q^{BF}(i,j)\,w^B(i,j)\right)}{\sum_{i,j}\left(w^A(i,j) + w^B(i,j)\right)}.$$

A value near 1 means the edges of both sources are kept with their strength and direction. The snippet below implements a simplified version, which we call Q_G: the strength preservation is the ratio of the smaller to the larger gradient magnitude, the orientation preservation falls linearly with the angle between the edges, and the weights are the source gradient magnitudes. Use a reference implementation when you report $Q^{AB/F}$.

Structure-based and perception-based metrics. Piella and Heijmans [23] proposed three variants of a fusion quality metric based on an image quality index of Wang and Bovik; they need

no ground-truth image because they compare F with the sources, and the authors found them consistent with subjective evaluations. VIFF [24] adapts the visual information fidelity criterion to fusion and computes it over several resolutions. For multi-exposure fusion, **MEF-SSIM** [25] was designed from a subjective study: its authors found that general fusion metrics predicted the perceived quality of multi-exposure fusion poorly, and built an SSIM-style index around a patch structure consistency measure that correlates much better with human judgments. Liu et al. [26] compared twelve fusion metrics over six multiresolution fusion algorithms for night-vision applications.

A toy that fools the metrics

The snippet builds an infrared–visible pair from the camera image: the visible image has a deep shadow on the right in which a person is invisible; the thermal image is smooth, with little texture, and the person glows. It scores four candidates: no fusion at all (the visible image), the average, pyramid fusion (max-abs details, and the brighter of the two coarse bases so that the warm target keeps its level), and the average with added noise. As an independent check of what a user cares about, it also measures the **person contrast**: the mean of F on the person minus the mean on a ring around them.

```
import numpy as np
import cv2
from scipy import ndimage as ndi
from skimage import data, img_as_float

rng = np.random.default_rng(1)
scene = img_as_float(data.camera())
yy, xx = np.mgrid[:512, :512]
person = ((yy - 330) / 60) ** 2 + ((xx - 420) / 22) ** 2 < 1        # a warm "person", 120
 ↪  px tall
shadow = ndi.gaussian_filter((xx > 340).astype(float), 8)          # a dark area on the right
vis = scene * (1 - 0.85 * shadow)                                  # visible: person hidden
 ↪  in shadow
ir = 0.25 + 0.3 * ndi.gaussian_filter(scene, 6)                    # thermal: smooth, little
 ↪  texture
ir = np.clip(ir + 0.6 * ndi.gaussian_filter(person.astype(float), 2)  # the warm body glows
             + 0.01 * rng.standard_normal(ir.shape), 0, 1)

def lap_pyr(img, n=5):
    G = [img.astype(np.float32)]
    for _ in range(n):
        G.append(cv2.pyrDown(G[-1]))
    return [G[i] - cv2.pyrUp(G[i + 1], dstsize=G[i].shape[::-1]) for i in range(n)] + [G[-1]]

def collapse(L):
    img = L[-1]
    for d in reversed(L[:-1]):
        img = cv2.pyrUp(img, dstsize=d.shape[::-1]) + d
    return img
```

```
def pyramid_fuse(a, b):                    # max-abs details, and the brighter coarse base
    La, Lb = lap_pyr(a), lap_pyr(b)
    out = [np.where(np.abs(p) >= np.abs(q), p, q) for p, q in zip(La[:-1], Lb[:-1])]
    out.append(np.maximum(La[-1], Lb[-1]))
    return np.clip(collapse(out), 0, 1).astype(float)

def entropy(F, bins=256):
    p = np.histogram(F, bins, (0, 1))[0] / F.size
    p = p[p > 0]
    return -np.sum(p * np.log2(p))

def spatial_frequency(F):
    rf = np.mean(np.diff(F, axis=1) ** 2)           # row frequency: horizontal differences
    cf = np.mean(np.diff(F, axis=0) ** 2)           # column frequency: vertical differences
    return np.sqrt(rf + cf)

def mutual_info(X, F, bins=64):
    pxy = np.histogram2d(X.ravel(), F.ravel(), bins, [[0, 1], [0, 1]])[0] / X.size
    px, pf = pxy.sum(1, keepdims=True), pxy.sum(0, keepdims=True)
    nz = pxy > 0
    return np.sum(pxy[nz] * np.log2(pxy[nz] / (px @ pf)[nz]))

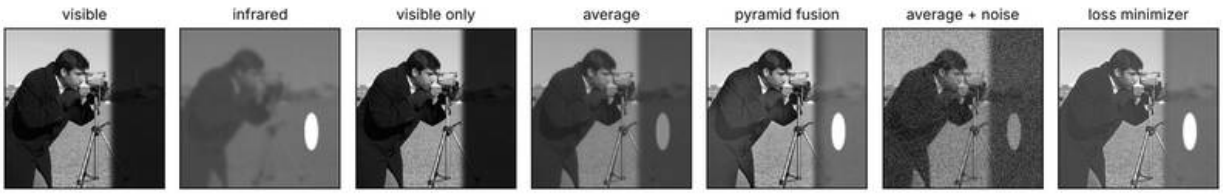
def q_gradient(A, B, F):
    """A simplified Q^{AB/F}: how well F keeps the edge strength and direction of A and B."""
    def grad(I):
        gx, gy = ndi.sobel(I, 1), ndi.sobel(I, 0)
        return np.hypot(gx, gy), np.arctan2(gy, gx)
    gF, aF = grad(F)
    num = den = 0.0
    for X in (A, B):
        gX, aX = grad(X)
        strength = np.minimum(gX, gF) / (np.maximum(gX, gF) + 1e-12)
        d = np.abs(aX - aF) % np.pi                 # edge direction is defined modulo pi
        direction = 1 - np.minimum(d, np.pi - d) / (np.pi / 2)
        num += np.sum(strength * direction * gX)    # weight each pixel by the source edge
         ↪  strength
        den += np.sum(gX)
    return num / den

ring = ndi.binary_dilation(person, iterations=15) & ~person
candidates = {"visible only": vis,
              "average": (vis + ir) / 2,
              "pyramid fusion": pyramid_fuse(vis, ir),
              "average + noise": np.clip((vis + ir) / 2 + 0.08 *
               ↪  rng.standard_normal(vis.shape), 0, 1)}
print(f"{'':16s}  EN    SF     SD     MI    Q_G   person contrast")
for name, F in candidates.items():
    mi = mutual_info(vis, F) + mutual_info(ir, F)
    tc = F[person].mean() - F[ring].mean()          # task check: does the person stand out?
    print(f"{name:16s} {entropy(F):4.2f}  {spatial_frequency(F):.3f}  {F.std():.3f}
     ↪  {mi:4.2f}  "
          f"{q_gradient(vis, ir, F):.3f}  {tc:6.3f}")
```

```
                 EN    SF     SD     MI    Q_G   person contrast
visible only   6.68  0.065  0.293  5.72  0.784  -0.000
average        6.69  0.034  0.172  4.33  0.455   0.257
pyramid fusion 7.48  0.067  0.221  3.75  0.779   0.513
average + noise 7.49 0.163  0.189  1.41  0.296   0.259
```

Read the table column by column (all numbers are from this toy):

- **Noise wins EN and SF.** Adding noise to the average gives the highest entropy and almost five times the average's spatial frequency, while making the image worse.
- **Not fusing at all wins SD, MI and Q_G.** The visible image alone has the most contrast, the highest mutual information (copying a source makes $\mathrm{MI}_{A,F}$ equal to that source's entropy) and even the highest edge score, because the strong visible edges carry most of the weight w^X. Yet the person is invisible: its contrast is zero (printed as -0.000).
- **Pyramid fusion wins only the task check.** It doubles the person's contrast relative to the average while keeping the visible texture (Q_G almost as high as the visible image), but it is not the best on any of the five standard metrics.



Higher is 'better' for every column; the shaded cell is each column's winner

candidate	EN (bits)	SF	SD	MI (bits)	Q_G	person contrast
visible only	6.68	0.065	0.293	5.72	0.784	0.000
average	6.69	0.034	0.172	4.33	0.455	0.257
pyramid fusion	7.48	0.067	0.221	3.75	0.779	0.513
average + noise	7.49	0.163	0.189	1.41	0.296	0.259
loss minimizer	6.99	0.065	0.200	4.24	0.718	0.513

Figure 5 — Five candidate fusions of the synthetic infrared–visible pair and their scores (our toy; the loss minimizer is computed in the deep-learning section). Every metric is 'higher is better', yet the winners are a noisy image and an image that ignores the infrared source. Only the task check identifies the useful results.

What to report

The toy is extreme on purpose, but each effect appears in practice: noise and over-sharpening raise EN, SF and SD; results that look like one source score well on MI; and metrics with source-dependent weights favor the source with stronger edges. Benchmark authors draw the same conclusion from real methods: researchers tend to pick the metrics that support their method [17], [7], and quantitative and qualitative comparisons may disagree [7]. Practical advice:

1. Report several metrics from **different families**, including at least one that compares with the sources ($Q^{AB/F}$, an SSIM-based score or VIFF), never only EN, SD or SF.
2. Use **reference-based** metrics where a reference can exist: synthetic multi-focus pairs, or multi-exposure datasets with selected references (with the caveat that the reference is one acceptable answer).
3. Add a **downstream task** when fusion feeds one, such as detection or segmentation accuracy on the fused images, as recent infrared–visible benchmarks do [27], [28].
4. Show **crops** of hard regions (depth edges, highlights, warm targets in shadow), and run a user study when the claim is about visual quality.

Deep image fusion

Plain version. Deep networks learn the weights, the features and the combination rule from data. The catch is that there are no examples of the correct output, so the network is trained to satisfy rules of the same kind as the metrics above.

The missing ground truth

A restoration network (DLCV 04) learns from pairs (degraded input, clean target). A fusion network has inputs but no target. The field uses four ways around this:

1. **Synthesize a task that has ground truth.** For multi-focus fusion, blur parts of sharp images: the sharp image is the target and the blur mask the true decision map. Most deep multi-focus methods are trained this way [7]; IFCNN trains a general fusion network on a synthetic multi-focus dataset built from an RGB-D dataset, so that ground-truth fused images exist [29].
2. **Train something else, then fuse features by a rule.** Train an autoencoder to reconstruct ordinary images, then at test time encode both sources, merge the features with a hand-made rule and decode [30].
3. **Unsupervised losses.** Define a loss that measures how well F keeps the intensity, gradients or structure of each source, and train the network to minimize it on unlabeled source pairs [31], [32].

4. **Let another task supervise.** Feed F to a detector or segmenter and add its loss, so that the fusion is judged by its usefulness [33], [27].

Unsupervised losses

A recent survey groups the losses of unsupervised infrared–visible fusion into pixel-level terms (L1 or MSE), terms built from evaluation metrics (above all SSIM) and terms that target data characteristics such as image gradients [2]. A typical combination is

$$\mathcal{L}(F) = \lambda_{\text{int}} \|F - T\|_1 + \lambda_{\text{grad}} \big\| |\nabla F| - |\nabla T_g| \big\|_1 + \lambda_{\text{ssim}} \sum_X \omega_X (1 - \text{SSIM}(F, I_X)),$$

where T is an intensity target built from the sources (for example a weighted average, or the pixel-wise maximum, which keeps the brighter thermal target), $|\nabla T_g|$ a gradient-magnitude target (for example the larger of the two sources' gradient magnitudes at each pixel), $\text{SSIM}(F, I_X)$ the structural similarity [16] between F and source X, ω_X per-source weights, and the λs weight the terms. Methods differ mainly in these choices: DeepFuse [31] trains with MEF-SSIM [25] alone, SwinFusion [34] with SSIM, texture and intensity terms, and U2Fusion [32] weights a similarity loss by an estimate of how much information each source holds.

Optional: Training signals of representative fusion networks (in the Supplementary notes). What DenseFuse, DeepFuse, U2Fusion, SwinFusion, FusionGAN, CDDFuse, DDFM and EMMA actually optimize, and what each choice implies for the result.

The loss defines the answer

Plain version. If nobody gives the network a target, the loss *is* the target. The best a network can do is to output, for each pair, the image that minimizes the loss. For a simple quadratic loss we can compute that image directly, without any network, and look at it.

The snippet continues from the toy of the evaluation section (it uses `vis`, `ir`, `person`, `ring` and the metric functions). It minimizes a squared-error version of the loss above, an intensity term toward the pixel-wise maximum of the sources plus a gradient term toward the stronger source gradient,

$$F^\star = \arg\min_F \|F - \max(I_A, I_B)\|_2^2 + \lambda \|\nabla F - \mathbf{g}\|_2^2,$$

where ∇F stacks the horizontal and vertical forward differences of F, $\mathbf{g}$ takes, at each position and in each direction, the difference of whichever source has the larger one in magnitude, and $\lambda > 0$ trades intensity against gradients. Setting the gradient of the objective to zero gives a sparse linear system,

$(\mathbf{I} + \lambda \mathbf{D}^\top \mathbf{D}) F = \max(I_A, I_B) + \lambda \mathbf{D}^\top \mathbf{g}$, where $\mathbf{D}$ is the difference operator; conjugate gradients solves it in seconds.

```
import scipy.sparse as sp
from scipy.sparse.linalg import cg

def diff_ops(H, W):
    d = lambda n: sp.diags([-np.ones(n - 1), np.ones(n - 1)], [0, 1], shape=(n - 1, n))
    return sp.kron(sp.eye(H), d(W)).tocsr(), sp.kron(d(H), sp.eye(W)).tocsr()  # Dx, Dy

def fuse_by_loss(a, b, lam=10.0):
    """Minimize ||F - max(a, b)||^2 + lam * ||grad F - g||^2, where g is the stronger source
     ↪  gradient."""
    H, W = a.shape
    Dx, Dy = diff_ops(H, W)
    gx = [Dx @ s.ravel() for s in (a, b)]
    gy = [Dy @ s.ravel() for s in (a, b)]
    pick_x = np.abs(gx[0]) >= np.abs(gx[1])
    pick_y = np.abs(gy[0]) >= np.abs(gy[1])
    tx, ty = np.where(pick_x, gx[0], gx[1]), np.where(pick_y, gy[0], gy[1])
    target = np.maximum(a, b).ravel()                # intensity term: the brighter source
    M = sp.eye(H * W) + lam * (Dx.T @ Dx + Dy.T @ Dy)  # normal equations of the quadratic
     ↪  loss
    rhs = target + lam * (Dx.T @ tx + Dy.T @ ty)
    f, info = cg(M, rhs, x0=target, rtol=1e-6, maxiter=2000)
    return np.clip(f.reshape(H, W), 0, 1)

for lam in (1.0, 10.0):
    F = fuse_by_loss(vis, ir, lam)
    mi = mutual_info(vis, F) + mutual_info(ir, F)
    print(f"loss minimizer, lam={lam:4.1f}: EN {entropy(F):.2f}  SF
     ↪  {spatial_frequency(F):.3f}  "
          f"SD {F.std():.3f}  MI {mi:.2f}  Q_G {q_gradient(vis, ir, F):.3f}  "
          f"person contrast {F[person].mean() - F[ring].mean():.3f}")
```

```
loss minimizer, lam= 1.0: EN 6.91  SF 0.060  SD 0.195  MI 4.68  Q_G 0.659  person contrast
 ↪  0.513
loss minimizer, lam=10.0: EN 6.99  SF 0.065  SD 0.200  MI 4.24  Q_G 0.718  person contrast
 ↪  0.513
```

The loss minimizer has the person as bright as pyramid fusion does (contrast 0.513) and keeps the visible texture (Figure 5, right). Raising λ from 1 to 10 raises Q_G and SF and lowers MI: the weight between loss terms moves the result along the metrics, and choosing it is choosing what the "correct" fusion is. This is the honest summary of unsupervised deep fusion. The network adds speed, the ability to generalize from many training pairs and features richer than raw gradients, but the definition of a good result is still written by hand, in the loss.

Families of deep fusion networks

The surveys organize the field by architecture and by training strategy [35], [2]. The table lists representative members of each family; the infrared–visible survey guide in the Paper guides

covers the taxonomy in more depth.

Family	Idea	Representative methods
Learned decision maps or weights	A network predicts which source is in focus, or a per-pixel weight; the output is still a selection or blend of source pixels	CNN focus map for multi-focus [36]; MEF-Net predicts weights at low resolution and upsamples them with a guided filter [37]
Autoencoder + fusion rule	Train an encoder–decoder to reconstruct images; fuse encoded features with a hand-made rule	DenseFuse [30]
End-to-end, unsupervised	A network maps the sources to F, trained with an intensity/gradient/SSIM-type loss	DeepFuse [31], U2Fusion [32], SwinFusion [34], CDDFuse [38]
Supervised on synthetic data	Train on synthesized pairs with a known fused image	IFCNN [29]
Adversarial (GAN)	A discriminator pushes F to look like one source's texture while a content loss keeps the other's intensity	FusionGAN [39], TarDAL [27]
Task-driven	A detection or segmentation loss on F guides the fusion network	SeAFusion [33], TarDAL [27], SegMiF [28], TDFusion [40]
Generative priors and language	A pretrained diffusion model or a text encoder guides the fusion	DDFM [41], Text-IF [42]

Two families rely on generative models explained elsewhere: GANs in ML 06 and diffusion models in ML 07. Across all families, the architecture matters less for the character of the result than the training signal, which is what the table's middle column really distinguishes.

Fusion for downstream tasks

Plain version. If the fused image feeds a detector, the best fusion is the one that makes the detector work, not the one with the best entropy.

SeAFusion [33] cascades a real-time fusion network with a segmentation network and adds a semantic loss, training the two in a loop [2]. **TarDAL** [27] writes fusion and detection as one bilevel optimization problem, unrolls it into a fusion network with one generator and two discriminators (one for the infrared targets, one for visible texture) followed by a detector, and introduces the M3FD benchmark captured with a synchronized infrared–visible system. **SegMiF** [28] exchanges intermediate features between a fusion and a segmentation sub-network through hierarchical attention, and collects a full-time benchmark with 15 annotated pixel-level classes. **TDFusion** [40] goes one step further and *learns the fusion loss itself*: a loss-generation module is trained by meta-learning so that the fusion network it supervises minimizes the downstream task loss. Task-driven training answers the "what is correct?" question with a measurable goal, but it optimizes for one task and one network, and task performance does not fully reflect visual quality [2].

Fusing images that are not aligned

Plain version. Real infrared and visible cameras are rarely perfectly aligned. Registration-robust methods estimate the misalignment and correct it inside the fusion network.

Pixel-wise fusion of misaligned pairs produces ghosts and double edges around objects [43], [44].

Two strategies appear in recent work [2]. The **style-transfer route** first translates one modality into the other, so that cross-modal registration becomes a single-modality one: UMF-CMGR [43] generates a pseudo-infrared image from the visible image, registers the real infrared image to it with a multi-level registration network, and then fuses. The **shared-feature route** maps both modalities into a common feature space and predicts deformation fields there. SuperFusion [44] combines registration (bidirectional deformation fields), fusion and a semantic constraint from a pretrained segmentation model in one framework. Multi-exposure fusion has the analogous problem with camera shake and moving objects (ghosting).

Datasets and benchmarks

Plain version. Infrared–visible fusion has several public collections of aligned pairs, some with object labels; multi-focus and multi-exposure fusion have smaller test sets and benchmark toolkits. None has a ground-truth fused image for real infrared–visible data.

Dataset	Task	Content (as described by its authors)
TNO [45]	IR–visible	Intensified visual, near-infrared and long-wave infrared night-time images of military and surveillance scenes
RoadScene [32]	IR–visible	221 aligned pairs of road scenes selected from FLIR videos, released with U2Fusion
MSRS [46]	IR–visible	1,444 aligned pairs of road scenes, built from the MFNet dataset [47] by removing 125 misaligned pairs and enhancing the infrared images
LLVIP [48]	IR–visible	15,488 strictly aligned pairs, mostly taken in very dark scenes, with pedestrian labels
M3FD [27]	IR–visible	4,200 registered pairs with detection labels, plus 300 pairs from independent scenes for fusion; images mostly 1024 × 768
FMB [28]	IR–visible	A full-time benchmark with 15 annotated pixel-level classes for fusion and segmentation
VIFB [17]	IR–visible	A test set of 21 pairs, a code library of 20 fusion algorithms and 13 metrics
Lytro [15]	Multi-focus	20 pairs of multi-focus images, long the most widely used test set [7]

Dataset	Task	Content (as described by its authors)
MFFW [6]	Multi-focus	19 real pairs collected from the Internet with a strong defocus spread effect
MFIFB [7]	Multi-focus	A benchmark of 105 test pairs, 30 algorithms and 20 metrics
SICE [49]	Multi-exposure	589 multi-exposure sequences (4,413 images), each with a reference chosen by subjective comparison of 13 fusion and HDR algorithms
MEFB [18]	Multi-exposure	A benchmark of 100 test image pairs, 16 algorithms and 20 metrics

Read the table with three cautions. **Preprocessing is part of the data:** RoadScene's infrared images had their background thermal noise processed, and MSRS's infrared images were contrast-enhanced with a dark-channel-prior-based method, so "the infrared source" is already processed.

Scenes are narrow: most infrared–visible data are urban roads and campuses, and the 2025 survey lists more diverse scenes and registration ground truth among the missing benchmarks [2].

Old test sets are saturated: on the 20 Lytro pairs, popular methods perform very similarly because the defocus spread effect is weak there [6], [7]. SICE is also used in DLCV 05 for exposure correction.

Failure modes

Plain version. Fusion fails where its assumptions fail: sources not aligned, sources that are not simply "sharp or blurred", weights that change too fast or too slowly, and rules that average what should be selected.

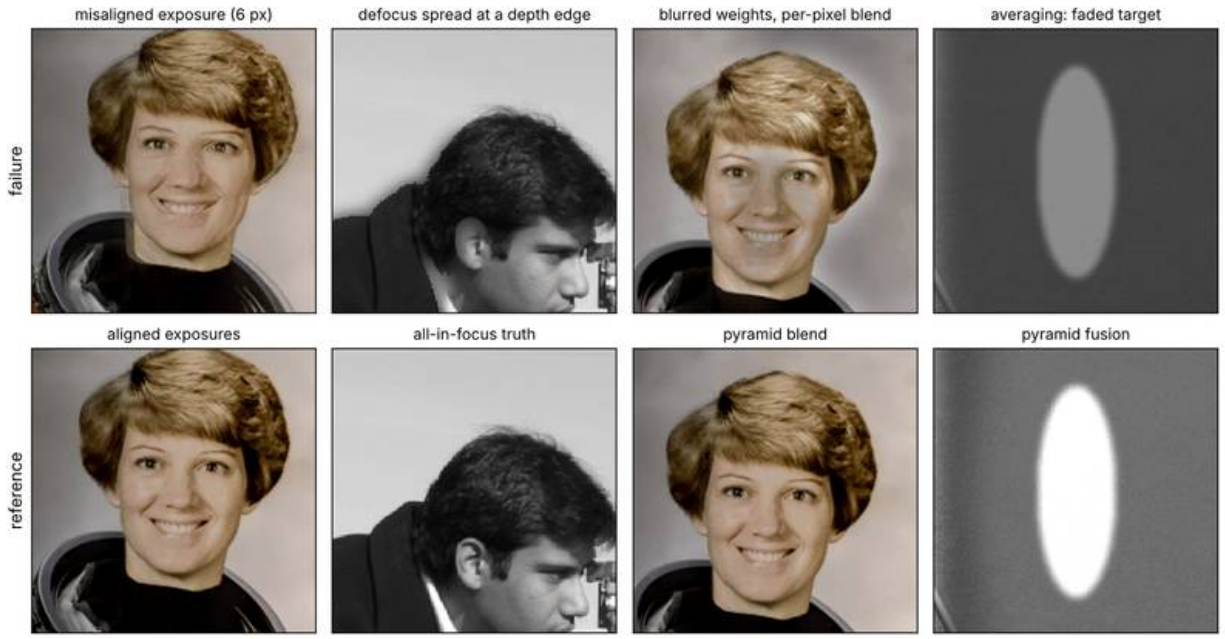


Figure 6 — Four failure modes on synthetic data (top) and a reference for each (bottom): misregistration ghosts, the defocus spread effect at a depth edge (a layered defocus model in which the blurred foreground spills over the background), halos from smoothed weights, and a washed-out thermal target.

- **Misregistration ghosts.** A shift of a few pixels between sources turns edges into double edges (Figure 6, first column). In multi-exposure fusion, moving objects cause the same effect. Check alignment before blaming the fusion rule.
- **Decision-map boundary artifacts.** Near depth edges, the real blur of the foreground spills over the background, so neither source is sharp and any binary decision leaves a soft ring or a blocky edge (second column). Hard switches also show small seams where the map is wrong.
- **Halos.** Weights that are smoothed too much, or a base layer fused at too fine a scale, create bright or dark bands along strong edges (third column) [5]. Multi-scale blending avoids them, but a pyramid that is too shallow brings back seams and low-frequency errors (Exercise 4).
- **Washed-out thermal targets.** Averaging, or a loss that weights the visible source heavily, halves the contrast of warm targets (fourth column); the opposite error makes the background look like an infrared image, flat and textureless.
- **Color shifts.** A common way to handle color is to fuse only the luminance (for example the Y channel of YCbCr) and to combine or copy the chroma separately, as DeepFuse does for exposures [31]. Where the fused luminance differs a lot from the source whose chroma is used, for example a bright thermal target on a dark visible background, the colors no longer fit and look washed out or tinted. For exposure stacks, selecting pyramid coefficients instead of blending them produced color shifts in the comparison of Mertens et al. [5].

- **Noise amplification.** Max-abs selection and gradient losses prefer strong high frequencies, and noise is the strongest high frequency in a dark visible image, so a fused night image can look sharper but noisier than either source. This is also why EN and SF reward such results.
- **Metric gaming.** A method tuned to EN, SF or SD can produce exactly the artifacts above while reporting better numbers (the toy of Figure 5).

Modern view

Surveys. Six surveys together cover the field:

- **Ma, Ma and Li (Information Fusion 2019) [3]** review classical and early deep infrared–visible fusion, introduce registration as a prerequisite, summarize applications and evaluation metrics, and compare eighteen methods with nine metrics. Takeaway: a reference point for classical methods and their evaluation habits before deep learning took over.
- **Liu et al. (Information Fusion 2018) [1]** wrote an early review (2018) of deep learning for pixel-level fusion across tasks. They list the difficulties of conventional fusion (hand-designed activity measures and fusion rules) and argue that learned features can replace them, grouping the early methods (CNNs, convolutional sparse representation, stacked autoencoders) into generic frameworks.
- **Zhang and Demiris (TPAMI 2023) [35]** review deep visible–infrared fusion by architecture (CNN-, autoencoder-, GAN- and transformer-based), with datasets and evaluation methods. Takeaway: deep methods have become the dominant approach for the task.
- **Liu et al. (TPAMI 2025) [2]** organize more than 180 learning-based infrared–visible methods along three dimensions, data compatibility (registration), fusion (architectures and losses) and task adaptation (detection, segmentation), and compare them on registration, fusion and downstream tasks.
- **Zhang (TPAMI 2022) [50]** surveys deep multi-focus fusion (methods, datasets, metrics) and compares deep and conventional methods experimentally.
- **Xu et al. (Remote Sensing 2022) [51]** review multi-exposure fusion for static and dynamic scenes and compare 18 methods with nine metrics.

Optional: The infrared–visible fusion survey of Liu et al. (in the Paper guides). How its taxonomy is built, how to read its comparison tables, and what it lists as open problems.

State of the art (2023–2026). Four trends stand out. First, **fusion is increasingly judged by downstream tasks**: methods are trained with detection or segmentation losses, benchmarks such as M3FD and FMB carry labels, and the fusion loss itself can be learned from the task loss [27],

[28], [40]. Second, **generative priors and language** have entered: diffusion-based fusion without task-specific training [41], self-supervised training from imaging priors [52], and text-guided fusion that also handles degraded sources [42]. Third, **registration is moving inside the model**, with joint registration–fusion networks for unaligned infrared–visible pairs [43], [44]. Fourth, **video fusion** is starting: a CVPR 2026 work introduces M3SVD, 220 synchronized and registered infrared–visible videos with 153,797 frames, and a model whose bi-temporal attention mitigates the temporal inconsistency of frame-by-frame image fusion [53]. For multi-focus fusion, a 2020 benchmark study found that deep methods trained on simulated data did not generalize well to challenging real pairs, while conventional methods remained competitive [7]. Treat claims of large improvements with care, and check which metrics they rest on.

Open problems.

- **Evaluation.** There is still no metric, or set of metrics, that agrees with human judgment and with task performance; the 2025 survey names better evaluation metrics as an essential direction [2].
- **Real misalignment and degradations.** Most methods assume registered, clean sources. Benchmarks with registration ground truth, and fusion that is robust to noise, blur, low light and adverse weather in either source, are open [2], [42].
- **Real defocus.** Training data that reproduces the defocus spread effect, and methods that handle depth edges, are needed for multi-focus fusion to generalize [6], [7].
- **Efficiency.** Many recent networks are large; fusion on drones, cars and handheld devices needs lightweight, hardware-friendly designs [2] (see DLCV 05A for the general techniques).
- **Video.** Research has fused images rather than videos, largely because large synchronized multi-sensor video datasets were missing [53]; temporal consistency at low cost is open.

Key takeaways

- Image fusion turns several registered images of one scene into one image; for real infrared–visible and multi-exposure data **there is no ground truth**, which shapes methods, training and evaluation.
- Averaging halves anything that only one source contains. Good rules weight sources by where they are informative: focus measures and **decision maps** for multi-focus, Mertens-style **quality weights** for exposures, salient targets for infrared.
- **Multi-scale blending** (Laplacian pyramids, wavelets) avoids seams: fine detail is chosen sharply, coarse brightness blended smoothly. Exposure fusion blends Laplacian levels with a

Gaussian pyramid of the weights.

- Deep fusion replaces hand-made measures with learned ones, but with no target the **loss defines the answer**: intensity, gradient and SSIM terms, adversarial terms, or a downstream task loss.
- Fusion metrics (EN, SD, SF, MI, $Q^{AB/F}$, SSIM- and VIF-based) each capture one aspect and **disagree**; source-free ones are inflated by noise, MI rewards copying a source. Report several families, references where possible, task results and crops.
- Most methods assume **pixel-aligned sources**; misregistration causes ghosts, and registration-robust fusion is an active topic.
- The main failure modes are ghosts, decision-map artifacts at depth edges (the defocus spread effect), halos, color shifts, amplified noise and washed-out thermal targets.

Exercises

1. **Contrast under averaging.** A warm target has intensity $b + a$ in the infrared image and b in the visible one, on a background of intensity b in both. (a) Show that the average has target contrast $a/2$ and that any per-pixel weighted average with weights $w_{\mathrm{IR}}, w_{\mathrm{VIS}}$ gives $w_{\mathrm{IR}}\, a$. (b) In our toy, the person contrast of the average is 0.257 and that of pyramid fusion 0.513. Which weight on the infrared source does pyramid fusion behave like on the target, and why does it achieve this without darkening the visible texture elsewhere?

   Hint

   (a) $F = w_{\mathrm{IR}}(b + a) + w_{\mathrm{VIS}}\, b = b + w_{\mathrm{IR}}\, a$, using $w_{\mathrm{IR}} + w_{\mathrm{VIS}} = 1$. (b) 0.513 is about twice 0.257, so on the target pyramid fusion behaves like $w_{\mathrm{IR}} \approx 1$. It can do so because the weights are not global: the coarse base takes the brighter source (the infrared target), while the fine levels take whichever detail is stronger, which is the visible texture away from the target. A global weight would have to trade the two against each other.

2. **Focus window and cleaning.** In the multi-focus snippet, vary the focus-measure smoothing $\sigma \in \{1, 2, 4\}$ and the median window $k \in \{1, 15, 31, 61\}$ ($k = 1$ means no cleaning), always followed by the soft blend with a Gaussian of standard deviation 2. Report PSNR. Which setting wins, and why might the answer differ on real photographs?

   Hint

   In our run all settings lie between 39.43 and 40.05 dB; the best is $\sigma = 2$ with no median cleaning (40.05 dB), and the largest window, $k = 61$, is the worst for every σ (about 39.4–39.5 dB). Without cleaning, the soft blend mixes A and B in the speckled flat regions, which averages their independent noise and slightly raises PSNR there; a large median window misplaces the map near the boundary of the disk. On real photographs, uncleaned speckle shows up as visible blotches where the sources differ (noise,

small misalignment, the defocus spread effect), so cleaning usually matters more for appearance than for PSNR on a toy.

3. **Area-based selection.** Replace `pick_stronger` in the pyramid snippet by a rule that compares *local energies*: smooth $(L_\ell^A)^2$ and $(L_\ell^B)^2$ with a Gaussian of standard deviation s and choose the coefficient whose energy is larger. Try $s \in \{1, 1.5, 3\}$ and compare with the per-coefficient rule.

Hint

Use `ndi.gaussian_filter(p**2, s)`. In our run PSNR rose from 37.60 dB (per coefficient) to 39.14, 39.42 and 39.62 dB for $s = 1, 1.5, 3$, close to the decision map. Smoothing the activity makes neighboring coefficients take the same decision, which is what the area-based rule and consistency verification of Li et al. aim for: in multi-focus fusion, focus is a property of regions, not of single coefficients.

4. **Pyramid depth and quality measures in exposure fusion.** (a) Rerun exposure fusion with pyramids of $n = 0, 1, 3, 5, 7, 9$ levels ($n = 0$ is the per-pixel blend) and report the well-exposed fraction; look at the images. (b) Use one quality measure at a time (contrast, saturation or well-exposedness only) at full depth. What do you observe?

Hint

(a) In our run the fraction was 72.3% for $n = 0$, then 72.2%, 70.9%, 69.6% and 68.5% for $n = 1, 3, 5, 7$, and 73.8% for $n = 9$; the mean absolute difference from the 9-level result was still 0.068 at $n = 7$. Shallow pyramids keep seams and low-frequency brightness errors, because the coarsest levels, where the weights are blended most smoothly, are missing; the result changes a lot even between 7 and 9 levels on a 512-pixel image. (b) Well-exposedness alone gave the highest fraction (77.5%), contrast alone 69.1% and saturation alone 71.5%. Mertens et al. also found that well-exposedness by itself gives fairly good images; contrast and saturation matter for detail and color, which this count does not measure.

5. **How noise inflates spatial frequency.** Let $F' = F + n$ where n is independent zero-mean noise with variance σ^2 at every pixel. (a) Show that the expected SF^2 grows by about $4\sigma^2$. (b) Check it on the toy: add noise with $\sigma = 0.02, 0.04, 0.08$ to the average and compare the measured SF with $\sqrt{\mathrm{SF}^2 + 4\sigma^2}$.

Hint

(a) A horizontal difference of F' is the difference of F plus $n(i,j) - n(i,j-1)$, whose variance is $2\sigma^2$ and which is uncorrelated with F; so $\mathbb{E}[\mathrm{RF}'^2] \approx \mathrm{RF}^2 + 2\sigma^2$, the same for CF, and $\mathbb{E}[\mathrm{SF}'^2] \approx \mathrm{SF}^2 + 4\sigma^2$. (b) In our run (with a different random seed from the snippet), the measured and predicted SF were 0.0528 and 0.0528 for $\sigma = 0.02$, 0.0871 and 0.0871 for 0.04, and 0.1631 and 0.1637 for 0.08; the small gap at 0.08 comes from clipping to $[0, 1]$. EN rose from 6.69 to 6.99, 7.20 and 7.49 bits.

6. **Mutual information rewards copying.** (a) Show that if $F = I_A$, then $\mathrm{MI}_{A,F} + \mathrm{MI}_{B,F} = H(I_A) + \mathrm{MI}(I_A, I_B)$, where H is the entropy of the binned image. (b) Verify it on the toy with 64 bins. (c) Compute the MI score of the average and of the visible image with 16, 64 and 256 bins. Does the ranking depend on the bin count? Does the value?

Hint

(a) $\mathrm{MI}(X, X) = H(X)$, because knowing X removes all uncertainty about X. (b) In our run $H(\mathrm{vis}) = 4.728$ bits and $\mathrm{MI}(\mathrm{ir}, \mathrm{vis}) = 0.995$ bits, which add up to 5.723, the MI score of "visible only" in the snippet. (c) The average scored 2.64, 4.33 and 5.00 bits and the visible image 3.42, 5.72 and 7.82 bits: the ranking stayed, but the values more than doubled from 16 to 256 bins for the visible image, so MI values from papers that use different bin counts or implementations cannot be compared.

Supplementary notes

These notes are optional. Each takes one point from the main text and goes into finer detail; nothing in the main text, code or exercises depends on them.

The defocus spread effect

This note explains why selecting the sharpest source cannot produce a correct all-in-focus image near depth edges, and what that means for methods and data.

The phenomenon. A real lens blurs an out-of-focus *object*, not an out-of-focus *region of the image*. When the near object is out of focus, its blurred image extends beyond its own outline and covers part of the far background. This is the **defocus spread effect**; Xu et al. describe it as a phenomenon of real-world multi-focus images that is not obvious in simulated data or in the Lytro pairs [6].

A layered model. Write the scene as a near layer with a mask $m(i, j) \in [0, 1]$ (1 on the object) and an image x, and a far layer with the same image x elsewhere (a simplification: in reality the far layer continues behind the object). With a defocus kernel h (here a Gaussian), the two shots are

$$I_{\mathrm{near}} = m\,x + (1 - m)\,(h * x), \qquad I_{\mathrm{far}} = h * (m\,x) + (1 - h * m)\,x,$$

where $*$ is convolution, I_{near} is focused on the near layer and I_{far} on the far one. In I_{far}, the term $h * (mx)$ is the blurred object, and $1 - h * m$ is the fraction of the background still visible through its blur. Just outside the outline, $0 < h * m < 1$: the far-focused shot shows the background *mixed with* a blurred copy of the object, while the near-focused shot shows the blurred background. Neither equals the truth x, so no decision map, however clever, can pick a correct pixel there. The width of the affected band is about the radius of the blur kernel.

A numerical check. The snippet continues from the multi-focus snippet (it uses `x`, `ndi`, `np`, `focus_measure` and `psnr`). It builds the layered model with the cameraman's silhouette as the near

layer and compares the decision map with an **oracle selection** that, at each pixel, picks whichever source is closer to the truth, the best result any per-pixel selection could achieve.

```
# Layered defocus: the cameraman's dark silhouette is the near layer, everything else the
 ↪  far layer.
dark = ndi.binary_opening(x < 0.2, iterations=2)
labels, n = ndi.label(dark)
fg = ndi.binary_fill_holes(labels == 1 + np.argmax(ndi.sum(dark, labels, range(1, n +
 ↪  1)))).astype(float)
blur = lambda img: ndi.gaussian_filter(img, 5.0)
A2 = fg * x + (1 - fg) * blur(x)              # focused near: the far layer is blurred
B2 = blur(fg * x) + (1 - blur(fg)) * x        # focused far: the blurred near layer spills
 ↪  outward
edge = ndi.binary_dilation(fg > 0.5, iterations=10) & ~ndi.binary_erosion(fg > 0.5,
 ↪  iterations=10)

clean = ndi.median_filter((focus_measure(A2) > focus_measure(B2)).astype(float), 31) > 0.5
w = ndi.gaussian_filter(clean.astype(float), 2.0)
decision = w * A2 + (1 - w) * B2
oracle = np.where(np.abs(A2 - x) <= np.abs(B2 - x), A2, B2)   # the best possible per-pixel
 ↪  selection
for name, F in [("decision map", decision), ("oracle selection", oracle)]:
    print(f"{name:16s} PSNR {psnr(x, F, data_range=1):.2f} dB overall, "
          f"{psnr(x[edge], F[edge], data_range=1):.2f} dB within 10 px of the depth edge "
          f"({edge.mean():.0%} of pixels)")
```

```
decision map     PSNR 31.08 dB overall, 21.47 dB within 10 px of the depth edge (11% of
 ↪  pixels)
oracle selection PSNR 32.65 dB overall, 23.04 dB within 10 px of the depth edge (11% of
 ↪  pixels)
```

Decision-map fusion falls from about 40 dB on the simple toy of the main text to about 31 dB on this layered model, and to about 21 dB within 10 pixels of the depth edge. The oracle does only slightly better (about 33 and 23 dB): no per-pixel selection can do much better, because near the edge neither source contains the right value. The error is a property of the data, not of the decision rule.

Consequences.

- **Training data.** Deep multi-focus methods trained on simulated pairs (blur pasted inside a mask) never see the effect; a benchmark study suggests this is why they generalize poorly to real pairs with strong defocus spread [7].
- **Evaluation.** Test sets without the effect, such as the 20 Lytro pairs, cannot separate methods on this point; MFFW was collected for that purpose and most methods fail to produce satisfactory results on it [6].

- **What a method can do.** Since selection cannot recover the hidden background, a method must either accept a soft band (and hide it with a smooth transition), generate the missing content (a generative prior), or use more information, such as more focus settings or depth.

Building and collapsing a Laplacian pyramid

This note writes out the Laplacian pyramid used throughout the chapter: the filter, the size bookkeeping, why the collapse is exact, and the pitfalls of a real implementation.

The two operators. REDUCE convolves the image with a small low-pass kernel k and keeps every second row and column; EXPAND inserts zeros between samples, convolves with $4k$ (the factor 4 restores the average brightness lost by the zeros) and returns an image twice as large. OpenCV's `pyrDown` and `pyrUp` use the separable kernel $k = \frac{1}{256}\mathbf{v}\mathbf{v}^\top$ with $\mathbf{v} = (1, 4, 6, 4, 1)^\top$, a binomial approximation of a Gaussian; you can check this on an impulse:

```
import numpy as np
import cv2

small = np.zeros((5, 5), np.float32)
small[2, 2] = 64.0
print(cv2.pyrUp(small)[2:7, 2:7])              # zeros inserted, then 4 x the kernel: 64 * 4
 ↪  * v v^T / 256
impulse = np.zeros((9, 9), np.float32)
impulse[4, 4] = 256.0
print(cv2.pyrDown(impulse)[1:4, 1:4])          # blur, then keep every 2nd pixel: taps at
 ↪  offsets 0 and +-2
print(cv2.pyrDown(np.zeros((7, 5), np.float32)).shape)   # odd sizes round up: (7, 5) -> (4,
 ↪  3)
```

```
[[ 1.  4.  6.  4.  1.]
 [ 4. 16. 24. 16.  4.]
 [ 6. 24. 36. 24.  6.]
 [ 4. 16. 24. 16.  4.]
 [ 1.  4.  6.  4.  1.]]
[[ 1.  6.  1.]
 [ 6. 36.  6.]
 [ 1.  6.  1.]]
(4, 3)
```

Sizes. An $H \times W$ level reduces to $\lceil H/2 \rceil \times \lceil W/2 \rceil$. For odd sizes, EXPAND of the smaller level is one pixel too large unless you tell it the target size, which is why the chapter's code passes `dstsize` to `pyrUp` (the last line of the output shows the $\lceil \cdot/2 \rceil$ rule). Forgetting it is a common bug in pyramid code: the subtraction $G_\ell - \text{EXPAND}(G_{\ell+1})$ fails or silently broadcasts.

Why the collapse is exact. By definition $L_\ell = G_\ell - \text{EXPAND}(G_{\ell+1})$, so $G_\ell = L_\ell + \text{EXPAND}(G_{\ell+1})$. Starting from $G_N = L_N$ and applying this identity for $\ell = N-1, \dots, 0$ rebuilds every G_ℓ and finally $G_0 = I$. Nothing about the kernel is used: the pyramid stores the prediction error of EXPAND, so any EXPAND works. What *does* depend on the kernel is what the levels mean: with a good low-pass kernel, L_ℓ is a band-pass image that responds to features of one size, which is what makes the max-abs rule meaningful.

Fused pyramids are not pyramids of any image. After fusion, the stack $\{L_\ell^F\}$ is generally not the Laplacian pyramid of its own collapse: collapsing and re-decomposing gives different levels. This is harmless (we only collapse once), but it is one reason fused results can contain small halos: coefficients chosen independently at neighboring levels need not be consistent with one another.

Storage. Level ℓ has about $HW/4^\ell$ values, so the whole pyramid has about $HW(1 + \frac{1}{4} + \frac{1}{16} + \dots) \approx \frac{4}{3}HW$ values: the Laplacian pyramid is slightly overcomplete, whereas the orthogonal wavelet transform of IP 07 is critically sampled (exactly HW values) but splits each scale into three orientation bands.

How many levels. Each level halves the size, so at most about $\log_2 \min(H, W)$ levels fit. For fusion rules that average the coarse base, more levels mean the averaging happens at a coarser scale, which reduces low-frequency artifacts; Mertens et al. note that a taller pyramid partly fixes their spurious low-frequency brightness changes, but that the height is limited by the size of the filter relative to the image [5]. Exercise 4 measures this effect on the toy.

Pitfalls.

- **Data type.** Laplacian levels are signed. Building them from `uint8` arrays clips or wraps negative values; convert to `float32` first, as the chapter's code does.
- **Color.** Decompose each channel separately but use *one* decision (or one weight map) for all channels, otherwise the channels can come from different sources and produce color fringes.
- **Borders.** Convolution near the image border needs an extension rule (OpenCV's default reflects the image about its border pixels). Different rules give slightly different coefficients near the border, a common source of small differences between implementations.

Sparse-representation fusion

This note describes the sparse-representation fusion scheme summarized in the main text: the coding model, the fusion steps and where its cost comes from.

The model. A **dictionary** $\mathbf{D} \in \mathbb{R}^{n \times m}$ has m columns ("atoms"), each a vectorized image patch of n pixels; it is **overcomplete** when $m > n$. A patch $\mathbf{p} \in \mathbb{R}^n$ has a **sparse representation** $\boldsymbol{\alpha} \in \mathbb{R}^m$ if $\mathbf{p} \approx \mathbf{D}\boldsymbol{\alpha}$ with few non-zero entries in $\boldsymbol{\alpha}$. A standard way to find one is to solve

$$\min_{\boldsymbol{\alpha}} \|\boldsymbol{\alpha}\|_0 \quad \text{subject to} \quad \|\mathbf{p} - \mathbf{D}\boldsymbol{\alpha}\|_2^2 \le \varepsilon,$$

where $\|\boldsymbol{\alpha}\|_0$ counts the non-zero entries and $\varepsilon \ge 0$ is the allowed approximation error; greedy pursuit algorithms, which add one atom at a time, solve it approximately. The dictionary can be fixed (for example DCT atoms) or learned from training patches or from the source images themselves [15].

Fusion steps. For two registered sources:

1. Cut both sources into overlapping patches at the same positions, typically with a sliding window, and vectorize each patch.
2. Code each patch of each source over the same dictionary, giving $\boldsymbol{\alpha}_A$ and $\boldsymbol{\alpha}_B$ for every position.
3. Measure the **activity** of each code, for example its ℓ_1 norm $\|\boldsymbol{\alpha}\|_1$, and keep the code with the larger activity: the **choose-max** rule [14]. Sharp, textured patches need more or larger coefficients than blurred ones, so in multi-focus fusion this selects the in-focus patch.
4. Reconstruct each fused patch as $\mathbf{D}\boldsymbol{\alpha}_F$, put the patches back, and average the overlapping pixels.

Fusion and restoration together. The tolerance ε decides what the codes represent. If the sources are noisy and ε is set near the noise energy of a patch, the codes describe the image structure but not the noise, so the reconstruction is denoised while it is fused; Yang and Li obtain fusion and restoration in one scheme by changing this approximation criterion [14].

Where the cost comes from. With a stride of one pixel there is a patch per pixel and per source, and each needs an iterative sparse coding over m atoms. This is why SR methods were the slowest family in the multi-focus benchmark of Zhang, at more than seven minutes per image pair [7]. Larger strides are faster but make the patch decisions coarser.

Pitfalls. The choose-max rule decides per patch, so neighboring patches can come from different sources; the averaging of overlaps softens the transitions but can also blur the boundary between an in-focus and an out-of-focus region. The patch mean (its brightness) is part of the activity unless it is removed before coding, in which case it must be fused separately, for example by averaging, as the base of a pyramid is.

Mutual information as a fusion metric

This note explains how the MI score of the main text is computed, why it favors copying a source, how strongly it depends on the histogram, and what normalization changes.

Computation. For a source X and the fused image F, both in $[0, 1]$, divide the range into b bins, count the pairs $(X(i,j), F(i,j))$ in a $b \times b$ joint histogram, divide by the number of pixels to get $p_{X,F}$, and sum $p \log_2(p/p_X p_F)$ over the non-empty cells. The fusion score adds the terms for the two sources. This is a **plug-in estimate**: it treats the histogram frequencies as the true probabilities,

so its value depends on the bin count and is biased upward when there are many bins and few pixels (IT 01 discusses estimation).

Why copying scores well. MI is bounded by the entropies, $\mathrm{MI}(X,F) \le \min\,(H(X),H(F))$, where H is the entropy of the binned image, and the bound is reached when F determines X, for example when $F = X$. If $F = I_A$, the score is $\mathrm{MI}(I_A,I_A) + \mathrm{MI}(I_B,I_A) = H(I_A) + \mathrm{MI}(I_A,I_B)$. A genuine fusion must change F relative to I_A wherever it brings in information from I_B, which lowers the first term; unless the second term rises by as much, the score falls. Copying the source with the larger entropy is therefore a strong baseline that a fusion method may not beat. Hossny, Nahavandi and Creighton gave an analytical argument that the classic MI sum is unsuitable as a fusion performance measure [21], and benchmark toolkits implement a normalized MI attributed to them [18]:

$$\mathrm{NMI} = 2\left[\frac{\mathrm{MI}(I_A,F)}{H(I_A)+H(F)} + \frac{\mathrm{MI}(I_B,F)}{H(I_B)+H(F)}\right],$$

where each term is a mutual information divided by the sum of the two entropies involved, so each lies between 0 and 1/2 (because $\mathrm{MI}(X,F) \le \min(H(X),H(F))$, which is at most their average) and NMI between 0 and 2.

What normalization does and does not fix. The snippet continues from the infrared–visible snippet and adds NMI to its candidates.

```
def H(X, bins=64):
    p = np.histogram(X, bins, (0, 1))[0] / X.size
    p = p[p > 0]
    return -np.sum(p * np.log2(p))

def nmi(A, B, F, bins=64):
    """Normalized MI: each term divided by the sum of the two entropies."""
    return 2 * (mutual_info(A, F, bins) / (H(A, bins) + H(F, bins))
                + mutual_info(B, F, bins) / (H(B, bins) + H(F, bins)))

for name, F in candidates.items():
    print(f"{name:16s} MI {mutual_info(vis, F) + mutual_info(ir, F):.2f}  NMI {nmi(vis, ir,
     ↪  F):.3f}")
```

```
visible only     MI 5.72  NMI 1.230
average          MI 4.33  NMI 0.948
pyramid fusion   MI 3.75  NMI 0.760
average + noise  MI 1.41  NMI 0.284
```

NMI puts the values on a bounded scale, but in our toy it ranks the candidates exactly as MI does: copying the visible image still wins, with $F = I_A$ giving $\mathrm{MI}(I_A,F)/(H(I_A)+H(F)) = 1/2$, the largest value a term can take. Neither MI nor NMI asks whether the information that was kept is the *useful* information (here, the warm person).

Practical rules. Report the bin count and the implementation; do not compare MI values across papers that may use different ones; and never use MI alone. If you use MI as a training loss, add a term that the copy-a-source solution violates.

Training signals of representative fusion networks

This note compares what representative deep fusion networks are trained to do: with no ground-truth fused image, the training signal, more than the architecture, decides what the output looks like.

Reconstruction, then a fixed rule (DenseFuse). The encoder–decoder of DenseFuse [30] is trained only to reconstruct visible images, with a loss $\mathcal{L} = \lambda \mathcal{L}_{\text{ssim}} + \mathcal{L}_p$, where $\mathcal{L}_p$ is a pixel loss between output and input, $\mathcal{L}_{\text{ssim}} = 1 - \text{SSIM}(O, I)$ the structural loss between output O and input I, and λ their balance. At test time both sources are encoded and their feature maps are merged either by addition or by an ℓ_1-norm-based strategy that weights each source's features by their activity, and the decoder turns the merged features into the fused image. *Implication:* the network never sees a fusion example; the fusion rule is as hand-made as in classical methods, only applied to learned features instead of pyramid coefficients.

A perceptual quality index as the loss (DeepFuse). DeepFuse [31] converts the exposures to YCbCr, fuses the luminance channels with a CNN whose loss is MEF-SSIM [25] computed between the fused luminance and the input stack, and fuses the chroma channels with a weighted rule outside the network. MEF-SSIM compares the structure of each fused patch with a desired structure built from the input patches, so the network needs no reference image. *Implication:* the network learns MEF-SSIM's notion of quality, including whatever MEF-SSIM does not see; a good MEF-SSIM score after training is partly a statement about the loss, not an independent confirmation.

Adaptive weights from the sources (U2Fusion). U2Fusion [32] extracts features of each source, measures how much information they contain, and turns the measurement into **adaptive preservation degrees** that weight a similarity loss between the fused image and each source. Because the degrees are computed per image pair, the same loss works for multi-modal, multi-exposure and multi-focus inputs, and one network is trained on the tasks in sequence while avoiding the loss of what it learned earlier. *Implication:* the definition of the answer moves from a fixed weight to a measurement; the result is as good as the information measure is at saying which source matters where.

Hand-balanced terms (SwinFusion). SwinFusion [34] uses an SSIM loss, a texture loss and an intensity loss, and models long-range dependencies with self-attention within each source and cross-attention between them. *Implication:* the three weights trade structure, texture and brightness against each other, exactly as λ did for the loss minimizer of the main text.

An adversary (FusionGAN). In FusionGAN [39] the generator is pushed by its content loss toward the infrared intensities plus the visible gradients, and a discriminator that compares the fused image with visible images pushes it to contain more visible detail. The authors describe their results as looking like sharpened infrared images with clearly highlighted targets [39]. *Implication:* the discriminator supplies a learned notion of "visible-looking detail", but adversarial training can also add detail that is in neither source; TarDAL later used two discriminators, one per modality [27].

A decomposition constraint (CDDFuse). CDDFuse [38] extracts low-frequency, global features with transformer blocks and high-frequency, local features with invertible blocks, and adds a correlation-driven loss that makes the low-frequency features of the two modalities correlated and their high-frequency features uncorrelated. *Implication:* the loss encodes a model of the data, shared base and modality-specific detail, which is the learned counterpart of the base/detail split of classical fusion.

No fusion training at all (DDFM). DDFM [41] formulates fusion as conditional generation under a denoising diffusion sampler: an unconditional diffusion model pretrained on natural images supplies the prior, and at each sampling step a maximum-likelihood subproblem, modeled with latent variables and solved by expectation–maximization, pulls the sample toward consistency with both sources. *Implication:* the "loss" becomes a statistical model of how the fused image relates to the sources, plus a generative prior that keeps the result natural-looking; no fine-tuning of the diffusion model is needed, but sampling is iterative and therefore much slower than one forward pass of a network.

A self-supervised imaging prior (EMMA). EMMA [52] trains a fusion module together with a pseudo-sensing module that imitates how the sensors would image the fused scene, and requires the whole loop to behave consistently under transformations to which natural imaging is equivariant. *Implication:* instead of hand-weighting terms, the training signal asks that the fused image could have produced the sources, an idea close to cycle consistency.

Summary.

Method	What is optimized	What decides the fused image
DenseFuse	Image reconstruction	A hand-made feature-merging rule
DeepFuse	MEF-SSIM on the luminance	MEF-SSIM's model of quality
U2Fusion	Similarity to sources, adaptively weighted	An information measure on features
SwinFusion	SSIM + texture + intensity	The three loss weights
FusionGAN	Content loss + adversarial loss	Infrared intensity, visible gradients and a discriminator
CDDFuse	Includes a correlation-driven decomposition loss	A shared-base / specific-detail model
DDFM	Nothing (no fusion training)	A pretrained diffusion prior + a likelihood model
EMMA	Self-supervised equivariant consistency	Consistency between fused image and sources

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Exposure fusion (Mertens, Kautz and Van Reeth)

Problem. Turning a bracketed exposure sequence into a good displayable photograph normally takes two steps: assemble an HDR radiance map, which needs the camera response curve, and tone-map it back to a low dynamic range. The authors ask whether the HDR step can be skipped [5].

Idea. Treat the exposures as a stack and collapse it with per-pixel weights from three simple quality measures (contrast, saturation, well-exposedness), combined by a product with exponents and normalized over the stack. Blend in a multiresolution fashion: Laplacian pyramids of the images, Gaussian pyramids of the weights. The method assumes aligned exposures, possibly after registration.

What it showed. Results whose quality the authors find comparable to existing tone-mapping operators, without camera calibration and at a computational cost of seconds in an unoptimized implementation; the same weights also allow a flash exposure to be included in the stack. Their comparison figures show why each design step matters: per-pixel blending gives seams, Gaussian-smoothed weights give halos, an edge-aware cross-bilateral smoothing removes only some of them, and multiresolution blending removes both. Compared with earlier pyramid fusion that selects the most salient coefficients, whose results on an exposure stack are too dark and show color shifts, their blending stays more faithful to the input sequence.

What to check when you read it. The comparison with tone mapping is explicitly informal, because tone mappers have implementation-specific parameters and settings; the paper's comparisons are visual, so judge it by its figures. Look at the limitations: a spurious low-frequency brightness change can appear when exposures differ strongly, only partly fixed by a taller pyramid, and flash shadows and reflections are not removed. Compare the figure that switches off two of the three measures with the text: well-exposedness alone gives fairly good but sometimes unnatural images because it always favors intensities near 0.5.

Why it matters here. It is the reference classical method for multi-exposure fusion, the source of the weighted-blending formulation that many learned methods (for example weight-predicting networks such as MEF-Net) keep, and the clearest demonstration of why fusion should blend in a multiresolution way. OpenCV ships an implementation, which the chapter uses as a check.

VIFB: a visible and infrared image fusion benchmark

Problem. Infrared–visible fusion papers used different test images, different metrics and their own implementations of competing methods, so their comparisons could not be checked or combined [17].

Idea. Collect a test set of 21 visible–infrared pairs from the Internet and from tracking datasets, put 20 fusion algorithms (conventional and deep) into one code library, implement 13 metrics in four families (information theory, image features, structural similarity, human perception) and publish all results.

What it showed. No method beat the others on all or most metrics. The three overall leaders each won a different family: one the information-theory metrics, one the feature-based metrics, one the structural metrics. The two methods with the best quantitative scores looked poor in the qualitative comparison: one introduced artifacts, and the other lost visible detail or failed to show a target that is invisible in the overexposed visible image. In 2020, the deep methods in the library were on average slightly worse than the best conventional ones, unlike in detection or tracking. Runtimes differed by orders of magnitude, which matters when fusion feeds a real-time tracker.

What to check when you read it. Read the metric table with the family of each metric in mind: a method "best on three metrics" may be best on three metrics of one family. Check which version you read: the arXiv version notes corrections to a table and a paragraph relative to the workshop version. With 21 pairs, small differences in average metric values are not meaningful; look at per-image results and at the images. The benchmark has no ground truth and no downstream task, so it measures agreement with the metrics, not usefulness.

Why it matters here. It is the cleanest demonstration, on real methods, of the chapter's main warning about evaluation: metrics from different families disagree, and the numerically best fusion can be the visually worst. Later benchmarks for multi-focus (MFIFB) and multi-exposure fusion (MEFB) follow the same design [7], [18].

The infrared–visible fusion survey of Liu et al. (TPAMI 2025)

Problem. Learning-based infrared–visible fusion grew quickly after 2018, and earlier surveys did not organize the newer work on registration and on fusion for downstream tasks [2].

Idea. A multi-dimensional taxonomy: methods for **visual enhancement** (autoencoder-, CNN-, GAN-, transformer-based and others, with their losses), methods for **data compatibility** (registration-free fusion of misaligned pairs, by style transfer or by a shared feature space) and methods for **task adaptation** (fusion for detection and for segmentation). A lookup table summarizes the core idea of each method, and the survey covers more than 180 approaches. It then compares methods quantitatively and qualitatively on registration, fusion and downstream

tasks.

What it showed. Its comparisons use nine fusion metrics, five that compare with the sources (MI, VIF, CC, SCD and $Q^{AB/F}$) and four that look only at the fused image (EN, SF, SD and average gradient), plus detection and segmentation results and efficiency. Its future-trends section lists misaligned and attacked data, benchmarks with registration ground truth and more diverse scenes, better evaluation metrics, lightweight design and combination with more tasks.

What to check when you read it. The survey calls the source-comparing metrics "reference-based"; the reference is a source image, not a ground-truth fusion. Several of the compared methods come from the authors' own group (for example TarDAL, SegMiF and UMF-CMGR share authors with the survey), which is common in surveys but a reason to read the comparison tables with the same care as a method paper's. Check, for each table, which dataset, which metric family and which downstream network is used, since a fusion method's rank changes with all three.

Why it matters here. It is the most recent broad map of the field this chapter could verify, and its three dimensions (data, fusion, task) are the same three questions the chapter asks: are the sources aligned, what should the fused image keep, and what is it for.

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DLCV 06 • Object Tracking: From Single-Object to Multi-Object

How computer vision follows objects through video: re-identification, single-object trackers from correlation filters to transformers and SAM 2, and multi-object tracking with Kalman filters, Hungarian matching, end-to-end models and the MOTA/IDF1/HOTA metrics.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-06-object-tracking/

What you'll learn

- Where tracking sits after classification and detection, and the two jobs every tracker does: follow motion across frames and keep each object's identity.
- A clear taxonomy: re-identification versus temporal tracking, single-object (SOT/VOT) versus multi-object tracking (MOT), and multi-target multi-camera tracking (MTMCT).
- How single-object trackers evolved: correlation filters (MOSSE, KCF), Siamese networks (SiamFC, SiamRPN), one-stream transformers (OSTrack, SeqTrack) and video segmentation models (SAM 2, SAMURAI, SAM 3).
- Multi-object tracking in depth: the tracking-by-detection pipeline (detector, Kalman filter, appearance embedding, Hungarian assignment), the SORT family (SORT, DeepSORT, ByteTrack, OC-SORT, BoT-SORT), joint detection-and-embedding (JDE, FairMOT), end-to-end query-based trackers (MOTR, MOTRv2, MOTIP) and open-vocabulary tracking.
- How trackers are measured: CMC and mAP for Re-ID, success and precision for SOT, and MOTA, IDF1 and HOTA for MOT, including why HOTA was introduced.
- A working NumPy tracker you can run, with measured ID switches and metrics on a synthetic scene.

The big picture

A detector looks at one picture and says "there are three people here". A tracker watches a video and says "this is the *same* person as one second ago, and that one just walked behind a pillar and will come out again". Tracking adds time and memory to detection.

> **Explain like I'm five**
>
> Imagine you are watching a soccer game with your friend, and you each pick a favorite player. Finding every player on the field in one photo is **detection**. Keeping your eyes on *your* player as she runs, turns and gets hidden behind others for a moment is **single-object tracking**. Putting a numbered sticker on *every* player and making sure the stickers never get swapped, even when two players crash into each other, is **multi-object tracking**. And recognizing your player again when the TV switches to a different camera is **re-identification**.

Why it matters: tracking is how a self-driving car knows that a pedestrian is *walking toward* the road rather than just standing near it, how sports analytics measures how far each player ran, how a warehouse counts forklifts and robots, and how a video editor keeps a mask glued to a moving object. Each of these needs not just *where* things are, but *which one is which* over time.

About this tutorial

This tutorial is expanded from an earlier Medium post (written in Traditional Chinese), "物件追蹤 Object Tracking 簡介" [1], keeping its structure and adding the technical depth, a runnable tracker and an up-to-date survey. You can read the original post here.

Related chapters. DLCV 06A estimates human pose and keypoints, which trackers often use as appearance-free cues, and DLCV 06B recognizes what people are doing over time (action recognition).

From classification to detection to tracking

Plain version. Classification answers *what* is in an image. Detection answers *what and where*. Tracking answers *what, where, and which one, over time*.

The progression in [1] is a useful map of computer vision tasks:

1. **Image classification** maps an image I to a label y.
2. **Object detection** maps an image to a set of boxes with labels and confidence scores, $\mathcal{D} = \{(b_j, c_j, s_j)\}_{j=1}^{N}$, where b_j is a bounding box, c_j a class and $s_j \in [0, 1]$ a score. Detection imitates the human ability to locate targets in a scene; it is covered in depth in DLCV 03 ·Object Detection.
3. **Object tracking** takes a sequence of frames $I_1, \dots, I_T$ and outputs **trajectories**: for each object k, a sequence of boxes (or masks) $(b_t^k)_{t \in \mathcal{T}_k}$ over the set of frames $\mathcal{T}_k$ in which it is visible, all carrying the same identity k.

The key new ingredient is the identity label. Detection is stateless: frame t and frame $t + 1$ are processed independently. Tracking is stateful: it carries information forward.

What tracking does

Plain version. Tracking imitates how people perceive moving things: we predict where something will go, look there, and keep treating it as the same object even when its appearance changes or it is briefly hidden.

More precisely, every tracker solves two coupled sub-problems:

- **Localization over time** (motion perception): estimate where each target is in each frame, ideally using the past to predict the future. This is what lets a tracker survive blur, missed detections and short occlusions.
- **Identity maintenance** (data association): decide which observation in frame t belongs to which target from frame $t-1$. This is what prevents two people who cross paths from swapping labels.

Single-object tracking puts most of its effort into the first problem, since there is only one identity. Multi-object tracking spends most of its effort on the second. Re-identification is the second problem with *no* motion information at all.

The difficulties that [1] lists for each family come from these two jobs: appearance changes (pose, lighting, scale, viewpoint), occlusion by other objects or the scene, similar-looking distractors, fast or non-linear motion, camera motion, and targets entering and leaving the scene.

A taxonomy of tracking

Plain version. You can sort trackers by *what* they follow (one object or many, one camera or many) and by *how* they keep identity (by appearance alone, or by following motion frame to frame).

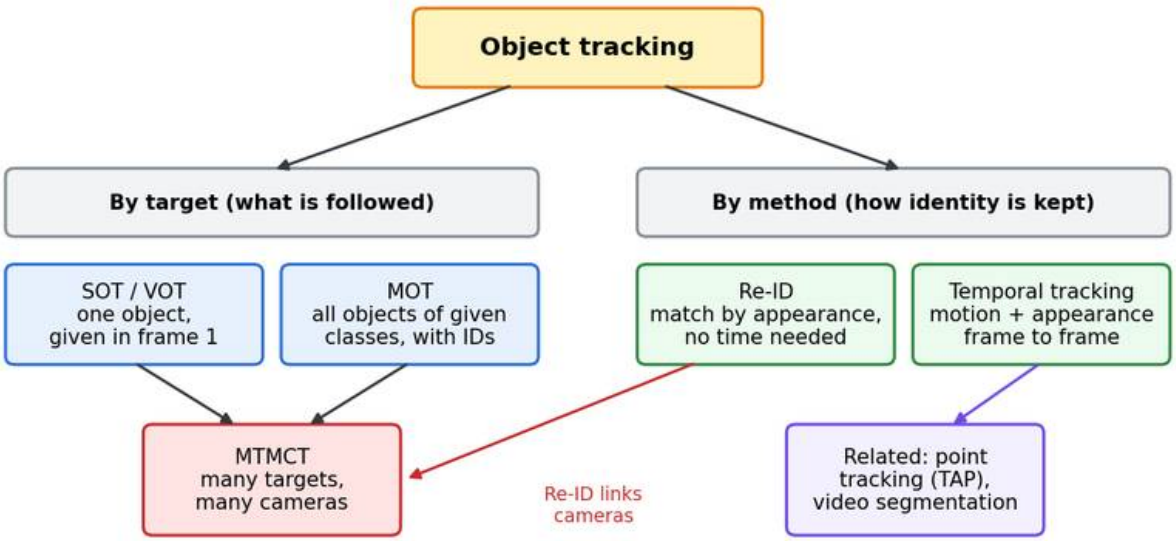


Figure 1 — A taxonomy of tracking. By target: single-object (SOT, also called VOT) and multi-object (MOT). By method: appearance-only re-identification and temporal tracking. The hardest combined setting is multi-target multi-camera tracking (MTMCT).

Following [1]:

- **By method.** *Re-identification* (*Re-ID*) matches a target across images that may be far apart in time or come from different cameras, using appearance (and sometimes pose) alone. *Temporal object tracking* follows targets frame by frame in one video stream, where motion continuity is available.
- **By target.** *Visual object tracking* (*VOT*), also called *single-object tracking* (*SOT*), follows one arbitrary object specified by a box in the first frame. *Multi-object tracking* (*MOT*) follows all objects of given classes (for example, every pedestrian) through consecutive frames and maintains a consistent ID for each.
- **The combined goal.** *Multi-target multi-camera tracking* (*MTMCT*) tracks many targets across a network of cameras and keeps the same identity when a person leaves one camera's view and appears in another. It combines MOT within each camera with Re-ID between cameras.

Most of the classical literature, and most benchmarks, use **pedestrians** as targets [1], because surveillance and autonomous driving made people the most valuable class to track. Modern benchmarks add dancers, athletes, vehicles and open-vocabulary categories.

Person re-identification

Plain version. Re-ID is a search problem: given a photo of a person from one camera, find the same person in photos taken by other cameras, ranked from most to least likely.

Problem formulation

Re-ID is posed as **retrieval** [7]. A *query* image q and a *gallery* $\mathcal{G} = \{g_1, \dots, g_M\}$ of detected person crops from other cameras are mapped by a network f_θ to embeddings. The gallery is ranked by a distance, usually cosine distance

$$d(q, g) = 1 - \frac{f_\theta(q)^\top f_\theta(g)}{\|f_\theta(q)\| \, \|f_\theta(g)\|},$$

where $f_\theta(\cdot) \in \mathbb{R}^D$ is the D-dimensional embedding produced by the network with parameters θ. Training makes embeddings of the same identity close and of different identities far apart. A typical recipe combines an identity classification loss (cross-entropy over the C training identities) with a **triplet loss**

$$\mathcal{L}_{\text{tri}} = \left[\, d(a, p) - d(a, n) + m \,\right]_+,$$

where a is an anchor image, p a positive (same identity), n a negative (different identity), $m > 0$ a margin and $[x]_+ = \max(x, 0)$. Test identities never appear in training, so Re-ID is an open-set problem: the network must learn a general notion of "same person".

Challenges

[1] groups the difficulties into **data factors** and **environment factors**. Data factors: limited labeled identities, noisy boxes from the detector, partial bodies, low resolution. Environment factors: different viewpoints and camera color responses, lighting, occlusion, background clutter, and people who change clothes. Re-ID also raises clear **privacy** concerns, which is part of why surveillance-style datasets have been withdrawn (below). The original post names city-scale surveillance systems as the main application; today Re-ID is just as common as the appearance module inside MOT trackers and in retail and sports analytics.

Representative models

The three models below all keep the retrieval setup above and change how the embedding network f_θ is trained. A plain ResNet-50 with a set of training "tricks" is a strong, reproducible baseline [14]; the centroid triplet loss compares images with identity *centroids* rather than with single images [15]; and CLIP-ReID adapts the vision-language model CLIP to Re-ID, where no text labels exist [16]. The point to remember: careful training of a standard backbone already goes a long way, so new Re-ID ideas should be compared against such a baseline.

Optional: More Re-ID models: a strong baseline, centroids and CLIP-ReID (in the Supplementary notes). The training tricks of the baseline, how the centroid loss is used at retrieval time, and CLIP-ReID's two training stages.

Datasets

Dataset	Content	Status
Market-1501 [12]	1,501 identities, 6 cameras, detector-cropped boxes	available
MSMT17 [13]	4,101 identities, 15 indoor and outdoor cameras	available
DukeMTMC [10] (and DukeMTMC-reID)	8 synchronized campus cameras	**withdrawn in 2019**

DukeMTMC was introduced with the IDF1 metric for multi-camera tracking [10], and a Re-ID subset was widely used. Its creators withdrew it in 2019 after privacy and ethics concerns about recording people on a campus, and it should no longer be downloaded or used; Peng et al. [11] trace how derived copies kept circulating afterwards and argue for better dataset stewardship.

Metrics: CMC rank-k and mAP

The **cumulative matching characteristic** at rank k is the fraction of queries whose first correct match appears in the top k of the ranked gallery:

$$\mathrm{CMC}(k) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}\left[\, r_q \le k \,\right],$$

where Q is the set of queries and r_q is the rank of the first true match for query q. *Rank-1 accuracy* is $\mathrm{CMC}(1)$. Because a query usually has several true matches in the gallery, rank-k alone ignores how well the *others* are ranked. **Mean average precision** fixes this:

$$\mathrm{AP}(q) = \frac{1}{G_q} \sum_{i=1}^{G_q} \frac{i}{r_{q,i}}, \qquad \mathrm{mAP} = \frac{1}{|Q|} \sum_{q} \mathrm{AP}(q),$$

where G_q is the number of true matches of q in the gallery and $r_{q,i}$ is the rank of the i-th one. So $i / r_{q,i}$ is the precision at the moment the i-th true match is found.

```
import numpy as np

def cmc_map(dist, q_ids, g_ids, max_rank=5):
    """dist: (Q, G) distances between query and gallery embeddings."""
    order = np.argsort(dist, axis=1)                # closest gallery items first
    match = g_ids[order] == q_ids[:, None]          # (Q, G) True where same person
    cmc = (np.cumsum(match, axis=1) > 0).mean(axis=0)[:max_rank]
    aps = []
    for row in match:
        hits = np.flatnonzero(row)                  # ranks (0-based) of true matches
        precision_at_hits = np.arange(1, len(hits) + 1) / (hits + 1)
        aps.append(precision_at_hits.mean())
    return cmc, np.mean(aps)

rng = np.random.default_rng(0)
centres = rng.normal(size=(50, 64))                 # 50 identities, 64-D embeddings
g_ids = np.repeat(np.arange(50), 4)                 # 4 gallery images per identity
q_ids = np.arange(50)
gallery = centres[g_ids] + 1.6 * rng.normal(size=(200, 64))
query = centres[q_ids] + 1.6 * rng.normal(size=(50, 64))
norm = lambda x: x / np.linalg.norm(x, axis=1, keepdims=True)
dist = 1 - norm(query) @ norm(gallery).T            # cosine distance
cmc, mAP = cmc_map(dist, q_ids, g_ids)
print("rank-1 %.2f  rank-5 %.2f  mAP %.2f" % (cmc[0], cmc[4], mAP))
```

On this synthetic gallery (4 images per identity) the script prints `rank-1 0.76  rank-5 0.92  mAP 0.57`: most queries find *one* correct image near the top, but the other three are scattered, which only mAP reveals.

Single-object tracking

Plain version. You draw a box around something in the first frame; the tracker must keep a box on that same thing for the rest of the video, without knowing in advance what kind of object it is.

Problem formulation

Given frames $I_1, \dots, I_T$ and an initial box b_1, a single-object tracker outputs $\hat{b}_2, \dots, \hat{b}_T$. The target is **class-agnostic**: it could be a person, a bird or a coffee cup. This makes SOT a form of *one-shot* learning: the only example of the target is the first-frame patch, the **template** z. In each new frame the tracker searches a region x around the last position. The challenges [1] lists are complex appearance changes, unusual motion patterns, and environmental change such as lighting and occlusion.

Classical: correlation filters

A correlation filter is a template slid over the search window; the place where it lines up best gives the highest score. Before deep learning, the fastest trackers learned such a **correlation filter** online. MOSSE [17] learns a filter H in the Fourier domain so that correlating it with training patches F_i produces sharp Gaussian peaks G_i at the target center:

$$\min_{H^*} \sum_i |F_i \odot H^* - G_i|^2 \quad \Rightarrow \quad H^* = \frac{\sum_i G_i \odot F_i^*}{\sum_i F_i \odot F_i^*},$$

where capital letters are 2-D discrete Fourier transforms, $\odot$ is element-wise multiplication, * is complex conjugation and the division is element-wise. Because correlation becomes multiplication in the frequency domain (see IP 04), training and detection cost only a few FFTs, so these trackers run at hundreds of frames per second on a CPU. KCF [18] showed that training on *all cyclic shifts* of a patch gives a circulant data matrix that the DFT diagonalizes, which turns kernel ridge regression into element-wise operations and allows multi-channel features such as HOG. Their weaknesses are a fixed search window, boundary effects from the cyclic assumption, and hand-crafted features.

Optional: MOSSE: a correlation filter learned in the Fourier domain (in the Supplementary notes). Why the closed form works one frequency at a time, how it is updated online, and a short tested example.

Siamese trackers

Plain version. Train one network to answer a single question — "where in this search image does the template appear?" — on huge amounts of video offline, then use it frozen at test time.

SiamFC [19] applies the *same* fully convolutional network φ to the template z and the search region x and cross-correlates the two feature maps:

$$f(z, x) = \varphi(z) \star \varphi(x) + b\,\mathbb{1},$$

where $\star$ is cross-correlation (the template features act as a convolution kernel), b is a learned bias and $\mathbb{1}$ is a map of ones. The output is a score map whose peak gives the new position. It is trained with a logistic loss on score-map positions labeled positive near the true center and negative elsewhere. Because nothing is updated online, it is fast and robust to drift, but it only predicts position, with scale handled by searching a small pyramid.

SiamRPN [20] adds a **region proposal network** after the Siamese features: one branch classifies anchors as target or background, another regresses box offsets. Tracking becomes "one-shot detection" with the template as the class definition, which yields accurate boxes with changing aspect ratios.

Transformer one-stream trackers

From 2022 the mainstream moved from Siamese CNNs to transformers. The key change is *where* the template and the search region interact (Figure 2). In a two-stream design they are encoded separately and compared once. In a **one-stream** design such as OSTrack [21], the image patches of z and x are concatenated into one token sequence and processed by a single Vision Transformer, so that every attention layer both extracts features and relates template to search. OSTrack also drops search tokens that are clearly background in early layers to save computation. SeqTrack [22] goes further in simplicity: an encoder-decoder transformer generates the four box coordinates as a sequence of discrete tokens, autoregressively, trained with plain cross-entropy — the same recipe as language modeling.

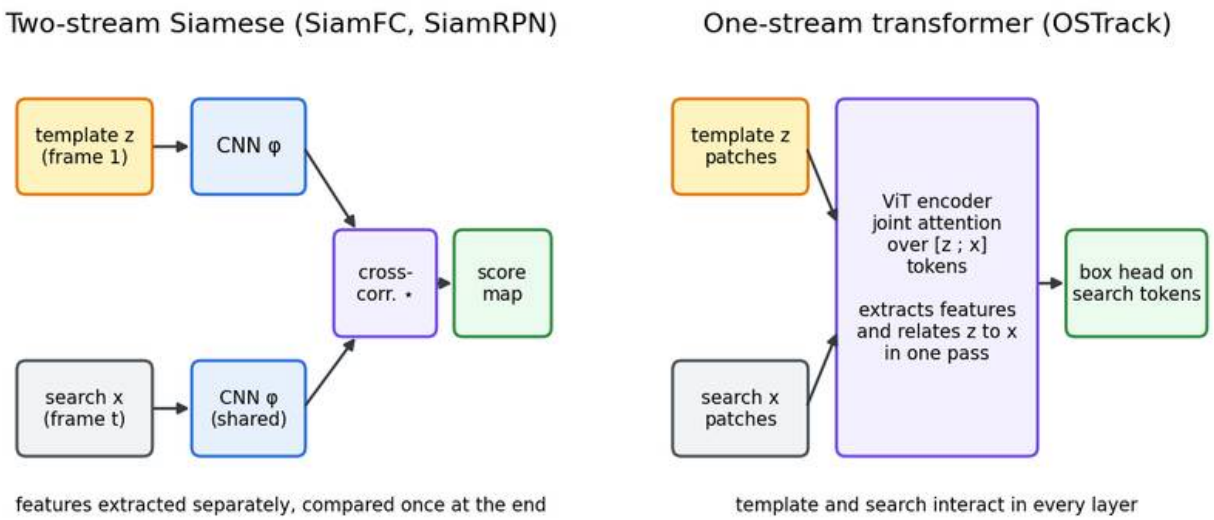


Figure 2 — Two-stream Siamese trackers compare template and search features once, at the end. One-stream transformer trackers let them interact in every layer.

Tracking with video segmentation models

Plain version. New "segment anything in video" models can be told "this object" with one click, and then follow it for the whole video as a pixel mask. That *is* single-object tracking, with a mask instead of a box.

SAM 2 [23] extends the Segment Anything model to video with a **streaming memory**: features and predicted masks from past frames, plus the prompted frames, are stored in a memory bank that the current frame attends to. Prompting one frame with a click, box or mask yields a "masklet" through the whole video, so it behaves like a mask-output SOT tracker. SAMURAI [24] adapts SAM 2 for tracking without any retraining: it adds a Kalman-filter motion model to choose among SAM 2's candidate masks and a motion-aware rule for which past frames enter memory, which helps in crowded scenes with similar distractors. SAM 3 [25] adds **promptable concept segmentation**: given a short noun phrase such as "yellow school bus" or an image exemplar, it detects, segments and tracks *all* matching instances in a video and gives each its own identity. That is a form of open-vocabulary MOT, and it blurs the old SOT/MOT boundary.

Benchmarks and metrics

SOT benchmarks differ in video length, scale and whether test object classes appear in training. The two classic scores ask a simple question each frame: does the predicted box overlap the true box enough, and is its center close enough?

Benchmark	What makes it distinct
OTB [26]	early standard benchmark with per-sequence attribute labels (occlusion, fast motion, ...); introduced the success and precision plots
VOT challenge [53]	yearly challenge series since 2013, with reset-based protocols; now VOTS, tracking *and* segmentation
LaSOT [27]	long-term: 1,400 sequences, more than 3.5 million frames, each frame manually annotated
GOT-10k [28]	one-shot protocol: training and test object classes do not overlap
TrackingNet [29]	large-scale, in-the-wild videos with a held-out evaluation server

The two classic measures, from OTB [26], are:

$$\text{Success}(\tau) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}[\,\text{IoU}(\hat{b}_t, b_t) > \tau\,], \qquad \text{AUC} = \int_0^1 \text{Success}(\tau)\, d\tau,$$

$$\text{Precision}(\delta) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}[\|\hat{c}_t - c_t\|_2 \le \delta],$$

where $\hat{b}_t, b_t$ are the predicted and ground-truth boxes, τ is an IoU threshold, $\hat{c}_t, c_t$ are their centers and δ is a pixel distance, conventionally 20. The AUC of the success plot is the headline number on OTB and LaSOT. GOT-10k reports **average overlap** (the mean IoU) and success rates at $\tau = 0.5$ and 0.75.

```
import numpy as np

def iou_xywh(a, b):
    """IoU between rows of a and b, boxes as [x, y, w, h] (top-left corner)."""
    x1 = np.maximum(a[:, 0], b[:, 0]); y1 = np.maximum(a[:, 1], b[:, 1])
    x2 = np.minimum(a[:, 0] + a[:, 2], b[:, 0] + b[:, 2])
    y2 = np.minimum(a[:, 1] + a[:, 3], b[:, 1] + b[:, 3])
    inter = np.clip(x2 - x1, 0, None) * np.clip(y2 - y1, 0, None)
    return inter / (a[:, 2] * a[:, 3] + b[:, 2] * b[:, 3] - inter)

def success_auc(pred, gt, thresholds=np.linspace(0, 1, 21)):
    ious = iou_xywh(pred, gt)
    curve = np.array([(ious > t).mean() for t in thresholds])
    return curve.mean()                                  # area under the success plot

def precision_at(pred, gt, px=20):
    c_pred = pred[:, :2] + pred[:, 2:] / 2; c_gt = gt[:, :2] + gt[:, 2:] / 2
    return (np.linalg.norm(c_pred - c_gt, axis=1) <= px).mean()

rng = np.random.default_rng(0)
gt = np.c_[np.linspace(50, 400, 300), np.full(300, 80.0), np.full(300, 40.0), np.full(300,
 ↪  60.0)]
pred = gt + rng.normal(0, 4, gt.shape)
pred[200:230, :2] += 120                                 # 30 frames lost on a distractor
print("success AUC %.3f   precision@20px %.3f" % (success_auc(pred, gt), precision_at(pred,
 ↪  gt)))
```

This prints `success AUC 0.647   precision@20px 0.900`: the 30 frames spent on a distractor cost precision 10 points, while the small jitter on the other frames costs AUC much more, because AUC also rewards tight boxes.

Multi-object tracking

Plain version. Find every object of interest in every frame, and give each one a number that stays the same for as long as it is in view — even when objects cross, hide behind each other, or leave and come back.

MOT tracks multiple targets **across consecutive frames** and maintains a consistent ID for each; a single frame on its own only supports detection. The challenges [1] lists are frequent occlusion,

deciding when trajectories start and end, similar appearance between targets, and interaction between targets (people in a crowd move together and occlude each other).

Formally, given detections $\mathcal{D}_t$ in each frame, MOT seeks a set of trajectories $\mathcal{T} = \{\tau_k\}$ that partitions the true detections, discards false positives, and fills in missed ones. Offline methods solve this over the whole video, often as a graph or network-flow problem; **online** methods, which this section focuses on, must decide at frame t using only frames $1..t$, which is what real-time applications require.

The tracking-by-detection pipeline

The dominant design for a decade has been **tracking-by-detection** [2], [4]: run a detector on every frame, then link detections over time (Figure 3). Each frame goes through four steps:

1. **Detect.** A detector (Faster R-CNN, YOLO-family, DETR-family; see DLCV 03) gives boxes and scores.
2. **Predict.** A motion model moves every existing track forward to where it should be now.
3. **Describe.** Optionally, a Re-ID network embeds each detection crop.
4. **Associate.** Build a cost matrix between predicted tracks and detections and solve the assignment problem. Matched tracks are updated; unmatched detections may start new tracks (*birth*); tracks unmatched for too long are deleted (*death*).

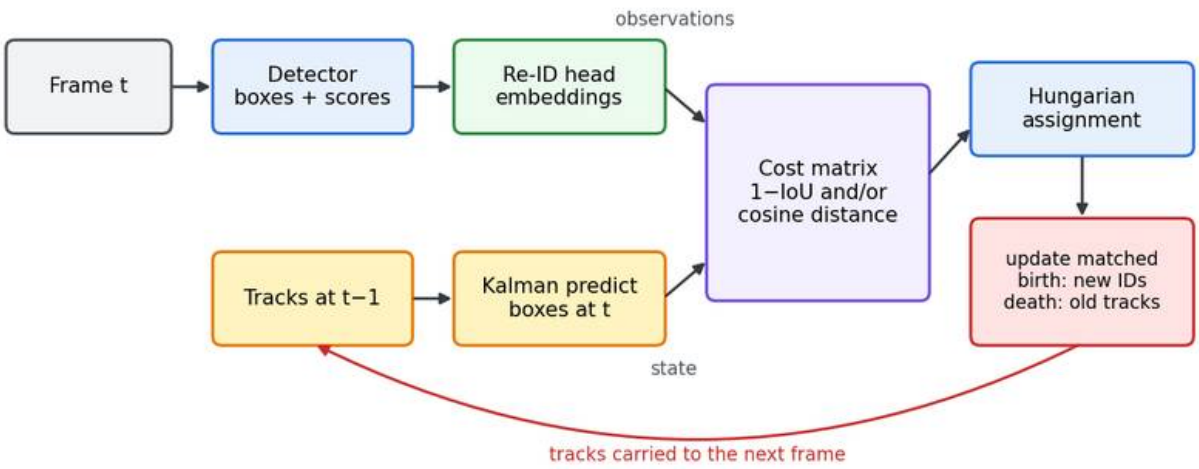


Figure 3 — Online tracking-by-detection. Observations (boxes, embeddings) and state (predicted tracks) meet in a cost matrix; the assignment result updates the state for the next frame.

Motion model: the Kalman filter

Plain version. Guess where the object is now from where it was and how fast it was moving; then nudge that guess toward the detection, trusting whichever of the two is less uncertain.

Nearly all tracking-by-detection systems use a **linear Kalman filter** [30] with a constant-velocity model. In our code the state of one track is

$$\mathbf{x} = [\, c_x,\ c_y,\ w,\ h,\ v_x,\ v_y \,]^\top,$$

where (c_x, c_y) is the box center, w, h its width and height, and (v_x, v_y) the center velocity in pixels per frame. (SORT uses the center, area and aspect ratio instead of w, h [32]; BoT-SORT returns to w and h [36].) The model has two steps.

Predict (time update):

$$\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\,\hat{\mathbf{x}}_{t-1|t-1},$$
$$\mathbf{P}_{t|t-1} = \mathbf{F}\,\mathbf{P}_{t-1|t-1}\,\mathbf{F}^\top + \mathbf{Q},$$

where $\mathbf{F}$ is the 6×6 transition matrix (identity plus 1s that add v_x, v_y to c_x, c_y, since $\Delta t = 1$ frame), $\mathbf{P}$ is the state covariance (our uncertainty) and $\mathbf{Q}$ is the process noise covariance (how much the true motion can deviate from constant velocity).

Update (measurement update) with a matched detection $\mathbf{z}_t = [c_x, c_y, w, h]^\top$:

$$\mathbf{S}_t = \mathbf{H}\,\mathbf{P}_{t|t-1}\,\mathbf{H}^\top + \mathbf{R}, \qquad \mathbf{K}_t = \mathbf{P}_{t|t-1}\,\mathbf{H}^\top\,\mathbf{S}_t^{-1},$$
$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t(\mathbf{z}_t - \mathbf{H}\,\hat{\mathbf{x}}_{t|t-1}), \qquad \mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\mathbf{H})\,\mathbf{P}_{t|t-1},$$

where $\mathbf{H} = [\,\mathbf{I}_4 \;\; \mathbf{0}\,]$ picks the observed part of the state, $\mathbf{R}$ is the detector's measurement noise covariance, $\mathbf{z}_t - \mathbf{H}\hat{\mathbf{x}}_{t|t-1}$ is the **innovation** (surprise), $\mathbf{S}_t$ its covariance, and $\mathbf{K}_t$ the **Kalman gain**. If the prediction is very uncertain ($\mathbf{P}$ large), the update moves the estimate almost all the way to the detection; if the detector is noisy ($\mathbf{R}$ large), it trusts the prediction. When a detection is missing, the tracker runs only the predict step ("coasting"), and $\mathbf{P}$ grows (Figure 4).

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# ---------- 1. constant-velocity Kalman filter ----------
class KalmanBox:
    """State x = [cx, cy, w, h, vx, vy]; measurement z = [cx, cy, w, h]."""
    def __init__(self, z, q=1.0, r=4.0):
        self.x = np.r_[z, 0.0, 0.0]
        self.P = np.diag([r, r, r, r, 100.0, 100.0])    # unknown velocity: large variance
        self.F = np.eye(6); self.F[0, 4] = self.F[1, 5] = 1.0   # dt = 1 frame
        self.H = np.eye(4, 6)
        self.Q = q * np.diag([1, 1, 0.1, 0.1, 0.5, 0.5])
        self.R = r * np.eye(4)
```

```
    def predict(self):
        self.x = self.F @ self.x
        self.P = self.F @ self.P @ self.F.T + self.Q
        return self.x[:4]

    def update(self, z):
        S = self.H @ self.P @ self.H.T + self.R          # innovation covariance
        K = self.P @ self.H.T @ np.linalg.inv(S)          # Kalman gain
        self.x = self.x + K @ (z - self.H @ self.x)
        self.P = (np.eye(6) - K @ self.H) @ self.P
```

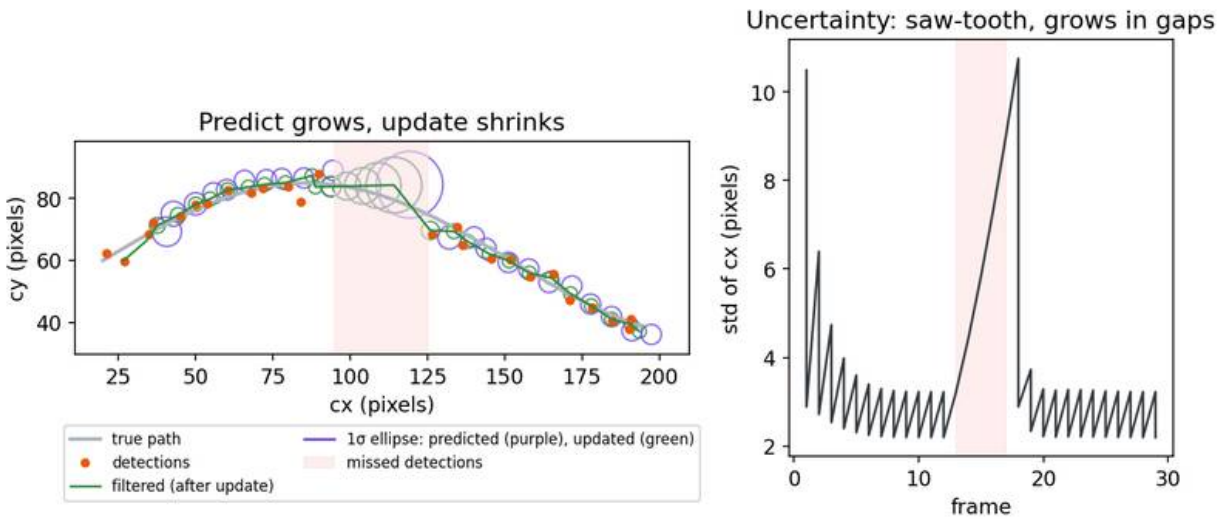


Figure 4 — Kalman predict and update on a synthetic track. Every predict step inflates the uncertainty, every update shrinks it. During five missed detections (pink) the filter coasts in a straight line and its uncertainty grows; this is exactly when identity errors happen.

The squared **Mahalanobis distance** $d_M^2 = \boldsymbol{\nu}^\top \mathbf{S}^{-1} \boldsymbol{\nu}$ of an innovation $\boldsymbol{\nu}$ follows a χ^2 distribution with 4 degrees of freedom under the model, so DeepSORT [33] uses it to *gate* (forbid) implausible matches. The constant-velocity assumption is also the main weakness: dancers, athletes and anything filmed by a moving camera violate it.

Appearance embeddings

A Re-ID network (previous section) turns each detection crop into a unit vector $\mathbf{e}_j$. Each track keeps an appearance memory, for example an exponential moving average $\mathbf{m}_k \leftarrow \alpha\, \mathbf{m}_k + (1-\alpha)\, \mathbf{e}_j$ followed by renormalization, with $\alpha \in [0, 1)$ (we use 0.9). The appearance cost is the cosine distance $1 - \mathbf{m}_k^\top \mathbf{e}_j$. Appearance is what lets a tracker recover an identity after a long occlusion, when motion prediction has become useless.

Association: cost matrices and the Hungarian algorithm

Plain version. Make a table of "how bad would it be to match track k with detection j" for every pair, then choose the one-to-one matching with the smallest total badness.

With K predicted tracks and N detections, build a cost matrix $\mathbf{C} \in \mathbb{R}^{K \times N}$, for example

$$C_{kj} = \lambda \, (1 - \mathrm{IoU}(\hat{b}_k, b_j)) + (1 - \lambda) \, (1 - \mathbf{m}_k^\top \mathbf{e}_j),$$

where $\hat{b}_k$ is the predicted box of track k, b_j detection j's box and $\lambda \in [0, 1]$ the weight of motion versus appearance ($\lambda = 1$ is pure IoU, as in SORT). Then solve the **linear assignment problem**

$$\min_{\mathbf{X} \in \{0,1\}^{K \times N}} \sum_{k,j} C_{kj} X_{kj} \quad \text{s.t.} \quad \sum_j X_{kj} \le 1, \ \sum_k X_{kj} \le 1,$$

so each track takes at most one detection and vice versa. The **Hungarian algorithm** [31] solves it exactly in polynomial time ($O(n^3)$ for $n = \max(K, N)$); SciPy's `linear_sum_assignment` implements a fast variant. Finally, any matched pair whose cost exceeds a threshold is rejected, which is how gating is applied in practice.

Optional: The Hungarian algorithm: optimal one-to-one assignment (in the Supplementary notes). Dual potentials and alternating paths, a worked 4×4 example checked against SciPy, and the rectangular and gated cases.

```
# ---------- 2. IoU and Hungarian association ----------
def iou_matrix(a, b):
    """a: (N,4), b: (M,4) boxes as [cx, cy, w, h] -> (N,M) IoU."""
    a1, a2 = a[:, None, :2] - a[:, None, 2:] / 2, a[:, None, :2] + a[:, None, 2:] / 2
    b1, b2 = b[None, :, :2] - b[None, :, 2:] / 2, b[None, :, :2] + b[None, :, 2:] / 2
    wh = np.clip(np.minimum(a2, b2) - np.maximum(a1, b1), 0, None)
    inter = wh[..., 0] * wh[..., 1]
    area = lambda x: x[..., 2] * x[..., 3]
    return inter / (area(a)[:, None] + area(b)[None, :] - inter + 1e-9)

def associate(cost, max_cost):
    """Hungarian assignment; pairs whose cost exceeds max_cost are rejected."""
    if cost.size == 0:
        return [], list(range(cost.shape[0])), list(range(cost.shape[1]))
    rows, cols = linear_sum_assignment(cost)
    pairs = [(r, c) for r, c in zip(rows, cols) if cost[r, c] <= max_cost]
    mr = {r for r, _ in pairs}; mc = {c for _, c in pairs}
    return (pairs, [r for r in range(cost.shape[0]) if r not in mr],
            [c for c in range(cost.shape[1]) if c not in mc])
```

A complete tracker on a synthetic scene

To see what each idea buys, we generate a scene with five pedestrians over 80 frames, designed to contain the classic failure cases:

- objects 0 and 1 **cross**, and while they overlap, object 1 (behind) gets only **low-confidence** detections;
- objects 2 and 3 meet and **bounce back** (non-linear motion, like dancers), with the rear one also low-confidence during the overlap;
- object 4 walks **behind a pole** for 8 frames, so its detections are low-confidence there;
- every detection is dropped with 5% probability, and there are occasional low-score false positives.

Each detection also carries a 16-D appearance vector: its object's fixed unit vector plus noise.

```
# ---------- 3. synthetic scene ----------
def make_scene(T=80, seed=0):
    rng = np.random.default_rng(seed)
    W, H = 30.0, 70.0
    t = np.arange(T, dtype=float)
    tr = {}
    # pair 1: plain crossing, B is behind A (occluded while they overlap)
    tr[0] = np.c_[80 + 5 * t, np.full(T, 120.0)]
    tr[1] = np.c_[480 - 5 * t, np.full(T, 128.0)]
    # pair 2: they meet and bounce back (non-linear motion, like dancers)
    m = 40
    x2 = np.where(t < m, 120 + 4 * t, 120 + 4 * m - 4 * (t - m))
    x3 = np.where(t < m, 440 - 4 * t, 440 - 4 * m + 4 * (t - m))
    tr[2] = np.c_[x2 + 10, np.full(T, 260.0)]
    tr[3] = np.c_[x3 - 10, np.full(T, 266.0)]
    # walker passing behind a pole (frames 30-37): detections become low-score
    tr[4] = np.c_[60 + 3 * t, 40 + 1.0 * t]
    gt = []                                    # per frame list of (id, box)
    dets = []                                  # per frame list of (box, score, gt_id)
    feats = {k: (lambda v: v / np.linalg.norm(v))(rng.normal(size=16)) for k in tr}
    for f in range(T):
        g, d = [], []
        boxes = {k: np.r_[tr[k][f], W, H] for k in tr}
        for k, b in boxes.items():
            g.append((k, b))
            score = rng.uniform(0.6, 0.95)
            if k == 1 and iou_matrix(b[None], boxes[0][None])[0, 0] > 0.2:
                score = rng.uniform(0.15, 0.45)       # occluded by object 0
            if k == 3 and iou_matrix(b[None], boxes[2][None])[0, 0] > 0.2:
                score = rng.uniform(0.15, 0.45)       # occluded by object 2
            if k == 4 and 30 <= f <= 37:
                score = rng.uniform(0.15, 0.45)       # behind the pole
            if rng.random() < 0.05:
                continue                               # random missed detection
            noisy = b + np.r_[rng.normal(0, 2.0, 2), rng.normal(0, 1.0, 2)]
```

```
            e = feats[k] + 0.15 * rng.normal(size=16)
            d.append((noisy, score, k, e / np.linalg.norm(e)))
        if rng.random() < 0.3:                          # low-score false positive
            fp = np.r_[rng.uniform(50, 550), rng.uniform(50, 320), W, H]
            e = rng.normal(size=16)
            d.append((fp, rng.uniform(0.1, 0.4), -1, e / np.linalg.norm(e)))
        gt.append(g); dets.append(d)
    return gt, dets
```

The tracker below has three switches. With both off it is **SORT-style**: Kalman prediction, IoU cost, Hungarian matching on high-score detections (score ≥ 0.5), tracks deleted after 2 missed frames. `use_low=True` adds a **BYTE**-style second pass: tracks left unmatched try the low-score detections, with IoU only. `use_app=True` adds the **appearance** cost to the first pass and a third pass that re-identifies tracks lost for up to 30 frames by appearance alone.

```
# ---------- 4. SORT-style tracker with optional BYTE and appearance ----------
class Track:
    def __init__(self, tid, z, emb):
        self.id, self.kf, self.emb = tid, KalmanBox(z), emb
        self.lost = 0

def run_tracker(dets, use_low=False, use_app=False, hi=0.5, max_age=2,
                app_age=30, lam=0.5):
    tracks, out, next_id = [], [], 0
    for d in dets:
        for tk in tracks:
            tk.kf.predict()
        boxes = np.array([x[0] for x in d]).reshape(-1, 4)
        scores = np.array([x[1] for x in d])
        embs = np.array([x[3] for x in d]).reshape(-1, 16)
        high = np.where(scores >= hi)[0]
        low = np.where((scores < hi) & (scores >= 0.1))[0]
        active = [tk for tk in tracks if tk.lost <= max_age]
        pred = np.array([tk.kf.x[:4] for tk in active]).reshape(-1, 4)
        # stage 1: high-score detections vs active tracks
        cost = 1 - iou_matrix(pred, boxes[high])
        gate = 0.7
        if use_app and len(active) and len(high):
            tE = np.array([tk.emb for tk in active])
            app = 1 - tE @ embs[high].T            # cosine distance
            cost = np.where(cost > 0.9, 9.0, lam * cost + (1 - lam) * app)
            gate = 0.6
        pairs, ut, ud = associate(cost, gate)
        matched = [(active[r], high[c]) for r, c in pairs]
        rest_tracks = [active[r] for r in ut]
        rest_high = [high[c] for c in ud]
        # stage 2 (BYTE): remaining tracks vs low-score detections, IoU only
        if use_low and rest_tracks and len(low):
            p2 = np.array([tk.kf.x[:4] for tk in rest_tracks])
            pairs2, ut2, _ = associate(1 - iou_matrix(p2, boxes[low]), 0.5)
            matched += [(rest_tracks[r], low[c]) for r, c in pairs2]
```

```
            rest_tracks = [rest_tracks[r] for r in ut2]
        # stage 3 (appearance only): long-lost tracks vs leftover high-score dets
        if use_app and rest_high:
            lost = [tk for tk in tracks if max_age < tk.lost <= app_age]
            if lost:
                app = 1 - np.array([tk.emb for tk in lost]) @ embs[rest_high].T
                pairs3, _, ud3 = associate(app, 0.3)
                matched += [(lost[r], rest_high[c]) for r, c in pairs3]
                rest_high = [rest_high[c] for c in ud3]
        for tk, j in matched:
            tk.kf.update(boxes[j]); tk.lost = 0
            tk.emb = 0.9 * tk.emb + 0.1 * embs[j]; tk.emb /= np.linalg.norm(tk.emb)
        hit = {id(tk) for tk, _ in matched}
        for tk in tracks:
            if id(tk) not in hit:
                tk.lost += 1
        for j in rest_high:                          # birth: unmatched high-score
            tracks.append(Track(next_id, boxes[j], embs[j])); next_id += 1
        tracks = [tk for tk in tracks if tk.lost <= (app_age if use_app else max_age)]
        out.append([(tk.id, tk.kf.x[:4].copy()) for tk in tracks if tk.lost == 0])
    return out
```

The metric code (MOTA, IDF1 and HOTA, explained below) and the experiment:

```
# ---------- 5. metrics: MOTA, IDF1, HOTA ----------
def frame_match(g, p, thr):
    if not g or not p:
        return []
    iou = iou_matrix(np.array([b for _, b in g]), np.array([b for _, b in p]))
    r, c = linear_sum_assignment(-iou)
    return [(g[i][0], p[j][0]) for i, j in zip(r, c) if iou[i, j] >= thr]

def mota(gt, hyp, thr=0.5):
    fn = fp = idsw = 0; ngt = 0; last = {}
    for g, p in zip(gt, hyp):
        m = frame_match(g, p, thr)
        ngt += len(g); fn += len(g) - len(m); fp += len(p) - len(m)
        for gi, pi in m:
            if gi in last and last[gi] != pi:
                idsw += 1
            last[gi] = pi
    return 1 - (fn + fp + idsw) / ngt, idsw

def idf1(gt, hyp, thr=0.5):
    gids = sorted({i for g in gt for i, _ in g}); pids = sorted({i for p in hyp for i, _ in
    ↪ p})
    gi = {k: n for n, k in enumerate(gids)}; pi = {k: n for n, k in enumerate(pids)}
    ov = np.zeros((len(gids), len(pids)))
    for g, p in zip(gt, hyp):
        if g and p:
            iou = iou_matrix(np.array([b for _, b in g]), np.array([b for _, b in p]))
            for a in range(len(g)):
```

```
                for b in range(len(p)):
                    ov[gi[g[a][0]], pi[p[b][0]]] += iou[a, b] >= thr
    r, c = linear_sum_assignment(-ov)
    idtp = ov[r, c].sum()
    n_gt = sum(len(g) for g in gt); n_p = sum(len(p) for p in hyp)
    return 2 * idtp / (n_gt + n_p)

def hota(gt, hyp, alphas=np.arange(0.05, 0.96, 0.05)):
    res = []
    for a in alphas:
        pairs = []; n_gt = n_p = 0
        for g, p in zip(gt, hyp):
            pairs += frame_match(g, p, a); n_gt += len(g); n_p += len(p)
        tp = len(pairs)
        if tp == 0:
            res.append((0, 0, 0)); continue
        cnt = {}
        for q in pairs:
            cnt[q] = cnt.get(q, 0) + 1
        gcount = {}; pcount = {}
        for g in gt:
            for i, _ in g: gcount[i] = gcount.get(i, 0) + 1
        for p in hyp:
            for i, _ in p: pcount[i] = pcount.get(i, 0) + 1
        ass = sum(cnt[q] / (gcount[q[0]] + pcount[q[1]] - cnt[q]) for q in pairs) / tp
        det = tp / (n_gt + n_p - tp)
        res.append((np.sqrt(det * ass), det, ass))
    return np.array(res).mean(axis=0)   # HOTA, DetA, AssA averaged over alpha

gt, dets = make_scene(seed=3)
for name, kw in [("IoU only", {}), ("+ low-score pass", dict(use_low=True)),
                 ("+ appearance", dict(use_low=True, use_app=True))]:
    hyp = run_tracker(dets, **kw)
    m, sw = mota(gt, hyp)
    print(f"{name:18s} MOTA {m:.2f}  IDF1 {idf1(gt, hyp):.2f}  "
          f"HOTA {hota(gt, hyp)[0]:.2f}  ID switches {sw}")
```

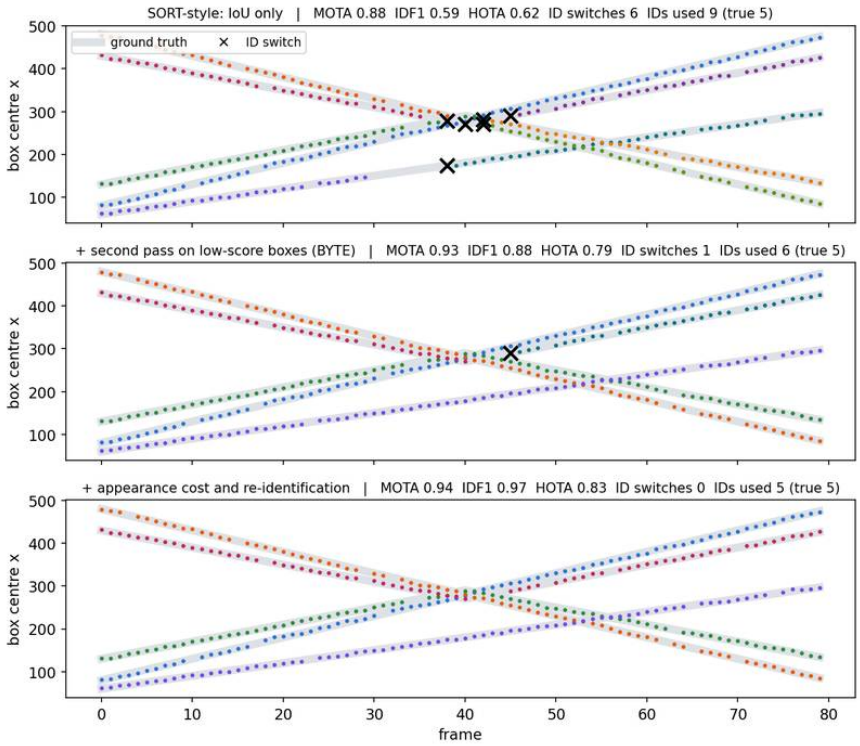


Figure 5 — The same synthetic detections tracked three ways. Colors are predicted IDs; black crosses mark ID switches. Low-score association removes fragmentation from occlusion; appearance fixes the bounce, where constant-velocity prediction points the wrong way.

The script prints:

Tracker	MOTA	IDF1	HOTA	ID switches	IDs used (true: 5)
SORT-style, IoU only	0.88	0.59	0.62	6	9
+ BYTE low-score pass	0.93	0.88	0.79	1	6
+ appearance and re-identification	0.94	0.97	0.83	0	5

Read it carefully. The IoU-only tracker loses the occluded objects because their detections fall below the score threshold, so their tracks die and come back with **new IDs** (fragmentation). The low-score pass keeps them alive, which fixes almost everything, but it cannot fix the bounce: the Kalman filter predicts the two dancers will keep going, so after the overlap one of them is not matched and later restarts with a new ID. Only appearance resolves that. Notice also how little MOTA moves (0.88 → 0.94) compared with IDF1 (0.59 → 0.97): MOTA is mostly a detection measure, a point we return to in the metrics section.

Interactive demonstration (online): open the online edition of this tutorial to use this demonstration.Open demonstration

The SORT family

These ideas map directly onto the most cited online trackers:

Tracker	Key idea
SORT [32]	Kalman filter + IoU cost + Hungarian assignment; deliberately minimal and very fast. Showed that detector quality dominates tracking quality.
DeepSORT [33]	Adds a CNN appearance descriptor trained on a person Re-ID dataset, Mahalanobis gating, and a *matching cascade* that gives priority to recently seen tracks. Fewer ID switches through longer occlusions.
ByteTrack [34]	Associates **every** detection box: high-score boxes first, then the remaining tracks against **low-score** boxes using IoU only. Occluded objects often get low scores, so they are recovered instead of discarded, while low-score background boxes stay unmatched and are thrown away.
OC-SORT [35]	"Observation-centric": when a lost track is re-found, it re-runs the Kalman updates along a virtual trajectory between the last and new observations (fixing error accumulated while coasting), and adds a motion-direction consistency term to the cost. Strong on non-linear motion such as DanceTrack.
BoT-SORT [36]	ByteTrack-style association plus camera-motion compensation (global image registration), a Kalman state with w, h instead of aspect ratio, and a fused IoU/Re-ID cost.

The lesson of this family is that *carefully engineered association on top of a strong detector* is very hard to beat on pedestrian benchmarks. The 2025 survey of Adžemović [4] reaches the same conclusion: heuristic methods lead on dense scenes with mostly linear motion, while learned association does better when motion is complex.

Joint detection and embedding, and detector-based tracking

Running a separate Re-ID network on every crop is expensive. **JDE** [37] adds an embedding head to a one-stage detector so that boxes and appearance vectors come out of one forward pass. **FairMOT** [38] argued that this sharing had been "unfair" to Re-ID: anchor-based detectors produce features that are ambiguous for identity, and detection dominates training. It uses an anchor-free, CenterNet-style detector with two equal branches (detection and Re-ID) on a high-resolution feature map. **Tracktor** [39], which the original post also cites, removes explicit association for most

cases: it feeds the previous frame's boxes to the detector's box-regression head, which pulls each box onto the object's new position and keeps its identity.

Tracking by query: end-to-end MOT

Plain version. Instead of hand-writing "predict, compare, match", give a transformer a memory slot per object and let it learn to update the slot each frame.

DETR-style detectors (see DLCV 03) represent objects as **queries**. MOTR [40] extends this to video: each detected object's output embedding becomes a **track query** that is passed to the next frame and keeps predicting the same object, while new *detect queries* find newborn objects. Identity is carried implicitly by the query; there is no Kalman filter and no Hungarian matching at inference. Training uses a tracklet-aware label assignment and a loss averaged over clips. The weakness was detection quality: one decoder must both detect and associate, and the two tasks conflict. MOTRv2 [41] bootstraps MOTR with proposals from a separate pretrained detector (YOLOX) used as anchors, which eases that conflict. MOTIP [42] (CVPR 2025) reformulates association as **ID prediction**: given the trajectories so far, each with an ID label drawn from a learnable ID dictionary, a decoder classifies each new detection into one of the existing IDs or "new". This is in-context, end-to-end trainable, and works with plain object-level features.

Open-vocabulary MOT

Classic MOT only tracks the classes in its training set. **OVTrack** [43] (CVPR 2023) tracks arbitrary categories by distilling knowledge from a vision-language model (CLIP) for both classification and association, and by learning robust appearance features from *hallucinated* image pairs produced with a diffusion model. SAM 3's concept prompts [25] approach the same goal from the segmentation side.

Datasets

Dataset	Domain	What it tests
MOT17 [44]	pedestrians, static and moving cameras	the standard pedestrian benchmark; the MOT16 sequences re-released as MOT17
MOT20 [45]	very crowded pedestrian scenes	heavy occlusion, small targets
DanceTrack [46]	group dancing	uniform appearance and diverse, non-linear motion; appearance cannot separate people
SportsMOT [47]	basketball, volleyball, football (240 sequences)	fast, variable-speed motion with similar but distinguishable appearance

Dataset	Domain	What it tests
BDD100K MOT [48]	driving video	multi-class tracking (cars, pedestrians, cyclists, ...) from a moving vehicle

The shift from MOT17 to DanceTrack and SportsMOT changed the field's priorities: on MOT17 a good detector and IoU matching go far, but on DanceTrack the motion model and association learning matter most.

Metrics: MOTA, IDF1, HOTA

Plain version. A tracker can fail in two ways: it can miss objects or invent them (detection errors), or it can mix up who is who (association errors). Different metrics weigh these two differently.

All three metrics first match predictions to ground truth in each frame at an IoU threshold. Let GT_t, FN_t, FP_t and IDSW_t be the number of ground-truth objects, misses, false positives and identity switches in frame t. An identity switch occurs when a ground-truth object is matched to a different predicted ID than at its previous match.

MOTA (CLEAR MOT [9]):

$$\mathrm{MOTA} = 1 - \frac{\sum_t (\mathrm{FN}_t + \mathrm{FP}_t + \mathrm{IDSW}_t)}{\sum_t \mathrm{GT}_t}.$$

It ranges from $-\infty$ to 1. Because there are usually far more detections than switches, FN and FP dominate it: MOTA is mostly a detection score. A switch is also counted only at the moment it happens, so a tracker that swaps two IDs once and keeps them swapped for 1,000 frames pays the same as one that swaps back after 1 frame.

IDF1 [10] computes a *global* one-to-one matching between ground-truth trajectories and predicted trajectories (again with the Hungarian algorithm) that maximizes the number of frames where they agree, then

$$\mathrm{IDF1} = \frac{2\,\mathrm{IDTP}}{2\,\mathrm{IDTP} + \mathrm{IDFP} + \mathrm{IDFN}},$$

where IDTP counts detections matched to the trajectory assigned to their ground-truth identity, and IDFP, IDFN the remaining predicted and ground-truth detections. IDF1 measures how long the tracker keeps the right identity, so it is mostly an association score — but because the matching is global, it can behave unintuitively, for example dropping when detection improves.

HOTA (Higher Order Tracking Accuracy [8]) was introduced because neither metric is balanced or decomposable. For a localization threshold α, with per-frame matches giving true positives (TP), FN and FP:

$$\mathrm{DetA}_\alpha = \frac{|\mathrm{TP}|}{|\mathrm{TP}| + |\mathrm{FN}| + |\mathrm{FP}|}, \qquad \mathrm{AssA}_\alpha = \frac{1}{|\mathrm{TP}|} \sum_{c \in \mathrm{TP}} \frac{|\mathrm{TPA}(c)|}{|\mathrm{TPA}(c)| + |\mathrm{FNA}(c)| + |\mathrm{FPA}(c)|},$$

$$\mathrm{HOTA}_\alpha = \sqrt{\mathrm{DetA}_\alpha \cdot \mathrm{AssA}_\alpha}, \qquad \mathrm{HOTA} = \frac{1}{19} \sum_{\alpha \in \{0.05, 0.10, \ldots, 0.95\}} \mathrm{HOTA}_\alpha .$$

For a matched pair c (ground-truth ID g, predicted ID p), $\mathrm{TPA}(c)$ is the set of all TPs in the video that also match g to p; $\mathrm{FNA}(c)$ the detections of g matched to another ID or missed; $\mathrm{FPA}(c)$ the detections of p matched to another object or to nothing. The geometric mean means a tracker must be good at *both* detection and association to score well, and DetA and AssA (and a localization accuracy, LocA) can be reported separately to diagnose failures. Averaging over α also rewards precise boxes. (Our code uses a simplified matching that maximizes IoU per frame; the official definition matches with a score that also favors pairs that have been associated before.)

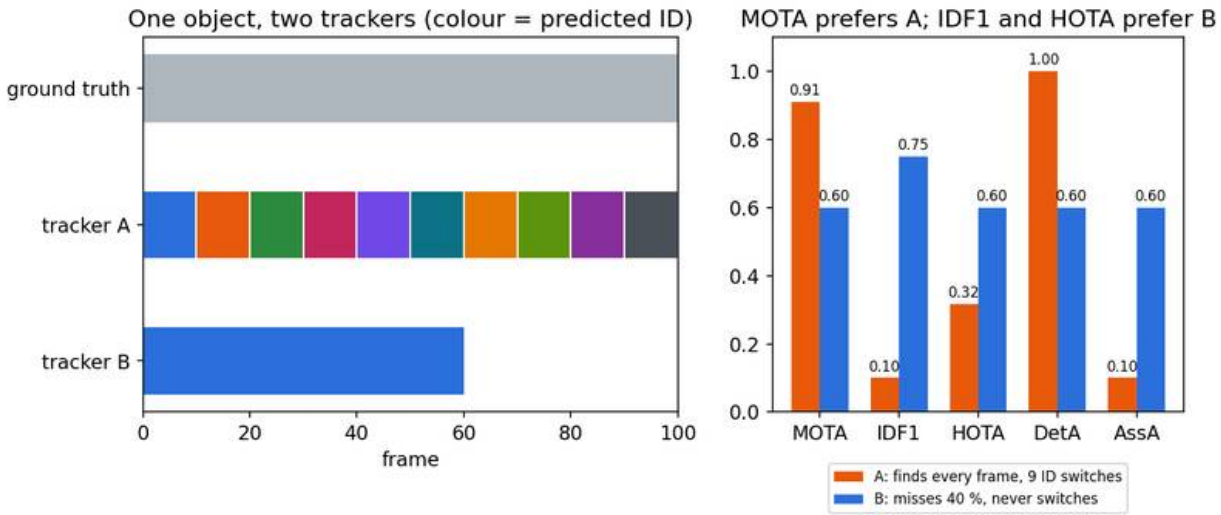


Figure 6 — Why HOTA was introduced. Tracker A detects every frame but switches identity nine times; tracker B misses 40% of frames but never switches. MOTA ranks A far ahead; IDF1 and HOTA rank B ahead, and DetA/AssA show exactly why.

Figure 6 is computed with the functions above: tracker A gets $\mathrm{MOTA} = 1 - 9/100 = 0.91$ and $\mathrm{AssA} = 0.10$, giving $\mathrm{HOTA} = \sqrt{1.0 \times 0.10} \approx 0.32$; tracker B gets 0.60 on MOTA, DetA, AssA and HOTA, and $\mathrm{IDF1} = 2 \cdot 60/(100 + 60) = 0.75$. Today, papers on MOT17, MOT20, DanceTrack and SportsMOT report HOTA, IDF1 and MOTA together, and HOTA is usually the primary ranking metric.

Point tracking: a related trend

Plain version. Instead of following whole objects, follow individual *points* — a freckle on a cheek, a corner of a box — through the whole video, even when they are briefly hidden.

Tracking any point (TAP) asks, for a query point (x, y) in frame t_0, for its position and visibility in every other frame. It generalizes optical flow (two frames) to long, occlusion-aware trajectories,

and is a natural successor to the classical keypoint tracking from IP 12. Representative models are TAPIR [49] and the CoTracker line [50], [51], which tracks many points *jointly*, so that a point that becomes occluded can be inferred from its visible neighbors. Point trackers are now used as motion cues inside object trackers, video editing and robotics.

Optional: Point tracking models: TAPIR and the CoTracker line (in the Supplementary notes). How each of the three models builds a trajectory and what CoTracker3 changed in training.

Multi-target multi-camera tracking

Plain version. Many cameras watch one place. Track everyone in each camera, then figure out that the woman leaving camera 3 is the same woman entering camera 5.

MTMCT is the ultimate goal named in [1]. A typical pipeline runs single-camera MOT to produce *tracklets*, embeds each tracklet with Re-ID features, and clusters tracklets across cameras using appearance, camera topology (which cameras connect, and travel times between them) and, with calibrated cameras, 3D positions on a ground plane. IDF1 was originally proposed for this setting [10]. The **AI City Challenge** is the main recurring benchmark. Its ninth edition (2025) [52] made Track 1 a multi-class **3D** multi-camera tracking task in warehouse-style environments, with people, humanoid robots, autonomous mobile robots and forklifts, calibrated cameras and 3D box annotations. That shift — from image boxes to calibrated 3D tracks across many views — is where MTMCT is heading.

Modern view

Plain version. Two kinds of tracker now coexist: carefully engineered detect-predict-match pipelines, and models that learn association end to end. Around them, tracking is reaching out to any category named in text, to individual points, and to many calibrated cameras.

What the surveys say

- **Luo et al. [3]** (*Artificial Intelligence*, 2021): the classical MOT review; a formal problem definition and a component-by-component analysis, independent of any network.
- **Ciaparrone et al. [2]** (*Neurocomputing*, 2020): deep MOT in four stages (detection, feature extraction/motion prediction, affinity, association); detection quality and learned appearance features drive most gains.
- **Adžemović [4]** (arXiv, 2025): tracking-by-detection families against end-to-end trackers; heuristic trackers lead on crowded, mostly linear scenes, learned association wins on complex motion.

- **Marvasti-Zadeh et al. [5]** (*IEEE T-ITS*, 2022) and **Thangavel (Kugarajeevan) et al. [6]** (*IEEE Access*, 2023): deep and transformer SOT; the one-stream (OSTrack-style) design emerges as the dominant pattern.
- **Ye et al. [7]** (*IEEE TPAMI*, 2022): closed-world versus open-world Re-ID, with the AGW baseline and the mINP metric.
- **Luiten et al. [8]** (*IJCV*, 2020): not a survey, but its re-examination of MOTA and IDF1 is the reason current leaderboards rank by HOTA.

Optional: The tracking surveys in more detail (in the Paper guides). Each review's taxonomy and conclusions, with full titles.

State of the art, 2024–2026

- **SOT** is dominated by one-stream transformer trackers in the OSTrack/SeqTrack lineage and, increasingly, by segmentation foundation models: SAM 2 with motion-aware memory (SAMURAI) gives strong zero-shot tracking without tracking-specific training [23], [24].
- **MOT** has two coexisting camps. Heuristic tracking-by-detection (ByteTrack, OC-SORT, BoT-SORT and descendants) remains the default in practice: simple, fast and modular, with a strong detector doing most of the work. End-to-end trackers (MOTR, MOTRv2, MOTIP) lead where motion is complex and appearance is ambiguous, such as DanceTrack and SportsMOT [4], [42].
- **Open-vocabulary and promptable tracking** is moving into mainstream models: OVTrack established the task, and SAM 3 detects, segments and tracks instances of a text concept with identities [25], [43].
- **Point tracking** (TAPIR, CoTracker, CoTracker3) matured into a general motion primitive [49]–[51].
- **MTMCT** is moving to calibrated, multi-class 3D tracking (AI City Challenge 2025) [52].

Open problems

- **Long-term occlusion and re-entry.** Motion models are useless after a few seconds; appearance must carry identity, but appearance is unreliable for uniform clothing (DanceTrack, sports) and across cameras.
- **Non-linear motion and camera motion.** Constant-velocity Kalman filters remain the default and are the weak point on dance, sports and driving data; learned motion models are not yet clearly better everywhere.
- **End-to-end versus heuristic association.** End-to-end trackers are elegant and win on complex motion, but need lots of video training data and compute and still lose to simple heuristics on crowded pedestrian scenes. A unified design that is best on both is open.

- **Open-vocabulary and long-tail tracking.** Rare classes, and appearance features that generalize to classes never seen in training.
- **Evaluation.** HOTA fixed much, but benchmarks are pedestrian-heavy, and ID-level evaluation over hours of video and many cameras remains hard. **Privacy and dataset ethics** (the DukeMTMC lesson) constrain what data can be collected at all.

Key takeaways

- Tracking = detection + time: every tracker must both follow motion and keep identities, and the balance between the two defines SOT, MOT and Re-ID.
- Re-ID is retrieval with learned embeddings; report both CMC rank-k and mAP; DukeMTMC was withdrawn in 2019 and should not be used.
- SOT went from correlation filters (fast, hand-crafted) to Siamese matching (offline-trained) to one-stream transformers; SAM 2-style video segmentation now does SOT with masks, and SAM 3 extends it to many instances from a text prompt.
- Online MOT is a loop of detect → Kalman predict → cost matrix → Hungarian assignment → update/birth/death. Most practical gains come from what goes into the cost matrix and which detections are allowed to match.
- ByteTrack's low-score pass fixes occlusion fragmentation; appearance fixes non-linear motion and long gaps; end-to-end query trackers learn association directly.
- MOTA mostly measures detection, IDF1 mostly association; HOTA = $\sqrt{\text{DetA} \cdot \text{AssA}}$ (averaged over IoU thresholds) balances and decomposes both.
- The open frontier is long-term identity, non-linear motion, open-vocabulary categories and multi-camera 3D tracking.

Exercises

1. **Kalman gain intuition.** For a 1-D constant-position model with prior variance P and measurement noise R, show that the Kalman gain is $K = P/(P + R)$ and that the posterior variance is $PR/(P + R)$. What happens to K after five frames without detections if each predict step adds Q to P?

   Hint

   With $H = 1$, $S = P + R$ and $K = P/S$. The posterior variance is $(1 - K)P = PR/(P + R)$, smaller than both P and R. After five coasting steps the prior is $P + 5Q$, so K moves toward 1: the filter will trust the next detection much more, which is why a wrong match after an occlusion moves the track a lot.

2. **Hungarian by hand.** Tracks A and B have predicted IoUs with detections 1 and 2 of $\begin{bmatrix} 0.6 & 0.5 \\ 0.55 & 0.0 \end{bmatrix}$.

What does greedy matching (highest IoU first) return, and what does the Hungarian algorithm on cost $1 - \text{IoU}$ return? Which total IoU is larger?

Hint

Greedy takes A–1 (0.6) first, leaving B–2 with IoU 0 (rejected), total 0.6. Hungarian picks A–2 and B–1, total $0.5 + 0.55 = 1.05$. Greedy leaves track B unmatched and starts a new ID for detection 2; this is one way ID switches appear.

3. **Break the tracker.** In the synthetic scene, change the bounce in `make_scene` so that objects 2 and 3 pass through each other instead (no velocity reversal). Which of the three trackers improves the most, and why?

Hint

With linear motion the constant-velocity prediction is correct through the overlap. With `seed=3`, the low-score tracker's IDF1 rises from 0.88 to about 0.96 and it uses the correct 5 IDs, almost matching the appearance tracker (about 0.96 as well); the IoU-only tracker still fragments occluded objects. Appearance helps most exactly when motion is non-linear, which is the design rationale of DanceTrack.

4. **Metric arithmetic.** A video has one object for 200 frames. A tracker detects it in all frames but uses ID 1 for frames 1–100 and ID 2 for frames 101–200. Compute MOTA, IDF1, DetA, AssA and HOTA at a single threshold, assuming perfect boxes.

Hint

$\text{MOTA} = 1 - 1/200 = 0.995$. The best global ID match covers 100 frames: $\text{IDF1} = 2 \cdot 100/(200 + 200) = 0.5$. $\text{DetA} = 1$. For each TP, $|\text{TPA}| = 100$, $|\text{FNA}| = 100$, $|\text{FPA}| = 0$, so $\text{AssA} = 0.5$ and $\text{HOTA} = \sqrt{0.5} \approx 0.71$. One swap barely moves MOTA but halves AssA.

5. **Re-ID metrics.** In the Re-ID snippet, change the noise level from 1.6 to 2.2 and then to 1.0. How do rank-1 and mAP change, and which one is more sensitive? Explain using the definition of AP.

Hint

mAP falls faster than rank-1 as noise grows, because rank-1 only needs the single easiest true match at the top, while AP penalizes every true match that is pushed down the list. With 4 true matches per query, the hardest one usually decides how much AP is lost.

6. **Design question.** You must track forklifts and workers across 12 calibrated cameras in a warehouse. Sketch a pipeline using components from this tutorial, and say which metric you would report and why.

Hint

Per-camera detection plus a ByteTrack/BoT-SORT-style tracker; project boxes to the ground plane using calibration; cross-camera association by 3D position and time first, Re-ID appearance second (workers may wear identical vests); handle the two classes separately. Report HOTA (balanced, decomposable into DetA/AssA) and IDF1 (long-term identity across cameras), as in the AI City Challenge setting.

Supplementary notes

These notes are optional background that the main text does not depend on.

More Re-ID models: a strong baseline, centroids and CLIP-ReID

The Re-ID section names three models that keep the retrieval setup (embed, then rank by distance) and change how the embedding network is trained. Here is what each one does.

- **Strong baselines.** Luo et al. [14] showed that a plain ResNet-50 with a set of training "tricks" — warm-up learning rate, random erasing augmentation, label smoothing, last-stride 1 for a larger feature map, and a **BNNeck** (a batch-norm layer that separates the triplet-loss feature from the classification feature) — is a strong, reproducible baseline.
- **Centroid triplet loss (CTL).** The model cited in that earlier post [1] (ResNet-50 at 256 × 128 input) [15] replaces instance-to-instance triplets with distances to *class centroids*, and at retrieval time compares a query to identity centroids rather than to every gallery image.
- **CLIP-ReID.** Li et al. [16] adapt the vision-language model CLIP to Re-ID, where no text labels exist. In a first stage they learn identity-specific text tokens with the encoders frozen; in a second stage they fine-tune the image encoder using those text features as extra supervision.

MOSSE: a correlation filter learned in the Fourier domain

A correlation filter is a template that is slid over the search window; the place where it lines up best gives the highest score. MOSSE [17] learns that template so that the score map is a single sharp bump on the target and almost flat elsewhere, and it computes every position at once with the FFT.

Let F_i be the 2-D DFT of the i-th preprocessed training window and G_i the DFT of the desired response g_i, a Gaussian bump at the target center. The correlation theorem turns correlation into an element-wise product, so the response to a window is $\mathcal{F}^{-1}(F \odot H^*)$, where H^* is the conjugated filter, $\odot$ is element-wise multiplication and $\mathcal{F}^{-1}$ the inverse DFT. The loss in the main text then

separates by frequency: for every frequency index (u, v) it is a function of one complex number $H^*(u, v)$ only,

$$\min_{H^*(u,v)} \sum_i \left| F_i(u,v)\, H^*(u,v) - G_i(u,v) \right|^2 \quad \Rightarrow \quad H^*(u,v) = \frac{\sum_i G_i(u,v)\, F_i^*(u,v)}{\sum_i F_i(u,v)\, F_i^*(u,v) + \varepsilon},$$

where * is complex conjugation and $\varepsilon > 0$ is a small regularizer. The numerator is the cross-spectrum between the inputs and the wanted output; the denominator is the energy spectrum of the inputs. Without ε, a frequency at which the training windows have almost no energy makes the division unstable; the paper therefore trains on several perturbed copies of the first window and adds ε [17]. To follow appearance changes, keep two running sums with learning rate $\eta \in (0, 1)$ and set $H_t^* = A_t / (B_t + \varepsilon)$:

$$A_t = \eta\, G \odot F_t^* + (1 - \eta)\, A_{t-1}, \qquad B_t = \eta\, F_t \odot F_t^* + (1 - \eta)\, B_{t-1},$$

where F_t is the DFT of the window cut at the new position in frame t. The paper reports $\eta = 0.125$ working well [17]. The **peak-to-sidelobe ratio** $\mathrm{PSR} = (g_{\max} - \mu_{\mathrm{sl}}) / \sigma_{\mathrm{sl}}$ measures how sharp the peak is: $g_{\max}$ is the response maximum, and $\mu_{\mathrm{sl}}, \sigma_{\mathrm{sl}}$ are the mean and standard deviation of the response outside an 11×11 window around the peak. The paper uses a PSR of about 7 as a sign that the target is occluded or lost [17], and then stops updating.

When it fails. The DFT treats the window as one period of an infinite periodic image, so correlation is *circular*: the filter implicitly sees every cyclic shift of the window, and the seam between the left and right edges looks like a real edge. That is why the window is preprocessed (logarithm, normalization to zero mean and a fixed scale, then a cosine window that fades the border to zero, following [17]). The filter also estimates translation only, from one fixed-size window, so fast scale or rotation changes break it. KCF [18] keeps the same circulant idea and adds kernels and multi-channel features.

The snippet trains on eight randomly rotated copies of one window of a scikit-image test photo, then follows it through a synthetic sequence in which the scene moves by $(2, 4)$ pixels per frame (row, column) with sensor noise added.

```
import numpy as np
from scipy import ndimage
from skimage import data

scene = data.camera().astype(float)
S = 64                                                  # window size (pixels)
yy, xx = np.mgrid[:S, :S]
g = np.exp(-((yy - S // 2) ** 2 + (xx - S // 2) ** 2) / (2 * 2.0 ** 2))   # Gaussian peak,
 ↪ sigma = 2
G = np.fft.fft2(g)
cos_win = np.outer(np.hanning(S), np.hanning(S))

def window(frame, cy, cx):                              # S x S crop centred on (cy, cx)
    return frame[cy - S // 2: cy + S // 2, cx - S // 2: cx + S // 2]

def prep(p):                                            # log, zero mean / unit std, cosine
 ↪ window
    p = np.log(p + 1.0)
    return (p - p.mean()) / (p.std() + 1e-5) * cos_win
```

```
def respond(Hs, patch):                                  # response = ifft2(F * H*)
    return np.fft.ifft2(np.fft.fft2(prep(patch)) * Hs).real

def psr(resp):                                           # peak-to-sidelobe ratio
    iy, ix = np.unravel_index(resp.argmax(), resp.shape)
    side = np.ones(resp.shape, bool); side[max(iy - 5, 0): iy + 6, max(ix - 5, 0): ix + 6] =
     ↪  False
    return (resp.max() - resp[side].mean()) / resp[side].std()

rng = np.random.default_rng(0)
cy, cx = 160, 250                                        # target centre in frame 0
A = np.zeros((S, S), complex); B = np.zeros((S, S), complex)
for _ in range(8):                                       # train on 8 small random rotations of
 ↪  the first window
    big = scene[cy - S: cy + S, cx - S: cx + S]      # 2S x 2S crop: rotate about the target
    warped = ndimage.rotate(big, rng.uniform(-5, 5), reshape=False, mode="nearest")
    F = np.fft.fft2(prep(window(warped, S, S)))
    A += G * F.conj(); B += F * F.conj()
eps, eta = 0.1, 0.125                                    # regularizer and learning rate

est = np.array([cy, cx])
for t in range(1, 9):                                    # the scene moves by (2, 4) pixels per
 ↪  frame
    frame = ndimage.shift(scene, (2 * t, 4 * t), order=1, mode="nearest")
    frame = np.clip(frame + rng.normal(0, 8, frame.shape), 0, 255)   # sensor noise
    resp = respond(A / (B + eps), window(frame, *est))
    iy, ix = np.unravel_index(resp.argmax(), resp.shape)
    est = est + np.array([iy - S // 2, ix - S // 2])             # peak offset = displacement
    print(f"frame {t}: estimate {est}  truth {[cy + 2 * t, cx + 4 * t]}  PSR
     ↪  {psr(resp):.1f}")
    F = np.fft.fft2(prep(window(frame, *est)))                   # online update at the new
     ↪  position
    A = eta * G * F.conj() + (1 - eta) * A
    B = eta * F * F.conj() + (1 - eta) * B

other = respond(A / (B + eps), window(scene, 60, 100))           # a window with no target
 ↪  in it
print("PSR on an unrelated window: %.1f" % psr(other))
```

The estimate equals the true position in all eight frames, with PSR between 36.9 and 39.9 on the target (frame 1: 37.2, frame 8: 39.9), against 2.8 on a window that contains no target. This is a toy: the motion is a pure translation of a still photo, so it only checks that the peak offset reads out the displacement and that the PSR separates "target present" from "target absent". It says nothing about real tracking accuracy; for that, see the benchmarks above.

The Hungarian algorithm: optimal one-to-one assignment

Four workers each quote a price for four jobs, and every job needs exactly one worker. Letting each worker in turn grab the cheapest remaining job can leave the last worker with an awful job.

The Hungarian method [31] finds the cheapest complete assignment. It does so by keeping a "price tag" on every worker and every job, and it accepts a pairing only when the tags prove that no cheaper assignment can exist.

Take a square cost matrix $\mathbf{C} \in \mathbb{R}^{n \times n}$ (n tracks, n detections), where C_{kj} is the cost of pairing track k with detection j. The assignment problem of the main text asks for a permutation matrix $\mathbf{X}$ minimizing $\sum_{k,j} C_{kj} X_{kj}$. Its **dual** problem assigns a *potential* u_k to every track and v_j to every detection:

$$\max_{\mathbf{u},\mathbf{v}} \sum_k u_k + \sum_j v_j \quad \text{s.t.} \quad u_k + v_j \le C_{kj} \ \text{ for all } k, j,$$

where $\mathbf{u} \in \mathbb{R}^n$ and $\mathbf{v} \in \mathbb{R}^n$ collect the potentials. Define the **reduced cost** $\bar{C}_{kj} = C_{kj} - u_k - v_j \ge 0$ and call an edge with $\bar{C}_{kj} = 0$ *tight*. For any assignment, $\sum C_{kj} X_{kj} \ge \sum_k u_k + \sum_j v_j$, because each of the n chosen pairs satisfies $C_{kj} \ge u_k + v_j$ and every track and detection appears exactly once. If an assignment uses only tight edges, equality holds and it must be optimal. The algorithm searches for exactly this certificate:

1. Keep feasible potentials and a partial matching on tight edges.
2. Pick an unmatched track and grow an **alternating tree**: from a track, follow tight edges to detections; from a matched detection, follow its matching edge back to its track.
3. If the tree reaches an unmatched detection, **flip** the edges along that path (matched become unmatched and the reverse). The matching grows by one pair.
4. Otherwise let δ be the smallest reduced cost from a reached track to an unreached detection. Add δ to the potentials of reached tracks and subtract it from those of reached detections. Feasibility is kept, and at least one new tight edge appears. Go back to step 2.

Each of the n augmentations costs $O(n^2)$ on a dense matrix, hence $O(n^3)$ in total. Tens to hundreds of objects per frame are therefore cheap. The method is exact for the cost you give it; it does not look at other frames, so a poor cost matrix still yields a poor matching.

Rectangular and gated cases. With $K \neq N$ the solver returns $\min(K, N)$ pairs and leaves the rest unmatched; this is equivalent to padding $\mathbf{C}$ with dummy rows or columns. SciPy's `linear_sum_assignment` accepts rectangular matrices directly and implements a modified Jonker–Volgenant shortest-augmenting-path algorithm [54], [55]. The `associate` function above rejects over-threshold pairs *after* solving. An alternative is to make forbidden pairs very expensive *inside* the problem, so that the solver prefers to keep as many valid pairs as possible; the two can disagree, as the last lines of the snippet show.

The snippet implements the four steps (one track added at a time) and checks it against SciPy on a 4×4 matrix whose entries are the numbers 1 to 16.

```
import numpy as np
from scipy.optimize import linear_sum_assignment
```

```
def hungarian(C):
    """Hungarian method (shortest augmenting paths, dual potentials); needs rows <= cols."""
    n, m = C.shape
    u, v = np.zeros(n + 1), np.zeros(m + 1)        # row and column potentials (index 0
     ↪  unused)
    p = np.zeros(m + 1, int)                        # p[j] = row currently matched to column
     ↪  j
    way = np.zeros(m + 1, int)                      # predecessor of column j in the search
     ↪  tree
    for i in range(1, n + 1):                       # add one row at a time
        p[0], j0 = i, 0
        minv, used = np.full(m + 1, np.inf), np.zeros(m + 1, bool)
        while True:                                 # grow the alternating tree
            used[j0] = True
            i0, delta = p[j0], np.inf
            for j in range(1, m + 1):
                if not used[j]:
                    cur = C[i0 - 1, j - 1] - u[i0] - v[j]      # reduced cost of edge (i0, j)
                    if cur < minv[j]:
                        minv[j], way[j] = cur, j0
                    if minv[j] < delta:
                        delta, j1 = minv[j], j
            for j in range(m + 1):                  # move the duals by delta
                if used[j]:
                    u[p[j]] += delta; v[j] -= delta
                else:
                    minv[j] -= delta
            j0 = j1
            if p[j0] == 0:
                break                               # reached a free column
        while j0:                                   # flip the alternating path
            j1 = way[j0]; p[j0] = p[j1]; j0 = j1
    return [(int(p[j]) - 1, j - 1) for j in range(1, m + 1) if p[j]], u[1:], v[1:]

C = np.array([[3., 12, 4, 11], [1, 5, 8, 6], [15, 13, 7, 10], [14, 9, 2, 16]])
pairs, u, v = hungarian(C)
rows, cols = linear_sum_assignment(C)
print("ours :", sorted(pairs), "cost", sum(C[i, j] for i, j in pairs))
print("scipy:", list(zip(rows.tolist(), cols.tolist())), "cost", C[rows, cols].sum())
print("duals: u", u, "v", v, "sum", u.sum() + v.sum())
print("min reduced cost", (C - u[:, None] - v[None, :]).min())

# greedy: repeatedly take the cheapest remaining pair
G, total = C.copy(), 0.0
for _ in range(4):
    i, j = np.unravel_index(np.argmin(G), G.shape)
    total += G[i, j]; G[i, :] = np.inf; G[:, j] = np.inf
print("greedy cost", total)

# rectangular: 4 tracks, 3 detections -> three pairs, one track left over
r3, c3 = linear_sum_assignment(C[:, :3])
print("rectangular:", list(zip(r3.tolist(), c3.tolist())), "cost", C[r3, c3].sum())

# gating inside the problem: forbidden pairs get a huge cost, then are dropped
```

```
Cg, gate = np.array([[0.50, 0.65], [0.30, 0.55]]), 0.6
r, c = linear_sum_assignment(Cg)
print("reject afterwards:", [(i, j) for i, j in zip(r.tolist(), c.tolist()) if Cg[i, j] <=
 ↪  gate])
r, c = linear_sum_assignment(np.where(Cg > gate, 1e6, Cg))
print("gate inside      :", [(i, j) for i, j in zip(r.tolist(), c.tolist()) if Cg[i, j] <=
 ↪  gate])
```

Both solvers return the pairs $(0,0)$, $(1,1)$, $(2,3)$, $(3,2)$ with total cost 20; the greedy rule costs 25. The potentials are $\mathbf{u} = (7,5,10,5)$ and $\mathbf{v} = (-4,0,-3,0)$, whose sum is also 20, and the smallest reduced cost is 0, so the dual certificate and the assignment meet. Dropping the last detection (4 tracks, 3 detections) gives three pairs of total cost 10 and leaves track 2 unmatched. In the 2×2 gating example with threshold 0.6, rejecting afterwards keeps one pair, $(1, 0)$, while the gate-inside version keeps two, $(0, 0)$ and $(1, 1)$: the cheapest full matching is $\{(0,1),(1,0)\}$ with total 0.95 against 1.05 for the other one, and rejecting afterwards then discards its over-threshold pair $(0, 1)$.

Point tracking models: TAPIR and the CoTracker line

Tracking any point (TAP) follows a query point through a whole video, with its visibility in every frame. Three models mark how the task developed.

TAPIR [49] first matches the query feature independently in every frame to get a rough trajectory, then refines it with temporal convolutions. CoTracker [50] tracks many points *jointly* with a transformer, so points help one another: a point that becomes occluded can be inferred from its visible neighbors. CoTracker3 [51] simplifies the architecture and trains on real videos pseudo-labeled by existing trackers, instead of only synthetic data.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

The tracking surveys in more detail

This note expands the short survey list in the Modern view: what each review covers, how it organizes the field, and what it concludes.

- **Luo et al., "Multiple object tracking: A literature review" [3]** (*Artificial Intelligence*, 2021; first released 2014). The classical reference. It defines the MOT problem formally, categorizes methods by initialization (detection-based versus detection-free), processing mode (online versus offline) and output type (deterministic versus probabilistic), and analyzes the components —

appearance model, motion model, interaction model, exclusion constraints, occlusion handling — and the inference methods. Read it to understand the problem independent of any network.

- **Ciaparrone et al., "Deep learning in video multi-object tracking: A survey" [2]** (*Neurocomputing*, 2020). Organizes deep MOT around four stages — detection, feature extraction/motion prediction, affinity computation and association — and reviews how deep learning entered each. Its main takeaway, from comparisons on MOTChallenge, is that detection quality and learned appearance features drive most gains.
- **Adžemović, "Deep learning-based multi-object tracking: A comprehensive survey from foundations to state-of-the-art" [4]** (arXiv, 2025). Splits tracking-by-detection into joint detection-and-embedding, heuristic-based, motion-based, affinity-learning and offline methods, and contrasts them with end-to-end trackers. It identifies 2022 (ByteTrack and MOTR) as the acceleration point and concludes that heuristic trackers lead on crowded, mostly linear scenes while learned association wins on complex motion.
- **Marvasti-Zadeh et al., "Deep learning for visual tracking: A comprehensive survey" [5]** (*IEEE T-ITS*, 2022). For SOT. Analyzes deep trackers along nine aspects (network architecture, exploitation, training, objective, output, use of correlation filters, aerial-view, long-term and online tracking) and compares them on OTB, VOT, LaSOT and aerial benchmarks.
- **Thangavel (Kugarajeevan) et al., "Transformers in single object tracking: An experimental survey" [6]** (*IEEE Access*, 2023). Classifies transformer trackers as CNN-Transformer hybrids, two-stream two-stage, and one-stream one-stage fully-transformer trackers, and evaluates their robustness and efficiency; the one-stream design (OSTrack-style) emerges as the dominant pattern.
- **Ye et al., "Deep learning for person re-identification: A survey and outlook" [7]** (*IEEE TPAMI*, 2022). Separates closed-world Re-ID (feature learning, metric learning, ranking) from open-world Re-ID (heterogeneous data, raw images, noisy labels, unsupervised and open-set settings), proposes the AGW baseline and the mINP metric, which measures the cost of finding *all* correct matches.
- **Luiten et al., "HOTA" [8]** (*IJCV*, 2020). Not a survey, but it re-examines the evaluation of the entire field, shows the biases of MOTA and IDF1 with worked examples and user studies, and is the reason current leaderboards rank by HOTA.

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DLCV 06A • Human Pose and Keypoint Estimation

How deep networks find the joints of every person in an image: skeletons and visibility, heatmaps versus regression and sub-pixel decoding, top-down and bottom-up multi-person pipelines, OKS-based AP and PCK, 3D pose from lifting, triangulation and body models, datasets and failure modes.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-06a-pose-keypoints/

What you'll learn

- What a pose is: a skeleton graph of named keypoints with coordinates and visibility flags, and how pose estimation differs from detection, segmentation and the generic keypoints of classical feature extraction.
- The two ways a network outputs a keypoint, direct regression and Gaussian heatmaps, and why decoding a heatmap needs sub-pixel refinement, with a tested experiment on quantization error versus heatmap stride.
- How multi-person pose is solved: top-down pipelines (a person detector, then SimpleBaseline, HRNet or ViTPose on each crop), bottom-up pipelines (all keypoints first, then grouping with part affinity fields or associative embedding) and one-stage or set-prediction models.
- How pose is measured: object keypoint similarity (OKS) and OKS-based AP with per-keypoint constants, PCK and PCKh, and MPJPE for 3D, with code.
- How 3D pose is obtained by lifting 2D keypoints, by triangulating them from several cameras or by fitting a parametric body model such as SMPL, and which datasets and failure modes (occlusion, crowds, left–right flips) shape the field.

The big picture

A detector puts a box around each person (DLCV 03). A box says *where* a person is, but not *what they are doing*: standing, waving, falling or kicking a ball all give similar boxes. **Pose estimation** goes one step further. It finds a small set of meaningful points on the body, the **keypoints** (nose, shoulders, elbows, wrists, hips, knees, ankles and so on), and connects them into a stick figure.

> **Explain like I'm five**
>
> Remember connect-the-dots pictures? Pose estimation is the opposite game. The computer looks at a photo of a person and has to *put the dots back*: one dot on each shoulder, elbow, wrist, hip, knee and ankle, each with the right name. Then it draws lines between the dots and you get a stick figure that moves the way the person moves. The hard parts are the ones you would also find hard: a hand hidden behind someone's back, two friends hugging so you cannot tell whose arm is whose, and remembering which side is *their* left.

Why it matters. A stick figure is a compact summary of what a body is doing. Sports analysis, physical therapy, fall detection and fitness apps measure joint angles from it. Animation, games and augmented reality drive avatars with it. Robots and cars use it to anticipate where people will move. Neuroscientists track the joints of mice and flies without attaching markers [1]. And the same ideas, heatmaps, keypoint grouping and keypoint metrics, are reused for faces, hands, animals and object parts.

This chapter builds on DLCV 03 for person detection, non-maximum suppression (NMS) and the AP machinery, and on DLCV 03B for dense prediction and upsampling; we link to them instead of repeating them. Tracking poses over time uses the association methods of DLCV 06, and 3D triangulation uses the camera geometry of DLCV 08. Two surveys frame the field: Zheng et al. [2] cover 2D and 3D pose with deep learning, and Chen et al. [3] organize monocular methods by task. We return to both in the Modern view.

What a pose is

Plain version. A pose is a list of named points on the body, each with a position and a flag that says whether it was visible, plus a fixed list of which points are joined by "bones".

Precise version. A **skeleton** is a graph $G = (V, E)$. The vertices $V = \{1, \dots, K\}$ are the K keypoint *types* (for example "left wrist"), and the edges $E \subset V \times V$ are the **limbs** that connect them. A 2D pose of one person is

$$P = \{(\mathbf{x}_k, v_k)\}_{k=1}^{K}, \qquad \mathbf{x}_k = (x_k, y_k) \in \mathbb{R}^2,$$

where $\mathbf{x}_k$ is the image position of keypoint k in pixels and v_k is its **visibility flag**. The COCO keypoint format uses $v_k = 0$ for "not labeled", $v_k = 1$ for "labeled but not visible" (occluded, position guessed by the annotator) and $v_k = 2$ for "labeled and visible" [4]. A 3D pose replaces $\mathbf{x}_k$ by a point $\mathbf{X}_k \in \mathbb{R}^3$, usually in millimetres in a camera or world frame.

The COCO person skeleton has $K = 17$ keypoints: the nose, both eyes, ears, shoulders, elbows, wrists, hips, knees and ankles [4]. Figure 1 shows them on a synthetic stick figure. Two conventions trip up every beginner:

- **Left and right are the person's, not the viewer's.** A person facing the camera has their left shoulder on the *right* side of the image. Flipping an image horizontally therefore turns every left keypoint into a right one, and data augmentation must swap the labels as well as mirror the coordinates.
- **Not visible is not the same as not labeled.** An occluded wrist ($v = 1$) still has a position and is still evaluated; an unlabeled keypoint ($v = 0$) is simply ignored.

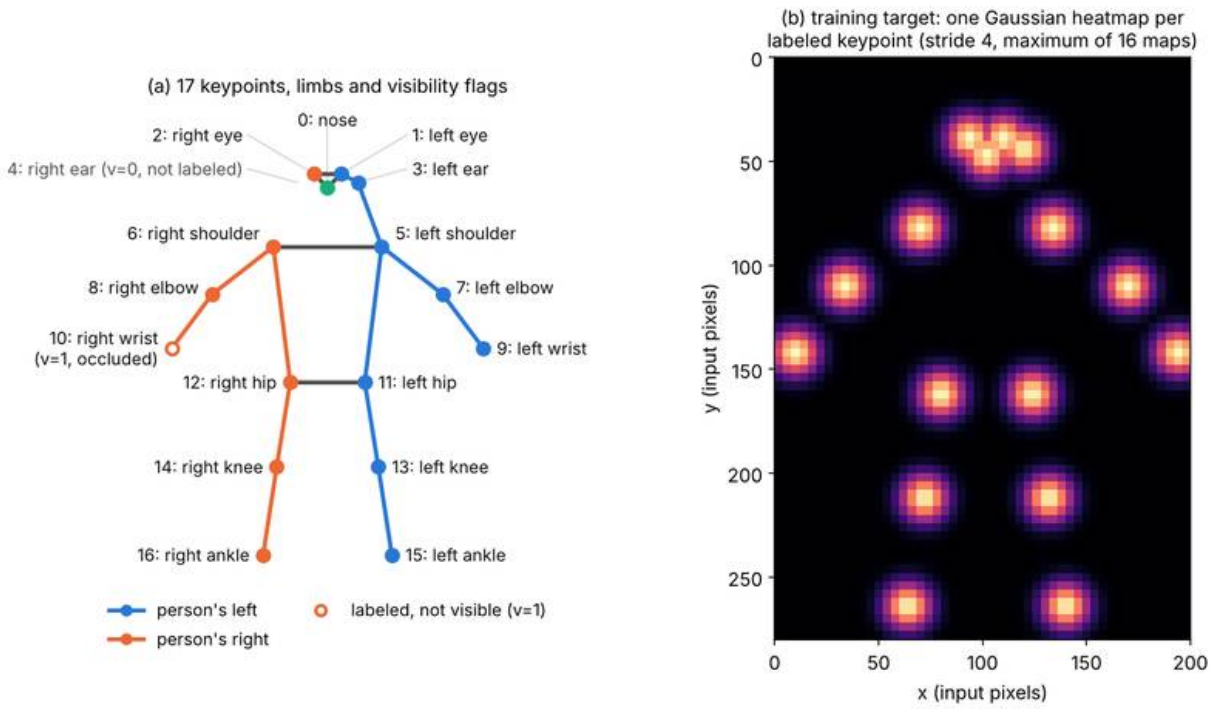


Figure 1 — (a) The 17-keypoint COCO-style skeleton on a synthetic person, with the person's left in blue and right in orange, and the two non-visible cases. (b) The corresponding training target of a heatmap network: one Gaussian blob per labeled keypoint at output stride 4 (the 16 maps are shown as their pixel-wise maximum).

Pose estimation versus its neighbours

Four related tasks are easy to confuse:

Task	Output	Where in this course
Generic keypoint detection	Corners and blobs that are easy to find again in another image, with descriptors; no semantic name	IP 12
Object detection	One box, class and score per object	DLCV 03
Instance segmentation	One pixel mask per object	DLCV 03B
Pose estimation	One set of *named* keypoints per object, connected by a fixed skeleton	this chapter

The keypoints of IP 12 are chosen *by the image*: wherever a corner or blob happens to be. The keypoints here are chosen *by the task*: the left elbow must be found even when it lies on a smooth sleeve with no corner at all. That is why semantic keypoints are learned from labeled examples rather than detected by a formula.

The task comes in several flavors. **Single-person 2D** estimation assumes one person per input crop. **Multi-person 2D** estimation must also decide how many people there are and which keypoint belongs to whom. **3D** estimation recovers depth. **Whole-body** estimation adds feet, hands and face landmarks, and **mesh recovery** outputs a full body surface instead of a sparse skeleton [5].

Two ways to output a keypoint

Plain version. A network can either *say the numbers* ("the wrist is at $x = 140$, $y = 128$"), which is regression, or *paint a picture* that glows where the wrist is, which is a heatmap. For most of the deep-learning era, painting won, because convolutional networks are better at painting than at saying numbers.

Direct regression

DeepPose [6] was an early deep pose estimator built on regression. A convolutional network maps the image to a vector of $2K$ numbers, the keypoint coordinates normalized by the person's box, and is trained with a squared error,

$$\mathcal{L}_{\text{reg}} = \sum_{k=1}^{K} \|\hat{\mathbf{x}}_k - \mathbf{x}_k\|_2^2,$$

where $\hat{\mathbf{x}}_k$ is the predicted and $\mathbf{x}_k$ the ground-truth position of keypoint k (in normalized coordinates). DeepPose refined its first guess with a cascade of further regressors applied to higher-resolution crops around each joint.

Regression is simple, fast and has no resolution limit, but the fully connected output throws away the spatial layout that convolutions are good at, and for years regression trailed heatmaps on accuracy.

Gaussian heatmaps

Tompson et al. [7] instead asked the network to output one image per keypoint type, a **heatmap** that is high near the keypoint and low elsewhere. The target for keypoint k is a Gaussian blob centred on the true position:

$$H_k^*(\mathbf{p}) = \exp\!\left(-\frac{\|\mathbf{p} - \mathbf{x}_k / s\|_2^2}{2\sigma^2}\right),$$

where $\mathbf{p}$ is a heatmap cell (integer coordinates), s is the **output stride** (how many input pixels one heatmap cell covers, typically 4), $\mathbf{x}_k/s$ is the true keypoint position in heatmap units and σ is the blob's width in cells. The network is trained with a pixel-wise mean squared error between the predicted heatmaps $\hat{H}_k$ and the targets,

$$\mathcal{L}_{\text{hm}} = \frac{1}{K\,|\Omega|} \sum_{k=1}^{K} \sum_{\mathbf{p}\in\Omega} w_k \big(\hat{H}_k(\mathbf{p}) - H_k^*(\mathbf{p})\big)^2,$$

where Ω is the set of heatmap cells and w_k is 0 for unlabeled keypoints and 1 otherwise. The stacked hourglass network uses this supervision, following Tompson et al., with $\sigma = 1$ pixel [8].

Why does this work so much better than regression? A heatmap **matches what convolutions do**: it is dense and translation-equivariant (if the person moves, the blob moves), so every output cell answers a local question, "is there a wrist *here*?". It **can express doubt**: two plausible wrist positions become two weaker blobs, where a single regressed coordinate would land in between.

And the peak height is a **free confidence score** for each keypoint.

The price is resolution: at stride 4 the peak can only sit on a grid of spacing 4 pixels, and K maps cost far more memory than $2K$ numbers. How dense prediction networks recover resolution is covered in DLCV 03B; the next section shows how much the grid costs and how to get most of it back.

Decoding a heatmap: argmax, quantization error and sub-pixel refinement

Plain version. Reading a heatmap is like reading a thermometer with coarse marks. Reporting the nearest mark (the brightest cell) is off by up to half a mark. Looking at how bright the *neighbouring* cells are tells you which way, and how far, to move between the marks.

The simplest decoder takes the brightest cell and maps it back to the image:

$$\hat{\mathbf{x}}_k = s \cdot \arg\max_{\mathbf{p}\in\Omega} \hat{H}_k(\mathbf{p}).$$

If the true position falls uniformly at random inside a cell, the error of this decoder in each axis is uniform on $[-s/2, s/2]$, and the expected Euclidean error is about $0.383\,s$ (the mean distance from the centre of a unit square to a random point in it, $(\sqrt{2} + \ln(1+\sqrt{2}))/6 \approx 0.383$, scaled by s). At stride 4 that is about 1.5 pixels of error *before the network makes any mistake*. On a person 100 pixels tall, that is enough to lose points on the strict thresholds of the OKS metric introduced later.

Three refinements are common:

- **The quarter offset.** Move the estimate a quarter of a cell from the peak toward its higher neighbour. The stacked hourglass paper added this to improve results at high precision thresholds [8], and SimpleBaseline uses it as standard practice [9].

- **A Taylor step on the log-heatmap.** The logarithm of a Gaussian is a quadratic (a parabola), so its maximum can be found exactly from the value, slope and curvature at the brightest cell: $\hat{\mathbf{p}} = \mathbf{m} - \mathbf{D}''(\mathbf{m})^{-1}\mathbf{D}'(\mathbf{m})$, where $\mathbf{m}$ is the brightest cell, and $\mathbf{D}'$ and $\mathbf{D}''$ are the gradient and Hessian of $\log \hat{H}_k$ there, estimated by finite differences. This is the decoding step of DARK [10]. DARK also smooths the predicted heatmap with a Gaussian before decoding, because real predictions are not perfect Gaussians, and centres the training targets on the exact (unrounded) keypoint position, as our formula for H_k^* already does.
- **Soft-argmax (integral regression).** Treat the heatmap as a probability distribution and output its *mean*: $\hat{\mathbf{x}}_k = s \sum_{\mathbf{p}} \mathbf{p}\, \tilde{H}_k(\mathbf{p})$, with $\tilde{H}_k = \mathrm{softmax}(\hat{H}_k)$ over all cells. Sun et al. [11] showed that this turns the heatmap into a differentiable regression output, so the network can be trained end to end on the coordinates, without quantization.

DARK's ablation makes the size of the effect concrete: for an HRNet-W32 model with a small 128×96 input on COCO validation, AP was 61.2 with plain argmax, 66.9 with the quarter offset and 68.4 with its Taylor decoding [10]. The decoder alone moved AP by more than 7 points.

Optional: Sub-pixel decoding by a Taylor step (in the Supplementary notes). The derivation of the Newton step, why it is exact for a Gaussian, and the edge cases (borders, flat and multi-peaked maps).

The code encodes 500 random keypoints into heatmaps at strides 1, 2, 4 and 8, adds a little noise to imitate an imperfect prediction, and decodes them with each method. All numbers come from this synthetic run.

```
import numpy as np

def encode(xy, shape, stride, sigma=2.0):
    """Gaussian heatmap of size shape=(H, W) cells for a keypoint at image position xy."""
    cx, cy = xy[0] / stride, xy[1] / stride           # exact (unrounded) center in cells
    ys, xs = np.mgrid[0:shape[0], 0:shape[1]]
    return np.exp(-((xs - cx) ** 2 + (ys - cy) ** 2) / (2 * sigma ** 2))

def peak(h):
    y, x = np.unravel_index(np.argmax(h), h.shape)
    return x, y

def decode_argmax(h, stride):
    x, y = peak(h)
    return np.array([x, y], float) * stride

def decode_quarter(h, stride):
    """Move a quarter cell from the peak toward the higher neighbour on each axis."""
    x, y = peak(h)
    p = np.array([x, y], float)
    if 0 < x < h.shape[1] - 1: p[0] += 0.25 * np.sign(h[y, x + 1] - h[y, x - 1])
    if 0 < y < h.shape[0] - 1: p[1] += 0.25 * np.sign(h[y + 1, x] - h[y - 1, x])
    return p * stride

def decode_taylor(h, stride):
    """One Newton step on log h at the peak: offset = -Hessian^-1 * gradient."""
```

```
    x, y = peak(h)
    p = np.array([x, y], float)
    if 1 <= x < h.shape[1] - 1 and 1 <= y < h.shape[0] - 1:
        L = np.log(np.maximum(h[y - 1:y + 2, x - 1:x + 2], 1e-10))
        g = np.array([L[1, 2] - L[1, 0], L[2, 1] - L[0, 1]]) / 2
        hxx = L[1, 2] - 2 * L[1, 1] + L[1, 0]
        hyy = L[2, 1] - 2 * L[1, 1] + L[0, 1]
        hxy = (L[2, 2] - L[2, 0] - L[0, 2] + L[0, 0]) / 4
        Hs = np.array([[hxx, hxy], [hxy, hyy]])
        if np.linalg.det(Hs) > 0 and hxx < 0:          # a proper maximum
            p += -np.linalg.solve(Hs, g)
    return p * stride

def decode_soft(logits, stride):
    """Soft-argmax: the expected position under softmax(logits)."""
    w = np.exp(logits - logits.max()); w /= w.sum()
    ys, xs = np.mgrid[0:w.shape[0], 0:w.shape[1]]
    return np.array([(w * xs).sum(), (w * ys).sum()]) * stride

rng = np.random.default_rng(0)
W, H = 192, 256                                    # input crop size in pixels (width,
 ↪  height)
pts = rng.uniform([16, 16], [W - 16, H - 16], (500, 2))

def mean_error(decoder, s, noise=0.01, use_log=False):
    shape, e = (H // s, W // s), 0.0
    for xy in pts:
        h = encode(xy, shape, s)
        h = np.clip(h + rng.normal(0, noise, h.shape), 1e-6, None)   # an imperfect
         ↪  "prediction"
        e += np.linalg.norm(decoder(np.log(h) if use_log else h, s) - xy)
    return e / len(pts)

print("stride  argmax  quarter  taylor   (mean error in input pixels)")
for s in (1, 2, 4, 8):
    print(f"{s:6d}  {mean_error(decode_argmax, s):6.2f}  {mean_error(decode_quarter,
     ↪  s):7.2f}"
          f"  {mean_error(decode_taylor, s):6.2f}")
print(f"soft-argmax, stride 4: clean map {mean_error(decode_soft, 4, 0.0, True):.2f},"
      f" noisy map {mean_error(decode_soft, 4, 0.01, True):.2f}")
```

```
stride  argmax  quarter  taylor   (mean error in input pixels)
     1    0.39     0.20    0.05
     2    0.80     0.40    0.11
     4    1.54     0.78    0.21
     8    3.20     1.59    0.43
soft-argmax, stride 4: clean map 0.04, noisy map 24.42
```

Four lessons are visible, and Figure 2 plots the first three.

1. **Argmax error grows linearly with the stride** and follows the $0.383\,s$ prediction closely (0.38,

0.77, 1.53, 3.06 for strides 1 to 8).

2. **The quarter offset halves it**, at no cost.
3. **The Taylor step removes most of what remains**, because the log of a Gaussian really is a parabola; what is left comes from the added noise.
4. **Soft-argmax is exact on a clean map but fails badly on a noisy one.** With logits equal to $\log \hat{H}$, the softmax weights are proportional to $\hat{H}$, and a small noise floor spread over thousands of cells outweighs the narrow peak and pulls the mean toward the centre of the map. Integral regression trains the network *through* the soft-argmax, which pushes it to put almost no mass away from the keypoint; applied after the fact to a heatmap trained with MSE, soft-argmax is a poor decoder.

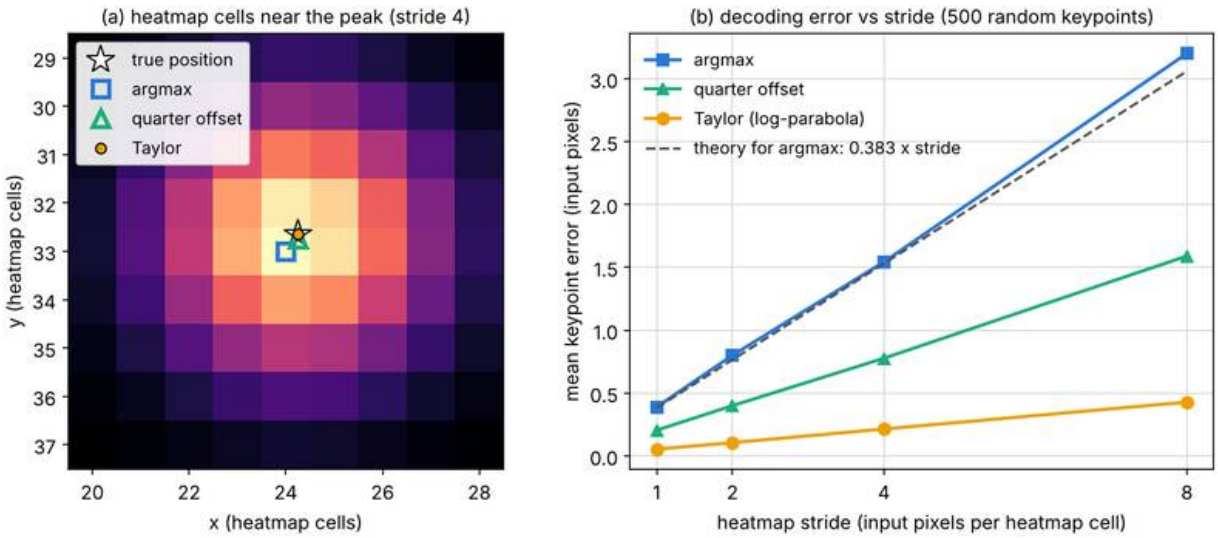


Figure 2 — Decoding a heatmap. (a) Near the peak at stride 4, the brightest cell (square) misses the true position (star); the quarter offset (triangle) moves partway; the Taylor step (dot) lands on it. (b) Mean error of each decoder over 500 random keypoints in the synthetic run of the code above; the dashed line is the argmax prediction of 0.383 times the stride.

Other output representations

Two newer representations sit between heatmaps and plain regression. **Coordinate classification** (SimCC [12]) predicts two one-dimensional distributions per keypoint, over x and over y, with several bins per pixel, which cuts quantization error without a large 2D map. **Likelihood-based regression** (RLE [13]) keeps direct regression but learns the shape of the error distribution during training, and its paper reports regression beating heatmap methods on multi-person pose. Both

are the natural outputs for the one-stage and query-based models later in this chapter.

Optional: Coordinate classification and likelihood-based regression (in the Supplementary notes). How SimCC's bins and splitting factor work, and why the choice of regression loss is an assumption about the error distribution.

Networks for single-person heatmaps

Plain version. To place a wrist precisely, the network needs fine detail (high resolution). To know that this blob *is* a left wrist and not a right one, it needs to see the whole body (a large receptive field). Every architecture below is a different way to have both at once.

The input is a **person crop**: a box around one person, enlarged a little, padded to a fixed aspect ratio and resized to a fixed size such as 256 × 192 (height × width). The output is K heatmaps at stride 4, 64 × 48 in this case.

- **Convolutional Pose Machines (CPM)** [14] stack several stages. Each stage sees the image features *and* the previous stage's heatmaps (belief maps), so later stages can use the positions of other joints as context, for example "an elbow lies between a shoulder and a wrist". Each stage is supervised (**intermediate supervision**), which counters vanishing gradients in a deep stack.
- **Stacked hourglass networks** [8] repeat a symmetric encoder–decoder ("hourglass") module several times, with intermediate supervision after each one. Every hourglass pools down to a very coarse resolution to gather whole-body context and upsamples back with skip connections, the same encoder–decoder idea described for segmentation in DLCV 03B.
- **SimpleBaseline** [9] showed that much of this machinery was unnecessary. It takes an ImageNet-pretrained ResNet (DLCV 02) and adds three deconvolution (transposed convolution) layers on top of its last stage to produce 64 × 48 heatmaps for a 256 × 192 input. Its ablation found that three deconvolution layers (heatmaps at 64 × 48) beat two (at 32 × 24) by 2.5 AP, a direct measurement of the resolution cost discussed above.
- **HRNet** [15] keeps a high-resolution branch alive through the whole network instead of recovering resolution at the end. It adds lower-resolution branches one by one, runs them **in parallel** and repeatedly exchanges information between all resolutions (multi-scale fusion). The heatmaps are read from the high-resolution branch, which has seen coarse context many times. Several later methods in this chapter build on HRNet: DARK reports its decoding results with HRNet models, and HigherHRNet and DEKR extend it to bottom-up pose.
- **ViTPose** [16] replaces the convolutional backbone by a plain, non-hierarchical vision transformer (ML 04) with 16×16 patches, followed by a lightweight decoder of a few deconvolution layers (or even a single upsampling layer and a convolution). The paper scales it from 100M to 1B parameters and shows that simple design plus large pretrained transformers sets a new

accuracy–throughput trade-off.

Optional: Simple Baselines (Xiao et al., 2018) (in the Paper guides). What the paper's ablations measure and what to check when you compare its numbers with other methods.

Two training details matter for every model in this list. **Data augmentation** rotates, scales and flips the crops; a horizontal flip must swap the left and right keypoint channels. **Flip testing** runs the model on the crop and on its mirror image, swaps the channels of the mirrored output back, and averages the heatmaps; the stacked hourglass paper reports about a 1% average improvement on validation from this [8], and later models adopted it as standard practice [9].

Optional: Flip testing and left–right bookkeeping (in the Supplementary notes). The index arithmetic of mirroring a heatmap and what goes wrong when the channels are not swapped.

Many people: top-down versus bottom-up

Plain version. With several people in a photo there are two orders of work. **Top-down**: first find each person, then find the joints inside each person's box. **Bottom-up**: first find every joint in the picture, then work out which joints belong together. A third family does both at once.

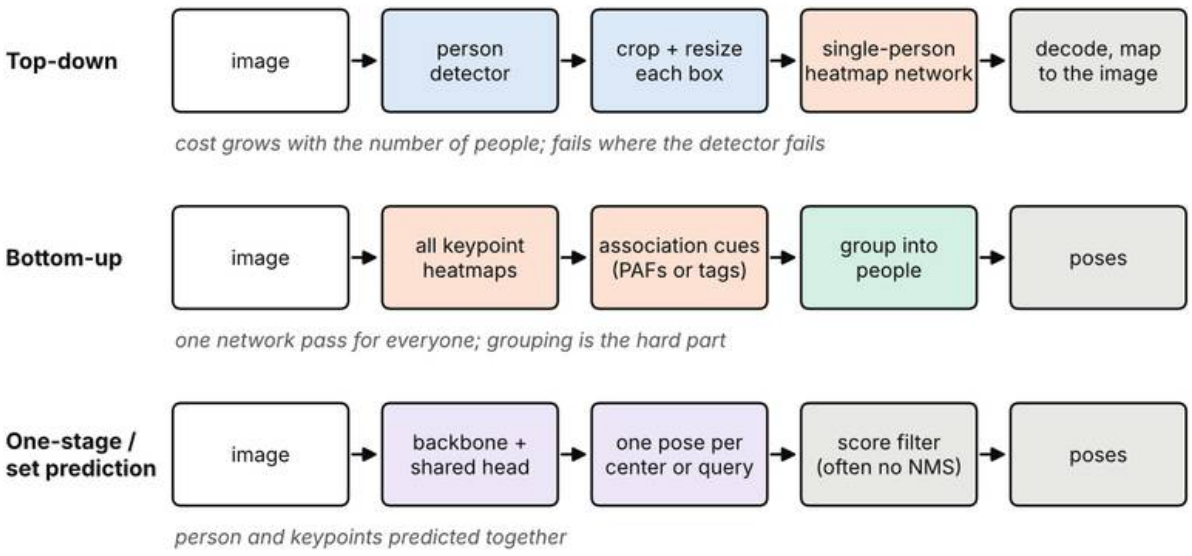


Figure 3 — Three multi-person pipelines. Top-down runs a single-person model per detected box; bottom-up finds all keypoints at once and groups them; one-stage and set-prediction models output a whole pose per person in one pass.

Top-down: detect, then estimate

A person detector (DLCV 03) proposes boxes; each box is cropped, resized and passed to a single-person network such as SimpleBaseline, HRNet or ViTPose; the heatmaps are decoded and the keypoints mapped back to image coordinates. Mask R-CNN [17] builds the same idea into the detector: its keypoint branch predicts, inside each region of interest, K masks of size $m \times m$, each trained as an m^2-way softmax whose target is a single "one-hot" pixel at the keypoint (see DLCV 03 for the RoI machinery).

Top-down methods are accurate because every person is resized to the same comfortable scale before the keypoint network sees them, which helps most for small and medium-sized people [18]. Their weaknesses follow from the pipeline:

- **Cost grows with the number of people**, since the keypoint network runs once per box.
- **They fail where the detector fails.** A missed person gets no pose; NMS (DLCV 03) may delete one of two overlapping people; and a box that contains two people gives the single-person network two left wrists to choose from. The OpenPose authors measured this on a COCO validation subset: the same single-person model scored 62.7 AP with ground-truth boxes but 52.7 AP with boxes from an SSD detector [18].

Bottom-up: detect all keypoints, then group

A bottom-up model runs once on the whole image and outputs K heatmaps that contain the keypoints of *all* people. Peaks are found with a local-maximum test, which is a per-keypoint NMS. The hard part is **grouping**: deciding which left elbow goes with which left wrist.

Part affinity fields (PAFs). OpenPose [18] predicts, besides the heatmaps, one 2D vector field per limb type. At a pixel on limb c of person k, the target is the unit vector pointing along that limb; elsewhere it is zero:

$$\mathbf{L}^*_{c,k}(\mathbf{p}) = \begin{cases} \mathbf{v} = \dfrac{\mathbf{x}_{j_2,k} - \mathbf{x}_{j_1,k}}{\|\mathbf{x}_{j_2,k} - \mathbf{x}_{j_1,k}\|_2} & \text{if } \mathbf{p} \text{ lies on limb } c \text{ of person } k, \\ \mathbf{0} & \text{otherwise,} \end{cases}$$

where j_1 and j_2 are the two keypoint types joined by limb c (for example left elbow and left wrist), $\mathbf{x}_{j,k}$ is the position of keypoint j of person k, and "on the limb" means within a distance σ_l (the limb width, in pixels) of the segment between them. Where limbs of several people overlap, the targets are averaged. To score a candidate connection between two detected keypoints $\mathbf{d}_{j_1}$ and $\mathbf{d}_{j_2}$, OpenPose integrates the predicted field $\mathbf{L}_c$ along the segment joining them:

$$E = \int_0^1 \mathbf{L}_c(\mathbf{p}(u)) \cdot \frac{\mathbf{d}_{j_2} - \mathbf{d}_{j_1}}{\|\mathbf{d}_{j_2} - \mathbf{d}_{j_1}\|_2} \, du, \qquad \mathbf{p}(u) = (1-u)\,\mathbf{d}_{j_1} + u\,\mathbf{d}_{j_2},$$

where u runs along the segment, and the integral is approximated by sampling a few evenly

spaced values of u. E is high when the field along the segment points the same way as the segment, which happens only if the two keypoints really are the ends of one limb. Finding the best assignment for all limbs at once is NP-hard, so OpenPose relaxes it: it keeps only the edges of a tree skeleton and solves one **bipartite matching** per limb type, for example with the Hungarian algorithm (see DLCV 06), then joins limbs that share keypoints into people.

Optional: OpenPose: part affinity fields (Cao et al., 2017) (in the Paper guides). The two-branch architecture, the greedy parse, and how to read the paper's top-down versus bottom-up comparison.

The code builds the ground-truth PAFs of two people standing side by side whose left forearms cross, adds small noise to the keypoint positions, and groups the keypoints limb by limb, once with the PAF score and once with plain distance (the shortest total limb length). The coordinates are synthetic.

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# 13 body keypoints and the 12 limbs of a tree that connects them
J = ["nose", "l_sho", "r_sho", "l_elb", "r_elb", "l_wri", "r_wri",
     "l_hip", "r_hip", "l_knee", "r_knee", "l_ank", "r_ank"]
LIMBS = [(0, 1), (0, 2), (1, 3), (3, 5), (2, 4), (4, 6),
         (1, 7), (2, 8), (7, 9), (9, 11), (8, 10), (10, 12)]

# two people side by side; their left forearms cross (synthetic coordinates)
A = np.array([[40, 20], [52, 38], [28, 38], [70, 50], [24, 62], [100, 74], [22, 86],
              [48, 80], [32, 80], [50, 110], [30, 110], [52, 140], [28, 140]], float)
B = np.array([[72, 24], [84, 42], [62, 42], [98, 58], [56, 64], [74, 76], [52, 86],
              [80, 84], [66, 84], [82, 114], [64, 114], [84, 144], [62, 144]], float)
people = [A, B]
H, W, width = 160, 120, 3.0

def render_paf(c):
    """Ground-truth part affinity field of limb c: unit vectors near each person's limb."""
    ys, xs = np.mgrid[0:H, 0:W]
    field, count = np.zeros((H, W, 2)), np.zeros((H, W))
    for P in people:
        a, b = P[LIMBS[c][0]], P[LIMBS[c][1]]
        v = (b - a) / np.linalg.norm(b - a)
        along = (xs - a[0]) * v[0] + (ys - a[1]) * v[1]
        across = np.abs(-(xs - a[0]) * v[1] + (ys - a[1]) * v[0])
        on = (along >= 0) & (along <= np.linalg.norm(b - a)) & (across <= width)
        field[on] += v; count[on] += 1
    return field / np.maximum(count, 1)[..., None]          # average where limbs overlap

def limb_score(field, p, q, n=10):
    """Line integral of the field along the segment p -> q, approximated by n samples."""
    d = (q - p) / np.linalg.norm(q - p)
    u = np.linspace(0, 1, n)[:, None]
    pts = np.round(p + u * (q - p)).astype(int)
    return np.mean(field[pts[:, 1], pts[:, 0]] @ d)

rng = np.random.default_rng(0)
```

```
cands = [np.stack([A[j], B[j]]) + rng.normal(0, 1.0, (2, 2)) for j in range(13)]  #
 ↪  detections

ok_paf = ok_near = 0
for c, (j1, j2) in enumerate(LIMBS):
    field = render_paf(c)
    S = np.array([[limb_score(field, p, q) for q in cands[j2]] for p in cands[j1]])
    D = np.linalg.norm(cands[j1][:, None] - cands[j2][None], axis=2)
    if (j1, j2) == (3, 5):                                  # the crossing forearms
        print("left elbow -> left wrist, PAF scores (rows: elbows A, B; cols: wrists A, B)")
        print(np.round(S, 2))
        print("distances in pixels"); print(np.round(D, 1))
    r, k = linear_sum_assignment(-S)                        # maximize total affinity
    ok_paf += np.sum(k == r)                                # candidate i belongs to
     ↪  person i
    r, k = linear_sum_assignment(D)                         # baseline: shortest total
     ↪  length
    ok_near += np.sum(k == r)
print(f"correct limbs with PAF scores:     {ok_paf}/{2 * len(LIMBS)}")
print(f"correct limbs with distance only:  {ok_near}/{2 * len(LIMBS)}")
```

```
left elbow -> left wrist, PAF scores (rows: elbows A, B; cols: wrists A, B)
[[0.67 0.04]
 [0.29 0.78]]
distances in pixels
[[41.1 27.2]
 [18.4 30.2]]
correct limbs with PAF scores:     24/24
correct limbs with distance only:  22/24
```

The printed matrices show the failure of geometry alone. Each elbow is *closer* to the other person's wrist (27.2 and 18.4 pixels) than to its own (41.1 and 30.2), so matching by distance connects both forearms wrongly. The PAF score is high only along the true limbs (0.67 and 0.78), because the field there points along the segment; along the wrong pairings it is low. Figure 4 draws both results. Note that the PAF here is the *ground truth*; a trained network predicts a blurrier field, and the method's accuracy depends on how well it does so.

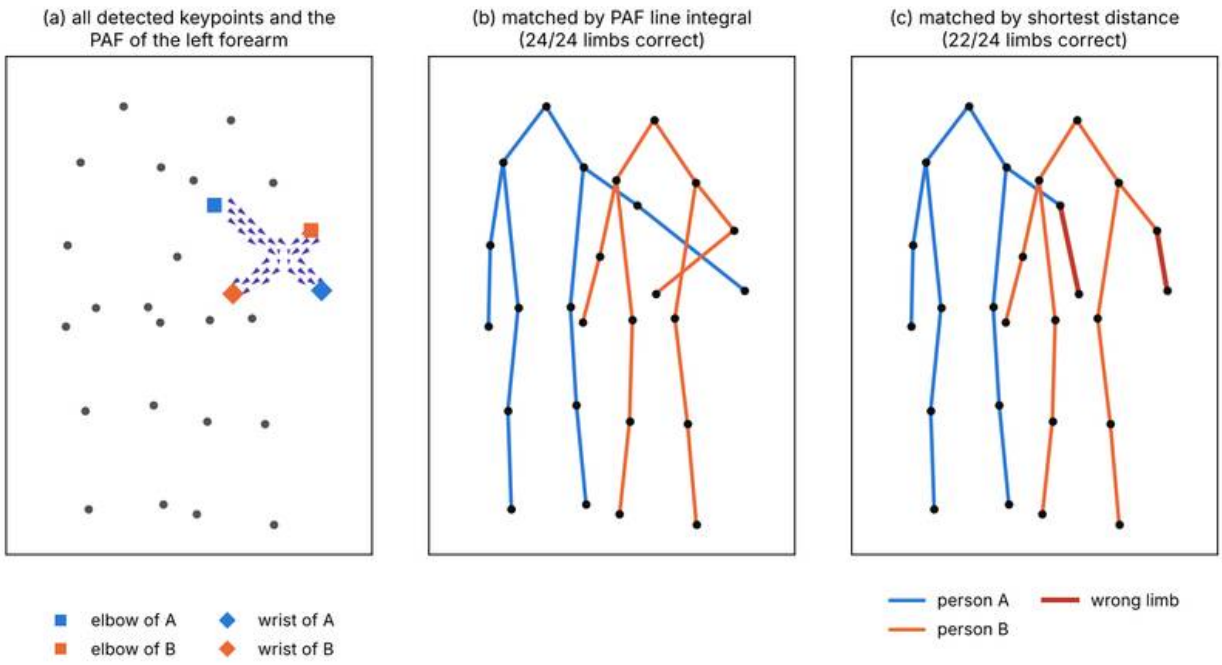


Figure 4 — Bottom-up grouping on the synthetic scene of the code above. (a) Detected keypoints and the left-forearm PAF. (b) Matching by the PAF line integral recovers every limb. (c) Matching by distance alone swaps the two crossing forearms (red).

Associative embedding. Newell et al. [19] avoid limb fields altogether. Alongside each keypoint heatmap the network outputs a **tag map**: one real number per pixel. Training pushes the tags of one person's keypoints to be equal and the tags of different people to be different. With $h_k(\mathbf{x})$ the tag of keypoint type k at pixel $\mathbf{x}$, $\mathbf{x}_{nk}$ the ground-truth position of keypoint k of person n, N people and $\bar{h}_n = \frac{1}{K}\sum_k h_k(\mathbf{x}_{nk})$ the mean tag of person n, the grouping loss is

$$\mathcal{L}_g = \frac{1}{N}\sum_{n}\sum_{k}\big(\bar{h}_n - h_k(\mathbf{x}_{nk})\big)^2 + \frac{1}{N^2}\sum_{n}\sum_{n'}\exp\Big(-\frac{1}{2\sigma^2}(\bar{h}_n - \bar{h}_{n'})^2\Big),$$

where the first term pulls each person's tags toward their mean and the second pushes the means of different people apart, with a penalty that fades as the distance between them grows (σ sets how far apart is "far enough"). The absolute tag values do not matter, only their differences. At test time, keypoints are added to people joint by joint, matching tags that are close and detections that are confident. CornerNet borrowed exactly this idea to pair box corners (DLCV 03).

Scale. Bottom-up methods see all people at the image's own scale, so small people get few heatmap cells; OpenPose was less accurate than top-down methods on medium-sized people for this reason [18]. HigherHRNet [20] adds higher-resolution heatmaps (HRNet's output upsampled by a transposed convolution) and supervises and aggregates several resolutions; its paper reports results above all top-down methods on the CrowdPose test set, where boxes are unreli-

able. DEKR [21] returns to **dense regression**, with one branch per keypoint whose adaptive convolutions focus on that keypoint's region, and reports better results than detection-and-grouping methods on COCO and CrowdPose.

One-stage and set-prediction pose

A third family predicts a whole pose per person directly, without a separate grouping step.

- **Center-based.** CenterNet ("Objects as Points") [22] detects each person as a centre point on a heatmap and regresses the $2K$ offsets from the centre to the keypoints; it then snaps each regressed keypoint to the nearest peak of a separate keypoint heatmap, so the centre acts as the grouping cue.
- **Real-time dense heads.** RTMO [23] puts a pose head into a YOLO-style one-stage detector and represents each keypoint with a pair of one-dimensional heatmaps (coordinate classification), aiming at top-down accuracy at one-stage speed.
- **Set prediction.** Transformer detectors in the DETR style (DLCV 03) output a fixed set of queries matched one-to-one to the ground truth, so no NMS or grouping is needed. ED-Pose [24] applies this to pose with two explicit box-detection stages: a human decoder detects people, and a human-to-keypoint decoder treats each keypoint as a small box to detect. Trained with an L1 regression loss and no heatmaps, it reports results above heatmap-based top-down methods with the same backbone on COCO.

Choosing a pipeline

	Top-down	Bottom-up	One-stage / set prediction
Main step	Detect people, then estimate each	Detect all keypoints, then group	Predict each person's pose directly
Cost vs number of people	Grows linearly	Nearly constant network cost; grouping grows	Nearly constant
Small people	Good (each crop is resized)	Harder (one image scale for all)	Depends on the feature pyramid
Crowds and overlap	Boxes overlap; NMS can remove people	Grouping can mix people	Queries or centres can collide
Typical use	High accuracy when people are few and well separated	Many people at once; real-time systems	End-to-end training; real-time

Measuring pose accuracy

Plain version. A predicted keypoint is "good" if it is close to the true one, but "close" must depend on the size of the person and on the keypoint: being 5 pixels off on a tiny person's nose is a big mistake; 5 pixels off on a large person's hip is not.

PCK and PCKh

The **percentage of correct keypoints** (**PCK**) counts a predicted keypoint as correct if it lies within a threshold distance of the ground truth, where the threshold is a fraction of the person's size. Yang and Ramanan used a fraction of the person's bounding-box size [25]. MPII's **PCKh** [26] uses 50% of the **head segment length** instead, because head size does not change when the person bends or stretches. Formally,

$$\mathrm{PCKh@}\alpha = \frac{1}{\sum_{n,k} \delta(v_{nk} > 0)} \sum_{n,k} \delta(v_{nk} > 0)\ \mathbb{1}\big[\|\hat{\mathbf{x}}_{nk} - \mathbf{x}_{nk}\|_2 \le \alpha\, \ell_n\big],$$

where n indexes people, k keypoints, ℓ_n is the head segment length of person n, α is the fraction ($\alpha = 0.5$ for the standard PCKh), $\delta(\cdot)$ and $\mathbb{1}[\cdot]$ are 1 when their condition holds and 0 otherwise, and only labeled keypoints count. PCK assumes the person has been found; it does not penalize missed or invented people, which is why multi-person benchmarks use an AP-style metric.

Object keypoint similarity and keypoint AP

COCO's metric mirrors box detection (DLCV 03). In box AP, a detection is matched to a ground-truth box if their IoU passes a threshold. In keypoint AP, a predicted pose is matched to a ground-truth person if their **object keypoint similarity** (**OKS**) passes a threshold [4]:

$$\mathrm{OKS} = \frac{\sum_i \exp\!\left(-d_i^2 / (2 s^2 \kappa_i^2)\right) \delta(v_i > 0)}{\sum_i \delta(v_i > 0)},$$

where i runs over the keypoints, d_i is the Euclidean distance between the predicted and ground-truth keypoint i, v_i is the ground-truth visibility flag, s is the object scale, defined as the square root of the person's segmentation area, and κ_i is a per-keypoint constant. Each term is an unnormalized Gaussian of the error with standard deviation $s\kappa_i$, so it is 1 for a perfect keypoint and falls toward 0 as the error grows; OKS averages it over the labeled keypoints. The predicted visibility flags are not used.

Where the constants come from. COCO had 5,000 validation images annotated twice and measured, for each keypoint type, how much human annotators disagree relative to the person's scale, $\sigma_i^2 = E[d_i^2/s^2]$. It then set $\kappa_i = 2\sigma_i$ [4]. The values of σ_i are 0.026 for the nose, 0.025 for the eyes,

0.035 for the ears, 0.079 for the shoulders, 0.072 for the elbows, 0.062 for the wrists, 0.107 for the hips, 0.087 for the knees and 0.089 for the ankles [4], [27]. Eyes are easy to place exactly; hips, hidden under clothing, are not, so the metric is more forgiving for hips. With $\kappa_i = 2\sigma_i$, errors of one, two and three annotator standard deviations give similarities of $e^{-1/8} \approx 0.88$, $e^{-1/2} \approx 0.61$ and $e^{-9/8} \approx 0.32$, and human annotators reach a median OKS of about 0.91 against each other [4].

From OKS to AP. The rest is the AP of DLCV 03: rank all predicted poses by their person-level score, match each to the unmatched ground-truth person with the highest OKS, and compute AP at OKS thresholds $0.50, 0.55, \ldots, 0.95$. The primary metric, AP, averages these ten values. Companion metrics are AP^{50}, AP^{75}, AP^{M} for medium people (area between 32^2 and 96^2) and AP^{L} for large ones (area above 96^2), plus the matching AR values. At most 20 poses per image are scored, and small people have no keypoint labels [4].

Optional: Inside OKS: unlabeled people, crowds and the scale s (in the Supplementary notes). How the evaluation code treats people without labeled keypoints, why every person counts equally, and what s means for a person who is partly hidden.

The code computes OKS with COCO's constants for a synthetic person about 200 pixels tall.

```
import numpy as np

NAMES = ["nose", "l_eye", "r_eye", "l_ear", "r_ear", "l_shoulder", "r_shoulder",
         "l_elbow", "r_elbow", "l_wrist", "r_wrist", "l_hip", "r_hip",
         "l_knee", "r_knee", "l_ankle", "r_ankle"]
SIGMAS = np.array([.26, .25, .25, .35, .35, .79, .79, .72, .72, .62, .62,
                   1.07, 1.07, .87, .87, .89, .89]) / 10.0       # COCO per-keypoint
                    ↪ constants

def oks(pred, gt, vis, area):
    """pred, gt: (17, 2) pixel coordinates; vis: (17,) COCO flags; area: segment area
     ↪ (s^2)."""
    d2 = ((pred - gt) ** 2).sum(1)
    kappa = 2 * SIGMAS
    ks = np.exp(-d2 / (2 * area * kappa ** 2))                # per-keypoint similarity
    labeled = vis > 0
    return ks[labeled].mean(), ks

# a standing person about 200 px tall, facing the camera (synthetic coordinates)
gt = np.array([[100, 30], [106, 25], [94, 25], [112, 28], [88, 28], [125, 60], [75, 60],
               [135, 95], [65, 95], [140, 128], [60, 128], [115, 130], [85, 130],
               [117, 175], [83, 175], [118, 220], [82, 220]], float)
vis = np.full(17, 2); vis[3] = 0                                  # left ear not labeled
area = 0.6 * (70 * 200)                                           # segment area: ~60% of its
 ↪ box

rng = np.random.default_rng(1)
jitter = gt + rng.normal(0, 3, gt.shape)                          # every keypoint off by ~3
 ↪ px per axis
flipped = jitter.copy(); flipped[[9, 10]] = flipped[[10, 9]]    # left/right wrists swapped
print(f"perfect          OKS = {oks(gt, gt, vis, area)[0]:.3f}")
print(f"3 px jitter      OKS = {oks(jitter, gt, vis, area)[0]:.3f}")
```

```
print(f"+ wrists swapped OKS = {oks(flipped, gt, vis, area)[0]:.3f}")

# the same 5-pixel error costs very different amounts on different keypoints
for k in (0, 5, 11):
    p = gt.copy(); p[k, 0] += 5
    print(f"5 px error on {NAMES[k]:10s}: similarity {oks(p, gt, vis, area)[1][k]:.3f}")
```

```
perfect          OKS = 1.000
3 px jitter      OKS = 0.933
+ wrists swapped OKS = 0.810
5 px error on nose      : similarity 0.577
5 px error on l_shoulder: similarity 0.942
5 px error on l_hip     : similarity 0.968
```

Three lessons follow from this synthetic example. Small jitter everywhere costs little (0.933). A single **left–right swap** of the wrists sets both wrist similarities to essentially zero and drops OKS to 0.810: this pose still counts as correct at thresholds up to 0.80 but fails at 0.85 and above, so it loses about a third of the ten AP thresholds. And the same 5-pixel error costs 0.42 of similarity on the nose but only 0.03 on a hip. Figure 5 shows the tolerance radius of each keypoint and the similarity curves.

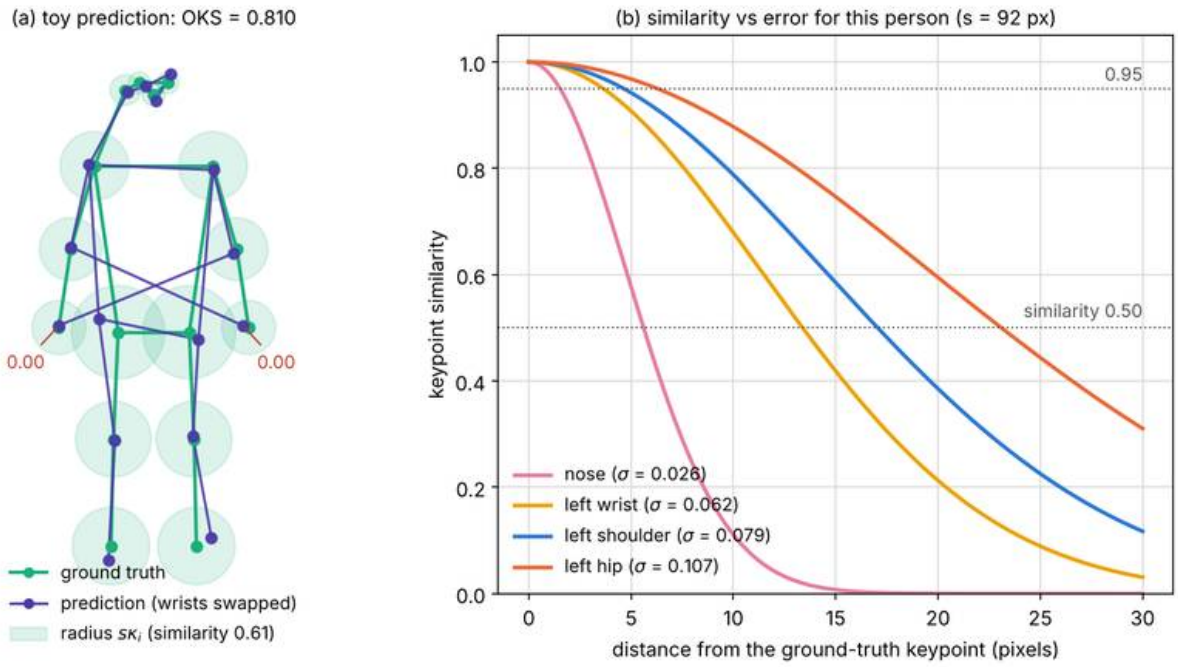


Figure 5 — OKS on the synthetic person of the code above. (a) Circles of radius $s\kappa_i$, where a keypoint's similarity is 0.61; the prediction is jittered by about 3 pixels and has its wrists swapped. (b) Keypoint similarity as a function of the error for four keypoint types, for this person's scale $s \approx 92$ pixels.

Metrics for 3D pose

For 3D pose, the standard metric is the **mean per-joint position error** (**MPJPE**), the average Euclidean distance between predicted and ground-truth 3D joints,

$$\text{MPJPE} = \frac{1}{K}\sum_{k=1}^{K} \|\hat{\mathbf{X}}_k - \mathbf{X}_k\|_2,$$

reported in millimetres and averaged over all test frames, where $\hat{\mathbf{X}}_k$ and $\mathbf{X}_k$ are the predicted and true 3D positions of joint k. Two protocols are common on Human3.6M: errors after aligning the root (central hip) joints of prediction and ground truth, called Protocol #1, and errors after an additional rigid alignment of the predicted skeleton to the ground truth, called Protocol #2 [28]; the second metric is also known as PA-MPJPE or reconstruction error [2]. The second hides errors in global rotation and scale, so the two numbers must never be compared with each other.

What the numbers do not tell you

Like box AP, keypoint AP mixes several kinds of error. Ronchi and Perona [29] split keypoint localization errors into four types: **jitter** (near the right keypoint but outside the human error margin), **inversion** (at the person's own left–right twin or another of their keypoints), **swap** (at a keypoint of a *different person*) and **miss** (near no keypoint). They found that errors depend most on the number of visible keypoints and on clutter from nearby people. Even COCO's doubly annotated data contain 1–2% outliers, mostly left–right confusions or the wrong person labeled [4]. Diagnose models by error type, not only by AP.

Datasets and benchmarks

Plain version. A handful of datasets defined what "good" means: COCO for many people in everyday photos, MPII for a wide range of activities, CrowdPose for heavy overlap, Human3.6M for 3D, and newer sets for whole bodies, video and animals.

Dataset	Content (from the papers or official pages)	Metric	Why it matters
COCO keypoints [30], [4]	17 body keypoints on people in everyday photos; small people unlabeled	OKS AP	The main multi-person benchmark
MPII Human Pose [26]	40,522 annotated people in images from YouTube videos, collected with queries for more than 800 activities	PCKh	Diverse activities; single- and multi-person tracks

Dataset	Content (from the papers or official pages)	Metric	Why it matters
CrowdPose [31]	20,000 images, about 80,000 people, 14 keypoints, selected for a uniform spread of crowding	OKS AP	Crowded scenes
COCO-WholeBody [32]	133 landmarks per person: 68 face, 42 hand, 23 body and feet	OKS AP	Face, hands and feet together
PoseTrack [33]	Videos with person tracks and articulated poses	AP, MOTA	Pose estimation and tracking in video
Human3.6M [34]	3.6 million 3D poses of 11 actors in 17 scenarios, recorded by 4 calibrated cameras with motion capture	MPJPE	The most widely used 3D benchmark
AP-10K [35]	10,015 images of 54 mammal species from 23 families, annotated in COCO style	AP	Animal pose and cross-species generalization

CrowdPose deserves a closer look because it defines crowding precisely. For person i, let N_a^i be the number of its own joints inside its box and N_b^i the number of *other* people's joints inside the same box; the **Crowd Index** of an image with n people is $\frac{1}{n}\sum_i N_b^i/N_a^i$ [31]. Common datasets are dominated by uncrowded images; CrowdPose was selected so that this index is spread nearly uniformly between 0 and 1, and its authors found the accuracy of strong models to be about 20 AP lower on crowded images than on uncrowded ones [31].

Two cautions apply when reading results. Human3.6M is an indoor dataset with a few actors [2], [34], so low MPJPE there does not imply good results on outdoor video. And on COCO, results depend on the person detector used by top-down methods (see the Simple Baselines guide), on input resolution and on extra training data, so compare numbers only under the same settings.

Beyond the human body

Plain version. Nothing in a heatmap network is specific to people. Change the list of keypoints and the training data, and the same machinery finds the corners of a face, the joints of a hand, or the paws of a dog.

- **Whole body.** COCO-WholeBody [32] adds feet, hands and dense face landmarks to the body keypoints, which creates a huge range of scales inside one person; its baseline model, ZoomNet, is built around the body's hierarchical structure to handle this. The journal version of OpenPose [36] added foot keypoints.
- **Animals.** AP-10K [35] benchmarks pose across 54 mammal species and tests generalization

to unseen animal families. In neuroscience, DeepLabCut [1] uses transfer learning from a pre-trained network to track user-defined body parts of mice and flies from about 200 labeled frames in its experiments, replacing reflective markers.

- **Objects.** The same formulation works for any category with consistent landmarks, such as the wheels and lights of cars, provided the keypoints are defined consistently and the category's left–right symmetries are handled like the body's.

From 2D to 3D pose

Plain version. A photo flattens the world: from one image, a hand pointing at the camera and a hand held at the hip can look almost the same. There are three ways to get the depth back: *guess* it from experience (lifting), *measure* it with two or more cameras (triangulation), or *fit* a model of the human body that only allows realistic shapes.

Lifting 2D keypoints

Martinez et al. [28] showed that a surprisingly large part of monocular 3D pose estimation is solved once the 2D keypoints are known. Their model takes the $2K$ coordinates of a detected 2D pose and maps them with a simple fully connected network (linear layers with batch normalization, dropout and residual connections) to $3K$ coordinates. Trained on Human3.6M, it beat the previous best methods by a large margin when given ground-truth 2D keypoints, which suggests that much of the remaining difficulty lies in the 2D detection rather than in the lifting. Lifting is also *ambiguous*: several 3D poses project to the same 2D pose, and a network with a single output can return only one of them.

VideoPose3D [37] lifts a whole *sequence* of 2D poses with dilated temporal convolutions, so motion over neighbouring frames helps resolve depth and jitter. It also introduced **back-projection**, a semi-supervised loss that estimates 3D poses for unlabeled video and requires them to project back onto the detected 2D keypoints.

Multi-view triangulation

With two or more calibrated cameras, depth is no longer a guess. Each camera's 2D keypoint defines a ray in space, and the 3D joint is the point where the rays (almost) meet. The geometry is the triangulation of DLCV 08, applied joint by joint: the camera matrices are the pinhole models of that chapter, and the linear (DLT) solution is derived in its Supplementary notes. Learnable triangulation [38] makes this step part of the network: it weights each view by a confidence predicted from the image, so that a view in which the joint is occluded contributes less, and its second variant aggregates 2D features from all views into a 3D volume refined by 3D convolutions.

The code projects a synthetic 13-joint skeleton, standing 4 m from two converging cameras, into both images, adds 2-pixel noise to the 2D keypoints, triangulates every joint with the DLT and

reports MPJPE for three distances between the cameras (the **baseline**). The numbers come from this synthetic run, without root alignment.

```
import numpy as np

def camera(K, angle_deg, center):
    """3x4 projection P = K [R | t] for a camera at `center` rotated about the y axis."""
    a = np.deg2rad(angle_deg)
    R = np.array([[np.cos(a), 0, -np.sin(a)], [0, 1, 0], [np.sin(a), 0, np.cos(a)]])
    return K @ np.hstack([R, -R @ center[:, None]])

def project(P, X):
    x = (P @ np.hstack([X, np.ones((len(X), 1))]).T).T
    return x[:, :2] / x[:, 2:]

def triangulate(Ps, xs):
    """Linear (DLT) triangulation of one point seen in several views."""
    A = []
    for P, (u, v) in zip(Ps, xs):
        A += [u * P[2] - P[0], v * P[2] - P[1]]
    X = np.linalg.svd(np.array(A))[2][-1]
    return X[:3] / X[3]

# a 13-joint 3D skeleton in millimetres, standing 4 m in front of the cameras (synthetic)
skel = np.array([[0, -1650, 0], [170, -1400, 0], [-170, -1400, 0], [250, -1120, 50], [-250,
 ↪  -1120, 50],
                 [280, -860, 120], [-280, -860, 120], [100, -900, 0], [-100, -900, 0],
                 [110, -480, 30], [-110, -480, 30], [110, -60, 0], [-110, -60, 0]], float)
skel += [0, 900, 4000]
K = np.array([[1000, 0, 640], [0, 1000, 360], [0, 0, 1]], float)

rng = np.random.default_rng(0)
for baseline in (200, 1000, 3000):                          # distance between the cameras
 ↪  (mm)
    half = baseline / 2
    Ps = [camera(K, np.degrees(np.arctan2(half, 4000)), np.array([-half, 0, 0.])),
          camera(K, -np.degrees(np.arctan2(half, 4000)), np.array([half, 0, 0.]))]
    errs = []
    for trial in range(200):                                    # 2D detections with 2 px noise
        obs = [project(P, skel) + rng.normal(0, 2.0, (13, 2)) for P in Ps]
        est = np.array([triangulate(Ps, [o[j] for o in obs]) for j in range(13)])
        errs.append(np.linalg.norm(est - skel, axis=1).mean())   # MPJPE of this trial
    print(f"baseline {baseline:4d} mm: MPJPE = {np.mean(errs):6.1f} mm")
```

```
baseline  200 mm: MPJPE =  183.7 mm
baseline 1000 mm: MPJPE =   37.9 mm
baseline 3000 mm: MPJPE =   17.4 mm
```

With cameras 20 cm apart, a 2-pixel image error becomes about 18 cm in 3D, almost all of it in depth (a per-axis check of the same run gives 182 mm in depth against 9 and 18 mm across):

the rays are nearly parallel, so a small shift moves their crossing point far along them. A wider baseline makes the rays cross at a larger angle and cuts the error by a factor of ten.

Parametric body models and mesh recovery

A skeleton says nothing about body shape, and nothing stops a lifting network from predicting a forearm that changes length from frame to frame. **Parametric body models** fix both problems by describing a body with a small set of parameters. The most widely used is SMPL [39]. It starts from a template mesh with $N = 6890$ vertices and a skeleton with $K = 23$ joints, and generates a posed body as

$$M(\boldsymbol{\beta}, \boldsymbol{\theta}) = W(T_P(\boldsymbol{\beta}, \boldsymbol{\theta}), J(\boldsymbol{\beta}), \boldsymbol{\theta}, \mathcal{W}), \qquad T_P(\boldsymbol{\beta}, \boldsymbol{\theta}) = \bar{\mathbf{T}} + B_S(\boldsymbol{\beta}) + B_P(\boldsymbol{\theta}),$$

where $\boldsymbol{\beta}$ are **shape** parameters (coefficients of the main modes of body-shape variation), $\boldsymbol{\theta}$ are **pose** parameters (an axis–angle rotation for each joint relative to its parent, plus the global orientation: $3 \times 23 + 3 = 72$ numbers), $\bar{\mathbf{T}}$ is the template mesh, B_S and B_P are vertex offsets that model identity and pose-dependent deformations such as bulging muscles, $J(\boldsymbol{\beta})$ are the joint locations regressed from the shaped mesh, $\mathcal{W}$ are the blend weights and W is a standard blend-skinning function that rotates each vertex around the joints. Keypoints are read off the mesh, so a fitted SMPL body always has consistent bone lengths.

Human mesh recovery (HMR) [40] regresses SMPL parameters and a camera directly from an image. Its main loss is the **reprojection error** of the model's joints against 2D keypoint labels, which allows training on in-the-wild images; an adversarial prior trained on real 3D bodies rejects implausible parameters. HMR 2.0, part of 4DHumans [41], makes the network fully transformer-based and tracks the reconstructed people through occlusions in video.

Optional: SMPL in more detail (in the Supplementary notes). What each term of the model does, how joints and keypoints are obtained from the mesh, and why the pose blend shapes vanish in the rest pose.

Failure modes and practical advice

Plain version. Pose models fail where people would hesitate too: hidden joints, tangled people, mirror-image confusion, bodies cut off by the frame, poses rarely seen in training, and blur.

- **Occlusion.** An occluded joint ($v = 1$) is labeled but not shown, so the network must infer it from context; self-occlusion makes legs especially prone to left–right confusion [29].
- **Crowds.** Overlapping people break detectors, confuse grouping and cause **swap** errors [29]; accuracy falls as the Crowd Index rises [31].

- **Left–right flips.** Front and back views look alike, so a model can mirror a limb or a whole person (an **inversion**); a forgotten channel swap in flip augmentation causes the same error systematically.
- **Truncation.** Joints outside the frame are unlabeled or guessed, and crops of a partial person are poorly centred.
- **Unusual poses.** Gymnastics, dancing or lying people are rare in training data; HMR 2.0's authors single them out as a case where earlier methods struggled [41].
- **Motion blur, low resolution and domain shift.** Fast limbs smear, small people have few pixels per limb, and models trained on adult photos transfer poorly to infants or artwork [2].

Practical advice that follows from this chapter: use an unbiased encoder and a sub-pixel decoder; check that flipping swaps left and right channels; report the detector and input size with top-down results; evaluate crowded and uncrowded images separately; break errors down by type; and in video, smooth or track keypoints over time. Tracking-by-detection with the association methods of DLCV 06 carries over directly: SimpleBaseline, for example, propagates poses to the next frame with optical flow and matches them with an OKS-based similarity [9], and PoseTrack provides the benchmark [33]. Following individual surface points instead of named joints is the point-tracking problem of DLCV 06.

Modern view

What the surveys say

Zheng et al. [2] review more than 250 deep-learning papers on 2D and 3D pose since 2014, organized as in this chapter: regression versus heatmaps, top-down versus bottom-up, and for 3D, lifting, mesh recovery and multi-view methods. Their future directions include domain adaptation, human–scene constraints and metrics for temporal smoothness.

Chen et al. [3] organize monocular methods into four tasks (2D or 3D, single- or multi-person), list the design tricks that proved effective (multi-stage structures, intermediate supervision, multi-scale fusion, body-structure constraints) and name occlusion and crowds as the main unresolved challenges.

Tian et al. [5] survey monocular 3D mesh recovery with body models, contrasting optimization-based fitting with regression networks, and point to the lack of in-the-wild 3D labels as the central difficulty.

Optional: Guide to the Modern view surveys (in the Paper guides). What each survey covers, how it is organized and how to use it as a map.

The state of the art, 2023–2026

- **Scaled transformers for top-down pose.** ViTPose [16] showed that plain vision transformers with simple decoders scale well (from 100M to 1B parameters in the paper), and its largest model set a new state of the art on the COCO keypoint benchmark at publication.
- **Human-centric foundation models.** Sapiens [42] pretrains vision transformers from 0.3 to 2 billion parameters on over 300 million images of humans with self-supervision, then fine-tunes them for 2D pose, body-part segmentation, depth and surface normals; its authors report that this human-specific pretraining improves all four tasks at the same compute budget.
- **Real-time systems.** RTMPose [43] (top-down, coordinate classification, engineered for CPU, GPU and mobile deployment) and RTMO [23] (one-stage) target the latency budgets of applications rather than leaderboard peaks.
- **End-to-end multi-person models.** Query-based models such as ED-Pose [24] remove NMS, grouping and heatmaps; ED-Pose reports higher COCO accuracy than heatmap-based top-down methods with the same backbone.
- **3D humans in video.** Transformer mesh recovery such as HMR 2.0 [41] reconstructs and tracks full 3D bodies from monocular video, moving the field from sparse skeletons toward complete body surfaces.

Open problems

1. **Crowds and heavy occlusion** remain the largest accuracy gap [31], [3].
2. **In-the-wild 3D.** Accurate 3D labels are scarce outside the lab [5], and the most widely used 3D benchmark, Human3.6M, is an indoor dataset [2], so monocular 3D methods are trained and tested far from where they are used.
3. **Temporal consistency.** Per-frame metrics such as MPJPE do not measure jitter or physical plausibility over time [2].
4. **Generalization across domains and species**: infants, artwork, unusual viewpoints and unseen animal families [2], [35].
5. **Evaluation.** OKS constants were measured from annotator disagreement on COCO people [4] and do not transfer automatically to other keypoint sets, and AP hides error types such as inversions and swaps [29].

Key takeaways

- A pose is a skeleton graph of named keypoints with positions and visibility flags; left and right are the person's own, and $v = 1$ (occluded) differs from $v = 0$ (unlabeled).

- Heatmaps (Gaussian blobs, MSE loss) dominate because they suit convolutions and express uncertainty; regression, coordinate classification and soft-argmax trade some of that for speed and end-to-end training.
- Heatmap decoding is limited by the stride: argmax loses about $0.383s$ pixels on average; the quarter offset halves that and a Taylor step on the log-heatmap removes most of the rest.
- Top-down pipelines (detector plus SimpleBaseline, HRNet or ViTPose) lead on accuracy but scale with the number of people and inherit detector failures; bottom-up pipelines detect all keypoints once and group them with part affinity fields or associative embedding; one-stage and query-based models predict whole poses directly.
- COCO's OKS is a scale- and keypoint-normalized Gaussian similarity, $\kappa_i = 2\sigma_i$ from annotator disagreement; keypoint AP averages over OKS 0.50–0.95. PCKh thresholds at half the head length; MPJPE measures 3D error in millimetres.
- 3D pose comes from lifting 2D keypoints (ambiguous), triangulating them across calibrated views (accurate with a wide baseline) or fitting a body model such as SMPL, whose 72 pose and few shape parameters keep bodies plausible.
- Occlusion, crowds, left–right inversions, truncation, unusual poses and blur cause most errors; diagnose by error type, not only by AP.

Exercises

1. **Quantization by hand.** A wrist lies at $x = 101.3$ pixels in a crop, and the heatmap stride is 4. Which heatmap column holds the peak? What does argmax decoding return, and what is its error? What does the quarter offset return? What would an exact Taylor step on a noise-free Gaussian return?

   Hint

   In heatmap units the wrist is at $101.3/4 = 25.325$, so the peak is in column 25 and argmax returns $25 \times 4 = 100$ pixels, an error of 1.3 pixels. The right neighbour (column 26) is closer to 25.325 than the left one, so the quarter offset moves to 25.25, which maps to 101.0 pixels (error 0.3). On a noise-free Gaussian, the log-heatmap is an exact parabola, so the Taylor step returns 25.325, that is 101.3 pixels.

2. **OKS by hand.** A person has segmentation area 10,000 pixels², so $s = 100$. A predicted knee is 10 pixels from the ground truth. Compute its keypoint similarity with COCO's constant. Repeat for a nose 10 pixels off. Which error would you rather make, and why does COCO treat them so differently?

   Hint

   Knee: $\kappa = 2 \times 0.087 = 0.174$, $s\kappa = 17.4$, similarity $= \exp(-100/(2 \times 17.4^2)) = \exp(-0.165) \approx 0.85$. Nose: $\kappa = 0.052$, $s\kappa = 5.2$, similarity $= \exp(-100/54.08) \approx 0.16$. Annotators place noses about three times more consistently than knees, and COCO scales the tolerance by that disagreement.

3. **When does top-down become too slow?** A person detector takes 20 ms per image and a single-person network takes 5 ms per crop. A bottom-up network takes 40 ms per image plus 1 ms of grouping per person. For how many people per image is bottom-up faster? Name two reasons, other than speed, to prefer one pipeline in a crowded train station.

   Hint

   Top-down costs $20 + 5N$ ms and bottom-up $40 + N$ ms, so bottom-up is faster when $4N > 20$, that is for more than 5 people. In a crowded station, top-down suffers from overlapping boxes and NMS removing people, but handles small distant people better because each crop is resized; bottom-up has a constant network cost but must group heavily overlapping limbs, and small people get few heatmap cells.

4. **Break the part affinity field.** In the grouping code, move person B so that its left forearm lies on the same line as person A's left forearm and overlaps it (one person standing behind the other). What happens to the PAF scores of the two correct and the two wrong pairings, and why? What extra cue could resolve the case?

   Hint

   Where the two limbs overlap, the target field is the average of two identical unit vectors, so it points along the shared line. Every candidate segment between an elbow and a wrist on that line, correct or not, is aligned with the field and gets a similar score, so the matching becomes ambiguous. Cues that can help: the lengths of the limbs relative to the rest of each body, tags as in associative embedding, the other limbs of each person (a global rather than per-limb parse), or temporal continuity in video.

5. **A swap the cameras cannot see.** Change the triangulation code so that both cameras aim at the centre of the skeleton, and in the second camera exchange the 2D positions of the left and right knees (an inversion error of the 2D detector). Triangulate the left knee and compute its mean reprojection error in the two views, with and without the swap. Then move the second camera 1 m *above* the first instead of beside it and repeat. Explain the difference.

   Hint

   In our run of this experiment, with the cameras side by side the reprojection error is about 1 pixel with or without the swap, yet the swapped knee lands more than half a metre from the truth. The two knees are at nearly the same height, and with a horizontal baseline the epipolar lines are nearly horizontal, so swapping the knees moves the observation *along* its epipolar line: the two rays still meet, just at the wrong depth. With the second camera above the first, the epipolar lines become nearly vertical, the swap moves the observation off its line, and the reprojection error jumps to tens of pixels, which flags the error. Camera layouts for multi-view pose should therefore not place all cameras along one horizontal line.

6. **Forgetting the channel swap.** You run flip testing but forget to swap the left and right channels of the mirrored heatmaps before averaging. Describe what the averaged left-wrist heatmap looks like for a person facing the camera with arms spread, and what argmax decoding returns.

Hint

Think about what the network sees in the mirrored image: a person facing the camera, so it labels the wrist on the image's right as "left wrist". That wrist is the person's *right* wrist, mirrored. Flipping the map back without swapping channels therefore puts a blob on the true *right* wrist in the left-wrist channel. The average has two half-height blobs, one on each wrist, and argmax picks whichever is slightly higher, so roughly half the time the left wrist is reported on the right wrist: an inversion error.

Supplementary notes

These notes are optional: each takes one point from the main text and treats it in more detail. Nothing in the main text, code or exercises depends on them.

Sub-pixel decoding by a Taylor step

This note derives the sub-pixel refinement used by DARK [10] and implemented in the decoding code, and lists the cases where it fails.

Setup. Let $\hat{H}$ be one predicted heatmap and $\mathbf{m}$ the cell with the largest value. Assume that near its peak the heatmap is a scaled Gaussian with an unknown centre $\boldsymbol{\mu}$ and covariance Σ:

$$\hat{H}(\mathbf{p}) \approx c \exp\Big(-\tfrac{1}{2}(\mathbf{p}-\boldsymbol{\mu})^{\top}\Sigma^{-1}(\mathbf{p}-\boldsymbol{\mu})\Big),$$

where $c > 0$ is the peak height and $\mathbf{p}$ is a position in heatmap units. Take the logarithm,

$D(\mathbf{p}) = \log \hat{H}(\mathbf{p}) = \log c - \tfrac{1}{2}(\mathbf{p}-\boldsymbol{\mu})^{\top}\Sigma^{-1}(\mathbf{p}-\boldsymbol{\mu})$. This is an exact quadratic in $\mathbf{p}$, with gradient $\mathbf{D}'(\mathbf{p}) = -\Sigma^{-1}(\mathbf{p}-\boldsymbol{\mu})$ and constant Hessian $\mathbf{D}'' = -\Sigma^{-1}$.

The step. Evaluate both at the brightest cell and solve for the centre:

$\mathbf{D}'(\mathbf{m}) = -\Sigma^{-1}(\mathbf{m}-\boldsymbol{\mu}) = \mathbf{D}''(\mathbf{m}-\boldsymbol{\mu})$, so

$$\boldsymbol{\mu} = \mathbf{m} - \mathbf{D}''(\mathbf{m})^{-1}\,\mathbf{D}'(\mathbf{m}).$$

This is one Newton step toward the maximum of D. Because D is exactly quadratic for a Gaussian, one step is exact; there is no need to iterate. DARK writes the same result as a second-order Taylor expansion of D around $\mathbf{m}$ [10]. SIFT refines its keypoints with the same kind of step on the difference-of-Gaussians function (see the keypoint localization note of IP 12).

Finite differences. On the grid, with $D_{x,y}$ the log-heatmap at column x and row y and $(x, y) = \mathbf{m}$:

$$D_x = \tfrac{1}{2}(D_{x+1,y} - D_{x-1,y}), \qquad D_{xx} = D_{x+1,y} - 2D_{x,y} + D_{x-1,y},$$
$$D_y = \tfrac{1}{2}(D_{x,y+1} - D_{x,y-1}), \qquad D_{yy} = D_{x,y+1} - 2D_{x,y} + D_{x,y-1},$$
$$D_{xy} = \tfrac{1}{4}(D_{x+1,y+1} - D_{x-1,y+1} - D_{x+1,y-1} + D_{x-1,y-1}),$$

where D_x, D_y form $\mathbf{D}'$ and D_{xx}, D_{xy}, D_{yy} form the 2×2 matrix $\mathbf{D}''$. For a quadratic function these central differences are exact, which is why the code's Taylor decoder recovers the true position on a noise-free Gaussian.

Why the logarithm matters. Fitting a parabola to $\hat{H}$ itself (not its log) also gives a sub-pixel estimate, but a Gaussian is not a parabola, so that estimate is biased toward the cell centre unless the blob is wide. The log turns the Gaussian into a parabola exactly.

Edge cases and pitfalls.

- *Peak on the border.* The central differences need all eight neighbours; at the map border the decoder must fall back to the cell centre (or a one-sided rule), as the code does.
- *Flat or saddle-shaped peaks.* If $\mathbf{D}''$ is not negative definite (determinant not positive, or $D_{xx} \geq 0$), the step can jump far away. The code checks this and keeps the cell centre.
- *Multiple peaks.* Trained networks often produce lumpy maps with several local maxima near the keypoint. DARK first smooths the predicted heatmap with a Gaussian kernel of the training width and rescales it to the original maximum [10], which makes the local shape closer to a single Gaussian.
- *Values near zero.* The log of tiny or negative predicted values is undefined or huge; clip the heatmap to a small positive floor before taking the log.
- *Biased targets.* If the training targets were centred on the *rounded* keypoint position, the network learns to put the peak on the cell centre, and no decoder can recover the lost fraction. DARK therefore centres the target Gaussian on the exact, unrounded position [10].

Coordinate classification and likelihood-based regression

This note gives the details of the two output representations that the main text mentions between heatmaps and plain regression.

SimCC: classify each coordinate. For an input crop of width W and height H pixels, SimCC [12] divides the horizontal axis into $N_x = W \cdot k$ bins and the vertical axis into $N_y = H \cdot k$ bins, where

$k \geq 1$ is the **splitting factor** (bins per pixel). A keypoint at (x, y) gets the class labels $c_x \approx kx$ and $c_y \approx ky$. The backbone (a CNN such as HRNet, or a transformer) produces one feature vector per keypoint (for a CNN, the keypoint's $H' \times W'$ feature map flattened), and two single linear layers turn it into scores over the N_x and N_y bins. Training uses a Kullback–Leibler divergence between the predicted and target distributions, and the paper replaces hard one-hot targets by smoothed ones (equal, Laplace or Gaussian label smoothing), because a bin next to the correct one is much less wrong than a bin far away. Decoding takes the highest-scoring bin on each axis and divides by k.

Quantization. With k bins per pixel the rounding error per axis is at most $1/(2k)$ pixels, compared with $s/2$ pixels for a heatmap at stride s. So $k = 2$ gives quarter-pixel resolution without any 2D map, and the cost grows only linearly with $W + H$ instead of with $W \cdot H$.

Edge cases. The two axes are classified independently, so when two candidate positions exist (two wrists of overlapping people inside one crop), the x of one can be combined with the y of the other, a failure a 2D heatmap does not have.

RLE: regression as maximum likelihood. Suppose the network predicts a mean $\hat{\mu}$ and a scale $\hat{\sigma}$ for a coordinate, and the label is μ_g. If the error is assumed Gaussian, maximizing the likelihood of the label means minimizing

$$\mathcal{L} = -\log P(\mu_g) \propto \log \hat{\sigma} + \frac{(\mu_g - \hat{\mu})^2}{2\hat{\sigma}^2},$$

which becomes the plain L2 loss when $\hat{\sigma}$ is held constant; a Laplace assumption with constant scale gives the L1 loss [13]. Every regression loss is therefore a hidden assumption about the shape of the error distribution. RLE [13] learns that shape instead: it rescales the error as $\bar{\mu} = (\mu_g - \hat{\mu})/\hat{\sigma}$ and models its density with a normalizing flow that learns only the *residual* (the change) relative to a simple base density, which the paper finds easier to optimize than learning the density from scratch. The flow is used only in the training loss; at test time the network outputs $\hat{\mu}$ as an ordinary regression, so inference costs nothing extra. The paper's ablation on COCO attributes 12.4 AP of improvement over standard regression to the method, of which 7.8 AP come from the residual formulation rather than direct likelihood estimation [13].

Why it matters for cost. The RLE paper notes that a standard three-layer deconvolution heatmap head costs 1.4 times the FLOPs of a ResNet-50 backbone, while a regression head costs about 1/20000 of them [13]. Making regression accurate therefore makes cheap heads viable.

Flip testing and left–right bookkeeping

This note works through the index arithmetic of mirroring heatmaps, the step that flip augmentation and flip testing share.

Mirroring the image. For a crop of width W_{in} pixels, a horizontal flip maps an image column x to $W_{\text{in}} - 1 - x$ (with pixel centres at integer coordinates). Keypoint coordinates are transformed the same way: $x_k \mapsto W_{\text{in}} - 1 - x_k$, with y_k unchanged.

Swapping the labels. After mirroring, the person's left side appears where the right side was. Let π be the permutation that exchanges each left keypoint with its right twin (left eye with right eye, left shoulder with right shoulder, and so on) and leaves the nose unchanged. A flipped training sample must use the label of keypoint $\pi(k)$ for channel k; otherwise the network is trained to call the person's right wrist "left" in half of the images, and it learns to hedge between the two.

Flip testing. Given heatmaps $\hat{H}_k$ for the crop and $\hat{H}^{\text{flip}}_k$ for its mirror image, the averaged map for channel k is

$$\bar{H}_k(x, y) = \tfrac{1}{2}\big(\hat{H}_k(x, y) + \hat{H}^{\text{flip}}_{\pi(k)}(W_{\text{h}} - 1 - x,\ y)\big),$$

where W_h is the heatmap width and (x, y) is a heatmap cell. Two operations happen: the mirrored map is flipped back column-wise, and its channels are permuted with π.

A subtle offset. The column mapping $x \mapsto W_h - 1 - x$ is exact only if heatmap cells and image pixels are aligned the same way after flipping. With stride s and image width $W_{in} = sW_h$, suppose cell x stands for image position sx (its first pixel). Flipping the image sends that position to $W_{in} - 1 - sx$, but the flipped cell $W_h - 1 - x$ stands for $s(W_h - 1 - x) = W_{in} - s - sx$. The two differ by $s - 1$ pixels, so averaging the two maps biases every keypoint by about $(s-1)/2$ pixels. If instead cell x stands for the centre of the pixels it covers, $sx + (s-1)/2$, the flipped cell stands for $W_{in} - 1 - (sx + (s-1)/2)$, exactly the flipped position, and the bias disappears. Whatever convention a codebase uses, test it with a synthetic heatmap: flip-test a perfect Gaussian and check that the decoded position does not move.

What goes wrong without the swap. Exercise 6 shows that forgetting π in flip testing averages a left-wrist blob with a right-wrist blob, which produces inversion errors. In augmentation the same mistake is quieter: it does not crash and only lowers accuracy, so it is a classic silent bug.

Inside OKS: unlabeled people, crowds and the scale s

This note describes the parts of COCO's keypoint evaluation that the OKS formula alone does not show, as specified on the evaluation page [4] and in the official code [27].

The scale s. It is the square root of the ground-truth person's **segmentation area**, not of the box area and not computed from the keypoints. A partly occluded person has a smaller visible segment, so the same pixel error counts for more. In the code, the term in the exponent is $d_i^2/(\kappa_i^2 \cdot \text{area})/2$, with $\kappa_i = 2\sigma_i$, which is the formula of the main text with $s^2 = \text{area}$.

Which keypoints count. Only ground-truth keypoints with $v_i > 0$ enter the average, so a predicted keypoint for an unlabeled body part neither helps nor hurts. Each person counts equally in AP, whether they have 3 labeled keypoints or 17; COCO does not weight people by the number of keypoints [4]. A person with only a few labeled keypoints therefore has a noisier OKS.

People with no labeled keypoints. Crowd regions and people without any labeled keypoint cannot be scored by distances to keypoints. For them, the code measures how far each predicted keypoint falls *outside* a region around the ground-truth box (the box extended by one box width to the left and to the right, and by one box height above and below), so that predictions inside this region cost nothing. This lenient rule lets such detections match without being counted as false positives, much as ignore regions work for boxes [4], [27].

Small people and the detection limit. Small people (segment area below 32^2) carry no keypoint labels, which is why keypoint AP has only medium and large size ranges. At most 20 predicted poses per image are scored, ranked by their person-level score [4]. The ranking uses the score the model outputs for the whole person, so a method's choice of person score (detector confi-

dence, mean keypoint confidence, or a product of both) can change AP without moving a single keypoint.

Interpreting thresholds. Since a keypoint at one annotator standard deviation scores 0.88, an OKS of 0.95 requires almost every keypoint to be well within human disagreement. COCO reports that human performance drops rapidly above OKS 0.95, so the strictest thresholds partly measure annotation noise [4].

SMPL in more detail

This note explains the terms of the SMPL body model [39] summarized in the main text.

Template and skinning. SMPL starts from an artist-made template mesh $\bar{\mathbf{T}} \in \mathbb{R}^{3N}$ with $N = 6890$ vertices in a rest pose, a skeleton of $K = 23$ joints plus a root, and **blend weights** $\mathcal{W} \in \mathbb{R}^{N \times K}$ that say how strongly each joint's rotation moves each vertex. Linear blend skinning $W(\cdot)$ moves every vertex by a weighted average of the rigid transformations of the joints that influence it. On its own, blend skinning produces well-known artifacts, such as collapsing elbows and shoulders when joints bend strongly.

Shape blend shapes. $B_S(\boldsymbol{\beta}) = \sum_{n=1}^{|\boldsymbol{\beta}|} \beta_n \mathbf{S}_n$ adds a linear combination of orthonormal principal components $\mathbf{S}_n \in \mathbb{R}^{3N}$ of body-shape variation, learned from registered scans of many people. A few coefficients already capture most of the variation in height, weight and proportions.

Pose blend shapes. $B_P(\boldsymbol{\theta})$ adds vertex offsets that depend linearly on the elements of the joints' rotation matrices, computed relative to the rest pose. These offsets correct the blend-skinning artifacts and model pose-dependent deformations. Because the rest-pose rotations are subtracted, B_P is exactly zero in the rest pose [39], which keeps shape and pose effects separate.

Joints from shape. People with different shapes have joints in different places, so SMPL computes the rest-pose joint locations from the shaped mesh with a learned linear regressor $\mathcal{J}$: $J(\boldsymbol{\beta}) = \mathcal{J}(\bar{\mathbf{T}} + B_S(\boldsymbol{\beta}))$ [39]. After posing, the 3D joints (and any keypoints defined as regressed from vertices) follow the mesh, so they always respect the person's bone lengths.

Why this helps pose estimation. A network that predicts $(\boldsymbol{\beta}, \boldsymbol{\theta})$ can only produce bodies the model can express, and the 72 pose numbers are rotations rather than free 3D positions. Projecting the model's joints into the image with an estimated camera gives 2D keypoints that can be compared with 2D labels, which is the reprojection loss HMR [40] uses to train on images without 3D ground truth.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Simple Baselines (Xiao et al., 2018)

The problem. By 2018, top pose estimators had become complicated: multi-stage hourglasses, cascaded pyramid networks, refinement modules. It was hard to tell which parts mattered. Xiao, Wu and Wei [9] asked how far the simplest reasonable design would go, and offered it as a baseline for evaluating new ideas, for both single-frame pose and pose tracking in video.

The idea. Take an ImageNet-pretrained ResNet, remove nothing, and add three deconvolution layers (with batch normalization and ReLU) on top of the last stage; predict K heatmaps with a 1×1 convolution; train with MSE against Gaussian targets. Use a standard top-down pipeline with a Faster R-CNN person detector, flip testing and the quarter-offset decoder. For tracking, propagate each person's keypoints to the next frame with optical flow to create extra boxes, and link people across frames with a greedy matching based on a flow-based OKS similarity.

What it showed. The simple network matched or beat the more complex models of the time on COCO; the paper's best single model reached 73.7 AP on COCO test-dev, and its tracking system was the state of the art on the PoseTrack benchmark at publication.

How to read its tables. The ablation (Table 2) changes one factor at a time on COCO val2017: heatmap resolution (three deconvolution layers giving 64×48 maps beat two giving 32×24 by 2.5 AP), deconvolution kernel size (a small effect, 0.3 AP between kernel sizes 4 and 2), backbone depth (deeper ResNets help steadily) and input size. The comparison with Hourglass and CPN (Table 3) uses numbers cited from the CPN paper, not re-implemented; the authors say so and note that differences could come from implementation details. Two other settings must match before you compare any top-down numbers: the person detector (the paper's detector has a person AP of 56.4 on COCO val2017, against 55.3 for the detector used by Hourglass and CPN) and the input size (256×192 versus 384×288).

Limits. It is a top-down method, so it inherits every detector failure and its cost grows with the number of people. Its heatmaps are still at stride 4, so decoding precision depends on the post-processing described in this chapter. HRNet, by some of the same authors, later showed that keeping high resolution throughout the network does better than recovering it at the end [15].

Why it matters here. It is the template of a modern top-down pose estimator, and later systems such as ViTPose explicitly keep its simple deconvolution decoder [16].

OpenPose: part affinity fields (Cao et al., 2017)

The problem. Top-down methods run one network per person, so their runtime grows with the crowd, and they cannot recover from a missed detection. Earlier bottom-up methods grouped keypoints by solving large graph problems that took minutes per image. Cao, Simon, Wei and Sheikh [18] wanted bottom-up grouping that was accurate *and* real-time.

The idea. One network with two branches predicts, in several refining stages, confidence maps for the keypoints and **part affinity fields** for the limbs: 2D vector fields that point along each limb. Each stage receives the image features plus both predictions of the previous stage, and every stage is supervised. At test time, candidate limbs are scored by the line integral of the field along them, and a greedy parse solves one bipartite matching per limb of a tree skeleton instead of the full NP-hard problem. The paper argues that the greedy parse works because the fields, computed with a large receptive field, already encode global context.

What it showed. The method won the first COCO keypoint challenge in 2016 and clearly improved the state of the art on the MPII multi-person benchmark, with a parsing step whose cost barely depends on the number of people.

How to read its tables. The self-comparison on a COCO validation subset (Table 4) is the most instructive: a top-down baseline that uses CPM as the single-person model scores 62.7 AP with ground-truth boxes but 52.7 AP with boxes from an SSD detector, which quantifies how much top-down accuracy depends on the detector; the PAF method alone scores 58.4 AP, and 61.0 AP when each person is refined by CPM. In the challenge table, note the medium-size column: the method is weaker than top-down entries on medium-sized people, which the authors attribute to seeing all people at a single image scale. The runtime figure compares parsing and network time as the number of people grows; the network cost is constant, the parsing cost small.

Limits. Accuracy on small and medium people, the greedy parse (limbs are matched locally, so an early mistake propagates), and ambiguous fields where limbs of different people overlap along the same line (Exercise 4). The journal version [36] refines the architecture and adds foot keypoints.

Why it matters here. PAFs are the clearest example of learning an explicit *association* signal next to the detections, the same idea that associative embedding implements with tags.

Guide to the Modern view surveys

This guide expands the survey summary in the Modern view: what each review covers, how it is organized and how to use it.

Deep Learning-Based Human Pose Estimation: A Survey (Zheng et al., *ACM Computing Surveys*,

2024) [2] is the broadest of the three. It covers more than 250 papers since 2014, for images and video, 2D and 3D. Its taxonomy follows the input and the inference procedure: 2D single-person (regression versus heatmaps), 2D multi-person (top-down versus bottom-up), and 3D split into single-view single-person (skeleton-only methods, either direct or lifting, versus human mesh recovery), single-view multi-person and multi-view. It also lists datasets and metrics and tabulates reported results. Use it as a map of the literature: find the branch your problem belongs to, then follow its references. Its future directions (domain adaptation, lighter body models, human–scene constraints, temporal metrics) are a good list of open questions.

Monocular Human Pose Estimation: A Survey of Deep Learning-based Methods (Chen et al., *Computer Vision and Image Understanding*, 2020) [3] is shorter and earlier. It sorts methods into four tasks (2D and 3D, single- and multi-person) and introduces the three kinds of body model used in the field: skeleton-based, contour-based and volume-based. Its value for a newcomer is its summary of design tricks that proved effective and of the gap between benchmarks and applications (occlusion, crowds, efficiency, data diversity).

Recovering 3D Human Mesh from Monocular Images: A Survey (Tian et al., *IEEE TPAMI*, 2023) [5] focuses on mesh recovery with body models such as SMPL. It contrasts optimization-based fitting (data terms plus priors, solved per image) with regression-based networks (trained end to end), reviews training objectives, datasets and metrics, and discusses the remaining difficulties: flexible motion, diverse appearance, complex environments and the shortage of in-the-wild 3D labels. Read it after the 3D section of this chapter if you want to go from skeletons to full bodies.

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DLCV 06B • Video Understanding and Action Recognition

How deep networks recognize what happens in a video: the tasks (clip classification, temporal and spatio-temporal action localization, retrieval and question answering), why time is hard to learn, two-stream, 3D, (2+1)D, SlowFast, TSN/TSM and video-transformer architectures, masked video pretraining, evaluation protocols, datasets and failure modes such as scene bias.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-06b-video-understanding/

What you'll learn

- The main video understanding tasks and what each outputs: clip classification, temporal action localization, spatio-temporal action detection, and (briefly) video–text retrieval and video question answering.
- Why video is hard: a model must use *time*, yet many benchmarks can be solved from a single frame (scene bias), and every extra frame multiplies the compute.
- How the architecture families add time to an image network, in math and in code: two-stream networks with optical flow, 3D convolutions (C3D, I3D inflation), factorized (2+1)D convolutions, SlowFast, TSN sampling and the temporal shift module, and video transformers (TimeSformer, ViViT, MViT, Video Swin).
- How video models are pretrained without labels (VideoMAE and its tube masking) and where video–language models fit.
- How video models are trained and evaluated: frame sampling, multi-view testing (clips × crops), top-1/top-5, temporal IoU and mAP, and the datasets behind the numbers (Kinetics, UCF101, HMDB51, Something-Something V2, AVA, Charades, ActivityNet, EPIC-KITCHENS).
- The typical failure modes, and simple tests that expose them.

The big picture

An image classifier looks at one picture and names what is in it. A video model must also understand what *happens*: who does what, in which order, for how long. The extra input is a time axis, and with it come two problems that images do not have. First, time is expensive: a 10-second clip at 30 frames per second is 300 images. Second, time is easy to ignore: a network can often guess "playing tennis" from the court alone, without ever looking at the swing.

> **Explain like I'm five**
>
> Look at one photo of a door. Is someone opening it or closing it? You cannot tell. Now flip through a few photos taken one after another, like a flip-book, and the answer is obvious. A video model has to learn to read the flip-book, not just one page. The tricky part is that it can often cheat: if every photo of "swimming" shows a pool, it can just look for blue water and never learn what swimming looks like.

Video understanding matters because most visual data is video: phone recordings, surveillance and traffic cameras, sports broadcasts, robots, cars and medical procedures. Recognizing actions is a building block for searching video archives, analyzing sports, monitoring safety, assisting people at home and controlling robots. The field also produced ideas that spread to the rest of deep learning, such as inflating a pretrained 2D network into 3D, factorizing space and time, and masked pretraining at very high masking ratios.

This tutorial assumes image classification, its training and its evaluation from DLCV 02. It reuses detection metrics from DLCV 03 and attention from ML 04. Five surveys frame the field: Zhu et al. on deep action recognition [1], Kong and Fu on recognition and prediction [2], Selva et al. on video transformers [3], Madan et al. on video foundation models [4] and Tang et al. on video understanding with large language models [5]. We return to them in the Modern view.

Related chapters. Following objects over time is DLCV 06 ·Object Tracking; body keypoints, which skeleton-based recognition uses as input, are DLCV 06A; optical flow is estimated in DLCV 08A; *generating* video with spatiotemporal attention and video autoencoders is DLCV 10A; and predicting the future of a scene for planning is DLCV 07 ·World Models.

Video understanding tasks

Plain version. The input is a stack of frames. The output depends on the question: one label for a short clip; the start, end and class of every action in a long video; a box and labels for every person at every moment; or an answer to a question written in words.

Precise version. A video is a tensor $V \in \mathbb{R}^{T \times H \times W \times 3}$, where T is the number of frames, $H \times W$ the frame size and 3 the color channels. A model rarely sees all of V. It sees a **clip** $x \in \mathbb{R}^{T' \times H' \times W' \times 3}$ of $T' \ll T$ sampled frames, resized and cropped to $H' \times W'$. Figure 1 shows the four task families.

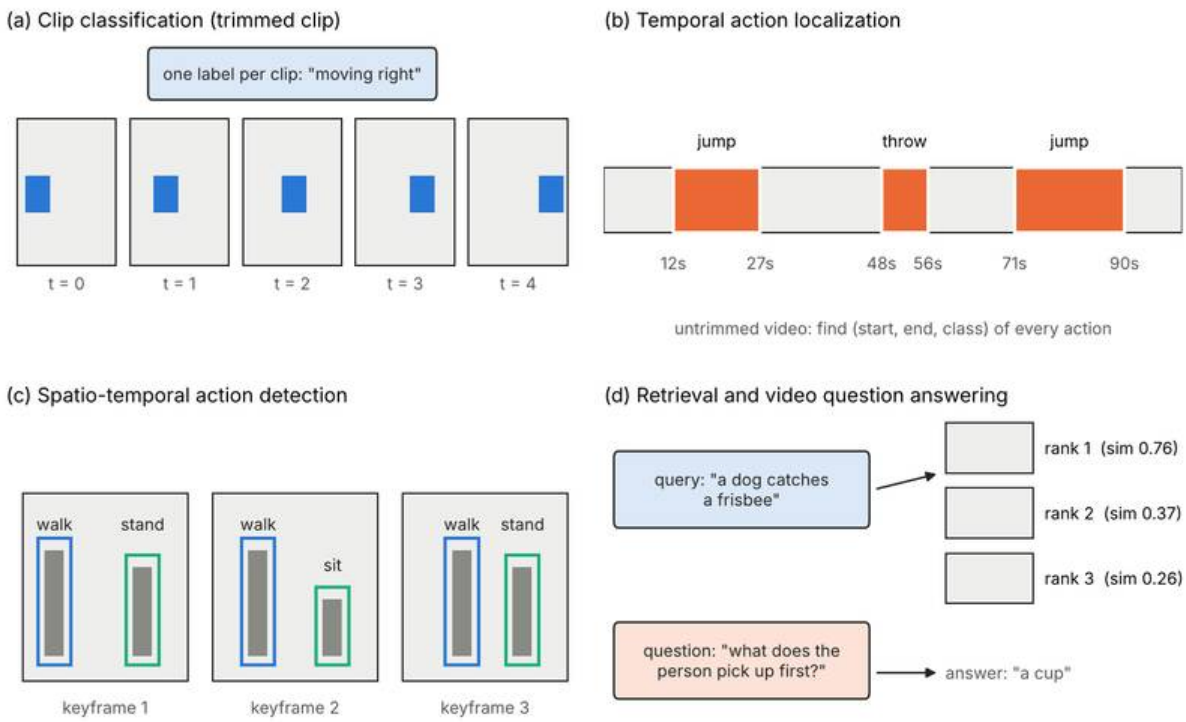


Figure 1 — The main video tasks: (a) one label per trimmed clip; (b) start, end and class of each action in an untrimmed video; (c) a box and action labels for every person in each keyframe; (d) ranking videos for a text query, or answering a question about a video.

1. **Clip (action) classification.** Given a *trimmed* clip that contains one action, predict its class $y \in \{1, \dots, K\}$. Trained with cross-entropy and evaluated with top-1 and top-5 accuracy, exactly as in DLCV 02. Some datasets are **multi-label** (several actions at once) and are evaluated with mean average precision (mAP) over classes.
2. **Temporal action localization** (also called temporal action detection). Given an *untrimmed* video, output a set $\{(s_i, e_i, c_i, p_i)\}$ of segments with start time s_i, end time e_i, class c_i and confidence p_i. It is object detection along one dimension, and it is evaluated with mAP at several **temporal IoU** thresholds.
3. **Spatio-temporal action detection.** Find *who* does *what*, *where* and *when*: a person box in each annotated keyframe, with one or more action labels per box. It is evaluated with frame-level mAP at a box-IoU threshold.
4. **Video–language tasks. Text-to-video retrieval** ranks a collection of videos by their similarity to a sentence, evaluated by Recall@k (the fraction of queries whose correct video is among the top k). **Video question answering** produces a short answer, often chosen from multiple

choices and scored by accuracy.

Neighboring problems are covered elsewhere. Linking the same object across frames is tracking (DLCV 06). Predicting what happens next is action anticipation or world modeling (DLCV 07). Labeling every pixel of every frame is video segmentation.

Why video is hard

Plain version. Three things make video harder than images: the label often depends on *order* and *motion*, the datasets let models skip that work, and the input is huge.

Time matters, but the data does not always force you to use it

Some actions can only be told apart by their motion: "opening a door" versus "closing a door", "pushing something left" versus "pushing something right", "sitting down" versus "standing up". A single frame of these is ambiguous. Many other actions are recognizable from one frame because of the **scene** and the **objects**: a swimming pool, a horse, a guitar. Datasets collected from the web contain many actions of the second kind.

This was visible early. Karpathy et al. trained CNNs on a dataset of 1 million YouTube videos in 487 sports classes and compared several ways of fusing information over time; their best spatio-temporal network improved only modestly over a single-frame model [6]. Later studies measured the problem directly. Li et al. defined the **representation bias** of a dataset and showed that common action datasets are biased toward static cues (objects, scenes and people); they also built Diving48, a dataset of competitive diving with 48 fine-grained dive classes, to reduce that bias [7]. Sevilla-Lara et al. asked people to recognize actions from videos whose frames had been shuffled, and kept only the classes that became unrecognizable; this "temporal" subset is where a model's ability to use time actually shows [8]. Something-Something was built to capture detailed physical aspects of actions rather than high-level concepts: its 174 classes are caption templates such as "pushing something from left to right" [9], which the VideoMAE authors describe as motion-centric [10].

The general phenomenon, a network solving the training task with the easiest cue available, is the **shortcut learning** of DLCV 02. In video the easiest cue is usually the scene.

A dataset where one frame says nothing

The code below makes this concrete with synthetic clips. A small square moves left or right; the label is the direction. The square's position in the middle frame has the same distribution for both labels, so **the middle frame alone carries no information about the label**. We then add a

"scene": a background brightness that agrees with the label in 90% of the training clips. Three logistic-regression classifiers see (1) the pixels of the middle frame, (2) a motion feature, and (3) both.

The motion feature is the classical brightness-constancy estimate (see DLCV 08A). If a pattern shifts by u pixels per frame, then $I_{t+1}(x) \approx I_t(x-u)$, so the temporal difference is $I_{t+1} - I_t \approx -u\,\partial_x I_t$. A least-squares fit of u over the whole frame gives

$$\hat{u}_t = -\frac{\sum_{x,y} \left(I_{t+1}(x,y) - I_t(x,y)\right) \partial_x I_t(x,y)}{\sum_{x,y} \left(\partial_x I_t(x,y)\right)^2},$$

where $I_t(x,y)$ is the brightness of frame t at column x and row y, $\partial_x I_t$ its horizontal derivative, and $\hat{u}_t$ the estimated horizontal motion between frames t and $t+1$ (positive means rightward). It is a product of a temporal derivative filter and a spatial derivative filter, which is exactly the kind of pattern a 3D convolution can learn.

```
import numpy as np
from sklearn.linear_model import LogisticRegression

def make_clips(n, T=8, S=20, noise=0.5, scene_bias=0.5, seed=0):
    """A 3x3 square moves 1 px/frame left (y=0) or right (y=1) on a noisy background.
    scene_bias = P(background brightness agrees with the label); 0.5 = no bias."""
    rng = np.random.default_rng(seed)
    y = rng.integers(0, 2, n)
    X = noise * rng.standard_normal((n, T, S, S))
    for i in range(n):
        step = 1 if y[i] == 1 else -1
        c_mid = rng.integers(T // 2, S - 3 - T // 2 + 1)   # middle-frame position: same for
        ↪ both labels
        r = rng.integers(0, S - 3 + 1)
        agree = rng.random() < scene_bias
        X[i] += 0.3 * (y[i] if agree else 1 - y[i])       # the "scene": background
        ↪ brightness
        for t in range(T):
            c = c_mid + step * (t - T // 2)
            X[i, t, r:r + 3, c:c + 3] += 1.0
    return X, y

def frame_features(X):     # appearance only: the pixels of the middle frame
    return X[:, X.shape[1] // 2].reshape(len(X), -1)

def motion_features(X):    # temporal difference times horizontal gradient, per frame pair
    dt = X[:, 1:] - X[:, :-1]
    dx = np.gradient(X[:, :-1], axis=3)
    u = -(dt * dx).sum(axis=(2, 3)) / ((dx ** 2).sum(axis=(2, 3)) + 1e-6)
    return u                                              # one motion estimate per frame pair

data = {"train": make_clips(2000, scene_bias=0.9, seed=1),  # scene agrees with the label 90%
        "same": make_clips(1000, scene_bias=0.9, seed=2),   # test: same distribution
        "no cue": make_clips(1000, scene_bias=0.5, seed=3), # test: scene says nothing
        "shift": make_clips(1000, scene_bias=0.1, seed=4)}  # test: scene misleads 90%
```

```
for name, f in [("one frame", frame_features), ("motion", motion_features),
                ("frame+motion", lambda X: np.c_[frame_features(X), motion_features(X)])]:
    clf = LogisticRegression(C=0.1, max_iter=5000).fit(f(data["train"][0]), data["train"][1])
    acc = {k: clf.score(f(X), y) for k, (X, y) in data.items() if k != "train"}
    print(f"{name:13s}", "  ".join(f"{k} {v:.3f}" for k, v in acc.items()))
```

```
one frame     same 0.867  no cue 0.495  shift 0.131
motion        same 0.945  no cue 0.929  shift 0.941
frame+motion  same 0.889  no cue 0.522  shift 0.163
```

(All numbers in this block and in Figure 2 come from this toy run.) On test data from the training distribution, the single-frame model looks good (0.867), because the scene predicts the label 90% of the time. Remove the scene cue and it drops to chance (0.495); make the scene misleading and it falls to 0.131. The motion model is about 0.93–0.94 in every case, because it uses the cue that actually defines the label. The combined model is the warning: it *has* the motion feature, yet it learned to rely on the hundreds of brightness pixels, and it fails under the shift (0.163). How strongly it leans on the scene depends on feature scaling and regularization (Exercise 1), but a benchmark drawn from the training distribution would never reveal the problem.

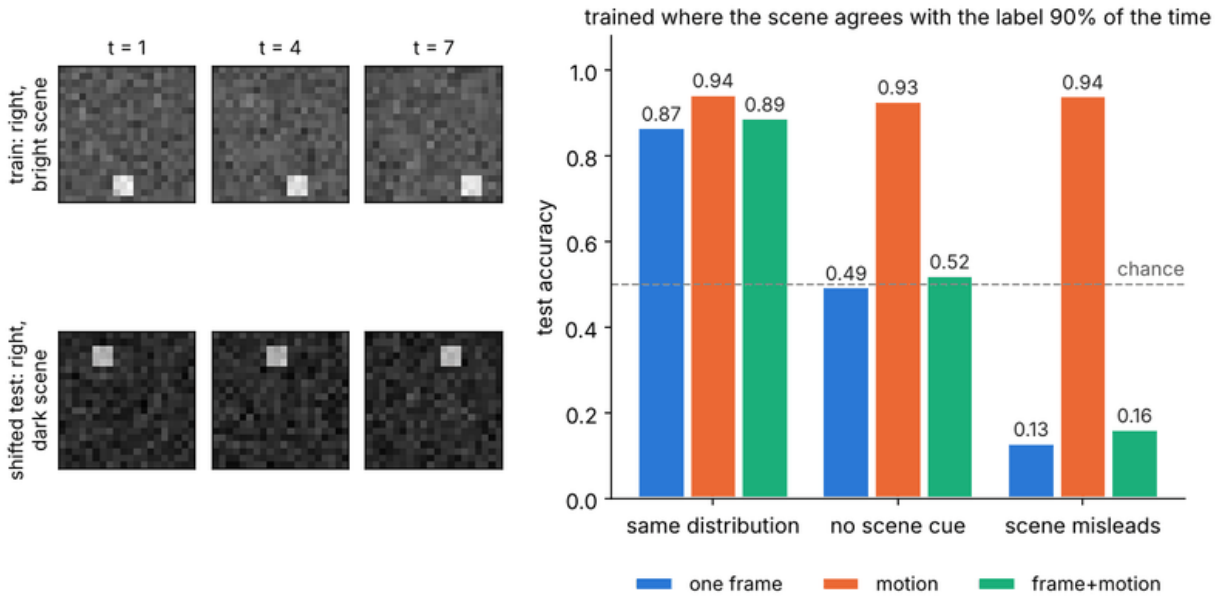


Figure 2 — The scene-bias toy. Left: frames of a training clip (bright scene) and a shifted test clip (dark scene); shown with less noise than in the experiment so the square is visible. Right: accuracy of the three classifiers on the three test sets. Only the motion model survives when the scene stops agreeing with the label.

Compute and memory

A clip of T' frames costs roughly T' times as much as one image for a per-frame network, and more for models that mix frames. For a convolution with C_{in} input channels, C_{out} output channels and a kernel of $k_t \times k_h \times k_w$ (time × height × width), stride 1 and "same" padding on a $T \times H \times W$ input,

$$\text{params} = C_{\text{in}}\, C_{\text{out}}\, k_t k_h k_w, \qquad \text{MACs} = \text{params} \times THW,$$

where MACs counts multiply–accumulate operations (one multiplication plus one addition each). A $3 \times 3 \times 3$ kernel therefore has 3 times the parameters and 3 times the work of a 3×3 kernel applied to the same frames. For a transformer the problem is the number of tokens N: self-attention compares N^2 query–key pairs, and N grows linearly with the number of frames. The code in the next section puts numbers on both.

How to add time to an image network

Plain version. There are five basic recipes (Figure 3): run an image network on a few frames and average its answers; add a second network that looks only at motion (optical flow); make every filter three-dimensional; run two pathways at different frame rates; or cut the clip into space–time patches and let a transformer attend across them. Most modern models mix these ideas.

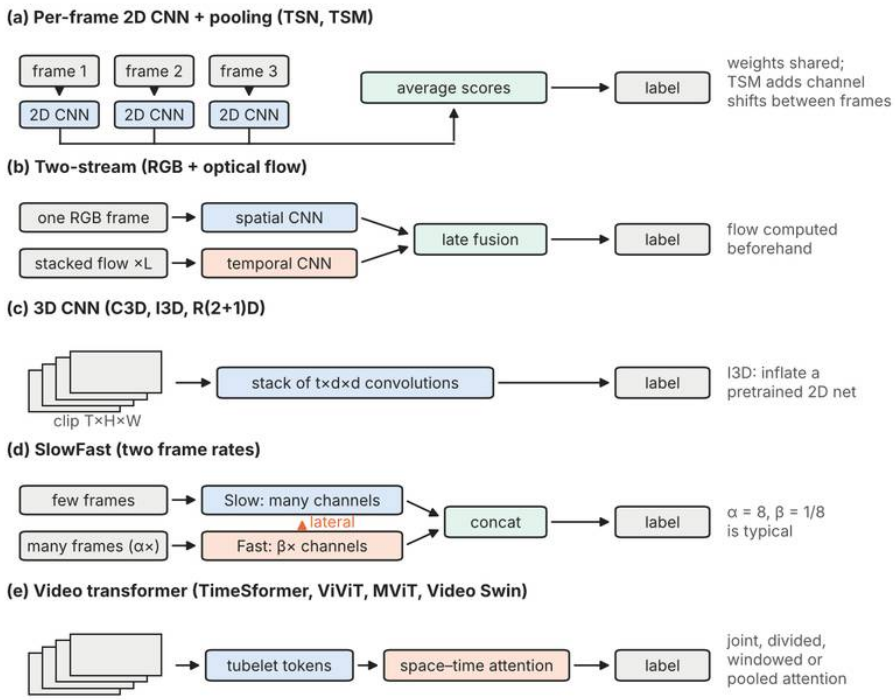


Figure 3 — Five ways to add time to an image network. Families (a)–(d) are convolutional; (e) is attention-

based. Real systems combine them, for example two-stream I3D or a transformer that uses SlowFast-style multiscale features.

Per-frame networks and late fusion

The simplest video model applies an image classifier f to K frames $x_{t_1}, \dots, x_{t_K}$ and averages the class scores:

$$\hat{y} = \operatorname{softmax}\Big(\frac{1}{K}\sum_{k=1}^{K} f(x_{t_k})\Big),$$

where $f(x) \in \mathbb{R}^C$ is the vector of class scores (logits) for one frame and C the number of classes. This is order-blind: shuffling the frames gives the same answer, so it cannot tell "opening" from "closing". It is still a strong baseline on scene-heavy datasets, and it is cheap and can start from any pretrained image backbone (DLCV 02). A recurrent network such as an LSTM [11] over per-frame features adds order; this design is discussed in ML 03A.

Two-stream networks: appearance plus optical flow

Plain version. Give the network the motion explicitly. One network looks at a single color frame (what is there); a second looks at a stack of optical-flow fields (how things move). Their predictions are combined at the end.

Simonyan and Zisserman's two-stream ConvNet [12] has a **spatial stream**, an image CNN on one RGB frame, and a **temporal stream**, a CNN whose input is the dense optical flow of L consecutive frame pairs. Each flow field d_t has a horizontal component d_t^x and a vertical component d_t^y, so the stacked input for frame τ has $2L$ channels:

$$I_\tau(u, v, 2k-1) = d^x_{\tau+k-1}(u, v), \qquad I_\tau(u, v, 2k) = d^y_{\tau+k-1}(u, v), \qquad k = 1, \dots, L,$$

where (u, v) is a pixel position and $d_t^x(u, v)$, $d_t^y(u, v)$ are the horizontal and vertical displacements of that pixel from frame t to frame $t+1$. The two streams are trained separately and fused late, by averaging their softmax scores or with a linear SVM on the stacked scores [12]. The paper showed that a ConvNet on stacked flow performs well despite limited training data, and that the two streams are complementary [12].

The weakness is cost: dense optical flow must be computed (and usually stored) for every frame before training, with a classical solver or a flow network (see DLCV 04B for warping with flow). The flow is also not learned for the recognition task. Temporal Segment Networks [13] and I3D [14] still used a flow stream; SlowFast (below) later learned motion end to end from raw frames, without flow, and its authors found it empirically more effective than the two-stream approach [15].

3D convolutions: C3D and I3D inflation

Plain version. Make every filter a small cube that slides over time as well as space. Then the network can learn motion detectors, such as "a bright edge that moves right", directly from the pixels.

A 3D convolution with kernel $w \in \mathbb{R}^{k_t \times k_h \times k_w}$ computes, for one input and one output channel,

$$y(t,i,j) = \sum_{a=0}^{k_t-1} \sum_{b=0}^{k_h-1} \sum_{c=0}^{k_w-1} w(a,b,c)\, x(t+a, i+b, j+c),$$

where $x(t,i,j)$ is the input at frame t, row i and column j (in practice summed over all input channels and with padding at the borders). **C3D** [16] trained a network of such layers on a large video dataset and found that a homogeneous architecture with small $3 \times 3 \times 3$ kernels in all layers was among the best; its features with a linear classifier were strong on several benchmarks.

Training 3D networks from scratch needs much more data than existed in 2015. Carreira and Zisserman solved both problems at once in 2017 [14]. They introduced **Kinetics**, a dataset of 400 human action classes with at least 400 YouTube clips per class, each about 10 seconds long [17], and the **Inflated 3D ConvNet (I3D)**: take a successful 2D image architecture and *inflate* every $N \times N$ filter and pooling kernel into $N \times N \times N$. To start from weights pretrained on ImageNet [18], they made the inflated network give the same response on a "boring video" (one image repeated in time) as the 2D network on the image: repeat each 2D filter N times along time and divide by N [14]. With w_{2D} the pretrained filter,

$$w_{3D}(a,b,c) = \frac{1}{N}\, w_{2D}(b,c), \qquad a = 0, \dots, N-1,$$

so on a constant-in-time input the N temporal taps add up to exactly the 2D response. The check below confirms it numerically.

```
import numpy as np
from scipy.ndimage import correlate

rng = np.random.default_rng(0)
w2d = rng.standard_normal((3, 3))            # a "pretrained" 2D filter
N = 3                                        # temporal size of the inflated filter
w3d = np.repeat(w2d[None] / N, N, axis=0)    # (N, 3, 3): copy N times, divide by N

img = rng.standard_normal((32, 32))
boring = np.repeat(img[None], 8, axis=0)     # an 8-frame "video" of one still image

r2d = correlate(img, w2d, mode="nearest")
r3d = correlate(boring, w3d, mode="nearest")  # 3D filtering over (t, y, x)
print("max |3D response on any frame - 2D response| =", np.abs(r3d - r2d[None]).max())
```

```
max |3D response on any frame - 2D response| = 1.7763568394002505e-15
```

The difference is at the level of floating-point rounding. Training then moves the filters away from

"constant in time" as they learn motion. The paper's final model is **two-stream I3D**, one I3D on RGB and one on flow with predictions averaged, because the authors still found the flow stream valuable [14]. After pretraining on Kinetics, I3D improved considerably on the small UCF101 and HMDB51 benchmarks [14], which helped make "pretrain on Kinetics, fine-tune on the target" the standard video recipe. The same inflation idea reappears in video generators (DLCV 10A), and I3D features are the feature network behind the Fréchet Video Distance [19] (DLCV 10A).

Optional: Quo Vadis: I3D and the Kinetics dataset (in the Paper guides). How the paper compares five architecture families and what its transfer experiments show.

Factorized (2+1)D convolutions

Plain version. A cube-shaped filter can be split into two cheaper steps: first a flat filter that looks within each frame, then a thin filter that looks across frames at each position. With the right width in between, the split has the same number of parameters but an extra nonlinearity, and it trains more easily.

Tran et al. [20] compared several spatiotemporal convolutions inside residual networks: 2D only,

3D everywhere, mixed (3D in early layers only, or late layers only) and a factorized **(2+1)D** block. The (2+1)D block replaces N_i filters of size $N_{i-1}\times t\times d\times d$ with M_i spatial filters of size $N_{i-1}\times 1\times d\times d$, a ReLU, and N_i temporal filters of size $M_i \times t \times 1 \times 1$. Here N_{i-1} and N_i are the input and output channel counts of layer i, t and d the temporal and spatial kernel sizes, and M_i the width of the intermediate space, chosen so that the parameter count matches the 3D layer [20]:

$$M_i = \left\lfloor \frac{t\, d^2\, N_{i-1}\, N_i}{d^2\, N_{i-1} + t\, N_i} \right\rfloor .$$

The paper names two benefits: the extra ReLU doubles the number of nonlinearities for the same parameter count, and the factorization makes optimization easier, visible as lower *training* error than an equally sized 3D network [20]. It is the video analogue of the depthwise-separable trick in image CNNs (ML 03): split one big operation into cheaper ones along different axes.

The code computes parameters and multiply–adds for one 64-to-64-channel layer on a 16-frame clip with 56 × 56 feature maps, and the token count of a video transformer for comparison.

```
import math

def conv_cost(c_in, c_out, kt, kh, kw, T, H, W):
    """Parameters and multiply-adds of one convolution (stride 1, 'same' padding, no
     ↪  bias)."""
    params = c_in * c_out * kt * kh * kw
```

```
    return params, params * T * H * W

c, t, d = 64, 3, 3                          # channels in = out, temporal and spatial
 ↪  kernel size
T, H, W = 16, 56, 56                        # a 16-frame clip, 56x56 feature maps
M = math.floor(t * d * d * c * c / (d * d * c + t * c))   # R(2+1)D's matched middle width

layers = {
    "2D  (1x3x3), per frame": [conv_cost(c, c, 1, d, d, T, H, W)],
    "3D  (3x3x3)":            [conv_cost(c, c, t, d, d, T, H, W)],
    f"(2+1)D, M={M}":         [conv_cost(c, M, 1, d, d, T, H, W), conv_cost(M, c, t, 1, 1,
     ↪  T, H, W)],
}
for name, parts in layers.items():
    p = sum(x[0] for x in parts); f = sum(x[1] for x in parts)
    print(f"{name:24s} params {p:>8,d}   GMACs {f / 1e9:5.2f}")

# a video transformer's token count with 2x16x16 tubelets on a 16x224x224 clip
n_tok = (16 // 2) * (224 // 16) ** 2
print(f"tokens {n_tok}, joint attention pairs {n_tok**2:,}, "
      f"divided (time, then space) pairs {n_tok * 8 + n_tok * 196:,}")
```

```
2D  (1x3x3), per frame   params   36,864   GMACs  1.85
3D  (3x3x3)              params  110,592   GMACs  5.55
(2+1)D, M=144            params  110,592   GMACs  5.55
tokens 1568, joint attention pairs 2,458,624, divided (time, then space) pairs 319,872
```

With equal input and output widths, $M = 144$ makes the $(2{+}1)$D block match the 3D layer exactly in parameters and work; the gain is in expressiveness and optimization, not in cost. A cheaper split is possible by choosing a smaller M. Figure 4 draws the factorization.

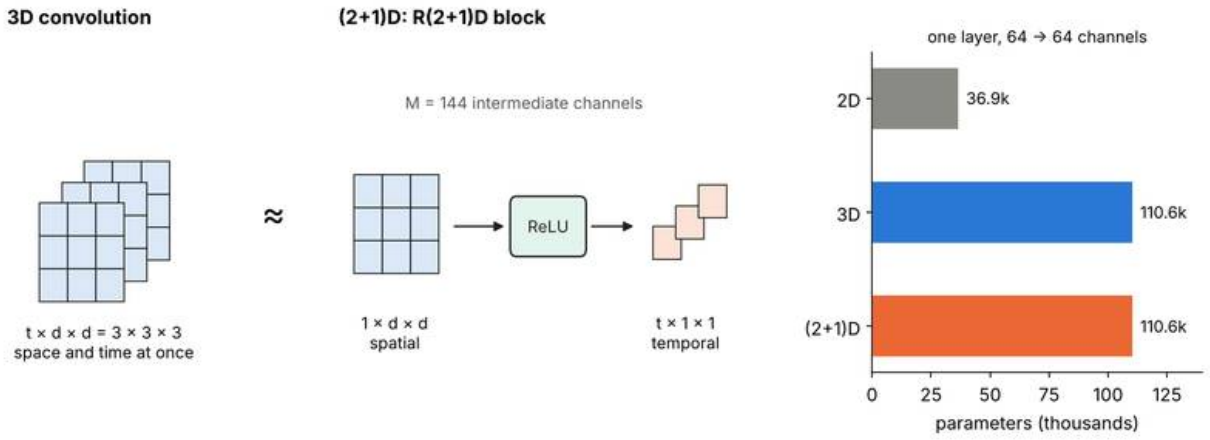


Figure 4 — A 3D convolution and its $(2{+}1)$D factorization. With the matched middle width M from the formula above, both have the same parameter count; the $(2{+}1)$D block adds a nonlinearity between the spatial and temporal steps.

Optional: Choosing the middle width of a (2+1)D block (in the Supplementary notes). Where the formula comes from, and what happens when input and output widths differ or when you want a cheaper block.

Cheap temporal modeling: TSN sampling and the temporal shift module

Plain version. Two tricks let a plain 2D network handle long videos cheaply. TSN looks at a few frames spread over the whole video instead of many frames from one moment. TSM lets neighboring frames swap a few feature channels inside the network, which mixes information across time at almost no cost.

Temporal Segment Networks (TSN) [13] divide a video into K equal segments, take one short snippet from each (a random position during training), score every snippet with the same network, and combine the scores with a **segmental consensus** such as averaging before the softmax. The loss is computed on the consensus, so the whole video supervises the network, not just one short window [13]. The paper's default was $K = 3$ segments [13]. This **sparse sampling** covers long-range structure with a fixed, small number of frames (Figure 5).

```
import numpy as np

def tsn_indices(n_frames, K, train, rng=None):
    """TSN-style sparse sampling: split the video into K equal segments, take one frame
    per segment (random position when training, the segment centre when testing)."""
    edges = np.linspace(0, n_frames, K + 1)
    if train:
        return np.floor(edges[:-1] + rng.random(K) * np.diff(edges)).astype(int)
    return np.floor((edges[:-1] + edges[1:]) / 2).astype(int)

def dense_clip(n_frames, length, stride, start):
    """A dense clip: `length` frames, `stride` apart, starting at `start`."""
    return start + stride * np.arange(length)

rng = np.random.default_rng(0)
n = 300                                          # a 10-second video at 30 fps
print("TSN train:", tsn_indices(n, 8, True, rng))
print("TSN test :", tsn_indices(n, 8, False))
print("dense 8x8:", dense_clip(n, 8, 8, start=100))   # 8 frames, every 8th frame
# multi-view testing: 10 clip start positions spread uniformly over the video
starts = np.linspace(0, n - 8 * 8, 10).astype(int)
print("10 clip starts:", starts)
```

```
TSN train: [ 23  47  76 113 180 221 247 289]
TSN test : [ 18  56  93 131 168 206 243 281]
dense 8x8: [100 108 116 124 132 140 148 156]
10 clip starts: [  0  26  52  78 104 131 157 183 209 236]
```

The eight sparse frames span the whole video; the dense clip covers about 2 seconds in detail. (Taking segment centres at test time is a common choice in later implementations; the original TSN paper tested on 25 frames with 10 crops each [13].) Dense clips are what 3D networks and SlowFast consume; the last line, ten clips spread over the video, is the multi-view testing protocol discussed under evaluation below.

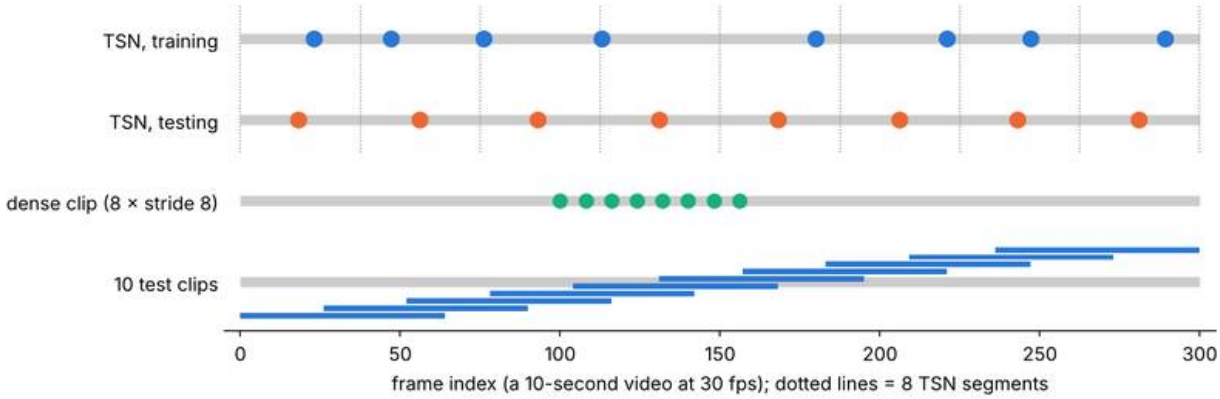


Figure 5 — Frame sampling, from the code above: TSN-style sampling takes one frame per segment (random when training, centred when testing); a dense clip takes consecutive frames at a fixed stride; multi-view testing averages predictions over several clips spread across the video.

The **Temporal Shift Module (TSM)** [21] adds real temporal interaction to a 2D network with zero extra parameters and zero extra multiply–adds. Inside each residual block, a fraction of the channels of the feature tensor is shifted by one frame forward in time, another fraction one frame backward, and the rest stay in place; the following 2D convolution then mixes channels that come from three different frames. Shifting most channels hurt both latency (data movement) and accuracy (less spatial information per frame), so the paper shifts 1/4 of the channels (1/8 in each direction) and places the shift on the residual branch [21]. For live video, a uni-directional version shifts only from past to current frames [21].

Optional: The temporal shift as a sparse temporal convolution (in the Supplementary notes). A short derivation and a tested snippet showing that shift plus a 1×1 convolution equals a temporal convolution with most weights fixed to zero.

SlowFast: two frame rates

Plain version. What things are changes slowly; how they move can change fast. So use two pathways: a heavy one that looks at few frames to understand the scene, and a light one that looks at many frames to catch the motion.

SlowFast networks [15] have a **Slow pathway** that samples T frames with a large temporal stride τ, and a **Fast pathway** that samples αT frames with stride τ/α, so it is α times denser in time.

The Fast pathway has only a fraction β of the Slow pathway's channels. Typical values are $\alpha = 8$ and $\beta = 1/8$ [15]. Because the cost of a layer grows roughly with the square of its channel count, the Fast pathway takes only about 20% of the total computation in the paper's instantiations [15].

Lateral connections feed Fast features into the Slow pathway at every stage; to match shapes they convert the αT Fast frames to T, for example with a time-strided 3D convolution, the paper's default [15].

Two design details carry a general lesson. The Slow pathway uses temporal kernels larger than 1 only in its last two stages, because the authors found that temporal convolutions in early layers *hurt* when the frame stride is large: fast-moving content has little correlation across a large time step unless the spatial receptive field is also large [15]. And SlowFast uses **no optical flow**: it learns motion end to end from raw frames [15]. It reported strong results on Kinetics, Charades and AVA classification and detection [15].

Optional: SlowFast networks (in the Paper guides). How to read its ablations on lateral connections and on the channel ratio β.

Two related ideas complete the CNN family. **Non-local blocks** [22] compute each position's response as a weighted sum over *all* space–time positions, an attention layer inserted into a 3D CNN to capture long-range dependencies. **X3D** [23] starts from a tiny 2D image network and expands it one axis at a time (in space, time, width and depth) to find accurate networks at low cost; one surprising finding was that networks with high spatiotemporal resolution can be extremely thin [23].

Video transformers: TimeSformer, ViViT, MViT and Video Swin

Plain version. Cut the clip into small space–time patches, turn each into a token, and let a transformer decide which patches should look at which. The hard part is that a clip has thousands of tokens, so the designs differ mainly in *which* tokens each token may attend to.

A Vision Transformer [24] (ML 04) turns an image into patches. A video transformer does the same in space–time. With **tubelet embedding** [25], non-overlapping tubes of $t \times p \times p$ pixels are linearly projected (equivalently, a 3D convolution with stride equal to its kernel), so a $T \times H \times W$ clip becomes

$$N = \frac{T}{t} \cdot \frac{H}{p} \cdot \frac{W}{p}$$

tokens, where t is the tubelet length in frames and p the patch side in pixels. VideoMAE's ViT-Base, for example, takes a $16 \times 224 \times 224$ clip with $2 \times 16 \times 16$ cubes, giving $8 \times 196 = 1568$ tokens [10]. **Joint** space–time attention compares all N^2 pairs, which the code above counted as about 2.46 million per layer. Four designs tame this:

- **TimeSformer** [26] compared several self-attention schemes and found **divided attention** best among those it tried: within each block, a token first attends to the tokens at the same spatial position in other frames (temporal attention) and then to the tokens of its own frame (spatial attention) [26]. In the configuration above that is about 0.32 million pairs instead of 2.46 million. It can also process clips over one minute long [26]. The cost ratio of joint versus divided attention is derived in DLCV 10A, which also explains why video *generators* often prefer full attention.
- **ViViT** [25] proposed four models: joint spatio-temporal attention; a **factorised encoder** (a spatial transformer per frame, then a temporal transformer over the frame embeddings, a form of late fusion); factorised self-attention inside each block; and factorised dot-product attention, where half of the heads attend over space and half over time. It also showed how to regularize these models and initialize them from pretrained image models so they can be trained on comparatively small datasets [25].
- **MViT** [27] brings back the multiscale hierarchy of CNNs: stages start at high spatiotemporal resolution with few channels and progressively expand channels while reducing resolution, using **pooling attention**, which pools the query, key and value sequences inside attention [27].
- **Video Swin Transformer** [28] argues for **locality**: attention is computed only inside local 3D windows, which are shifted between consecutive layers so information can cross window borders. It extends the image Swin Transformer [29] (DLCV 02) and starts from pretrained image weights, giving a better speed–accuracy trade-off than global attention [28].

Attention is not the only way to model long clips. **VideoMamba** [30] adapts the Mamba state-space model [31], whose cost grows linearly with the number of tokens, to video; the underlying selective scan is explained in ML 03A.

Learning video representations without labels

Plain version. Labeling video is slow and expensive, but unlabeled video is plentiful. So hide most of a clip and train the network to fill it in; to do that well, it must learn how things look and how they move.

VideoMAE [10] applies masked autoencoding (MAE [32], see ML 05) to video. It masks space–time cubes, encodes only the visible ones with a ViT, and reconstructs the pixels of the masked ones with a light decoder. Two things are specific to video. First, consecutive frames are highly redundant, so the masking ratio can be extremely high: 90% to 95% still works well [10]. Second, if masks are drawn independently per frame, a masked patch can often be copied from the same position in a neighboring frame, an **information leakage** that makes the task too easy. VideoMAE therefore uses **tube masking**: the same spatial mask in every frame [10]. The snippet measures

the leakage.

```
import numpy as np

rng = np.random.default_rng(0)
T, P, ratio = 8, 196, 0.9               # 8 token frames of 14x14 patches, 90% masked

def leak(mask):
    """Fraction of masked tokens whose same-position token in a neighbouring frame is
     ↪  visible."""
    vis = ~mask
    nb = np.zeros_like(mask)
    nb[1:] |= vis[:-1]; nb[:-1] |= vis[1:]
    return (mask & nb).sum() / mask.sum()

random_mask = rng.random((T, P)) < ratio                  # independent per token
tube_mask = np.repeat((rng.random(P) < ratio)[None], T, 0) # same spatial mask in every frame
print(f"random masking: {random_mask.mean():.2f} masked, leak {leak(random_mask):.2f}")
print(f"tube masking  : {tube_mask.mean():.2f} masked, leak {leak(tube_mask):.2f}")
```

```
random masking: 0.90 masked, leak 0.16
tube masking  : 0.89 masked, leak 0.00
```

With independent masks, about one masked token in six has a visible twin in an adjacent frame (for interior frames the expected value is $1 - 0.9^2 = 0.19$; the first and last frames have only one neighbor). Tube masking removes that shortcut entirely, which is the point: a pretext task is only useful if its shortcuts are closed. VideoMAE also reported strong results on small datasets of a few thousand videos without extra data, found that data quality matters more than data quantity for this kind of pretraining, and noted that the domain shift between pretraining and target data is an important issue [10]. **VideoMAE V2** [33] scaled the approach to a billion-parameter ViT with **dual masking**, which also masks the decoder's input to save computation.

A second line predicts *features* instead of pixels. **V-JEPA** [34] trains video models only with a feature-prediction objective, without pretrained image encoders, text, negative examples or reconstruction, on about 2 million videos, and evaluates them with a frozen backbone on both motion- and appearance-based tasks [34]. Its successor V-JEPA 2 [35] and the JEPA idea are covered in ML 05 and DLCV 07.

Video and language, briefly

Plain version. Instead of a fixed list of action names, describe videos with free text. A model that places videos and sentences in the same space can search videos by description; a model that feeds video tokens into a language model can answer questions about them.

Retrieval. CLIP-style image–text models [36] (ML 09) transfer to video: **CLIP4Clip** [37] transfers CLIP to text–video retrieval end to end and studies whether image features are enough, how post-pretraining on video–text data helps and how to model the dependency between frames, on benchmarks such as **MSR-VTT**, a collection of 10K web video clips with 200K clip–sentence pairs [38]. **Video foundation models** such as **InternVideo2** [39] combine masked video modeling, video–text contrastive learning and next-token prediction in one progressive training scheme, with a video encoder of up to 6 billion parameters [39].

Question answering. Multimodal large language models (ML 09) accept sampled frames as visual tokens. Long videos stress them: **Video-MME** [40] evaluates them on 900 manually annotated videos from 11 seconds to 1 hour long (2,700 question–answer pairs), with subtitles and audio as optional inputs, and found that handling longer sequences remained a weakness [40]. A practical caution, by analogy with scene bias: check whether a question can be answered from a single frame, or from the question text alone, before crediting a model with temporal understanding; a single-frame baseline is a cheap sanity check.

Localizing actions in time and space

Plain version. Real videos are not trimmed. A cooking video contains "cut", "stir" and "pour" with long boring stretches in between. Temporal localization finds where each action starts and ends; spatio-temporal detection also says which person is doing it.

Temporal action localization

The task mirrors object detection along the time axis (DLCV 03), and so do its methods. A clip-level backbone (often frozen, such as I3D or SlowFast features) produces a feature sequence along time; a localization head turns it into scored segments.

- **Proposal-based (two-stage).** First generate class-agnostic candidate segments, then classify them. **BMN** [41] represents a proposal as a pair of start and end boundaries and scores all densely distributed pairs at once in a **boundary-matching confidence map**; combined with an action classifier it gives detections [41].
- **Anchor-free, single-stage. ActionFormer** [42] classifies *every moment* in time and regresses the distances to the action's start and end, with no proposals and no predefined anchor windows. It uses a multiscale transformer with local self-attention and a light decoder [42], the temporal counterpart of the anchor-free detectors of DLCV 03. Duplicate segments are removed with non-maximum suppression in time, as for boxes (ActionFormer uses a soft variant [42]).

Temporal IoU and mAP. For a predicted segment $\hat{g} = [\hat{s}, \hat{e}]$ and a ground-truth segment $g = [s, e]$,

$$\mathrm{tIoU}(\hat{g}, g) = \frac{\max\left(0, \min(\hat{e}, e) - \max(\hat{s}, s)\right)}{(\hat{e} - \hat{s}) + (e - s) - \max\left(0, \min(\hat{e}, e) - \max(\hat{s}, s)\right)},$$

where the numerator is the length of the overlap and the denominator the length of the union. For one class and threshold θ, predictions are sorted by score; a prediction is a true positive if its tIoU with a not-yet-matched ground truth of that class is at least θ. Average precision (AP) is the area under the resulting precision–recall curve, mAP averages AP over classes, and an **average mAP** averages over several thresholds. Two conventions are common: THUMOS14 averages over tIoU $\{0.3, 0.4, \dots, 0.7\}$ and ActivityNet 1.3 over $\{0.5, 0.55, \dots, 0.95\}$ [42]. The code computes AP for one class on synthetic detections (85% of actions found, with boundary errors, plus random false alarms).

```
import numpy as np

def tiou(seg, segs):
    """Temporal IoU of one segment [s, e] with an array of segments (n, 2)."""
    inter = np.clip(np.minimum(seg[1], segs[:, 1]) - np.maximum(seg[0], segs[:, 0]), 0, None)
    union = (seg[1] - seg[0]) + (segs[:, 1] - segs[:, 0]) - inter
    return inter / union

def average_precision(preds, gts, thr):
    """preds: list of (video, start, end, score); gts: dict video -> (m, 2) array.
    Greedy matching in score order; each ground truth can be matched once."""
    n_gt = sum(len(g) for g in gts.values())
    used = {v: np.zeros(len(g), bool) for v, g in gts.items()}
    tp = []
    for v, s, e, _ in sorted(preds, key=lambda p: -p[3]):
        ious = tiou(np.array([s, e]), gts[v])
        ious[used[v]] = -1                       # already matched ground truths are off
        ↪ limits
        j = int(np.argmax(ious))
        hit = ious[j] >= thr
        if hit:
            used[v][j] = True
        tp.append(hit)
    tp = np.array(tp, float)
    recall = np.cumsum(tp) / n_gt
    precision = np.cumsum(tp) / np.arange(1, len(tp) + 1)
    precision = np.maximum.accumulate(precision[::-1])[::-1]   # monotone envelope
    return np.sum(np.diff(np.r_[0, recall]) * precision)

rng = np.random.default_rng(0)
gts, preds = {}, []
for v in range(20):                              # 20 untrimmed videos, 1-3 actions each
    starts = np.sort(rng.uniform(0, 100, rng.integers(1, 4)))
    g = np.c_[starts, starts + rng.uniform(4, 15, len(starts))]
    gts[v] = g
    for s, e in g:
```

```
        if rng.random() < 0.85:                  # the detector finds 85% of actions...
            j = rng.normal(0, 0.2 * (e - s), 2)  # ...with boundary errors
            preds.append((v, s + j[0], e + j[1], rng.uniform(0.5, 1.0)))
    for _ in range(rng.integers(0, 3)):          # plus some false alarms
        s = rng.uniform(0, 100)
        preds.append((v, s, s + rng.uniform(3, 12), rng.uniform(0.0, 0.7)))

thrs = [0.3, 0.4, 0.5, 0.6, 0.7]
aps = [average_precision(preds, gts, t) for t in thrs]
print("AP at tIoU:", "  ".join(f"{t}: {a:.3f}" for t, a in zip(thrs, aps)))
print(f"average over [0.3:0.1:0.7]: {np.mean(aps):.3f}")
```

```
AP at tIoU: 0.3: 0.731  0.4: 0.696  0.5: 0.696  0.6: 0.604  0.7: 0.284
average over [0.3:0.1:0.7]: 0.602
```

(Toy numbers.) Up to tIoU 0.5 the AP is limited mainly by the 15% missed actions and the false alarms; at 0.7 it collapses because boundary errors of about 20% of the action length are no longer good enough. Boundaries of actions are genuinely ambiguous (when does "pour" start?), which is why strict thresholds are hard even for people, and why benchmarks report several.

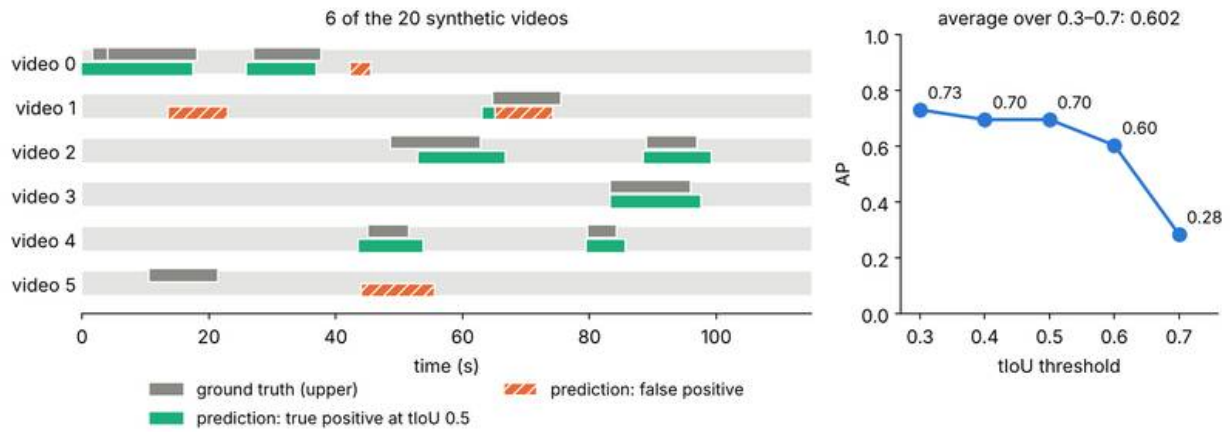


Figure 6 — The temporal-AP toy from the code above. Left: ground truth (gray) and predictions on six of the 20 videos, colored by whether they match at tIoU 0.5. Right: AP falls as the tIoU threshold becomes stricter.

Optional: Average precision for temporal localization, step by step (in the Supplementary notes). A hand-worked example of the matching rule, duplicates and the precision envelope.

Spatio-temporal action detection

Here the model must say which person does which action. In **AVA** [43], every person in keyframes sampled at 1 Hz from 15-minute movie clips gets a box and one or more labels from

80 atomic actions (14 pose classes, 49 person–object and 17 person–person interactions) [43]. A common pipeline is detection plus video classification, as in SlowFast's AVA detector [15]: an off-the-shelf person detector (DLCV 03) proposes boxes on the keyframe, a video backbone computes a spatiotemporal feature map for a clip around it, the features inside each box are pooled, and a per-class sigmoid classifier gives multi-label predictions, because one person can "stand", "talk to" and "hold" at once. The metric is **frame-mAP at IoU 0.5**: AP per action class over all keyframe detections, a detection counting only if its box overlaps a ground-truth box of that class with IoU at least 0.5, averaged over classes [43]. Linking detections into tubes over time borrows the association methods of tracking (DLCV 06).

Skeleton-based action recognition

Instead of pixels, a model can read the **pose**: the 2D or 3D coordinates of body joints over time, estimated as in DLCV 06A or captured with depth sensors, as in the NTU RGB+D dataset of 60 action classes [44]. **ST-GCN** [45] builds a graph whose nodes are the joints in every frame, with *spatial* edges along the bones and *temporal* edges connecting the same joint in consecutive frames, and applies graph convolutions over it [45]. Skeletons are compact, ignore clothing and background, and are therefore less prone to scene bias: on the **Mimetics** dataset of mimed actions without the usual objects and scenes, a pose-based model held up much better than 3D CNNs [46]. The price is that skeletons miss objects ("drinking" versus "eating") and depend on the pose estimator's errors.

Optional: Graph convolution on a skeleton sequence (ST-GCN) (in the Supplementary notes). The normalized adjacency, the partition of neighbors and how the convolution is implemented.

Training and evaluating video classifiers

Plain version. How you pick frames, how many views you average at test time and which dataset you report on change the numbers as much as many architecture changes. Read the protocol before comparing two papers.

Sampling and training

Three choices define the input: the **number of frames** T', the **stride** between them (frame rate), and dense versus sparse (TSN-style) sampling. Motion-centric datasets reward dense, fine-grained sampling; scene-centric datasets tolerate sparse sampling. VideoMAE, for example, uses TSN-style uniform sampling on Something-Something V2 and dense sampling on Kinetics-400 [10]. Spatial augmentation is the same as for images (random resized crops, flips), with one video-specific trap: **horizontal flipping changes the label** of direction-dependent classes such

as "pushing something from left to right". Training otherwise follows DLCV 02. Initialization usually comes from an image model (inflation, or a pretrained ViT) or from a self-supervised video model.

Multi-view testing

At test time, one clip rarely covers the whole video, so predictions are averaged over V_t temporal clips and V_s spatial crops:

$$\hat{p}(y \mid \text{video}) = \frac{1}{V_t V_s} \sum_{i=1}^{V_t} \sum_{j=1}^{V_s} \text{softmax}\left(f(x_{ij})\right)_y,$$

where x_{ij} is the i-th clip cropped at the j-th spatial position. SlowFast, for example, samples 10 clips uniformly along the video and takes 3 crops of 256 × 256 from each, 30 views in total, and reports cost as **GFLOPs per view × number of views**, noting that some earlier works used more than 100 views [15]. VideoMAE evaluated with 2 clips × 3 crops on Something-Something V2 and 5 × 3 on Kinetics-400 [10]. The total inference cost is the per-view cost times the number of views, so a model that looks cheap per view may be expensive in practice. Report both, and compare models only under the same protocol.

Datasets and benchmarks

Dataset	Task	Content (as described by its authors)
UCF101 [47]	clip classification	101 action classes, over 13k clips, 27 hours of user-uploaded video
HMDB51 [48]	clip classification	51 action classes from movies and web videos; small and deliberately hard
Kinetics-400 [17]	clip classification	400 human action classes, at least 400 clips each, about 10 s per clip, from YouTube
Something-Something V2 [9], [10]	clip classification	174 motion-centric classes defined as caption templates; about 169k training and 20k validation videos
Diving48 [7]	clip classification	over 18,000 clips of 48 fine-grained dive classes, built to reduce static bias
Charades [49]	multi-label classification, localization	9,848 videos of everyday indoor activities recorded by 267 people, 157 action classes
THUMOS14 [50]	temporal localization	untrimmed videos, including background videos without the target actions
ActivityNet [51]	untrimmed classification, localization	a large-scale benchmark of human activities in untrimmed web videos
AVA [43]	spatio-temporal detection	80 atomic actions in 430 15-minute movie clips, 1.58M action labels

Dataset	Task	Content (as described by its authors)
EPIC-KITCHENS-100 [52], [53]	egocentric recognition, detection, anticipation	100 hours of unscripted head-mounted video in 45 environments, 90K actions
NTU RGB+D [44]	skeleton / RGB-D recognition	over 56 thousand samples, 60 action classes, 40 subjects
MSR-VTT [38]	retrieval, captioning	10K web clips, 200K clip–sentence pairs
Video-MME [40]	video QA for multimodal LLMs	900 videos from 11 s to 1 h, 2,700 question–answer pairs

UCF101 and HMDB51 report average accuracy over **three official train/test splits** [12], [13]; Kinetics and Something-Something report top-1 and top-5 on their validation sets; Charades and AVA report mAP. Two warnings. First, Kinetics and Something-Something reward different things: the VideoMAE authors describe Kinetics videos as mostly stationary and scene-related, while Something-Something V2 is motion-centric [10]. A model's ranking can change between them. Second, the authors of EPIC-KITCHENS-100 built in a "test of time": testing models trained on footage collected in 2018 on new footage collected two years later, and an unsupervised domain adaptation challenge [53]. Benchmarks like these measure robustness that a single held-out split cannot.

Failure modes

Plain version. Video models fail in a few predictable ways. Most of them come from the model using a cue that happened to work on the training data.

1. **Scene and object bias.** The model predicts "playing tennis" from a tennis court. Evidence: representation bias toward static cues in common datasets [7]; models that fail on mimed actions without the usual objects and scenes [46]. Remedies include debiasing losses, such as an adversarial scene classifier and a loss that penalizes confident predictions on videos with the person masked out [54], motion-centric training data [9] and pose input [46].
2. **Temporal shortcuts.** The model ignores frame order. Test: shuffle or reverse the frames. If accuracy barely moves, the model (or the benchmark) does not use time; Sevilla-Lara et al. built their temporal subset of classes exactly this way [8].
3. **Short temporal context.** Most clip models see only a few seconds at a time. Actions defined over minutes ("assembling furniture") or questions about a whole film need long-range memory, which remains a weakness even for large multimodal models [40].
4. **Domain shift.** Third-person YouTube video, egocentric kitchens, surveillance at night and medical endoscopy look very different. Models trained on one transfer imperfectly to another; EPIC-KITCHENS-100 measures this with unseen kitchens and footage recorded later

[52], [53]. Distribution-shift tests from DLCV 02 apply.

5. **Ambiguous labels and boundaries.** Actions overlap, have fuzzy starts and ends and depend on intent. Multi-label datasets (Charades, AVA) and several tIoU thresholds acknowledge this; single-label accuracy hides it.
6. **Unfair comparisons.** Different numbers of views, frame counts, resolutions and pretraining data make reported numbers incomparable [1]. Always check the protocol line of a results table.

Two cheap diagnostics catch most of these: a **single-frame baseline** (how much of the score needs no time at all?) and a **shuffled-frames test** (does order matter to the model?). Run both before trusting a video model.

Modern view

What the surveys say

Five surveys cover the field from different angles. **Zhu et al.** [1] review over 200 papers on deep action recognition, introduce 17 datasets, present the models in chronological order (early attempts, two-stream networks, 3D convolutions, compute-efficient models), benchmark popular methods under one protocol with released code, and name the field's main challenges: long-range temporal modeling, compute cost and results made incomparable by different datasets and protocols. **Kong and Fu** [2] cover both recognition (labeling a complete action) and prediction (anticipating an action from its first part), including datasets and evaluation protocols. **Selva et al.** [3] organize video transformers by how they tokenize the input, how they reduce redundancy and reintroduce inductive biases, how they capture long-term dynamics and how they are trained, and find that they outperform 3D CNNs on action classification even at lower computational cost. **Madan et al.** [4] analyze over 200 video foundation models in three groups (image-based models adapted to video, video-based models, and universal models across modalities) and report that image-based models outperform video-based ones on most video understanding tasks, while universal multimodal models do best. **Tang et al.** [5] categorize video understanding with large language models by how the video reaches the language model (video analyzer, video embedder or both) and by the role of the language model (summarizer, manager, text decoder, regressor or hidden layer).

Optional: Guide to the Modern view surveys (in the Paper guides). How each survey is organized and what to read first.

The state of the art, 2023–2026

- **Self-supervised pretraining is a common starting point.** Masked video modeling (VideoMAE [10], VideoMAE V2 [33]) and feature prediction (V-JEPA [34]) produce encoders that are fine-tuned or probed for downstream tasks; the VideoMAE V2 authors call scale the primary factor for such foundation models [33].
- **Video foundation models unify tasks.** Models such as InternVideo2 [39] train one encoder with masked modeling, video–text contrast and next-token prediction, and are evaluated across recognition, retrieval and dialogue. The finding of Madan et al. that adapted image models remain very competitive [4] is a reminder that much of video understanding is still appearance.
- **Multimodal LLMs take over open-ended understanding.** Video question answering, captioning and temporal grounding are increasingly handled by language models with visual tokens, evaluated on benchmarks such as Video-MME [40]. The bottleneck is the number of visual tokens for long videos.
- **Efficient long-range models.** Linear-cost sequence models such as VideoMamba [30] and streaming designs (in the spirit of online TSM [21]) target long and live video.
- **Localization builds on pretrained encoders.** Temporal localizers run on features from pretrained video models; ActionFormer, for example, uses two-stream I3D features pretrained on Kinetics for THUMOS14 [42], so better encoders and better localization heads improve results together.

Open problems

1. **Genuine temporal reasoning.** Benchmarks where one frame or the language prior is not enough, and models that use order, causality and duration.
2. **Long video.** Memory and token budgets for hours of video, and evaluation of long-range understanding.
3. **Fine-grained and egocentric actions.** Hands, small objects, and intent; transfer across viewpoints and environments.
4. **Efficient and online recognition.** Low latency on devices, with honest cost reporting (FLOPs × views, real latency).
5. **Evaluation.** Comparable protocols, contamination of benchmarks by web-scale pretraining data, and metrics that reward temporal understanding rather than appearance.

Key takeaways

- Video tasks range from one label per trimmed clip to (start, end, class) segments, person-level action boxes and free-text answers; the metrics follow the task (top-1/top-5, mAP at temporal IoU, frame-mAP, Recall@k).
- The central difficulty is that labels often require motion, but many datasets can be solved from appearance; a model will take the scene shortcut if it can, as the toy experiment showed.
- Two-stream networks add motion through optical flow; 3D CNNs learn it from pixels; I3D inflates a pretrained 2D network by repeating each filter over time and dividing by its length.
- (2+1)D factorization splits a 3D convolution into spatial and temporal steps with a matched middle width: same parameters, an extra nonlinearity and easier optimization.
- TSN covers a whole video with a few sparsely sampled frames; TSM mixes neighboring frames with a channel shift at zero extra computation; SlowFast pairs a heavy low-frame-rate pathway with a light high-frame-rate one.
- Video transformers tokenize clips into tubelets and differ in which tokens attend to which: divided (TimeSformer), factorized (ViViT), pooled and multiscale (MViT) or windowed (Video Swin).
- VideoMAE pretrains by reconstructing 90–95% masked tubes; tube masking closes the leakage that independent masks leave open.
- Compare results only under the same protocol (frames, views, resolution, pretraining), and test every model with a single-frame baseline and shuffled frames.

Exercises

1. **How much does the combined model cheat?** In the scene-bias code, standardize the features (`sklearn.preprocessing.StandardScaler` in a pipeline) and vary `C` over 0.01, 0.1 and 1. How do the "no cue" and "shift" accuracies of the frame+motion model change? Why does standardization help it use the motion feature?

   Hint

   The motion features have a small numeric range compared with the 400 pixel features, so with an L2 penalty the model finds it cheaper to spread weight over many pixels. In our runs, standardizing gave the frame+motion model "no cue"/"shift" accuracies of 0.752/0.547 at `C = 0.01`, 0.824/0.685 at `C = 0.1` and 0.842/0.708 at `C = 1`, still well below the motion-only model. Scaling reduces the shortcut but does not remove it while the scene remains predictive in training.

2. **Inflation by hand.** A 2D filter $w_{2D} = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}$ (a horizontal derivative) is inflated to $N = 3$ frames. Write w_{3D}, and compute its response at one position for (a) a boring video where every frame has the row $[5, 3, 1]$ and (b) a video whose three frames have rows $[5, 3, 1]$, $[3, 1, 0]$ and $[1, 0, 0]$. What does the difference tell you about what training must change?

Hint

w_{3D} has three temporal slices, each $\frac{1}{3}[1, 0, -1]$. (a) Each slice gives $\frac{1}{3}(5 - 1) = \frac{4}{3}$, total 4, the same as the 2D response. (b) The slices give $\frac{4}{3}$, $\frac{3}{3}$ and $\frac{1}{3}$, total $\frac{8}{3}$. The inflated filter averages the frames, so it responds to motion only through blurring; to become a motion detector, training must make the temporal slices *different* (for example, opposite signs on the first and last).

3. **(2+1)D with unequal widths.** For a layer with $N_{i-1} = 64$ input and $N_i = 128$ output channels, $t = 3$, $d = 3$, compute M_i, the 3D parameter count and the (2+1)D parameter count. Then choose $M = 64$ and compute the savings.

Hint

$M_i = \lfloor 3 \cdot 9 \cdot 64 \cdot 128/(9 \cdot 64 + 3 \cdot 128) \rfloor = \lfloor 221184/960 \rfloor = 230$. 3D: $64 \cdot 128 \cdot 27 = 221{,}184$. (2+1)D with $M = 230$: $64 \cdot 230 \cdot 9 + 230 \cdot 128 \cdot 3 = 132{,}480 + 88{,}320 = 220{,}800$ (slightly less, because of the floor). With $M = 64$: 36,864 + 24,576 = 61,440, about 28% of the 3D layer.

4. **Views and cost.** Model A costs 36 GFLOPs per view and is tested with 10 × 3 views; model B costs 180 GFLOPs per view and is tested with 1 × 3 views. Which is more expensive per video? Design a fair comparison.

Hint

A: $36 \times 30 = 1080$ GFLOPs; B: $180 \times 3 = 540$ GFLOPs, so A costs twice as much per video despite being five times cheaper per view. A fair comparison plots accuracy against total inference cost for several view counts of each model, on the same frames and resolution.

5. **A shuffled-frames test.** Extend the scene-bias code: shuffle the frame order of each test clip, with a new permutation per clip (`rng = np.random.default_rng(5)`, then `np.stack([x[rng.permutation(8)] for x in X])`), and re-evaluate the three models trained on ordered clips. Predict the result before you run it.

Hint

The one-frame model still reads only the scene, so it barely changes (0.868 / 0.499 / 0.127 on the three test sets in our run). The motion model's per-pair estimates now have random signs, and it drops to chance (0.522 / 0.500 / 0.466). The frame+motion model does not move (0.864 / 0.497 / 0.133), which exposes that it never relied on order. A model whose accuracy survives shuffling is not using time.

6. **tIoU by hand.** A ground-truth action runs from 10 s to 20 s. Predictions: P1 [12, 22] with score 0.9, P2 [9, 19] with score 0.8, P3 [30, 35] with score 0.7. Compute each tIoU, mark TP/FP at $\theta = 0.5$ and $\theta = 0.75$, and compute AP at both thresholds.

Hint

P1: overlap 8, union 12, tIoU 0.667. P2: overlap 9, union 11, tIoU 0.818. P3: 0. At 0.5: P1 is TP (first in score order), P2 is a duplicate (FP), P3 FP; precision at recall 1 is 1, so AP = 1. At 0.75: P1 is FP, P2 is TP (precision 1/2 at recall 1), P3 FP; AP = 0.5. A worse box with a higher score can cost half the AP at a strict threshold.

Supplementary notes

These notes are optional: each takes one point from the main text and treats it in more detail. Nothing in the main text, the code or the exercises depends on them.

Choosing the middle width of a (2+1)D block

This note derives the R(2+1)D middle width and looks at its edge cases.

Setup. A full 3D layer maps N_{i-1} input channels to N_i output channels with kernels of size $t \times d \times d$. Its parameter count (without bias) is

$$P_{3D} = N_{i-1} N_i t d^2.$$

The (2+1)D block uses M spatial filters of size $N_{i-1} \times 1 \times d \times d$ and then N_i temporal filters of size $M \times t \times 1 \times 1$:

$$P_{2+1}(M) = M N_{i-1} d^2 + N_i M t = M (d^2 N_{i-1} + t N_i).$$

Matching. Setting $P_{2+1}(M) = P_{3D}$ and solving for M gives $M = t d^2 N_{i-1} N_i / (d^2 N_{i-1} + t N_i)$; the floor makes it an integer [20]. Because P_{2+1} is linear in M, the floor makes the block slightly *smaller* than the 3D layer whenever the division is not exact (Exercise 3 loses 384 parameters out of 221,184). The multiply–adds follow the same ratio, since both steps run at the same $T \times H \times W$ resolution (with stride 1).

Edge cases.

- *Equal widths* $N_{i-1} = N_i = N$: $M = t d^2 N / (d^2 + t)$. For $t = d = 3$, $M = 27N/12 = 2.25N$; the middle layer is *wider* than the input. This is why R(2+1)D is not a compression trick at matched size.
- *Temporal size 1* ($t = 1$): $M = d^2 N_{i-1} N_i / (d^2 N_{i-1} + N_i)$, which is smaller than N_i; the block becomes a spatial convolution followed by a 1×1 channel mixer, with an extra ReLU. In practice such layers are left as plain 2D convolutions.
- *Strides.* If the 3D layer downsamples in space or time, the stride is split accordingly: spatial stride in the spatial step, temporal stride in the temporal step [20].
- *Cheaper blocks.* Choosing M below the matched value trades accuracy for cost. With $M = N$ and $t = d = 3$, the block costs $N^2(9 + 3) = 12N^2$ parameters instead of $27N^2$, about 44% of the 3D layer.

Why the extra nonlinearity matters. Two linear maps in a row are equivalent to one linear map, but with a ReLU between them the block can represent functions that a single 3D convolution cannot, in the same way that two stacked 3×3 convolutions with a ReLU are more expressive than one 5×5 convolution. R(2+1)D attributes part of its gain to this, and part to easier optimization, visible as lower training error at equal depth [20].

The temporal shift as a sparse temporal convolution

This note shows why shifting channels in time and then applying a 1×1 convolution is a temporal convolution in disguise.

The shift. Let $x_t \in \mathbb{R}^C$ be the feature vector at one spatial position in frame t, and split the channels into three groups: A (the first $C/8$), B (the next $C/8$) and R (the rest). The bi-directional shift produces

$$\tilde{x}_t[A] = x_{t+1}[A], \qquad \tilde{x}_t[B] = x_{t-1}[B], \qquad \tilde{x}_t[R] = x_t[R],$$

with zeros where $t+1$ or $t-1$ falls outside the clip. No multiplications are involved, only data movement.

The following convolution. A 1×1 convolution with weight matrix $W \in \mathbb{R}^{C\times C}$ computes $y_t = W\tilde{x}_t$. Writing $W[:,A]$ for the columns of W that belong to group A, and so on,

$$y_t = W[:,A]\,x_{t+1}[A] + W[:,B]\,x_{t-1}[B] + W[:,R]\,x_t[R].$$

This is a temporal convolution of kernel size 3, $y_t = K_{+1}x_{t+1} + K_{-1}x_{t-1} + K_0 x_t$, whose three kernels are copies of W with most columns set to zero: K_{+1} keeps only the columns of group A, K_{-1} only those of B, and K_0 only those of R [21]. A full temporal convolution would have $3C^2$ free weights; the shifted version has C^2, the cost of the 1×1 convolution that was already there. The same argument holds for a 3×3 spatial convolution after the shift, applied position by position.

```
import numpy as np

rng = np.random.default_rng(0)
T, C = 8, 16
x = rng.standard_normal((T, C))           # one spatial position: T frames, C channels
fold = C // 8                             # 1/8 of the channels in each direction

def temporal_shift(x):
    out = np.zeros_like(x)
    out[:-1, :fold] = x[1:, :fold]                 # these channels see the next frame
    out[1:, fold:2 * fold] = x[:-1, fold:2 * fold] # these channels see the previous frame
    out[:, 2 * fold:] = x[:, 2 * fold:]            # the rest stay in place
    return out

W = rng.standard_normal((C, C))           # the channel mixing of the following 1x1
 ↪  convolution
y_tsm = temporal_shift(x) @ W.T
```

```
# the same result written as a temporal convolution with three sparse kernels
K_next, K_prev, K_now = np.zeros_like(W), np.zeros_like(W), np.zeros_like(W)
K_next[:, :fold] = W[:, :fold]
K_prev[:, fold:2 * fold] = W[:, fold:2 * fold]
K_now[:, 2 * fold:] = W[:, 2 * fold:]
xp = np.vstack([np.zeros((1, C)), x, np.zeros((1, C))])   # zero padding in time
y_conv = xp[2:] @ K_next.T + xp[:-2] @ K_prev.T + xp[1:-1] @ K_now.T
print("max difference:", np.abs(y_tsm - y_conv).max())
print("nonzero kernel weights:", sum((K != 0).sum() for K in (K_next, K_prev, K_now)),
      "of", 3 * C * C, "in a full temporal convolution")
```

```
max difference: 3.552713678800501e-15
nonzero kernel weights: 256 of 768 in a full temporal convolution
```

Design choices and pitfalls. (1) *Proportion.* Too few shifted channels give too little temporal mixing; too many leave each frame with too little of its own information. The paper found 1/4 (1/8 per direction) best for the residual variant [21]. (2) *Residual versus in-place.* Placing the shift on the residual branch keeps the unshifted features available through the skip connection, which worked better than shifting in place [21]. (3) *Online use.* For live video, future frames are not available, so only the backward shift (past to present) is used; the shifted channels of the previous frame can be cached, so each new frame costs one 2D forward pass [21]. (4) *Receptive field.* One shift reaches one frame in each direction; stacking L shifted blocks reaches L frames, so temporal context grows with depth, as with stacked temporal convolutions.

Average precision for temporal localization, step by step

This note works through the matching rule behind temporal AP on a small example, including the cases that commonly cause bugs.

Example. One video, one class, two ground-truth actions $g_1 = [10, 20]$ and $g_2 = [40, 50]$, and four predictions sorted by score:

rank	segment	score	tIoU with g_1	tIoU with g_2
1	$[11, 21]$	0.95	0.818	0
2	$[12, 20]$	0.90	0.800	0
3	$[60, 70]$	0.70	0	0
4	$[42, 54]$	0.60	0	0.571

Matching at $\theta = 0.5$. Rank 1 matches g_1 (TP). Rank 2 also overlaps g_1 well, but g_1 is already matched, and its overlap with g_2 is 0, so it is a **duplicate** and counts as FP. Rank 3 matches nothing (FP). Rank 4 matches g_2 (TP). The cumulative counts after each rank are TP = 1, 1, 1, 2, so precision is $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{2}$ and recall is $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1$.

The envelope. Interpolated precision at each rank is the maximum precision at that rank or later: $1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$. AP sums precision times the recall increase at each TP: $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = 0.75$.

At $\theta = 0.6$. Rank 4 now fails ($0.571 < 0.6$), so g_2 is never found; recall stops at $\frac{1}{2}$ and AP $= \frac{1}{2} \cdot 1 = 0.5$. Missed ground truths lower AP through the recall axis; false positives lower it through the precision axis.

Pitfalls.

- *Match to the best unmatched ground truth.* The rule in the main-text code takes the highest tIoU among ground truths not yet matched. If an implementation instead takes the best ground truth overall and counts the prediction as FP when that one is already matched, a prediction that also overlaps a second, nearby action is lost. Check which rule a benchmark's official evaluation code uses before comparing numbers.
- *Ties in score.* Sorting is not unique when scores tie; results can differ slightly between implementations. Use a stable sort.
- *Classes without ground truth* in the test set are usually skipped in the mean over classes; including them as AP 0 or 1 changes mAP.
- *NMS.* AP punishes duplicates, so localizers apply (soft) non-maximum suppression in time before evaluation, as detectors do for boxes (DLCV 03).

Graph convolution on a skeleton sequence (ST-GCN)

This note explains the graph convolution used by ST-GCN for skeleton-based recognition.

The graph. A skeleton sequence has V joints over T frames. ST-GCN connects joints along the bones within each frame (spatial edges) and each joint to itself in the next frame (temporal edges) [45]. Let $A \in \{0,1\}^{V \times V}$ be the bone adjacency matrix ($A_{ij} = 1$ if joints i and j share a bone) and I the identity (self-connections).

One spatial layer. The paper adopts the standard degree-normalized graph convolution; the single-frame layer with one neighbor group is [45]

$$f_{\text{out}} = \Lambda^{-\frac{1}{2}} (A + I) \, \Lambda^{-\frac{1}{2}} f_{\text{in}} \, W, \qquad \Lambda_{ii} = \sum_{j} (A + I)_{ij},$$

where $f_{\text{in}} \in \mathbb{R}^{V \times C_{\text{in}}}$ holds one feature vector per joint (initially its coordinates and the pose estimator's confidence [45]), $W \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}}}$ is the learned weight matrix shared by all joints, and Λ is the diagonal degree matrix that normalizes by the number of neighbors. Each joint's output is a degree-normalized average of its own and its neighbors' features, followed by a linear map.

Partitioning the neighbors. A single W treats all neighbors alike, which is like a convolution kernel with one weight. ST-GCN splits each joint's neighbors into subsets and gives each subset

its own weights, $f_{\text{out}} = \sum_j \Lambda_j^{-1/2} A_j \Lambda_j^{-1/2} f_{\text{in}} W_j$ with $\sum_j A_j = A + I$ [45]. Its **spatial configuration** partition uses three subsets: the joint itself; the neighbors closer to the skeleton's center of gravity (centripetal); and those farther away (centrifugal) [45]. This lets the layer distinguish "the elbow moves toward the body" from "away from it". A learnable mask M multiplies each A_j element-wise to weight the importance of individual edges [45].

Time. The temporal edges are handled by an ordinary 1D convolution along the frame axis for each joint (the paper uses a temporal kernel of 9 frames), so one ST-GCN unit is a graph convolution over joints followed by a temporal convolution over frames, the same space-then-time factorization as (2+1)D [45].

Pitfalls. Coordinates must be normalized (the paper applies batch normalization to the input skeletons) so that the shared weights see similar scales for all joints [45]; missing joints from the pose estimator must be filled or masked; and two-person actions need a convention for which body comes first.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Quo Vadis: I3D and the Kinetics dataset (Carreira and Zisserman, 2017)

The problem. In 2017, video architectures were compared on UCF101 and HMDB51, which are small enough that most methods performed similarly, so it was hard to tell which designs were actually better [14].

The idea. Two contributions. First, re-evaluate the main architecture families on the much larger Kinetics dataset [17]: a CNN with an LSTM on top, a C3D-like 3D CNN trained from scratch [16], a two-stream network, a "3D-fused" two-stream network, and the new two-stream I3D. Second, build I3D by inflating an ImageNet-pretrained Inception-V1 [55], bootstrapping its 3D filters from the 2D ones with the boring-video rule described in the main text [14].

What it showed. I3D did best on all three datasets with RGB, flow or both; two-stream versions were better than single-stream ones everywhere; and pretraining on Kinetics followed by fine-tuning gave large gains on UCF101 and HMDB51 [14]. Interestingly, flow alone was stronger than RGB on HMDB51 but weaker on Kinetics, which the authors linked to the larger camera motion in Kinetics [14], an early sign that what a benchmark rewards shapes the conclusions.

How to read it. Start with the architecture comparison table, which trains and tests each model on each dataset separately, then the transfer table, which compares training from scratch, frozen Kinetics features and fine-tuning. Check what each model was pretrained on (all but the 3D CNN

start from ImageNet), how many frames each sees at training and test time, and whether flow was used: these differences explain much of the ranking.

Limits. The flow stream still needs precomputed optical flow; Kinetics clips are short, and the authors themselves found the RGB stream more discriminative than flow there [14]; and inflation inherits the image network's design rather than asking what a video network should look like, a question later answered by R(2+1)D, SlowFast and X3D [20], [15], [23].

Why it matters here. I3D made "inflate a pretrained image model and pretrain on Kinetics" the standard recipe, and its features became a common video representation, including the feature network of the Fréchet Video Distance discussed in DLCV 10A.

SlowFast networks (Feichtenhofer et al., 2019)

The problem. Should a video network treat space and time symmetrically? Two-stream networks used optical flow for motion; 3D CNNs used one frame rate for everything [15].

The idea. Use two frame rates. A Slow pathway with many channels and few frames captures spatial semantics; a Fast pathway with few channels (ratio β) and many frames (α times more) captures motion; lateral connections from Fast to Slow fuse them at every stage. No optical flow is used [15].

What it showed. In the ablations on Kinetics, every SlowFast variant beat the Slow-only pathway; the Fast pathway alone was weak, yet adding it improved the Slow pathway, which the authors read as evidence that the two represent largely complementary information [15]. Among lateral connections, a time-strided convolution worked best. Every channel ratio from $\beta = 1/32$ to $1/4$ improved over Slow-only, with $1/6$ and $1/8$ best; even $\beta = 1/32$, adding only about 5% computation, helped [15]. The same backbone gave strong results on Charades and on AVA detection [15].

How to read it. Read Table 1 (the instantiation with $\alpha = 8$, $\beta = 1/8$, and where temporal kernels are larger than 1), then the ablation table: lateral connection types, the channel ratio, and the variants that remove or modify pathways. When reading the main results tables, note the "GFLOPs × views" column, which the paper introduced to make inference cost comparable, and its observation that some earlier methods used more than 100 views [15].

Limits. Two fixed frame rates are a hand-designed prior; the model still sees only a few seconds; and later transformer models such as MViT, from overlapping authors, revisited the multiscale idea with attention [27].

Why it matters here. SlowFast is the clearest example of an architecture built around the asymmetry between "what" and "how it moves", and its evaluation conventions (30 views, FLOPs per view) are still used.

Guide to the Modern view surveys

This guide expands the short survey summary in the Modern view: what each review covers, how it is organized and what to read it for.

A Comprehensive Study of Deep Video Action Recognition (Zhu et al., 2020) [1] is the best entry point for the CNN era. It introduces 17 datasets that influenced model design, then follows the models chronologically: early deep models, two-stream networks, 3D CNNs and compute-efficient models. Its distinctive part is a benchmark of popular methods under a common setting with released code, which addresses the problem it names at the start: results that cannot be compared because of different datasets and protocols. Read its dataset section and its benchmark tables first.

Human Action Recognition and Prediction: A Survey (Kong and Fu, *IJCV*, 2022) [2] places recognition next to *prediction*, inferring an action from its incomplete beginning, which matters for driving and surveillance. It covers models, algorithms, technical difficulties, datasets and evaluation protocols for both, and discusses future directions.

Video Transformers: A Survey (Selva et al., *IEEE TPAMI*, 2023) [3] analyzes video transformers by design question: how video is handled at the input (tokenization and embedding), architectural changes to reduce redundancy, reintroduce inductive biases and capture long-term dynamics, and training regimes including self-supervision. Its performance comparison on action classification finds transformers ahead of 3D CNNs even at lower computational cost. Read it to understand *why* the attention variants in the main text exist.

Foundation Models for Video Understanding: A Survey (Madan et al., 2024) [4] analyzes over 200 video foundation models and benchmarks across 14 video tasks in three categories. Its taxonomy (image-based models adapted to video, video-based models with video-specific encoding, and universal models across modalities) is a useful map, and its main finding is the one to remember: adapted image models outperformed video-based models on most tasks, while universal multimodal models were strongest.

Video Understanding with Large Language Models: A Survey (Tang et al., 2023, accepted to *IEEE TCSVT* according to its arXiv record) [5] organizes video LLMs in two ways: by how video information reaches the language model (a video analyzer that produces text, a video embedder that produces tokens, or both) and by the role the language model plays (summarizer, manager, text decoder, regressor, hidden layer). It also surveys tasks, datasets, benchmarks and evaluation methods for these models. Read it after ML 09.

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DLCV 07 • World Models: Video, Latent Dynamics and Planning

What a world model is, how model-based reinforcement learning uses one (World Models, the Dreamer line, MuZero), why video generators are proposed as world simulators, how latent dynamics and JEPA-style predictive representations work, and how to evaluate all of them, with complete CPU experiments.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-07-world-models/

What you'll learn

- What a world model is, and the difference between an observation, a hidden state, a belief and an action.
- How model-based reinforcement learning uses a learned model in three different ways: planning online, learning a policy in imagination, and tree search with a value-equivalent model, with World Models, PlaNet, the Dreamer line, TD-MPC2 and MuZero as examples.
- Why video prediction and generation are proposed as "world simulators", why squared-error predictors produce blur, and what interactive generators such as Genie can and cannot show.
- How recurrent state-space models and JEPA-style predictive representations learn latent dynamics without (or with) reconstructing pixels.
- How to fit a small dynamics model, measure multi-step error, plan with it, and detect when a planner exploits model errors, all on a CPU.
- How to evaluate a world model, and which problems remain open.

The big picture

Tracking asks where an object went. A world model asks what could happen next, and especially what would happen **if an agent took a particular action**. A robot approaching a cup needs more than an object label: it needs to predict how its gripper and the cup will move, which outcomes are uncertain, and which action is likely to reach its goal.

> **Explain like I'm five**
>
> Before pushing a toy car, you imagine where it will go. You can imagine a gentle push and a strong push, pick the better one, push, and then look again to see what really happened. A world model is a learned version of that imagination. A planner is the part that tries out pushes in imagination and picks one. Looking again after each push is what keeps small imagination mistakes from piling up.

The term "world model" covers several research traditions. Ding et al. organize the literature around two functions: building internal representations that capture how the world works, and predicting future states to simulate and guide decisions [1]. This chapter uses an operational definition: **a world model is a learned model that represents enough of an environment to predict relevant changes under specified conditions, usually including the agent's own actions.** Its usefulness depends on the task. A model for rendering a scene, a model for learning representations, and a model for controlling a robot may need very different information.

Observations, states, beliefs and actions

Plain version. A camera image is not the world; it is a partial view of it. A world model has to keep track of what it cannot currently see, like the speed of a ball or an object behind an occluder.

Let s_t denote the underlying environment state at time t, o_t the observation and a_t the executed action:

$$s_{t+1} \sim p(s_{t+1} \mid s_t, a_t), \qquad o_t \sim p(o_t \mid s_t).$$

The first distribution is the **transition** (dynamics) model; the second is the **observation** model. The state might contain object positions, velocities and contact conditions. An image shows only a projection, with occlusions and sensor noise; Szeliski covers the geometry and motion side of this observation process [3]. Two identical images can have different futures: a ball at the same position might be moving left or right.

A learned model therefore builds a representation from history:

$$z_t = E_\theta(o_{0:t}, a_{0:t-1}), \qquad p_\theta(z_{t+1} \mid z_t, a_t),$$

where E_θ is an encoder with parameters θ, $o_{0:t}$ is the sequence of observations up to t, and $p_\theta(z_{t+1} \mid z_t, a_t)$ is a learned latent transition. An encoder does not automatically make z_t a sufficient state. If it discards velocity, hidden objects or a relevant material property, the transition model cannot recover that information. Recurrent memory or a short video context helps, but must be tested under the actual occlusions and delays.

When the state is only partially observed, the principled object is a **belief**, a probability distribution over possible states given the history. Bishop's treatment of sequential latent-variable models (hidden Markov models and linear dynamical systems) gives the probabilistic background [2]. A compact latent vector can approximate useful aspects of this distribution; it need not map one-to-one onto human-readable physical variables.

A prediction factorization

Predicting the future splits into three questions: where are we now, how do things change under the actions, and what would the camera see?

For a proposed action sequence, the probability of future observations factorizes as

$$p_\theta(o_{t+1:t+H} \mid o_{\le t}, a_{0:t+H-1}) = \int p_\theta(z_t \mid o_{\le t}, a_{<t}) \prod_{k=t}^{t+H-1} p_\theta(z_{k+1} \mid z_k, a_k)\, p_\theta(o_{k+1} \mid z_{k+1})\, dz_{t:t+H},$$

where H is the prediction horizon. The three factors are **inference** (what is the current latent state given the past?), **transition** (how does it evolve under the actions?) and **rendering** (what would the camera see?). A planner may work directly with z and skip the pixel decoder if its goal and cost are defined in latent space. Much of the design space in this chapter consists of choosing which of these factors to learn, and how.

Model-based reinforcement learning

Plain version. Reinforcement learning (RL) learns to act by trial and error. If the agent also learns how the world responds to its actions, it can practice in its head, which needs far fewer real trials.

The setting

An RL agent interacts with an environment over discrete steps. At each step it observes, chooses a_t from a **policy** $\pi(a_t \mid s_t)$, receives a reward r_t, and the environment moves on. The goal is to maximize the expected **return**

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k},$$

where $\gamma \in [0, 1)$ is a discount factor that weights near rewards more than distant ones. The **value** $V^\pi(s) = \mathbb{E}_\pi[G_t \mid s_t = s]$ is the expected return from state s when following π.

Model-free methods learn a policy or value directly from real transitions. **Model-based** methods also learn a model of transitions and rewards and use it to reduce real interaction. The idea is old: Sutton's Dyna architecture (1991) already combined learning from real experience with planning on experience simulated by a learned model, using the same value updates for both [4]. Moerland et al. survey the field and organize it along two axes: how the dynamics model is learned (handling stochasticity, uncertainty, partial observability, temporal abstraction) and how planning and learning are integrated (where to start planning, how much budget to give it, how to plan, and how to feed the results back into acting and learning) [5].

Three ways to use a model

Once a model is learned, there are three main ways to turn its predictions into decisions.

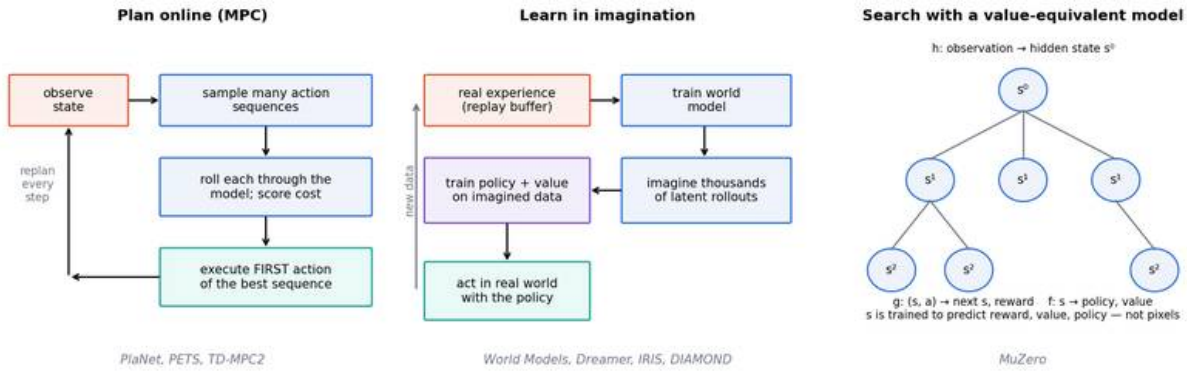


Figure 1 — Three ways to use a learned model. The same model can serve more than one role; the difference is where the decision is computed.

1. **Plan online (model-predictive control).** At every real step, search for a good action sequence by simulating candidates in the model, execute only the first action, observe, and repeat. PlaNet [7], PETS [29] and TD-MPC2 [11] work this way.
2. **Learn in imagination.** Train a policy (and usually a value function) on trajectories generated by the model, then act with that policy without searching at run time. World Models [6], the Dreamer line [8], [9], [10], IRIS [12] and DIAMOND [13] work this way.
3. **Search with a value-equivalent model.** Learn a model that predicts only what planning needs (rewards, values, policy), not observations, and run tree search in its latent space. MuZero [14] is the canonical example.

All three share one danger: the model is wrong somewhere, and an optimizer is very good at finding exactly those places. We return to this in the section on uncertainty and exploitation.

Why errors compound

Each imagined step starts from the previous imagined step, so small prediction mistakes feed into the next prediction and can grow with every step.

Suppose the learned transition $\hat{f}$ has one-step error at most ε, and the true dynamics f are Lipschitz with constant L (a small change in state changes the next state by at most L times as much). Starting from the same state and applying the same actions, the gap e_k between predicted and true states after k steps obeys $e_{k+1} \le L e_k + \varepsilon$, so

$$e_H \le \varepsilon \sum_{k=0}^{H-1} L^k = \varepsilon \, \frac{L^H - 1}{L - 1} \quad (L \neq 1).$$

For $L > 1$ the bound grows exponentially with the horizon H; for $L < 1$ it saturates at $\varepsilon/(1 - L)$. This is a worst-case bound, but it explains two common design choices: replanning often (MPC) and using short model rollouts. Janner et al. made the second idea concrete: their MBPO algorithm branches short model rollouts from real states rather than imagining long trajectories from the start, and they argue both theoretically and empirically that this keeps the benefit of model data while limiting its bias [30].

World Models: compress, predict, control

Plain version. First learn to summarize each frame in a few numbers, then learn to predict the next summary, then train a tiny controller on top. The controller can even be trained entirely inside the model's "dream".

Ha and Schmidhuber's *World Models* (published at NeurIPS 2018 as "Recurrent World Models Facilitate Policy Evolution", with an interactive online version) made the idea popular in deep learning [6]. Their agent has three parts:

- **V (vision):** a variational autoencoder that compresses each 64×64 frame into a small latent vector z_t.
- **M (memory):** a recurrent network with a mixture-density output that predicts a distribution over z_{t+1} given z_t, the action and its hidden state h_t.
- **C (controller):** a deliberately tiny linear policy $a_t = W_c[z_t; h_t] + b_c$, trained with an evolution strategy rather than backpropagation.

In a car-racing task, the controller using V and M features solved the task; in a VizDoom task, the controller was trained entirely inside the environment simulated by M and then transferred to the real game [6]. The design separates *perception*, *prediction* and *decision*, and shows that a controller can be very small when the representation already carries the relevant dynamics. It also showed the central risk: a controller trained inside an imperfect dream can learn to exploit the dream's mistakes.

Latent dynamics: from PlaNet to the Dreamer line

Plain version. Instead of predicting future images, predict a compact hidden state and only decode images when training. A hidden state that mixes a deterministic memory with a random part can represent both what the agent remembers and what it is unsure about.

Recurrent state-space models

Hafner et al.'s PlaNet learned environment dynamics from images and chose actions by fast online planning in latent space; its key modeling idea is a latent dynamics model with both deterministic and stochastic components, the **recurrent state-space model** (RSSM) [7]. The RSSM keeps a deterministic memory h_t and a stochastic state z_t:

$$h_t = f_\theta(h_{t-1}, z_{t-1}, a_{t-1}), \qquad z_t \sim p_\theta(z_t \mid h_t) \ \text{(prior)}, \qquad z_t \sim q_\theta(z_t \mid h_t, o_t) \ \text{(posterior)}.$$

Here f_θ is a recurrent cell, the **prior** predicts z_t without seeing the current image (this is what is used when imagining), and the **posterior** also sees the image o_t (used during training and when filtering real observations). Heads read the model state (h_t, z_t) to predict the image $\hat{o}_t$, the reward $\hat{r}_t$ and a continuation flag (whether the episode goes on).

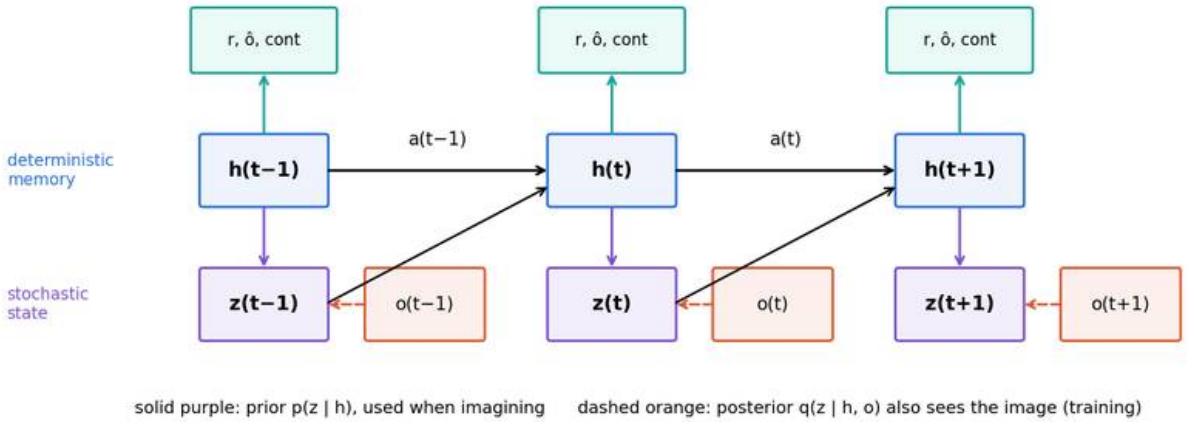


Figure 2 — An unrolled RSSM. Imagination uses only the prior path (solid); training additionally conditions z on the real image (dashed).

Training maximizes a variational lower bound on the likelihood of observed sequences. A simplified per-step loss is

$$\mathcal{L}_t = -\log p_\theta(o_t \mid h_t, z_t) - \log p_\theta(r_t \mid h_t, z_t) + \beta\, D_{\mathrm{KL}}(q_\theta(z_t \mid h_t, o_t) \,\|\, p_\theta(z_t \mid h_t)),$$

with $\beta > 0$ weighting the KL term. The reconstruction terms make the state informative; the KL term teaches the prior to predict what the posterior will see, which is exactly the skill needed for imagination. Real implementations add loss balancing and other stabilizers, so treat this as a teaching outline, not the exact loss of any one paper.

Optional: The ELBO and the reparameterization trick (in the Supplementary notes). Where this loss comes from and how gradients pass through the sampled state, with a one-variable example.

Dreamer: learning behavior in imagination

Dreamer (2020) kept the RSSM but replaced online planning with an actor–critic trained entirely on imagined latent trajectories. Its key move was to propagate analytic gradients of learned state values back through trajectories imagined in the compact latent space [8]. Starting from latent states inferred from real data, the agent imagines H steps with its actor, scores them with a learned critic, and trains the critic on bootstrapped targets such as the λ-return

$$V_t^\lambda = \hat{r}_t + \gamma\, \hat{c}_t \big[(1-\lambda)\, v_\psi(s_{t+1}) + \lambda\, V_{t+1}^\lambda \big],$$

where $\hat{r}_t$ and $\hat{c}_t$ are the predicted reward and continuation, v_ψ is the critic, and $\lambda \in [0,1]$ trades off one-step bootstrapping ($\lambda = 0$) against longer imagined returns ($\lambda = 1$). The actor is updated to increase these values.

The line continued:

- **DreamerV2** (2021) switched to *discrete* (categorical) latent variables and became the first agent to reach human-level performance on the 55-game Atari benchmark by learning behaviors inside a separately trained world model [9].
- **DreamerV3** (Nature, 2025) used robustness techniques based on normalization, balancing and transformations so that a single configuration outperformed specialized methods across more than 150 tasks, and was the first algorithm to collect diamonds in Minecraft from scratch without human data or curricula [10].

Two related lines show that the RSSM is one choice among several. **TD-MPC2** plans in the latent space of an *implicit*, decoder-free world model (no image reconstruction), reported strong results across 104 online RL tasks with a single set of hyperparameters, and trained a single 317M-parameter agent on 80 tasks [11]. **IRIS** replaced the RSSM with a discrete autoencoder plus an autoregressive Transformer over image tokens and learned from the equivalent of two hours of

Atari gameplay [12]. **DIAMOND** went further toward pixels: it trains the agent inside a *diffusion* world model, arguing that discrete latents can drop visual details that matter for control, and also demonstrated the model as an interactive neural game engine for a 3D game [13].

MuZero: a model that predicts only what planning needs

Plain version. You do not have to predict what the screen will look like to play a good game. You only need to predict how good positions are, which moves look promising, and what rewards you will get. MuZero learns exactly that, and then searches.

AlphaZero-style tree search needs a simulator of the game rules. MuZero removes that requirement by learning a model that, applied repeatedly, predicts the quantities most directly relevant to planning: reward, action-selection policy and value [14]. It has three functions:

$$s^0 = h_\theta(o_1, \dots, o_t), \qquad (r^k, s^k) = g_\theta(s^{k-1}, a^k), \qquad (\mathbf{p}^k, v^k) = f_\theta(s^k),$$

where h_θ (representation) encodes the observation history into a hidden state s^0, g_θ (dynamics) maps a hidden state and a hypothetical action a^k to a predicted reward r^k and next hidden state, and f_θ (prediction) outputs a policy vector $\mathbf{p}^k$ and a value v^k. A Monte Carlo tree search unrolls g_θ to choose actions. Training unrolls the model for K steps along real trajectories and minimizes

$$\mathcal{L} = \sum_{k=0}^{K} \Big[\ell^r(u_{t+k}, r^k) + \ell^v(z_{t+k}, v^k) + \ell^p(\boldsymbol{\pi}_{t+k}, \mathbf{p}^k) \Big] + c\|\theta\|^2,$$

where u are observed rewards, z are value targets (bootstrapped returns), $\boldsymbol{\pi}$ are the visit-count distributions produced by the search, ℓ^r, ℓ^v, ℓ^p are per-quantity losses and c weights an L2 penalty.

The hidden state s^k is never asked to reconstruct an observation. It only has to be **value-equivalent**: planning with it must give the same rewards, values and policies as planning in the real environment would. With this design MuZero reached a new state of the art on 57 Atari games and, without being given the rules, matched AlphaZero's superhuman performance in Go, chess and shogi [14]. The lesson for vision is that a world model's target should be chosen by its use; predicting every pixel is one option, not a requirement.

Video prediction and generation as world simulators

Plain version. If a model can generate realistic future video, perhaps it has learned how the world works. Sometimes it has; often it has only learned what videos usually look like. Testing which is the case is the hard part.

Action-conditioned video prediction

Finn et al. trained an **action-conditioned video prediction** model on a dataset of 59,000 robot pushing interactions; by predicting pixel motion rather than raw pixels, the model generalized partly to unseen objects, and conditioning on different future actions produced a "visual imagination" of different futures [15]. This is a world model in the most literal sense: image in, action in, image out.

Why squared-error predictors blur

Real futures are uncertain. Suppose that from the same past, a ball goes left with probability $\frac{1}{2}$ and right with probability $\frac{1}{2}$. For squared error, the best single prediction is the **conditional mean**:

$$\hat{o}^\star = \arg\min_{\hat{o}} \; \mathbb{E}[\|o_{t+1} - \hat{o}\|^2 \mid o_{\le t}] = \mathbb{E}[o_{t+1} \mid o_{\le t}],$$

which here is two half-bright balls, an image that never happens. A generative model that **samples** one of the two futures produces a realistic image but, on average, a *higher* squared error.

```
import numpy as np

x = np.arange(64)
def frame(center):                      # a 1-D "image" with one bright ball
    return np.exp(-0.5 * ((x - center) / 2.0) ** 2)

# Same past, two equally likely futures: the ball bounces left or right.
futures = np.stack([frame(20), frame(44)])
mean_pred = futures.mean(axis=0)        # what an MSE-trained predictor converges to
sample_pred = futures[0]                # what a good generative sampler might output

def expected_mse(pred):                 # average over the two possible futures
    return np.mean([(pred - f) ** 2 for f in futures])

print(f"peak brightness: true future 1.00, mean prediction {mean_pred.max():.2f}")
print(f"expected MSE of mean prediction:   {expected_mse(mean_pred):.4f}")
print(f"expected MSE of a sharp sample:    {expected_mse(sample_pred):.4f}")
```

Output:

```
peak brightness: true future 1.00, mean prediction 0.50
expected MSE of mean prediction:   0.0277
expected MSE of a sharp sample:    0.0554
```

The sharp sample has exactly twice the expected error of the blurry mean. This is why pixel MSE rewards blur, why stochastic video models were introduced, and why evaluating a world model with a single squared-error number can prefer the wrong model.

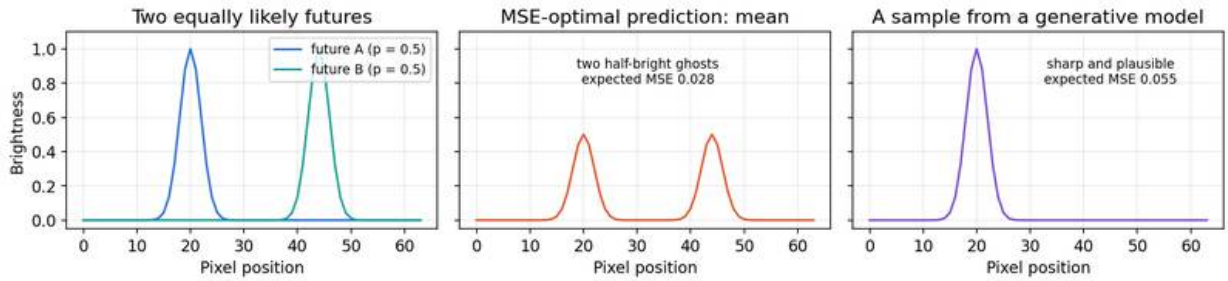


Figure 3 — Output of the code above. The MSE-optimal prediction averages incompatible futures; a sample is plausible but scores worse on MSE.

Babaeizadeh et al.'s SV2P addressed this by predicting a different possible future for each sample of a latent variable, and found substantially better predictions than the same model without stochasticity [16]. Modern video generators use the same principle at a much larger scale, with autoregressive token models or diffusion and flow models (see ML 07 and ML 08).

Large generative world models

Several large models now generate video conditioned on actions, text or actions inferred from the video itself, and are presented as world simulators.

- **GAIA-1** casts driving world modeling as next-token prediction over discrete tokens of video, text and action, producing realistic driving scenarios with control over ego-vehicle behavior and scene features [17].
- **UniSim** learns a "universal simulator" of real-world interaction by combining datasets that are rich along different axes (objects in images, dense actions in robot data, motion in navigation data); it simulates both high-level instructions and low-level controls, and policies trained purely in it were deployed zero-shot on real robots [18].
- **Genie** is a generative interactive environment trained without action labels on unlabelled Internet video. It consists of a video tokenizer, an autoregressive dynamics model and a *latent action model* that infers actions from changes between frames, so users can act frame by frame in generated worlds even though no action labels were used in training [19]. Such latent actions need not correspond to calibrated physical actuators.
- **Genie 3**, announced by Google DeepMind in August 2025, generates worlds from text that can be navigated in real time at 24 frames per second and 720p, keeping consistency for a few minutes. The announcement lists limitations: a constrained action space, difficulty modeling several independent agents, inaccurate real-world locations, and interaction limited to minutes rather than hours [20].
- **Cosmos** is NVIDIA's open platform of "world foundation models": pretrained video models, video tokenizers, a data-curation pipeline and post-training examples meant to be fine-tuned into task-specific world models for robots and vehicles [21].

Does realistic video mean physical understanding?

Two careful studies say: not necessarily. Motamed et al.'s **Physics-IQ** benchmark tests fluid dynamics, optics, solid mechanics, magnetism and thermodynamics; across current video models (including Sora, Runway, Pika, Lumiere, Stable Video Diffusion and VideoPoet), physical understanding was severely limited and unrelated to visual realism [22]. Kang et al. trained diffusion video models on a 2D simulator governed by classical mechanics and found perfect in-distribution generalization, measurable scaling for combinatorial generalization, but failure out of distribution; the models behaved as if retrieving the closest training example rather than abstracting a rule [23].

For robot control, also ask what an "action" means: force, joint target, gripper command, camera movement or an inferred token. A wrong unit, timing offset or control frequency changes the conditional problem even if the video looks convincing.

Predictive representations: the JEPA family

Plain version. Predicting every pixel wastes effort on details that do not matter, like leaf texture. Instead, predict an abstract *description* of the missing or future part. If the description is good, the prediction is useful for planning.

A **joint-embedding predictive architecture** (JEPA) encodes a visible *context* x and a hidden *target* y with encoders, and trains a predictor to map the context embedding to the target embedding:

$$\mathcal{L} = \| P_\phi(E_\theta(x),\, m) - \mathrm{sg}[\bar{E}(y)] \|^2,$$

where E_θ is the context encoder, $\bar{E}$ the target encoder, P_ϕ a predictor conditioned on information m about where the target is (for example its position in the image or time), and sg (stop-gradient) blocks gradients into the target branch. Nothing is reconstructed in pixel space. The danger is **collapse**: if both encoders output a constant, the loss is zero. Asymmetric designs (a predictor on one side only, stop-gradient, and a target encoder that is not trained by the same gradient) are used to prevent it; ML 05 discusses these self-supervised mechanisms in more detail.

The JEPA line began as representation learning, on images (I-JEPA [24]) and then on video (V-JEPA [25]), and V-JEPA 2.1 [27] later refined its dense features. Two models show what turns it into a world model: action-conditioned prediction, and planning toward a goal in feature space.

- **V-JEPA 2** pretrains an action-free JEPA on over one million hours of Internet video, then post-trains a latent **action-conditioned** world model, V-JEPA 2-AC, on less than 62 hours of unlabeled robot video; planning toward image goals with it enabled zero-shot pick-and-place on Franka arms in two labs without data from those environments, task-specific training or reward [26]. The split between passive video and action data makes the observation/action distinction of this chapter concrete.
- **DINO-WM** shows the same idea with an off-the-shelf encoder: it predicts future DINOv2 patch features from offline trajectories and plans by optimizing action sequences toward the features of a goal image, without reconstructing pixels or learning a reward model [28].

Optional: More of the JEPA line: I-JEPA, V-JEPA and V-JEPA 2.1 (in the Supplementary notes). What I-JEPA learned about masking, how V-JEPA trains on video alone, and what V-JEPA 2.1 adds for dense features.

Goals in feature space need care. A distance $\|z - z_{\text{goal}}\|^2$ rewards only what the representation preserves. If the representation ignores gripper closure, a planner may reach a visually similar state without actually grasping the object. Better feature quality alone does not establish that an encoder predicts the consequences of arbitrary actions.

How the families relate

The families differ mainly in what they predict, and each prediction target raises its own question to test.

Family	Prediction target	Main benefit	Question to test
Latent dynamics (RSSM, TD-MPC2)	compact state, reward, value	cheap, repeated rollouts	does the latent retain task-relevant dynamics?
Value-equivalent (MuZero)	reward, value, policy only	no effort on irrelevant detail	does it transfer when the reward changes?
Autoregressive video (GAIA-1, Genie, IRIS)	next frame or token	explicit sequential factorization	do errors drift over long rollouts?
Diffusion/flow video (DIAMOND, Cosmos)	jointly generated frames or chunks	expressive, uncertain futures	can actions steer outcomes at control latency?
Predictive embeddings (JEPA, DINO-WM)	target features	structure without pixel reconstruction	are actions represented and goals meaningful?

These are compatible choices. A Transformer can parameterize any of them; a latent world model can decode through diffusion; a video generator can use an autoregressive outer loop and a diffusion inner loop. Architecture, prediction target, temporal factorization and training objective are separate design decisions.

Learning dynamics from data

Plain version. A dynamics model is only as good as its data: frames, actions and rewards must stay in order and correctly aligned, and test episodes must be truly separate from training ones.

Training examples should retain ordering and action alignment:

$$(o_t, a_t, o_{t+1}, r_t, \text{done}_{t+1}),$$

where done_{t+1} marks the end of an episode. Record whether each action happens before or after each camera timestamp. Split by whole episodes, environment instances and relevant subjects; randomly splitting adjacent frames leaks nearly identical scenes and makes a transition predictor look better than it is (see ML 10 for leakage in general).

The data-collection policy matters. A model trained on slow, gentle interactions may fail for abrupt motion. Observational correlations are especially problematic when actions are absent: a model may learn that a cup moves when a hand approaches, yet have no evidence about what alternative pushes would do. This is the difference between passive video and interventional data.

A **one-step loss** measures prediction from real history. A **multi-step loss** unrolls the model's own predictions and penalizes their accumulated error. Unrolling changes training difficulty and gradient paths; it does not remove distribution shift outside the observed interactions.

Complete CPU experiment: fit, predict, plan

Plain version. We now run the whole loop on a toy system: collect data, fit a model, check its predictions one step and many steps ahead, and plan with it.

We use a two-dimensional state $x_t = (\text{position}, \text{velocity})$, a scalar action u_t and a damped linear system:

$$x_{t+1} = Ax_t + Bu_t + \epsilon_t,$$

where A is the 2×2 state-transition matrix, B the action-input vector and ϵ_t small Gaussian noise. The system is deliberately fully observed. It isolates dynamics and planning before adding image encoders and partial observability. Training and test episodes are generated independently. We fit only on training transitions and evaluate both one-step prediction and open-loop rollouts with the same held-out actions.

The planner uses the **cross-entropy method** (CEM): sample candidate action sequences from a Gaussian, keep the lowest-cost "elite" sequences under the model, refit the Gaussian's mean and spread to them, and repeat. It executes one action, receives a fresh real state and replans. This is model-predictive control.

Optional: The cross-entropy method (in the Supplementary notes). The sample, rank and refit loop written out, with a small run on a known model.

```
import numpy as np

rng = np.random.default_rng(7)
A = np.array([[1.0, 0.1], [0.0, 0.94]])
B = np.array([0.005, 0.1])

def collect(n, steps=40):
    x = np.zeros((n, steps + 1, 2))
    x[:, 0] = rng.normal(size=(n, 2)) * [0.8, 0.3]
    u = rng.uniform(-1, 1, size=(n, steps))
    for t in range(steps):
        x[:, t + 1] = (x[:, t] @ A.T + u[:, t, None] * B
                       + rng.normal(scale=0.002, size=(n, 2)))
    return x, u

train, train_u = collect(120)
test, test_u = collect(80)
phi = np.column_stack([train[:, :-1].reshape(-1, 2), train_u.ravel()])
target = train[:, 1:].reshape(-1, 2)
W = np.linalg.lstsq(phi, target, rcond=None)[0]
W_no_action = np.linalg.lstsq(phi[:, :2], target, rcond=None)[0]

def predict(x, u):
    return np.concatenate([x, np.asarray(u)[..., None]], axis=-1) @ W

one_step = predict(test[:, :-1], test_u)
```

```
rollout = np.zeros_like(test)
rollout[:, 0] = test[:, 0]
for t in range(test_u.shape[1]):
    rollout[:, t + 1] = predict(rollout[:, t], test_u[:, t])
rmse = lambda a, b: np.sqrt(np.mean((a - b) ** 2))
print("One-step:", rmse(one_step, test[:, 1:]))
print("Without actions:", rmse(test[:, :-1] @ W_no_action, test[:, 1:]))
print("Final-horizon rollout:", rmse(rollout[:, -1], test[:, -1]))

def plan(x, mean, horizon=20, population=256):
    std = np.ones(horizon)
    for _ in range(4):
        candidates = np.clip(rng.normal(mean, std, (population, horizon)), -1, 1)
        imagined = np.broadcast_to(x, (population, 2)).copy()
        costs = np.zeros(population)
        for t in range(horizon):
            imagined = predict(imagined, candidates[:, t])
            costs += (imagined[:, 0] ** 2 + 0.15 * imagined[:, 1] ** 2
                      + 0.02 * candidates[:, t] ** 2)
        costs += 20 * imagined[:, 0] ** 2
        elite = candidates[np.argsort(costs)[:32]]
        mean = elite.mean(axis=0)
        std = elite.std(axis=0) + 0.05
    return mean

real = np.array([1.5, 0.0])
mean = np.zeros(20)
for _ in range(40):
    sequence = plan(real, mean)
    real = A @ real + B * sequence[0]  # real system; never used inside plan()
    mean = np.r_[sequence[1:], 0.0]
print("Controlled final state:", real)
assert rmse(one_step, test[:, 1:]) < 0.01
assert abs(real[0]) < 0.1
```

Output:

```
One-step: 0.0019866110218121832
Without actions: 0.04106321931941812
Final-horizon rollout: 0.013772759824274517
Controlled final state: [-0.00453985 -0.0210205 ]
```

The one-step RMSE (0.0020) is at the noise level; dropping the action input raises it twentyfold (0.0411), so the action genuinely matters. The open-loop rollout error at horizon 40 (0.0138) is about seven times the one-step error even for this nearly perfect linear model, a mild case of the compounding derived above. MPC drives the position from 1.5 to about -0.0045, compared with 1.5 under zero action. These are synthetic results from this script, not published comparisons.

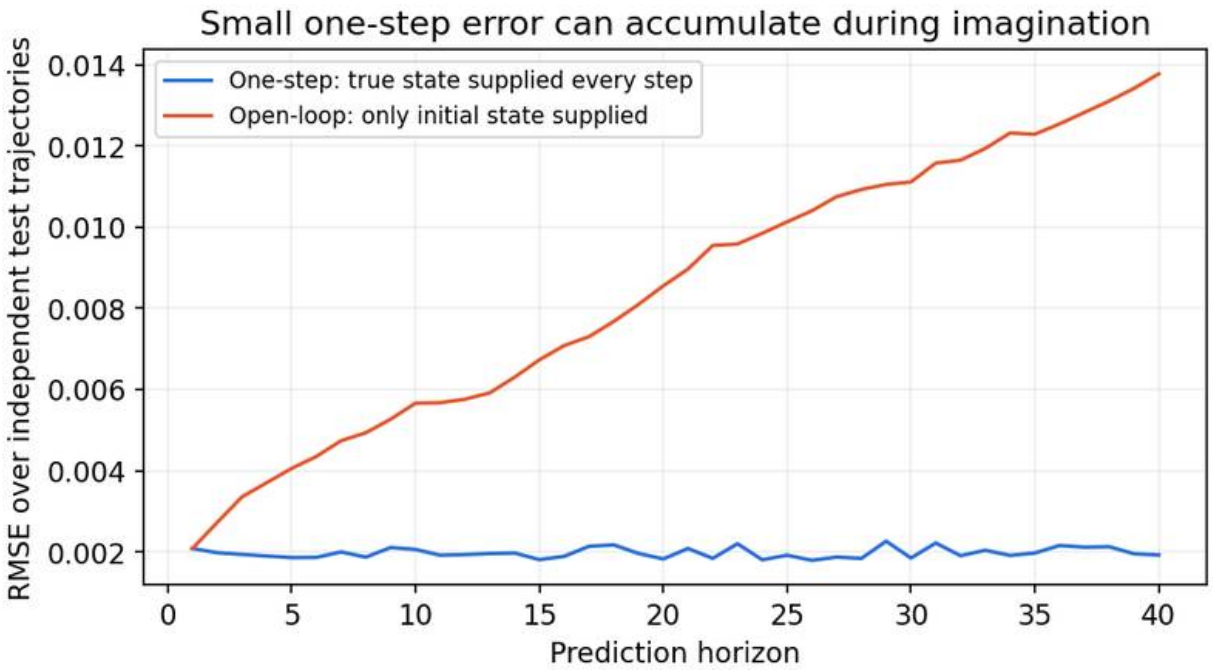


Figure 4 — Independent synthetic test episodes. One-step evaluation receives the true state each time; open-loop evaluation receives only the initial state and the held-out actions.

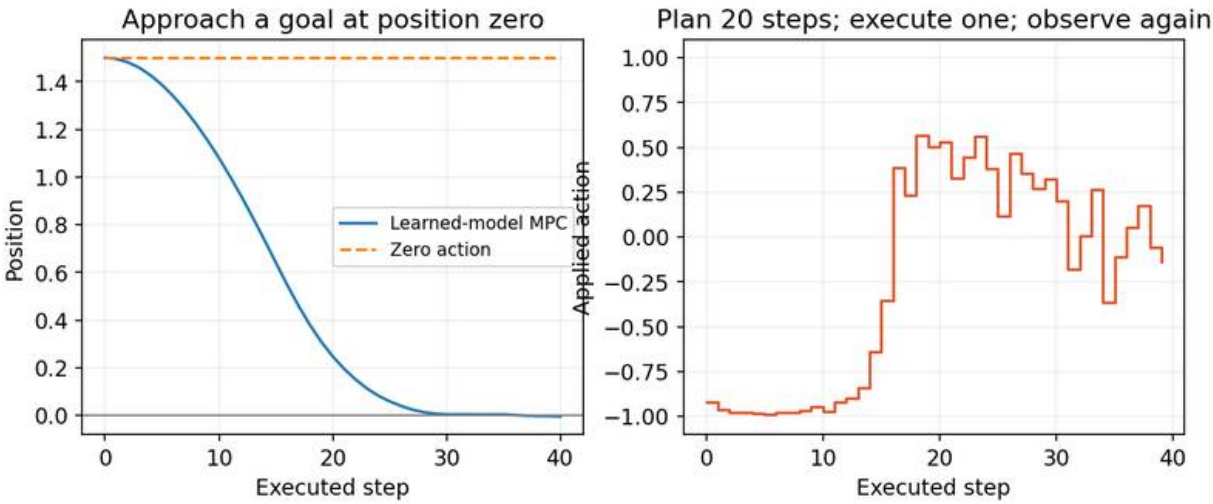


Figure 5 — CEM plans with the learned transition model. Only its first action is executed in the synthetic environment before replanning from a new observation.

Replacing x_t with an image embedding does not reproduce Dreamer or V-JEPA 2-AC: it adds an encoder-learning problem, latent uncertainty and the need for a valid goal or reward.

Planning, uncertainty and model exploitation

Plain version. A planner looks for the action the model likes best. If the model is wrong somewhere, the planner will happily go there. Knowing *where* the model is unsure is what keeps it honest.

A finite-horizon planning objective can be written as

$$a^{*}_{t:t+H-1} = \arg\min_{a_{t:t+H-1}} \mathbb{E}_{\theta}\Big[\sum_{k=1}^{H} c(z_{t+k}, a_{t+k-1}) + \lambda_{\mathrm{T}}\, c_{\mathrm{T}}(z_{t+H})\Big],$$

where c is a per-step cost, c_{T} a terminal cost weighted by λ_{T}, and the expectation is over the model's predicted latent states z. **Model-predictive control** executes the first action, observes again and replans. Feedback corrects drift, but cannot rescue an action whose immediate consequence was badly modeled.

Two kinds of uncertainty

Aleatoric uncertainty is inherent randomness (a coin flip, the ball that may go left or right); more data does not remove it. **Epistemic** uncertainty comes from lack of data (the model has never seen this situation); more data does remove it. Sampling a generative model can represent aleatoric uncertainty, but sample diversity is not proof of calibration. Epistemic uncertainty is commonly approximated with **ensembles**: several models trained on bootstrap resamples or with different initializations agree where data is dense and tend to disagree where it is not. Chua et al.'s PETS combined probabilistic networks (for aleatoric noise) in an ensemble (for epistemic uncertainty) with sampling-based uncertainty propagation and CEM planning, matching the final performance of strong model-free methods on several benchmarks with far fewer samples [29].

Code: a planner exploiting a learned model

The agent has logged data only for actions in $[-1, 1]$, but its planner may choose any action in $[-3, 3]$.

```
import numpy as np
from sklearn.neural_network import MLPRegressor

rng = np.random.default_rng(0)
true_reward = lambda a: np.sin(3 * a) * np.exp(-0.3 * a ** 2)   # unknown to the agent

# Logged interactions only cover actions in [-1, 1].
a_data = rng.uniform(-1, 1, 200)
r_data = true_reward(a_data) + 0.05 * rng.normal(size=200)
```

```
ensemble = []
for k in range(5):                        # bootstrap + different initialization
    idx = rng.integers(0, 200, 200)
    m = MLPRegressor(hidden_layer_sizes=(64, 64), max_iter=3000, random_state=k)
    ensemble.append(m.fit(a_data[idx, None], r_data[idx]))

grid = np.linspace(-3, 3, 601)
preds = np.stack([m.predict(grid[:, None]) for m in ensemble])
mean, std = preds.mean(0), preds.std(0)

for name, score in [("single model", preds[0]),
                    ("ensemble mean", mean),
                    ("pessimistic: mean - 2 std", mean - 2 * std)]:
    a_star = grid[np.argmax(score)]       # the "planner": pick the best-looking action
    print(f"{name:26s} picks a = {a_star:+.2f}, predicted {score.max():.2f}, "
          f"true {true_reward(a_star):.2f}")
print(f"mean std inside data {std[np.abs(grid) <= 1].mean():.3f}, "
      f"outside {std[np.abs(grid) > 1.5].mean():.3f}")
```

Output:

```
single model               picks a = -3.00, predicted 1.79, true -0.03
ensemble mean              picks a = -3.00, predicted 1.58, true -0.03
pessimistic: mean - 2 std  picks a = +0.43, predicted 0.87, true 0.91
mean std inside data 0.025, outside 0.321
```

Both the single model and the ensemble mean extrapolate a rising line outside the data, and the planner runs to the edge of the action space, where the true reward is about zero. Ensemble disagreement is thirteen times larger outside the data than inside, and a *pessimistic* score (mean minus two standard deviations) keeps the planner inside the data, where it finds an action with true reward 0.91. Restricting actions, shortening horizons, penalizing disagreement and testing with real feedback are the practical responses. Agreement among ensemble members is not proof of correctness, though: all five can share the same bias.

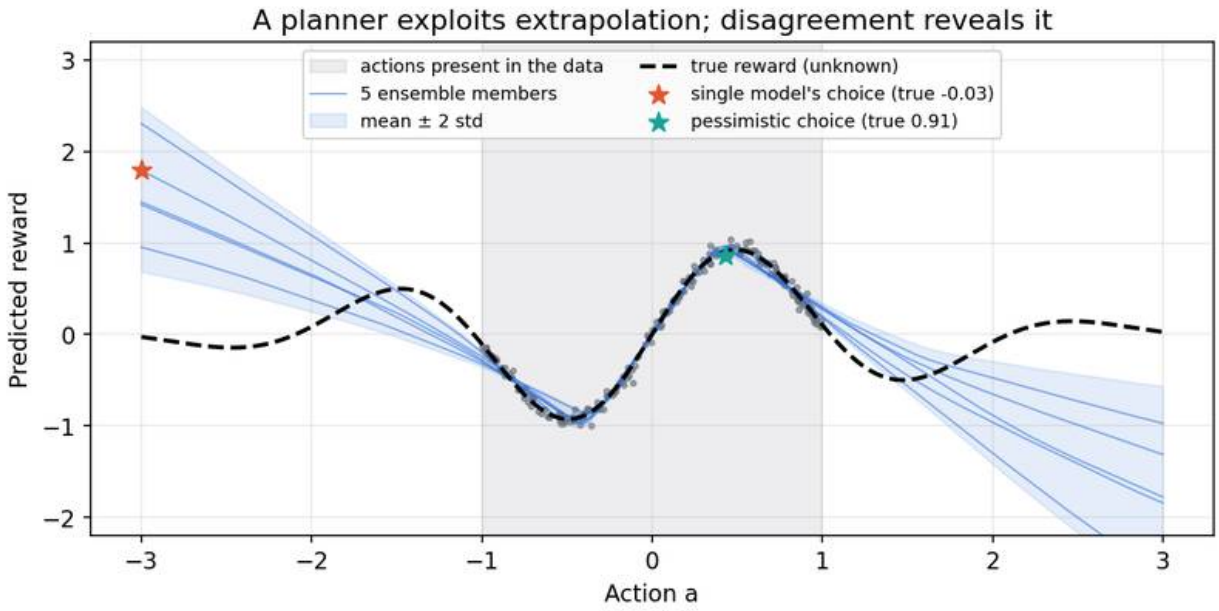


Figure 6 — Output of the code above. Optimizing a learned model drives the planner to where the model is least reliable; ensemble spread exposes those regions.

How to evaluate a world model

Plain version. No single number says whether a world model is good. Ask several questions, each with its own measurement and its own control.

Question	Measurement	Important control
Does it predict known dynamics?	one-step and horizon-indexed state or feature error	independent episodes; fixed action sequence
Does appearance stay coherent?	reconstruction and video-quality metrics, identity consistency	compare camera motion and occlusions
Do actions matter correctly?	matched initial context with different specified actions	real action-conditioned reference trajectories
Are physical constraints respected?	collisions, object permanence, conservation; benchmarks such as Physics-IQ [22]	task-specific tolerances; out-of-distribution cases [23]
Does it help decisions?	success rate, return, interventions, control latency	test in the real or a trusted environment

Pixel fidelity and decision usefulness can disagree, as the blur example showed. An averaged prediction can have low squared error while showing an impossible scene; an accurate latent predictor can omit visually irrelevant texture. Report the target, horizon, conditions and output resolution with every metric.

For comparison, include a persistence baseline (predict "nothing changes"), an action-free predictor and a simple task-appropriate dynamics model. Evaluate held-out objects, viewpoints, action ranges and environment properties separately. Report closed-loop control results in addition to imagined ones, and account for encoder and sampler latency: a planner that needs ten seconds per action may be useless for a robot running at 10 Hz.

Modern view

Plain version. Three lines of work now meet under the name "world model": model-based reinforcement learning, video generators reframed as simulators, and predictive representations. The open questions are mostly about whether their predictions are reliable enough to act on.

Surveys. Moerland et al. give the most systematic treatment of model-based RL: dynamics-model learning (stochasticity, uncertainty, partial observability, temporal abstraction), a taxonomy of how planning and learning are integrated, and implicit (value-equivalent, MuZero-style) approaches [5]. Ding et al. (ACM Computing Surveys) organize the broader "world model" literature by function, understanding the world versus predicting its future, and review applications in generative games, autonomous driving, robotics and social simulation [1]. Zhu et al. survey "general world models" from the video-generation side, covering video generators, autonomous-driving world models and world models inside autonomous agents, and discuss their limitations [31]. Guan et al. give an initial survey of world models specifically for autonomous driving [32].

State of the art, 2023–2026.

- *Model-based RL matured.* DreamerV3 showed a single configuration working across more than 150 tasks and was published in Nature in 2025 [10]; TD-MPC2 showed decoder-free latent planning scaling to a multi-task 317M-parameter agent [11]; IRIS and DIAMOND showed Transformer and diffusion world models competitive on the Atari 100k benchmark [12], [13].
- *Video generators were reframed as simulators.* GAIA-1, UniSim, Genie, Genie 3 and Cosmos trained large models on video with actions, text or inferred latent actions [17], [18], [19], [20], [21]. The strongest evidence that they help agents is UniSim's zero-shot transfer of policies trained inside the simulator [18]; the strongest caution is that realism does not imply physics [22], [23].
- *Predictive representations moved toward control.* The JEPA line went from images (I-JEPA) to video (V-JEPA) to action-conditioned planning on real robots (V-JEPA 2-AC) and denser features (V-JEPA 2.1) [24], [25], [26], [27]; DINO-WM showed that frozen self-supervised features can already support zero-shot planning [28].

Open problems.

- *Long-horizon consistency.* Errors compound; generated worlds drift; object identity and off-screen state are hard to keep for minutes, let alone hours [20].
- *Causal and physical correctness.* Models trained on passive video learn correlations; out-of-distribution physics fails [22], [23]. Interventional (action) data is scarce and expensive.

- *Action interfaces.* Latent actions inferred from video, robot joint commands and text instructions are different things; mapping between them reliably is unsolved.
- *Exploitation and uncertainty.* Planners and policies trained in imagination exploit model errors; calibrated epistemic uncertainty for large video models is largely missing.
- *Evaluation.* There is no agreed benchmark that ties prediction quality to downstream decision quality across domains; most papers still report either video metrics or task returns, rarely both.
- *Cost.* Sampling a large video model can be far slower than real time, which conflicts with closed-loop control.

Key takeaways

- A world model predicts how an environment changes, usually under the agent's actions; its prediction target and action interface determine what it can support.
- Observations are not states. Partial observability requires memory or belief estimation.
- A model can be used to plan online (MPC), to train a policy in imagination (World Models, Dreamer), or to search with a value-equivalent latent (MuZero).
- RSSMs combine deterministic memory with stochastic latents; the KL term trains the prior to imagine what the posterior will see.
- Squared-error video prediction averages possible futures into blur; stochastic and generative models trade lower likelihood of blur for realistic samples, but realism is not physical understanding.
- JEPA-style models predict features instead of pixels; they need anti-collapse design and meaningful goals in feature space.
- One-step errors compound over long horizons, and optimizers exploit model mistakes; ensembles, pessimism, short rollouts and replanning are the standard defenses.
- The strongest evidence connects controlled action interventions and held-out predictions to successful behavior in the real environment.

Exercises

1. **Action ablation.** In the CPU experiment, remove the action column. Compare one-step error, horizon-40 error and control behavior. Why can a short-term action-free predictor still look reasonable?

   Hint

   Imagined actions no longer affect the action-free predictor, so a planner cannot rank their consequences at all: every candidate gets the same predicted cost. Over one step the damped dynamics change the state only slightly, so persistence-like predictions still score moderately well, which is why one-step error alone is a weak test.

2. **Partial observations.** Hide velocity and fit position-to-position dynamics. Then use the last two positions and the previous action as features. When does this history help, and where does observation noise hurt?

Hint

The difference of two positions estimates velocity (a finite difference), so the two-position model recovers most of the state. Noise in each position is amplified when divided by the time step, so the benefit shrinks as observation noise grows.

3. **Planning horizon.** Compare horizons 5, 20 and 40, adjusting the initial mean to the chosen length. Measure final distance, action effort and runtime across several independent seeds.

Hint

Keep the cost definition fixed and count all model calls. Very short horizons cannot "see" that braking early avoids overshoot; very long horizons cost more and accumulate more model error, although for this linear system the error is small.

4. **Domain shift.** Collect training episodes with velocity damping 0.94, then execute control in an environment with damping 0.75. Compare open-loop execution of a plan with replanning at every step. Identify an action range where the training data provide weak coverage.

Hint

Separate the learned dynamics (fitted on 0.94) from the true execution dynamics (0.75); do not refit on evaluation transitions. Replanning uses fresh observations, so it partially corrects the biased model; open-loop execution cannot.

5. **Mean versus sample.** Extend the blur example to three equally likely futures at positions 10, 32 and 54. Compute the expected MSE of the mean prediction and of a single sharp sample. Then propose a metric that would prefer the sample.

Hint

The mean now has three one-third-height bumps; the expected MSEs are about 0.037 (mean) and 0.074 (sample), again a factor of two. That factor is general: if X and Y are independent draws from the same distribution, $\mathbb{E}\|X-Y\|^2 = 2\,\mathbb{E}\|X-\mathbb{E}X\|^2$, so a perfect sampler always has twice the MSE of the mean. A distribution-level metric (for example, comparing sets of generated samples with sets of real futures, as FID-style metrics do for images) or a best-of-N error over several samples would reward sharp, plausible predictions.

6. **Pessimism trade-off.** In the ensemble code, replace the factor 2 in "mean − 2 std" by 0, 0.5, 1 and 4. Plot the chosen action and its true reward for each. What goes wrong when the factor is too large?

Hint

At 0 the planner exploits extrapolation; as the factor grows it stays inside the data. Too much pessimism makes the planner avoid any action with even slight disagreement, so it may miss the best action inside the data and, in an RL setting, stop exploring.

Supplementary notes

These notes are optional background that the main text does not depend on.

The ELBO and the reparameterization trick: training a latent state

Plain version. The camera never shows the hidden state z, so we cannot train it directly. The ELBO is a score for a *guess* about z: the guess earns points when it explains the image well and loses points when it strays from what the model could have predicted before looking. The reparameterization trick lets gradients flow through a random guess.

Precise version. A latent-variable model has a prior $p(z)$ and a likelihood $p(o \mid z)$. The log-evidence $\log p(o) = \log \int p(z)\, p(o \mid z)\, dz$ is usually intractable. For any distribution $q(z)$ over the latent,

$$\log p(o) = \underbrace{\mathbb{E}_q[\log p(o \mid z)] - D_{\mathrm{KL}}(q(z) \,\|\, p(z))}_{\mathrm{ELBO}(q)} + D_{\mathrm{KL}}(q(z) \,\|\, p(z \mid o)),$$

where $p(z \mid o)$ is the true posterior and $D_{\mathrm{KL}} \geq 0$. So the ELBO (evidence lower bound) never exceeds $\log p(o)$, and the gap is exactly how far q is from the true posterior. Maximizing the ELBO over q closes the gap; maximizing it over the model parameters raises the evidence. The per-step loss $\mathcal{L}_t$ in the RSSM section has this shape: reconstruction terms, plus a KL between the posterior $q_\theta(z_t \mid h_t, o_t)$ and a prior $p_\theta(z_t \mid h_t)$ that is itself predicted by the model; $\beta = 1$ gives the plain bound, other values reweight it.

To learn q by gradient descent we must differentiate $\mathbb{E}_q[\cdot]$ with respect to the parameters of q. If $q = \mathcal{N}(\mu, s^2)$, write

$$z = \mu + s\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, 1),$$

where μ and s are the mean and standard deviation of q and ε is noise that does not depend on them. Then $\mathbb{E}_q[f(z)] = \mathbb{E}_\varepsilon[f(\mu + s\varepsilon)]$, and the gradient moves inside the expectation. This is the reparameterization trick of Kingma and Welling [33]; Rezende et al. [34] developed the same idea under the name stochastic backpropagation. It needs a continuous z.

Worked example. Take $p(z) = \mathcal{N}(0, 1)$, $p(o \mid z) = \mathcal{N}(o; z, \sigma^2)$ with $\sigma^2 = 0.5$, and one observation $o = 2$. Everything is Gaussian, so the true posterior is $\mathcal{N}(4/3,\ 1/3)$ (precisions add; this is a hand calculation) and the evidence is $\mathcal{N}(o; 0, 1 + \sigma^2)$ in closed form.

```
import numpy as np

o, var = 2.0, 0.5                       # observation and likelihood variance sigma^2

def elbo(mu, s):                        # closed form for q = N(mu, s^2), prior N(0, 1)
    recon = -0.5 * np.log(2 * np.pi * var) - ((o - mu) ** 2 + s ** 2) / (2 * var)
    kl = 0.5 * (mu ** 2 + s ** 2 - 1 - np.log(s ** 2))
    return recon - kl

evidence = -0.5 * np.log(2 * np.pi * (1 + var)) - o ** 2 / (2 * (1 + var))  # log N(o; 0,
 ↪  1+var)
mu_star, s_star = (o / var) / (1 + 1 / var), np.sqrt(1 / (1 + 1 / var))     # exact posterior
print(f"log evidence log p(o):        {evidence:.4f}")
print(f"ELBO at the exact posterior:  {elbo(mu_star, s_star):.4f}")
print(f"ELBO at q = N(0, 1):          {elbo(0.0, 1.0):.4f}")
gap = np.log(s_star / 1.0) + (1.0 + (0.0 - mu_star) ** 2) / (2 * s_star ** 2) - 0.5  # KL(q
 ↪  || posterior)
print(f"evidence minus that ELBO:     {evidence - elbo(0.0, 1.0):.4f}  (= KL(q || posterior)
 ↪  = {gap:.4f})")

# Gradient of the ELBO with respect to mu, estimated from samples of z ~ q.
rng = np.random.default_rng(0)
mu, s, n = 0.0, 1.0, 100_000
true_grad = (o - mu) / var - mu                       # analytic value
eps = rng.standard_normal(n)
z = mu + s * eps                                      # reparameterization: z = mu + s * eps
g_rep = (o - z) / var - mu                            # pathwise gradient (KL term analytic)
logp = -0.5 * np.log(2 * np.pi * var) - (o - z) ** 2 / (2 * var)
g_score = logp * (z - mu) / s ** 2 - mu               # score-function (REINFORCE) estimator
print(f"true gradient {true_grad:.3f}")
print(f"reparameterized: mean {g_rep.mean():.3f}, per-sample std {g_rep.std():.3f}")
print(f"score function:  mean {g_score.mean():.3f}, per-sample std {g_score.std():.3f}")
```

Output:

```
log evidence log p(o):        -2.4550
ELBO at the exact posterior:  -2.4550
ELBO at q = N(0, 1):          -5.5724
evidence minus that ELBO:     3.1174  (= KL(q || posterior) = 3.1174)
true gradient 4.000
reparameterized: mean 4.002, per-sample std 2.000
score function:  mean 4.013, per-sample std 9.840
```

At the exact posterior the ELBO equals the log-evidence (-2.4550). A poor guess, $q = \mathcal{N}(0, 1)$, scores -5.5724, and the shortfall 3.1174 is exactly its KL divergence from the posterior. Both gradient estimators are centered on the true value 4.000 with 10^5 samples, but a single reparameterized sample is far less noisy (standard deviation 2.000 against 9.840 for the score-function estimator), which is why it is the default for continuous latents.

Common mistakes. Reading the ELBO as the likelihood (it is a lower bound, and a loose one if q is poor); forgetting that the trick needs the noise ε to be independent of the parameters; and applying it to a discrete latent, where $\mu + s\varepsilon$ has no meaning and a different gradient estimator is required.

More of the JEPA line: I-JEPA, V-JEPA and V-JEPA 2.1

The JEPA section keeps the two models that act as world models; three more members of the family show how the idea of predicting features instead of pixels developed on images and video.

- **I-JEPA** predicts the representations of several large target blocks of an image from one informative context block, and found the masking strategy (large, semantic targets; spatially distributed context) crucial for learning semantic features without hand-crafted augmentations [24].
- **V-JEPA** applies feature prediction as a stand-alone objective to video, trained on 2 million videos without pretrained image encoders, text, negatives or reconstruction, and produced frozen features that work for both motion- and appearance-based tasks [25].
- **V-JEPA 2.1** adds a dense predictive loss in which both visible and masked tokens contribute, deep self-supervision across intermediate layers, and multi-modal tokenizers for images and videos, aiming at spatially structured, temporally consistent dense features [27].

The cross-entropy method: planning by sampling and refitting

Plain version. Throw 64 darts around a rough guess, see which 8 land closest to the bullseye, move your aim to the middle of those 8, tighten your spread, and throw again. After a few rounds the darts cluster near a good spot, and you never needed to know the slope of the board.

Precise version. The planner minimizes a cost $J(a)$ over an action sequence $a = (a_0, \dots, a_{H-1})$, where J is computed by rolling the model forward from the current state. It keeps a Gaussian $\mathcal{N}(\mu, \operatorname{diag} \sigma^2)$ over sequences and repeats four steps:

1. sample N sequences $a^{(1)}, \dots, a^{(N)}$ from the Gaussian (and clip them to the allowed action range);
2. evaluate $J(a^{(i)})$ for each by a model rollout;
3. keep the K cheapest, the **elites**, forming the set $\mathcal{E}$;
4. refit: $\mu \leftarrow \frac{1}{K} \sum_{a \in \mathcal{E}} a$ and $\sigma \leftarrow \operatorname{std}_{\mathcal{E}}(a)$ (computed per time step).

Here H is the horizon, N the number of samples per round and $K < N$ the number of elites. Step 4 is the maximum-likelihood fit of the Gaussian to the elites, which is the choice that minimizes the cross-entropy between the elite samples and the Gaussian family; this is where the method gets its name. Rubinstein presented it as an optimizer for combinatorial and continuous problems built on Kullback–Leibler cross-entropy and importance sampling [35]; de Boer et al. give a tutorial

that also covers rare-event simulation [36]. PETS used CEM-based planning inside MPC [29]. The cost is RNH model steps per plan for R rounds, with no gradients required, so J may be non-differentiable and all N rollouts run in parallel. The chapter's CPU experiment adds a floor of 0.05 to σ to stop the spread from collapsing too early (the snippet below does the same), and in MPC the mean is shifted by one step and reused as the starting guess at the next time step.

```
import numpy as np

rng = np.random.default_rng(0)
A = np.array([[1.0, 0.1], [0.0, 0.94]])       # known model: position, velocity
B = np.array([0.005, 0.1])
H, N, K = 15, 64, 8                           # horizon, samples per round, elites

def cost(x0, U):                              # U: (N, H) action sequences -> (N,) costs
    x = np.broadcast_to(x0, (len(U), 2)).copy()
    c = np.zeros(len(U))
    for t in range(H):
        x = x @ A.T + U[:, t, None] * B
        c += x[:, 0] ** 2 + 0.15 * x[:, 1] ** 2 + 0.02 * U[:, t] ** 2
    return c + 20 * x[:, 0] ** 2

x0 = np.array([1.5, 0.0])
mean, std = np.zeros(H), np.ones(H)
for i in range(5):
    U = np.clip(rng.normal(mean, std, (N, H)), -1, 1)
    c = cost(x0, U)
    elite = U[np.argsort(c)[:K]]
    mean, std = elite.mean(axis=0), elite.std(axis=0) + 0.05   # refit to the elites
    print(f"round {i}: best {c.min():.3f}, mean of elites {np.sort(c)[:K].mean():.3f}, "
          f"mean std {std.mean():.3f}")

U = np.clip(rng.normal(0.0, 1.0, (5 * N, H)), -1, 1)           # same budget, no refitting
print(f"random shooting, {5 * N} samples: best {cost(x0, U).min():.3f}")
print(f"zero actions: {cost(x0, np.zeros((1, H)))[0]:.3f}")
print(f"cost of the final mean sequence: {cost(x0, mean[None])[0]:.3f}")
```

Output:

```
round 0: best 47.314, mean of elites 53.939, mean std 0.570
round 1: best 42.572, mean of elites 45.558, mean std 0.408
round 2: best 37.952, mean of elites 39.610, mean std 0.325
round 3: best 35.143, mean of elites 37.146, mean std 0.231
round 4: best 34.089, mean of elites 35.691, mean std 0.210
random shooting, 320 samples: best 46.672
zero actions: 78.750
cost of the final mean sequence: 35.669
```

The first round is plain random shooting. Each later round samples from a distribution that has moved toward the cheap region and narrowed (mean standard deviation 0.570 → 0.210), so the

best cost falls from 47.314 to 34.089. Random shooting with the same 320 samples reaches only 46.672, and doing nothing costs 78.750. These are numbers from this toy, not a benchmark.

When it fails. A single Gaussian covers one basin, so it can commit to the wrong one when the cost has several; too few samples or too small a spread stops it exploring; and the search space grows with the horizon H. Most importantly, CEM optimizes the *model's* cost with great energy, so it finds model errors as readily as good actions (see the section on uncertainty and exploitation).

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DLCV 08 • 3D Reconstruction: From Multi-View Geometry to NeRF and Gaussian Splatting

How a set of photos becomes a 3D model: camera geometry, structure from motion and multi-view stereo, neural radiance fields and signed distance fields, 3D Gaussian Splatting, mesh extraction, feed-forward reconstruction models, and how reconstructions are evaluated, with tested NumPy and OpenCV code.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-08-3d-reconstruction/

What you'll learn

- What 3D reconstruction produces, and the common ways to store a 3D scene (points, depth maps, voxels, meshes, signed distance functions, radiance fields, 3D Gaussians) with their trade-offs.
- The camera geometry behind every method: the pinhole model, epipolar geometry, the essential and fundamental matrices and triangulation, with a tested OpenCV example.
- The classical pipeline: structure from motion (matching, RANSAC, bundle adjustment, COLMAP) and multi-view stereo (plane sweep, depth-map fusion, MVSNet).
- How neural fields reconstruct a scene by rendering it: NeRF's volume rendering, hash-grid speed-ups (Instant-NGP) and SDF surfaces (NeuS).
- How 3D Gaussian Splatting renders and grows a scene in real time, how meshes are extracted (marching cubes, Poisson, SuGaR, 2DGS), and what feed-forward models (DUSt3R, MASt3R, VGGT) change.
- How to evaluate views and geometry (PSNR/SSIM/LPIPS, Chamfer distance, F-score; DTU, Tanks and Temples, Mip-NeRF 360), and how reconstruction fails.

The big picture

You walk around a statue and take thirty photos with your phone. 3D reconstruction turns those photos into a 3D model you can spin, measure, 3D-print or place in a game. Two questions must be answered together: *where was the camera for each photo*, and *where is each piece of the statue's surface*. Neither is given, and each helps answer the other.

> **Explain like I'm five**
>
> Close one eye and hold up a finger: it jumps when you switch eyes. Things that are close jump a lot and things far away hardly move. Your brain uses this jump to feel depth. A computer does the same with photos taken from different places: it finds the same spot in two photos, measures how much it "jumped", and works out how far away it is. Do that for millions of spots and you have the shape. The

newer methods add one more trick: they keep guessing the 3D scene, *draw* what each photo should look like, and fix the guess wherever the drawing does not match the real photo.

Why it matters. Reconstruction is how maps, film sets, cultural-heritage archives, robot environments, augmented-reality anchors and digital twins of factories are built. It is also a clean example of how deep learning changed computer vision: the classical pipeline is pure geometry and optimization; neural radiance fields (NeRF) [5] made the scene itself a trained network; 3D Gaussian Splatting (3DGS) [6] replaced the network with millions of small, explicit blobs that render in real time; and the newest models predict cameras and geometry in a single forward pass.

Figure 1 shows the roadmap of this tutorial. The classical stages on the left still feed almost every modern method: NeRF and 3DGS are trained on camera poses computed by structure-from-motion software such as COLMAP [2], [4].

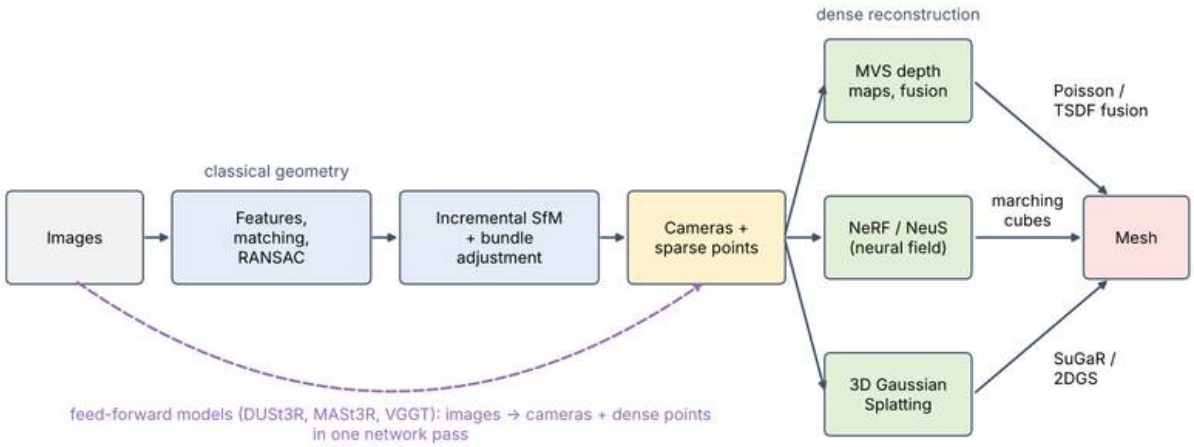


Figure 1 — The reconstruction pipeline covered in this tutorial. Classical geometry (left) gives cameras and sparse points; dense methods (right) give a full model; feed-forward networks (dashed) shortcut the classical stages.

This tutorial assumes you know keypoints and descriptor matching from IP 12 and the basics of deep networks from DLCV 01. The standard textbook for the geometry is Hartley and Zisserman [1].

Related chapters. DLCV 08A treats the dense correspondence problems behind reconstruction — stereo matching, monocular depth and optical flow — in depth, and DLCV 09 uses 3D geometry to detect objects.

What 3D reconstruction is

Plain version. The input is a set of photos of a scene that does not move. The output is the camera position and orientation for every photo, plus a description of the scene's shape (and usually its colour) in 3D.

Precise version. We are given N images $I_1, \dots, I_N$ of a static scene. We want camera parameters $(K_i, R_i, \mathbf{t}_i)$ for each image (defined in the next section) and a scene model $\mathcal{S}$ such that rendering $\mathcal{S}$ with camera i reproduces I_i. Two variants matter. **Geometric reconstruction** asks for an accurate surface; it is judged against laser or structured-light scans. **Novel-view synthesis** asks for realistic images from camera positions that were *not* photographed; it is judged against held-out photos. NeRF and 3DGS were designed for the second goal, which is why extracting a good surface from them took extra work.

The problem has a built-in limitation: from photos alone, the whole scene can be scaled up or down without changing any image. A toy house photographed from close by and a real house photographed from far away can produce the same pictures. Reconstruction from images is therefore defined only **up to scale** (and up to the choice of the world coordinate frame) unless something else, such as a known distance or a calibrated stereo rig, fixes it [1].

Seven ways to store a 3D scene

Plain version. There is no single "3D format". Each representation makes some operations easy (rendering, measuring, editing, learning) and others hard. Figure 2 shows the same torus in each of the seven common ones, plus an eighth form, the pointmap, that we meet with the feed-forward models later.

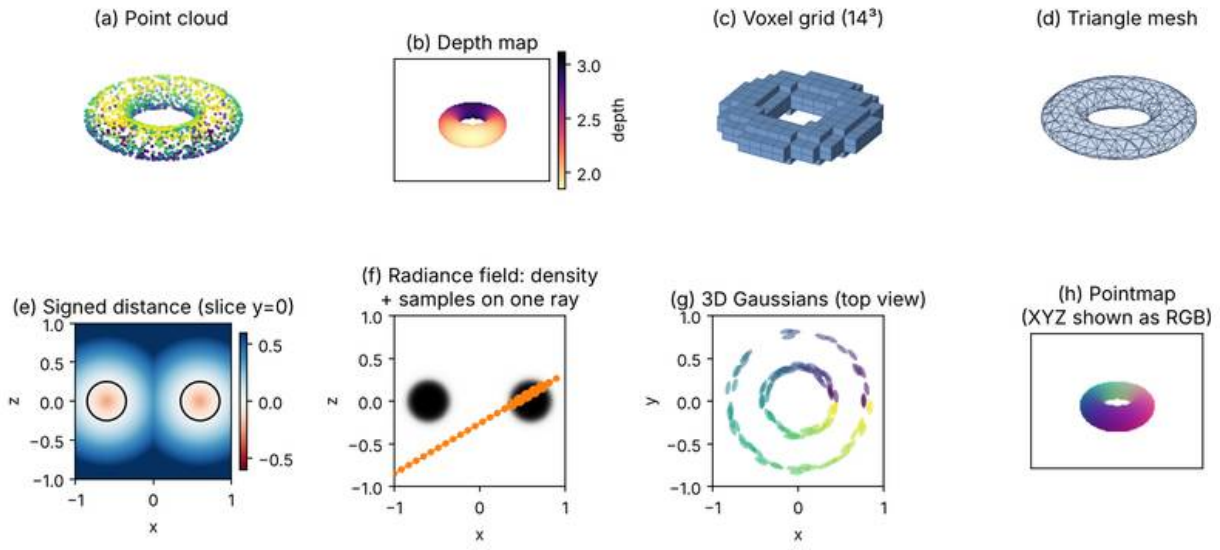


Figure 2 — One shape, eight encodings: (a) point cloud, (b) depth map, (c) voxel grid, (d) triangle mesh, (e) signed distance function, (f) radiance field (density slice and samples along a ray), (g) 3D Gaussians, (h) a pointmap, the per-pixel 3D-point image predicted by feed-forward models.

Representation	What is stored	Strengths	Weaknesses
Point cloud	A list of 3D points, often with colour and normal	Simple; direct output of triangulation and depth sensors	No surface or connectivity; holes; hard to render nicely
Depth map	One distance per pixel of one camera	Fits image networks; easy to estimate per view	Covers one viewpoint only; must be fused
Voxel grid	A value (occupancy, colour) per cell of a 3D grid	Regular, easy to process with 3D convolutions	Memory grows with the cube of resolution
Mesh	Vertices and triangles	Compact, editable, what graphics engines and printers use	Fixed topology; hard to optimize directly from images
Signed distance function (SDF)	Distance to the nearest surface, negative inside	Watertight surface as the zero level set; normals from the gradient	Needs a closed surface; must be converted for rendering
Radiance field	Density and view-dependent colour at every 3D point	Photorealistic novel views, including thin and fuzzy structures	Slow to render with ray marching; surface is implicit
3D Gaussians	Many small ellipsoids with colour and opacity	Real-time rendering; explicit and editable	Many primitives (memory); surfaces are not well defined

An SDF can be a stored grid or a function computed by a network: DeepSDF [20] learns a network that outputs signed distance, and Occupancy Networks [21] learn the probability that a point is inside the object. Both are called **neural implicit** representations, because the surface is given

implicitly as a level set of a learned function.

Camera geometry essentials

Plain version. Before we can reason about 3D, we need a precise model of how a camera turns a 3D point into a pixel. The pinhole model does this with one matrix multiplication and one division.

The pinhole camera: intrinsics and extrinsics

A camera does two things to a 3D point. It first describes the point relative to itself (how far to the left, up and ahead of the camera it is), then flattens it onto the sensor by dividing by its distance. In symbols, a world point $\mathbf{X} \in \mathbb{R}^3$ is first moved into the camera's coordinate frame, then projected:

$$\mathbf{X}_c = R\,\mathbf{X} + \mathbf{t}, \qquad \lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K\mathbf{X}_c, \qquad K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix},$$

where R (a 3×3 rotation) and $\mathbf{t}$ (a translation) are the **extrinsics**: they say where the camera is and which way it points; $\mathbf{X}_c$ is the point in camera coordinates, with the z axis along the viewing direction; K holds the **intrinsics**: focal lengths f_x, f_y in pixels, the principal point (c_x, c_y) where the optical axis meets the image, and a skew s that is zero for real sensors; (u, v) is the pixel; and λ equals the depth z_c of the point. Dividing by depth is what makes far things look small. The camera centre in world coordinates is $C = -R^\top \mathbf{t}$. Combining both steps gives the 3×4 **projection matrix** $P = K[R \mid \mathbf{t}]$.

Real lenses also bend straight lines (radial distortion); calibration tools estimate extra distortion coefficients and undistort the images first, so the rest of this tutorial assumes an ideal pinhole. The homogeneous coordinates used here, with their trailing 1, are the same as in the homography note of IP 02 (Homography: the projective transform).

Two views: epipolar geometry

Plain version. If you see a point in one photo, you do not know how far away it is, so in 3D it could lie anywhere on a ray. In the second photo, that ray appears as a *line*. The true match must sit on that line. This turns a 2D search into a 1D search and, in reverse, lets us recover how the two cameras are placed from matched points alone.

Figure 3 (a) shows the geometry. The two camera centres and the 3D point $\mathbf{X}$ span a plane, the **epipolar plane**. It cuts each image in an **epipolar line**. Any point $\mathbf{X}'$ on the same ray of camera 1

has the same image **x** in camera 1, and its image in camera 2 moves along that line.

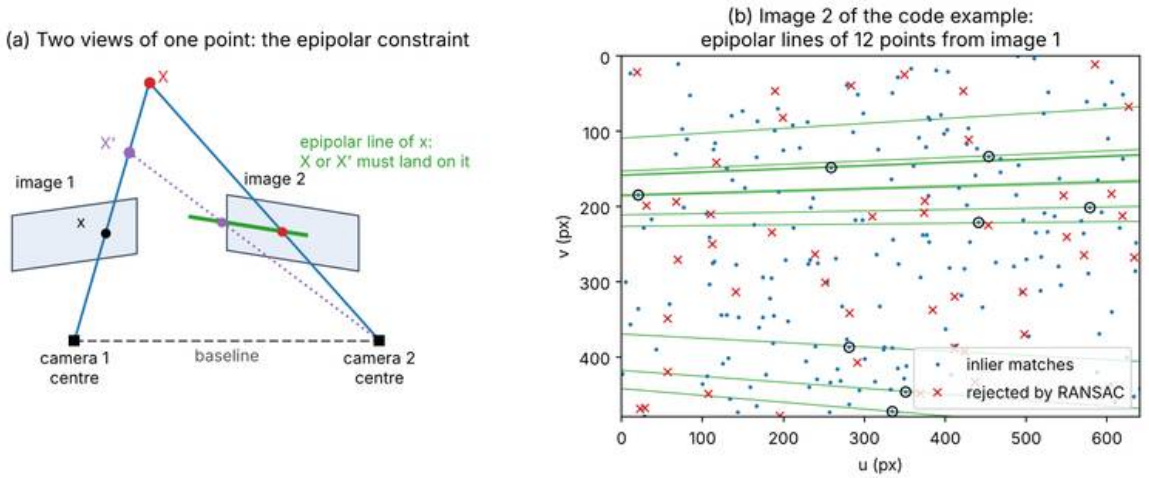


Figure 3 — (a) The epipolar constraint: all points on the ray through x project onto one line in image 2. (b) Image 2 of the code example below: epipolar lines computed from the estimated essential matrix pass through the matching points (circled).

In algebra, let $\hat{\mathbf{x}}_1 = K_1^{-1}(u_1, v_1, 1)^\top$ and $\hat{\mathbf{x}}_2 = K_2^{-1}(u_2, v_2, 1)^\top$ be the *normalized* coordinates of a matched pair (pixels with the intrinsics removed). If camera 2 sits at rotation R and translation $\mathbf{t}$ relative to camera 1, every correct match satisfies

$$\hat{\mathbf{x}}_2^\top E\,\hat{\mathbf{x}}_1 = 0, \qquad E = [\mathbf{t}]_\times R, \qquad [\mathbf{t}]_\times = \begin{bmatrix} 0 & -t_3 & t_2 \\ t_3 & 0 & -t_1 \\ -t_2 & t_1 & 0 \end{bmatrix},$$

where E is the 3×3 **essential matrix** [7] and $[\mathbf{t}]_\times$ is the matrix that computes the cross product with $\mathbf{t}$. The equation says that the two viewing rays and the baseline lie in one plane. In pixel coordinates the same constraint reads $\mathbf{x}_2^\top F \mathbf{x}_1 = 0$ with the **fundamental matrix** $F = K_2^{-\top} E K_1^{-1}$, where $\mathbf{x}_1, \mathbf{x}_2$ are homogeneous pixel coordinates. $F\mathbf{x}_1$ is exactly the epipolar line in image 2 drawn in Figure 3 (b).

In plain words: E **encodes the relative pose of two calibrated cameras, and** F **does the same when the intrinsics are unknown.** E has five degrees of freedom: three for the rotation and two for the *direction* of the translation, whose length is lost to the scale ambiguity. Five matches therefore suffice, the minimal case solved by Nistér's five-point algorithm [9]. With eight or more matches the constraint is linear in the nine entries of E or F: the eight-point algorithm [7], [8] stacks one equation per match and takes the null vector with the singular value decomposition.

Optional: The normalized eight-point algorithm (in the Supplementary notes). Why the pixel coordinates must be rescaled first, the rank-2 step, and a test of both versions.

Once E is known, it can be factored into R and the direction of $\mathbf{t}$. The factorization gives four

candidate poses; only one places the points in front of both cameras, which is called the **cheirality** (or chirality) check [1].

Triangulation

Plain version. With both cameras known, a matched pair gives two rays. Where they (almost) cross is the 3D point. Noise means they never meet exactly, so we pick the point that best explains both observations.

The simplest method is linear. Each image gives two linear equations in the unknown point (the pinhole equation with the unknown depth eliminated), and the four equations are solved, like the eight-point system, as the null vector of a small matrix with the SVD [1]. The quality of a triangulated point is judged by its **reprojection error**: project it back into each image and measure the pixel distance to the observed point.

Optional: Linear triangulation (DLT) (in the Supplementary notes). The four equations written out, why the linear answer is not the optimal one, and a test of how the baseline controls the error.

Accuracy depends strongly on the angle between the two rays. For two parallel cameras with focal length f and baseline B, a point at depth Z shifts by the **disparity** $d = fB/Z$ pixels between the images, so a small disparity error δd gives a depth error $\delta Z \approx Z^2 \delta d/(fB)$. Depth error grows with the square of distance and shrinks with a wider baseline; Exercise 1 puts numbers on this.

Code: relative pose and triangulation

The snippet builds a synthetic scene, projects it into two cameras with 0.5-pixel noise, replaces 15% of the matches by random wrong ones, and then recovers the relative pose with OpenCV. `cv2.findEssentialMat` runs RANSAC around the five-point solver (its documentation names Nistér's algorithm [9]); `cv2.recoverPose` performs the cheirality check. RANSAC's answer comes from a minimal sample, so we refit E linearly on all inliers, the "refit on the consensus set" step described in the RANSAC note of IP 12 (RANSAC: fitting a model despite outliers).

```
import numpy as np
import cv2

rng = np.random.default_rng(0)
K = np.array([[800.0, 0, 320], [0, 800.0, 240], [0, 0, 1]])   # focal length 800 px,
 ↪  principal point (320, 240)

def look_at(C, target):
    """Rotation of a camera at centre C that looks at target (rows: camera x, y, z axes)."""
    z = (target - C) / np.linalg.norm(target - C)
    x = np.cross([0.0, 1.0, 0.0], z); x /= np.linalg.norm(x)
    return np.stack([x, np.cross(z, x), z])
```

```
def project(K, R, t, X):
    """Pinhole projection of world points X (N, 3) to pixels (N, 2)."""
    x = (X @ R.T + t) @ K.T          # world -> camera -> homogeneous pixels
    return x[:, :2] / x[:, 2:]       # divide by depth

def linear_E(x1, x2, K):
    """Least-squares E from many matches: each match gives one linear equation in E's 9
     ↪  entries."""
    Ki = np.linalg.inv(K)
    a = np.c_[x1, np.ones(len(x1))] @ Ki.T              # normalized camera coordinates
    b = np.c_[x2, np.ones(len(x2))] @ Ki.T
    A = np.einsum('ni,nj->nij', b, a).reshape(-1, 9)     # row n encodes b_n^T E a_n = 0
    E = np.linalg.svd(A)[2][-1].reshape(3, 3)            # best null vector of A
    U, _, Vt = np.linalg.svd(E)
    return U @ np.diag([1.0, 1.0, 0.0]) @ Vt             # nearest valid essential matrix

X = rng.uniform([-3, -2, 4], [3, 2, 8], size=(300, 3))          # 300 scene points
C2 = np.array([2.0, 0.0, 0.0])                                   # camera 2 centre: 2 units
 ↪  to the right
R2 = look_at(C2, np.array([0.0, 0.0, 6.0])); t2 = -R2 @ C2       # camera 1 is the world
 ↪  frame
x1 = project(K, np.eye(3), np.zeros(3), X) + rng.normal(0, 0.5, (300, 2))   # 0.5 px noise
x2 = project(K, R2, t2, X) + rng.normal(0, 0.5, (300, 2))
x2[:45] = rng.uniform([0, 0], [640, 480], (45, 2))               # 45 wrong matches (15 %)

def pose_errors(R, t):
    rot = np.degrees(np.arccos(np.clip((np.trace(R.T @ R2) - 1) / 2, -1, 1)))
    return rot, np.degrees(np.arccos(t.ravel() @ t2 / np.linalg.norm(t2)))

E, mask = cv2.findEssentialMat(x1, x2, K, method=cv2.RANSAC, prob=0.999, threshold=1.5)
_, R, t, mask = cv2.recoverPose(E, x1, x2, K, mask=mask)
inl = mask.ravel() > 0
print(f"inliers kept: {inl.sum()} of 300 | wrong matches rejected: {(~inl[:45]).sum()} of
 ↪  45")
print("RANSAC estimate      : rotation error %.2f deg, translation direction error %.2f deg"
 ↪  % pose_errors(R, t))
_, R, t, _ = cv2.recoverPose(linear_E(x1[inl], x2[inl], K), x1[inl], x2[inl], K)
print("refit on all inliers : rotation error %.2f deg, translation direction error %.2f deg"
 ↪  % pose_errors(R, t))
print(f"length of estimated t: {np.linalg.norm(t):.2f} (true baseline
 ↪  {np.linalg.norm(C2):.2f})")

P1 = K @ np.hstack([np.eye(3), np.zeros((3, 1))])     # P1 = K [I | 0]
P2 = K @ np.hstack([R, t])                            # P2 = K [R | t]
Xh = cv2.triangulatePoints(P1, P2, x1[inl].T, x2[inl].T)
Xe = (Xh[:3] / Xh[3]).T                               # homogeneous -> Euclidean
res = np.r_[np.linalg.norm(project(K, np.eye(3), np.zeros(3), Xe) - x1[inl], axis=1),
            np.linalg.norm(project(K, R, t.ravel(), Xe) - x2[inl], axis=1)]
print(f"reprojection error: RMS {np.sqrt(np.mean(res ** 2)):.2f} px")
rel = np.linalg.norm(2.0 * Xe - X[inl], axis=1) / np.linalg.norm(X[inl], axis=1)   # scale
 ↪  fixed by the true baseline
print(f"3D error after fixing the scale: median {100 * np.median(rel):.2f} % of the
 ↪  distance")
```

```
inliers kept: 250 of 300 | wrong matches rejected: 44 of 45
RANSAC estimate      : rotation error 0.30 deg, translation direction error 0.42 deg
refit on all inliers : rotation error 0.07 deg, translation direction error 0.46 deg
length of estimated t: 1.00 (true baseline 2.00)
reprojection error: RMS 0.40 px
3D error after fixing the scale: median 0.31 % of the distance
```

These numbers are from this synthetic run only. RANSAC rejects 44 of the 45 wrong matches. Refitting on all inliers reduces the rotation error from 0.30° to 0.07°; the translation direction stays within half a degree either way. The estimated translation has length exactly 1: OpenCV returns only its direction, because the images cannot tell the true baseline of 2. Once the scale is supplied from outside, the triangulated points are within about 0.3% of their true distance. The reprojection error of 0.40 px is below the 0.5 px noise level, as expected when the points are fitted to their own observations.

Structure from motion

Plain version. Structure from motion (SfM) does the two-view trick for hundreds of photos at once: it finds matching points across all of them, places the cameras one by one, triangulates the points, and keeps polishing everything so that all 3D points project onto their observed pixels.

Features, matching and geometric verification

First, SfM must know which pixels in different photos show the same physical point. It starts with local features. Each image yields keypoints with descriptors, such as SIFT [11] (IP 12), and descriptors are matched between image pairs (IP 12, Matching). Many matches are wrong, so each pair is **geometrically verified**: RANSAC [10] fits a fundamental matrix (or an essential matrix or a homography) and keeps the pair only if enough matches agree [2]. The verified pairs form a **scene graph**: images are nodes, and an edge means "these two images see the same points".

Incremental reconstruction

COLMAP's paper describes the standard incremental pipeline [2], which grew out of earlier systems for Internet photo collections such as Photo Tourism [12]:

1. **Initialize** with a carefully chosen pair of images that share many matches and have a wide baseline; estimate their relative pose and triangulate the shared points.

2. **Register** a new image by solving the Perspective-n-Point (PnP) problem: find the camera pose from matches between its keypoints and already triangulated 3D points, inside RANSAC.
3. **Triangulate** new points seen by the new image and at least one registered image.
4. **Bundle adjust** (below) to refine all cameras and points together, then **filter** points with large reprojection error.
5. Repeat from step 2 until no more images can be registered.

The COLMAP paper improves each of these steps, including how the next image is chosen and how points are triangulated. Its central lesson is in step 4: registration and triangulation must be interleaved with repeated bundle adjustment, re-triangulation and outlier filtering, because without that refinement small errors accumulate as images are added and the reconstruction drifts [2].

Bundle adjustment

Plain version. Bundle adjustment (BA) is the final polish: move every camera and every 3D point a little until all the "bundles" of rays line up with the pixels where the points were actually seen.

BA solves the nonlinear least-squares problem

$$\min_{\{P_c\},\{\mathbf{X}_k\}} \sum_{(c,k)\in O} \rho\Big(\|\pi(P_c,\mathbf{X}_k)-\mathbf{x}_{ck}\|^2\Big),$$

where P_c are the parameters of camera c (pose, and intrinsics if unknown), $\mathbf{X}_k$ are the 3D points, O is the set of observations (point k seen in image c), $\mathbf{x}_{ck}$ is the observed pixel, π is the pinhole projection, and ρ is a robust loss that down-weights outliers [2], [13]. The standard solver is Levenberg–Marquardt, a damped Gauss–Newton method. It is fast because each residual depends on only one camera and one point, so the Jacobian is almost all zeros [2], [13]. BA cannot fix the global scale and coordinate frame, and it finds only a local minimum, so it needs the good starting point that the incremental steps provide.

Optional: Bundle adjustment in code (in the Supplementary notes). A complete small BA with `scipy.optimize.least_squares`*, its sparsity pattern, and the gauge problem.*

COLMAP in practice

COLMAP is the open-source SfM and MVS pipeline released with [2] and [3], with a graphical and a command-line interface [4]. It outputs the camera intrinsics and poses plus a sparse point cloud, the standard input of the dense methods below: the NeRF paper takes the poses of its real scenes from COLMAP [5], and 3DGS initializes its Gaussians from the SfM points [6]. Every later stage inherits the errors made here (see *Pose errors* under failure modes).

Multi-view stereo

Plain version. SfM gives cameras and a sparse set of points. Multi-view stereo (MVS) fills in the rest: for every pixel of a reference image it searches, along the epipolar line in the other images, for the depth at which all views agree on what the surface looks like.

Photo-consistency and plane sweep

The core test is **photo-consistency**: if a pixel's surface point is at depth Z, then the small patch around it should look the same in every image that sees that point. A depth hypothesis is scored by a matching cost such as the sum of squared differences (SSD) or normalized cross-correlation between patches.

The **plane sweep** algorithm [14] tests many depths efficiently. Sweep a plane, parallel to the reference image, through a range of depths $Z_1, \dots, Z_D$. For each depth, warp every source image onto the reference view as if the whole scene lay on that plane; the warp for one plane is a homography.

Compute the cost per pixel, stack the costs into a **cost volume** of size $D \times H \times W$, and choose for each pixel the depth with the lowest cost:

$$\hat{Z}(u,v) = \arg\min_{k} \sum_{(u',v') \in \mathcal{W}(u,v)} \left(I_{\text{ref}}(u',v') - \tilde{I}^{(k)}_{\text{src}}(u',v') \right)^2,$$

where $\tilde{I}^{(k)}_{\text{src}}$ is the source image warped by the plane at depth Z_k and $\mathcal{W}(u,v)$ is a small window around the pixel. For two parallel cameras (a *rectified* pair), the plane homography reduces to a horizontal shift by the disparity fB/Z_k, which keeps the code short.

Optional: The plane-induced homography (in the Supplementary notes). The warp for general camera poses and tilted planes.

```
import numpy as np
from scipy import ndimage as ndi

rng = np.random.default_rng(0)
H, W, f, B = 120, 160, 150.0, 0.5                    # image size, focal length (px),
 ↪  baseline
tex = ndi.gaussian_filter(rng.random((400, 400)), 1.5)   # random texture painted on the
 ↪  world
tex = (tex - tex.min()) / np.ptp(tex)

def render(cx):
    """Ray-cast a camera at (cx, 0, 0) looking down +Z: a square at Z=5 in front of a wall
     ↪  at Z=8."""
    v, u = np.mgrid[0:H, 0:W].astype(float)
    dx, dy = (u - W / 2) / f, (v - H / 2) / f          # ray direction (dx, dy, 1)
    Xs, Ys = cx + 5 * dx, 5 * dy                        # hit point on the plane Z = 5
    front = (np.abs(Xs) < 1) & (np.abs(Ys) < 1)         # inside the square?
```

```
    Z = np.where(front, 5.0, 8.0)
    X, Y = cx + Z * dx, Z * dy
    img = ndi.map_coordinates(tex, [200 + 40 * Y + 100 * front, 200 + 40 * X], order=1)
    img[~front & (X > 1.5) & (Y < -0.5)] = 0.5          # a textureless patch on the wall
    return img, Z

ref, Z_true = render(0.0)                                # reference camera
src, _ = render(B)                                       # source camera, B to the right

depths = 1 / np.linspace(1 / 10, 1 / 3.5, 80)            # depth hypotheses, uniform in 1/Z
cost = np.empty((len(depths), H, W))
for k, Z in enumerate(depths):
    d = f * B / Z                                        # the plane Z induces a horizontal
     ↪  shift d
    warped = ndi.shift(src, (0, d), order=1, mode='nearest')   # src pixel u - d lands on u
    cost[k] = ndi.uniform_filter((ref - warped) ** 2, 7)       # windowed SSD
     ↪  photo-consistency
Z_est = depths[cost.argmin(0)]                           # winner-take-all depth per pixel

ok = np.abs(Z_est - Z_true) / Z_true < 0.05
flat = (Z_true == 8) & (ref == 0.5)
inner = np.zeros_like(ok); inner[8:-8, 20:-8] = True    # ignore borders the source camera
 ↪  cannot see
print(f"pixels within 5% of the true depth: {100 * ok[inner & ~flat].mean():.1f}%
 ↪  (textured), "
      f"{100 * ok[inner & flat].mean():.1f}% (textureless patch)")
```

```
pixels within 5% of the true depth: 95.3% (textured), 14.8% (textureless patch)
```

On textured surfaces 95.3% of the evaluated pixels get a depth within 5% of the truth in this synthetic run. In the textureless patch only 14.8% do, because every depth hypothesis matches a flat grey patch equally well. Figure 4 (d) shows this directly: the cost curve of the textured pixel has one sharp minimum at the true depth; the curve of the textureless pixel is flat. The remaining errors on textured surfaces sit at the left border (seen by one camera only) and next to the square, where part of the background is hidden from the second camera (occlusion).

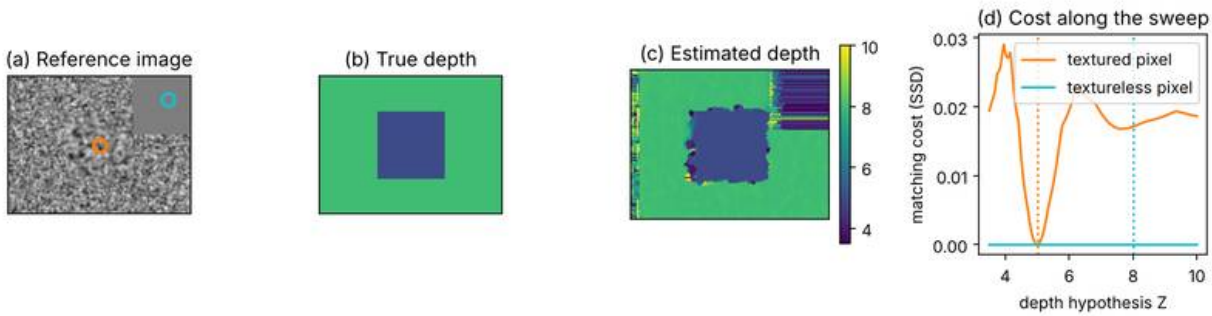


Figure 4 — Plane-sweep stereo on a synthetic rectified pair. (a) Reference image with a textureless patch (top right). (b) True depth. (c) Winner-take-all depth from a 7 × 7 SSD window. (d) Cost curves for the

two circled pixels; dotted lines mark the true depths.

From depth maps to one model

A depth map per image is not yet one model. **Depth-map fusion** merges them. The volumetric method of Curless and Levoy [15] stores, in a voxel grid, a running weighted average of the signed distance to the surface measured by each depth map:

$$D_{i+1}(\mathbf{x}) = \frac{W_i(\mathbf{x})D_i(\mathbf{x}) + w_{i+1}(\mathbf{x})\, d_{i+1}(\mathbf{x})}{W_i(\mathbf{x}) + w_{i+1}(\mathbf{x})}, \qquad W_{i+1}(\mathbf{x}) = W_i(\mathbf{x}) + w_{i+1}(\mathbf{x}),$$

where $d_{i+1}(\mathbf{x})$ is the signed distance from voxel $\mathbf{x}$ to the surface seen in depth map $i+1$ along the viewing ray, w_{i+1} is its confidence weight, and D_i, W_i are the accumulated distance and weight after i maps. The surface is the zero level set of D, extracted by marching cubes (below). In practice the distances are truncated near the surface, hence the name **TSDF** (truncated signed distance function) fusion. COLMAP's MVS [3] instead estimates a depth and a normal per pixel with per-pixel view selection and a multi-view geometric consistency check, then fuses the consistent depths into a dense point cloud.

Learning-based MVS

Plain version. The hand-made parts of MVS, the patch cost and the rules that clean up the cost volume, can be learned from data.

MVSNet [16] keeps the plane-sweep structure but makes every step differentiable. A shared CNN extracts features from all images, and the features of each source view are warped onto fronto-parallel planes of the reference camera (differentiable homography warping). A **variance-based cost** merges any number of views into one cost volume: low variance means the views agree. A 3D CNN regularizes the volume, a softmax along the depth axis gives a probability per depth, and the expected depth under that distribution (the "soft argmin") is a continuous output that can be trained with a regression loss [16]. MVSNet was trained on DTU [37] and evaluated on Tanks and Temples [38] without fine-tuning [16]. Its authors motivate learning with the low-textured, specular and reflective regions where hand-made costs fail [16].

Neural radiance fields (NeRF)

Plain version. NeRF stores a scene as a small neural network that answers one question for any point in space and any viewing direction: "how much stuff is here, and what colour does it send toward me?" To make an image, it shoots a ray through each pixel, asks the network at many

points along the ray, and blends the answers. Training adjusts the network until the rendered images match the photos.

The scene as a function

NeRF [5] is a multilayer perceptron $F_\Theta : (\mathbf{x}, \mathbf{d}) \mapsto (\sigma, \mathbf{c})$, where $\mathbf{x} \in \mathbb{R}^3$ is a position, $\mathbf{d}$ a viewing direction, $\sigma \geq 0$ the **volume density** at $\mathbf{x}$ and $\mathbf{c}$ the RGB colour emitted toward $\mathbf{d}$. The network is built so that density depends on position only, while colour also depends on direction, which lets NeRF reproduce shiny highlights that change with the viewpoint [5]. One network represents one scene; there is no training set of many scenes.

Optional: NeRF (Mildenhall et al., 2020) (in the Paper guides). What the paper set out to solve, what its tables and ablations show, and what it does not claim.

Volume rendering along a ray

Plain version. Picture the ray as passing through fog of varying thickness. Each bit of fog both glows with its own colour and blocks the light from behind it. The pixel colour is the sum of all the glows, each dimmed by the fog in front of it.

A camera ray is $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ with origin $\mathbf{o}$ (the camera centre), direction $\mathbf{d}$ and distance t between near and far bounds t_n and t_f. Following classical volume rendering [17], the expected colour is

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\, \sigma(\mathbf{r}(t))\, \mathbf{c}(\mathbf{r}(t), \mathbf{d})\, dt, \qquad T(t) = \exp\!\left(- \int_{t_n}^{t} \sigma(\mathbf{r}(s))\, ds \right),$$

where $T(t)$ is the **transmittance**, the probability that the ray travels from t_n to t without being stopped [5]. NeRF evaluates it with N samples $t_1 < \ldots < t_N$ along the ray:

$$\hat{C}(\mathbf{r}) = \sum_{i=1}^{N} w_i\, \mathbf{c}_i, \qquad w_i = T_i\, \alpha_i, \qquad \alpha_i = 1 - e^{-\sigma_i \delta_i}, \qquad T_i = \prod_{j=1}^{i-1} (1 - \alpha_j),$$

where $\sigma_i, \mathbf{c}_i$ are the network outputs at sample i, $\delta_i = t_{i+1} - t_i$ is the spacing, α_i is the **opacity** of segment i, T_i is the light that survives the segments before it, and w_i is the weight of sample i in the pixel. This is exactly front-to-back **alpha compositing** of semi-transparent layers [5]. The sum of the weights, $\sum_i w_i$, is the pixel's total opacity, and $\sum_i w_i t_i$ is an estimate of depth. Every step is differentiable, so the photo error can be pushed back into the network weights.

The snippet checks the quadrature against an exact answer. The ray crosses blue fog, then a red surface layer, then reaches a white background; for piecewise-constant density the integrals have closed forms.

```
import numpy as np

# One ray from t = 0 to t = 6: blue fog on [1, 2] (density 0.5), a red surface on [3, 3.5]
 ↪  (density 4),
# white background behind everything.
slabs = [(1.0, 2.0, 0.5, np.array([0.2, 0.4, 1.0])),
         (3.0, 3.5, 4.0, np.array([1.0, 0.1, 0.1]))]
bg = np.ones(3)

def field(t):
    """Density sigma(t) and colour c(t) at sample positions t."""
    sigma, c = np.zeros_like(t), np.zeros(t.shape + (3,))
    for a, b, s, col in slabs:
        inside = (t >= a) & (t < b)
        sigma[inside], c[inside] = s, col
    return sigma, c

def render(n):
    """NeRF quadrature: alpha_i = 1 - exp(-sigma_i delta_i), T_i = prod_{j<i} (1 - alpha_j),
     ↪  w_i = T_i alpha_i."""
    edges = np.linspace(0, 6, n + 1)
    t, delta = (edges[:-1] + edges[1:]) / 2, np.diff(edges)
    sigma, c = field(t)
    alpha = 1 - np.exp(-sigma * delta)
    T = np.concatenate([[1.0], np.cumprod(1 - alpha)[:-1]])
    w = T * alpha
    T_end = T[-1] * (1 - alpha[-1])                    # light that reaches the background
    return w @ c + T_end * bg, w.sum(), w @ t

# Analytic answer for piecewise-constant density (exact integrals of the volume rendering
 ↪  equation)
C, acc, depth, T0 = np.zeros(3), 0.0, 0.0, 1.0
for a, b, s, col in slabs:
    L = b - a
    absorbed = T0 * (1 - np.exp(-s * L))
    C, acc = C + absorbed * col, acc + absorbed
    depth += T0 * (a * (1 - np.exp(-s * L)) + (1 - np.exp(-s * L)) / s - L * np.exp(-s * L))
    T0 *= np.exp(-s * L)
C += T0 * bg
print(f"analytic : colour {np.round(C, 4)}, opacity {acc:.4f}, depth {depth:.4f}")
for n in [16, 64, 256, 1024]:
    Cq, accq, dq = render(n)
    print(f"N = {n:4d} : colour {np.round(Cq, 4)}, opacity {accq:.4f}, depth {dq:.4f}")
```

```
analytic : colour [0.6852 0.2919 0.528 ], opacity 0.9179, depth 2.2373
N =   16 : colour [0.7498 0.3318 0.5195], opacity 0.8466, depth 2.1655
N =   64 : colour [0.7006 0.2986 0.5232], opacity 0.9040, depth 2.2263
N =  256 : colour [0.689  0.2934 0.5267], opacity 0.9146, depth 2.2348
N = 1024 : colour [0.6862 0.2923 0.5277], opacity 0.9171, depth 2.2367
```

The quadrature converges to the analytic colour, opacity and depth as the number of samples

grows. With 16 samples it is visibly off, because the thin surface layer falls into one or two coarse bins. This is why NeRF does not sample uniformly: a coarse network with 64 samples per ray first finds where the weights are large, and a fine network adds 128 samples concentrated there (hierarchical sampling) [5]. Note also the depth: 2.24 (2.23 with the 256 samples of Figure 5), between the fog and the surface, where no surface exists. Semi-transparent "fog" in the wrong place is the origin of the floater artifacts discussed under failure modes. Figure 5 plots the density, transmittance and weights for this ray.

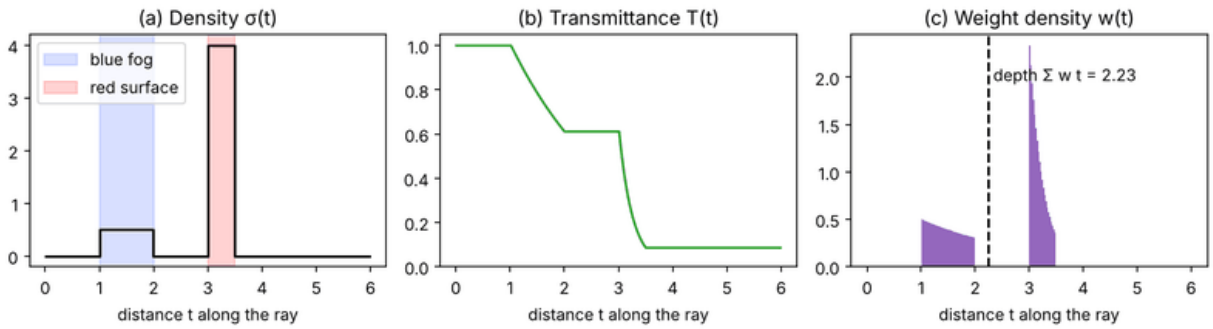


Figure 5 — Volume rendering of the ray in the code: (a) density, (b) transmittance, (c) weights per unit length. The expected depth (dashed) lies between the fog and the surface, where nothing is.

Positional encoding

Plain version. A plain MLP fed with raw (x, y, z) learns smooth, blurry functions. Feeding it sines and cosines of the coordinates at many frequencies lets it represent sharp edges and fine texture.

NeRF maps each coordinate p (normalized to $[-1, 1]$) to

$$\gamma(p) = (\sin(2^0\pi p), \cos(2^0\pi p), \ldots, \sin(2^{L-1}\pi p), \cos(2^{L-1}\pi p)),$$

where L is the number of frequency bands: $L = 10$ for position and $L = 4$ for direction [5]. Without the encoding, the paper's results lose their fine detail [5]. Tancik et al. [18] explained why: coordinate MLPs are biased toward low frequencies (a spectral bias), and a Fourier feature mapping of the input removes this bias. The sinusoids resemble the positional information of Transformers (ML 04), but here they encode continuous coordinates rather than token positions.

Training, and why it is slow

Training needs posed images and nothing else. Each step samples a batch of pixels (rays) from the training images, renders them with the coarse and fine networks, and minimizes the squared colour error [5]. The cost is the problem. Every pixel needs hundreds of network evaluations, and a scene needs a few hundred thousand steps. The paper reports 100–300k iterations, about 1–2 days on one NVIDIA V100 GPU, per scene [5]. Rendering is slow for the same reason: the

paper uses 256 network queries per ray, between 150 and 200 million per image, and reports about 30 seconds per frame on a V100 [5].

Making it fast: hash grids

Plain version. Instead of asking one big network to remember everything, store learned features in lookup tables at several resolutions, and use a tiny network to decode them. Looking up a table is far cheaper than running many network layers.

Instant-NGP [19] replaces the frequency encoding with a **multiresolution hash encoding**. Space is covered by L grids, from coarse to fine. Each grid stores trainable feature vectors at its corners, but in a table of fixed size T; on fine grids, where there are far more corners than table entries, a hash function decides which entry a corner uses, so several corners share one entry. To encode a point, the features of the corners around it are interpolated on every level, concatenated and passed to a small MLP. The paper does not resolve the shared entries explicitly; it relies on the coarse levels and the MLP to sort them out [19]. The authors report training of high-quality results "in a matter of seconds" and rendering in tens of milliseconds at 1920 × 1080 on one GPU [19]. Grid-based encodings of this kind are now common in fast NeRF variants.

Optional: The multiresolution hash encoding (in the Supplementary notes). The level resolutions, the hash function, the parameter count, and a worked example.

Neural implicit surfaces: NeuS

Plain version. NeRF's density does not have to form a clean surface: it can be a cloud. If you want a surface, let the network output a *signed distance* instead, and convert distance to opacity in a way that makes the rendered weight peak exactly on the surface.

NeuS [22] represents the scene as an SDF network $f(\mathbf{x})$ whose zero level set is the surface. It converts the SDF into opacity through the sigmoid $\Phi_s(x) = (1 + e^{-sx})^{-1}$, where s is a learned sharpness that grows during training. The authors show that the obvious choice, using the logistic density $\phi_s(f) = \Phi_s'(f)$ directly as the volume density, gives weights that peak *before* the surface (the bias), and they derive an opacity that is unbiased to first order and still handles occlusion [22]. In discrete form,

$$\alpha_i = \max\left(\frac{\Phi_s(f(\mathbf{p}_i)) - \Phi_s(f(\mathbf{p}_{i+1}))}{\Phi_s(f(\mathbf{p}_i))},\ 0 \right),$$

where $\mathbf{p}_i$ is the i-th sample on the ray; the weights then follow the same compositing formula as NeRF. An **Eikonal** term $(\|\nabla f\| - 1)^2$ keeps f a valid distance function [22]. Because the surface is a level set, a mesh follows directly from marching cubes. NeuS reconstructs objects without foreground masks, and it was evaluated on DTU [37] among other data [22].

Optional: Why NeuS weights peak on the surface (in the Supplementary notes). The bias of the naive density, measured on one ray.

3D Gaussian Splatting

Plain version. Instead of a network you must query along every ray, 3DGS represents the scene with millions of small, coloured, semi-transparent ellipsoids, like a cloud of soft paint drops. To make an image it projects each drop onto the screen as a soft ellipse and stacks them from front to back. No ray marching, no network: just sorting and blending, which a GPU does very fast.

The Gaussian primitive

Each blob is a 3D bell curve: dense at its centre, fading smoothly outward, and stretched more in some directions than in others. Precisely, each primitive [6] has a centre $\boldsymbol{\mu} \in \mathbb{R}^3$, a covariance Σ that sets its size and orientation, an opacity $o \in [0,1]$ and a colour stored as spherical-harmonic coefficients (a small set of smooth functions on the sphere, so the colour can change with the viewing direction). Its footprint in space is

$$G(\mathbf{x}) = \exp\!\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\top}\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big), \qquad \Sigma = R\,S\,S^{\top}R^{\top},$$

where $S = \mathrm{diag}(s_1, s_2, s_3)$ holds the three axis lengths and R is a rotation stored as a unit quaternion [6]. The factorization guarantees that Σ stays a valid (positive semi-definite) covariance during gradient descent, which optimizing the six entries of Σ directly would not [6].

Projection to the image

Plain version. A 3D ellipsoid seen through a camera looks like a 2D ellipse. 3DGS computes that ellipse with a local linear approximation of the camera projection, which is accurate for small Gaussians.

Following EWA splatting [23], the 2D covariance of the projected Gaussian is

$$\Sigma' = J\,W\,\Sigma\,W^{\top}J^{\top},$$

where W is the rotation part of the world-to-camera transform, J is the Jacobian of the perspective projection evaluated at the Gaussian's centre, and keeping only the top-left 2×2 block of Σ' gives the image-space ellipse [6], [23]. The projected centre is simply the pinhole projection of $\boldsymbol{\mu}$.

Optional: Projecting a 3D Gaussian: the local affine approximation (in the Supplementary notes). The Jacobian written out, and when the approximation breaks.

Alpha compositing and real-time rendering

The pixel colour uses the same compositing rule as NeRF, but over the Gaussians that cover the pixel, sorted by depth:

$$C = \sum_i \mathbf{c}_i \, \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \qquad \alpha_i = o_i \exp\!\left(-\tfrac{1}{2} (\mathbf{u} - \boldsymbol{\mu}'_i)^\top \Sigma'^{-1}_i (\mathbf{u} - \boldsymbol{\mu}'_i)\right),$$

where $\mathbf{u}$ is the pixel position, $\boldsymbol{\mu}'_i$ and Σ'_i are the projected centre and covariance of Gaussian i, and $\mathbf{c}_i$ its view-dependent colour. Why is it fast? The renderer [6] splits the screen into 16×16-pixel tiles, assigns each visible Gaussian to the tiles it touches, sorts all (tile, depth) keys once with a GPU radix sort, and then blends each tile's list front to back in parallel, stopping a pixel once it is nearly opaque. There is no network to evaluate and no empty space to sample. The paper reports real-time (at least 30 frames per second) novel-view synthesis at 1080p with quality comparable to the best NeRF variant of the time [6].

Optional: 3D Gaussian Splatting (Kerbl et al., 2023) (in the Paper guides). How to separate the representation from the engineering when you read the paper, and its stated limitations.

Optimization and adaptive density control

Plain version. Training starts from the sparse SfM points, one Gaussian per point, and then grows and prunes the set: where detail is missing it adds Gaussians, where one big Gaussian covers too much it splits it, and invisible ones are removed.

All parameters are optimized by gradient descent on the loss $(1 - \lambda)\mathcal{L}_1 + \lambda \mathcal{L}_{\text{D-SSIM}}$ between rendered and real images, with $\lambda = 0.2$ [6]. Every 100 iterations the method densifies Gaussians whose view-space position gradient is large, a sign that the region is not yet well explained [6]:

- a *small* Gaussian in an under-reconstructed region is **cloned** (copied and moved along the gradient);
- a *large* Gaussian in a region of high variance is **split** into two smaller ones, with scales divided by 1.6;
- Gaussians with opacity below a threshold are **pruned**; in addition, all opacities are reset close to zero every 3000 iterations, which limits the growth of Gaussians caused by floaters close to the input cameras [6].

Code: a toy 2D splatting fit

The snippet fits 40 two-dimensional Gaussians to a 48×48 image with the Adam optimizer [49], using exactly the compositing rule above (with a fixed depth order and no densification). The backward pass is written out by hand with the chain rule (the 3DGS paper likewise derives its gradients explicitly instead of relying on automatic differentiation [6]); the gradient was checked against finite differences while writing this tutorial.

```
import numpy as np
from skimage.draw import disk, polygon, ellipse

# Target: a small synthetic RGB image (48 x 48) on a black background
H = W = 48
target = np.zeros((H, W, 3))
target[disk((16, 16), 9)] = [0.9, 0.2, 0.2]
target[ellipse(30, 30, 6, 14, rotation=np.deg2rad(30))] = [0.2, 0.4, 0.9]
target[polygon([40, 44, 34], [6, 22, 14])] = [0.2, 0.8, 0.3]
ys, xs = np.mgrid[0:H, 0:W].astype(float)

def sig(z):
    return 1 / (1 + np.exp(-z))

def render(p, target=None):
    """Front-to-back alpha compositing of K 2D Gaussians (index order = depth order).
    p: mu (K,2), log_s (K,2), theta (K,), op (K,) opacity logits, col (K,3) colour logits."""
    c, s, th = sig(p['col']), np.exp(p['log_s']), p['theta']
    dx = xs[None] - p['mu'][:, 0, None, None]; dy = ys[None] - p['mu'][:, 1, None, None]
    ct, st = np.cos(th)[:, None, None], np.sin(th)[:, None, None]
    u1, u2 = ct * dx + st * dy, -st * dx + ct * dy                  # offsets in the
     ↪  Gaussian's own axes
    sx2, sy2 = (s[:, 0] ** 2)[:, None, None], (s[:, 1] ** 2)[:, None, None]
    g = np.exp(-0.5 * (u1 ** 2 / sx2 + u2 ** 2 / sy2))              # Gaussian footprint
    o = sig(p['op'])[:, None, None]
    a = np.minimum(o * g, 0.99)                                      # alpha of each
     ↪  Gaussian at each pixel
    T = np.cumprod(np.concatenate([np.ones((1, H, W)), 1 - a[:-1]]), axis=0)   #
     ↪  transmittance in front
    w = a * T                                                        # compositing weights
    img = np.einsum('khw,kc->hwc', w, c)                            # black background
     ↪  adds nothing
    if target is None:
        return img
    # backward pass for the loss L = mean squared error (the chain rule written out by hand)
    G = 2 * (img - target) / target.size                             # dL/dimg, shape (H,
     ↪  W, 3)
    contrib = w[..., None] * c[:, None, None, :]                    # (K, H, W, 3)
    behind = np.cumsum(contrib[::-1], axis=0)[::-1] - contrib        # colour from
     ↪  Gaussians behind k
    dC_da = T[..., None] * c[:, None, None, :] - behind / (1 - a[..., None])
    da = np.einsum('khwc,hwc->khw', dC_da, G) * (o * g < 0.99)
    dq = da * o * g * -0.5                                           # through g = exp(-q
     ↪  / 2)
    grads = {
        'col': np.einsum('khw,hwc->kc', w, G) * c * (1 - c),
        'op': (da * g).sum((1, 2)) * sig(p['op']) * (1 - sig(p['op'])),
        'log_s': np.stack([(dq * -2 * u1 ** 2 / sx2).sum((1, 2)), (dq * -2 * u2 ** 2 /
         ↪  sy2).sum((1, 2))], 1),
        'theta': (dq * 2 * u1 * u2 * (1 / sx2 - 1 / sy2)).sum((1, 2)),
        'mu': -np.stack([(dq * (2 * u1 * ct / sx2 - 2 * u2 * st / sy2)).sum((1, 2)),
                         (dq * (2 * u1 * st / sx2 + 2 * u2 * ct / sy2)).sum((1, 2))], 1),
    }
```

```
    return img, grads

rng = np.random.default_rng(0)
K = 40
p = {'mu': rng.uniform(4, 44, (K, 2)), 'log_s': np.full((K, 2), np.log(3.0)), 'theta':
 ↪  rng.uniform(0, np.pi, K),
     'op': np.zeros(K), 'col': np.zeros((K, 3))}
lr = {'mu': 0.5, 'log_s': 0.05, 'theta': 0.05, 'op': 0.1, 'col': 0.1}
m = {k: np.zeros_like(v) for k, v in p.items()}; v2 = {k: np.zeros_like(v) for k, v in
 ↪  p.items()}

def psnr(img):
    return 10 * np.log10(1 / np.mean((img - target) ** 2))

for it in range(1, 501):
    img, gr = render(p, target)
    for k in p:                                                    # Adam update
        m[k] = 0.9 * m[k] + 0.1 * gr[k]; v2[k] = 0.999 * v2[k] + 0.001 * gr[k] ** 2
        p[k] -= lr[k] * (m[k] / (1 - 0.9 ** it)) / (np.sqrt(v2[k] / (1 - 0.999 ** it)) +
         ↪  1e-8)
    if it in (1, 50, 100, 200, 500):
        print(f"iteration {it:3d}: PSNR {psnr(img):.2f} dB")
```

```
iteration   1: PSNR 11.32 dB
iteration  50: PSNR 20.88 dB
iteration 100: PSNR 21.85 dB
iteration 200: PSNR 23.12 dB
iteration 500: PSNR 23.57 dB
```

PSNR (defined in DLCV 04) rises quickly and then levels off near 23.6 dB in this run. The plateau is the point of the exercise: with a fixed number of soft blobs, sharp corners and the thin tip of the triangle cannot be fitted, and blobs that started in the wrong place stay there. That is exactly what adaptive density control fixes, and Exercise 4 asks you to add it. Figure 6 shows the fit and the final ellipses.

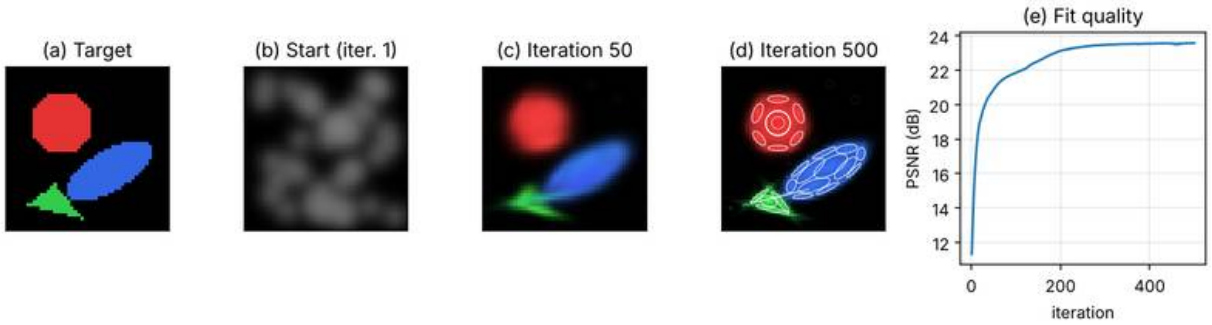


Figure 6 — Toy splatting with 40 Gaussians and alpha compositing: (a) target, (b) start, (c) after 50 and (d) 500 iterations, with each Gaussian's one-standard-deviation ellipse; (e) PSNR against iteration. Numbers are from this toy run only.

Getting meshes

Plain version. Graphics engines, physics simulators and 3D printers want triangles. Every representation above needs a conversion step to produce them, and the conversion can add its own errors.

Marching cubes

Marching cubes [24] turns a sampled scalar field (an SDF, an occupancy or a fused TSDF) into a triangle mesh of one level set. It visits each cube of eight neighbouring grid samples, looks at which corners are inside and which are outside, places vertices on the cube edges where the sign changes (by linear interpolation of the field values), and connects them with triangles from a lookup table. scikit-image provides an implementation [46]. The snippet extracts the zero level set of an exact torus SDF at several grid resolutions and measures the result with the geometric metrics defined in the evaluation section below.

```
import numpy as np
from scipy.spatial import cKDTree
from skimage.measure import marching_cubes

R, r = 0.6, 0.25                                    # torus: ring radius R, tube radius r

def sdf_torus(x, y, z):
    """Exact signed distance to the torus (negative inside)."""
    return np.hypot(np.hypot(x, y) - R, z) - r

def sample_mesh(verts, faces, n, rng):
    """n points spread uniformly over the area of a triangle mesh."""
    tri = verts[faces]
    area = 0.5 * np.linalg.norm(np.cross(tri[:, 1] - tri[:, 0], tri[:, 2] - tri[:, 0]),
     ↪  axis=1)
    k = rng.choice(len(faces), n, p=area / area.sum())
    a, b = rng.random((2, n, 1))
    flip = a + b > 1; a[flip], b[flip] = 1 - a[flip], 1 - b[flip]
    return tri[k, 0] + a * (tri[k, 1] - tri[k, 0]) + b * (tri[k, 2] - tri[k, 0])

def sample_torus(n, rng):
    """n points uniform over the true torus surface (rejection sampling on the area
     ↪  element)."""
    u, v = rng.uniform(0, 2 * np.pi, (2, 3 * n))
    ok = rng.random(3 * n) * (R + r) < R + r * np.cos(v)   # keep with probability
     ↪  proportional to R + r cos v
    u, v = u[ok][:n], v[ok][:n]
    return np.c_[(R + r * np.cos(v)) * np.cos(u), (R + r * np.cos(v)) * np.sin(u), r *
     ↪  np.sin(v)]

def geometry_metrics(P, G, tau):
    """Accuracy, completeness, Chamfer (their mean) and F-score at threshold tau."""
```

```
    d_pg = cKDTree(G).query(P)[0]                       # each reconstructed point -> nearest
     ↪  true point
    d_gp = cKDTree(P).query(G)[0]                       # each true point -> nearest
     ↪  reconstructed point
    prec, rec = (d_pg < tau).mean(), (d_gp < tau).mean()
    return d_pg.mean(), d_gp.mean(), (d_pg.mean() + d_gp.mean()) / 2, 2 * prec * rec / (prec
     ↪  + rec)

rng = np.random.default_rng(0)
G = sample_torus(100_000, rng)
acc, comp, cd, fs = geometry_metrics(sample_torus(100_000, rng), G, tau=0.01)
print(f"true surface vs itself: Chamfer {cd:.4f} | F-score@0.01 {fs:.3f}   (the floor set by
 ↪  sampling)")
for n in [12, 24, 48, 96]:
    ax = np.linspace(-1, 1, n)
    vol = sdf_torus(*np.meshgrid(ax, ax, ax, indexing='ij'))
    verts, faces, _, _ = marching_cubes(vol, level=0.0, spacing=(ax[1] - ax[0],) * 3)
    verts -= 1.0                                        # grid index space -> world
     ↪  coordinates [-1, 1]
    P = sample_mesh(verts, faces, 100_000, rng)
    acc, comp, cd, fs = geometry_metrics(P, G, tau=0.01)
    print(f"grid {n:2d}^3: {len(faces):6d} triangles | acc {acc:.4f} comp {comp:.4f} "
          f"Chamfer {cd:.4f} | F-score@0.01 {fs:.3f} | max |SDF| on vertices
           ↪  {np.abs(sdf_torus(*verts.T)).max():.4f}")
```

```
true surface vs itself: Chamfer 0.0039 | F-score@0.01 0.995   (the floor set by sampling)
grid 12^3:    448 triangles | acc 0.0154 comp 0.0154 Chamfer 0.0154 | F-score@0.01 0.299 |
 ↪  max |SDF| on vertices 0.0084
grid 24^3:   2272 triangles | acc 0.0052 comp 0.0052 Chamfer 0.0052 | F-score@0.01 0.988 |
 ↪  max |SDF| on vertices 0.0032
grid 48^3:   9568 triangles | acc 0.0040 comp 0.0039 Chamfer 0.0040 | F-score@0.01 0.995 |
 ↪  max |SDF| on vertices 0.0008
grid 96^3:  38832 triangles | acc 0.0039 comp 0.0039 Chamfer 0.0039 | F-score@0.01 0.995 |
 ↪  max |SDF| on vertices 0.0002
```

Two lessons come from this run. First, the vertices are almost exactly on the surface even on the 12^3 grid (the largest |SDF| is 0.0084), yet the Chamfer distance is 0.0154: the flat triangles *between* the vertices cut across the curved tube. Second, the metric stops improving at about 0.0039, because comparing two independent random samplings of the *same* true surface already gives that value. A measured error is meaningful only relative to this floor, which depends on how many points are sampled. Figure 7 shows the meshes and the curves.

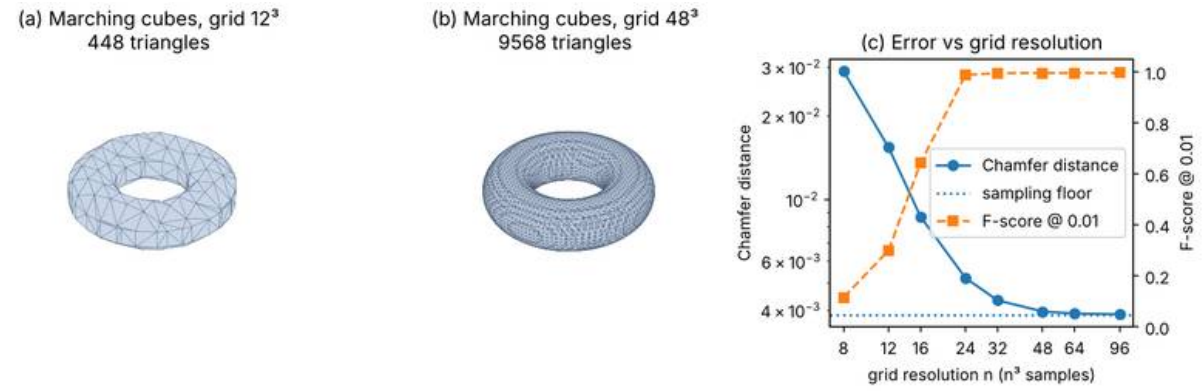


Figure 7 — Marching cubes on an exact torus SDF: (a) 12^3 and (b) 48^3 grids; (c) Chamfer distance and F-score at $\tau = 0.01$ against grid resolution, with the floor obtained by comparing two samplings of the true surface.

Poisson surface reconstruction

When the input is a point cloud with normals, as from MVS, Poisson surface reconstruction [25] is the standard tool. It seeks an indicator function χ (1 inside, 0 outside) whose gradient matches the normals, which leads to the Poisson equation $\Delta\chi = \nabla \cdot \mathbf{V}$, where Δ is the Laplacian and $\mathbf{V}$ the vector field built from the normals; the surface is a level set of χ [25]. Because it uses all points at once, it is robust to noise [25]; the screened variant [26] also pulls the surface toward the points. The result is watertight, which is convenient but means that holes in the data are filled with invented surface.

Meshes from Gaussians

3D Gaussians do not define a surface: after optimization they are unorganized, overlapping and often not aligned with the true geometry. Two lines of work address this. **SuGaR** [27] adds a regularization term that encourages the Gaussians to lie flat on the surface, then extracts a mesh with Poisson reconstruction from points sampled on the level set of the Gaussians' density, and optionally binds new Gaussians to the mesh and refines both together [27]. **2D Gaussian Splatting** [28] changes the primitive: each Gaussian is a flat, oriented 2D disk (a surface element), rendered with an exact ray–disk intersection, plus depth-distortion and normal-consistency losses. Meshes are then obtained by fusing the rendered depth maps with TSDF fusion [28], the same idea as Curless and Levoy's method above.

Feed-forward reconstruction

Plain version. Everything so far optimizes one scene at a time. The newest models are trained once on large collections of 3D data and then, given new photos, output the cameras and a dense 3D point for every pixel in a single pass, without matching, RANSAC or bundle adjustment.

DUSt3R [29] takes two images with unknown intrinsics and poses and regresses two **pointmaps**: for each pixel of each image, the 3D point it sees, both expressed in the coordinate frame of the first camera [29]. A pointmap is an image with three channels (X, Y, Z) (Figure 2 (h)). From it, depth, focal length, relative pose and pixel correspondences all follow. The network is a Transformer [48] encoder–decoder: both images go through the same Vision Transformer encoder [47], and two decoders exchange information through cross-attention before regression heads output the pointmaps and a confidence per pixel [29]. The output is defined only up to an unknown scale [29]. For more than two images, DUSt3R predicts pairwise pointmaps and then runs a *global alignment* that solves for one consistent set of pointmaps, poses and per-pair scales, a learned-geometry counterpart of bundle adjustment [29].

MASt3R [30] adds a head that outputs dense local features trained with a matching loss, and a fast reciprocal matching scheme, so that the pointmap network also yields accurate pixel correspondences [30]. **VGGT** [31] goes further: a large Transformer with alternating frame-wise and global attention takes one, a few or hundreds of images and predicts camera parameters, depth maps, pointmaps and point tracks for all of them at once; the authors report that this takes less than a second and often beats optimization-based alternatives without further processing [31].

These models do not make geometry obsolete. Their predictions can still be refined: the VGGT authors report that its cameras and tracks are a good initialization for bundle adjustment, which improves accuracy further [31]. Their accuracy also depends on how close a scene is to their training data. They are, however, the best tools available when images are few, overlap little or have no calibration.

Dynamic scenes (brief)

Plain version. If the scene moves, a single 3D model is not enough: the model must also depend on time.

The common recipe is a canonical (static) scene plus a deformation over time. D-NeRF [32] adds time as an input and learns two networks together: one encodes the scene in a canonical space, the other maps that canonical scene to its deformed state at a given time; it works from a single camera moving around the scene [32]. 4D Gaussian Splatting [33] keeps one set of 3D Gaussians and predicts their deformation at each timestamp with a lightweight MLP fed by a factorized space–time feature grid; the paper reports real-time rendering, for example 82 frames per second at 800 × 800 on an RTX 3090 GPU [33]. Dynamic reconstruction is much less constrained than

static reconstruction, because each moment is seen from fewer viewpoints.

Evaluating reconstructions

Plain version. Two different things can be measured: whether the *pictures* rendered from new viewpoints look like real photos, and whether the *shape* is geometrically correct. A method can do well on one and badly on the other.

Novel-view quality

Hold out some photos, render the reconstruction from their camera poses, and compare. The standard metrics are PSNR, SSIM [34] and LPIPS [35], all defined in DLCV 04. PSNR and SSIM measure pixel and structural agreement; LPIPS measures perceptual similarity in deep-feature space. A common protocol, which 3DGS adopts from Mip-NeRF 360, holds out every 8th photo of a capture as the test set [6]. Novel-view metrics depend on the held-out poses being correct, so they also reward good SfM.

Geometric accuracy

Let $\mathcal{P}$ be points sampled from the reconstruction and $\mathcal{G}$ points from the ground-truth scan. Define the two directed distances

$$\text{acc} = \frac{1}{|\mathcal{P}|} \sum_{\mathbf{p} \in \mathcal{P}} \min_{\mathbf{g} \in \mathcal{G}} \|\mathbf{p} - \mathbf{g}\|, \qquad \text{comp} = \frac{1}{|\mathcal{G}|} \sum_{\mathbf{g} \in \mathcal{G}} \min_{\mathbf{p} \in \mathcal{P}} \|\mathbf{g} - \mathbf{p}\|,$$

where $|\cdot|$ counts points. **Accuracy** asks "are my points near the truth?"; **completeness** asks "is all of the truth covered?". Their sum or mean is the **Chamfer distance**, widely used as a loss and metric for point sets since Fan et al. [36], who define it with squared distances; papers differ in squaring and in sum versus mean, so check the definition before comparing numbers. On DTU, MVSNet reports the mean of accuracy and completeness as an "overall" score [16].

Tanks and Temples [38] uses a threshold τ instead of averages. With $e_{\mathbf{p}}$ the distance from a reconstructed point to the ground truth and $e_{\mathbf{g}}$ the distance from a ground-truth point to the reconstruction,

$$P(\tau) = \frac{100}{|\mathcal{P}|} \sum_{\mathbf{p} \in \mathcal{P}} [\, e_{\mathbf{p}} < \tau \,], \qquad R(\tau) = \frac{100}{|\mathcal{G}|} \sum_{\mathbf{g} \in \mathcal{G}} [\, e_{\mathbf{g}} < \tau \,], \qquad F(\tau) = \frac{2P(\tau)R(\tau)}{P(\tau) + R(\tau)},$$

where $[\cdot]$ is 1 if the condition holds and 0 otherwise, P is precision (in percent), R is recall and F is their harmonic mean, the **F-score** [38]. Because it is a harmonic mean, F is high only if the reconstruction is both accurate and complete; a sparse set of perfect points or a cloud that fills all

space both score poorly [38]. The F-score is also robust to a few far outliers, which can dominate a Chamfer average (Exercise 5).

Datasets and benchmarks

- **DTU** [37]: a multi-view stereo dataset of 80 tabletop scenes, each with 49 or 64 precisely known camera positions from a six-axis industrial robot and reference geometry from a structured-light scanner [37]. Standard for MVS and neural surface methods (NeuS uses 15 of its scenes [22]).
- **Tanks and Temples** [38]: high-resolution video of large indoor and outdoor scenes captured outside the lab, with ground truth from an industrial laser scanner, split into an *intermediate* and an *advanced* group, and evaluated by the F-score above on an online server [38].
- **Mip-NeRF 360** [39]: 9 unbounded scenes (5 outdoor, 4 indoor), each with a central object and a detailed background, captured for novel-view synthesis [39]. 3DGS evaluates on it together with Tanks and Temples scenes, among others [6].

Failure modes

Plain version. Every method above rests on the same assumptions: the scene is static, its surfaces look the same from every direction, there is texture to match, the views overlap, and the cameras are known. Each failure mode below breaks one of them.

- **Textureless surfaces** (white walls, sky). Photo-consistency is equal at every depth, as in the plane-sweep run; reconstructions show holes or noise. Learned MVS and feed-forward models fill them with learned priors, which may be wrong.
- **Reflective and transparent surfaces.** A mirror or a glossy car shows a reflected scene that moves differently from the surface, so matching and density both go wrong. NeRF tends to fake a highlight with light sources hidden *inside* the object under a semi-transparent surface; Ref-NeRF [40] instead models the radiance reflected about the surface normal, which improves specular surfaces [40].
- **Few views.** With two or three images, many scenes explain the photos equally well, and per-scene optimization overfits (good training views, broken new views). Feed-forward models such as DUSt3R [29] are the usual remedy.
- **Floaters.** Semi-transparent blobs of density in free space, often close to the cameras, that look right from training views but appear as fog from new views. Mip-NeRF 360 names them and adds a distortion-based regularizer to suppress them [39]; 3DGS resets opacities periodically for the same reason [6].
- **Scale ambiguity.** Monocular image collections fix geometry only up to scale; metric size needs

a known distance, a calibrated rig or other sensors [1].

- **Pose errors.** NeRF and 3DGS assume accurate poses; small errors blur fine detail. BARF [41] jointly optimizes the poses and the radiance field, using a coarse-to-fine schedule on the positional encoding, because the high frequencies otherwise hurt registration [41].
- **Moving objects and lighting changes**, such as people walking through a capture or the sun moving, break the static-scene assumption itself.

Modern view

What the surveys say

Four reviews frame this tutorial. Seitz et al. [42] gave multi-view stereo its classical taxonomy, which still describes every method here: the scene representation, the photo-consistency measure, the visibility model, the shape prior, the reconstruction algorithm and the initialization requirements. Furukawa and Hernández [43] present MVS as an image–geometry consistency optimization built from two ingredients, robust photometric consistency measures and efficient optimization, and survey the successful algorithms and remaining challenges. Tewari et al. [44] review neural rendering, from scene representations and differentiable image formation to applications such as novel-view synthesis, generalization across scenes, dynamic content, editing and relighting, and list open problems such as integration with classical graphics tools and interpretability. Chen and Wang [45] survey 3D Gaussian Splatting: its principles and optimization, seven directions that extend it, its applications, and performance comparisons; they identify a shift from per-scene optimization toward feed-forward Gaussian prediction.

Optional: Guide to the Modern view surveys (in the Paper guides). What each review covers, how it organizes the field and what to take from it.

The state of the art, 2023–2026

- **Explicit, rasterizable representations** won the speed race. Since 3DGS [6], a large share of view-synthesis work builds on Gaussian primitives. The 3DGS survey [45] groups the follow-up work into sparse-input, memory-efficient and photorealistic variants, better optimization, Gaussians with extra properties, hybrid representations and new rendering algorithms; geometry (SuGaR [27], 2DGS [28]) and dynamics (4DGS [33]) are active directions of their own.
- **Neural implicit surfaces** in the NeuS line [22] remain the choice when a watertight, accurate surface matters more than rendering speed.
- **Feed-forward geometry** is the largest change. DUSt3R [29], MASt3R [30] and VGGT [31] estimate cameras and dense geometry directly from images; they are increasingly used to initialize or replace parts of the classical SfM pipeline, and the 3DGS survey [45] describes the same trend for Gaussians.

- **The classical pipeline remains the reference.** COLMAP [2], [3] is still the usual source of camera poses for training and evaluation, and bundle adjustment is still used to refine learned predictions.

Open problems

1. **Geometry and appearance together.** Methods tuned for novel views produce poor surfaces, and accurate-surface methods render less realistically. One representation that does both, fast, is still open.
2. **Reflections, transparency and lighting.** Separating surface, material and illumination (inverse rendering) is unsolved for real captures.
3. **Generalization of feed-forward models**: accuracy far from the training distribution, metric scale, and very large scenes.
4. **Dynamic and in-the-wild captures** with moving people, changing light and few viewpoints.
5. **Evaluation.** Pose errors in the "ground truth", sampling-dependent Chamfer numbers and inconsistent metric definitions make cross-paper comparisons fragile.

Key takeaways

- 3D reconstruction recovers both cameras and scene; from images alone the result is defined only up to scale.
- The pinhole model $\lambda \mathbf{x} = K(R\mathbf{X} + \mathbf{t})$, the epipolar constraint $\hat{\mathbf{x}}_2^\top E \hat{\mathbf{x}}_1 = 0$ and triangulation are the geometric core of every method, old or new.
- Structure from motion chains matching, RANSAC, incremental registration and bundle adjustment (reprojection-error minimization); COLMAP's output is the standard input to dense methods.
- Multi-view stereo finds per-pixel depth by photo-consistency over a plane sweep and fuses depth maps; it fails where there is no texture, and MVSNet learns the cost and its regularization.
- NeRF renders a learned density and colour field with alpha compositing along rays; positional encoding gives detail, hash grids give speed, and NeuS turns the field into an unbiased SDF surface.
- 3D Gaussian Splatting replaces ray marching by sorting and blending projected anisotropic Gaussians, and grows them with clone, split and prune; it renders in real time but needs extra work (SuGaR, 2DGS) for surfaces.
- Feed-forward models (DUSt3R, MASt3R, VGGT) predict cameras and pointmaps in one pass and are reshaping the classical pipeline.
- Evaluate views with PSNR/SSIM/LPIPS on held-out images and geometry with accuracy, completeness, Chamfer distance and F-score, always stating the definition and sampling.

Exercises

1. **Depth error by hand.** A rectified stereo pair has $f = 800$ px and baseline $B = 0.1$ m. Using $\delta Z \approx Z^2 \delta d/(fB)$, how large is the depth error for a point at $Z = 5$ m if the disparity is off by 0.5 px? How large must B be to bring it under 2 cm?

   Hint

   $\delta Z \approx 25 \cdot 0.5/(800 \cdot 0.1) = 0.156$ m. Solving $25 \cdot 0.5/(800B) < 0.02$ gives $B > 0.78$ m. A wider baseline helps, but it also makes the two views look more different, which makes matching harder.

2. **See the scale ambiguity.** In the two-view snippet, multiply all scene points `X` and the camera centre `C2` by 3 before projecting. Show that the pixel coordinates do not change and that the estimated pose is the same. Name three pieces of outside information that would fix the scale.

   Hint

   Projection divides by depth, so scaling $\mathbf{X}$ and $\mathbf{t}$ together cancels: $K(R(3\mathbf{X}) + 3\mathbf{t}) = 3K(R\mathbf{X} + \mathbf{t})$. Outside information: a measured distance in the scene (a ruler or a marker of known size), the known baseline of a stereo rig, or another sensor such as an inertial measurement unit or GPS.

3. **Read the weights.** In the volume-rendering snippet, prove that $\sum_i w_i = 1 - \prod_i (1 - \alpha_i)$. Then remove the fog slab and recompute the depth. Why is the "depth" of a ray that passes through fog not the depth of any surface?

   Hint

   By induction: $T_{i+1} = T_i(1 - \alpha_i)$ so $w_i = T_i - T_{i+1}$, and the sum telescopes to $T_1 - T_{N+1} = 1 - T_{N+1}$. Without fog, all weight sits on the red layer and the depth lands near 3. With fog, $\sum_i w_i t_i$ averages two separate places, so it can fall in empty space; this is how a floater corrupts a depth map.

4. **Add adaptive density control.** Extend the toy splatting snippet: every 100 iterations, find the Gaussians with the largest accumulated position gradient; clone the small ones and split the large ones (divide the scales by 1.6), and remove those whose opacity falls below 0.02. Compare the final PSNR with the fixed 40 Gaussians, and with simply starting from 80 Gaussians.

   Hint

   When the arrays grow, the Adam moment arrays `m` and `v2` must grow with them (initialize the new rows at zero). For reference, simply starting from 80 Gaussians instead of 40, with nothing else changed, reached 27.00 dB at iteration 500 in our run (23.57 dB with 40). Densification should place the new Gaussians where the error is, for example at the triangle's corners.

5. **Chamfer versus F-score.** In the marching-cubes snippet (grid 48), first add 1,000 "floater" points near $(0, 0, 0.6)$, then instead delete the half of the mesh samples with $x < 0$. Recompute accuracy, completeness, Chamfer distance and F-score. Which metric reacts to which defect?

Hint

In our run, the floaters (1% of the points) raised accuracy from 0.0040 to 0.0097 and the Chamfer distance from 0.0040 to 0.0068, while the F-score at 0.01 barely moved (0.995 to 0.990). Deleting half the surface left accuracy at 0.0040 but raised completeness to 0.2226, the Chamfer distance to 0.1133, and lowered the F-score to 0.665. Averages are sensitive to a few far outliers; thresholded scores are not, but they report missing parts clearly through recall.

6. **Window size in plane sweep.** Run the plane-sweep snippet with SSD windows of 3, 7 and 15 pixels. How do the textured and textureless scores change, and what happens to the edges of the square?

Hint

In our run the textured score was 95.2%, 95.3% and 96.6%, and the textureless score 4.9%, 14.8% and 27.7%. Large windows reach texture outside the flat patch, so its pixels borrow depth from their surroundings; the price is that depth edges are blurred and the square appears "fattened" into the background. Learned MVS uses a 3D CNN to get this context without the blur.

The scene models built here are also a building block of the world models in DLCV 07, where an agent must predict how a 3D environment changes.

Supplementary notes

These notes are optional: each one takes a single point from the main text and goes into finer detail. Nothing in the main text, code or exercises depends on them.

The normalized eight-point algorithm

This note explains how to estimate the fundamental matrix linearly from eight or more matches, and why rescaling the pixel coordinates first is essential.

The linear system. Write F as a vector $\mathbf{f} = (F_{11}, F_{12}, \dots, F_{33})$. A match $\mathbf{x}_1 = (u_1, v_1, 1)$, $\mathbf{x}_2 = (u_2, v_2, 1)$ satisfies $\mathbf{x}_2^\top F \mathbf{x}_1 = 0$, which is one linear equation $\mathbf{a}^\top \mathbf{f} = 0$ with $\mathbf{a} = \mathbf{x}_2 \otimes \mathbf{x}_1$ (all nine products $u_2u_1, u_2v_1, u_2, v_2u_1, \dots, 1$). Stacking $n \geq 8$ matches gives $A\mathbf{f} = \mathbf{0}$ with $A \in \mathbb{R}^{n \times 9}$. The least-squares solution with $\|\mathbf{f}\| = 1$ is the right singular vector of A for its smallest singular value [7], [8].

The rank-2 step. A true fundamental matrix has rank 2 (all epipolar lines pass through one point, the epipole). The linear solution usually has rank 3. Replace it by the closest rank-2 matrix in the

Frobenius norm: compute $F = U\,\mathrm{diag}(\sigma_1, \sigma_2, \sigma_3)V^\top$ and set $\sigma_3 = 0$ [8]. For an essential matrix the two non-zero singular values must also be equal, which is why the main-text code uses $\mathrm{diag}(1, 1, 0)$.

Why normalize. In pixel coordinates the entries of $\mathbf{a}$ range from about 1 to about 10^5 (u_2u_1 for pixels in the hundreds). The matrix $A^\top A$ is then badly conditioned (see condition number in ML 01c), and noise is amplified in the small singular vector. Hartley [8] showed that a simple fix makes the linear algorithm competitive: translate each image's points so that their centroid is at the origin and scale them so that their mean distance from the origin is $\sqrt{2}$, with a similarity T_1 for image 1 and T_2 for image 2. Solve for $\hat{F}$ in the normalized coordinates, enforce rank 2, then undo the normalization with $F = T_2^\top \hat{F}\, T_1$.

A test. The snippet compares both versions on noisy matches (1 px noise) and measures the Sampson error, a first-order approximation of the geometric distance of a match to the epipolar constraint [1], on 1,000 fresh noise-free matches.

```
import numpy as np

rng = np.random.default_rng(1)
K = np.array([[800.0, 0, 320], [0, 800.0, 240], [0, 0, 1]])
a = np.deg2rad(12)
R = np.array([[np.cos(a), 0, -np.sin(a)], [0, 1, 0], [np.sin(a), 0, np.cos(a)]])  # camera 2
 ↪  rotation
t = np.array([-1.0, 0.1, 0.1])                                                    # camera
 ↪  2 translation
X = rng.uniform([-3, -2, 4], [3, 2, 8], (100, 3))
proj = lambda P: (P @ K.T)[:, :2] / (P @ K.T)[:, 2:]
x1 = proj(X) + rng.normal(0, 1.0, (100, 2))           # 1 px noise
x2 = proj(X @ R.T + t) + rng.normal(0, 1.0, (100, 2))

def eight_point(x1, x2, normalize=True):
    """Fundamental matrix from >= 8 matches (pixels); x2h^T F x1h = 0."""
    def T_of(x):                                      # move centroid to 0, mean distance to
     ↪  sqrt(2)
        c = x.mean(0); s = np.sqrt(2) / np.linalg.norm(x - c, axis=1).mean()
        return np.array([[s, 0, -s * c[0]], [0, s, -s * c[1]], [0, 0, 1]])
    T1, T2 = (T_of(x1), T_of(x2)) if normalize else (np.eye(3), np.eye(3))
    a = np.c_[x1, np.ones(len(x1))] @ T1.T
    b = np.c_[x2, np.ones(len(x2))] @ T2.T
    A = np.einsum('ni,nj->nij', b, a).reshape(-1, 9)
    F = np.linalg.svd(A)[2][-1].reshape(3, 3)
    U, S, Vt = np.linalg.svd(F)
    F = U @ np.diag([S[0], S[1], 0.0]) @ Vt           # enforce rank 2
    F = T2.T @ F @ T1                                 # undo the normalization
    return F / np.linalg.norm(F)

def sampson(F, x1, x2):
    """First-order geometric distance of each match to the epipolar constraint (pixels)."""
    a, b = np.c_[x1, np.ones(len(x1))], np.c_[x2, np.ones(len(x2))]
    Fa, Ftb = a @ F.T, b @ F
    num = np.sum(b * Fa, axis=1) ** 2
```

```
    return np.sqrt(num / (Fa[:, 0] ** 2 + Fa[:, 1] ** 2 + Ftb[:, 0] ** 2 + Ftb[:, 1] ** 2))

tx = np.array([[0, -t[2], t[1]], [t[2], 0, -t[0]], [-t[1], t[0], 0]])
F_true = np.linalg.inv(K).T @ tx @ R @ np.linalg.inv(K)       # F = K^-T [t]x R K^-1
F_true /= np.linalg.norm(F_true)
X_test = rng.uniform([-3, -2, 4], [3, 2, 8], (1000, 3))       # fresh, noise-free test
 ↪  matches
y1, y2 = proj(X_test), proj(X_test @ R.T + t)
for n in [8, 15, 100]:
    for norm in [False, True]:
        F = eight_point(x1[:n], x2[:n], norm)
        print(f"{n:3d} matches, normalized={str(norm):5s}: mean Sampson error on test
         ↪  matches "
              f"{sampson(F, y1, y2).mean():8.3f} px")
print(f"true F on the same test matches: {sampson(F_true, y1, y2).mean():.2e} px")
```

```
  8 matches, normalized=False: mean Sampson error on test matches    3.186 px
  8 matches, normalized=True : mean Sampson error on test matches    3.016 px
 15 matches, normalized=False: mean Sampson error on test matches    6.091 px
 15 matches, normalized=True : mean Sampson error on test matches    1.083 px
100 matches, normalized=False: mean Sampson error on test matches    0.632 px
100 matches, normalized=True : mean Sampson error on test matches    0.181 px
true F on the same test matches: 2.34e-14 px
```

With 15 or 100 matches the normalized version is several times more accurate in this run; with the minimum of 8 noisy matches both are poor, which is one reason RANSAC samples are followed by a refit on all inliers. Pitfalls: the normalization must be computed per image, the rank-2 step must happen *before* denormalization, and degenerate configurations (all points on one plane, or no translation between the cameras) leave F undetermined whatever the algorithm.

Linear triangulation (DLT)

This note writes out the linear triangulation used in the main text, explains why its answer is not the optimal one, and measures how the baseline controls its error.

The equations. Write the unknown point in homogeneous form $\tilde{\mathbf{X}} = (X, Y, Z, 1)^\top$ and let $\mathbf{p}_i^{k\top}$ be the k-th row of the projection matrix P_i of camera i. The pinhole model says $\lambda_i (u_i, v_i, 1)^\top = P_i \tilde{\mathbf{X}}$, so the depth is $\lambda_i = \mathbf{p}_i^{3\top} \tilde{\mathbf{X}}$. Substituting it into the first two rows removes the unknown depth and leaves two linear equations per image:

$$A\tilde{\mathbf{X}} = \mathbf{0}, \qquad A = \begin{bmatrix} u_1 \mathbf{p}_1^{3\top} - \mathbf{p}_1^{1\top} \\ v_1 \mathbf{p}_1^{3\top} - \mathbf{p}_1^{2\top} \\ u_2 \mathbf{p}_2^{3\top} - \mathbf{p}_2^{1\top} \\ v_2 \mathbf{p}_2^{3\top} - \mathbf{p}_2^{2\top} \end{bmatrix},$$

where (u_i, v_i) is the observed pixel in image i and A is a 4×4 matrix. This is the direct linear transformation (DLT) method [1]. With noise, A has no exact null vector. The unit vector that

minimizes $\|A\tilde{\mathbf{X}}\|$ is the right singular vector of A for its smallest singular value; dividing it by its last entry gives the Euclidean point. Each additional view adds two more rows to A and nothing else changes.

Why it is not optimal. The quantity minimized, $\|A\tilde{\mathbf{X}}\|$, is an *algebraic* error: it has no direct meaning in pixels, and it changes if the image coordinates are rescaled. With Gaussian pixel noise, the sensible goal is the reprojection error, the sum of squared pixel distances between the projections of $\mathbf{X}$ and the observations. Hartley and Zisserman [1] describe methods that minimize it directly. In practice the DLT answer is the starting point, and bundle adjustment then refines all points and cameras together.

Edge cases. (i) When the baseline is short or the point is far away, the two rays are almost parallel, A is close to rank 2, and the depth becomes very uncertain. (ii) A point at infinity has last homogeneous coordinate 0, so the final division fails; such points must be kept in homogeneous form or removed. (iii) A wrong match can produce a point behind a camera, so after dividing by the last entry, check that the depth $\mathbf{p}_i^{3\top}\tilde{\mathbf{X}}$ is positive in every view (the cheirality check of the main text).

A test. The snippet triangulates one point 6 units in front of camera 1, seen by two cameras with 0.5 px noise, for three baselines. It compares the spread of the depth error with the first-order formula $\delta Z \approx Z^2\delta d/(fB)$ of the main text; the disparity error, a difference of two independent 0.5 px errors, has standard deviation $0.5\sqrt{2}$ px.

```
import numpy as np
import cv2

rng = np.random.default_rng(0)
K = np.array([[800.0, 0, 320], [0, 800.0, 240], [0, 0, 1]])
X_true = np.array([0.3, -0.2, 6.0])                    # one point, 6 units in front of
 ↪  camera 1

def triangulate_dlt(P1, P2, x1, x2):
    """Linear triangulation: four equations A X = 0 in the homogeneous point X (4-vector)."""
    A = np.stack([x1[0] * P1[2] - P1[0], x1[1] * P1[2] - P1[1],
                  x2[0] * P2[2] - P2[0], x2[1] * P2[2] - P2[1]])
    Xh = np.linalg.svd(A)[2][-1]                       # right singular vector, smallest
     ↪  singular value
    return Xh[:3] / Xh[3]                              # homogeneous -> Euclidean

def proj(P, X):
    x = P @ np.append(X, 1.0)
    return x[:2] / x[2]

P1 = K @ np.hstack([np.eye(3), np.zeros((3, 1))])
for B in [2.0, 0.5, 0.1]:                              # camera 2 sits B to the right,
 ↪  same orientation
    P2 = K @ np.hstack([np.eye(3), [[-B], [0.0], [0.0]]])
    dz = []
    for _ in range(2000):
```

```
        x1 = proj(P1, X_true) + rng.normal(0, 0.5, 2)      # 0.5 px noise in each image
        x2 = proj(P2, X_true) + rng.normal(0, 0.5, 2)
        dz.append(triangulate_dlt(P1, P2, x1, x2)[2] - X_true[2])
    predicted = X_true[2] ** 2 * (0.5 * np.sqrt(2)) / (800 * B)   # Z^2 * std(disparity
     ↪  error) / (f B)
    print(f"baseline {B:3.1f}: ray angle {np.degrees(np.arctan2(B, 6)):5.2f} deg | "
          f"depth error std {np.std(dz):.4f} (first-order prediction {predicted:.4f})")
Xcv = cv2.triangulatePoints(P1, P2, x1.reshape(2, 1), x2.reshape(2, 1)).ravel()
print(f"difference from cv2.triangulatePoints: {np.abs(Xcv[:3] / Xcv[3] -
 ↪  triangulate_dlt(P1, P2, x1, x2)).max():.1e}")
```

```
baseline 2.0: ray angle 18.43 deg | depth error std 0.0158 (first-order prediction 0.0159)
baseline 0.5: ray angle  4.76 deg | depth error std 0.0617 (first-order prediction 0.0636)
baseline 0.1: ray angle  0.95 deg | depth error std 0.3166 (first-order prediction 0.3182)
difference from cv2.triangulatePoints: 2.0e-13
```

The measured spread matches the first-order prediction. Shrinking the baseline from 2 to 0.1 units, which makes the angle between the rays smaller than one degree, multiplies the depth error by 20. The last line shows that OpenCV's `cv2.triangulatePoints` returns the same point up to rounding.

Bundle adjustment in code

This note builds a complete small bundle adjustment with SciPy, to show the residual function, the sparsity of the Jacobian and the gauge freedom.

Parameters. Each camera has a rotation vector $\boldsymbol{\omega} \in \mathbb{R}^3$ (axis times angle) and a translation $\mathbf{t} \in \mathbb{R}^3$; each point has three coordinates. With n_c cameras and n_p points, there are $6n_c + 3n_p$ unknowns and two residuals (the u and v errors) per observation.

Gauge. Reprojection errors do not change if the whole scene is rotated, translated or scaled together: seven directions (three for rotation, three for translation, one for scale) leave the cost unchanged. Without fixing them, the normal equations are singular. Here camera 0 is held fixed at the identity, which removes six; the scale stays free and is effectively pinned by the starting point. Real systems fix it explicitly, for example by fixing the distance between two cameras.

Sparsity. A residual of observation (c, k) depends only on camera c and point k, so each pair of Jacobian rows has at most $6+3$ non-zeros. Passing this pattern to `least_squares` lets it use sparse finite differences and a sparse solver.

```
import numpy as np
from scipy.optimize import least_squares
from scipy.sparse import lil_matrix
from scipy.spatial.transform import Rotation
```

```
rng = np.random.default_rng(0)
f, c = 800.0, np.array([320.0, 240.0])
n_cam, n_pts = 4, 80
X_true = rng.uniform([-3, -2, 4], [3, 2, 8], (n_pts, 3))
rvec_true = np.array([[0, 0, 0], [0, -0.15, 0], [0, -0.3, 0.02], [0.05, -0.45, 0]])  #
 ↪  rotation vectors
tvec_true = np.array([[0, 0, 0], [1.0, 0, 0.1], [2.0, 0.1, 0.3], [3.0, 0, 0.6]])

def project(rvec, tvec, X):
    Xc = Rotation.from_rotvec(rvec).apply(X) + tvec
    return f * Xc[:, :2] / Xc[:, 2:] + c

cam_idx = np.repeat(np.arange(n_cam), n_pts)           # every camera sees every point
pt_idx = np.tile(np.arange(n_pts), n_cam)
obs = np.concatenate([project(rvec_true[i], tvec_true[i], X_true) for i in range(n_cam)])
obs += rng.normal(0, 0.5, obs.shape)                    # 0.5 px measurement noise

def residuals(params):
    """Reprojection errors. Camera 0 is held fixed (it defines the world frame)."""
    cams = np.vstack([np.zeros(6), params[:6 * (n_cam - 1)].reshape(-1, 6)])
    X = params[6 * (n_cam - 1):].reshape(-1, 3)
    pred = np.empty_like(obs)
    for i in range(n_cam):
        m = cam_idx == i
        pred[m] = project(cams[i, :3], cams[i, 3:], X[pt_idx[m]])
    return (pred - obs).ravel()

def sparsity():
    """Each residual depends on one camera (6 numbers) and one point (3 numbers)."""
    S = lil_matrix((2 * len(obs), 6 * (n_cam - 1) + 3 * n_pts), dtype=int)
    for k, (i, j) in enumerate(zip(cam_idx, pt_idx)):
        rows = [2 * k, 2 * k + 1]
        if i > 0:
            S[np.ix_(rows, range(6 * (i - 1), 6 * i))] = 1
        S[np.ix_(rows, range(6 * (n_cam - 1) + 3 * j, 6 * (n_cam - 1) + 3 * j + 3))] = 1
    return S

# A rough start, as incremental SfM would provide: perturbed poses and points
x0 = np.concatenate([np.c_[rvec_true[1:] + rng.normal(0, 0.02, (3, 3)),
                           tvec_true[1:] + rng.normal(0, 0.1, (3, 3))].ravel(),
                     (X_true + rng.normal(0, 0.2, X_true.shape)).ravel()])
rms = lambda p: np.sqrt(np.mean(residuals(p) ** 2))
sol = least_squares(residuals, x0, jac_sparsity=sparsity(), x_scale='jac', method='trf')
print(f"unknowns {x0.size}, residuals {2 * len(obs)}, Jacobian non-zeros {sparsity().nnz} "
      f"({100 * sparsity().nnz / (x0.size * 2 * len(obs)):.1f}%)")
print(f"reprojection RMS: start {rms(x0):.2f} px -> after bundle adjustment {rms(sol.x):.3f}
 ↪  px")
X_hat = sol.x[6 * (n_cam - 1):].reshape(-1, 3)
print(f"median 3D point error after BA: {np.median(np.linalg.norm(X_hat - X_true,
 ↪  axis=1)):.4f} units")
```

```
unknowns 258, residuals 640, Jacobian non-zeros 4800 (2.9%)
reprojection RMS: start 32.31 px -> after bundle adjustment 0.388 px
```

```
median 3D point error after BA: 0.0122 units
```

Only 2.9% of the Jacobian is non-zero even in this tiny problem, and the fraction shrinks as problems grow, because each new point adds columns that touch only its own observations. The reprojection error falls from 32.31 px to 0.388 px, just below the 0.5 px noise, as for the two-view fit in the main text. Production solvers go further: they eliminate the points with the Schur complement and solve a reduced system in the camera parameters only [13], and they use a robust loss ρ so that a few wrong matches cannot pull the solution (here `least_squares(..., loss='huber')` would do that).

The plane-induced homography

This note gives the image warp used by plane sweep for general cameras, of which the horizontal shift in the main text is a special case.

Let a plane be $\mathbf{n}^\top \mathbf{X} = d$ in the reference camera's coordinates, with unit normal $\mathbf{n}$ and distance d from the camera centre. Let the source camera be at rotation R and translation $\mathbf{t}$ relative to the reference camera, and let K_r, K_s be the two intrinsic matrices. Then every pixel $\mathbf{x}_r$ of the reference image whose ray hits the plane maps to the source pixel

$$\mathbf{x}_s \sim H\mathbf{x}_r, \qquad H = K_s \left(R + \frac{\mathbf{t}\mathbf{n}^\top}{d} \right) K_r^{-1},$$

where $\sim$ means equality up to a scale factor (homogeneous coordinates) [1]. Derivation in one line: a point on the plane satisfies $\mathbf{n}^\top \mathbf{X}/d = 1$, so $R\mathbf{X} + \mathbf{t} = (R + \mathbf{t}\mathbf{n}^\top/d)\mathbf{X}$, and $\mathbf{X}$ is proportional to $K_r^{-1}\mathbf{x}_r$.

Special cases: for a fronto-parallel plane $\mathbf{n} = (0, 0, 1)$ at depth $d = Z$, identical intrinsics K, no rotation and a sideways translation $\mathbf{t} = (-B, 0, 0)$ (the source camera B to the right), H shifts every pixel left by fB/Z, the disparity used in the snippet. As $d \to \infty$ the homography tends to $K_s R K_r^{-1}$, the *infinite homography*, which depends on rotation only: very distant points do not reveal depth. Pitfalls: the homography direction (reference to source or the reverse) must match how the warp samples the image, and depth hypotheses are best spaced uniformly in $1/Z$, because disparity, not depth, is what changes linearly in the image.

The multiresolution hash encoding

This note gives the details of the Instant-NGP encoding summarized in the main text: how the grid resolutions are chosen, how grid corners are hashed, and how many parameters the encoding has.

Levels. The encoding has L levels. Level l, for $l = 0, \dots, L-1$, is a grid of resolution

$$N_l = \left\lfloor N_{\min} b^l \right\rfloor, \qquad b = \exp\!\left(\frac{\ln N_{\max} - \ln N_{\min}}{L - 1} \right),$$

where $N_{\min}$ and $N_{\max}$ are the coarsest and finest resolutions, $\lfloor \cdot \rfloor$ rounds down, and b is the growth factor, so the resolutions form a geometric progression [19]. The paper's typical values are $L = 16$ levels, $F = 2$ features per table entry, T between 2^{14} and 2^{24} entries per level, $N_{\min} = 16$, and $N_{\max}$ between 512 and 524,288, chosen to match the finest detail in the data [19].

Lookup. On level l, the input point is scaled by N_l, and the grid cell that contains it has 2^d corners with integer coordinates ($d = 3$ in 3D). If the whole grid fits in the table, that is $(N_l + 1)^d \le T$, every corner gets its own entry. Otherwise a corner $\mathbf{x} = (x_1, \dots, x_d)$ uses the entry

$$h(\mathbf{x}) = \left(\bigoplus_{i=1}^{d} x_i \pi_i \right) \bmod T,$$

where $\oplus$ is bitwise XOR and the constants are $\pi_1 = 1$, $\pi_2 = 2\,654\,435\,761$ and $\pi_3 = 805\,459\,861$ [19]. Only $d - 1$ of the dimensions need a large prime to decorrelate them, and $\pi_1 = 1$ keeps corners that are neighbours along the first axis next to each other in memory [19]. The features of the 2^d corners are interpolated linearly in each dimension (trilinearly in 3D) according to the point's position inside the cell. The interpolated vectors of all L levels are concatenated, together with any extra inputs such as the encoded viewing direction, and passed to the MLP [19].

Collisions. On fine levels many corners share one entry, and the encoding does not detect or resolve this. The gradients of colliding corners are averaged, so the corners with the largest gradients, those most relevant to the loss, dominate the shared entry; the authors argue that the table therefore concentrates on the sparse regions with the most important fine detail, and that the multiresolution structure lets the network tell collisions apart [19].

Parameter count. The tables hold at most $T \cdot L \cdot F$ trainable numbers, whatever $N_{\max}$ is [19]. Memory is fixed by T, not by the finest resolution.

A worked example. The snippet computes the levels for $N_{\max} = 2048$ and $T = 2^{19}$, values inside the paper's typical ranges, and hashes three neighbouring corners.

```
import numpy as np

L, F, T = 16, 2, 2 ** 19                                  # levels, features per entry, table
 ↪  size
N_min, N_max = 16, 2048                                   # coarsest and finest grid
 ↪  resolution
b = np.exp((np.log(N_max) - np.log(N_min)) / (L - 1))   # growth factor between levels
N = np.floor(N_min * b ** np.arange(L)).astype(int)     # resolution of each level
dense = (N + 1) ** 3 <= T                                 # does the full grid fit in the
 ↪  table?
print(f"growth factor b = {b:.3f}; resolutions {N.tolist()}")
print(f"levels stored 1:1 (no hashing): {dense.sum()}, hashed levels: {(~dense).sum()}")
print(f"finest level: {(N[-1] + 1) ** 3:,} grid vertices share {T:,} entries "
      f"(about {(N[-1] + 1) ** 3 / T:,.0f} per entry)")
print(f"trainable encoding parameters: at most T * L * F = {T * L * F:,}")

PI = np.array([1, 2654435761, 805459861], dtype=np.uint64)     # the primes of the paper
 ↪  (pi_1 = 1)
```

```
def spatial_hash(corner):
    """h(x) = (x_1 pi_1  XOR  x_2 pi_2  XOR  x_3 pi_3) mod T for an integer grid corner."""
    return int(np.bitwise_xor.reduce(np.asarray(corner, dtype=np.uint64) * PI) %
     ↪  np.uint64(T))
for corner in [(100, 200, 300), (101, 200, 300), (100, 201, 300)]:
    print(f"corner {corner} -> table entry {spatial_hash(corner)}")
```

```
growth factor b = 1.382; resolutions [16, 22, 30, 42, 58, 80, 111, 153, 212, 294, 406, 561,
 ↪  776, 1072, 1482, 2048]
levels stored 1:1 (no hashing): 5, hashed levels: 11
finest level: 8,602,523,649 grid vertices share 524,288 entries (about 16,408 per entry)
trainable encoding parameters: at most T * L * F = 16,777,216
corner (100, 200, 300) -> table entry 110768
corner (101, 200, 300) -> table entry 110769
corner (100, 201, 300) -> table entry 10497
```

With these settings the five coarsest levels fit their tables directly and the other eleven are hashed. On the finest level, about 16,000 grid corners share each table entry, which is workable only because most of them never receive a large gradient. The last three lines show the effect of $\pi_1 = 1$: a step along the first axis moves to the next entry, while a step along the second axis jumps far away.

Why NeuS weights peak on the surface

This note measures, on one ray, the bias that NeuS was designed to remove.

Take a ray that meets a plane at $t^* = 3$ at an angle, so the signed distance along the ray is $f(t) = -\cos\theta\,(t - t^*)$ with $\cos\theta = 0.5$ (positive in front of the surface, negative behind). The *naive* choice uses the logistic density $\phi_s(f(t))$, the derivative of the sigmoid Φ_s, as NeRF's volume density σ. The weight $w(t) = T(t)\sigma(t)$ then peaks before t^*: transmittance T is already falling while σ is still rising, and their product peaks early. NeuS defines instead an opaque density $\rho(t) = \max(-\frac{d\Phi_s(f)/dt}{\Phi_s(f)}, 0)$, whose discrete form is the α_i formula of the main text; for a plane the resulting weight is exactly the normalized logistic density centred on the surface [22].

```
import numpy as np

t = np.linspace(0, 6, 6001)                              # dense samples along one ray
t_star, cos_theta = 3.0, 0.5                             # the ray meets a plane at t* = 3, at
 ↪  60 degrees
f = -cos_theta * (t - t_star)                            # signed distance along the ray
 ↪  (positive outside)

def weights_from_alpha(alpha):
    T = np.concatenate([[1.0], np.cumprod(1 - alpha)[:-1]])
    return T * alpha

for s in [10.0, 40.0]:
```

```
    Phi = 1 / (1 + np.exp(-s * f))                    # sigmoid of the SDF
    phi = s * np.exp(-s * f) / (1 + np.exp(-s * f)) ** 2   # logistic density (derivative of
     ↪  Phi)
    delta = np.diff(t, append=t[-1] + (t[1] - t[0]))
    w_naive = weights_from_alpha(1 - np.exp(-phi * delta))                    # density =
     ↪  phi(SDF)
    alpha_neus = np.clip((Phi[:-1] - Phi[1:]) / Phi[:-1], 0, 1)              # NeuS
     ↪  discrete opacity
    w_neus = weights_from_alpha(np.append(alpha_neus, 0.0))
    for name, w in [("naive", w_naive), ("NeuS ", w_neus)]:
        print(f"s = {s:4.0f}, {name}: weight peak at t = {t[w.argmax()]:.3f}, "
              f"mean depth {np.sum(w * t) / w.sum():.3f} (surface at 3.000)")
```

```
s =   10, naive: weight peak at t = 2.824, mean depth 2.810 (surface at 3.000)
s =   10, NeuS : weight peak at t = 2.999, mean depth 2.999 (surface at 3.000)
s =   40, naive: weight peak at t = 2.956, mean depth 2.953 (surface at 3.000)
s =   40, NeuS : weight peak at t = 3.000, mean depth 2.999 (surface at 3.000)
```

The naive weights peak 0.18 units early at $s = 10$ and 0.04 units early at $s = 40$: the bias shrinks as the transition sharpens, but never vanishes. The NeuS weights peak on the surface for both. The paper proves unbiasedness to first order, that is, for locally planar surfaces [22]; on strongly curved surfaces, or where a ray grazes a surface, small biases remain. A practical pitfall: s is learned, and if it is initialized too large, the weights are concentrated in a very narrow band that rays rarely sample, so gradients are weak early in training.

Projecting a 3D Gaussian: the local affine approximation

This note writes out the Jacobian used by EWA splatting and 3DGS, and measures how good the approximation is.

The perspective projection $\pi(x, y, z) = (f_x x/z,\ f_y y/z)$ is not linear, so the image of a Gaussian is not exactly a Gaussian. EWA splatting [23] replaces π near the Gaussian's centre $\boldsymbol{\mu}_c = (x, y, z)$ (in camera coordinates) by its first-order Taylor expansion. The Jacobian is

$$J = \frac{\partial \pi}{\partial (x, y, z)} = \begin{bmatrix} f_x/z & 0 & -f_x x/z^2 \\ 0 & f_y/z & -f_y y/z^2 \end{bmatrix},$$

where f_x, f_y are the focal lengths in pixels. A linear map sends a Gaussian with covariance Σ_c to a Gaussian with covariance $J\Sigma_c J^\top$, which is the 2×2 block of $\Sigma' = JW\Sigma W^\top J^\top$ in the main text, with $\Sigma_c = W\Sigma W^\top$ [6], [23]. The snippet compares this with the exact covariance of many projected samples, for the same Gaussian at three depths.

```
import numpy as np
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(0)
fx = fy = 500.0                                       # focal lengths (pixels)
```

```
def project_cov(mu_c, Sigma_c):
    """2D covariance of a 3D Gaussian (already in camera coordinates) under the local affine
    approximation: Sigma' = J Sigma J^T, with J the Jacobian of (x, y, z) -> (fx x/z, fy
     ↪  y/z) at the mean."""
    x, y, z = mu_c
    J = np.array([[fx / z, 0, -fx * x / z ** 2],
                  [0, fy / z, -fy * y / z ** 2]])
    return J @ Sigma_c @ J.T

S = np.diag([0.30, 0.10, 0.05])                          # axis lengths (standard deviations)
Rm = Rotation.from_euler('xyz', [20, 40, 10], degrees=True).as_matrix()
Sigma = Rm @ S @ S.T @ Rm.T                              # Sigma = R S S^T R^T
for z in [2.0, 4.0, 8.0]:
    mu = np.array([0.5, -0.3, z])
    approx = project_cov(mu, Sigma)
    P = rng.multivariate_normal(mu, Sigma, 200_000)    # Monte Carlo: project samples exactly
    uv = np.c_[fx * P[:, 0] / P[:, 2], fy * P[:, 1] / P[:, 2]]
    exact = np.cov(uv.T)
    err = np.linalg.norm(approx - exact) / np.linalg.norm(exact)
    print(f"depth {z:.0f}: std along axes (approx)
     ↪  {np.sqrt(np.linalg.eigvalsh(approx))[::-1].round(2)} px, "
          f"(sampled) {np.sqrt(np.linalg.eigvalsh(exact))[::-1].round(2)} px, relative error
           ↪  {100 * err:.1f}%")
```

```
depth 2: std along axes (approx) [68.94 25.13] px, (sampled) [71.9  25.53] px, relative
 ↪  error 8.0%
depth 4: std along axes (approx) [31.69 12.28] px, (sampled) [31.93 12.3 ] px, relative
 ↪  error 1.5%
depth 8: std along axes (approx) [15.19  6.05] px, (sampled) [15.22  6.05] px, relative
 ↪  error 0.5%
```

The error is small when the Gaussian is small compared with its distance (0.5% at depth 8) and grows as it comes closer (8.0% at depth 2), because the projection is less linear over a larger angle. In 3DGS this matters for large Gaussians near the camera and at wide fields of view; the paper's limitations section mentions popping and "splotchy" artifacts from large Gaussians [6], and the 2DGS paper points out that the affine approximation is accurate only near the image centre and that 3D Gaussians give depth that is inconsistent across viewpoints; it replaces both with an explicit ray–splat intersection [28]. Two pitfalls when implementing it: J must be evaluated in camera coordinates (after applying W), and a Gaussian whose centre is behind or very close to the camera ($z \to 0$) must be culled, since J blows up.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

NeRF (Mildenhall et al., 2020)

The problem. Given a set of photos with known cameras, synthesize photorealistic images from new viewpoints of a complex real scene. Earlier methods either blended nearby photos (artifacts away from the inputs) or optimized discrete voxel grids (memory-limited resolution) [5].

The idea. Represent the scene as a continuous 5D function (3D position plus 2D viewing direction) stored in an MLP, render it with classical, differentiable volume rendering, and fit it to the photos with a plain squared colour loss. Two ingredients make it work: the positional encoding of the inputs, and hierarchical sampling with a coarse and a fine network [5].

What it showed. On synthetic objects and on real forward-facing scenes, NeRF outperformed the neural and image-based rendering methods it compared against in PSNR and SSIM; on LPIPS it was best except on the real scenes, where one image-based baseline was slightly better [5]. It also reproduced view-dependent effects such as specular highlights [5]. The ablation table shows the contribution of each component: removing the positional encoding or the view dependence clearly lowers quality [5].

How to read it. Check what the inputs are: real scenes use COLMAP poses [5], so results include SfM quality. Note the cost: about 1–2 days of training per scene on one GPU [5], which the paper states openly; most follow-up work attacks this. Notice what is *not* claimed: NeRF targets images, not accurate surfaces, and it assumes a static scene with constant lighting. The communications version [5] (CACM) summarizes the same method for a broad audience.

Why it matters here. NeRF set the template used by nearly everything after it in this tutorial: a differentiable scene representation trained by comparing renderings with photos.

3D Gaussian Splatting (Kerbl et al., 2023)

The problem. Radiance fields gave the best image quality, but either trained and rendered slowly (high quality) or traded quality for speed; none reached real-time rendering at 1080p for complete, unbounded scenes [6].

The idea. Start from the SfM points, represent the scene with anisotropic 3D Gaussians that keep the useful properties of a volumetric field, optimize them with interleaved densification and pruning, and render them with a tile-based, visibility-aware rasterizer that sorts once per image [6].

What it showed. Quality comparable to the best previous method (Mip-NeRF 360) with training times competitive with the fastest methods, and real-time rendering, on the Mip-NeRF 360 and Tanks and Temples scenes among others [6]. The ablations isolate initialization from SfM, densification by cloning and splitting, anisotropic covariances, letting an unlimited number of

Gaussians per pixel receive gradients, and spherical harmonics [6].

How to read it. Separate the representation (Gaussians with spherical-harmonic colour) from the engineering (the rasterizer and explicit gradients). Look at the train/test protocol (every 8th image) and at which baseline numbers were re-run by the authors and which were copied [6]. Read the limitations: artifacts in poorly observed regions, elongated or "splotchy" Gaussians, popping, no regularization of the optimization, and memory consumption significantly higher than NeRF-based methods [6].

Why it matters here. It replaced ray marching with sorting and blending, and it is the starting point of the mesh-extraction, dynamic-scene and feed-forward Gaussian work in this tutorial.

Guide to the Modern view surveys

This guide expands the survey summary in the Modern view: what each review covers, how it is organized and what to take from it.

A Comparison and Evaluation of Multi-View Stereo Reconstruction Algorithms (Seitz, Curless, Diebel, Scharstein and Szeliski, CVPR 2006) [42] surveys MVS algorithms with a six-property taxonomy (scene representation, photo-consistency measure, visibility model, shape prior, reconstruction algorithm, initialization requirements), and introduces calibrated datasets with high-accuracy ground truth and an evaluation methodology [42]. Read it for the vocabulary: almost every later method, including neural ones, can be placed in its six columns.

Multi-View Stereo: A Tutorial (Furukawa and Hernández, *Foundations and Trends in Computer Graphics and Vision*, 2015) [43] is a practical, algorithm-focused monograph. It frames MVS as image–geometry consistency optimization, explains photometric consistency measures and optimization methods in detail, shows how successful algorithms and products combine them, and ends with structural priors and open challenges [43]. Read it to understand what learned MVS replaced and why.

Advances in Neural Rendering (Tewari et al., *Computer Graphics Forum*, 2022, Eurographics state-of-the-art report) [44] organizes neural rendering into fundamentals (scene representations, differentiable image formation, optimization) and applications (novel-view synthesis of static content, generalization over scenes, non-static content, compositionality and editing, relighting and materials, light fields), followed by open challenges and social implications [44]. It predates 3DGS, so read it for the NeRF era and its open questions.

A Survey on 3D Gaussian Splatting (Chen and Wang, *ACM Computing Surveys*, 2026) [45] covers the background of radiance fields, the principles and optimization of 3DGS, seven directions that improve it (sparse input, memory efficiency, photorealism, optimization, extra properties, hybrid representations, new rendering algorithms), application areas and a performance comparison, followed by open questions [45]. Its future-directions section describes the move from per-scene

optimization to feed-forward Gaussian prediction and toward Gaussian-based world models [45]. Use its taxonomy to place new 3DGS papers.

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DLCV 08A • Depth Estimation and Optical Flow

Dense correspondence for depth and motion: rectified stereo and disparity, matching costs, semi-global matching and learned cost volumes, monocular relative and metric depth with scale-and-shift-invariant losses, self-supervised depth from video, optical flow from FlowNet to RAFT, scene flow, and the metrics, benchmarks and failure modes of all three, with tested NumPy and OpenCV code.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-08a-depth-optical-flow/

What you'll learn

- Why stereo depth, monocular depth and optical flow are three versions of one problem, **dense correspondence**, and what each one can and cannot recover.
- How a rectified stereo pair turns depth into a horizontal shift, $Z = fB/d$, and why depth error grows with the square of distance.
- The classical stereo pipeline (matching cost, cost volume, aggregation with semi-global matching, winner-take-all, left–right check) and the learned versions (MC-CNN, GC-Net, PSMNet, RAFT-Stereo, FoundationStereo), with a tested block-matching experiment.
- Why depth from one image is ambiguous in scale, the difference between relative and metric depth, how scale-and-shift-invariant losses work, and what MiDaS, Depth Anything, Marigold, Metric3D and Depth Pro do; plus self-supervised depth from video (Monodepth2).
- How optical flow became a learning problem (FlowNet, PWC-Net, RAFT and its successors), scene flow in brief, and how depth, stereo and flow are evaluated (AbsRel, $\delta < 1.25$, bad-τ, D1, EPE, Fl-all) on KITTI, Middlebury, ETH3D, Sintel and NYU Depth v2.

The big picture

A depth map gives, for every pixel, how far away the surface is. An optical flow field gives, for every pixel, where that point moves in the next frame. Robots, cars, phones and video editors need both: depth to know where things are, flow to know how they move. Both are answers to the same question: *which pixel in the other image shows the same point?*

> **Explain like I'm five**
>
> Hold a pencil in front of your face and close one eye, then the other. The pencil jumps sideways, while the window far behind it hardly moves. How far something jumps tells you how close it is. A stereo camera does the same with two lenses, and a computer measures the jump for every pixel. If you have only one eye, you can still guess distances, because you know how big a car or a door usually is. But you can be fooled by a toy car close up. Optical flow is the same jumping game played over time: compare this video frame with the next one and measure how far each bit moved.

Why it matters. Stereo depth lets robots and cars measure distance without a laser scanner; single-image depth blurs portrait backgrounds on phones and places objects into video; optical flow drives video super-resolution, frame interpolation and stabilization.

How this tutorial relates to others. DLCV 08 covers camera geometry, triangulation and multi-view stereo; DLCV 04B introduces brightness constancy and warping; long-range point tracking is in DLCV 06 and sparse keypoint matching in IP 12. We link to them instead of repeating them. Our focus is the *dense* case: one answer per pixel.

Figure 1 shows the four model families of this tutorial: three match two images, the fourth relies on what it has learned.

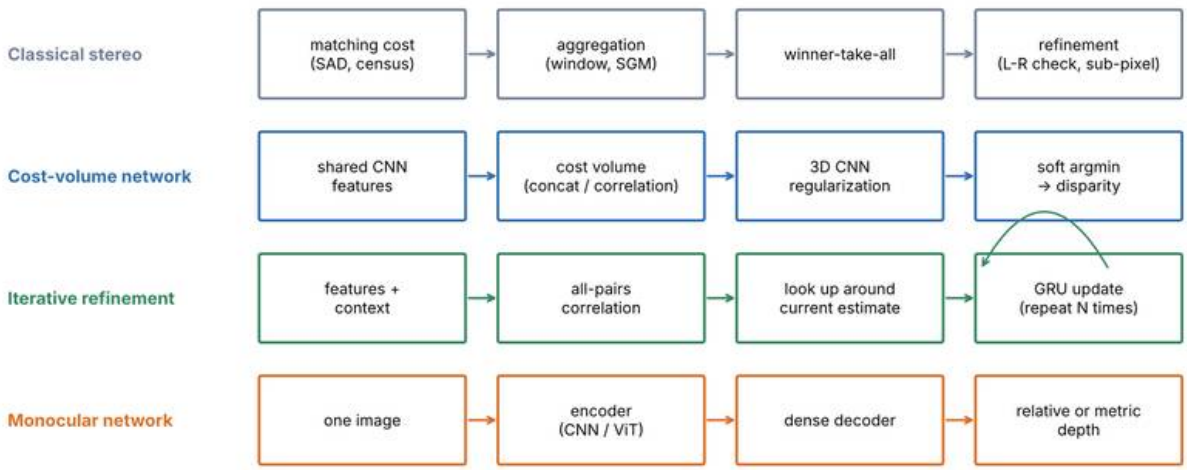


Figure 1 — Four families of dense correspondence and depth models. Classical stereo (grey) and cost-volume networks (blue) follow the same four stages; iterative models (green) repeat a lookup-and-update step; monocular networks (orange) predict depth from a single image.

Three dense correspondence problems

Plain version. Stereo matching finds, for each pixel of the left image, the matching pixel in the right image, which lies on the same row. Optical flow does the same between two frames of a video, but the match can be anywhere nearby. Monocular depth has no second image at all and must predict depth from appearance.

Precise version. Let I_1, I_2 be two images of $H \times W$ pixels and $\mathbf{p} = (u, v)$ a pixel, with column u and row v.

- **Optical flow** is a vector field $\mathbf{F}(\mathbf{p}) = (\Delta u, \Delta v)$ such that the point seen at $\mathbf{p}$ in I_1 is seen at $\mathbf{p}+\mathbf{F}(\mathbf{p})$ in I_2. Two unknowns per pixel; the cameras and the scene may both move.

- **Stereo disparity** is the special case of two cameras that see the scene at the same instant and have been *rectified* (next section), so that every match lies on the same row: $\mathbf{F}(\mathbf{p}) = (-d(\mathbf{p}), 0)$. One unknown per pixel, the disparity $d \geq 0$, which converts to depth.
- **Monocular depth** asks for a depth map $Z(\mathbf{p})$ from I_1 alone. No correspondence is available, so the answer comes from learned regularities, and it is only defined up to an unknown scale unless the network has learned metric cues.
- **Scene flow** asks for the 3D motion of the point seen at each pixel; it combines depth at two times with optical flow (see Scene flow below).

The two-image problems share one structure: a **cost** that scores a candidate match, a search over candidates, and a way to resolve pixels where the cost is ambiguous.

Stereo geometry: rectification and disparity

Plain version. Two cameras side by side, pointing the same way, see a nearby point at noticeably different columns and a far point at almost the same column. The column difference, the disparity, is all we need to compute depth.

Rectification

For a general pair of cameras, the match of a pixel lies on its **epipolar line** in the other image (DLCV 08, epipolar geometry). Rectification warps both images with a homography each so that all epipolar lines become horizontal and corresponding lines have the same row index [1]. After rectification the two virtual cameras have the same intrinsics, parallel optical axes and a purely horizontal baseline. Stereo rigs are calibrated once, and every frame is rectified before matching; all the stereo benchmarks below distribute rectified pairs. In the rest of this tutorial the pair is always rectified.

Disparity and depth

Place the left camera centre at the origin and the right camera at distance B along the x axis (Figure 2a). A point at depth Z and horizontal position X projects to column $x_L = fX/Z$ in the left image and $x_R = f(X - B)/Z$ in the right image, both measured from the principal point. Subtracting gives

$$d = x_L - x_R = \frac{fB}{Z}, \qquad Z = \frac{fB}{d},$$

where f is the focal length in pixels, B is the **baseline** (distance between the camera centres, in

metres), Z is the depth along the optical axis and d is the **disparity** in pixels. Depth and disparity are inversely proportional: disparity is an *inverse depth*, scaled by fB. Large disparities belong to near objects, and a disparity of zero means a point at infinity.

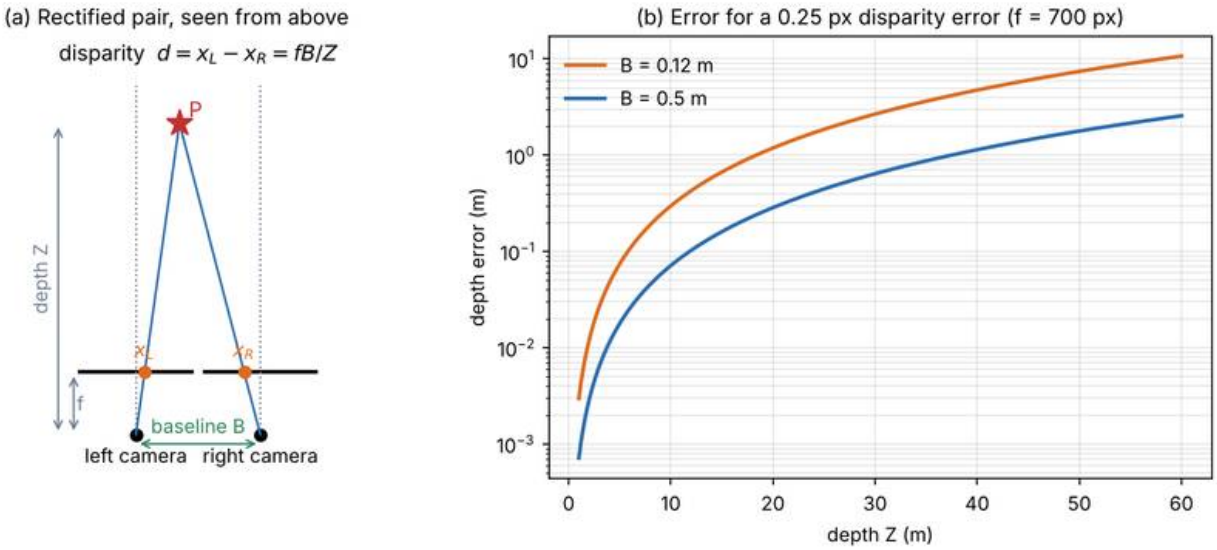


Figure 2 — (a) A rectified stereo pair seen from above: a point shifts by the disparity d between the two images. (b) Depth error for a disparity error of 0.25 px with a focal length of 700 px; the error grows with the square of the distance and shrinks with a wider baseline.

Because $Z = fB/d$, a small disparity error δd becomes a depth error of about $\delta Z \approx Z^2 \delta d/(fB)$, the same relation that DLCV 08, Triangulation derives. The snippet puts numbers on it for a hypothetical camera with $f = 700$ px.

```
import numpy as np

f = 700.0                                   # focal length in pixels (a hypothetical rig)
for B in [0.12, 0.5]:                       # baselines in metres
    print(f"baseline {B} m: depth at a disparity of 1 px = {f * B:.0f} m")
    for Z in [2, 10, 30, 60]:
        d = f * B / Z                       # disparity in pixels
        dZ = Z ** 2 * 0.25 / (f * B)        # depth error for a 0.25 px disparity error
        print(f"   Z = {Z:2d} m  d = {d:6.2f} px  depth error for 0.25 px: {dZ:6.3f} m ({100
         ↪  * dZ / Z:4.1f}%)")
```

```
baseline 0.12 m: depth at a disparity of 1 px = 84 m
   Z =  2 m  d =  42.00 px  depth error for 0.25 px:  0.012 m ( 0.6%)
   Z = 10 m  d =   8.40 px  depth error for 0.25 px:  0.298 m ( 3.0%)
   Z = 30 m  d =   2.80 px  depth error for 0.25 px:  2.679 m ( 8.9%)
   Z = 60 m  d =   1.40 px  depth error for 0.25 px: 10.714 m (17.9%)
baseline 0.5 m: depth at a disparity of 1 px = 350 m
   Z =  2 m  d = 175.00 px  depth error for 0.25 px:  0.003 m ( 0.1%)
```

```
Z = 10 m  d =  35.00 px  depth error for 0.25 px:  0.071 m ( 0.7%)
Z = 30 m  d =  11.67 px  depth error for 0.25 px:  0.643 m ( 2.1%)
Z = 60 m  d =   5.83 px  depth error for 0.25 px:  2.571 m ( 4.3%)
```

At 10 m, a quarter-pixel disparity error costs 30 cm with a 12 cm baseline and 7 cm with a 50 cm baseline. At 60 m, the short baseline sees a disparity of only 1.4 px and its depth is uncertain by 18%. This is why stereo cameras for driving use wide baselines and high resolution, and why sub-pixel accuracy matters: the last fraction of a pixel decides the far range.

Classical stereo matching

Plain version. For each pixel of the left image, slide along the same row of the right image, compare small patches, and keep the shift that matches best. Then clean up the answers that are clearly wrong.

Scharstein and Szeliski organized dense two-frame stereo algorithms into four steps that most methods perform in some form [2]:

1. **Matching cost computation**: a per-pixel, per-disparity dissimilarity.
2. **Cost (support) aggregation**: combine costs over a neighbourhood to reduce noise.
3. **Disparity computation or optimization**: choose a disparity per pixel, either locally or by optimizing an energy over the whole image.
4. **Disparity refinement**: sub-pixel estimates, consistency checks, filling and smoothing.

The Middlebury stereo website, which accompanies that paper, still hosts an online evaluation [2], [3]. The plane sweep in DLCV 08 is exactly steps 1–3 with windowed SSD; here we look at each step more closely and add the refinement.

Matching costs and the cost volume

The simplest cost compares intensities in a small window. The **sum of absolute differences** (SAD) at pixel (u, v) and disparity d is

$$C(u, v, d) = \sum_{(i,j) \in \mathcal{W}} |I_L(u + i, v + j) - I_R(u + i - d, v + j)|,$$

where I_L, I_R are the left and right images, $\mathcal{W}$ is the set of offsets of a $w \times w$ window centred on the pixel, and d ranges over integer hypotheses $0, 1, \ldots, D - 1$. Squared differences give SSD. Computing C for every pixel and every d fills a **cost volume** of size $D \times H \times W$. The **winner-take-all** (WTA) rule picks $\hat{d}(u, v) = \arg\min_d C(u, v, d)$.

SAD assumes both cameras record the same brightness for the same point. Real cameras differ in gain and exposure, so practical systems use **zero-mean normalized cross-correlation** (ZNCC) or the **census transform**, which encodes which neighbours are darker than the pixel as a bit string and compares bit strings [4].

Optional: Robust matching costs: ZNCC and census (in the Supplementary notes). The definitions, and a test in which a brightness change between the views breaks SAD but not the other two.

Figure 3a shows one row of a cost volume: dark valleys trace the surfaces. Figure 3b shows four typical cost curves. A textured pixel has one sharp minimum; a textureless pixel a flat curve; a repetitive pattern several equal minima, one per period; and an occluded pixel, hidden in the right image, no correct minimum at all.

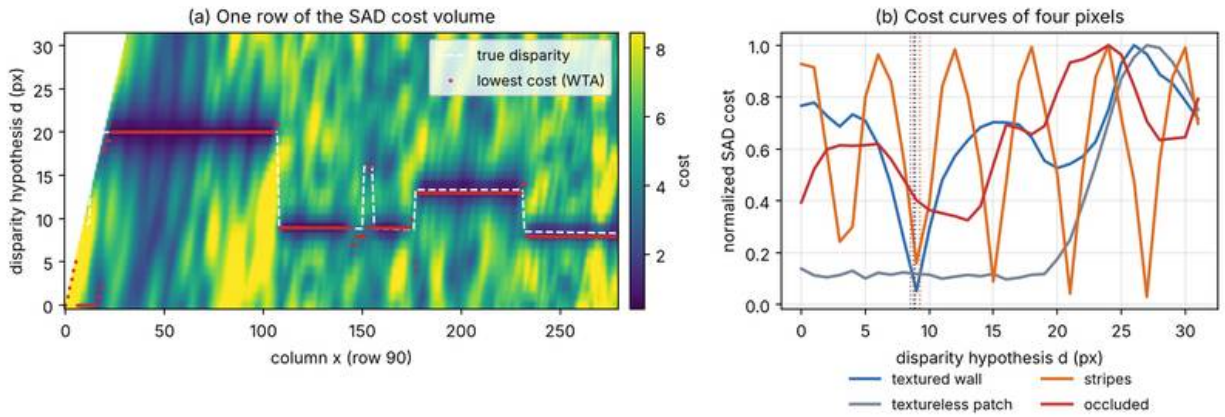


Figure 3 — (a) One row of the SAD cost volume of the synthetic scene below; low cost (dark) follows the true disparity (dashed). (b) Cost curves of four pixels; dotted lines mark their true disparities.

Code: block matching with a left–right check

The snippet ray-casts a rectified pair of a synthetic scene with known depth: a slanted textured wall, a box, a disc and a thin pole in front of it, plus a textureless patch and a patch of fine stripes on the wall. It builds the SAD cost volume, takes the WTA disparity, and runs the **left–right consistency check**: match again with the right image as reference, and reject pixels where the two answers disagree by more than one pixel. Occluded pixels usually fail this check, because their "match" in the right image belongs to a different surface whose own match points elsewhere. For comparison it also runs OpenCV's semi-global matcher (next subsection). Errors are reported as **bad-1**, the percentage of pixels whose disparity is off by more than 1 px, separately for each kind of region.

```
import numpy as np
import cv2
```

```
from scipy import ndimage as ndi

rng = np.random.default_rng(0)
H, W, f, B = 180, 280, 320.0, 0.25                    # image size, focal length (px),
 ↪  baseline
tex = ndi.gaussian_filter(rng.random((700, 700)), 1.2)
tex = (tex - tex.min()) / np.ptp(tex)                 # random texture painted on every
 ↪  surface

def render(cx):
    """Ray-cast a rectified camera at (cx, 0, 0) looking down +Z: image, depth Z, region
     ↪  labels."""
    v, u = np.mgrid[0:H, 0:W].astype(float)
    dx, dy = (u - W / 2) / f, (v - H / 2) / f           # ray direction (dx, dy, 1)
    Z = (9 + 0.15 * cx) / (1 - 0.15 * dx)               # slanted wall Z = 9 + 0.15 X
    lab = np.zeros((H, W), int)                         # 0 wall, 1 box, 2 disc, 3 thin pole
    for k, z, inside in [(2, 6.0, lambda X, Y: (X - 1.2) ** 2 + (Y - 0.3) ** 2 < 0.36),
                         (3, 5.0, lambda X, Y: np.abs(X - 0.2) < 0.04),
                         (1, 4.0, lambda X, Y: (np.abs(X + 1.0) < 0.6) & (np.abs(Y) < 0.6))]:
        hit = inside(cx + z * dx, z * dy) & (z < Z)
        Z, lab = np.where(hit, z, Z), np.where(hit, k, lab)
    X, Y = cx + Z * dx, Z * dy
    img = ndi.map_coordinates(tex, [350 + 40 * Y + 60 * lab, 350 + 40 * X], order=1)
    flat = (lab == 0) & (X > 2.2) & (X < 3.6) & (Y > -2.0) & (Y < -0.9)
    rep = (lab == 0) & (X > -3.6) & (X < -2.2) & (Y > 0.9) & (Y < 2.0)
    img[flat] = 0.5                                      # textureless patch:
     ↪  label 4
    img[rep] = 0.5 + 0.4 * np.sin(2 * np.pi * X[rep] / 0.15)       # repetitive stripes:
     ↪  label 5
    lab[flat], lab[rep] = 4, 5
    return img + rng.normal(0, 0.01, img.shape), Z, lab

left, Z_left, lab = render(0.0)
right, Z_right, _ = render(B)
d_true = f * B / Z_left                                # ground-truth disparity of the left
 ↪  view
rows, cols = np.mgrid[0:H, 0:W]
xr = np.round(cols - d_true).astype(int)               # where each left pixel lands in the
 ↪  right view
occluded = (xr < 0) | (np.abs(Z_right[rows, np.clip(xr, 0, W - 1)] - Z_left) > 0.05)

def sad_volume(ref, other, D, win, sign):
    """cost[d](y, x) = sum over a win x win window of |ref(y, x) - other(y, x - sign*d)|."""
    cost = np.full((D, H, W), np.inf)
    for d in range(D):
        c = ndi.uniform_filter(np.abs(ref - np.roll(other, sign * d, axis=1)), win) * win **
         ↪  2
        if sign > 0: c[:, :d] = np.inf                 # no partner pixel exists there
        else:        c[:, W - d:] = np.inf
        cost[d] = c
    return cost

D, win = 32, 7
d_left = sad_volume(left, right, D, win, +1).argmin(0)     # winner-take-all, left image as
 ↪  reference
```

```
d_right = sad_volume(right, left, D, win, -1).argmin(0)    # winner-take-all, right image as
 ↪  reference
lr_ok = np.abs(d_left - d_right[rows, np.clip(cols - d_left, 0, W - 1)]) <= 1   # left-right
 ↪  check

sgbm = cv2.StereoSGBM_create(minDisparity=0, numDisparities=D, blockSize=5,
                             P1=8 * 25, P2=32 * 25, uniquenessRatio=0, disp12MaxDiff=-1)
u8 = lambda a: np.clip(a * 255, 0, 255).astype(np.uint8)
d_sgbm = sgbm.compute(u8(left), u8(right)).astype(float) / 16    # OpenCV stores 16 x
 ↪  disparity

valid = ~occluded & (cols >= D)                        # visible in both views, away from
 ↪  the left strip
regions = {"textured": valid & np.isin(lab, [0, 1, 2]), "textureless": valid & (lab == 4),
           "stripes": valid & (lab == 5), "thin pole": valid & (lab == 3)}
bad = lambda d, m: 100 * np.mean(np.abs(d - d_true)[m] > 1)      # % of pixels off by > 1 px
print(f"disparity {d_true.min():.1f}-{d_true.max():.1f} px; occluded: {100 *
 ↪  occluded.mean():.1f}% of pixels")
print("bad-1 (%)          " + "".join(f"{k:>12s}" for k in regions))
for name, d in [("SAD + WTA", d_left), ("OpenCV SGBM", d_sgbm)]:
    print(f"{name:19s}" + "".join(f"{bad(d, m):12.1f}" for m in regions.values()))
print(f"left-right check: keeps {100 * lr_ok[valid].mean():.1f}% of visible pixels, "
      f"rejects {100 * (~lr_ok[occluded]).mean():.1f}% of occluded ones")
kept = valid & lr_ok
print(f"SAD + WTA bad-1 on all visible pixels: {bad(d_left, valid):.1f}% -> after the check
 ↪  {bad(d_left, kept):.1f}%")
```

```
disparity 8.3-20.0 px; occluded: 7.7% of pixels
bad-1 (%)              textured textureless     stripes   thin pole
SAD + WTA                   1.8        67.7        89.6        17.4
OpenCV SGBM                 1.7        18.2        53.3        67.2
left-right check: keeps 96.6% of visible pixels, rejects 93.0% of occluded ones
SAD + WTA bad-1 on all visible pixels: 6.1% -> after the check 3.6%
```

These numbers come from this synthetic scene only. On textured surfaces, SAD block matching is right on 98% of the pixels; in the textureless patch it is wrong on two thirds of them and on the stripes on nine out of ten, as the cost curves predicted. The left–right check rejects 93% of the occluded pixels while keeping 96.6% of the visible ones, and bad-1 on the kept pixels drops from 6.1% to 3.6%. Rejected pixels are *holes*, not errors: later stages know not to trust them and can fill them from neighbours.

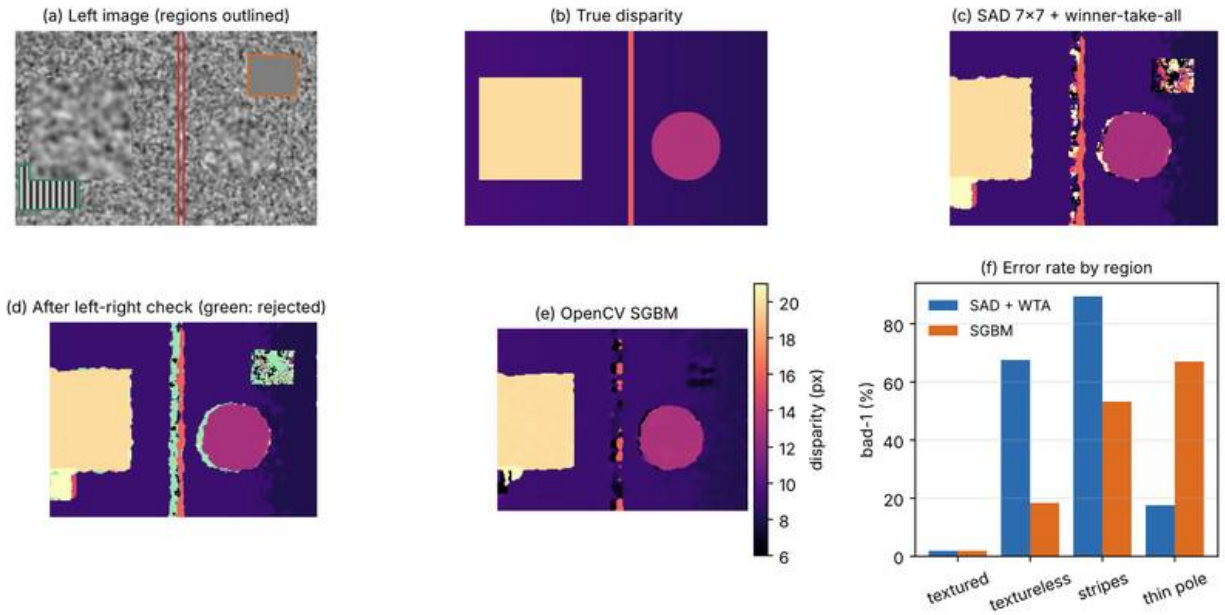


Figure 4 — Block matching on the synthetic rectified pair. (c) SAD 7 × 7 with winner-take-all; (d) after the left–right check (rejected pixels in green); (e) OpenCV's semi-global matcher; (f) bad-1 per region from the snippet. The left strip, which has no partner pixels, is not evaluated.

Cost aggregation and semi-global matching

Plain version. Window matching decides each pixel alone. A better rule lets neighbouring pixels "vote": a pixel in a blank region should take the disparity of its textured neighbours, unless there is good evidence for a depth jump.

The classical way to say this is an energy over the whole disparity map D,

$$E(D) = \sum_{\mathbf{p}} C(\mathbf{p}, D(\mathbf{p})) + \sum_{(\mathbf{p},\mathbf{q})} P_1 \, [\, |D(\mathbf{p}) - D(\mathbf{q})| = 1 \,] + P_2 \, [\, |D(\mathbf{p}) - D(\mathbf{q})| > 1 \,],$$

where the second sum runs over pairs of neighbouring pixels $\mathbf{p}, \mathbf{q}$, $[\cdot]$ is 1 when the condition holds and 0 otherwise, P_1 is a small penalty for a change of one disparity level (slanted surfaces) and $P_2 > P_1$ is a larger penalty for a jump (depth edges) [5], [6]. Minimizing this energy exactly over a 2D image is intractable in general.

Semi-global matching (SGM) by Hirschmüller [5] approximates it by minimizing the energy along many straight 1D paths through each pixel, each of which can be solved exactly by dynamic programming, and summing the results. Along a path with direction $\mathbf{r}$ the aggregated cost is

$$L_{\mathbf{r}}(\mathbf{p}, d) = C(\mathbf{p}, d) + \min\Big(L_{\mathbf{r}}(\mathbf{p} - \mathbf{r}, d),\; L_{\mathbf{r}}(\mathbf{p} - \mathbf{r}, d \pm 1) + P_1,\; \min_k L_{\mathbf{r}}(\mathbf{p} - \mathbf{r}, k) + P_2\Big) - \min_k L_{\mathbf{r}}(\mathbf{p} - \mathbf{r}, k),$$

where $\mathbf{p} - \mathbf{r}$ is the previous pixel on the path and the last term only keeps the numbers bounded

[6]. The costs of all directions are summed and WTA picks the disparity, so each pixel receives evidence from far away in many directions at a cost linear in the size of the cost volume. OpenCV's `StereoSGBM` is a modified version: by default 5 directions instead of the full algorithm's 8, small blocks instead of single pixels, and the Birchfield–Tomasi cost instead of mutual information [7].

The snippet shows SGM's trade-off. In the textureless patch, bad pixels drop from 67.7% to 18.2%, because the smoothness term carries the wall's disparity into it. But the thin pole, about five pixels wide, is mostly erased (17.4% → 67.2%): it costs two P_2 jumps for little evidence (Figure 4e). Neither method resolves the stripes, a genuine ambiguity of the data. Smoothness priors bet that the world is made of large surfaces, and that bet loses on thin structures.

Optional: Semi-global matching step by step (in the Supplementary notes). The recursion along one scanline in code, why $P_1 < P_2$, and how two opposite paths fill a textureless gap without blurring a depth step.

Refinement: sub-pixel disparity and post-processing

WTA returns integer disparities, but Figure 2b showed that fractions of a pixel matter. The usual fix fits a parabola through the cost at the best integer disparity and its two neighbours and takes the parabola's minimum. Further post-processing steps include the left–right check, a median filter and filling of rejected pixels from their neighbours; MC-CNN's pipeline, for example, combines cross-based aggregation, SGM, a left–right check, sub-pixel enhancement, a median filter and a bilateral filter [6].

Optional: Sub-pixel disparity and the soft argmin (in the Supplementary notes). The parabola formula, and why a learned soft argmin can land between two surfaces at a depth edge.

Learned stereo matching

Plain version. Every step of the classical pipeline has a hand-made rule. Deep stereo replaces them, one at a time, by learned ones: first the cost, then the aggregation, and finally the whole pipeline trained end to end on large synthetic datasets.

The surveys of Laga et al. [8] and Tosi et al. [9] follow the same order as this section: learned parts of the pipeline, then end-to-end networks, then the architectures of the 2020s.

Learning the matching cost

MC-CNN by Žbontar and LeCun [6] trains a convolutional network to decide whether two small patches show the same point, using pairs built from ground-truth disparities as a binary classification set. The network's output replaces SAD as the matching cost, and the rest of the classical pipeline (cross-based aggregation, SGM, left–right check, sub-pixel enhancement, filtering) follows. The lesson: a learned similarity is far more discriminative than raw intensities, but the hand-made aggregation remained.

End-to-end networks with cost volumes

End-to-end training needs dense ground truth for many images, which real scans rarely give. Mayer et al. [10] rendered three datasets (FlyingThings3D, Monkaa, Driving); FlyingThings3D alone has about 25,000 stereo frames of random objects flying along random 3D paths, and with it they trained DispNet, an encoder–decoder that regresses disparity [10]. Together these "Scene Flow" datasets became the usual pre-training data for stereo networks.

GC-Net by Kendall et al. [11] made the cost volume explicit and learnable. A shared 2D CNN extracts features from both images. For every disparity level the left feature is **concatenated** with the right feature shifted by that level, which gives a 4D volume of size height × width × (max disparity + 1) × feature size [11]. 3D convolutions over this volume learn the aggregation, using context in space and in disparity. Finally, a **soft argmin** turns the regularized costs c_d into a sub-pixel disparity:

$$\hat{d} = \sum_{d=0}^{D_{\max}} d \cdot \sigma(-c_d), \qquad \sigma(-c_d) = \frac{e^{-c_d}}{\sum_{d'} e^{-c_{d'}}},$$

where σ is the softmax over the disparity dimension and $D_{\max}$ the largest disparity [11]. Unlike the argmin it is differentiable, so the network trains end to end with a regression loss on $\hat{d}$. **PSM-Net** [12] added a spatial pyramid pooling module that gathers context at several scales before the cost volume and a stacked-hourglass 3D CNN with intermediate supervision to regularize it, trained with the smooth L1 loss [12]. Both target ill-posed regions such as occlusions, repeated patterns, textureless and reflective surfaces, where local patches carry no answer [12].

The price of 3D convolutions is memory and computation proportional to $D \times H \times W$, so these networks work at reduced resolution and a fixed maximum disparity.

Iterative refinement and zero-shot stereo

RAFT-Stereo [13] adapts the RAFT optical flow network (described in the optical flow section below) to rectified stereo: it replaces RAFT's 4D all-pairs volume by a 3D volume that only compares pixels on the same row, and refines a disparity field step by step with recurrent units that look up the volume around the current estimate, adding multi-level GRUs that pass information across the image more efficiently [13]. There is no 3D convolution and no fixed disparity range to regularize. Tosi et al. call this design "a game changer", noting that it brought much higher robustness to domain shift and that most recent frameworks follow it [9]. IGEV-Stereo [14] combines the two ideas: a geometry encoding volume that encodes geometry and context gives the recurrent updates a good starting point and is indexed by them at every iteration [14].

The open problem these surveys emphasized was **generalization**: networks fine-tuned on one benchmark often fail on another [9], [15]. FoundationStereo [16] targets zero-shot use: it is

trained on a large synthetic dataset of about one million stereo pairs with automatic removal of ambiguous samples, and it adapts features from a monocular vision foundation model through a side-tuning backbone to narrow the gap between synthetic and real images [16].

Monocular depth estimation

Plain version. With one image there is nothing to match. A network has to *recognize* depth the way people do with one eye closed: from perspective, the known sizes of things, occlusion, shading, haze and texture that gets finer with distance. These cues tell you which things are nearer, but they cannot tell a toy car from a real one.

Why one image is not enough: scale ambiguity

Scale the whole scene by a factor s and the image does not change, because projection divides by depth (Figure 5a; see also DLCV 08 and, for 3D detection, DLCV 09). Eigen et al. called the global scale "a fundamental ambiguity in depth prediction" and showed that much of their error came from the mean depth alone [17].

A network can still learn *metric* cues, such as typical object sizes or the camera height of a driving dataset, but they fail when the camera or scene type changes. So two kinds of output are used:

- **Relative (affine-invariant) depth.** The prediction is correct up to an unknown scale and shift. Many such models, MiDaS and Depth Anything among them, predict *inverse* depth (disparity-like values) [18], [19], so the output $\hat{y}$ satisfies $1/Z \approx s\hat{y} + t$ for unknown $s > 0$ and t. This is enough for ordering, segmentation, editing and many robot tasks when combined with a few metric measurements.
- **Metric depth.** The prediction is in metres. It requires either a known camera or a model that estimates the camera from the image.

Losses that ignore scale and shift

To train on datasets whose depth is known only up to scale, or in different units, the loss must ignore the unknown factors. Eigen et al. introduced a **scale-invariant log error** [17]:

$$D(y, y^*) = \frac{1}{n}\sum_{i=1}^{n}(\log y_i - \log y_i^* + \alpha(y, y^*))^2, \qquad \alpha(y, y^*) = \frac{1}{n}\sum_{i=1}^{n}(\log y_i^* - \log y_i),$$

where y_i and y_i^* are the predicted and true depths at pixel i, n is the number of pixels with ground truth, and α is the log-scale offset that best aligns the two maps; multiplying y by any constant leaves D unchanged [17]. For training they used a blend of this error and the ordinary log-space L2 error, weighted by $\lambda = 0.5$, which they found gave good absolute-scale predictions [17].

MiDaS by Ranftl et al. [18] went one step further: it predicts inverse depth (disparity) up to scale *and* shift, and aligns each prediction to the ground truth before computing the loss. With $\mathbf{d} \in \mathbb{R}^M$ the predicted disparities and $\mathbf{d}^*$ the ground truth at the M valid pixels, the alignment solves

$$(s, t) = \arg\min_{s,t} \sum_{i=1}^{M} (s d_i + t - d_i^*)^2, \qquad \hat{d}_i = s d_i + t,$$

which is an ordinary least-squares fit of a line with a closed-form solution [18]. The **scale-and-shift-invariant loss** is then $\mathcal{L}_{\text{ssi}} = \frac{1}{2M} \sum_i \rho(\hat{d}_i - d_i^*)$ with a per-pixel loss ρ, here the squared error [18]. MiDaS also has a robust variant (median-based alignment, trimmed absolute loss), a gradient-matching term that keeps depth edges sharp, and mixes five training datasets, including one built from 3D films [18]. It is evaluated by **zero-shot cross-dataset transfer**, on datasets never seen in training [18].

Optional: MiDaS (Ranftl et al., 2022) (in the Paper guides). The problem the paper poses, how its tables are aligned, and what to check before comparing its numbers with others.

The snippet builds a toy depth map (a ground plane, a far wall and a box), simulates a relative model whose output is inverse depth with an unknown scale and shift, and compares three ways of reading it.

```
import numpy as np

rng = np.random.default_rng(1)
H, W, f, cam_h = 120, 160, 150.0, 1.5
v, u = np.mgrid[0:H, 0:W].astype(float)
Z = np.full((H, W), 30.0)                                        # far wall
below = v > H / 2 + 1
Z[below] = np.minimum(30.0, f * cam_h / (v[below] - H / 2))    # ground plane seen by a
 ↪  camera 1.5 m high
box = (np.abs(u - 50) < 15) & (v > 45) & (v < 80)
Z[box] = 5.0                                                     # an object 5 m away

# A "relative" network outputs inverse depth up to an unknown scale and shift, plus some
 ↪  noise.
pred = 0.37 / Z + 0.05 + rng.normal(0, 0.001, Z.shape)

def metrics(Zp, Zt):
    absrel = np.mean(np.abs(Zp - Zt) / Zt)
    rmse = np.sqrt(np.mean((Zp - Zt) ** 2))
    delta1 = np.mean(np.maximum(Zp / Zt, Zt / Zp) < 1.25)
    return f"AbsRel {absrel:6.3f}   RMSE {rmse:6.2f} m   delta<1.25 {100 * delta1:5.1f}%"

Z_raw = 1 / pred                                                 # read the output as if it
 ↪  were metric
Z_med = Z_raw * np.median(Z) / np.median(Z_raw)                  # median scaling (scale only)
A = np.c_[pred.ravel(), np.ones(pred.size)]                      # least squares for (s, t) in
 ↪  inverse depth
s, t = np.linalg.lstsq(A, (1 / Z).ravel(), rcond=None)[0]
```

```
Z_ssi = 1 / (s * pred + t)
print("raw output        ", metrics(Z_raw, Z))
print("median scaling    ", metrics(Z_med, Z))
print("scale+shift (LSQ) ", metrics(Z_ssi, Z), f"  s = {s:.3f}, t = {t:.4f}")
```

```
raw output         AbsRel  0.460   RMSE  10.46 m   delta<1.25  10.2%
median scaling     AbsRel  0.785   RMSE   7.24 m   delta<1.25  54.8%
scale+shift (LSQ)  AbsRel  0.043   RMSE   1.84 m   delta<1.25  99.5%   s = 2.699, t = -0.1348
```

These are toy numbers. Read as metric depth, the output is useless ($\delta < 1.25$ on 10% of the pixels).

Median scaling multiplies the depth by one factor; it helps δ but makes AbsRel *worse*, because no single factor can undo a shift in inverse depth. The least-squares fit finds $s = 2.699$, $t = -0.1348$, almost exactly the simulated $1/Z = 2.703\,\hat{y} - 0.135$, and the map becomes nearly perfect. The same prediction can look poor or excellent depending on the alignment, so always check which one a paper used.

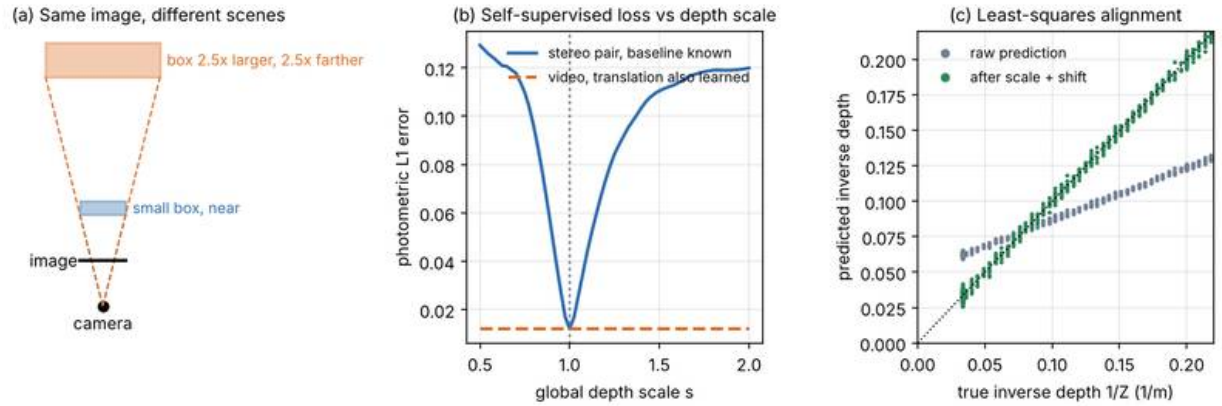


Figure 5 — Scale ambiguity. (a) A box 2.5 times larger and farther gives the same image. (b) A self-supervised photometric loss pins the scale only when the baseline is known; if the camera translation is also learned, every scale fits equally well. (c) An affine-invariant prediction becomes accurate after a least-squares scale and shift in inverse depth (data from the snippet).

From relative to metric foundation models

Modern monocular models use standard dense-prediction networks; the progress came from training data and from how scale is handled. On the **relative** side, DPT [20] brought a Vision Transformer encoder to dense prediction, Depth Anything [19], [21] scaled training with tens of millions of pseudo-labelled images, and Marigold [22] turned an image diffusion model into a depth estimator. On the **metric** side, ZoeDepth [23] adds metric heads to a relative model, while Metric3D [24], [25], UniDepth [26] and Depth Pro [27] handle the camera explicitly, by mapping

images to a canonical focal length or by estimating the camera from the image. The one idea to remember: relative depth is now robust across many kinds of images, and metric depth becomes usable once the model knows, or can infer, the camera. Dense pointmap models such as DUSt3R and VGGT, which predict geometry and cameras jointly, are covered in DLCV 08, Feed-forward reconstruction.

Optional: Monocular depth foundation models: data and the camera (in the Supplementary notes). What each of these seven models changes, model by model.

Self-supervised depth from stereo pairs and video

Plain version. Instead of ground-truth depth, use a second view as the teacher. If the network's depth were right, you could warp the second view onto the first and get the same picture. The difference between the warped and the real image is the training signal.

View synthesis as supervision

Garg et al. [28] trained a monocular network on stereo pairs with a known, small camera displacement: the predicted depth of the left image, together with the known baseline, warps the right image into the left view, and the photometric error of this reconstruction is the loss, with no depth labels at all [28]. Godard et al. [29] added a **left–right consistency** loss that makes the disparities predicted for the two views agree, which improved quality [29]. Zhou et al. [30] removed the second camera: from unlabelled monocular video they trained a depth network and a **pose network** together, with view synthesis between neighbouring frames as the only supervision [30].

For a target frame I_t and a source frame $I_{t'}$, each target pixel $\mathbf{p}$ (in homogeneous coordinates) is mapped to the source by

$$\mathbf{p}' \sim K\, T_{t\to t'}\, D_t(\mathbf{p})\, K^{-1}\mathbf{p},$$

where K is the camera intrinsic matrix, $D_t(\mathbf{p})$ is the predicted depth of the pixel, $T_{t\to t'}$ is the predicted relative camera pose (rotation and translation) and $\sim$ means equality up to the homogeneous scale. Sampling $I_{t'}$ at $\mathbf{p}'$ with a bilinear sampler (DLCV 04B) gives the synthesized image $I_{t'\to t}$; gradients flow through it into both networks.

Monodepth2

Monodepth2 by Godard et al. [31] showed that a simple model with careful design choices beats more complex ones. Its photometric error between two images combines SSIM (DLCV 04, full-reference metrics) and L1:

$$\mathrm{pe}(I_a, I_b) = \frac{\alpha}{2}\left(1 - \mathrm{SSIM}(I_a, I_b)\right) + (1-\alpha)\|I_a - I_b\|_1, \qquad \alpha = 0.85,$$

where I_a, I_b are the target image and a synthesized image, compared per pixel [31], [32]. Three ideas make it robust [31]:

1. **Per-pixel minimum reprojection.** With several source frames (previous and next), take the *minimum* error over sources instead of the average, $\mathcal{L}_p = \min_{t'} \mathrm{pe}(I_t, I_{t'\to t})$, because a pixel occluded in one source is usually visible in the other.
2. **Auto-masking.** Ignore pixels whose error *without* warping is already lower than with warping. This removes a static camera, and objects that move with the camera, which would otherwise teach the network "infinite depth" for cars driving ahead.
3. **Full-resolution multi-scale loss.** Upsample lower-resolution depth predictions to the input size before computing the error, which reduces artefacts.

An edge-aware smoothness term, weighted by $\lambda = 0.001$, discourages depth changes where the image itself is smooth [31].

Scale, again

Trained from monocular video, a depth network never sees a metric quantity: depth and translation can be scaled together without changing $\mathbf{p}'$. Figure 5b shows it on the synthetic stereo scene: with the baseline fixed, the photometric loss has a sharp minimum at the true scale; if the translation scales with the depth, the loss is flat. This is why Monodepth2 reports monocular-trained results after per-image **median ground-truth scaling**, following SfMLearner, but not for stereo-trained ones, "as scale can be inferred from the known camera baseline" [31]. The other limits are those of matching itself: textureless and reflective surfaces, moving objects and occlusions.

Optical flow as a learning problem

Plain version. Flow is stereo without the rulebook: the match can move in any direction, by a fraction of a pixel or by a hundred pixels, and objects move independently. Classical methods solved an energy for every image pair; deep methods learn to match and refine from synthetic videos.

From energies to networks

The classical methods start from brightness constancy, $I_1(\mathbf{p}) = I_2(\mathbf{p} + \mathbf{F}(\mathbf{p}))$, which gives one equation for two unknowns per pixel, and add an assumption to resolve it: global smoothness of the flow field (Horn and Schunck [33]), constant flow in a small window solved by least squares (Lucas and Kanade [34]), or a local polynomial model of the image (Farnebäck [35]). DLCV 04B introduces these ideas and the aperture problem.

Optional: Brightness constancy and the classical flow solvers (in DLCV 04B's Supplementary notes). The linearized flow equation, the Horn–Schunck energy and the polynomial expansion behind Farnebäck's method.

Hur and Roth [36] describe the move to deep learning in two families: CNNs used as feature extractors inside classical energy formulations, and end-to-end networks that regress the flow directly. The second family won. Four milestones lead to RAFT:

- **FlowNet** [37] regressed flow end to end with a CNN. Lacking real training data, its authors built **Flying Chairs**, 22,872 image pairs of rendered 3D chair models moved by random affine transformations over photographs, and found that networks trained on it still generalized to Sintel and KITTI [37]. Its FlowNetC variant added a **correlation layer** that compares the two images' features at a range of displacements.
- **FlowNet 2.0** [38] stacked several FlowNets, each warping the second image with the current estimate, and showed that the order of the training datasets matters a lot [38].
- **SPyNet** [39] used a classical image pyramid: at each level one image is warped by the current flow and a small network predicts an update of less than a pixel [39].
- **PWC-Net** [40] combined **p**yramid, **w**arping and **c**ost volume on *learned* features, and is 17 times smaller than FlowNet2 [40].

RAFT: all-pairs correlation and iterative updates

Plain version. RAFT compares every pixel of the first image with every pixel of the second, once. Then it starts from zero motion and, like a person adjusting a sketch, repeatedly looks at the match scores near its current guess and nudges the guess, the same small update applied again and again.

RAFT by Teed and Deng [41] has three parts [41]:

1. **Features.** A CNN encoder maps both images to features at 1/8 resolution with $D = 256$ channels; a separate context network encodes the first image only.
2. **All-pairs correlation.** The correlation volume holds the dot product of every feature pair,

$$C_{ijkl} = \sum_{h} g(I_1)_{ijh}\, g(I_2)_{klh},$$

where $g(I)_{ijh}$ is channel h of the feature at row i, column j, and $C \in \mathbb{R}^{H\times W\times H\times W}$ at feature resolution. Average pooling of the last two dimensions with kernels 1, 2, 4 and 8 gives a four-level **correlation pyramid**, which keeps full resolution for the first image and covers large displacements in the coarse levels. 3. **Iterative updates.** Starting from $\mathbf{f}_0 = \mathbf{0}$, a convolutional GRU (gated

recurrent unit) repeatedly reads the correlation values in a small neighbourhood (radius $r = 4$) around the position each pixel currently points to, at all pyramid levels, together with the context features and the current flow, and outputs an update: $\mathbf{f}_{k+1} = \mathbf{f}_k + \Delta\mathbf{f}$. The weights are shared across iterations. A learned convex upsampling turns the 1/8-resolution flow into full resolution.

The loss supervises every iterate with exponentially increasing weight,

$$\mathcal{L} = \sum_{i=1}^{N} \gamma^{N-i} \, \|\mathbf{f}_{\text{gt}} - \mathbf{f}_i\|_1, \qquad \gamma = 0.8,$$

where $\mathbf{f}_i$ is the upsampled flow after update i, N is the number of updates and $\mathbf{f}_{\text{gt}}$ the ground truth [41]. RAFT unrolls 12 updates in training and can run more at test time; the paper evaluates after 32 updates on Sintel and 24 on KITTI [41]. Training follows the standard schedule: Flying Chairs, then FlyingThings3D, then fine-tuning on the target benchmark [41].

Why did this design win? The all-pairs volume avoids a weakness of coarse-to-fine schemes, which can lose a small fast object at a coarse level for good [41]; the shared update behaves like a learned optimizer; and the recipe transferred to stereo [13] and scene flow [42].

Optional: RAFT (Teed and Deng, 2020) (in the Paper guides). How to read RAFT's ablations and its benchmark tables.

After RAFT

Later models change one part of the recipe. GMFlow [43] and Unimatch [44] replace iterative lookups by **global matching** with Transformer features: every pixel is compared with all pixels of the other image and the flow is the softmax-weighted average position; Unimatch uses one model for flow, stereo and depth from posed images [44]. FlowFormer [45] encodes RAFT's cost volume with transformer layers. SEA-RAFT [46] keeps RAFT but adds a mixture-of-Laplace loss, a directly regressed initial flow and rigid-motion pre-training, for speed and better generalization [46].

Code: measuring flow errors

The snippet builds a pair of frames with known flow: the background (the camera test image) rotates by 2° and zooms by 3% about the centre, and a textured patch moves by (14, 6) pixels on top of it. It runs OpenCV's Farnebäck flow with one and with four pyramid levels and reports the **end-point error** (EPE, the length of the error vector) and the **Fl** outlier rate used by KITTI (defined in Evaluation) on different kinds of pixels.

```
import numpy as np
import cv2
from scipy import ndimage as ndi
from skimage import color, data, transform
```

```
I0 = transform.resize(data.camera() / 255.0, (240, 320), anti_aliasing=True)
H, W = I0.shape
rows, cols = np.mgrid[0:H, 0:W].astype(float)
cy, cx = H / 2, W / 2

# Ground-truth motion from frame 0 to frame 1: the background rotates by 2 degrees and zooms
 ↪  by 3 %
# about the image centre; a 50 x 50 textured patch moves by (dx, dy) = (14, 6) pixels.
a, s = np.deg2rad(2.0), 1.03
bx = cx + s * (np.cos(a) * (cols - cx) - np.sin(a) * (rows - cy))     # where each
 ↪  background pixel goes
by = cy + s * (np.sin(a) * (cols - cx) + np.cos(a) * (rows - cy))
flow = np.stack([bx - cols, by - rows], -1)
patch = transform.resize(color.rgb2gray(data.astronaut())[30:230, 150:350], (50, 50),
 ↪  anti_aliasing=True)
obj0 = (rows >= 90) & (rows < 140) & (cols >= 120) & (cols < 170)
flow[obj0] = (14.0, 6.0)

# Render frame 1 by backward mapping (inverse rotation and zoom), then paste the moved patch.
xs = cx + (np.cos(a) * (cols - cx) + np.sin(a) * (rows - cy)) / s
ys = cy + (-np.sin(a) * (cols - cx) + np.cos(a) * (rows - cy)) / s
I1 = ndi.map_coordinates(I0, [ys, xs], order=3, mode="reflect")
I0 = I0.copy(); I0[90:140, 120:170] = patch
I1[96:146, 134:184] = patch
# occluded: background points of frame 0 that land under the patch in frame 1
occ = ~obj0 & (by >= 96) & (by < 146) & (bx >= 134) & (bx < 184)

def evaluate(est, mask):
    epe = np.linalg.norm(est - flow, axis=-1)[mask]
    mag = np.linalg.norm(flow, axis=-1)[mask]
    fl = np.mean((epe > 3) & (epe > 0.05 * mag))       # KITTI outlier: > 3 px and > 5 % of
     ↪  the true motion
    return f"EPE {epe.mean():5.2f} px  Fl {100 * fl:5.1f}%"

u8 = lambda im: (np.clip(im, 0, 1) * 255).round().astype(np.uint8)
inner = np.zeros((H, W), bool); inner[10:-10, 10:-10] = True
grad = ndi.gaussian_filter(np.hypot(ndi.sobel(I0, 0), ndi.sobel(I0, 1)), 4)
bg = inner & ~obj0 & ~occ
flat = grad < 0.05                                     # almost no texture (mostly the sky)
for levels in [1, 4]:
    est = cv2.calcOpticalFlowFarneback(u8(I0), u8(I1), None, pyr_scale=0.5, levels=levels,
     ↪  winsize=15,
                                       iterations=5, poly_n=7, poly_sigma=1.5, flags=0)
    print(f"Farneback, {levels} pyramid level(s):")
    for name, m in [("background, textured", bg & ~flat), ("background, flat", bg & flat),
                    ("moving patch", obj0), ("occluded", occ), ("all (Fl-all)", inner)]:
        print(f"   {name:21s} {evaluate(est, m)}")
print(f"true motion: background up to {np.linalg.norm(flow[~obj0], axis=-1).max():.1f} px,
 ↪  patch "
      f"{np.hypot(14, 6):.1f} px; occluded pixels: {occ.sum()}")
```

```
Farneback, 1 pyramid level(s):
```

```
   background, textured  EPE  0.87 px  Fl   9.3%
   background, flat      EPE  5.51 px  Fl  90.3%
   moving patch          EPE 18.24 px  Fl 100.0%
   occluded              EPE  5.07 px  Fl  68.0%
   all (Fl-all)          EPE  2.86 px  Fl  35.6%
Farneback, 4 pyramid level(s):
   background, textured  EPE  1.18 px  Fl  11.9%
   background, flat      EPE  5.52 px  Fl  90.5%
   moving patch          EPE  2.30 px  Fl  30.4%
   occluded              EPE 12.20 px  Fl  99.8%
   all (Fl-all)          EPE  2.55 px  Fl  35.2%
true motion: background up to 9.3 px, patch 15.2 px; occluded pixels: 871
```

These numbers are from this toy pair only. With one level, the patch's 15 px motion is out of reach and every patch pixel is an outlier; with four levels the patch is found (EPE 2.30 px), the coarse-to-fine idea of SPyNet and PWC-Net in classical form. Two failures remain. In flat regions (mostly the sky) 90% of the pixels are outliers: there is nothing to match. On **occluded** pixels, the strip the patch covers in frame 1, the pyramid makes things worse (EPE 12.2 px), because the coarse levels drag them along with the patch (Figure 6). Learned methods are trained to handle both cases from data, but neither disappears.

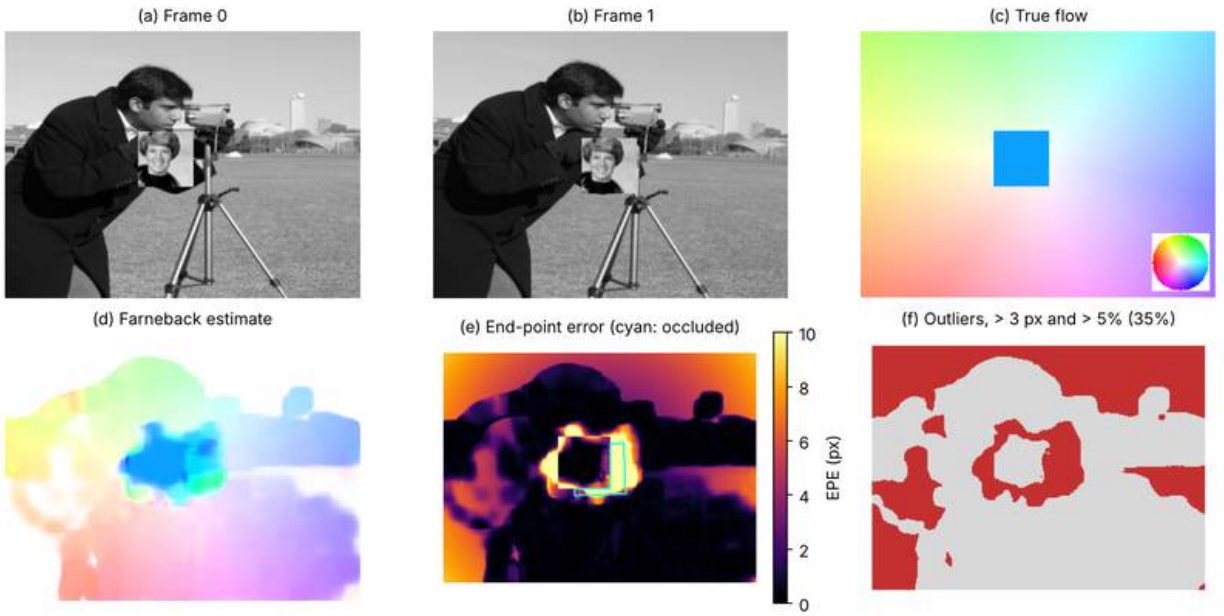


Figure 6 — Farnebäck flow (4 pyramid levels) on the toy pair. Flow is drawn with hue for direction and saturation for length. Errors concentrate in textureless regions, around the moving patch and in the occluded strip (cyan). A 10 px border is not evaluated.

Scene flow in brief

Plain version. Optical flow says how a point moves in the image. Scene flow says how it moves in 3D: closer, farther, left or right, in metres.

Scene flow is the dense 3D motion field of the scene, a term that goes back to Vedula et al. [47]; optical flow is its projection onto the image. RAFT-3D states the task as: given a pair of stereo or RGB-D video frames, estimate the 3D motion of every pixel [42]. With a stereo camera it can be estimated from the disparity at two times plus the optical flow between them. The KITTI 2015 benchmark by Menze and Geiger [48] evaluates exactly these parts: D1 (disparity in the first frame), D2 (disparity in the second frame), Fl (optical flow) and SF, which counts a pixel as wrong if any of the three is wrong [49]. Menze and Geiger modelled the scene as a small number of rigidly moving objects, with each superpixel a 3D plane assigned to one of them [48]; RAFT-3D [42] instead estimates a dense field of rigid 3D motions with RAFT-style iterations and groups pixels softly into rigid objects with learned embeddings [42]. Dynamic reconstruction (DLCV 08) and point tracking (DLCV 06) are its long-range relatives.

Evaluation

Plain version. All three tasks are scored per pixel against ground truth, with an average error and a "percentage of bad pixels". The details, especially how monocular predictions are aligned and which pixels count, change the numbers more than you would expect.

Depth metrics

The standard set goes back to Eigen et al. [17]. With y_i the predicted and y_i^* the true depth at the n evaluated pixels,

$$\text{AbsRel} = \frac{1}{n}\sum_i \frac{|y_i - y_i^*|}{y_i^*}, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - y_i^*)^2},$$

$$\text{SqRel} = \frac{1}{n}\sum_i \frac{(y_i - y_i^*)^2}{y_i^*}, \qquad \delta_k = \frac{1}{n}\Big|\Big\{i : \max\Big(\tfrac{y_i}{y_i^*}, \tfrac{y_i^*}{y_i}\Big) < 1.25^k\Big\}\Big|,$$

plus RMSE of $\log y$ [17]. AbsRel, SqRel and RMSE are errors (lower is better); $\delta_1, \delta_2, \delta_3$ with $k = 1, 2, 3$ are accuracies (higher is better). AbsRel weights near and far pixels by their relative error; RMSE is dominated by far pixels; δ_1 counts pixels within 25% of the truth.

State three conventions with every result: the **alignment** (none for metric models, per-image median scaling for self-supervised monocular ones [31], least-squares scale and shift for affine-invariant ones [18]); the **depth cap** (80 m is standard on KITTI [31]); and the **split** (often that

of Eigen et al. on KITTI [17], [31]). The alignment snippet showed how much the first one alone changes.

Stereo metrics

- **EPE** (mean absolute disparity error, in pixels).
- **bad-τ**: percentage of pixels with a disparity error larger than τ pixels. Middlebury's Version 3 evaluation reports bad 2.0 by default, on all or only non-occluded pixels [3]; KITTI 2012 reports the percentage of erroneous pixels at thresholds from 2 to 5 pixels, in non-occluded areas (Out-Noc) and in total (Out-All) [50].
- **D1** (KITTI 2015): a pixel's disparity is correct if its error is below 3 px *or* below 5% of the true value; D1 is the percentage of outliers, reported over background, foreground or all pixels (D1-bg, D1-fg, D1-all) [49]. The relative clause keeps near objects, with large disparities, from being judged too harshly.

Flow metrics

- **EPE**: the mean Euclidean distance between estimated and true flow vectors, $\frac{1}{n}\sum_i \|\mathbf{F}_i - \mathbf{F}_i^*\|_2$, where $\mathbf{F}_i$ and $\mathbf{F}_i^*$ are the estimated and true flow at pixel i; FlowNet calls it the standard error measure for optical flow and also trains with it [37]. Sintel results are reported as average EPE (often written AEPE).
- **Fl-all** (KITTI 2015): the percentage of pixels whose end-point error is at least 3 px *and* at least 5% of the true flow length, over all pixels with ground truth [49]. The snippet's `evaluate` function computes exactly this.

Datasets and benchmarks

Dataset	Task	What it contains	Ground truth
KITTI 2012 [51], [50]	stereo, flow	194 training and 195 test pairs of real road scenes	sparse, from a laser scanner
KITTI 2015 [48], [49]	stereo, flow, scene flow	200 training and 200 test scenes with moving cars	sparse, semi-automatic, with 3D CAD models fitted to moving vehicles
Middlebury 2014 [52]	stereo	33 high-resolution indoor scenes; 20 used in the Version 3 evaluation (10 training, 10 test)	dense, subpixel-accurate
ETH3D [53]	stereo, multi-view stereo	indoor and outdoor scenes with DSLR images and low-resolution stereo videos; includes a low-resolution two-view benchmark	high-precision laser scans

Dataset	Task	What it contains	Ground truth
MPI Sintel [54]	flow (also stereo, depth)	frames of an open-source animated film; "clean" and "final" passes, the final one with motion blur and atmospheric effects	dense, rendered
Flying Chairs [37]	flow (training)	22,872 synthetic image pairs	dense, rendered
FlyingThings3D [10]	stereo, flow, scene flow (training)	about 25,000 synthetic stereo frames at 960 × 540	dense, rendered
NYU Depth v2 [55]	monocular depth (indoor)	Kinect video of 464 indoor scenes; 1,449 densely labelled RGB-D pairs	dense, from the Kinect depth camera, missing values filled in

Sizes and ground-truth types come from the cited pages and papers; the description of Sintel's two passes is from [37], and KITTI 2015's semi-automatic ground truth from [49], [48]. The synthetic sets are used for pre-training, the real ones for fine-tuning and testing.

Failure modes

Plain version. Every method in this tutorial relies on one of two things: something to match, or a prior learned from training data. Each failure below breaks one of them.

- **Textureless regions** (white walls, sky, road): flat matching costs (Figure 3b), filled by smoothness or learned context, both possibly wrong.
- **Repetitive patterns** (fences, tiles): several disparities fit equally well, as the stripes showed.
- **Reflective and transparent surfaces**: the matched content is a reflection or the scene behind. Tosi et al. count transparent and reflective objects among the four main open challenges of deep stereo, with domain shift, over-smoothing and asymmetric stereo pairs [9].
- **Occlusions**: no correct match exists; they need detection (left–right check) and filling, and in flow they are dragged along with neighbouring motion.
- **Thin structures and depth edges**: windows fatten foreground objects, smoothness erases poles and wires, and a soft argmin over a two-peaked cost lands between surfaces.
- **Large motion of small objects**: coarse-to-fine schemes can lose them; RAFT's all-pairs volume targets this [41].
- **Domain gap**: models pre-trained on synthetic data and fine-tuned on one benchmark degrade on new cameras, weather and scenes [9], [36].
- **Scale and the camera** (monocular): relative depth has no scale, and metric depth fails when the focal length is unknown or the image has been cropped unexpectedly.

- **Moving objects in self-supervision** break the static-scene assumption; auto-masking handles only the simplest cases [31]. Rajapaksha et al. add thin, small and distant objects to the open gaps of monocular depth [56].

Modern view

What the surveys say

Five reviews cover the ground of this tutorial. **Laga et al.** [8] survey deep stereo (two-view and multi-view) up to about 2019, split into methods that learn parts of the classical pipeline and end-to-end pipelines, with datasets and supervision modes; monocular depth is outside their scope.

Poggi et al. [15] stress that learning and stereo help each other: learning improves stereo, and stereo supervises monocular networks; they also cover confidence estimation and domain shift.

Tosi et al. [9] review the 2020s: architectures grouped into cost volume aggregation, neural architecture search, iterative optimization, Vision Transformers and Markov random fields, and four open challenges (domain shift, over-smoothing, transparent and reflective objects, asymmetric stereo); in 2024 they noted that a stereo foundation model was still missing. **Rajapaksha et al.** [56] organize monocular depth by input and output, architecture and degree of supervision. **Hur and Roth** [36] trace optical flow from energy minimization to CNNs and name domain overfitting and complex training schedules as the main limits.

Optional: Guide to the Modern view surveys (in the Paper guides). Which survey to read for which question, and what each one does not cover.

The state of the art, 2023–2026

- **Foundation models for monocular depth.** Large-scale data (Depth Anything [19], [21]), generative priors (Marigold [22]) and explicit camera handling (Metric3D [24], [25], UniDepth [26], Depth Pro [27]) made zero-shot depth from arbitrary photos practical, first relative and now increasingly metric.
- **Iterative matching for stereo and flow.** The RAFT recipe of a correlation volume plus recurrent updates dominates both tasks [9]; recent variants add geometry volumes (IGEV-Stereo [14]), global matching with attention (GMFlow, Unimatch [43], [44]) or simpler training (SEA-RAFT [46]).
- **Zero-shot stereo.** FoundationStereo [16] answered the missing-foundation-model point of Tosi et al. by combining a million synthetic pairs with monocular priors.

- **Convergence with 3D reconstruction.** Feed-forward models that predict pointmaps and cameras (DLCV 08) blur the line between depth estimation, stereo and structure from motion.

Open problems

1. **Metric depth in the wild**: unknown, cropped or unusual cameras, and a reliable uncertainty for each prediction.
2. **Non-Lambertian scenes**: glass, mirrors and water for both stereo and flow [9].
3. **Sharp boundaries and thin structures** without over-smoothing or "flying pixels" between surfaces.
4. **Temporal consistency**: per-frame depth flickers in video; flow and depth over long sequences should agree with each other.
5. **Evaluation**: sparse or semi-automatic ground truth, differing alignment conventions and benchmark saturation make small differences hard to interpret.

Key takeaways

- Stereo, optical flow and monocular depth are dense correspondence problems; stereo is flow restricted to one row, and monocular depth replaces matching by learned priors.
- In a rectified pair, $Z = fB/d$; depth error grows as $Z^2\delta d/(fB)$, so baseline, resolution and subpixel accuracy set the usable range.
- The classical stereo pipeline is cost, aggregation, optimization and refinement; SGM's 1D-path dynamic programming fills textureless regions but erases thin structures, and the left–right check finds most occlusions.
- Learned stereo moved from learned costs (MC-CNN) to 3D-convolved cost volumes with soft argmin (GC-Net, PSMNet) to iterative updates (RAFT-Stereo) and zero-shot models (FoundationStereo), trained on synthetic data first.
- One image fixes depth only up to scale; relative models predict affine-invariant inverse depth trained with scale-and-shift-invariant losses, and metric models must infer the camera.
- Self-supervised depth uses view synthesis with a photometric loss; Monodepth2's minimum reprojection and auto-masking handle occlusion and moving objects, and monocular training needs median scaling at test time.
- RAFT's all-pairs correlation pyramid and shared recurrent update define modern optical flow; EPE and Fl-all (> 3 px and $> 5\%$) are its metrics on Sintel and KITTI.
- Always report the alignment, the evaluated pixels (occluded or not, depth cap) and the split; they change the numbers as much as the method does.

Exercises

1. **Choose a baseline.** A stereo camera with $f = 1000$ px must measure depth at 40 m with an error below 1 m, assuming a disparity error of 0.3 px. What is the smallest baseline? What disparity does a point at 40 m then have, and what happens at 80 m?

   Hint

   From $\delta Z \approx Z^2 \delta d/(fB) < 1$: $B > 1600 \cdot 0.3/1000 = 0.48$ m. The disparity at 40 m is $fB/Z = 12$ px. At 80 m the error is four times larger (about 4 m) and the disparity is 6 px.

2. **Window size and fattening.** In the block-matching snippet, change `win` to 3, 7 and 15 and compare bad-1 on the four regions. Why does a large window help in one place and hurt in another?

   Hint

   In our run, textureless bad-1 fell from 84.4% (3×3) to 67.7% (7×7) and 39.9% (15×15), while the thin pole rose from 15.3% to 17.4% and 74.6%. A large window gathers texture from outside the flat patch, but near depth edges it mixes two surfaces and spreads the foreground disparity into the background ("foreground fattening"), which swallows the pole.

3. **Stripes and the period.** The stripes on the wall have a period of 0.15 world units, and the wall there is about 8.6 units away. Estimate their period in pixels (use $f = 320$ px), then explain why block matching gives wrong disparities there. Which change to the scene would make the stripes matchable, and which change to the algorithm might help?

   Hint

   The period is about $320 \cdot 0.15/8.6 \approx 5.6$ px, smaller than the true disparity there (about 9.3 px), so shifts that differ by one period fit equally well. A wider search does not help. Non-periodic texture would; a smoothness term that prefers the surrounding wall's disparity reduces the errors (in our run SGBM's bad-1 on the stripes was 53.3% against 89.6% for block matching) but does not remove them.

4. **Scale-invariant losses by hand.** Show that Eigen's error $D(y, y^*)$ does not change when y is multiplied by a constant $c > 0$. Then show that MiDaS's least-squares (s, t) makes $\mathcal{L}_{\text{ssi}}$ unchanged under $d \mapsto ad + b$ for any $a \neq 0$ and b.

   Hint

   $\log(cy_i) = \log c + \log y_i$, and α absorbs $-\log c$, so every residual is unchanged. For MiDaS, if (s, t) is optimal for d, then $(s/a, t - sb/a)$ gives the same aligned values for $ad + b$.

5. **Pyramids and large motion.** In the flow snippet, move the patch by $(24, 10)$ instead of $(14, 6)$ and run with 1, 4 and 6 pyramid levels; then move it by only $(2, 1)$. Update the pasted position in frame 1 and the occlusion mask too. When does the pyramid help?

Hint

In our run, at $(24, 10)$ the patch was lost with every setting (EPE about 26 px), and 6 levels gave exactly the same result as 4, so the extra levels added nothing here. At $(2, 1)$ all three settings gave a patch EPE of 0.27 px. A pyramid helps only for motions that its coarse levels can bring within reach of the local model; beyond that, a different search is needed, such as RAFT's all-pairs correlation.

Supplementary notes

These notes are optional: each one takes a single point from the main text and goes into finer detail. Nothing in the main text, code or exercises depends on them.

Robust matching costs: ZNCC and census

This note defines two matching costs that tolerate brightness differences between the two cameras, and tests them against SAD.

ZNCC. For windows $\mathbf{a}$ and $\mathbf{b}$ of m pixels each (from the left image at (u, v) and the right image at $(u - d, v)$), with means $\bar{a}, \bar{b}$,

$$\mathrm{ZNCC}(\mathbf{a}, \mathbf{b}) = \frac{\sum_{k=1}^{m} (a_k - \bar{a})(b_k - \bar{b})}{\sqrt{\sum_k (a_k - \bar{a})^2 \, \sum_k (b_k - \bar{b})^2}} \in [-1, 1],$$

and the cost is $-\mathrm{ZNCC}$ (or $1 - \mathrm{ZNCC}$). Subtracting the means removes an offset and dividing by the norms removes a gain, so a change $b_k \mapsto g b_k + o$ with $g > 0$ leaves ZNCC unchanged. The denominator is close to zero in textureless windows, where ZNCC becomes unstable; implementations add a small constant.

Census. For a pixel $\mathbf{p}$ and a set of neighbour offsets $\mathcal{N}$, the census transform is the bit string $b_{\mathbf{q}}(\mathbf{p}) = [\, I(\mathbf{p} + \mathbf{q}) < I(\mathbf{p}) \,]$ for all offsets $\mathbf{q} \in \mathcal{N}$, where I is the image and $[\cdot]$ is 1 when the condition holds [4]. Two pixels are compared by the Hamming distance of their bit strings (the number of differing bits), usually summed over a window. Census depends only on the *order* of intensities, so any strictly increasing change of brightness leaves it unchanged; and a single outlier pixel in the neighbourhood changes at most one bit of the centre's string. Its weakness is that it throws away contrast: a faint pattern and a strong one give the same bits, and image noise flips bits in flat regions.

A test. The snippet builds a rectified pair of a single plane at disparity 10 and applies a gain of 1.3 and an offset of 0.1 to the right image.

```
import numpy as np
from scipy import ndimage as ndi

rng = np.random.default_rng(3)
H, W, D, win = 100, 160, 24, 7
left = ndi.gaussian_filter(rng.random((H, W + D)), 1.0)
d_true = 10                                          # one fronto-parallel plane,
 ↪  disparity 10 px
right = np.roll(left, -d_true, axis=1)[:, :W]        # right(x) = left(x + 10)
left = left[:, :W]
right = 1.3 * right + 0.1                            # the right camera has a different
 ↪  gain and offset

def census(img, r=2):
    """Bit string per pixel: is each neighbour in a (2r+1)^2 window darker than the
     ↪  centre?"""
    bits = [np.roll(img, (dy, dx), (0, 1)) < img for dy in range(-r, r + 1) for dx in
     ↪  range(-r, r + 1)
            if (dy, dx) != (0, 0)]
    return np.stack(bits, -1)

def zncc_cost(a, b):
    box = lambda x: ndi.uniform_filter(x, win)
    ma, mb = box(a), box(b)
    va, vb = box(a * a) - ma ** 2, box(b * b) - mb ** 2
    return -(box(a * b) - ma * mb) / np.sqrt(va * vb + 1e-12)      # negative correlation =
     ↪  cost

cl, cr = census(left), census(right)
costs = {"SAD": [], "ZNCC": [], "census": []}
for d in range(D):
    r_d = np.roll(right, d, axis=1)                  # right pixel x - d moved to column x
    costs["SAD"].append(ndi.uniform_filter(np.abs(left - r_d), win))
    costs["ZNCC"].append(zncc_cost(left, r_d))
    ham = (cl != np.roll(cr, d, axis=1)).sum(-1).astype(float)     # Hamming distance of bit
     ↪  strings
    costs["census"].append(ndi.uniform_filter(ham, win))
inner = (slice(10, -10), slice(D + 5, -10))
for name, c in costs.items():
    d_hat = np.argmin(np.stack(c), axis=0)[inner]
    print(f"{name:7s} correct disparity on {100 * np.mean(d_hat == d_true):5.1f}% of pixels")
```

```
SAD     correct disparity on   3.9% of pixels
ZNCC    correct disparity on 100.0% of pixels
census  correct disparity on 100.0% of pixels
```

In this test, SAD finds the right disparity on only 3.9% of the pixels, because the brightness change dominates the cost; ZNCC and census are correct everywhere. This is why classical systems use census or ZNCC, and why learned features, which can learn the same invariances, replaced them.

Semi-global matching step by step

This note walks through SGM's recursion on one scanline, the part that the main text states in one equation.

The recursion. On a path with direction $\mathbf{r}$ (for a horizontal scanline, $\mathbf{r} = (1, 0)$, left to right), the aggregated cost at pixel x and disparity d is

$$L(x,d) = C(x,d) + \min(L(x-1,d),\ L(x-1,d-1) + P_1,\ L(x-1,d+1) + P_1,\ m_{x-1} + P_2) - m_{x-1},$$

where $C(x,d)$ is the matching cost, $L(x-1,\cdot)$ are the aggregated costs of the previous pixel, $m_{x-1} = \min_k L(x-1,k)$, P_1 is the penalty for a change of one level and P_2 the penalty for a larger jump [5], [6]. The four options are: keep the previous disparity (free), change by one level (cost P_1), or jump from the best previous disparity to any other (cost P_2). Subtracting m_{x-1} does not change the minimizer; it only keeps L bounded.

Why $P_1 < P_2$. A slanted surface changes disparity gradually, one level every few pixels, so small steps must be cheap. A depth edge is a large jump that happens rarely, so it should cost more, but a fixed amount regardless of size, so that real edges are not smoothed into ramps. MC-CNN, for example, lowers both penalties where the image gradient is large, because depth edges often coincide with intensity edges [6].

Why several paths. One left-to-right path only propagates information from the left; a pixel at the left end of a textureless region gets help, one at the right end does not. Summing paths from several directions gives each pixel support from all around. The full algorithm uses 8 directions; OpenCV's default uses 5 [7].

A test on one scanline. The snippet builds a cost array for 60 pixels with a disparity step from 5 to 9 at column 30 and a textureless stretch (columns 10–23) where all costs are nearly equal, and compares WTA with the sum of a left-to-right and a right-to-left path.

```
import numpy as np

rng = np.random.default_rng(0)
Wd, D = 60, 16
d_true = np.where(np.arange(Wd) < 30, 5, 9)             # a depth step at column 30
cost = rng.uniform(0.6, 1.0, (Wd, D))                   # matching cost C(x, d) along one
 ↪  scanline
textured = np.ones(Wd, bool); textured[10:24] = False    # columns 10..23 have no texture
cost[textured, d_true[textured]] = 0.1                   # textured pixels: a clear minimum
cost[~textured] = 0.6 + 0.02 * rng.standard_normal((14, D))   # textureless: flat and noisy

def aggregate(C, P1=0.05, P2=0.3):
    """One SGM path from left to right: L(x, d) = C(x, d) + min(...) - min_k L(x-1, k)."""
    L = np.empty_like(C); L[0] = C[0]
    for x in range(1, len(C)):
        prev = L[x - 1]; m = prev.min()
        up = np.r_[prev[1:], np.inf] + P1               # neighbour had disparity d + 1
```

```
        down = np.r_[np.inf, prev[:-1]] + P1            # neighbour had disparity d - 1
        L[x] = C[x] + np.minimum.reduce([prev, up, down, np.full(D, m + P2)]) - m
    return L

S = aggregate(cost) + aggregate(cost[::-1])[::-1]       # two paths: left-to-right and
 ↪  right-to-left
for name, c in [("winner-take-all", cost), ("SGM, 2 paths", S)]:
    d_hat = c.argmin(1)
    print(f"{name:16s} correct: textured {np.mean(d_hat[textured] == d_true[textured]):.2f},
     ↪  "
          f"textureless {np.mean(d_hat[~textured] == d_true[~textured]):.2f}, "
          f"step kept at x=29/30: {d_hat[29]}/{d_hat[30]}")
```

```
winner-take-all  correct: textured 1.00, textureless 0.00, step kept at x=29/30: 5/9
SGM, 2 paths     correct: textured 1.00, textureless 1.00, step kept at x=29/30: 5/9
```

WTA gets every textured pixel right and every textureless one wrong. Two SGM paths fill the textureless stretch with the disparity of its neighbours, and the depth step at columns 29/30 survives: a jump costs P_2 only once, while keeping the wrong disparity beyond the step would pay the large matching cost at every following textured pixel.

Sub-pixel disparity and the soft argmin

This note gives the two common ways to get a sub-pixel disparity from a cost curve, and shows where the learned one goes wrong.

Parabola fitting. Let k be the integer disparity with the lowest cost and c_-, c_0, c_+ the costs at $k-1$, $k, k+1$. The parabola through these three points has its minimum at

$$\hat{d} = k + \frac{c_- - c_+}{2\,(c_- - 2c_0 + c_+)},$$

where the denominator is positive because c_0 is the minimum. The correction lies within ± 0.5, so the estimate stays next to the winning integer. For a cost that is exactly quadratic near its minimum the formula is exact; for other cost shapes it is an approximation.

Soft argmin. GC-Net's soft argmin [11] computes the expectation of d under the softmax of $-c_d$. When the cost has one clear minimum, the softmax is concentrated and the expectation is a good sub-pixel estimate. When the cost has two minima, as for a window that straddles a depth edge, the expectation lands *between* them, at a disparity that belongs to neither surface. This is one source of the over-smoothed depth discontinuities that Tosi et al. list as a challenge of deep stereo [9]. GC-Net's authors note that the network can learn to pre-scale the costs so that the softmax becomes sharper [11].

```
import numpy as np
```

```
d = np.arange(32, dtype=float)                              # integer disparity hypotheses

def parabola(c):
    """Sub-pixel minimum from the costs at the best integer d and its two neighbours."""
    k = int(np.argmin(c)); cm, c0, cp = c[k - 1], c[k], c[k + 1]
    return k + (cm - cp) / (2 * (cm - 2 * c0 + cp))

def soft_argmin(c, beta=1.0):
    p = np.exp(-beta * (c - c.min())); p /= p.sum()     # softmax of the negative cost
    return (p * d).sum()

one = (d - 7.3) ** 2 / 4                                    # one surface at disparity 7.3
two = np.minimum((d - 6.0) ** 2, (d - 15.0) ** 2 + 0.2) / 4   # window straddles an edge: 6
 ↪ or 15
for name, c in [("one surface at 7.3", one), ("edge between 6 and 15", two)]:
    print(f"{name:22s} argmin {np.argmin(c):2d} | parabola {parabola(c):5.2f} | soft argmin
     ↪ {soft_argmin(c):5.2f}")
```

```
one surface at 7.3     argmin  7 | parabola  7.30 | soft argmin  7.30
edge between 6 and 15  argmin  6 | parabola  6.00 | soft argmin 10.39
```

For one surface at 7.3, the parabola and the soft argmin both return 7.30. For a cost with minima at 6 and 15, the parabola stays at the winning minimum (6.00) while the soft argmin returns 10.39, a depth halfway between the foreground and the background.

Monocular depth foundation models: data and the camera

This note describes, model by model, how the monocular depth models named in the main text obtain their training signal and how they deal with scale and the camera.

Relative (affine-invariant) models and their backbone. These predict depth or inverse depth up to an unknown scale and shift, and are trained with an alignment-based loss such as MiDaS's [18].

- **DPT** [20] replaced the convolutional encoder with a Vision Transformer: tokens from several transformer stages are reassembled into image-like maps at several resolutions and fused by a convolutional decoder. Because every stage has a global receptive field, predictions are more globally coherent, especially with much training data [20].
- **Depth Anything** [19] keeps the relative, affine-invariant formulation and scales the data: a teacher trained on 1.5M labelled images pseudo-labels about 62M unlabelled images, and a student learns from both under strong augmentation, with a feature-alignment loss to a frozen

  DINOv2 encoder that preserves semantic priors [19]. **Depth Anything V2** [21] replaces all labelled real images by precise synthetic ones, scales up the teacher, and teaches smaller students

through large-scale pseudo-labelled real images; fine-tuning with metric labels gives metric variants [21].

- **Marigold** [22] fine-tunes the Stable Diffusion image generator into an affine-invariant depth estimator, using only synthetic training data, to reuse the generator's prior about the visual world (diffusion models are covered in ML 07).

Metric models. These must produce metres, which requires knowing or inferring the focal length, because the same object looks larger in the image with a longer focal length.

- **ZoeDepth** [23] pre-trains on twelve datasets with relative depth and fine-tunes metric heads on NYU Depth v2 and KITTI, routing each image to the right head with a classifier [23].
- **Metric3D** [24] identifies the camera as the source of metric ambiguity. It transforms training images into a **canonical camera space** (one fixed focal length) and maps predictions back, which allows training on over 8 million images from thousands of camera models [24]. **Metric3D v2** [25] adds joint depth and surface-normal estimation.
- **UniDepth** [26] predicts metric 3D points directly, with a camera module that predicts a dense camera representation to condition the depth features, so no intrinsics are needed at test time [26].
- **Depth Pro** [27] predicts metric depth with sharp boundaries without camera metadata, using a multi-scale ViT and its own focal-length estimation; its authors report a 2.25-megapixel depth map in 0.3 seconds on a standard GPU [27].

What to check when comparing them. Relative models are evaluated after a per-image scale-and-shift alignment, metric models without it; a zero-shot result means the test dataset was not used in training; and metric results depend on whether the true intrinsics were given to the model or estimated by it.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

RAFT (Teed and Deng, 2020)

The problem. In 2020, learned optical flow used coarse-to-fine pyramids (PWC-Net) or stacks of networks (FlowNet 2.0). Coarse-to-fine schemes can miss small, fast objects that disappear at

low resolution, and stacked networks are large and limited to a fixed number of refinement stages [41].

The idea. Compute visual similarity once, for all pairs of pixels, and then refine a single flow field with one small, shared recurrent update operator that looks up the similarity volume around its current estimate. The paper's sections follow the three parts: feature extraction, the 4D correlation pyramid with its lookup operator, and the ConvGRU update with convex upsampling, followed by the sequence loss [41].

What it showed. RAFT reported state-of-the-art accuracy on Sintel and KITTI at the time, strong cross-dataset generalization when trained only on synthetic data, and efficiency in inference time, training and parameter count [41]. Its iteration study shows that results improve quickly in the first few updates and continue to improve with more, without diverging even after 200 updates [41].

How to read its tables. Separate the rows by training data: "C+T" (Flying Chairs then FlyingThings3D) measures generalization to Sintel and KITTI training sets, while the fine-tuned rows measure benchmark test accuracy. Sintel has two columns, clean and final; the final pass, with motion blur and atmospheric effects, is the harder one. KITTI is reported as EPE and Fl-all. The ablation table varies one design choice at a time (update operator, weight tying, context, feature scale, lookup radius, correlation pooling, correlation range, features for refinement, upsampling); read it to see which pieces matter most for each dataset.

Limits. The all-pairs volume has memory quadratic in the number of pixels at 1/8 resolution; the paper describes an equivalent on-demand computation that trades memory for time [41]. Like all supervised flow models, RAFT relies on synthetic pre-training and target-domain fine-tuning for its best numbers.

Why it matters here. RAFT's recipe became the template for RAFT-Stereo, RAFT-3D and most current flow and stereo networks.

MiDaS (Ranftl et al., 2022)

The problem. Monocular depth networks trained on one dataset generalize poorly, and existing datasets are incompatible: some give metric depth, some depth up to scale, some disparity up to scale and shift [18].

The idea. Predict inverse depth, and make the loss blind to each image's unknown scale and shift by aligning prediction and ground truth before comparing them. Mix many datasets with a principled multi-objective strategy, and add a large new source of stereo data from 3D films [18].

What it showed. Mixing complementary datasets greatly improves monocular depth, and the authors evaluate with zero-shot cross-dataset transfer, on datasets never seen in training, which

became the standard protocol for "depth foundation models" [18].

How to read its tables. Every number is computed after aligning the prediction to the ground truth in scale and shift, so they measure relative depth quality only. The loss comparison varies the alignment and the robust loss; the mixing experiments vary the datasets and the mixing strategy. When comparing with a later paper, check that it uses the same alignment, the same test datasets and the same error definitions.

Limits. The output has no metric scale; downstream use needs a few metric measurements or a separate scale estimate.

Why it matters here. The scale-and-shift-invariant loss in the main text, and the relative-depth branch of every later foundation model, come from this paper.

Guide to the Modern view surveys

Which survey for which question. For deep stereo up to about 2019 and the vocabulary of the field (pipelines, cost volumes, training and supervision), start with Laga et al. [8]. For the link between stereo and self-supervised monocular depth, confidence estimation and domain adaptation, read Poggi et al. [15]. For the 2020s architectures (iterative, transformer-based, efficient) and the current challenges, read Tosi et al. [9], which also has benchmark comparison tables. For monocular depth from images and video, read Rajapaksha et al. [56]. For optical flow, read Hur and Roth [36].

What they do not cover. None of these surveys covers the newest monocular foundation models or zero-shot stereo in depth: Tosi et al. and Rajapaksha et al. appeared in 2024, before Depth Pro and FoundationStereo, and Hur and Roth's chapter does not cover RAFT. For those, read the model papers in the state-of-the-art list directly.

How to use them. Read the taxonomy figure and the discussion section of each survey first; they summarize the authors' view of the field in a few pages. Use the benchmark tables to find representative methods, but treat numbers copied from different papers with the caution described in Evaluation: alignment, evaluated pixels and splits differ.

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DLCV 09 • 3D Object Detection: LiDAR, Cameras and Bird's-Eye View

How machines find cars, people and cyclists in 3D: point clouds and coordinate frames, LiDAR detectors from PointNet to CenterPoint, camera-only bird's-eye-view methods, sensor fusion, rotated 3D boxes and IoU, and how KITTI, nuScenes and Waymo measure the results.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-09-3d-object-detection/

What you'll learn

- What a 3D detector outputs (a box with centre, size and yaw, a class, a score and often a velocity), which sensors feed it, and how coordinate frames connect them.
- Why point clouds need special networks: they are unordered, sparse, and their density falls quickly with range; and how far a classical pipeline (ground removal, clustering, box fitting) gets on its own.
- How the main LiDAR families work: point-based (PointNet, PointNet++, PointRCNN), voxel and pillar grids (VoxelNet, SECOND, PointPillars), centre-based heads (CenterPoint), point–voxel hybrids (PV-RCNN) and sparse transformers.
- How cameras alone can detect in 3D despite the missing depth (lift-splat, BEVDet, BEVDepth, DETR3D, PETR, BEVFormer), and how LiDAR, camera and radar are fused.
- How 3D boxes are encoded and compared (yaw encoding, rotated BEV IoU, 3D IoU, rotated NMS), and how KITTI 3D AP, nuScenes mAP and NDS, and Waymo mAPH are defined.
- Why 3D detectors fail: far and sparse objects, occlusion, weather, calibration errors and the shift from one dataset to another.

The big picture

A 2D detector draws rectangles on an image. A **3D detector** places boxes in the real world: where each object is in metres, how big it is, which way it faces and, for moving things, how fast it goes. A self-driving car does not brake for a rectangle of pixels; it brakes because a pedestrian is 14 metres ahead and walking into its lane.

> **Explain like I'm five**
>
> Close your eyes in a dark room and wave a torch that can only tell you how far away the wall is in each direction. After a while you have a cloud of dots that outline the furniture. A 3D detector looks at such a cloud of dots and says: "there is a sofa here, 2 metres long, turned a bit to the left". Dots on things far away are few and far between, so faraway sofas are hard. If you only have a photo instead of dots, you see the sofa clearly but have to guess how far away it is.

Why it matters. 3D detection is the core of the perception stack in autonomous driving, where it feeds tracking, motion prediction and planning [1]. Robots need it to grasp and avoid objects, and augmented reality needs it to anchor virtual content to real things. The task also forced new ideas that 2D vision never needed: networks for unordered point sets, sparse convolution, the bird's-eye-view representation, and ways to "lift" flat images into 3D.

This chapter assumes the 2D detection vocabulary of DLCV 03 ·Object Detection: box regression targets, IoU, anchors and label assignment, NMS and the focal loss (its core building blocks), centre-based detection (anchor-free detectors), DETR-style set prediction (set prediction with transformers) and average precision (evaluating detectors). We reuse those ideas and only explain what changes in 3D. Two surveys frame the field: Mao et al. [1] and Qian et al. [2]; we return to them in the Modern view.

The 3D detection problem

Plain version. The input is one moment of sensor data: a LiDAR sweep, a ring of camera images, radar returns, or a mix. The output is a list of objects, each with a 3D box, a class name and a confidence. The box is "upright": it can turn left or right, but it never tips over.

Precise version. A 3D detector outputs a set

$$\hat{Y} = \{(\hat{\mathbf{b}}_i, \hat{c}_i, \hat{s}_i, \hat{\mathbf{v}}_i)\}_{i=1}^{N}, \qquad \hat{\mathbf{b}}_i = (x, y, z, l, w, h, \theta),$$

where (x, y, z) is the box centre in metres, (l, w, h) are its length, width and height, θ is the **yaw** (heading angle about the vertical axis), $\hat{c}_i$ is one of K classes, $\hat{s}_i \in [0, 1]$ is a confidence, $\hat{\mathbf{v}}_i = (v_x, v_y)$ is an optional ground-plane velocity in metres per second, and N varies from scene to scene.

This 7-number box is called a **7-DoF** (degrees of freedom) box; the Waymo Open Dataset, for example, labels every object this way, with length, width and height measured along the box's own x, y and z axes [3]. Roll and pitch are assumed to be zero, which is reasonable for road users on a roughly flat road.

Compared with 2D detection, three things change:

1. **The box lives in metric space.** Sizes do not shrink with distance, so a car is about 4.5 m long whether it is 5 m or 80 m away. This makes size priors (anchors, class-average sizes) far more useful than in images.
2. **Orientation matters.** A 2D box is axis-aligned. A 3D box must turn with the object, and a wrong yaw can ruin the overlap even when the centre is perfect.
3. **The data are sparse and uneven.** A LiDAR sees surfaces, not volumes, mostly the sides that face it, and far objects get only a few points.

Sensors and coordinate frames

Plain version. Each sensor measures in its own coordinates. To combine them, we move everything into one shared frame attached to the car, using the known position and orientation of each sensor, its **calibration**.

Three sensors dominate (Figure 1, left):

- **LiDAR** sends laser pulses and times their echoes. A spinning unit returns a **point cloud**: tens of thousands of points (x, y, z) per sweep, each with an **intensity** (reflectance). Depth is measured directly and accurately, but there is no colour, and the points are sparse.
- **Cameras** give dense colour and texture at low cost but no depth: a pixel tells you the *direction* of a ray, not how far along the ray the object is.
- **Radar** returns a handful of points per object with range and, through the Doppler effect, radial velocity. It works in fog and rain, but its angular resolution is poor.

The nuScenes car, for example, carries 6 cameras, 5 radars and 1 LiDAR covering 360 degrees [4]; the Waymo car carries five LiDARs and five cameras [3].

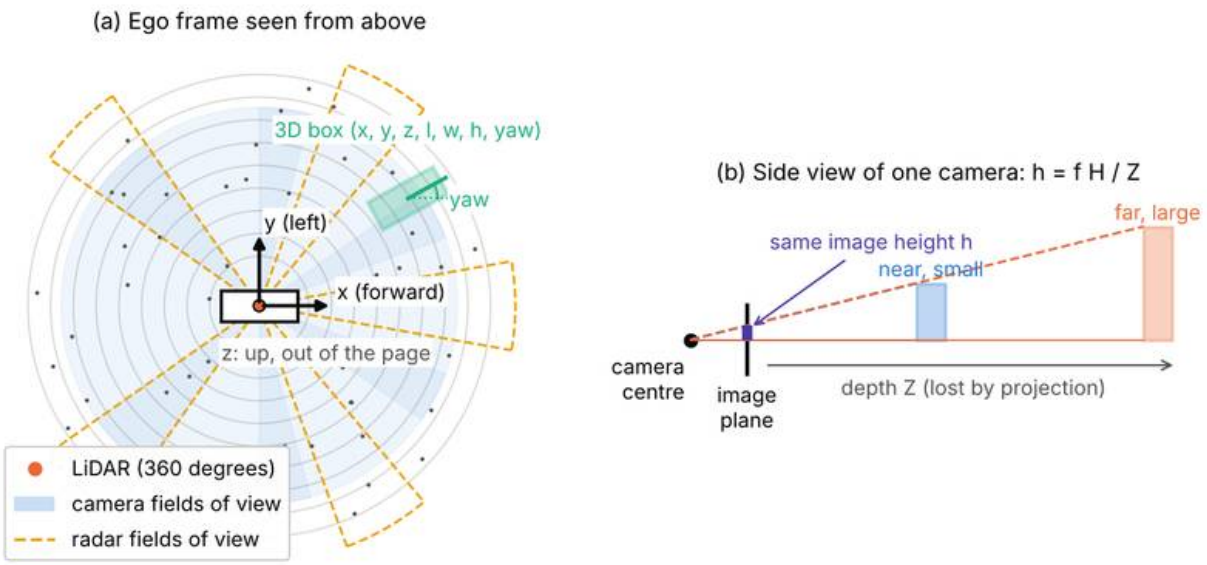


Figure 1 — (a) The ego (vehicle) frame seen from above: x forward, y left, z up, with the fields of view of a LiDAR, six cameras and several radars, and one 7-DoF box. (b) Why a single camera loses depth: a near small object and a far large object project to the same image height.

A convention used by the Waymo dataset, and by this chapter, is a right-handed **vehicle frame** with x forward, y to the left and z up; each sensor has its own frame, and a 4×4 matrix called the

extrinsics maps sensor coordinates into the vehicle frame [3]. In formulas, a point $\mathbf{p}_s$ measured by a sensor becomes

$$\mathbf{p}_v = R\,\mathbf{p}_s + \mathbf{t},$$

where R is a 3×3 rotation matrix and $\mathbf{t}$ a 3-vector translation, together the sensor's extrinsics. A camera additionally has **intrinsics** K, a 3×3 matrix holding the focal lengths and the image centre in pixels. A point $\mathbf{p}_c = (X, Y, Z)$ in camera coordinates (with Z pointing out of the lens) projects to the pixel

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \frac{1}{Z} K \begin{bmatrix} X \\ Y \\ Z \end{bmatrix},$$

where (u, v) are pixel coordinates. The division by Z is the heart of the matter: every point on the ray $\lambda(X, Y, Z)$, $\lambda > 0$, lands on the same pixel, so the depth is lost (Figure 1, right). Calibration matters in practice: if R is off by half a degree, a point 60 m away moves by about half a metre, which is enough to paint a pedestrian's pixels onto the wall behind them.

The bird's-eye view

Plain version. Look at the scene from straight above. On a road, objects rarely stand on top of each other, so from above they do not overlap, and a car has the same size wherever it is. That top-down map, the **bird's-eye view** (BEV), is where most modern detectors do their work.

In the BEV we drop z and keep a 2D grid over (x, y), for example 0.16 m cells. Height becomes a feature channel instead of an axis. Because objects keep their metric size and do not occlude each other in this view, an ordinary 2D convolutional detector can run on the BEV grid, and the planner downstream also thinks in this map. The price is that height detail is compressed, which matters little for cars on a road and more for indoor scenes with tables and shelves.

Point clouds: what the network actually sees

Plain version. A point cloud is a bag of points, not an image. The order of the points carries no meaning, most of space is empty, and the number of points on an object falls roughly with the square of its distance.

A LiDAR sweep is a set $P = \{\mathbf{p}_j\}_{j=1}^{M}$ with $\mathbf{p}_j = (x_j, y_j, z_j, r_j)$, where r_j is the intensity and M is the number of points, often tens of thousands. Three properties shape every network in this chapter:

1. **Unordered.** Any permutation of the list describes the same scene, so a network f that consumes the points must be **permutation invariant**: $f(\mathbf{p}_1, \dots, \mathbf{p}_M) = f(\mathbf{p}_{\pi(1)}, \dots, \mathbf{p}_{\pi(M)})$ for every permutation π of the indices.

2. **Sparse.** Points lie on surfaces. Divide the space around a car into 16 cm columns and more than 9 in 10 of them are empty (we measure this below).
3. **Range-dependent density.** A spinning LiDAR fires rays at fixed angular steps $\Delta\phi$ (horizontal) and $\Delta\epsilon$ (vertical). At range r, neighbouring rays are about $r\,\Delta\phi$ and $r\,\Delta\epsilon$ apart, so an object of fixed face area A receives about

$$n(r) \approx \frac{A}{r^2\,\Delta\phi\,\Delta\epsilon}$$

points, where $n(r)$ is the expected number of points and the angles are in radians. Doubling the distance cuts the points by about four.

The snippet below builds a toy spinning LiDAR (32 beams, one ray every 0.2 degrees) and counts the rays that hit a car-sized box placed at increasing range. It uses the **slab method**: a ray hits a box if the intervals where it lies between each pair of parallel faces overlap.

```
import numpy as np

# A toy spinning LiDAR mounted 1.8 m above the ground: 32 beams between -16 and +10 degrees
# elevation, one ray every 0.2 degrees of azimuth (1800 rays per beam per turn).
elev = np.deg2rad(np.linspace(-16, 10, 32))
azim = np.deg2rad(np.arange(0, 360, 0.2))
E, A = np.meshgrid(elev, azim, indexing="ij")
dirs = np.stack([np.cos(E) * np.cos(A), np.cos(E) * np.sin(A), np.sin(E)], -1).reshape(-1, 3)
origin = np.array([0.0, 0.0, 1.8])

def hits_on_box(center, size):
    """Count rays whose first hit is an axis-aligned box (slab method)."""
    lo, hi = center - size / 2, center + size / 2
    with np.errstate(divide="ignore", invalid="ignore"):
        t1, t2 = (lo - origin) / dirs, (hi - origin) / dirs
    t_near = np.nanmax(np.minimum(t1, t2), axis=1)
    t_far = np.nanmin(np.maximum(t1, t2), axis=1)
    return int(np.sum((t_far >= t_near) & (t_far > 0)))

car = np.array([4.5, 1.8, 1.5])                      # length, width, height in metres
ranges = np.array([5, 10, 20, 40, 80])
counts = np.array([hits_on_box(np.array([r, 0, car[2] / 2]), car) for r in ranges])
for r, c in zip(ranges, counts):
    print(f"car at {r:2d} m: {c:5d} points")
slope = np.polyfit(np.log(ranges[1:]), np.log(counts[1:]), 1)[0]
print(f"log-log slope (10-80 m): {slope:.2f}")
```

```
car at  5 m:  2817 points
car at 10 m:   924 points
car at 20 m:   172 points
car at 40 m:    39 points
car at 80 m:     7 points
log-log slope (10-80 m): -2.33
```

The count drops from thousands of points at 5 m to 7 points at 80 m. The fitted slope is a bit steeper than −2 because only a handful of the 32 beams cross a 1.5 m tall object far away, so the vertical sampling becomes coarse; real sensors with more beams follow $1/r^2$ more closely. Figure 2 shows the same effect on a whole scene: identical cars, very different evidence.

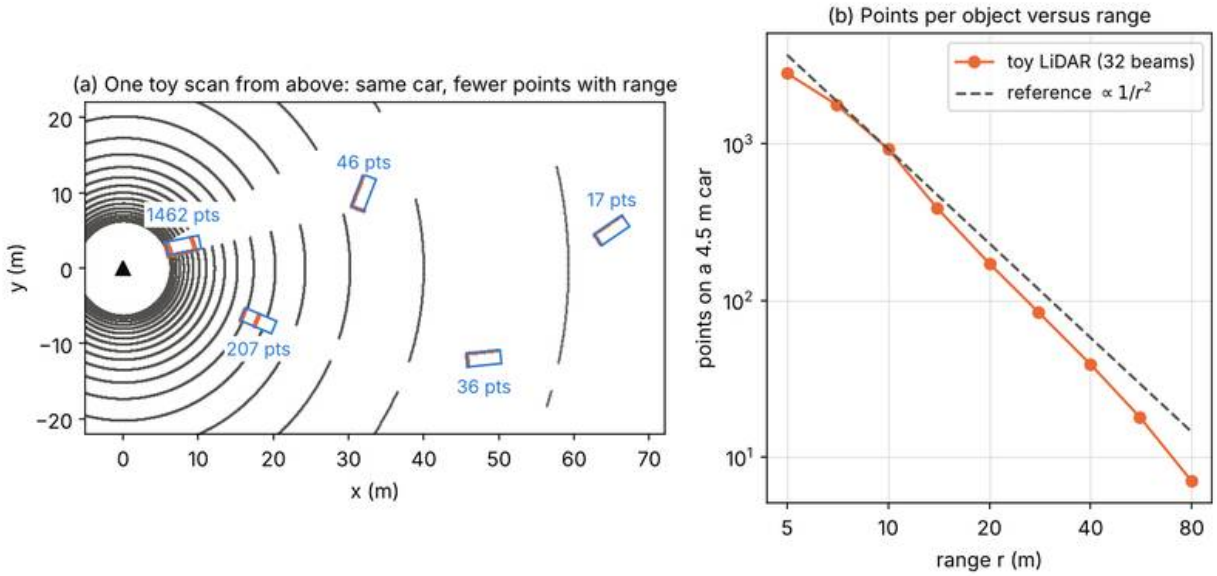


Figure 2 — (a) A toy ray-cast LiDAR scan seen from above: five identical cars, labelled with the number of points that hit each. (b) Points on a 4.5 m car versus range, with a $1/r^2$ reference line.

A classical baseline: ground removal, clustering and box fitting

Plain version. Before deep learning, a common LiDAR pipeline was: remove the road, group the remaining points into blobs, and draw a tight rectangle around each blob. It needs no training data, and it tells you exactly which problems the networks later had to solve.

The three steps are:

1. **Ground removal with RANSAC** [5]. Fit a plane $\mathbf{n}^\top \mathbf{p} = d$ (unit normal $\mathbf{n}$, offset d) by repeatedly picking three random points, building the plane through them, and counting the **inliers**, points with $|\mathbf{n}^\top \mathbf{p} - d| < \tau$ for a tolerance τ. The plane with the most inliers is the ground. (RANSAC is explained in more detail in the Supplementary notes of IP 12 ·Feature Extraction.)

2. **Clustering with DBSCAN** [6]. Group the remaining points in the BEV: two points are neighbours if they are closer than ε, a point with at least m neighbours is a core point, and clusters grow from core points; isolated points are labelled noise. DBSCAN needs no number of clusters, which suits a scene with an unknown number of objects. (See PR 07 ·Clustering, Mixtures and EM for clustering in general.)

3. **Oriented box fitting.** For each cluster, try yaw angles θ, rotate the points into the box frame, take the tight axis-aligned rectangle, and score it. Searching over angles with a fitting criterion is the idea of search-based L-shape fitting [7]. We use a simple criterion: the mean distance from each point to the nearest rectangle edge. A LiDAR usually sees an "L" (two sides of a car), and the right yaw puts those points on two edges.

```
import numpy as np
from sklearn.cluster import DBSCAN

rng = np.random.default_rng(0)

def box_surface(cx, cy, l, w, h, yaw, n):
    """n points on the sides of a yawed box that face the sensor at the origin."""
    c, s = np.cos(yaw), np.sin(yaw)
    pts = []
    for nx, ny, half, along in [(1, 0, l / 2, w), (-1, 0, l / 2, w), (0, 1, w / 2, l), (0,
     ↪  -1, w / 2, l)]:
        normal = np.array([c * nx - s * ny, s * nx + c * ny])
        mid = np.array([cx, cy]) + half * normal
        if normal @ mid >= 0:                       # face points away from the sensor: hidden
            continue
        t = rng.uniform(-along / 2, along / 2, n)
        tangent = np.array([-normal[1], normal[0]])
        xy = mid + t[:, None] * tangent
        pts.append(np.column_stack([xy, rng.uniform(0, h, n)]))
    return np.vstack(pts)

# ground plane z = 0 with slight noise, sampled uniformly in range and angle (density ~ 1/r)
r, a = rng.uniform(2, 60, 20000), rng.uniform(-np.pi / 2, np.pi / 2, 20000)
ground = np.column_stack([r * np.cos(a), r * np.sin(a), rng.normal(0, 0.03, r.size)])
objects = [(12, -4, 4.5, 1.8, 1.5, 0.3), (25, 6, 4.2, 1.7, 1.4, -1.0), (40, -10, 0.8, 0.6,
 ↪  1.7, 0.0)]
cloud = [ground] + [box_surface(*o, n=int(40000 / np.hypot(o[0], o[1]) ** 2) + 5) for o in
 ↪  objects]
pts = np.vstack(cloud) + rng.normal(0, 0.02, (sum(len(c) for c in cloud), 3))

# 1) RANSAC plane fit: pick 3 points, count inliers within 0.1 m, keep the best plane
best = None
for _ in range(100):
    p = pts[rng.choice(len(pts), 3, replace=False)]
    n = np.cross(p[1] - p[0], p[2] - p[0])
    if np.linalg.norm(n) < 1e-9:
        continue
    n /= np.linalg.norm(n)
    inl = np.abs((pts - p[0]) @ n) < 0.1
    if best is None or inl.sum() > best.sum():
        best = inl
above = pts[~best]
print(f"{len(pts)} points, {best.sum()} ground inliers, {len(above)} left")

# 2) DBSCAN in the bird's-eye view
labels = DBSCAN(eps=0.7, min_samples=5).fit_predict(above[:, :2])
```

```
# 3) oriented box per cluster: search the yaw; score = mean distance of points to the
 ↪  nearest edge
for k in range(labels.max() + 1):
    q = above[labels == k]
    best_score, best_fit = np.inf, None
    for yaw in np.deg2rad(np.arange(0, 180, 1)):
        R = np.array([[np.cos(yaw), -np.sin(yaw)], [np.sin(yaw), np.cos(yaw)]])
        uv = q[:, :2] @ R                              # coordinates along the box axes
        lo, hi = uv.min(0), uv.max(0)
        edge_dist = np.minimum(np.minimum(uv - lo, hi - uv)[:, 0], np.minimum(uv - lo, hi -
         ↪  uv)[:, 1])
        if edge_dist.mean() < best_score and hi[0] - lo[0] >= hi[1] - lo[1]:   # x axis =
         ↪  long side
            best_score, best_fit = edge_dist.mean(), (R @ ((lo + hi) / 2), hi - lo,
             ↪  np.rad2deg(yaw))
    ctr, size, yaw = best_fit
    print(f"cluster {k}: {len(q):4d} pts, centre ({ctr[0]:5.1f}, {ctr[1]:5.1f}), "
          f"size {size[0]:.1f} x {size[1]:.1f} m, yaw {yaw:4.0f} deg, height {q[:,
           ↪  2].max():.1f} m")
```

```
20694 points, 19944 ground inliers, 750 left
cluster 0:  471 pts, centre ( 12.0,  -4.0), size 4.6 x 1.9 m, yaw   17 deg, height 1.5 m
cluster 1:  124 pts, centre ( 25.0,   6.1), size 4.1 x 1.9 m, yaw  123 deg, height 1.4 m
cluster 2:   53 pts, centre ( 40.0,  -9.9), size 0.8 x 0.6 m, yaw    0 deg, height 1.7 m
```

On this synthetic scene (true objects: a car at $(12, -4)$ with yaw 17°, a car at $(25, 6)$ with yaw $-57°$, which is 123° modulo 180°, and a 0.8×0.6 m pedestrian at $(40, -10)$) the pipeline recovers centres, sizes and yaws well. Figure 3 shows the result. A tempting simpler criterion, the smallest rectangle area, fails here: the hull of an L-shaped point set is roughly a triangle, and a rectangle along the triangle's long diagonal encloses it with the same area, so the area criterion can pick a yaw about 20° off.

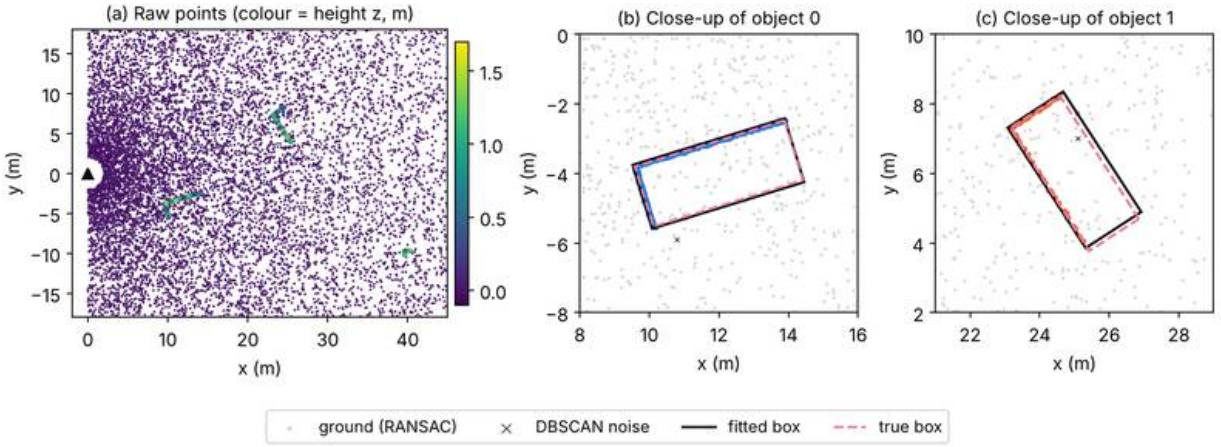


Figure 3 — The classical pipeline on a synthetic scene. (a) Raw points coloured by height. (b, c) After

RANSAC ground removal and DBSCAN clustering: object points (colour), the fitted box (black) and the true box (dashed). The LiDAR sees only the two sides facing it.

The baseline also shows its limits, and each one is a reason for learning:

- **No semantics.** A cluster is "something". Telling a pedestrian from a pole, or a car from a bush, needs learned features.
- **Partial views.** The box is fitted to visible points only. A far car seen from the back gives a short line, so its length is unknown. Learned detectors use class-level size priors to complete what is not seen.
- **Merging and splitting.** Two people walking together merge into one cluster; a truck seen through a gap splits into two.
- **Heading.** The fit gives yaw only modulo 180°: it cannot tell the front of a car from its back.

3D boxes: encoding, IoU and NMS

These are the 3D versions of the building blocks of DLCV 03. We explain only what is new.

Box encoding and the yaw problem

Plain version. As in 2D, a network predicts small corrections to a reference box (an anchor or a point). The new part is the angle: angles wrap around, and a box turned by 180° covers exactly the same space, so a naive angle loss punishes the network for answers that are geometrically correct.

VoxelNet [8], SECOND [9] and PointPillars [10] use the same residuals between a ground-truth box (superscript g) and an anchor (superscript a):

$$\Delta x = \frac{x^g - x^a}{d^a}, \quad \Delta y = \frac{y^g - y^a}{d^a}, \quad \Delta z = \frac{z^g - z^a}{h^a}, \quad \Delta l = \log \frac{l^g}{l^a}, \quad \Delta w = \log \frac{w^g}{w^a}, \quad \Delta h = \log \frac{h^g}{h^a},$$

where $d^a = \sqrt{(l^a)^2 + (w^a)^2}$ is the diagonal of the anchor's base, which normalizes horizontal offsets in both directions by one length. These are the 2D targets of DLCV 03 with one more dimension.

For the angle, VoxelNet regressed the raw difference $\theta^g - \theta^a$. SECOND [9] pointed out that this creates a conflict: a prediction off by exactly π gives the same box but a large loss. Its **sine-error loss**

$$L_\theta = \text{SmoothL1}(\sin(\theta^p - \theta^t))$$

(with θ^p the predicted and θ^t the target angle, and SmoothL1 the smooth L1 loss of DLCV 03: quadratic for errors below 1, linear above) is zero both for a perfect prediction and for one flipped

by π. Because it cannot tell front from back, SECOND adds a small **direction classifier** with two bins that decides which way the box points [9]; PointPillars uses the same pair of losses [10]. CenterPoint instead regresses $(\sin\theta, \cos\theta)$ and recovers the angle with $\theta = \operatorname{atan2}(\sin\theta, \cos\theta)$, which is continuous and unambiguous [11].

Optional: Yaw encodings compared (in the Supplementary notes). The four encodings side by side on worked angle pairs, including the flipped-box case.

BEV IoU and 3D IoU of rotated boxes

Plain version. IoU still means "overlap divided by total area", but the boxes are now rotated, so the overlap is a polygon, not a rectangle. In 3D, multiply that polygon's area by the overlap of the two height intervals.

For two upright boxes A and B, let $A_{\text{bev}} \cap B_{\text{bev}}$ be the intersection of their BEV rectangles (a convex polygon with up to 8 corners) and let

$$o_z = \max\Big(0,\ \min\left(z_A + \tfrac{h_A}{2},\, z_B + \tfrac{h_B}{2}\right) - \max\left(z_A - \tfrac{h_A}{2},\, z_B - \tfrac{h_B}{2}\right)\Big)$$

be the overlap of their vertical extents, where z is a box's centre height and h its height. Then

$$\text{IoU}_{\text{bev}} = \frac{|A_{\text{bev}} \cap B_{\text{bev}}|}{l_A w_A + l_B w_B - |A_{\text{bev}} \cap B_{\text{bev}}|}, \qquad \text{IoU}_{\text{3D}} = \frac{|A_{\text{bev}} \cap B_{\text{bev}}|\, o_z}{l_A w_A h_A + l_B w_B h_B - |A_{\text{bev}} \cap B_{\text{bev}}|\, o_z},$$

where $|\cdot|$ is area and l, w, h are the box sizes. The intersection polygon of two convex polygons can be computed with **Sutherland–Hodgman clipping** [12]: cut the first polygon by each edge line of the second in turn, keeping only the part on the inner side. The polygon's area then follows from the **shoelace formula** $\frac{1}{2}\left|\sum_k (x_k y_{k+1} - x_{k+1} y_k)\right|$ over its corners (x_k, y_k), with indices taken cyclically.

```
import numpy as np

def corners(box):
    """BEV corners (counter-clockwise) of box = (x, y, z, l, w, h, yaw); z is the box
     ↪  centre."""
    x, y, _, l, w, _, yaw = box
    c, s = np.cos(yaw), np.sin(yaw)
    local = np.array([[l, w], [-l, w], [-l, -w], [l, -w]]) / 2
    return local @ np.array([[c, s], [-s, c]]) + [x, y]

def clip(subject, clipper):
    """Sutherland-Hodgman: keep the part of polygon `subject` inside convex CCW polygon
     ↪  `clipper`."""
    out = list(subject)
    for i in range(len(clipper)):
        a, b = clipper[i], clipper[(i + 1) % len(clipper)]
        inside = lambda p: (b[0] - a[0]) * (p[1] - a[1]) - (b[1] - a[1]) * (p[0] - a[0]) >= 0
```

```
        pts, out = out, []
        for j in range(len(pts)):
            p, q = pts[j], pts[(j + 1) % len(pts)]
            if inside(q):
                if not inside(p):
                    out.append(cross_point(p, q, a, b))
                out.append(q)
            elif inside(p):
                out.append(cross_point(p, q, a, b))
        if not out:
            break
    return np.array(out)

def cross_point(p, q, a, b):
    """Intersection of segment p-q with the infinite line through a and b."""
    d1, d2 = q - p, b - a
    t = ((a[0] - p[0]) * d2[1] - (a[1] - p[1]) * d2[0]) / (d1[0] * d2[1] - d1[1] * d2[0])
    return p + t * d1

def area(poly):                                  # shoelace formula
    if len(poly) < 3:
        return 0.0
    x, y = poly[:, 0], poly[:, 1]
    return 0.5 * abs(x @ np.roll(y, -1) - y @ np.roll(x, -1))

def iou_bev_3d(a, b):
    inter = area(clip(corners(a), corners(b)))
    bev = inter / (a[3] * a[4] + b[3] * b[4] - inter)
    dz = max(0.0, min(a[2] + a[5] / 2, b[2] + b[5] / 2) - max(a[2] - a[5] / 2, b[2] - b[5] /
     ↪  2))
    inter3 = inter * dz
    return bev, inter3 / (a[3] * a[4] * a[5] + b[3] * b[4] * b[5] - inter3)

car = (0, 0, 0, 4, 2, 1.5, 0)
cases = {"identical":                (car, car),
         "rotated by 180 deg":       (car, (0, 0, 0, 4, 2, 1.5, np.pi)),
         "rotated by 90 deg":        (car, (0, 0, 0, 4, 2, 1.5, np.pi / 2)),
         "unit square vs 45 deg":    ((0, 0, 0, 1, 1, 1, 0), (0, 0, 0, 1, 1, 1, np.pi / 4)),
         "shifted 1 m along length": (car, (1, 0, 0, 4, 2, 1.5, 0)),
         "same BEV, raised 0.5 m":   (car, (0, 0, 0.5, 4, 2, 1.5, 0)),
         "far apart":                (car, (10, 0, 0, 4, 2, 1.5, 0.3))}
for name, (a, b) in cases.items():
    bev, iou3 = iou_bev_3d(np.array(a, float), np.array(b, float))
    print(f"{name:26s} BEV IoU {bev:.4f}   3D IoU {iou3:.4f}")
```

```
identical                  BEV IoU 1.0000   3D IoU 1.0000
rotated by 180 deg         BEV IoU 1.0000   3D IoU 1.0000
rotated by 90 deg          BEV IoU 0.3333   3D IoU 0.3333
unit square vs 45 deg      BEV IoU 0.7071   3D IoU 0.7071
shifted 1 m along length   BEV IoU 0.6000   3D IoU 0.6000
same BEV, raised 0.5 m     BEV IoU 1.0000   3D IoU 0.5000
far apart                  BEV IoU 0.0000   3D IoU 0.0000
```

Every value matches a hand calculation: a 4×2 box against itself rotated by 90° overlaps in a 2×2 square, so IoU $= 4/(8 + 8 - 4) = 1/3$; a unit square against itself rotated by 45° overlaps in a regular octagon of area $2(\sqrt{2} - 1)$, which gives $\sqrt{2}/2 \approx 0.7071$; a box raised by a third of its height keeps the full BEV overlap but only half of the 3D one. Note the second line: a box turned by 180° has IoU 1. IoU is blind to the heading direction, which is why the Waymo metric adds a heading term (see below).

Figure 4 (right) shows how harsh yaw errors are for elongated objects. With a perfect centre and size, a car-sized box falls below IoU 0.7 at a yaw error of about 17°, and a bus-sized box at about 8.5°, while a square pedestrian box never drops below 0.7.

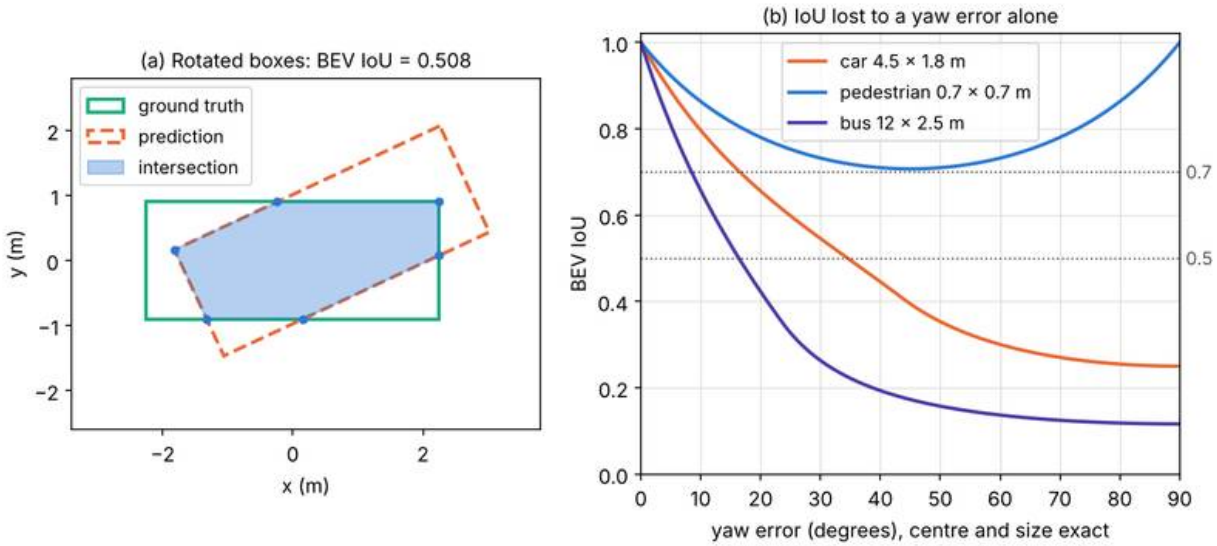


Figure 4 — (a) Two rotated boxes; their intersection is a polygon found by clipping. (b) BEV IoU lost to a yaw error alone, for three object shapes with exact centre and size.

Rotated NMS and its alternatives

Plain version. Removing duplicate boxes works as in 2D, but "overlap" must now be measured between rotated boxes, which is slower. Two shortcuts are common: approximate the overlap with axis-aligned boxes, or avoid NMS altogether with centre peaks or set prediction.

Greedy NMS in 3D sorts boxes by score and removes those whose BEV IoU with a kept box exceeds a threshold, using the rotated IoU above. Because objects on a road rarely overlap in the BEV, the threshold can be low. PointPillars found that NMS on axis-aligned BEV boxes with threshold 0.5 gave similar results to rotated NMS and was much faster [10]. CenterPoint reads objects off the local maxima of a centre heatmap, like CenterNet [13], which removes most duplicates by construction; the paper still applies NMS to the peaks before its second stage [11]. Camera methods in the DETR family [14], such as DETR3D, predict a set of boxes and need no NMS at all [15].

LiDAR-based detectors

Plain version. LiDAR detectors differ mainly in how they turn an unordered point set into something a network can process: keep the points (point-based), drop them into a 3D grid (voxels), drop them into vertical columns (pillars), or mix the two. After that, most use a 2D detection head on the BEV.

Point-based: PointNet, PointNet++ and PointRCNN

Plain version. Process every point with the same small network, then summarize all points with a maximum. A maximum does not care about the order of its inputs, so the result is permutation invariant.

PointNet [16] computes

$$f(\mathbf{p}_1, \dots, \mathbf{p}_M) = \gamma\Big(\max_{j=1,\dots,M} h(\mathbf{p}_j) \Big),$$

where h is a shared multilayer perceptron applied to each point, max is taken element-wise over the points (a symmetric function, so the order does not matter), and γ is a second network that maps the pooled feature to the output. It sees each point alone and then the whole set at once, so it misses local structure. **PointNet++** [17] fixes this with **set abstraction** layers: choose a subset of centre points by **farthest point sampling** (each new centre is the point farthest from those already chosen), gather the neighbours within a radius of each centre (a **ball query**), and run a small PointNet on each neighbourhood. Stacking these layers gives a hierarchy of local features, like a CNN's growing receptive fields. Both are mainly backbones: they produce features, and a detection head is built on top.

PointRCNN [18] is a two-stage detector that works directly on points. Stage 1 segments the scene into foreground and background points with a PointNet++ backbone and proposes a 3D box from each foreground point, bottom-up. Stage 2 pools the points inside each proposal, transforms them into the proposal's own **canonical coordinates** (centred and rotated so the box is axis-aligned), and refines the box. Point-based detectors are accurate on small objects, but sampling and neighbour search on large clouds are slow.

Voxel-based: VoxelNet and SECOND

Plain version. Divide space into small 3D cells (voxels), summarize the points in each cell with a tiny PointNet, and then run convolutions on the resulting 3D grid. Because almost all voxels are empty, compute only where there are points.

VoxelNet [8] groups points by voxel and augments each point with its offset from the voxel's centroid, giving a 7-D input $(x, y, z, r, x - v_x, y - v_y, z - v_z)$, where (v_x, v_y, v_z) is the mean of the points in the voxel. A stack of **voxel feature encoding** (VFE) layers, each a shared fully connected layer followed by element-wise max pooling within the voxel, turns each voxel into one feature vector. 3D convolutions then process the grid, and a region proposal network in the style of Faster R-CNN [19] predicts boxes on the BEV feature map. It was end-to-end and strong, but dense 3D convolutions made it slow.

SECOND [9] replaced the dense 3D convolutions by **sparse convolutions**, which store and compute only the occupied voxels. It uses mostly **submanifold** sparse convolutions [20], whose outputs are active only where the inputs are active, so the set of occupied sites does not grow layer after layer. SECOND also introduced the sine-error angle loss and ground-truth sampling augmentation (both described in this chapter), and sparse voxel backbones of this kind were adopted by later detectors such as PV-RCNN [21].

Optional: Sparse and submanifold convolution (in the Supplementary notes). Rule tables, why ordinary sparse convolution "dilates" the active set, and a small measurement.

Pillar-based: PointPillars

Plain version. Make the cells infinitely tall. Each vertical column (a **pillar**) is summarized by a tiny PointNet, and the result is a flat "pseudo-image" that an ordinary 2D CNN can process quickly.

PointPillars [10] has three stages: a pillar feature network, a 2D convolutional backbone and an SSD-style detection head [22]. Each point in a pillar is decorated to $D = 9$ dimensions: (x, y, z, r), its offsets (x_c, y_c, z_c) from the mean of the points in the pillar, and its offsets (x_p, y_p) from the pillar's centre in the x–y plane. The non-empty pillars are packed into a dense tensor of shape (D, P, N), with at most P pillars per sample and N points per pillar (12,000 and 100 in the paper, at a 0.16 m grid), zero-padded or randomly subsampled. A linear layer, batch normalization and ReLU are applied to every point, a max over the N points gives one C-dimensional feature per pillar, and the features are **scattered** back to their grid positions to form a $C \times H \times W$ pseudo-image. The paper reports 62 Hz for its full pipeline, and a faster variant at 105 Hz [10].

Optional: PointPillars: a reading guide (in the Paper guides). How to read the paper's speed and accuracy tables, and what changed after KITTI switched to 40 recall positions.

The snippet performs the input side of this pipeline on a toy cloud: binning, the 9-D decoration, the dense (P, N, D) tensor and the scatter (the PointNet weights are random here; in a real model they are learned).

```
import numpy as np
```

```
rng = np.random.default_rng(0)
# toy cloud: ground sampled uniformly in range and angle (so density falls as 1/r) + two
 ↪  "cars"
r, a = rng.uniform(2, 70, 10000), rng.uniform(-0.5, 0.5, 10000)
ground = np.column_stack([r * np.cos(a), r * np.sin(a), rng.normal(-1.7, 0.03, r.size)])
cars = [c + rng.uniform([-2.2, -0.9, -1.7], [2.2, 0.9, -0.2], (n, 3))
        for c, n in [(np.array([10.0, 2.0, 0.0]), 3000), (np.array([35.0, -6.0, 0.0]), 60)]]
xyz = np.vstack([ground] + cars)
pts = np.column_stack([xyz, rng.uniform(0, 1, len(xyz))])          # x, y, z, reflectance

# 1) bin into a bird's-eye-view grid of 0.16 m pillars over x in [0, 70.4), y in [-40, 40)
vx, x0, y0, W, H = 0.16, 0.0, -40.0, 440, 500
ix = np.floor((pts[:, 0] - x0) / vx).astype(int)
iy = np.floor((pts[:, 1] - y0) / vx).astype(int)
ok = (ix >= 0) & (ix < W) & (iy >= 0) & (iy < H)
pts, ix, iy = pts[ok], ix[ok], iy[ok]
pillar_id, inv, counts = np.unique(iy * W + ix, return_inverse=True, return_counts=True)
print(f"{len(pts)} points -> {len(pillar_id)} non-empty pillars out of {W * H} "
      f"({100 * (1 - len(pillar_id) / (W * H)):.1f}% empty); max {counts.max()} points in
       ↪  one pillar")

# 2) per-point 9-D features: x, y, z, r, offsets to the pillar mean, offsets to the pillar
 ↪  centre
mean = np.zeros((len(pillar_id), 3))
np.add.at(mean, inv, pts[:, :3])
mean /= counts[:, None]
centre = np.column_stack([(ix + 0.5) * vx + x0, (iy + 0.5) * vx + y0])
feat = np.column_stack([pts, pts[:, :3] - mean[inv], pts[:, :2] - centre])

# 3) dense tensor (P pillars, N points, D=9): pad with zeros, keep at most N points per
 ↪  pillar
P, N = len(pillar_id), 16          # PointPillars caps P at 12000 and N at 100, sampling at
 ↪  random
order = np.argsort(inv, kind="stable")
slot = np.arange(len(inv)) - np.searchsorted(inv[order], inv[order])   # rank inside its
 ↪  pillar
keep = slot < N
dense = np.zeros((P, N, 9))
dense[inv[order][keep], slot[keep]] = feat[order][keep]
print("dense pillar tensor:", dense.shape, "| points dropped by the cap N:",
 ↪  int((~keep).sum()))

# 4) a one-layer 'PointNet' (random weights here): linear + ReLU per point, max over points,
 ↪  scatter
Wt = rng.normal(0, 0.3, (9, 64))
pillar_feat = np.maximum(dense @ Wt, 0).max(axis=1)                    # (P, 64)
canvas = np.zeros((64, H * W))
canvas[:, pillar_id] = pillar_feat.T
print("pseudo-image:", canvas.reshape(64, H, W).shape)
```

```
13060 points -> 9481 non-empty pillars out of 220000 (95.7% empty); max 24 points in one
 ↪  pillar
dense pillar tensor: (9481, 16, 9) | points dropped by the cap N: 23
```

```
pseudo-image: (64, 500, 440)
```

More than 95% of the 220,000 cells are empty in this toy, close to the roughly 97% sparsity the PointPillars paper reports for a real 64-beam sweep at the same grid size [10]. This is why pillarization and scattering, not the convolutions, decide how much memory the input step needs. Figure 5 shows the grid on a ray-cast scan.

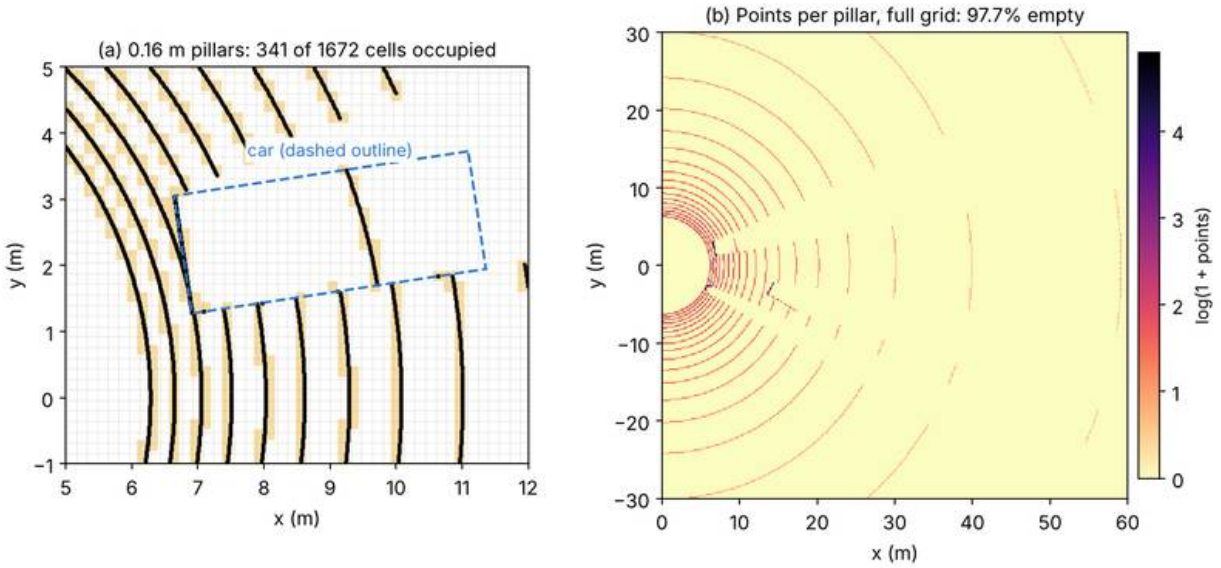


Figure 5 — Pillarization of a toy ray-cast scan. (a) Occupied 0.16 m pillars (shaded) in a close-up. (b) Points per pillar over the whole grid: 97.7% of the pillars are empty, and occupied ones thin out with range.

Centre-based heads: CenterPoint

Plain version. Instead of anchors at several orientations, find object *centres* as peaks in a BEV heatmap and read every other property from the feature at the peak. A centre has no orientation, so the rotation problem of anchors disappears.

CenterPoint [11] keeps a standard voxel or pillar backbone and replaces the anchor head by a centre head in the spirit of CenterNet [13]. The heatmap has one channel per class and is trained with a focal loss [23] against Gaussian peaks rendered at the projected box centres; because objects are small in the BEV, the Gaussian radius is enlarged to at least 2 cells. At each centre, regression heads predict a sub-voxel offset, the height above ground, the 3D size, the yaw as $(\sin\theta, \cos\theta)$ and a 2D velocity. The velocity comes from feeding two consecutive sweeps, and it turns tracking into greedy closest-point matching of centres between frames. A light second stage samples features at the centres of the box faces and predicts a refined box and a confidence trained toward the box's 3D IoU with the ground truth. Centre-based heads spread beyond LiDAR: the camera detector BEVDet, for example, uses CenterPoint's head [24].

Point–voxel hybrids: PV-RCNN

Plain version. Voxels are fast and give good proposals; points keep exact positions. Use voxels to propose and points to refine.

PV-RCNN [21] runs a sparse 3D voxel CNN to get features and proposals, then summarizes the multi-scale voxel features into a small set of **keypoints** (voxel set abstraction). For each proposal,

RoI-grid pooling gathers keypoint features at a regular grid of points inside the box, with several radii, and a refinement head predicts the final box and confidence. Two-stage hybrids of this kind were among the most accurate LiDAR detectors before transformer backbones.

Transformers and sequence models on points

Plain version. A car covers only a tiny part of a 150 m scene, so shrinking the feature map, as 2D networks do, throws away detail. Attention lets features interact over long distances without shrinking, if it is restricted to the occupied voxels.

SST [25] observed that downsampling brings little benefit in LiDAR detection and kept a single stride throughout, using attention among nearby occupied voxels to regain the receptive field that downsampling would have provided; the paper reports that this helps most on small objects such as pedestrians. **DSVT** [26] reorganized this attention so that it runs fully in parallel on a GPU and is easy to deploy. **LION** [27] and **Voxel Mamba** [28] replace attention by linear-time sequence models (a linear recurrent network and a state space model) that read the voxels as long sequences. The one thing to remember: these backbones buy long-range context at a cost that grows with the number of occupied voxels, not with the size of the scene.

Optional: Attention on sparse voxels (in the Supplementary notes). How windows, equal-size sets and space-filling-curve serialization make attention and sequence models practical on sparse voxels.

Camera-only 3D detection

Plain version. Cameras are cheap and see colour, texture and signs, but a photo is flat. Every camera detector must answer the same question: how far away is each thing it sees? This section first shows why that is hard, then the two main ways to build a 3D map from images: push pixels out into 3D, or pull image features into 3D positions.

Why one camera is not enough: depth ambiguity

Plain version. A photo of a toy car close up and a real car far away can look identical. To get depth from one image, a network must rely on clues such as the object's typical size, where it touches the road and perspective, and small errors in those clues become large errors in metres far away.

For a pinhole camera with focal length f (in pixels), an object of real height H at depth Z appears $h = fH/Z$ pixels tall, so $Z = fH/h$. Differentiating gives

$$\left|\frac{\partial Z}{\partial h}\right| = \frac{fH}{h^2} = \frac{Z^2}{fH},$$

so a one-pixel measurement error costs a depth error that grows with the square of the distance, and an error in the assumed size H scales the depth by the same factor.

```
import numpy as np

f, H = 1000.0, 1.5                      # focal length in pixels, true car height in metres
for Z in [10, 20, 40, 80]:
    h = f * H / Z                       # image height of the car in pixels
    z_pix = f * H / (h - 1)             # depth if we mis-measure the height by one pixel
    z_size = f * (1.1 * H) / h          # depth if we assume a car 10% taller than it is
    print(f"Z = {Z:2d} m: car is {h:6.2f} px tall | 1 px error -> {z_pix - Z:5.2f} m | "
          f"10% size error -> {z_size - Z:4.1f} m")
```

```
Z = 10 m: car is 150.00 px tall | 1 px error ->  0.07 m | 10% size error ->  1.0 m
Z = 20 m: car is  75.00 px tall | 1 px error ->  0.27 m | 10% size error ->  2.0 m
Z = 40 m: car is  37.50 px tall | 1 px error ->  1.10 m | 10% size error ->  4.0 m
Z = 80 m: car is  18.75 px tall | 1 px error ->  4.51 m | 10% size error ->  8.0 m
```

At 80 m a single pixel already moves the car by about 4.5 m, far outside a 3D IoU of 0.7. This is why monocular 3D detectors trail LiDAR so far on far objects. Monocular methods such as **FCOS3D** [29] adapt a 2D one-stage detector (FCOS [30]) to regress depth, size and orientation at the projected 3D centre. **Pseudo-LiDAR** [31] made an influential point: much of the gap between image-based and LiDAR-based detectors came from the *representation*, not only from depth quality; turning an estimated depth map into a 3D point cloud and running a LiDAR detector on it worked far better than processing the depth map as an image.

Lifting images into the bird's-eye view

Plain version. Each pixel does not know its depth, so give it a probability distribution over depths, place a copy of its feature at every possible depth weighted by that probability, and then drop all

those 3D features onto the BEV grid. The result can be fused across cameras and processed like a LiDAR BEV map.

Lift-Splat-Shoot (LSS) [32] "lifts" each image into a frustum of features. At each pixel, the network predicts a context vector $\mathbf{c} \in \mathbb{R}^C$ and a distribution $\boldsymbol{\alpha}$ over a set $\mathcal{D}$ of discrete depths ($\alpha_d \geq 0$, $\sum_{d \in \mathcal{D}} \alpha_d = 1$). The feature placed at depth d along the pixel's ray is

$$\mathbf{c}_d = \alpha_d \, \mathbf{c},$$

an outer product of depth probabilities and context. A confident depth puts the feature at one place; an uncertain depth spreads it along the ray. The "splat" step transforms all frustum points into the vehicle frame with the camera extrinsics and intrinsics and sum-pools them into BEV pillars, using a cumulative-sum trick to make the pooling fast [32]. The paper evaluated BEV segmentation and planning; **BEVDet** [24] turned the same view transform into a 3D detector with a BEV encoder and a centre-based head, plus data augmentation applied in both image and BEV space. **BEVDepth** [33] found that the depth distribution learned only from the detection loss was poor, and supervised it explicitly with depth from LiDAR points projected into the images, together with a depth network aware of the camera parameters.

Optional: Lift-Splat-Shoot: a reading guide (in the Paper guides). What the LSS paper actually evaluated, its robustness experiments and its limits.

Querying images from 3D: DETR3D, PETR and BEVFormer

Plain version. Go the other way round. Start from 3D guesses, either object queries or a grid of BEV cells, project them into the images with the known camera matrices, and fetch the image features they land on. No depth map is needed.

DETR3D [15] decodes each object query to a 3D reference point, projects it into all cameras, samples the image features there by bilinear interpolation and uses them to refine the query; training uses DETR's set-to-set loss [14], so no NMS is needed. **PETR** [34] instead encodes 3D position into the image features themselves: the camera frustum is discretized into points at several depths, transformed to 3D with the camera parameters, and passed through a small network to produce a **3D position embedding** that is added to the image features; ordinary DETR-style queries then attend to these position-aware features. **BEVFormer** [35] keeps a grid of learnable **BEV queries**, one per BEV cell. Spatial cross-attention lets each BEV query sample features in the image regions where its location projects, using deformable attention [36], and temporal self-attention lets it look at the BEV features of the previous frame, which helps with velocity and with objects that are briefly hidden. Its BEV map supports detection and map segmentation heads.

Time turned out to be a strong source of depth for cameras: motion between frames is a form of stereo. **StreamPETR** [37] propagates object queries from frame to frame online and reports

camera-only results comparable to LiDAR-based methods on nuScenes, and **Far3D** [38] targets long-range camera detection with 3D queries generated from 2D detections.

Fusing LiDAR, cameras and radar

Plain version. LiDAR is good at "where", cameras at "what". Fusion combines them, and the main design question is *where* in the network to combine: at the raw points, at the proposals, in a shared BEV map, or through attention.

We group methods by where fusion happens; Wang et al. [39] give a finer taxonomy.

- **Point-level (early) fusion. PointPainting** [40] projects each LiDAR point into an image semantic segmentation and appends the class scores to the point; any LiDAR detector can then run on the "painted" cloud.
- **Proposal-level fusion. MV3D** [41] generates 3D proposals from a BEV map of the point cloud and fuses region features from the BEV, the LiDAR front view and the image. **Frustum PointNets** [42] go from 2D to 3D: a 2D detector finds the object in the image, and a PointNet processes only the points inside the frustum behind that 2D box.
- **BEV-level fusion.** Two methods named **BEVFusion** turn camera features into a BEV map (with an LSS-style view transform) and combine them with the LiDAR BEV map. One, from MIT [43], fuses with concatenation and a convolutional encoder, supports detection and map segmentation, and speeds up the camera-to-BEV pooling by more than 40 times. The other [44] keeps the camera stream independent of the LiDAR, so the detector still works when LiDAR fails, and reports large gains under simulated LiDAR malfunctions.
- **Query-level fusion. TransFusion** [45] first predicts object queries from LiDAR, then lets each query attend to image features with a transformer decoder. This "soft" association by attention, rather than a hard projection of points to pixels, makes it more robust to poor image conditions and calibration errors. **UniTR** [46] goes further and processes camera and LiDAR tokens with one shared transformer backbone.

Radar. Radar adds velocity and robustness to weather. **CenterFusion** [47] detects object centres in the image, associates radar returns with each detection through a frustum around it, and uses the radar features to improve depth, rotation and velocity, raising the nuScenes detection score of a camera detector substantially.

Training: assignment and augmentation

Plain version. Training a 3D detector resembles 2D training, with two twists: assignment uses rotated BEV overlaps or centre heatmaps, and data augmentation can cut objects out of one scene and paste them into another, which is impossible to do convincingly with photos.

Assignment. Anchor-based heads place a few anchors per BEV cell (one size per class, usually at yaw 0° and 90°) and assign them by BEV IoU, like the 2D rules of DLCV 03. VoxelNet's car anchors, for instance, are positive above BEV IoU 0.6 and negative below 0.45 [8]. Because 3D scenes contain many more anchors than objects, these heads use the focal loss [23]. Centre-based heads replace IoU assignment by Gaussian heatmap targets [11].

Augmentation. LiDAR scenes are few (KITTI has 7,481 training scans [48]) and objects per scene are fewer still. Standard augmentations are [8], [9], [10]:

- **Ground-truth sampling.** Build a database of all labelled boxes and the points inside them, then paste randomly chosen objects into each training scene [9]. PointPillars samples 15 cars, 0 pedestrians and 8 cyclists per scene on KITTI [10]. VoxelNet already used a collision test to reject physically impossible results when perturbing boxes [8], and the same check is needed when pasting.
- **Per-object perturbation.** Rotate and translate each box with its points; VoxelNet draws rotations from $[-\pi/10, \pi/10]$ and translations from a zero-mean Gaussian with standard deviation 1 m [8].
- **Global transforms.** Rotate the whole scene about the vertical axis (VoxelNet draws the angle from $[-\pi/4, \pi/4]$, simulating the car making a turn), scale it by a factor from $[0.95, 1.05]$, and flip it left–right [8], [10].

These are cheap because a point cloud is just coordinates: rotating points and boxes together keeps the labels exact. The catch is consistency across sensors: an object pasted into a LiDAR scan is not in the camera images, so fusion models need extra care.

Optional: Ground-truth sampling in detail (in the Supplementary notes). The database, the collision check and the pitfalls (floating objects, inconsistent occlusion, camera mismatch).

Datasets and metrics

Plain version. Three public benchmarks shaped the field. KITTI is small and front-facing, with a strict 3D IoU metric. nuScenes and Waymo are large and 360-degree; nuScenes matches boxes by centre distance and adds error terms, and Waymo weights each correct box by how well its heading matches.

Dataset	Sensors (from the papers)	Scale (from the papers)	Main detection metric
KITTI 3D [49], [48]	Two stereo camera pairs (grey and colour), a Velodyne HDL-64E LiDAR	7,481 training and 7,518 test images with point clouds; 80,256 labelled objects	3D AP at IoU 0.7 (car) / 0.5 (pedestrian, cyclist), easy / moderate / hard
nuScenes [4], [50]	6 cameras, 5 radars, 1 LiDAR, 360°	1,000 scenes of 20 s; 23 classes and 8 attributes (10 classes for detection)	mAP with centre-distance matching; NDS
Waymo Open [3], [51]	5 LiDARs, 5 cameras	1,150 scenes of 20 s; about 12M labelled 3D LiDAR objects	mAPH at LEVEL_1 / LEVEL_2

KITTI: 3D AP with difficulty levels

On the KITTI 3D benchmark a detection is correct if its 3D IoU with a ground-truth box is at least 0.7 for cars and 0.5 for pedestrians and cyclists [48]. Objects are split into three cumulative difficulty levels by their image appearance [48]:

- **Easy:** 2D box at least 40 px tall, fully visible, at most 15% truncated.
- **Moderate:** at least 25 px tall, at most partly occluded, at most 30% truncated.
- **Hard:** at least 25 px tall, at most "difficult to see", at most 50% truncated.

Methods are ranked by moderate AP. Since 8 October 2019 the benchmark computes AP from 40 recall positions instead of the original 11, following the analysis of Simonelli et al. [48], [52]; papers write this as $\mathrm{AP}|_{R_{40}}$, and numbers computed with the two rules must not be compared. The labels cover only objects visible in the front camera, so KITTI LiDAR detectors usually keep only the points that project into the image [10].

nuScenes: centre-distance mAP and NDS

Matching by distance. nuScenes does not use IoU to decide whether a detection is correct. A prediction matches the ground-truth object with the smallest 2D centre distance on the ground plane, if that distance is below a threshold [50]. AP is computed per class for thresholds $\{0.5, 1, 2, 4\}$ m, integrating the precision–recall curve only where recall and precision exceed 0.1, and

$$\mathrm{mAP} = \frac{1}{|\mathbb{C}|\,|\mathbb{D}|} \sum_{c \in \mathbb{C}} \sum_{d \in \mathbb{D}} \mathrm{AP}_{c,d},$$

where $\mathbb{C}$ is the set of 10 detection classes and $\mathbb{D} = \{0.5, 1, 2, 4\}$ the set of distance thresholds in metres [50]. Each class is evaluated only within its own range from the ego vehicle (30 m for barriers and traffic cones, 40 m for bicycles, motorcycles and pedestrians, 50 m for the vehicle classes), and at most 500 boxes per sample may be submitted [50].

Why distance? IoU punishes the same centre error far more on small objects than on large ones (Figure 6). A 0.25 m error, for example, takes a 0.7 m pedestrian box to about IoU 0.47, below 0.5,

while a car shifted by the same amount along its length keeps an IoU near 0.9.

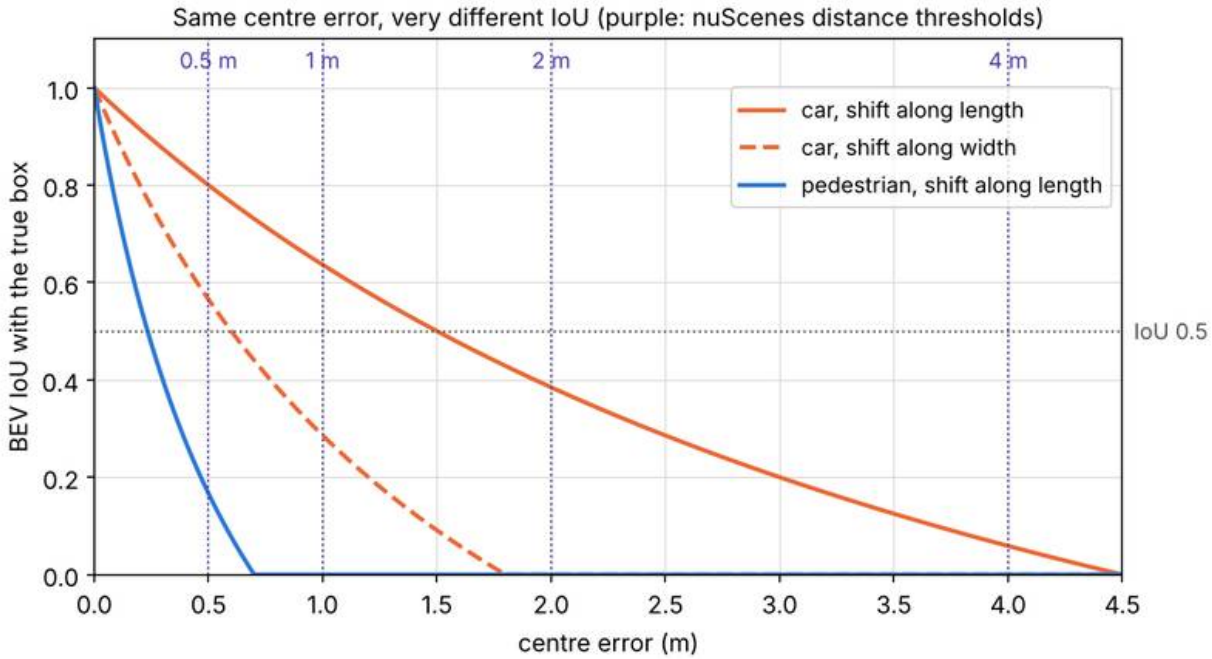


Figure 6 — BEV IoU against centre error for a car (4.5×1.8 m) and a pedestrian (0.7×0.7 m), with sizes and yaw exact. The purple lines are the nuScenes matching thresholds.

The snippet makes the same point with AP. Two classes get identical centre errors (Gaussian, standard deviation 0.25 m per axis) and the same number of false positives; only their sizes differ. (We compute plain 101-point interpolated AP and skip nuScenes' 0.1 clipping, which does not change the comparison.)

```
import numpy as np

rng = np.random.default_rng(0)
size = {"car": (4.5, 1.8), "pedestrian": (0.7, 0.7)}     # BEV length x width in metres

def bev_iou(a, b):
    """Axis-aligned BEV IoU of boxes (x, y, l, w); yaw is 0 in this toy."""
    ox = max(0, min(a[0] + a[2] / 2, b[0] + b[2] / 2) - max(a[0] - a[2] / 2, b[0] - b[2] /
     ↪  2))
    oy = max(0, min(a[1] + a[3] / 2, b[1] + b[3] / 2) - max(a[1] - a[3] / 2, b[1] - b[3] /
     ↪  2))
    return ox * oy / (a[2] * a[3] + b[2] * b[3] - ox * oy)

def average_precision(gts, preds, is_match):
    """Greedy matching in score order; AP = mean interpolated precision at 101 recall
     ↪  points."""
    used, tp = np.zeros(len(gts), bool), []
    for p in sorted(preds, key=lambda p: -p[-1]):
        ok = [j for j in range(len(gts)) if not used[j] and is_match(p, gts[j])]
```

```
        if ok:
            j = min(ok, key=lambda j: np.hypot(p[0] - gts[j][0], p[1] - gts[j][1]))
            used[j] = True
        tp.append(bool(ok))
    tp = np.array(tp)
    recall = np.cumsum(tp) / len(gts)
    precision = np.cumsum(tp) / np.arange(1, len(tp) + 1)
    envelope = np.maximum.accumulate(precision[::-1])[::-1]
    return np.mean([envelope[recall >= r].max() if np.any(recall >= r) else 0
                    for r in np.linspace(0, 1, 101)])

for cls, (l, w) in size.items():
    gts = [(x, y, l, w) for x, y in rng.uniform(-50, 50, (100, 2))]
    preds = [(g[0] + dx, g[1] + dy, l, w, s) for g, (dx, dy), s in
             zip(gts, rng.normal(0, 0.25, (100, 2)), rng.uniform(0.3, 1, 100))]
    preds += [(x, y, l, w, s) for (x, y), s in zip(rng.uniform(-50, 50, (20, 2)),
     ↪  rng.uniform(0, 0.6, 20))]
    ap_iou = average_precision(gts, preds, lambda p, g: bev_iou(p, g) >= 0.5)
    ap_d = [average_precision(gts, preds, lambda p, g, t=t: np.hypot(p[0] - g[0], p[1] -
     ↪  g[1]) <= t)
            for t in (0.5, 1, 2, 4)]
    print(f"{cls:10s} AP(BEV IoU>=0.5) = {ap_iou:.3f} | AP(centre dist <= 0.5/1/2/4 m) = "
          f"{' / '.join(f'{a:.3f}' for a in ap_d)} | mean = {np.mean(ap_d):.3f}")
```

```
car        AP(BEV IoU>=0.5) = 0.951 | AP(centre dist <= 0.5/1/2/4 m) = 0.789 / 0.988 / 0.988
 ↪  / 0.990 | mean = 0.939
pedestrian AP(BEV IoU>=0.5) = 0.126 | AP(centre dist <= 0.5/1/2/4 m) = 0.791 / 0.979 / 0.979
 ↪  / 0.979 | mean = 0.932
```

In this toy run, IoU matching rates the pedestrian detector as nearly useless (AP 0.126) and the car detector as excellent (0.951), although both made identical centre errors. Centre-distance matching gives both about the same score. Distance matching does not ignore box quality; it moves it into separate error terms.

True-positive errors and NDS. For matched detections at the 2 m threshold, nuScenes measures five **TP errors**: translation error ATE (centre distance in metres), scale error ASE ($1-$IoU after aligning centre and orientation), orientation error AOE (smallest yaw difference in radians), velocity error AVE (m/s) and attribute error AAE ($1-$ attribute accuracy) [50]. Their class means are combined with mAP into the **nuScenes detection score**:

$$\mathrm{NDS} = \frac{1}{10}\Big[\,5\,\mathrm{mAP} + \sum_{e \in \mathbb{E}} \big(1 - \min(1, e)\big)\Big],$$

where $\mathbb{E} = \{\mathrm{mATE}, \mathrm{mASE}, \mathrm{mAOE}, \mathrm{mAVE}, \mathrm{mAAE}\}$ is the set of mean TP errors [50]. Half of NDS comes from mAP, the other half from box quality, orientation, velocity and attributes, which is why methods that estimate velocity well (with multiple sweeps or frames) gain a lot of NDS.

Optional: Computing NDS step by step (in the Supplementary notes). How TP errors are averaged, which classes skip which terms, and a worked comparison of two detectors.

Waymo: AP weighted by heading

The Waymo Open Dataset uses 3D IoU matching with thresholds 0.7 for vehicles and 0.5 for pedestrians and cyclists, and its primary metric is **APH**, average precision weighted by heading [51]. Each true positive counts not as 1 but as a heading accuracy

$$a_h = 1 - \frac{\min\left(|\tilde{\theta} - \theta|,\ 2\pi - |\tilde{\theta} - \theta|\right)}{\pi} \in [0, 1],$$

where $\tilde{\theta}$ and θ are the predicted and ground-truth headings in $[-\pi, \pi]$ [3], [53]. A correct box facing exactly backwards ($|\tilde{\theta} - \theta| = \pi$) therefore counts as 0, which closes the loophole that IoU leaves open. Ground-truth boxes are split into two difficulty levels: **LEVEL_2** for boxes with 1 to 5 LiDAR points (or marked as LEVEL_2 in the released labels), **LEVEL_1** for the rest (boxes with no LiDAR points are not evaluated), and LEVEL_2 metrics include both [51]. Its 2020 3D detection challenge ranks entries by **mAPH at LEVEL_2**, the mean over vehicles, pedestrians and cyclists, and break results down by range: $[0, 35)$, $[35, 50)$ and beyond 50 m [51].

Efficiency and deployment

Plain version. A car's computer must finish perception well before the next sweep arrives, on embedded hardware, every time. The representation decides most of the cost: how many cells, how many points per cell, and whether the operators run fast on the target chip.

- **Grid size and range set the budget.** Halving the cell size quadruples the BEV cells. PointPillars traded accuracy for speed exactly this way, from 62 Hz at its default grid to 105 Hz with coarser pillars [10].
- **Sparse operators need support.** Sparse convolution and point sampling are efficient in principle, but they rely on custom kernels; pillar networks run on dense 2D convolutions that every inference engine supports. DSVT made deployment an explicit design goal, building its pooling without custom CUDA operators and reporting real-time inference after deployment [26].
- **Camera view transforms are expensive.** Lifting every pixel to many depths creates huge intermediate point sets; BEVFusion's optimized BEV pooling reduced that step's latency by more than 40 times [43].
- **Benchmark speed is not vehicle speed.** PointPillars' paper notes that KITTI uses only the points inside the front camera's view, about 10% of the sweep, and that timings on desktop GPUs do not transfer to embedded computers [10]. Measure end to end, on the target hardware, with full 360-degree input.
- **Post-processing and temporal input count too.** Rotated NMS runs over many boxes; multi-sweep inputs multiply the points. nuScenes limits the past sensor data a detection method

may use to about 0.5 s [50].

Failure modes

Plain version. 3D detectors fail where the evidence is thin (far, small or hidden objects), where the sensors are disturbed (weather, dirt, miscalibration), and where the test data differ from the training data (another city, another sensor).

- **Far and sparse objects.** Points fall roughly with $1/r^2$ (Figure 2), so far objects have a few points or none. Benchmarks acknowledge this: Waymo's LEVEL_2 collects boxes with at most 5 points [51], and nuScenes evaluates each class only within 30–50 m [50]. Camera depth errors grow with Z^2, which hurts the same objects.
- **Occlusion and truncation.** The sensor sees only surfaces facing it, and a parked car hides the pedestrian behind it. KITTI's difficulty levels are defined by occlusion and truncation [48]; temporal models help by remembering briefly hidden objects [35].
- **Confusable classes.** PointPillars reports that pedestrians and cyclists are often confused with each other, and pedestrians with poles and tree trunks [10].
- **Weather.** Fog, rain and snow scatter laser pulses, creating spurious near returns and losing far ones. Hahner et al. simulate physically based fog on clear-weather LiDAR scans and show that training with it improves detection in real fog [54].
- **Calibration and synchronization errors.** Fusion that hard-projects points to pixels breaks when extrinsics drift; TransFusion's soft association was designed for this [45], and LSS showed that training with noisy extrinsics or dropped cameras makes a camera BEV model more robust to them [32].
- **Common corruptions.** A benchmark of 27 LiDAR and camera corruptions applied to KITTI, nuScenes and Waymo, testing 24 detectors, found that motion-related corruptions hurt all models most, that LiDAR–camera fusion models were more robust, and that camera-only models were extremely vulnerable to image corruptions [55].
- **Domain shift between datasets.** A detector trained on one dataset can drop sharply on another. Wang et al. found that the main hurdle between German and US driving datasets was simply the different average car size, and that correcting for it closes much of the gap [56]. The Waymo paper itself reports a pronounced gap between data from different cities [3]. Different LiDAR models (beam count, mounting height) shift the point distribution too.

Modern view

Figure 7 places the methods of this chapter on a timeline, by the year of each cited publication.

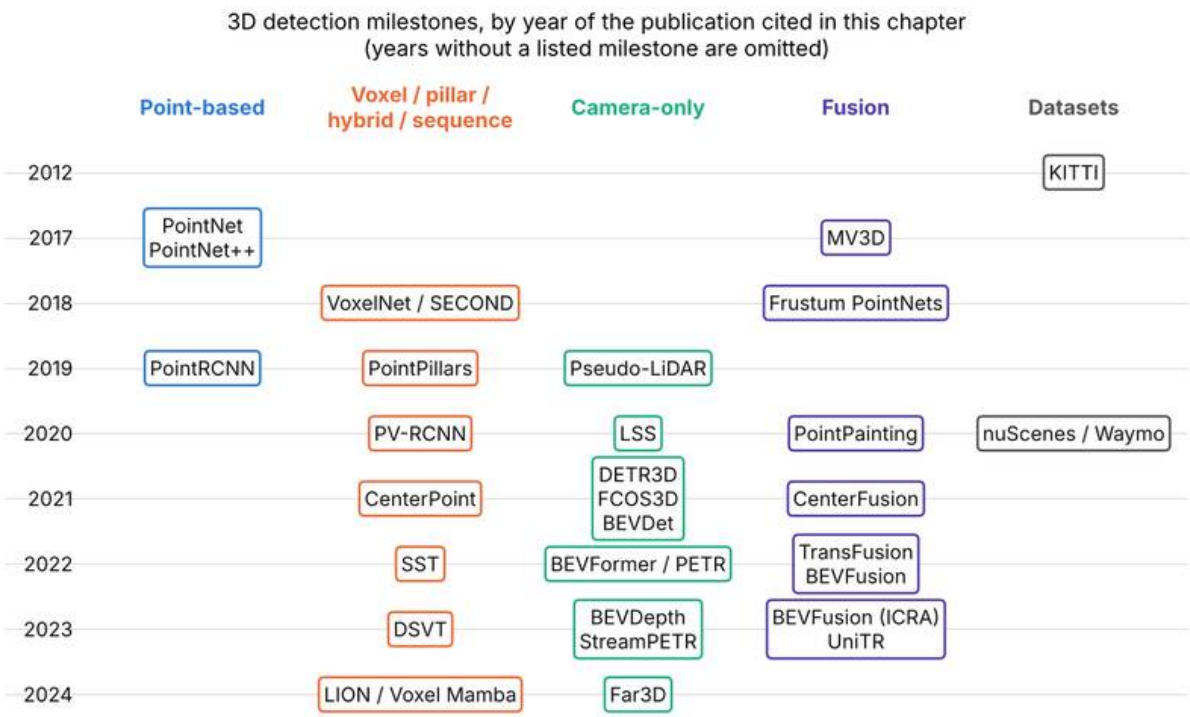


Figure 7 — Milestones by year of the publication cited in this chapter, grouped by family. Years without a listed milestone are omitted.

What the surveys say

Six surveys frame the field, each with its own taxonomy. Mao et al. [1] give the most complete map of 3D detection for driving: LiDAR methods by data representation (points, grids, point–voxel, range images), camera methods (monocular, stereo, multi-view), fusion, transformers, temporal and label-efficient detection, with an analysis of accuracy and speed trends. Qian et al. [2] cover sensors, datasets and metrics and include a case study with runtime, error and robustness analyses. Guo et al. [57] review deep learning on point clouds as a whole, of which detection is one task. For cameras, Ma et al. [58] and Li et al. [59] survey bird's-eye-view perception; the latter frames the core problems as the view transformation, BEV ground truth, multi-sensor pipelines and generalization across sensor setups, and adds a practical "recipe" of tricks. Wang et al. [39] organize fusion methods by fusion stage, inputs and granularity. Two shared takeaways: the BEV has become the common meeting place of sensors and tasks, and accuracy gains on the standard benchmarks have outpaced gains in robustness and deployability.

Optional: Guide to the Modern view surveys (in the Paper guides). What each survey covers, how it organizes the field and how to use it.

The state of the art, 2023–2026

- **LiDAR backbones** have moved from sparse convolutions toward sparse transformers and linear-time sequence models over serialized voxels (DSVT [26], LION [27], Voxel Mamba [28]). Point Transformer V3 [60] argues that scale matters more than intricate design, and improves further with joint training on several datasets.
- **Camera-only detection** has narrowed the gap to LiDAR, mainly through temporal modelling and sparse queries (StreamPETR [37]) and better depth supervision (BEVDepth [33]); long-range camera detection is an active direction (Far3D [38]).
- **Fusion** in a shared BEV or token space (BEVFusion [43], [44], UniTR [46]) is the standard recipe for the strongest systems, and robustness to missing or degraded sensors is now a design goal rather than an afterthought.
- **End-to-end driving.** Detection is increasingly one stage of a single network trained for planning, as in UniAD [61], where perception, prediction and planning tasks communicate through unified query interfaces.

Open problems

1. **Far, small and rare objects.** Sparse evidence at range, and classes that are rare in training data (animals, debris, unusual vehicles), remain weak points; closed-set datasets do not cover the long tail [1].
2. **Robustness.** Weather, sensor faults, calibration drift and motion corruptions still cause large drops [55], [54].
3. **Generalization across sensors and places.** Models trained on one LiDAR model, camera rig or city often fail on another [56], [3].
4. **Metrics that match driving.** AP treats a missed pedestrian at 10 m and one at 45 m alike; nuScenes and Waymo add heading, velocity and range breakdowns, but, as Mao et al. note, maximizing AP alone may not be optimal for the driving system as a whole [1].
5. **Labels.** 3D boxes are expensive to annotate, which motivates semi-supervised, self-supervised and auto-labelling methods [1].

Key takeaways

- A 3D detector outputs upright 7-DoF boxes $(x, y, z, l, w, h, \theta)$ in metres with class, score and often velocity; calibration (extrinsics and intrinsics) ties the sensors to one vehicle frame.
- Point clouds are unordered, sparse and thin out roughly as $1/r^2$; networks must be permutation invariant and exploit sparsity.

- LiDAR families differ in representation: points (PointNet++, PointRCNN), voxels with sparse convolution (VoxelNet, SECOND), pillars (PointPillars), centre heads (CenterPoint), hybrids (PV-RCNN) and sparse transformers; most detect in the bird's-eye view.
- Cameras lose depth; camera detectors either lift pixels with a depth distribution (LSS, BEVDet, BEVDepth) or query images from 3D (DETR3D, PETR, BEVFormer), and time helps a lot.
- Yaw needs care: the sine-error loss plus a direction bin, or $(\sin\theta, \cos\theta)$; rotated BEV and 3D IoU use polygon clipping, and IoU ignores a 180° flip.
- KITTI uses 3D IoU (0.7 cars) with easy/moderate/hard levels and 40 recall points; nuScenes uses centre-distance mAP plus TP errors in NDS; Waymo uses heading-weighted APH at LEVEL_1/LEVEL_2.
- Failures concentrate on far, small, occluded objects, bad weather, miscalibration and domain shift; benchmark speed seldom equals on-vehicle speed.

Exercises

1. **Points at range.** A LiDAR has a horizontal step of 0.1° and a vertical step of 0.4°. Estimate how many points fall on the 1.8 m × 1.5 m back of a car at 20 m and at 60 m, using $n(r) \approx A/(r^2 \Delta\phi \Delta\epsilon)$. How many beams cross the car vertically at 60 m?

   Hint

   Convert to radians: $\Delta\phi = 1.75{\times}10^{-3}$, $\Delta\epsilon = 6.98{\times}10^{-3}$. With $A = 2.7\ \text{m}^2$, $n(20) \approx 2.7/(400{\cdot}1.22{\times}10^{-5}) \approx 550$ and $n(60) \approx 62$. Vertically, rays are $r\,\Delta\epsilon \approx 0.42$ m apart at 60 m, so only about 3–4 beams cross a 1.5 m tall object; the rest of the points come from the fine horizontal step.

2. **IoU by hand.** Two 4 m × 2 m × 1.5 m boxes share a centre; one is rotated by 90° about the vertical axis. Compute the BEV IoU and the 3D IoU. Then raise the second box by 0.75 m and recompute the 3D IoU. Check your answers with the IoU snippet.

   Hint

   BEV intersection is a 2 m × 2 m square: $\text{IoU}_{\text{bev}} = 4/(8+8-4) = 1/3$. With equal heights the 3D IoU is also $6/(12+12-6) = 1/3$. Raised by 0.75 m, the height overlap is 0.75 m, so the intersection is $3\ \text{m}^3$ and $\text{IoU}_{\text{3D}} = 3/(24-3) \approx 0.143$.

3. **Break the classical pipeline.** In the classical-baseline snippet, move the pedestrian to $(13.5, -2.2)$, next to the first car (in our run its points come within about 0.15 m of the car's points), and run it again. What does DBSCAN do? Change `eps` to fix it, and explain what new failure a small `eps` causes for the far objects.

   Hint

In our run, every `eps` from 0.7 down to 0.15 merges the pedestrian and the car into one cluster with a wrong box (5.1 × 1.8 m at yaw 28° instead of 4.5 × 1.8 m at 17°). At `eps=0.12` the pedestrian separates, but the car at 25 m already breaks into several fragments; at `eps=0.1` even the near car splits into its two visible sides. Far objects have fewer points, so the gaps between their points are larger (on a real scanner they grow like $r\,\Delta\phi$). A fixed `eps` cannot suit all ranges, a direct consequence of the $1/r^2$ density.

4. **Matching protocols.** Using the matching snippet, find the standard deviation of the centre error at which the car's AP under IoU ≥ 0.5 matches the pedestrian's AP at standard deviation 0.25 m. What does this say about comparing per-class APs computed with IoU matching?

   Hint

   In our runs the car's IoU-based AP falls to 0.117 at a standard deviation of 0.75 m, about three times the pedestrian's 0.25 m (roughly the ratio of the car's width to the pedestrian's), in line with Figure 6. Per-class IoU-based APs therefore mix detection quality with object size, so a low pedestrian AP does not mean the pedestrian detector localizes worse in metres.

5. **Heading-weighted AP.** A detector finds all 10 cars in a scene with no false positives and perfect IoU, but 3 of them face backwards and 2 are off by 30°. What are its AP and APH (assume all detections have the same score)? Which nuScenes TP error would reveal the same problem?

   Hint

   AP is 1, since every box matches. Heading accuracies: $5 \times 1 + 3 \times 0 + 2 \times (1 - 30/180) = 6.67$, so APH ≈ 0.667. In nuScenes the orientation error AOE (in radians) would rise, lowering NDS through the term $1 - \min(1, \text{mAOE})$.

6. **Pillar budget.** In the pillarization snippet, change the grid to 0.32 m and to 0.08 m (adjust `W` and `H` so the grid still covers 70.4 m × 80 m). Report the number of non-empty pillars, the share of empty cells and the number of points dropped by the cap N. Relate the trend to the speed–accuracy trade-off reported by PointPillars.

   Hint

   In our runs, 0.32 m gives 7,567 non-empty pillars (86.2% empty) and drops 1,734 points at $N = 16$; 0.08 m gives 10,908 pillars (98.8% empty) and drops none. Coarser pillars mean fewer, fuller pillars: a smaller pseudo-image and more points dropped by the cap, so faster but less precise. Finer pillars mean more pillars with one or two points each, a larger pseudo-image and more empty cells. PointPillars reports higher speed with coarser grids and a loss mainly for pedestrians and cyclists.

Detections are the input to tracking, which links them over time; see DLCV 06 ·Object Tracking.

Supplementary notes

These notes are optional: each one takes a single point from the main text and goes into finer detail. Nothing in the main text or the exercises depends on them.

Yaw encodings compared: raw angle, sine error and (sin, cos)

This note compares the ways of regressing a heading angle and shows exactly where each one fails.

Let θ^t be the target yaw and θ^p the prediction, both in radians. Four choices are common:

1. **Raw residual** $|\theta^p - \theta^t|$. Wrong across the $\pm\pi$ boundary: 179° and −179° are 2° apart, but the residual is 358°.
2. **Wrapped residual** $|\operatorname{wrap}(\theta^p - \theta^t)|$, with $\operatorname{wrap}(\alpha) = ((\alpha + \pi) \bmod 2\pi) - \pi$ mapping any angle into $[-\pi, \pi)$. Correct for headings, but a box flipped by π, which covers exactly the same space, receives the maximum penalty.
3. **Sine error** $|\sin(\theta^p - \theta^t)|$, the term inside SECOND's loss [9]. It is 0 at $\Delta = 0$ and at $\Delta = \pi$, so flipped boxes are not punished; the direction then comes from a separate two-bin classifier. Near $\Delta = \pi/2$ its slope is zero, so the gradient vanishes for boxes turned sideways.
4. **Vector target** $(\cos\theta, \sin\theta)$, used by CenterPoint [11]: regress two numbers and decode with atan2. The Euclidean distance between the unit vectors is $2|\sin(\Delta/2)|$, which grows smoothly from 0 to 2 and distinguishes front from back.

The snippet evaluates all four on a target of 170°. The "direction bin" column is a simplified stand-in for a two-bin classifier: it asks whether the prediction points into the same half of the circle as the target ($\cos\Delta > 0$).

```
import numpy as np

def wrap(a):                                        # angle difference into (-pi, pi]
    return (a + np.pi) % (2 * np.pi) - np.pi

gt = np.deg2rad(170.0)
for pred_deg in [175, -175, -10, 100]:
    pred = np.deg2rad(pred_deg)
    raw = abs(pred - gt)                            # naive residual
    sin_err = abs(np.sin(pred - gt))                # SECOND-style residual sin(pred - gt)
    vec = np.hypot(np.cos(pred) - np.cos(gt), np.sin(pred) - np.sin(gt))   # (cos, sin)
    ↪ target
    print(f"pred {pred_deg:5d} deg: |raw| {raw:5.2f}  |wrapped| {abs(wrap(pred - gt)):5.2f}
    ↪ "
          f"|sin| {sin_err:4.2f}  (cos,sin) dist {vec:4.2f}  same direction bin:
           ↪ {np.cos(pred - gt) > 0}")
```

```
pred   175 deg: |raw|  0.09  |wrapped|  0.09  |sin| 0.09  (cos,sin) dist 0.09  same
 ↪  direction bin: True
pred  -175 deg: |raw|  6.02  |wrapped|  0.26  |sin| 0.26  (cos,sin) dist 0.26  same
 ↪  direction bin: True
pred   -10 deg: |raw|  3.14  |wrapped|  3.14  |sin| 0.00  (cos,sin) dist 2.00  same
 ↪  direction bin: False
pred   100 deg: |raw|  1.22  |wrapped|  1.22  |sin| 0.94  (cos,sin) dist 1.15  same
 ↪  direction bin: True
```

The second row shows the boundary problem of the raw residual (6.02 rad for a 10° error). The third row is the flipped box: the sine error is 0 and only the direction bin catches the mistake; the vector target sees it directly (distance 2). The fourth row shows the price of the sine error: a 70° error gives 0.94, close to the maximum of 1, so errors beyond 90° would start to look *better*.

Pitfall. When combining the sine-error loss with a direction classifier at inference, decode the angle from the regression first, then add π if the classifier disagrees with the decoded direction. Forgetting this step produces boxes with correct IoU but backwards heading, which IoU-based AP does not notice and APH punishes fully.

Sparse and submanifold convolution

This note explains how sparse convolution saves computation and why the submanifold variant is the one used in most layers of LiDAR backbones.

Setting. A voxel grid has a set $\mathcal{A}$ of **active sites** (voxels that contain points or features); elsewhere the features are zero. A dense 3D convolution with kernel size k computes an output at every site, at a cost proportional to the full grid volume, although almost all inputs are zero.

Ordinary sparse convolution computes outputs only where at least one input in the kernel window is active, so the output active set is

$$\mathcal{A}_{\text{out}} = \{\, \mathbf{u} : \exists\, \mathbf{a} \in \mathcal{A},\ \|\mathbf{u} - \mathbf{a}\|_\infty \le \lfloor k/2 \rfloor \,\},$$

where $\mathbf{u}$ and $\mathbf{a}$ are integer grid positions and $\|\cdot\|_\infty$ is the largest absolute coordinate difference. Each layer dilates the active set by the kernel radius, so after a few layers the "sparse" tensor is no longer sparse.

Submanifold sparse convolution [20] restricts outputs to the input active set, $\mathcal{A}_{\text{out}} = \mathcal{A}$. It cannot spread information into empty space, so backbones interleave many submanifold layers with a few ordinary strided sparse convolutions that downsample. SECOND's middle extractor follows this pattern: several submanifold layers per stage and one ordinary sparse convolution to downsample along z, before converting to a dense BEV map [9].

Implementation. Both variants are computed with a **rule book**: for each kernel offset, a list of

(input index, output index) pairs. The layer gathers the input features for each offset, multiplies them by that offset's weight matrix and scatters the results into the outputs. Building the rule book is a hashing problem; SECOND's contribution included a GPU rule-generation algorithm to make it fast [9].

The snippet measures the dilation on a 2D slice with 1% active sites plus one dense object.

```
import numpy as np
from scipy.ndimage import binary_dilation

rng = np.random.default_rng(0)
active = np.zeros((200, 200), bool)                 # a BEV slice of a voxel grid
active[rng.integers(0, 200, 400), rng.integers(0, 200, 400)] = True   # 1% of sites hold
 ↪  points
active[90:94, 50:70] = True                         # plus one dense "car"
kernel = np.ones((3, 3), bool)

regular = active.copy()
for layer in range(1, 4):
    regular = binary_dilation(regular, kernel)      # a 3x3 sparse conv writes to every
     ↪  neighbour
    print(f"after {layer} layer(s): sparse conv {regular.mean():6.1%} active | "
          f"submanifold conv {active.mean():6.1%} active")
```

```
after 1 layer(s): sparse conv   8.9% active | submanifold conv   1.2% active
after 2 layer(s): sparse conv  22.2% active | submanifold conv   1.2% active
after 3 layer(s): sparse conv  38.4% active | submanifold conv   1.2% active
```

After three ordinary 3×3 sparse convolutions, the active share grows from 1.2% to 38.4%; the submanifold version stays at 1.2%. Since cost scales with the number of active outputs, the difference compounds over the dozens of layers in a backbone.

Edge case. Because submanifold layers never fill gaps, two parts of one object separated by an empty voxel cannot exchange information through them. Strided ordinary sparse convolutions, or attention over windows as in SST and DSVT [25], [26], provide that connectivity.

Attention on sparse voxels: windows, sets and sequences

This note explains how transformer and sequence-model backbones organize the occupied voxels of a LiDAR scan so that the computation stays affordable.

The problem. A sweep has N occupied voxels, often tens of thousands. Full self-attention compares every voxel with every other, so its cost grows like N^2. All practical designs therefore let each voxel interact only with a group of others, and they differ in how the groups are formed.

Windows. Divide the grid into non-overlapping windows of fixed size (for example 12×12 voxels) and run attention among the occupied voxels inside each window. The catch is that windows hold

very different numbers of voxels: many near the sensor, a handful far away. GPUs want equal-size batches, so a naive implementation pads every window to the largest one, and the padding dominates the cost. SST [25] groups voxels by region in this way and also uses regions shifted by half their size, so that objects cut by a region border still exchange information.

Equal-size sets. DSVT [26] splits each window into sets that all hold the same number τ of voxels. For a window with N_w occupied voxels, the number of sets is

$$S_w = \left\lceil \frac{N_w}{\tau} \right\rceil,$$

where $\lceil \cdot \rceil$ rounds up. The voxels are sorted by an index inside the window and spread evenly over the S_w sets; a set that would be short is filled with repeated voxels, which are masked out of the attention. Every set now has exactly τ tokens, so all sets of all windows are processed in one parallel batch, and a dense window simply receives more sets. Because attention now stays inside a set, DSVT alternates two sortings in consecutive layers, along the x axis and along the y axis, so that voxels in different sets of one layer meet in the next.

The snippet counts the attention pairs (the cost of attention is the number of token pairs) on a toy scan, for padding every window to the largest versus equal-size sets.

```
import numpy as np

rng = np.random.default_rng(0)
# occupied BEV voxels of a toy scan: density falls with range, as for a spinning LiDAR
r = 2 + 70 * rng.random(20000) ** 2                  # more points close to the sensor
a = rng.uniform(-np.pi, np.pi, r.size)
cells = np.unique(np.floor(np.column_stack([r * np.cos(a), r * np.sin(a)]) /
 ↪  0.32).astype(int), axis=0)

win = 12                                             # windows of 12 x 12 voxels
_, n = np.unique(cells // win, axis=0, return_counts=True)   # occupied voxels per window
tau = 36                                             # voxels per set
S = -(-n // tau)                                     # ceil(N / tau) sets per window

print(f"{len(cells)} occupied voxels in {len(n)} non-empty windows; "
      f"per window: min {n.min()}, median {int(np.median(n))}, max {n.max()}")
print(f"pad every window to {n.max()}:  {len(n) * n.max() ** 2 / 1e6:5.1f} M attention
 ↪  pairs")
print(f"equal-size sets, tau = {tau}: {S.sum() * tau ** 2 / 1e6:5.1f} M attention pairs in
 ↪  {S.sum()} sets")
```

```
14431 occupied voxels in 1136 non-empty windows; per window: min 1, median 8, max 120
pad every window to 120:   16.4 M attention pairs
equal-size sets, tau = 36:   1.6 M attention pairs in 1234 sets
```

Most windows are nearly empty (median 8 voxels) while a few near the sensor are full, so padding to the largest window costs ten times more pairs than equal-size sets. The comparison is not

quite like for like: with sets, each voxel attends only within its set, which is why the alternating partitions are needed.

Sequences. A second family first **serializes** the voxels: it orders them along a **space-filling curve**, a path that visits every cell of the grid once and keeps cells that are close in space mostly close in the order. Point Transformer V3 builds its backbone around such serialization [60]. Linear-time sequence models then make very long groups affordable: LION [27] applies linear recurrent networks to large groups of voxels, and Voxel Mamba [28] uses a Hilbert curve to turn all voxels of a scene into a single sequence for a state space model, with no grouping at all.

Pitfall. No ordering of a 2D or 3D grid keeps all neighbours adjacent: wherever the curve turns back, two voxels side by side in space can end up far apart in the sequence. Serialization-based designs therefore add ways to mix information across the curve's jumps: Point Transformer V3 switches between several serialization patterns [60], and Voxel Mamba adds a second, coarser scale to its sequence model [28].

Ground-truth sampling in detail

This note describes the ground-truth sampling augmentation step by step, with the pitfalls that matter in practice.

Steps.

1. **Build the database once.** For every labelled training box, store its class, its 7-DoF parameters and the points inside it (in the box frame or in sensor coordinates) [9].
2. **Sample per scene.** For each training scene, draw a fixed number of objects per class from the database; PointPillars uses 15 cars, 0 pedestrians and 8 cyclists on KITTI [10]. Classes that are rare in the data can be sampled more often, which counters the class imbalance.
3. **Reject collisions.** Discard any sampled box whose BEV footprint overlaps an existing box or another sampled box. VoxelNet used the same kind of collision test for its per-box perturbations [8]. A BEV IoU greater than 0 with any box is a simple test.
4. **Paste.** Add the sampled points and boxes to the scene. Then apply the per-object and global augmentations.

Pitfalls.

- **Floating or buried objects.** The database box keeps its original height. On a sloped road or with a different sensor height, pasted cars float or sink. A simple fix is to re-estimate the local ground height (for example with the RANSAC plane of the classical baseline) and shift the box.
- **Inconsistent occlusion.** A pasted car does not cast a LiDAR shadow on the points behind it, and it may sit behind a wall yet keep all its points. The network can learn to spot pasted objects

by these artefacts.

- **Range consistency.** An object captured at 10 m has the point density of 10 m. Pasted at 50 m it is far too dense. Pasting objects near their original range, or subsampling points according to the new range ($n \propto 1/r^2$), keeps the density realistic.
- **Multi-sensor consistency.** The pasted object is absent from the camera images, so a fusion model sees LiDAR evidence without image evidence. For fusion training, the augmentation must either be switched off or paste matching image content as well.

Computing NDS step by step

This note follows the nuScenes devkit definitions [50] from per-class errors to the final score.

1. **Match at 2 m.** For TP errors, predictions are matched to ground truth by centre distance with a 2 m threshold, per class.
2. **Average over recall.** For each TP error type, the error is averaged over the cumulative mean at each achieved recall level above 10%. If a class does not reach 10% recall, all its TP errors are set to 1.
3. **Skip undefined terms.** Orientation error is ignored for traffic cones and computed modulo 180° for barriers; velocity and attribute errors are ignored for barriers and cones.
4. **Average over classes** to get mATE, mASE, mAOE, mAVE and mAAE.
5. **Convert to scores and combine.** Each TP score is $\max(1 - e, 0)$; NDS gives mAP a weight of 5 and each of the five TP scores a weight of 1, normalized by 10.

Translation and velocity errors are unbounded (metres and metres per second), so clipping at 1 matters: a model with a velocity error above 1 m/s gets no credit for velocity at all. The snippet compares two made-up detectors (the numbers are invented for illustration).

```
def nds(mAP, tp_errors):
    """nuScenes detection score from mAP and the five mean TP errors (mATE, mASE, mAOE,
     ↪  mAVE, mAAE)."""
    tp_scores = [max(1 - e, 0.0) for e in tp_errors]
    return (5 * mAP + sum(tp_scores)) / 10

# two made-up detectors: B localizes and estimates velocity better, A has the higher mAP
A = dict(mAP=0.45, tp=[0.60, 0.27, 0.45, 1.20, 0.20])
B = dict(mAP=0.40, tp=[0.35, 0.25, 0.30, 0.30, 0.18])
for name, d in [("A", A), ("B", B)]:
    print(f"detector {name}: mAP {d['mAP']:.2f}  NDS {nds(d['mAP'], d['tp']):.3f}")
```

```
detector A: mAP 0.45  NDS 0.473
detector B: mAP 0.40  NDS 0.562
```

Detector A has the higher mAP, but B wins on NDS by almost 0.09, mostly because its velocity error (0.30 m/s against 1.20 m/s) earns 0.7 points instead of 0. Velocity needs more than one sweep or frame, which is why the temporal input matters so much for NDS. When you compare papers, check whether they report NDS on the validation or the test split and whether they used past sweeps.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Guide to the Modern view surveys

This guide expands the short survey summary in the Modern view: what each review covers, how it is organized and when to read it.

3D Object Detection for Autonomous Driving: A Comprehensive Survey (Mao, Shi, Wang and Li, *IJCV*, 2023) [1]. The most complete single map of the field. It defines the task, sensors, datasets and metrics, then organizes LiDAR detectors by data representation: point-based, grid-based (voxels, pillars, BEV maps), point–voxel and range-image methods, followed by learning objectives. Camera methods are split into monocular, stereo and multi-view; then come multi-modal fusion, transformer-based, temporal and label-efficient detection, and related topics such as simulation and robustness. Its analysis section collects accuracy and speed statistics on KITTI, nuScenes and Waymo, and observes that LiDAR results on KITTI converged quickly, pushing research toward the larger datasets. Read it to place any method in the landscape.

3D Object Detection for Autonomous Driving: A Survey (Qian, Lai and Li, *Pattern Recognition*, 2022) [2]. Shorter and more practical. It identifies three core challenges (recovering 3D from images without depth, learning from occluded and unstructured point clouds, and aligning heterogeneous features across modalities) and adds a case study of fifteen representative methods with runtime, error and robustness analyses. Read it for a compact introduction with quantitative comparisons.

Deep Learning for 3D Point Clouds: A Survey (Guo et al., *IEEE TPAMI*, 2021) [57]. A survey of point-cloud learning in general, covering shape classification, detection and tracking, and segmentation. Read it for the backbone side: how point-based, voxel-based and graph-based operators relate, beyond driving.

Vision-Centric BEV Perception: A Survey (Ma et al., *IEEE TPAMI*, 2024) [58] and **Delving into the Devils of Bird's-Eye-View Perception: A Review, Evaluation and Recipe** (Li et al., *IEEE TPAMI*, 2024) [59]. Both focus on bird's-eye-view perception from cameras and its extensions. Li

et al. organize the field around four questions: how to recover 3D through the view transform, how to obtain BEV ground truth, how to build pipelines that fuse different sources and views, and how to generalize when the sensor setup changes. They also describe industrial system designs and a practical guide of tricks for camera, LiDAR and fusion inputs. Read them before implementing a BEV model; they collect the details that papers leave out.

Multi-Modal 3D Object Detection in Autonomous Driving: A Survey (Wang et al., *IJCV*, 2023) [39]. Introduces the sensors and their data properties, the datasets suitable for fusion, and then reviews fusion networks by three design choices: the fusion stage, the fusion inputs and the fusion granularity, and how these choices evolved. Read it when designing a fusion model.

How to read these surveys. Survey tables compile numbers from many papers with different training data, splits, test-time augmentation and hardware. Use them to see trends, not to rank two methods; for a fair comparison, look for papers that re-run baselines under one protocol.

PointPillars: a reading guide

The problem. In 2018, LiDAR detectors faced a trade-off: fixed, hand-designed BEV encodings were fast but less accurate, and learned encoders (VoxelNet's VFE with 3D convolutions) were accurate but slow [10].

The idea. Learn the encoding with a PointNet, but organize the points in vertical pillars so that everything after the encoder is a dense 2D CNN. No binning along z means no 3D convolutions and no z resolution to tune.

What it showed. The paper reports that, using LiDAR only, PointPillars outperformed the published state of the art on the KITTI BEV and 3D test benchmarks, including fusion methods, while running at 62 Hz (Tables 1 and 2). For example, its Table 2 lists moderate car 3D AP of 74.99 for PointPillars at 62 Hz, 73.66 for SECOND at 20 Hz and 65.11 for VoxelNet at 4.4 Hz. Section 6 breaks down its 16.2 ms runtime: the pillar encoder takes 1.3 ms, the backbone and heads 7.7 ms, and the remaining steps loading, decorating, uploading, scattering and NMS. Table 4 compares encoders inside one network at several grid sizes, which isolates the encoder's contribution.

How to read its tables. First, these KITTI numbers predate the switch to 40 recall positions in October 2019 [48], so they cannot be compared directly with later $\mathrm{AP}|_{R_{40}}$ results. Second, the speeds in Tables 1 and 2 come from different papers and hardware; the paper's own timings were measured on a desktop CPU and a GTX 1080 Ti GPU. Table 4 is the controlled comparison: the same network and training, with only the encoder and the grid resolution changed, scored by BEV mAP on the KITTI validation set. Third, the paper notes that only about 10% of the sweep, the part visible in the front camera, is processed on KITTI, so a full 360-degree system is slower. Fourth, the speed–accuracy curve (Figure 5) was obtained by varying the pillar size, and the losses at coarse grids came mainly from pedestrians and cyclists.

Limits. Collapsing height into features works for road scenes but loses vertical detail; small objects suffer at coarse grids; the anchor head with two orientations per class inherits the yaw issues discussed in this chapter, which later centre-based heads avoided [11].

Why it matters here. PointPillars made the "points → BEV pseudo-image → 2D detector" design standard. It is the reference baseline of the Waymo dataset paper [3], and LSS borrowed its pillar pooling for cameras [32].

Lift-Splat-Shoot: a reading guide

The problem. A planner wants one BEV map around the car, but six cameras give six perspective images with no depth. Processing each camera on its own and merging the outputs afterwards leaves the fusion to hand-written rules; LSS wants one model that learns from the entire camera rig how to fuse it [32].

The idea. Lift each pixel to a frustum of features using a predicted depth distribution ($\mathbf{c}_d = \alpha_d \mathbf{c}$), splat all frustums from all cameras into one BEV grid with sum pooling, and train end to end from BEV labels. The architecture uses the camera calibration explicitly, so it adapts when intrinsics and extrinsics change between cars.

What it showed. On nuScenes BEV segmentation of cars and vehicles, and of the drivable area and lane boundaries, the method outperforms a CNN baseline without 3D structure, a version with a frozen image encoder, and earlier BEV baselines (its Tables 1 and 2). Its robustness experiments show that training with noisy extrinsics helps under strong test-time noise, and that randomly dropping a camera during training improves results both with dropped cameras and with all six present (Figure 6). It also adds cameras unseen in training at test time and improves without retraining (Table 3).

How to read it. The paper's grid is 100 m × 100 m at 0.5 m cells, with depth bins from 4 to 45 m in 1 m steps [32]. The tasks are BEV segmentation and planning, not 3D box detection, so its numbers are IoUs of BEV masks; detection results with this view transform come from follow-up work such as BEVDet [24]. Some of the compared numbers in its tables come from concurrent works that use different BEV grids and splits, and the paper itself warns that those comparisons are not exact.

Limits. Depth is supervised only indirectly by the BEV task, which BEVDepth later showed to be a weakness [33]; the frustum point sets are large, which made efficient BEV pooling a research topic of its own [43]; and the fixed depth range limits far detection.

Why it matters here. LSS is the "lift" in most camera BEV detectors and in BEV fusion; understanding $\mathbf{c}_d = \alpha_d \mathbf{c}$ is understanding how a 2D image becomes a 3D map.

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DLCV 10 • Image Generation: From GANs to Text-to-Image Systems

How image generators are built and judged as vision systems: the task and what good means, the model families on one map, text encoders, cross-attention and guidance in text-to-image models, control, personalization and editing, autoregressive and unified multimodal generators, evaluation, data, risks and failure modes.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-10-image-generation/

What you'll learn

- The kinds of image generation (unconditional, class-conditional, text-to-image, image-to-image, editing, inpainting) and the three questions behind "good": **fidelity**, **diversity** and **alignment** with the condition.
- One map of the model families (GANs, VAEs and tokenizers, diffusion, flow matching, autoregressive and unified multimodal models), the key idea of each in plain words, and where the theory is derived on this site.
- How a text-to-image system is put together: a text encoder (CLIP or T5), **cross-attention** from image features to words, classifier-free guidance as it is used in practice, and a U-Net or transformer backbone.
- How a pretrained generator is controlled, personalized and used for editing (ControlNet, textual inversion, DreamBooth, LoRA, SDEdit, Prompt-to-Prompt, InstructPix2Pix, RePaint).
- How text-to-image models are evaluated (FID and its pitfalls, CLIPScore, compositional benchmarks such as GenEval, human-preference models), what data they are trained on, which risks they carry (memorization, bias, deepfakes, provenance) and where they still fail (counting, text, hands, spatial relations, attribute binding).

The big picture

An image generator turns a request into a new picture. The request can be nothing at all ("any face"), a class label ("a goldfinch"), a sentence ("a red ball on the grass"), or another image plus an instruction ("make it winter"). The answer should look real, should vary from one try to the next, and should actually show what was asked.

> **Explain like I'm five**
>
> Imagine a sketch artist who works for the police. You describe a face; the artist starts with a rough smudge and keeps refining it, glancing back at your words after every stroke ("big eyes, short hair").

A generator does the same thing many times per second. The "words" are turned into numbers by a reading model, the "glancing back" is a step called cross-attention, and the "smudge-to-picture" refinement is the generator itself. Some artists work differently, placing one mosaic tile after another, but they all listen to the description while they draw.

Why it matters. Image generation went from small, blurry faces to photorealistic, prompt-following pictures in less than a decade, and it is now part of design tools, photo editors, data pipelines and restoration systems (DLCV 04C uses the same models to restore damaged photos). As a computer-vision **system**, a modern generator combines a language model, an image compressor, a generative network, a sampler and an evaluation pipeline. Each part can fail on its own, and each part has its own literature.

Where this chapter sits. The theory lives in the ML track, and this chapter does not repeat it. ML 06 derives VAEs, GANs, normalizing flows and the classic metrics (Inception Score, FID, precision and recall) with code. ML 07 derives diffusion, DDIM, classifier-free guidance, latent diffusion, diffusion transformers, flow matching and few-step samplers. ML 08 covers autoregressive models over pixels and tokens, and ML 09 covers CLIP and parameter-efficient fine-tuning. Here we take the **system view**: how these pieces are assembled into text-to-image products, how they are steered, how they are judged, and what can go wrong. The two code examples run in seconds on a CPU with NumPy and scikit-image, and use no network weights.

Next. DLCV 10A extends these systems from images to video.

The task: what to generate, and what "good" means

Plain version. Every generation task is "draw a sample that fits this condition". The tasks differ only in what the condition is. And "good" always means three things at once: realistic, varied and faithful to the request.

Six kinds of generation

All the tasks below are sampling from a conditional distribution $p(x \mid c)$, where x is an image and c is the condition; the unconditional case uses an empty condition $c = \emptyset$.

Task	Condition c	Output	Example method in this chapter
Unconditional	nothing	an image from the training domain	StyleGAN on faces [1]
Class-conditional	a class label	an image of that class	BigGAN [2], DiT [3]
Text-to-image	a text prompt	an image matching the prompt	Imagen [4], latent diffusion [5]

Task	Condition c	Output	Example method in this chapter
Image-to-image	an image (often plus text)	a new image that keeps its structure	SDEdit [6], ControlNet [7]
Editing	an image plus an edit request	the edited image	Prompt-to-Prompt [8], InstructPix2Pix [9]
Inpainting / outpainting	an image plus a mask	the image with the masked region filled	RePaint [10]

Because the form is shared, one trained text-to-image model can serve most rows of the table: edits, masks and control images are extra conditions or a different starting point for the sampler.

Three questions: fidelity, diversity, alignment

- **Fidelity (quality).** Does each sample look like a real image of the domain, without artifacts?
- **Diversity (coverage).** Do many samples together cover the variety of real images, or does the model repeat a few favorites (mode collapse)?
- **Alignment.** Does the sample show what the condition asked for: the right objects, counts, colors, positions and text?

Add a fourth, practical question: **cost**, the number of network evaluations and the seconds per image. The first three questions pull against each other. Most generators expose a dial that trades diversity for fidelity and alignment: the **truncation trick** of BigGAN shrinks the spread of the input noise [2], and **classifier-free guidance** in diffusion models pushes samples toward the condition [11]. Turning such a dial up makes images cleaner and more on-prompt, but more alike. Keep this trade-off in mind whenever you read a single "quality" number. The metrics that measure fidelity and diversity separately are derived in ML 06; this chapter adds the alignment side in the evaluation section below.

The model families on one map

Plain version. There are only a few ways to turn noise into a picture: in one shot (GANs), by repeated denoising (diffusion), by following a learned flow (flow matching), or by writing the picture piece by piece (autoregressive models). Separately, each of them can work on raw pixels, on a compressed continuous code, or on discrete tokens.

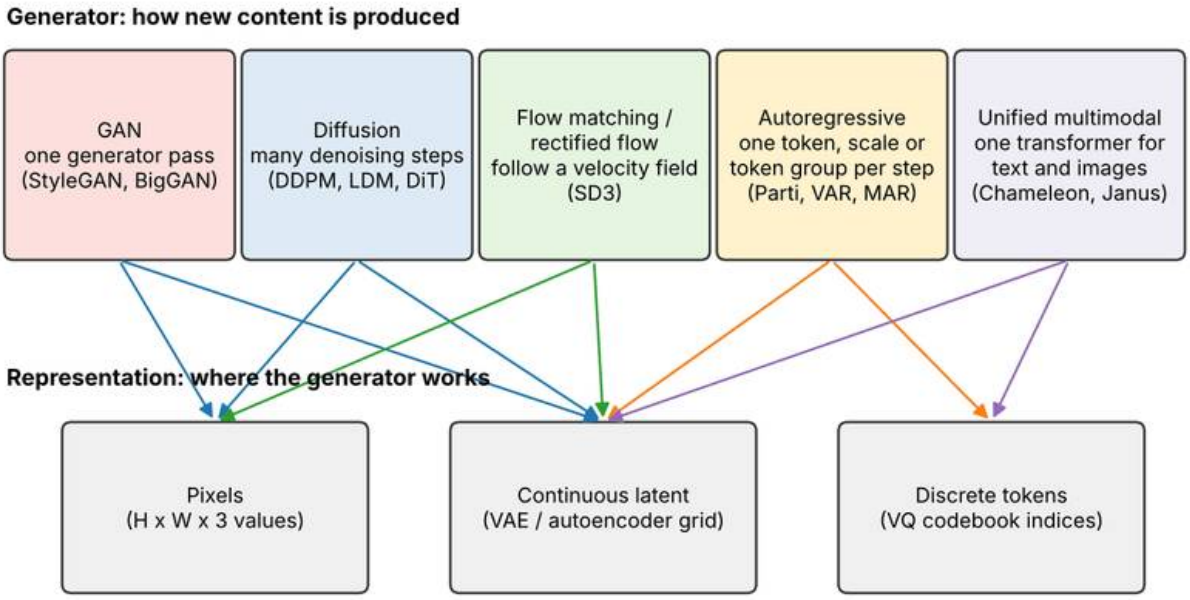


Figure 1 — The families on one map. The top row is how new content is produced; the bottom row is where the generator works. Arrows show common pairings; the representation is a separate design choice from the generator.

GANs: one pass through a generator. A generator maps random noise to an image, and a discriminator, trained at the same time, tries to tell generated images from real ones; the generator improves by fooling it [12]. Sampling is a single forward pass, so GANs are fast. StyleGAN [1] maps the noise to an intermediate "style" vector and injects it at every resolution of the generator, which separates coarse attributes (pose, identity) from fine ones (freckles, hair) and allows scale-specific control. BigGAN [2] scaled class-conditional GANs on ImageNet and introduced the truncation trick mentioned above. Training is unstable and can miss modes, which made GANs hard to scale to open-ended prompts. Theory and code: ML 06, GANs.

VAEs and tokenizers: compress first. A variational autoencoder (VAE) learns an encoder to a compact latent code, a decoder back to pixels, and a simple prior over codes [13]. In today's systems the autoencoder's main job is **compression in front of another generator**: latent diffusion runs its denoiser on a continuous latent grid that is several times smaller than the image [5], and token-based models use a **vector-quantized** autoencoder (VQ-VAE [14], VQGAN [15]) that turns an image into a grid of integer indices from a learned codebook, like words from a dictionary. The decoder sets a ceiling: no generator can produce detail the decoder cannot reconstruct. Details: ML 06, VAEs and ML 08, visual tokens.

Diffusion: many small denoising steps. Train a network to remove a little Gaussian noise from a noisy image; to generate, start from pure noise and apply it many times [16]. Training is a stable

regression, and with better architectures and classifier guidance diffusion models overtook GANs on ImageNet [17]. Running the same process on an autoencoder's latent grid, **latent diffusion**, made high-resolution text-to-image generation affordable [5]. Theory and code: ML 07 and its section on latent diffusion.

Flow matching and rectified flow: follow a velocity field. Draw a noise sample and a data sample, connect them with a straight line, and train a network to predict the direction of travel along such lines; to generate, start from noise and follow the predicted directions [18], [19]. It is closely related to diffusion (a different path between noise and data) and is now used at scale: Stable Diffusion 3 trains a large latent transformer with a rectified-flow objective [20]. Theory: ML 07, flow matching.

Autoregressive and unified models: write the picture piece by piece. Turn the image into a sequence of tokens and predict the next token (or the next group of tokens, or the next, sharper version of the whole token map) given everything generated so far. This is the language-model recipe applied to images, and its newest form is a single transformer that reads and writes both text and images. It gets its own section below.

Family	How it samples	Sequential network calls	Main strength	Main thing to watch
GAN	one generator pass	one	speed, smooth editable latent space	training stability, mode coverage
Diffusion	repeated denoising from noise	tens to hundreds; fewer with distillation	quality, coverage, stable training	sampling cost
Flow matching	integrate a learned velocity	tens; fewer with distillation	simple objective, straighter paths	still needs several steps unless distilled
Autoregressive	predict tokens in some order	one per token, group or scale	reuses language-model machinery, exact likelihood of tokens	tokenizer ceiling, error accumulation

Two points from the map are easy to forget. The generator and the representation are separate choices: a diffusion model can work on pixels (Imagen [4]) or on latents (latent diffusion [5]), and an autoregressive model can use discrete or continuous tokens [21]. And real systems mix families: an adversarial loss trains the tokenizer of VQGAN [15], and adversarial distillation turns a diffusion model into a one-to-four-step generator [22].

The text-to-image system

Plain version. A text-to-image model has three parts that are trained or chosen separately: a **reader** that turns the prompt into a list of vectors, a **painter** that repeatedly refines a noisy image

while looking at those vectors, and a **decoder** that turns the painter's compact result into pixels. Most of the engineering is about how the painter looks at the words.

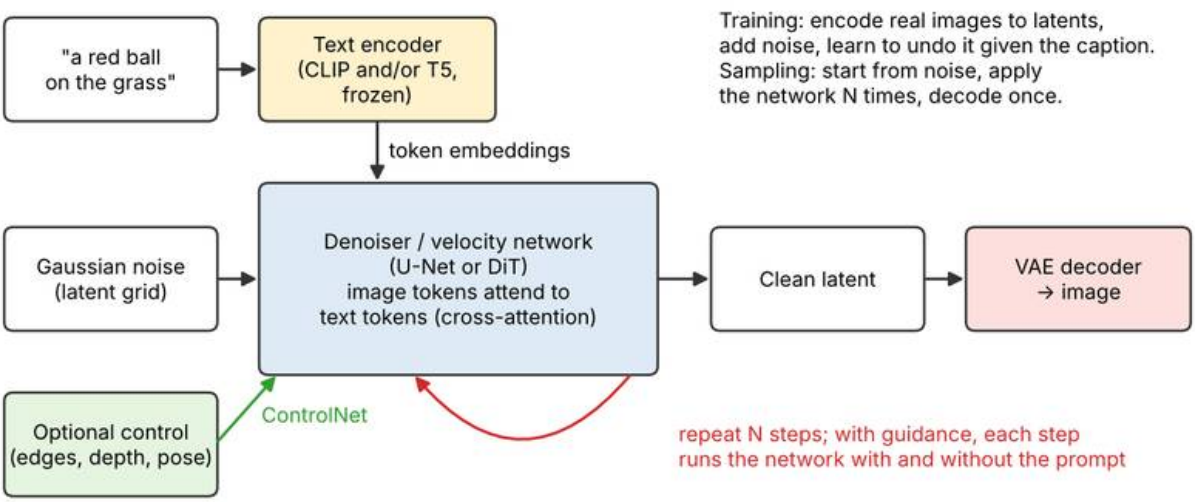


Figure 2 — A latent text-to-image pipeline. The text encoder and the autoencoder are usually trained separately and frozen; only the denoiser (or velocity network) is trained on image–caption pairs. Sampling calls the denoiser N times and the decoder once.

In training, each image of an image–caption pair is encoded to a latent, noised to a random level, and the denoiser learns to undo the noise **given the caption's embeddings**. At sampling time, the denoiser starts from pure noise and runs for N steps, then the decoder produces pixels. The diffusion and sampling equations behind these two sentences are in ML 07; below we look at the parts that are specific to text.

Text encoders: CLIP or a language model

Plain version. The text encoder decides what the generator can "understand". A model trained to match images and captions knows visual concepts; a large language model trained on text alone knows grammar, word order and spelling-related structure better. Many systems use both.

Two kinds of encoder dominate:

- **CLIP text encoders** are trained contrastively so that an image and its caption land near each other in a shared embedding space [23] (see ML 09). Their embeddings are visually grounded.
- **Language-model encoders** such as T5 [24] are trained on text only. Imagen's key finding was that a large **frozen** T5 encoder works surprisingly well, and that making the text encoder larger improved both fidelity and image–text alignment much more than making the image diffusion

model larger [4].

Recent systems combine them. SDXL adds a second text encoder to the design of earlier Stable Diffusion models [25]. Stable Diffusion 3 uses three frozen encoders, two CLIP models and T5-XXL; its authors report that dropping T5 at inference time costs little on most prompts but clearly hurts prompts with long written text or many details [20]. A related finding concerns spelling: text encoders that see whole word pieces rather than characters ("character-blind") make it hard to draw a word letter by letter, and character-aware encoders render visual text much more accurately [26]. DALL·E 2 takes yet another route: a "prior" network first generates a CLIP **image** embedding from the caption, and a diffusion decoder then generates the image from that embedding [27].

Cross-attention: how words reach pixels

Plain version. At every layer, each position of the image asks a question ("what should be here?") and looks over all the words of the prompt to find the ones that answer it. The answer it collects is a weighted mix of the words, with large weights on the words that match.

Latent diffusion introduced cross-attention layers into the U-Net denoiser for this purpose [5]. Let $\Phi \in \mathbb{R}^{N \times d_\phi}$ be the image features at one layer (N spatial positions, each a vector of width d_ϕ), and $E \in \mathbb{R}^{L \times d_e}$ the text-encoder outputs (L tokens of width d_e). Learned matrices $W_Q \in \mathbb{R}^{d_\phi \times d}$, $W_K, W_V \in \mathbb{R}^{d_e \times d}$ project them to a common width d:

$$Q = \Phi W_Q, \qquad K = E W_K, \qquad V = E W_V, \qquad A = \mathrm{softmax}\Big(\frac{Q K^\top}{\sqrt{d}}\Big), \qquad \mathrm{out} = AV.$$

Here $Q \in \mathbb{R}^{N \times d}$ holds one **query** per image position, $K, V \in \mathbb{R}^{L \times d}$ hold one **key** and one **value** per text token, the softmax is taken along each row, and $A \in \mathbb{R}^{N \times L}$ is the **attention map**: row i is a probability distribution over the L tokens that says how much position i listens to each word. The output $\mathrm{out} \in \mathbb{R}^{N \times d}$ gives every image position a mix of word values; it is added back to the image features. Queries come from the image and keys and values come from the text, which is exactly what makes it *cross*-attention rather than self-attention (compare NLP 07). Reading column j of A as an image gives a heat map of **where word j is being drawn**, a fact that Prompt-to-Prompt turned into an editing method [8].

Transformer backbones offer an alternative. Stable Diffusion 3's MM-DiT concatenates image tokens and text tokens into one sequence and lets them attend to each other in both directions, with separate weights for the two modalities [20]. The text then also "reads" the image, not only the other way round.

The toy below builds a 16×16 grid of image features in which a disk region resembles the words "red" and "ball" and the background resembles "grass", then runs one cross-attention step. The projection matrices are identities to keep it readable; in a trained model they are learned.

```
import numpy as np

rng = np.random.default_rng(0)
d = 32                                          # feature width
words = ["a", "red", "ball", "on", "grass"]  # the prompt, one text token per word
E = rng.normal(size=(len(words), d))         # text-encoder outputs, one row per word

# A 16 x 16 grid of image tokens: a disk ("the ball") on a background ("the grass").
yy, xx = np.mgrid[:16, :16]
ball = ((yy - 6) ** 2 + (xx - 9) ** 2 <= 9).ravel()   # 29 of the 256 positions
X = 1.0 * rng.normal(size=(256, d))                   # noisy image features
X[ball] += 0.5 * (E[1] + E[2])                        # ball region looks like "red ball"
X[~ball] += 0.5 * E[4]                                # background looks like "grass"

# Cross-attention: queries from the image, keys and values from the text.
# Real models learn W_q, W_k, W_v; identity matrices keep the toy readable.
Q, K, V = X, E, E
scores = Q @ K.T / np.sqrt(d)                         # 256 image tokens x 5 words
A = np.exp(scores - scores.max(axis=1, keepdims=True))
A /= A.sum(axis=1, keepdims=True)                     # softmax over the words
out = A @ V                                           # each image token gets a mix of word
 ↪  values

print("each row sums to 1:", np.allclose(A.sum(axis=1), 1), "| output shape:", out.shape)
for j, w in enumerate(words):
    print(f"{w:6s} mean weight inside the ball={A[ball, j].mean():.2f}  "
          f"outside={A[~ball, j].mean():.2f}")
```

It prints:

```
each row sums to 1: True | output shape: (256, 32)
a      mean weight inside the ball=0.02  outside=0.06
red    mean weight inside the ball=0.43  outside=0.07
ball   mean weight inside the ball=0.52  outside=0.04
on     mean weight inside the ball=0.02  outside=0.08
grass  mean weight inside the ball=0.01  outside=0.74
```

(These numbers come from our toy run, not from any trained model.) Inside the disk, almost all the attention goes to “red” and “ball”; outside, three quarters of it goes to “grass”. The function words “a” and “on” get little weight anywhere. Figure 3 shows the five maps. Notice what the mechanism does **not** guarantee: “red” and “ball” share the same region because the features say so, and nothing in the attention formula forces a color word to attach to the right object. When two objects are present, a color can attend to the wrong one, which is one root of the attribute-binding failures discussed later [28].

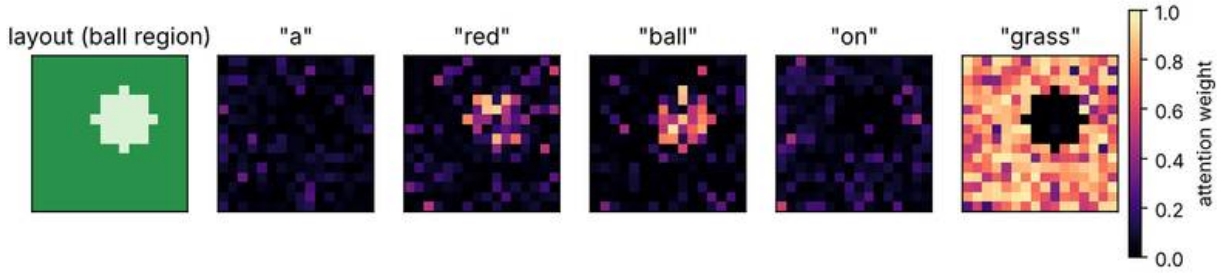


Figure 3 — Cross-attention maps in the toy. Each map is one column of the attention matrix A reshaped to the image grid: where each word is being 'drawn'. Prompt-to-Prompt edits images by reusing such maps [8].

Guidance as used in practice

Classifier-free guidance (CFG) is derived and tested in ML 07; here is the reminder. During training the caption is dropped (replaced by an empty prompt) for a random fraction of examples, so one network learns both a conditional and an unconditional prediction [11]. At sampling time the two are combined, $\hat{\epsilon} = \epsilon_\emptyset + w(\epsilon_c - \epsilon_\emptyset)$, where ϵ_c and $\epsilon_\emptyset$ are the predictions with and without the prompt and w is the guidance scale; $w > 1$ moves past the conditional prediction toward the prompt.

For a system builder, four practical facts matter:

- **It is a dial, not a measurement.** Higher w usually gives more prompt-faithful, higher-contrast images with less variety; very high w gives over-saturated, distorted images. Report the value and the convention (some code writes $1 + w$).
- **It doubles the work per step** unless the two predictions are batched, so a 30-step guided sampler costs about 60 network evaluations.
- **It also works for token models.** Autoregressive and masked token generators apply the same formula to next-token logits (ML 08).
- **Distilled models may bake it in.** Some distilled checkpoints are trained to reproduce guided outputs, so the guidance scale becomes an input or disappears; read the release documentation before comparing settings (ML 07).

Backbones and some landmark systems

The denoiser was first a **U-Net** [29], a convolutional encoder–decoder with skip connections, with cross-attention layers inserted at several resolutions [5]. The **diffusion transformer (DiT)** replaces it with a Vision Transformer over latent patches and showed steady gains as transformer

compute grew [3] (details in ML 07). Large recent systems mostly use transformers. The table lists a few systems whose design is documented in papers; it is a map of design choices, not a ranking.

System	Where it generates	Generator and backbone	Text conditioning
Latent diffusion / Stable Diffusion [5]	autoencoder latent	diffusion, U-Net	cross-attention to a text encoder
DALL·E 2 [27]	pixels	diffusion decoder from a CLIP image embedding	a prior maps the caption to a CLIP image embedding
Imagen [4]	pixels, 64 → 256 → 1024 cascade	diffusion	frozen T5-XXL
Parti [30]	discrete tokens (ViT-VQGAN)	autoregressive encoder–decoder transformer	the text is the encoder's input
Muse [31]	discrete tokens	masked parallel decoding, transformer	embeddings from a pretrained language model
SDXL [25]	autoencoder latent	diffusion, larger U-Net, optional refiner	two text encoders
Stable Diffusion 3 [20]	autoencoder latent	rectified flow, MM-DiT transformer	two CLIP encoders and T5-XXL

The resolutions of the Imagen cascade are from the project page listed with its reference.

Optional: Stable Diffusion 3 (in the Paper guides). How one paper combines a rectified-flow objective, a joint text–image transformer, three text encoders and curated captions, and how to read its ablations.

Control, personalization and editing

Plain version. A prompt is a weak steering wheel: words cannot pin down an exact pose, a particular person's dog, or "change only the hat". Three groups of methods add stronger steering: extra **control images** (edges, depth, pose), small **personalization** updates that teach the model a new concept, and **editing** procedures that change a given image while keeping the rest.

Spatial control: ControlNet

ControlNet adds a spatial condition (an edge map, a depth map, a segmentation, a human pose) to a large pretrained text-to-image model without damaging it [7]. It **locks** the original network, makes a **trainable copy** of its encoding layers that receives the condition, and connects the copy back to the original through **zero convolutions**: 1×1 convolutions whose weights and biases start at zero. At the first training step the added branch outputs exactly zero, so the model behaves exactly like the original and learning starts from a working generator. The paper reports robust training with both small (under 50k) and large (over 1M) datasets [7].

Optional: ControlNet's zero convolutions (in the Supplementary notes). The equation of one controlled block and why gradients still reach the zero-initialized layers.

Personalization: teaching the model a new concept

Three methods trade storage, speed and fidelity differently:

- **Textual inversion** keeps the whole model frozen and learns only a new word embedding (a new "pseudo-word") from 3–5 images of a concept, by minimizing the usual denoising loss on those images [32]. The stored result is a single vector, but its expressiveness is limited by what the frozen model can already draw.
- **DreamBooth** fine-tunes the whole model on a few images of a subject captioned with a rare identifier plus the class name, such as "a [V] dog", and adds a **class-specific prior-preservation loss** that keeps the model able to draw ordinary dogs; without it, the model drifts and loses diversity [33].
- **LoRA** (low-rank adaptation) freezes each weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and learns an additive update $\Delta W = BA$ with $B \in \mathbb{R}^{d_{\text{out}} \times r}$, $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and a small rank r [34]. It was introduced for language models; the same update can be attached to the attention layers of a text-to-image denoiser, which makes a personalized style or subject a small file that can be switched on and off (ML 09, PEFT).

Optional: Personalization objectives side by side (in the Supplementary notes). The exact losses of textual inversion and DreamBooth, and what each method stores.

Editing and inpainting

- **SDEdit** adds noise to an input image (for example a rough color sketch) up to an intermediate noise level, then runs the ordinary reverse process [6]. A low starting noise level keeps the input's layout and colors (faithful); a high one gives the model more freedom (realistic). One number sets the balance.
- **Prompt-to-Prompt** edits a generated image by changing the prompt ("a cat on a bike" → "a dog on a bike") while re-using the original cross-attention maps during sampling, so the layout stays the same and only the changed word's content changes [8]. Editing a *real* photo this way first requires finding a noise that reproduces it, for example with deterministic DDIM inversion (ML 07).
- **InstructPix2Pix** follows written instructions ("make it look like winter"). Its training data was generated automatically by combining a language model (GPT-3) and a text-to-image model (Stable Diffusion) into a large set of before-and-after editing examples; a conditional diffusion model trained on these pairs then edits in a single sampling run, without per-image fine-tuning or inversion [9].

- **RePaint** inpaints with an unconditional diffusion model: at every reverse step it replaces the known region with a correctly noised copy of the input and lets the model generate only the masked region [10].

A survey by Huang et al. organizes the many diffusion-based editing methods by learning strategy, input condition and task [35].

Fast sampling, briefly

A text-to-image model is used interactively, so the number of network calls matters. The routes are covered in ML 07, few-step generation: deterministic samplers such as DDIM need fewer steps with the same network [36]; progressive distillation halves the step count repeatedly [37]; consistency models learn to jump from any point of a trajectory to its end [38]; and adversarial diffusion distillation (ADD) combines a diffusion teacher with an adversarial loss to sample a large model in 1–4 steps [22]. When you compare systems, count every network evaluation (both guidance passes included) plus the decoder.

Autoregressive and unified multimodal generators

Plain version. An autoregressive generator writes an image the way a language model writes a sentence: one token, or one group of tokens, at a time, each conditioned on what came before. The newest step is to use **one** transformer for both text and images, so the same model can describe a picture, answer questions about it and draw a new one.

The autoregressive line in one paragraph

The theory is in ML 08; here is the summary. A tokenizer (VQ-VAE or VQGAN, ML 08) turns the image into a grid of integers, and a transformer models them with the chain rule, trained by next-token cross-entropy with teacher forcing (ML 08). Raster-order models such as Parti [30] and LlamaGen [39] predict one token per step; masked models such as MaskGIT [40] and Muse [31] predict many tokens in parallel and refine them over a few rounds (ML 08); VAR predicts a whole, sharper token map per step, coarse to fine [41] (ML 08); and MAR drops the discrete codebook and models each continuous token with a small diffusion head [21] (ML 08). Sampling cost, KV caching and the comparison with diffusion are in ML 08.

Autoregressive text-to-image in practice

From the system side, four facts are worth knowing.

- **The text enters as context, not through cross-attention to a separate denoiser.** Parti treats text-to-image as translation: a transformer encoder reads the prompt and a decoder writes image tokens; the paper scales this encoder–decoder model to 20 billion parameters and introduces the PartiPrompts benchmark of over 1,600 English prompts [30]. Muse instead feeds a text embedding from a pretrained large language model to a masked token transformer, and it supports inpainting, outpainting and mask-free editing without fine-tuning [31].
- **Language-model infrastructure transfers.** LlamaGen uses a plain Llama-style decoder; besides class-conditional models from 111M to 3.1B parameters it trains a 775M text-conditional model, and it reports that an LLM serving framework sped up image generation by 326%–414% [39].
- **The tokenizer is decisive.** MAGVIT-v2 argues that the tokenizer is the key component: with its new "lookup-free" quantizer, which lets the vocabulary grow large, a language model outperformed diffusion models on ImageNet when given the same training data, a similar model size and a similar training budget [42]. Small details such as text and thin lines are typical casualties of a coarse tokenizer (ML 08).
- **Guidance and sampling knobs carry over.** CFG on logits, temperature and top-k sampling set the fidelity–diversity balance, and every guided step costs two forward passes.

Unified multimodal models

Plain version. Instead of a text model that hands a prompt to a separate image model, a unified model puts text and image pieces into **one sequence** and trains one transformer on it. The design questions are: are image pieces discrete tokens or continuous vectors, is the image part generated by next-token prediction or by diffusion, and do understanding and generation share one image encoder?

The models below are documented in papers and illustrate the main design choices:

- **Chameleon** puts discrete image tokens and text tokens in one sequence from the start ("early fusion") and handles captioning, question answering and the generation of text, images and mixed documents with one model [43].
- **Emu3** trains one transformer from scratch with next-token prediction **only**, over discrete tokens of images, text and video [44].
- **Transfusion** keeps images continuous and trains one transformer with a next-token loss on text and a diffusion loss on image patches; its authors report that this scales better than quantizing images into tokens [45].
- **Show-o** combines autoregressive text modeling with discrete (masked) diffusion over image tokens [46].
- **Janus** shares one transformer but uses **separate** visual encoders for understanding and for gen-

eration, because the two tasks need different levels of image detail [47].

Model	Image representation	Image objective	Visual encoders
Chameleon [43]	discrete tokens	next token	one, shared
Emu3 [44]	discrete tokens	next token	one, shared
Transfusion [45]	continuous patches	diffusion	one, shared
Show-o [46]	discrete tokens	masked (discrete) diffusion	one, shared
Janus [47]	discrete tokens for generation	next token	separate for understanding and generation

Optional: Unified multimodal models, one by one (in the Supplementary notes). What each paper changes in the transformer, the tokenizer and the training data.

What is still open: whether one model can be state-of-the-art at both understanding and generation at the same size, how much a shared visual tokenizer costs each task (the motivation for Janus), and how evaluation should compare a unified model with a specialized diffusion system. Claims about scaling in these papers are each paper's own measurements, on its own data; they are not yet a settled law.

Evaluating text-to-image models

Plain version. You need four kinds of evidence: do the images look real and varied (distribution metrics), do they match their prompts overall (alignment scores), do they get the details right (compositional checks), and do people like them (human preference). Each kind catches failures the others miss.

Distribution metrics and their pitfalls

FID compares feature statistics of generated and real images; precision and recall separate fidelity from coverage. Both are defined and implemented in ML 06, and ML 07 covers the newer feature spaces. For text-to-image models the common protocol is a **zero-shot COCO FID**: generate images from COCO captions [48] with a model that was not trained on COCO and compare with the real COCO images, as Imagen did [4]. Five pitfalls matter in practice:

- **Finite-sample bias.** FID is biased for finite sample sets, and the bias depends on the model being evaluated, so even equal sample counts do not make a comparison fully fair; Chong and Forsyth propose extrapolating to an infinite sample count [49].
- **Preprocessing.** The resizing filter and JPEG compression used before feature extraction change FID substantially; Parmar et al. show this and provide a reference implementation [50].
- **What the features see.** Inception features are so close to ImageNet classes that matching the class histogram of generated images can lower FID without making images better [51].

- **Agreement with people.** In a large human study no common metric correlated strongly with human judgments, and FID under-rated diffusion models; the authors recommend richer feature extractors such as DINOv2 [52]. CMMD replaces Inception features and the Gaussian assumption with CLIP embeddings and a kernel distance [53].
- **No alignment.** FID does not look at the prompt at all: a model that ignores every prompt but draws beautiful COCO-like images can score well.

Prompt alignment: CLIPScore

CLIPScore measures how well an image and a text agree in CLIP's embedding space [54]. With $\mathbf{c}$ the CLIP embedding of the text and $\mathbf{v}$ the CLIP embedding of the image,

$$\text{CLIP-S}(\mathbf{c}, \mathbf{v}) = w \cdot \max\big(\cos(\mathbf{c}, \mathbf{v}), 0\big),$$

where cos is the cosine similarity and w is a constant that only rescales the range ($w = 2.5$ in the original paper, which proposed the score for image captioning). For text-to-image evaluation the score is averaged over prompt–image pairs. It is cheap and needs no reference image, but it inherits CLIP's blind spots: an embedding that summarizes the whole image is weak at counts, spatial relations and which attribute belongs to which object. The GenEval study found that its object-level checks agree with human judgments much better than CLIPScore on such compositional tasks [55].

Compositional benchmarks

Plain version. Instead of one overall similarity, ask concrete yes-or-no questions about the image: is there a dog, are there exactly three, is the dog left of the cat, is the car red? A detector or a vision–language model answers them.

GenEval generates prompts from templates for six tasks (single object, two objects, counting, colors, position and attribute binding), detects the objects with an instance-segmentation model, classifies object colors with CLIP, and scores each image as correct or not [55]. **T2I-CompBench** provides 6,000 compositional prompts in three categories (attribute binding, object relationships and complex compositions) with metrics tailored to each [56]. Imagen's **DrawBench** [4] and Parti's **PartiPrompts** [30] are prompt sets for side-by-side human comparison. Object detection itself is the subject of DLCV 03; an open-vocabulary detector (DLCV 03) can check objects outside a fixed class list.

Optional: GenEval (in the Paper guides). How the benchmark is built, how its scores are computed and how to read its tables.

The toy below shows why object-level checks matter. It draws synthetic "generated images" made of colored disks and squares for the prompt "two red circles and a blue square", detects the shapes

with connected components, and compares two scores: a **presence score** that asks which prompt words appear anywhere in the image (a deliberately simple stand-in for a global similarity score, not CLIP itself) and an **exact check** that requires every (color, shape) pair with the right count, as GenEval does.

```
import numpy as np
from skimage.draw import disk, rectangle
from skimage.measure import label, regionprops

COLORS = {"red": (0.85, 0.1, 0.1), "blue": (0.1, 0.2, 0.85)}

def render(objects, size=96):
    """Draw (color, shape) objects at fixed slots on a gray canvas: a stand-in for a
    ↪ generated image."""
    img = np.full((size, size, 3), 0.8)
    for k, (col, shape) in enumerate(objects):
        cy, cx = 24 + 48 * (k // 2), 24 + 48 * (k % 2)
        if shape == "circle":
            rr, cc = disk((cy, cx), 14)
        else:
            rr, cc = rectangle((cy - 12, cx - 12), extent=(24, 24))
        img[rr, cc] = COLORS[col]
    return img

def detect(img):
    """Toy 'detector': connected colored regions -> (color, shape) pairs."""
    found = []
    for region in regionprops(label(np.abs(img - 0.8).sum(-1) > 0.3)):
        r, g, b = img[tuple(region.coords.T)].mean(0)
        col = "red" if r > b else "blue"
        shape = "square" if region.extent > 0.9 else "circle"   # filled bbox fraction
        found.append((col, shape))
    return found

def presence_score(prompt, img):
    """Bag-of-words check: which prompt words appear anywhere (ignores counts and
    ↪ binding)."""
    words = {w for obj in prompt for w in obj}
    seen = {w for obj in detect(img) for w in obj}
    return len(words & seen) / len(words)

def exact_check(prompt, img):
    """GenEval-style check: every (color, shape) pair with the right count."""
    return sorted(detect(img)) == sorted(prompt)

prompt = [("red", "circle"), ("red", "circle"), ("blue", "square")]   # "two red circles and
↪ a blue square"
candidates = {
    "correct":          prompt,
    "wrong count":      prompt + [("red", "circle")],
    "colors swapped":   [("blue", "circle"), ("blue", "circle"), ("red", "square")],
    "object missing":   prompt[:2],
```

```
}
for name, objs in candidates.items():
    img = render(objs)
    print(f"{name:15s} presence score={presence_score(prompt, img):.2f}  "
          f"exact check={'pass' if exact_check(prompt, img) else 'fail'}")
```

It prints:

```
correct         presence score=1.00  exact check=pass
wrong count     presence score=1.00  exact check=fail
colors swapped  presence score=1.00  exact check=fail
object missing  presence score=0.50  exact check=fail
```

The presence score gives full marks to the wrong count and to the swapped colors, because every word of the prompt is present *somewhere*; only the exact check catches them. Real benchmarks replace our connected-component "detector" with a trained detector and a color classifier, and inherit their errors: the check is only as good as the detector, which is why GenEval reports its agreement with human annotators [55]. Figure 4 shows the four images.

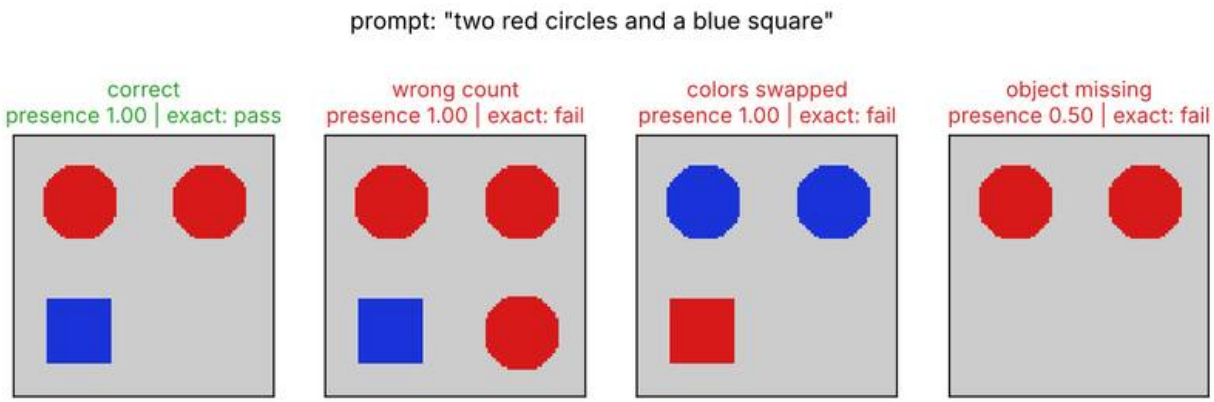


Figure 4 — A word-presence score cannot see counting or binding errors; an object-level check can. Synthetic shapes stand in for generated images.

Human preference and holistic evaluation

People remain the reference. Because human studies are slow, several groups trained **preference models** that predict which of two images people prefer for a prompt: ImageReward was trained on 137k expert comparisons [57], and PickScore on Pick-a-Pic, an open dataset of real users' choices collected through a web app [58]. **HEIM** evaluates models along 12 aspects (including alignment, quality, aesthetics, originality, reasoning, knowledge, bias, toxicity, fairness,

robustness, multilinguality and efficiency) over 62 scenarios and 26 models, and finds that no single model is best at everything [59]. A learned preference model is itself a model with biases; if a generator is tuned to maximize it, the score can rise faster than real quality.

A reporting checklist. For any claim about a text-to-image model, state: the prompts (and how they were sampled), the number of images per prompt and seeds, the sampler, the number of steps and guidance scale, the resolution, the metric implementations and feature extractors, and show uncurated samples. Report distribution, alignment, compositional and preference results side by side, and say which numbers you ran yourself and which were copied from papers.

Data

Plain version. A generator can only learn what its training images and captions show, so the dataset decides its knowledge, its style and many of its biases. Large text-to-image datasets are scraped from the web, which raises questions of quality, copyright and consent.

- **Class-conditional benchmarks** use ImageNet [60]; text-to-image papers often evaluate zero-shot on COCO captions [48]; face generators use datasets such as FFHQ, the high-quality face dataset introduced with StyleGAN [1].
- **LAION-5B** is an openly released dataset of 5.85 billion CLIP-filtered image–text pairs, of which 2.32 billion have English text [61]. It distributes URLs and alt-text rather than the images themselves, and its datasheet states that LAION releases the metadata under CC-BY-4.0 but does not own the copyright of the images or text, that people depicted in the images may not have consented to their upload, and that the dataset should be used for academic research in its current form; it describes ways to request removal of links [61].
- **Content audits.** Birhane et al. examined the earlier LAION-400M dataset and found explicit and problematic image–text pairs, including pornography, malign stereotypes and slurs [62].
- **Captions are data too.** Stable Diffusion 3 trains on a 50/50 mix of original captions and synthetic captions written by a vision–language model, and filters its data for sexual content, low aesthetic scores and near-duplicates (to reduce regurgitation of training images) [20].

Risks and responsible use

Plain version. A model that can draw anything can also copy, stereotype and deceive. The practical responses are to measure these behaviors, to reduce them in the data and the model, and to label generated images so that viewers can check where they came from.

Memorization and copying

Carlini et al. showed that diffusion models memorize individual training images and can emit them: with a generate-and-filter pipeline they extracted over a thousand training examples, from photos of individual people to trademarked logos, and they found diffusion models less private than GANs in their experiments [63]. Somepalli et al. built image-retrieval tools to compare generated images with training images and found cases where models, including Stable Diffusion, copied content from their training data [64]. Deduplicating training data [20] and checking generated images against the training set are the standard first defenses.

Bias

Bianchi et al. found that ordinary prompts (traits, occupations, objects) produce and amplify demographic stereotypes, and that neither counter-stereotypical prompts nor system guardrails removed them in their tests [65]. Luccioni et al. proposed a way to measure social bias in generated people by comparing the variation caused by gender and ethnicity words with the variation caused by professions, and found under-representation of marginalized identities in the three systems they tested [66]. Bias evaluation needs its own prompts and analysis; overall quality metrics do not reveal it. See also ML 09, data provenance, bias and privacy.

Deepfakes, provenance and watermarks

Two complementary tools exist for labeling generated images:

- **Signed provenance (C2PA).** The C2PA specification defines a "manifest" of provenance claims (who made the asset and how) that is cryptographically signed and bound to the asset, either by a cryptographic hash (a hard binding) or by a fingerprint or invisible watermark (a soft binding) [67]. A valid signature shows that the claims were not altered; it does not prove that the content is true, and a manifest stored in metadata can simply be removed.
- **Invisible watermarks.** Tree-Ring watermarking hides a pattern in the Fourier transform of the **initial noise** of a diffusion sampler and detects it by inverting the sampler [68]. Stable Signature fine-tunes the latent decoder so that every generated image carries a binary signature that a pretrained extractor can read [69]. Classical watermarking is covered in IP 08.

Neither is a complete answer. Zhao et al. prove and show that pixel-level invisible watermarks can be removed by a "regeneration attack" (add noise, then reconstruct with a generative model) [70], and Saberi et al. show a fundamental trade-off for low-perturbation watermarks between failing to detect watermarked images and falsely flagging real ones, and demonstrate spoofing attacks that make real images look watermarked [71]. Treat provenance and watermarks as evidence to weigh, not as proof.

Optional: Hard bindings, soft bindings and watermarks (in the Supplementary notes). How each mechanism survives cropping, re-encoding and metadata stripping.

Failure modes

Plain version. Today's generators are excellent at "a photo of X" and still unreliable at the precise parts of a request: how many, which one has which color, where things are, what the text says. Knowing the typical failures tells you what to test.

- **Counting.** "Four apples" often gives three or five. GenEval's counting task (2, 3 or 4 objects) measures it directly [55].
- **Spatial relations.** "A cup to the left of a laptop" is often drawn in the wrong arrangement; in GenEval's original study the best model produced the correct position in only 15% of images [55].
- **Attribute binding and leakage.** In "a red car and a blue bench" the colors can swap between objects or leak onto the background; GenEval's best model at the time got attribute binding right in 35% of images [55]. Attend-and-Excite also documents **catastrophic neglect**, where one of the subjects in the prompt is not drawn at all, and steers the cross-attention maps at sampling time so that every subject token gets attention [28].
- **Text rendering.** Misspelled signs and labels are common; character-blind text encoders are one documented cause [26], and Stable Diffusion 3's ablation shows that the T5 encoder matters most for prompts with written text [20].
- **Hands.** Generated hands often have the wrong number of fingers or irregular shapes; HandRefiner repairs them afterwards by inpainting conditioned on a reconstructed hand mesh [72]. Note that object-level benchmarks cannot check this: GenEval's authors point out that their detector can count people but not fingers [55].
- **Copying of content and style.** A prompt that names a famous image, product or artist can pull out a near-copy of training content [64], [63].

How to test. Build a small prompt set per failure type, generate several images per prompt with fixed seeds, check them with an object-level tool where possible and by eye where not, and track the failure rate across model versions.

Modern view

Surveys worth reading.

- **Zhang et al. [73]** (2023) is a self-contained review of text-to-image diffusion models: a short introduction to diffusion, the background of text-conditioned synthesis, the pioneering systems

and their improvements, applications beyond images (video, other modalities, text-guided editing) and open challenges.

- **Croitoru et al. [74]** (IEEE TPAMI 2023) survey diffusion models in vision, grouped by formulation (DDPM, score networks, SDEs) and by application (generation, super-resolution, inpainting, editing, translation, segmentation).
- **Huang et al. [35]** (IEEE TPAMI 2025) classify diffusion-based image editing by learning strategy, user-input condition and editing task, give special attention to inpainting and outpainting, and propose a benchmark (EditEval) with a metric scored by a large multimodal model.
- **Xiong et al. [75]** (TMLR 2025) survey autoregressive vision models, split by representation into pixel-based, token-based and scale-based models, and discuss their links to other generative families and their uses in image, video, 3-D and multimodal generation.

State of the art, 2023–2026. (1) **Latent transformers with flow objectives and several text encoders** became the standard large-scale recipe, as in Stable Diffusion 3 [20], building on DiT [3]. (2) **Autoregressive models returned** through better tokenizers [42], next-scale prediction [41] and continuous tokens [21]. (3) **Unified multimodal models** that understand and generate with one transformer appeared in quick succession [43], [45], [46], [47], [44]. (4) **Few-step generation** made interactive use practical [22], [38]. (5) **Evaluation moved beyond FID** toward compositional checks [55], [56], learned preference models [57], [58] and holistic benchmarks [59]. Recent work also revisits pixel-space generation and generation in representation spaces; see ML 07, pixel space, latent space and representation autoencoders.

Open problems. Reliable counting, spatial reasoning and attribute binding; legible text in any language; evaluation metrics that agree with people at the level of single images and cannot be gamed; preventing memorization without losing quality; measuring and reducing bias; provenance that survives editing; consent and licensing for training data; and the cost of high-resolution and video generation.

Key takeaways

- Every generation task samples from $p(x \mid c)$; tasks differ only in the condition c, and "good" always means fidelity, diversity and alignment, plus cost.
- The generator (GAN, diffusion, flow, autoregressive) and the representation (pixels, continuous latent, discrete tokens) are separate design choices, and real systems mix families.
- A text-to-image model is a frozen text encoder (CLIP and/or T5), a denoiser or token model that reads the text through cross-attention or joint attention, a sampler with guidance, and a decoder.
- Cross-attention maps show where each word is drawn; they power editing methods such as Prompt-to-Prompt, but nothing in them forces attributes to bind to the right object.

- ControlNet adds spatial control through a trainable copy joined by zero convolutions; textual inversion, DreamBooth and LoRA personalize at different costs; SDEdit, Prompt-to-Prompt, InstructPix2Pix and RePaint edit.
- Unified multimodal models put text and image pieces in one sequence; they differ in discrete versus continuous images, next-token versus diffusion objectives, and shared versus separate visual encoders.
- Evaluate with distribution metrics, alignment scores, object-level compositional checks and human preference together, and report the full protocol; FID alone ignores the prompt.
- Memorization, bias, deepfakes and data consent are measurable risks; watermarks and signed provenance help but can be removed or spoofed.

Exercises

1. **Map the tasks.** For each task in the table of six kinds of generation, write down what the condition c is as a tensor (shape and type) for a latent text-to-image model, and which part of the pipeline in Figure 2 receives it.

   Hint

   The prompt becomes L token embeddings that enter through cross-attention. A class label can be written as a text prompt or as an embedding added to the time embedding. For image-to-image and editing, the input image is encoded to a latent and either noised to an intermediate level (SDEdit) or concatenated to the denoiser's input channels (InstructPix2Pix); a control image enters through ControlNet's trainable copy; a mask for inpainting decides which latent positions are regenerated.

2. **Neglect in the toy.** In the cross-attention snippet, add a sixth word "dog" whose embedding is a fresh random vector that no image region resembles. What does its attention map look like, and what does it do to the weights of the other words? Then make the ball region resemble only "ball" (not "red"). Which map changes?

   Hint

   The "dog" column gets small, unstructured weights everywhere, because no query matches its key: nothing tells the generator where a dog should go, which is the situation Attend-and-Excite fixes by strengthening a neglected token's attention during sampling. The softmax still sums to one per row, so the other words lose a little weight. Removing "red" from the ball features makes the "red" map dark: a color word attends only where the features already suggest it.

3. **Break the metric.** Extend the compositional-check snippet with a candidate that has the right objects but in an image where the blue square overlaps one red circle. What does the toy detector report, and what would a real detector-based benchmark need to handle it?

Hint

Overlapping shapes merge into one connected component, so the toy detector sees fewer objects and the wrong shape; the exact check fails even though a person might accept the image. Real benchmarks use instance segmentation, which can separate overlapping objects, and GenEval adds rules for overlapping boxes when judging positions. Every automatic check has such blind spots, which is why benchmarks report agreement with human annotators.

4. **Design an evaluation.** You must decide whether a new model is better than the current one at "product photos with a short brand name on the package". Write an evaluation plan: prompts, number of images, automatic metrics, human study, and what result would change your decision.

Hint

Use a fixed prompt set that varies product type, text length and background; several seeds per prompt; an OCR check of the rendered text against the requested string; an object-level check of the product type; FID or CMMD against real product photos for realism (with a fixed implementation and sample count); and a blind side-by-side human comparison on alignment and quality. Decide in advance which difference you would act on, and include a memorization check against the training set for brand logos.

5. **Provenance under editing.** An image carries a C2PA manifest with a hard binding and was generated by a model with a Stable Signature watermark. It is then cropped, re-saved as JPEG and uploaded to a site that strips metadata. Which evidence of its origin is likely to survive, and why?

Hint

The manifest is lost with the metadata, and even if it were kept, the hash of the hard binding would no longer match the cropped, re-encoded pixels. A soft binding or a robust invisible watermark can survive moderate cropping and compression (Stable Signature reports detection after strong cropping), but a regeneration attack can remove pixel-level watermarks. The honest answer is "maybe the watermark, not the manifest", which is why the two are designed to work together.

Supplementary notes

These notes are optional background that the main text does not depend on.

Unified multimodal models, one by one

This note lists, for each unified model in the main text, what it changes relative to a standard language model; all statements are from the papers' abstracts and method descriptions.

- **Chameleon** [43] converts images into discrete tokens and places them in the same sequence as text tokens from the start ("early fusion"), so a single token-based transformer can produce text, images, or documents that interleave both. The paper's contributions include a stable training approach from scratch, an alignment recipe and an architectural parameterization tailored to this early-fusion setting; it evaluates on visual question answering, captioning, text tasks, image generation and long mixed-modal generation judged by people.
- **Emu3** [44] tokenizes images, text and **video** into one discrete vocabulary and trains a single transformer from scratch on mixtures of such sequences with next-token prediction only, without diffusion and without combining a CLIP-style encoder with a language model. Video generation is then "predict the next token in a video sequence". The journal version appeared in *Nature* in 2026.
- **Transfusion** [45] trains one transformer on mixed sequences with two losses: next-token cross-entropy on text tokens and a diffusion (denoising) loss on continuous image patches. The paper reports scaling experiments up to 7B parameters and 2T multimodal tokens, finds that this recipe scales better than quantizing images into discrete tokens, and shows that modality-specific encoding and decoding layers let it compress each image to as few as 16 patches.
- **Show-o** [46] mixes autoregressive prediction (for text) with discrete diffusion, that is masked-token prediction, for image tokens inside one transformer, so the same model handles question answering, text-to-image generation, text-guided inpainting and extrapolation, and mixed-modality generation.
- **Janus** [47] argues that understanding needs high-level semantic features while generation needs fine detail, so a single visual encoder (as in Chameleon) serves both poorly, especially understanding. It keeps one autoregressive transformer but gives understanding and generation **separate visual encoding pathways** (in the paper, a SigLIP encoder for understanding and a VQ tokenizer that turns images into discrete IDs for generation), which also lets each pathway choose its own encoder.

How to compare them. Hold three axes in mind: (a) discrete versus continuous image representation (a discrete tokenizer caps detail; continuous patches need a diffusion-style loss), (b) the image objective (next token, masked tokens or diffusion), and (c) whether understanding and generation share the visual encoder. Then check, for each reported result, the model size, the training tokens and whether the comparison baselines had similar budgets.

ControlNet's zero convolutions: why training starts from the original model

This note writes out one ControlNet block and explains why zero-initialized layers still learn.

Let $\mathcal{F}(\cdot;\Theta)$ be a block of the pretrained network with frozen parameters Θ, mapping a feature map x to $y = \mathcal{F}(x;\Theta)$. ControlNet makes a trainable copy with parameters Θ_c (initialized from Θ) and two zero convolutions $\mathcal{Z}(\cdot;\Theta_{z1})$ and $\mathcal{Z}(\cdot;\Theta_{z2})$: 1×1 convolutions whose weights and biases are initialized to zero. With a condition feature map c, the controlled block computes [7]

$$y_c = \mathcal{F}(x;\Theta) + \mathcal{Z}\Big(\mathcal{F}(x + \mathcal{Z}(c;\Theta_{z1});\ \Theta_c);\ \Theta_{z2}\Big).$$

At initialization both zero convolutions output zero, so $y_c = y$: the controlled network is exactly the pretrained one, and no random noise from new layers disturbs it. The trainable copy also receives x, so it starts as a full working copy of the block.

Why the zeros do not stay zero. For a 1×1 convolution $\mathcal{Z}(h) = Wh + b$, the gradient of the loss $\mathcal{L}$ with respect to the weight is $\partial\mathcal{L}/\partial W = (\partial\mathcal{L}/\partial\mathcal{Z})\, h^\top$, which depends on the input h, not on W. So even with $W = 0$ the weight receives a nonzero gradient whenever its input h and the upstream gradient are nonzero, and after the first update the branch starts to contribute. (What *is* zero at the first step is the gradient flowing *through* $\mathcal{Z}$ to earlier layers, because that gradient is multiplied by W.) The paper details this gradient argument in its supplementary material.

Practical consequences. Training cannot damage the pretrained model at the start, works with modest datasets, and several ControlNets (for example depth and pose) can be added to the same frozen model, each with its own scale.

Personalization objectives side by side

This note gives the exact training objectives of textual inversion and DreamBooth and compares what the three personalization methods store.

Textual inversion [32]. Let $\mathcal{E}$ be the frozen image encoder of a latent diffusion model, c_θ the frozen text encoder, ϵ_θ the frozen denoiser, and y a prompt that contains a placeholder word S_* whose embedding v is the only trainable quantity. With $z = \mathcal{E}(x)$ for a training image x, a random time t, noise $\epsilon \sim \mathcal{N}(0, I)$ and the noised latent z_t,

$$v_* = \arg\min_v\ \mathbb{E}_{z,\,y,\,\epsilon,\,t}\ \|\epsilon - \epsilon_\theta(z_t,\ t,\ c_\theta(y))\|_2^2,$$

which is the ordinary latent-diffusion loss with everything frozen except the new word's embedding. The paper initializes v from the embedding of a coarse one-word description such as "cat".

DreamBooth [33]. Let $\hat{x}_\theta$ be the (trainable) model's prediction of the clean image, c the conditioning for a prompt with the identifier ("a [V] dog"), c_{pr} the conditioning for the plain class prompt ("a dog"), and x_{pr} images generated by the frozen original model from c_{pr}. With noise-schedule coefficients α_t, σ_t, weights w_t, and a weight λ for the second term,

$$\mathbb{E}_{x,c,\epsilon,\epsilon',t}\Big[w_t\|\hat{x}_\theta(\alpha_t x + \sigma_t\epsilon,\ c) - x\|_2^2 + \lambda\, w_{t'}\|\hat{x}_\theta(\alpha_{t'}x_{\text{pr}} + \sigma_{t'}\epsilon',\ c_{\text{pr}}) - x_{\text{pr}}\|_2^2\Big].$$

The first term fits the 3–5 subject images; the second, the **prior-preservation** term, asks the model to keep reproducing its own samples of the class, which counters "language drift" (forgetting what an ordinary dog looks like) and loss of diversity. The paper reports good results with about 1,000 iterations and $\lambda = 1$.

LoRA [34] can be combined with either objective: instead of updating a weight matrix W, it trains B and A in $W + BA$, so only $r(d_{\text{in}} + d_{\text{out}})$ numbers per adapted matrix are stored.

Method	What is trained	What is stored per concept	Typical weakness
Textual inversion	one word embedding	one vector	limited to what the frozen model can draw
DreamBooth	all denoiser weights	a full model copy (or a LoRA, if combined)	overfitting and drift without the prior term
LoRA	low-rank updates of chosen layers	small matrices A, B	rank and layer choice need tuning

Hard bindings, soft bindings and watermarks

This note explains how each labeling mechanism ties a label to an image and what typical edits do to it.

Hard binding. A C2PA manifest can include a cryptographic hash of the asset's bytes (or of parts of it) [67]. Any change to those bytes (cropping, resizing, re-encoding) changes the hash, so validation then reports a mismatch. This is the point: a hard binding detects tampering, it is not meant to survive editing. An editing tool that supports the standard can instead write a **new** manifest that records the edit and refers to the earlier asset as an "ingredient".

Soft binding. The specification also allows content identifiers that are not exact hashes: a perceptual **fingerprint** (not statistically unique) or an **invisible watermark** embedded in the pixels [67]. These can survive moderate edits, and they can be used to look up a manifest that was stripped from the file.

Generation-time watermarks. Tree-Ring [68] writes a key pattern into the Fourier transform of the sampler's initial noise; detection inverts the diffusion sampler to recover an estimate of that noise and tests for the key, and the Fourier construction makes it invariant to convolutions, crops, dilations, flips and rotations, according to the paper. Stable Signature [69] fine-tunes the latent decoder so that every decoded image carries a fixed binary signature; an extractor reads the bits and a statistical test decides; the paper reports detection of images cropped to 10% of their content with over 90% accuracy at a false-positive rate below 10^{-6}.

What breaks them.

Edit	Hash (hard binding)	Manifest in metadata	Robust invisible watermark
Metadata stripped	not checkable (manifest gone)	lost	survives
Crop or JPEG re-save	mismatch	kept but invalid for the new pixels	usually survives moderate edits
Regeneration (noise, then reconstruct)	mismatch	lost if re-saved without it	pixel-level marks can be removed [70]

Edit	Hash (hard binding)	Manifest in metadata	Robust invisible watermark
Adding a watermark pattern to a real image	—	—	spoofing is possible [71]

The lesson is the one in the main text: combine mechanisms and treat each as evidence with known failure modes.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Esser et al., "Scaling Rectified Flow Transformers for High-Resolution Image Synthesis" (ICML 2024)

The problem. Rectified flow (straight paths between noise and data) is simple and has attractive properties, but in 2024 it was not established as better than standard diffusion formulations for large text-to-image models, and text-to-image backbones still mostly used a U-Net with one-way cross-attention [20].

The idea. Two changes, tested at scale. First, train rectified-flow models with timestep distributions that sample intermediate noise levels more often (for example a logit-normal density), because the endpoints are easy and the middle is where learning happens. Second, a transformer backbone, MM-DiT, in which image tokens and text tokens are concatenated and attend to each other in both directions, with separate weights for each modality. Text comes from three frozen encoders (CLIP-L/14, CLIP-G/14 and T5-XXL).

What it showed. In a large comparison of formulations, rectified flow with the tuned timestep sampling performed best for high-resolution text-to-image synthesis. MM-DiT improved over variants with shared weights or with cross-attention only. Validation loss followed predictable scaling trends with model size and correlated with metrics and human ratings; the largest model (8B parameters) was compared with open and proprietary systems in human preference studies and on GenEval. Dropping T5 at inference cost little on most prompts but hurt prompts with written text. The data section describes filtering (sexual content, aesthetics, deduplication) and a 50/50 mix of original and synthetic captions.

How to read it. Read Section 2–3 for the time-sampling densities (they replace the "noise schedule" of diffusion), Section 4 with the architecture figure, then Section 5's ablations one table at a time. Check: (a) which comparisons use small models and few training steps (most ablations) and which use the full model; (b) that GenEval and human-preference comparisons involve different prompt sets and protocols; (c) which improvements come from data (captions, filtering) rather than from the architecture or objective.

Limits. Many conclusions come from ablations at smaller scale; the human studies compare against specific system versions at one point in time; the paper's training data is not public, so data effects cannot be reproduced exactly.

Why it matters here. It is the clearest documented example of the modern text-to-image recipe in this chapter: latent space, rectified-flow objective, joint text–image transformer, several text encoders, curated and recaptioned data, and evaluation by GenEval and human preference.

Ghosh et al., "GenEval: An Object-Focused Framework for Evaluating Text-to-Image Alignment" (NeurIPS 2023)

The problem. FID measures overall image statistics and CLIPScore overall image–text similarity; neither tells you whether a model drew *two* dogs, a *red* car, or a cup *left of* a laptop. Human evaluation does, but it is slow and expensive [55].

The idea. Turn alignment into object-level checks performed by existing vision models. Prompts are generated from templates for six tasks: single object (80 prompts), two objects (99), counting (80), colors (94), position (100) and attribute binding (100), with objects drawn from the 80 COCO classes, colors from 11 basic color terms, counts of 2, 3 or 4, and four relative positions. A Mask2Former instance-segmentation model trained on COCO detects objects (confidence threshold 0.3, raised to 0.9 for counting); positions are judged from bounding-box centroids; colors are classified by CLIP ViT-L/14 on the object's crop with the background replaced by gray. Each image is scored correct or incorrect, and four images are generated per prompt.

What it showed. In a human study of 6,000 annotations on 1,200 images, GenEval agreed with annotators 83% of the time, against an inter-annotator agreement of 88%, and agreed with humans much better than CLIPScore on the compositional tasks. Single objects were easy for every model tested, but even the best model at the time produced correct relative positions in 15% and correct attribute binding in 35% of images.

How to read it. Start with Table 1 (tasks and templates) and Section 3.2 (the scoring pipeline), then the human-agreement analysis, then the benchmark table. Check: (a) that the overall score is an average over tasks, so a model can hide a weak task behind strong easy tasks; read per-task columns; (b) the detector thresholds and their effect on counting; (c) which models were tested (open-source models of 2023); later papers report GenEval for newer models, so compare scores only within one table and one version of the benchmark code.

Limits. The authors note that GenEval is bounded by its detector: COCO-trained detectors know a fixed set of classes, cannot check fine details such as the number of fingers, and generalize poorly to artistic styles. Template prompts are short and unlike real user prompts, and once a benchmark is popular, models may be tuned toward it.

Why it matters here. It is the model for the object-level checks of the evaluation section and the

source of several of the failure-mode numbers; it is now reported by many text-to-image papers, including Stable Diffusion 3 [20].

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DLCV 10A • Video Generation

How video generators are built and judged: video autoencoders and tokenizers, factorized and full spatiotemporal attention, text-to-video and image-to-video, camera and motion control, long videos, temporal consistency, autoregressive versus diffusion generation, FVD and benchmark suites, data, risks and failure modes.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-10a-video-generation/

What you'll learn

- Why video generation is much harder than image generation: the size of the data, the need for believable motion, and **temporal consistency** (no flicker, no identity drift, objects that persist).
- How video is compressed before generation: **3D and causal video VAEs**, discrete **video tokenizers**, and how compression factors set the number of tokens a model must handle.
- How the main architectures work: temporal layers **inflated** into a pretrained image model, **factorized** space–time attention versus **full** spatiotemporal attention, cascades versus single-stage models.
- How the condition enters: **text-to-video**, **image-to-video**, camera and motion control, and how long videos are produced by extension or hierarchy.
- How **autoregressive** and **diffusion** video generators differ, and how hybrids make diffusion causal and streamable.
- How video generators are evaluated (**FVD** and its content bias, **VBench**-style dimension suites, human studies), what data they use, which risks they carry and where they still fail.

The big picture

A video generator makes a short moving picture from a request: a sentence ("a dog runs along a beach at sunset"), a still image to bring to life, a camera path, or the first seconds of a clip to continue. The result must look good in every frame, the motion must be believable, and things must stay the same thing from one frame to the next.

> **Explain like I'm five**
>
> Think of a flipbook. If you draw every page separately from memory, the dog's spots move around, its ears change size and the whole book seems to shimmer when you flip it. A good flipbook artist first

sketches the rough motion through all the pages at once, then adds the details so that each spot stays on the same part of the dog as it runs. A video generator does the same thing: it plans many frames together, in a compressed sketch space, so that the details move with the objects instead of jumping around.

Why it matters. Video generation is used in creative tools, for animating photos, for generating training data and simulation scenes, and as a candidate "world simulator" for robots and vehicles. It is also the most expensive kind of visual generation: a five-second clip contains dozens of images that must all agree. Most of the engineering is about one trade-off: modeling all frames jointly, for consistency, without paying for every pixel of every frame to interact with every other.

Where this chapter sits. The image side is covered elsewhere and is not repeated. DLCV 10 explains text encoders, cross-attention, guidance in practice, control and editing, text-to-image evaluation and risks. ML 07 derives diffusion, classifier-free guidance, latent diffusion, diffusion transformers, flow matching and few-step samplers, and its section Video adds temporal compression and a consistency problem is the short preview that this chapter expands. ML 08 covers autoregressive generation and visual tokenizers. DLCV 07 discusses action-conditioned video prediction, large generative world models and whether realistic video implies physical understanding. DLCV 04B introduces optical flow, warping and temporal-consistency measures. Here we focus on what is specific to **generating** video. All code runs in seconds on a CPU with NumPy, SciPy and scikit-image, without network weights.

What makes video harder than images

Plain version. A video is a stack of images in time, so everything about image generation still applies, plus three new problems: there is far more data, the motion itself must be right, and the frames must agree with each other. A model that draws each frame well but independently produces a shimmering, unstable clip.

The tasks

Every task below samples a video $x \in \mathbb{R}^{F \times H \times W \times 3}$ (with F frames of height H and width W, three color channels) from a conditional distribution $p(x \mid c)$; the tasks differ only in the condition c.

Task	Condition c	Examples in this chapter
Text-to-video (T2V)	a text prompt	Make-A-Video [1], CogVideoX [2], Wan [3]
Image-to-video (I2V)	a first frame, often plus a prompt	Stable Video Diffusion [4], DynamiCrafter [5]
Video extension or prediction	earlier frames of the same clip	VDM [6], Video LDM [7], StreamingT2V [8]

Task	Condition c	Examples in this chapter
Frame interpolation	keyframes to fill in between	Make-A-Video [1], Stable Video Diffusion [4]
Video editing	an input video plus an edit request	Tune-A-Video [9], Text2Video-Zero [10]
Controlled generation	a camera path or object trajectories	MotionCtrl [11], CameraCtrl [12], DragNUWA [13]

As with images (DLCV 10), one trained model can often serve several rows: a first frame, earlier frames or keyframes are extra conditions, and masking decides which frames are generated.

Three demands on every clip

- **Per-frame quality.** Each frame on its own should look like a real image: sharp, free of artifacts, well composed.
- **Motion quality.** The clip should move, and the motion should be plausible: speed, direction and deformation should look like the real thing. A clip that barely moves is a failure even if every frame is perfect.
- **Temporal consistency.** What is the same object should stay the same object. Three failures have names. **Flicker** is fast, frame-to-frame change of brightness or fine texture that no real motion explains. **Identity drift** is a slow change of a subject's appearance (a face, a logo, the pattern on a shirt) over the clip. A failure of **object permanence** is an object that disappears, changes or reappears wrongly after being hidden.

The demands pull against each other. The easiest way to remove flicker is to generate almost no motion; the easiest way to get dramatic motion is to let details change freely. Evaluation, later in this chapter, has to check all three together.

The cost

A video is large. A clip of $F = 81$ frames at 480×832 pixels (about five seconds at 16 frames per second) holds $81 \cdot 480 \cdot 832 \cdot 3 \approx 9.7 \times 10^7$ numbers, the size of about 81 images. A generator also cannot simply treat the frames one after another: consistency requires information to flow between distant frames. With attention, the cost of letting every position see every other position grows with the **square** of the number of positions. Two design levers control it: **compress** the video before generating (the next section), and **restrict** which positions attend to which (the section after it).

Compressing video: 3D VAEs and video tokenizers

Plain version. Like latent image diffusion, almost every modern video generator works on a compressed version of the video, produced by an autoencoder. The new step is compressing **time** as well as space: neighboring frames are very similar, so a group of four frames can share one latent frame. Compression makes generation affordable, but the decoder then limits how much detail and fast motion the final video can contain.

Why not compress each frame with an image autoencoder?

The first latent video models reused the autoencoder of an image latent diffusion model [14] and encoded every frame separately. The Video LDM authors report that this causes flickering, because the image autoencoder was trained on images only and does not know that consecutive frames should decode consistently; they fine-tuned the **decoder** with added temporal layers and a 3D-convolutional discriminator on video, and found this step critical for good results [7]. Stable Video Diffusion also builds on a pretrained image latent model and compresses only in space [4], [3]. A frame-wise autoencoder also keeps one latent frame per video frame, so it does nothing for the cost in time.

3D causal VAEs

A **3D VAE** uses convolutions over time as well as space, so it can downsample both [2]. Most recent systems make it **causal in time**: every temporal convolution is padded only at the beginning of the clip, so a latent frame depends only on the current and earlier video frames, never on later ones [15], [2]. A useful consequence is that the very first frame is encoded on its own. The usual frame layout is therefore "$1+T$": the first frame becomes the first latent frame, and every following group of f_t frames becomes one more latent frame. A still image is then just a one-frame video, which lets one autoencoder, and one generator, train on images and videos together [15], [16], [3].

Let a clip have $F = 1 + T$ frames of size $H \times W$, compressed by a factor f_t in time and f_s along each spatial axis into C latent channels. The latent grid and the number of transformer tokens are

$$T' = 1 + \frac{T}{f_t}, \qquad H' = \frac{H}{f_s}, \qquad W' = \frac{W}{f_s}, \qquad L = T' \cdot \frac{H'}{p} \cdot \frac{W'}{p},$$

where T', H' and W' are the numbers of latent frames, rows and columns, p is the side of the square patch that the transformer groups into one token (often $p = 2$), and L is the sequence length. Wan's report gives exactly this count, $L = (1+T/4)\cdot H/16\cdot W/16$, for its 4×8×8 autoencoder and 2×2 patches [3]. Two numbers summarize a design: the **compression of numbers**, $\frac{3FHW}{CT'H'W'}$,

and the **pixels per token**, about $f_t f_s^2 p^2$. Because the latent has more channels than the 3 of RGB, a video VAE compresses tokens much more than it compresses numbers.

Figure 1 shows the arithmetic for a 17-frame clip.

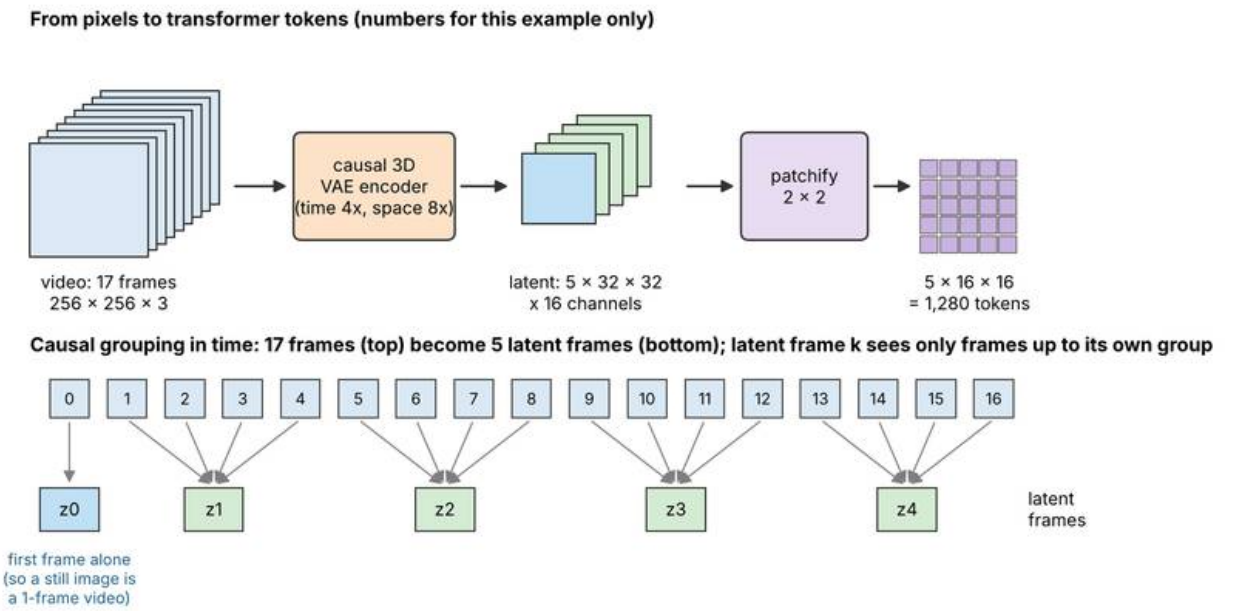


Figure 1 — A causal video VAE in numbers. The first frame is encoded alone; each later latent frame summarizes four frames and sees no future frames. A 2 × 2 patchify step then sets the transformer's sequence length.

Published systems document their factors. The table lists a few (a map of choices, not a ranking).

System	Compression (time × height × width)	Latent channels	Notes
Stable Video Diffusion [4]	1 × 8 × 8	4	image autoencoder, no temporal compression (factors as listed in [3])
CogVideoX [2]	4 × 8 × 8	16	3D causal VAE
HunyuanVideo [16]	4 × 8 × 8	16	causal 3D VAE, $(T+1)$-frame input
Wan [3]	4 × 8 × 8	16	causal; first frame compressed only in space
Movie Gen [17]	8 × 8 × 8	16	temporal autoencoder inflated from an image autoencoder
LTX-Video [18]	8 × 32 × 32	128	about 1:192 overall; no separate patchify step
MAGVIT-v2 in VideoPoet [19]	17 frames of 128 × 128 → 5 × 16 × 16 discrete tokens	vocabulary of 2^{18}	discrete tokens for a language model

LTX-Video frames the comparison in pixels per token: a $4 \times 8 \times 8$ autoencoder followed by 2×2 patches gives 1,024 pixels per token, while its own design gives 8,192 [18]. Pushing compression further has a price. CogVideoX's ablation found that a $8 \times 16 \times 16$ setting made the generator's training converge with great difficulty even with more latent channels [2], and LTX-Video asks its decoder to perform the final denoising step in pixel space to recover fine detail lost by the strong compression [18].

Optional: Causal 3D convolutions and the 1 + T frame layout (in the Supplementary notes). How causal padding works, why the first frame is independent, and how strided convolutions map $1+s\,t$ frames to $1+t$.

Continuous latents or discrete tokens

Like image models (DLCV 10), video generators either work on a **continuous** latent grid (a VAE, used by diffusion and flow models) or on **discrete tokens** from a learned codebook (used by masked and autoregressive token models). MAGVIT introduced a 3D tokenizer that quantizes a video into spatiotemporal tokens for masked generation [20]. MAGVIT-v2 replaced it with a causal 3D CNN tokenizer and a **lookup-free quantizer** whose vocabulary can grow large, and with it a language model outperformed diffusion models on image and video benchmarks in the authors' tests [15]; codebooks and their problems are covered in ML 08.

Code: latent size and attention cost

The snippet computes the latent grid, the number of tokens and the number of query–key pairs in one attention layer for an 81-frame, 480×832 clip, for three autoencoders. The pair counts anticipate the next section: "full" lets every token attend to every token, "factorized" splits attention into a spatial part within each latent frame and a temporal part at each position.

```
def latent_grid(frames, height, width, ft, fs, channels=16):
    """Latent shape of a causal video VAE: the first frame alone, then groups of ft
     ↪  frames."""
    t = 1 + (frames - 1) // ft
    return t, height // fs, width // fs, channels

def attention_pairs(t, s):
    """Query-key pairs in one attention layer over t latent frames of s tokens each."""
    full = (t * s) ** 2                 # every token attends to every token
    factorized = t * s**2 + s * t**2    # spatial attention per frame + temporal attention
     ↪  per position
    return full, factorized

frames, H, W, patch = 81, 480, 832, 2   # about 5 s at 16 fps; DiT patches of 2 x 2 latent
 ↪  positions
pixels = frames * H * W * 3
print(f"pixel values: {pixels:,}")
for name, ft, fs, c in [("image VAE, every frame", 1, 8, 4),
```

```
                      ("video VAE 4x8x8", 4, 8, 16),
                      ("video VAE 8x16x16", 8, 16, 64)]:
    t, h, w, c = latent_grid(frames, H, W, ft, fs, c)
    s = (h // patch) * (w // patch)     # tokens per latent frame after patchify
    full, fact = attention_pairs(t, s)
    print(f"{name:23s} latent {t}x{h}x{w}x{c} ({pixels / (t*h*w*c):5.1f}x fewer numbers)  "
          f"tokens {t*s:7,}  full {full:.1e}  factorized {fact:.1e}  ratio {full/fact:4.0f}")
```

It prints:

```
pixel values: 97,044,480
image VAE, every frame  latent 81x60x104x4 ( 48.0x fewer numbers)  tokens 126,360  full
 ↪  1.6e+10  factorized 2.1e+08  ratio   77
video VAE 4x8x8         latent 21x60x104x16 ( 46.3x fewer numbers)  tokens  32,760  full
 ↪  1.1e+09  factorized 5.2e+07  ratio   21
video VAE 8x16x16       latent 11x30x52x64 ( 88.4x fewer numbers)  tokens   4,290  full
 ↪  1.8e+07  factorized 1.7e+06  ratio   11
```

(All numbers are our own arithmetic; the $8 \times 16 \times 16$ row with 64 channels is a hypothetical configuration, not a specific model.) Three lessons. First, the $4 \times 8 \times 8$ video VAE keeps about as many numbers as the frame-wise image VAE, but it cuts the **tokens** by a factor of four, and the attention pairs by a factor of sixteen, because each latent position carries more channels. Second, the 32,760 tokens match Wan's formula for this clip size. Third, full attention over 32,760 tokens means about 10^9 query–key pairs per layer, which is why the choice between full and factorized attention, and the compression factor, decide what a model can afford.

Architectures: from inflated image models to video transformers

Plain version. Early video generators started from a good image generator and **inflated** it: they kept its layers, which treat each frame as a separate image, and inserted new temporal layers that let frames exchange information. Later systems were built as video transformers from the start, and the main design question became how attention should connect positions across space and time.

Inflating a pretrained image model

The idea is to reuse what an image model already knows about appearance and to learn only motion. Video Diffusion Models (VDM) turned the image diffusion U-Net into a **3D U-Net factorized over space and time**: every 3×3 convolution became a $1 \times 3 \times 3$ convolution that acts on each frame separately, each spatial attention block kept attending within a frame, and a new temporal attention block after it attended over frames at each spatial position [6]. Because temporal

attention can be switched off, the same network can be trained on single images and on videos at once, and the authors found this joint training important for sample quality [6].

Later systems keep the pretrained image layers and insert new temporal layers that are **initialized to do nothing**, so the inflated model starts exactly as the image model. Let h be the features of one block, reshaped so that frames are treated as a batch of images. The inflated block computes

$$h_{\text{out}} = h + Z(\,\text{TempAttn}(h)),$$

where TempAttn attends along the frame axis at each spatial position and Z is a linear projection whose weights start at zero, so that $h_{\text{out}} = h$ at the start of training (the same idea as ControlNet's zero convolutions in DLCV 10). Training then moves the model gradually from "a batch of independent images" toward "a video". Make-A-Video [1], Video LDM [7], AnimateDiff [21] and Open-Sora [22] all use a version of this idea: identity-initialized layers, a learned mixing weight or zero-initialized projections. Make-A-Video also shows the payoff: it learns appearance and its link to text from image–text pairs and only motion from unlabeled video, so it needs no paired text–video data [1].

Optional: Four ways to start a video model from an image model (in the Supplementary notes). What Make-A-Video, Video LDM, AnimateDiff and Open-Sora each freeze, add and initialize.

Factorized or full spatiotemporal attention

Plain version. A token can look at every other token in the clip (full attention), or first at the tokens of its own frame and then at the tokens in the same place in other frames (factorized attention). Factorized attention is much cheaper; full attention can follow an object that moves far between frames.

Let a clip have T latent frames of S tokens each, so $N = TS$ tokens in total. One full attention layer compares $N^2 = T^2S^2$ query–key pairs. Factorized attention runs spatial attention inside each frame (T problems of size S, so TS^2 pairs) and temporal attention at each position (S problems of size T, so ST^2 pairs). The ratio is

$$\frac{T^2S^2}{TS^2 + ST^2} = \frac{TS}{S+T},$$

which is close to T when $S \gg T$, as in the snippet above (ratio 21 for 21 latent frames). The factorized pattern comes from video understanding: TimeSformer compared self-attention schemes and found "divided" space–time attention best among those it tried for video classification [23], and ViViT proposed several variants that factorize the spatial and temporal dimensions [24]. Latte compared four such decompositions inside a latent video diffusion transformer (interleaved spatial and temporal blocks, late fusion, and two ways of splitting attention inside a block) and found interleaved spatial and temporal transformer blocks performed best in its experiments [25].

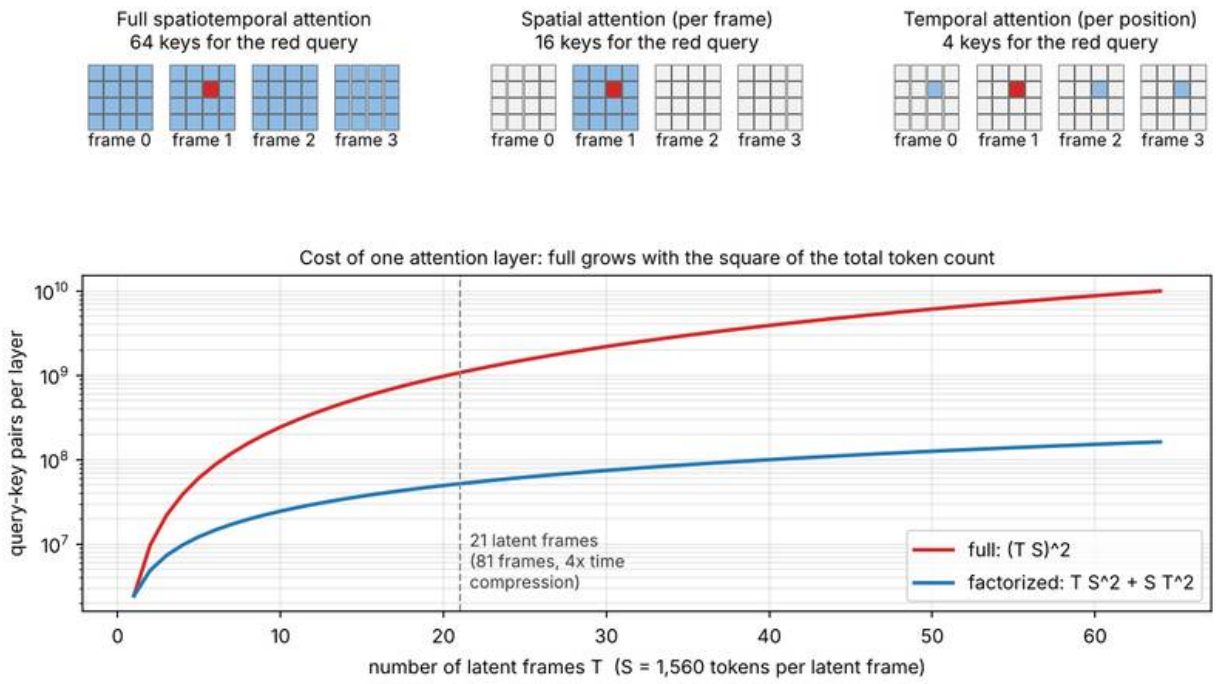


Figure 2 — Which keys a query sees under full, spatial and temporal attention, and what that costs. Factorized attention sends information between distant positions in different frames only indirectly, through several layers.

The case for full attention is about motion. CogVideoX argues that with separated spatial and temporal attention, an object that moves far between adjacent frames cannot attend directly to itself in the next frame; information must travel indirectly through other patches, which makes consistency of large motions hard to learn; it therefore uses full 3D attention over text and video tokens together [2]. HunyuanVideo chose full attention, citing better results than divided attention, a single design for images and videos, and the reuse of acceleration methods developed for language models [16]. Wan also uses full spatiotemporal attention [3]; Open-Sora Plan moved from a factorized "2+1D" denoiser to 3D full attention and added a cheaper sparse attention operation [26]; and LTX-Video uses its very high compression precisely so that full attention stays affordable [18]. The practical rule: full attention is preferred when the token count allows it, and the autoencoder is designed to make the token count allow it.

Cascades or one model for the whole clip

Cascades generate a short, small, low-frame-rate video and then improve it with separate models. Imagen Video uses seven sub-models (one base model, three spatial super-resolution and three temporal super-resolution models) to reach 128 frames of 1280 × 768 at 24 frames per second, about 5.3 seconds [27]. Make-A-Video combines a base model, a frame-interpolation network and two super-resolution models [1]; Video LDM generates sparse keyframes and fills in frames with an interpolation model [7]. A cascade keeps each stage small, but the base stage fixes the

global motion from few frames. Lumiere argued that generating distant keyframes and then interpolating makes global temporal consistency difficult, and introduced a **Space-Time U-Net** that downsamples and upsamples in time as well as space and generates the full-frame-rate clip in one pass [28]. Large latent transformers with temporal compression take the same single-stage route; Movie Gen notes that its 8× temporal compression lets it forgo the frame-interpolation models of earlier work [17].

A map of landmark systems

The table lists systems whose design is documented in a paper or technical report. It is a map of design choices, not a ranking; Figure 3 places them in time.

System	Generator and backbone	Where it generates	Note
VDM [6]	diffusion, factorized 3D U-Net	pixels	joint image–video training
Make-A-Video [1]	diffusion, inflated U-Net, cascade	pixels	no paired text–video data
Imagen Video [27]	diffusion, cascade of seven models	pixels	spatial and temporal super-resolution
Video LDM [7]	diffusion, frozen image LDM plus temporal layers	image-autoencoder latent	fine-tuned temporal decoder
Stable Video Diffusion [4]	diffusion, inflated latent U-Net	image-autoencoder latent	staged training with data curation
Lumiere [28]	diffusion, Space-Time U-Net	pixels	whole clip in one pass
Latte [25]	diffusion transformer, factorized	latent	compares four factorizations
CogVideoX [2]	diffusion transformer, full 3D attention	3D causal VAE latent	text and video tokens concatenated
Open-Sora [22]	diffusion transformer, factorized (STDiT)	3D autoencoder latent	fully open training code and weights
Open-Sora Plan [26]	diffusion transformer, 3D attention with a cheaper sparse variant (Skiparse)	wavelet-flow VAE latent	open project with data pipeline
HunyuanVideo [16]	flow-matching transformer, full attention	causal 3D VAE latent	over 13 billion parameters
Movie Gen [17]	flow-matching transformer	temporal-autoencoder latent	30B parameters, synchronized audio
Wan [3]	flow-matching diffusion transformer, full attention	causal 3D VAE latent	1.3B and 14B models
LTX-Video [18]	latent diffusion transformer, full attention	highly compressed latent	decoder does the last denoising step
MAGVIT [20], VideoPoet [19]	masked or autoregressive token transformer	discrete tokens	see the section on autoregressive generation

Flow matching and diffusion transformers are derived in ML 07 and ML 07, diffusion transformers [29], [30]. Parameter counts above are the ones stated in the papers.

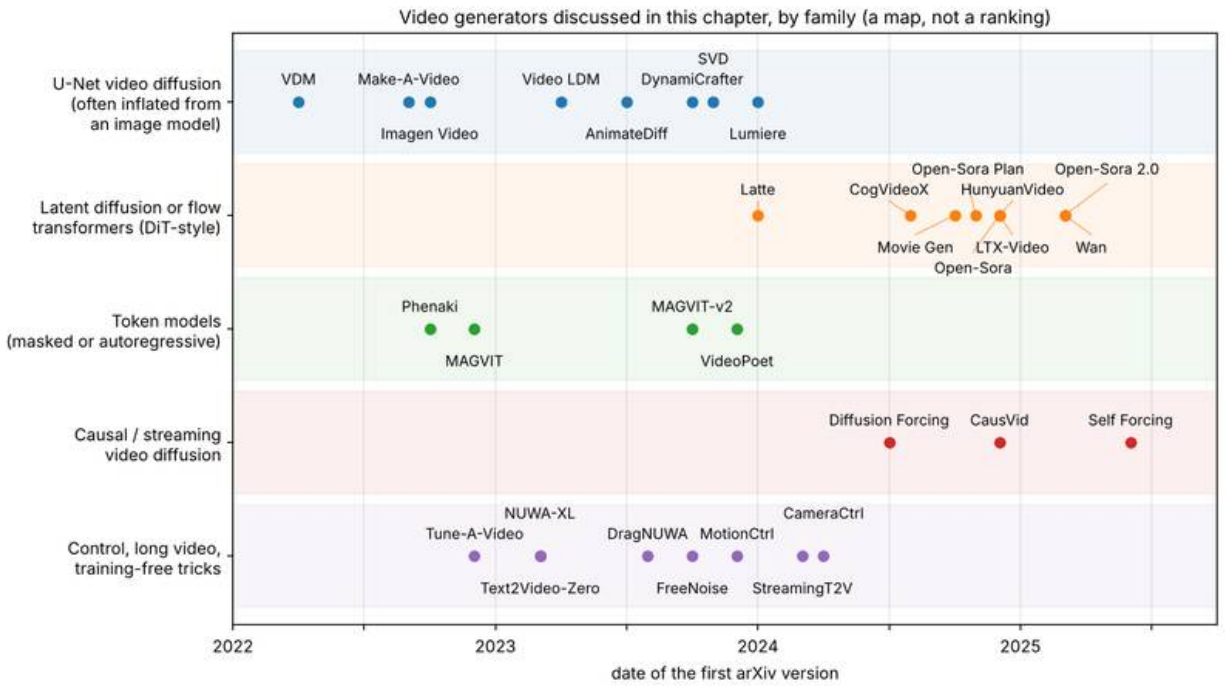


Figure 3 — The systems in this chapter by family and by the date of their first arXiv version. The center of gravity moved from inflated U-Nets (2022–2023) to latent transformers with full attention (2024–2025), with token models and causal diffusion as parallel lines.

Conditioning: text-to-video and image-to-video

Plain version. The prompt enters a video model the same way it enters an image model, through attention to text-encoder outputs. A first frame for image-to-video enters as an extra input that every generated frame can see. The difference from images is that the condition must hold for the whole clip, and that "how much it moves" is itself something to control.

Text-to-video

Text encoders, cross-attention and joint text–image attention are explained in DLCV 10. Video systems use the same mechanisms. Wan embeds text with **cross-attention** from video tokens to text-encoder outputs [3]. CogVideoX concatenates text and video tokens in one sequence and uses an **expert adaptive LayerNorm** with separate modulation for the two modalities, because their feature spaces and numerical scales differ [2]. HunyuanVideo processes video and text tokens in separate streams first and then concatenates them for joint processing ("dual-stream to single-stream"), with a multimodal language model as its main text encoder [16]. Captions must describe motion and events, not only appearance (see Data).

Make-A-Video also feeds the **frame rate** to the model, as data augmentation and as a control of motion speed [1]. Guidance works as for images (DLCV 10).

Image-to-video

Image-to-video ("animate this photo") must keep the input frame's content while inventing motion. Three documented designs:

- **Stable Video Diffusion** replaces the text embedding with the image embedding of the conditioning frame from a pretrained image–text model, and concatenates a noise-augmented copy of the frame to the denoiser input along the channel axis, repeated for every frame. The authors found that a constant guidance scale caused either drift away from the conditioning frame or oversaturation, and instead increased the guidance scale linearly across frames [4].
- **DynamiCrafter** projects the image into a text-aligned context space with a query transformer, so that the video model can attend to it like a prompt, and also concatenates the full image with the initial noise to preserve visual detail [5].
- **HunyuanVideo** uses the latent of the reference image directly as the first latent frame [16], a natural choice when the causal VAE encodes the first frame alone.

A common failure of image-to-video is **too little motion**: the model can score well on fidelity to the input by barely changing it. StreamingT2V reports that competing image-to-video models tend toward "video stagnation" when used repeatedly to extend a clip [8].

Camera and motion control

Plain version. A prompt such as "the camera slowly orbits the statue" is a weak way to describe motion. Control methods give the generator the motion directly: a camera path (where the camera is in each frame) or trajectories (where an object or a dragged point should go). The challenge is to obey the motion without changing the content.

- **Camera motion as a small adapter.** Stable Video Diffusion trained separate camera-motion LoRA modules for horizontal movement, zooming and a static camera on top of its base model [4]; AnimateDiff's **MotionLoRA** adapts its motion module to new motion patterns such as shot types at a low cost [21].
- **Camera poses as a dense input. CameraCtrl** trains a plug-and-play camera module on top of a video diffusion model, leaving the base model untouched. Instead of feeding raw camera matrices, it represents each frame's camera as a per-pixel **Plücker embedding**, a 6-number description of the viewing ray through each pixel, which ties camera geometry to image positions;

the authors found that training videos with diverse camera paths and an appearance similar to the base model's data gave the best balance of control and generalization [12].

- **Camera and objects together. MotionCtrl** controls camera motion and object motion independently, with camera poses and object trajectories as conditions that carry no appearance information [11]. **DragNUWA** combines text, an image and trajectories for fine-grained control of open-domain video [13].

A camera path is the same pose sequence that 3D reconstruction estimates (DLCV 08). Following it does not make the generated scene geometrically consistent, and a control signal is only as reliable as the poses or tracks it was trained on.

Optional: Camera poses as Plücker embeddings (in the Supplementary notes). What the six numbers per pixel mean, and how translation and rotation change them, with a tested snippet.

Long videos

Plain version. Models are trained on short clips, a few seconds long, because memory grows with length. To make a longer video you must either keep extending a clip piece by piece, which tends to drift, or plan the whole video coarsely first and fill in the details, which needs long training videos. Either way, keeping the story and the identities straight over a minute is still hard.

Four strategies are documented:

1. **Extend autoregressively, chunk by chunk.** Generate a clip, then generate the next clip conditioned on the last frames of the previous one. VDM showed that simply replacing the known frames during sampling gives continuations that look good alone but do not match the earlier frames, and proposed **reconstruction guidance**, which adds a guidance term from the model's reconstruction of the conditioning frames [6]. Video LDM trains prediction models that receive a few context frames through a mask [7]. **StreamingT2V** adds a short-term memory (attention to features of the previous chunk) for smooth transitions and a long-term memory (features of the first chunk) so the model does not forget the initial scene [8].
2. **Plan coarse, then fill in. NUWA-XL** applies a global diffusion model to generate keyframes over the entire time range and local diffusion models that recursively fill in frames between them, which lets it train directly on long videos and generate segments in parallel [31].
3. **Change the noise, not the model. FreeNoise** extends a pretrained short-clip model without tuning: it reschedules a sequence of initial noises with long-range correlation and applies temporal attention over windows, and adds a motion-injection method for prompts that change over time [32].
4. **Variable-length generation from a story. Phenaki** compresses video into discrete tokens with

a tokenizer that is causal in time, so it handles variable lengths, and generates long videos from a sequence of prompts that change over time [33].

The central difficulty is **error accumulation**: each chunk is conditioned on generated, slightly wrong frames, and errors compound, exactly as in the rollouts of a learned world model (DLCV 07). The causal-diffusion methods in the section on autoregressive generation address the same problem from the training side.

Optional: Reconstruction guidance for extending a video (in the Supplementary notes). Why replacing the known frames is not enough, and the correction term VDM adds.

Temporal consistency

Plain version. A consistent video keeps each detail attached to the thing it belongs to: when the dog moves, its spots move with it; when the camera pans, the texture of the wall slides along. The simplest test is to undo the motion and check that nothing else changed. The trap is that a video that does not move at all passes this test perfectly.

Where inconsistency comes from

A generator invents detail that the condition does not specify. If that invented detail is drawn afresh for every frame, it flickers. If it is fixed to the screen instead of to the objects, it "sticks" while the scene moves under it. Consistency means the invented detail must be **propagated along the motion**. Architectures do this through temporal layers and joint generation of all frames; several training-free methods enforce it in pretrained image models. Text2Video-Zero makes the self-attention of every frame attend to the **first frame** (cross-frame attention), to preserve the appearance and identity of the foreground object, and adds motion dynamics to the latent codes to keep the background consistent [10]. Make-A-Video uses the **same noise initialization** for every frame in its final per-frame super-resolution stage, to encourage consistent invented detail across frames [1]. Tune-A-Video adds a spatiotemporal attention mechanism to an image model and tunes it on a single video [9].

Measuring consistency

Three measures are common, and each is defined in terms of quantities you can compute from the generated video alone:

- **Frame difference** $D = \frac{1}{(F-1)\,P} \sum_{t=2}^{F} \|x_t - x_{t-1}\|_1$, where x_t is frame t and P the number of pixel values per frame. It is cheap but mixes real motion with flicker; VBench measures temporal

flickering this way only on static scenes, where all change is flicker [34].

- **Warping error** $E = \frac{1}{(F-1)\,P}\sum_{t=2}^{F}\|x_t - \mathcal{W}(x_{t-1}, u_t)\|_1$, where u_t is the motion (optical flow) from frame $t-1$ to frame t and $\mathcal{W}$ warps the previous frame along it. It removes the real motion before comparing; flow, warping and its masked form are explained in DLCV 04B. For a generated video the flow must be **estimated from the video itself**.
- **Feature similarity over time.** VBench's subject consistency compares the features of each frame with those of the first and the previous frame, $S = \frac{1}{F-1}\sum_{t=2}^{F}\frac{1}{2}(\langle d_1, d_t\rangle + \langle d_{t-1}, d_t\rangle)$, where d_t is a unit-length feature vector of frame t from a self-supervised image model and $\langle\cdot,\cdot\rangle$ is the dot product (cosine similarity) [34].

All three reward a still video. For that reason VBench pairs them with a **dynamic degree** dimension that measures the amount of motion with an optical-flow network [34].

Code: flicker, sticking and freezing

The toy builds a "coarse video" that pans an image 2 pixels per frame, as a prompt might request, and adds invented fine detail in four ways: new noise every frame, the same noise fixed on the screen, the same noise moved along with the scene, and a frozen video that repeats the first frame. It estimates each video's own motion with phase correlation and reports the three quantities.

```
import numpy as np
from scipy import ndimage as ndi
from skimage import data, transform
from skimage.registration import phase_cross_correlation

rng = np.random.default_rng(0)
T, v = 16, 2                                    # 16 frames; the prompt asks for a pan of 2
 ↪  px per frame
scene = transform.resize(data.camera() / 255.0, (128, 128), anti_aliasing=True)
content = np.stack([np.roll(scene, v * t, axis=1) for t in range(T)])   # the "coarse" video

def detail():                                   # invented fine texture, as a generator would
 ↪  add
    n = ndi.gaussian_filter(rng.standard_normal(scene.shape), 0.7)
    return 0.04 * n / n.std()

d0 = detail()
videos = {
    "independent noise": content + np.stack([detail() for _ in range(T)]),  # new detail
     ↪  every frame
    "shared noise":      content + d0,                                       # same detail,
     ↪  fixed on screen
    "propagated noise":  content + np.stack([np.roll(d0, v * t, axis=1) for t in range(T)]),
     ↪   # detail moves with the scene
}
```

```
videos["frozen"] = np.repeat(videos["propagated noise"][:1], T, axis=0)      # a still image
 ↪  repeated

c = (slice(None), slice(8, -8), slice(8, -8))  # ignore borders (np.roll wraps around)
def measure(vid):
    shifts = [phase_cross_correlation(a, b, upsample_factor=10)[0][1] for a, b in
     ↪  zip(vid[1:], vid[:-1])]
    flicker = np.abs(np.diff(vid, axis=0))[c].mean()                         #
     ↪  frame-difference energy
    warped = np.stack([np.roll(f, int(round(s)), axis=1) for f, s in zip(vid[:-1], shifts)])
    warp_err = np.abs(vid[1:] - warped)[c].mean()                            # error after
     ↪  the video's own motion
    return flicker, warp_err, np.mean(np.abs(shifts))

for name, vid in videos.items():
    f, w, m = measure(vid)
    print(f"{name:18s} frame difference {f:.4f}   warping error {w:.4f}   estimated motion
     ↪  {m:.2f} px/frame")
```

It prints:

```
independent noise  frame difference 0.0808   warping error 0.0446   estimated motion 2.00
 ↪  px/frame
shared noise       frame difference 0.0528   warping error 0.0419   estimated motion 2.01
 ↪  px/frame
propagated noise   frame difference 0.0796   warping error 0.0000   estimated motion 2.00
 ↪  px/frame
frozen             frame difference 0.0000   warping error 0.0000   estimated motion 0.00
 ↪  px/frame
```

(All numbers are from our toy run.) Read the four lines together:

- **Frame difference alone is misleading.** It rates the screen-fixed noise (0.0528) as *more* stable than the correct, propagated detail (0.0796), because the propagated detail moves, and it gives the frozen video a perfect zero.
- **Warping error separates the cases that move.** After undoing the estimated motion, only the propagated detail leaves nothing behind; both new-every-frame noise and screen-fixed noise remain visible.
- **A motion measure is needed as well.** The frozen video also has zero warping error. Only the estimated motion (0.00 instead of the requested 2 pixels per frame) exposes it, which is the role of VBench's dynamic degree.

Figure 4 shows the same effects as motion-compensated x–t slices: correct, propagated detail forms straight vertical lines.

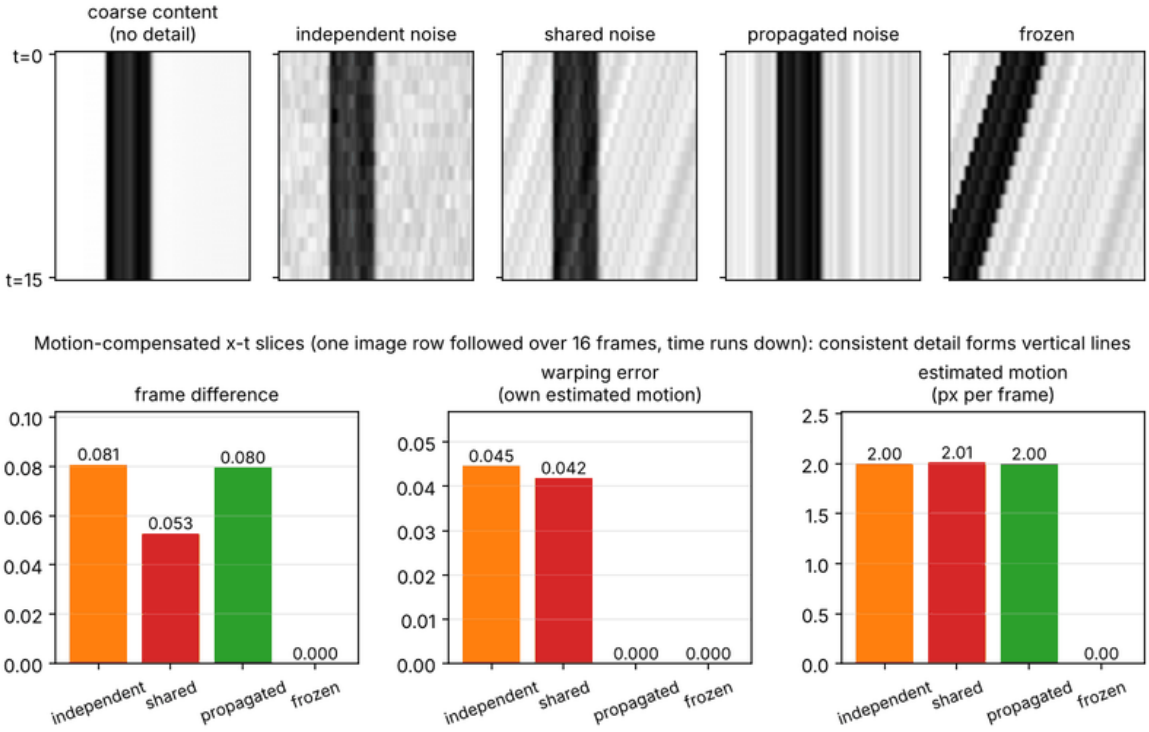


Figure 4 — The consistency toy. Undoing the requested motion turns correct detail into vertical lines; flicker shows as speckle, screen-fixed detail as slanted streaks, and a frozen video as a slanted band. No single measure ranks all four correctly.

Autoregressive and diffusion video generation

Plain version. There are two ways to produce a clip. A **diffusion** (or flow) model refines all frames of a fixed-length clip together, from noise to video, so every frame can see the future as well as the past. An **autoregressive** model writes the video piece by piece, each piece conditioned on what came before, like a language model writing text; it can go on as long as you like and can stream, but early mistakes stay. Recent hybrids try to get both.

Token models: masked and autoregressive

With a discrete video tokenizer, video generation becomes sequence modeling (ML 08). **MAGVIT** uses masked token modeling, predicting many tokens in parallel, and supports many generation tasks (such as frame prediction, interpolation and inpainting) with a single model [20]. **Phenaki** generates its causal video tokens with a bidirectional masked transformer conditioned on text

[33]. **VideoPoet** is a decoder-only transformer that processes text, image, video and audio tokens; MAGVIT-v2 turns 17 frames of 128×128 video at 8 frames per second into a $5 \times 16 \times 16$ grid, that is 1,280 tokens from a vocabulary of 2^{18}, and the model is trained like a language model, with pretraining on a mixture of generative tasks followed by task-specific adaptation [19]. Token models reuse language-model infrastructure (KV caching, ML 08) and can extend a clip naturally, but every generated token is a commitment, and errors accumulate (ML 08).

Diffusion: the whole clip at once

A standard video diffusion transformer denoises all latent frames jointly with **bidirectional** attention. CausVid states the consequence plainly: generating a single frame requires processing the entire sequence, including the future, which makes such models hard to use interactively [35]. Their strengths are the ones that made diffusion dominant for images (stable training, high quality, guidance), plus global consistency within the clip. Their limits are a fixed maximum length and high latency.

Hybrids: causal, streamable diffusion

- **Diffusion Forcing** trains a diffusion model to denoise a sequence of tokens with **independent noise levels per token**, and uses it as a causal next-token model that generates future frames without fully diffusing past ones; this allows rolling out video beyond the training horizon, where its baselines diverge [36].
- **CausVid** adapts a pretrained bidirectional video diffusion transformer into a causal, autoregressive one that generates frames on the fly, distills its 50-step sampler into a 4-step one with a bidirectional teacher supervising the causal student, and uses KV caching for streaming generation; the authors report that this mitigates error accumulation and allows long videos despite training on short clips [35].
- **Self Forcing** attacks **exposure bias** directly: during training, each frame is generated conditioned on the model's own previous outputs (an autoregressive rollout with KV caching), not on ground-truth frames, and a loss is applied to the whole generated video; it reports real-time streaming generation with sub-second latency on a single GPU [37].

Aspect	Autoregressive or causal	Bidirectional diffusion
Length	open-ended, chunk by chunk	fixed by the training clip length, unless extended
Latency	first frames available early, can stream	whole clip must be denoised first
Future information	none (causal)	every frame sees past and future
Main risk	error accumulation, exposure bias	memory and compute for long clips
Typical use	interactive generation, world models, live video	offline text-to-video and image-to-video

The same causal-versus-bidirectional distinction matters for world models, where actions arrive one step at a time (DLCV 07). Remember also, from ML 07, that a causal **autoencoder** does not make a generator autoregressive: the generator's attention and sampling decide that.

Evaluating video generators

Plain version. You need evidence of four things: that the videos as a set look like real videos (distribution metrics such as FVD), that they match their prompts, that their motion and consistency are good (dimension-by-dimension suites such as VBench), and that people prefer them (human studies). The standard distribution metric turns out to look mostly at single frames, so it can miss exactly the temporal problems this chapter is about.

Fréchet Video Distance

FVD extends FID (ML 07) to video [38]. Each video, real or generated, is passed through a pretrained video network, an I3D action-recognition model [39] trained on Kinetics-400 [40], and its output (the logits) is taken as a feature vector. A Gaussian is fitted to each set of features, and

$$\mathrm{FVD} = \|\mu_r - \mu_g\|_2^2 + \mathrm{Tr}\big(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\big),$$

where μ_r, Σ_r are the mean and covariance of the real-video features, μ_g, Σ_g those of the generated videos, Tr is the trace and $(\cdot)^{1/2}$ the matrix square root. Lower is better. The original paper introduced FVD with a human study in which it correlated well with judgments of generated videos [38]. Class-conditional and unconditional benchmarks often report FVD on UCF-101, a dataset of 101 human-action classes [41], [1], [25].

Its main limit: content bias. Ge et al. found that FVD increases only slightly under large temporal corruption at matched frame quality, and that a well-chosen subset of generated videos **without motion** can lower FVD drastically [42]. They trace the bias to the features: an action classifier trained on Kinetics learns object and scene appearance, because many actions can be recognized from one frame; features from a self-supervised video model, VideoMAE v2 [43], were much less biased [42]. The pitfalls of FID also apply (DLCV 10), plus video-specific ones: clip length, frame rate and resizing to a low, square resolution [42].

Optional: Ge et al., "On the Content Bias in Fréchet Video Distance" (in the Paper guides). How the paper separates frame quality from motion quality, and how to read its sensitivity plots.

Code: a Fréchet distance that cannot see time

The toy makes "real" clips of a blob moving in a straight line and two corrupted sets: one with **blurred frames** (worse frames, correct motion) and one with **shuffled frames** (perfect frames, broken motion). It computes a Fréchet distance with two hand-made feature extractors: one that averages per-frame appearance statistics over time, like an extractor that mostly sees content, and one that adds two temporal statistics.

```
import numpy as np
from scipy import linalg, ndimage as ndi

rng = np.random.default_rng(0)
T, S = 16, 32
yy, xx = np.mgrid[:S, :S]

def real_videos(n):
    """Toy 'real' clips: a bright blob moving in a straight line, plus mild sensor noise."""
    vids = np.empty((n, T, S, S))
    for i in range(n):
        p, vel, r = rng.uniform(10, 22, 2), rng.uniform(-0.7, 0.7, 2), rng.uniform(2.5, 4)
        for t in range(T):
            cy, cx = p + vel * t
            vids[i, t] = np.exp(-((yy - cy) ** 2 + (xx - cx) ** 2) / (2 * r**2))
    return vids + 0.02 * rng.standard_normal(vids.shape)

def frame_stats(f):                       # appearance of one frame
    gy, gx = np.gradient(f)
    return [f.mean(), f.std(), f.max(), np.abs(gx).mean() + np.abs(gy).mean()]

def appearance_features(v):               # frame statistics averaged over time: blind to
 ↪  frame order
    return np.mean([frame_stats(f) for f in v], axis=0)

def motion_features(v):                   # add how the content changes from frame to frame
    d = np.abs(np.diff(v, axis=0)).mean(axis=(1, 2))
    return np.concatenate([appearance_features(v), [d.mean(), d.std()]])

def frechet(a, b):                        # Fréchet distance between Gaussian fits of two
 ↪  feature sets
    mu1, mu2 = a.mean(0), b.mean(0)
    s1, s2 = np.cov(a, rowvar=False), np.cov(b, rowvar=False)
    covmean = linalg.sqrtm(s1 @ s2).real
    return float(((mu1 - mu2) ** 2).sum() + np.trace(s1 + s2 - 2 * covmean))

real, other_real = real_videos(400), real_videos(400)
blurred = ndi.gaussian_filter(other_real, sigma=(0, 0, 0.7, 0.7))          # worse frames,
 ↪  same motion
shuffled = np.stack([v[rng.permutation(T)] for v in other_real])          # perfect frames,
 ↪  broken motion

for name, feats in [("appearance features", appearance_features), ("+ motion features",
 ↪  motion_features)]:
```

```
    f = lambda vids: np.stack([feats(v) for v in vids])
    ref = f(real)
    scale = ref.std(0)                    # put all features on a comparable scale
    fd = {k: frechet(ref / scale, f(v) / scale) for k, v in
          [("second real set", other_real), ("blurred frames", blurred), ("shuffled frames",
           ↪  shuffled)]}
    print(f"{name:20s} " + "   ".join(f"{k} {d:6.2f}" for k, d in fd.items()))
```

It prints:

```
appearance features  second real set   0.01   blurred frames 115.10   shuffled frames   0.01
+ motion features    second real set   0.07   blurred frames 141.87   shuffled frames
 ↪  12397.17
```

(All numbers are from our toy run, with toy features, not I3D.) With appearance-only features, the shuffled set, whose motion is destroyed, is exactly as close to the real set as a second real set (0.01), while mild blur is heavily penalized. Adding two temporal statistics makes the shuffled set by far the worst. The real FVD is not this extreme, since I3D does see some motion, but this is the shape of the content bias that Ge et al. measured [42]: **the metric is only as temporal as its features**. Figure 5 shows the clips and the distances.

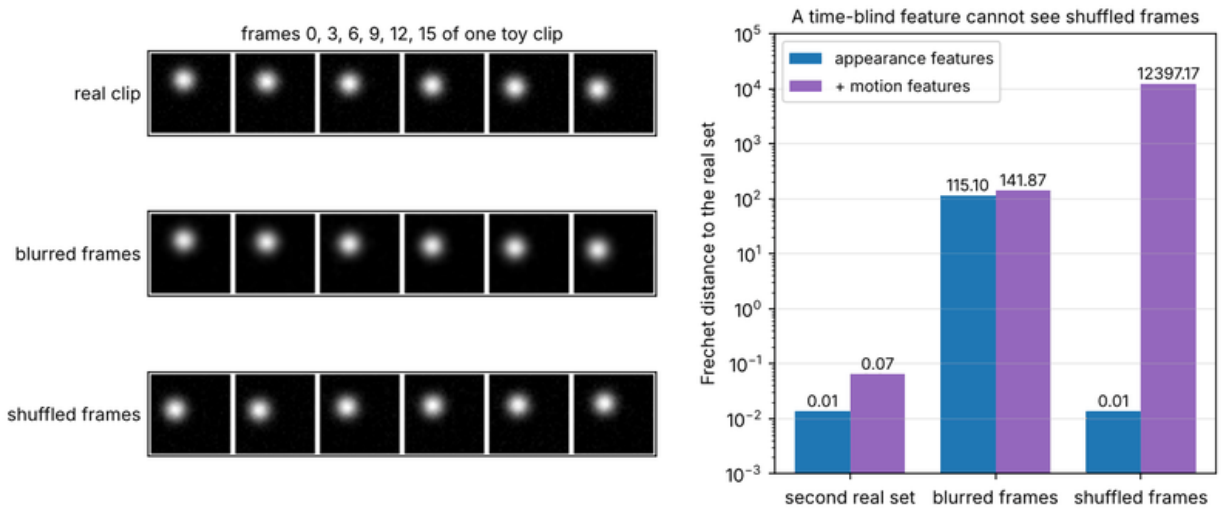


Figure 5 — A Fréchet distance computed on time-blind features cannot tell a shuffled clip from a real one. Toy data and toy features.

Benchmark suites and human evaluation

Plain version. Instead of one number, modern benchmarks ask many narrow questions, each with its own prompts and its own automatic check, and confirm that the checks agree with people.

- **VBench** splits "video generation quality" into 16 dimensions, each with tailored prompts and an evaluation method: temporal quality (subject consistency, background consistency, temporal flickering, motion smoothness, dynamic degree), frame-wise quality (aesthetic and imaging quality), semantics (object class, multiple objects, human action, color, spatial relationship, scene), style (appearance and temporal style) and overall consistency; it provides human preference annotations for each dimension to validate the automatic scores [34]. **VBench++** extends it to image-to-video, with an image suite of adaptive aspect ratio, and to trustworthiness [44].
- **EvalCrafter** builds 700 prompts from an analysis of real user data, evaluates visual quality, content quality, motion quality and text–video alignment with 17 objective metrics, and fits coefficients so that the combined score agrees with users' opinions [45].
- **Human evaluation** remains the reference. Stable Video Diffusion compares its training choices with human preference studies [4]; HunyuanVideo reports evaluations by professionals [16]; Movie Gen evaluates with pairwise A/B tests by expert annotators along text alignment, visual quality, and realness and aesthetics, over a public set of 1,003 prompts, and reports that FVD and Inception Score did not correlate with its human evaluation of video quality [17]. OpenSora 2.0 reports both human evaluation and VBench scores [46].

Physical plausibility needs its own tests; DLCV 07 describes benchmarks that found physical understanding unrelated to visual realism.

A reporting checklist. State the prompts and how they were chosen, the clip length, frame rate and resolution, the number of samples and seeds, the sampler, steps and guidance (including any schedule across frames), the feature extractor and preprocessing of every distribution metric, the dimension suite and its version, and the human-study protocol (pairwise or absolute, number of annotators, majority rule). Report a temporal measure **together with** a motion measure and a per-frame quality measure, and show uncurated clips.

Data

Plain version. Video models learn appearance, motion and the link to language from their training clips. Video with good captions is much scarcer than captioned images, so systems combine image and video data and spend great effort on filtering and recaptioning clips.

- **Images plus videos.** VDM [6], Phenaki [33] and many later systems train jointly on images and videos; Make-A-Video learns text–visual links only from image–text pairs [1]; the causal "1 + T" autoencoders make images one-frame videos [16], [3].
- **Web video collections.** WebVid, web videos with weak captions, was introduced with the Frozen in Time retrieval model (WebVid-2M in the paper; a 10M version was released later, according to its arXiv record) [47], and Make-A-Video was trained on WebVid-10M [1]. **Panda-70M** curates 3.8M high-resolution videos, splits them into semantically consistent clips and captions them with several cross-modality teacher models plus a retrieval model that selects the best caption, giving 70M clips with captions [48].
- **Curation matters.** Stable Video Diffusion's Large Video Dataset starts with 580M annotated clips, forming 212 years of content, then removes nearly static clips with optical-flow scores, clips with much written text by OCR, and clips with low aesthetic scores; models trained on curated subsets were preferred by people over models trained on uncurated data [4]. CogVideoX describes its own video-captioning model as a key part of its data pipeline [2].
- **Evaluation datasets.** UCF-101 [41] and Kinetics [40] are human-action datasets that supply FVD benchmarks and FVD's feature network.

Optional: Stable Video Diffusion (in the Paper guides). How one paper turns data curation and staged training into controlled experiments, and how to read its human-preference plots.

Web video raises the same questions of copyright, consent and content safety as web images; see DLCV 10, Data.

Risks and responsible use

Plain version. Realistic generated video can put words in someone's mouth or show events that never happened. The defenses are the same three as for images (detect fakes, label generated content, and limit misuse at the source), but video is harder: it is re-encoded on every upload, and a few convincing seconds are enough to mislead.

- **Deepfakes and detection.** FaceForensics++ is a benchmark for detecting facial manipulations (four manipulation methods at random compression levels and sizes, over 1.8 million manipulated images, with a hidden test set); its authors found that domain-specific knowledge improves detection even under strong compression and that the detectors outperform human observers on their benchmark [49]. Detectors trained on one generation method can fail on newer ones, so detection is an arms race.
- **Watermarks for video. Video Seal** jointly trains a watermark embedder and extractor with transformations such as video codecs applied in between, and propagates the watermark over time so that not every high-resolution frame has to be watermarked [50]. Signed provenance manifests, invisible watermarks and their failure modes are explained in DLCV 10, Deepfakes,

provenance and watermarks and its note Hard bindings, soft bindings and watermarks; the same lessons apply to video, where re-encoding and cropping are routine.

- **Bias, memorization and consent** follow from the data as for images (DLCV 10, Risks); people appear in web video far more often than in many image collections, which raises the stakes of consent.
- **False authority.** A generated video that looks physically right is easily taken as evidence. Treat generated clips as illustrations, not records, and see DLCV 07 on why realism is not physical correctness.

Failure modes

Plain version. Today's video generators are good at short, single-shot clips of common scenes. They still fail at keeping things the same over time, at physics, at fine-grained motion and at anything long.

- **Flicker and texture sticking.** Fine detail changes from frame to frame or stays fixed while the scene moves (Figure 4).
- **Identity drift and morphing.** Faces, text and patterns change slowly; two objects merge or one object splits.
- **Object permanence.** Objects vanish when occluded or come back different.
- **Too little motion.** Near-static clips that score well on consistency; dynamic-degree measures exist because of this [34], and image-to-video extension can stagnate [8].
- **Implausible physics and dynamics.** Motion looks fluid but is physically wrong (see DLCV 07).
- **Drift in long videos.** Error accumulation when extending clip by clip [35], [37]; forgetting of the initial scene [8].
- **Weak control.** Camera paths that are only roughly followed, trajectories that change object shape; MotionCtrl's design goal of appearance-free motion conditions targets exactly this [11].
- **Inherited image failures.** Counting, attribute binding, text rendering and hands fail as in images (DLCV 10, Failure modes), now with the extra chance of changing over time.

How to test. Build small prompt sets per failure type (fast motion, occlusion, a person turning around, written text, a requested camera move), generate several seeds per prompt, measure consistency **and** motion, inspect motion-compensated x–t slices for flicker and sticking, and watch the clips; a frame grid hides most temporal failures.

Modern view

Surveys worth reading.

- **Melnik et al. [51]** (TMLR 2024) survey video diffusion models: the mathematical formulation, architectural choices, methods for temporal consistency, a taxonomy of applications by input modality (text, images, video, audio), training and evaluation practice with datasets and metrics, and open challenges such as long videos and computational cost.
- **Xing et al. [52]** (2023) review video diffusion models in three areas, video generation, video editing and other video understanding tasks, with further categories in each, and discuss challenges and future trends.
- **Liu et al. [53]** (2024) review OpenAI's Sora, released in February 2024, from public technical reports and reverse engineering: background, related technologies, applications and limitations such as safe and unbiased generation. Because it relies on reverse engineering, read its technical claims as informed hypotheses.

State of the art, 2023–2026. Five trends stand out. (1) **Latent transformers with full attention and causal 3D VAEs** became the dominant recipe for large open models (CogVideoX [2], HunyuanVideo [16], Wan [3]), with flow-matching objectives in several of them [16], [17], [3]. (2) **Open models report closing much of the gap** with closed systems in their own human evaluations and VBench scores, and training cost became a research topic: Open-Sora 2.0 reports training a commercial-level model for about $200k [46]. (3) **Higher compression** reduces tokens so that full attention stays affordable, with the decoder taking over more of the detail [18]. (4) **Causal, few-step, streaming generation** brought interactive and real-time use [35], [37], [36]. (5) **Evaluation moved beyond FVD** toward dimension suites [34], [44], [45] and large human studies [17], with a recognized need for temporally sensitive features [42]. Synchronized audio generation also appeared [17], [19].

Open problems. Consistency and identity over minutes rather than seconds; physically correct dynamics and interaction (DLCV 07); precise, disentangled control of camera, objects and timing; metrics that measure motion and consistency without rewarding stillness and that agree with people; the cost of training and sampling long, high-resolution clips; caption quality for events and actions; and provenance that survives re-encoding.

Key takeaways

- Video generation adds three demands to image generation: per-frame quality, believable motion and temporal consistency; easy fixes for one (a frozen video) fail another.
- Generators work in a compressed latent. Causal 3D VAEs compress time (often 4×) and space (often 8×) with a "$1 + T$" layout so that images are one-frame videos; compression factors and patch size set the token count, and the decoder caps detail.
- Inflating an image model (temporal layers initialized to do nothing at first) reuses appearance knowledge; factorized attention is cheap, full spatiotemporal attention follows large motion, and stronger compression makes full attention affordable.
- Text enters through cross-attention or joint attention as for images; image-to-video adds the first frame as a condition; camera and motion control add pose or trajectory inputs such as Plücker embeddings.
- Long videos come from chunk-wise extension, coarse-to-fine hierarchies or noise rescheduling; all fight error accumulation.
- Bidirectional diffusion generates a fixed clip jointly; autoregressive and causal-diffusion models stream and extend but must handle exposure bias.
- FVD is biased toward per-frame content; evaluate with a temporal measure, a motion measure, dimension suites such as VBench and human studies together.
- Data curation (motion, text and aesthetic filters, recaptioning) is decisive; deepfakes and provenance are real risks, and watermarks for video must survive re-encoding.

Exercises

1. **Token budgets.** A model uses a causal $4\times8\times8$ VAE and 2×2 patches. How many tokens does a 121-frame clip at 720×1280 produce? How many query–key pairs does one full attention layer have, and by what factor does factorized attention reduce it? Which single change, doubling the temporal compression or doubling the patch size, reduces the full-attention cost more?

   Hint

   $T' = 1 + 120/4 = 31$, and each latent frame has $(720/16)\cdot(1280/16) = 45\cdot80 = 3600$ tokens, so $L = 111{,}600$ tokens and about 1.25×10^{10} pairs. The factorized ratio is $TS/(T+S) = 31\cdot3600/3631 \approx 30.7$. Doubling f_t gives $T' = 16$ (pairs shrink about 3.8×); doubling p divides the tokens per frame by 4 (pairs shrink 16×), at the cost of coarser spatial tokens. Use the snippet's functions to check.

2. **Break the consistency measures.** In the consistency toy, add a fifth video in which the propagated detail is shifted by 2 pixels per frame but the coarse content does **not** move (a "moving texture on a still scene"). What do the three measures report, and which of them would you trust to detect this failure?

   Hint

Phase correlation now sees two competing motions: the texture moves, the content does not. Depending on their relative strength, the estimated shift is 0 or 2, and whichever layer is not compensated leaves a warping error. A single global motion estimate is not enough for real videos with several moving parts; dense optical flow with occlusion masks is used in practice (DLCV 04B).

3. **Content bias in your own features.** Modify the Fréchet toy so that the "generated" set has correct frames but the blob moves twice as fast as in the real set. Which feature extractor detects it? Then design a third feature that would also detect a blob that moves at the right speed but reverses direction halfway.

Hint

Speed changes the mean frame difference, so only the motion features detect it. A reversal keeps the mean absolute difference similar; a feature that compares the direction of motion over time (for example the correlation between the first and second halves of the centroid trajectory, or the net displacement divided by the path length) is needed. Real metrics face the same question: which temporal properties do the features encode?

4. **Choose an architecture.** You must build an image-to-video model for product photos (a slow 3-second camera orbit around an object) with a small video dataset and a strong pretrained image model. Choose between inflating the image model with factorized temporal layers and training a full-attention video transformer from scratch, and say how you would add camera control.

Hint

With little video and a strong image model, inflation with zero- or identity-initialized temporal layers reuses appearance knowledge and needs less data; camera motion is slow, so factorized attention's weakness with large motion matters less. Condition on the photo (concatenated latent plus an image embedding, as in Stable Video Diffusion) and add a camera module with per-pixel Plücker embeddings, trained on clips with diverse, known camera paths. Evaluate identity preservation of the product across frames, camera-path accuracy and a dynamic-degree measure.

5. **Evaluation plan.** A team claims its new model has "better temporal consistency" because its VBench subject-consistency score is higher. List three alternative explanations and the additional evidence you would ask for.

Hint

The model might move less (check dynamic degree and motion smoothness), might generate simpler or shorter clips (check prompts, clip length and frame rate), or might differ in frame quality that affects the feature extractor (check frame-wise quality). Ask for the full dimension table with the same VBench version, uncurated clips per prompt, a human pairwise study on consistency with motion held comparable, and FVD with a temporally sensitive feature extractor.

Supplementary notes

These notes are optional background that the main text does not depend on.

Causal 3D convolutions and the 1 + T frame layout

This note explains how a causal temporal convolution is padded, why the first frame ends up encoded on its own, and how strided layers produce the "$1 + T/f_t$" latent length.

Ordinary versus causal padding. Consider one temporal convolution with kernel size k_t along the frame axis (the spatial part is unchanged). An ordinary "same" convolution pads $\lfloor (k_t - 1)/2 \rfloor$ frames before the clip and $\lfloor k_t/2 \rfloor$ after it, so output frame t depends on input frames $t - \lfloor (k_t - 1)/2 \rfloor$ to $t + \lfloor k_t/2 \rfloor$, including future frames. A **causal** convolution pads $k_t - 1$ frames before the clip and none after, so output t depends only on frames $t - k_t + 1$ to t [15], [2].

Why the first frame is independent. With causal padding, output frame 0 sees only padding and input frame 0. Stacking causal layers keeps this property, so the latent of the first frame is a function of the first frame alone. That is what lets the same encoder tokenize a single image [15].

Strided downsampling. A causal convolution with stride s maps $1 + s\,t$ input frames to $1 + t$ outputs; the decoder upsamples by s and drops the first $s - 1$ resulting frames to map $1 + t$ back to $1 + s\,t$ [15]. Two such stages with $s = 2$ give $f_t = 4$: 17 frames become 5 latent frames, 81 become 21.

Chunked encoding of long videos. Because a causal layer needs only the last $k_t - 1$ frames of context, a long video can be encoded or decoded chunk by chunk while caching those frames. Wan's VAE processes one chunk per latent frame (at most 4 frames) and keeps, for a kernel of size 3, the features of the last two frames of the previous chunk, which supports encoding and decoding of arbitrarily long videos with bounded memory [3]. CogVideoX uses the same property to split long videos across devices, each device sending only $k_t - 1$ frames to the next [2].

A tested check. The snippet implements one causal temporal convolution with kernel 4 and stride 4 on 17 "frames" (scalars, for clarity) and confirms the output length and that no output depends on a later frame.

```
import numpy as np

def causal_conv1d(x, w, stride):
    """Causal temporal convolution: pad len(w) - 1 zeros in front, none behind."""
    k = len(w)
    xp = np.concatenate([np.zeros(k - 1), x])
    return np.array([xp[i * stride:i * stride + k] @ w for i in range((len(x) - 1) // stride
     ↪  + 1)])

rng = np.random.default_rng(0)
```

```
x, w = rng.normal(size=17), rng.normal(size=4)
y = causal_conv1d(x, w, stride=4)
print("17 frames ->", len(y), "latent frames")
for j in [0, 1, 4, 5, 16]:                    # perturb one frame, see which outputs change
    x2 = x.copy(); x2[j] += 1.0
    changed = np.flatnonzero(~np.isclose(causal_conv1d(x2, w, 4), y))
    print(f"frame {j:2d} changes latent frames {changed.tolist()}")
```

It prints:

```
17 frames -> 5 latent frames
frame  0 changes latent frames [0]
frame  1 changes latent frames [1]
frame  4 changes latent frames [1]
frame  5 changes latent frames [2]
frame 16 changes latent frames [4]
```

Frame 0 influences only latent frame 0, and each later group of four frames influences only its own latent frame, as in Figure 1. Real encoders stack several layers with overlapping kernels, so a latent frame also sees some earlier groups, but never later ones.

Pitfalls. Frame counts that do not fit $1 + st$ must be padded or trimmed; Open-Sora reports that training its autoencoder only on 17-frame clips caused blurriness at other lengths, which mixed-length training fixed [22]. Normalization layers that pool statistics over time break causality; Wan replaced group normalization with RMS normalization for this reason [3].

Four ways to start a video model from an image model

This note compares, system by system, how four video generators reuse a pretrained image model: what they freeze, what they add and how the new parts are initialized.

All four follow the main text's rule: the new temporal parts must leave the image model's behavior unchanged at the start of training. They differ in what they keep frozen and in how the "do nothing" start is implemented.

- **Make-A-Video** [1] stacks a 1D temporal convolution after each pretrained 2D convolution ("pseudo-3D convolution") and a temporal attention layer after each pretrained spatial attention layer. The new 1D layers are initialized as the **identity** function, so at initialization the network generates separate images that each match the text but are not coherent in time. Spatial and temporal layers are then trained together on video.
- **Video LDM** [7] **freezes** the spatial layers of a pretrained latent image model and trains only the inserted temporal layers (temporal attention and 3D-convolutional residual blocks). After each temporal layer, its output z' is mixed with the spatial output z as $\alpha z + (1 - \alpha)z'$, where $\alpha \in [0,$

1] is learned; with $\alpha = 1$ the temporal layers are skipped and the image model is recovered exactly. Because the spatial layers are untouched, the trained temporal layers also transfer to fine-tuned versions of the same image model.

- **AnimateDiff** [21] trains in stages. First a **domain adapter** (LoRA layers in the attention of the image model, trained on single frames of the video data) absorbs the visual quality gap between the image training data and the video data, such as motion blur, compression artifacts and watermarks, so that this gap is not learned as part of the motion; it is dropped at inference.

  Then the model is inflated and a **motion module** (temporal transformer blocks with sinusoidal position encoding along time) is trained with the image weights frozen; its output projection is **zero-initialized** and added through a residual connection. Optionally, **MotionLoRA** adapts the motion module to a new motion pattern. Because the image layers never change, the motion module can be plugged into personalized models derived from the same base model.
- **Open-Sora** [22] starts from a pretrained image diffusion transformer and adds a temporal attention layer to each block; the projections of the new layers are initialized to zero to preserve the image generation ability at the start; the added temporal attention doubles the parameter count.

System	Image layers during video training	New temporal parts	"Do nothing" start
Make-A-Video	trained	1D temporal convolution and temporal attention per layer	identity initialization
Video LDM	frozen	temporal attention and 3D-convolution blocks	learned mix α, skip with $\alpha = 1$
AnimateDiff	frozen	motion module (temporal transformer)	zero-initialized output projection
Open-Sora	trained	temporal attention per block	zero-initialized projections

Pitfalls. Freezing the image layers keeps the model compatible with the image ecosystem (personalized checkpoints, LoRAs) but limits how much appearance can adapt to video, such as motion blur or video-specific framing. Training everything adapts better but loses that compatibility. In both cases a frame-wise image autoencoder may still cause flicker at decoding time, which Video LDM addressed by fine-tuning the decoder on video (main text).

Reconstruction guidance for extending a video

This note derives the correction that VDM adds when a clip is extended by sampling new frames conditioned on known ones.

Setting. Split a clip into known frames x^a (for example the last frames of the previous chunk) and frames to generate x^b. A diffusion model trained on whole clips $x = [x^a, x^b]$ gives a denoising

estimate $\hat{x}_\theta(z_t) = [\hat{x}^a_\theta(z_t), \hat{x}^b_\theta(z_t)]$ of the clean clip from the noisy clip z_t at noise level t, where $z_t = \alpha_t x + \sigma_t \epsilon$ with schedule coefficients α_t, σ_t and Gaussian noise ϵ (notation of ML 07).

The replacement method and its flaw. The simple approach replaces, at every sampling step, the noisy known part z^a_t by a freshly noised copy of x^a, and updates z^b_t as usual. The update of z^b then follows $\hat{x}^b_\theta(z_t) \approx \mathbb{E}[x^b \mid z_t]$, while the correct target is $\mathbb{E}[x^b \mid z_t, x^a]$. In the VDM experiments, the generated frames looked good alone but were often not coherent with the known frames [6].

The correction. VDM writes the missing part with the score of $q(x^a \mid z_t)$ and approximates that distribution by a Gaussian centered at the model's own reconstruction of the known frames, $q(x^a \mid z_t) \approx \mathcal{N}(\hat{x}^a_\theta(z_t), (\sigma_t^2/\alpha_t^2)I)$. With a weighting factor w_r the adjusted estimate is [6]

$$\tilde{x}^b_\theta(z_t) = \hat{x}^b_\theta(z_t) - \frac{w_r \alpha_t}{2} \nabla_{z^b_t} \|x^a - \hat{x}^a_\theta(z_t)\|_2^2.$$

The gradient asks: how should the noisy new frames change so that the model's reconstruction of the **known** frames gets closer to the real known frames? It flows through the network, because the model reconstructs all frames jointly. The authors found that a weight $w_r > 1$ tends to improve sample quality, as with other forms of guidance [6].

Uses. The same formula extends a clip in time (known frames before), interpolates in time (known frames at a lower frame rate) and, with the error measured on a downsampled prediction, upsamples in space [6]. It needs one extra backward pass through the network per step.

Edge cases. The Gaussian approximation is exact only as $t \to 0$ for a perfect model; at high noise, the reconstruction is blurry and the guidance is weak or misleading, so the weight matters. Conditioning on many known frames costs memory, because they are processed together with the new frames.

Camera poses as Plücker embeddings

This note explains the per-pixel camera representation used by CameraCtrl [12] and what its six numbers encode.

Camera parameters. A pinhole camera has intrinsics $K \in \mathbb{R}^{3\times3}$ (focal lengths and principal point) and extrinsics: a rotation $R \in \mathbb{R}^{3\times3}$ and a translation; let $o \in \mathbb{R}^3$ be the camera center in world coordinates. CameraCtrl argues that feeding these raw numbers to a generator works poorly: the translation has no fixed scale while the rotation is constrained, and raw matrices are hard to relate to image positions [12].

One ray per pixel. Pixel (u, v) sees the world along a ray from o in the direction $d_{u,v} \in \mathbb{R}^3$, obtained by back-projecting the pixel with K^{-1}, rotating it into world coordinates and normalizing it to unit length. The **Plücker embedding** of that pixel is

$$p_{u,v} = (o \times d_{u,v},\ d_{u,v}) \in \mathbb{R}^6,$$

where × is the cross product. The second half is the ray's direction; the first half, the **moment**, encodes where the ray is (it is perpendicular to the plane containing the ray and the world origin, and its length is the ray's distance from the origin). Two rays with the same direction but different positions have different moments. Stacking $p_{u,v}$ over all pixels gives a $6 \times h \times w$ map per frame and an $n \times 6 \times h \times w$ tensor for a clip of n frames, which a convolutional or attention module can align with the video latents pixel by pixel [12].

A tested check. The snippet computes the embedding for a small pinhole camera, then moves the camera forward and rotates it, and reports which half of the embedding changes.

```
import numpy as np

def plucker(K, R, o, h=4, w=6):
    """Per-pixel Plücker embedding (moment, direction) for a camera with world rotation R
     ↪  and center o."""
    v, u = np.mgrid[:h, :w] + 0.5
    pix = np.stack([u, v, np.ones_like(u)], -1).reshape(-1, 3)
    d = (R @ np.linalg.inv(K) @ pix.T).T             # ray directions in world coordinates
    d /= np.linalg.norm(d, axis=1, keepdims=True)
    return np.cross(o, d), d                         # (h*w, 3) moments and (h*w, 3)
     ↪  directions

K = np.array([[5.0, 0, 3], [0, 5.0, 2], [0, 0, 1]])
I, o = np.eye(3), np.array([1.0, 0.0, 0.0])
yaw = np.deg2rad(10)
Ry = np.array([[np.cos(yaw), 0, np.sin(yaw)], [0, 1, 0], [-np.sin(yaw), 0, np.cos(yaw)]])

m0, d0 = plucker(K, I, o)
m1, d1 = plucker(K, I, o + np.array([0, 0, 0.5]))   # move forward along the viewing axis
m2, d2 = plucker(K, Ry, o)                           # turn the camera by 10 degrees
print(f"move forward: direction change {np.abs(d1 - d0).max():.3f}, moment change {np.abs(m1
 ↪  - m0).max():.3f}")
print(f"turn 10 deg:  direction change {np.abs(d2 - d0).max():.3f}, moment change {np.abs(m2
 ↪  - m0).max():.3f}")
```

It prints:

```
move forward: direction change 0.000, moment change 0.223
turn 10 deg:  direction change 0.173, moment change 0.091
```

Moving the camera changes only the moments (where the rays are); turning it changes the directions (and, because the center is off the origin, the moments too). The embedding therefore separates the two kinds of camera motion in a form that is attached to every pixel. Note that the overall scale of the scene is still undetermined from video alone, so camera paths estimated from real clips carry an arbitrary scale that the training data must normalize.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Blattmann et al., "Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets" (2023)

The problem. By 2023 many groups turned image latent diffusion models into video models by inserting temporal layers and fine-tuning on small, high-quality video sets, but training recipes varied widely and there was no agreed strategy for curating video data [4].

The idea. Treat training as three explicit stages and study each: (I) text-to-image pretraining (the model starts from a pretrained image latent diffusion model [14]); (II) video pretraining on large amounts of curated video; (III) fine-tuning on a small subset of high-quality videos at higher resolution. For stage II, build the Large Video Dataset (580M annotated clip pairs, 212 years of content) and annotate every clip with dense optical flow, OCR text detection, aesthetic scores and image–text similarities, plus synthetic captions, so that filters can be chosen systematically.

What it showed. An image-pretrained video model was preferred by people over the same model trained from scratch, in both quality and prompt following. Curated pretraining subsets (static clips, text-heavy clips and low-aesthetic clips removed; thresholds selected by human-preference ranking of small models) beat uncurated data of the same size, even when four times smaller, and the advantage persisted after stage III fine-tuning. The resulting base model gave a text-to-video model competitive with closed models in human studies, a strong image-to-video model (14 and 25 frames; image embedding instead of text, noise-augmented frame concatenated to the input, guidance increasing linearly over frames), camera-motion LoRAs, a frame-interpolation model and a multi-view 3D prior.

How to read it. Read Section 3 (the three stages and the curation study) first; most of its evidence is in plots of human preference (Elo or win rates), not tables of FVD. Check: (a) which comparisons are made at small scale (most curation ablations) and which with the final model; (b) that human studies compare against specific closed systems at one point in time; (c) the appendix, which holds the filter thresholds, model details and the evaluation protocol.

Limits. The dataset is not released, so the curation results cannot be reproduced exactly; human-preference studies depend on prompts and annotators; the autoencoder compresses only in space, which keeps the number of latent frames equal to the number of frames.

Why it matters here. It is the clearest documented case for the chapter's point that video data curation (especially removing static clips) and staged training are as important as architecture, and it is the reference design for image-to-video conditioning.

Ge et al., "On the Content Bias in Fréchet Video Distance" (CVPR 2024)

The problem. FVD is the most common distribution metric for video generation, but several papers had reported cases where it contradicted human judgment, with a recurring suspicion that it values the quality of individual frames over the realism of motion [42].

The idea. Measure the bias in controlled settings. Starting from real videos, create two distorted sets with matched per-frame quality: one with the **same** distortion applied to every frame (spatial only) and one with a **different** distortion per frame (spatiotemporal, which adds temporal inconsistency). Matched frame quality is confirmed with FID on the extracted frames; the temporal sensitivity of FVD is then the ratio of its change to the change in FID. Second, probe the metric's "perceptual null space": choose, from many generated videos without motion, a subset that minimizes FVD. Third, swap the I3D features for features of self-supervised video models (VideoMAE v2 [43]) and repeat.

What it showed. With I3D features, FVD increases only slightly under large temporal corruption, and selecting static generated videos can decrease FVD drastically without improving temporal quality, both signs of a bias toward per-frame content. The bias was attributed to the features of a supervised action classifier trained on Kinetics, a dataset in which many actions can be recognized from appearance alone. FVD computed with self-supervised VideoMAE v2 features was much more sensitive to temporal quality in all experiments.

How to read it. Start with Figure 1 (the motivating example) and the experimental design figure, then the sensitivity plots, comparing I3D with VideoMAE v2 variants per distortion type and dataset. Check: (a) that the conclusions are about relative sensitivity, not absolute FVD values, which are not comparable across feature extractors; (b) which VideoMAE v2 variant (pretrained only, or fine-tuned on an action dataset) each curve uses; (c) the real-world examples at the end, where the analysis is applied to published models.

Limits. The authors note that FVD still requires resizing to a low, square resolution, that long videos remain hard to evaluate because the transformer features have quadratic cost, and that their study covers a set of distortions and datasets, not every failure of generated video.

Why it matters here. It explains why this chapter insists on temporal and motion measures next to FVD, and why a feature extractor must be reported with every FVD number.

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DLCV 11 • Robustness: Distribution Shift, Adversarial Examples and Anomalies

Why vision models fail outside their test set and what to do about it: kinds of distribution shift and the benchmarks that measure them, augmentation, pre-training, test-time and domain adaptation; adversarial threat models, FGSM, PGD, adversarial training, honest evaluation and certified robustness; out-of-distribution and industrial anomaly detection with their scores, metrics and pitfalls.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-11-robustness/

What you'll learn

- How to tell apart the three robustness problems of a vision model: **distribution shift** (the data changed), **adversarial examples** (someone changed the input on purpose) and **out-of-distribution or anomalous inputs** (the input does not belong to the task at all), and why each needs its own test.
- How shift is described (covariate, label and concept shift), which benchmarks measure it (corruptions, renditions, natural adversarial images, controlled and in-the-wild datasets), and what the evidence says helps: augmentation, more diverse pre-training, test-time adaptation, domain adaptation and domain generalization.
- How an adversarial threat model is stated (an ℓ_p ball, white-box or black-box access), how **FGSM** and **PGD** work in equations and NumPy, what **adversarial training** buys and costs, why weak evaluations overstate robustness (gradient masking), and what a **certificate** from randomized smoothing guarantees.
- How out-of-distribution detection and anomaly detection are defined, how the common scores work (**maximum softmax probability, energy, Mahalanobis distance, k-nearest-neighbour distance**), how they are measured (**AUROC, FPR at 95% TPR, AUPR**), and how industrial defect detection with a **patch memory bank** (the idea behind PatchCore) works.
- Which pitfalls make robustness numbers misleading: near- versus far-OOD, contaminated OOD test sets, model selection on the test distribution, and gains that hold for one kind of shift only.

The big picture

A model that scores 98% on its test set has only been checked on data that looks like its training data. Deployed models meet other data: photos from a new camera, images edited by an attacker, and objects that were never part of the label set. Robustness is the study of what happens then, and of how to make the answer less alarming.

> **Explain like I'm five**
>
> Think of a student who learned everything from one textbook. Three things can go wrong in the exam. First, the exam is printed in a smudged, faded font: the questions are the same kind, but they look different (**distribution shift**). Second, a prankster changes a few letters in a question so that the student misreads it, choosing exactly the letters that confuse this student most (**adversarial example**). Third, a question from a completely different subject appears: the right answer is "this is not my subject", but the student writes an answer anyway, confidently (**out-of-distribution input**). A good student copes with the smudges, is hard to trick, and knows when to say "I don't know".

Why it matters. The three failures cause different damage. Shift silently lowers accuracy in the field, so a model that looked finished keeps making more mistakes after release. Adversarial examples are a security problem: any system whose decision is worth manipulating (content filters, identity checks, traffic-sign recognition) will be attacked. Unanswerable inputs produce confident nonsense, which is the most dangerous output a safety-critical system can give. The fixes differ too, so the first practical skill is to name which of the three problems you face.

Where this chapter sits. DLCV 02 already introduces ImageNet-C, -R and -A, the one-step FGSM attack, shortcut learning and calibration with temperature scaling; here we build on them rather than repeat them. ML 10 defines covariate, label and concept shift with importance weights and covers monitoring after deployment. PR 01 derives the reject option, and PR 10 separates selective classification, open-set recognition and anomaly detection on a small digits project. ML 01 and ML 05A cover reconstruction-based anomaly scores, and ML 09 explains why fine-tuning can erode a foundation model's robustness. All code in this chapter runs in seconds on a CPU with NumPy, SciPy, scikit-image and scikit-learn; the "networks" are small multilayer perceptrons on 8×8 digits, so every number we print is from a toy and is marked as such.

Three robustness problems, one map

Plain version. Ask three questions about a failure. Did the data change while the task stayed the same? Did someone change this one input on purpose, by a tiny amount? Is the input outside the task, so that no label is correct? These are distribution shift, adversarial examples and out-of-distribution (OOD) inputs.

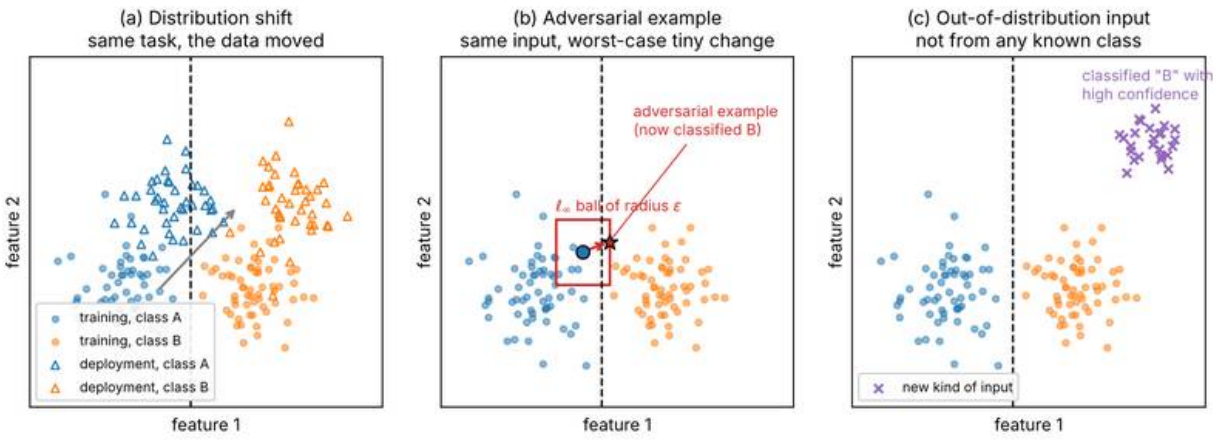


Figure 1 — The three robustness problems on one synthetic two-class problem. (a) Distribution shift: deployment data (triangles) moved away from the training data, and a boundary that was fine for training data now makes errors. (b) Adversarial example: a worst-case point inside a small box around a correct input crosses the boundary. (c) Out-of-distribution input: points from a kind of input never seen in training still receive a confident class label.

Precise version. Let a classifier f be trained on samples from a distribution $p_{\text{train}}(\mathbf{x}, y)$ over images $\mathbf{x}$ and labels $y \in \mathcal{Y} = \{1, \dots, K\}$. Write $\text{acc}_p(f) = \Pr_{(\mathbf{x},y)\sim p}[f(\mathbf{x}) = y]$ for its accuracy on a distribution p.

1. **Distribution shift.** Test data come from $p_{\text{test}} \neq p_{\text{train}}$, but the label set $\mathcal{Y}$ is the same. The question is how much $\text{acc}_{p_{\text{test}}}(f)$ falls below $\text{acc}_{p_{\text{train}}}(f)$, and how to reduce the fall.
2. **Adversarial robustness.** For each test pair $(\mathbf{x}, y)$, an adversary may replace $\mathbf{x}$ by any $\mathbf{x}'$ in an allowed set $\mathcal{S}(\mathbf{x})$, usually a small ball around $\mathbf{x}$. The model is robust at $(\mathbf{x}, y)$ if $f(\mathbf{x}') = y$ for **every** $\mathbf{x}' \in \mathcal{S}(\mathbf{x})$. The question is about the worst case, not the average.
3. **OOD and anomaly detection.** Some test inputs have no correct label in $\mathcal{Y}$ (a new object category), or come from a region of input space the training data never covered (a defective part, a corrupted frame). We want a score $s(\mathbf{x})$ that is high for in-distribution (ID) inputs and low for these, so that the system can abstain or raise an alarm.

	Distribution shift	Adversarial example	OOD / anomaly
What changed	the data distribution	one input, on purpose	the input is outside the task
Who causes it	the world (camera, place, time)	an adversary with a goal	the world (new classes, defects)
What we want	keep accuracy	keep the label under every allowed change	flag the input, do not answer
Typical metric	accuracy on shifted test sets	robust accuracy at a budget ε	AUROC, FPR at 95% TPR
In this chapter	Distribution shift	Adversarial examples	OOD and anomaly detection

The three overlap at the edges. A heavy corruption can make an input effectively OOD; an adversarial example is a shift chosen by an opponent; and ImageNet-A, a set of natural photos selected because they fool a classifier [1], sits between shift and adversarial examples. Still, the table shows why one robustness number can never cover all three: they ask different questions and need different test data.

Distribution shift

Plain version. The test data are of the same kind as the training data (the same classes, the same task) but look different: other cameras, lighting, styles, places or times. Accuracy drops, often a lot, and the model gives no warning.

Kinds of shift

The usual vocabulary splits the joint distribution in two ways, $p(\mathbf{x}, y) = p(y \mid \mathbf{x})\, p(\mathbf{x}) = p(\mathbf{x} \mid y)\, p(y)$, and names the shift by which factor changes:

- **Covariate shift:** $p(\mathbf{x})$ changes, $p(y \mid \mathbf{x})$ does not. A new scanner, fog, a different hospital's staining.
- **Label (prior) shift:** $p(y)$ changes, $p(\mathbf{x} \mid y)$ does not. A rare disease becomes common during an outbreak.
- **Concept shift:** $p(y \mid \mathbf{x})$ itself changes. The definition of a class is revised.

ML 10 develops these with importance weights and the label-shift correction, so we do not repeat them. Two more terms are common in vision. **Domain shift** is covariate shift between named data sources (synthetic versus real, one hospital versus another), and **subpopulation shift** keeps the same kinds of images but changes how often each group appears; WILDS contains both [2]. The survey by Yang et al. adds a distinction that becomes important later in this chapter: a **covariate shift** changes how inputs look while the labels stay valid, whereas a **semantic shift** introduces inputs whose labels are outside $\mathcal{Y}$ [3]. We want models to be *robust* to covariate shift and to *detect* semantic shift.

Benchmarks that measure robustness to shift

Plain version. Because the real deployment data are rarely available in advance, researchers built test sets that change one thing at a time, or that collect naturally different photos of the same classes.

The main families, all labelled with ImageNet classes unless noted:

- **Synthetic corruptions. ImageNet-C** applies noise, blur, weather and digital corruptions at five severities, and its authors also proposed ImageNet-P for perturbation sequences [4]. The details and the mean corruption error are in DLCV 02.
- **Renditions and style. ImageNet-R** collects art, cartoons, sculptures and other renditions; the same paper introduced further real-world shift datasets and found that methods that helped with texture and local image statistics did not help with others, such as geographic change [5].
- **Natural adversarial images. ImageNet-A** keeps unmodified photos that a ResNet-50 got wrong; they transfer to other models, showing shared weaknesses [1].
- **Controlled collection. ObjectNet** photographs objects with randomized backgrounds, rotations and viewpoints, as a test set only, with no training split [6].
- **Re-collection. ImageNetV2** rebuilt the test set by following the original collection process; accuracy dropped although the procedure was copied as closely as possible [7] (see ML 10).
- **In the wild. WILDS** gathers ten datasets with shifts that arose in real applications, such as tissue slides from different hospitals and camera traps at new locations, and reports that standard training gives much lower out-of-distribution than in-distribution performance, with existing robustness methods not closing the gap [2].

How to read the numbers. A model with higher clean accuracy usually also has higher shifted accuracy, so "it scores better on ImageNet-R" may only mean "it is a better model". Taori et al. evaluated 204 ImageNet models in 213 test conditions and proposed **effective robustness**: accuracy on the shifted set *beyond* what a fitted trend over standard models predicts from the clean accuracy [8]. Miller et al. found that, for many shifts, shifted and clean accuracy lie close to a straight line across architectures and training choices ("accuracy on the line"), with weaker correlation for some synthetic shifts and for one medical dataset [9]. The consequence for practice: compare robustness interventions at matched clean accuracy, or report both numbers side by side.

Optional: Effective robustness and accuracy on the line (in the Supplementary notes). How the baseline trend is fitted, why logit or probit axes are used, and how to read a robustness scatter plot.

Code: corruptions, augmentation and test-time normalization

The toy trains a small MLP on scikit-learn's 8×8 digits and corrupts the *test* images with three corruptions at severities 1 to 5. The network standardizes each pixel with statistics computed on the training set, the way a batch-normalization layer uses running statistics at test time. We compare three models: the plain one; one trained with noise augmentation (a copy of every training image with random-severity noise); and the plain model with **test-batch normalization**, which recomputes the per-pixel mean and standard deviation on the corrupted test batch instead of using the

training statistics.

```
import numpy as np
from scipy.ndimage import gaussian_filter
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier

X, y = load_digits(return_X_y=True)
X = X / 16.0                                          # 8x8 images, pixels in [0, 1]
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=0)
rng = np.random.default_rng(0)

corruptions = {                                       # severity s = 1..5
    "noise":    lambda X, s: np.clip(X + rng.normal(0, 0.1 * s, X.shape), 0, 1),
    "blur":     lambda X, s: gaussian_filter(X.reshape(-1, 8, 8), (0, 0.25 * s, 0.25 *
     ↪  s)).reshape(-1, 64),
    "contrast": lambda X, s: 0.5 + (X - 0.5) * (1 - 0.15 * s),
}

def norm(X, ref):                                     # standardize each pixel with ref's
 ↪  statistics
    return (X - ref.mean(0)) / (ref.std(0) + 0.05)

def fit(X, y):
    return MLPClassifier(hidden_layer_sizes=(64,), alpha=1e-3, max_iter=400,
     ↪  random_state=0).fit(norm(X, Xtr), y)

plain = fit(Xtr, ytr)
s_rand = rng.integers(1, 6, (len(Xtr), 1))            # random noise severity per training
 ↪  image
augmented = fit(np.vstack([Xtr, corruptions["noise"](Xtr, s_rand)]), np.r_[ytr, ytr])

for name, corrupt in corruptions.items():
    rows = {"plain": [], "noise-aug": [], "test-norm": []}
    for s in range(6):
        Xs = Xte if s == 0 else corrupt(Xte, s)
        rows["plain"].append(plain.score(norm(Xs, Xtr), yte))
        rows["noise-aug"].append(augmented.score(norm(Xs, Xtr), yte))
        rows["test-norm"].append(plain.score(norm(Xs, Xs), yte))   # statistics of the test
         ↪  batch
    for k, v in rows.items():
        print(f"{name:8s} {k:9s}", " ".join(f"{a:.2f}" for a in v))
```

It prints the test accuracy at severities 0 (clean) to 5:

```
noise    plain     0.98 0.94 0.75 0.54 0.41 0.31
noise    noise-aug 0.97 0.96 0.93 0.82 0.74 0.62
noise    test-norm 0.98 0.97 0.90 0.83 0.73 0.61
blur     plain     0.98 0.98 0.96 0.82 0.46 0.21
blur     noise-aug 0.97 0.97 0.97 0.90 0.78 0.58
```

```
blur     test-norm 0.98 0.98 0.97 0.95 0.89 0.81
contrast plain     0.98 0.92 0.46 0.20 0.19 0.19
contrast noise-aug 0.97 0.98 0.96 0.93 0.84 0.60
contrast test-norm 0.98 0.98 0.98 0.98 0.98 0.98
```

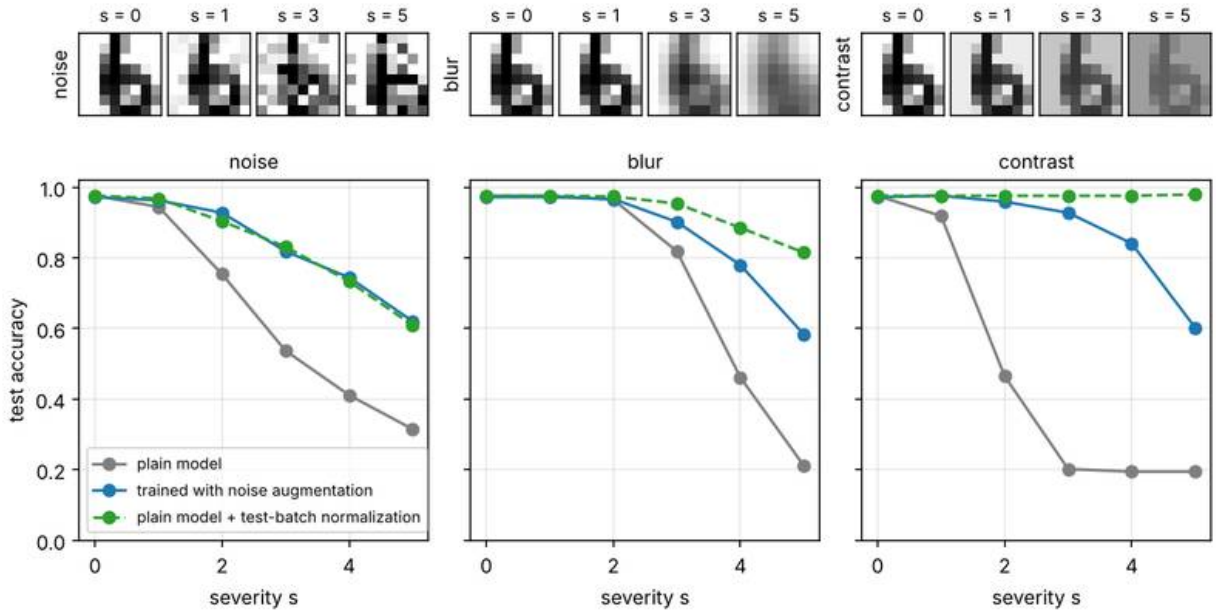


Figure 2 — Accuracy of a small MLP on corrupted digits (our toy run). Top: one test image per corruption and severity. Bottom: plain model, model trained with noise augmentation, and plain model with per-pixel statistics recomputed on the corrupted test batch.

(All numbers are from our toy run.) Four lessons carry over to real networks:

- **Mild-looking changes are enough.** A contrast reduction to 70% of the original range (severity 2) halves the plain model's accuracy, although a person reads the digits easily.
- **Augmentation helps, and not only for the corruption it used.** Training with noise also helped with blur and contrast here, which is the hope behind diverse augmentation. It cost a little clean accuracy (0.98 to 0.97), and real benchmarks show that the transfer to *other* shifts is far from guaranteed [5], [8].
- **Re-estimating normalization statistics is a cheap and strong fix when its assumption holds.** The contrast corruption is an affine change of every pixel, which per-pixel standardization undoes exactly, so accuracy stays at 0.98. Noise changes the statistics in a way standardization cannot undo, so the gain is smaller.
- **Test-batch normalization needs a batch.** It used all 540 test images of one corruption at once. With a few images, or with a batch that mixes several conditions, the statistics are wrong (Exercise 1).

What helps, and what does not

Plain version. Five families of fixes exist. Train on more varied images; start from a model pretrained on very diverse data; adapt the model to the test data while it runs; adapt it in advance with unlabeled data from the target domain; or train on several domains and hope to generalize to new ones. Each has a cost and a condition under which it works.

Data augmentation. Augmentation makes the training distribution wider so that the test distribution is more likely to fall inside it. **AugMix** mixes several chains of random augmentations of one image, and adds a Jensen–Shannon consistency loss that asks the network to give similar predictions for the original and the augmented versions [10]. Its authors deliberately removed operations that overlap with ImageNet-C corruptions, so the benchmark measures transfer to unseen corruptions rather than memorization of the test corruptions [10]. Stylized training, which replaces image textures with painting styles, made a ResNet-50 rely more on shape and improved robustness to many distortions [11]. The caution comes from the ImageNet-R study: augmentation and larger models helped on some real-world shifts (style, blur) but not on others (geography), and no evaluated method consistently improved robustness across all of them [5]. General augmentation recipes are in DLCV 02.

Pre-training on more diverse data. In the testbed of Taori et al., most robustness interventions gave little or no effective robustness on natural shifts; the main exception was training on larger and more diverse data [8]. Zero-shot CLIP models, trained on image–text pairs from the web, improved effective robustness by a large amount on natural ImageNet shifts, but the CLIP paper also showed that the benefit almost vanished when a classifier was fitted on ImageNet in a fully supervised way [12]. Fang et al. tested five possible causes (dataset size, training distribution, language supervision during training, language supervision at test time, contrastive loss) and found that the **more diverse training distribution** was the main cause [13]. Fine-tuning can therefore erode robustness; WiSE-FT averages the weights of the zero-shot and fine-tuned models to keep most of both [14] (see also LP-FT in ML 09).

Test-time adaptation (TTA). If unlabeled test images arrive in batches, the model can adjust itself before predicting. The simplest form is the one in our toy: Schneider et al. replaced the batch-normalization statistics of 25 pretrained models with statistics of the corrupted test images and improved corruption robustness consistently, even when adapting with a single sample [15]. **TENT** goes one step further: it re-estimates the normalization statistics *and* updates the channel-wise affine parameters of the normalization layers online, by minimizing the entropy of the model's own predictions on each test batch [16]. TTA assumes a stream of test data from a reasonably stable shift; it adds computation at test time and can drift if the stream changes.

Optional: Test-time entropy minimization (in the Supplementary notes). The TENT objective, which parameters it changes and why, and the failure cases to watch for.

Unsupervised domain adaptation (UDA). If unlabeled images from the target domain are

available *during training*, the model can be trained to work there without target labels. **Domain-adversarial training (DANN)** adds a small domain classifier on top of the features and a **gradient-reversal layer** between them, so that the features become good for the label task on the labelled source domain and, at the same time, useless for telling source from target [17]. The idea rests on domain-adaptation theory: if a classifier cannot tell the domains apart in feature space, a good source classifier should transfer.

Optional: Domain-adversarial training and the gradient-reversal layer (in the Supplementary notes). The DANN objective as a min-max problem, the layer that implements it in one line, and when aligning features hurts.

Domain generalization (DG). Without any target data, the model is trained on several source domains in the hope that it works on an unseen one. The survey by Zhou et al. groups DG methods into domain alignment, meta-learning, data augmentation and ensemble learning, among others [18]. Gulrajani and Lopez-Paz then reran many DG algorithms in one testbed, DomainBed, with equal tuning effort and explicit model-selection rules, and found that plain empirical risk minimization (ERM, ordinary training on the pooled sources) with a strong network and augmentation matched or beat them [19]. Their key point is methodological: a DG method is incomplete without a rule for choosing hyperparameters that does not peek at the test domain.

Optional: Gulrajani and Lopez-Paz, "In Search of Lost Domain Generalization" (in the Paper guides). How DomainBed's three model-selection rules work and how to read its tables.

What does not help, or not reliably. Robustness on synthetic corruptions was only weakly predictive of robustness on natural shifts in the testbed of Taori et al., and the same held for ℓ_p adversarial robustness [8]. A method tuned on one benchmark often helps only that benchmark [5]. Architecture changes at fixed data have mattered less than the data itself [8], [13]. The practical rule follows: test on several kinds of shift, including at least one real shift from your own deployment.

Confidence under shift

Plain version. Under shift, a model is wrong more often, but it is usually not *less confident* by the same amount, so its probabilities stop meaning what they say.

Ovadia et al. benchmarked uncertainty methods under dataset shift on several tasks and found that calibration degrades as shift grows, that post-hoc calibration such as temperature scaling fitted on in-distribution validation data falls short under shift, and that methods that average over several models (such as deep ensembles) held up best [20]. Temperature scaling is explained in DLCV 02, and ML 10 shows on a toy how a fitted temperature stops working under shift. The practical consequence: a confidence threshold chosen on clean validation data will accept more wrong answers in the field, so monitor the distribution of confidences after deployment (ML 10).

Adversarial examples

Plain version. For almost any correctly classified image, there is a nearby image, often indistinguishable to a person, that the network classifies wrongly. An attacker finds it by asking the network itself which direction to move each pixel.

What an adversarial example is

Szegedy et al. reported that deep networks can be made to misclassify an image by an imperceptible perturbation found by maximizing the network's prediction error, and that the same perturbation often fools a different network trained on a different subset of the data [21]. Goodfellow et al. explained the effect by the near-linearity of networks in high dimensions and introduced the one-step fast gradient sign method [22], which DLCV 02 derives and tests. This section adds the parts needed to *evaluate* and *improve* robustness: threat models, the multi-step attack, adversarial training, honest evaluation and certificates.

Threat models

Plain version. "Robust" means nothing until you say robust to what: how much may the attacker change, and what does the attacker know?

A **threat model** fixes three things.

1. The allowed perturbations. Most work uses an ℓ_p ball of radius ε around the clean image $\mathbf{x} \in [0,1]^d$:

$$\mathcal{S}_p(\mathbf{x}) = \{\mathbf{x}' \in [0,1]^d : \|\mathbf{x}' - \mathbf{x}\|_p \le \varepsilon\}, \qquad \|\boldsymbol{\delta}\|_\infty = \max_i |\delta_i|, \qquad \|\boldsymbol{\delta}\|_2 = \Big(\sum_i \delta_i^2\Big)^{1/2},$$

where d is the number of pixel values, $\boldsymbol{\delta} = \mathbf{x}' - \mathbf{x}$ is the perturbation and δ_i its i-th entry. The ℓ_∞ ball lets every pixel change by at most ε (a small change everywhere); the ℓ_2 ball bounds the total energy of the change; the ℓ_0 "norm" counts how many pixels change (a few pixels, changed a lot). On CIFAR-10, $\varepsilon = 8/255$ in ℓ_∞ is a standard budget: Madry et al. trained and attacked with $\varepsilon = 8$ on the 0–255 pixel scale [23], and RobustBench keeps leaderboards for ℓ_∞ and ℓ_2 threat models and for common corruptions [24]. A small ℓ_p norm is a convenient proxy for "imperceptible", not a definition of it.

2. The attacker's knowledge. In a **white-box** attack, the attacker knows the model and its weights and can compute gradients. In a **black-box** attack, the attacker only queries the model. Papernot et al. showed a practical black-box attack: label inputs by querying the target model, train a local substitute model on them, craft adversarial examples on the substitute, and rely on **transfer** to

the target [25]. Other black-box attacks estimate the decision boundary from queries alone. Robustness claims should be made against white-box attacks, because what one attacker cannot do today, a better-informed one may do tomorrow.

3. The goal. An **untargeted** attack wants any wrong label; a **targeted** attack wants a specific one (a stop sign read as a speed limit). Targeted attacks are harder, so the two give different numbers.

FGSM and PGD, precisely

Plain version. FGSM takes one big step uphill on the loss. PGD takes many small steps uphill and, after each step, pulls the image back into the allowed ball. PGD finds stronger adversarial examples and is the standard first-order attack.

Let $\mathcal{L}(\boldsymbol{\theta}, \mathbf{x}, y)$ be the cross-entropy loss of the network with parameters $\boldsymbol{\theta}$ on input $\mathbf{x}$ with label y.

FGSM [22] for the ℓ_∞ ball is

$$\mathbf{x}_{\text{adv}} = \Pi_{[0,1]^d}(\mathbf{x} + \varepsilon \ \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\boldsymbol{\theta}, \mathbf{x}, y))),$$

where $\nabla_{\mathbf{x}} \mathcal{L}$ is the gradient of the loss with respect to the input pixels, sign is applied to each entry (giving -1, 0 or $+1$), and $\Pi_{[0,1]^d}$ clips each pixel to the valid range. Moving every pixel by exactly $\pm\varepsilon$ in the direction that increases the loss is the best a *linear* approximation of the loss can do inside the ℓ_∞ ball.

Projected gradient descent (PGD) [23] repeats a smaller step T times, starting from a random point in the ball:

$$\mathbf{x}^{(0)} = \mathbf{x} + \mathbf{u}, \quad \mathbf{u} \sim \mathcal{U}[-\varepsilon, \varepsilon]^d, \qquad \mathbf{x}^{(t+1)} = \Pi_{\mathcal{S}(\mathbf{x})}(\mathbf{x}^{(t)} + \alpha \ \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\boldsymbol{\theta}, \mathbf{x}^{(t)}, y))),$$

where $\mathbf{u}$ is uniform noise in the ball (the random start), α is the step size, $t = 0, \dots, T-1$ counts the steps, and $\Pi_{\mathcal{S}(\mathbf{x})}$ is the projection onto the allowed set. For the ℓ_∞ ball intersected with the valid pixel range, the projection is two clips: first $x'_i \leftarrow \min(\max(x'_i, x_i - \varepsilon), x_i + \varepsilon)$, then $x'_i \leftarrow \min(\max(x'_i, 0), 1)$. The final iterate $\mathbf{x}^{(T)}$ is the adversarial example. PGD is "gradient *ascent* on the loss"; the name comes from optimization, where the same algorithm minimizes. Because the loss surface is not concave, the random start and a few restarts help; Madry et al. observed that the losses reached from many random starts were well concentrated, which supported PGD as a reliable first-order adversary [23].

A second common loss is the **margin loss** of Carlini and Wagner, which directly maximizes the gap between the largest wrong logit and the correct logit instead of the cross-entropy [26]. It matters when the cross-entropy saturates, and it is one reason a serious evaluation runs several attacks.

Optional: Attack settings that decide an evaluation (in the Supplementary notes). Step size, number of steps, restarts, the margin loss, ℓ_2 PGD, and what AutoAttack fixes.

Adversarial training

Plain version. If the model fails on the worst nearby image, train it on the worst nearby image. During training, attack every batch with PGD and fit the attacked images instead of the clean ones.

Madry et al. wrote robust training as a min-max (saddle point) problem [23]:

$$\min_{\boldsymbol{\theta}} \rho(\boldsymbol{\theta}), \qquad \rho(\boldsymbol{\theta}) = \mathbb{E}_{(\mathbf{x},y)\sim\mathcal{D}}\Big[\max_{\delta\in\mathcal{S}} \mathcal{L}(\boldsymbol{\theta}, \mathbf{x}+\boldsymbol{\delta}, y)\Big],$$

where $\mathcal{D}$ is the data distribution, $\mathcal{S}$ the set of allowed perturbations (for example, $\|\boldsymbol{\delta}\|_\infty \leq \varepsilon$), the inner maximum is the attacker's best perturbation and the outer minimum is the defender's training. In practice, the inner maximum is approximated by PGD with a few steps and the outer minimum by ordinary stochastic gradient descent on the resulting adversarial batch.

What it costs:

- **Compute.** Every training step needs a T-step attack, so training is roughly $T + 1$ times as expensive.
- **Capacity.** Madry et al. found that robust classification needs larger networks than clean classification; small networks trained against PGD could collapse to predicting one class, and training against one-step FGSM overfitted to its narrow set of perturbations and did not give robustness to PGD at large ε [23].
- **Clean accuracy.** Robust models are usually less accurate on clean data. Tsipras et al. proved that this trade-off exists in a simple, natural setting and argued that robust models learn different features, which happen to align better with human perception [27]. **TRADES** splits the robust error into the natural error plus a boundary error and trains with a loss that trades the two explicitly through one weight [28].

Optional: Madry et al., "Towards Deep Learning Models Resistant to Adversarial Attacks" (in the Paper guides). The saddle-point view, the capacity experiments and how to read its attack tables.

Code: FGSM, PGD and adversarial training on digits

NumPy computes the input gradient of a one-hidden-layer MLP by hand: for softmax outputs $\mathbf{p}$ and a one-hot label $\mathbf{e}_y$, the gradient of the cross-entropy with respect to the logits is $\mathbf{p} - \mathbf{e}_y$, which we pass back through the output layer, the ReLU and the first layer. Both models get the same 50 passes over the training data; the robust one is fitted on PGD examples ($\varepsilon = 0.15$, 5 steps) made against its current weights.

```
import numpy as np
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
```

```
from sklearn.neural_network import MLPClassifier

X, y = load_digits(return_X_y=True)
X = X / 16.0                                          # pixels in [0, 1]
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=0)
rng = np.random.default_rng(0)

def input_grad(m, X, y):
    """Gradient of the cross-entropy loss with respect to the input pixels."""
    (W1, W2), (b1, b2) = m.coefs_, m.intercepts_
    h = np.maximum(X @ W1 + b1, 0)                    # hidden ReLU layer
    z = h @ W2 + b2                                   # logits
    g = np.exp(z - z.max(1, keepdims=True)); g /= g.sum(1, keepdims=True)
    g[np.arange(len(y)), y] -= 1                      # dL/dz = softmax(z) - one-hot(y)
    return ((g @ W2.T) * (h > 0)) @ W1.T              # back through ReLU and first layer

def fgsm(m, X, y, eps):
    return np.clip(X + eps * np.sign(input_grad(m, X, y)), 0, 1)

def pgd(m, X, y, eps, steps=10):
    alpha = 2.5 * eps / steps                         # step size
    Xa = np.clip(X + rng.uniform(-eps, eps, X.shape), 0, 1)        # random start in the ball
    for _ in range(steps):
        Xa = Xa + alpha * np.sign(input_grad(m, Xa, y))             # ascent step on the loss
        Xa = np.clip(np.clip(Xa, X - eps, X + eps), 0, 1)           # project: eps-ball,
         ↪  then [0, 1]
    return Xa

def train(adversarial, epochs=50, eps=0.15):
    m = MLPClassifier(hidden_layer_sizes=(128,), random_state=0)
    m.partial_fit(Xtr, ytr, classes=np.arange(10))    # first pass on clean images
    for _ in range(epochs - 1):                       # Madry et al.: fit the worst case
     ↪  found by PGD
        m.partial_fit(pgd(m, Xtr, ytr, eps, steps=5) if adversarial else Xtr, ytr)
    return m

standard, robust = train(False), train(True)
for eps in [0, 0.05, 0.1, 0.15, 0.2]:
    acc = [m.score(atk(m, Xte, yte, eps), yte) for m in (standard, robust) for atk in (fgsm,
     ↪  pgd)]
    print(f"eps={eps:.2f}  standard: FGSM {acc[0]:.3f} PGD {acc[1]:.3f}"
          f"   adversarially trained: FGSM {acc[2]:.3f} PGD {acc[3]:.3f}")
```

It prints (about 10–20 seconds on a laptop CPU):

```
eps=0.00  standard: FGSM 0.967 PGD 0.967   adversarially trained: FGSM 0.956 PGD 0.956
eps=0.05  standard: FGSM 0.861 PGD 0.854   adversarially trained: FGSM 0.891 PGD 0.889
eps=0.10  standard: FGSM 0.574 PGD 0.528   adversarially trained: FGSM 0.757 PGD 0.746
eps=0.15  standard: FGSM 0.237 PGD 0.178   adversarially trained: FGSM 0.626 PGD 0.593
eps=0.20  standard: FGSM 0.057 PGD 0.007   adversarially trained: FGSM 0.415 PGD 0.361
```

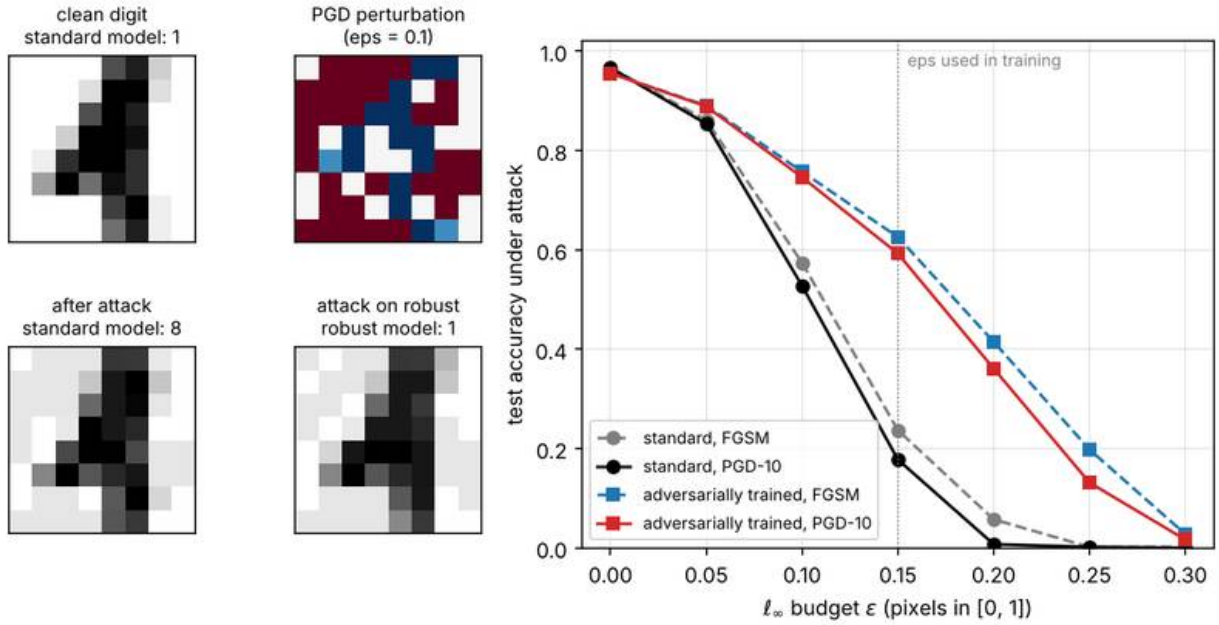


Figure 3 — FGSM and PGD on two MLPs (our toy run). Left: a PGD perturbation with $\varepsilon = 0.1$ changes the standard model's answer from 1 to 8 but not the adversarially trained model's. Right: accuracy under attack against the budget ε; PGD is always at least as strong as FGSM, and adversarial training raises robust accuracy at every ε at a small cost in clean accuracy.

(All numbers are from our toy run.) Read the table in three directions:

- **Down a column:** accuracy falls quickly with ε. At $\varepsilon = 0.1$, each pixel may change by 1.6 of the dataset's 16 gray levels, and the standard model drops from 0.967 to 0.528 under PGD.
- **Across an attack pair:** PGD is never weaker than FGSM, and the gap grows with ε (0.057 versus 0.007 at $\varepsilon = 0.2$ for the standard model). An evaluation with FGSM alone overstates robustness.
- **Across models:** adversarial training raises PGD accuracy at $\varepsilon = 0.15$ from 0.178 to 0.593, and costs about one point of clean accuracy (0.967 to 0.956), a small version of the trade-off. Beyond the training budget, robustness fades (0.131 at $\varepsilon = 0.25$ in the figure): robustness is guaranteed for nothing outside the threat model it was trained for.

Evaluating robustness honestly

Plain version. It is easy to build a "defense" that only breaks the attacker's tools, for example by making gradients useless, while the model stays just as easy to fool. The robustness then exists only on paper, until someone uses a better attack.

Athalye, Carlini and Wagner named this **obfuscated gradients**, a kind of gradient masking, and

described three types [29]: **shattered gradients** (non-differentiable or numerically unstable operations, such as quantizing the input, give gradients that are zero or wrong); **stochastic gradients** (random input transformations at test time make a single gradient misleading); and **vanishing or exploding gradients** (defenses that run an iterative procedure, such as projecting the input onto a learned manifold, behave like an extremely deep network). For each type they gave a counter-attack: **BPDA** (backward pass differentiable approximation) runs the defense normally in the forward pass but replaces it by a differentiable approximation, often the identity, in the backward pass; **EOT** (expectation over transformation) averages gradients over the random transformations; and **reparameterization** changes variables so that the iterative step disappears from the gradient path. In a case study of nine white-box defenses accepted at ICLR 2018, seven relied on obfuscated gradients, and their attacks fully circumvented six and partially one [29].

The same paper lists warning signs that an evaluation is broken: one-step attacks work better than iterative ones; black-box attacks work better than white-box ones; unbounded attacks do not reach 100% success; random sampling finds adversarial examples that gradient attacks miss; and increasing the budget does not increase success [29]. Our toy passes the checks it can: PGD beats FGSM, and accuracy reaches zero as ε grows.

To make evaluations less dependent on the evaluator's care, Croce and Hein proposed **AutoAttack**, a parameter-free ensemble of two improved PGD variants (with an automatic step-size schedule and a different loss) and two complementary existing attacks; applied to over 50 published defenses, it gave lower robust accuracy than reported in all but one case [30]. **RobustBench** builds a public leaderboard on AutoAttack with rules on allowed models, and invites adaptive attacks where AutoAttack may overestimate robustness [24]. A reported robust accuracy should therefore state the threat model, ε, the attacks (at least AutoAttack or an adaptive attack designed for the defense), the number of steps and restarts, and the test set.

Optional: Athalye et al., "Obfuscated Gradients Give a False Sense of Security" (in the Paper guides). The three types of masking, the attacks that undo them, and what its case-study table shows.

Certified robustness

Plain version. An attack can only show that a model is *not* robust at a point. A certificate proves that *no* perturbation within a radius changes the answer, whatever attack is used.

Randomized smoothing [31] was, according to its authors, the only certified defense shown to be feasible on ImageNet at the time, and it remains the standard reference point. Take any base classifier f and define a smoothed classifier g that returns the class f predicts most often under Gaussian noise:

$$g(\mathbf{x}) = \arg\max_{c \in \mathcal{Y}} \Pr_{\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})} [f(\mathbf{x} + \boldsymbol{\eta}) = c],$$

where $\boldsymbol{\eta}$ is isotropic Gaussian noise with standard deviation σ per pixel and $\mathbf{I}$ is the identity matrix. Cohen, Rosenfeld and Kolter proved that if the top class c_A has probability at least $\underline{p_A}$ under the noise and every other class at most $\overline{p_B}$, then g predicts c_A for every perturbation δ with

$$\|\delta\|_2 < R = \frac{\sigma}{2}\big(\Phi^{-1}(\underline{p_A}) - \Phi^{-1}(\overline{p_B})\big),$$

where Φ^{-1} is the inverse of the standard normal cumulative distribution function [31]. In practice, $\underline{p_A}$ is a lower confidence bound estimated from Monte Carlo samples, and the method abstains when it cannot certify. The guarantee holds for the ℓ_2 threat model, with a chosen failure probability, and for the smoothed classifier g, not for f. Larger σ allows larger radii but lowers clean accuracy, the same trade-off in another form.

Optional: Randomized smoothing in practice (in the Supplementary notes). The certification algorithm step by step, why the radius is capped by the number of samples, and a tested NumPy snippet.

Physical and patch attacks

Plain version. An attack does not have to be invisible or digital. A printed sticker can fool a camera.

Eykholt et al. generated perturbations that survive changes of viewpoint, distance and lighting, printed them as black-and-white stickers on a real stop sign, and caused a road-sign classifier to output the attacker's target class in most lab images and in many frames of a drive-by video [32].

Brown et al. made **adversarial patches**: universal, printable image regions that, placed anywhere in a scene, make a classifier report a chosen class [33]. Such attacks leave the ℓ_p threat model: they are large but localized. They matter for any camera-based system, and they connect to the presentation attacks on face recognition discussed in DLCV 02A.

Out-of-distribution and anomaly detection

Plain version. A classifier always picks one of its known classes, even for a photo of something it has never seen. OOD detection adds a second output, a score that says "this input looks like my training data" or "it does not", so the system can refuse to answer.

Which task are you solving?

Several closely related tasks go by different names. Yang et al. place them in one framework, **generalized OOD detection**, and distinguish them by what kind of shift must be detected and by whether ID data has one class or many [3]:

Task	In-distribution data	Must flag	Typical example
OOD detection	a K-class training set	inputs from other classes (semantic shift), while still classifying ID inputs	a pet classifier sees a car
Open-set recognition	K known classes	unknown classes, while classifying known ones	PR 10, DLCV 02A
Sensory anomaly detection	normal images of one kind	inputs that look different (covariate shift)	a scratched part on a production line
Semantic anomaly / novelty detection	one or a few classes	inputs from any other class	a new species in camera-trap images

OOD detection and open-set recognition differ mainly in their history and benchmarks; the survey treats open-set recognition as multi-class novelty detection [3]. All of them differ from the **reject option** of PR 01, which abstains on *ambiguous* inputs between known classes rather than on *novel* ones.

Scores

Plain version. Each detector boils down to one number per input. Output-based scores ask how confident the classifier is; feature-based scores ask how far the input's features lie from the training features.

Let $\mathbf{z}(\mathbf{x}) \in \mathbb{R}^K$ be the logits of a trained classifier, $\mathbf{h}(\mathbf{x}) \in \mathbb{R}^m$ its penultimate features (the input to the last linear layer), and adopt the convention that a **larger score means more in-distribution**. Four standard post-hoc scores, which need no retraining:

Maximum softmax probability (MSP) [34]:

$$s_{\text{MSP}}(\mathbf{x}) = \max_k \frac{e^{z_k(\mathbf{x})}}{\sum_{j=1}^{K} e^{z_j(\mathbf{x})}},$$

where z_k is the k-th logit. Hendrycks and Gimpel showed that correctly classified examples tend to have higher MSP than misclassified and OOD ones, which made MSP the standard baseline [34].

Energy score [35]:

$$E(\mathbf{x}) = -T \log \sum_{k=1}^{K} e^{z_k(\mathbf{x})/T}, \qquad s_{\text{energy}}(\mathbf{x}) = -E(\mathbf{x}),$$

where $T > 0$ is a temperature (usually $T = 1$). The negative energy is a log-sum-exp of the logits, a smooth maximum that keeps the *size* of the logits, which the softmax normalizes away. Liu et al. argued that energy is better aligned with the input density than the softmax confidence, and reported lower FPR at 95% TPR than MSP on their CIFAR benchmarks [35].

Mahalanobis distance [36]: fit one Gaussian per class to the training features, with class means $\boldsymbol{\mu}_c$ and one shared ("tied") covariance $\boldsymbol{\Sigma}$, and score

$$s_{\text{Maha}}(\mathbf{x}) = -\min_c \, (\mathbf{h}(\mathbf{x}) - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{h}(\mathbf{x}) - \boldsymbol{\mu}_c),$$

where the minimum runs over the K classes. This is the linear discriminant analysis of PR 01 placed on learned features. Lee et al. also added a small input perturbation and combined several layers; we use the last hidden layer only.

k-nearest-neighbour (kNN) distance [37]: normalize features to unit length, $\hat{\mathbf{h}} = \mathbf{h}/\|\mathbf{h}\|_2$, and score

$$s_{\text{kNN}}(\mathbf{x}) = -\|\hat{\mathbf{h}}(\mathbf{x}) - \hat{\mathbf{h}}_{(k)}\|_2,$$

where $\hat{\mathbf{h}}_{(k)}$ is the k-th nearest normalized training feature to $\hat{\mathbf{h}}(\mathbf{x})$. Sun et al. found that this non-parametric score, with no Gaussian assumption, works well and that feature normalization is critical [37].

These scores leave the classifier unchanged. Training-time methods change it: **Outlier Exposure** trains the classifier to give uniform (low-confidence) predictions on a large auxiliary set of outlier images, which improved detection of unseen OOD data in the original experiments [38].

Metrics

Plain version. Treat detection as a binary decision, ID versus OOD, and ask how well the score separates the two groups, either over all thresholds or at one threshold that keeps most ID inputs.

Treat ID test inputs as positives and OOD inputs as negatives, and accept an input when $s(\mathbf{x}) \geq \tau$. Then $\text{TPR}(\tau)$ is the fraction of ID inputs accepted and $\text{FPR}(\tau)$ the fraction of OOD inputs accepted. Three metrics are standard [35]:

- **AUROC**, the area under the curve of TPR against FPR as τ varies. It equals the probability that a random ID input gets a higher score than a random OOD input; 0.5 is chance, 1 is perfect separation.
- **FPR@95TPR** $= \text{FPR}(\tau_{95})$, where τ_{95} is the threshold with $\text{TPR}(\tau_{95}) = 0.95$: the fraction of OOD inputs still accepted when 95% of ID inputs are kept. Lower is better. It describes one operating point, the kind a deployed system actually uses.
- **AUPR**, the area under the precision–recall curve, reported with ID inputs as the positive class (AUPR-In) or with OOD inputs as the positive class (AUPR-Out) [34]. It depends on the ratio of ID to OOD test inputs, which should therefore be reported.

ROC and precision–recall curves themselves are explained in DLCV 02.

Code: four scores, near and far OOD

ID data are the digits 0–4. **Near-OOD** inputs are the unseen digits 5–9: the same kind of image, new classes. **Far-OOD** inputs are 8×8 patches of photographs from scikit-image, rescaled to the same pixel range: a different kind of image altogether. The classifier never sees either during training.

```
import numpy as np
from scipy.special import logsumexp, softmax
from skimage import data, transform
from sklearn.datasets import load_digits
from sklearn.metrics import roc_auc_score
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier

X, y = load_digits(return_X_y=True)
X = X / 16.0
known = y < 5                                          # in-distribution (ID): digits 0-4
Xtr, Xid, ytr, yid = train_test_split(X[known], y[known], test_size=0.3, stratify=y[known],
 ↪  random_state=0)
X_near = X[~known]                                     # near-OOD: unseen digits 5-9
rng = np.random.default_rng(0)
photos = [transform.resize(im / 255.0, (256, 256)) for im in (data.camera(), data.moon(),
 ↪  data.coins(), data.text())]
X_far = []                                             # far-OOD: 8x8 patches of photographs
for _ in range(600):
    im = photos[rng.integers(4)]; r, c = rng.integers(0, 224, 2)
    p = transform.resize(im[r:r + 32, c:c + 32], (8, 8), anti_aliasing=True)
    X_far.append(((p - p.min()) / (np.ptp(p) + 1e-8)).ravel())
X_far = np.array(X_far)

m = MLPClassifier(hidden_layer_sizes=(64,), alpha=1e-3, max_iter=500,
 ↪  random_state=0).fit(Xtr, ytr)
(W1, W2), (b1, b2) = m.coefs_, m.intercepts_
feat = lambda X: np.maximum(X @ W1 + b1, 0)            # penultimate (hidden) features
unit = lambda F: F / (np.linalg.norm(F, axis=1, keepdims=True) + 1e-12)
F = feat(Xtr)
mu = np.stack([F[ytr == c].mean(0) for c in range(5)])           # class means
cov = sum(np.cov(F[ytr == c].T, bias=True) * np.sum(ytr == c) for c in range(5)) / len(F)
P = np.linalg.inv(cov + 1e-3 * np.eye(len(cov)))                # shared precision matrix

def scores(X):                                         # every score: larger = "more
 ↪  in-distribution"
    H = feat(X); z = H @ W2 + b2; d = H[:, None] - mu[None]
    return {"MSP": softmax(z, axis=1).max(1),
            "energy": logsumexp(z, axis=1),            # = -E(x) with temperature 1
            "Mahalanobis": -np.einsum("ncd,de,nce->nc", d, P, d).min(1),
            "kNN": -np.sort(np.linalg.norm(unit(H)[:, None] - unit(F)[None], axis=2), 1)[:,
             ↪  9]}

s_id = scores(Xid)
```

```
print(f"ID test accuracy {m.score(Xid, yid):.3f}")
for name, X_ood in [("near", X_near), ("far", X_far)]:
    s_ood = scores(X_ood)
    for k in s_id:
        auroc = roc_auc_score(np.r_[np.ones(len(Xid)), np.zeros(len(X_ood))], np.r_[s_id[k],
        ↪  s_ood[k]])
        fpr95 = np.mean(s_ood[k] >= np.quantile(s_id[k], 0.05))  # OOD accepted when 95% of
        ↪  ID is kept
        print(f"{name:4s} {k:11s} AUROC {auroc:.3f}  FPR@95TPR {fpr95:.3f}")
```

It prints:

```
ID test accuracy 1.000
near MSP         AUROC 0.941  FPR@95TPR 0.291
near energy      AUROC 0.955  FPR@95TPR 0.258
near Mahalanobis AUROC 0.961  FPR@95TPR 0.209
near kNN         AUROC 0.970  FPR@95TPR 0.152
far  MSP         AUROC 0.870  FPR@95TPR 0.448
far  energy      AUROC 0.811  FPR@95TPR 0.660
far  Mahalanobis AUROC 1.000  FPR@95TPR 0.000
far  kNN         AUROC 0.995  FPR@95TPR 0.010
```

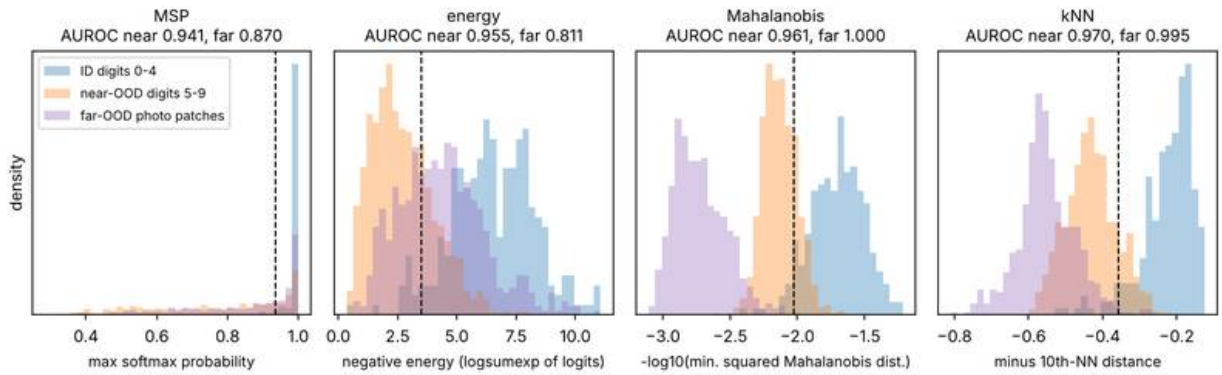


Figure 4 — Score distributions for ID digits 0–4, near-OOD digits 5–9 and far-OOD photograph patches (our toy run). The dashed line is the threshold that accepts 95% of ID test inputs; everything to its right is accepted. The Mahalanobis panel uses a logarithmic scale for display only, which does not change any ranking.

(All numbers are from our toy run.) Three lessons:

- **"Far" in content is not "easy" for every score.** The photo patches are nothing like digits, yet MSP accepts 45% and energy 66% of them at the 95%-TPR threshold. The classifier maps them confidently onto some digit, because confidence measures distance from the decision boundary, not distance from the data. Feature-distance scores see that the patches' features are far from all training features and reject almost all of them.

- **The ranking of scores changes with the OOD set.** Energy beats MSP on the unseen digits but loses to it on the patches. A large benchmark study, OpenOOD v1.5, likewise found no single method that wins across datasets [39].
- **AUROC and FPR@95TPR disagree in size.** An AUROC of 0.941 sounds good, but at a deployable threshold MSP still accepts 29% of unseen digits.

PR 10 runs a related open-set experiment with held-out digits on a different classifier and shows how a proxy-unknown set can mislead (PR 10).

Industrial anomaly detection

Plain version. On a production line, defects are rare and varied, so you cannot collect examples of every defect. Instead, you learn only what *normal* parts look like, patch by patch, and flag any patch that looks unlike every normal patch you have seen.

MVTec AD made this setting a standard benchmark: 5354 high-resolution images of object and texture categories, defect-free images for training, test images with more than 70 types of defects (scratches, dents, contaminations, structural changes), and pixel-precise masks of the defect regions [40]. Methods are scored at two levels: **image-level** (is this part defective?) and **pixel-level** (where is the defect?), usually with AUROC; region-overlap measures such as AU-PRO are also used, for example by MVTec AD 2 [41]. This is *sensory* anomaly detection in the terms of the table above.

PatchCore [42] is a strong, simple method built on the kNN idea. It passes defect-free training images through a network pre-trained on ImageNet and keeps **mid-level patch features** (the last layers are too specific to ImageNet classes and too coarse for small defects), each averaged over a small neighbourhood for context. All these features go into a **memory bank**, which **greedy coreset subsampling** shrinks to a small subset that still covers rare normal patterns. At test time, each patch's **distance to its nearest neighbour** in the memory bank forms the anomaly map, and the map's maximum (slightly re-weighted) is the image score. The method was designed for the "cold-start" setting in which only nominal (defect-free) images are available, and it reached state-of-the-art detection and localization on MVTec AD [42].

Optional: PatchCore step by step (in the Supplementary notes). The neighbourhood aggregation, the min-imax coreset objective and its greedy solution, and the score re-weighting, with every symbol defined.

Code: a patch memory bank on a brick wall

Without network weights we use raw 7×7 pixel patches as features, which keeps the mechanics of PatchCore (memory bank, coreset, nearest-neighbour distance map, max-pooled image score) and drops the learned features. "Products" are 64×64 crops of scikit-image's brick-wall texture: normal training crops from the top half of the image, test crops from the bottom half, and half of the test crops receive a thin synthetic scratch.

```
import numpy as np
from skimage import data, draw, transform
from skimage.util import view_as_windows
from sklearn.metrics import roc_auc_score
from sklearn.neighbors import NearestNeighbors

rng = np.random.default_rng(0)
tex = transform.rescale(data.brick() / 255.0, 0.5)   # 256x256 brick-wall "product surface"
S, P = 64, 7                                          # image size, patch size

def crop(top):                                        # normal images: top half; test
 ↪  images: bottom half
    r = rng.integers(0, 128 - S) + (0 if top else 128); c = rng.integers(0, 256 - S)
    return tex[r:r + S, c:c + S].copy()

def scratch(im):                                      # a thin line, 0.5 brighter or darker
    r, c = rng.integers(16, S - 16, 2); a = rng.uniform(0, np.pi); L = rng.integers(8, 16)
    rr, cc = draw.line(r, c, int(r + L * np.sin(a)), int(c + L * np.cos(a)))
    mask = np.zeros_like(im, bool); mask[rr, cc] = True
    im[mask] = np.clip(im[mask] + rng.choice([-0.5, 0.5]), 0, 1)
    return im, mask

def patches(im, step=1):                              # one 7x7 patch centred on each pixel
    return view_as_windows(np.pad(im, P // 2, mode="reflect"), (P, P), step).reshape(-1, P *
     ↪  P)

def greedy_coreset(M, n):                             # keep n patches that cover the bank
 ↪  well
    keep, d = [0], np.linalg.norm(M - M[0], axis=1)
    for _ in range(n - 1):
        i = int(d.argmax()); keep.append(i)
        d = np.minimum(d, np.linalg.norm(M - M[i], axis=1))
    return M[keep]

bank = np.vstack([patches(crop(top=True), step=2) for _ in range(16)])
memory = NearestNeighbors(n_neighbors=1).fit(greedy_coreset(bank, len(bank) // 10))

is_defect, img_score, pix_score, pix_true = [], [], [], []
for k in range(40):
    im, mask = scratch(crop(top=False)) if k % 2 else (crop(top=False), np.zeros((S, S),
     ↪  bool))
    amap = memory.kneighbors(patches(im))[0].reshape(S, S)   # nearest-neighbour distance
     ↪  per pixel
    is_defect.append(k % 2); img_score.append(amap.max())
```

```
    pix_score.append(amap.ravel()); pix_true.append(mask.ravel())
print(f"memory bank {len(bank)} -> {len(bank) // 10} patches")
print(f"image-level AUROC {roc_auc_score(is_defect, img_score):.3f}")
print(f"pixel-level AUROC {roc_auc_score(np.concatenate(pix_true),
 ↪  np.concatenate(pix_score)):.3f}")
```

It prints:

```
memory bank 16384 -> 1638 patches
image-level AUROC 0.955
pixel-level AUROC 0.996
```

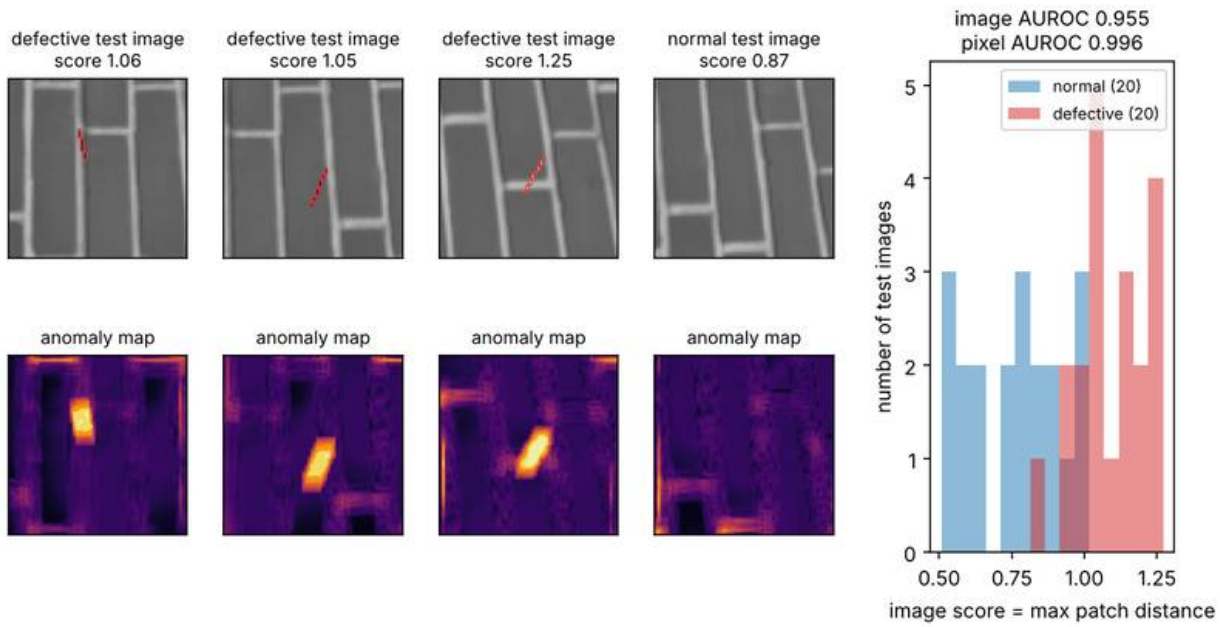


Figure 5 — A patch memory bank on brick-wall crops (our toy run). Top: test images (red outline: the synthetic scratch). Bottom: nearest-neighbour distance maps. Right: image scores (maximum of the map) for 20 normal and 20 defective test images.

(All numbers are from our toy run.) The pixel-level AUROC is almost perfect, but the image-level AUROC is lower: one image score is the *maximum* over 4096 patch distances, so a single unusual but normal patch (a mortar corner that the memory bank lacks) raises a normal image's score as much as a real defect does. More normal training images reduce such false alarms; fainter defects increase them (Exercise 4). In the anomaly maps, mortar edges glow a little: they are the rarest normal structures, which is exactly why PatchCore keeps a coreset that covers them instead of a random subsample.

Pitfalls in OOD and anomaly evaluation

Plain version. OOD numbers depend more on *which* OOD data was chosen than on the method, and many published OOD test sets are flawed. Read every OOD result together with its test sets.

- **Near- versus far-OOD.** OpenOOD groups OOD test sets into near-OOD (similar to the ID data) and far-OOD (clearly different), and finds a large gap in detection performance between the two groups for all detectors [39]. Our toy shows that the label can mislead: for output-based scores, "far" photo patches were harder than "near" digits.
- **Contaminated OOD test sets.** Bitterwolf et al. found that many OOD test sets used with ImageNet-1K contain a large share of images that actually show ID classes, in some cases more than half, which distorts the evaluation; they built NINCO, an OOD test set with every image checked to be free of ID objects [43].
- **Pre-training overlap.** A detector built on a model pre-trained on web-scale data may have seen the "unknown" classes during pre-training. Then the test is no longer about the unknown, and the setting must be described accordingly.
- **Thresholds and model selection.** Like any hyperparameter, the threshold and the choice of score must be set on validation data. Tuning them on the OOD test set (or on a proxy that is too similar to it) gives the same optimism as the oracle model selection that DomainBed warns against [19].
- **Covariate shift inside the ID set.** In deployment, an ID image may arrive blurred or under new lighting. OpenOOD v1.5 evaluates "full-spectrum" detection, in which covariate-shifted ID images must be *accepted* while semantic OOD images are rejected, and finds it a significant challenge for all current approaches [39]. This is where the first and third problems of this chapter meet.

A robustness evaluation plan

Plain version. Before claiming a model is robust, write down what it is robust *to*, test that with the strongest tools available, and report the result next to the clean number.

1. **Name the problem.** Shift (which kind?), adversary (which threat model?), or unknown inputs (which kind of novelty?). If several, evaluate each separately.
2. **Build the test data first.** At least one real shifted set from the expected deployment (new site, device or time), synthetic corruptions as a cheap supplement, and a held-out OOD set that is checked for ID contamination.
3. **Use strong evaluation.** For adversarial claims: white-box attacks, PGD with restarts and AutoAttack or an adaptive attack, the warning-sign checks above, and the exact ε and norm. For

shift: effective robustness or matched clean accuracy. For OOD: AUROC and FPR@95TPR on near- and far-OOD sets.

4. **Keep the test data out of the loop.** Choose hyperparameters, thresholds and checkpoints on validation data from the training distribution, or state clearly that an oracle was used.
5. **Report the trade-offs.** Clean accuracy, robust accuracy, calibration, and compute cost together; a gain in one is not free.
6. **Monitor after release.** Track input statistics and confidence distributions, as in ML 10, and in multi-sensor systems the sensor faults listed in DLCV 09.

Modern view

Surveys worth reading.

- **Yang, Zhou, Li and Liu [3]** (IJCV 2024) unify anomaly detection, novelty detection, open-set recognition, OOD detection and outlier detection as sub-tasks of *generalized OOD detection*, organized by whether covariate or semantic shift must be detected and by whether ID data has one or several classes; they then review each area, with most detail on OOD detection methods (classification-based, density-based and distance-based). Read it first to sort out the vocabulary.
- **Zhou et al. [18]** (TPAMI) define domain generalization, relate it to domain adaptation and transfer learning, and review methods grouped into domain alignment, meta-learning, data augmentation, ensemble learning and others, together with theory and applications beyond vision.
- **Liang, He and Tan [44]** (IJCV) categorize test-time adaptation by the form of test data into test-time domain adaptation, test-time batch adaptation and online test-time adaptation, with a taxonomy of algorithms in each group and a discussion of open challenges.
- **Akhtar, Mian, Kardan and Shah [45]** (IEEE Access 2021) review adversarial attacks and defenses in computer vision since 2018: attacks on classifiers, attacks beyond classification, physical-world attacks, theories of why adversarial examples exist, and defenses grouped into model modification, input modification and external detectors, with certified defenses treated separately.
- **Pang, Shen, Cao and van den Hengel [46]** (ACM Computing Surveys 2021) organize deep anomaly detection into three frameworks: deep learning for generic feature extraction, learning representations of normality, and end-to-end anomaly score learning, with eleven finer categories, and discuss each one's assumptions, advantages and disadvantages.
- **Liu et al. [47]** (Machine Intelligence Research 2024) review industrial image anomaly detection by network architecture, level of supervision, loss function, metrics and datasets, and propose

a setting closer to real manufacturing.

The takeaways they share. Distribution shift, adversarial robustness and OOD detection grew as separate communities with separate benchmarks, and the surveys repeatedly find that benchmark choice, model selection and evaluation protocol drive results as much as methods do. Simple baselines (ERM with good training, MSP or kNN scores, adversarial training with PGD) remain hard to beat when compared fairly.

State of the art, 2023–2026. Five trends are visible. (1) **Foundation models changed the starting point.** Large, diverse pre-training gives much of the natural-shift robustness that specialized methods could not [12], [13], and robust fine-tuning such as weight interpolation tries to keep it [14]; OOD benchmarks now include CLIP and DINOv2 backbones [39]. (2) **Generated data for adversarial training.** Training on images from a modern diffusion model, without external real data, set new results on RobustBench for CIFAR-10 and CIFAR-100 [48]; robust accuracy on these benchmarks still lies far below clean accuracy. (3) **Better OOD evaluation.** Unified benchmarks with near- and far-OOD splits and full-spectrum settings [49], [39] and cleaned test sets [43] replaced ad hoc comparisons. (4) **Zero- and few-shot anomaly detection.** WinCLIP uses CLIP with prompt ensembles over state words and window-level features to classify and segment anomalies with no or very few normal images [50]. (5) **Harder industrial benchmarks.** Because results on MVTec AD saturated, MVTec AD 2 adds transparent and overlapping objects, unusual illumination, high variance of normal data, very small defects and test scenarios with lighting changes, and reports that state-of-the-art methods remain far from solving it [41].

Open problems. Robustness to natural shifts that no benchmark anticipated; adversarial robustness beyond small ℓ_p balls (patches, physical changes, semantic edits) and at the scale of foundation models; certificates that are tight at useful radii for large images; OOD detection when the backbone may have seen everything during pre-training; detectors that accept covariate-shifted ID inputs while rejecting semantic novelty; and a common evaluation protocol that tests all three problems of this chapter on the same model.

Key takeaways

- Robustness has three different questions: does accuracy survive a change of data (shift), does the label survive a worst-case small change (adversarial), and can the model say "not my task" (OOD and anomalies). Each needs its own test data and metric.
- Shifts are covariate, label or concept shifts; benchmarks probe corruptions, renditions, natural hard images, controlled collection, re-collection and real deployments. Compare interventions at matched clean accuracy (effective robustness).
- What helps against shift: diverse augmentation, more diverse pre-training data, and, when test

data arrive in batches, re-estimated normalization statistics or entropy-based test-time adaptation. Domain adaptation needs unlabeled target data; domain generalization methods rarely beat well-tuned ERM.
- An adversarial claim must state its threat model (ℓ_p norm, ε, white- or black-box). PGD with a random start is the standard first-order attack; adversarial training (training on PGD examples) is the most reliable defense and costs compute, capacity and some clean accuracy.
- Weak evaluations overstate robustness: watch for gradient masking, use AutoAttack or adaptive attacks, and remember that certificates such as randomized smoothing prove robustness only for their own norm and radius.
- OOD scores come from outputs (MSP, energy) or features (Mahalanobis, kNN); none wins everywhere. Report AUROC and FPR@95TPR on several OOD sets, near and far, checked for ID contamination.
- Industrial anomaly detection learns normal patches only; a memory bank with nearest-neighbour distance (PatchCore) gives image scores and defect maps, and image-level decisions are harder than pixel-level ranking.

Exercises

1. **Test-batch normalization with small or mixed batches.** In the corruption snippet, apply test-batch normalization to batches of 8 test images instead of all 540, and then to one batch that mixes all three corruptions at severity 3. How does accuracy change, compared with the plain model and the noise-augmented one? What does this say about deploying test-time adaptation on a camera that switches between day and night?

   Hint

   Split `Xs` into chunks with `np.array_split(Xs, len(Xs) // 8)`, normalize each chunk with its own statistics, and concatenate the predictions. In our run at severity 3, batches of 8 kept part of the gain (noise: 0.86 with the full batch, 0.74 with batches of 8, 0.56 for the plain model; blur: 0.95, 0.87, 0.82). A batch mixing the three corruptions still helped here (0.91 against 0.55 for the plain model), because all three corruptions change the per-pixel statistics in a similar direction (less contrast, more spread); do not expect that in general. Small batches give noisy statistics, and a batch that mixes very different conditions (day and night) gives statistics that fit neither, so real TTA methods need reasonably large batches from one condition, or a way to detect a change in the stream and reset.

2. **A defense that only breaks the gradient.** Add a "defense" to the adversarial snippet that rounds every input pixel to the nearest multiple of 0.25 before the standard model sees it, and recompute the input gradient through the rounding (it is zero almost everywhere). What do FGSM and PGD report? Then attack again with BPDA, treating the rounding as the identity in the backward pass only. Which warning signs from the evaluation section did the first result show?

   Hint

With a zero gradient, `np.sign` returns zeros and PGD never moves from its random start. In our run the "defended" model kept 0.96 accuracy under this PGD at $\varepsilon = 0.15$, while random $\pm\varepsilon$ sign noise did better as an attack (0.93): random sampling beating a gradient attack is one of the warning signs. With BPDA, use `input_grad(m, quantize(Xa), y)` as the gradient at `Xa`. At $\varepsilon = 0.1$ the attack is weaker than on the undefended model (0.79 against 0.53), because rounding to steps of 0.25 undoes perturbations smaller than half a step; at $\varepsilon = 0.15$ BPDA drives accuracy to 0.01, below the undefended model's 0.18. This is the shattered-gradient case of Athalye et al.

3. **Accuracy–robustness trade-off.** Train the robust model with training budgets $\varepsilon \in \{0.05, 0.1, 0.15, 0.2\}$ and plot clean accuracy against PGD accuracy at $\varepsilon = 0.1$. Where is the knee of the curve? What changes if you increase the hidden layer from 128 to 512 units?

   Hint

   In our runs with 128 hidden units, clean accuracy fell from 0.972 to 0.941 as the training budget grew from 0.05 to 0.2, while PGD accuracy at $\varepsilon = 0.1$ rose from 0.706 to 0.754 (at 0.1 and 0.15) and then dropped to 0.733: training far beyond the test budget costs clean accuracy without further gains. With 512 hidden units, both numbers improved at every budget (for example 0.980 clean and 0.806 robust at a training budget of 0.1), in line with Madry et al.'s finding that robustness needs capacity. Use the same number of epochs and the same evaluation attack for every setting; exact values change a little with the random seed.

4. **Faint defects and more normal data.** In the brick-wall snippet, change the scratch strength from 0.5 to 0.35, and then also increase the number of normal training crops from 16 to 32. Report image- and pixel-level AUROC in each case, and explain the difference between the two metrics.

   Hint

   In our runs, the image-level AUROC dropped clearly at strength 0.35 with 16 crops (to about 0.70) and recovered partly with 32 crops (to about 0.88), while the pixel-level AUROC stayed above 0.99 in both cases. Image-level scores take the maximum over 4096 patches, so they are hurt by the rarest normal patches; more normal data puts more of those patches into the memory bank. Pixel-level AUROC pools millions of pixels, almost all normal, and ranks most scratch pixels above them even when a few normal patches are high. Expect some variation with the random seed, because there are only 40 test images.

5. **Choose the right tool.** For each scenario, say which of the three robustness problems it is, which evaluation you would run first, and which method you would try first: (a) a skin-lesion classifier moved to a clinic with a different dermatoscope; (b) a content filter that users deliberately try to bypass with small image edits; (c) a wildlife camera-trap classifier that starts seeing a species not in its label set; (d) a weld-inspection camera with no images of defective welds.

   Hint

(a) Covariate (domain) shift: collect a labelled test set from the new clinic; try normalization-statistics adaptation or fine-tuning on a few labelled target images, and compare with a more diverse pre-trained backbone at matched clean accuracy. (b) Adversarial, with an explicit threat model (which edits are allowed?); evaluate with white-box and transfer attacks, and try adversarial training; for patch-like edits, ℓ_p robustness is not enough. (c) Semantic shift: OOD or open-set detection; evaluate AUROC and FPR@95TPR with held-out species, using feature-distance scores on a good backbone. (d) Sensory anomaly detection: a patch memory bank on normal welds; evaluate with a small set of real defects collected later, at image and pixel level.

Supplementary notes

These notes are optional background that the main text does not depend on.

Effective robustness and accuracy on the line

This note explains how a robustness scatter plot is read, how the baseline trend is fitted, and what "effective robustness" measures beyond plain accuracy.

The scatter plot. Evaluate many models $f_1, \dots, f_n$ on a clean test set and on a shifted test set. Plot one point per model, with clean accuracy $\mathrm{acc}_1(f)$ on the horizontal axis and shifted accuracy $\mathrm{acc}_2(f)$ on the vertical axis. A model on the diagonal $\mathrm{acc}_2 = \mathrm{acc}_1$ is unaffected by the shift; real models lie below it.

The baseline. Taori et al. fitted a trend line $\beta(\cdot)$ through standard models without any robustness intervention, with both axes in logit scale, $\log(a/(1-a))$ for an accuracy a [8]. Miller et al. used the probit transform $\Phi^{-1}(\mathrm{acc})$, where Φ^{-1} is the inverse standard normal cumulative distribution function, and found that on many shifts the points of very different models fall close to a straight line in that scale [9]. The transform spreads out accuracies near 0 and 1, where differences are compressed on a linear scale.

Two numbers per intervention. For a model f, the **effective robustness** is

$$\rho(f) = \mathrm{acc}_2(f) - \beta(\mathrm{acc}_1(f)),$$

the vertical distance of its point above the baseline trend [8]. A model with $\rho = 0$ is exactly as robust as its clean accuracy predicts. For a model f' obtained from f by a robustness intervention (an augmentation, a loss), the **relative robustness** is

$$\tau(f') = \mathrm{acc}_2(f') - \mathrm{acc}_2(f),$$

the change in shifted accuracy itself [8]. An intervention can raise ρ while lowering both accuracies, which would be no use; Taori et al. therefore ask for positive values of both.

Edge cases. The baseline depends on which models define it (for example, ImageNet-trained convolutional networks), so ρ is relative to a population. The linear relation is an empirical regularity,

not a law: Miller et al. reported weaker correlations for some synthetic corruptions of CIFAR-10 and for a medical tissue dataset in WILDS [9]. And a high ρ on one shift says nothing about another; zero-shot CLIP's large effective robustness on natural ImageNet shifts largely disappeared after supervised fitting on ImageNet [12].

How to use it. When a paper reports "better robustness", look for the scatter plot. If the new model lies on the same line as the baselines, the gain is explained by its clean accuracy.

Test-time entropy minimization

This note describes what TENT optimizes during testing, which parameters it touches and why, and when it can fail.

Setting. A model with parameters $\boldsymbol{\theta}$ was trained on a source distribution. At test time, unlabeled batches $\mathbf{X}_t = \{\mathbf{x}_{t,1}, \dots, \mathbf{x}_{t,B}\}$ arrive from a shifted distribution, one batch at a time (t indexes the batch, B is the batch size). No source data and no labels are available; this is *fully* test-time adaptation [16].

Objective. For a prediction $\hat{\mathbf{p}} = \mathrm{softmax}(\mathbf{z})$ over K classes, the Shannon entropy is

$$H(\hat{\mathbf{p}}) = -\sum_{k=1}^{K} \hat{p}_k \log \hat{p}_k,$$

which is 0 for a one-hot prediction and $\log K$ for a uniform one. TENT minimizes the mean entropy of the predictions on the current batch, $\frac{1}{B}\sum_b H(\hat{\mathbf{p}}_{t,b})$, with one gradient step per batch. In the paper's default online scheme, the current batch's normalization statistics are used at once, while the affine update follows the prediction for the current batch and so takes effect from the next batch on [16]. The motivation in the paper is empirical: lower-entropy predictions tend to be more accurate, and stronger corruption tends to raise entropy [16].

Which parameters. Only the normalization layers change. For each channel j, a batch-normalization layer computes $\gamma_j(h_j - \mu_j)/\sigma_j + \beta_j$, where h_j is the channel's activation, μ_j and σ_j its mean and standard deviation, and γ_j, β_j a learned scale and shift. TENT (i) replaces μ_j and σ_j by the statistics of the current test batch, as Schneider et al. do [15], and (ii) updates only γ_j and β_j by the entropy gradient, keeping all convolution weights fixed [16]. These channel-wise parameters are a tiny fraction of the model, which makes online updates cheap and limits how far the model can move.

Failure cases. Entropy minimization rewards confidence, not correctness. If the batch is small or mixes several conditions, the statistics are poor (Exercise 1); if the shift is so strong that most predictions are wrong, minimizing entropy makes the model confidently wrong; and if the stream changes, accumulated updates can drift. Label shift inside a batch (one class dominating) also biases the batch statistics. The TTA survey discusses such settings and the methods proposed for them [44].

Domain-adversarial training and the gradient-reversal layer

This note writes DANN's objective, shows how one layer implements its min-max structure, and lists when feature alignment can hurt.

Model. A feature extractor $G_f(\cdot;\boldsymbol{\theta}_f)$ maps an image to features; a label predictor $G_y(\cdot;\boldsymbol{\theta}_y)$ maps features to class probabilities; a domain classifier $G_d(\cdot;\boldsymbol{\theta}_d)$ maps features to the probability that the image comes from the target domain [17]. Training data are labelled source images $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ and unlabeled target images $\{\mathbf{x}_i\}_{i=n+1}^{N}$, with domain labels $d_i = 0$ for source and $d_i = 1$ for target.

Objective. With $\mathcal{L}_y$ the classification loss on source images and $\mathcal{L}_d$ the domain-classification loss on all images,

$$E(\boldsymbol{\theta}_f, \boldsymbol{\theta}_y, \boldsymbol{\theta}_d) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}_y^i(\boldsymbol{\theta}_f, \boldsymbol{\theta}_y) - \lambda \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}_d^i(\boldsymbol{\theta}_f, \boldsymbol{\theta}_d),$$

where $\lambda \geq 0$ weights the two terms. DANN seeks a saddle point: $(\boldsymbol{\theta}_f, \boldsymbol{\theta}_y)$ minimize E (good labels, and features that make the domain classifier *fail*), while $\boldsymbol{\theta}_d$ maximizes E, that is, minimizes its own domain loss [17]. (We write the loss in this simplified form; the paper's normalization of the terms differs in detail.)

The gradient-reversal layer. Place a layer R between G_f and G_d that is the identity in the forward pass and multiplies the gradient by -1 in the backward pass:

$$R(\mathbf{v}) = \mathbf{v}, \qquad \frac{\partial R}{\partial \mathbf{v}} = -\mathbf{I},$$

where $\mathbf{v}$ is the feature vector and $\mathbf{I}$ the identity matrix. Then ordinary backpropagation on the label loss plus λ times the domain loss trains G_d to classify domains and, through the reversed gradient, trains G_f to confuse it [17]. No alternating optimization is needed; in a modern framework the layer is a two-line custom function.

Edge cases and pitfalls.

- *Label shift between domains.* If class proportions differ between source and target, perfectly aligned feature distributions force some target images of one class onto source regions of another class. Alignment of marginal feature distributions is not alignment of class-conditional ones.
- *Trivial alignment.* Features can become domain-invariant by discarding information the label task needs. The label loss counteracts this only on the source domain.
- *Model selection.* Without target labels, choosing λ and the stopping point is hard, the same problem that DomainBed raises for domain generalization [19].

Attack settings that decide an evaluation

This note lists the choices inside an attack that change the reported robust accuracy, and what AutoAttack automates.

Step size and number of steps. PGD with T steps of size α can move at most $T\alpha$ in each coordinate. With $T\alpha < \varepsilon$, it cannot reach the corners of the ℓ_∞ ball, where FGSM-like solutions lie. A common rule, used in our snippet, is $\alpha = 2.5\,\varepsilon/T$: Madry et al. used it with $T = 100$ in their evaluations at other budgets and in ℓ_2, so that the boundary of the ball can be reached from any starting point inside it [23]. Too large a step makes PGD oscillate between corners; too small a step stalls. AutoAttack's PGD variant replaces the fixed step by a schedule: at preset checkpoints it halves the step size if the loss has not increased often enough since the last checkpoint [30].

Random restarts. The loss is not concave in δ, so different random starts reach different local maxima. An evaluation should take, for each test image, the best result over several restarts; an image counts as robust only if every restart fails.

The loss. Cross-entropy can saturate: when the model is very confident, its gradient is tiny and the attack makes little progress, which is one mechanism of gradient masking. The Carlini–Wagner margin loss,

$$\mathcal{L}_{\mathrm{CW}}(\mathbf{x}', y) = \max_{k \neq y} z_k(\mathbf{x}') - z_y(\mathbf{x}'),$$

where z_k are the logits, becomes positive exactly when the attack succeeds and does not saturate in the same way [26]. AutoAttack adds the Difference of Logits Ratio loss, which is invariant to shifting and rescaling the logits, whereas cross-entropy is invariant only to shifting [30].

ℓ_2 PGD. For the ℓ_2 ball, the step follows the gradient direction normalized to a fixed length instead of its sign [23], $\mathbf{x}' \leftarrow \mathbf{x}' + \alpha\,\mathbf{g}/\|\mathbf{g}\|_2$ with $\mathbf{g} = \nabla_{\mathbf{x}}\mathcal{L}$, and the projection rescales δ to length ε when it is longer:

$\delta \leftarrow \delta \cdot \min(1, \varepsilon/\|\delta\|_2)$, followed by clipping to $[0, 1]$. (Clipping after rescaling keeps the perturbation inside the ball because clipping can only shorten it.)

Targeted versus untargeted, and which test images. Untargeted robust accuracy is the fraction of test images for which no attack in the ensemble finds any wrong label. Report it on the full test set (or a stated subset), together with clean accuracy on the same images; robust accuracy cannot exceed clean accuracy, because a misclassified clean image counts as already broken.

Checklist. Norm and ε (and the pixel scale it refers to); attacks used; steps, step size and restarts; losses; whether the attack was adapted to the defense; number of test images. AutoAttack's value is that most of these choices are fixed and the same for everyone [30], which is why RobustBench uses it [24].

Randomized smoothing in practice

This note goes through the certification procedure of Cohen et al., explains why the certified radius is capped, and gives a tested snippet on digits.

The procedure. Given a base classifier f, noise level σ, an input $\mathbf{x}$, sample sizes n_0 and n, and a failure probability α, the CERTIFY procedure is [31]:

1. Draw n_0 noisy copies $\mathbf{x} + \boldsymbol{\eta}$, classify them with f, and take the most frequent class $\hat{c}_A$ (a guess at the top class).
2. Draw n fresh noisy copies and count how many are classified $\hat{c}_A$; call the count k.
3. Compute a one-sided $(1-\alpha)$ lower confidence bound $\underline{p_A}$ on the binomial probability from k successes in n trials; Cohen et al. use the Clopper–Pearson bound.
4. If $\underline{p_A} > 1/2$, return $\hat{c}_A$ and the radius $R = \sigma\,\Phi^{-1}(\underline{p_A})$; otherwise abstain.

Step 4 uses the bound $\overline{p_B} = 1 - \underline{p_A}$ on all other classes, which turns the radius of the main text into $\frac{\sigma}{2}(\Phi^{-1}(\underline{p_A}) - \Phi^{-1}(1-\underline{p_A})) = \sigma\,\Phi^{-1}(\underline{p_A})$, because $\Phi^{-1}(1-p) = -\Phi^{-1}(p)$. With probability at least $1-\alpha$, a returned class is what the smoothed classifier g predicts everywhere within radius R [31]. Using separate samples for steps 1 and 2 keeps the bound valid: selecting the class and estimating its probability from the same samples would bias the estimate.

Why the radius is capped. Even if all n noisy copies agree ($k = n$), the Clopper–Pearson lower bound is $\underline{p_A} = \alpha^{1/n}$, so $R \le \sigma\,\Phi^{-1}(\alpha^{1/n})$. With $n = 2000$, $\alpha = 0.001$ and $\sigma = 0.5$, this cap is about 1.35 (our calculation). Larger radii need many more samples, and the paper notes that R grows only very slowly with n [31].

Training the base classifier. The certificate holds for any f, but it is useful only if f classifies noisy inputs well, so f is trained with Gaussian noise of the same σ added to its inputs.

```
import numpy as np
from scipy.stats import beta, norm
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier

X, y = load_digits(return_X_y=True)
X = X / 16.0
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=0)
rng = np.random.default_rng(0)
sigma = 0.5
Xnoisy = np.vstack([Xtr + rng.normal(0, sigma, Xtr.shape) for _ in range(4)])   # train f
 ↪  under noise
f = MLPClassifier(hidden_layer_sizes=(128,), alpha=1e-3, max_iter=200, early_stopping=True,
 ↪  random_state=0).fit(Xnoisy, np.tile(ytr, 4))
```

```
def certify(x, n0=100, n=2000, alpha=0.001):
    votes0 = np.bincount(f.predict(x + rng.normal(0, sigma, (n0, 64))), minlength=10)
    c = votes0.argmax()                                  # guess the top class with a small
     ↪  sample
    k = np.sum(f.predict(x + rng.normal(0, sigma, (n, 64))) == c)
    p_low = beta.ppf(alpha, k, n - k + 1)                # one-sided Clopper-Pearson lower
     ↪  bound
    return (c, sigma * norm.ppf(p_low)) if p_low > 0.5 else (-1, 0.0)   # -1 means abstain

res = [certify(x) for x in Xte[:200]]
for r in [0, 0.25, 0.5, 0.75, 1.0]:
    ok = np.mean([(c == t) and (R >= r) for (c, R), t in zip(res, yte[:200])])
    print(f"certified accuracy at l2 radius {r:.2f}: {ok:.3f}")
print("abstained:", sum(c == -1 for c, _ in res), "of", len(res))
```

It prints:

```
certified accuracy at l2 radius 0.00: 0.825
certified accuracy at l2 radius 0.25: 0.590
certified accuracy at l2 radius 0.50: 0.290
certified accuracy at l2 radius 0.75: 0.070
certified accuracy at l2 radius 1.00: 0.000
abstained: 32 of 200
```

(Our toy run.) **Certified accuracy** at radius r is the fraction of test images that are certified with $R \geq r$ *and* correctly classified. At $r = 0$ it is the accuracy of the smoothed classifier counting abstentions as errors. To put the radii in scale: an ℓ_2 change of 0.5 on these $[0,1]$ images equals, for example, a change of 0.0625 in every one of the 64 pixels. Note that the noise is not clipped to $[0,1]$; the theorem does not need it.

Edge cases. The certificate is for ℓ_2; an ℓ_∞ ball of radius ε in d dimensions contains ℓ_2 perturbations of length up to $\varepsilon\sqrt{d}$, so ℓ_2 certificates translate into weak ℓ_∞ ones for large images. The guarantee is about g, which is only available through sampling, so prediction is also randomized and costs n forward passes per image.

PatchCore step by step

This note gives the details of PatchCore's patch features, coreset selection and scoring that the main text summarized [42].

Patch features with local context. Let $\phi_j(\mathbf{x}) \in \mathbb{R}^{c^* \times h^* \times w^*}$ be the feature map at level j of an ImageNet-pre-trained network, with c^* channels on an $h^* \times w^*$ grid, and $\phi_j(\mathbf{x}, h, w) \in \mathbb{R}^{c^*}$ the feature vector at grid position (h, w). For an odd neighbourhood size p, the neighbourhood of (h, w) is

$$\mathcal{N}_p^{(h,w)} = \{(a,b) : |a-h| \leq \lfloor p/2 \rfloor,\ |b-w| \leq \lfloor p/2 \rfloor\},$$

and the locally aware feature is an aggregation, adaptive average pooling in the paper, of the feature vectors in that neighbourhood. By default, features from blocks 2 and 3 of the network are combined, so that each patch feature mixes finer and coarser information [42]. The set of all such features over all normal training images forms the memory bank $\mathcal{M}$.

Coreset selection. The goal is a subset $\mathcal{M}_C \subset \mathcal{M}$ of a chosen size such that every feature in $\mathcal{M}$ is close to some feature in $\mathcal{M}_C$. PatchCore uses the minimax facility location objective

$$\mathcal{M}_C^* = \arg\min_{\mathcal{M}_C \subset \mathcal{M}} \max_{\mathbf{m} \in \mathcal{M}} \min_{\mathbf{n} \in \mathcal{M}_C} \|\mathbf{m} - \mathbf{n}\|_2,$$

the largest distance from any stored feature to its nearest kept feature [42]. This problem is NP-hard, so it is approximated greedily: start from one feature, then repeatedly add the feature whose distance to the current subset is largest (the `greedy_coreset` function in the main snippet). The paper also applies a random linear projection to reduce the dimension during selection, to save time. A random subsample of the same size would keep mostly common patches and drop rare normal ones, which then look anomalous at test time.

Scoring. For a test image with patch features $\mathcal{P}(\mathbf{x}_{\text{test}})$, find the test patch that is farthest from its nearest memory feature,

$$\mathbf{m}^{\text{test},*}, \mathbf{m}^* = \arg\max_{\mathbf{m}^{\text{test}} \in \mathcal{P}(\mathbf{x}_{\text{test}})} \arg\min_{\mathbf{m} \in \mathcal{M}_C} \|\mathbf{m}^{\text{test}} - \mathbf{m}\|_2, \qquad s^* = \|\mathbf{m}^{\text{test},*} - \mathbf{m}^*\|_2.$$

The image score s re-weights s^* by how isolated $\mathbf{m}^*$ is within the memory bank: if the nearest memory feature is itself far from its own neighbours $\mathcal{N}_b(\mathbf{m}^*)$ (the b nearest memory features), it is a rare normal pattern, and the score is increased [42]:

$$s = \Big(1 - \frac{\exp\|\mathbf{m}^{\text{test},*} - \mathbf{m}^*\|_2}{\sum_{\mathbf{m} \in \mathcal{N}_b(\mathbf{m}^*)} \exp\|\mathbf{m}^{\text{test},*} - \mathbf{m}\|_2}\Big)\, s^*.$$

The per-patch distances, placed back at their grid positions, upsampled to the input resolution by bilinear interpolation and smoothed with a Gaussian, give the segmentation map [42].

Pitfalls. The memory bank grows with the number of training images and the feature resolution, so the coreset size trades speed against coverage. Normal variation that never appears in training (a new batch of material, new lighting) is flagged as anomalous, which is correct for a sensory anomaly detector but may not be what the factory wants; MVTec AD 2 adds lighting-change test scenarios for this reason [41]. And the ImageNet features may not suit very different imaging (X-ray, dark-field illumination).

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Madry et al., "Towards Deep Learning Models Resistant to Adversarial Attacks" (ICLR 2018)

The problem. By 2017 many defenses against adversarial examples had been proposed and quickly broken, and there was no agreed definition of what a defense should achieve [23].

The idea. State robustness as an optimization problem. The adversarial (robust) risk is the expected loss when an adversary may perturb each input within a set $\mathcal{S}$, and training minimizes it: the min-max problem in the main text. This one formula defines the attack (the inner maximization), the defense (the outer minimization) and the guarantee (a small value of the objective means low loss against every perturbation in $\mathcal{S}$). The authors argue that PGD with random starts is a strong, in some sense "universal", first-order adversary, and train against it.

What it showed. (1) The inner problem is tractable in practice: PGD from many random starts reaches loss values that are well concentrated, with no extreme outliers, suggesting that PGD finds near-maximal values within the reach of first-order methods. (2) Training against PGD reliably reduced the adversarial loss during training. (3) Capacity matters: larger networks were more robust, small networks trained against PGD could collapse, and FGSM-based training overfitted to its own narrow perturbations ("label leaking") without robustness to PGD at large ε. (4) The resulting MNIST ($\varepsilon = 0.3$) and CIFAR-10 ($\varepsilon = 8$ on the 0–255 scale) models resisted a range of white-box and transfer attacks, and the authors published them as open challenges.

How to read it. Read Section 2 (the saddle-point formulation) and Section 3 (the loss-landscape experiments) first. In the result tables, each row is an attack: its source (white-box on the same model, or black-box from an independently trained copy, a naturally trained copy, or a different architecture), its method (FGSM, PGD with given steps and restarts, or PGD with the Carlini–Wagner loss), and the resulting accuracy. Check: (a) the threat model, ℓ_∞ with the stated ε only; (b) that the strongest attack per column is the number to quote; (c) the additional experiments at other ε values and with ℓ_2-bounded attacks, which show how far robustness extends beyond the training threat model.

Limits. Robustness is shown only against the attacks tried and only for one norm and radius; it is empirical, not certified. Training cost grows with the number of PGD steps, and clean accuracy drops. Later work (TRADES, generated-data training, AutoAttack evaluations) refined both training and evaluation, but the formulation remained the reference point.

Why it matters here. The chapter's adversarial-training code is exactly this recipe on a small scale, and the min-max formula is the common language of almost every later robustness paper.

Athalye et al., "Obfuscated Gradients Give a False Sense of Security" (ICML 2018)

The problem. Many defenses reported high robustness to white-box iterative attacks. Were they robust, or did they only prevent the attacker's gradient from being useful [29]?

The idea. Identify the mechanism, gradient masking in the form of *obfuscated gradients*, and build attacks that do not rely on the masked gradient. The paper names three types (shattered, stochastic, and vanishing or exploding gradients), lists behaviors that reveal them (the warning signs in the main text), and for each type gives a counter-attack: BPDA, EOT and reparameterization.

What it showed. In a case study of the non-certified white-box defenses accepted at ICLR 2018, seven of nine relied on obfuscated gradients; the new attacks circumvented six completely and one partially, in the threat model each paper had considered [29]. Two certified defenses and one with only black-box claims were left out of the case study. The paper also stresses an asymmetry: to break a defense one attack suffices, while a defender must show that no attack within the threat model succeeds.

How to read it. Read Section 3 (the three types and their symptoms) and Section 4 (the attacks) first, then Table 1, which lists each defense with its dataset, its perturbation bound and the accuracy left under the new attacks. Note the two adversarial-training defenses at the bottom of the table: the authors found that Madry et al.'s PGD training does *not* rely on obfuscated gradients. Check: (a) that each attack keeps the original paper's threat model and ε, so the numbers are a fair re-evaluation; (b) the per-defense sections, which show how BPDA's approximation was chosen in each case; (c) the evaluation recommendations in the discussion.

Limits. The analysis is a case study of one conference's defenses; it shows that these defenses fail, not that all defenses of a type must fail. Adaptive attacks require effort and expertise, which is the gap that AutoAttack later tried to close for standard settings [30].

Why it matters here. It is the reason this chapter asks for white-box, adaptive and ensemble attacks before any robustness claim is believed, and the source of the warning signs used in Exercise 2.

Gulrajani and Lopez-Paz, "In Search of Lost Domain Generalization" (ICLR 2021)

The problem. Many domain generalization algorithms reported gains over plain training, but under different datasets, architectures and, above all, different (often unstated) rules for choosing hyperparameters, sometimes using the test domain [19].

The idea. Treat model selection as part of the problem. A DG algorithm must come with a rule for choosing hyperparameters and checkpoints that uses only training domains. The paper defines three rules: a **training-domain validation set** (hold out part of each training domain), **leave-one-domain-out cross-validation** (train with one training domain held out, in turn, and average), and **test-domain validation**, an *oracle* that peeks at labelled test-domain data and must be disclosed as such. It then implements DomainBed, a testbed with seven multi-domain datasets (including Colored MNIST, Rotated MNIST, VLCS, PACS, Office-Home, TerraIncognita and DomainNet),

nine algorithms and the three selection rules, with a random hyperparameter search of 20 trials per algorithm and test domain, and repeats the whole study three times.

What it showed. With a ResNet-50, strong data augmentation and careful tuning, ERM matched or beat all previously published results under training-domain validation, and no algorithm beat ERM by more than about a point on average under any selection rule. Training-domain validation worked better than leave-one-domain-out. The oracle rule gave higher numbers, which shows both possible headroom and how optimistic oracle-selected results are.

How to read it. The main table reports, for each selection rule, the average accuracy over test domains per dataset, with standard errors over the three repetitions. Read across a row to compare an algorithm with ERM on the same footing, and compare the same algorithm between tables to see the effect of the selection rule. Check: (a) the standard errors before believing any difference; (b) the appendix, which collects results reported in the literature for more than thirty algorithms, to see how far published numbers depart from the controlled ones.

Limits. The conclusion is about these datasets, these algorithms and this tuning budget; a method may still help elsewhere. The datasets are small-to-medium image benchmarks; foundation-model backbones, which changed the field later, are not part of the study.

Why it matters here. It is the clearest example of the chapter's recurring lesson: robustness gains disappear when baselines are tuned well and when test data stay out of model selection.

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DLCV 11A • Text in Images: OCR and Document Understanding

How machines read text in photos, scans and screenshots: the detection, recognition and layout pipeline and where each stage fails; regression and segmentation text detectors, CTC, attention and transformer recognizers, text spotting, LayoutLM-style and OCR-free document models; CER, WER and H-mean; the standard benchmarks and the failure modes that matter.

Online edition: https://blog.shyandram.dev/en/learn/deep-learning/dl-11a-text-in-images/

What you'll learn

- How a text-reading system is organized — **detection** (where is the text?), **recognition** (what does it say?) and **layout and understanding** (how do the pieces fit together?) — and which errors start at each stage.
- Why text needs its own detectors: quadrilateral regression (EAST), segmentation with shrunk text kernels (PSENet) and **differentiable binarization** (DBNet), with the formulas checked in NumPy.
- How a cropped word becomes a character string: CRNN with **CTC** decoding, attention decoders with rectification, transformer recognizers such as TrOCR, and the help and harm of a language model.
- How document models use layout: LayoutLM's 2-D position embeddings, key information extraction, and **OCR-free** models (Donut, Pix2Struct, vision-language models) that write the answer directly, sometimes including text that is not there.
- How OCR is measured — **CER and WER** from edit distance, detection **precision, recall and H-mean at IoU 0.5**, end-to-end and document metrics — on which benchmarks, and which failure modes the numbers hide.

The big picture

Text is everywhere in images: street signs, product labels, license plates, receipts, forms, slides and screenshots. Reading it turns pixels into a string that a program can search, translate, check or store. This is **optical character recognition** (**OCR**). The hard version, text in ordinary photos, is called **scene text** reading; the structured version, scans and PDFs of pages, is **document** analysis.

Explain like I'm five

Think of how you read a poster on a wall. First your eyes find the patches that have writing on them (that is **detection**). Then you look closely at one patch and sound out the letters, using what words usually look like to guess a smudged one (that is **recognition**). Finally you work out what the poster means: the big line at the top is the title, the small line at the bottom is the date, and the two columns are read one after the other (that is **layout and understanding**). A computer does the same three jobs, and it can make a mistake at each one.

Why a separate chapter? A word is long, thin, often rotated or curved, and packed close to its neighbors; its "class" is not a label from a fixed list but any string over an alphabet that may hold tens of thousands of symbols; and one wrong character can change the meaning ("1046" versus "1O46" on an invoice). The field therefore has its own detectors, sequence recognizers, metrics and benchmarks. It is now being reshaped by models that read a whole page and write structured text directly, which makes the pipeline optional but adds a new risk: fluent output that the image does not support. Figure 1 shows the pipeline and where each stage typically fails.

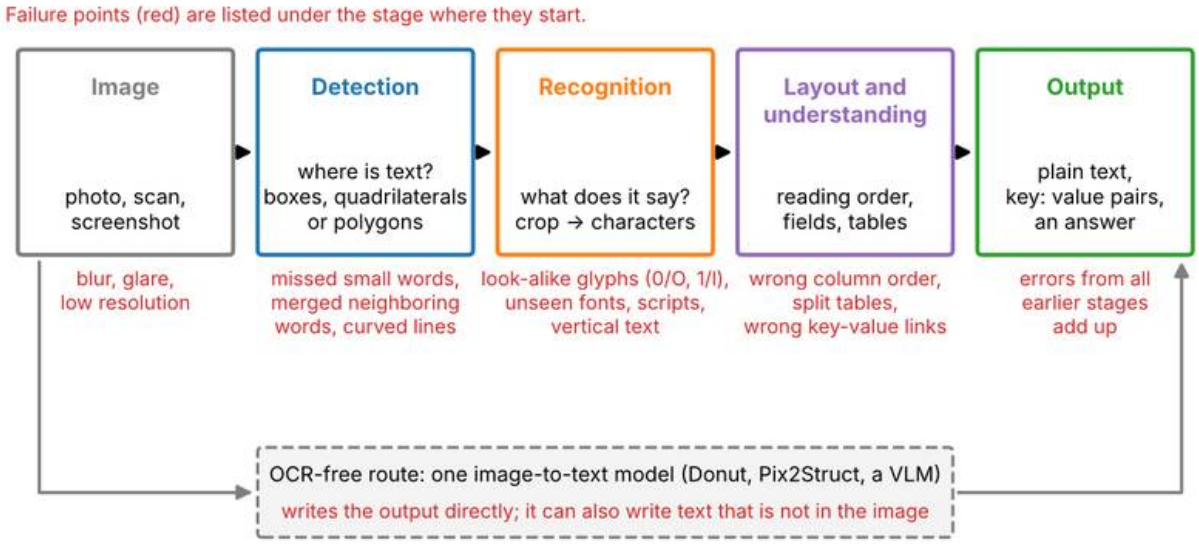


Figure 1 — The reading pipeline. Each stage has its own typical failures (red), and errors made early cannot be repaired later. OCR-free models replace the middle stages with one image-to-text model.

This chapter assumes DLCV 03 for boxes, IoU, NMS and precision–recall. Other ideas are linked rather than repeated: the CTC algorithm (PR 06's CTC note), edit distance (PR 06), attention (NLP 07) and vision-language models (ML 09). Surveys by Long et al. [1] and Chen et al. [2] frame scene text; Cui et al. [3] and Zhang et al. [4] cover documents.

The reading pipeline and where it breaks

Plain version. A classical OCR system is a chain: find text regions, read each one, then order the readings and attach meaning. Every link passes its mistakes to the next.

Scene text versus document text

The two settings differ in what is hard.

	Scene text	Document text
Source	photos, video, street views	scans, PDFs, screenshots, phone photos of pages
Amount	a few words per image	hundreds to thousands of words per page
Geometry	any orientation, perspective, curves	mostly horizontal lines; columns, tables
Appearance	decorative fonts, clutter, glare, blur	small print, noise, stamps, handwriting
Main difficulty	finding and reading each word	reading order, structure and meaning

Benchmarks follow the split. The Robust Reading competitions held with ICDAR (the International Conference on Document Analysis and Recognition) defined many of the standard scene-text benchmarks: ICDAR 2013 with focused, mostly horizontal English text [5], and ICDAR 2015's *Incidental Scene Text* challenge [6], recorded with wearable cameras (Google Glass), where words are often small, blurred and rotated [7]. Document benchmarks ask for layout regions, form fields or answers about a page (Document understanding).

Four places where errors start

Errors are easier to fix when you know which stage produced them.

1. **Image.** Blur, glare, low resolution and compression remove the evidence. No later stage can recover a character that is not in the pixels; a generative model that "recovers" it is guessing (see Failure modes).
2. **Detection.** A missed word is lost for good; two words merged into one box are read as one string; a box that cuts off the last letter produces a deletion.
3. **Recognition.** Look-alike glyphs (0/O, 1/l/I, rn/m), unseen fonts and scripts, and vertical text cause substitutions.
4. **Layout and understanding.** Correct words in the wrong order, a table read across instead of down, or a value attached to the wrong key give a wrong answer from perfect characters.

A useful habit: evaluate each stage on its own (recognition on ground-truth crops, for example) before the whole chain, so you know which stage to improve.

Before deep learning: thresholds and connected components

Plain version. On a clean page, text is just dark ink on light paper. Choose a gray level that separates the two, glue nearby ink pixels into blobs, and draw a box around each blob. This works surprisingly often, and seeing where it fails explains what learned detectors had to fix.

The classical recipe has three steps. **Binarization** labels each pixel as ink or paper by comparing it with a threshold. A single global threshold, for example the one chosen by Otsu's method [8] (derived in IP 10), fails under uneven lighting, so document systems use a local threshold computed in a window around each pixel. Sauvola's method [9], designed for document images, is a common choice:

$$T(x,y) = m(x,y)\left[1 + k\left(\frac{s(x,y)}{R} - 1\right)\right],$$

where $T(x,y)$ is the threshold at pixel (x,y), $m(x,y)$ and $s(x,y)$ are the mean and standard deviation of the gray levels in a $w \times w$ window centered on the pixel, R is the dynamic range of the standard deviation, and $k > 0$ (scikit-image's default is 0.2) controls how much local contrast lowers the threshold. A pixel darker than T is ink. **Grouping** then joins ink pixels into **connected components** (IP 09), usually after a horizontal dilation that bridges the gaps between the letters of a word. **Box extraction** takes the bounding box of each component.

The snippet renders a small invoice with matplotlib, darkens its left side, adds noise, and runs this recipe with a global and a local threshold. It scores the boxes with the precision, recall and H-mean used by text benchmarks (defined precisely in Measuring OCR): a predicted box counts as correct if its IoU with a still-unmatched ground-truth word is at least 0.5.

```
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from scipy import ndimage as ndi
from skimage.filters import threshold_otsu, threshold_sauvola

H, W = 200, 560
WORDS = [("INVOICE", 20, 18, 22), ("No.", 200, 24, 14), ("1046", 245, 24, 14),
         ("Date:", 330, 24, 14), ("2026-11-01", 385, 24, 14),
         ("Item", 20, 80, 13), ("Qty", 260, 80, 13), ("Price", 400, 80, 13),
         ("Paper", 20, 110, 13), ("12", 265, 110, 13), ("3.50", 405, 110, 13),
         ("Total", 20, 160, 15), ("42.00", 395, 160, 15)]     # (word, x, y, font size)

def render(words):
```

```
    """Draw words with matplotlib; return a gray image in [0, 1] and word boxes."""
    fig = plt.figure(figsize=(W / 100, H / 100), dpi=100)
    texts = [fig.text(x / W, 1 - y / H, s, fontsize=fs, va="top") for s, x, y, fs in words]
    fig.canvas.draw()
    img = np.asarray(fig.canvas.buffer_rgba())[..., :3].mean(axis=2) / 255.0
    r = fig.canvas.get_renderer()
    boxes = [[b.x0, H - b.y1, b.x1, H - b.y0]
             for b in (t.get_window_extent(r) for t in texts)]
    plt.close(fig)
    return img, np.array(boxes)

def iou(a, b):
    w = min(a[2], b[2]) - max(a[0], b[0]); h = min(a[3], b[3]) - max(a[1], b[1])
    inter = max(w, 0) * max(h, 0)
    area = lambda r: (r[2] - r[0]) * (r[3] - r[1])
    return inter / (area(a) + area(b) - inter)

def detect(img, thresh, gap=9):
    ink = img < thresh                                      # dark pixels = ink
    ink = ndi.binary_opening(ink, np.ones((2, 2)))          # drop isolated noise pixels
    words = ndi.binary_dilation(ink, np.ones((3, gap)))  # bridge gaps between letters
    lab, _ = ndi.label(words)
    out = []
    for y, x in ndi.find_objects(lab):
        if (y.stop - y.start) * (x.stop - x.start) > 40:   # ignore tiny blobs
            out.append([x.start, y.start, x.stop, y.stop])
    return np.array(out, float)

def h_mean(pred, gt, thr=0.5):
    used, tp = set(), 0
    for p in pred:                                          # greedy one-to-one matching
        cand = [(iou(p, g), j) for j, g in enumerate(gt) if j not in used]
        best = max(cand, default=(0, -1))
        if best[0] >= thr:
            used.add(best[1]); tp += 1
    P, R = tp / max(len(pred), 1), tp / len(gt)
    return P, R, 2 * P * R / max(P + R, 1e-9)

rng = np.random.default_rng(0)
clean, gt = render(WORDS)
shade = 0.45 + 0.55 * np.linspace(0, 1, W)[None, :]      # light falls off to the left
img = np.clip(clean * shade + rng.normal(0, 0.04, clean.shape), 0, 1)

t_sau = threshold_sauvola(img, window_size=31)
for name, t, gap in [("Otsu, gap 9", threshold_otsu(img), 9), ("Sauvola, gap 3", t_sau, 3),
                     ("Sauvola, gap 9", t_sau, 9), ("Sauvola, gap 25", t_sau, 25)]:
    pred = detect(img, t, gap)
    P, R, F = h_mean(pred, gt)
    print(f"{name:16s} boxes={len(pred):2d}  P={P:.2f}  R={R:.2f}  H-mean={F:.2f}")
```

```
Otsu, gap 9      boxes= 8  P=0.62  R=0.38  H-mean=0.48
Sauvola, gap 3   boxes=33  P=0.15  R=0.38  H-mean=0.22
Sauvola, gap 9   boxes=13  P=1.00  R=1.00  H-mean=1.00
```

```
Sauvola, gap 25  boxes=11  P=0.64  R=0.54  H-mean=0.58
```

Figure 2 shows what happened. The global threshold marks the whole dark left side as ink, so the words there vanish into one large false box. The local threshold recovers every word, but only for one hand-chosen gap: a narrow dilation splits words into letters, a wide one merges "No." with "1046" and "Date:" with the date. All numbers here come from this synthetic toy.

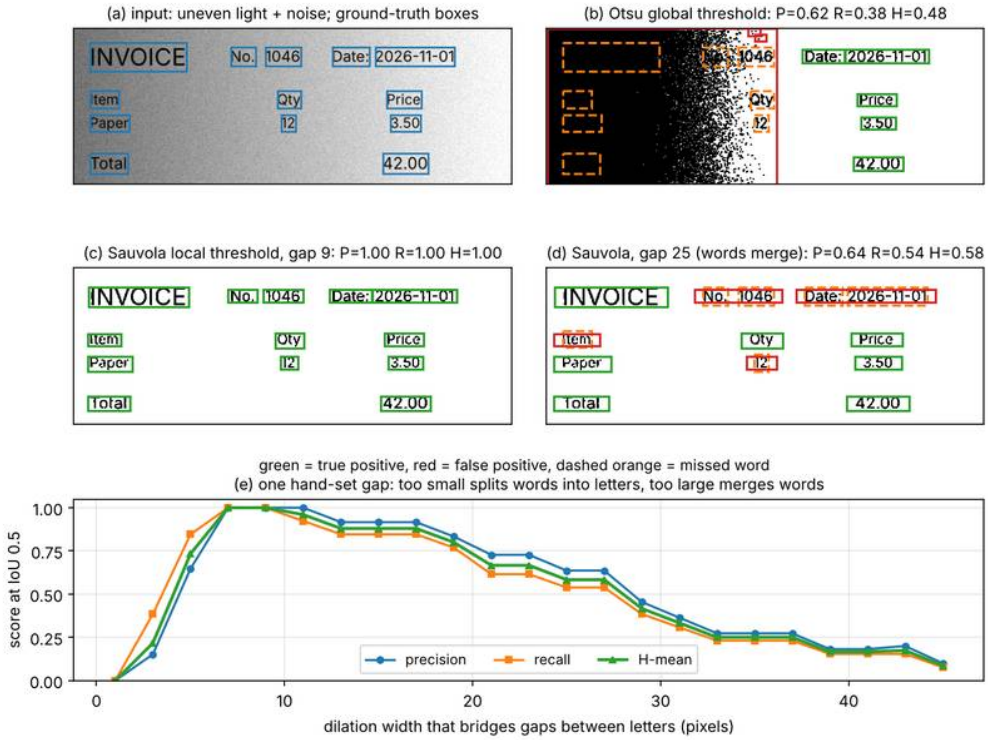


Figure 2 — Classical detection on a synthetic invoice (toy data). (a) Input with ground truth. (b) A global threshold fails under uneven light. (c) A local (Sauvola) threshold with a 9-pixel gap finds every word. (d) A 25-pixel gap merges neighbors. (e) Scores at IoU 0.5 against the gap: one fixed rule cannot suit every font size and spacing.

This experiment is the whole motivation for learned text detectors. Real images have every font size at once, text on textured backgrounds, rotation, and words closer together than letters inside other words. A network can learn *where text is* from context and appearance, and it can learn to keep neighbors apart.

Text detection

Plain version. A text detector is an object detector with one class, "text", and special care for three things: long thin shapes, any orientation (even curves), and words that nearly touch. Modern

detectors either regress the corners of each word from every pixel, or segment a text mask and then split it into words.

Why text is not just another object

The detectors of DLCV 03 predict upright (axis-aligned) boxes. For text this breaks in three ways.

- **Extreme aspect ratios.** A text line can be many times wider than it is tall, far outside anchor shapes tuned for ordinary objects.
- **Orientation and curvature.** Words on signs and packages are rotated, seen in perspective, or follow a curve. Benchmarks therefore annotate quadrilaterals (ICDAR 2015 [6]) or polygons (Total-Text [10], CTW1500 [11]), and compute IoU between polygons instead of rectangles.
- **Dense neighbors.** Words on a line are a few pixels apart; merging them, or letting NMS delete one, loses recall.

The first two problems combine badly. The snippet computes the IoU between a long thin word and the tightest upright box around it, as the word rotates.

```
import numpy as np
from skimage.draw import polygon

def rotated_word(cx, cy, w, h, deg):
    """Corners of a w x h word box rotated by deg around its center (x, y)."""
    t = np.deg2rad(deg)
    R = np.array([[np.cos(t), -np.sin(t)], [np.sin(t), np.cos(t)]])
    corners = np.array([[-w, -h], [w, -h], [w, h], [-w, h]]) / 2
    return corners @ R.T + [cx, cy]

def mask(quad, shape=(300, 300)):
    m = np.zeros(shape, bool)
    rr, cc = polygon(quad[:, 1], quad[:, 0], shape)
    m[rr, cc] = True
    return m

for deg in [0, 10, 20, 30, 45]:
    quad = rotated_word(150, 150, 160, 24, deg)          # a long, thin word
    x0, y0 = quad.min(axis=0); x1, y1 = quad.max(axis=0)
    aabb = np.array([[x0, y0], [x1, y0], [x1, y1], [x0, y1]])  # tightest upright box
    a, b = mask(quad), mask(aabb)
    print(f"angle {deg:2d} deg: IoU(word, its upright box) = {(a & b).sum() / (a |
     ↪  b).sum():.2f}")
```

```
angle  0 deg: IoU(word, its upright box) = 1.00
angle 10 deg: IoU(word, its upright box) = 0.47
angle 20 deg: IoU(word, its upright box) = 0.31
```

```
angle 30 deg: IoU(word, its upright box) = 0.25
angle 45 deg: IoU(word, its upright box) = 0.22
```

For a $w \times h$ rectangle rotated by θ the exact value is $wh/[(w\cos\theta + h\sin\theta)(w\sin\theta + h\cos\theta)]$, where w and h are the word's width and height and the denominator is the area of its upright bounding box. Already at 10 degrees the *best possible* upright box fails the usual IoU 0.5 test (Figure 3). Text detectors therefore output rotated rectangles, quadrilaterals or polygons; how IoU is computed for rotated shapes is the same polygon-clipping problem solved for bird's-eye-view boxes in DLCV 09.

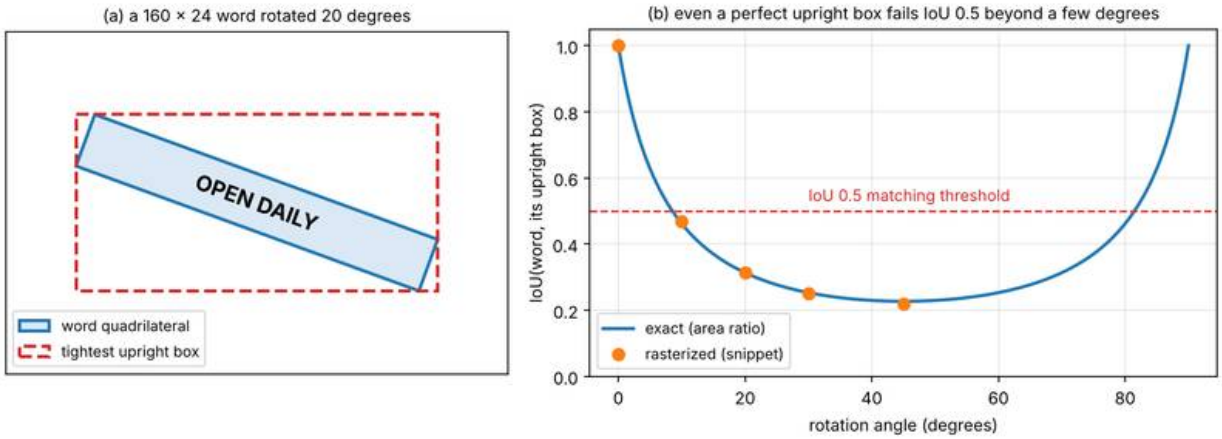


Figure 3 — Why text detectors predict quadrilaterals. (a) A 160 × 24 word rotated 20° and its tightest upright box. (b) Their IoU against the angle: beyond about 8.5° even a perfect upright box is counted as a miss at IoU 0.5.

Regression-based detectors: EAST

Plain version. Let every pixel inside a word vote for the word's shape: "I am text, and the word I belong to has its edges this far from me, at this angle." Keep the confident votes and merge the ones that agree.

EAST (Efficient and Accurate Scene Text detector) [12] is a fully convolutional network with a U-shaped feature-merging branch that outputs, at each location of a downsampled map, a **score** in $[0,1]$ (is this pixel inside a shrunk version of a word?) and a **geometry**. Two geometries were proposed. **RBOX** has five channels: the distances from the pixel to the four sides of a rotated rectangle and the rectangle's rotation angle. **QUAD** has eight channels: the offsets $(\Delta x_i, \Delta y_i)$, $i = 1, \ldots, 4$, from the pixel to the four corners of a quadrilateral. Because tens of thousands of pixels vote, EAST first merges neighboring geometries row by row, averaging them with their scores as weights (**locality-aware NMS**), and only then runs standard NMS [12]. One network

pass replaces the candidate-grouping steps of earlier systems. Its weakness is reach: a pixel must predict edges that may be hundreds of pixels away, and a quadrilateral cannot follow a curved word.

Segmentation-based detectors: PSENet and DBNet

Plain version. Instead of asking each pixel for the whole word's shape, ask only "is this pixel text?" That gives a mask that can follow any curve. The difficulty is splitting a mask into words when the words touch, so these detectors predict a *thinner* core for each word, keep the cores apart, and grow them back afterwards.

The network is an encoder–decoder like those of DLCV 03B, predicting a per-pixel text **probability map** P; the trick is in the labels. Each ground-truth polygon G with area A and perimeter L is shrunk inward by an offset

$$D = \frac{A\,(1 - r^2)}{L},$$

where $r \in (0, 1)$ is a shrink ratio; the shrunk polygon (the word's **kernel**) is the positive region. Kernels of adjacent words are separated by a margin even when the words touch, so connected components of the predicted kernel map are word instances; each is then dilated back to full size.

PSENet (Progressive Scale Expansion Network) [13] predicts several kernels per word at different shrink ratios and grows the smallest, well-separated ones into the larger ones with a breadth-first search, so a pixel claimed by one word cannot be taken by its neighbor. The price is a slow, sequential post-process.

DBNet [14] keeps a single kernel and makes the binarization step itself trainable. A standard binarization $B = \mathbf{1}[P \geq t]$ with a fixed threshold t has zero gradient almost everywhere, so it cannot be trained. DBNet predicts a second map, a **threshold map** T, and uses the approximate step

$$\hat{B}_{i,j} = \frac{1}{1 + e^{-k\,(P_{i,j} - T_{i,j})}},$$

where $\hat{B}_{i,j}$ is the approximate binary value at pixel (i, j), $P_{i,j}$ the predicted text probability, $T_{i,j}$ the learned threshold and k an amplifying factor set to 50 [14]. Writing $x = P_{i,j} - T_{i,j}$ and $f(x) = \hat{B}_{i,j}$, the binary cross-entropy for a text pixel, $\ell_+ = -\log f(x)$, has derivative $\partial \ell_+ / \partial x = -k\,(1 - f(x))$, and for a background pixel, $\ell_- = -\log(1 - f(x))$, $\partial \ell_- / \partial x = k\, f(x)$. The gradient is scaled by k and is largest where a pixel is on the wrong side of its threshold. The threshold map is supervised to be high in a band around each word's border, so the binarization is strict where words meet and relaxed inside them.

The snippet checks the derivative numerically and shows, on a 1-D profile, a threshold that is high between two words separating them where a fixed threshold merges them.

```
import numpy as np

def db(P, T, k=50.0):
    """Differentiable binarization: an approximate step at the learned threshold T."""
    return 1.0 / (1.0 + np.exp(-k * (P - T)))

x = np.array([-0.10, -0.02, 0.0, 0.02, 0.10])            # x = P - T at five pixels
for k in [1.0, 50.0]:
    f = db(x, 0.0, k)
    g_pos = -k * (1 - f)       # d/dx of -log f       (pixel labeled text)
    g_neg = k * f              # d/dx of -log(1 - f)  (pixel labeled background)
    # check the text-pixel gradient by finite differences
    eps = 1e-6
    num = (-np.log(db(x + eps, 0, k)) + np.log(db(x - eps, 0, k))) / (2 * eps)
    assert np.allclose(num, g_pos, rtol=1e-4)
    print(f"k={k:4.0f}  B_hat={np.round(f, 3)}  dl+/dx={np.round(g_pos, 2)}
     ↪  dl-/dx={np.round(g_neg, 2)}")

# Two words whose probability map P does not drop to 0 between them.
P = np.array([0.1, 0.9, 0.95, 0.9, 0.55, 0.5, 0.55, 0.9, 0.95, 0.9, 0.1])
T = np.array([0.3, 0.3, 0.3, 0.3, 0.7, 0.7, 0.7, 0.3, 0.3, 0.3, 0.3])   # high at the gap
print("fixed t=0.3:", (P >= 0.3).astype(int))
print("DB, k=50   :", (db(P, T) > 0.5).astype(int))
```

```
k=   1  B_hat=[0.475 0.495 0.5   0.505 0.525]  dl+/dx=[-0.52 -0.5  -0.5  -0.5  -0.48]
 ↪  dl-/dx=[0.48 0.5  0.5  0.5  0.52]
k=  50  B_hat=[0.007 0.269 0.5   0.731 0.993]  dl+/dx=[-49.67 -36.55 -25.   -13.45  -0.33]
 ↪  dl-/dx=[ 0.33 13.45 25.   36.55 49.67]
fixed t=0.3: [0 1 1 1 1 1 1 1 1 1 0]
DB, k=50   : [0 1 1 1 0 0 0 1 1 1 0]
```

With $k = 1$ the "binary" map is a gray blur near 0.5 and every pixel gets the same small gradient; with $k = 50$ it is nearly binary, and a text pixel 0.1 below its threshold gets a gradient of about -50 while a correct one gets almost none. Figure 4 repeats the separation in 2-D.

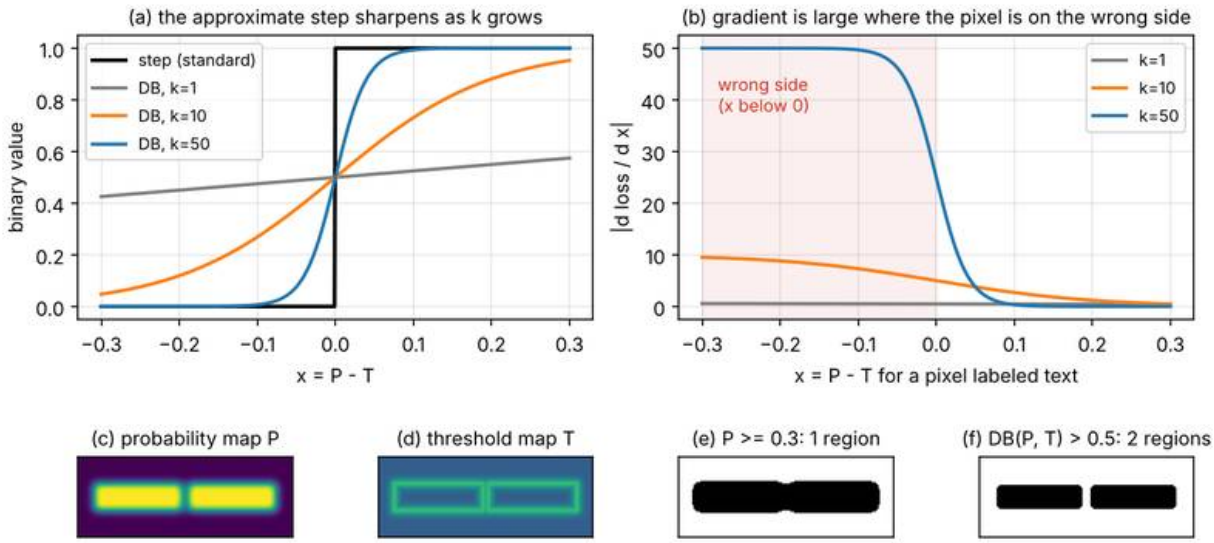


Figure 4 — Differentiable binarization. (a) The approximate step for several k. (b) The text-pixel gradient is amplified by k on the wrong side of the threshold. (c)–(f) Toy maps: a probability map that blurs across a 4-pixel gap merges two words at a fixed threshold, while a threshold map that is high at borders (hand-made here, learned in DBNet) keeps them apart.

At inference DBNet drops the threshold branch: it binarizes P at a constant 0.2, takes connected components, and expands each region by $D' = A'r'/L'$, with A' and L' the region's area and perimeter and $r' = 1.5$ [14]. The threshold map is thus mainly a training device that makes P itself sharp, and the simple post-processing keeps the detector fast.

Optional: Shrinking and expanding text polygons (in the Supplementary notes). The shrink offset worked out for rectangles, why it never erases a word, and why one expansion ratio cannot restore every aspect ratio exactly.

Optional: DBNet: a reading guide (in the Paper guides). How the paper's ablation and benchmark tables are organized and what to check in them.

Character-level and curved-text detectors

Some detectors find letters first and glue neighbors into words. CRAFT [15] predicts a **region score** (center of a character) and an **affinity score** (between two characters of one word) and links characters into words of any shape; because real datasets label only words, it estimates character labels on real images with an interim model. Polygon detectors such as the one introduced with CTW1500 [11] instead regress many boundary points along a curved line. Richer shape models follow irregular text better but need more complex outputs or post-processing.

Text recognition

Plain version. Recognition takes one cropped word or line and outputs its characters. The modern approach reads the crop like a sequence: slide along it from left to right, describe each thin slice, and turn the sequence of slices into a sequence of characters, without ever cutting the word into separate letters.

Recognition as sequence prediction

Early systems cut a word into characters and classified each, which fails when letters touch or break. **CRNN** (Convolutional Recurrent Neural Network) [16] made recognition segmentation-free and end-to-end trainable. A crop is resized to a fixed height; a CNN turns it into a feature map whose columns form a **feature sequence**, each column describing a thin vertical strip of the image; a deep bidirectional LSTM (see ML 03a) adds left and right context; and a transcription layer turns per-column predictions into a label sequence. With rectangular pooling windows, a 100×32 input yields a sequence of 25 frames, longer than most English words [16]. Nothing tells the network which frame belongs to which character; CTC handles that missing alignment.

CTC: training without alignments

Plain version. At every frame the network may say a character or "nothing new" (a **blank**). To read the output, merge repeated characters and delete the blanks. To train, add up the probability of every frame-by-frame story that reads out as the correct word.

Connectionist temporal classification (CTC) [17] defines, for per-frame probabilities $q_t(s)$ of each symbol s (a character of the alphabet or the blank $\emptyset$) at frame t, a collapsing map $\mathcal{B}$ (merge repeats, then remove blanks) and the probability of a label sequence y given the input x,

$$p(y \mid x) = \sum_{z:\ \mathcal{B}(z)=y} \prod_{t=1}^{T} q_t(z_t),$$

where $z = (z_1, \dots, z_T)$ is an alignment (one symbol per frame), T the number of frames, and the sum runs over every alignment that collapses to y. Training minimizes $-\log p(y \mid x)$, computed by a forward recursion in $O(TL)$ for a label of length L. The full derivation, with a brute-force check, is in PR 06's CTC note; here we only need what it implies for decoding.

The cheapest decoder is **greedy** (best-path) decoding: take the most likely symbol at each frame and collapse. CRNN uses exactly this approximation in its lexicon-free mode [16]. But the most likely *path* need not collapse to the most likely *text*, because a text's probability is spread over many paths. The snippet builds a five-frame toy where this matters.

```
import itertools
import numpy as np

alphabet = ["-", "c", "a", "t"]                # index 0 is the CTC blank
# Per-frame probabilities from a recognizer (rows: 5 frames, columns: symbols).
probs = np.array([[0.10, 0.80, 0.05, 0.05],
                  [0.20, 0.05, 0.70, 0.05],
                  [0.60, 0.05, 0.30, 0.05],
                  [0.55, 0.02, 0.03, 0.40],
                  [0.55, 0.02, 0.03, 0.40]])

def collapse(path):
    """CTC map B: merge repeated symbols, then remove blanks."""
    merged = [s for i, s in enumerate(path) if i == 0 or s != path[i - 1]]
    return "".join(alphabet[s] for s in merged if s != 0)

greedy = probs.argmax(axis=1)                  # best symbol at every frame
print("greedy path:", "".join(alphabet[s] for s in greedy), "->", repr(collapse(greedy)))

# Exact: add up every alignment that collapses to the same text (4**5 = 1024 paths).
total = {}
for path in itertools.product(range(4), repeat=len(probs)):
    p = np.prod(probs[np.arange(len(probs)), path])
    total[collapse(path)] = total.get(collapse(path), 0.0) + p
for text, p in sorted(total.items(), key=lambda kv: -kv[1])[:3]:
    print(f"p({text!r}) = {p:.3f}")
print("collapse('caa-at') =", repr(collapse([1, 2, 2, 0, 2, 3])),
      " collapse('caaat') =", repr(collapse([1, 2, 2, 2, 3])))
```

```
greedy path: ca--- -> 'ca'
p('cat') = 0.363
p('ca') = 0.179
p('ct') = 0.091
collapse('caa-at') = 'caat'  collapse('caaat') = 'cat'
```

The “t” is never the single most likely symbol in any frame, so greedy decoding drops it, yet the alignments that contain a “t” in frame 4 or 5 add up to twice the probability of “ca” (Figure 5). The last line shows the blank’s second job: a doubled letter survives only with a blank between its copies. Real decoders use **prefix beam search**, which keeps the best few partial transcriptions and merges alignments that collapse to the same prefix.

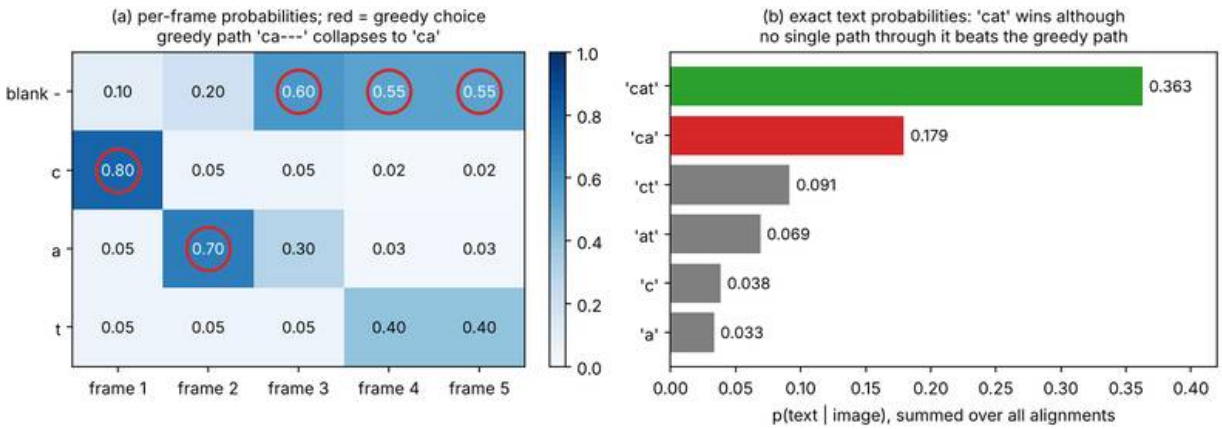


Figure 5 — CTC decoding on a toy matrix. (a) Greedy decoding picks 'ca—', which collapses to 'ca'. (b) Summing over all 1,024 alignments, 'cat' is twice as likely as 'ca', although every single path that reads 'cat' is less likely than the greedy path.

Optional: CTC prefix beam search, step by step (in the Supplementary notes). The two probabilities each prefix keeps, the three update rules, and a tested implementation that finds 'cat' on this toy.

CTC has two built-in limits (see PR 06's note): frame outputs are independent given the image, so the model cannot condition on characters it has already emitted, and the alignment is monotonic, so reading order must match frame order. Vertical or curved text breaks the second unless it is rectified first.

Attention decoders and rectification

Plain version. An attention decoder writes the word one character at a time. Before each character it looks back at the image features and chooses where to look; it also remembers what it has already written, so it behaves like a small language model of spelling.

An **attention decoder** is the encoder–decoder of NLP 07 applied to image features: at step i it attends to the feature sequence and predicts character y_i from the result and from $y_1, \dots, y_{i-1}$, until an end token. It drops CTC's independence assumption at the price of sequential decoding.

ASTER [18] pairs such a decoder with a **rectification network**: a thin-plate spline (TPS) transformation, predicted from the image and trained only through the recognition loss, warps a curved or perspective word into a straight horizontal one before reading it.

Baek et al. [7] showed that most recognizers of the time fit one four-stage template — **transformation** (none or TPS), **feature extraction**, **sequence modeling** (none or BiLSTM) and **prediction** (CTC or attention) — and, retraining all combinations on the same data, that attention mainly costs speed and that earlier comparisons had been distorted by different training sets and test subsets.

Optional: Baek et al., "What Is Wrong With Scene Text Recognition Model Comparisons?" (in the Paper guides). The four-stage framework, the dataset audit and how to read the trade-off plots.

Transformer recognizers

TrOCR [19] replaces both the CNN and the RNN with pre-trained transformers. The image is resized to 384 × 384 and split into 16 × 16 patches, as in a Vision Transformer (ML 04); an image encoder initialized from DeiT or BEiT encodes the patches, and a text decoder initialized from RoBERTa or MiniLM generates **wordpieces** (subword tokens) autoregressively, decoded with beam search [19]. It is pre-trained on synthetic text images and fine-tuned on printed, handwritten and scene text: recognition now borrows its backbones and decoder from general vision and language pre-training.

PARSeq [20] trains one transformer decoder on many random orders of the output positions (**permutation language modeling**), so the same weights can decode left to right, decode all characters in parallel without context, or refine a first guess using context on both sides.

Language models: help and harm

A recognizer that "knows English" can fix a smudged letter: "PLEA_E" is almost certainly "PLEASE". The knowledge can be **implicit** (an autoregressive decoder conditions on its previous characters), a **lexicon** (CRNN can restrict the output to a word list [16]) or an **explicit language model**: in ABINet [21] a separate bidirectional language model iteratively corrects the vision model's output.

The same prior can hurt. Text that is not language — serial numbers, codes, prices, names, mixed scripts — is exactly where OCR matters most and where a language model "corrects" correct readings. PARSeq's authors note this risk for external language models that do not see the image [20], and OCRBench reports that large multimodal models are weak on **non-semantic text** (random character strings) as well as on handwriting, multilingual text and mathematical expressions [22]. When the content is not natural language, measure with and without the language prior.

Optional: Where the language model sits in a recognizer (in the Supplementary notes). Lexicon search, implicit and explicit language models, and how to test whether a prior is over-correcting.

Training data: synthetic text

Labeled real crops are scarce, and recognizers trained only on **synthetic text** generalize well across scene-text datasets [1]. MJSynth [23] renders single words with varied fonts, shadows, distortions

and noise (8.9 million word images); SynthText [24] pastes text into natural photos following the scene geometry (5.5 million cropped words) [7]. Training on both beats either alone, and a combined 20% sample of each beats either full set, so diversity can matter more than size [7].

End-to-end text spotting

Plain version. Instead of running a detector and then a separate recognizer on each crop, one network does both and shares its features. Reading helps finding (a region that reads as a word is probably text), and finding helps reading.

A **text spotter** outputs (shape, transcription) pairs directly. FOTS [25] shares convolutional features between a dense per-pixel detection branch and a recognition branch decoded with CTC, through **RoIRotate**, an operator that crops and rotates the feature region of each detected quadrilateral into an upright, fixed-height strip. Mask TextSpotter [26] extends Mask R-CNN (see instance segmentation in DLCV 03B) with masks for whole words and for individual characters, so it can read curved text without a sequence model. ABCNet [27] fits each curved text instance with parametric Bézier curves and samples features along them with **BezierAlign**, a curved analogue of RoIAlign. End-to-end evaluation then asks whether each ground-truth word was both located and read correctly, which we define in Measuring OCR.

Document understanding

Plain version. On a page, reading the words is often the easy part. The questions are which words belong together, in which order to read them, and what they mean: which number is the total, which line is the title, which cell is in which column.

Layout analysis and reading order

Layout analysis finds the regions of a page — paragraphs, titles, lists, tables, figures — usually by object detection or instance segmentation on page images. PubLayNet [28] matched the XML and PDF versions of over a million PubMed Central articles to label over 360 thousand page images automatically with five classes (text, title, list, table, figure). DocLayNet [29] has 80,863 manually annotated pages from diverse sources with 11 classes, in COCO format, because models trained only on scientific articles transfer poorly to reports, manuals and forms.

Given regions and words, the system must also produce a **reading order**. The obvious rule, "top to bottom, then left to right", fails on the simplest multi-column page, as the snippet shows on a two-column notice.

```
import numpy as np

# Word boxes (x0, y0, x1, y1) on a two-column page: each column is one sentence.
left = "Pay the invoice within thirty days of receipt".split()
right = "Late payments carry a two percent monthly fee".split()
words, boxes = [], []
for col, text in [(0, left), (1, right)]:
    for k, w in enumerate(text):
        row, pos = divmod(k, 3)                       # three words per line
        x0 = 40 + 300 * col + 80 * pos
        y0 = 100 + 30 * row + 3 * col                 # columns not perfectly aligned
        words.append(w); boxes.append([x0, y0, x0 + 70, y0 + 20])
boxes = np.array(boxes, float)

def naive_order(boxes, line_tol=10):
    """Top-to-bottom, then left-to-right, treating boxes within line_tol as one line."""
    line = np.round(boxes[:, 1] / line_tol)
    return np.lexsort((boxes[:, 0], line))

def column_order(boxes, line_tol=10):
    """Split the page at the widest empty vertical strip, then read column by column."""
    x_min, x_max = int(boxes[:, 0].min()), int(boxes[:, 2].max())
    covered = np.zeros(x_max - x_min, bool)
    for x0, _, x1, _ in boxes.astype(int):
        covered[x0 - x_min:x1 - x_min] = True
    edges = np.flatnonzero(np.diff(np.r_[1, covered.astype(int), 1]))  # empty-run bounds
    starts, ends = edges[::2], edges[1::2]
    widest = np.argmax(ends - starts)
    split = x_min + (starts[widest] + ends[widest]) / 2
    col = (boxes[:, 0] > split).astype(int)
    line = np.round(boxes[:, 1] / line_tol)
    return np.lexsort((boxes[:, 0], line, col))

print("naive :", " ".join(words[i] for i in naive_order(boxes)))
print("column:", " ".join(words[i] for i in column_order(boxes)))
```

```
naive : Pay the invoice Late payments carry within thirty days a two percent of receipt
 ↪  monthly fee
column: Pay the invoice within thirty days of receipt Late payments carry a two percent
 ↪  monthly fee
```

Every word in the naive output is correct, and the text is still useless. The column rule fixes this page but is another hand-made rule that fails on tables or figures (Figure 6); learned layout and document models exist largely to replace such rules.

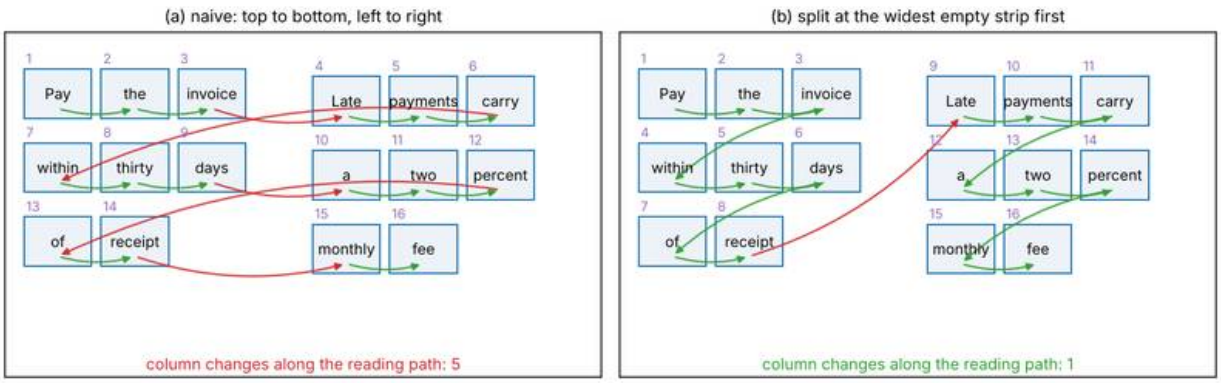


Figure 6 — Reading order on a synthetic two-column page. (a) Sorting by line then by x interleaves the columns (5 column changes). (b) Splitting at the widest empty vertical strip first gives the intended order (1 change), but only for this kind of layout.

Text plus layout: the LayoutLM family

Plain version. Give a language model not only the words of a page but also where each word sits. Then "Total" followed by a number at the bottom right of a receipt means something different from the same word in the middle of a paragraph.

LayoutLM [30] starts from BERT (NLP 10) and adds a **2-D position embedding** to each word's token embedding. Each word comes with its OCR box (x_0, y_0, x_1, y_1) — upper-left and lower-right corners — rescaled to integer "virtual" coordinates from 0 to 1,000 so that pages of different sizes share one scale. Four embedding lookups are added, with one table shared by x_0 and x_1 and another by y_0 and y_1:

$$\mathbf{h}_i = \mathbf{e}_{\text{tok}}(w_i) + \mathbf{e}_{\text{pos}}(i) + \mathbf{X}(x_{0,i}) + \mathbf{Y}(y_{0,i}) + \mathbf{X}(x_{1,i}) + \mathbf{Y}(y_{1,i}),$$

where $\mathbf{h}_i$ is the input vector of the i-th word w_i, $\mathbf{e}_{\text{tok}}$ and $\mathbf{e}_{\text{pos}}$ are BERT's token and 1-D position embeddings, and $\mathbf{X}$, $\mathbf{Y}$ are the learned tables for horizontal and vertical coordinates. Pre-training masks some words but keeps their 2-D positions, and the model predicts them from text and layout context (a **masked visual-language model**), on about 11 million scanned pages OCRed with the open-source Tesseract engine [30], [31]. LayoutLM also adds visual features of each word's image region, computed with a Faster R-CNN model, for downstream tasks [30]. LayoutLMv3 [32] instead feeds the page image as patches, masks both text tokens and image patches and adds a word–patch alignment objective, so one model serves text-centric tasks (forms, receipts, question answering) and image-centric ones (document classification, layout analysis).

Key information extraction — pulling "invoice number", "date" and "total" out of a page — is then token classification over the OCR words, as in NLP 06's sequence labeling, plus linking keys to values. FUNSD [33] is a small benchmark of this kind: 199 real, noisy scanned forms whose semantic entities are labeled *question*, *answer*, *header* or *other*, with links between them.

The family's weakness is its input: it cannot fix a word the OCR engine missed or misread, and it inherits the engine's reading order.

OCR-free models: Donut and Pix2Struct

Plain version. Skip the OCR engine. Train one model to look at the page image and *write* the desired output — the full text, a JSON record, or an answer — token by token, the way an image captioner writes a caption.

Donut (Document understanding transformer) [34] is a Swin Transformer encoder (ML 04) and a BART decoder. It is pre-trained to read all the text of a page in reading order — a "pseudo-OCR" task on 11 million scanned English documents plus synthetic pages in Chinese, Japanese, Korean and English from its SynthDoG generator — and fine-tuned to emit a token sequence that converts one-to-one into JSON, for classification, information extraction or question answering [34]. It avoids the cost of running OCR, the inflexibility of OCR engines across languages and document types, and the propagation of OCR errors. **Pix2Struct** [35] is pre-trained to parse masked screenshots of web pages into simplified HTML, which mixes OCR, language modeling and captioning in one objective; it accepts variable-resolution inputs and renders the question directly onto the image.

The price is resolution and verifiability. Donut uses 2560 × 1920 inputs and reports that accuracy rises quickly with input size on documents with tiny text, at a growing compute cost [34]; and because the model *generates* rather than *transcribes*, nothing ties each output character to a region of the image.

Optional: Donut: a reading guide (in the Paper guides). The two training stages, the JSON output format, the TED-based accuracy, and the resolution analysis.

Vision-language models for documents

General multimodal large language models (ML 09) now answer questions about documents and transcribe pages; reading small text depends on tiling or dynamic resolution (What the diagram hides). Three cautions apply. First, a fluent answer is not a transcription: check it against detections or ask for coordinates and verify them. Second, generative readers have their own failure modes: the academic-document model Nougat [36] reports that it sometimes degenerates into repeating a sentence over and over (on 1.5% of its test pages, more often out of domain), and that numbers in reference lists and section titles are sometimes skipped or hallucinated. Third, measure the specific skill: OCRBench [22] gathers 29 datasets to test text recognition, scene-text and document question answering, key information extraction and handwritten mathematical expressions separately, instead of hiding OCR inside one aggregate score.

Handwriting, many scripts and CJK text

Plain version. Handwriting varies from writer to writer and joins letters together; many scripts have far more symbols than the Latin alphabet; and some, such as Chinese, can also be written vertically. Each of these breaks an assumption that English scene-text models make quietly.

Handwriting

Offline handwriting recognition reads lines of handwritten text from scans. The IAM database [37] of handwritten English sentences is, in the TrOCR authors' words, the most popular dataset for handwritten text recognition; TrOCR uses a split of 6,161 training, 966 validation and 2,915 test lines and reports case-sensitive CER [19]. Line-level recognizers with CTC or attention apply unchanged; the difficulty is variation between writers, connected letters and limited labeled data, which is why synthetic pre-training and large pre-trained decoders help. For Chinese handwriting, the CASIA online and offline databases [38] provide data for both pen-trajectory (online) and scanned (offline) recognition.

Large character sets and vertical text

Chinese and Japanese use alphabets of tens of thousands of symbols, against a few dozen for Latin script [1]. RCTW-17 lists what makes Chinese text different: a much larger character set, no spaces between words, and many characters made of several unconnected parts [39]. Each has a consequence. The recognizer's output layer grows with the alphabet and most characters are rare: the Chinese Text in the Wild dataset labels 1,018,402 character instances in 32,285 street-view images, covering 3,850 distinct characters, most of them rarely used [40]. Unconnected parts break connected-component grouping. And without spaces, systems work on lines and characters rather than words (see NLP 01).

Vertical text is the other difficulty: Chinese can be set vertically as well as horizontally, and street-view datasets such as CTW contain vertical text [40]. The W3C's *Requirements for Chinese Text Layout* describes vertical composition: characters run from top to bottom and lines from right to left, while embedded Latin words are either set upright or rotated 90° clockwise [41]. A recognizer that assumes left-to-right frames on a horizontal crop reads such a column as garbage; Baek et al. list vertical text among the failure cases common to all 24 recognizers they tested [7]. The remedy is to decide each detected line's orientation first and rotate or rearrange it before reading.

Optional: Reading vertical and mixed-direction text (in the Supplementary notes). The layout rules that matter for OCR, how orientation is decided per line, and what a crop must look like before a horizontal recognizer can read it.

Many scripts at once

Multilingual systems must also decide *which* script they are looking at. The ICDAR 2019 multilingual challenge (MLT-2019) [42] has 20,000 real images with text in 10 languages, and includes a script-identification task; its languages cover seven scripts [2].

Measuring OCR

Plain version. Count mistakes the way a proofreader would: how many characters must be inserted, deleted or changed to turn the output into the truth. For detection, count how many found boxes are right and how many real words were found, as in object detection.

CER and WER from edit distance

The **edit distance** (Levenshtein distance) between a reference string r and a recognized string h is the minimum number of insertions, deletions and substitutions that turn one into the other; the Wagner–Fischer dynamic program computes it [43] (derived in PR 06). Following an optimal alignment back through the table splits the total into S substitutions, D deletions and I insertions. Then

$$\mathrm{CER} = \frac{S + D + I}{N}, \qquad \mathrm{WER} = \frac{S_w + D_w + I_w}{N_w},$$

where N is the number of characters in the reference, and S_w, D_w, I_w and N_w are the same quantities counted over words (sequences split at spaces). Both are error rates: 0 is perfect, and they can exceed 1 when the output contains many insertions. Other conventions exist: scene-text recognition on cropped words usually reports **word accuracy** (the fraction of crops read exactly), some papers call one minus that the word error rate, and multilingual competitions report one minus a **normalized edit distance** (NED) [2]. Always state which one you report.

```
import numpy as np

def edit_ops(ref, hyp):
    """Levenshtein alignment of two sequences; returns (S, D, I) counts."""
    n, m = len(ref), len(hyp)
    E = np.zeros((n + 1, m + 1), int)
    E[:, 0], E[0, :] = np.arange(n + 1), np.arange(m + 1)
    for i in range(1, n + 1):
        for j in range(1, m + 1):
            E[i, j] = min(E[i - 1, j] + 1, E[i, j - 1] + 1,
                          E[i - 1, j - 1] + (ref[i - 1] != hyp[j - 1]))
    S = D = I = 0
    i, j = n, m
    while i > 0 or j > 0:                       # walk back along one optimal path
```

```
        if i > 0 and j > 0 and E[i, j] == E[i - 1, j - 1] + (ref[i - 1] != hyp[j - 1]):
            S += ref[i - 1] != hyp[j - 1]; i, j = i - 1, j - 1
        elif i > 0 and E[i, j] == E[i - 1, j] + 1:
            D += 1; i -= 1
        else:
            I += 1; j -= 1
    return S, D, I

def rates(ref, hyp):
    c = edit_ops(ref, hyp)
    w = edit_ops(ref.split(), hyp.split())
    return c, sum(c) / len(ref), w, sum(w) / len(ref.split())

ref = "Total due 25.80 EUR by 1 May"
for hyp in ["Total due 25.80 EUR by 1 May",
            "Tota1 due 25.8O EUR by l May",      # three look-alike glyphs
            "Total due25.80 EUR by 1 May",       # one missing space
            "Total due 25.80 EUR"]:              # last words not read
    c, cer, w, wer = rates(ref, hyp)
    print(f"{hyp:30s} chars S,D,I={c} CER={cer:.3f} | words S,D,I={w} WER={wer:.3f}")
```

```
Total due 25.80 EUR by 1 May   chars S,D,I=(0, 0, 0) CER=0.000 | words S,D,I=(0, 0, 0)
 ↪  WER=0.000
Tota1 due 25.8O EUR by l May   chars S,D,I=(3, 0, 0) CER=0.107 | words S,D,I=(3, 0, 0)
 ↪  WER=0.429
Total due25.80 EUR by 1 May    chars S,D,I=(0, 1, 0) CER=0.036 | words S,D,I=(1, 1, 0)
 ↪  WER=0.286
Total due 25.80 EUR            chars S,D,I=(0, 9, 0) CER=0.321 | words S,D,I=(0, 3, 0)
 ↪  WER=0.429
```

Three look-alike substitutions cost 3 of 28 characters (CER 0.107) but ruin 3 of 7 words (WER 0.429), and one turns an amount into a non-number. One missing space costs one character but two word errors; losing the last three words costs nine characters, spaces included. CER says how much must be fixed; WER, or better a task check such as “is the total correct?”, says whether the output is usable (Figure 7).

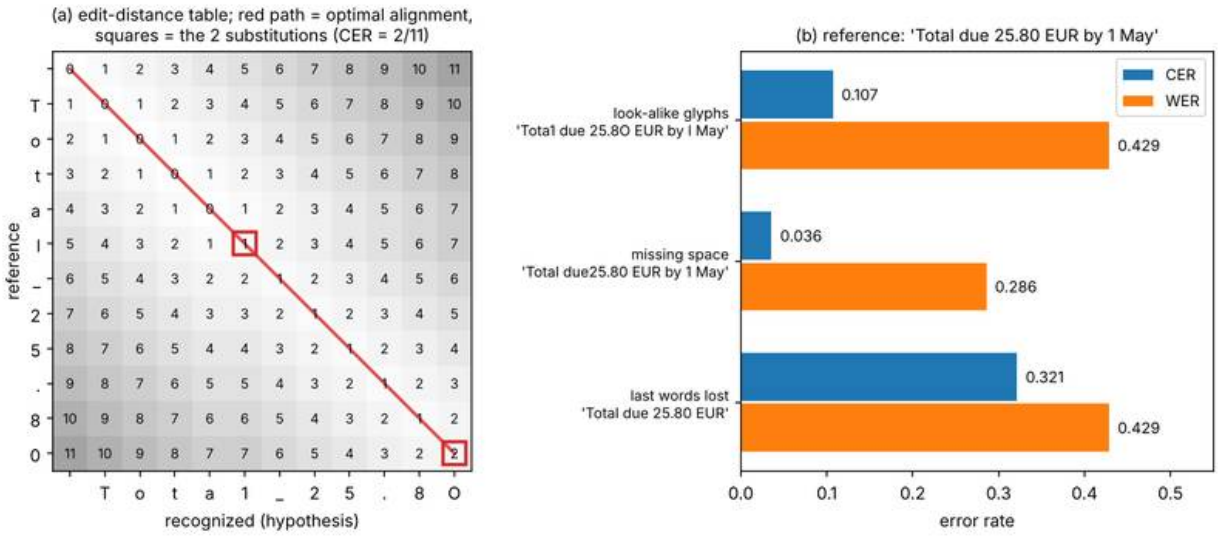


Figure 7 — CER and WER. (a) The edit-distance table for a short pair; the red path is an optimal alignment with two substitutions, so CER = 2/11. (b) For the same reference, small character errors can produce large word error rates.

Detection: precision, recall and H-mean at IoU 0.5

Text detection uses the precision and recall of DLCV 03 with one matching rule and one summary number. In the protocol used by the Robust Reading competitions, as described for MLT-2019, a detection is a true positive if its IoU with a ground-truth quadrilateral exceeds 0.5, each ground truth and each detection may be matched at most once, and detections that overlap a region marked "do not care" (illegible or out-of-scope text) by more than half of that region's area are removed before matching [42]. Counts are accumulated over the whole test set, and

$$P = \frac{\mathrm{TP}}{|\mathcal{D}|}, \qquad R = \frac{\mathrm{TP}}{|\mathcal{G}|}, \qquad H = \frac{2PR}{P + R},$$

where TP is the number of matched pairs, $|\mathcal{D}|$ the number of detections after filtering, $|\mathcal{G}|$ the number of ground-truth words that are not "do not care", and H the **H-mean** (harmonic mean of precision and recall, the F-score). Unlike COCO AP, H-mean is a single point of the precision–recall curve, at a confidence threshold chosen by the submitter; Long et al. recommend also reporting AP [1]. ICDAR 2013 used a different, overlap-based protocol, so its numbers are not comparable with IoU-based ones [1].

End-to-end and document metrics

End-to-end spotting first matches detections to ground truth as above and then compares the transcriptions, either for every word (**end-to-end recognition**) or only for words in a given vocabulary

(**word spotting**); results depend strongly on the size of that vocabulary [2]. Document benchmarks use task metrics: **ANLS** (average normalized Levenshtein similarity), an edit-distance-based score for DocVQA answers, chosen so that small OCR slips are not scored as total failures [44]; field-level F1 and a tree-edit-distance accuracy for extracted JSON records [34]; and COCO-style mAP for layout regions [29].

Optional: Evaluation protocols in detail (in the Supplementary notes). "Do not care" filtering, one-to-many matching in the ICDAR 2013 protocol, vocabularies, NED and AED, ANLS and TED accuracy, with their edge cases.

Datasets and benchmarks

The table lists the benchmarks named in this chapter. Sizes are as stated in the cited papers.

Benchmark	Task	Content
ICDAR 2013 [5]	detection, recognition, end-to-end	focused scene text, mostly horizontal English; 462 images [10]
ICDAR 2015 Incidental [6]	detection, recognition, end-to-end	wearable-camera images; 1,000 train / 500 test for detection [12]; quadrilateral labels
Total-Text [10]	detection, spotting	1,555 images (1,255 / 300), 9,330 words, polygons; horizontal, multi-oriented and curved
CTW1500 [11]	curved-text detection	1,500 images (1,000 / 500), over 10,000 text annotations as polygons
IIIT5K [45]	cropped-word recognition	words from web image search; 2,000 train / 3,000 test [7]
SVT [46]	cropped-word recognition	Google Street View; 257 train / 647 test words [7]
MLT-2019 [42]	multilingual detection, script ID, end-to-end	20,000 images, 10 languages
RCTW-17 [39], CTW [40]	Chinese detection and recognition	12,263 images; 32,285 images with character-level labels
IAM [37], CASIA-HWDB [38]	handwriting	English sentences; Chinese online and offline handwriting
FUNSD [33]	form understanding	199 scanned forms; question / answer / header / other entities with links
PubLayNet [28], DocLayNet [29]	layout analysis	over 360k auto-labeled article pages (5 classes); 80,863 hand-labeled pages (11 classes)
DocVQA [44]	document question answering	50,000 questions on over 12,000 document images; ANLS
OCRBench [22], OmniDocBench [47]	OCR skills of multimodal models; page parsing	29 datasets; 9 document sources, 19 layout categories

Two cautions apply to all of them. Different papers have evaluated on different subsets of the same benchmark (Baek et al. found two versions each of IC03 and IC15 in use), so check the

subset before comparing numbers [7]. And the labels are not perfect: Baek et al. estimated the mislabeling rate of the common recognition benchmarks at 1.3% ignoring special characters, rising to 6.1% when special characters count and 24.1% when case counts [7].

Failure modes

Plain version. OCR fails where the pixels stop carrying the answer, where the shape or script differs from training, and where a model fills a gap with something plausible. The last kind is the most dangerous, because the output looks perfect.

Look-alike glyphs and low resolution. Many characters differ by a few pixels: 0 and O, 1, l and I, 5 and S, 8 and B, "rn" and "m". The snippet makes a tiny template reader for 15 characters and shows how it degrades as the number of pixels per character drops.

```
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from skimage.transform import resize

CHARS = "0O8B5S1lIZ2aekx"

def glyph(c):
    """Render one character (ink = 1, paper = 0) on a 48 x 48 canvas."""
    fig = plt.figure(figsize=(0.48, 0.48), dpi=100)
    fig.text(0.5, 0.5, c, fontsize=26, ha="center", va="center")
    fig.canvas.draw()
    img = 1 - np.asarray(fig.canvas.buffer_rgba())[..., :3].mean(axis=2) / 255.0
    plt.close(fig)
    return img

def ncc(a, b):
    a, b = a - a.mean(), b - b.mean()
    return (a * b).sum() / np.sqrt((a * a).sum() * (b * b).sum())

templates = {c: glyph(c) for c in CHARS}
rng = np.random.default_rng(0)
for h in [48, 12, 8, 6]:                          # pixels per character cell
    errors, total = {}, 0
    for c, img in templates.items():
        for _ in range(20):
            low = resize(img, (h, h), anti_aliasing=True)       # camera sees few pixels
            low = low + rng.normal(0, 0.05, low.shape)          # sensor noise
            back = resize(low, (48, 48), order=1)               # what the reader gets
            guess = max(CHARS, key=lambda t: ncc(back, templates[t]))
            total += 1
            if guess != c:
                errors[(c, guess)] = errors.get((c, guess), 0) + 1
    top = sorted(errors.items(), key=lambda kv: -kv[1])[:3]
```

```
    print(f"{h:2d}px  accuracy={1 - sum(errors.values()) / total:.2f}  most confused: "
          + ", ".join(f"{a}->{b} ({n})" for (a, b), n in top))
```

```
48px  accuracy=1.00  most confused:
12px  accuracy=0.93  most confused: l->I (20)
 8px  accuracy=0.84  most confused: I->1 (13), l->1 (12), 0->O (6)
 6px  accuracy=0.53  most confused: B->8 (20), 0->8 (15), S->8 (12)
```

Lowercase "l" and capital "I" are confused already at 12 pixels, because this sans-serif font draws both as a plain bar (Figure 8); at 6 pixels half the readings are wrong. A deep recognizer does far better than template matching, but no recognizer can separate glyphs that the font draws identically; only context can. Super-resolution can help (DLCV 04A), but see the last item below.

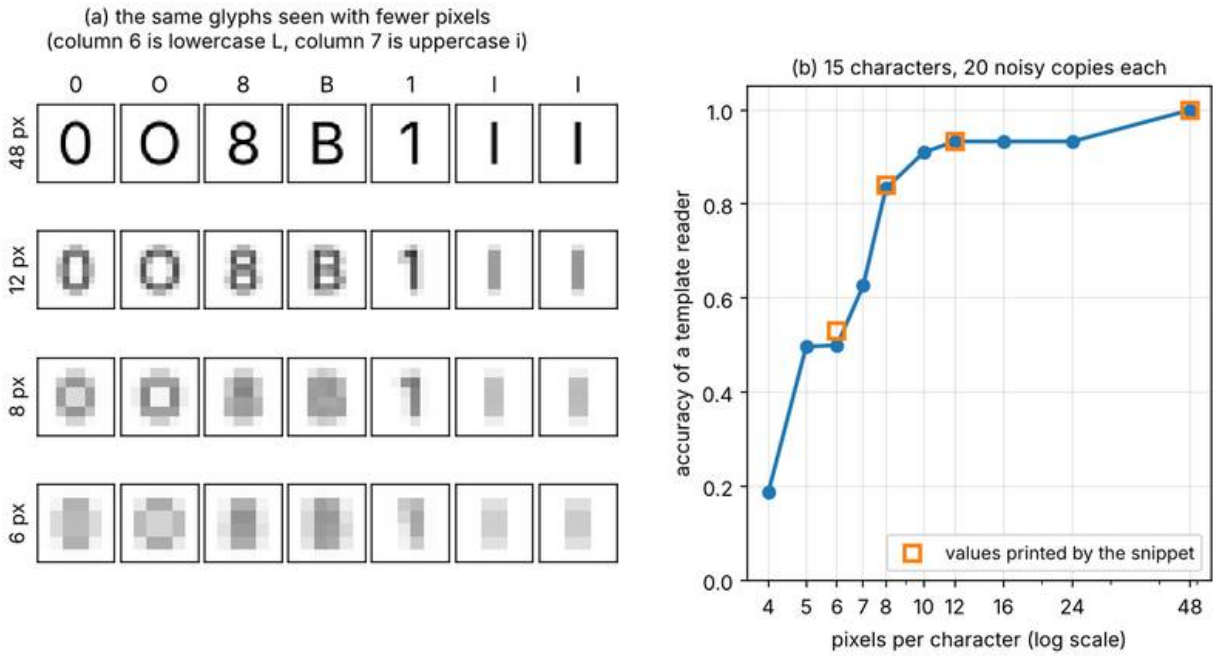


Figure 8 — Look-alike glyphs at low resolution (toy template reader, 15 characters, 20 noisy copies each). (a) The same glyphs with fewer pixels. (b) Accuracy against resolution; squares mark the values printed by the snippet.

Other common failures, each with the stage where it starts (Figure 1):

- **Curved, rotated and perspective text** (detection and recognition): upright boxes miss it (Figure 3); horizontal recognizers misread it unless it is rectified.
- **Unusual fonts and styles** (recognition): calligraphic brand names and shop signs were among the cases that all 24 models in Baek et al.'s study failed on, along with vertical text, special characters, heavy occlusion and low resolution [7].

- **Merged or split instances** (detection): Figure 2's gap trade-off in learned form; segmentation detectors fight it with shrunk kernels and learned thresholds.
- **Wrong layout** (understanding): correct words in the wrong order (Figure 6), tables read across rows, a value linked to the wrong key.
- **Distribution shift** (all stages): layout models trained only on article pages degrade on more diverse documents [29]; DLCV 11 treats shift in general.
- **Hallucinated text** (any generative stage): a generative reader can output text that is not in the image — repeated sentences, invented section numbers [36], a "corrected" serial number. The same happens upstream: a generative restoration or super-resolution model can turn unreadable letters into clean, legible, wrong ones, as discussed in DLCV 04C; and text in generated images is often misspelled (DLCV 10). Never run OCR on a generatively enhanced image for anything that must be true, such as an amount, a name or a license plate, without checking against the original.

Modern view

What the surveys say

Four reviews cover this chapter's ground. **Long et al.** [1] organize scene-text detection by how much of a word each prediction covers (whole words versus characters, segments or kernels) and recognition into CTC-based and encoder–decoder methods; they see generalization across datasets and large alphabets as the open problems. **Chen et al.** [2] focus on recognition, organized by pipeline stage, and collect the datasets and evaluation protocols. **Cui et al.** [3] review document tasks from rule-based analysis to pre-trained LayoutLM-style models, and **Zhang et al.** [4] split document parsing into modular pipelines and unified VLM-based models, naming complex layouts, the reliability of VLM parsing and inference cost as the main challenges. OCRBench [22] complements them with an evaluation of large multimodal models on text-related tasks.

Optional: Guide to the Modern view surveys (in the Paper guides). What each review covers, how it organizes the field and what it concludes.

The state of the art, 2023–2026

- **Pipelines remain the natural choice** when coordinates, speed or auditability matter: a DBNet-style detector, an attention or transformer recognizer and a layout model, each evaluated and corrected separately.

- **End-to-end page readers are maturing fast.** Models that turn a page into Markdown or structured text in one pass now cover formulas, tables and charts: Nougat for academic papers [36]; GOT, which treats plain text, formulas, tables, charts and sheet music as "characters" for one encoder–decoder model [48]; olmOCR, a fine-tuned vision-language model for linearizing PDFs at scale, released with its own benchmark of hard pages [49]; and DeepSeek-OCR, which studies how few vision tokens a page can be compressed into while remaining readable [50]. All of these reports come from their authors.
- **Evaluation is catching up.** OmniDocBench [47] (CVPR 2025) evaluates pipelines and end-to-end models on pages from nine document sources with 19 layout categories and 15 attribute labels, so that results can be broken down by document type and difficulty instead of one score.

Open problems

1. **Faithfulness.** Generative readers can produce text that is not on the page, and their errors look like correct text. Character-level grounding (which pixels support each output token) and calibrated confidence are not yet standard.
2. **Non-semantic and rare text.** Codes, numbers, names, rare characters in large alphabets and mixed scripts are where language priors help least and errors cost most [22].
3. **Layout and reading order** on complex pages — multi-column articles with floating figures, nested tables, forms with handwritten entries — remain hard for pipelines and end-to-end models alike; robustness to complex layouts is among the main challenges named by Zhang et al. [4].
4. **Generalization across domains and scripts**, including vertical and curved text, without collecting new labels for each domain [1].
5. **Comparable evaluation.** Different subsets, metrics (CER, word accuracy, NED, ANLS) and output normalizations make published numbers hard to compare [7].

Key takeaways

- Reading text is a pipeline — detection, recognition, layout and understanding — and each stage has its own failures; evaluate stages separately before the whole chain.
- Text breaks generic detection: long, thin, rotated, curved and densely packed words need quadrilateral or polygon outputs and polygon IoU; even a perfect upright box fails IoU 0.5 beyond about 8.5° for a 160 × 24 word.
- Segmentation detectors (PSENet, DBNet) predict shrunk word kernels so that touching words stay separate; differentiable binarization makes the threshold learnable with a k-amplified gradient, and at inference DBNet needs only a fixed threshold and connected components.

- Recognition is sequence prediction: CTC trains without alignments but greedy decoding can miss the most probable text; attention and transformer decoders model context and spelling, which helps on natural words and can hurt on codes and numbers.
- Document models add layout: LayoutLM's 2-D position embeddings on OCR boxes; OCR-free models (Donut, Pix2Struct, VLMs) write the output directly, trading the OCR error chain for resolution limits and possible hallucinated text.
- Report the right metric: CER and WER from edit distance (and say which variant), detection P/R/H-mean at IoU 0.5 with "do not care" filtering, task metrics such as ANLS; check benchmark subsets and label noise before comparing numbers.

Exercises

1. **Gap and font size.** In the detection snippet, scale every font size in `WORDS` by 0.6 and then by 1.6, and rerun the sweep over the dilation gap. How does the best gap move, and what does that tell you about one fixed post-processing rule for an image that contains both sizes?

   Hint

   The best gap scales with letter spacing, hence with font size, so an image with both sizes has no single good gap. Learned detectors avoid the problem by predicting word regions, or shrunk kernels, directly.

2. **Upright boxes by hand.** Show that the IoU between a $w \times h$ rectangle rotated by θ and its tightest upright box is $wh/[(w\cos\theta + h\sin\theta)(w\sin\theta + h\cos\theta)]$. For $w = 160$ and $h = 24$, find the angle at which it drops to 0.5, and compare with a square word ($w = h$).

   Hint

   The rotated rectangle lies inside its upright box, so the intersection is wh and the union is the box area. Solving numerically for $w/h = 160/24$ gives about 8.5°. For a square, the minimum at 45° is $1/2$ exactly, so a square never falls below 0.5: the problem is specific to elongated shapes such as text lines.

3. **Greedy versus exact CTC.** Change the last two rows of the CTC matrix so that the blank and "t" have probabilities 0.65 and 0.30 in both frames. Is "cat" still the most probable text? Then let the "t" probability p_t in frames 4 and 5 run from 0.05 to 0.45, with the blank taking $0.95 - p_t$. For which values do greedy and exact decoding agree?

   Hint

   Reuse the brute-force loop. At $p_t = 0.30$, "cat" (about 0.294) still beats "ca" (about 0.249). Greedy decoding returns "ca" until p_t exceeds the blank (above 0.475), but the exact answer switches to "cat" already at $p_t \approx 0.27$; between those values the two disagree. Below the crossover they agree on "ca".

4. **CER that hides a disaster.** Construct a reference of 40 characters and two recognitions with the same CER of 0.05 such that one is harmless for an invoice-processing system and the other is fatal. Propose a field-level metric that separates them.

Hint

Two errors in a long description word ("Paper A4 recyled") versus two errors in the total ("42.00" read as "4Z.0O"). A field-level exact match on the key fields (invoice number, date, total), or the F1 over extracted fields that Donut reports, distinguishes them; CER and WER weight every character or word equally.

5. **Reading order breaks again.** Add a heading "PAYMENT TERMS" that spans both columns above the two-column notice of the reading-order snippet. What does `column_order` produce? Modify it so that full-width lines are read first, and describe a layout that still defeats your rule.

Hint

The heading crosses the split and is assigned to the left column by its left edge, which happens to work at the top but not for a full-width line in the middle. Treat boxes that overlap the split as a separate band and read band by band. A table, or a figure inside one column, still breaks rules based on empty strips.

Supplementary notes

These notes are optional: each takes one point from the chapter and works through it in more detail. Nothing in the main text, the code or the exercises depends on them.

CTC prefix beam search, step by step

This note explains how a CTC decoder finds a likely *text* rather than a likely *path*, by searching over prefixes instead of alignments.

What is searched. A **prefix** ℓ is a partial transcription, such as "ca". After reading frames $1, \dots, t$, many alignments collapse to the same prefix. Prefix beam search keeps, for each prefix, two numbers:

- $p_b(\ell, t)$, the total probability of alignments of frames $1..t$ that collapse to ℓ and **end in a blank**;
- $p_{nb}(\ell, t)$, the total probability of those that **end in the last label of** ℓ.

Their sum is the probability that the first t frames read as ℓ. The split is needed because the next symbol behaves differently in the two cases.

Update rules. For each kept prefix ℓ and each symbol s at frame $t+1$, with $q = q_{t+1}(s)$ the network's probability of s:

1. **Blank** ($s = \emptyset$): the prefix does not change, and the alignment now ends in a blank: add $(p_b + p_{nb})\, q$ to $p_b(\ell, t+1)$.
2. **Repeat of the last label** (s equals the last label of ℓ): if the previous symbol was that label, the repeat merges and the prefix stays ℓ: add $p_{nb}\, q$ to $p_{nb}(\ell, t+1)$. If the previous symbol was a blank, the repeat starts a new copy: add $p_b\, q$ to $p_{nb}(\ell s, t+1)$. This is how "aa" can be written as "a-a".
3. **Any other label**: extend the prefix: add $(p_b + p_{nb})\, q$ to $p_{nb}(\ell s, t+1)$.

After each frame, keep only the B prefixes with the largest $p_b + p_{nb}$ (the **beam width**). At the end, return the best prefix. With an unlimited beam the result is exact for the CTC model; with $B = 1$ it behaves almost like greedy decoding.

```
import numpy as np

alphabet = ["-", "c", "a", "t"]
probs = np.array([[0.10, 0.80, 0.05, 0.05],
                  [0.20, 0.05, 0.70, 0.05],
                  [0.60, 0.05, 0.30, 0.05],
                  [0.55, 0.02, 0.03, 0.40],
                  [0.55, 0.02, 0.03, 0.40]])

def prefix_beam_search(probs, beam=3):
    # Each prefix keeps two numbers: p_b (all its alignments end in blank)
    # and p_nb (they end in the prefix's last label).
    beams = {(): (1.0, 0.0)}
    for frame in probs:
        nxt = {}
        def add(prefix, pb, pnb):
            ob, onb = nxt.get(prefix, (0.0, 0.0))
            nxt[prefix] = (ob + pb, onb + pnb)
        for prefix, (pb, pnb) in beams.items():
            add(prefix, (pb + pnb) * frame[0], 0.0)              # emit blank
            for s in range(1, len(frame)):
                p = frame[s]
                if prefix and prefix[-1] == s:
                    add(prefix, 0.0, pnb * p)                    # repeat merges
                    add(prefix + (s,), 0.0, pb * p)              # blank separated
                else:
                    add(prefix + (s,), 0.0, (pb + pnb) * p)      # new label
        beams = dict(sorted(nxt.items(), key=lambda kv: -sum(kv[1]))[:beam])
    return [("".join(alphabet[s] for s in k), sum(v)) for k, v in beams.items()]

for B in [1, 3, 20]:
    top = prefix_beam_search(probs, beam=B)[:2]
    print(f"beam={B:2d}: " + ", ".join(f"{t!r} p={p:.3f}" for t, p in top))
```

```
beam= 1: 'ca' p=0.155
beam= 3: 'cat' p=0.344, 'ca' p=0.179
beam=20: 'cat' p=0.363, 'ca' p=0.179
```

On the chapter's toy, a beam of 1 returns "ca", like greedy decoding. A beam of 3 already finds "cat", with a slightly low probability (0.344) because some of its alignments passed through prefixes that were pruned. A beam of 20 keeps every prefix here and reproduces the exact value 0.363 from the brute-force sum.

Pitfalls. Work in log space for long sequences, since the products underflow. Do not deduplicate by path, only by prefix. When a lexicon or language model is added, its score is usually applied when a prefix is extended by a label, weighted by a tuned factor, and often combined with a bonus per emitted character so that the search does not favor short outputs; the weights are tuned on held-out data, and the same over-correction risk described in the main text applies.

Shrinking and expanding text polygons

This note works through the label geometry of segmentation-based text detectors: how much a word is shrunk for training, and how a detection is grown back.

Shrinking. For a ground-truth polygon with area A and perimeter L, PSENet and DBNet move every edge inward by $D = A(1 - r^2)/L$ with a shrink ratio r (DBNet uses $r = 0.4$) [14], [13]. For a polygon this offset is applied with a polygon-clipping algorithm (DBNet uses Vatti clipping). For an axis-aligned $w \times h$ rectangle the formula becomes

$$D = \frac{wh\,(1 - r^2)}{2(w + h)}, \qquad w_s = w - 2D, \qquad h_s = h - 2D,$$

where $w_s \times h_s$ is the shrunk kernel. Since $w/(w+h) < 1$, we have $2D < h(1-r^2) < h$, so the kernel never vanishes, however thin the word: the offset adapts to the shape through A/L. For a long word, $A/L \approx h/2$ and the kernel height is about hr^2; for $r = 0.4$, about 16% of the word's height.

Expanding. At inference DBNet expands each detected region by $D' = A'r'/L'$ with $r' = 1.5$, where A' and L' are the region's area and perimeter [14]. If the detected region were exactly the shrunk kernel of a rectangle, the result would be:

```
def shrink_offset(w, h, r=0.4):
    """DB/PSENet-style inward offset D = A (1 - r^2) / L for a w x h rectangle."""
    return w * h * (1 - r**2) / (2 * (w + h))

def unclip_offset(w, h, r_prime=1.5):
    """Outward offset D' = A' r' / L' applied to a detected (shrunk) rectangle."""
    return w * h * r_prime / (2 * (w + h))
```

```
for w, h in [(100, 20), (300, 20), (40, 40)]:
    D = shrink_offset(w, h)
    ws, hs = w - 2 * D, h - 2 * D                      # the training target ("kernel")
    Dp = unclip_offset(ws, hs)
    print(f"word {w:3d}x{h:2d}: D={D:4.1f} -> kernel {ws:5.1f}x{hs:4.1f}; "
          f"D'={Dp:4.1f} -> box {ws + 2 * Dp:5.1f}x{hs + 2 * Dp:4.1f}")
```

```
word 100x20: D= 7.0 -> kernel  86.0x 6.0; D'= 4.2 -> box  94.4x14.4
word 300x20: D= 7.9 -> kernel 284.2x 4.2; D'= 3.1 -> box 290.5x10.5
word  40x40: D= 8.4 -> kernel  23.2x23.2; D'= 8.7 -> box  40.6x40.6
```

In this idealized calculation, $r' = 1.5$ restores a square word almost exactly but leaves thin words too short in height: the expansion is not the inverse of the shrink, because both depend on the shape through A/L and the kernel's A'/L' shrinks faster for thin shapes. In a real network the binarized region at threshold 0.2 is not exactly the kernel (the probability map has soft edges), and r' is a constant tuned on data [14]. Two practical consequences: check the expansion ratio on your own data's aspect ratios, and expect height errors on long text lines to lower IoU even when the word is found.

Edge cases. Very small words have kernels of a pixel or two that a downsampled output map may not resolve, so detectors often ignore or down-weight ground-truth words below a minimum size. Self-intersecting or badly ordered annotation polygons make A meaningless and must be cleaned before label generation. The threshold-map target uses the band between the shrunk and the dilated polygon (both at offset D), with values derived from the distance to the original border [14].

Where the language model sits in a recognizer

This note compares the ways linguistic knowledge enters a text recognizer and how to test whether it is helping.

Four placements.

1. **None (pure CTC on visual features).** Each frame's output depends only on the image features, through the encoder's receptive field. Errors are visual (look-alike glyphs) and the output can be any string, which is right for codes and serial numbers.
2. **Lexicon constraint.** The output must be a word from a list $\mathcal{V}$. CRNN computes $p(y \mid x)$ only for words within edit distance δ of the greedy output, found quickly with a BK-tree (a metric tree for discrete distances), and returns the most probable one [16]. Strong when the vocabulary is known (a product catalog, a street-name list); wrong for anything outside it.
3. **Implicit language model.** An autoregressive attention or transformer decoder predicts y_i from the image and $y_{<i}$, so spelling statistics of the training text are learned inside the decoder.

Baek et al. attribute attention's ability to recover occluded letters to this implicit character-level modeling [7]. The statistics come from the training labels; models trained on synthetic data rendered from a word list learn that list's spelling.

4. **Explicit language model.** A separate model scores or corrects the visual output. ABINet blocks the gradient from the language model to the vision model so that the language model is trained as a real language model, uses a bidirectional cloze network, and iterates the correction to reduce the effect of noisy input [21]. PARSeq instead learns an ensemble of internal autoregressive models with shared weights by training on permutations of the output order, and argues that an external model that never sees the image may "rectify" correct predictions [20].

How to test for over-correction. Build a held-out set of non-semantic strings — random letters, digits, mixed case — rendered like your real data, and compare CER with and without the language component; OCRBench's non-semantic subset does the same for multimodal models [22]. If CER on random strings is much worse than on dictionary words with the same glyph statistics, the model is reading partly from its prior. For fields such as amounts and identifiers, use a format grammar (digits, separators, check digits) instead of a natural-language prior.

Reading vertical and mixed-direction text

This note collects the layout facts that matter when an OCR system meets Chinese, Japanese or Korean text set vertically.

The layout rules. In vertical composition, characters are arranged from top to bottom and lines from right to left; columns of text blocks are arranged from top to bottom [41]. Embedded Latin text has three conventions: single letters and acronyms are set upright, one per character cell, in the same direction as the Han characters; general words, sentences and long numbers are set in proportional fonts rotated 90° clockwise; and two- or three-digit numbers are often set horizontally inside the vertical line ("horizontal-in-vertical") [41]. Quotation marks are usually replaced by corner brackets in vertical text [41].

Consequences for OCR.

- **Orientation is a per-line decision.** A detector must output the reading direction of each line, not only its shape. A tall, narrow box could be a vertical line of Chinese or a rotated horizontal line; only the content tells. One practical design adds an orientation classifier (0°, 90°, 180°, 270°, or horizontal/vertical) between detection and recognition.
- **The crop must match what the recognizer was trained on.** A recognizer trained on horizontal lines can read a vertical line only after the line is transformed so that reading order runs left to right in the frames. For upright CJK characters this means stacking the character cells side by side (each character keeps its orientation); simply rotating the whole crop by 90° turns every character on its side. Rotated Latin words inside the line need the opposite treatment.

- **Mixed lines need segment-level direction.** A vertical line containing a rotated English word has two directions inside one line, which a single CTC or attention pass over one frame order cannot read correctly.
- **Evaluation must follow reading order.** CER compares strings, so a correct set of characters in the wrong order is scored as many errors; ground truth must be stored in the intended reading order, which for vertical layouts is not the order of increasing x.

Baek et al. found vertical text among the failure cases shared by all the recognizers they compared, because the models assume horizontal input by construction [7]; street-view data such as the CTW dataset contain it [40].

Evaluation protocols in detail

This note fills in the details of text-detection and recognition metrics that change reported numbers.

"Do not care" regions. Ground truth marks illegible text and other text outside the task as "do not care". In the protocol described for MLT-2019, a detection t is discarded if its intersection with a "do not care" region d covers more than half of d, that is if $|t \cap d|/|d| > 0.5$, where $|\cdot|$ is area; remaining detections are matched one-to-one to the remaining ground truth with IoU > 0.5 [42]. Total-Text likewise asks that detections of "do not care" areas be filtered out before scoring [10]. A method that detects illegible text is therefore neither rewarded nor punished, which is the intent; but a system that merges a legible word with a neighboring "do not care" region can lose the legible word.

One-to-many matching (ICDAR 2013). The ICDAR 2011/2013 protocol follows the overlap-based method of Wolf and Jolion [51]: a match requires both the fraction of the ground truth covered and the fraction of the detection covered to exceed thresholds, and one-to-many or many-to-one matches (a line detected as several words, or several words as one line) are accepted with a reduced score, usually 0.8, instead of failing [1]. Numbers from this protocol and from the IoU protocol are not comparable.

Single-threshold H-mean. H-mean is computed at the confidence threshold chosen by the method's authors, usually the one that maximizes it on the test set itself. This is optimistic and hides the precision–recall trade-off; AP, as used by RCTW-17 [39], integrates over all thresholds [1].

End-to-end vocabularies. End-to-end and word-spotting results are reported with a strongly contextualised list (100 words per image, including all words in the image plus distractors), a weakly contextualised list (all words in the test set), or a generic list of about 90,000 words [2]. With the 100-word list, recognition reduces to choosing among a few candidates; numbers with different lists cannot be compared.

Normalized and average edit distance. NED divides each line's edit distance by the longer of the two strings, so every line contributes at most 1 regardless of length; $1 - \text{NED}$ is reported as a score [2]. RCTW-17's end-to-end task also used the **average edit distance (AED)**: each detection is matched to the ground-truth polygon of maximum IoU (or to nothing if no IoU exceeds 0.5), the edit distances of matched pairs are summed — an unmatched detection or ground truth counts as an edit distance against the empty string — and the sum is divided by the number of test images [39]. AED punishes missed and spurious text by its length, which NED alone does not.

Document metrics. DocVQA's primary metric, ANLS, is an edit-distance-based similarity between predicted and ground-truth answers, averaged over questions; it was first proposed for scene-text question answering, and the DocVQA authors adopted it because exact-match accuracy gives zero credit to an answer with one OCR slip [44]. Read the metric's definition on the benchmark's evaluation page before comparing with other similarity scores. Donut's TED-based accuracy for structured outputs is $\max(0,\ 1 - \text{TED}(\text{pr}, \text{gt})/\text{TED}(\phi, \text{gt}))$, where TED is the tree edit distance between predicted and ground-truth trees and ϕ is the empty tree; unlike field-level F1, it gives partial credit and measures the nesting of groups [34].

Normalization. Many recognition benchmarks evaluate only case-insensitive letters and digits; TrOCR, for example, filters its scene-text outputs to a 36-character set of lowercase letters and digits, while it scores IAM handwriting with case-sensitive CER [19]. Always state case sensitivity, punctuation handling and whitespace rules alongside the number.

Paper guides

These guides are optional: each walks through one paper the chapter builds on — the problem it poses, the idea, what it showed and what to check when you read it.

Guide to the Modern view surveys

This guide expands the short survey summary in the Modern view: for each review, what it covers, how it organizes the field and what it concludes.

Long, He and Yao, "Scene Text Detection and Recognition: The Deep Learning Era" (IJCV 2021) [1] organizes detection into early deep methods, methods inspired by generic object detection, and methods based on sub-text components (characters, segments, kernels); recognition into CTC-based and encoder–decoder methods; then end-to-end systems and auxiliary techniques such as synthetic data and weak or semi-supervision. It also reviews the benchmarks and the evaluation protocols (IoU-based and DetEval-style matching, end-to-end vocabularies). Its conclusions: deep learning made oriented and curved text tractable, but most methods and datasets focus on English, languages with alphabets of tens of thousands of symbols remain hard, robustness to imperfect detection and generalization across datasets are open, and single-threshold

F-scores should be complemented with AP.

Chen, Jin, Zhu, Luo and Wang, "Text Recognition in the Wild: A Survey" (ACM Computing Surveys 2021) [2] concentrates on recognition. It separates segmentation-based methods (preprocessing, character segmentation, character recognition) from segmentation-free ones, and describes the latter by stage: image preprocessing, feature representation, sequence modeling and prediction (CTC or attention). Its most practical parts are the catalogue of synthetic and real datasets and the evaluation protocols for Latin text (word accuracy, end-to-end recognition and word spotting with three vocabularies) and for multilingual text (normalized edit distance).

Cui, Xu, Lv and Wei, "Document AI: Benchmarks, Models and Applications" (arXiv 2021) [3] is a short review of document tasks — layout analysis, visual information extraction, document question answering and document image classification — with their benchmarks, tracing the methods from rule-based heuristics through statistical learning to deep, pre-trained multimodal models.

Zhang et al., "Document Parsing Unveiled" (arXiv 2024) [4] reviews document parsing, the conversion of pages into structured, machine-readable form. Its taxonomy has two branches: modular pipelines (layout analysis followed by recognizers for text, tables, mathematical expressions and visual elements) and unified models driven by vision-language models. It summarizes metrics and benchmarks and names robustness to complex layouts, the reliability of VLM-based parsing and inference efficiency as the open challenges.

Liu et al., "OCRBench" (Science China Information Sciences 2024) [22] is an evaluation rather than a survey. It tests large multimodal models on text recognition, scene-text and document question answering, key information extraction and handwritten mathematical expressions across 29 datasets, and finds them weakest on multilingual, handwritten, non-semantic and mathematical text.

Baek et al., "What Is Wrong With Scene Text Recognition Model Comparisons?" (ICCV 2019)

The problem. By 2019 many scene-text recognizers each claimed state-of-the-art accuracy, but they had been trained on different synthetic datasets and tested on different versions of the same benchmarks, so it was impossible to tell which ideas helped [7].

The idea. Fix the data, then decompose the models. The paper fixes training to the union of MJSynth (8.9 M images) and SynthText (5.5 M) and evaluates on the union of seven benchmarks, 8,539 images in total. It then describes recognizers as four stages — transformation (none or TPS), feature extraction (VGG, RCNN or ResNet), sequence modeling (none or BiLSTM) and prediction (CTC or attention) — and trains all $2 \times 3 \times 2 \times 2 = 24$ combinations, each with five random seeds, measuring accuracy, time per image and number of parameters.

What it showed.

- **Dataset inconsistency was real.** Two versions of IC03 (860 and 867 images) and of IC15 (1,811 and 2,077) were in use, and on IC03 the seven missing examples alone can move accuracy by 0.8 points. Training on MJSynth only, SynthText only or both gave clearly different accuracies for the same model.
- **Modules have different costs.** Attention mainly costs time; the feature extractor mainly costs memory. Accuracy gains from each upgrade were about twice as large on irregular benchmarks (curved, rotated, perspective) as on regular ones.
- **Some failures are shared by all 24 models**: 644 of 8,539 images (7.5%). The paper groups them into calligraphic fonts, vertical text, special characters, heavy occlusion, low resolution and label noise.

How to read the tables and plots. The main figure plots accuracy against time and against parameters for all combinations, with the frontier of best trade-offs; read along a frontier to see which module upgrade buys the most accuracy per millisecond or per parameter. The module table reports *marginal* accuracy (averaged over all combinations containing a module), which shows a module's typical effect but not its best case. Check which benchmark subsets every reported number uses.

Limits. All models are trained on synthetic data only and evaluated on English letters and digits; the framework predates transformer recognizers and large pre-training.

Why it matters here. It is the source of the four-stage view of recognition in the main text, of the "diversity over size" lesson for synthetic data, and of the warning to check subsets and label noise before comparing numbers.

DBNet: a reading guide

The problem. Segmentation-based detectors handle curved text, but turning a probability map into separate word instances needed heuristic, slow post-processing (PSENet's progressive expansion, for example), and the fixed binarization threshold was outside training [14].

The idea. Put the binarization inside the network. Predict a probability map P and a threshold map T, combine them with the approximate step $\hat{B} = 1/(1 + e^{-k(P-T)})$, $k = 50$, and supervise P and $\hat{B}$ with binary cross-entropy on shrunk text kernels and T with an L1 loss in a band around each border, weighting the three losses $1 : 1 : 10$. Hard negative mining keeps positives and negatives at a 1:3 ratio in the cross-entropy terms. Deformable convolutions in the backbone give a flexible receptive field for very long or very short words. At inference, binarize P at 0.2, take connected components and expand them.

What it showed. Ablations on MSRA-TD500 and CTW1500 compare a plain segmentation baseline with and without DB and deformable convolution, for ResNet-18 and ResNet-50 backbones, and DB improves F-measure for both backbones on both datasets; a second ablation, on MLT-2017, tests whether supervising the threshold map helps. The paper then reports precision, recall, F-measure and frames per second on five benchmarks (Total-Text, CTW1500, ICDAR 2015, MSRA-TD500 and MLT-2017), at several input sizes.

How to read the tables. Each benchmark table lists P, R, F and FPS; the number in brackets is the height of the input image, which changes both accuracy and speed, so compare rows at matching sizes. Speeds are measured with batch size 1 on one GPU and include post-processing, about 30% of the inference time; check whether competing methods' speeds were measured on the same hardware or copied from their papers. Because the threshold branch is dropped at inference, a model trained with DB runs exactly as fast as the same model trained without it. The threshold-map ablation is the cleanest evidence for the "adaptive threshold" claim; the DB-versus-baseline ablation mixes the effect of the learned threshold with that of the extra supervision.

Limits. The learned threshold is not used at inference, so the separation of touching words at test time rests on how sharp P has become. A word whose kernel is lost (very small or very thin text) cannot be recovered by expansion. The expansion ratio r' is a fixed constant (see the Supplementary note on polygons).

Why it matters here. DBNet is the reference design for the segmentation branch of text detection in the main text, and it shows how moving a hand-tuned step into training can simplify inference.

Donut: a reading guide

The problem. Document-understanding systems relied on an external OCR engine whose cost, language coverage and errors limited everything downstream [34].

The idea. A single image-to-sequence transformer (a Swin encoder and a BART decoder) trained in two stages. **Reading**: predict all the text of a page in reading order, using 11 million scanned English documents with pseudo-labels from a commercial OCR service, plus synthetic pages from SynthDoG in Chinese, Japanese, Korean and English. **Understanding**: fine-tune the decoder to emit token sequences that map one-to-one to JSON — a class token for classification, nested fields for information extraction, an answer for question answering.

What it showed. Experiments cover document classification (RVL-CDIP), information extraction (the public CORD receipt and Chinese train-ticket datasets, plus in-service Japanese business-card and Korean receipt datasets) and DocVQA, comparing against OCR-based models in accuracy and in speed including OCR time. Analyses vary the backbone, the pre-training data and the input resolution, visualize the decoder's cross-attention as a rough text locator, and show that

OCR-based baselines depend strongly on which OCR engine they use.

How to read the tables. Two metrics are used for information extraction: field-level F1 (exact field match) and TED-based accuracy (tree edit distance, with partial credit). They can rank methods differently, so read both. Speed columns for OCR-based methods include the OCR engine; check which engine, because the paper shows that this choice changes both speed and accuracy. For DocVQA, the resolution study is the key result: accuracy grows quickly with input size because documents contain tiny text, and the larger input makes the model slower.

Limits. The paper's own example shows Donut missing tiny text in a large image, and generation offers no character-level grounding, so errors cannot be traced to pixels as with an OCR box. The pre-training labels on real documents come from an OCR engine, so OCR is not absent from the pipeline, only from inference.

Why it matters here. Donut is the clearest early statement of the OCR-free design that today's VLM-based document readers follow, and of its two costs: resolution and verifiability.

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